inorder_cache:

```
---------- Begin Simulation Statistics ----------
simSeconds                    0.000030          # Number of seconds simulated (Second)
simTicks                      29955015          # Number of ticks simulated (Tick)
finalTick                     29955015          # Number of ticks from beginning of simulation (restored from checkpoints and never reset) (Tick)
simFreq                       1000000000000     # The number of ticks per simulated second ((Tick/Second))
hostSeconds                   0.03              # Real time elapsed on the host (Second)
hostTickRate                  963896612         # The number of ticks simulated per host second (ticks/s) ((Tick/Second))
hostMemory                    1202992           # Number of bytes of host memory used (Byte)
simInsts                      6555              # Number of instructions simulated (Count)
simOps                        12992             # Number of ops (including micro ops) simulated (Count)
hostInstRate                  210211            # Simulator instruction rate (inst/s) ((Count/Second))
hostOpRate                    416584            # Simulator op (including micro ops) rate (op/s) ((Count/Second))
board.cache_hierarchy.dptw_caches.blockedCycles::no_mshrs          0          # number of cycles access was blocked (Cycle)
board.cache_hierarchy.dptw_caches.blockedCycles::no_targets        0          # number of cycles access was blocked (Cycle)
board.cache_hierarchy.dptw_caches.blockedCauses::no_mshrs          0          # number of times access was blocked (Count)
board.cache_hierarchy.dptw_caches.blockedCauses::no_targets        0          # number of times access was blocked (Count)
board.cache_hierarchy.dptw_caches.avgBlocked::no_mshrs             nan        # average number of cycles each access was blocked ((Cycle/Count))
board.cache_hierarchy.dptw_caches.avgBlocked::no_targets           nan        # average number of cycles each access was blocked ((Cycle/Count))
board.cache_hierarchy.dptw_caches.replacements                     0          # number of replacements (Count)
board.cache_hierarchy.dptw_caches.power_state.pwrStateResidencyTicks::UNDEFINED    29955015    # Cumulative time (in ticks) in various power states (Tick)
board.cache_hierarchy.dptw_caches.tags.tagsInUse                   0          # Average ticks per tags in use ((Tick/Count))
board.cache_hierarchy.dptw_caches.tags.totalRefs                   0          # Total number of references to valid blocks. (Count)
board.cache_hierarchy.dptw_caches.tags.sampledRefs                 0          # Sample count of references to valid blocks. (Count)
board.cache_hierarchy.dptw_caches.tags.avgRefs                     nan        # Average number of references to valid blocks. ((Count/Count))
board.cache_hierarchy.dptw_caches.tags.warmupTick                  0          # The tick when the warmup percentage was hit. (Tick)
board.cache_hierarchy.dptw_caches.tags.tagAccesses                 0          # Number of tag accesses (Count)
board.cache_hierarchy.dptw_caches.tags.dataAccesses                0          # Number of data accesses (Count)
board.cache_hierarchy.dptw_caches.tags.power_state.pwrStateResidencyTicks::UNDEFINED    29955015    # Cumulative time (in ticks) in various power states (Tick)
board.cache_hierarchy.iptw_caches.blockedCycles::no_mshrs          0          # number of cycles access was blocked (Cycle)
board.cache_hierarchy.iptw_caches.blockedCycles::no_targets        0          # number of cycles access was blocked (Cycle)
board.cache_hierarchy.iptw_caches.blockedCauses::no_mshrs          0          # number of times access was blocked (Count)
board.cache_hierarchy.iptw_caches.blockedCauses::no_targets        0          # number of times access was blocked (Count)
board.cache_hierarchy.iptw_caches.avgBlocked::no_mshrs             nan        # average number of cycles each access was blocked ((Cycle/Count))
board.cache_hierarchy.iptw_caches.avgBlocked::no_targets           nan        # average number of cycles each access was blocked ((Cycle/Count))
board.cache_hierarchy.iptw_caches.replacements                     0          # number of replacements (Count)
board.cache_hierarchy.iptw_caches.power_state.pwrStateResidencyTicks::UNDEFINED    29955015    # Cumulative time (in ticks) in various power states (Tick)
board.cache_hierarchy.iptw_caches.tags.tagsInUse                   0          # Average ticks per tags in use ((Tick/Count))
board.cache_hierarchy.iptw_caches.tags.totalRefs                   0          # Total number of references to valid blocks. (Count)
board.cache_hierarchy.iptw_caches.tags.sampledRefs                 0          # Sample count of references to valid blocks. (Count)
board.cache_hierarchy.iptw_caches.tags.avgRefs                     nan        # Average number of references to valid blocks. ((Count/Count))
board.cache_hierarchy.iptw_caches.tags.warmupTick                  0          # The tick when the warmup percentage was hit. (Tick)
board.cache_hierarchy.iptw_caches.tags.tagAccesses                 0          # Number of tag accesses (Count)
board.cache_hierarchy.iptw_caches.tags.dataAccesses                0          # Number of data accesses (Count)
board.cache_hierarchy.iptw_caches.tags.power_state.pwrStateResidencyTicks::UNDEFINED    29955015    # Cumulative time (in ticks) in various power states (Tick)
board.cache_hierarchy.l1dcaches.demandHits::processor.cores.core.data    2489    # number of demand (read+write) hits (Count)
```

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board.cache_hierarchy.l1dcaches.demandHits::total 2489 # number of demand (read+write) hits (Count)
board.cache_hierarchy.l1dcaches.overallHits::processor.cores.core.data 2489 # number of overall hits (Count)
board.cache_hierarchy.l1dcaches.overallHits::total 2489 # number of overall hits (Count)
board.cache_hierarchy.l1dcaches.demandMisses::processor.cores.core.data 138 # number of demand (read+write) misses (Count)
board.cache_hierarchy.l1dcaches.demandMisses::total 138 # number of demand (read+write) misses (Count)
board.cache_hierarchy.l1dcaches.overallMisses::processor.cores.core.data 138 # number of overall misses (Count)
board.cache_hierarchy.l1dcaches.overallMisses::total 138 # number of overall misses (Count)
board.cache_hierarchy.l1dcaches.demandMissLatency::processor.cores.core.data 7332327 # number of demand (read+write) miss ticks (Tick)
board.cache_hierarchy.l1dcaches.demandMissLatency::total 7332327 # number of demand (read+write) miss ticks (Tick)
board.cache_hierarchy.l1dcaches.overallMissLatency::processor.cores.core.data 7332327 # number of overall miss ticks (Tick)
board.cache_hierarchy.l1dcaches.overallMissLatency::total 7332327 # number of overall miss ticks (Tick)
board.cache_hierarchy.l1dcaches.demandAccesses::processor.cores.core.data 2627 # number of demand (read+write) accesses (Count)
board.cache_hierarchy.l1dcaches.demandAccesses::total 2627 # number of demand (read+write) accesses (Count)
board.cache_hierarchy.l1dcaches.overallAccesses::processor.cores.core.data 2627 # number of overall (read+write) accesses (Count)
board.cache_hierarchy.l1dcaches.overallAccesses::total 2627 # number of overall (read+write) accesses (Count)
board.cache_hierarchy.l1dcaches.demandMissRate::processor.cores.core.data 0.052531 # miss rate for demand accesses (Ratio)
board.cache_hierarchy.l1dcaches.demandMissRate::total 0.052531 # miss rate for demand accesses (Ratio)
board.cache_hierarchy.l1dcaches.overallMissRate::processor.cores.core.data 0.052531 # miss rate for overall accesses (Ratio)
board.cache_hierarchy.l1dcaches.overallMissRate::total 0.052531 # miss rate for overall accesses (Ratio)
board.cache_hierarchy.l1dcaches.demandAvgMissLatency::processor.cores.core.data 53132.804348 # average overall miss latency in ticks ((Tick/Count))
board.cache_hierarchy.l1dcaches.demandAvgMissLatency::total 53132.804348 # average overall miss latency in ticks ((Tick/Count))
board.cache_hierarchy.l1dcaches.overallAvgMissLatency::processor.cores.core.data 53132.804348 # average overall miss latency ((Tick/Count))
board.cache_hierarchy.l1dcaches.overallAvgMissLatency::total 53132.804348 # average overall miss latency ((Tick/Count))
board.cache_hierarchy.l1dcaches.blockedCycles::no_mshrs 0 # number of cycles access was blocked (Cycle)
board.cache_hierarchy.l1dcaches.blockedCycles::no_targets 0 # number of cycles access was blocked (Cycle)
board.cache_hierarchy.l1dcaches.blockedCauses::no_mshrs 0 # number of times access was blocked (Count)
board.cache_hierarchy.l1dcaches.blockedCauses::no_targets 0 # number of times access was blocked (Count)
board.cache_hierarchy.l1dcaches.avgBlocked::no_mshrs nan # average number of cycles each access was blocked ((Cycle/Count))
board.cache_hierarchy.l1dcaches.avgBlocked::no_targets nan # average number of cycles each access was blocked ((Cycle/Count))
board.cache_hierarchy.l1dcaches.demandMshrMisses::processor.cores.core.data 138 # number of demand (read+write) MSHR misses (Count)
board.cache_hierarchy.l1dcaches.demandMshrMisses::total 138 # number of demand (read+write) MSHR misses (Count)
board.cache_hierarchy.l1dcaches.overallMshrMisses::cache_hierarchy.l1dcaches.prefetcher 47 # number of overall MSHR misses (Count)
board.cache_hierarchy.l1dcaches.overallMshrMisses::processor.cores.core.data 138 # number of overall MSHR misses (Count)
board.cache_hierarchy.l1dcaches.overallMshrMisses::total 185 # number of overall MSHR misses (Count)
board.cache_hierarchy.l1dcaches.demandMshrMissLatency::processor.cores.core.data 7286373 # number of demand (read+write) MSHR miss ticks (Tick)
board.cache_hierarchy.l1dcaches.demandMshrMissLatency::total 7286373 # number of demand (read+write) MSHR miss ticks (Tick)
board.cache_hierarchy.l1dcaches.overallMshrMissLatency::cache_hierarchy.l1dcaches.prefetcher 1101786 # number of overall MSHR miss ticks (Tick)
board.cache_hierarchy.l1dcaches.overallMshrMissLatency::processor.cores.core.data 7286373 # number of overall MSHR miss ticks (Tick)
board.cache_hierarchy.l1dcaches.overallMshrMissLatency::total 8388159 # number of overall MSHR miss ticks (Tick)
board.cache_hierarchy.l1dcaches.demandMshrMissRate::processor.cores.core.data 0.052531 # mshr miss ratio for demand accesses (Ratio)
board.cache_hierarchy.l1dcaches.demandMshrMissRate::total 0.052531 # mshr miss ratio for demand accesses (Ratio)
board.cache_hierarchy.l1dcaches.overallMshrMissRate::cache_hierarchy.l1dcaches.prefetcher inf # mshr miss ratio for overall accesses (Ratio)
board.cache_hierarchy.l1dcaches.overallMshrMissRate::processor.cores.core.data 0.052531 # mshr miss ratio for overall accesses (Ratio)
board.cache_hierarchy.l1dcaches.overallMshrMissRate::total 0.070423 # mshr miss ratio for overall accesses (Ratio)
board.cache_hierarchy.l1dcaches.demandAvgMshrMissLatency::processor.cores.core.data 52799.804348 # average overall mshr miss latency ((Tick/Count))
board.cache_hierarchy.l1dcaches.demandAvgMshrMissLatency::total 52799.804348 # average overall mshr miss latency ((Tick/Count))
board.cache_hierarchy.l1dcaches.overallAvgMshrMissLatency::cache_hierarchy.l1dcaches.prefetcher 23442.255319 # average overall mshr miss latency ((Tick/Count))
board.cache_hierarchy.l1dcaches.overallAvgMshrMissLatency::processor.cores.core.data 52799.804348 # average overall mshr miss latency ((Tick/Count))
board.cache_hierarchy.l1dcaches.overallAvgMshrMissLatency::total 45341.400000 # average overall mshr miss latency ((Tick/Count))
```

board.cache_hierarchy.l1dcaches.replacements 0 # number of replacements (Count)

board.cache_hierarchy.l1dcaches.HardPFReq.mshrMisses::cache_hierarchy.l1dcaches.prefetcher 47 # number of HardPFReq MSHR misses (Count)

board.cache_hierarchy.l1dcaches.HardPFReq.mshrMisses::total 47 # number of HardPFReq MSHR misses (Count)

board.cache_hierarchy.l1dcaches.HardPFReq.mshrMissLatency::cache_hierarchy.l1dcaches.prefetcher 1101786 # number of HardPFReq MSHR miss ticks (Tick)

board.cache_hierarchy.l1dcaches.HardPFReq.mshrMissLatency::total 1101786 # number of HardPFReq MSHR miss ticks (Tick)

board.cache_hierarchy.l1dcaches.HardPFReq.mshrMissRate::cache_hierarchy.l1dcaches.prefetcher inf # mshr miss rate for HardPFReq accesses (Ratio)

board.cache_hierarchy.l1dcaches.HardPFReq.mshrMissRate::total inf # mshr miss rate for HardPFReq accesses (Ratio)

board.cache_hierarchy.l1dcaches.HardPFReq.avgMshrMissLatency::cache_hierarchy.l1dcaches.prefetcher 23442.255319 # average HardPFReq mshr miss latency ((Tick/Count))

board.cache_hierarchy.l1dcaches.HardPFReq.avgMshrMissLatency::total 23442.255319 # average HardPFReq mshr miss latency ((Tick/Count))

board.cache_hierarchy.l1dcaches.ReadReq.hits::processor.cores.core.data 1197 # number of ReadReq hits (Count)

board.cache_hierarchy.l1dcaches.ReadReq.hits::total 1197 # number of ReadReq hits (Count)

board.cache_hierarchy.l1dcaches.ReadReq.misses::processor.cores.core.data 79 # number of ReadReq misses (Count)

board.cache_hierarchy.l1dcaches.ReadReq.misses::total 79 # number of ReadReq misses (Count)

board.cache_hierarchy.l1dcaches.ReadReq.missLatency::processor.cores.core.data 4402593 # number of ReadReq miss ticks (Tick)

board.cache_hierarchy.l1dcaches.ReadReq.missLatency::total 4402593 # number of ReadReq miss ticks (Tick)

board.cache_hierarchy.l1dcaches.ReadReq.accesses::processor.cores.core.data 1276 # number of ReadReq accesses(hits+misses) (Count)

board.cache_hierarchy.l1dcaches.ReadReq.accesses::total 1276 # number of ReadReq accesses(hits+misses) (Count)

board.cache_hierarchy.l1dcaches.ReadReq.missRate::processor.cores.core.data 0.061912 # miss rate for ReadReq accesses (Ratio)

board.cache_hierarchy.l1dcaches.ReadReq.missRate::total 0.061912 # miss rate for ReadReq accesses (Ratio)

board.cache_hierarchy.l1dcaches.ReadReq.avgMissLatency::processor.cores.core.data 55729.025316 # average ReadReq miss latency ((Tick/Count))

board.cache_hierarchy.l1dcaches.ReadReq.avgMissLatency::total 55729.025316 # average ReadReq miss latency ((Tick/Count))

board.cache_hierarchy.l1dcaches.ReadReq.mshrMisses::processor.cores.core.data 79 # number of ReadReq MSHR misses (Count)

board.cache_hierarchy.l1dcaches.ReadReq.mshrMisses::total 79 # number of ReadReq MSHR misses (Count)

board.cache_hierarchy.l1dcaches.ReadReq.mshrMissLatency::processor.cores.core.data 4376286 # number of ReadReq MSHR miss ticks (Tick)

board.cache_hierarchy.l1dcaches.ReadReq.mshrMissLatency::total 4376286 # number of ReadReq MSHR miss ticks (Tick)

board.cache_hierarchy.l1dcaches.ReadReq.mshrMissRate::processor.cores.core.data 0.061912 # mshr miss rate for ReadReq accesses (Ratio)

board.cache_hierarchy.l1dcaches.ReadReq.mshrMissRate::total 0.061912 # mshr miss rate for ReadReq accesses (Ratio)

board.cache_hierarchy.l1dcaches.ReadReq.avgMshrMissLatency::processor.cores.core.data 55396.025316 # average ReadReq mshr miss latency ((Tick/Count))

board.cache_hierarchy.l1dcaches.ReadReq.avgMshrMissLatency::total 55396.025316 # average ReadReq mshr miss latency ((Tick/Count))

board.cache_hierarchy.l1dcaches.WriteReq.hits::processor.cores.core.data 1292 # number of WriteReq hits (Count)

board.cache_hierarchy.l1dcaches.WriteReq.hits::total 1292 # number of WriteReq hits (Count)

board.cache_hierarchy.l1dcaches.WriteReq.misses::processor.cores.core.data 59 # number of WriteReq misses (Count)

board.cache_hierarchy.l1dcaches.WriteReq.misses::total 59 # number of WriteReq misses (Count)

board.cache_hierarchy.l1dcaches.WriteReq.missLatency::processor.cores.core.data 2929734 # number of WriteReq miss ticks (Tick)

board.cache_hierarchy.l1dcaches.WriteReq.missLatency::total 2929734 # number of WriteReq miss ticks (Tick)

board.cache_hierarchy.l1dcaches.WriteReq.accesses::processor.cores.core.data 1351 # number of WriteReq accesses(hits+misses) (Count)

board.cache_hierarchy.l1dcaches.WriteReq.accesses::total 1351 # number of WriteReq accesses(hits+misses) (Count)

board.cache_hierarchy.l1dcaches.WriteReq.missRate::processor.cores.core.data 0.043671 # miss rate for WriteReq accesses (Ratio)

board.cache_hierarchy.l1dcaches.WriteReq.missRate::total 0.043671 # miss rate for WriteReq accesses (Ratio)

board.cache_hierarchy.l1dcaches.WriteReq.avgMissLatency::processor.cores.core.data 49656.508475 # average WriteReq miss latency ((Tick/Count))

board.cache_hierarchy.l1dcaches.WriteReq.avgMissLatency::total 49656.508475 # average WriteReq miss latency ((Tick/Count))

board.cache_hierarchy.l1dcaches.WriteReq.mshrMisses::processor.cores.core.data 59 # number of WriteReq MSHR misses (Count)

board.cache_hierarchy.l1dcaches.WriteReq.mshrMisses::total 59 # number of WriteReq MSHR misses (Count)

board.cache_hierarchy.l1dcaches.WriteReq.mshrMissLatency::processor.cores.core.data 2910087 # number of WriteReq MSHR miss ticks (Tick)

board.cache_hierarchy.l1dcaches.WriteReq.mshrMissLatency::total 2910087 # number of WriteReq MSHR miss ticks (Tick)

board.cache_hierarchy.l1dcaches.WriteReq.mshrMissRate::processor.cores.core.data 0.043671 # mshr miss rate for WriteReq accesses (Ratio)

board.cache_hierarchy.l1dcaches.WriteReq.mshrMissRate::total 0.043671 # mshr miss rate for WriteReq accesses (Ratio)

board.cache_hierarchy.l1dcaches.WriteReq.avgMshrMissLatency::processor.cores.core.data 49323.508475 # average WriteReq mshr miss latency ((Tick/Count))

board.cache_hierarchy.l1dcaches.WriteReq.avgMshrMissLatency::total 49323.508475 # average WriteReq mshr miss latency ((Tick/Count))

board.cache_hierarchy.l1dcaches.power_state.pwrStateResidencyTicks::UNDEFINED 29955015 # Cumulative time (in ticks) in various power states (Tick)

board.cache_hierarchy.l1dcaches.prefetcher.demandMshrMisses 138 # demands not covered by prefetchs (Count)

board.cache_hierarchy.l1dcaches.prefetcher.pfIssued 163 # number of hwpf issued (Count)

board.cache_hierarchy.l1dcaches.prefetcher.pfUseful 46 # number of useful prefetch (Count)

board.cache_hierarchy.l1dcaches.prefetcher.pfUsefulButMiss 0 # number of hit on prefetch but cache block is not in an usable state (Count)

board.cache_hierarchy.l1dcaches.prefetcher.accuracy 0.282209 # accuracy of the prefetcher (Count)

board.cache_hierarchy.l1dcaches.prefetcher.coverage 0.250000 # coverage brought by this prefetcher (Count)

board.cache_hierarchy.l1dcaches.prefetcher.pfHitInCache 96 # number of prefetches hitting in cache (Count)

board.cache_hierarchy.l1dcaches.prefetcher.pfHitInMSHR 20 # number of prefetches hitting in a MSHR (Count)

board.cache_hierarchy.l1dcaches.prefetcher.pfHitInWB 0 # number of prefetches hit in the Write Buffer (Count)

board.cache_hierarchy.l1dcaches.prefetcher.pfLate 116 # number of late prefetches (hitting in cache, MSHR or WB) (Count)

board.cache_hierarchy.l1dcaches.prefetcher.pfIdentified 163 # number of prefetch candidates identified (Count)

board.cache_hierarchy.l1dcaches.prefetcher.pfBufferHit 0 # number of redundant prefetches already in prefetch queue (Count)

board.cache_hierarchy.l1dcaches.prefetcher.pfInCache 0 # number of redundant prefetches already in cache/mshr dropped (Count)

board.cache_hierarchy.l1dcaches.prefetcher.pfRemovedDemand 0 # number of prefetches dropped due to a demand for the same address (Count)

board.cache_hierarchy.l1dcaches.prefetcher.pfRemovedFull 0 # number of prefetches dropped due to prefetch queue size (Count)

board.cache_hierarchy.l1dcaches.prefetcher.pfSpanPage 13 # number of prefetches that crossed the page (Count)

board.cache_hierarchy.l1dcaches.prefetcher.pfUsefulSpanPage 12 # number of prefetches that is useful and crossed the page (Count)

board.cache_hierarchy.l1dcaches.prefetcher.power_state.pwrStateResidencyTicks::UNDEFINED 29955015 # Cumulative time (in ticks) in various power states (Tick)

board.cache_hierarchy.l1dcaches.tags.tagsInUse 111.876919 # Average ticks per tags in use ((Tick/Count))

board.cache_hierarchy.l1dcaches.tags.totalRefs 2674 # Total number of references to valid blocks. (Count)

board.cache_hierarchy.l1dcaches.tags.sampledRefs 185 # Sample count of references to valid blocks. (Count)

board.cache_hierarchy.l1dcaches.tags.avgRefs 14.454054 # Average number of references to valid blocks. ((Count/Count))

board.cache_hierarchy.l1dcaches.tags.warmupTick 132534 # The tick when the warmup percentage was hit. (Tick)

board.cache_hierarchy.l1dcaches.tags.occupancies::cache_hierarchy.l1dcaches.prefetcher 29.704353 # Average occupied blocks per tick, per requestor ((Count/Tick))

board.cache_hierarchy.l1dcaches.tags.occupancies::processor.cores.core.data 82.172566 # Average occupied blocks per tick, per requestor ((Count/Tick))

board.cache_hierarchy.l1dcaches.tags.avgOccs::cache_hierarchy.l1dcaches.prefetcher 0.058016 # Average percentage of cache occupancy ((Ratio/Tick))

board.cache_hierarchy.l1dcaches.tags.avgOccs::processor.cores.core.data 0.160493 # Average percentage of cache occupancy ((Ratio/Tick))

board.cache_hierarchy.l1dcaches.tags.avgOccs::total 0.218510 # Average percentage of cache occupancy ((Ratio/Tick))

board.cache_hierarchy.l1dcaches.tags.occupanciesTaskId::1022 47 # Occupied blocks per task id (Count)

board.cache_hierarchy.l1dcaches.tags.occupanciesTaskId::1024 138 # Occupied blocks per task id (Count)

board.cache_hierarchy.l1dcaches.tags.ageTaskId_1022::1 47 # Occupied blocks per task id, per block age (Count)

board.cache_hierarchy.l1dcaches.tags.ageTaskId_1024::0 32 # Occupied blocks per task id, per block age (Count)

board.cache_hierarchy.l1dcaches.tags.ageTaskId_1024::1 106 # Occupied blocks per task id, per block age (Count)

board.cache_hierarchy.l1dcaches.tags.ratioOccsTaskId::1022 0.091797 # Ratio of occupied blocks and all blocks, per task id (Ratio)

board.cache_hierarchy.l1dcaches.tags.ratioOccsTaskId::1024 0.269531 # Ratio of occupied blocks and all blocks, per task id (Ratio)

board.cache_hierarchy.l1dcaches.tags.tagAccesses 21201 # Number of tag accesses (Count)

board.cache_hierarchy.l1dcaches.tags.dataAccesses 21201 # Number of data accesses (Count)

board.cache_hierarchy.l1dcaches.tags.power_state.pwrStateResidencyTicks::UNDEFINED 29955015 # Cumulative time (in ticks) in various power states (Tick)

board.cache_hierarchy.l1icaches.demandHits::processor.cores.core.inst 8214 # number of demand (read+write) hits (Count)

board.cache_hierarchy.l1icaches.demandHits::total 8214 # number of demand (read+write) hits (Count)

board.cache_hierarchy.l1icaches.overallHits::processor.cores.core.inst 8214 # number of overall hits (Count)

board.cache_hierarchy.l1icaches.overallHits::total 8214 # number of overall hits (Count)

board.cache_hierarchy.l1icaches.demandMisses::processor.cores.core.inst 353 # number of demand (read+write) misses (Count)

board.cache_hierarchy.l1icaches.demandMisses::total 353 # number of demand (read+write) misses (Count)

board.cache_hierarchy.l1icaches.overallMisses::processor.cores.core.inst 353 # number of overall misses (Count)

board.cache_hierarchy.l1icaches.overallMisses::total 353 # number of overall misses (Count)

board.cache_hierarchy.l1icaches.demandMissLatency::processor.cores.core.inst 18987327 # number of demand (read+write) miss ticks (Tick)

board.cache_hierarchy.l1icaches.demandMissLatency::total 18987327 # number of demand (read+write) miss ticks (Tick)

board.cache_hierarchy.l1icaches.overallMissLatency::processor.cores.core.inst 18987327 # number of overall miss ticks (Tick)

board.cache_hierarchy.l1icaches.overallMissLatency::total 18987327 # number of overall miss ticks (Tick)

board.cache_hierarchy.l1icaches.demandAccesses::processor.cores.core.inst 8567 # number of demand (read+write) accesses (Count)

board.cache_hierarchy.l1icaches.demandAccesses::total 8567 # number of demand (read+write) accesses (Count)

board.cache_hierarchy.l1icaches.overallAccesses::processor.cores.core.inst 8567 # number of overall (read+write) accesses (Count)

board.cache_hierarchy.l1icaches.overallAccesses::total 8567 # number of overall (read+write) accesses (Count)

board.cache_hierarchy.l1icaches.demandMissRate::processor.cores.core.inst 0.041205 # miss rate for demand accesses (Ratio)

board.cache_hierarchy.l1icaches.demandMissRate::total 0.041205 # miss rate for demand accesses (Ratio)

board.cache_hierarchy.l1icaches.overallMissRate::processor.cores.core.inst 0.041205 # miss rate for overall accesses (Ratio)

board.cache_hierarchy.l1icaches.overallMissRate::total 0.041205 # miss rate for overall accesses (Ratio)

board.cache_hierarchy.l1icaches.demandAvgMissLatency::processor.cores.core.inst 53788.461756 # average overall miss latency in ticks ((Tick/Count))

board.cache_hierarchy.l1icaches.demandAvgMissLatency::total 53788.461756 # average overall miss latency in ticks ((Tick/Count))

board.cache_hierarchy.l1icaches.overallAvgMissLatency::processor.cores.core.inst 53788.461756 # average overall miss latency ((Tick/Count))

board.cache_hierarchy.l1icaches.overallAvgMissLatency::total 53788.461756 # average overall miss latency ((Tick/Count))

board.cache_hierarchy.l1icaches.blockedCycles::no_mshrs 0 # number of cycles access was blocked (Cycle)

board.cache_hierarchy.l1icaches.blockedCycles::no_targets 0 # number of cycles access was blocked (Cycle)

board.cache_hierarchy.l1icaches.blockedCauses::no_mshrs 0 # number of times access was blocked (Count)

board.cache_hierarchy.l1icaches.blockedCauses::no_targets 0 # number of times access was blocked (Count)

board.cache_hierarchy.l1icaches.avgBlocked::no_mshrs nan # average number of cycles each access was blocked ((Cycle/Count))

board.cache_hierarchy.l1icaches.avgBlocked::no_targets nan # average number of cycles each access was blocked ((Cycle/Count))

board.cache_hierarchy.l1icaches.demandMshrMisses::processor.cores.core.inst 353 # number of demand (read+write) MSHR misses (Count)

board.cache_hierarchy.l1icaches.demandMshrMisses::total 353 # number of demand (read+write) MSHR misses (Count)

board.cache_hierarchy.l1icaches.overallMshrMisses::processor.cores.core.inst 353 # number of overall MSHR misses (Count)

board.cache_hierarchy.l1icaches.overallMshrMisses::total 353 # number of overall MSHR misses (Count)

board.cache_hierarchy.l1icaches.demandMshrMissLatency::processor.cores.core.inst 18869778 # number of demand (read+write) MSHR miss ticks (Tick)

board.cache_hierarchy.l1icaches.demandMshrMissLatency::total 18869778 # number of demand (read+write) MSHR miss ticks (Tick)

board.cache_hierarchy.l1icaches.overallMshrMissLatency::processor.cores.core.inst 18869778 # number of overall MSHR miss ticks (Tick)

board.cache_hierarchy.l1icaches.overallMshrMissLatency::total 18869778 # number of overall MSHR miss ticks (Tick)

board.cache_hierarchy.l1icaches.demandMshrMissRate::processor.cores.core.inst 0.041205 # mshr miss ratio for demand accesses (Ratio)

board.cache_hierarchy.l1icaches.demandMshrMissRate::total 0.041205 # mshr miss ratio for demand accesses (Ratio)

board.cache_hierarchy.l1icaches.overallMshrMissRate::processor.cores.core.inst 0.041205 # mshr miss ratio for overall accesses (Ratio)

board.cache_hierarchy.l1icaches.overallMshrMissRate::total 0.041205 # mshr miss ratio for overall accesses (Ratio)

board.cache_hierarchy.l1icaches.demandAvgMshrMissLatency::processor.cores.core.inst 53455.461756 # average overall mshr miss latency ((Tick/Count))

board.cache_hierarchy.l1icaches.demandAvgMshrMissLatency::total 53455.461756 # average overall mshr miss latency ((Tick/Count))

board.cache_hierarchy.l1icaches.overallAvgMshrMissLatency::processor.cores.core.inst 53455.461756 # average overall mshr miss latency ((Tick/Count))

board.cache_hierarchy.l1icaches.overallAvgMshrMissLatency::total 53455.461756 # average overall mshr miss latency ((Tick/Count))

board.cache_hierarchy.l1icaches.replacements 10 # number of replacements (Count)

board.cache_hierarchy.l1icaches.ReadReq.hits::processor.cores.core.inst 8214 # number of ReadReq hits (Count)

board.cache_hierarchy.l1icaches.ReadReq.hits::total 8214 # number of ReadReq hits (Count)

board.cache_hierarchy.l1icaches.ReadReq.misses::processor.cores.core.inst 353 # number of ReadReq misses (Count)

board.cache_hierarchy.l1icaches.ReadReq.misses::total 353 # number of ReadReq misses (Count)

board.cache_hierarchy.l1icaches.ReadReq.missLatency::processor.cores.core.inst 18987327 # number of ReadReq miss ticks (Tick)

board.cache_hierarchy.l1icaches.ReadReq.missLatency::total 18987327 # number of ReadReq miss ticks (Tick)

board.cache_hierarchy.l1icaches.ReadReq.accesses::processor.cores.core.inst 8567 # number of ReadReq accesses(hits+misses) (Count)

board.cache_hierarchy.l1icaches.ReadReq.accesses::total 8567 # number of ReadReq accesses(hits+misses) (Count)

board.cache_hierarchy.l1icaches.ReadReq.missRate::processor.cores.core.inst 0.041205 # miss rate for ReadReq accesses (Ratio)

board.cache_hierarchy.l1icaches.ReadReq.missRate::total 0.041205 # miss rate for ReadReq accesses (Ratio)

board.cache_hierarchy.l1icaches.ReadReq.avgMissLatency::processor.cores.core.inst 53788.461756 # average ReadReq miss latency ((Tick/Count))

board.cache_hierarchy.l1icaches.ReadReq.avgMissLatency::total 53788.461756 # average ReadReq miss latency ((Tick/Count))

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board.cache_hierarchy.l1icaches.ReadReq.mshrMisses::processor.cores.core.inst 353 # number of ReadReq MSHR misses (Count)
board.cache_hierarchy.l1icaches.ReadReq.mshrMisses::total 353 # number of ReadReq MSHR misses (Count)
board.cache_hierarchy.l1icaches.ReadReq.mshrMissLatency::processor.cores.core.inst 18869778 # number of ReadReq MSHR miss ticks (Tick)
board.cache_hierarchy.l1icaches.ReadReq.mshrMissLatency::total 18869778 # number of ReadReq MSHR miss ticks (Tick)
board.cache_hierarchy.l1icaches.ReadReq.mshrMissRate::processor.cores.core.inst 0.041205 # mshr miss rate for ReadReq accesses (Ratio)
board.cache_hierarchy.l1icaches.ReadReq.mshrMissRate::total 0.041205 # mshr miss rate for ReadReq accesses (Ratio)
board.cache_hierarchy.l1icaches.ReadReq.avgMshrMissLatency::processor.cores.core.inst 53455.461756 # average ReadReq mshr miss latency ((Tick/Count))
board.cache_hierarchy.l1icaches.ReadReq.avgMshrMissLatency::total 53455.461756 # average ReadReq mshr miss latency ((Tick/Count))
board.cache_hierarchy.l1icaches.power_state.pwrStateResidencyTicks::UNDEFINED 29955015 # Cumulative time (in ticks) in various power states (Tick)
board.cache_hierarchy.l1icaches.prefetcher.demandMshrMisses 353 # demands not covered by prefetchs (Count)
board.cache_hierarchy.l1icaches.prefetcher.pfIssued 0 # number of hwpf issued (Count)
board.cache_hierarchy.l1icaches.prefetcher.pfUseful 0 # number of useful prefetch (Count)
board.cache_hierarchy.l1icaches.prefetcher.pfUsefulButMiss 0 # number of hit on prefetch but cache block is not in an usable state (Count)
board.cache_hierarchy.l1icaches.prefetcher.accuracy nan # accuracy of the prefetcher (Count)
board.cache_hierarchy.l1icaches.prefetcher.coverage 0 # coverage brought by this prefetcher (Count)
board.cache_hierarchy.l1icaches.prefetcher.pfHitInCache 0 # number of prefetches hitting in cache (Count)
board.cache_hierarchy.l1icaches.prefetcher.pfHitInMSHR 0 # number of prefetches hitting in a MSHR (Count)
board.cache_hierarchy.l1icaches.prefetcher.pfHitInWB 0 # number of prefetches hit in the Write Buffer (Count)
board.cache_hierarchy.l1icaches.prefetcher.pfLate 0 # number of late prefetches (hitting in cache, MSHR or WB) (Count)
board.cache_hierarchy.l1icaches.prefetcher.pfIdentified 0 # number of prefetch candidates identified (Count)
board.cache_hierarchy.l1icaches.prefetcher.pfBufferHit 0 # number of redundant prefetches already in prefetch queue (Count)
board.cache_hierarchy.l1icaches.prefetcher.pfInCache 0 # number of redundant prefetches already in cache/mshr dropped (Count)
board.cache_hierarchy.l1icaches.prefetcher.pfRemovedDemand 0 # number of prefetches dropped due to a demand for the same address (Count)
board.cache_hierarchy.l1icaches.prefetcher.pfRemovedFull 0 # number of prefetches dropped due to prefetch queue size (Count)
board.cache_hierarchy.l1icaches.prefetcher.pfSpanPage 0 # number of prefetches that crossed the page (Count)
board.cache_hierarchy.l1icaches.prefetcher.pfUsefulSpanPage 0 # number of prefetches that is useful and crossed the page (Count)
board.cache_hierarchy.l1icaches.prefetcher.power_state.pwrStateResidencyTicks::UNDEFINED 29955015 # Cumulative time (in ticks) in various power states (Tick)
board.cache_hierarchy.l1icaches.tags.tagsInUse 164.298300 # Average ticks per tags in use ((Tick/Count))
board.cache_hierarchy.l1icaches.tags.totalRefs 8567 # Total number of references to valid blocks. (Count)
board.cache_hierarchy.l1icaches.tags.sampledRefs 353 # Sample count of references to valid blocks. (Count)
board.cache_hierarchy.l1icaches.tags.avgRefs 24.269122 # Average number of references to valid blocks. ((Count/Count))
board.cache_hierarchy.l1icaches.tags.warmupTick 70263 # The tick when the warmup percentage was hit. (Tick)
board.cache_hierarchy.l1icaches.tags.occupancies::processor.cores.core.inst 164.298300 # Average occupied blocks per tick, per requestor ((Count/Tick))
board.cache_hierarchy.l1icaches.tags.avgOccs::processor.cores.core.inst 0.320895 # Average percentage of cache occupancy ((Ratio/Tick))
board.cache_hierarchy.l1icaches.tags.avgOccs::total 0.320895 # Average percentage of cache occupancy ((Ratio/Tick))
board.cache_hierarchy.l1icaches.tags.occupanciesTaskId::1024 343 # Occupied blocks per task id (Count)
board.cache_hierarchy.l1icaches.tags.ageTaskId_1024::0 133 # Occupied blocks per task id, per block age (Count)
board.cache_hierarchy.l1icaches.tags.ageTaskId_1024::1 210 # Occupied blocks per task id, per block age (Count)
board.cache_hierarchy.l1icaches.tags.ratioOccsTaskId::1024 0.669922 # Ratio of occupied blocks and all blocks, per task id (Ratio)
board.cache_hierarchy.l1icaches.tags.tagAccesses 68889 # Number of tag accesses (Count)
board.cache_hierarchy.l1icaches.tags.dataAccesses 68889 # Number of data accesses (Count)
board.cache_hierarchy.l1icaches.tags.power_state.pwrStateResidencyTicks::UNDEFINED 29955015 # Cumulative time (in ticks) in various power states (Tick)
board.cache_hierarchy.l2bus.transDist::ReadResp 479 # Transaction distribution (Count)
board.cache_hierarchy.l2bus.transDist::CleanEvict 10 # Transaction distribution (Count)
board.cache_hierarchy.l2bus.transDist::HardPFReq 71 # Transaction distribution (Count)
board.cache_hierarchy.l2bus.transDist::ReadExReq 59 # Transaction distribution (Count)
board.cache_hierarchy.l2bus.transDist::ReadExResp 59 # Transaction distribution (Count)
board.cache_hierarchy.l2bus.transDist::ReadSharedReq 479 # Transaction distribution (Count)
```

board.cache_hierarchy.l2bus.pktCount_board.cache_hierarchy.l1icaches.mem_side_port::board.cache_hierarchy.l2cache.cpu_side_port 716 # Packet count per connected requestor and responder (Count)

board.cache_hierarchy.l2bus.pktCount_board.cache_hierarchy.l1dcaches.mem_side_port::board.cache_hierarchy.l2cache.cpu_side_port 370 # Packet count per connected requestor and responder (Count)

board.cache_hierarchy.l2bus.pktCount::total 1086 # Packet count per connected requestor and responder (Count)

board.cache_hierarchy.l2bus.pktSize_board.cache_hierarchy.l1icaches.mem_side_port::board.cache_hierarchy.l2cache.cpu_side_port 22592 # Cumulative packet size per connected requestor and responder (Byte)

board.cache_hierarchy.l2bus.pktSize_board.cache_hierarchy.l1dcaches.mem_side_port::board.cache_hierarchy.l2cache.cpu_side_port 11840 # Cumulative packet size per connected requestor and responder (Byte)

board.cache_hierarchy.l2bus.pktSize::total 34432 # Cumulative packet size per connected requestor and responder (Byte)

board.cache_hierarchy.l2bus.snoops 71 # Total snoops (Count)

board.cache_hierarchy.l2bus.snoopTraffic 0 # Total snoop traffic (Byte)

board.cache_hierarchy.l2bus.snoopFanout::samples 609 # Request fanout histogram (Count)

board.cache_hierarchy.l2bus.snoopFanout::mean 0.001642 # Request fanout histogram (Count)

board.cache_hierarchy.l2bus.snoopFanout::stdev 0.040522 # Request fanout histogram (Count)

board.cache_hierarchy.l2bus.snoopFanout::underflows 0 0.00% 0.00% # Request fanout histogram (Count)

board.cache_hierarchy.l2bus.snoopFanout::0 608 99.84% 99.84% # Request fanout histogram (Count)

board.cache_hierarchy.l2bus.snoopFanout::1 1 0.16% 100.00% # Request fanout histogram (Count)

board.cache_hierarchy.l2bus.snoopFanout::2 0 0.00% 100.00% # Request fanout histogram (Count)

board.cache_hierarchy.l2bus.snoopFanout::3 0 0.00% 100.00% # Request fanout histogram (Count)

board.cache_hierarchy.l2bus.snoopFanout::4 0 0.00% 100.00% # Request fanout histogram (Count)

board.cache_hierarchy.l2bus.snoopFanout::overflows 0 0.00% 100.00% # Request fanout histogram (Count)

board.cache_hierarchy.l2bus.snoopFanout::min_value 0 # Request fanout histogram (Count)

board.cache_hierarchy.l2bus.snoopFanout::max_value 1 # Request fanout histogram (Count)

board.cache_hierarchy.l2bus.snoopFanout::total 609 # Request fanout histogram (Count)

board.cache_hierarchy.l2bus.power_state.pwrStateResidencyTicks::UNDEFINED 29955015 # Cumulative time (in ticks) in various power states (Tick)

board.cache_hierarchy.l2bus.reqLayer0.occupancy 193044 # Layer occupancy (ticks) (Tick)

board.cache_hierarchy.l2bus.reqLayer0.utilization 0.0 # Layer utilization (Ratio)

board.cache_hierarchy.l2bus.respLayer0.occupancy 352647 # Layer occupancy (ticks) (Tick)

board.cache_hierarchy.l2bus.respLayer0.utilization 0.0 # Layer utilization (Ratio)

board.cache_hierarchy.l2bus.respLayer1.occupancy 184815 # Layer occupancy (ticks) (Tick)

board.cache_hierarchy.l2bus.respLayer1.utilization 0.0 # Layer utilization (Ratio)

board.cache_hierarchy.l2bus.snoop_filter.totRequests 548 # Total number of requests made to the snoop filter. (Count)

board.cache_hierarchy.l2bus.snoop_filter.hitSingleRequests 11 # Number of requests hitting in the snoop filter with a single holder of the requested data. (Count)

board.cache_hierarchy.l2bus.snoop_filter.hitMultiRequests 0 # Number of requests hitting in the snoop filter with multiple (>1) holders of the requested data. (Count)

board.cache_hierarchy.l2bus.snoop_filter.totSnoops 0 # Total number of snoops made to the snoop filter. (Count)

board.cache_hierarchy.l2bus.snoop_filter.hitSingleSnoops 0 # Number of snoops hitting in the snoop filter with a single holder of the requested data. (Count)

board.cache_hierarchy.l2bus.snoop_filter.hitMultiSnoops 0 # Number of snoops hitting in the snoop filter with multiple (>1) holders of the requested data. (Count)

board.cache_hierarchy.l2cache.demandHits::cache_hierarchy.l1dcaches.prefetcher 8 # number of demand (read+write) hits (Count)

board.cache_hierarchy.l2cache.demandHits::processor.cores.core.inst 1 # number of demand (read+write) hits (Count)

board.cache_hierarchy.l2cache.demandHits::processor.cores.core.data 5 # number of demand (read+write) hits (Count)

board.cache_hierarchy.l2cache.demandHits::total 14 # number of demand (read+write) hits (Count)

board.cache_hierarchy.l2cache.overallHits::cache_hierarchy.l1dcaches.prefetcher 8 # number of overall hits (Count)

board.cache_hierarchy.l2cache.overallHits::processor.cores.core.inst 1 # number of overall hits (Count)

board.cache_hierarchy.l2cache.overallHits::processor.cores.core.data 5 # number of overall hits (Count)

board.cache_hierarchy.l2cache.overallHits::total 14 # number of overall hits (Count)

board.cache_hierarchy.l2cache.demandMisses::cache_hierarchy.l1dcaches.prefetcher 39 # number of demand (read+write) misses (Count)

board.cache_hierarchy.l2cache.demandMisses::processor.cores.core.inst 352 # number of demand (read+write) misses (Count)

board.cache_hierarchy.l2cache.demandMisses::processor.cores.core.data 133 # number of demand (read+write) misses (Count)

board.cache_hierarchy.l2cache.demandMisses::total 524 # number of demand (read+write) misses (Count)

board.cache_hierarchy.l2cache.overallMisses::cache_hierarchy.l1dcaches.prefetcher 39 # number of overall misses (Count)

board.cache_hierarchy.l2cache.overallMisses::processor.cores.core.inst 352 # number of overall misses (Count)

board.cache_hierarchy.l2cache.overallMisses::processor.cores.core.data 133 # number of overall misses (Count)

board.cache_hierarchy.l2cache.overallMisses::total 524 # number of overall misses (Count)

board.cache_hierarchy.l2cache.demandMissLatency::cache_hierarchy.l1dcaches.prefetcher 1046547 # number of demand (read+write) miss ticks (Tick)

board.cache_hierarchy.l2cache.demandMissLatency::processor.cores.core.inst 18630684 # number of demand (read+write) miss ticks (Tick)

board.cache_hierarchy.l2cache.demandMissLatency::processor.cores.core.data 7175817 # number of demand (read+write) miss ticks (Tick)

board.cache_hierarchy.l2cache.demandMissLatency::total 26853048 # number of demand (read+write) miss ticks (Tick)

board.cache_hierarchy.l2cache.overallMissLatency::cache_hierarchy.l1dcaches.prefetcher 1046547 # number of overall miss ticks (Tick)

board.cache_hierarchy.l2cache.overallMissLatency::processor.cores.core.inst 18630684 # number of overall miss ticks (Tick)

board.cache_hierarchy.l2cache.overallMissLatency::processor.cores.core.data 7175817 # number of overall miss ticks (Tick)

board.cache_hierarchy.l2cache.overallMissLatency::total 26853048 # number of overall miss ticks (Tick)

board.cache_hierarchy.l2cache.demandAccesses::cache_hierarchy.l1dcaches.prefetcher 47 # number of demand (read+write) accesses (Count)

board.cache_hierarchy.l2cache.demandAccesses::processor.cores.core.inst 353 # number of demand (read+write) accesses (Count)

board.cache_hierarchy.l2cache.demandAccesses::processor.cores.core.data 138 # number of demand (read+write) accesses (Count)

board.cache_hierarchy.l2cache.demandAccesses::total 538 # number of demand (read+write) accesses (Count)

board.cache_hierarchy.l2cache.overallAccesses::cache_hierarchy.l1dcaches.prefetcher 47 # number of overall (read+write) accesses (Count)

board.cache_hierarchy.l2cache.overallAccesses::processor.cores.core.inst 353 # number of overall (read+write) accesses (Count)

board.cache_hierarchy.l2cache.overallAccesses::processor.cores.core.data 138 # number of overall (read+write) accesses (Count)

board.cache_hierarchy.l2cache.overallAccesses::total 538 # number of overall (read+write) accesses (Count)

board.cache_hierarchy.l2cache.demandMissRate::cache_hierarchy.l1dcaches.prefetcher 0.829787 # miss rate for demand accesses (Ratio)

board.cache_hierarchy.l2cache.demandMissRate::processor.cores.core.inst 0.997167 # miss rate for demand accesses (Ratio)

board.cache_hierarchy.l2cache.demandMissRate::processor.cores.core.data 0.963768 # miss rate for demand accesses (Ratio)

board.cache_hierarchy.l2cache.demandMissRate::total 0.973978 # miss rate for demand accesses (Ratio)

board.cache_hierarchy.l2cache.overallMissRate::cache_hierarchy.l1dcaches.prefetcher 0.829787 # miss rate for overall accesses (Ratio)

board.cache_hierarchy.l2cache.overallMissRate::processor.cores.core.inst 0.997167 # miss rate for overall accesses (Ratio)

board.cache_hierarchy.l2cache.overallMissRate::processor.cores.core.data 0.963768 # miss rate for overall accesses (Ratio)

board.cache_hierarchy.l2cache.overallMissRate::total 0.973978 # miss rate for overall accesses (Ratio)

board.cache_hierarchy.l2cache.demandAvgMissLatency::cache_hierarchy.l1dcaches.prefetcher 26834.538462 # average overall miss latency in ticks ((Tick/Count))

board.cache_hierarchy.l2cache.demandAvgMissLatency::processor.cores.core.inst 52928.079545 # average overall miss latency in ticks ((Tick/Count))

board.cache_hierarchy.l2cache.demandAvgMissLatency::processor.cores.core.data 53953.511278 # average overall miss latency in ticks ((Tick/Count))

board.cache_hierarchy.l2cache.demandAvgMissLatency::total 51246.274809 # average overall miss latency in ticks ((Tick/Count))

board.cache_hierarchy.l2cache.overallAvgMissLatency::cache_hierarchy.l1dcaches.prefetcher 26834.538462 # average overall miss latency ((Tick/Count))

board.cache_hierarchy.l2cache.overallAvgMissLatency::processor.cores.core.inst 52928.079545 # average overall miss latency ((Tick/Count))

board.cache_hierarchy.l2cache.overallAvgMissLatency::processor.cores.core.data 53953.511278 # average overall miss latency ((Tick/Count))

board.cache_hierarchy.l2cache.overallAvgMissLatency::total 51246.274809 # average overall miss latency ((Tick/Count))

board.cache_hierarchy.l2cache.blockedCycles::no_mshrs 0 # number of cycles access was blocked (Cycle)

board.cache_hierarchy.l2cache.blockedCycles::no_targets 0 # number of cycles access was blocked (Cycle)

board.cache_hierarchy.l2cache.blockedCauses::no_mshrs 0 # number of times access was blocked (Count)

board.cache_hierarchy.l2cache.blockedCauses::no_targets 0 # number of times access was blocked (Count)

board.cache_hierarchy.l2cache.avgBlocked::no_mshrs nan # average number of cycles each access was blocked ((Cycle/Count))

board.cache_hierarchy.l2cache.avgBlocked::no_targets nan # average number of cycles each access was blocked ((Cycle/Count))

board.cache_hierarchy.l2cache.demandMshrHits::cache_hierarchy.l1dcaches.prefetcher 35 # number of demand (read+write) MSHR hits (Count)

board.cache_hierarchy.l2cache.demandMshrHits::total 35 # number of demand (read+write) MSHR hits (Count)

board.cache_hierarchy.l2cache.overallMshrHits::cache_hierarchy.l1dcaches.prefetcher 35 # number of overall MSHR hits (Count)

board.cache_hierarchy.l2cache.overallMshrHits::total 35 # number of overall MSHR hits (Count)

board.cache_hierarchy.l2cache.demandMshrMisses::cache_hierarchy.l1dcaches.prefetcher 4 # number of demand (read+write) MSHR misses (Count)

board.cache_hierarchy.l2cache.demandMshrMisses::processor.cores.core.inst 352 # number of demand (read+write) MSHR misses (Count)

board.cache_hierarchy.l2cache.demandMshrMisses::processor.cores.core.data 133 # number of demand (read+write) MSHR misses (Count)

board.cache_hierarchy.l2cache.demandMshrMisses::total 489 # number of demand (read+write) MSHR misses (Count)

board.cache_hierarchy.l2cache.overallMshrMisses::cache_hierarchy.l1dcaches.prefetcher 4 # number of overall MSHR misses (Count)

board.cache_hierarchy.l2cache.overallMshrMisses::cache_hierarchy.l2cache.prefetcher 51 # number of overall MSHR misses (Count)

board.cache_hierarchy.l2cache.overallMshrMisses::processor.cores.core.inst 352 # number of overall MSHR misses (Count)

board.cache_hierarchy.l2cache.overallMshrMisses::processor.cores.core.data 133 # number of overall MSHR misses (Count)

board.cache_hierarchy.l2cache.overallMshrMisses::total 540 # number of overall MSHR misses (Count)

board.cache_hierarchy.l2cache.demandMshrMissLatency::cache_hierarchy.l1dcaches.prefetcher 289044 # number of demand (read+write) MSHR miss ticks (Tick)

board.cache_hierarchy.l2cache.demandMshrMissLatency::processor.cores.core.inst 18513468 # number of demand (read+write) MSHR miss ticks (Tick)

board.cache_hierarchy.l2cache.demandMshrMissLatency::processor.cores.core.data 7131528 # number of demand (read+write) MSHR miss ticks (Tick)

board.cache_hierarchy.l2cache.demandMshrMissLatency::total 25934040 # number of demand (read+write) MSHR miss ticks (Tick)

board.cache_hierarchy.l2cache.overallMshrMissLatency::cache_hierarchy.l1dcaches.prefetcher 289044 # number of overall MSHR miss ticks (Tick)

board.cache_hierarchy.l2cache.overallMshrMissLatency::cache_hierarchy.l2cache.prefetcher 2602214 # number of overall MSHR miss ticks (Tick)

board.cache_hierarchy.l2cache.overallMshrMissLatency::processor.cores.core.inst 18513468 # number of overall MSHR miss ticks (Tick)

board.cache_hierarchy.l2cache.overallMshrMissLatency::processor.cores.core.data 7131528 # number of overall MSHR miss ticks (Tick)

board.cache_hierarchy.l2cache.overallMshrMissLatency::total 28536254 # number of overall MSHR miss ticks (Tick)

board.cache_hierarchy.l2cache.demandMshrMissRate::cache_hierarchy.l1dcaches.prefetcher 0.085106 # mshr miss ratio for demand accesses (Ratio)

board.cache_hierarchy.l2cache.demandMshrMissRate::processor.cores.core.inst 0.997167 # mshr miss ratio for demand accesses (Ratio)

board.cache_hierarchy.l2cache.demandMshrMissRate::processor.cores.core.data 0.963768 # mshr miss ratio for demand accesses (Ratio)

board.cache_hierarchy.l2cache.demandMshrMissRate::total 0.908922 # mshr miss ratio for demand accesses (Ratio)

board.cache_hierarchy.l2cache.overallMshrMissRate::cache_hierarchy.l1dcaches.prefetcher 0.085106 # mshr miss ratio for overall accesses (Ratio)

board.cache_hierarchy.l2cache.overallMshrMissRate::cache_hierarchy.l2cache.prefetcher inf # mshr miss ratio for overall accesses (Ratio)

board.cache_hierarchy.l2cache.overallMshrMissRate::processor.cores.core.inst 0.997167 # mshr miss ratio for overall accesses (Ratio)

board.cache_hierarchy.l2cache.overallMshrMissRate::processor.cores.core.data 0.963768 # mshr miss ratio for overall accesses (Ratio)

board.cache_hierarchy.l2cache.overallMshrMissRate::total 1.003717 # mshr miss ratio for overall accesses (Ratio)

board.cache_hierarchy.l2cache.demandAvgMshrMissLatency::cache_hierarchy.l1dcaches.prefetcher 72261 # average overall mshr miss latency ((Tick/Count))

board.cache_hierarchy.l2cache.demandAvgMshrMissLatency::processor.cores.core.inst 52595.079545 # average overall mshr miss latency ((Tick/Count))

board.cache_hierarchy.l2cache.demandAvgMshrMissLatency::processor.cores.core.data 53620.511278 # average overall mshr miss latency ((Tick/Count))

board.cache_hierarchy.l2cache.demandAvgMshrMissLatency::total 53034.846626 # average overall mshr miss latency ((Tick/Count))

board.cache_hierarchy.l2cache.overallAvgMshrMissLatency::cache_hierarchy.l1dcaches.prefetcher 72261 # average overall mshr miss latency ((Tick/Count))

board.cache_hierarchy.l2cache.overallAvgMshrMissLatency::cache_hierarchy.l2cache.prefetcher 51023.803922 # average overall mshr miss latency ((Tick/Count))

board.cache_hierarchy.l2cache.overallAvgMshrMissLatency::processor.cores.core.inst 52595.079545 # average overall mshr miss latency ((Tick/Count))

board.cache_hierarchy.l2cache.overallAvgMshrMissLatency::processor.cores.core.data 53620.511278 # average overall mshr miss latency ((Tick/Count))

board.cache_hierarchy.l2cache.overallAvgMshrMissLatency::total 52844.914815 # average overall mshr miss latency ((Tick/Count))

board.cache_hierarchy.l2cache.replacements 0 # number of replacements (Count)

board.cache_hierarchy.l2cache.HardPFReq.mshrMisses::cache_hierarchy.l2cache.prefetcher 51 # number of HardPFReq MSHR misses (Count)

board.cache_hierarchy.l2cache.HardPFReq.mshrMisses::total 51 # number of HardPFReq MSHR misses (Count)

board.cache_hierarchy.l2cache.HardPFReq.mshrMissLatency::cache_hierarchy.l2cache.prefetcher 2602214 # number of HardPFReq MSHR miss ticks (Tick)

board.cache_hierarchy.l2cache.HardPFReq.mshrMissLatency::total 2602214 # number of HardPFReq MSHR miss ticks (Tick)

board.cache_hierarchy.l2cache.HardPFReq.mshrMissRate::cache_hierarchy.l2cache.prefetcher inf # mshr miss rate for HardPFReq accesses (Ratio)

board.cache_hierarchy.l2cache.HardPFReq.mshrMissRate::total inf # mshr miss rate for HardPFReq accesses (Ratio)

board.cache_hierarchy.l2cache.HardPFReq.avgMshrMissLatency::cache_hierarchy.l2cache.prefetcher 51023.803922 # average HardPFReq mshr miss latency ((Tick/Count))

board.cache_hierarchy.l2cache.HardPFReq.avgMshrMissLatency::total 51023.803922 # average HardPFReq mshr miss latency ((Tick/Count))

board.cache_hierarchy.l2cache.ReadExReq.hits::processor.cores.core.data 5 # number of ReadExReq hits (Count)

board.cache_hierarchy.l2cache.ReadExReq.hits::total 5 # number of ReadExReq hits (Count)

board.cache_hierarchy.l2cache.ReadExReq.misses::processor.cores.core.data 54 # number of ReadExReq misses (Count)

board.cache_hierarchy.l2cache.ReadExReq.misses::total 54 # number of ReadExReq misses (Count)

board.cache_hierarchy.l2cache.ReadExReq.missLatency::processor.cores.core.data 2852478 # number of ReadExReq miss ticks (Tick)

board.cache_hierarchy.l2cache.ReadExReq.missLatency::total 2852478 # number of ReadExReq miss ticks (Tick)

board.cache_hierarchy.l2cache.ReadExReq.accesses::processor.cores.core.data 59 # number of ReadExReq accesses(hits+misses) (Count)

board.cache_hierarchy.l2cache.ReadExReq.accesses::total 59 # number of ReadExReq accesses(hits+misses) (Count)

board.cache_hierarchy.l2cache.ReadExReq.missRate::processor.cores.core.data 0.915254 # miss rate for ReadExReq accesses (Ratio)

board.cache_hierarchy.l2cache.ReadExReq.missRate::total 0.915254 # miss rate for ReadExReq accesses (Ratio)

board.cache_hierarchy.l2cache.ReadExReq.avgMissLatency::processor.cores.core.data 52823.666667 # average ReadExReq miss latency ((Tick/Count))

board.cache_hierarchy.l2cache.ReadExReq.avgMissLatency::total 52823.666667 # average ReadExReq miss latency ((Tick/Count))

board.cache_hierarchy.l2cache.ReadExReq.mshrMisses::processor.cores.core.data 54 # number of ReadExReq MSHR misses (Count)

board.cache_hierarchy.l2cache.ReadExReq.mshrMisses::total 54 # number of ReadExReq MSHR misses (Count)

board.cache_hierarchy.l2cache.ReadExReq.mshrMissLatency::processor.cores.core.data 2834496 # number of ReadExReq MSHR miss ticks (Tick)

board.cache_hierarchy.l2cache.ReadExReq.mshrMissLatency::total 2834496 # number of ReadExReq MSHR miss ticks (Tick)

board.cache_hierarchy.l2cache.ReadExReq.mshrMissRate::processor.cores.core.data 0.915254 # mshr miss rate for ReadExReq accesses (Ratio)

board.cache_hierarchy.l2cache.ReadExReq.mshrMissRate::total 0.915254 # mshr miss rate for ReadExReq accesses (Ratio)

board.cache_hierarchy.l2cache.ReadExReq.avgMshrMissLatency::processor.cores.core.data 52490.666667 # average ReadExReq mshr miss latency ((Tick/Count))

board.cache_hierarchy.l2cache.ReadExReq.avgMshrMissLatency::total 52490.666667 # average ReadExReq mshr miss latency ((Tick/Count))

board.cache_hierarchy.l2cache.ReadSharedReq.hits::cache_hierarchy.l1dcaches.prefetcher 8 # number of ReadSharedReq hits (Count)

board.cache_hierarchy.l2cache.ReadSharedReq.hits::processor.cores.core.inst 1 # number of ReadSharedReq hits (Count)

board.cache_hierarchy.l2cache.ReadSharedReq.hits::total 9 # number of ReadSharedReq hits (Count)

board.cache_hierarchy.l2cache.ReadSharedReq.misses::cache_hierarchy.l1dcaches.prefetcher 39 # number of ReadSharedReq misses (Count)

board.cache_hierarchy.l2cache.ReadSharedReq.misses::processor.cores.core.inst 352 # number of ReadSharedReq misses (Count)

board.cache_hierarchy.l2cache.ReadSharedReq.misses::processor.cores.core.data 79 # number of ReadSharedReq misses (Count)

board.cache_hierarchy.l2cache.ReadSharedReq.misses::total 470 # number of ReadSharedReq misses (Count)

board.cache_hierarchy.l2cache.ReadSharedReq.missLatency::cache_hierarchy.l1dcaches.prefetcher 1046547 # number of ReadSharedReq miss ticks (Tick)

board.cache_hierarchy.l2cache.ReadSharedReq.missLatency::processor.cores.core.inst 18630684 # number of ReadSharedReq miss ticks (Tick)

board.cache_hierarchy.l2cache.ReadSharedReq.missLatency::processor.cores.core.data 4323339 # number of ReadSharedReq miss ticks (Tick)

board.cache_hierarchy.l2cache.ReadSharedReq.missLatency::total 24000570 # number of ReadSharedReq miss ticks (Tick)

board.cache_hierarchy.l2cache.ReadSharedReq.accesses::cache_hierarchy.l1dcaches.prefetcher 47 # number of ReadSharedReq accesses(hits+misses) (Count)

board.cache_hierarchy.l2cache.ReadSharedReq.accesses::processor.cores.core.inst 353 # number of ReadSharedReq accesses(hits+misses) (Count)

board.cache_hierarchy.l2cache.ReadSharedReq.accesses::processor.cores.core.data 79 # number of ReadSharedReq accesses(hits+misses) (Count)

board.cache_hierarchy.l2cache.ReadSharedReq.accesses::total 479 # number of ReadSharedReq accesses(hits+misses) (Count)

board.cache_hierarchy.l2cache.ReadSharedReq.missRate::cache_hierarchy.l1dcaches.prefetcher 0.829787 # miss rate for ReadSharedReq accesses (Ratio)

board.cache_hierarchy.l2cache.ReadSharedReq.missRate::processor.cores.core.inst 0.997167 # miss rate for ReadSharedReq accesses (Ratio)

board.cache_hierarchy.l2cache.ReadSharedReq.missRate::processor.cores.core.data 1 # miss rate for ReadSharedReq accesses (Ratio)

board.cache_hierarchy.l2cache.ReadSharedReq.missRate::total 0.981211 # miss rate for ReadSharedReq accesses (Ratio)

board.cache_hierarchy.l2cache.ReadSharedReq.avgMissLatency::cache_hierarchy.l1dcaches.prefetcher 26834.538462 # average ReadSharedReq miss latency ((Tick/Count))

board.cache_hierarchy.l2cache.ReadSharedReq.avgMissLatency::processor.cores.core.inst 52928.079545 # average ReadSharedReq miss latency ((Tick/Count))

board.cache_hierarchy.l2cache.ReadSharedReq.avgMissLatency::processor.cores.core.data 54725.810127 # average ReadSharedReq miss latency ((Tick/Count))

board.cache_hierarchy.l2cache.ReadSharedReq.avgMissLatency::total 51065.042553 # average ReadSharedReq miss latency ((Tick/Count))

board.cache_hierarchy.l2cache.ReadSharedReq.mshrHits::cache_hierarchy.l1dcaches.prefetcher 35 # number of ReadSharedReq MSHR hits (Count)

board.cache_hierarchy.l2cache.ReadSharedReq.mshrHits::total 35 # number of ReadSharedReq MSHR hits (Count)

board.cache_hierarchy.l2cache.ReadSharedReq.mshrMisses::cache_hierarchy.l1dcaches.prefetcher 4 # number of ReadSharedReq MSHR misses (Count)

board.cache_hierarchy.l2cache.ReadSharedReq.mshrMisses::processor.cores.core.inst 352 # number of ReadSharedReq MSHR misses (Count)

board.cache_hierarchy.l2cache.ReadSharedReq.mshrMisses::processor.cores.core.data 79 # number of ReadSharedReq MSHR misses (Count)

board.cache_hierarchy.l2cache.ReadSharedReq.mshrMisses::total 435 # number of ReadSharedReq MSHR misses (Count)

board.cache_hierarchy.l2cache.ReadSharedReq.mshrMissLatency::cache_hierarchy.l1dcaches.prefetcher 289044 # number of ReadSharedReq MSHR miss ticks (Tick)

board.cache_hierarchy.l2cache.ReadSharedReq.mshrMissLatency::processor.cores.core.inst 18513468 # number of ReadSharedReq MSHR miss ticks (Tick)

board.cache_hierarchy.l2cache.ReadSharedReq.mshrMissLatency::processor.cores.core.data 4297032 # number of ReadSharedReq MSHR miss ticks (Tick)

board.cache_hierarchy.l2cache.ReadSharedReq.mshrMissLatency::total 23099544 # number of ReadSharedReq MSHR miss ticks (Tick)

board.cache_hierarchy.l2cache.ReadSharedReq.mshrMissRate::cache_hierarchy.l1dcaches.prefetcher 0.085106 # mshr miss rate for ReadSharedReq accesses (Ratio)

board.cache_hierarchy.l2cache.ReadSharedReq.mshrMissRate::processor.cores.core.inst 0.997167 # mshr miss rate for ReadSharedReq accesses (Ratio)

board.cache_hierarchy.l2cache.ReadSharedReq.mshrMissRate::processor.cores.core.data 1 # mshr miss rate for ReadSharedReq accesses (Ratio)

board.cache_hierarchy.l2cache.ReadSharedReq.mshrMissRate::total 0.908142 # mshr miss rate for ReadSharedReq accesses (Ratio)

board.cache_hierarchy.l2cache.ReadSharedReq.avgMshrMissLatency::cache_hierarchy.l1dcaches.prefetcher 72261 # average ReadSharedReq mshr miss latency ((Tick/Count))

board.cache_hierarchy.l2cache.ReadSharedReq.avgMshrMissLatency::processor.cores.core.inst 52595.079545 # average ReadSharedReq mshr miss latency ((Tick/Count))

board.cache_hierarchy.l2cache.ReadSharedReq.avgMshrMissLatency::processor.cores.core.data 54392.810127 # average ReadSharedReq mshr miss latency ((Tick/Count))

board.cache_hierarchy.l2cache.ReadSharedReq.avgMshrMissLatency::total 53102.400000 # average ReadSharedReq mshr miss latency ((Tick/Count))

board.cache_hierarchy.l2cache.power_state.pwrStateResidencyTicks::UNDEFINED 29955015 # Cumulative time (in ticks) in various power states (Tick)

board.cache_hierarchy.l2cache.prefetcher.demandMshrMisses 489 # demands not covered by prefetchs (Count)

board.cache_hierarchy.l2cache.prefetcher.pfIssued 146 # number of hwpf issued (Count)

board.cache_hierarchy.l2cache.prefetcher.pfUseful 13 # number of useful prefetch (Count)

board.cache_hierarchy.l2cache.prefetcher.pfUsefulButMiss 0 # number of hit on prefetch but cache block is not in an usable state (Count)

board.cache_hierarchy.l2cache.prefetcher.accuracy 0.089041 # accuracy of the prefetcher (Count)

board.cache_hierarchy.l2cache.prefetcher.coverage 0.025896 # coverage brought by this prefetcher (Count)

board.cache_hierarchy.l2cache.prefetcher.pfHitInCache 24 # number of prefetches hitting in cache (Count)

board.cache_hierarchy.l2cache.prefetcher.pfHitInMSHR 71 # number of prefetches hitting in a MSHR (Count)

board.cache_hierarchy.l2cache.prefetcher.pfHitInWB 0 # number of prefetches hit in the Write Buffer (Count)

board.cache_hierarchy.l2cache.prefetcher.pfLate 95 # number of late prefetches (hitting in cache, MSHR or WB) (Count)

board.cache_hierarchy.l2cache.prefetcher.pfIdentified 181 # number of prefetch candidates identified (Count)

board.cache_hierarchy.l2cache.prefetcher.pfBufferHit 23 # number of redundant prefetches already in prefetch queue (Count)

board.cache_hierarchy.l2cache.prefetcher.pfInCache 0 # number of redundant prefetches already in cache/mshr dropped (Count)

board.cache_hierarchy.l2cache.prefetcher.pfRemovedDemand 9 # number of prefetches dropped due to a demand for the same address (Count)

board.cache_hierarchy.l2cache.prefetcher.pfRemovedFull 0 # number of prefetches dropped due to prefetch queue size (Count)

board.cache_hierarchy.l2cache.prefetcher.pfSpanPage 15 # number of prefetches that crossed the page (Count)

board.cache_hierarchy.l2cache.prefetcher.pfUsefulSpanPage 0 # number of prefetches that is useful and crossed the page (Count)

board.cache_hierarchy.l2cache.prefetcher.power_state.pwrStateResidencyTicks::UNDEFINED 29955015 # Cumulative time (in ticks) in various power states (Tick)

board.cache_hierarchy.l2cache.tags.tagsInUse 279.481983 # Average ticks per tags in use ((Tick/Count))

board.cache_hierarchy.l2cache.tags.totalRefs 564 # Total number of references to valid blocks. (Count)

board.cache_hierarchy.l2cache.tags.sampledRefs 540 # Sample count of references to valid blocks. (Count)

board.cache_hierarchy.l2cache.tags.avgRefs 1.044444 # Average number of references to valid blocks. ((Count/Count))

board.cache_hierarchy.l2cache.tags.warmupTick 69597 # The tick when the warmup percentage was hit. (Tick)

board.cache_hierarchy.l2cache.tags.occupancies::cache_hierarchy.l1dcaches.prefetcher 2.755945 # Average occupied blocks per tick, per requestor ((Count/Tick))

board.cache_hierarchy.l2cache.tags.occupancies::cache_hierarchy.l2cache.prefetcher 31.175133 # Average occupied blocks per tick, per requestor ((Count/Tick))

board.cache_hierarchy.l2cache.tags.occupancies::processor.cores.core.inst 165.154072 # Average occupied blocks per tick, per requestor ((Count/Tick))

board.cache_hierarchy.l2cache.tags.occupancies::processor.cores.core.data 80.396834 # Average occupied blocks per tick, per requestor ((Count/Tick))

board.cache_hierarchy.l2cache.tags.avgOccs::cache_hierarchy.l1dcaches.prefetcher 0.000673 # Average percentage of cache occupancy ((Ratio/Tick))

board.cache_hierarchy.l2cache.tags.avgOccs::cache_hierarchy.l2cache.prefetcher 0.007611 # Average percentage of cache occupancy ((Ratio/Tick))

board.cache_hierarchy.l2cache.tags.avgOccs::processor.cores.core.inst 0.040321 # Average percentage of cache occupancy ((Ratio/Tick))

board.cache_hierarchy.l2cache.tags.avgOccs::processor.cores.core.data 0.019628 # Average percentage of cache occupancy ((Ratio/Tick))

board.cache_hierarchy.l2cache.tags.avgOccs::total 0.068233 # Average percentage of cache occupancy ((Ratio/Tick))

board.cache_hierarchy.l2cache.tags.occupanciesTaskId::1022 55 # Occupied blocks per task id (Count)

board.cache_hierarchy.l2cache.tags.occupanciesTaskId::1024 485 # Occupied blocks per task id (Count)

board.cache_hierarchy.l2cache.tags.ageTaskId_1022::1 55 # Occupied blocks per task id, per block age (Count)

board.cache_hierarchy.l2cache.tags.ageTaskId_1024::0 164 # Occupied blocks per task id, per block age (Count)

board.cache_hierarchy.l2cache.tags.ageTaskId_1024::1 321 # Occupied blocks per task id, per block age (Count)

board.cache_hierarchy.l2cache.tags.ratioOccsTaskId::1022 0.013428 # Ratio of occupied blocks and all blocks, per task id (Ratio)

board.cache_hierarchy.l2cache.tags.ratioOccsTaskId::1024 0.118408 # Ratio of occupied blocks and all blocks, per task id (Ratio)

board.cache_hierarchy.l2cache.tags.tagAccesses 9308 # Number of tag accesses (Count)

board.cache_hierarchy.l2cache.tags.dataAccesses 9308 # Number of data accesses (Count)

board.cache_hierarchy.l2cache.tags.power_state.pwrStateResidencyTicks::UNDEFINED 29955015 # Cumulative time (in ticks) in various power states (Tick)

board.cache_hierarchy.membus.transDist::ReadResp 486 # Transaction distribution (Count)

board.cache_hierarchy.membus.transDist::ReadExReq 54 # Transaction distribution (Count)

board.cache_hierarchy.membus.transDist::ReadExResp 54 # Transaction distribution (Count)

board.cache_hierarchy.membus.transDist::ReadSharedReq 486 # Transaction distribution (Count)

board.cache_hierarchy.membus.pktCount_board.cache_hierarchy.l2cache.mem_side_port::board.memory.mem_ctrl.port 1080 # Packet count per connected requestor and responder (Count)

board.cache_hierarchy.membus.pktCount_board.cache_hierarchy.l2cache.mem_side_port::total 1080 # Packet count per connected requestor and responder (Count)

board.cache_hierarchy.membus.pktCount::total 1080 # Packet count per connected requestor and responder (Count)

board.cache_hierarchy.membus.pktSize_board.cache_hierarchy.l2cache.mem_side_port::board.memory.mem_ctrl.port 34560 # Cumulative packet size per connected requestor and responder (Byte)

board.cache_hierarchy.membus.pktSize_board.cache_hierarchy.l2cache.mem_side_port::total 34560 # Cumulative packet size per connected requestor and responder (Byte)

board.cache_hierarchy.membus.pktSize::total 34560 # Cumulative packet size per connected requestor and responder (Byte)

board.cache_hierarchy.membus.snoops 0 # Total snoops (Count)

board.cache_hierarchy.membus.snoopTraffic 0 # Total snoop traffic (Byte)

board.cache_hierarchy.membus.snoopFanout::samples 540 # Request fanout histogram (Count)

board.cache_hierarchy.membus.snoopFanout::mean 0 # Request fanout histogram (Count)

board.cache_hierarchy.membus.snoopFanout::stdev 0 # Request fanout histogram (Count)

board.cache_hierarchy.membus.snoopFanout::underflows 0 0.00% 0.00% # Request fanout histogram (Count)

board.cache_hierarchy.membus.snoopFanout::0 540 100.00% 100.00% # Request fanout histogram (Count)

board.cache_hierarchy.membus.snoopFanout::1 0 0.00% 100.00% # Request fanout histogram (Count)

board.cache_hierarchy.membus.snoopFanout::overflows 0 0.00% 100.00% # Request fanout histogram (Count)

board.cache_hierarchy.membus.snoopFanout::min_value 0 # Request fanout histogram (Count)

board.cache_hierarchy.membus.snoopFanout::max_value 0 # Request fanout histogram (Count)

board.cache_hierarchy.membus.snoopFanout::total 540 # Request fanout histogram (Count)

board.cache_hierarchy.membus.badaddr_responder.power_state.pwrStateResidencyTicks::UNDEFINED 29955015 # Cumulative time (in ticks) in various power states (Tick)

board.cache_hierarchy.membus.power_state.pwrStateResidencyTicks::UNDEFINED 29955015 # Cumulative time (in ticks) in various power states (Tick)

board.cache_hierarchy.membus.reqLayer0.occupancy 188659 # Layer occupancy (ticks) (Tick)

board.cache_hierarchy.membus.reqLayer0.utilization 0.0 # Layer utilization (Ratio)

board.cache_hierarchy.membus.respLayer2.occupancy 454567 # Layer occupancy (ticks) (Tick)

board.cache_hierarchy.membus.respLayer2.utilization 0.0 # Layer utilization (Ratio)

board.cache_hierarchy.membus.snoop_filter.totRequests 540 # Total number of requests made to the snoop filter. (Count)

board.cache_hierarchy.membus.snoop_filter.hitSingleRequests 0 # Number of requests hitting in the snoop filter with a single holder of the requested data. (Count)

board.cache_hierarchy.membus.snoop_filter.hitMultiRequests 0 # Number of requests hitting in the snoop filter with multiple (>1) holders of the requested data. (Count)

board.cache_hierarchy.membus.snoop_filter.totSnoops 0 # Total number of snoops made to the snoop filter. (Count)

board.cache_hierarchy.membus.snoop_filter.hitSingleSnoops 0 # Number of snoops hitting in the snoop filter with a single holder of the requested data. (Count)

board.cache_hierarchy.membus.snoop_filter.hitMultiSnoops 0 # Number of snoops hitting in the snoop filter with multiple (>1) holders of the requested data. (Count)

board.clk_domain.clock 333 # Clock period in ticks (Tick)

board.clk_domain.voltage_domain.voltage 1 # Voltage in Volts (Volt)

board.memory.mem_ctrl.avgPriority_cache_hierarchy.l1dcaches.prefetcher::samples 4.00 # Average QoS priority value for accepted requests (Count)

board.memory.mem_ctrl.avgPriority_cache_hierarchy.l2cache.prefetcher::samples 51.00 # Average QoS priority value for accepted requests (Count)

board.memory.mem_ctrl.avgPriority_processor.cores.core.inst::samples 352.00 # Average QoS priority value for accepted requests (Count)

board.memory.mem_ctrl.avgPriority_processor.cores.core.data::samples 133.00 # Average QoS priority value for accepted requests (Count)

board.memory.mem_ctrl.priorityMinLatency 0.000000018750 # per QoS priority minimum request to response latency (Second)

board.memory.mem_ctrl.priorityMaxLatency 0.000000551626 # per QoS priority maximum request to response latency (Second)

board.memory.mem_ctrl.numReadWriteTurnArounds 0 # Number of turnarounds from READ to WRITE (Count)

board.memory.mem_ctrl.numWriteReadTurnArounds 0 # Number of turnarounds from WRITE to READ (Count)

board.memory.mem_ctrl.numStayReadState 1080 # Number of times bus staying in READ state (Count)

board.memory.mem_ctrl.numStayWriteState 0 # Number of times bus staying in WRITE state (Count)

board.memory.mem_ctrl.readReqs 540 # Number of read requests accepted (Count)

board.memory.mem_ctrl.writeReqs 0 # Number of write requests accepted (Count)

board.memory.mem_ctrl.readBursts 540 # Number of controller read bursts, including those serviced by the write queue (Count)

board.memory.mem_ctrl.writeBursts 0 # Number of controller write bursts, including those merged in the write queue (Count)

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board.memory.mem_ctrl.servicedByWrQ 0 # Number of controller read bursts serviced by the write queue (Count)
board.memory.mem_ctrl.mergedWrBursts 0 # Number of controller write bursts merged with an existing one (Count)
board.memory.mem_ctrl.neitherReadNorWriteReqs 0 # Number of requests that are neither read nor write (Count)
board.memory.mem_ctrl.avgRdQLen 1.10 # Average read queue length when enqueuing ((Count/Tick))
board.memory.mem_ctrl.avgWrQLen 0.00 # Average write queue length when enqueuing ((Count/Tick))
board.memory.mem_ctrl.numRdRetry 0 # Number of times read queue was full causing retry (Count)
board.memory.mem_ctrl.numWrRetry 0 # Number of times write queue was full causing retry (Count)
board.memory.mem_ctrl.readPktSize::0 0 # Read request sizes (log2) (Count)
board.memory.mem_ctrl.readPktSize::1 0 # Read request sizes (log2) (Count)
board.memory.mem_ctrl.readPktSize::2 0 # Read request sizes (log2) (Count)
board.memory.mem_ctrl.readPktSize::3 0 # Read request sizes (log2) (Count)
board.memory.mem_ctrl.readPktSize::4 0 # Read request sizes (log2) (Count)
board.memory.mem_ctrl.readPktSize::5 0 # Read request sizes (log2) (Count)
board.memory.mem_ctrl.readPktSize::6 540 # Read request sizes (log2) (Count)
board.memory.mem_ctrl.writePktSize::0 0 # Write request sizes (log2) (Count)
board.memory.mem_ctrl.writePktSize::1 0 # Write request sizes (log2) (Count)
board.memory.mem_ctrl.writePktSize::2 0 # Write request sizes (log2) (Count)
board.memory.mem_ctrl.writePktSize::3 0 # Write request sizes (log2) (Count)
board.memory.mem_ctrl.writePktSize::4 0 # Write request sizes (log2) (Count)
board.memory.mem_ctrl.writePktSize::5 0 # Write request sizes (log2) (Count)
board.memory.mem_ctrl.writePktSize::6 0 # Write request sizes (log2) (Count)
board.memory.mem_ctrl.rdQLenPdf::0 485 # What read queue length does an incoming req see (Count)
board.memory.mem_ctrl.rdQLenPdf::1 20 # What read queue length does an incoming req see (Count)
board.memory.mem_ctrl.rdQLenPdf::2 8 # What read queue length does an incoming req see (Count)
board.memory.mem_ctrl.rdQLenPdf::3 8 # What read queue length does an incoming req see (Count)
board.memory.mem_ctrl.rdQLenPdf::4 5 # What read queue length does an incoming req see (Count)
board.memory.mem_ctrl.rdQLenPdf::5 5 # What read queue length does an incoming req see (Count)
board.memory.mem_ctrl.rdQLenPdf::6 3 # What read queue length does an incoming req see (Count)
board.memory.mem_ctrl.rdQLenPdf::7 3 # What read queue length does an incoming req see (Count)
board.memory.mem_ctrl.rdQLenPdf::8 3 # What read queue length does an incoming req see (Count)
board.memory.mem_ctrl.rdQLenPdf::9 0 # What read queue length does an incoming req see (Count)
board.memory.mem_ctrl.rdQLenPdf::10 0 # What read queue length does an incoming req see (Count)
board.memory.mem_ctrl.rdQLenPdf::11 0 # What read queue length does an incoming req see (Count)
board.memory.mem_ctrl.rdQLenPdf::12 0 # What read queue length does an incoming req see (Count)
board.memory.mem_ctrl.rdQLenPdf::13 0 # What read queue length does an incoming req see (Count)
board.memory.mem_ctrl.rdQLenPdf::14 0 # What read queue length does an incoming req see (Count)
board.memory.mem_ctrl.rdQLenPdf::15 0 # What read queue length does an incoming req see (Count)
board.memory.mem_ctrl.rdQLenPdf::16 0 # What read queue length does an incoming req see (Count)
board.memory.mem_ctrl.rdQLenPdf::17 0 # What read queue length does an incoming req see (Count)
board.memory.mem_ctrl.rdQLenPdf::18 0 # What read queue length does an incoming req see (Count)
board.memory.mem_ctrl.rdQLenPdf::19 0 # What read queue length does an incoming req see (Count)
board.memory.mem_ctrl.rdQLenPdf::20 0 # What read queue length does an incoming req see (Count)
board.memory.mem_ctrl.rdQLenPdf::21 0 # What read queue length does an incoming req see (Count)
board.memory.mem_ctrl.rdQLenPdf::22 0 # What read queue length does an incoming req see (Count)
board.memory.mem_ctrl.rdQLenPdf::23 0 # What read queue length does an incoming req see (Count)
board.memory.mem_ctrl.rdQLenPdf::24 0 # What read queue length does an incoming req see (Count)
board.memory.mem_ctrl.rdQLenPdf::25 0 # What read queue length does an incoming req see (Count)
board.memory.mem_ctrl.rdQLenPdf::26 0 # What read queue length does an incoming req see (Count)
board.memory.mem_ctrl.rdQLenPdf::27 0 # What read queue length does an incoming req see (Count)
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board.memory.mem_ctrl.rdQLenPdf::28	0	# What read queue length does an incoming req see (Count)
board.memory.mem_ctrl.rdQLenPdf::29	0	# What read queue length does an incoming req see (Count)
board.memory.mem_ctrl.rdQLenPdf::30	0	# What read queue length does an incoming req see (Count)
board.memory.mem_ctrl.rdQLenPdf::31	0	# What read queue length does an incoming req see (Count)
board.memory.mem_ctrl.wrQLenPdf::0	0	# What write queue length does an incoming req see (Count)
board.memory.mem_ctrl.wrQLenPdf::1	0	# What write queue length does an incoming req see (Count)
board.memory.mem_ctrl.wrQLenPdf::2	0	# What write queue length does an incoming req see (Count)
board.memory.mem_ctrl.wrQLenPdf::3	0	# What write queue length does an incoming req see (Count)
board.memory.mem_ctrl.wrQLenPdf::4	0	# What write queue length does an incoming req see (Count)
board.memory.mem_ctrl.wrQLenPdf::5	0	# What write queue length does an incoming req see (Count)
board.memory.mem_ctrl.wrQLenPdf::6	0	# What write queue length does an incoming req see (Count)
board.memory.mem_ctrl.wrQLenPdf::7	0	# What write queue length does an incoming req see (Count)
board.memory.mem_ctrl.wrQLenPdf::8	0	# What write queue length does an incoming req see (Count)
board.memory.mem_ctrl.wrQLenPdf::9	0	# What write queue length does an incoming req see (Count)
board.memory.mem_ctrl.wrQLenPdf::10	0	# What write queue length does an incoming req see (Count)
board.memory.mem_ctrl.wrQLenPdf::11	0	# What write queue length does an incoming req see (Count)
board.memory.mem_ctrl.wrQLenPdf::12	0	# What write queue length does an incoming req see (Count)
board.memory.mem_ctrl.wrQLenPdf::13	0	# What write queue length does an incoming req see (Count)
board.memory.mem_ctrl.wrQLenPdf::14	0	# What write queue length does an incoming req see (Count)
board.memory.mem_ctrl.wrQLenPdf::15	0	# What write queue length does an incoming req see (Count)
board.memory.mem_ctrl.wrQLenPdf::16	0	# What write queue length does an incoming req see (Count)
board.memory.mem_ctrl.wrQLenPdf::17	0	# What write queue length does an incoming req see (Count)
board.memory.mem_ctrl.wrQLenPdf::18	0	# What write queue length does an incoming req see (Count)
board.memory.mem_ctrl.wrQLenPdf::19	0	# What write queue length does an incoming req see (Count)
board.memory.mem_ctrl.wrQLenPdf::20	0	# What write queue length does an incoming req see (Count)
board.memory.mem_ctrl.wrQLenPdf::21	0	# What write queue length does an incoming req see (Count)
board.memory.mem_ctrl.wrQLenPdf::22	0	# What write queue length does an incoming req see (Count)
board.memory.mem_ctrl.wrQLenPdf::23	0	# What write queue length does an incoming req see (Count)
board.memory.mem_ctrl.wrQLenPdf::24	0	# What write queue length does an incoming req see (Count)
board.memory.mem_ctrl.wrQLenPdf::25	0	# What write queue length does an incoming req see (Count)
board.memory.mem_ctrl.wrQLenPdf::26	0	# What write queue length does an incoming req see (Count)
board.memory.mem_ctrl.wrQLenPdf::27	0	# What write queue length does an incoming req see (Count)
board.memory.mem_ctrl.wrQLenPdf::28	0	# What write queue length does an incoming req see (Count)
board.memory.mem_ctrl.wrQLenPdf::29	0	# What write queue length does an incoming req see (Count)
board.memory.mem_ctrl.wrQLenPdf::30	0	# What write queue length does an incoming req see (Count)
board.memory.mem_ctrl.wrQLenPdf::31	0	# What write queue length does an incoming req see (Count)
board.memory.mem_ctrl.wrQLenPdf::32	0	# What write queue length does an incoming req see (Count)
board.memory.mem_ctrl.wrQLenPdf::33	0	# What write queue length does an incoming req see (Count)
board.memory.mem_ctrl.wrQLenPdf::34	0	# What write queue length does an incoming req see (Count)
board.memory.mem_ctrl.wrQLenPdf::35	0	# What write queue length does an incoming req see (Count)
board.memory.mem_ctrl.wrQLenPdf::36	0	# What write queue length does an incoming req see (Count)
board.memory.mem_ctrl.wrQLenPdf::37	0	# What write queue length does an incoming req see (Count)
board.memory.mem_ctrl.wrQLenPdf::38	0	# What write queue length does an incoming req see (Count)
board.memory.mem_ctrl.wrQLenPdf::39	0	# What write queue length does an incoming req see (Count)
board.memory.mem_ctrl.wrQLenPdf::40	0	# What write queue length does an incoming req see (Count)
board.memory.mem_ctrl.wrQLenPdf::41	0	# What write queue length does an incoming req see (Count)
board.memory.mem_ctrl.wrQLenPdf::42	0	# What write queue length does an incoming req see (Count)
board.memory.mem_ctrl.wrQLenPdf::43	0	# What write queue length does an incoming req see (Count)
board.memory.mem_ctrl.wrQLenPdf::44	0	# What write queue length does an incoming req see (Count)

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board.memory.mem_ctrl.wrQLenPdf::45 0 # What write queue length does an incoming req see (Count)
board.memory.mem_ctrl.wrQLenPdf::46 0 # What write queue length does an incoming req see (Count)
board.memory.mem_ctrl.wrQLenPdf::47 0 # What write queue length does an incoming req see (Count)
board.memory.mem_ctrl.wrQLenPdf::48 0 # What write queue length does an incoming req see (Count)
board.memory.mem_ctrl.wrQLenPdf::49 0 # What write queue length does an incoming req see (Count)
board.memory.mem_ctrl.wrQLenPdf::50 0 # What write queue length does an incoming req see (Count)
board.memory.mem_ctrl.wrQLenPdf::51 0 # What write queue length does an incoming req see (Count)
board.memory.mem_ctrl.wrQLenPdf::52 0 # What write queue length does an incoming req see (Count)
board.memory.mem_ctrl.wrQLenPdf::53 0 # What write queue length does an incoming req see (Count)
board.memory.mem_ctrl.wrQLenPdf::54 0 # What write queue length does an incoming req see (Count)
board.memory.mem_ctrl.wrQLenPdf::55 0 # What write queue length does an incoming req see (Count)
board.memory.mem_ctrl.wrQLenPdf::56 0 # What write queue length does an incoming req see (Count)
board.memory.mem_ctrl.wrQLenPdf::57 0 # What write queue length does an incoming req see (Count)
board.memory.mem_ctrl.wrQLenPdf::58 0 # What write queue length does an incoming req see (Count)
board.memory.mem_ctrl.wrQLenPdf::59 0 # What write queue length does an incoming req see (Count)
board.memory.mem_ctrl.wrQLenPdf::60 0 # What write queue length does an incoming req see (Count)
board.memory.mem_ctrl.wrQLenPdf::61 0 # What write queue length does an incoming req see (Count)
board.memory.mem_ctrl.wrQLenPdf::62 0 # What write queue length does an incoming req see (Count)
board.memory.mem_ctrl.wrQLenPdf::63 0 # What write queue length does an incoming req see (Count)
board.memory.mem_ctrl.bytesReadWrQ 0 # Total number of bytes read from write queue (Byte)
board.memory.mem_ctrl.bytesReadSys 34560 # Total read bytes from the system interface side (Byte)
board.memory.mem_ctrl.bytesWrittenSys 0 # Total written bytes from the system interface side (Byte)
board.memory.mem_ctrl.avgRdBWSys 1153730018.16223431 # Average system read bandwidth in Byte/s ((Byte/Second))
board.memory.mem_ctrl.avgWrBWSys 0.00000000 # Average system write bandwidth in Byte/s ((Byte/Second))
board.memory.mem_ctrl.totGap 29905731 # Total gap between requests (Tick)
board.memory.mem_ctrl.avgGap 55380.98 # Average gap between requests ((Tick/Count))
board.memory.mem_ctrl.requestorReadBytes::cache_hierarchy.l1dcaches.prefetcher 256 # Per-requestor bytes read from memory (Byte)
board.memory.mem_ctrl.requestorReadBytes::cache_hierarchy.l2cache.prefetcher 3264 # Per-requestor bytes read from memory (Byte)
board.memory.mem_ctrl.requestorReadBytes::processor.cores.core.inst 22528 # Per-requestor bytes read from memory (Byte)
board.memory.mem_ctrl.requestorReadBytes::processor.cores.core.data 8512 # Per-requestor bytes read from memory (Byte)
board.memory.mem_ctrl.requestorReadRate::cache_hierarchy.l1dcaches.prefetcher 8546148.282683216035 # Per-requestor bytes read from memory rate ((Byte/Second))
board.memory.mem_ctrl.requestorReadRate::cache_hierarchy.l2cache.prefetcher 108963390.604211002588 # Per-requestor bytes read from memory rate ((Byte/Second))
board.memory.mem_ctrl.requestorReadRate::processor.cores.core.inst 752061048.876123070717 # Per-requestor bytes read from memory rate ((Byte/Second))
board.memory.mem_ctrl.requestorReadRate::processor.cores.core.data 284159430.399216949940 # Per-requestor bytes read from memory rate ((Byte/Second))
board.memory.mem_ctrl.requestorReadAccesses::cache_hierarchy.l1dcaches.prefetcher 4 # Per-requestor read serviced memory accesses (Count)
board.memory.mem_ctrl.requestorReadAccesses::cache_hierarchy.l2cache.prefetcher 51 # Per-requestor read serviced memory accesses (Count)
board.memory.mem_ctrl.requestorReadAccesses::processor.cores.core.inst 352 # Per-requestor read serviced memory accesses (Count)
board.memory.mem_ctrl.requestorReadAccesses::processor.cores.core.data 133 # Per-requestor read serviced memory accesses (Count)
board.memory.mem_ctrl.requestorReadTotalLat::cache_hierarchy.l1dcaches.prefetcher 180348 # Per-requestor read total memory access latency (Tick)
board.memory.mem_ctrl.requestorReadTotalLat::cache_hierarchy.l2cache.prefetcher 1386486 # Per-requestor read total memory access latency (Tick)
board.memory.mem_ctrl.requestorReadTotalLat::processor.cores.core.inst 8948853 # Per-requestor read total memory access latency (Tick)
board.memory.mem_ctrl.requestorReadTotalLat::processor.cores.core.data 3520097 # Per-requestor read total memory access latency (Tick)
board.memory.mem_ctrl.requestorReadAvgLat::cache_hierarchy.l1dcaches.prefetcher 45087.00 # Per-requestor read average memory access latency ((Tick/Count))
board.memory.mem_ctrl.requestorReadAvgLat::cache_hierarchy.l2cache.prefetcher 27186.00 # Per-requestor read average memory access latency ((Tick/Count))
board.memory.mem_ctrl.requestorReadAvgLat::processor.cores.core.inst 25422.88 # Per-requestor read average memory access latency ((Tick/Count))
board.memory.mem_ctrl.requestorReadAvgLat::processor.cores.core.data 26466.89 # Per-requestor read average memory access latency ((Tick/Count))
board.memory.mem_ctrl.dram.bytesRead::cache_hierarchy.l1dcaches.prefetcher 256 # Number of bytes read from this memory (Byte)
board.memory.mem_ctrl.dram.bytesRead::cache_hierarchy.l2cache.prefetcher 3264 # Number of bytes read from this memory (Byte)
board.memory.mem_ctrl.dram.bytesRead::processor.cores.core.inst 22528 # Number of bytes read from this memory (Byte)
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board.memory.mem_ctrl.dram.bytesRead::processor.cores.core.data        8512                    # Number of bytes read from this memory (Byte)
board.memory.mem_ctrl.dram.bytesRead::total        34560                    # Number of bytes read from this memory (Byte)
board.memory.mem_ctrl.dram.bytesInstRead::processor.cores.core.inst        22528                    # Number of instructions bytes read from this memory (Byte)
board.memory.mem_ctrl.dram.bytesInstRead::total        22528                    # Number of instructions bytes read from this memory (Byte)
board.memory.mem_ctrl.dram.numReads::cache_hierarchy.l1dcaches.prefetcher        4                    # Number of read requests responded to by this memory (Count)
board.memory.mem_ctrl.dram.numReads::cache_hierarchy.l2cache.prefetcher        51                    # Number of read requests responded to by this memory (Count)
board.memory.mem_ctrl.dram.numReads::processor.cores.core.inst        352                    # Number of read requests responded to by this memory (Count)
board.memory.mem_ctrl.dram.numReads::processor.cores.core.data        133                    # Number of read requests responded to by this memory (Count)
board.memory.mem_ctrl.dram.numReads::total        540                    # Number of read requests responded to by this memory (Count)
board.memory.mem_ctrl.dram.bwRead::cache_hierarchy.l1dcaches.prefetcher        8546148                    # Total read bandwidth from this memory ((Byte/Second))
board.memory.mem_ctrl.dram.bwRead::cache_hierarchy.l2cache.prefetcher        108963391                    # Total read bandwidth from this memory ((Byte/Second))
board.memory.mem_ctrl.dram.bwRead::processor.cores.core.inst        752061049                    # Total read bandwidth from this memory ((Byte/Second))
board.memory.mem_ctrl.dram.bwRead::processor.cores.core.data        284159430                    # Total read bandwidth from this memory ((Byte/Second))
board.memory.mem_ctrl.dram.bwRead::total        1153730018                    # Total read bandwidth from this memory ((Byte/Second))
board.memory.mem_ctrl.dram.bwInstRead::processor.cores.core.inst        752061049                    # Instruction read bandwidth from this memory ((Byte/Second))
board.memory.mem_ctrl.dram.bwInstRead::total        752061049                    # Instruction read bandwidth from this memory ((Byte/Second))
board.memory.mem_ctrl.dram.bwTotal::cache_hierarchy.l1dcaches.prefetcher        8546148                    # Total bandwidth to/from this memory ((Byte/Second))
board.memory.mem_ctrl.dram.bwTotal::cache_hierarchy.l2cache.prefetcher        108963391                    # Total bandwidth to/from this memory ((Byte/Second))
board.memory.mem_ctrl.dram.bwTotal::processor.cores.core.inst        752061049                    # Total bandwidth to/from this memory ((Byte/Second))
board.memory.mem_ctrl.dram.bwTotal::processor.cores.core.data        284159430                    # Total bandwidth to/from this memory ((Byte/Second))
board.memory.mem_ctrl.dram.bwTotal::total        1153730018                    # Total bandwidth to/from this memory ((Byte/Second))
board.memory.mem_ctrl.dram.readBursts        540                    # Number of DRAM read bursts (Count)
board.memory.mem_ctrl.dram.writeBursts        0                    # Number of DRAM write bursts (Count)
board.memory.mem_ctrl.dram.perBankRdBursts::0        74                    # Per bank write bursts (Count)
board.memory.mem_ctrl.dram.perBankRdBursts::1        0                    # Per bank write bursts (Count)
board.memory.mem_ctrl.dram.perBankRdBursts::2        0                    # Per bank write bursts (Count)
board.memory.mem_ctrl.dram.perBankRdBursts::3        29                    # Per bank write bursts (Count)
board.memory.mem_ctrl.dram.perBankRdBursts::4        14                    # Per bank write bursts (Count)
board.memory.mem_ctrl.dram.perBankRdBursts::5        14                    # Per bank write bursts (Count)
board.memory.mem_ctrl.dram.perBankRdBursts::6        54                    # Per bank write bursts (Count)
board.memory.mem_ctrl.dram.perBankRdBursts::7        5                    # Per bank write bursts (Count)
board.memory.mem_ctrl.dram.perBankRdBursts::8        39                    # Per bank write bursts (Count)
board.memory.mem_ctrl.dram.perBankRdBursts::9        76                    # Per bank write bursts (Count)
board.memory.mem_ctrl.dram.perBankRdBursts::10        99                    # Per bank write bursts (Count)
board.memory.mem_ctrl.dram.perBankRdBursts::11        35                    # Per bank write bursts (Count)
board.memory.mem_ctrl.dram.perBankRdBursts::12        11                    # Per bank write bursts (Count)
board.memory.mem_ctrl.dram.perBankRdBursts::13        63                    # Per bank write bursts (Count)
board.memory.mem_ctrl.dram.perBankRdBursts::14        6                    # Per bank write bursts (Count)
board.memory.mem_ctrl.dram.perBankRdBursts::15        21                    # Per bank write bursts (Count)
board.memory.mem_ctrl.dram.perBankWrBursts::0        0                    # Per bank write bursts (Count)
board.memory.mem_ctrl.dram.perBankWrBursts::1        0                    # Per bank write bursts (Count)
board.memory.mem_ctrl.dram.perBankWrBursts::2        0                    # Per bank write bursts (Count)
board.memory.mem_ctrl.dram.perBankWrBursts::3        0                    # Per bank write bursts (Count)
board.memory.mem_ctrl.dram.perBankWrBursts::4        0                    # Per bank write bursts (Count)
board.memory.mem_ctrl.dram.perBankWrBursts::5        0                    # Per bank write bursts (Count)
board.memory.mem_ctrl.dram.perBankWrBursts::6        0                    # Per bank write bursts (Count)
board.memory.mem_ctrl.dram.perBankWrBursts::7        0                    # Per bank write bursts (Count)
board.memory.mem_ctrl.dram.perBankWrBursts::8        0                    # Per bank write bursts (Count)
board.memory.mem_ctrl.dram.perBankWrBursts::9        0                    # Per bank write bursts (Count)
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board.memory.mem_ctrl.dram.perBankWrBursts::10 0 # Per bank write bursts (Count)
board.memory.mem_ctrl.dram.perBankWrBursts::11 0 # Per bank write bursts (Count)
board.memory.mem_ctrl.dram.perBankWrBursts::12 0 # Per bank write bursts (Count)
board.memory.mem_ctrl.dram.perBankWrBursts::13 0 # Per bank write bursts (Count)
board.memory.mem_ctrl.dram.perBankWrBursts::14 0 # Per bank write bursts (Count)
board.memory.mem_ctrl.dram.perBankWrBursts::15 0 # Per bank write bursts (Count)
board.memory.mem_ctrl.dram.totQLat 3910784 # Total ticks spent queuing (Tick)
board.memory.mem_ctrl.dram.totBusLat 2700000 # Total ticks spent in databus transfers (Tick)
board.memory.mem_ctrl.dram.totMemAccLat 14035784 # Total ticks spent from burst creation until serviced by the DRAM (Tick)
board.memory.mem_ctrl.dram.avgQLat 7242.19 # Average queueing delay per DRAM burst ((Tick/Count))
board.memory.mem_ctrl.dram.avgBusLat 5000.00 # Average bus latency per DRAM burst ((Tick/Count))
board.memory.mem_ctrl.dram.avgMemAccLat 25992.19 # Average memory access latency per DRAM burst ((Tick/Count))
board.memory.mem_ctrl.dram.readRowHits 423 # Number of row buffer hits during reads (Count)
board.memory.mem_ctrl.dram.writeRowHits 0 # Number of row buffer hits during writes (Count)
board.memory.mem_ctrl.dram.readRowHitRate 78.33 # Row buffer hit rate for reads (Ratio)
board.memory.mem_ctrl.dram.writeRowHitRate nan # Row buffer hit rate for writes (Ratio)
board.memory.mem_ctrl.dram.bytesPerActivate::samples 108 # Bytes accessed per row activation (Byte)
board.memory.mem_ctrl.dram.bytesPerActivate::mean 295.703704 # Bytes accessed per row activation (Byte)
board.memory.mem_ctrl.dram.bytesPerActivate::gmean 185.491620 # Bytes accessed per row activation (Byte)
board.memory.mem_ctrl.dram.bytesPerActivate::stdev 306.211193 # Bytes accessed per row activation (Byte)
board.memory.mem_ctrl.dram.bytesPerActivate::0-127 33 30.56% 30.56% # Bytes accessed per row activation (Byte)
board.memory.mem_ctrl.dram.bytesPerActivate::128-255 34 31.48% 62.04% # Bytes accessed per row activation (Byte)
board.memory.mem_ctrl.dram.bytesPerActivate::256-383 13 12.04% 74.07% # Bytes accessed per row activation (Byte)
board.memory.mem_ctrl.dram.bytesPerActivate::384-511 4 3.70% 77.78% # Bytes accessed per row activation (Byte)
board.memory.mem_ctrl.dram.bytesPerActivate::512-639 6 5.56% 83.33% # Bytes accessed per row activation (Byte)
board.memory.mem_ctrl.dram.bytesPerActivate::640-767 4 3.70% 87.04% # Bytes accessed per row activation (Byte)
board.memory.mem_ctrl.dram.bytesPerActivate::768-895 2 1.85% 88.89% # Bytes accessed per row activation (Byte)
board.memory.mem_ctrl.dram.bytesPerActivate::896-1023 2 1.85% 90.74% # Bytes accessed per row activation (Byte)
board.memory.mem_ctrl.dram.bytesPerActivate::1024-1151 10 9.26% 100.00% # Bytes accessed per row activation (Byte)
board.memory.mem_ctrl.dram.bytesPerActivate::total 108 # Bytes accessed per row activation (Byte)
board.memory.mem_ctrl.dram.dramBytesRead 34560 # Total bytes read (Byte)
board.memory.mem_ctrl.dram.dramBytesWritten 0 # Total bytes written (Byte)
board.memory.mem_ctrl.dram.avgRdBW 1153.730018 # Average DRAM read bandwidth in MiBytes/s ((Byte/Second))
board.memory.mem_ctrl.dram.avgWrBW 0 # Average DRAM write bandwidth in MiBytes/s ((Byte/Second))
board.memory.mem_ctrl.dram.peakBW 12800.00 # Theoretical peak bandwidth in MiByte/s ((Byte/Second))
board.memory.mem_ctrl.dram.busUtil 9.01 # Data bus utilization in percentage (Ratio)
board.memory.mem_ctrl.dram.busUtilRead 9.01 # Data bus utilization in percentage for reads (Ratio)
board.memory.mem_ctrl.dram.busUtilWrite 0.00 # Data bus utilization in percentage for writes (Ratio)
board.memory.mem_ctrl.dram.pageHitRate 78.33 # Row buffer hit rate, read and write combined (Ratio)
board.memory.mem_ctrl.dram.power_state.pwrStateResidencyTicks::UNDEFINED 29955015 # Cumulative time (in ticks) in various power states (Tick)
board.memory.mem_ctrl.dram.rank0.actEnergy 249900 # Energy for activate commands per rank (pJ) (Joule)
board.memory.mem_ctrl.dram.rank0.preEnergy 113850 # Energy for precharge commands per rank (pJ) (Joule)
board.memory.mem_ctrl.dram.rank0.readEnergy 1356600 # Energy for read commands per rank (pJ) (Joule)
board.memory.mem_ctrl.dram.rank0.writeEnergy 0 # Energy for write commands per rank (pJ) (Joule)
board.memory.mem_ctrl.dram.rank0.refreshEnergy 1843920.000000 # Energy for refresh commands per rank (pJ) (Joule)
board.memory.mem_ctrl.dram.rank0.actBackEnergy 12555960 # Energy for active background per rank (pJ) (Joule)
board.memory.mem_ctrl.dram.rank0.preBackEnergy 929760 # Energy for precharge background per rank (pJ) (Joule)
board.memory.mem_ctrl.dram.rank0.actPowerDownEnergy 0 # Energy for active power-down per rank (pJ) (Joule)
board.memory.mem_ctrl.dram.rank0.prePowerDownEnergy 0 # Energy for precharge power-down per rank (pJ) (Joule)
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board.memory.mem_ctrl.dram.rank0.selfRefreshEnergy 0 # Energy for self refresh per rank (pJ) (Joule)
board.memory.mem_ctrl.dram.rank0.totalEnergy 17049990 # Total energy per rank (pJ) (Joule)
board.memory.mem_ctrl.dram.rank0.averagePower 569.186495 # Core power per rank (mW) (Watt)
board.memory.mem_ctrl.dram.rank0.totalIdleTime 0 # Total Idle time Per DRAM Rank (Tick)
board.memory.mem_ctrl.dram.rank0.pwrStateTime::IDLE 2322495 # Time in different power states (Tick)
board.memory.mem_ctrl.dram.rank0.pwrStateTime::REF 780000 # Time in different power states (Tick)
board.memory.mem_ctrl.dram.rank0.pwrStateTime::SREF 0 # Time in different power states (Tick)
board.memory.mem_ctrl.dram.rank0.pwrStateTime::PRE_PDN 0 # Time in different power states (Tick)
board.memory.mem_ctrl.dram.rank0.pwrStateTime::ACT 26852520 # Time in different power states (Tick)
board.memory.mem_ctrl.dram.rank0.pwrStateTime::ACT_PDN 0 # Time in different power states (Tick)
board.memory.mem_ctrl.dram.rank1.actEnergy 585480 # Energy for activate commands per rank (pJ) (Joule)
board.memory.mem_ctrl.dram.rank1.preEnergy 296010 # Energy for precharge commands per rank (pJ) (Joule)
board.memory.mem_ctrl.dram.rank1.readEnergy 2499000 # Energy for read commands per rank (pJ) (Joule)
board.memory.mem_ctrl.dram.rank1.writeEnergy 0 # Energy for write commands per rank (pJ) (Joule)
board.memory.mem_ctrl.dram.rank1.refreshEnergy 1843920.000000 # Energy for refresh commands per rank (pJ) (Joule)
board.memory.mem_ctrl.dram.rank1.actBackEnergy 13280430 # Energy for active background per rank (pJ) (Joule)
board.memory.mem_ctrl.dram.rank1.preBackEnergy 319680 # Energy for precharge background per rank (pJ) (Joule)
board.memory.mem_ctrl.dram.rank1.actPowerDownEnergy 0 # Energy for active power-down per rank (pJ) (Joule)
board.memory.mem_ctrl.dram.rank1.prePowerDownEnergy 0 # Energy for precharge power-down per rank (pJ) (Joule)
board.memory.mem_ctrl.dram.rank1.selfRefreshEnergy 0 # Energy for self refresh per rank (pJ) (Joule)
board.memory.mem_ctrl.dram.rank1.totalEnergy 18824520 # Total energy per rank (pJ) (Joule)
board.memory.mem_ctrl.dram.rank1.averagePower 628.426325 # Core power per rank (mW) (Watt)
board.memory.mem_ctrl.dram.rank1.totalIdleTime 0 # Total Idle time Per DRAM Rank (Tick)
board.memory.mem_ctrl.dram.rank1.pwrStateTime::IDLE 749094 # Time in different power states (Tick)
board.memory.mem_ctrl.dram.rank1.pwrStateTime::REF 780000 # Time in different power states (Tick)
board.memory.mem_ctrl.dram.rank1.pwrStateTime::SREF 0 # Time in different power states (Tick)
board.memory.mem_ctrl.dram.rank1.pwrStateTime::PRE_PDN 0 # Time in different power states (Tick)
board.memory.mem_ctrl.dram.rank1.pwrStateTime::ACT 28425921 # Time in different power states (Tick)
board.memory.mem_ctrl.dram.rank1.pwrStateTime::ACT_PDN 0 # Time in different power states (Tick)
board.memory.mem_ctrl.power_state.pwrStateResidencyTicks::UNDEFINED 29955015 # Cumulative time (in ticks) in various power states (Tick)
board.processor.cores.core.numCycles 89955 # Number of cpu cycles simulated (Cycle)
board.processor.cores.core.cpi 13.695950 # CPI: cycles per instruction (core level) ((Cycle/Count))
board.processor.cores.core.ipc 0.073014 # IPC: instructions per cycle (core level) ((Count/Cycle))
board.processor.cores.core.numWorkItemsStarted 0 # Number of work items this cpu started (Count)
board.processor.cores.core.numWorkItemsCompleted 0 # Number of work items this cpu completed (Count)
board.processor.cores.core.commitStats0.numInsts 6568 # Number of instructions committed (thread level) (Count)
board.processor.cores.core.commitStats0.numOps 13006 # Number of ops (including micro ops) committed (thread level) (Count)
board.processor.cores.core.commitStats0.numInstsNotNOP 0 # Number of instructions committed excluding NOPs or prefetches (Count)
board.processor.cores.core.commitStats0.numOpsNotNOP 0 # Number of Ops (including micro ops) Simulated (Count)
board.processor.cores.core.commitStats0.cpi 13.695950 # CPI: cycles per instruction (thread level) ((Cycle/Count))
board.processor.cores.core.commitStats0.ipc 0.073014 # IPC: instructions per cycle (thread level) ((Count/Cycle))
board.processor.cores.core.commitStats0.numMemRefs 0 # Number of memory references committed (Count)
board.processor.cores.core.commitStats0.numFpInsts 411 # Number of float instructions (Count)
board.processor.cores.core.commitStats0.numIntInsts 12799 # Number of integer instructions (Count)
board.processor.cores.core.commitStats0.numLoadInsts 1272 # Number of load instructions (Count)
board.processor.cores.core.commitStats0.numStoreInsts 1354 # Number of store instructions (Count)
board.processor.cores.core.commitStats0.numVecInsts 0 # Number of vector instructions (Count)
board.processor.cores.core.commitStats0.committedInstType::No_OpClass 27 0.21% 0.21% # Class of committed instruction. (Count)
board.processor.cores.core.commitStats0.committedInstType::IntAlu 10134 77.92% 78.13% # Class of committed instruction. (Count)
```

board.processor.cores.core.commitStats0.committedInstType::IntMult 4 0.03% 78.16% # Class of committed instruction. (Count)

board.processor.cores.core.commitStats0.committedInstType::IntDiv 21 0.16% 78.32% # Class of committed instruction. (Count)

board.processor.cores.core.commitStats0.committedInstType::FloatAdd 2 0.02% 78.33% # Class of committed instruction. (Count)

board.processor.cores.core.commitStats0.committedInstType::FloatCmp 0 0.00% 78.33% # Class of committed instruction. (Count)

board.processor.cores.core.commitStats0.committedInstType::FloatCvt 0 0.00% 78.33% # Class of committed instruction. (Count)

board.processor.cores.core.commitStats0.committedInstType::FloatMult 0 0.00% 78.33% # Class of committed instruction. (Count)

board.processor.cores.core.commitStats0.committedInstType::FloatMultAcc 0 0.00% 78.33% # Class of committed instruction. (Count)

board.processor.cores.core.commitStats0.committedInstType::FloatDiv 0 0.00% 78.33% # Class of committed instruction. (Count)

board.processor.cores.core.commitStats0.committedInstType::FloatMisc 0 0.00% 78.33% # Class of committed instruction. (Count)

board.processor.cores.core.commitStats0.committedInstType::FloatSqrt 0 0.00% 78.33% # Class of committed instruction. (Count)

board.processor.cores.core.commitStats0.committedInstType::SimdAdd 0 0.00% 78.33% # Class of committed instruction. (Count)

board.processor.cores.core.commitStats0.committedInstType::SimdAddAcc 0 0.00% 78.33% # Class of committed instruction. (Count)

board.processor.cores.core.commitStats0.committedInstType::SimdAlu 82 0.63% 78.96% # Class of committed instruction. (Count)

board.processor.cores.core.commitStats0.committedInstType::SimdCmp 0 0.00% 78.96% # Class of committed instruction. (Count)

board.processor.cores.core.commitStats0.committedInstType::SimdCvt 48 0.37% 79.33% # Class of committed instruction. (Count)

board.processor.cores.core.commitStats0.committedInstType::SimdMisc 62 0.48% 79.81% # Class of committed instruction. (Count)

board.processor.cores.core.commitStats0.committedInstType::SimdMult 0 0.00% 79.81% # Class of committed instruction. (Count)

board.processor.cores.core.commitStats0.committedInstType::SimdMultAcc 0 0.00% 79.81% # Class of committed instruction. (Count)

board.processor.cores.core.commitStats0.committedInstType::SimdMatMultAcc 0 0.00% 79.81% # Class of committed instruction. (Count)

board.processor.cores.core.commitStats0.committedInstType::SimdShift 0 0.00% 79.81% # Class of committed instruction. (Count)

board.processor.cores.core.commitStats0.committedInstType::SimdShiftAcc 0 0.00% 79.81% # Class of committed instruction. (Count)

board.processor.cores.core.commitStats0.committedInstType::SimdDiv 0 0.00% 79.81% # Class of committed instruction. (Count)

board.processor.cores.core.commitStats0.committedInstType::SimdSqrt 0 0.00% 79.81% # Class of committed instruction. (Count)

board.processor.cores.core.commitStats0.committedInstType::SimdFloatAdd 0 0.00% 79.81% # Class of committed instruction. (Count)

board.processor.cores.core.commitStats0.committedInstType::SimdFloatAlu 0 0.00% 79.81% # Class of committed instruction. (Count)

board.processor.cores.core.commitStats0.committedInstType::SimdFloatCmp 0 0.00% 79.81% # Class of committed instruction. (Count)

board.processor.cores.core.commitStats0.committedInstType::SimdFloatCvt 0 0.00% 79.81% # Class of committed instruction. (Count)

board.processor.cores.core.commitStats0.committedInstType::SimdFloatDiv 0 0.00% 79.81% # Class of committed instruction. (Count)

board.processor.cores.core.commitStats0.committedInstType::SimdFloatMisc 0 0.00% 79.81% # Class of committed instruction. (Count)

board.processor.cores.core.commitStats0.committedInstType::SimdFloatMult 0 0.00% 79.81% # Class of committed instruction. (Count)

board.processor.cores.core.commitStats0.committedInstType::SimdFloatMultAcc 0 0.00% 79.81% # Class of committed instruction. (Count)

board.processor.cores.core.commitStats0.committedInstType::SimdFloatMatMultAcc 0 0.00% 79.81% # Class of committed instruction. (Count)

board.processor.cores.core.commitStats0.committedInstType::SimdFloatSqrt 0 0.00% 79.81% # Class of committed instruction. (Count)

board.processor.cores.core.commitStats0.committedInstType::SimdReduceAdd 0 0.00% 79.81% # Class of committed instruction. (Count)

board.processor.cores.core.commitStats0.committedInstType::SimdReduceAlu 0 0.00% 79.81% # Class of committed instruction. (Count)

board.processor.cores.core.commitStats0.committedInstType::SimdReduceCmp 0 0.00% 79.81% # Class of committed instruction. (Count)

board.processor.cores.core.commitStats0.committedInstType::SimdFloatReduceAdd 0 0.00% 79.81% # Class of committed instruction. (Count)

board.processor.cores.core.commitStats0.committedInstType::SimdFloatReduceCmp 0 0.00% 79.81% # Class of committed instruction. (Count)

board.processor.cores.core.commitStats0.committedInstType::SimdAes 0 0.00% 79.81% # Class of committed instruction. (Count)

board.processor.cores.core.commitStats0.committedInstType::SimdAesMix 0 0.00% 79.81% # Class of committed instruction. (Count)

board.processor.cores.core.commitStats0.committedInstType::SimdSha1Hash 0 0.00% 79.81% # Class of committed instruction. (Count)

board.processor.cores.core.commitStats0.committedInstType::SimdSha1Hash2 0 0.00% 79.81% # Class of committed instruction. (Count)

board.processor.cores.core.commitStats0.committedInstType::SimdSha256Hash 0 0.00% 79.81% # Class of committed instruction. (Count)

board.processor.cores.core.commitStats0.committedInstType::SimdSha256Hash2 0 0.00% 79.81% # Class of committed instruction. (Count)

board.processor.cores.core.commitStats0.committedInstType::SimdShaSigma2 0 0.00% 79.81% # Class of committed instruction. (Count)

board.processor.cores.core.commitStats0.committedInstType::SimdShaSigma3 0 0.00% 79.81% # Class of committed instruction. (Count)

board.processor.cores.core.commitStats0.committedInstType::SimdPredAlu 0 0.00% 79.81% # Class of committed instruction. (Count)

board.processor.cores.core.commitStats0.committedInstType::Matrix 0 0.00% 79.81% # Class of committed instruction. (Count)

board.processor.cores.core.commitStats0.committedInstType::MatrixMov 0 0.00% 79.81% # Class of committed instruction. (Count)

board.processor.cores.core.commitStats0.committedInstType::MatrixOP 0 0.00% 79.81% # Class of committed instruction. (Count)

board.processor.cores.core.commitStats0.committedInstType::MemRead 1224 9.41% 89.22% # Class of committed instruction. (Count)

board.processor.cores.core.commitStats0.committedInstType::MemWrite 1193 9.17% 98.39% # Class of committed instruction. (Count)

board.processor.cores.core.commitStats0.committedInstType::FloatMemRead 48 0.37% 98.76% # Class of committed instruction. (Count)

board.processor.cores.core.commitStats0.committedInstType::FloatMemWrite 161 1.24% 100.00% # Class of committed instruction. (Count)

board.processor.cores.core.commitStats0.committedInstType::IprAccess 0 0.00% 100.00% # Class of committed instruction. (Count)

board.processor.cores.core.commitStats0.committedInstType::InstPrefetch 0 0.00% 100.00% # Class of committed instruction. (Count)

board.processor.cores.core.commitStats0.committedInstType::SimdUnitStrideLoad 0 0.00% 100.00% # Class of committed instruction. (Count)

board.processor.cores.core.commitStats0.committedInstType::SimdUnitStrideStore 0 0.00% 100.00% # Class of committed instruction. (Count)

board.processor.cores.core.commitStats0.committedInstType::SimdUnitStrideMaskLoad 0 0.00% 100.00% # Class of committed instruction. (Count)

board.processor.cores.core.commitStats0.committedInstType::SimdUnitStrideMaskStore 0 0.00% 100.00% # Class of committed instruction. (Count)

board.processor.cores.core.commitStats0.committedInstType::SimdStridedLoad 0 0.00% 100.00% # Class of committed instruction. (Count)

board.processor.cores.core.commitStats0.committedInstType::SimdStridedStore 0 0.00% 100.00% # Class of committed instruction. (Count)

board.processor.cores.core.commitStats0.committedInstType::SimdIndexedLoad 0 0.00% 100.00% # Class of committed instruction. (Count)

board.processor.cores.core.commitStats0.committedInstType::SimdIndexedStore 0 0.00% 100.00% # Class of committed instruction. (Count)

board.processor.cores.core.commitStats0.committedInstType::SimdWholeRegisterLoad 0 0.00% 100.00% # Class of committed instruction. (Count)

board.processor.cores.core.commitStats0.committedInstType::SimdWholeRegisterStore 0 0.00% 100.00% # Class of committed instruction. (Count)

board.processor.cores.core.commitStats0.committedInstType::SimdUnitStrideFaultOnlyFirstLoad 0 0.00% 100.00% # Class of committed instruction. (Count)

board.processor.cores.core.commitStats0.committedInstType::SimdUnitStrideSegmentedLoad 0 0.00% 100.00% # Class of committed instruction. (Count)

board.processor.cores.core.commitStats0.committedInstType::SimdUnitStrideSegmentedStore 0 0.00% 100.00% # Class of committed instruction. (Count)

board.processor.cores.core.commitStats0.committedInstType::SimdExt 0 0.00% 100.00% # Class of committed instruction. (Count)

board.processor.cores.core.commitStats0.committedInstType::SimdFloatExt 0 0.00% 100.00% # Class of committed instruction. (Count)

board.processor.cores.core.commitStats0.committedInstType::SimdConfig 0 0.00% 100.00% # Class of committed instruction. (Count)

board.processor.cores.core.commitStats0.committedInstType::total 13006 # Class of committed instruction. (Count)

board.processor.cores.core.commitStats0.committedControl::IsControl 1483 # Class of control type instructions committed (Count)

board.processor.cores.core.commitStats0.committedControl::IsDirectControl 1300 # Class of control type instructions committed (Count)

board.processor.cores.core.commitStats0.committedControl::IsIndirectControl 183 # Class of control type instructions committed (Count)

board.processor.cores.core.commitStats0.committedControl::IsCondControl 1108 # Class of control type instructions committed (Count)

board.processor.cores.core.commitStats0.committedControl::IsUncondControl 375 # Class of control type instructions committed (Count)

board.processor.cores.core.commitStats0.committedControl::IsCall 127 # Class of control type instructions committed (Count)

board.processor.cores.core.commitStats0.committedControl::IsReturn 122 # Class of control type instructions committed (Count)

board.processor.cores.core.exec_context.thread_0.numMatAluAccesses 0 # Number of matrix alu accesses (Count)

board.processor.cores.core.exec_context.thread_0.numCallsReturns 249 # Number of times a function call or return occured (Count)

board.processor.cores.core.exec_context.thread_0.numMatInsts 0 # Number of matrix instructions (Count)

board.processor.cores.core.exec_context.thread_0.numIdleCycles 0.003003 # Number of idle cycles (Cycle)

board.processor.cores.core.exec_context.thread_0.numBusyCycles 89954.996997 # Number of busy cycles (Cycle)

board.processor.cores.core.exec_context.thread_0.notIdleFraction 1.000000 # Percentage of non-idle cycles (Ratio)

board.processor.cores.core.exec_context.thread_0.idleFraction 0.000000 # Percentage of idle cycles (Ratio)

board.processor.cores.core.executeStats0.numInsts 0 # Number of executed instructions (Count)

board.processor.cores.core.executeStats0.numNop 0 # Number of nop insts executed (Count)

board.processor.cores.core.executeStats0.numBranches 0 # Number of branches executed (Count)

board.processor.cores.core.executeStats0.numLoadInsts 0 # Number of load instructions executed (Count)

board.processor.cores.core.executeStats0.numStoreInsts 2626 # Number of stores executed (Count)

board.processor.cores.core.executeStats0.instRate 0 # Inst execution rate ((Count/Cycle))

board.processor.cores.core.executeStats0.numFpAluAccesses 411 # Number of float alu accesses (Count)

board.processor.cores.core.executeStats0.numFpRegReads 563 # Number of times the floating registers were read (Count)

board.processor.cores.core.executeStats0.numFpRegWrites 202 # Number of times the floating registers were written (Count)

board.processor.cores.core.executeStats0.numIntAluAccesses 12799 # Number of integer alu accesses (Count)

board.processor.cores.core.executeStats0.numIntRegReads 14442 # Number of times the integer registers were read (Count)

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board.processor.cores.core.executeStats0.numIntRegWrites        8883                    # Number of times the integer registers were written (Count)
board.processor.cores.core.executeStats0.numMemRefs        2626                    # Number of memory refs (Count)
board.processor.cores.core.executeStats0.numMiscRegReads        5696                    # Number of times the Misc registers were read (Count)
board.processor.cores.core.executeStats0.numVecAluAccesses        0                    # Number of vector alu accesses (Count)
board.processor.cores.core.executeStats0.numDiscardedOps        0                    # Number of ops (including micro ops) which were discarded before commit (Count)
board.processor.cores.core.fetchStats0.numInsts        6568                    # Number of instructions fetched (thread level) (Count)
board.processor.cores.core.fetchStats0.numOps        13006                    # Number of ops (including micro ops) fetched (thread level) (Count)
board.processor.cores.core.fetchStats0.fetchRate        0.073014                    # Number of inst fetches per cycle ((Count/Cycle))
board.processor.cores.core.fetchStats0.numBranches        1483                    # Number of branches fetched (Count)
board.processor.cores.core.fetchStats0.branchRate        0.016486                    # Number of branch fetches per cycle (Ratio)
board.processor.cores.core.fetchStats0.numFetchSuspends        0                    # Number of times Execute suspended instruction fetching (Count)
board.processor.cores.core.interrupts.clk_domain.clock        5328                    # Clock period in ticks (Tick)
board.processor.cores.core.mmu.dtb.rdAccesses        1276                    # TLB accesses on read requests (Count)
board.processor.cores.core.mmu.dtb.wrAccesses        1354                    # TLB accesses on write requests (Count)
board.processor.cores.core.mmu.dtb.rdMisses        11                    # TLB misses on read requests (Count)
board.processor.cores.core.mmu.dtb.wrMisses        11                    # TLB misses on write requests (Count)
board.processor.cores.core.mmu.dtb.walker.power_state.pwrStateResidencyTicks::UNDEFINED        29955015                    # Cumulative time (in ticks) in various power states (Tick)
board.processor.cores.core.mmu.itb.rdAccesses        0                    # TLB accesses on read requests (Count)
board.processor.cores.core.mmu.itb.wrAccesses        8567                    # TLB accesses on write requests (Count)
board.processor.cores.core.mmu.itb.rdMisses        0                    # TLB misses on read requests (Count)
board.processor.cores.core.mmu.itb.wrMisses        39                    # TLB misses on write requests (Count)
board.processor.cores.core.mmu.itb.walker.power_state.pwrStateResidencyTicks::UNDEFINED        29955015                    # Cumulative time (in ticks) in various power states (Tick)
board.processor.cores.core.power_state.pwrStateResidencyTicks::ON        29955015                    # Cumulative time (in ticks) in various power states (Tick)
board.processor.cores.core.thread_0.numInsts        0                    # Number of Instructions committed (Count)
board.processor.cores.core.thread_0.numOps        0                    # Number of Ops committed (Count)
board.processor.cores.core.thread_0.numMemRefs        0                    # Number of Memory References (Count)
board.processor.cores.core.workload.numSyscalls        11                    # Number of system calls (Count)
board.workload.inst.arm        0                    # number of arm instructions executed (Count)
board.workload.inst.quiesce        0                    # number of quiesce instructions executed (Count)

---------- End Simulation Statistics   ----------
```

inorder_nocache:

```
------ Begin Simulation Statistics ----------
simSeconds                                   0.000461                       # Number of seconds simulated (Second)
simTicks                                     460759779                      # Number of ticks simulated (Tick)
finalTick                                    460759779                      # Number of ticks from beginning of simulation (restored from checkpoints and never reset) (Tick)
simFreq                                      1000000000000                  # The number of ticks per simulated second ((Tick/Second))
hostSeconds                                  0.03                           # Real time elapsed on the host (Second)
hostTickRate                                 15057017058                    # The number of ticks simulated per host second (ticks/s) ((Tick/Second))
hostMemory                                   1192752                        # Number of bytes of host memory used (Byte)
simInsts                                     6555                           # Number of instructions simulated (Count)
simOps                                       12992                          # Number of ops (including micro ops) simulated (Count)
hostInstRate                                 213518                         # Simulator instruction rate (inst/s) ((Count/Second))
hostOpRate                                   423137                         # Simulator op (including micro ops) rate (op/s) ((Count/Second))
board.cache_hierarchy.membus.transDist::ReadReq        9843                 # Transaction distribution (Count)
board.cache_hierarchy.membus.transDist::ReadResp       9843                 # Transaction distribution (Count)
board.cache_hierarchy.membus.transDist::WriteReq       1351                 # Transaction distribution (Count)
board.cache_hierarchy.membus.transDist::WriteResp      1351                 # Transaction distribution (Count)
board.cache_hierarchy.membus.pktCount_board.processor.cores.core.icache_port::board.memory.mem_ctrl.port       17134                 # Packet count per connected requestor and responder (Count)
board.cache_hierarchy.membus.pktCount_board.processor.cores.core.icache_port::total       17134                 # Packet count per connected requestor and responder (Count)
board.cache_hierarchy.membus.pktCount_board.processor.cores.core.dcache_port::board.memory.mem_ctrl.port       5254                 # Packet count per connected requestor and responder (Count)
board.cache_hierarchy.membus.pktCount_board.processor.cores.core.dcache_port::total       5254                 # Packet count per connected requestor and responder (Count)
board.cache_hierarchy.membus.pktCount::total       22388                 # Packet count per connected requestor and responder (Count)
board.cache_hierarchy.membus.pktSize_board.processor.cores.core.icache_port::board.memory.mem_ctrl.port       68536                 # Cumulative packet size per connected requestor and responder (Byte)
board.cache_hierarchy.membus.pktSize_board.processor.cores.core.icache_port::total       68536                 # Cumulative packet size per connected requestor and responder (Byte)
board.cache_hierarchy.membus.pktSize_board.processor.cores.core.dcache_port::board.memory.mem_ctrl.port       18600                 # Cumulative packet size per connected requestor and responder (Byte)
board.cache_hierarchy.membus.pktSize_board.processor.cores.core.dcache_port::total       18600                 # Cumulative packet size per connected requestor and responder (Byte)
board.cache_hierarchy.membus.pktSize::total       87136                 # Cumulative packet size per connected requestor and responder (Byte)
board.cache_hierarchy.membus.snoops                 0                    # Total snoops (Count)
board.cache_hierarchy.membus.snoopTraffic           0                    # Total snoop traffic (Byte)
board.cache_hierarchy.membus.snoopFanout::samples       11194                 # Request fanout histogram (Count)
board.cache_hierarchy.membus.snoopFanout::mean          0                 # Request fanout histogram (Count)
board.cache_hierarchy.membus.snoopFanout::stdev         0                 # Request fanout histogram (Count)
board.cache_hierarchy.membus.snoopFanout::underflows          0      0.00%      0.00% # Request fanout histogram (Count)
board.cache_hierarchy.membus.snoopFanout::0       11194    100.00%    100.00% # Request fanout histogram (Count)
board.cache_hierarchy.membus.snoopFanout::1          0      0.00%    100.00% # Request fanout histogram (Count)
board.cache_hierarchy.membus.snoopFanout::overflows          0      0.00%    100.00% # Request fanout histogram (Count)
board.cache_hierarchy.membus.snoopFanout::min_value          0                 # Request fanout histogram (Count)
board.cache_hierarchy.membus.snoopFanout::max_value          0                 # Request fanout histogram (Count)
board.cache_hierarchy.membus.snoopFanout::total       11194                 # Request fanout histogram (Count)
board.cache_hierarchy.membus.badaddr_responder.power_state.pwrStateResidencyTicks::UNDEFINED    460759779                 # Cumulative time (in ticks) in various power states (Tick)
board.cache_hierarchy.membus.power_state.pwrStateResidencyTicks::UNDEFINED    460759779                 # Cumulative time (in ticks) in various power states (Tick)
board.cache_hierarchy.membus.reqLayer2.occupancy      4177485                 # Layer occupancy (ticks) (Tick)
board.cache_hierarchy.membus.reqLayer2.utilization          0.0                 # Layer utilization (Ratio)
board.cache_hierarchy.membus.respLayer0.occupancy      7444528                 # Layer occupancy (ticks) (Tick)
board.cache_hierarchy.membus.respLayer0.utilization          0.0                 # Layer utilization (Ratio)
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board.cache_hierarchy.membus.respLayer1.occupancy 1992504 # Layer occupancy (ticks) (Tick)
board.cache_hierarchy.membus.respLayer1.utilization 0.0 # Layer utilization (Ratio)
board.cache_hierarchy.membus.snoop_filter.totRequests 0 # Total number of requests made to the snoop filter. (Count)
board.cache_hierarchy.membus.snoop_filter.hitSingleRequests 0 # Number of requests hitting in the snoop filter with a single holder of the requested data. (Count)
board.cache_hierarchy.membus.snoop_filter.hitMultiRequests 0 # Number of requests hitting in the snoop filter with multiple (>1) holders of the requested data. (Count)
board.cache_hierarchy.membus.snoop_filter.totSnoops 0 # Total number of snoops made to the snoop filter. (Count)
board.cache_hierarchy.membus.snoop_filter.hitSingleSnoops 0 # Number of snoops hitting in the snoop filter with a single holder of the requested data. (Count)
board.cache_hierarchy.membus.snoop_filter.hitMultiSnoops 0 # Number of snoops hitting in the snoop filter with multiple (>1) holders of the requested data. (Count)
board.clk_domain.clock 333 # Clock period in ticks (Tick)
board.clk_domain.voltage_domain.voltage 1 # Voltage in Volts (Volt)
board.memory.mem_ctrl.avgPriority_processor.cores.core.inst::samples 8567.00 # Average QoS priority value for accepted requests (Count)
board.memory.mem_ctrl.avgPriority_processor.cores.core.data::samples 1295.00 # Average QoS priority value for accepted requests (Count)
board.memory.mem_ctrl.priorityMinLatency 0.000000018750 # per QoS priority minimum request to response latency (Second)
board.memory.mem_ctrl.priorityMaxLatency 0.000271477686 # per QoS priority maximum request to response latency (Second)
board.memory.mem_ctrl.numReadWriteTurnArounds 10 # Number of turnarounds from READ to WRITE (Count)
board.memory.mem_ctrl.numWriteReadTurnArounds 10 # Number of turnarounds from WRITE to READ (Count)
board.memory.mem_ctrl.numStayReadState 20801 # Number of times bus staying in READ state (Count)
board.memory.mem_ctrl.numStayWriteState 150 # Number of times bus staying in WRITE state (Count)
board.memory.mem_ctrl.readReqs 9843 # Number of read requests accepted (Count)
board.memory.mem_ctrl.writeReqs 1351 # Number of write requests accepted (Count)
board.memory.mem_ctrl.readBursts 9843 # Number of controller read bursts, including those serviced by the write queue (Count)
board.memory.mem_ctrl.writeBursts 1351 # Number of controller write bursts, including those merged in the write queue (Count)
board.memory.mem_ctrl.servicedByWrQ 171 # Number of controller read bursts serviced by the write queue (Count)
board.memory.mem_ctrl.mergedWrBursts 1161 # Number of controller write bursts merged with an existing one (Count)
board.memory.mem_ctrl.neitherReadNorWriteReqs 0 # Number of requests that are neither read nor write (Count)
board.memory.mem_ctrl.avgRdQLen 1.00 # Average read queue length when enqueuing ((Count/Tick))
board.memory.mem_ctrl.avgWrQLen 22.87 # Average write queue length when enqueuing ((Count/Tick))
board.memory.mem_ctrl.numRdRetry 0 # Number of times read queue was full causing retry (Count)
board.memory.mem_ctrl.numWrRetry 0 # Number of times write queue was full causing retry (Count)
board.memory.mem_ctrl.readPktSize::0 87 # Read request sizes (log2) (Count)
board.memory.mem_ctrl.readPktSize::1 15 # Read request sizes (log2) (Count)
board.memory.mem_ctrl.readPktSize::2 222 # Read request sizes (log2) (Count)
board.memory.mem_ctrl.readPktSize::3 9519 # Read request sizes (log2) (Count)
board.memory.mem_ctrl.readPktSize::4 0 # Read request sizes (log2) (Count)
board.memory.mem_ctrl.readPktSize::5 0 # Read request sizes (log2) (Count)
board.memory.mem_ctrl.readPktSize::6 0 # Read request sizes (log2) (Count)
board.memory.mem_ctrl.writePktSize::0 34 # Write request sizes (log2) (Count)
board.memory.mem_ctrl.writePktSize::1 6 # Write request sizes (log2) (Count)
board.memory.mem_ctrl.writePktSize::2 137 # Write request sizes (log2) (Count)
board.memory.mem_ctrl.writePktSize::3 1174 # Write request sizes (log2) (Count)
board.memory.mem_ctrl.writePktSize::4 0 # Write request sizes (log2) (Count)
board.memory.mem_ctrl.writePktSize::5 0 # Write request sizes (log2) (Count)
board.memory.mem_ctrl.writePktSize::6 0 # Write request sizes (log2) (Count)
board.memory.mem_ctrl.rdQLenPdf::0 9668 # What read queue length does an incoming req see (Count)
board.memory.mem_ctrl.rdQLenPdf::1 4 # What read queue length does an incoming req see (Count)
board.memory.mem_ctrl.rdQLenPdf::2 0 # What read queue length does an incoming req see (Count)
board.memory.mem_ctrl.rdQLenPdf::3 0 # What read queue length does an incoming req see (Count)
board.memory.mem_ctrl.rdQLenPdf::4 0 # What read queue length does an incoming req see (Count)
board.memory.mem_ctrl.rdQLenPdf::5 0 # What read queue length does an incoming req see (Count)
```

board.memory.mem_ctrl.rdQLenPdf::6	0	# What read queue length does an incoming req see (Count)
board.memory.mem_ctrl.rdQLenPdf::7	0	# What read queue length does an incoming req see (Count)
board.memory.mem_ctrl.rdQLenPdf::8	0	# What read queue length does an incoming req see (Count)
board.memory.mem_ctrl.rdQLenPdf::9	0	# What read queue length does an incoming req see (Count)
board.memory.mem_ctrl.rdQLenPdf::10	0	# What read queue length does an incoming req see (Count)
board.memory.mem_ctrl.rdQLenPdf::11	0	# What read queue length does an incoming req see (Count)
board.memory.mem_ctrl.rdQLenPdf::12	0	# What read queue length does an incoming req see (Count)
board.memory.mem_ctrl.rdQLenPdf::13	0	# What read queue length does an incoming req see (Count)
board.memory.mem_ctrl.rdQLenPdf::14	0	# What read queue length does an incoming req see (Count)
board.memory.mem_ctrl.rdQLenPdf::15	0	# What read queue length does an incoming req see (Count)
board.memory.mem_ctrl.rdQLenPdf::16	0	# What read queue length does an incoming req see (Count)
board.memory.mem_ctrl.rdQLenPdf::17	0	# What read queue length does an incoming req see (Count)
board.memory.mem_ctrl.rdQLenPdf::18	0	# What read queue length does an incoming req see (Count)
board.memory.mem_ctrl.rdQLenPdf::19	0	# What read queue length does an incoming req see (Count)
board.memory.mem_ctrl.rdQLenPdf::20	0	# What read queue length does an incoming req see (Count)
board.memory.mem_ctrl.rdQLenPdf::21	0	# What read queue length does an incoming req see (Count)
board.memory.mem_ctrl.rdQLenPdf::22	0	# What read queue length does an incoming req see (Count)
board.memory.mem_ctrl.rdQLenPdf::23	0	# What read queue length does an incoming req see (Count)
board.memory.mem_ctrl.rdQLenPdf::24	0	# What read queue length does an incoming req see (Count)
board.memory.mem_ctrl.rdQLenPdf::25	0	# What read queue length does an incoming req see (Count)
board.memory.mem_ctrl.rdQLenPdf::26	0	# What read queue length does an incoming req see (Count)
board.memory.mem_ctrl.rdQLenPdf::27	0	# What read queue length does an incoming req see (Count)
board.memory.mem_ctrl.rdQLenPdf::28	0	# What read queue length does an incoming req see (Count)
board.memory.mem_ctrl.rdQLenPdf::29	0	# What read queue length does an incoming req see (Count)
board.memory.mem_ctrl.rdQLenPdf::30	0	# What read queue length does an incoming req see (Count)
board.memory.mem_ctrl.rdQLenPdf::31	0	# What read queue length does an incoming req see (Count)
board.memory.mem_ctrl.wrQLenPdf::0	1	# What write queue length does an incoming req see (Count)
board.memory.mem_ctrl.wrQLenPdf::1	1	# What write queue length does an incoming req see (Count)
board.memory.mem_ctrl.wrQLenPdf::2	1	# What write queue length does an incoming req see (Count)
board.memory.mem_ctrl.wrQLenPdf::3	1	# What write queue length does an incoming req see (Count)
board.memory.mem_ctrl.wrQLenPdf::4	1	# What write queue length does an incoming req see (Count)
board.memory.mem_ctrl.wrQLenPdf::5	1	# What write queue length does an incoming req see (Count)
board.memory.mem_ctrl.wrQLenPdf::6	1	# What write queue length does an incoming req see (Count)
board.memory.mem_ctrl.wrQLenPdf::7	1	# What write queue length does an incoming req see (Count)
board.memory.mem_ctrl.wrQLenPdf::8	1	# What write queue length does an incoming req see (Count)
board.memory.mem_ctrl.wrQLenPdf::9	1	# What write queue length does an incoming req see (Count)
board.memory.mem_ctrl.wrQLenPdf::10	1	# What write queue length does an incoming req see (Count)
board.memory.mem_ctrl.wrQLenPdf::11	1	# What write queue length does an incoming req see (Count)
board.memory.mem_ctrl.wrQLenPdf::12	1	# What write queue length does an incoming req see (Count)
board.memory.mem_ctrl.wrQLenPdf::13	1	# What write queue length does an incoming req see (Count)
board.memory.mem_ctrl.wrQLenPdf::14	1	# What write queue length does an incoming req see (Count)
board.memory.mem_ctrl.wrQLenPdf::15	1	# What write queue length does an incoming req see (Count)
board.memory.mem_ctrl.wrQLenPdf::16	1	# What write queue length does an incoming req see (Count)
board.memory.mem_ctrl.wrQLenPdf::17	11	# What write queue length does an incoming req see (Count)
board.memory.mem_ctrl.wrQLenPdf::18	11	# What write queue length does an incoming req see (Count)
board.memory.mem_ctrl.wrQLenPdf::19	11	# What write queue length does an incoming req see (Count)
board.memory.mem_ctrl.wrQLenPdf::20	11	# What write queue length does an incoming req see (Count)
board.memory.mem_ctrl.wrQLenPdf::21	11	# What write queue length does an incoming req see (Count)
board.memory.mem_ctrl.wrQLenPdf::22	11	# What write queue length does an incoming req see (Count)

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board.memory.mem_ctrl.wrQLenPdf::23          11                # What write queue length does an incoming req see (Count)
board.memory.mem_ctrl.wrQLenPdf::24          11                # What write queue length does an incoming req see (Count)
board.memory.mem_ctrl.wrQLenPdf::25          11                # What write queue length does an incoming req see (Count)
board.memory.mem_ctrl.wrQLenPdf::26          11                # What write queue length does an incoming req see (Count)
board.memory.mem_ctrl.wrQLenPdf::27          11                # What write queue length does an incoming req see (Count)
board.memory.mem_ctrl.wrQLenPdf::28          11                # What write queue length does an incoming req see (Count)
board.memory.mem_ctrl.wrQLenPdf::29          11                # What write queue length does an incoming req see (Count)
board.memory.mem_ctrl.wrQLenPdf::30          10                # What write queue length does an incoming req see (Count)
board.memory.mem_ctrl.wrQLenPdf::31          10                # What write queue length does an incoming req see (Count)
board.memory.mem_ctrl.wrQLenPdf::32          10                # What write queue length does an incoming req see (Count)
board.memory.mem_ctrl.wrQLenPdf::33           0                # What write queue length does an incoming req see (Count)
board.memory.mem_ctrl.wrQLenPdf::34           0                # What write queue length does an incoming req see (Count)
board.memory.mem_ctrl.wrQLenPdf::35           0                # What write queue length does an incoming req see (Count)
board.memory.mem_ctrl.wrQLenPdf::36           0                # What write queue length does an incoming req see (Count)
board.memory.mem_ctrl.wrQLenPdf::37           0                # What write queue length does an incoming req see (Count)
board.memory.mem_ctrl.wrQLenPdf::38           0                # What write queue length does an incoming req see (Count)
board.memory.mem_ctrl.wrQLenPdf::39           0                # What write queue length does an incoming req see (Count)
board.memory.mem_ctrl.wrQLenPdf::40           0                # What write queue length does an incoming req see (Count)
board.memory.mem_ctrl.wrQLenPdf::41           0                # What write queue length does an incoming req see (Count)
board.memory.mem_ctrl.wrQLenPdf::42           0                # What write queue length does an incoming req see (Count)
board.memory.mem_ctrl.wrQLenPdf::43           0                # What write queue length does an incoming req see (Count)
board.memory.mem_ctrl.wrQLenPdf::44           0                # What write queue length does an incoming req see (Count)
board.memory.mem_ctrl.wrQLenPdf::45           0                # What write queue length does an incoming req see (Count)
board.memory.mem_ctrl.wrQLenPdf::46           0                # What write queue length does an incoming req see (Count)
board.memory.mem_ctrl.wrQLenPdf::47           0                # What write queue length does an incoming req see (Count)
board.memory.mem_ctrl.wrQLenPdf::48           0                # What write queue length does an incoming req see (Count)
board.memory.mem_ctrl.wrQLenPdf::49           0                # What write queue length does an incoming req see (Count)
board.memory.mem_ctrl.wrQLenPdf::50           0                # What write queue length does an incoming req see (Count)
board.memory.mem_ctrl.wrQLenPdf::51           0                # What write queue length does an incoming req see (Count)
board.memory.mem_ctrl.wrQLenPdf::52           0                # What write queue length does an incoming req see (Count)
board.memory.mem_ctrl.wrQLenPdf::53           0                # What write queue length does an incoming req see (Count)
board.memory.mem_ctrl.wrQLenPdf::54           0                # What write queue length does an incoming req see (Count)
board.memory.mem_ctrl.wrQLenPdf::55           0                # What write queue length does an incoming req see (Count)
board.memory.mem_ctrl.wrQLenPdf::56           0                # What write queue length does an incoming req see (Count)
board.memory.mem_ctrl.wrQLenPdf::57           0                # What write queue length does an incoming req see (Count)
board.memory.mem_ctrl.wrQLenPdf::58           0                # What write queue length does an incoming req see (Count)
board.memory.mem_ctrl.wrQLenPdf::59           0                # What write queue length does an incoming req see (Count)
board.memory.mem_ctrl.wrQLenPdf::60           0                # What write queue length does an incoming req see (Count)
board.memory.mem_ctrl.wrQLenPdf::61           0                # What write queue length does an incoming req see (Count)
board.memory.mem_ctrl.wrQLenPdf::62           0                # What write queue length does an incoming req see (Count)
board.memory.mem_ctrl.wrQLenPdf::63           0                # What write queue length does an incoming req see (Count)
board.memory.mem_ctrl.rdPerTurnAround::samples          10                # Reads before turning the bus around for writes (Count)
board.memory.mem_ctrl.rdPerTurnAround::mean   756.100000                # Reads before turning the bus around for writes (Count)
board.memory.mem_ctrl.rdPerTurnAround::gmean   618.742320                # Reads before turning the bus around for writes (Count)
board.memory.mem_ctrl.rdPerTurnAround::stdev   479.052178                # Reads before turning the bus around for writes (Count)
board.memory.mem_ctrl.rdPerTurnAround::128-191           1     10.00%     10.00% # Reads before turning the bus around for writes (Count)
board.memory.mem_ctrl.rdPerTurnAround::320-383           1     10.00%     20.00% # Reads before turning the bus around for writes (Count)
board.memory.mem_ctrl.rdPerTurnAround::384-447           1     10.00%     30.00% # Reads before turning the bus around for writes (Count)
board.memory.mem_ctrl.rdPerTurnAround::448-511           1     10.00%     40.00% # Reads before turning the bus around for writes (Count)
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board.memory.mem_ctrl.rdPerTurnAround::512-575          1     10.00%     50.00% # Reads before turning the bus around for writes (Count)
board.memory.mem_ctrl.rdPerTurnAround::704-767          1     10.00%     60.00% # Reads before turning the bus around for writes (Count)
board.memory.mem_ctrl.rdPerTurnAround::768-831          1     10.00%     70.00% # Reads before turning the bus around for writes (Count)
board.memory.mem_ctrl.rdPerTurnAround::1152-1215        1     10.00%     80.00% # Reads before turning the bus around for writes (Count)
board.memory.mem_ctrl.rdPerTurnAround::1344-1407        1     10.00%     90.00% # Reads before turning the bus around for writes (Count)
board.memory.mem_ctrl.rdPerTurnAround::1600-1663        1     10.00%    100.00% # Reads before turning the bus around for writes (Count)
board.memory.mem_ctrl.rdPerTurnAround::total           10                     # Reads before turning the bus around for writes (Count)
board.memory.mem_ctrl.wrPerTurnAround::samples         10                     # Writes before turning the bus around for reads (Count)
board.memory.mem_ctrl.wrPerTurnAround::mean            16                     # Writes before turning the bus around for reads (Count)
board.memory.mem_ctrl.wrPerTurnAround::gmean    16.000000                     # Writes before turning the bus around for reads (Count)
board.memory.mem_ctrl.wrPerTurnAround::16              10    100.00%    100.00% # Writes before turning the bus around for reads (Count)
board.memory.mem_ctrl.wrPerTurnAround::total           10                     # Writes before turning the bus around for reads (Count)
board.memory.mem_ctrl.bytesReadWrQ                 10944                     # Total number of bytes read from write queue (Byte)
board.memory.mem_ctrl.bytesReadSys                 77150                     # Total read bytes from the system interface side (Byte)
board.memory.mem_ctrl.bytesWrittenSys               9986                     # Total written bytes from the system interface side (Byte)
board.memory.mem_ctrl.avgRdBWSys        167440830.37681985                   # Average system read bandwidth in Byte/s ((Byte/Second))
board.memory.mem_ctrl.avgWrBWSys         21672898.66679097                   # Average system write bandwidth in Byte/s ((Byte/Second))
board.memory.mem_ctrl.totGap                   460647891                     # Total gap between requests (Tick)
board.memory.mem_ctrl.avgGap                    41151.32                     # Average gap between requests ((Tick/Count))
board.memory.mem_ctrl.requestorReadBytes::processor.cores.core.inst       68536                     # Per-requestor bytes read from memory (Byte)
board.memory.mem_ctrl.requestorReadBytes::processor.cores.core.data        7477                     # Per-requestor bytes read from memory (Byte)
board.memory.mem_ctrl.requestorWriteBytes::processor.cores.core.data        1171                     # Per-requestor bytes write to memory (Byte)
board.memory.mem_ctrl.requestorReadRate::processor.cores.core.inst 148745622.173761814833                     # Per-requestor bytes read from memory rate ((Byte/Second))
board.memory.mem_ctrl.requestorReadRate::processor.cores.core.data 16227544.896014023572                     # Per-requestor bytes read from memory rate ((Byte/Second))
board.memory.mem_ctrl.requestorWriteRate::processor.cores.core.data 2541454.470139417332                     # Per-requestor bytes write to memory rate ((Byte/Second))
board.memory.mem_ctrl.requestorReadAccesses::processor.cores.core.inst        8567                     # Per-requestor read serviced memory accesses (Count)
board.memory.mem_ctrl.requestorReadAccesses::processor.cores.core.data        1276                     # Per-requestor read serviced memory accesses (Count)
board.memory.mem_ctrl.requestorWriteAccesses::processor.cores.core.data        1351                     # Per-requestor write serviced memory accesses (Count)
board.memory.mem_ctrl.requestorReadTotalLat::processor.cores.core.inst    185265724                     # Per-requestor read total memory access latency (Tick)
board.memory.mem_ctrl.requestorReadTotalLat::processor.cores.core.data     26871579                     # Per-requestor read total memory access latency (Tick)
board.memory.mem_ctrl.requestorWriteTotalLat::processor.cores.core.data   7586762200                     # Per-requestor write total memory access latency (Tick)
board.memory.mem_ctrl.requestorReadAvgLat::processor.cores.core.inst     21625.51                     # Per-requestor read average memory access latency ((Tick/Count))
board.memory.mem_ctrl.requestorReadAvgLat::processor.cores.core.data     21059.23                     # Per-requestor read average memory access latency ((Tick/Count))
board.memory.mem_ctrl.requestorWriteAvgLat::processor.cores.core.data   5615664.10                     # Per-requestor write average memory access latency ((Tick/Count))
board.memory.mem_ctrl.dram.bytesRead::processor.cores.core.inst       68536                     # Number of bytes read from this memory (Byte)
board.memory.mem_ctrl.dram.bytesRead::processor.cores.core.data        8614                     # Number of bytes read from this memory (Byte)
board.memory.mem_ctrl.dram.bytesRead::total        77150                     # Number of bytes read from this memory (Byte)
board.memory.mem_ctrl.dram.bytesInstRead::processor.cores.core.inst       68536                     # Number of instructions bytes read from this memory (Byte)
board.memory.mem_ctrl.dram.bytesInstRead::total        68536                     # Number of instructions bytes read from this memory (Byte)
board.memory.mem_ctrl.dram.bytesWritten::processor.cores.core.data        9986                     # Number of bytes written to this memory (Byte)
board.memory.mem_ctrl.dram.bytesWritten::total        9986                     # Number of bytes written to this memory (Byte)
board.memory.mem_ctrl.dram.numReads::processor.cores.core.inst        8567                     # Number of read requests responded to by this memory (Count)
board.memory.mem_ctrl.dram.numReads::processor.cores.core.data        1276                     # Number of read requests responded to by this memory (Count)
board.memory.mem_ctrl.dram.numReads::total        9843                     # Number of read requests responded to by this memory (Count)
board.memory.mem_ctrl.dram.numWrites::processor.cores.core.data        1351                     # Number of write requests responded to by this memory (Count)
board.memory.mem_ctrl.dram.numWrites::total        1351                     # Number of write requests responded to by this memory (Count)
board.memory.mem_ctrl.dram.bwRead::processor.cores.core.inst    148745622                     # Total read bandwidth from this memory ((Byte/Second))
board.memory.mem_ctrl.dram.bwRead::processor.cores.core.data     18695208                     # Total read bandwidth from this memory ((Byte/Second))
board.memory.mem_ctrl.dram.bwRead::total    167440830                     # Total read bandwidth from this memory ((Byte/Second))
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board.memory.mem_ctrl.dram.bwInstRead::processor.cores.core.inst    148745622                    # Instruction read bandwidth from this memory ((Byte/Second))
board.memory.mem_ctrl.dram.bwInstRead::total    148745622                    # Instruction read bandwidth from this memory ((Byte/Second))
board.memory.mem_ctrl.dram.bwWrite::processor.cores.core.data    21672899                    # Write bandwidth from this memory ((Byte/Second))
board.memory.mem_ctrl.dram.bwWrite::total    21672899                    # Write bandwidth from this memory ((Byte/Second))
board.memory.mem_ctrl.dram.bwTotal::processor.cores.core.inst    148745622                    # Total bandwidth to/from this memory ((Byte/Second))
board.memory.mem_ctrl.dram.bwTotal::processor.cores.core.data    40368107                    # Total bandwidth to/from this memory ((Byte/Second))
board.memory.mem_ctrl.dram.bwTotal::total    189113729                    # Total bandwidth to/from this memory ((Byte/Second))
board.memory.mem_ctrl.dram.readBursts    9672                    # Number of DRAM read bursts (Count)
board.memory.mem_ctrl.dram.writeBursts    160                    # Number of DRAM write bursts (Count)
board.memory.mem_ctrl.dram.perBankRdBursts::0    1325                    # Per bank write bursts (Count)
board.memory.mem_ctrl.dram.perBankRdBursts::1    0                    # Per bank write bursts (Count)
board.memory.mem_ctrl.dram.perBankRdBursts::2    0                    # Per bank write bursts (Count)
board.memory.mem_ctrl.dram.perBankRdBursts::3    572                    # Per bank write bursts (Count)
board.memory.mem_ctrl.dram.perBankRdBursts::4    130                    # Per bank write bursts (Count)
board.memory.mem_ctrl.dram.perBankRdBursts::5    197                    # Per bank write bursts (Count)
board.memory.mem_ctrl.dram.perBankRdBursts::6    1404                    # Per bank write bursts (Count)
board.memory.mem_ctrl.dram.perBankRdBursts::7    784                    # Per bank write bursts (Count)
board.memory.mem_ctrl.dram.perBankRdBursts::8    1431                    # Per bank write bursts (Count)
board.memory.mem_ctrl.dram.perBankRdBursts::9    729                    # Per bank write bursts (Count)
board.memory.mem_ctrl.dram.perBankRdBursts::10    1184                    # Per bank write bursts (Count)
board.memory.mem_ctrl.dram.perBankRdBursts::11    794                    # Per bank write bursts (Count)
board.memory.mem_ctrl.dram.perBankRdBursts::12    97                    # Per bank write bursts (Count)
board.memory.mem_ctrl.dram.perBankRdBursts::13    770                    # Per bank write bursts (Count)
board.memory.mem_ctrl.dram.perBankRdBursts::14    14                    # Per bank write bursts (Count)
board.memory.mem_ctrl.dram.perBankRdBursts::15    241                    # Per bank write bursts (Count)
board.memory.mem_ctrl.dram.perBankWrBursts::0    0                    # Per bank write bursts (Count)
board.memory.mem_ctrl.dram.perBankWrBursts::1    0                    # Per bank write bursts (Count)
board.memory.mem_ctrl.dram.perBankWrBursts::2    0                    # Per bank write bursts (Count)
board.memory.mem_ctrl.dram.perBankWrBursts::3    0                    # Per bank write bursts (Count)
board.memory.mem_ctrl.dram.perBankWrBursts::4    0                    # Per bank write bursts (Count)
board.memory.mem_ctrl.dram.perBankWrBursts::5    0                    # Per bank write bursts (Count)
board.memory.mem_ctrl.dram.perBankWrBursts::6    0                    # Per bank write bursts (Count)
board.memory.mem_ctrl.dram.perBankWrBursts::7    0                    # Per bank write bursts (Count)
board.memory.mem_ctrl.dram.perBankWrBursts::8    0                    # Per bank write bursts (Count)
board.memory.mem_ctrl.dram.perBankWrBursts::9    47                    # Per bank write bursts (Count)
board.memory.mem_ctrl.dram.perBankWrBursts::10    29                    # Per bank write bursts (Count)
board.memory.mem_ctrl.dram.perBankWrBursts::11    46                    # Per bank write bursts (Count)
board.memory.mem_ctrl.dram.perBankWrBursts::12    4                    # Per bank write bursts (Count)
board.memory.mem_ctrl.dram.perBankWrBursts::13    34                    # Per bank write bursts (Count)
board.memory.mem_ctrl.dram.perBankWrBursts::14    0                    # Per bank write bursts (Count)
board.memory.mem_ctrl.dram.perBankWrBursts::15    0                    # Per bank write bursts (Count)
board.memory.mem_ctrl.dram.totQLat    30787303                    # Total ticks spent queuing (Tick)
board.memory.mem_ctrl.dram.totBusLat    48360000                    # Total ticks spent in databus transfers (Tick)
board.memory.mem_ctrl.dram.totMemAccLat    212137303                    # Total ticks spent from burst creation until serviced by the DRAM (Tick)
board.memory.mem_ctrl.dram.avgQLat    3183.14                    # Average queueing delay per DRAM burst ((Tick/Count))
board.memory.mem_ctrl.dram.avgBusLat    5000.00                    # Average bus latency per DRAM burst ((Tick/Count))
board.memory.mem_ctrl.dram.avgMemAccLat    21933.14                    # Average memory access latency per DRAM burst ((Tick/Count))
board.memory.mem_ctrl.dram.readRowHits    8758                    # Number of row buffer hits during reads (Count)
board.memory.mem_ctrl.dram.writeRowHits    146                    # Number of row buffer hits during writes (Count)
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board.memory.mem_ctrl.dram.readRowHitRate        90.55                    # Row buffer hit rate for reads (Ratio)
board.memory.mem_ctrl.dram.writeRowHitRate        91.25                    # Row buffer hit rate for writes (Ratio)
board.memory.mem_ctrl.dram.bytesPerActivate::samples          924                    # Bytes accessed per row activation (Byte)
board.memory.mem_ctrl.dram.bytesPerActivate::mean   679.341991                    # Bytes accessed per row activation (Byte)
board.memory.mem_ctrl.dram.bytesPerActivate::gmean   490.908481                    # Bytes accessed per row activation (Byte)
board.memory.mem_ctrl.dram.bytesPerActivate::stdev   386.424070                    # Bytes accessed per row activation (Byte)
board.memory.mem_ctrl.dram.bytesPerActivate::0-127          112     12.12%     12.12% # Bytes accessed per row activation (Byte)
board.memory.mem_ctrl.dram.bytesPerActivate::128-255          83      8.98%     21.10% # Bytes accessed per row activation (Byte)
board.memory.mem_ctrl.dram.bytesPerActivate::256-383          82      8.87%     29.98% # Bytes accessed per row activation (Byte)
board.memory.mem_ctrl.dram.bytesPerActivate::384-511          46      4.98%     34.96% # Bytes accessed per row activation (Byte)
board.memory.mem_ctrl.dram.bytesPerActivate::512-639          48      5.19%     40.15% # Bytes accessed per row activation (Byte)
board.memory.mem_ctrl.dram.bytesPerActivate::640-767          48      5.19%     45.35% # Bytes accessed per row activation (Byte)
board.memory.mem_ctrl.dram.bytesPerActivate::768-895          34      3.68%     49.03% # Bytes accessed per row activation (Byte)
board.memory.mem_ctrl.dram.bytesPerActivate::896-1023          20      2.16%     51.19% # Bytes accessed per row activation (Byte)
board.memory.mem_ctrl.dram.bytesPerActivate::1024-1151          451     48.81%    100.00% # Bytes accessed per row activation (Byte)
board.memory.mem_ctrl.dram.bytesPerActivate::total          924                    # Bytes accessed per row activation (Byte)
board.memory.mem_ctrl.dram.dramBytesRead        619008                    # Total bytes read (Byte)
board.memory.mem_ctrl.dram.dramBytesWritten        10240                    # Total bytes written (Byte)
board.memory.mem_ctrl.dram.avgRdBW        1343.450597                    # Average DRAM read bandwidth in MiBytes/s ((Byte/Second))
board.memory.mem_ctrl.dram.avgWrBW          22.224162                    # Average DRAM write bandwidth in MiBytes/s ((Byte/Second))
board.memory.mem_ctrl.dram.peakBW           12800.00                    # Theoretical peak bandwidth in MiByte/s ((Byte/Second))
board.memory.mem_ctrl.dram.busUtil           10.67                    # Data bus utilization in percentage (Ratio)
board.memory.mem_ctrl.dram.busUtilRead          10.50                    # Data bus utilization in percentage for reads (Ratio)
board.memory.mem_ctrl.dram.busUtilWrite          0.17                    # Data bus utilization in percentage for writes (Ratio)
board.memory.mem_ctrl.dram.pageHitRate          90.56                    # Row buffer hit rate, read and write combined (Ratio)
board.memory.mem_ctrl.dram.power_state.pwrStateResidencyTicks::UNDEFINED    460759779                    # Cumulative time (in ticks) in various power states (Tick)
board.memory.mem_ctrl.dram.rank0.actEnergy      2391900                    # Energy for activate commands per rank (pJ) (Joule)
board.memory.mem_ctrl.dram.rank0.preEnergy      1263735                    # Energy for precharge commands per rank (pJ) (Joule)
board.memory.mem_ctrl.dram.rank0.readEnergy     31501680                    # Energy for read commands per rank (pJ) (Joule)
board.memory.mem_ctrl.dram.rank0.writeEnergy           0                    # Energy for write commands per rank (pJ) (Joule)
board.memory.mem_ctrl.dram.rank0.refreshEnergy 36263760.000000                    # Energy for refresh commands per rank (pJ) (Joule)
board.memory.mem_ctrl.dram.rank0.actBackEnergy    146368020                    # Energy for active background per rank (pJ) (Joule)
board.memory.mem_ctrl.dram.rank0.preBackEnergy     53674560                    # Energy for precharge background per rank (pJ) (Joule)
board.memory.mem_ctrl.dram.rank0.actPowerDownEnergy           0                    # Energy for active power-down per rank (pJ) (Joule)
board.memory.mem_ctrl.dram.rank0.prePowerDownEnergy           0                    # Energy for precharge power-down per rank (pJ) (Joule)
board.memory.mem_ctrl.dram.rank0.selfRefreshEnergy           0                    # Energy for self refresh per rank (pJ) (Joule)
board.memory.mem_ctrl.dram.rank0.totalEnergy    271463655                    # Total energy per rank (pJ) (Joule)
board.memory.mem_ctrl.dram.rank0.averagePower   589.165260                    # Core power per rank (mW) (Watt)
board.memory.mem_ctrl.dram.rank0.totalIdleTime           0                    # Total Idle time Per DRAM Rank (Tick)
board.memory.mem_ctrl.dram.rank0.pwrStateTime::IDLE    136590115                    # Time in different power states (Tick)
board.memory.mem_ctrl.dram.rank0.pwrStateTime::REF     15340000                    # Time in different power states (Tick)
board.memory.mem_ctrl.dram.rank0.pwrStateTime::SREF           0                    # Time in different power states (Tick)
board.memory.mem_ctrl.dram.rank0.pwrStateTime::PRE_PDN           0                    # Time in different power states (Tick)
board.memory.mem_ctrl.dram.rank0.pwrStateTime::ACT    308829664                    # Time in different power states (Tick)
board.memory.mem_ctrl.dram.rank0.pwrStateTime::ACT_PDN           0                    # Time in different power states (Tick)
board.memory.mem_ctrl.dram.rank1.actEnergy      4234020                    # Energy for activate commands per rank (pJ) (Joule)
board.memory.mem_ctrl.dram.rank1.preEnergy      2242845                    # Energy for precharge commands per rank (pJ) (Joule)
board.memory.mem_ctrl.dram.rank1.readEnergy     37556400                    # Energy for read commands per rank (pJ) (Joule)
board.memory.mem_ctrl.dram.rank1.writeEnergy       835200                    # Energy for write commands per rank (pJ) (Joule)
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board.memory.mem_ctrl.dram.rank1.refreshEnergy 36263760.000000 # Energy for refresh commands per rank (pJ) (Joule)
board.memory.mem_ctrl.dram.rank1.actBackEnergy 194889270 # Energy for active background per rank (pJ) (Joule)
board.memory.mem_ctrl.dram.rank1.preBackEnergy 12814560 # Energy for precharge background per rank (pJ) (Joule)
board.memory.mem_ctrl.dram.rank1.actPowerDownEnergy 0 # Energy for active power-down per rank (pJ) (Joule)
board.memory.mem_ctrl.dram.rank1.prePowerDownEnergy 0 # Energy for precharge power-down per rank (pJ) (Joule)
board.memory.mem_ctrl.dram.rank1.selfRefreshEnergy 0 # Energy for self refresh per rank (pJ) (Joule)
board.memory.mem_ctrl.dram.rank1.totalEnergy 288836055 # Total energy per rank (pJ) (Joule)
board.memory.mem_ctrl.dram.rank1.averagePower 626.869072 # Core power per rank (mW) (Watt)
board.memory.mem_ctrl.dram.rank1.totalIdleTime 0 # Total Idle time Per DRAM Rank (Tick)
board.memory.mem_ctrl.dram.rank1.pwrStateTime::IDLE 31404571 # Time in different power states (Tick)
board.memory.mem_ctrl.dram.rank1.pwrStateTime::REF 15340000 # Time in different power states (Tick)
board.memory.mem_ctrl.dram.rank1.pwrStateTime::SREF 0 # Time in different power states (Tick)
board.memory.mem_ctrl.dram.rank1.pwrStateTime::PRE_PDN 0 # Time in different power states (Tick)
board.memory.mem_ctrl.dram.rank1.pwrStateTime::ACT 414015208 # Time in different power states (Tick)
board.memory.mem_ctrl.dram.rank1.pwrStateTime::ACT_PDN 0 # Time in different power states (Tick)
board.memory.mem_ctrl.power_state.pwrStateResidencyTicks::UNDEFINED 460759779 # Cumulative time (in ticks) in various power states (Tick)
board.processor.cores.core.numCycles 1383663 # Number of cpu cycles simulated (Cycle)
board.processor.cores.core.cpi 210.667326 # CPI: cycles per instruction (core level) ((Cycle/Count))
board.processor.cores.core.ipc 0.004747 # IPC: instructions per cycle (core level) ((Count/Cycle))
board.processor.cores.core.numWorkItemsStarted 0 # Number of work items this cpu started (Count)
board.processor.cores.core.numWorkItemsCompleted 0 # Number of work items this cpu completed (Count)
board.processor.cores.core.commitStats0.numInsts 6568 # Number of instructions committed (thread level) (Count)
board.processor.cores.core.commitStats0.numOps 13006 # Number of ops (including micro ops) committed (thread level) (Count)
board.processor.cores.core.commitStats0.numInstsNotNOP 0 # Number of instructions committed excluding NOPs or prefetches (Count)
board.processor.cores.core.commitStats0.numOpsNotNOP 0 # Number of Ops (including micro ops) Simulated (Count)
board.processor.cores.core.commitStats0.cpi 210.667326 # CPI: cycles per instruction (thread level) ((Cycle/Count))
board.processor.cores.core.commitStats0.ipc 0.004747 # IPC: instructions per cycle (thread level) ((Count/Cycle))
board.processor.cores.core.commitStats0.numMemRefs 0 # Number of memory references committed (Count)
board.processor.cores.core.commitStats0.numFpInsts 411 # Number of float instructions (Count)
board.processor.cores.core.commitStats0.numIntInsts 12799 # Number of integer instructions (Count)
board.processor.cores.core.commitStats0.numLoadInsts 1272 # Number of load instructions (Count)
board.processor.cores.core.commitStats0.numStoreInsts 1354 # Number of store instructions (Count)
board.processor.cores.core.commitStats0.numVecInsts 0 # Number of vector instructions (Count)
board.processor.cores.core.commitStats0.committedInstType::No_OpClass 27 0.21% 0.21% # Class of committed instruction. (Count)
board.processor.cores.core.commitStats0.committedInstType::IntAlu 10134 77.92% 78.13% # Class of committed instruction. (Count)
board.processor.cores.core.commitStats0.committedInstType::IntMult 4 0.03% 78.16% # Class of committed instruction. (Count)
board.processor.cores.core.commitStats0.committedInstType::IntDiv 21 0.16% 78.32% # Class of committed instruction. (Count)
board.processor.cores.core.commitStats0.committedInstType::FloatAdd 2 0.02% 78.33% # Class of committed instruction. (Count)
board.processor.cores.core.commitStats0.committedInstType::FloatCmp 0 0.00% 78.33% # Class of committed instruction. (Count)
board.processor.cores.core.commitStats0.committedInstType::FloatCvt 0 0.00% 78.33% # Class of committed instruction. (Count)
board.processor.cores.core.commitStats0.committedInstType::FloatMult 0 0.00% 78.33% # Class of committed instruction. (Count)
board.processor.cores.core.commitStats0.committedInstType::FloatMultAcc 0 0.00% 78.33% # Class of committed instruction. (Count)
board.processor.cores.core.commitStats0.committedInstType::FloatDiv 0 0.00% 78.33% # Class of committed instruction. (Count)
board.processor.cores.core.commitStats0.committedInstType::FloatMisc 0 0.00% 78.33% # Class of committed instruction. (Count)
board.processor.cores.core.commitStats0.committedInstType::FloatSqrt 0 0.00% 78.33% # Class of committed instruction. (Count)
board.processor.cores.core.commitStats0.committedInstType::SimdAdd 0 0.00% 78.33% # Class of committed instruction. (Count)
board.processor.cores.core.commitStats0.committedInstType::SimdAddAcc 0 0.00% 78.33% # Class of committed instruction. (Count)
board.processor.cores.core.commitStats0.committedInstType::SimdAlu 82 0.63% 78.96% # Class of committed instruction. (Count)
board.processor.cores.core.commitStats0.committedInstType::SimdCmp 0 0.00% 78.96% # Class of committed instruction. (Count)
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board.processor.cores.core.commitStats0.committedInstType::SimdCvt 48 0.37% 79.33% # Class of committed instruction. (Count)

board.processor.cores.core.commitStats0.committedInstType::SimdMisc 62 0.48% 79.81% # Class of committed instruction. (Count)

board.processor.cores.core.commitStats0.committedInstType::SimdMult 0 0.00% 79.81% # Class of committed instruction. (Count)

board.processor.cores.core.commitStats0.committedInstType::SimdMultAcc 0 0.00% 79.81% # Class of committed instruction. (Count)

board.processor.cores.core.commitStats0.committedInstType::SimdMatMultAcc 0 0.00% 79.81% # Class of committed instruction. (Count)

board.processor.cores.core.commitStats0.committedInstType::SimdShift 0 0.00% 79.81% # Class of committed instruction. (Count)

board.processor.cores.core.commitStats0.committedInstType::SimdShiftAcc 0 0.00% 79.81% # Class of committed instruction. (Count)

board.processor.cores.core.commitStats0.committedInstType::SimdDiv 0 0.00% 79.81% # Class of committed instruction. (Count)

board.processor.cores.core.commitStats0.committedInstType::SimdSqrt 0 0.00% 79.81% # Class of committed instruction. (Count)

board.processor.cores.core.commitStats0.committedInstType::SimdFloatAdd 0 0.00% 79.81% # Class of committed instruction. (Count)

board.processor.cores.core.commitStats0.committedInstType::SimdFloatAlu 0 0.00% 79.81% # Class of committed instruction. (Count)

board.processor.cores.core.commitStats0.committedInstType::SimdFloatCmp 0 0.00% 79.81% # Class of committed instruction. (Count)

board.processor.cores.core.commitStats0.committedInstType::SimdFloatCvt 0 0.00% 79.81% # Class of committed instruction. (Count)

board.processor.cores.core.commitStats0.committedInstType::SimdFloatDiv 0 0.00% 79.81% # Class of committed instruction. (Count)

board.processor.cores.core.commitStats0.committedInstType::SimdFloatMisc 0 0.00% 79.81% # Class of committed instruction. (Count)

board.processor.cores.core.commitStats0.committedInstType::SimdFloatMult 0 0.00% 79.81% # Class of committed instruction. (Count)

board.processor.cores.core.commitStats0.committedInstType::SimdFloatMultAcc 0 0.00% 79.81% # Class of committed instruction. (Count)

board.processor.cores.core.commitStats0.committedInstType::SimdFloatMatMultAcc 0 0.00% 79.81% # Class of committed instruction. (Count)

board.processor.cores.core.commitStats0.committedInstType::SimdFloatSqrt 0 0.00% 79.81% # Class of committed instruction. (Count)

board.processor.cores.core.commitStats0.committedInstType::SimdReduceAdd 0 0.00% 79.81% # Class of committed instruction. (Count)

board.processor.cores.core.commitStats0.committedInstType::SimdReduceAlu 0 0.00% 79.81% # Class of committed instruction. (Count)

board.processor.cores.core.commitStats0.committedInstType::SimdReduceCmp 0 0.00% 79.81% # Class of committed instruction. (Count)

board.processor.cores.core.commitStats0.committedInstType::SimdFloatReduceAdd 0 0.00% 79.81% # Class of committed instruction. (Count)

board.processor.cores.core.commitStats0.committedInstType::SimdFloatReduceCmp 0 0.00% 79.81% # Class of committed instruction. (Count)

board.processor.cores.core.commitStats0.committedInstType::SimdAes 0 0.00% 79.81% # Class of committed instruction. (Count)

board.processor.cores.core.commitStats0.committedInstType::SimdAesMix 0 0.00% 79.81% # Class of committed instruction. (Count)

board.processor.cores.core.commitStats0.committedInstType::SimdSha1Hash 0 0.00% 79.81% # Class of committed instruction. (Count)

board.processor.cores.core.commitStats0.committedInstType::SimdSha1Hash2 0 0.00% 79.81% # Class of committed instruction. (Count)

board.processor.cores.core.commitStats0.committedInstType::SimdSha256Hash 0 0.00% 79.81% # Class of committed instruction. (Count)

board.processor.cores.core.commitStats0.committedInstType::SimdSha256Hash2 0 0.00% 79.81% # Class of committed instruction. (Count)

board.processor.cores.core.commitStats0.committedInstType::SimdShaSigma2 0 0.00% 79.81% # Class of committed instruction. (Count)

board.processor.cores.core.commitStats0.committedInstType::SimdShaSigma3 0 0.00% 79.81% # Class of committed instruction. (Count)

board.processor.cores.core.commitStats0.committedInstType::SimdPredAlu 0 0.00% 79.81% # Class of committed instruction. (Count)

board.processor.cores.core.commitStats0.committedInstType::Matrix 0 0.00% 79.81% # Class of committed instruction. (Count)

board.processor.cores.core.commitStats0.committedInstType::MatrixMov 0 0.00% 79.81% # Class of committed instruction. (Count)

board.processor.cores.core.commitStats0.committedInstType::MatrixOP 0 0.00% 79.81% # Class of committed instruction. (Count)

board.processor.cores.core.commitStats0.committedInstType::MemRead 1224 9.41% 89.22% # Class of committed instruction. (Count)

board.processor.cores.core.commitStats0.committedInstType::MemWrite 1193 9.17% 98.39% # Class of committed instruction. (Count)

board.processor.cores.core.commitStats0.committedInstType::FloatMemRead 48 0.37% 98.76% # Class of committed instruction. (Count)

board.processor.cores.core.commitStats0.committedInstType::FloatMemWrite 161 1.24% 100.00% # Class of committed instruction. (Count)

board.processor.cores.core.commitStats0.committedInstType::IprAccess 0 0.00% 100.00% # Class of committed instruction. (Count)

board.processor.cores.core.commitStats0.committedInstType::InstPrefetch 0 0.00% 100.00% # Class of committed instruction. (Count)

board.processor.cores.core.commitStats0.committedInstType::SimdUnitStrideLoad 0 0.00% 100.00% # Class of committed instruction. (Count)

board.processor.cores.core.commitStats0.committedInstType::SimdUnitStrideStore 0 0.00% 100.00% # Class of committed instruction. (Count)

board.processor.cores.core.commitStats0.committedInstType::SimdUnitStrideMaskLoad 0 0.00% 100.00% # Class of committed instruction. (Count)

board.processor.cores.core.commitStats0.committedInstType::SimdUnitStrideMaskStore 0 0.00% 100.00% # Class of committed instruction. (Count)

board.processor.cores.core.commitStats0.committedInstType::SimdStridedLoad 0 0.00% 100.00% # Class of committed instruction. (Count)

board.processor.cores.core.commitStats0.committedInstType::SimdStridedStore 0 0.00% 100.00% # Class of committed instruction. (Count)

board.processor.cores.core.commitStats0.committedInstType::SimdIndexedLoad 0 0.00% 100.00% # Class of committed instruction. (Count)

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board.processor.cores.core.commitStats0.committedInstType::SimdIndexedStore 0 0.00% 100.00% # Class of committed instruction. (Count)
board.processor.cores.core.commitStats0.committedInstType::SimdWholeRegisterLoad 0 0.00% 100.00% # Class of committed instruction. (Count)
board.processor.cores.core.commitStats0.committedInstType::SimdWholeRegisterStore 0 0.00% 100.00% # Class of committed instruction. (Count)
board.processor.cores.core.commitStats0.committedInstType::SimdUnitStrideFaultOnlyFirstLoad 0 0.00% 100.00% # Class of committed instruction. (Count)
board.processor.cores.core.commitStats0.committedInstType::SimdUnitStrideSegmentedLoad 0 0.00% 100.00% # Class of committed instruction. (Count)
board.processor.cores.core.commitStats0.committedInstType::SimdUnitStrideSegmentedStore 0 0.00% 100.00% # Class of committed instruction. (Count)
board.processor.cores.core.commitStats0.committedInstType::SimdExt 0 0.00% 100.00% # Class of committed instruction. (Count)
board.processor.cores.core.commitStats0.committedInstType::SimdFloatExt 0 0.00% 100.00% # Class of committed instruction. (Count)
board.processor.cores.core.commitStats0.committedInstType::SimdConfig 0 0.00% 100.00% # Class of committed instruction. (Count)
board.processor.cores.core.commitStats0.committedInstType::total 13006 # Class of committed instruction. (Count)
board.processor.cores.core.commitStats0.committedControl::IsControl 1483 # Class of control type instructions committed (Count)
board.processor.cores.core.commitStats0.committedControl::IsDirectControl 1300 # Class of control type instructions committed (Count)
board.processor.cores.core.commitStats0.committedControl::IsIndirectControl 183 # Class of control type instructions committed (Count)
board.processor.cores.core.commitStats0.committedControl::IsCondControl 1108 # Class of control type instructions committed (Count)
board.processor.cores.core.commitStats0.committedControl::IsUncondControl 375 # Class of control type instructions committed (Count)
board.processor.cores.core.commitStats0.committedControl::IsCall 127 # Class of control type instructions committed (Count)
board.processor.cores.core.commitStats0.committedControl::IsReturn 122 # Class of control type instructions committed (Count)
board.processor.cores.core.exec_context.thread_0.numMatAluAccesses 0 # Number of matrix alu accesses (Count)
board.processor.cores.core.exec_context.thread_0.numCallsReturns 249 # Number of times a function call or return occured (Count)
board.processor.cores.core.exec_context.thread_0.numMatInsts 0 # Number of matrix instructions (Count)
board.processor.cores.core.exec_context.thread_0.numIdleCycles 0.003003 # Number of idle cycles (Cycle)
board.processor.cores.core.exec_context.thread_0.numBusyCycles 1383662.996997 # Number of busy cycles (Cycle)
board.processor.cores.core.exec_context.thread_0.notIdleFraction 1.000000 # Percentage of non-idle cycles (Ratio)
board.processor.cores.core.exec_context.thread_0.idleFraction 0.000000 # Percentage of idle cycles (Ratio)
board.processor.cores.core.executeStats0.numInsts 0 # Number of executed instructions (Count)
board.processor.cores.core.executeStats0.numNop 0 # Number of nop insts executed (Count)
board.processor.cores.core.executeStats0.numBranches 0 # Number of branches executed (Count)
board.processor.cores.core.executeStats0.numLoadInsts 0 # Number of load instructions executed (Count)
board.processor.cores.core.executeStats0.numStoreInsts 2626 # Number of stores executed (Count)
board.processor.cores.core.executeStats0.instRate 0 # Inst execution rate ((Count/Cycle))
board.processor.cores.core.executeStats0.numFpAluAccesses 411 # Number of float alu accesses (Count)
board.processor.cores.core.executeStats0.numFpRegReads 563 # Number of times the floating registers were read (Count)
board.processor.cores.core.executeStats0.numFpRegWrites 202 # Number of times the floating registers were written (Count)
board.processor.cores.core.executeStats0.numIntAluAccesses 12799 # Number of integer alu accesses (Count)
board.processor.cores.core.executeStats0.numIntRegReads 14442 # Number of times the integer registers were read (Count)
board.processor.cores.core.executeStats0.numIntRegWrites 8883 # Number of times the integer registers were written (Count)
board.processor.cores.core.executeStats0.numMemRefs 2626 # Number of memory refs (Count)
board.processor.cores.core.executeStats0.numMiscRegReads 5696 # Number of times the Misc registers were read (Count)
board.processor.cores.core.executeStats0.numVecAluAccesses 0 # Number of vector alu accesses (Count)
board.processor.cores.core.executeStats0.numDiscardedOps 0 # Number of ops (including micro ops) which were discarded before commit (Count)
board.processor.cores.core.fetchStats0.numInsts 6568 # Number of instructions fetched (thread level) (Count)
board.processor.cores.core.fetchStats0.numOps 13006 # Number of ops (including micro ops) fetched (thread level) (Count)
board.processor.cores.core.fetchStats0.fetchRate 0.004747 # Number of inst fetches per cycle ((Count/Cycle))
board.processor.cores.core.fetchStats0.numBranches 1483 # Number of branches fetched (Count)
board.processor.cores.core.fetchStats0.branchRate 0.001072 # Number of branch fetches per cycle (Ratio)
board.processor.cores.core.fetchStats0.numFetchSuspends 0 # Number of times Execute suspended instruction fetching (Count)
board.processor.cores.core.interrupts.clk_domain.clock 5328 # Clock period in ticks (Tick)
board.processor.cores.core.mmu.dtb.rdAccesses 1276 # TLB accesses on read requests (Count)
board.processor.cores.core.mmu.dtb.wrAccesses 1354 # TLB accesses on write requests (Count)
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board.processor.cores.core.mmu.dtb.rdMisses          11                    # TLB misses on read requests (Count)
board.processor.cores.core.mmu.dtb.wrMisses          11                    # TLB misses on write requests (Count)
board.processor.cores.core.mmu.dtb.walker.power_state.pwrStateResidencyTicks::UNDEFINED    460759779                    # Cumulative time (in ticks) in various power states (Tick)
board.processor.cores.core.mmu.itb.rdAccesses          0                    # TLB accesses on read requests (Count)
board.processor.cores.core.mmu.itb.wrAccesses          8567                    # TLB accesses on write requests (Count)
board.processor.cores.core.mmu.itb.rdMisses          0                    # TLB misses on read requests (Count)
board.processor.cores.core.mmu.itb.wrMisses          39                    # TLB misses on write requests (Count)
board.processor.cores.core.mmu.itb.walker.power_state.pwrStateResidencyTicks::UNDEFINED    460759779                    # Cumulative time (in ticks) in various power states (Tick)
board.processor.cores.core.power_state.pwrStateResidencyTicks::ON    460759779                    # Cumulative time (in ticks) in various power states (Tick)
board.processor.cores.core.thread_0.numInsts          0                    # Number of Instructions committed (Count)
board.processor.cores.core.thread_0.numOps          0                    # Number of Ops committed (Count)
board.processor.cores.core.thread_0.numMemRefs          0                    # Number of Memory References (Count)
board.processor.cores.core.workload.numSyscalls          11                    # Number of system calls (Count)
board.workload.inst.arm          0                    # number of arm instructions executed (Count)
board.workload.inst.quiesce          0                    # number of quiesce instructions executed (Count)

---------- End Simulation Statistics   ----------
```

ooo_cache:

```
---------- Begin Simulation Statistics ----------
simSeconds                    0.000020          # Number of seconds simulated (Second)
simTicks                      19514466          # Number of ticks simulated (Tick)
finalTick                     19514466          # Number of ticks from beginning of simulation (restored from checkpoints and never reset) (Tick)
simFreq                       1000000000000     # The number of ticks per simulated second ((Tick/Second))
hostSeconds                   0.08              # Real time elapsed on the host (Second)
hostTickRate                  241405124         # The number of ticks simulated per host second (ticks/s) ((Tick/Second))
hostMemory                    1207096           # Number of bytes of host memory used (Byte)
simInsts                      6555              # Number of instructions simulated (Count)
simOps                        12992             # Number of ops (including micro ops) simulated (Count)
hostInstRate                  80989             # Simulator instruction rate (inst/s) ((Count/Second))
hostOpRate                    160512            # Simulator op (including micro ops) rate (op/s) ((Count/Second))
board.cache_hierarchy.dptw_caches.blockedCycles::no_mshrs      0      # number of cycles access was blocked (Cycle)
board.cache_hierarchy.dptw_caches.blockedCycles::no_targets    0      # number of cycles access was blocked (Cycle)
board.cache_hierarchy.dptw_caches.blockedCauses::no_mshrs      0      # number of times access was blocked (Count)
board.cache_hierarchy.dptw_caches.blockedCauses::no_targets    0      # number of times access was blocked (Count)
board.cache_hierarchy.dptw_caches.avgBlocked::no_mshrs      nan       # average number of cycles each access was blocked ((Cycle/Count))
board.cache_hierarchy.dptw_caches.avgBlocked::no_targets    nan       # average number of cycles each access was blocked ((Cycle/Count))
board.cache_hierarchy.dptw_caches.replacements      0      # number of replacements (Count)
board.cache_hierarchy.dptw_caches.power_state.pwrStateResidencyTicks::UNDEFINED    19514466    # Cumulative time (in ticks) in various power states (Tick)
board.cache_hierarchy.dptw_caches.tags.tagsInUse      0      # Average ticks per tags in use ((Tick/Count))
board.cache_hierarchy.dptw_caches.tags.totalRefs      0      # Total number of references to valid blocks. (Count)
board.cache_hierarchy.dptw_caches.tags.sampledRefs      0      # Sample count of references to valid blocks. (Count)
board.cache_hierarchy.dptw_caches.tags.avgRefs      nan      # Average number of references to valid blocks. ((Count/Count))
board.cache_hierarchy.dptw_caches.tags.warmupTick      0      # The tick when the warmup percentage was hit. (Tick)
board.cache_hierarchy.dptw_caches.tags.tagAccesses      0      # Number of tag accesses (Count)
board.cache_hierarchy.dptw_caches.tags.dataAccesses      0      # Number of data accesses (Count)
board.cache_hierarchy.dptw_caches.tags.power_state.pwrStateResidencyTicks::UNDEFINED    19514466    # Cumulative time (in ticks) in various power states (Tick)
board.cache_hierarchy.iptw_caches.blockedCycles::no_mshrs      0      # number of cycles access was blocked (Cycle)
board.cache_hierarchy.iptw_caches.blockedCycles::no_targets    0      # number of cycles access was blocked (Cycle)
board.cache_hierarchy.iptw_caches.blockedCauses::no_mshrs      0      # number of times access was blocked (Count)
board.cache_hierarchy.iptw_caches.blockedCauses::no_targets    0      # number of times access was blocked (Count)
board.cache_hierarchy.iptw_caches.avgBlocked::no_mshrs      nan       # average number of cycles each access was blocked ((Cycle/Count))
board.cache_hierarchy.iptw_caches.avgBlocked::no_targets    nan       # average number of cycles each access was blocked ((Cycle/Count))
board.cache_hierarchy.iptw_caches.replacements      0      # number of replacements (Count)
board.cache_hierarchy.iptw_caches.power_state.pwrStateResidencyTicks::UNDEFINED    19514466    # Cumulative time (in ticks) in various power states (Tick)
board.cache_hierarchy.iptw_caches.tags.tagsInUse      0      # Average ticks per tags in use ((Tick/Count))
board.cache_hierarchy.iptw_caches.tags.totalRefs      0      # Total number of references to valid blocks. (Count)
board.cache_hierarchy.iptw_caches.tags.sampledRefs      0      # Sample count of references to valid blocks. (Count)
board.cache_hierarchy.iptw_caches.tags.avgRefs      nan      # Average number of references to valid blocks. ((Count/Count))
board.cache_hierarchy.iptw_caches.tags.warmupTick      0      # The tick when the warmup percentage was hit. (Tick)
board.cache_hierarchy.iptw_caches.tags.tagAccesses      0      # Number of tag accesses (Count)
board.cache_hierarchy.iptw_caches.tags.dataAccesses      0      # Number of data accesses (Count)
board.cache_hierarchy.iptw_caches.tags.power_state.pwrStateResidencyTicks::UNDEFINED    19514466    # Cumulative time (in ticks) in various power states (Tick)
board.cache_hierarchy.l1dcaches.demandHits::processor.cores.core.data    2949    # number of demand (read+write) hits (Count)
board.cache_hierarchy.l1dcaches.demandHits::total    2949    # number of demand (read+write) hits (Count)
```

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board.cache_hierarchy.l1dcaches.overallHits::processor.cores.core.data        2949                    # number of overall hits (Count)
board.cache_hierarchy.l1dcaches.overallHits::total        2949                    # number of overall hits (Count)
board.cache_hierarchy.l1dcaches.demandMisses::processor.cores.core.data        269                    # number of demand (read+write) misses (Count)
board.cache_hierarchy.l1dcaches.demandMisses::total        269                    # number of demand (read+write) misses (Count)
board.cache_hierarchy.l1dcaches.overallMisses::processor.cores.core.data        269                    # number of overall misses (Count)
board.cache_hierarchy.l1dcaches.overallMisses::total        269                    # number of overall misses (Count)
board.cache_hierarchy.l1dcaches.demandMissLatency::processor.cores.core.data     13785867                    # number of demand (read+write) miss ticks (Tick)
board.cache_hierarchy.l1dcaches.demandMissLatency::total     13785867                    # number of demand (read+write) miss ticks (Tick)
board.cache_hierarchy.l1dcaches.overallMissLatency::processor.cores.core.data     13785867                    # number of overall miss ticks (Tick)
board.cache_hierarchy.l1dcaches.overallMissLatency::total     13785867                    # number of overall miss ticks (Tick)
board.cache_hierarchy.l1dcaches.demandAccesses::processor.cores.core.data        3218                    # number of demand (read+write) accesses (Count)
board.cache_hierarchy.l1dcaches.demandAccesses::total        3218                    # number of demand (read+write) accesses (Count)
board.cache_hierarchy.l1dcaches.overallAccesses::processor.cores.core.data        3218                    # number of overall (read+write) accesses (Count)
board.cache_hierarchy.l1dcaches.overallAccesses::total        3218                    # number of overall (read+write) accesses (Count)
board.cache_hierarchy.l1dcaches.demandMissRate::processor.cores.core.data     0.083592                    # miss rate for demand accesses (Ratio)
board.cache_hierarchy.l1dcaches.demandMissRate::total     0.083592                    # miss rate for demand accesses (Ratio)
board.cache_hierarchy.l1dcaches.overallMissRate::processor.cores.core.data     0.083592                    # miss rate for overall accesses (Ratio)
board.cache_hierarchy.l1dcaches.overallMissRate::total     0.083592                    # miss rate for overall accesses (Ratio)
board.cache_hierarchy.l1dcaches.demandAvgMissLatency::processor.cores.core.data 51248.576208                    # average overall miss latency in ticks ((Tick/Count))
board.cache_hierarchy.l1dcaches.demandAvgMissLatency::total 51248.576208                    # average overall miss latency in ticks ((Tick/Count))
board.cache_hierarchy.l1dcaches.overallAvgMissLatency::processor.cores.core.data 51248.576208                    # average overall miss latency ((Tick/Count))
board.cache_hierarchy.l1dcaches.overallAvgMissLatency::total 51248.576208                    # average overall miss latency ((Tick/Count))
board.cache_hierarchy.l1dcaches.blockedCycles::no_mshrs        0                    # number of cycles access was blocked (Cycle)
board.cache_hierarchy.l1dcaches.blockedCycles::no_targets        0                    # number of cycles access was blocked (Cycle)
board.cache_hierarchy.l1dcaches.blockedCauses::no_mshrs        0                    # number of times access was blocked (Count)
board.cache_hierarchy.l1dcaches.blockedCauses::no_targets        0                    # number of times access was blocked (Count)
board.cache_hierarchy.l1dcaches.avgBlocked::no_mshrs        nan                    # average number of cycles each access was blocked ((Cycle/Count))
board.cache_hierarchy.l1dcaches.avgBlocked::no_targets        nan                    # average number of cycles each access was blocked ((Cycle/Count))
board.cache_hierarchy.l1dcaches.demandMshrHits::processor.cores.core.data        126                    # number of demand (read+write) MSHR hits (Count)
board.cache_hierarchy.l1dcaches.demandMshrHits::total        126                    # number of demand (read+write) MSHR hits (Count)
board.cache_hierarchy.l1dcaches.overallMshrHits::processor.cores.core.data        126                    # number of overall MSHR hits (Count)
board.cache_hierarchy.l1dcaches.overallMshrHits::total        126                    # number of overall MSHR hits (Count)
board.cache_hierarchy.l1dcaches.demandMshrMisses::processor.cores.core.data        143                    # number of demand (read+write) MSHR misses (Count)
board.cache_hierarchy.l1dcaches.demandMshrMisses::total        143                    # number of demand (read+write) MSHR misses (Count)
board.cache_hierarchy.l1dcaches.overallMshrMisses::cache_hierarchy.l1dcaches.prefetcher        56                    # number of overall MSHR misses (Count)
board.cache_hierarchy.l1dcaches.overallMshrMisses::processor.cores.core.data        143                    # number of overall MSHR misses (Count)
board.cache_hierarchy.l1dcaches.overallMshrMisses::total        199                    # number of overall MSHR misses (Count)
board.cache_hierarchy.l1dcaches.demandMshrMissLatency::processor.cores.core.data     9150507                    # number of demand (read+write) MSHR miss ticks (Tick)
board.cache_hierarchy.l1dcaches.demandMshrMissLatency::total     9150507                    # number of demand (read+write) MSHR miss ticks (Tick)
board.cache_hierarchy.l1dcaches.overallMshrMissLatency::cache_hierarchy.l1dcaches.prefetcher     2393481                    # number of overall MSHR miss ticks (Tick)
board.cache_hierarchy.l1dcaches.overallMshrMissLatency::processor.cores.core.data     9150507                    # number of overall MSHR miss ticks (Tick)
board.cache_hierarchy.l1dcaches.overallMshrMissLatency::total     11543988                    # number of overall MSHR miss ticks (Tick)
board.cache_hierarchy.l1dcaches.demandMshrMissRate::processor.cores.core.data     0.044438                    # mshr miss ratio for demand accesses (Ratio)
board.cache_hierarchy.l1dcaches.demandMshrMissRate::total     0.044438                    # mshr miss ratio for demand accesses (Ratio)
board.cache_hierarchy.l1dcaches.overallMshrMissRate::cache_hierarchy.l1dcaches.prefetcher        inf                    # mshr miss ratio for overall accesses (Ratio)
board.cache_hierarchy.l1dcaches.overallMshrMissRate::processor.cores.core.data     0.044438                    # mshr miss ratio for overall accesses (Ratio)
board.cache_hierarchy.l1dcaches.overallMshrMissRate::total     0.061840                    # mshr miss ratio for overall accesses (Ratio)
board.cache_hierarchy.l1dcaches.demandAvgMshrMissLatency::processor.cores.core.data 63989.559441                    # average overall mshr miss latency ((Tick/Count))
board.cache_hierarchy.l1dcaches.demandAvgMshrMissLatency::total 63989.559441                    # average overall mshr miss latency ((Tick/Count))
```

board.cache_hierarchy.l1dcaches.overallAvgMshrMissLatency::cache_hierarchy.l1dcaches.prefetcher 42740.732143 # average overall mshr miss latency ((Tick/Count))

board.cache_hierarchy.l1dcaches.overallAvgMshrMissLatency::processor.cores.core.data 63989.559441 # average overall mshr miss latency ((Tick/Count))

board.cache_hierarchy.l1dcaches.overallAvgMshrMissLatency::total 58009.989950 # average overall mshr miss latency ((Tick/Count))

board.cache_hierarchy.l1dcaches.replacements 0 # number of replacements (Count)

board.cache_hierarchy.l1dcaches.HardPFReq.mshrMisses::cache_hierarchy.l1dcaches.prefetcher 56 # number of HardPFReq MSHR misses (Count)

board.cache_hierarchy.l1dcaches.HardPFReq.mshrMisses::total 56 # number of HardPFReq MSHR misses (Count)

board.cache_hierarchy.l1dcaches.HardPFReq.mshrMissLatency::cache_hierarchy.l1dcaches.prefetcher 2393481 # number of HardPFReq MSHR miss ticks (Tick)

board.cache_hierarchy.l1dcaches.HardPFReq.mshrMissLatency::total 2393481 # number of HardPFReq MSHR miss ticks (Tick)

board.cache_hierarchy.l1dcaches.HardPFReq.mshrMissRate::cache_hierarchy.l1dcaches.prefetcher inf # mshr miss rate for HardPFReq accesses (Ratio)

board.cache_hierarchy.l1dcaches.HardPFReq.mshrMissRate::total inf # mshr miss rate for HardPFReq accesses (Ratio)

board.cache_hierarchy.l1dcaches.HardPFReq.avgMshrMissLatency::cache_hierarchy.l1dcaches.prefetcher 42740.732143 # average HardPFReq mshr miss latency ((Tick/Count))

board.cache_hierarchy.l1dcaches.HardPFReq.avgMshrMissLatency::total 42740.732143 # average HardPFReq mshr miss latency ((Tick/Count))

board.cache_hierarchy.l1dcaches.ReadReq.hits::processor.cores.core.data 1681 # number of ReadReq hits (Count)

board.cache_hierarchy.l1dcaches.ReadReq.hits::total 1681 # number of ReadReq hits (Count)

board.cache_hierarchy.l1dcaches.ReadReq.misses::processor.cores.core.data 186 # number of ReadReq misses (Count)

board.cache_hierarchy.l1dcaches.ReadReq.misses::total 186 # number of ReadReq misses (Count)

board.cache_hierarchy.l1dcaches.ReadReq.missLatency::processor.cores.core.data 10158498 # number of ReadReq miss ticks (Tick)

board.cache_hierarchy.l1dcaches.ReadReq.missLatency::total 10158498 # number of ReadReq miss ticks (Tick)

board.cache_hierarchy.l1dcaches.ReadReq.accesses::processor.cores.core.data 1867 # number of ReadReq accesses(hits+misses) (Count)

board.cache_hierarchy.l1dcaches.ReadReq.accesses::total 1867 # number of ReadReq accesses(hits+misses) (Count)

board.cache_hierarchy.l1dcaches.ReadReq.missRate::processor.cores.core.data 0.099625 # miss rate for ReadReq accesses (Ratio)

board.cache_hierarchy.l1dcaches.ReadReq.missRate::total 0.099625 # miss rate for ReadReq accesses (Ratio)

board.cache_hierarchy.l1dcaches.ReadReq.avgMissLatency::processor.cores.core.data 54615.580645 # average ReadReq miss latency ((Tick/Count))

board.cache_hierarchy.l1dcaches.ReadReq.avgMissLatency::total 54615.580645 # average ReadReq miss latency ((Tick/Count))

board.cache_hierarchy.l1dcaches.ReadReq.mshrHits::processor.cores.core.data 99 # number of ReadReq MSHR hits (Count)

board.cache_hierarchy.l1dcaches.ReadReq.mshrHits::total 99 # number of ReadReq MSHR hits (Count)

board.cache_hierarchy.l1dcaches.ReadReq.mshrMisses::processor.cores.core.data 87 # number of ReadReq MSHR misses (Count)

board.cache_hierarchy.l1dcaches.ReadReq.mshrMisses::total 87 # number of ReadReq MSHR misses (Count)

board.cache_hierarchy.l1dcaches.ReadReq.mshrMissLatency::processor.cores.core.data 5809185 # number of ReadReq MSHR miss ticks (Tick)

board.cache_hierarchy.l1dcaches.ReadReq.mshrMissLatency::total 5809185 # number of ReadReq MSHR miss ticks (Tick)

board.cache_hierarchy.l1dcaches.ReadReq.mshrMissRate::processor.cores.core.data 0.046599 # mshr miss rate for ReadReq accesses (Ratio)

board.cache_hierarchy.l1dcaches.ReadReq.mshrMissRate::total 0.046599 # mshr miss rate for ReadReq accesses (Ratio)

board.cache_hierarchy.l1dcaches.ReadReq.avgMshrMissLatency::processor.cores.core.data 66772.241379 # average ReadReq mshr miss latency ((Tick/Count))

board.cache_hierarchy.l1dcaches.ReadReq.avgMshrMissLatency::total 66772.241379 # average ReadReq mshr miss latency ((Tick/Count))

board.cache_hierarchy.l1dcaches.WriteReq.hits::processor.cores.core.data 1268 # number of WriteReq hits (Count)

board.cache_hierarchy.l1dcaches.WriteReq.hits::total 1268 # number of WriteReq hits (Count)

board.cache_hierarchy.l1dcaches.WriteReq.misses::processor.cores.core.data 83 # number of WriteReq misses (Count)

board.cache_hierarchy.l1dcaches.WriteReq.misses::total 83 # number of WriteReq misses (Count)

board.cache_hierarchy.l1dcaches.WriteReq.missLatency::processor.cores.core.data 3627369 # number of WriteReq miss ticks (Tick)

board.cache_hierarchy.l1dcaches.WriteReq.missLatency::total 3627369 # number of WriteReq miss ticks (Tick)

board.cache_hierarchy.l1dcaches.WriteReq.accesses::processor.cores.core.data 1351 # number of WriteReq accesses(hits+misses) (Count)

board.cache_hierarchy.l1dcaches.WriteReq.accesses::total 1351 # number of WriteReq accesses(hits+misses) (Count)

board.cache_hierarchy.l1dcaches.WriteReq.missRate::processor.cores.core.data 0.061436 # miss rate for WriteReq accesses (Ratio)

board.cache_hierarchy.l1dcaches.WriteReq.missRate::total 0.061436 # miss rate for WriteReq accesses (Ratio)

board.cache_hierarchy.l1dcaches.WriteReq.avgMissLatency::processor.cores.core.data 43703.240964 # average WriteReq miss latency ((Tick/Count))

board.cache_hierarchy.l1dcaches.WriteReq.avgMissLatency::total 43703.240964 # average WriteReq miss latency ((Tick/Count))

board.cache_hierarchy.l1dcaches.WriteReq.mshrHits::processor.cores.core.data 27 # number of WriteReq MSHR hits (Count)

board.cache_hierarchy.l1dcaches.WriteReq.mshrHits::total 27 # number of WriteReq MSHR hits (Count)

board.cache_hierarchy.l1dcaches.WriteReq.mshrMisses::processor.cores.core.data 56 # number of WriteReq MSHR misses (Count)

board.cache_hierarchy.l1dcaches.WriteReq.mshrMisses::total 56 # number of WriteReq MSHR misses (Count)

board.cache_hierarchy.l1dcaches.WriteReq.mshrMissLatency::processor.cores.core.data 3341322 # number of WriteReq MSHR miss ticks (Tick)

board.cache_hierarchy.l1dcaches.WriteReq.mshrMissLatency::total 3341322 # number of WriteReq MSHR miss ticks (Tick)

board.cache_hierarchy.l1dcaches.WriteReq.mshrMissRate::processor.cores.core.data 0.041451 # mshr miss rate for WriteReq accesses (Ratio)

board.cache_hierarchy.l1dcaches.WriteReq.mshrMissRate::total 0.041451 # mshr miss rate for WriteReq accesses (Ratio)

board.cache_hierarchy.l1dcaches.WriteReq.avgMshrMissLatency::processor.cores.core.data 59666.464286 # average WriteReq mshr miss latency ((Tick/Count))

board.cache_hierarchy.l1dcaches.WriteReq.avgMshrMissLatency::total 59666.464286 # average WriteReq mshr miss latency ((Tick/Count))

board.cache_hierarchy.l1dcaches.power_state.pwrStateResidencyTicks::UNDEFINED 19514466 # Cumulative time (in ticks) in various power states (Tick)

board.cache_hierarchy.l1dcaches.prefetcher.demandMshrMisses 143 # demands not covered by prefetchs (Count)

board.cache_hierarchy.l1dcaches.prefetcher.pfIssued 188 # number of hwpf issued (Count)

board.cache_hierarchy.l1dcaches.prefetcher.pfUseful 21 # number of useful prefetch (Count)

board.cache_hierarchy.l1dcaches.prefetcher.pfUsefulButMiss 0 # number of hit on prefetch but cache block is not in an usable state (Count)

board.cache_hierarchy.l1dcaches.prefetcher.accuracy 0.111702 # accuracy of the prefetcher (Count)

board.cache_hierarchy.l1dcaches.prefetcher.coverage 0.128049 # coverage brought by this prefetcher (Count)

board.cache_hierarchy.l1dcaches.prefetcher.pfHitInCache 49 # number of prefetches hitting in cache (Count)

board.cache_hierarchy.l1dcaches.prefetcher.pfHitInMSHR 83 # number of prefetches hitting in a MSHR (Count)

board.cache_hierarchy.l1dcaches.prefetcher.pfHitInWB 0 # number of prefetches hit in the Write Buffer (Count)

board.cache_hierarchy.l1dcaches.prefetcher.pfLate 132 # number of late prefetches (hitting in cache, MSHR or WB) (Count)

board.cache_hierarchy.l1dcaches.prefetcher.pfIdentified 203 # number of prefetch candidates identified (Count)

board.cache_hierarchy.l1dcaches.prefetcher.pfBufferHit 12 # number of redundant prefetches already in prefetch queue (Count)

board.cache_hierarchy.l1dcaches.prefetcher.pfInCache 0 # number of redundant prefetches already in cache/mshr dropped (Count)

board.cache_hierarchy.l1dcaches.prefetcher.pfRemovedDemand 0 # number of prefetches dropped due to a demand for the same address (Count)

board.cache_hierarchy.l1dcaches.prefetcher.pfRemovedFull 0 # number of prefetches dropped due to prefetch queue size (Count)

board.cache_hierarchy.l1dcaches.prefetcher.pfSpanPage 17 # number of prefetches that crossed the page (Count)

board.cache_hierarchy.l1dcaches.prefetcher.pfUsefulSpanPage 0 # number of prefetches that is useful and crossed the page (Count)

board.cache_hierarchy.l1dcaches.prefetcher.power_state.pwrStateResidencyTicks::UNDEFINED 19514466 # Cumulative time (in ticks) in various power states (Tick)

board.cache_hierarchy.l1dcaches.tags.tagsInUse 120.574524 # Average ticks per tags in use ((Tick/Count))

board.cache_hierarchy.l1dcaches.tags.totalRefs 3148 # Total number of references to valid blocks. (Count)

board.cache_hierarchy.l1dcaches.tags.sampledRefs 199 # Sample count of references to valid blocks. (Count)

board.cache_hierarchy.l1dcaches.tags.avgRefs 15.819095 # Average number of references to valid blocks. ((Count/Count))

board.cache_hierarchy.l1dcaches.tags.warmupTick 133200 # The tick when the warmup percentage was hit. (Tick)

board.cache_hierarchy.l1dcaches.tags.occupancies::cache_hierarchy.l1dcaches.prefetcher 37.228577 # Average occupied blocks per tick, per requestor ((Count/Tick))

board.cache_hierarchy.l1dcaches.tags.occupancies::processor.cores.core.data 83.345947 # Average occupied blocks per tick, per requestor ((Count/Tick))

board.cache_hierarchy.l1dcaches.tags.avgOccs::cache_hierarchy.l1dcaches.prefetcher 0.072712 # Average percentage of cache occupancy ((Ratio/Tick))

board.cache_hierarchy.l1dcaches.tags.avgOccs::processor.cores.core.data 0.162785 # Average percentage of cache occupancy ((Ratio/Tick))

board.cache_hierarchy.l1dcaches.tags.avgOccs::total 0.235497 # Average percentage of cache occupancy ((Ratio/Tick))

board.cache_hierarchy.l1dcaches.tags.occupanciesTaskId::1022 56 # Occupied blocks per task id (Count)

board.cache_hierarchy.l1dcaches.tags.occupanciesTaskId::1024 143 # Occupied blocks per task id (Count)

board.cache_hierarchy.l1dcaches.tags.ageTaskId_1022::0 15 # Occupied blocks per task id, per block age (Count)

board.cache_hierarchy.l1dcaches.tags.ageTaskId_1022::1 41 # Occupied blocks per task id, per block age (Count)

board.cache_hierarchy.l1dcaches.tags.ageTaskId_1024::0 53 # Occupied blocks per task id, per block age (Count)

board.cache_hierarchy.l1dcaches.tags.ageTaskId_1024::1 90 # Occupied blocks per task id, per block age (Count)

board.cache_hierarchy.l1dcaches.tags.ratioOccsTaskId::1022 0.109375 # Ratio of occupied blocks and all blocks, per task id (Ratio)

board.cache_hierarchy.l1dcaches.tags.ratioOccsTaskId::1024 0.279297 # Ratio of occupied blocks and all blocks, per task id (Ratio)

board.cache_hierarchy.l1dcaches.tags.tagAccesses 25943 # Number of tag accesses (Count)

board.cache_hierarchy.l1dcaches.tags.dataAccesses 25943 # Number of data accesses (Count)

board.cache_hierarchy.l1dcaches.tags.power_state.pwrStateResidencyTicks::UNDEFINED 19514466 # Cumulative time (in ticks) in various power states (Tick)

board.cache_hierarchy.l1icaches.demandHits::processor.cores.core.inst 2025 # number of demand (read+write) hits (Count)

board.cache_hierarchy.l1icaches.demandHits::total 2025 # number of demand (read+write) hits (Count)

board.cache_hierarchy.l1icaches.overallHits::processor.cores.core.inst 2025 # number of overall hits (Count)

board.cache_hierarchy.l1icaches.overallHits::total 2025 # number of overall hits (Count)

board.cache_hierarchy.l1icaches.demandMisses::processor.cores.core.inst 573 # number of demand (read+write) misses (Count)

board.cache_hierarchy.l1icaches.demandMisses::total 573 # number of demand (read+write) misses (Count)

board.cache_hierarchy.l1icaches.overallMisses::processor.cores.core.inst 573 # number of overall misses (Count)

board.cache_hierarchy.l1icaches.overallMisses::total 573 # number of overall misses (Count)

board.cache_hierarchy.l1icaches.demandMissLatency::processor.cores.core.inst 29277027 # number of demand (read+write) miss ticks (Tick)

board.cache_hierarchy.l1icaches.demandMissLatency::total 29277027 # number of demand (read+write) miss ticks (Tick)

board.cache_hierarchy.l1icaches.overallMissLatency::processor.cores.core.inst 29277027 # number of overall miss ticks (Tick)

board.cache_hierarchy.l1icaches.overallMissLatency::total 29277027 # number of overall miss ticks (Tick)

board.cache_hierarchy.l1icaches.demandAccesses::processor.cores.core.inst 2598 # number of demand (read+write) accesses (Count)

board.cache_hierarchy.l1icaches.demandAccesses::total 2598 # number of demand (read+write) accesses (Count)

board.cache_hierarchy.l1icaches.overallAccesses::processor.cores.core.inst 2598 # number of overall (read+write) accesses (Count)

board.cache_hierarchy.l1icaches.overallAccesses::total 2598 # number of overall (read+write) accesses (Count)

board.cache_hierarchy.l1icaches.demandMissRate::processor.cores.core.inst 0.220554 # miss rate for demand accesses (Ratio)

board.cache_hierarchy.l1icaches.demandMissRate::total 0.220554 # miss rate for demand accesses (Ratio)

board.cache_hierarchy.l1icaches.overallMissRate::processor.cores.core.inst 0.220554 # miss rate for overall accesses (Ratio)

board.cache_hierarchy.l1icaches.overallMissRate::total 0.220554 # miss rate for overall accesses (Ratio)

board.cache_hierarchy.l1icaches.demandAvgMissLatency::processor.cores.core.inst 51094.287958 # average overall miss latency in ticks ((Tick/Count))

board.cache_hierarchy.l1icaches.demandAvgMissLatency::total 51094.287958 # average overall miss latency in ticks ((Tick/Count))

board.cache_hierarchy.l1icaches.overallAvgMissLatency::processor.cores.core.inst 51094.287958 # average overall miss latency ((Tick/Count))

board.cache_hierarchy.l1icaches.overallAvgMissLatency::total 51094.287958 # average overall miss latency ((Tick/Count))

board.cache_hierarchy.l1icaches.blockedCycles::no_mshrs 0 # number of cycles access was blocked (Cycle)

board.cache_hierarchy.l1icaches.blockedCycles::no_targets 0 # number of cycles access was blocked (Cycle)

board.cache_hierarchy.l1icaches.blockedCauses::no_mshrs 0 # number of times access was blocked (Count)

board.cache_hierarchy.l1icaches.blockedCauses::no_targets 0 # number of times access was blocked (Count)

board.cache_hierarchy.l1icaches.avgBlocked::no_mshrs nan # average number of cycles each access was blocked ((Cycle/Count))

board.cache_hierarchy.l1icaches.avgBlocked::no_targets nan # average number of cycles each access was blocked ((Cycle/Count))

board.cache_hierarchy.l1icaches.demandMshrHits::processor.cores.core.inst 149 # number of demand (read+write) MSHR hits (Count)

board.cache_hierarchy.l1icaches.demandMshrHits::total 149 # number of demand (read+write) MSHR hits (Count)

board.cache_hierarchy.l1icaches.overallMshrHits::processor.cores.core.inst 149 # number of overall MSHR hits (Count)

board.cache_hierarchy.l1icaches.overallMshrHits::total 149 # number of overall MSHR hits (Count)

board.cache_hierarchy.l1icaches.demandMshrMisses::processor.cores.core.inst 424 # number of demand (read+write) MSHR misses (Count)

board.cache_hierarchy.l1icaches.demandMshrMisses::total 424 # number of demand (read+write) MSHR misses (Count)

board.cache_hierarchy.l1icaches.overallMshrMisses::processor.cores.core.inst 424 # number of overall MSHR misses (Count)

board.cache_hierarchy.l1icaches.overallMshrMisses::total 424 # number of overall MSHR misses (Count)

board.cache_hierarchy.l1icaches.demandMshrMissLatency::processor.cores.core.inst 23078898 # number of demand (read+write) MSHR miss ticks (Tick)

board.cache_hierarchy.l1icaches.demandMshrMissLatency::total 23078898 # number of demand (read+write) MSHR miss ticks (Tick)

board.cache_hierarchy.l1icaches.overallMshrMissLatency::processor.cores.core.inst 23078898 # number of overall MSHR miss ticks (Tick)

board.cache_hierarchy.l1icaches.overallMshrMissLatency::total 23078898 # number of overall MSHR miss ticks (Tick)

board.cache_hierarchy.l1icaches.demandMshrMissRate::processor.cores.core.inst 0.163202 # mshr miss ratio for demand accesses (Ratio)

board.cache_hierarchy.l1icaches.demandMshrMissRate::total 0.163202 # mshr miss ratio for demand accesses (Ratio)

board.cache_hierarchy.l1icaches.overallMshrMissRate::processor.cores.core.inst 0.163202 # mshr miss ratio for overall accesses (Ratio)

board.cache_hierarchy.l1icaches.overallMshrMissRate::total 0.163202 # mshr miss ratio for overall accesses (Ratio)

board.cache_hierarchy.l1icaches.demandAvgMshrMissLatency::processor.cores.core.inst 54431.363208 # average overall mshr miss latency ((Tick/Count))

board.cache_hierarchy.l1icaches.demandAvgMshrMissLatency::total 54431.363208 # average overall mshr miss latency ((Tick/Count))

board.cache_hierarchy.l1icaches.overallAvgMshrMissLatency::processor.cores.core.inst 54431.363208 # average overall mshr miss latency ((Tick/Count))

board.cache_hierarchy.l1icaches.overallAvgMshrMissLatency::total 54431.363208 # average overall mshr miss latency ((Tick/Count))

board.cache_hierarchy.l1icaches.replacements 32 # number of replacements (Count)

board.cache_hierarchy.l1icaches.ReadReq.hits::processor.cores.core.inst 2025 # number of ReadReq hits (Count)

board.cache_hierarchy.l1icaches.ReadReq.hits::total 2025 # number of ReadReq hits (Count)

board.cache_hierarchy.l1icaches.ReadReq.misses::processor.cores.core.inst 573 # number of ReadReq misses (Count)

board.cache_hierarchy.l1icaches.ReadReq.misses::total 573 # number of ReadReq misses (Count)

board.cache_hierarchy.l1icaches.ReadReq.missLatency::processor.cores.core.inst 29277027 # number of ReadReq miss ticks (Tick)

board.cache_hierarchy.l1icaches.ReadReq.missLatency::total 29277027 # number of ReadReq miss ticks (Tick)

board.cache_hierarchy.l1icaches.ReadReq.accesses::processor.cores.core.inst 2598 # number of ReadReq accesses(hits+misses) (Count)

board.cache_hierarchy.l1icaches.ReadReq.accesses::total 2598 # number of ReadReq accesses(hits+misses) (Count)

board.cache_hierarchy.l1icaches.ReadReq.missRate::processor.cores.core.inst 0.220554 # miss rate for ReadReq accesses (Ratio)

board.cache_hierarchy.l1icaches.ReadReq.missRate::total 0.220554 # miss rate for ReadReq accesses (Ratio)

board.cache_hierarchy.l1icaches.ReadReq.avgMissLatency::processor.cores.core.inst 51094.287958 # average ReadReq miss latency ((Tick/Count))

board.cache_hierarchy.l1icaches.ReadReq.avgMissLatency::total 51094.287958 # average ReadReq miss latency ((Tick/Count))

board.cache_hierarchy.l1icaches.ReadReq.mshrHits::processor.cores.core.inst 149 # number of ReadReq MSHR hits (Count)

board.cache_hierarchy.l1icaches.ReadReq.mshrHits::total 149 # number of ReadReq MSHR hits (Count)

board.cache_hierarchy.l1icaches.ReadReq.mshrMisses::processor.cores.core.inst 424 # number of ReadReq MSHR misses (Count)

board.cache_hierarchy.l1icaches.ReadReq.mshrMisses::total 424 # number of ReadReq MSHR misses (Count)

board.cache_hierarchy.l1icaches.ReadReq.mshrMissLatency::processor.cores.core.inst 23078898 # number of ReadReq MSHR miss ticks (Tick)

board.cache_hierarchy.l1icaches.ReadReq.mshrMissLatency::total 23078898 # number of ReadReq MSHR miss ticks (Tick)

board.cache_hierarchy.l1icaches.ReadReq.mshrMissRate::processor.cores.core.inst 0.163202 # mshr miss rate for ReadReq accesses (Ratio)

board.cache_hierarchy.l1icaches.ReadReq.mshrMissRate::total 0.163202 # mshr miss rate for ReadReq accesses (Ratio)

board.cache_hierarchy.l1icaches.ReadReq.avgMshrMissLatency::processor.cores.core.inst 54431.363208 # average ReadReq mshr miss latency ((Tick/Count))

board.cache_hierarchy.l1icaches.ReadReq.avgMshrMissLatency::total 54431.363208 # average ReadReq mshr miss latency ((Tick/Count))

board.cache_hierarchy.l1icaches.power_state.pwrStateResidencyTicks::UNDEFINED 19514466 # Cumulative time (in ticks) in various power states (Tick)

board.cache_hierarchy.l1icaches.prefetcher.demandMshrMisses 424 # demands not covered by prefetchs (Count)

board.cache_hierarchy.l1icaches.prefetcher.pfIssued 0 # number of hwpf issued (Count)

board.cache_hierarchy.l1icaches.prefetcher.pfUseful 0 # number of useful prefetch (Count)

board.cache_hierarchy.l1icaches.prefetcher.pfUsefulButMiss 0 # number of hit on prefetch but cache block is not in an usable state (Count)

board.cache_hierarchy.l1icaches.prefetcher.accuracy nan # accuracy of the prefetcher (Count)

board.cache_hierarchy.l1icaches.prefetcher.coverage 0 # coverage brought by this prefetcher (Count)

board.cache_hierarchy.l1icaches.prefetcher.pfHitInCache 0 # number of prefetches hitting in cache (Count)

board.cache_hierarchy.l1icaches.prefetcher.pfHitInMSHR 0 # number of prefetches hitting in a MSHR (Count)

board.cache_hierarchy.l1icaches.prefetcher.pfHitInWB 0 # number of prefetches hit in the Write Buffer (Count)

board.cache_hierarchy.l1icaches.prefetcher.pfLate 0 # number of late prefetches (hitting in cache, MSHR or WB) (Count)

board.cache_hierarchy.l1icaches.prefetcher.pfIdentified 0 # number of prefetch candidates identified (Count)

board.cache_hierarchy.l1icaches.prefetcher.pfBufferHit 0 # number of redundant prefetches already in prefetch queue (Count)

board.cache_hierarchy.l1icaches.prefetcher.pfInCache 0 # number of redundant prefetches already in cache/mshr dropped (Count)

board.cache_hierarchy.l1icaches.prefetcher.pfRemovedDemand 0 # number of prefetches dropped due to a demand for the same address (Count)

board.cache_hierarchy.l1icaches.prefetcher.pfRemovedFull 0 # number of prefetches dropped due to prefetch queue size (Count)

board.cache_hierarchy.l1icaches.prefetcher.pfSpanPage 0 # number of prefetches that crossed the page (Count)

board.cache_hierarchy.l1icaches.prefetcher.pfUsefulSpanPage 0 # number of prefetches that is useful and crossed the page (Count)

board.cache_hierarchy.l1icaches.prefetcher.power_state.pwrStateResidencyTicks::UNDEFINED 19514466 # Cumulative time (in ticks) in various power states (Tick)

board.cache_hierarchy.l1icaches.tags.tagsInUse 199.685823 # Average ticks per tags in use ((Tick/Count))

board.cache_hierarchy.l1icaches.tags.totalRefs 2448 # Total number of references to valid blocks. (Count)

board.cache_hierarchy.l1icaches.tags.sampledRefs 423 # Sample count of references to valid blocks. (Count)

board.cache_hierarchy.l1icaches.tags.avgRefs 5.787234 # Average number of references to valid blocks. ((Count/Count))

board.cache_hierarchy.l1icaches.tags.warmupTick 70263 # The tick when the warmup percentage was hit. (Tick)

board.cache_hierarchy.l1icaches.tags.occupancies::processor.cores.core.inst 199.685823 # Average occupied blocks per tick, per requestor ((Count/Tick))

board.cache_hierarchy.l1icaches.tags.avgOccs::processor.cores.core.inst 0.390011 # Average percentage of cache occupancy ((Ratio/Tick))

board.cache_hierarchy.l1icaches.tags.avgOccs::total 0.390011 # Average percentage of cache occupancy ((Ratio/Tick))

board.cache_hierarchy.l1icaches.tags.occupanciesTaskId::1024 391 # Occupied blocks per task id (Count)

board.cache_hierarchy.l1icaches.tags.ageTaskId_1024::0 226 # Occupied blocks per task id, per block age (Count)

board.cache_hierarchy.l1icaches.tags.ageTaskId_1024::1 165 # Occupied blocks per task id, per block age (Count)

board.cache_hierarchy.l1icaches.tags.ratioOccsTaskId::1024 0.763672 # Ratio of occupied blocks and all blocks, per task id (Ratio)

board.cache_hierarchy.l1icaches.tags.tagAccesses 21207 # Number of tag accesses (Count)

board.cache_hierarchy.l1icaches.tags.dataAccesses 21207 # Number of data accesses (Count)

board.cache_hierarchy.l1icaches.tags.power_state.pwrStateResidencyTicks::UNDEFINED 19514466 # Cumulative time (in ticks) in various power states (Tick)

board.cache_hierarchy.l2bus.transDist::ReadResp 566 # Transaction distribution (Count)

board.cache_hierarchy.l2bus.transDist::CleanEvict 32 # Transaction distribution (Count)

board.cache_hierarchy.l2bus.transDist::HardPFReq 95 # Transaction distribution (Count)

board.cache_hierarchy.l2bus.transDist::ReadExReq 56 # Transaction distribution (Count)

board.cache_hierarchy.l2bus.transDist::ReadExResp 56 # Transaction distribution (Count)

board.cache_hierarchy.l2bus.transDist::ReadSharedReq 567 # Transaction distribution (Count)

board.cache_hierarchy.l2bus.pktCount_board.cache_hierarchy.l1icaches.mem_side_port::board.cache_hierarchy.l2cache.cpu_side_port 879 # Packet count per connected requestor and responder (Count)

board.cache_hierarchy.l2bus.pktCount_board.cache_hierarchy.l1dcaches.mem_side_port::board.cache_hierarchy.l2cache.cpu_side_port 398 # Packet count per connected requestor and responder (Count)

board.cache_hierarchy.l2bus.pktCount::total 1277 # Packet count per connected requestor and responder (Count)

board.cache_hierarchy.l2bus.pktSize_board.cache_hierarchy.l1icaches.mem_side_port::board.cache_hierarchy.l2cache.cpu_side_port 27072 # Cumulative packet size per connected requestor and responder (Byte)

board.cache_hierarchy.l2bus.pktSize_board.cache_hierarchy.l1dcaches.mem_side_port::board.cache_hierarchy.l2cache.cpu_side_port 12736 # Cumulative packet size per connected requestor and responder (Byte)

board.cache_hierarchy.l2bus.pktSize::total 39808 # Cumulative packet size per connected requestor and responder (Byte)

board.cache_hierarchy.l2bus.snoops 95 # Total snoops (Count)

board.cache_hierarchy.l2bus.snoopTraffic 0 # Total snoop traffic (Byte)

board.cache_hierarchy.l2bus.snoopFanout::samples 718 # Request fanout histogram (Count)

board.cache_hierarchy.l2bus.snoopFanout::mean 0.005571 # Request fanout histogram (Count)

board.cache_hierarchy.l2bus.snoopFanout::stdev 0.074483 # Request fanout histogram (Count)

board.cache_hierarchy.l2bus.snoopFanout::underflows 0 0.00% 0.00% # Request fanout histogram (Count)

board.cache_hierarchy.l2bus.snoopFanout::0 714 99.44% 99.44% # Request fanout histogram (Count)

board.cache_hierarchy.l2bus.snoopFanout::1 4 0.56% 100.00% # Request fanout histogram (Count)

board.cache_hierarchy.l2bus.snoopFanout::2 0 0.00% 100.00% # Request fanout histogram (Count)

board.cache_hierarchy.l2bus.snoopFanout::3 0 0.00% 100.00% # Request fanout histogram (Count)

board.cache_hierarchy.l2bus.snoopFanout::4 0 0.00% 100.00% # Request fanout histogram (Count)

board.cache_hierarchy.l2bus.snoopFanout::overflows 0 0.00% 100.00% # Request fanout histogram (Count)

board.cache_hierarchy.l2bus.snoopFanout::min_value 0 # Request fanout histogram (Count)

board.cache_hierarchy.l2bus.snoopFanout::max_value 1 # Request fanout histogram (Count)

board.cache_hierarchy.l2bus.snoopFanout::total 718 # Request fanout histogram (Count)

board.cache_hierarchy.l2bus.power_state.pwrStateResidencyTicks::UNDEFINED 19514466 # Cumulative time (in ticks) in various power states (Tick)

board.cache_hierarchy.l2bus.reqLayer0.occupancy 229667 # Layer occupancy (ticks) (Tick)

board.cache_hierarchy.l2bus.reqLayer0.utilization 0.0 # Layer utilization (Ratio)

board.cache_hierarchy.l2bus.respLayer0.occupancy 422577 # Layer occupancy (ticks) (Tick)

board.cache_hierarchy.l2bus.respLayer0.utilization 0.0 # Layer utilization (Ratio)

board.cache_hierarchy.l2bus.respLayer1.occupancy 198801 # Layer occupancy (ticks) (Tick)

board.cache_hierarchy.l2bus.respLayer1.utilization 0.0 # Layer utilization (Ratio)

board.cache_hierarchy.l2bus.snoop_filter.totRequests 655 # Total number of requests made to the snoop filter. (Count)

board.cache_hierarchy.l2bus.snoop_filter.hitSingleRequests 36 # Number of requests hitting in the snoop filter with a single holder of the requested data. (Count)

board.cache_hierarchy.l2bus.snoop_filter.hitMultiRequests 0 # Number of requests hitting in the snoop filter with multiple (>1) holders of the requested data. (Count)

board.cache_hierarchy.l2bus.snoop_filter.totSnoops 0 # Total number of snoops made to the snoop filter. (Count)

board.cache_hierarchy.l2bus.snoop_filter.hitSingleSnoops 0 # Number of snoops hitting in the snoop filter with a single holder of the requested data. (Count)

board.cache_hierarchy.l2bus.snoop_filter.hitMultiSnoops 0 # Number of snoops hitting in the snoop filter with multiple (>1) holders of the requested data. (Count)

board.cache_hierarchy.l2cache.demandHits::cache_hierarchy.l1dcaches.prefetcher 12 # number of demand (read+write) hits (Count)

board.cache_hierarchy.l2cache.demandHits::processor.cores.core.inst 5 # number of demand (read+write) hits (Count)

board.cache_hierarchy.l2cache.demandHits::processor.cores.core.data 6 # number of demand (read+write) hits (Count)

board.cache_hierarchy.l2cache.demandHits::total 23 # number of demand (read+write) hits (Count)

board.cache_hierarchy.l2cache.overallHits::cache_hierarchy.l1dcaches.prefetcher 12 # number of overall hits (Count)

board.cache_hierarchy.l2cache.overallHits::processor.cores.core.inst 5 # number of overall hits (Count)

board.cache_hierarchy.l2cache.overallHits::processor.cores.core.data 6 # number of overall hits (Count)

board.cache_hierarchy.l2cache.overallHits::total 23 # number of overall hits (Count)

board.cache_hierarchy.l2cache.demandMisses::cache_hierarchy.l1dcaches.prefetcher 44 # number of demand (read+write) misses (Count)

board.cache_hierarchy.l2cache.demandMisses::processor.cores.core.inst 419 # number of demand (read+write) misses (Count)

board.cache_hierarchy.l2cache.demandMisses::processor.cores.core.data 137 # number of demand (read+write) misses (Count)

board.cache_hierarchy.l2cache.demandMisses::total 600 # number of demand (read+write) misses (Count)

board.cache_hierarchy.l2cache.overallMisses::cache_hierarchy.l1dcaches.prefetcher 44 # number of overall misses (Count)

board.cache_hierarchy.l2cache.overallMisses::processor.cores.core.inst 419 # number of overall misses (Count)

board.cache_hierarchy.l2cache.overallMisses::processor.cores.core.data 137 # number of overall misses (Count)

board.cache_hierarchy.l2cache.overallMisses::total 600 # number of overall misses (Count)

board.cache_hierarchy.l2cache.demandMissLatency::cache_hierarchy.l1dcaches.prefetcher 2316939 # number of demand (read+write) miss ticks (Tick)

board.cache_hierarchy.l2cache.demandMissLatency::processor.cores.core.inst 22775868 # number of demand (read+write) miss ticks (Tick)

board.cache_hierarchy.l2cache.demandMissLatency::processor.cores.core.data 9029295 # number of demand (read+write) miss ticks (Tick)

board.cache_hierarchy.l2cache.demandMissLatency::total 34122102 # number of demand (read+write) miss ticks (Tick)

board.cache_hierarchy.l2cache.overallMissLatency::cache_hierarchy.l1dcaches.prefetcher 2316939 # number of overall miss ticks (Tick)

board.cache_hierarchy.l2cache.overallMissLatency::processor.cores.core.inst 22775868 # number of overall miss ticks (Tick)

board.cache_hierarchy.l2cache.overallMissLatency::processor.cores.core.data 9029295 # number of overall miss ticks (Tick)

board.cache_hierarchy.l2cache.overallMissLatency::total 34122102 # number of overall miss ticks (Tick)

board.cache_hierarchy.l2cache.demandAccesses::cache_hierarchy.l1dcaches.prefetcher 56 # number of demand (read+write) accesses (Count)

board.cache_hierarchy.l2cache.demandAccesses::processor.cores.core.inst 424 # number of demand (read+write) accesses (Count)

board.cache_hierarchy.l2cache.demandAccesses::processor.cores.core.data 143 # number of demand (read+write) accesses (Count)

board.cache_hierarchy.l2cache.demandAccesses::total 623 # number of demand (read+write) accesses (Count)

board.cache_hierarchy.l2cache.overallAccesses::cache_hierarchy.l1dcaches.prefetcher 56 # number of overall (read+write) accesses (Count)

board.cache_hierarchy.l2cache.overallAccesses::processor.cores.core.inst 424 # number of overall (read+write) accesses (Count)

board.cache_hierarchy.l2cache.overallAccesses::processor.cores.core.data 143 # number of overall (read+write) accesses (Count)

board.cache_hierarchy.l2cache.overallAccesses::total 623 # number of overall (read+write) accesses (Count)

board.cache_hierarchy.l2cache.demandMissRate::cache_hierarchy.l1dcaches.prefetcher 0.785714 # miss rate for demand accesses (Ratio)

board.cache_hierarchy.l2cache.demandMissRate::processor.cores.core.inst 0.988208 # miss rate for demand accesses (Ratio)

board.cache_hierarchy.l2cache.demandMissRate::processor.cores.core.data 0.958042 # miss rate for demand accesses (Ratio)

board.cache_hierarchy.l2cache.demandMissRate::total 0.963082 # miss rate for demand accesses (Ratio)

board.cache_hierarchy.l2cache.overallMissRate::cache_hierarchy.l1dcaches.prefetcher 0.785714 # miss rate for overall accesses (Ratio)

board.cache_hierarchy.l2cache.overallMissRate::processor.cores.core.inst 0.988208 # miss rate for overall accesses (Ratio)

board.cache_hierarchy.l2cache.overallMissRate::processor.cores.core.data 0.958042 # miss rate for overall accesses (Ratio)

board.cache_hierarchy.l2cache.overallMissRate::total 0.963082 # miss rate for overall accesses (Ratio)

board.cache_hierarchy.l2cache.demandAvgMissLatency::cache_hierarchy.l1dcaches.prefetcher 52657.704545 # average overall miss latency in ticks ((Tick/Count))

board.cache_hierarchy.l2cache.demandAvgMissLatency::processor.cores.core.inst 54357.680191 # average overall miss latency in ticks ((Tick/Count))

board.cache_hierarchy.l2cache.demandAvgMissLatency::processor.cores.core.data 65907.262774 # average overall miss latency in ticks ((Tick/Count))

board.cache_hierarchy.l2cache.demandAvgMissLatency::total 56870.170000 # average overall miss latency in ticks ((Tick/Count))

board.cache_hierarchy.l2cache.overallAvgMissLatency::cache_hierarchy.l1dcaches.prefetcher 52657.704545 # average overall miss latency ((Tick/Count))

board.cache_hierarchy.l2cache.overallAvgMissLatency::processor.cores.core.inst 54357.680191 # average overall miss latency ((Tick/Count))

board.cache_hierarchy.l2cache.overallAvgMissLatency::processor.cores.core.data 65907.262774 # average overall miss latency ((Tick/Count))

board.cache_hierarchy.l2cache.overallAvgMissLatency::total 56870.170000 # average overall miss latency ((Tick/Count))

board.cache_hierarchy.l2cache.blockedCycles::no_mshrs 0 # number of cycles access was blocked (Cycle)

board.cache_hierarchy.l2cache.blockedCycles::no_targets 0 # number of cycles access was blocked (Cycle)

board.cache_hierarchy.l2cache.blockedCauses::no_mshrs 0 # number of times access was blocked (Count)

board.cache_hierarchy.l2cache.blockedCauses::no_targets 0 # number of times access was blocked (Count)

board.cache_hierarchy.l2cache.avgBlocked::no_mshrs nan # average number of cycles each access was blocked ((Cycle/Count))

board.cache_hierarchy.l2cache.avgBlocked::no_targets nan # average number of cycles each access was blocked ((Cycle/Count))

board.cache_hierarchy.l2cache.demandMshrHits::cache_hierarchy.l1dcaches.prefetcher 35 # number of demand (read+write) MSHR hits (Count)

board.cache_hierarchy.l2cache.demandMshrHits::total 35 # number of demand (read+write) MSHR hits (Count)

board.cache_hierarchy.l2cache.overallMshrHits::cache_hierarchy.l1dcaches.prefetcher 35 # number of overall MSHR hits (Count)

board.cache_hierarchy.l2cache.overallMshrHits::total 35 # number of overall MSHR hits (Count)

board.cache_hierarchy.l2cache.demandMshrMisses::cache_hierarchy.l1dcaches.prefetcher 9 # number of demand (read+write) MSHR misses (Count)

board.cache_hierarchy.l2cache.demandMshrMisses::processor.cores.core.inst 419 # number of demand (read+write) MSHR misses (Count)

board.cache_hierarchy.l2cache.demandMshrMisses::processor.cores.core.data 137 # number of demand (read+write) MSHR misses (Count)

board.cache_hierarchy.l2cache.demandMshrMisses::total 565 # number of demand (read+write) MSHR misses (Count)

board.cache_hierarchy.l2cache.overallMshrMisses::cache_hierarchy.l1dcaches.prefetcher 9 # number of overall MSHR misses (Count)

board.cache_hierarchy.l2cache.overallMshrMisses::cache_hierarchy.l2cache.prefetcher 64 # number of overall MSHR misses (Count)

board.cache_hierarchy.l2cache.overallMshrMisses::processor.cores.core.inst 419 # number of overall MSHR misses (Count)

board.cache_hierarchy.l2cache.overallMshrMisses::processor.cores.core.data 137 # number of overall MSHR misses (Count)

board.cache_hierarchy.l2cache.overallMshrMisses::total 629 # number of overall MSHR misses (Count)

board.cache_hierarchy.l2cache.demandMshrMissLatency::cache_hierarchy.l1dcaches.prefetcher 827838 # number of demand (read+write) MSHR miss ticks (Tick)

board.cache_hierarchy.l2cache.demandMshrMissLatency::processor.cores.core.inst 22636674 # number of demand (read+write) MSHR miss ticks (Tick)

board.cache_hierarchy.l2cache.demandMshrMissLatency::processor.cores.core.data 8983674 # number of demand (read+write) MSHR miss ticks (Tick)

board.cache_hierarchy.l2cache.demandMshrMissLatency::total 32448186 # number of demand (read+write) MSHR miss ticks (Tick)

board.cache_hierarchy.l2cache.overallMshrMissLatency::cache_hierarchy.l1dcaches.prefetcher 827838 # number of overall MSHR miss ticks (Tick)

board.cache_hierarchy.l2cache.overallMshrMissLatency::cache_hierarchy.l2cache.prefetcher 4087712 # number of overall MSHR miss ticks (Tick)

board.cache_hierarchy.l2cache.overallMshrMissLatency::processor.cores.core.inst 22636674 # number of overall MSHR miss ticks (Tick)

board.cache_hierarchy.l2cache.overallMshrMissLatency::processor.cores.core.data 8983674 # number of overall MSHR miss ticks (Tick)

board.cache_hierarchy.l2cache.overallMshrMissLatency::total 36535898 # number of overall MSHR miss ticks (Tick)

board.cache_hierarchy.l2cache.demandMshrMissRate::cache_hierarchy.l1dcaches.prefetcher 0.160714 # mshr miss ratio for demand accesses (Ratio)

board.cache_hierarchy.l2cache.demandMshrMissRate::processor.cores.core.inst 0.988208 # mshr miss ratio for demand accesses (Ratio)

board.cache_hierarchy.l2cache.demandMshrMissRate::processor.cores.core.data 0.958042 # mshr miss ratio for demand accesses (Ratio)

board.cache_hierarchy.l2cache.demandMshrMissRate::total 0.906902 # mshr miss ratio for demand accesses (Ratio)

board.cache_hierarchy.l2cache.overallMshrMissRate::cache_hierarchy.l1dcaches.prefetcher 0.160714 # mshr miss ratio for overall accesses (Ratio)

board.cache_hierarchy.l2cache.overallMshrMissRate::cache_hierarchy.l2cache.prefetcher inf # mshr miss ratio for overall accesses (Ratio)

board.cache_hierarchy.l2cache.overallMshrMissRate::processor.cores.core.inst 0.988208 # mshr miss ratio for overall accesses (Ratio)

board.cache_hierarchy.l2cache.overallMshrMissRate::processor.cores.core.data 0.958042 # mshr miss ratio for overall accesses (Ratio)

board.cache_hierarchy.l2cache.overallMshrMissRate::total 1.009631 # mshr miss ratio for overall accesses (Ratio)

board.cache_hierarchy.l2cache.demandAvgMshrMissLatency::cache_hierarchy.l1dcaches.prefetcher 91982 # average overall mshr miss latency ((Tick/Count))

board.cache_hierarchy.l2cache.demandAvgMshrMissLatency::processor.cores.core.inst 54025.474940 # average overall mshr miss latency ((Tick/Count))

board.cache_hierarchy.l2cache.demandAvgMshrMissLatency::processor.cores.core.data 65574.262774 # average overall mshr miss latency ((Tick/Count))

board.cache_hierarchy.l2cache.demandAvgMshrMissLatency::total 57430.417699 # average overall mshr miss latency ((Tick/Count))

board.cache_hierarchy.l2cache.overallAvgMshrMissLatency::cache_hierarchy.l1dcaches.prefetcher 91982 # average overall mshr miss latency ((Tick/Count))

board.cache_hierarchy.l2cache.overallAvgMshrMissLatency::cache_hierarchy.l2cache.prefetcher 63870.500000 # average overall mshr miss latency ((Tick/Count))

board.cache_hierarchy.l2cache.overallAvgMshrMissLatency::processor.cores.core.inst 54025.474940 # average overall mshr miss latency ((Tick/Count))

board.cache_hierarchy.l2cache.overallAvgMshrMissLatency::processor.cores.core.data 65574.262774 # average overall mshr miss latency ((Tick/Count))

board.cache_hierarchy.l2cache.overallAvgMshrMissLatency::total 58085.688394 # average overall mshr miss latency ((Tick/Count))

board.cache_hierarchy.l2cache.replacements 0 # number of replacements (Count)

board.cache_hierarchy.l2cache.HardPFReq.mshrMisses::cache_hierarchy.l2cache.prefetcher 64 # number of HardPFReq MSHR misses (Count)

board.cache_hierarchy.l2cache.HardPFReq.mshrMisses::total 64 # number of HardPFReq MSHR misses (Count)

board.cache_hierarchy.l2cache.HardPFReq.mshrMissLatency::cache_hierarchy.l2cache.prefetcher 4087712 # number of HardPFReq MSHR miss ticks (Tick)

board.cache_hierarchy.l2cache.HardPFReq.mshrMissLatency::total 4087712 # number of HardPFReq MSHR miss ticks (Tick)

board.cache_hierarchy.l2cache.HardPFReq.mshrMissRate::cache_hierarchy.l2cache.prefetcher inf # mshr miss rate for HardPFReq accesses (Ratio)

board.cache_hierarchy.l2cache.HardPFReq.mshrMissRate::total inf # mshr miss rate for HardPFReq accesses (Ratio)

board.cache_hierarchy.l2cache.HardPFReq.avgMshrMissLatency::cache_hierarchy.l2cache.prefetcher 63870.500000 # average HardPFReq mshr miss latency ((Tick/Count))

board.cache_hierarchy.l2cache.HardPFReq.avgMshrMissLatency::total 63870.500000 # average HardPFReq mshr miss latency ((Tick/Count))

board.cache_hierarchy.l2cache.ReadExReq.hits::processor.cores.core.data 5 # number of ReadExReq hits (Count)

board.cache_hierarchy.l2cache.ReadExReq.hits::total 5 # number of ReadExReq hits (Count)

board.cache_hierarchy.l2cache.ReadExReq.misses::processor.cores.core.data 51 # number of ReadExReq misses (Count)

board.cache_hierarchy.l2cache.ReadExReq.misses::total 51 # number of ReadExReq misses (Count)

board.cache_hierarchy.l2cache.ReadExReq.missLatency::processor.cores.core.data 3285045 # number of ReadExReq miss ticks (Tick)

board.cache_hierarchy.l2cache.ReadExReq.missLatency::total 3285045 # number of ReadExReq miss ticks (Tick)

board.cache_hierarchy.l2cache.ReadExReq.accesses::processor.cores.core.data 56 # number of ReadExReq accesses(hits+misses) (Count)

board.cache_hierarchy.l2cache.ReadExReq.accesses::total 56 # number of ReadExReq accesses(hits+misses) (Count)

board.cache_hierarchy.l2cache.ReadExReq.missRate::processor.cores.core.data 0.910714 # miss rate for ReadExReq accesses (Ratio)

board.cache_hierarchy.l2cache.ReadExReq.missRate::total 0.910714 # miss rate for ReadExReq accesses (Ratio)

board.cache_hierarchy.l2cache.ReadExReq.avgMissLatency::processor.cores.core.data 64412.647059 # average ReadExReq miss latency ((Tick/Count))

board.cache_hierarchy.l2cache.ReadExReq.avgMissLatency::total 64412.647059 # average ReadExReq miss latency ((Tick/Count))

board.cache_hierarchy.l2cache.ReadExReq.mshrMisses::processor.cores.core.data 51 # number of ReadExReq MSHR misses (Count)

board.cache_hierarchy.l2cache.ReadExReq.mshrMisses::total 51 # number of ReadExReq MSHR misses (Count)

board.cache_hierarchy.l2cache.ReadExReq.mshrMissLatency::processor.cores.core.data 3268062 # number of ReadExReq MSHR miss ticks (Tick)

board.cache_hierarchy.l2cache.ReadExReq.mshrMissLatency::total 3268062 # number of ReadExReq MSHR miss ticks (Tick)

board.cache_hierarchy.l2cache.ReadExReq.mshrMissRate::processor.cores.core.data 0.910714 # mshr miss rate for ReadExReq accesses (Ratio)

board.cache_hierarchy.l2cache.ReadExReq.mshrMissRate::total 0.910714 # mshr miss rate for ReadExReq accesses (Ratio)

board.cache_hierarchy.l2cache.ReadExReq.avgMshrMissLatency::processor.cores.core.data 64079.647059 # average ReadExReq mshr miss latency ((Tick/Count))

board.cache_hierarchy.l2cache.ReadExReq.avgMshrMissLatency::total 64079.647059 # average ReadExReq mshr miss latency ((Tick/Count))

board.cache_hierarchy.l2cache.ReadSharedReq.hits::cache_hierarchy.l1dcaches.prefetcher 12 # number of ReadSharedReq hits (Count)

board.cache_hierarchy.l2cache.ReadSharedReq.hits::processor.cores.core.inst 5 # number of ReadSharedReq hits (Count)

board.cache_hierarchy.l2cache.ReadSharedReq.hits::processor.cores.core.data 1 # number of ReadSharedReq hits (Count)

board.cache_hierarchy.l2cache.ReadSharedReq.hits::total 18 # number of ReadSharedReq hits (Count)

board.cache_hierarchy.l2cache.ReadSharedReq.misses::cache_hierarchy.l1dcaches.prefetcher 44 # number of ReadSharedReq misses (Count)

board.cache_hierarchy.l2cache.ReadSharedReq.misses::processor.cores.core.inst 419 # number of ReadSharedReq misses (Count)

board.cache_hierarchy.l2cache.ReadSharedReq.misses::processor.cores.core.data 86 # number of ReadSharedReq misses (Count)

board.cache_hierarchy.l2cache.ReadSharedReq.misses::total 549 # number of ReadSharedReq misses (Count)

board.cache_hierarchy.l2cache.ReadSharedReq.missLatency::cache_hierarchy.l1dcaches.prefetcher 2316939 # number of ReadSharedReq miss ticks (Tick)

board.cache_hierarchy.l2cache.ReadSharedReq.missLatency::processor.cores.core.inst 22775868 # number of ReadSharedReq miss ticks (Tick)

board.cache_hierarchy.l2cache.ReadSharedReq.missLatency::processor.cores.core.data 5744250 # number of ReadSharedReq miss ticks (Tick)

board.cache_hierarchy.l2cache.ReadSharedReq.missLatency::total 30837057 # number of ReadSharedReq miss ticks (Tick)

board.cache_hierarchy.l2cache.ReadSharedReq.accesses::cache_hierarchy.l1dcaches.prefetcher 56 # number of ReadSharedReq accesses(hits+misses) (Count)

board.cache_hierarchy.l2cache.ReadSharedReq.accesses::processor.cores.core.inst 424 # number of ReadSharedReq accesses(hits+misses) (Count)

board.cache_hierarchy.l2cache.ReadSharedReq.accesses::processor.cores.core.data 87 # number of ReadSharedReq accesses(hits+misses) (Count)

board.cache_hierarchy.l2cache.ReadSharedReq.accesses::total 567 # number of ReadSharedReq accesses(hits+misses) (Count)

board.cache_hierarchy.l2cache.ReadSharedReq.missRate::cache_hierarchy.l1dcaches.prefetcher 0.785714 # miss rate for ReadSharedReq accesses (Ratio)

board.cache_hierarchy.l2cache.ReadSharedReq.missRate::processor.cores.core.inst 0.988208 # miss rate for ReadSharedReq accesses (Ratio)

board.cache_hierarchy.l2cache.ReadSharedReq.missRate::processor.cores.core.data 0.988506 # miss rate for ReadSharedReq accesses (Ratio)

board.cache_hierarchy.l2cache.ReadSharedReq.missRate::total 0.968254 # miss rate for ReadSharedReq accesses (Ratio)

board.cache_hierarchy.l2cache.ReadSharedReq.avgMissLatency::cache_hierarchy.l1dcaches.prefetcher 52657.704545 # average ReadSharedReq miss latency ((Tick/Count))

board.cache_hierarchy.l2cache.ReadSharedReq.avgMissLatency::processor.cores.core.inst 54357.680191 # average ReadSharedReq miss latency ((Tick/Count))

board.cache_hierarchy.l2cache.ReadSharedReq.avgMissLatency::processor.cores.core.data 66793.604651 # average ReadSharedReq miss latency ((Tick/Count))

board.cache_hierarchy.l2cache.ReadSharedReq.avgMissLatency::total 56169.502732 # average ReadSharedReq miss latency ((Tick/Count))

board.cache_hierarchy.l2cache.ReadSharedReq.mshrHits::cache_hierarchy.l1dcaches.prefetcher 35 # number of ReadSharedReq MSHR hits (Count)

board.cache_hierarchy.l2cache.ReadSharedReq.mshrHits::total 35 # number of ReadSharedReq MSHR hits (Count)

board.cache_hierarchy.l2cache.ReadSharedReq.mshrMisses::cache_hierarchy.l1dcaches.prefetcher 9 # number of ReadSharedReq MSHR misses (Count)

board.cache_hierarchy.l2cache.ReadSharedReq.mshrMisses::processor.cores.core.inst 419 # number of ReadSharedReq MSHR misses (Count)

board.cache_hierarchy.l2cache.ReadSharedReq.mshrMisses::processor.cores.core.data 86 # number of ReadSharedReq MSHR misses (Count)

board.cache_hierarchy.l2cache.ReadSharedReq.mshrMisses::total 514 # number of ReadSharedReq MSHR misses (Count)

board.cache_hierarchy.l2cache.ReadSharedReq.mshrMissLatency::cache_hierarchy.l1dcaches.prefetcher 827838 # number of ReadSharedReq MSHR miss ticks (Tick)

board.cache_hierarchy.l2cache.ReadSharedReq.mshrMissLatency::processor.cores.core.inst 22636674 # number of ReadSharedReq MSHR miss ticks (Tick)

board.cache_hierarchy.l2cache.ReadSharedReq.mshrMissLatency::processor.cores.core.data 5715612 # number of ReadSharedReq MSHR miss ticks (Tick)

board.cache_hierarchy.l2cache.ReadSharedReq.mshrMissLatency::total 29180124 # number of ReadSharedReq MSHR miss ticks (Tick)

board.cache_hierarchy.l2cache.ReadSharedReq.mshrMissRate::cache_hierarchy.l1dcaches.prefetcher 0.160714 # mshr miss rate for ReadSharedReq accesses (Ratio)

board.cache_hierarchy.l2cache.ReadSharedReq.mshrMissRate::processor.cores.core.inst 0.988208 # mshr miss rate for ReadSharedReq accesses (Ratio)

board.cache_hierarchy.l2cache.ReadSharedReq.mshrMissRate::processor.cores.core.data 0.988506 # mshr miss rate for ReadSharedReq accesses (Ratio)

board.cache_hierarchy.l2cache.ReadSharedReq.mshrMissRate::total 0.906526 # mshr miss rate for ReadSharedReq accesses (Ratio)

board.cache_hierarchy.l2cache.ReadSharedReq.avgMshrMissLatency::cache_hierarchy.l1dcaches.prefetcher 91982 # average ReadSharedReq mshr miss latency ((Tick/Count))

board.cache_hierarchy.l2cache.ReadSharedReq.avgMshrMissLatency::processor.cores.core.inst 54025.474940 # average ReadSharedReq mshr miss latency ((Tick/Count))

board.cache_hierarchy.l2cache.ReadSharedReq.avgMshrMissLatency::processor.cores.core.data 66460.604651 # average ReadSharedReq mshr miss latency ((Tick/Count))

board.cache_hierarchy.l2cache.ReadSharedReq.avgMshrMissLatency::total 56770.669261 # average ReadSharedReq mshr miss latency ((Tick/Count))

board.cache_hierarchy.l2cache.power_state.pwrStateResidencyTicks::UNDEFINED 19514466 # Cumulative time (in ticks) in various power states (Tick)

board.cache_hierarchy.l2cache.prefetcher.demandMshrMisses 565 # demands not covered by prefetchs (Count)

board.cache_hierarchy.l2cache.prefetcher.pfIssued 166 # number of hwpf issued (Count)

board.cache_hierarchy.l2cache.prefetcher.pfUseful 19 # number of useful prefetch (Count)

board.cache_hierarchy.l2cache.prefetcher.pfUsefulButMiss 0 # number of hit on prefetch but cache block is not in an usable state (Count)

board.cache_hierarchy.l2cache.prefetcher.accuracy 0.114458 # accuracy of the prefetcher (Count)

board.cache_hierarchy.l2cache.prefetcher.coverage 0.032534 # coverage brought by this prefetcher (Count)

board.cache_hierarchy.l2cache.prefetcher.pfHitInCache 29 # number of prefetches hitting in cache (Count)

board.cache_hierarchy.l2cache.prefetcher.pfHitInMSHR 73 # number of prefetches hitting in a MSHR (Count)

board.cache_hierarchy.l2cache.prefetcher.pfHitInWB 0 # number of prefetches hit in the Write Buffer (Count)

board.cache_hierarchy.l2cache.prefetcher.pfLate 102 # number of late prefetches (hitting in cache, MSHR or WB) (Count)

board.cache_hierarchy.l2cache.prefetcher.pfIdentified 207 # number of prefetch candidates identified (Count)

board.cache_hierarchy.l2cache.prefetcher.pfBufferHit 27 # number of redundant prefetches already in prefetch queue (Count)

board.cache_hierarchy.l2cache.prefetcher.pfInCache 0 # number of redundant prefetches already in cache/mshr dropped (Count)

board.cache_hierarchy.l2cache.prefetcher.pfRemovedDemand 10 # number of prefetches dropped due to a demand for the same address (Count)

board.cache_hierarchy.l2cache.prefetcher.pfRemovedFull 0 # number of prefetches dropped due to prefetch queue size (Count)

board.cache_hierarchy.l2cache.prefetcher.pfSpanPage 13 # number of prefetches that crossed the page (Count)

board.cache_hierarchy.l2cache.prefetcher.pfUsefulSpanPage 2 # number of prefetches that is useful and crossed the page (Count)

board.cache_hierarchy.l2cache.prefetcher.power_state.pwrStateResidencyTicks::UNDEFINED 19514466 # Cumulative time (in ticks) in various power states (Tick)

board.cache_hierarchy.l2cache.tags.tagsInUse 332.342904 # Average ticks per tags in use ((Tick/Count))

board.cache_hierarchy.l2cache.tags.totalRefs 683 # Total number of references to valid blocks. (Count)

board.cache_hierarchy.l2cache.tags.sampledRefs 628 # Sample count of references to valid blocks. (Count)

board.cache_hierarchy.l2cache.tags.avgRefs 1.087580 # Average number of references to valid blocks. ((Count/Count))

board.cache_hierarchy.l2cache.tags.warmupTick 69597 # The tick when the warmup percentage was hit. (Tick)

board.cache_hierarchy.l2cache.tags.occupancies::cache_hierarchy.l1dcaches.prefetcher 6.956077 # Average occupied blocks per tick, per requestor ((Count/Tick))

board.cache_hierarchy.l2cache.tags.occupancies::cache_hierarchy.l2cache.prefetcher 42.177179 # Average occupied blocks per tick, per requestor ((Count/Tick))

board.cache_hierarchy.l2cache.tags.occupancies::processor.cores.core.inst 202.496547 # Average occupied blocks per tick, per requestor ((Count/Tick))

board.cache_hierarchy.l2cache.tags.occupancies::processor.cores.core.data 80.713101 # Average occupied blocks per tick, per requestor ((Count/Tick))

board.cache_hierarchy.l2cache.tags.avgOccs::cache_hierarchy.l1dcaches.prefetcher 0.001698 # Average percentage of cache occupancy ((Ratio/Tick))

board.cache_hierarchy.l2cache.tags.avgOccs::cache_hierarchy.l2cache.prefetcher 0.010297 # Average percentage of cache occupancy ((Ratio/Tick))

board.cache_hierarchy.l2cache.tags.avgOccs::processor.cores.core.inst 0.049438 # Average percentage of cache occupancy ((Ratio/Tick))

board.cache_hierarchy.l2cache.tags.avgOccs::processor.cores.core.data 0.019705 # Average percentage of cache occupancy ((Ratio/Tick))

board.cache_hierarchy.l2cache.tags.avgOccs::total 0.081138 # Average percentage of cache occupancy ((Ratio/Tick))

board.cache_hierarchy.l2cache.tags.occupanciesTaskId::1022 73 # Occupied blocks per task id (Count)

board.cache_hierarchy.l2cache.tags.occupanciesTaskId::1024 555 # Occupied blocks per task id (Count)

board.cache_hierarchy.l2cache.tags.ageTaskId_1022::0 22 # Occupied blocks per task id, per block age (Count)

board.cache_hierarchy.l2cache.tags.ageTaskId_1022::1 51 # Occupied blocks per task id, per block age (Count)

board.cache_hierarchy.l2cache.tags.ageTaskId_1024::0 272 # Occupied blocks per task id, per block age (Count)

board.cache_hierarchy.l2cache.tags.ageTaskId_1024::1 283 # Occupied blocks per task id, per block age (Count)

board.cache_hierarchy.l2cache.tags.ratioOccsTaskId::1022 0.017822 # Ratio of occupied blocks and all blocks, per task id (Ratio)

board.cache_hierarchy.l2cache.tags.ratioOccsTaskId::1024 0.135498 # Ratio of occupied blocks and all blocks, per task id (Ratio)

board.cache_hierarchy.l2cache.tags.tagAccesses 11108 # Number of tag accesses (Count)

board.cache_hierarchy.l2cache.tags.dataAccesses 11108 # Number of data accesses (Count)

board.cache_hierarchy.l2cache.tags.power_state.pwrStateResidencyTicks::UNDEFINED 19514466 # Cumulative time (in ticks) in various power states (Tick)

board.cache_hierarchy.membus.transDist::ReadResp 577 # Transaction distribution (Count)

board.cache_hierarchy.membus.transDist::ReadExReq 51 # Transaction distribution (Count)

board.cache_hierarchy.membus.transDist::ReadExResp 51 # Transaction distribution (Count)

board.cache_hierarchy.membus.transDist::ReadSharedReq 578 # Transaction distribution (Count)

board.cache_hierarchy.membus.pktCount_board.cache_hierarchy.l2cache.mem_side_port::board.memory.mem_ctrl.port 1257 # Packet count per connected requestor and responder (Count)

board.cache_hierarchy.membus.pktCount_board.cache_hierarchy.l2cache.mem_side_port::total 1257 # Packet count per connected requestor and responder (Count)

board.cache_hierarchy.membus.pktCount::total 1257 # Packet count per connected requestor and responder (Count)

board.cache_hierarchy.membus.pktSize_board.cache_hierarchy.l2cache.mem_side_port::board.memory.mem_ctrl.port 40192 # Cumulative packet size per connected requestor and responder (Byte)

board.cache_hierarchy.membus.pktSize_board.cache_hierarchy.l2cache.mem_side_port::total 40192 # Cumulative packet size per connected requestor and responder (Byte)

board.cache_hierarchy.membus.pktSize::total 40192 # Cumulative packet size per connected requestor and responder (Byte)

board.cache_hierarchy.membus.snoops 0 # Total snoops (Count)

board.cache_hierarchy.membus.snoopTraffic 0 # Total snoop traffic (Byte)

board.cache_hierarchy.membus.snoopFanout::samples 629 # Request fanout histogram (Count)

board.cache_hierarchy.membus.snoopFanout::mean 0 # Request fanout histogram (Count)

board.cache_hierarchy.membus.snoopFanout::stdev 0 # Request fanout histogram (Count)

board.cache_hierarchy.membus.snoopFanout::underflows 0 0.00% 0.00% # Request fanout histogram (Count)

board.cache_hierarchy.membus.snoopFanout::0 629 100.00% 100.00% # Request fanout histogram (Count)

board.cache_hierarchy.membus.snoopFanout::1 0 0.00% 100.00% # Request fanout histogram (Count)

board.cache_hierarchy.membus.snoopFanout::overflows 0 0.00% 100.00% # Request fanout histogram (Count)

board.cache_hierarchy.membus.snoopFanout::min_value 0 # Request fanout histogram (Count)

board.cache_hierarchy.membus.snoopFanout::max_value 0 # Request fanout histogram (Count)

board.cache_hierarchy.membus.snoopFanout::total 629 # Request fanout histogram (Count)

board.cache_hierarchy.membus.badaddr_responder.power_state.pwrStateResidencyTicks::UNDEFINED 19514466 # Cumulative time (in ticks) in various power states (Tick)

board.cache_hierarchy.membus.power_state.pwrStateResidencyTicks::UNDEFINED 19514466 # Cumulative time (in ticks) in various power states (Tick)

board.cache_hierarchy.membus.reqLayer0.occupancy 218628 # Layer occupancy (ticks) (Tick)

board.cache_hierarchy.membus.reqLayer0.utilization 0.0 # Layer utilization (Ratio)

board.cache_hierarchy.membus.respLayer2.occupancy 521679 # Layer occupancy (ticks) (Tick)

board.cache_hierarchy.membus.respLayer2.utilization 0.0 # Layer utilization (Ratio)

board.cache_hierarchy.membus.snoop_filter.totRequests 629 # Total number of requests made to the snoop filter. (Count)

board.cache_hierarchy.membus.snoop_filter.hitSingleRequests 0 # Number of requests hitting in the snoop filter with a single holder of the requested data. (Count)

board.cache_hierarchy.membus.snoop_filter.hitMultiRequests 0 # Number of requests hitting in the snoop filter with multiple (>1) holders of the requested data. (Count)

board.cache_hierarchy.membus.snoop_filter.totSnoops 0 # Total number of snoops made to the snoop filter. (Count)

board.cache_hierarchy.membus.snoop_filter.hitSingleSnoops 0 # Number of snoops hitting in the snoop filter with a single holder of the requested data. (Count)

board.cache_hierarchy.membus.snoop_filter.hitMultiSnoops 0 # Number of snoops hitting in the snoop filter with multiple (>1) holders of the requested data. (Count)

board.clk_domain.clock 333 # Clock period in ticks (Tick)

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board.clk_domain.voltage_domain.voltage 1 # Voltage in Volts (Volt)
board.memory.mem_ctrl.avgPriority_cache_hierarchy.l1dcaches.prefetcher::samples 9.00 # Average QoS priority value for accepted requests (Count)
board.memory.mem_ctrl.avgPriority_cache_hierarchy.l2cache.prefetcher::samples 64.00 # Average QoS priority value for accepted requests (Count)
board.memory.mem_ctrl.avgPriority_processor.cores.core.inst::samples 419.00 # Average QoS priority value for accepted requests (Count)
board.memory.mem_ctrl.avgPriority_processor.cores.core.data::samples 137.00 # Average QoS priority value for accepted requests (Count)
board.memory.mem_ctrl.priorityMinLatency 0.000000018750 # per QoS priority minimum request to response latency (Second)
board.memory.mem_ctrl.priorityMaxLatency 0.000000503586 # per QoS priority maximum request to response latency (Second)
board.memory.mem_ctrl.numReadWriteTurnArounds 0 # Number of turnarounds from READ to WRITE (Count)
board.memory.mem_ctrl.numWriteReadTurnArounds 0 # Number of turnarounds from WRITE to READ (Count)
board.memory.mem_ctrl.numStayReadState 1175 # Number of times bus staying in READ state (Count)
board.memory.mem_ctrl.numStayWriteState 0 # Number of times bus staying in WRITE state (Count)
board.memory.mem_ctrl.readReqs 629 # Number of read requests accepted (Count)
board.memory.mem_ctrl.writeReqs 0 # Number of write requests accepted (Count)
board.memory.mem_ctrl.readBursts 629 # Number of controller read bursts, including those serviced by the write queue (Count)
board.memory.mem_ctrl.writeBursts 0 # Number of controller write bursts, including those merged in the write queue (Count)
board.memory.mem_ctrl.servicedByWrQ 0 # Number of controller read bursts serviced by the write queue (Count)
board.memory.mem_ctrl.mergedWrBursts 0 # Number of controller write bursts merged with an existing one (Count)
board.memory.mem_ctrl.neitherReadNorWriteReqs 0 # Number of requests that are neither read nor write (Count)
board.memory.mem_ctrl.avgRdQLen 2.11 # Average read queue length when enqueuing ((Count/Tick))
board.memory.mem_ctrl.avgWrQLen 0.00 # Average write queue length when enqueuing ((Count/Tick))
board.memory.mem_ctrl.numRdRetry 0 # Number of times read queue was full causing retry (Count)
board.memory.mem_ctrl.numWrRetry 0 # Number of times write queue was full causing retry (Count)
board.memory.mem_ctrl.readPktSize::0 0 # Read request sizes (log2) (Count)
board.memory.mem_ctrl.readPktSize::1 0 # Read request sizes (log2) (Count)
board.memory.mem_ctrl.readPktSize::2 0 # Read request sizes (log2) (Count)
board.memory.mem_ctrl.readPktSize::3 0 # Read request sizes (log2) (Count)
board.memory.mem_ctrl.readPktSize::4 0 # Read request sizes (log2) (Count)
board.memory.mem_ctrl.readPktSize::5 0 # Read request sizes (log2) (Count)
board.memory.mem_ctrl.readPktSize::6 629 # Read request sizes (log2) (Count)
board.memory.mem_ctrl.writePktSize::0 0 # Write request sizes (log2) (Count)
board.memory.mem_ctrl.writePktSize::1 0 # Write request sizes (log2) (Count)
board.memory.mem_ctrl.writePktSize::2 0 # Write request sizes (log2) (Count)
board.memory.mem_ctrl.writePktSize::3 0 # Write request sizes (log2) (Count)
board.memory.mem_ctrl.writePktSize::4 0 # Write request sizes (log2) (Count)
board.memory.mem_ctrl.writePktSize::5 0 # Write request sizes (log2) (Count)
board.memory.mem_ctrl.writePktSize::6 0 # Write request sizes (log2) (Count)
board.memory.mem_ctrl.rdQLenPdf::0 304 # What read queue length does an incoming req see (Count)
board.memory.mem_ctrl.rdQLenPdf::1 165 # What read queue length does an incoming req see (Count)
board.memory.mem_ctrl.rdQLenPdf::2 66 # What read queue length does an incoming req see (Count)
board.memory.mem_ctrl.rdQLenPdf::3 36 # What read queue length does an incoming req see (Count)
board.memory.mem_ctrl.rdQLenPdf::4 13 # What read queue length does an incoming req see (Count)
board.memory.mem_ctrl.rdQLenPdf::5 13 # What read queue length does an incoming req see (Count)
board.memory.mem_ctrl.rdQLenPdf::6 11 # What read queue length does an incoming req see (Count)
board.memory.mem_ctrl.rdQLenPdf::7 9 # What read queue length does an incoming req see (Count)
board.memory.mem_ctrl.rdQLenPdf::8 6 # What read queue length does an incoming req see (Count)
board.memory.mem_ctrl.rdQLenPdf::9 4 # What read queue length does an incoming req see (Count)
board.memory.mem_ctrl.rdQLenPdf::10 2 # What read queue length does an incoming req see (Count)
board.memory.mem_ctrl.rdQLenPdf::11 0 # What read queue length does an incoming req see (Count)
board.memory.mem_ctrl.rdQLenPdf::12 0 # What read queue length does an incoming req see (Count)
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board.memory.mem_ctrl.rdQLenPdf::13	0	# What read queue length does an incoming req see (Count)
board.memory.mem_ctrl.rdQLenPdf::14	0	# What read queue length does an incoming req see (Count)
board.memory.mem_ctrl.rdQLenPdf::15	0	# What read queue length does an incoming req see (Count)
board.memory.mem_ctrl.rdQLenPdf::16	0	# What read queue length does an incoming req see (Count)
board.memory.mem_ctrl.rdQLenPdf::17	0	# What read queue length does an incoming req see (Count)
board.memory.mem_ctrl.rdQLenPdf::18	0	# What read queue length does an incoming req see (Count)
board.memory.mem_ctrl.rdQLenPdf::19	0	# What read queue length does an incoming req see (Count)
board.memory.mem_ctrl.rdQLenPdf::20	0	# What read queue length does an incoming req see (Count)
board.memory.mem_ctrl.rdQLenPdf::21	0	# What read queue length does an incoming req see (Count)
board.memory.mem_ctrl.rdQLenPdf::22	0	# What read queue length does an incoming req see (Count)
board.memory.mem_ctrl.rdQLenPdf::23	0	# What read queue length does an incoming req see (Count)
board.memory.mem_ctrl.rdQLenPdf::24	0	# What read queue length does an incoming req see (Count)
board.memory.mem_ctrl.rdQLenPdf::25	0	# What read queue length does an incoming req see (Count)
board.memory.mem_ctrl.rdQLenPdf::26	0	# What read queue length does an incoming req see (Count)
board.memory.mem_ctrl.rdQLenPdf::27	0	# What read queue length does an incoming req see (Count)
board.memory.mem_ctrl.rdQLenPdf::28	0	# What read queue length does an incoming req see (Count)
board.memory.mem_ctrl.rdQLenPdf::29	0	# What read queue length does an incoming req see (Count)
board.memory.mem_ctrl.rdQLenPdf::30	0	# What read queue length does an incoming req see (Count)
board.memory.mem_ctrl.rdQLenPdf::31	0	# What read queue length does an incoming req see (Count)
board.memory.mem_ctrl.wrQLenPdf::0	0	# What write queue length does an incoming req see (Count)
board.memory.mem_ctrl.wrQLenPdf::1	0	# What write queue length does an incoming req see (Count)
board.memory.mem_ctrl.wrQLenPdf::2	0	# What write queue length does an incoming req see (Count)
board.memory.mem_ctrl.wrQLenPdf::3	0	# What write queue length does an incoming req see (Count)
board.memory.mem_ctrl.wrQLenPdf::4	0	# What write queue length does an incoming req see (Count)
board.memory.mem_ctrl.wrQLenPdf::5	0	# What write queue length does an incoming req see (Count)
board.memory.mem_ctrl.wrQLenPdf::6	0	# What write queue length does an incoming req see (Count)
board.memory.mem_ctrl.wrQLenPdf::7	0	# What write queue length does an incoming req see (Count)
board.memory.mem_ctrl.wrQLenPdf::8	0	# What write queue length does an incoming req see (Count)
board.memory.mem_ctrl.wrQLenPdf::9	0	# What write queue length does an incoming req see (Count)
board.memory.mem_ctrl.wrQLenPdf::10	0	# What write queue length does an incoming req see (Count)
board.memory.mem_ctrl.wrQLenPdf::11	0	# What write queue length does an incoming req see (Count)
board.memory.mem_ctrl.wrQLenPdf::12	0	# What write queue length does an incoming req see (Count)
board.memory.mem_ctrl.wrQLenPdf::13	0	# What write queue length does an incoming req see (Count)
board.memory.mem_ctrl.wrQLenPdf::14	0	# What write queue length does an incoming req see (Count)
board.memory.mem_ctrl.wrQLenPdf::15	0	# What write queue length does an incoming req see (Count)
board.memory.mem_ctrl.wrQLenPdf::16	0	# What write queue length does an incoming req see (Count)
board.memory.mem_ctrl.wrQLenPdf::17	0	# What write queue length does an incoming req see (Count)
board.memory.mem_ctrl.wrQLenPdf::18	0	# What write queue length does an incoming req see (Count)
board.memory.mem_ctrl.wrQLenPdf::19	0	# What write queue length does an incoming req see (Count)
board.memory.mem_ctrl.wrQLenPdf::20	0	# What write queue length does an incoming req see (Count)
board.memory.mem_ctrl.wrQLenPdf::21	0	# What write queue length does an incoming req see (Count)
board.memory.mem_ctrl.wrQLenPdf::22	0	# What write queue length does an incoming req see (Count)
board.memory.mem_ctrl.wrQLenPdf::23	0	# What write queue length does an incoming req see (Count)
board.memory.mem_ctrl.wrQLenPdf::24	0	# What write queue length does an incoming req see (Count)
board.memory.mem_ctrl.wrQLenPdf::25	0	# What write queue length does an incoming req see (Count)
board.memory.mem_ctrl.wrQLenPdf::26	0	# What write queue length does an incoming req see (Count)
board.memory.mem_ctrl.wrQLenPdf::27	0	# What write queue length does an incoming req see (Count)
board.memory.mem_ctrl.wrQLenPdf::28	0	# What write queue length does an incoming req see (Count)
board.memory.mem_ctrl.wrQLenPdf::29	0	# What write queue length does an incoming req see (Count)

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board.memory.mem_ctrl.wrQLenPdf::30 0 # What write queue length does an incoming req see (Count)
board.memory.mem_ctrl.wrQLenPdf::31 0 # What write queue length does an incoming req see (Count)
board.memory.mem_ctrl.wrQLenPdf::32 0 # What write queue length does an incoming req see (Count)
board.memory.mem_ctrl.wrQLenPdf::33 0 # What write queue length does an incoming req see (Count)
board.memory.mem_ctrl.wrQLenPdf::34 0 # What write queue length does an incoming req see (Count)
board.memory.mem_ctrl.wrQLenPdf::35 0 # What write queue length does an incoming req see (Count)
board.memory.mem_ctrl.wrQLenPdf::36 0 # What write queue length does an incoming req see (Count)
board.memory.mem_ctrl.wrQLenPdf::37 0 # What write queue length does an incoming req see (Count)
board.memory.mem_ctrl.wrQLenPdf::38 0 # What write queue length does an incoming req see (Count)
board.memory.mem_ctrl.wrQLenPdf::39 0 # What write queue length does an incoming req see (Count)
board.memory.mem_ctrl.wrQLenPdf::40 0 # What write queue length does an incoming req see (Count)
board.memory.mem_ctrl.wrQLenPdf::41 0 # What write queue length does an incoming req see (Count)
board.memory.mem_ctrl.wrQLenPdf::42 0 # What write queue length does an incoming req see (Count)
board.memory.mem_ctrl.wrQLenPdf::43 0 # What write queue length does an incoming req see (Count)
board.memory.mem_ctrl.wrQLenPdf::44 0 # What write queue length does an incoming req see (Count)
board.memory.mem_ctrl.wrQLenPdf::45 0 # What write queue length does an incoming req see (Count)
board.memory.mem_ctrl.wrQLenPdf::46 0 # What write queue length does an incoming req see (Count)
board.memory.mem_ctrl.wrQLenPdf::47 0 # What write queue length does an incoming req see (Count)
board.memory.mem_ctrl.wrQLenPdf::48 0 # What write queue length does an incoming req see (Count)
board.memory.mem_ctrl.wrQLenPdf::49 0 # What write queue length does an incoming req see (Count)
board.memory.mem_ctrl.wrQLenPdf::50 0 # What write queue length does an incoming req see (Count)
board.memory.mem_ctrl.wrQLenPdf::51 0 # What write queue length does an incoming req see (Count)
board.memory.mem_ctrl.wrQLenPdf::52 0 # What write queue length does an incoming req see (Count)
board.memory.mem_ctrl.wrQLenPdf::53 0 # What write queue length does an incoming req see (Count)
board.memory.mem_ctrl.wrQLenPdf::54 0 # What write queue length does an incoming req see (Count)
board.memory.mem_ctrl.wrQLenPdf::55 0 # What write queue length does an incoming req see (Count)
board.memory.mem_ctrl.wrQLenPdf::56 0 # What write queue length does an incoming req see (Count)
board.memory.mem_ctrl.wrQLenPdf::57 0 # What write queue length does an incoming req see (Count)
board.memory.mem_ctrl.wrQLenPdf::58 0 # What write queue length does an incoming req see (Count)
board.memory.mem_ctrl.wrQLenPdf::59 0 # What write queue length does an incoming req see (Count)
board.memory.mem_ctrl.wrQLenPdf::60 0 # What write queue length does an incoming req see (Count)
board.memory.mem_ctrl.wrQLenPdf::61 0 # What write queue length does an incoming req see (Count)
board.memory.mem_ctrl.wrQLenPdf::62 0 # What write queue length does an incoming req see (Count)
board.memory.mem_ctrl.wrQLenPdf::63 0 # What write queue length does an incoming req see (Count)
board.memory.mem_ctrl.bytesReadWrQ 0 # Total number of bytes read from write queue (Byte)
board.memory.mem_ctrl.bytesReadSys 40256 # Total read bytes from the system interface side (Byte)
board.memory.mem_ctrl.bytesWrittenSys 0 # Total written bytes from the system interface side (Byte)
board.memory.mem_ctrl.avgRdBWSys 2062879916.87807369 # Average system read bandwidth in Byte/s ((Byte/Second))
board.memory.mem_ctrl.avgWrBWSys 0.00000000 # Average system write bandwidth in Byte/s ((Byte/Second))
board.memory.mem_ctrl.totGap 19509471 # Total gap between requests (Tick)
board.memory.mem_ctrl.avgGap 31016.65 # Average gap between requests ((Tick/Count))
board.memory.mem_ctrl.requestorReadBytes::cache_hierarchy.l1dcaches.prefetcher 576 # Per-requestor bytes read from memory (Byte)
board.memory.mem_ctrl.requestorReadBytes::cache_hierarchy.l2cache.prefetcher 4096 # Per-requestor bytes read from memory (Byte)
board.memory.mem_ctrl.requestorReadBytes::processor.cores.core.inst 26816 # Per-requestor bytes read from memory (Byte)
board.memory.mem_ctrl.requestorReadBytes::processor.cores.core.data 8768 # Per-requestor bytes read from memory (Byte)
board.memory.mem_ctrl.requestorReadRate::cache_hierarchy.l1dcaches.prefetcher 29516564.788398511708 # Per-requestor bytes read from memory rate ((Byte/Second))
board.memory.mem_ctrl.requestorReadRate::cache_hierarchy.l2cache.prefetcher 209895571.828611642122 # Per-requestor bytes read from memory rate ((Byte/Second))
board.memory.mem_ctrl.requestorReadRate::processor.cores.core.inst 1374160071.815441846848 # Per-requestor bytes read from memory rate ((Byte/Second))
board.memory.mem_ctrl.requestorReadRate::processor.cores.core.data 449307708.445621788502 # Per-requestor bytes read from memory rate ((Byte/Second))
```

board.memory.mem_ctrl.requestorReadAccesses::cache_hierarchy.l1dcaches.prefetcher 9 # Per-requestor read serviced memory accesses (Count)

board.memory.mem_ctrl.requestorReadAccesses::cache_hierarchy.l2cache.prefetcher 64 # Per-requestor read serviced memory accesses (Count)

board.memory.mem_ctrl.requestorReadAccesses::processor.cores.core.inst 419 # Per-requestor read serviced memory accesses (Count)

board.memory.mem_ctrl.requestorReadAccesses::processor.cores.core.data 137 # Per-requestor read serviced memory accesses (Count)

board.memory.mem_ctrl.requestorReadTotalLat::cache_hierarchy.l1dcaches.prefetcher 582987 # Per-requestor read total memory access latency (Tick)

board.memory.mem_ctrl.requestorReadTotalLat::cache_hierarchy.l2cache.prefetcher 2563823 # Per-requestor read total memory access latency (Tick)

board.memory.mem_ctrl.requestorReadTotalLat::processor.cores.core.inst 11326899 # Per-requestor read total memory access latency (Tick)

board.memory.mem_ctrl.requestorReadTotalLat::processor.cores.core.data 5267386 # Per-requestor read total memory access latency (Tick)

board.memory.mem_ctrl.requestorReadAvgLat::cache_hierarchy.l1dcaches.prefetcher 64776.33 # Per-requestor read average memory access latency ((Tick/Count))

board.memory.mem_ctrl.requestorReadAvgLat::cache_hierarchy.l2cache.prefetcher 40059.73 # Per-requestor read average memory access latency ((Tick/Count))

board.memory.mem_ctrl.requestorReadAvgLat::processor.cores.core.inst 27033.17 # Per-requestor read average memory access latency ((Tick/Count))

board.memory.mem_ctrl.requestorReadAvgLat::processor.cores.core.data 38448.07 # Per-requestor read average memory access latency ((Tick/Count))

board.memory.mem_ctrl.dram.bytesRead::cache_hierarchy.l1dcaches.prefetcher 576 # Number of bytes read from this memory (Byte)

board.memory.mem_ctrl.dram.bytesRead::cache_hierarchy.l2cache.prefetcher 4096 # Number of bytes read from this memory (Byte)

board.memory.mem_ctrl.dram.bytesRead::processor.cores.core.inst 26752 # Number of bytes read from this memory (Byte)

board.memory.mem_ctrl.dram.bytesRead::processor.cores.core.data 8768 # Number of bytes read from this memory (Byte)

board.memory.mem_ctrl.dram.bytesRead::total 40192 # Number of bytes read from this memory (Byte)

board.memory.mem_ctrl.dram.bytesInstRead::processor.cores.core.inst 26752 # Number of instructions bytes read from this memory (Byte)

board.memory.mem_ctrl.dram.bytesInstRead::total 26752 # Number of instructions bytes read from this memory (Byte)

board.memory.mem_ctrl.dram.numReads::cache_hierarchy.l1dcaches.prefetcher 9 # Number of read requests responded to by this memory (Count)

board.memory.mem_ctrl.dram.numReads::cache_hierarchy.l2cache.prefetcher 64 # Number of read requests responded to by this memory (Count)

board.memory.mem_ctrl.dram.numReads::processor.cores.core.inst 418 # Number of read requests responded to by this memory (Count)

board.memory.mem_ctrl.dram.numReads::processor.cores.core.data 137 # Number of read requests responded to by this memory (Count)

board.memory.mem_ctrl.dram.numReads::total 628 # Number of read requests responded to by this memory (Count)

board.memory.mem_ctrl.dram.bwRead::cache_hierarchy.l1dcaches.prefetcher 29516565 # Total read bandwidth from this memory ((Byte/Second))

board.memory.mem_ctrl.dram.bwRead::cache_hierarchy.l2cache.prefetcher 209895572 # Total read bandwidth from this memory ((Byte/Second))

board.memory.mem_ctrl.dram.bwRead::processor.cores.core.inst 1370880454 # Total read bandwidth from this memory ((Byte/Second))

board.memory.mem_ctrl.dram.bwRead::processor.cores.core.data 449307708 # Total read bandwidth from this memory ((Byte/Second))

board.memory.mem_ctrl.dram.bwRead::total 2059600299 # Total read bandwidth from this memory ((Byte/Second))

board.memory.mem_ctrl.dram.bwInstRead::processor.cores.core.inst 1370880454 # Instruction read bandwidth from this memory ((Byte/Second))

board.memory.mem_ctrl.dram.bwInstRead::total 1370880454 # Instruction read bandwidth from this memory ((Byte/Second))

board.memory.mem_ctrl.dram.bwTotal::cache_hierarchy.l1dcaches.prefetcher 29516565 # Total bandwidth to/from this memory ((Byte/Second))

board.memory.mem_ctrl.dram.bwTotal::cache_hierarchy.l2cache.prefetcher 209895572 # Total bandwidth to/from this memory ((Byte/Second))

board.memory.mem_ctrl.dram.bwTotal::processor.cores.core.inst 1370880454 # Total bandwidth to/from this memory ((Byte/Second))

board.memory.mem_ctrl.dram.bwTotal::processor.cores.core.data 449307708 # Total bandwidth to/from this memory ((Byte/Second))

board.memory.mem_ctrl.dram.bwTotal::total 2059600299 # Total bandwidth to/from this memory ((Byte/Second))

board.memory.mem_ctrl.dram.readBursts 629 # Number of DRAM read bursts (Count)

board.memory.mem_ctrl.dram.writeBursts 0 # Number of DRAM write bursts (Count)

board.memory.mem_ctrl.dram.perBankRdBursts::0 98 # Per bank write bursts (Count)

board.memory.mem_ctrl.dram.perBankRdBursts::1 0 # Per bank write bursts (Count)

board.memory.mem_ctrl.dram.perBankRdBursts::2 1 # Per bank write bursts (Count)

board.memory.mem_ctrl.dram.perBankRdBursts::3 31 # Per bank write bursts (Count)

board.memory.mem_ctrl.dram.perBankRdBursts::4 23 # Per bank write bursts (Count)

board.memory.mem_ctrl.dram.perBankRdBursts::5 18 # Per bank write bursts (Count)

board.memory.mem_ctrl.dram.perBankRdBursts::6 59 # Per bank write bursts (Count)

board.memory.mem_ctrl.dram.perBankRdBursts::7 15 # Per bank write bursts (Count)

board.memory.mem_ctrl.dram.perBankRdBursts::8 51 # Per bank write bursts (Count)

board.memory.mem_ctrl.dram.perBankRdBursts::9 83 # Per bank write bursts (Count)

board.memory.mem_ctrl.dram.perBankRdBursts::10 107 # Per bank write bursts (Count)

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board.memory.mem_ctrl.dram.perBankRdBursts::11          35                    # Per bank write bursts (Count)
board.memory.mem_ctrl.dram.perBankRdBursts::12          12                    # Per bank write bursts (Count)
board.memory.mem_ctrl.dram.perBankRdBursts::13          63                    # Per bank write bursts (Count)
board.memory.mem_ctrl.dram.perBankRdBursts::14           6                    # Per bank write bursts (Count)
board.memory.mem_ctrl.dram.perBankRdBursts::15          27                    # Per bank write bursts (Count)
board.memory.mem_ctrl.dram.perBankWrBursts::0           0                     # Per bank write bursts (Count)
board.memory.mem_ctrl.dram.perBankWrBursts::1           0                     # Per bank write bursts (Count)
board.memory.mem_ctrl.dram.perBankWrBursts::2           0                     # Per bank write bursts (Count)
board.memory.mem_ctrl.dram.perBankWrBursts::3           0                     # Per bank write bursts (Count)
board.memory.mem_ctrl.dram.perBankWrBursts::4           0                     # Per bank write bursts (Count)
board.memory.mem_ctrl.dram.perBankWrBursts::5           0                     # Per bank write bursts (Count)
board.memory.mem_ctrl.dram.perBankWrBursts::6           0                     # Per bank write bursts (Count)
board.memory.mem_ctrl.dram.perBankWrBursts::7           0                     # Per bank write bursts (Count)
board.memory.mem_ctrl.dram.perBankWrBursts::8           0                     # Per bank write bursts (Count)
board.memory.mem_ctrl.dram.perBankWrBursts::9           0                     # Per bank write bursts (Count)
board.memory.mem_ctrl.dram.perBankWrBursts::10          0                     # Per bank write bursts (Count)
board.memory.mem_ctrl.dram.perBankWrBursts::11          0                     # Per bank write bursts (Count)
board.memory.mem_ctrl.dram.perBankWrBursts::12          0                     # Per bank write bursts (Count)
board.memory.mem_ctrl.dram.perBankWrBursts::13          0                     # Per bank write bursts (Count)
board.memory.mem_ctrl.dram.perBankWrBursts::14          0                     # Per bank write bursts (Count)
board.memory.mem_ctrl.dram.perBankWrBursts::15          0                     # Per bank write bursts (Count)
board.memory.mem_ctrl.dram.totQLat          7947345                 # Total ticks spent queuing (Tick)
board.memory.mem_ctrl.dram.totBusLat        3145000                 # Total ticks spent in databus transfers (Tick)
board.memory.mem_ctrl.dram.totMemAccLat     19741095                # Total ticks spent from burst creation until serviced by the DRAM (Tick)
board.memory.mem_ctrl.dram.avgQLat          12634.89                # Average queueing delay per DRAM burst ((Tick/Count))
board.memory.mem_ctrl.dram.avgBusLat        5000.00                 # Average bus latency per DRAM burst ((Tick/Count))
board.memory.mem_ctrl.dram.avgMemAccLat     31384.89                # Average memory access latency per DRAM burst ((Tick/Count))
board.memory.mem_ctrl.dram.readRowHits          498                 # Number of row buffer hits during reads (Count)
board.memory.mem_ctrl.dram.writeRowHits         0                   # Number of row buffer hits during writes (Count)
board.memory.mem_ctrl.dram.readRowHitRate       79.17               # Row buffer hit rate for reads (Ratio)
board.memory.mem_ctrl.dram.writeRowHitRate      nan                 # Row buffer hit rate for writes (Ratio)
board.memory.mem_ctrl.dram.bytesPerActivate::samples        121                 # Bytes accessed per row activation (Byte)
board.memory.mem_ctrl.dram.bytesPerActivate::mean   312.595041                  # Bytes accessed per row activation (Byte)
board.memory.mem_ctrl.dram.bytesPerActivate::gmean   193.942074                 # Bytes accessed per row activation (Byte)
board.memory.mem_ctrl.dram.bytesPerActivate::stdev   316.049220                  # Bytes accessed per row activation (Byte)
board.memory.mem_ctrl.dram.bytesPerActivate::0-127          38     31.40%     31.40% # Bytes accessed per row activation (Byte)
board.memory.mem_ctrl.dram.bytesPerActivate::128-255        28     23.14%     54.55% # Bytes accessed per row activation (Byte)
board.memory.mem_ctrl.dram.bytesPerActivate::256-383        22     18.18%     72.73% # Bytes accessed per row activation (Byte)
board.memory.mem_ctrl.dram.bytesPerActivate::384-511         6      4.96%     77.69% # Bytes accessed per row activation (Byte)
board.memory.mem_ctrl.dram.bytesPerActivate::512-639         6      4.96%     82.64% # Bytes accessed per row activation (Byte)
board.memory.mem_ctrl.dram.bytesPerActivate::640-767         4      3.31%     85.95% # Bytes accessed per row activation (Byte)
board.memory.mem_ctrl.dram.bytesPerActivate::768-895         2      1.65%     87.60% # Bytes accessed per row activation (Byte)
board.memory.mem_ctrl.dram.bytesPerActivate::896-1023        1      0.83%     88.43% # Bytes accessed per row activation (Byte)
board.memory.mem_ctrl.dram.bytesPerActivate::1024-1151      14     11.57%    100.00% # Bytes accessed per row activation (Byte)
board.memory.mem_ctrl.dram.bytesPerActivate::total          121                 # Bytes accessed per row activation (Byte)
board.memory.mem_ctrl.dram.dramBytesRead        40256               # Total bytes read (Byte)
board.memory.mem_ctrl.dram.dramBytesWritten         0               # Total bytes written (Byte)
board.memory.mem_ctrl.dram.avgRdBW          2062.879917             # Average DRAM read bandwidth in MiBytes/s ((Byte/Second))
board.memory.mem_ctrl.dram.avgWrBW                  0               # Average DRAM write bandwidth in MiBytes/s ((Byte/Second))
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board.memory.mem_ctrl.dram.peakBW 12800.00 # Theoretical peak bandwidth in MiByte/s ((Byte/Second))
board.memory.mem_ctrl.dram.busUtil 16.12 # Data bus utilization in percentage (Ratio)
board.memory.mem_ctrl.dram.busUtilRead 16.12 # Data bus utilization in percentage for reads (Ratio)
board.memory.mem_ctrl.dram.busUtilWrite 0.00 # Data bus utilization in percentage for writes (Ratio)
board.memory.mem_ctrl.dram.pageHitRate 79.17 # Row buffer hit rate, read and write combined (Ratio)
board.memory.mem_ctrl.dram.power_state.pwrStateResidencyTicks::UNDEFINED 19514466 # Cumulative time (in ticks) in various power states (Tick)
board.memory.mem_ctrl.dram.rank0.actEnergy 299880 # Energy for activate commands per rank (pJ) (Joule)
board.memory.mem_ctrl.dram.rank0.preEnergy 136620 # Energy for precharge commands per rank (pJ) (Joule)
board.memory.mem_ctrl.dram.rank0.readEnergy 1749300 # Energy for read commands per rank (pJ) (Joule)
board.memory.mem_ctrl.dram.rank0.writeEnergy 0 # Energy for write commands per rank (pJ) (Joule)
board.memory.mem_ctrl.dram.rank0.refreshEnergy 1229280.000000 # Energy for refresh commands per rank (pJ) (Joule)
board.memory.mem_ctrl.dram.rank0.actBackEnergy 8810490 # Energy for active background per rank (pJ) (Joule)
board.memory.mem_ctrl.dram.rank0.preBackEnergy 74400 # Energy for precharge background per rank (pJ) (Joule)
board.memory.mem_ctrl.dram.rank0.actPowerDownEnergy 0 # Energy for active power-down per rank (pJ) (Joule)
board.memory.mem_ctrl.dram.rank0.prePowerDownEnergy 0 # Energy for precharge power-down per rank (pJ) (Joule)
board.memory.mem_ctrl.dram.rank0.selfRefreshEnergy 0 # Energy for self refresh per rank (pJ) (Joule)
board.memory.mem_ctrl.dram.rank0.totalEnergy 12299970 # Total energy per rank (pJ) (Joule)
board.memory.mem_ctrl.dram.rank0.averagePower 630.300107 # Core power per rank (mW) (Watt)
board.memory.mem_ctrl.dram.rank0.totalIdleTime 0 # Total Idle time Per DRAM Rank (Tick)
board.memory.mem_ctrl.dram.rank0.pwrStateTime::IDLE 123310 # Time in different power states (Tick)
board.memory.mem_ctrl.dram.rank0.pwrStateTime::REF 520000 # Time in different power states (Tick)
board.memory.mem_ctrl.dram.rank0.pwrStateTime::SREF 0 # Time in different power states (Tick)
board.memory.mem_ctrl.dram.rank0.pwrStateTime::PRE_PDN 0 # Time in different power states (Tick)
board.memory.mem_ctrl.dram.rank0.pwrStateTime::ACT 18871156 # Time in different power states (Tick)
board.memory.mem_ctrl.dram.rank0.pwrStateTime::ACT_PDN 0 # Time in different power states (Tick)
board.memory.mem_ctrl.dram.rank1.actEnergy 628320 # Energy for activate commands per rank (pJ) (Joule)
board.memory.mem_ctrl.dram.rank1.preEnergy 322575 # Energy for precharge commands per rank (pJ) (Joule)
board.memory.mem_ctrl.dram.rank1.readEnergy 2734620 # Energy for read commands per rank (pJ) (Joule)
board.memory.mem_ctrl.dram.rank1.writeEnergy 0 # Energy for write commands per rank (pJ) (Joule)
board.memory.mem_ctrl.dram.rank1.refreshEnergy 1229280.000000 # Energy for refresh commands per rank (pJ) (Joule)
board.memory.mem_ctrl.dram.rank1.actBackEnergy 8757480 # Energy for active background per rank (pJ) (Joule)
board.memory.mem_ctrl.dram.rank1.preBackEnergy 119040 # Energy for precharge background per rank (pJ) (Joule)
board.memory.mem_ctrl.dram.rank1.actPowerDownEnergy 0 # Energy for active power-down per rank (pJ) (Joule)
board.memory.mem_ctrl.dram.rank1.prePowerDownEnergy 0 # Energy for precharge power-down per rank (pJ) (Joule)
board.memory.mem_ctrl.dram.rank1.selfRefreshEnergy 0 # Energy for self refresh per rank (pJ) (Joule)
board.memory.mem_ctrl.dram.rank1.totalEnergy 13791315 # Total energy per rank (pJ) (Joule)
board.memory.mem_ctrl.dram.rank1.averagePower 706.722644 # Core power per rank (mW) (Watt)
board.memory.mem_ctrl.dram.rank1.totalIdleTime 0 # Total Idle time Per DRAM Rank (Tick)
board.memory.mem_ctrl.dram.rank1.pwrStateTime::IDLE 253634 # Time in different power states (Tick)
board.memory.mem_ctrl.dram.rank1.pwrStateTime::REF 520000 # Time in different power states (Tick)
board.memory.mem_ctrl.dram.rank1.pwrStateTime::SREF 0 # Time in different power states (Tick)
board.memory.mem_ctrl.dram.rank1.pwrStateTime::PRE_PDN 0 # Time in different power states (Tick)
board.memory.mem_ctrl.dram.rank1.pwrStateTime::ACT 18740832 # Time in different power states (Tick)
board.memory.mem_ctrl.dram.rank1.pwrStateTime::ACT_PDN 0 # Time in different power states (Tick)
board.memory.mem_ctrl.power_state.pwrStateResidencyTicks::UNDEFINED 19514466 # Cumulative time (in ticks) in various power states (Tick)
board.processor.cores.core.numCycles 58603 # Number of cpu cycles simulated (Cycle)
board.processor.cores.core.cpi 8.940198 # CPI: cycles per instruction (core level) ((Cycle/Count))
board.processor.cores.core.ipc 0.111854 # IPC: instructions per cycle (core level) ((Count/Cycle))
board.processor.cores.core.numWorkItemsStarted 0 # Number of work items this cpu started (Count)
```

```
board.processor.cores.core.numWorkItemsCompleted 0 # Number of work items this cpu completed (Count)
board.processor.cores.core.instsAdded 23749 # Number of instructions added to the IQ (excludes non-spec) (Count)
board.processor.cores.core.nonSpecInstsAdded 76 # Number of non-speculative instructions added to the IQ (Count)
board.processor.cores.core.instsIssued 21102 # Number of instructions issued (Count)
board.processor.cores.core.squashedInstsIssued 238 # Number of squashed instructions issued (Count)
board.processor.cores.core.squashedInstsExamined 10827 # Number of squashed instructions iterated over during squash; mainly for profiling (Count)
board.processor.cores.core.squashedOperandsExamined 14782 # Number of squashed operands that are examined and possibly removed from graph (Count)
board.processor.cores.core.squashedNonSpecRemoved 76 # Number of squashed non-spec instructions that were removed (Count)
board.processor.cores.core.numIssuedDist::samples 22765 # Number of insts issued each cycle (Count)
board.processor.cores.core.numIssuedDist::mean 0.926949 # Number of insts issued each cycle (Count)
board.processor.cores.core.numIssuedDist::stdev 2.007361 # Number of insts issued each cycle (Count)
board.processor.cores.core.numIssuedDist::underflows 0 0.00% 0.00% # Number of insts issued each cycle (Count)
board.processor.cores.core.numIssuedDist::0 17541 77.05% 77.05% # Number of insts issued each cycle (Count)
board.processor.cores.core.numIssuedDist::1 922 4.05% 81.10% # Number of insts issued each cycle (Count)
board.processor.cores.core.numIssuedDist::2 793 3.48% 84.59% # Number of insts issued each cycle (Count)
board.processor.cores.core.numIssuedDist::3 565 2.48% 87.07% # Number of insts issued each cycle (Count)
board.processor.cores.core.numIssuedDist::4 705 3.10% 90.16% # Number of insts issued each cycle (Count)
board.processor.cores.core.numIssuedDist::5 596 2.62% 92.78% # Number of insts issued each cycle (Count)
board.processor.cores.core.numIssuedDist::6 726 3.19% 95.97% # Number of insts issued each cycle (Count)
board.processor.cores.core.numIssuedDist::7 593 2.60% 98.58% # Number of insts issued each cycle (Count)
board.processor.cores.core.numIssuedDist::8 324 1.42% 100.00% # Number of insts issued each cycle (Count)
board.processor.cores.core.numIssuedDist::overflows 0 0.00% 100.00% # Number of insts issued each cycle (Count)
board.processor.cores.core.numIssuedDist::min_value 0 # Number of insts issued each cycle (Count)
board.processor.cores.core.numIssuedDist::max_value 8 # Number of insts issued each cycle (Count)
board.processor.cores.core.numIssuedDist::total 22765 # Number of insts issued each cycle (Count)
board.processor.cores.core.statFuBusy::No_OpClass 0 0.00% 0.00% # attempts to use FU when none available (Count)
board.processor.cores.core.statFuBusy::IntAlu 522 93.05% 93.05% # attempts to use FU when none available (Count)
board.processor.cores.core.statFuBusy::IntMult 0 0.00% 93.05% # attempts to use FU when none available (Count)
board.processor.cores.core.statFuBusy::IntDiv 0 0.00% 93.05% # attempts to use FU when none available (Count)
board.processor.cores.core.statFuBusy::FloatAdd 0 0.00% 93.05% # attempts to use FU when none available (Count)
board.processor.cores.core.statFuBusy::FloatCmp 0 0.00% 93.05% # attempts to use FU when none available (Count)
board.processor.cores.core.statFuBusy::FloatCvt 0 0.00% 93.05% # attempts to use FU when none available (Count)
board.processor.cores.core.statFuBusy::FloatMult 0 0.00% 93.05% # attempts to use FU when none available (Count)
board.processor.cores.core.statFuBusy::FloatMultAcc 0 0.00% 93.05% # attempts to use FU when none available (Count)
board.processor.cores.core.statFuBusy::FloatDiv 0 0.00% 93.05% # attempts to use FU when none available (Count)
board.processor.cores.core.statFuBusy::FloatMisc 0 0.00% 93.05% # attempts to use FU when none available (Count)
board.processor.cores.core.statFuBusy::FloatSqrt 0 0.00% 93.05% # attempts to use FU when none available (Count)
board.processor.cores.core.statFuBusy::SimdAdd 0 0.00% 93.05% # attempts to use FU when none available (Count)
board.processor.cores.core.statFuBusy::SimdAddAcc 0 0.00% 93.05% # attempts to use FU when none available (Count)
board.processor.cores.core.statFuBusy::SimdAlu 8 1.43% 94.47% # attempts to use FU when none available (Count)
board.processor.cores.core.statFuBusy::SimdCmp 0 0.00% 94.47% # attempts to use FU when none available (Count)
board.processor.cores.core.statFuBusy::SimdCvt 1 0.18% 94.65% # attempts to use FU when none available (Count)
board.processor.cores.core.statFuBusy::SimdMisc 1 0.18% 94.83% # attempts to use FU when none available (Count)
board.processor.cores.core.statFuBusy::SimdMult 0 0.00% 94.83% # attempts to use FU when none available (Count)
board.processor.cores.core.statFuBusy::SimdMultAcc 0 0.00% 94.83% # attempts to use FU when none available (Count)
board.processor.cores.core.statFuBusy::SimdMatMultAcc 0 0.00% 94.83% # attempts to use FU when none available (Count)
board.processor.cores.core.statFuBusy::SimdShift 0 0.00% 94.83% # attempts to use FU when none available (Count)
board.processor.cores.core.statFuBusy::SimdShiftAcc 0 0.00% 94.83% # attempts to use FU when none available (Count)
board.processor.cores.core.statFuBusy::SimdDiv 0 0.00% 94.83% # attempts to use FU when none available (Count)
```

board.processor.cores.core.statFuBusy::SimdSqrt 0 0.00% 94.83% # attempts to use FU when none available (Count)

board.processor.cores.core.statFuBusy::SimdFloatAdd 0 0.00% 94.83% # attempts to use FU when none available (Count)

board.processor.cores.core.statFuBusy::SimdFloatAlu 0 0.00% 94.83% # attempts to use FU when none available (Count)

board.processor.cores.core.statFuBusy::SimdFloatCmp 0 0.00% 94.83% # attempts to use FU when none available (Count)

board.processor.cores.core.statFuBusy::SimdFloatCvt 0 0.00% 94.83% # attempts to use FU when none available (Count)

board.processor.cores.core.statFuBusy::SimdFloatDiv 0 0.00% 94.83% # attempts to use FU when none available (Count)

board.processor.cores.core.statFuBusy::SimdFloatMisc 0 0.00% 94.83% # attempts to use FU when none available (Count)

board.processor.cores.core.statFuBusy::SimdFloatMult 0 0.00% 94.83% # attempts to use FU when none available (Count)

board.processor.cores.core.statFuBusy::SimdFloatMultAcc 0 0.00% 94.83% # attempts to use FU when none available (Count)

board.processor.cores.core.statFuBusy::SimdFloatMatMultAcc 0 0.00% 94.83% # attempts to use FU when none available (Count)

board.processor.cores.core.statFuBusy::SimdFloatSqrt 0 0.00% 94.83% # attempts to use FU when none available (Count)

board.processor.cores.core.statFuBusy::SimdReduceAdd 0 0.00% 94.83% # attempts to use FU when none available (Count)

board.processor.cores.core.statFuBusy::SimdReduceAlu 0 0.00% 94.83% # attempts to use FU when none available (Count)

board.processor.cores.core.statFuBusy::SimdReduceCmp 0 0.00% 94.83% # attempts to use FU when none available (Count)

board.processor.cores.core.statFuBusy::SimdFloatReduceAdd 0 0.00% 94.83% # attempts to use FU when none available (Count)

board.processor.cores.core.statFuBusy::SimdFloatReduceCmp 0 0.00% 94.83% # attempts to use FU when none available (Count)

board.processor.cores.core.statFuBusy::SimdAes 0 0.00% 94.83% # attempts to use FU when none available (Count)

board.processor.cores.core.statFuBusy::SimdAesMix 0 0.00% 94.83% # attempts to use FU when none available (Count)

board.processor.cores.core.statFuBusy::SimdSha1Hash 0 0.00% 94.83% # attempts to use FU when none available (Count)

board.processor.cores.core.statFuBusy::SimdSha1Hash2 0 0.00% 94.83% # attempts to use FU when none available (Count)

board.processor.cores.core.statFuBusy::SimdSha256Hash 0 0.00% 94.83% # attempts to use FU when none available (Count)

board.processor.cores.core.statFuBusy::SimdSha256Hash2 0 0.00% 94.83% # attempts to use FU when none available (Count)

board.processor.cores.core.statFuBusy::SimdShaSigma2 0 0.00% 94.83% # attempts to use FU when none available (Count)

board.processor.cores.core.statFuBusy::SimdShaSigma3 0 0.00% 94.83% # attempts to use FU when none available (Count)

board.processor.cores.core.statFuBusy::SimdPredAlu 0 0.00% 94.83% # attempts to use FU when none available (Count)

board.processor.cores.core.statFuBusy::Matrix 0 0.00% 94.83% # attempts to use FU when none available (Count)

board.processor.cores.core.statFuBusy::MatrixMov 0 0.00% 94.83% # attempts to use FU when none available (Count)

board.processor.cores.core.statFuBusy::MatrixOP 0 0.00% 94.83% # attempts to use FU when none available (Count)

board.processor.cores.core.statFuBusy::MemRead 10 1.78% 96.61% # attempts to use FU when none available (Count)

board.processor.cores.core.statFuBusy::MemWrite 12 2.14% 98.75% # attempts to use FU when none available (Count)

board.processor.cores.core.statFuBusy::FloatMemRead 3 0.53% 99.29% # attempts to use FU when none available (Count)

board.processor.cores.core.statFuBusy::FloatMemWrite 4 0.71% 100.00% # attempts to use FU when none available (Count)

board.processor.cores.core.statFuBusy::IprAccess 0 0.00% 100.00% # attempts to use FU when none available (Count)

board.processor.cores.core.statFuBusy::InstPrefetch 0 0.00% 100.00% # attempts to use FU when none available (Count)

board.processor.cores.core.statFuBusy::SimdUnitStrideLoad 0 0.00% 100.00% # attempts to use FU when none available (Count)

board.processor.cores.core.statFuBusy::SimdUnitStrideStore 0 0.00% 100.00% # attempts to use FU when none available (Count)

board.processor.cores.core.statFuBusy::SimdUnitStrideMaskLoad 0 0.00% 100.00% # attempts to use FU when none available (Count)

board.processor.cores.core.statFuBusy::SimdUnitStrideMaskStore 0 0.00% 100.00% # attempts to use FU when none available (Count)

board.processor.cores.core.statFuBusy::SimdStridedLoad 0 0.00% 100.00% # attempts to use FU when none available (Count)

board.processor.cores.core.statFuBusy::SimdStridedStore 0 0.00% 100.00% # attempts to use FU when none available (Count)

board.processor.cores.core.statFuBusy::SimdIndexedLoad 0 0.00% 100.00% # attempts to use FU when none available (Count)

board.processor.cores.core.statFuBusy::SimdIndexedStore 0 0.00% 100.00% # attempts to use FU when none available (Count)

board.processor.cores.core.statFuBusy::SimdWholeRegisterLoad 0 0.00% 100.00% # attempts to use FU when none available (Count)

board.processor.cores.core.statFuBusy::SimdWholeRegisterStore 0 0.00% 100.00% # attempts to use FU when none available (Count)

board.processor.cores.core.statFuBusy::SimdUnitStrideFaultOnlyFirstLoad 0 0.00% 100.00% # attempts to use FU when none available (Count)

board.processor.cores.core.statFuBusy::SimdUnitStrideSegmentedLoad 0 0.00% 100.00% # attempts to use FU when none available (Count)

board.processor.cores.core.statFuBusy::SimdUnitStrideSegmentedStore 0 0.00% 100.00% # attempts to use FU when none available (Count)

board.processor.cores.core.statFuBusy::SimdExt 0 0.00% 100.00% # attempts to use FU when none available (Count)

board.processor.cores.core.statFuBusy::SimdFloatExt 0 0.00% 100.00% # attempts to use FU when none available (Count)

board.processor.cores.core.statFuBusy::SimdConfig 0 0.00% 100.00% # attempts to use FU when none available (Count)

board.processor.cores.core.statIssuedInstType_0::No_OpClass 156 0.74% 0.74% # Number of instructions issued per FU type, per thread (Count)

board.processor.cores.core.statIssuedInstType_0::IntAlu 16570 78.52% 79.26% # Number of instructions issued per FU type, per thread (Count)

board.processor.cores.core.statIssuedInstType_0::IntMult 5 0.02% 79.29% # Number of instructions issued per FU type, per thread (Count)

board.processor.cores.core.statIssuedInstType_0::IntDiv 43 0.20% 79.49% # Number of instructions issued per FU type, per thread (Count)

board.processor.cores.core.statIssuedInstType_0::FloatAdd 2 0.01% 79.50% # Number of instructions issued per FU type, per thread (Count)

board.processor.cores.core.statIssuedInstType_0::FloatCmp 0 0.00% 79.50% # Number of instructions issued per FU type, per thread (Count)

board.processor.cores.core.statIssuedInstType_0::FloatCvt 0 0.00% 79.50% # Number of instructions issued per FU type, per thread (Count)

board.processor.cores.core.statIssuedInstType_0::FloatMult 0 0.00% 79.50% # Number of instructions issued per FU type, per thread (Count)

board.processor.cores.core.statIssuedInstType_0::FloatMultAcc 0 0.00% 79.50% # Number of instructions issued per FU type, per thread (Count)

board.processor.cores.core.statIssuedInstType_0::FloatDiv 0 0.00% 79.50% # Number of instructions issued per FU type, per thread (Count)

board.processor.cores.core.statIssuedInstType_0::FloatMisc 0 0.00% 79.50% # Number of instructions issued per FU type, per thread (Count)

board.processor.cores.core.statIssuedInstType_0::FloatSqrt 0 0.00% 79.50% # Number of instructions issued per FU type, per thread (Count)

board.processor.cores.core.statIssuedInstType_0::SimdAdd 0 0.00% 79.50% # Number of instructions issued per FU type, per thread (Count)

board.processor.cores.core.statIssuedInstType_0::SimdAddAcc 0 0.00% 79.50% # Number of instructions issued per FU type, per thread (Count)

board.processor.cores.core.statIssuedInstType_0::SimdAlu 90 0.43% 79.93% # Number of instructions issued per FU type, per thread (Count)

board.processor.cores.core.statIssuedInstType_0::SimdCmp 0 0.00% 79.93% # Number of instructions issued per FU type, per thread (Count)

board.processor.cores.core.statIssuedInstType_0::SimdCvt 53 0.25% 80.18% # Number of instructions issued per FU type, per thread (Count)

board.processor.cores.core.statIssuedInstType_0::SimdMisc 65 0.31% 80.49% # Number of instructions issued per FU type, per thread (Count)

board.processor.cores.core.statIssuedInstType_0::SimdMult 0 0.00% 80.49% # Number of instructions issued per FU type, per thread (Count)

board.processor.cores.core.statIssuedInstType_0::SimdMultAcc 0 0.00% 80.49% # Number of instructions issued per FU type, per thread (Count)

board.processor.cores.core.statIssuedInstType_0::SimdMatMultAcc 0 0.00% 80.49% # Number of instructions issued per FU type, per thread (Count)

board.processor.cores.core.statIssuedInstType_0::SimdShift 0 0.00% 80.49% # Number of instructions issued per FU type, per thread (Count)

board.processor.cores.core.statIssuedInstType_0::SimdShiftAcc 0 0.00% 80.49% # Number of instructions issued per FU type, per thread (Count)

board.processor.cores.core.statIssuedInstType_0::SimdDiv 0 0.00% 80.49% # Number of instructions issued per FU type, per thread (Count)

board.processor.cores.core.statIssuedInstType_0::SimdSqrt 0 0.00% 80.49% # Number of instructions issued per FU type, per thread (Count)

board.processor.cores.core.statIssuedInstType_0::SimdFloatAdd 0 0.00% 80.49% # Number of instructions issued per FU type, per thread (Count)

board.processor.cores.core.statIssuedInstType_0::SimdFloatAlu 0 0.00% 80.49% # Number of instructions issued per FU type, per thread (Count)

board.processor.cores.core.statIssuedInstType_0::SimdFloatCmp 0 0.00% 80.49% # Number of instructions issued per FU type, per thread (Count)

board.processor.cores.core.statIssuedInstType_0::SimdFloatCvt 0 0.00% 80.49% # Number of instructions issued per FU type, per thread (Count)

board.processor.cores.core.statIssuedInstType_0::SimdFloatDiv 0 0.00% 80.49% # Number of instructions issued per FU type, per thread (Count)

board.processor.cores.core.statIssuedInstType_0::SimdFloatMisc 0 0.00% 80.49% # Number of instructions issued per FU type, per thread (Count)

board.processor.cores.core.statIssuedInstType_0::SimdFloatMult 0 0.00% 80.49% # Number of instructions issued per FU type, per thread (Count)

board.processor.cores.core.statIssuedInstType_0::SimdFloatMultAcc 0 0.00% 80.49% # Number of instructions issued per FU type, per thread (Count)

board.processor.cores.core.statIssuedInstType_0::SimdFloatMatMultAcc 0 0.00% 80.49% # Number of instructions issued per FU type, per thread (Count)

board.processor.cores.core.statIssuedInstType_0::SimdFloatSqrt 0 0.00% 80.49% # Number of instructions issued per FU type, per thread (Count)

board.processor.cores.core.statIssuedInstType_0::SimdReduceAdd 0 0.00% 80.49% # Number of instructions issued per FU type, per thread (Count)

board.processor.cores.core.statIssuedInstType_0::SimdReduceAlu 0 0.00% 80.49% # Number of instructions issued per FU type, per thread (Count)

board.processor.cores.core.statIssuedInstType_0::SimdReduceCmp 0 0.00% 80.49% # Number of instructions issued per FU type, per thread (Count)

board.processor.cores.core.statIssuedInstType_0::SimdFloatReduceAdd 0 0.00% 80.49% # Number of instructions issued per FU type, per thread (Count)

board.processor.cores.core.statIssuedInstType_0::SimdFloatReduceCmp 0 0.00% 80.49% # Number of instructions issued per FU type, per thread (Count)

board.processor.cores.core.statIssuedInstType_0::SimdAes 0 0.00% 80.49% # Number of instructions issued per FU type, per thread (Count)

board.processor.cores.core.statIssuedInstType_0::SimdAesMix 0 0.00% 80.49% # Number of instructions issued per FU type, per thread (Count)

board.processor.cores.core.statIssuedInstType_0::SimdSha1Hash 0 0.00% 80.49% # Number of instructions issued per FU type, per thread (Count)

board.processor.cores.core.statIssuedInstType_0::SimdSha1Hash2 0 0.00% 80.49% # Number of instructions issued per FU type, per thread (Count)

board.processor.cores.core.statIssuedInstType_0::SimdSha256Hash 0 0.00% 80.49% # Number of instructions issued per FU type, per thread (Count)

board.processor.cores.core.statIssuedInstType_0::SimdSha256Hash2 0 0.00% 80.49% # Number of instructions issued per FU type, per thread (Count)

board.processor.cores.core.statIssuedInstType_0::SimdShaSigma2 0 0.00% 80.49% # Number of instructions issued per FU type, per thread (Count)

board.processor.cores.core.statIssuedInstType_0::SimdShaSigma3 0 0.00% 80.49% # Number of instructions issued per FU type, per thread (Count)

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board.processor.cores.core.statIssuedInstType_0::SimdPredAlu 0 0.00% 80.49% # Number of instructions issued per FU type, per thread (Count)
board.processor.cores.core.statIssuedInstType_0::Matrix 0 0.00% 80.49% # Number of instructions issued per FU type, per thread (Count)
board.processor.cores.core.statIssuedInstType_0::MatrixMov 0 0.00% 80.49% # Number of instructions issued per FU type, per thread (Count)
board.processor.cores.core.statIssuedInstType_0::MatrixOP 0 0.00% 80.49% # Number of instructions issued per FU type, per thread (Count)
board.processor.cores.core.statIssuedInstType_0::MemRead 2090 9.90% 90.39% # Number of instructions issued per FU type, per thread (Count)
board.processor.cores.core.statIssuedInstType_0::MemWrite 1798 8.52% 98.91% # Number of instructions issued per FU type, per thread (Count)
board.processor.cores.core.statIssuedInstType_0::FloatMemRead 60 0.28% 99.19% # Number of instructions issued per FU type, per thread (Count)
board.processor.cores.core.statIssuedInstType_0::FloatMemWrite 170 0.81% 100.00% # Number of instructions issued per FU type, per thread (Count)
board.processor.cores.core.statIssuedInstType_0::IprAccess 0 0.00% 100.00% # Number of instructions issued per FU type, per thread (Count)
board.processor.cores.core.statIssuedInstType_0::InstPrefetch 0 0.00% 100.00% # Number of instructions issued per FU type, per thread (Count)
board.processor.cores.core.statIssuedInstType_0::SimdUnitStrideLoad 0 0.00% 100.00% # Number of instructions issued per FU type, per thread (Count)
board.processor.cores.core.statIssuedInstType_0::SimdUnitStrideStore 0 0.00% 100.00% # Number of instructions issued per FU type, per thread (Count)
board.processor.cores.core.statIssuedInstType_0::SimdUnitStrideMaskLoad 0 0.00% 100.00% # Number of instructions issued per FU type, per thread (Count)
board.processor.cores.core.statIssuedInstType_0::SimdUnitStrideMaskStore 0 0.00% 100.00% # Number of instructions issued per FU type, per thread (Count)
board.processor.cores.core.statIssuedInstType_0::SimdStridedLoad 0 0.00% 100.00% # Number of instructions issued per FU type, per thread (Count)
board.processor.cores.core.statIssuedInstType_0::SimdStridedStore 0 0.00% 100.00% # Number of instructions issued per FU type, per thread (Count)
board.processor.cores.core.statIssuedInstType_0::SimdIndexedLoad 0 0.00% 100.00% # Number of instructions issued per FU type, per thread (Count)
board.processor.cores.core.statIssuedInstType_0::SimdIndexedStore 0 0.00% 100.00% # Number of instructions issued per FU type, per thread (Count)
board.processor.cores.core.statIssuedInstType_0::SimdWholeRegisterLoad 0 0.00% 100.00% # Number of instructions issued per FU type, per thread (Count)
board.processor.cores.core.statIssuedInstType_0::SimdWholeRegisterStore 0 0.00% 100.00% # Number of instructions issued per FU type, per thread (Count)
board.processor.cores.core.statIssuedInstType_0::SimdUnitStrideFaultOnlyFirstLoad 0 0.00% 100.00% # Number of instructions issued per FU type, per thread (Count)
board.processor.cores.core.statIssuedInstType_0::SimdUnitStrideSegmentedLoad 0 0.00% 100.00% # Number of instructions issued per FU type, per thread (Count)
board.processor.cores.core.statIssuedInstType_0::SimdUnitStrideSegmentedStore 0 0.00% 100.00% # Number of instructions issued per FU type, per thread (Count)
board.processor.cores.core.statIssuedInstType_0::SimdExt 0 0.00% 100.00% # Number of instructions issued per FU type, per thread (Count)
board.processor.cores.core.statIssuedInstType_0::SimdFloatExt 0 0.00% 100.00% # Number of instructions issued per FU type, per thread (Count)
board.processor.cores.core.statIssuedInstType_0::SimdConfig 0 0.00% 100.00% # Number of instructions issued per FU type, per thread (Count)
board.processor.cores.core.statIssuedInstType_0::total 21102 # Number of instructions issued per FU type, per thread (Count)
board.processor.cores.core.issueRate 0.360084 # Inst issue rate ((Count/Cycle))
board.processor.cores.core.fuBusy 561 # FU busy when requested (Count)
board.processor.cores.core.fuBusyRate 0.026585 # FU busy rate (busy events/executed inst) ((Count/Count))
board.processor.cores.core.intInstQueueReads 64855 # Number of integer instruction queue reads (Count)
board.processor.cores.core.intInstQueueWrites 34150 # Number of integer instruction queue writes (Count)
board.processor.cores.core.intInstQueueWakeupAccesses 19532 # Number of integer instruction queue wakeup accesses (Count)
board.processor.cores.core.fpInstQueueReads 913 # Number of floating instruction queue reads (Count)
board.processor.cores.core.fpInstQueueWrites 506 # Number of floating instruction queue writes (Count)
board.processor.cores.core.fpInstQueueWakeupAccesses 437 # Number of floating instruction queue wakeup accesses (Count)
board.processor.cores.core.vecInstQueueReads 0 # Number of vector instruction queue reads (Count)
board.processor.cores.core.vecInstQueueWrites 0 # Number of vector instruction queue writes (Count)
board.processor.cores.core.vecInstQueueWakeupAccesses 0 # Number of vector instruction queue wakeup accesses (Count)
board.processor.cores.core.intAluAccesses 21042 # Number of integer alu accesses (Count)
board.processor.cores.core.fpAluAccesses 465 # Number of floating point alu accesses (Count)
board.processor.cores.core.vecAluAccesses 0 # Number of vector alu accesses (Count)
board.processor.cores.core.numSquashedInsts 843 # Number of squashed instructions skipped in execute (Count)
board.processor.cores.core.numSwp 0 # Number of swp insts executed (Count)
board.processor.cores.core.timesIdled 286 # Number of times that the entire CPU went into an idle state and unscheduled itself (Count)
board.processor.cores.core.idleCycles 35838 # Total number of cycles that the CPU has spent unscheduled due to idling (Cycle)
board.processor.cores.core.MemDepUnit__0.insertedLoads 2416 # Number of loads inserted to the mem dependence unit. (Count)
board.processor.cores.core.MemDepUnit__0.insertedStores 2332 # Number of stores inserted to the mem dependence unit. (Count)
board.processor.cores.core.MemDepUnit__0.conflictingLoads 38 # Number of conflicting loads. (Count)
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board.processor.cores.core.MemDepUnit__0.conflictingStores 33 # Number of conflicting stores. (Count)

board.processor.cores.core.MemDepUnit__1.insertedLoads 0 # Number of loads inserted to the mem dependence unit. (Count)

board.processor.cores.core.MemDepUnit__1.insertedStores 0 # Number of stores inserted to the mem dependence unit. (Count)

board.processor.cores.core.MemDepUnit__1.conflictingLoads 0 # Number of conflicting loads. (Count)

board.processor.cores.core.MemDepUnit__1.conflictingStores 0 # Number of conflicting stores. (Count)

board.processor.cores.core.MemDepUnit__2.insertedLoads 0 # Number of loads inserted to the mem dependence unit. (Count)

board.processor.cores.core.MemDepUnit__2.insertedStores 0 # Number of stores inserted to the mem dependence unit. (Count)

board.processor.cores.core.MemDepUnit__2.conflictingLoads 0 # Number of conflicting loads. (Count)

board.processor.cores.core.MemDepUnit__2.conflictingStores 0 # Number of conflicting stores. (Count)

board.processor.cores.core.MemDepUnit__3.insertedLoads 0 # Number of loads inserted to the mem dependence unit. (Count)

board.processor.cores.core.MemDepUnit__3.insertedStores 0 # Number of stores inserted to the mem dependence unit. (Count)

board.processor.cores.core.MemDepUnit__3.conflictingLoads 0 # Number of conflicting loads. (Count)

board.processor.cores.core.MemDepUnit__3.conflictingStores 0 # Number of conflicting stores. (Count)

board.processor.cores.core.branchPred.lookups_0::NoBranch 0 0.00% 0.00% # Number of BP lookups (Count)

board.processor.cores.core.branchPred.lookups_0::Return 267 7.08% 7.08% # Number of BP lookups (Count)

board.processor.cores.core.branchPred.lookups_0::CallDirect 307 8.14% 15.22% # Number of BP lookups (Count)

board.processor.cores.core.branchPred.lookups_0::CallIndirect 75 1.99% 17.21% # Number of BP lookups (Count)

board.processor.cores.core.branchPred.lookups_0::DirectCond 2737 72.56% 89.77% # Number of BP lookups (Count)

board.processor.cores.core.branchPred.lookups_0::DirectUncond 325 8.62% 98.38% # Number of BP lookups (Count)

board.processor.cores.core.branchPred.lookups_0::IndirectCond 0 0.00% 98.38% # Number of BP lookups (Count)

board.processor.cores.core.branchPred.lookups_0::IndirectUncond 61 1.62% 100.00% # Number of BP lookups (Count)

board.processor.cores.core.branchPred.lookups_0::total 3772 # Number of BP lookups (Count)

board.processor.cores.core.branchPred.squashes_0::NoBranch 0 0.00% 0.00% # Number of branches that got squashed (completely removed) as an earlier branch was mispredicted. (Count)

board.processor.cores.core.branchPred.squashes_0::Return 145 6.34% 6.34% # Number of branches that got squashed (completely removed) as an earlier branch was mispredicted. (Count)

board.processor.cores.core.branchPred.squashes_0::CallDirect 220 9.62% 15.97% # Number of branches that got squashed (completely removed) as an earlier branch was mispredicted. (Count)

board.processor.cores.core.branchPred.squashes_0::CallIndirect 35 1.53% 17.50% # Number of branches that got squashed (completely removed) as an earlier branch was mispredicted. (Count)

board.processor.cores.core.branchPred.squashes_0::DirectCond 1626 71.13% 88.63% # Number of branches that got squashed (completely removed) as an earlier branch was mispredicted. (Count)

board.processor.cores.core.branchPred.squashes_0::DirectUncond 220 9.62% 98.25% # Number of branches that got squashed (completely removed) as an earlier branch was mispredicted. (Count)

board.processor.cores.core.branchPred.squashes_0::IndirectCond 0 0.00% 98.25% # Number of branches that got squashed (completely removed) as an earlier branch was mispredicted. (Count)

board.processor.cores.core.branchPred.squashes_0::IndirectUncond 40 1.75% 100.00% # Number of branches that got squashed (completely removed) as an earlier branch was mispredicted. (Count)

board.processor.cores.core.branchPred.squashes_0::total 2286 # Number of branches that got squashed (completely removed) as an earlier branch was mispredicted. (Count)

board.processor.cores.core.branchPred.corrected_0::NoBranch 0 0.00% 0.00% # Number of branches that got corrected but not yet commited. Branches get corrected by decode or after execute. Also a branch misprediction can be detected out-of-order. Therefore, a corrected branch might not end up beeing committed in case an even earlier branch was mispredicted (Count)

board.processor.cores.core.branchPred.corrected_0::Return 1 0.16% 0.16% # Number of branches that got corrected but not yet commited. Branches get corrected by decode or after execute. Also a branch misprediction can be detected out-of-order. Therefore, a corrected branch might not end up beeing committed in case an even earlier branch was mispredicted (Count)

board.processor.cores.core.branchPred.corrected_0::CallDirect 106 16.59% 16.74% # Number of branches that got corrected but not yet commited. Branches get corrected by decode or after execute. Also a branch misprediction can be detected out-of-order. Therefore, a corrected branch might not end up beeing committed in case an even earlier branch was mispredicted (Count)

board.processor.cores.core.branchPred.corrected_0::CallIndirect 34 5.32% 22.07% # Number of branches that got corrected but not yet commited. Branches get corrected by decode or after execute. Also a branch misprediction can be detected out-of-order. Therefore, a corrected branch might not end up beeing committed in case an even earlier branch was mispredicted (Count)

board.processor.cores.core.branchPred.corrected_0::DirectCond 359 56.18% 78.25% # Number of branches that got corrected but not yet commited. Branches get corrected by decode or after execute. Also a branch misprediction can be detected out-of-order. Therefore, a corrected branch might not end up beeing committed in case an even earlier branch was mispredicted (Count)

board.processor.cores.core.branchPred.corrected_0::DirectUncond 113 17.68% 95.93% # Number of branches that got corrected but not yet commited. Branches get corrected by decode or after execute. Also a branch misprediction can be detected out-of-order. Therefore, a corrected branch might not end up beeing committed in case an even earlier branch was mispredicted (Count)

board.processor.cores.core.branchPred.corrected_0::IndirectCond 0 0.00% 95.93% # Number of branches that got corrected but not yet commited. Branches get corrected by decode or after execute. Also a branch misprediction can be detected out-of-order. Therefore, a corrected branch might not end up beeing committed in case an even earlier branch was mispredicted (Count)

board.processor.cores.core.branchPred.corrected_0::IndirectUncond 26 4.07% 100.00% # Number of branches that got corrected but not yet commited. Branches get corrected by decode or after execute. Also a branch misprediction can be detected out-of-order. Therefore, a corrected branch might not end up beeing committed in case an even earlier branch was mispredicted (Count)

board.processor.cores.core.branchPred.corrected_0::total 639 # Number of branches that got corrected but not yet commited. Branches get corrected by decode or after execute. Also a branch misprediction can be detected out-of-order. Therefore, a corrected branch might not end up beeing committed in case an even earlier branch was mispredicted (Count)

```
board.processor.cores.core.branchPred.earlyResteers_0::NoBranch          0                    # Number of branches that got redirected after decode. (Count)
board.processor.cores.core.branchPred.earlyResteers_0::Return            0                    # Number of branches that got redirected after decode. (Count)
board.processor.cores.core.branchPred.earlyResteers_0::CallDirect         0                    # Number of branches that got redirected after decode. (Count)
board.processor.cores.core.branchPred.earlyResteers_0::CallIndirect       0                    # Number of branches that got redirected after decode. (Count)
board.processor.cores.core.branchPred.earlyResteers_0::DirectCond         0                    # Number of branches that got redirected after decode. (Count)
board.processor.cores.core.branchPred.earlyResteers_0::DirectUncond       0                    # Number of branches that got redirected after decode. (Count)
board.processor.cores.core.branchPred.earlyResteers_0::IndirectCond       0                    # Number of branches that got redirected after decode. (Count)
board.processor.cores.core.branchPred.earlyResteers_0::IndirectUncond     0                    # Number of branches that got redirected after decode. (Count)
board.processor.cores.core.branchPred.earlyResteers_0::total              0                    # Number of branches that got redirected after decode. (Count)
board.processor.cores.core.branchPred.committed_0::NoBranch          0      0.00%      0.00% # Number of branches finally committed  (Count)
board.processor.cores.core.branchPred.committed_0::Return           122      8.23%      8.23% # Number of branches finally committed  (Count)
board.processor.cores.core.branchPred.committed_0::CallDirect         87      5.87%     14.09% # Number of branches finally committed  (Count)
board.processor.cores.core.branchPred.committed_0::CallIndirect       40      2.70%     16.79% # Number of branches finally committed  (Count)
board.processor.cores.core.branchPred.committed_0::DirectCond       1108     74.71%     91.50% # Number of branches finally committed  (Count)
board.processor.cores.core.branchPred.committed_0::DirectUncond      105      7.08%     98.58% # Number of branches finally committed  (Count)
board.processor.cores.core.branchPred.committed_0::IndirectCond        0      0.00%     98.58% # Number of branches finally committed  (Count)
board.processor.cores.core.branchPred.committed_0::IndirectUncond      21      1.42%    100.00% # Number of branches finally committed  (Count)
board.processor.cores.core.branchPred.committed_0::total             1483                    # Number of branches finally committed  (Count)
board.processor.cores.core.branchPred.mispredicted_0::NoBranch          0      0.00%      0.00% # Number of committed branches that were mispredicted. (Count)
board.processor.cores.core.branchPred.mispredicted_0::Return            0      0.00%      0.00% # Number of committed branches that were mispredicted. (Count)
board.processor.cores.core.branchPred.mispredicted_0::CallDirect        56     12.15%     12.15% # Number of committed branches that were mispredicted. (Count)
board.processor.cores.core.branchPred.mispredicted_0::CallIndirect       32      6.94%     19.09% # Number of committed branches that were mispredicted. (Count)
board.processor.cores.core.branchPred.mispredicted_0::DirectCond        298     64.64%     83.73% # Number of committed branches that were mispredicted. (Count)
board.processor.cores.core.branchPred.mispredicted_0::DirectUncond       54     11.71%     95.44% # Number of committed branches that were mispredicted. (Count)
board.processor.cores.core.branchPred.mispredicted_0::IndirectCond        0      0.00%     95.44% # Number of committed branches that were mispredicted. (Count)
board.processor.cores.core.branchPred.mispredicted_0::IndirectUncond      21      4.56%    100.00% # Number of committed branches that were mispredicted. (Count)
board.processor.cores.core.branchPred.mispredicted_0::total             461                    # Number of committed branches that were mispredicted. (Count)
board.processor.cores.core.branchPred.targetProvider_0::NoTarget       2649     70.23%     70.23% # The component providing the target for taken branches (Count)
board.processor.cores.core.branchPred.targetProvider_0::BTB            845     22.40%     92.63% # The component providing the target for taken branches (Count)
board.processor.cores.core.branchPred.targetProvider_0::RAS            267      7.08%     99.71% # The component providing the target for taken branches (Count)
board.processor.cores.core.branchPred.targetProvider_0::Indirect        11      0.29%    100.00% # The component providing the target for taken branches (Count)
board.processor.cores.core.branchPred.targetProvider_0::total          3772                    # The component providing the target for taken branches (Count)
board.processor.cores.core.branchPred.targetWrong_0::NoBranch          606     94.84%     94.84% # Number of branches where the target was incorrect or not available at prediction time. (Count)
board.processor.cores.core.branchPred.targetWrong_0::Return             32      5.01%     99.84% # Number of branches where the target was incorrect or not available at prediction time. (Count)
board.processor.cores.core.branchPred.targetWrong_0::CallDirect          1      0.16%    100.00% # Number of branches where the target was incorrect or not available at prediction time. (Count)
board.processor.cores.core.branchPred.targetWrong_0::CallIndirect        0      0.00%    100.00% # Number of branches where the target was incorrect or not available at prediction time. (Count)
board.processor.cores.core.branchPred.targetWrong_0::DirectCond           0      0.00%    100.00% # Number of branches where the target was incorrect or not available at prediction time. (Count)
board.processor.cores.core.branchPred.targetWrong_0::DirectUncond         0      0.00%    100.00% # Number of branches where the target was incorrect or not available at prediction time. (Count)
board.processor.cores.core.branchPred.targetWrong_0::IndirectCond         0      0.00%    100.00% # Number of branches where the target was incorrect or not available at prediction time. (Count)
board.processor.cores.core.branchPred.targetWrong_0::IndirectUncond       0      0.00%    100.00% # Number of branches where the target was incorrect or not available at prediction time. (Count)
board.processor.cores.core.branchPred.targetWrong_0::total             639                    # Number of branches where the target was incorrect or not available at prediction time. (Count)
board.processor.cores.core.branchPred.condPredicted          2737                    # Number of conditional branches predicted (Count)
board.processor.cores.core.branchPred.condPredictedTaken      982                    # Number of conditional branches predicted as taken (Count)
board.processor.cores.core.branchPred.condIncorrect           639                    # Number of conditional branches incorrect (Count)
board.processor.cores.core.branchPred.predTakenBTBMiss        325                    # Number of branches predicted taken but missed in BTB (Count)
board.processor.cores.core.branchPred.NotTakenMispredicted     609                    # Number branches predicted 'not taken' but turned out to be taken (Count)
board.processor.cores.core.branchPred.TakenMispredicted         30                    # Number branches predicted taken but are actually not taken (Count)
board.processor.cores.core.branchPred.BTBLookups             3772                    # Number of BTB lookups (Count)
board.processor.cores.core.branchPred.BTBUpdates              548                    # Number of BTB updates (Count)
```

board.processor.cores.core.branchPred.BTBHits 1337 # Number of BTB hits (Count)

board.processor.cores.core.branchPred.BTBHitRatio 0.354454 # BTB Hit Ratio (Ratio)

board.processor.cores.core.branchPred.BTBMispredicted 447 # Number BTB mispredictions. No target found or target wrong (Count)

board.processor.cores.core.branchPred.indirectLookups 136 # Number of indirect predictor lookups. (Count)

board.processor.cores.core.branchPred.indirectHits 11 # Number of indirect target hits. (Count)

board.processor.cores.core.branchPred.indirectMisses 125 # Number of indirect misses. (Count)

board.processor.cores.core.branchPred.indirectMispredicted 0 # Number of mispredicted indirect branches. (Count)

board.processor.cores.core.branchPred.btb.lookups::NoBranch 0 0.00% 0.00% # Number of BTB lookups (Count)

board.processor.cores.core.branchPred.btb.lookups::Return 267 7.08% 7.08% # Number of BTB lookups (Count)

board.processor.cores.core.branchPred.btb.lookups::CallDirect 307 8.14% 15.22% # Number of BTB lookups (Count)

board.processor.cores.core.branchPred.btb.lookups::CallIndirect 75 1.99% 17.21% # Number of BTB lookups (Count)

board.processor.cores.core.branchPred.btb.lookups::DirectCond 2737 72.56% 89.77% # Number of BTB lookups (Count)

board.processor.cores.core.branchPred.btb.lookups::DirectUncond 325 8.62% 98.38% # Number of BTB lookups (Count)

board.processor.cores.core.branchPred.btb.lookups::IndirectCond 0 0.00% 98.38% # Number of BTB lookups (Count)

board.processor.cores.core.branchPred.btb.lookups::IndirectUncond 61 1.62% 100.00% # Number of BTB lookups (Count)

board.processor.cores.core.branchPred.btb.lookups::total 3772 # Number of BTB lookups (Count)

board.processor.cores.core.branchPred.btb.misses::NoBranch 0 0.00% 0.00% # Number of BTB misses (Count)

board.processor.cores.core.branchPred.btb.misses::Return 264 10.84% 10.84% # Number of BTB misses (Count)

board.processor.cores.core.branchPred.btb.misses::CallDirect 145 5.95% 16.80% # Number of BTB misses (Count)

board.processor.cores.core.branchPred.btb.misses::CallIndirect 75 3.08% 19.88% # Number of BTB misses (Count)

board.processor.cores.core.branchPred.btb.misses::DirectCond 1735 71.25% 91.13% # Number of BTB misses (Count)

board.processor.cores.core.branchPred.btb.misses::DirectUncond 155 6.37% 97.49% # Number of BTB misses (Count)

board.processor.cores.core.branchPred.btb.misses::IndirectCond 0 0.00% 97.49% # Number of BTB misses (Count)

board.processor.cores.core.branchPred.btb.misses::IndirectUncond 61 2.51% 100.00% # Number of BTB misses (Count)

board.processor.cores.core.branchPred.btb.misses::total 2435 # Number of BTB misses (Count)

board.processor.cores.core.branchPred.btb.updates::NoBranch 0 0.00% 0.00% # Number of BTB updates (Count)

board.processor.cores.core.branchPred.btb.updates::Return 0 0.00% 0.00% # Number of BTB updates (Count)

board.processor.cores.core.branchPred.btb.updates::CallDirect 106 19.34% 19.34% # Number of BTB updates (Count)

board.processor.cores.core.branchPred.btb.updates::CallIndirect 0 0.00% 19.34% # Number of BTB updates (Count)

board.processor.cores.core.branchPred.btb.updates::DirectCond 329 60.04% 79.38% # Number of BTB updates (Count)

board.processor.cores.core.branchPred.btb.updates::DirectUncond 113 20.62% 100.00% # Number of BTB updates (Count)

board.processor.cores.core.branchPred.btb.updates::IndirectCond 0 0.00% 100.00% # Number of BTB updates (Count)

board.processor.cores.core.branchPred.btb.updates::IndirectUncond 0 0.00% 100.00% # Number of BTB updates (Count)

board.processor.cores.core.branchPred.btb.updates::total 548 # Number of BTB updates (Count)

board.processor.cores.core.branchPred.btb.mispredict::NoBranch 0 0.00% 0.00% # Number of BTB mispredictions. No target found or target wrong. (Count)

board.processor.cores.core.branchPred.btb.mispredict::Return 0 0.00% 0.00% # Number of BTB mispredictions. No target found or target wrong. (Count)

board.processor.cores.core.branchPred.btb.mispredict::CallDirect 106 19.34% 19.34% # Number of BTB mispredictions. No target found or target wrong. (Count)

board.processor.cores.core.branchPred.btb.mispredict::CallIndirect 0 0.00% 19.34% # Number of BTB mispredictions. No target found or target wrong. (Count)

board.processor.cores.core.branchPred.btb.mispredict::DirectCond 329 60.04% 79.38% # Number of BTB mispredictions. No target found or target wrong. (Count)

board.processor.cores.core.branchPred.btb.mispredict::DirectUncond 113 20.62% 100.00% # Number of BTB mispredictions. No target found or target wrong. (Count)

board.processor.cores.core.branchPred.btb.mispredict::IndirectCond 0 0.00% 100.00% # Number of BTB mispredictions. No target found or target wrong. (Count)

board.processor.cores.core.branchPred.btb.mispredict::IndirectUncond 0 0.00% 100.00% # Number of BTB mispredictions. No target found or target wrong. (Count)

board.processor.cores.core.branchPred.btb.mispredict::total 548 # Number of BTB mispredictions. No target found or target wrong. (Count)

board.processor.cores.core.branchPred.btb.power_state.pwrStateResidencyTicks::UNDEFINED 19514466 # Cumulative time (in ticks) in various power states (Tick)

board.processor.cores.core.branchPred.indirectBranchPred.lookups 136 # Number of lookups (Count)

board.processor.cores.core.branchPred.indirectBranchPred.hits 11 # Number of hits of a tag (Count)

board.processor.cores.core.branchPred.indirectBranchPred.misses 125 # Number of misses (Count)

board.processor.cores.core.branchPred.indirectBranchPred.targetRecords 60 # Number of targets that where recorded/installed in the cache (Count)

board.processor.cores.core.branchPred.indirectBranchPred.indirectRecords 196 # Number of indirect branches/calls recorded in the indirect hist (Count)

board.processor.cores.core.branchPred.indirectBranchPred.speculativeOverflows 1 # Number of times more than the allowed capacity for speculative branches/calls where in flight and destroy the path history (Count)

board.processor.cores.core.branchPred.ras.pushes 527 # Number of times a PC was pushed onto the RAS (Count)

board.processor.cores.core.branchPred.ras.pops 522 # Number of times a PC was poped from the RAS (Count)

board.processor.cores.core.branchPred.ras.squashes 400 # Number of times the stack operation was squashed due to wrong speculation. (Count)

board.processor.cores.core.branchPred.ras.used 122 # Number of times the RAS is the provider (Count)

board.processor.cores.core.branchPred.ras.correct 122 # Number of times the RAS is the provider and the prediction is correct (Count)

board.processor.cores.core.branchPred.ras.incorrect 0 # Number of times the RAS is the provider and the prediction is wrong (Count)

board.processor.cores.core.commit.commitSquashedInsts 10583 # The number of squashed insts skipped by commit (Count)

board.processor.cores.core.commit.branchMispredicts 420 # The number of times a branch was mispredicted (Count)

board.processor.cores.core.commit.numCommittedDist::samples 21064 # Number of insts commited each cycle (Count)

board.processor.cores.core.commit.numCommittedDist::mean 0.616787 # Number of insts commited each cycle (Count)

board.processor.cores.core.commit.numCommittedDist::stdev 1.694722 # Number of insts commited each cycle (Count)

board.processor.cores.core.commit.numCommittedDist::underflows 0 0.00% 0.00% # Number of insts commited each cycle (Count)

board.processor.cores.core.commit.numCommittedDist::0 17468 82.93% 82.93% # Number of insts commited each cycle (Count)

board.processor.cores.core.commit.numCommittedDist::1 862 4.09% 87.02% # Number of insts commited each cycle (Count)

board.processor.cores.core.commit.numCommittedDist::2 587 2.79% 89.81% # Number of insts commited each cycle (Count)

board.processor.cores.core.commit.numCommittedDist::3 794 3.77% 93.58% # Number of insts commited each cycle (Count)

board.processor.cores.core.commit.numCommittedDist::4 302 1.43% 95.01% # Number of insts commited each cycle (Count)

board.processor.cores.core.commit.numCommittedDist::5 241 1.14% 96.15% # Number of insts commited each cycle (Count)

board.processor.cores.core.commit.numCommittedDist::6 120 0.57% 96.72% # Number of insts commited each cycle (Count)

board.processor.cores.core.commit.numCommittedDist::7 79 0.38% 97.10% # Number of insts commited each cycle (Count)

board.processor.cores.core.commit.numCommittedDist::8 611 2.90% 100.00% # Number of insts commited each cycle (Count)

board.processor.cores.core.commit.numCommittedDist::overflows 0 0.00% 100.00% # Number of insts commited each cycle (Count)

board.processor.cores.core.commit.numCommittedDist::min_value 0 # Number of insts commited each cycle (Count)

board.processor.cores.core.commit.numCommittedDist::max_value 8 # Number of insts commited each cycle (Count)

board.processor.cores.core.commit.numCommittedDist::total 21064 # Number of insts commited each cycle (Count)

board.processor.cores.core.commit.amos 0 # Number of atomic instructions committed (Count)

board.processor.cores.core.commit.membars 0 # Number of memory barriers committed (Count)

board.processor.cores.core.commit.functionCalls 127 # Number of function calls committed. (Count)

board.processor.cores.core.commit.committedInstType_0::No_OpClass 27 0.21% 0.21% # Class of committed instruction (Count)

board.processor.cores.core.commit.committedInstType_0::IntAlu 10123 77.92% 78.12% # Class of committed instruction (Count)

board.processor.cores.core.commit.committedInstType_0::IntMult 4 0.03% 78.16% # Class of committed instruction (Count)

board.processor.cores.core.commit.committedInstType_0::IntDiv 21 0.16% 78.32% # Class of committed instruction (Count)

board.processor.cores.core.commit.committedInstType_0::FloatAdd 2 0.02% 78.33% # Class of committed instruction (Count)

board.processor.cores.core.commit.committedInstType_0::FloatCmp 0 0.00% 78.33% # Class of committed instruction (Count)

board.processor.cores.core.commit.committedInstType_0::FloatCvt 0 0.00% 78.33% # Class of committed instruction (Count)

board.processor.cores.core.commit.committedInstType_0::FloatMult 0 0.00% 78.33% # Class of committed instruction (Count)

board.processor.cores.core.commit.committedInstType_0::FloatMultAcc 0 0.00% 78.33% # Class of committed instruction (Count)

board.processor.cores.core.commit.committedInstType_0::FloatDiv 0 0.00% 78.33% # Class of committed instruction (Count)

board.processor.cores.core.commit.committedInstType_0::FloatMisc 0 0.00% 78.33% # Class of committed instruction (Count)

board.processor.cores.core.commit.committedInstType_0::FloatSqrt 0 0.00% 78.33% # Class of committed instruction (Count)

board.processor.cores.core.commit.committedInstType_0::SimdAdd 0 0.00% 78.33% # Class of committed instruction (Count)

board.processor.cores.core.commit.committedInstType_0::SimdAddAcc 0 0.00% 78.33% # Class of committed instruction (Count)

board.processor.cores.core.commit.committedInstType_0::SimdAlu 82 0.63% 78.96% # Class of committed instruction (Count)

board.processor.cores.core.commit.committedInstType_0::SimdCmp 0 0.00% 78.96% # Class of committed instruction (Count)

board.processor.cores.core.commit.committedInstType_0::SimdCvt 48 0.37% 79.33% # Class of committed instruction (Count)

board.processor.cores.core.commit.committedInstType_0::SimdMisc 62 0.48% 79.81% # Class of committed instruction (Count)

board.processor.cores.core.commit.committedInstType_0::SimdMult 0 0.00% 79.81% # Class of committed instruction (Count)

board.processor.cores.core.commit.committedInstType_0::SimdMultAcc 0 0.00% 79.81% # Class of committed instruction (Count)

board.processor.cores.core.commit.committedInstType_0::SimdMatMultAcc 0 0.00% 79.81% # Class of committed instruction (Count)

board.processor.cores.core.commit.committedInstType_0::SimdShift 0 0.00% 79.81% # Class of committed instruction (Count)

board.processor.cores.core.commit.committedInstType_0::SimdShiftAcc 0 0.00% 79.81% # Class of committed instruction (Count)

board.processor.cores.core.commit.committedInstType_0::SimdDiv 0 0.00% 79.81% # Class of committed instruction (Count)

board.processor.cores.core.commit.committedInstType_0::SimdSqrt 0 0.00% 79.81% # Class of committed instruction (Count)

board.processor.cores.core.commit.committedInstType_0::SimdFloatAdd 0 0.00% 79.81% # Class of committed instruction (Count)

board.processor.cores.core.commit.committedInstType_0::SimdFloatAlu 0 0.00% 79.81% # Class of committed instruction (Count)

board.processor.cores.core.commit.committedInstType_0::SimdFloatCmp 0 0.00% 79.81% # Class of committed instruction (Count)

board.processor.cores.core.commit.committedInstType_0::SimdFloatCvt 0 0.00% 79.81% # Class of committed instruction (Count)

board.processor.cores.core.commit.committedInstType_0::SimdFloatDiv 0 0.00% 79.81% # Class of committed instruction (Count)

board.processor.cores.core.commit.committedInstType_0::SimdFloatMisc 0 0.00% 79.81% # Class of committed instruction (Count)

board.processor.cores.core.commit.committedInstType_0::SimdFloatMult 0 0.00% 79.81% # Class of committed instruction (Count)

board.processor.cores.core.commit.committedInstType_0::SimdFloatMultAcc 0 0.00% 79.81% # Class of committed instruction (Count)

board.processor.cores.core.commit.committedInstType_0::SimdFloatMatMultAcc 0 0.00% 79.81% # Class of committed instruction (Count)

board.processor.cores.core.commit.committedInstType_0::SimdFloatSqrt 0 0.00% 79.81% # Class of committed instruction (Count)

board.processor.cores.core.commit.committedInstType_0::SimdReduceAdd 0 0.00% 79.81% # Class of committed instruction (Count)

board.processor.cores.core.commit.committedInstType_0::SimdReduceAlu 0 0.00% 79.81% # Class of committed instruction (Count)

board.processor.cores.core.commit.committedInstType_0::SimdReduceCmp 0 0.00% 79.81% # Class of committed instruction (Count)

board.processor.cores.core.commit.committedInstType_0::SimdFloatReduceAdd 0 0.00% 79.81% # Class of committed instruction (Count)

board.processor.cores.core.commit.committedInstType_0::SimdFloatReduceCmp 0 0.00% 79.81% # Class of committed instruction (Count)

board.processor.cores.core.commit.committedInstType_0::SimdAes 0 0.00% 79.81% # Class of committed instruction (Count)

board.processor.cores.core.commit.committedInstType_0::SimdAesMix 0 0.00% 79.81% # Class of committed instruction (Count)

board.processor.cores.core.commit.committedInstType_0::SimdSha1Hash 0 0.00% 79.81% # Class of committed instruction (Count)

board.processor.cores.core.commit.committedInstType_0::SimdSha1Hash2 0 0.00% 79.81% # Class of committed instruction (Count)

board.processor.cores.core.commit.committedInstType_0::SimdSha256Hash 0 0.00% 79.81% # Class of committed instruction (Count)

board.processor.cores.core.commit.committedInstType_0::SimdSha256Hash2 0 0.00% 79.81% # Class of committed instruction (Count)

board.processor.cores.core.commit.committedInstType_0::SimdShaSigma2 0 0.00% 79.81% # Class of committed instruction (Count)

board.processor.cores.core.commit.committedInstType_0::SimdShaSigma3 0 0.00% 79.81% # Class of committed instruction (Count)

board.processor.cores.core.commit.committedInstType_0::SimdPredAlu 0 0.00% 79.81% # Class of committed instruction (Count)

board.processor.cores.core.commit.committedInstType_0::Matrix 0 0.00% 79.81% # Class of committed instruction (Count)

board.processor.cores.core.commit.committedInstType_0::MatrixMov 0 0.00% 79.81% # Class of committed instruction (Count)

board.processor.cores.core.commit.committedInstType_0::MatrixOP 0 0.00% 79.81% # Class of committed instruction (Count)

board.processor.cores.core.commit.committedInstType_0::MemRead 1224 9.42% 89.23% # Class of committed instruction (Count)

board.processor.cores.core.commit.committedInstType_0::MemWrite 1191 9.17% 98.40% # Class of committed instruction (Count)

board.processor.cores.core.commit.committedInstType_0::FloatMemRead 48 0.37% 98.77% # Class of committed instruction (Count)

board.processor.cores.core.commit.committedInstType_0::FloatMemWrite 160 1.23% 100.00% # Class of committed instruction (Count)

board.processor.cores.core.commit.committedInstType_0::IprAccess 0 0.00% 100.00% # Class of committed instruction (Count)

board.processor.cores.core.commit.committedInstType_0::InstPrefetch 0 0.00% 100.00% # Class of committed instruction (Count)

board.processor.cores.core.commit.committedInstType_0::SimdUnitStrideLoad 0 0.00% 100.00% # Class of committed instruction (Count)

board.processor.cores.core.commit.committedInstType_0::SimdUnitStrideStore 0 0.00% 100.00% # Class of committed instruction (Count)

board.processor.cores.core.commit.committedInstType_0::SimdUnitStrideMaskLoad 0 0.00% 100.00% # Class of committed instruction (Count)

board.processor.cores.core.commit.committedInstType_0::SimdUnitStrideMaskStore 0 0.00% 100.00% # Class of committed instruction (Count)

board.processor.cores.core.commit.committedInstType_0::SimdStridedLoad 0 0.00% 100.00% # Class of committed instruction (Count)

board.processor.cores.core.commit.committedInstType_0::SimdStridedStore 0 0.00% 100.00% # Class of committed instruction (Count)

board.processor.cores.core.commit.committedInstType_0::SimdIndexedLoad 0 0.00% 100.00% # Class of committed instruction (Count)

board.processor.cores.core.commit.committedInstType_0::SimdIndexedStore 0 0.00% 100.00% # Class of committed instruction (Count)

board.processor.cores.core.commit.committedInstType_0::SimdWholeRegisterLoad 0 0.00% 100.00% # Class of committed instruction (Count)

board.processor.cores.core.commit.committedInstType_0::SimdWholeRegisterStore 0 0.00% 100.00% # Class of committed instruction (Count)

board.processor.cores.core.commit.committedInstType_0::SimdUnitStrideFaultOnlyFirstLoad 0 0.00% 100.00% # Class of committed instruction (Count)

board.processor.cores.core.commit.committedInstType_0::SimdUnitStrideSegmentedLoad 0 0.00% 100.00% # Class of committed instruction (Count)

board.processor.cores.core.commit.committedInstType_0::SimdUnitStrideSegmentedStore 0 0.00% 100.00% # Class of committed instruction (Count)

board.processor.cores.core.commit.committedInstType_0::SimdExt 0 0.00% 100.00% # Class of committed instruction (Count)

board.processor.cores.core.commit.committedInstType_0::SimdFloatExt 0 0.00% 100.00% # Class of committed instruction (Count)

board.processor.cores.core.commit.committedInstType_0::SimdConfig 0 0.00% 100.00% # Class of committed instruction (Count)

board.processor.cores.core.commit.committedInstType_0::total 12992 # Class of committed instruction (Count)

board.processor.cores.core.commit.commitEligibleSamples 611 # number cycles where commit BW limit reached (Cycle)

board.processor.cores.core.commitStats0.numInsts 6555 # Number of instructions committed (thread level) (Count)

board.processor.cores.core.commitStats0.numOps 12992 # Number of ops (including micro ops) committed (thread level) (Count)

board.processor.cores.core.commitStats0.numInstsNotNOP 6555 # Number of instructions committed excluding NOPs or prefetches (Count)

board.processor.cores.core.commitStats0.numOpsNotNOP 12992 # Number of Ops (including micro ops) Simulated (Count)

board.processor.cores.core.commitStats0.cpi 8.940198 # CPI: cycles per instruction (thread level) ((Cycle/Count))

board.processor.cores.core.commitStats0.ipc 0.111854 # IPC: instructions per cycle (thread level) ((Count/Cycle))

board.processor.cores.core.commitStats0.numMemRefs 2623 # Number of memory references committed (Count)

board.processor.cores.core.commitStats0.numFpInsts 410 # Number of float instructions (Count)

board.processor.cores.core.commitStats0.numIntInsts 12796 # Number of integer instructions (Count)

board.processor.cores.core.commitStats0.numLoadInsts 1272 # Number of load instructions (Count)

board.processor.cores.core.commitStats0.numStoreInsts 1351 # Number of store instructions (Count)

board.processor.cores.core.commitStats0.numVecInsts 0 # Number of vector instructions (Count)

board.processor.cores.core.commitStats0.committedInstType::No_OpClass 27 0.21% 0.21% # Class of committed instruction. (Count)

board.processor.cores.core.commitStats0.committedInstType::IntAlu 10123 77.92% 78.12% # Class of committed instruction. (Count)

board.processor.cores.core.commitStats0.committedInstType::IntMult 4 0.03% 78.16% # Class of committed instruction. (Count)

board.processor.cores.core.commitStats0.committedInstType::IntDiv 21 0.16% 78.32% # Class of committed instruction. (Count)

board.processor.cores.core.commitStats0.committedInstType::FloatAdd 2 0.02% 78.33% # Class of committed instruction. (Count)

board.processor.cores.core.commitStats0.committedInstType::FloatCmp 0 0.00% 78.33% # Class of committed instruction. (Count)

board.processor.cores.core.commitStats0.committedInstType::FloatCvt 0 0.00% 78.33% # Class of committed instruction. (Count)

board.processor.cores.core.commitStats0.committedInstType::FloatMult 0 0.00% 78.33% # Class of committed instruction. (Count)

board.processor.cores.core.commitStats0.committedInstType::FloatMultAcc 0 0.00% 78.33% # Class of committed instruction. (Count)

board.processor.cores.core.commitStats0.committedInstType::FloatDiv 0 0.00% 78.33% # Class of committed instruction. (Count)

board.processor.cores.core.commitStats0.committedInstType::FloatMisc 0 0.00% 78.33% # Class of committed instruction. (Count)

board.processor.cores.core.commitStats0.committedInstType::FloatSqrt 0 0.00% 78.33% # Class of committed instruction. (Count)

board.processor.cores.core.commitStats0.committedInstType::SimdAdd 0 0.00% 78.33% # Class of committed instruction. (Count)

board.processor.cores.core.commitStats0.committedInstType::SimdAddAcc 0 0.00% 78.33% # Class of committed instruction. (Count)

board.processor.cores.core.commitStats0.committedInstType::SimdAlu 82 0.63% 78.96% # Class of committed instruction. (Count)

board.processor.cores.core.commitStats0.committedInstType::SimdCmp 0 0.00% 78.96% # Class of committed instruction. (Count)

board.processor.cores.core.commitStats0.committedInstType::SimdCvt 48 0.37% 79.33% # Class of committed instruction. (Count)

board.processor.cores.core.commitStats0.committedInstType::SimdMisc 62 0.48% 79.81% # Class of committed instruction. (Count)

board.processor.cores.core.commitStats0.committedInstType::SimdMult 0 0.00% 79.81% # Class of committed instruction. (Count)

board.processor.cores.core.commitStats0.committedInstType::SimdMultAcc 0 0.00% 79.81% # Class of committed instruction. (Count)

board.processor.cores.core.commitStats0.committedInstType::SimdMatMultAcc 0 0.00% 79.81% # Class of committed instruction. (Count)

board.processor.cores.core.commitStats0.committedInstType::SimdShift 0 0.00% 79.81% # Class of committed instruction. (Count)

board.processor.cores.core.commitStats0.committedInstType::SimdShiftAcc 0 0.00% 79.81% # Class of committed instruction. (Count)

board.processor.cores.core.commitStats0.committedInstType::SimdDiv 0 0.00% 79.81% # Class of committed instruction. (Count)

board.processor.cores.core.commitStats0.committedInstType::SimdSqrt 0 0.00% 79.81% # Class of committed instruction. (Count)

board.processor.cores.core.commitStats0.committedInstType::SimdFloatAdd 0 0.00% 79.81% # Class of committed instruction. (Count)

board.processor.cores.core.commitStats0.committedInstType::SimdFloatAlu 0 0.00% 79.81% # Class of committed instruction. (Count)

board.processor.cores.core.commitStats0.committedInstType::SimdFloatCmp 0 0.00% 79.81% # Class of committed instruction. (Count)

board.processor.cores.core.commitStats0.committedInstType::SimdFloatCvt 0 0.00% 79.81% # Class of committed instruction. (Count)

board.processor.cores.core.commitStats0.committedInstType::SimdFloatDiv 0 0.00% 79.81% # Class of committed instruction. (Count)

board.processor.cores.core.commitStats0.committedInstType::SimdFloatMisc 0 0.00% 79.81% # Class of committed instruction. (Count)

board.processor.cores.core.commitStats0.committedInstType::SimdFloatMult 0 0.00% 79.81% # Class of committed instruction. (Count)

board.processor.cores.core.commitStats0.committedInstType::SimdFloatMultAcc 0 0.00% 79.81% # Class of committed instruction. (Count)

board.processor.cores.core.commitStats0.committedInstType::SimdFloatMatMultAcc 0 0.00% 79.81% # Class of committed instruction. (Count)

board.processor.cores.core.commitStats0.committedInstType::SimdFloatSqrt 0 0.00% 79.81% # Class of committed instruction. (Count)

board.processor.cores.core.commitStats0.committedInstType::SimdReduceAdd 0 0.00% 79.81% # Class of committed instruction. (Count)

board.processor.cores.core.commitStats0.committedInstType::SimdReduceAlu 0 0.00% 79.81% # Class of committed instruction. (Count)

board.processor.cores.core.commitStats0.committedInstType::SimdReduceCmp 0 0.00% 79.81% # Class of committed instruction. (Count)

board.processor.cores.core.commitStats0.committedInstType::SimdFloatReduceAdd 0 0.00% 79.81% # Class of committed instruction. (Count)

board.processor.cores.core.commitStats0.committedInstType::SimdFloatReduceCmp 0 0.00% 79.81% # Class of committed instruction. (Count)

board.processor.cores.core.commitStats0.committedInstType::SimdAes 0 0.00% 79.81% # Class of committed instruction. (Count)

board.processor.cores.core.commitStats0.committedInstType::SimdAesMix 0 0.00% 79.81% # Class of committed instruction. (Count)

board.processor.cores.core.commitStats0.committedInstType::SimdSha1Hash 0 0.00% 79.81% # Class of committed instruction. (Count)

board.processor.cores.core.commitStats0.committedInstType::SimdSha1Hash2 0 0.00% 79.81% # Class of committed instruction. (Count)

board.processor.cores.core.commitStats0.committedInstType::SimdSha256Hash 0 0.00% 79.81% # Class of committed instruction. (Count)

board.processor.cores.core.commitStats0.committedInstType::SimdSha256Hash2 0 0.00% 79.81% # Class of committed instruction. (Count)

board.processor.cores.core.commitStats0.committedInstType::SimdShaSigma2 0 0.00% 79.81% # Class of committed instruction. (Count)

board.processor.cores.core.commitStats0.committedInstType::SimdShaSigma3 0 0.00% 79.81% # Class of committed instruction. (Count)

board.processor.cores.core.commitStats0.committedInstType::SimdPredAlu 0 0.00% 79.81% # Class of committed instruction. (Count)

board.processor.cores.core.commitStats0.committedInstType::Matrix 0 0.00% 79.81% # Class of committed instruction. (Count)

board.processor.cores.core.commitStats0.committedInstType::MatrixMov 0 0.00% 79.81% # Class of committed instruction. (Count)

board.processor.cores.core.commitStats0.committedInstType::MatrixOP 0 0.00% 79.81% # Class of committed instruction. (Count)

board.processor.cores.core.commitStats0.committedInstType::MemRead 1224 9.42% 89.23% # Class of committed instruction. (Count)

board.processor.cores.core.commitStats0.committedInstType::MemWrite 1191 9.17% 98.40% # Class of committed instruction. (Count)

board.processor.cores.core.commitStats0.committedInstType::FloatMemRead 48 0.37% 98.77% # Class of committed instruction. (Count)

board.processor.cores.core.commitStats0.committedInstType::FloatMemWrite 160 1.23% 100.00% # Class of committed instruction. (Count)

board.processor.cores.core.commitStats0.committedInstType::IprAccess 0 0.00% 100.00% # Class of committed instruction. (Count)

board.processor.cores.core.commitStats0.committedInstType::InstPrefetch 0 0.00% 100.00% # Class of committed instruction. (Count)

board.processor.cores.core.commitStats0.committedInstType::SimdUnitStrideLoad 0 0.00% 100.00% # Class of committed instruction. (Count)

board.processor.cores.core.commitStats0.committedInstType::SimdUnitStrideStore 0 0.00% 100.00% # Class of committed instruction. (Count)

board.processor.cores.core.commitStats0.committedInstType::SimdUnitStrideMaskLoad 0 0.00% 100.00% # Class of committed instruction. (Count)

board.processor.cores.core.commitStats0.committedInstType::SimdUnitStrideMaskStore 0 0.00% 100.00% # Class of committed instruction. (Count)

board.processor.cores.core.commitStats0.committedInstType::SimdStridedLoad 0 0.00% 100.00% # Class of committed instruction. (Count)

board.processor.cores.core.commitStats0.committedInstType::SimdStridedStore 0 0.00% 100.00% # Class of committed instruction. (Count)

board.processor.cores.core.commitStats0.committedInstType::SimdIndexedLoad 0 0.00% 100.00% # Class of committed instruction. (Count)

board.processor.cores.core.commitStats0.committedInstType::SimdIndexedStore 0 0.00% 100.00% # Class of committed instruction. (Count)

board.processor.cores.core.commitStats0.committedInstType::SimdWholeRegisterLoad 0 0.00% 100.00% # Class of committed instruction. (Count)

board.processor.cores.core.commitStats0.committedInstType::SimdWholeRegisterStore 0 0.00% 100.00% # Class of committed instruction. (Count)

board.processor.cores.core.commitStats0.committedInstType::SimdUnitStrideFaultOnlyFirstLoad 0 0.00% 100.00% # Class of committed instruction. (Count)

board.processor.cores.core.commitStats0.committedInstType::SimdUnitStrideSegmentedLoad 0 0.00% 100.00% # Class of committed instruction. (Count)

board.processor.cores.core.commitStats0.committedInstType::SimdUnitStrideSegmentedStore 0 0.00% 100.00% # Class of committed instruction. (Count)

board.processor.cores.core.commitStats0.committedInstType::SimdExt 0 0.00% 100.00% # Class of committed instruction. (Count)

board.processor.cores.core.commitStats0.committedInstType::SimdFloatExt 0 0.00% 100.00% # Class of committed instruction. (Count)

board.processor.cores.core.commitStats0.committedInstType::SimdConfig 0 0.00% 100.00% # Class of committed instruction. (Count)

board.processor.cores.core.commitStats0.committedInstType::total 12992 # Class of committed instruction. (Count)

board.processor.cores.core.commitStats0.committedControl::IsControl 1483 # Class of control type instructions committed (Count)

board.processor.cores.core.commitStats0.committedControl::IsDirectControl 1300 # Class of control type instructions committed (Count)

board.processor.cores.core.commitStats0.committedControl::IsIndirectControl 183 # Class of control type instructions committed (Count)

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board.processor.cores.core.commitStats0.committedControl::IsCondControl 1108 # Class of control type instructions committed (Count)
board.processor.cores.core.commitStats0.committedControl::IsUncondControl 375 # Class of control type instructions committed (Count)
board.processor.cores.core.commitStats0.committedControl::IsCall 127 # Class of control type instructions committed (Count)
board.processor.cores.core.commitStats0.committedControl::IsReturn 122 # Class of control type instructions committed (Count)
board.processor.cores.core.decode.idleCycles 13370 # Number of cycles decode is idle (Cycle)
board.processor.cores.core.decode.blockedCycles 4777 # Number of cycles decode is blocked (Cycle)
board.processor.cores.core.decode.runCycles 3589 # Number of cycles decode is running (Cycle)
board.processor.cores.core.decode.unblockCycles 592 # Number of cycles decode is unblocking (Cycle)
board.processor.cores.core.decode.squashCycles 437 # Number of cycles decode is squashing (Cycle)
board.processor.cores.core.decode.branchResolved 969 # Number of times decode resolved a branch (Count)
board.processor.cores.core.decode.branchMispred 227 # Number of times decode detected a branch misprediction (Count)
board.processor.cores.core.decode.decodedInsts 27890 # Number of instructions handled by decode (Count)
board.processor.cores.core.decode.squashedInsts 1071 # Number of squashed instructions handled by decode (Count)
board.processor.cores.core.executeStats0.numInsts 20259 # Number of executed instructions (Count)
board.processor.cores.core.executeStats0.numNop 0 # Number of nop insts executed (Count)
board.processor.cores.core.executeStats0.numBranches 2136 # Number of branches executed (Count)
board.processor.cores.core.executeStats0.numLoadInsts 2023 # Number of load instructions executed (Count)
board.processor.cores.core.executeStats0.numStoreInsts 1824 # Number of stores executed (Count)
board.processor.cores.core.executeStats0.instRate 0.345699 # Inst execution rate ((Count/Cycle))
board.processor.cores.core.executeStats0.numCCRegReads 9090 # Number of times the CC registers were read (Count)
board.processor.cores.core.executeStats0.numCCRegWrites 6193 # Number of times the CC registers were written (Count)
board.processor.cores.core.executeStats0.numFpRegReads 604 # Number of times the floating registers were read (Count)
board.processor.cores.core.executeStats0.numFpRegWrites 223 # Number of times the floating registers were written (Count)
board.processor.cores.core.executeStats0.numIntRegReads 20886 # Number of times the integer registers were read (Count)
board.processor.cores.core.executeStats0.numIntRegWrites 14218 # Number of times the integer registers were written (Count)
board.processor.cores.core.executeStats0.numMemRefs 3847 # Number of memory refs (Count)
board.processor.cores.core.executeStats0.numMiscRegReads 8818 # Number of times the Misc registers were read (Count)
board.processor.cores.core.executeStats0.numVecAluAccesses 0 # Number of vector alu accesses (Count)
board.processor.cores.core.executeStats0.numDiscardedOps 0 # Number of ops (including micro ops) which were discarded before commit (Count)
board.processor.cores.core.fetch.predictedBranches 1123 # Number of branches that fetch has predicted taken (Count)
board.processor.cores.core.fetch.cycles 8600 # Number of cycles fetch has run and was not squashing or blocked (Cycle)
board.processor.cores.core.fetch.squashCycles 1314 # Number of cycles fetch has spent squashing (Cycle)
board.processor.cores.core.fetch.miscStallCycles 4 # Number of cycles fetch has spent waiting on interrupts, or bad addresses, or out of MSHRs (Cycle)
board.processor.cores.core.fetch.pendingTrapStallCycles 52 # Number of stall cycles due to pending traps (Cycle)
board.processor.cores.core.fetch.cacheLines 2598 # Number of cache lines fetched (Count)
board.processor.cores.core.fetch.icacheSquashes 289 # Number of outstanding Icache misses that were squashed (Count)
board.processor.cores.core.fetch.nisnDist::samples 22765 # Number of instructions fetched each cycle (Total) (Count)
board.processor.cores.core.fetch.nisnDist::mean 1.452800 # Number of instructions fetched each cycle (Total) (Count)
board.processor.cores.core.fetch.nisnDist::stdev 2.896600 # Number of instructions fetched each cycle (Total) (Count)
board.processor.cores.core.fetch.nisnDist::underflows 0 0.00% 0.00% # Number of instructions fetched each cycle (Total) (Count)
board.processor.cores.core.fetch.nisnDist::0 17650 77.53% 77.53% # Number of instructions fetched each cycle (Total) (Count)
board.processor.cores.core.fetch.nisnDist::1 186 0.82% 78.35% # Number of instructions fetched each cycle (Total) (Count)
board.processor.cores.core.fetch.nisnDist::2 214 0.94% 79.29% # Number of instructions fetched each cycle (Total) (Count)
board.processor.cores.core.fetch.nisnDist::3 446 1.96% 81.25% # Number of instructions fetched each cycle (Total) (Count)
board.processor.cores.core.fetch.nisnDist::4 263 1.16% 82.40% # Number of instructions fetched each cycle (Total) (Count)
board.processor.cores.core.fetch.nisnDist::5 418 1.84% 84.24% # Number of instructions fetched each cycle (Total) (Count)
board.processor.cores.core.fetch.nisnDist::6 287 1.26% 85.50% # Number of instructions fetched each cycle (Total) (Count)
board.processor.cores.core.fetch.nisnDist::7 151 0.66% 86.16% # Number of instructions fetched each cycle (Total) (Count)
board.processor.cores.core.fetch.nisnDist::8 3150 13.84% 100.00% # Number of instructions fetched each cycle (Total) (Count)
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board.processor.cores.core.fetch.nisnDist::overflows 0 0.00% 100.00% # Number of instructions fetched each cycle (Total) (Count)
board.processor.cores.core.fetch.nisnDist::min_value 0 # Number of instructions fetched each cycle (Total) (Count)
board.processor.cores.core.fetch.nisnDist::max_value 8 # Number of instructions fetched each cycle (Total) (Count)
board.processor.cores.core.fetch.nisnDist::total 22765 # Number of instructions fetched each cycle (Total) (Count)
board.processor.cores.core.fetchStats0.numInsts 17010 # Number of instructions fetched (thread level) (Count)
board.processor.cores.core.fetchStats0.numOps 0 # Number of ops (including micro ops) fetched (thread level) (Count)
board.processor.cores.core.fetchStats0.fetchRate 0.290258 # Number of inst fetches per cycle ((Count/Cycle))
board.processor.cores.core.fetchStats0.numBranches 3772 # Number of branches fetched (Count)
board.processor.cores.core.fetchStats0.branchRate 0.064365 # Number of branch fetches per cycle (Ratio)
board.processor.cores.core.fetchStats0.icacheStallCycles 13452 # ICache total stall cycles (Cycle)
board.processor.cores.core.fetchStats0.numFetchSuspends 0 # Number of times Execute suspended instruction fetching (Count)
board.processor.cores.core.iew.idleCycles 0 # Number of cycles IEW is idle (Cycle)
board.processor.cores.core.iew.squashCycles 437 # Number of cycles IEW is squashing (Cycle)
board.processor.cores.core.iew.blockCycles 1290 # Number of cycles IEW is blocking (Cycle)
board.processor.cores.core.iew.unblockCycles 232 # Number of cycles IEW is unblocking (Cycle)
board.processor.cores.core.iew.dispatchedInsts 23825 # Number of instructions dispatched to IQ (Count)
board.processor.cores.core.iew.dispSquashedInsts 60 # Number of squashed instructions skipped by dispatch (Count)
board.processor.cores.core.iew.dispLoadInsts 2416 # Number of dispatched load instructions (Count)
board.processor.cores.core.iew.dispStoreInsts 2332 # Number of dispatched store instructions (Count)
board.processor.cores.core.iew.dispNonSpecInsts 26 # Number of dispatched non-speculative instructions (Count)
board.processor.cores.core.iew.iqFullEvents 3 # Number of times the IQ has become full, causing a stall (Count)
board.processor.cores.core.iew.lsqFullEvents 229 # Number of times the LSQ has become full, causing a stall (Count)
board.processor.cores.core.iew.memOrderViolationEvents 4 # Number of memory order violations (Count)
board.processor.cores.core.iew.predictedTakenIncorrect 38 # Number of branches that were predicted taken incorrectly (Count)
board.processor.cores.core.iew.predictedNotTakenIncorrect 473 # Number of branches that were predicted not taken incorrectly (Count)
board.processor.cores.core.iew.branchMispredicts 511 # Number of branch mispredicts detected at execute (Count)
board.processor.cores.core.iew.instsToCommit 20182 # Cumulative count of insts sent to commit (Count)
board.processor.cores.core.iew.writebackCount 19969 # Cumulative count of insts written-back (Count)
board.processor.cores.core.iew.producerInst 13584 # Number of instructions producing a value (Count)
board.processor.cores.core.iew.consumerInst 20629 # Number of instructions consuming a value (Count)
board.processor.cores.core.iew.wbRate 0.340750 # Insts written-back per cycle ((Count/Cycle))
board.processor.cores.core.iew.wbFanout 0.658490 # Average fanout of values written-back ((Count/Count))
board.processor.cores.core.interrupts.clk_domain.clock 5328 # Clock period in ticks (Tick)
board.processor.cores.core.lsq0.forwLoads 114 # Number of loads that had data forwarded from stores (Count)
board.processor.cores.core.lsq0.squashedLoads 1144 # Number of loads squashed (Count)
board.processor.cores.core.lsq0.ignoredResponses 0 # Number of memory responses ignored because the instruction is squashed (Count)
board.processor.cores.core.lsq0.memOrderViolation 4 # Number of memory ordering violations (Count)
board.processor.cores.core.lsq0.squashedStores 981 # Number of stores squashed (Count)
board.processor.cores.core.lsq0.rescheduledLoads 0 # Number of loads that were rescheduled (Count)
board.processor.cores.core.lsq0.blockedByCache 0 # Number of times an access to memory failed due to the cache being blocked (Count)
board.processor.cores.core.lsq0.loadToUse::samples 1272 # Distribution of cycle latency between the first time a load is issued and its completion (Unspecified)
board.processor.cores.core.lsq0.loadToUse::mean 16.420597 # Distribution of cycle latency between the first time a load is issued and its completion (Unspecified)
board.processor.cores.core.lsq0.loadToUse::stdev 53.618432 # Distribution of cycle latency between the first time a load is issued and its completion (Unspecified)
board.processor.cores.core.lsq0.loadToUse::0-9 1174 92.30% 92.30% # Distribution of cycle latency between the first time a load is issued and its completion (Unspecified)
board.processor.cores.core.lsq0.loadToUse::10-19 4 0.31% 92.61% # Distribution of cycle latency between the first time a load is issued and its completion (Unspecified)
board.processor.cores.core.lsq0.loadToUse::40-49 1 0.08% 92.69% # Distribution of cycle latency between the first time a load is issued and its completion (Unspecified)
board.processor.cores.core.lsq0.loadToUse::60-69 1 0.08% 92.77% # Distribution of cycle latency between the first time a load is issued and its completion (Unspecified)
board.processor.cores.core.lsq0.loadToUse::80-89 1 0.08% 92.85% # Distribution of cycle latency between the first time a load is issued and its completion (Unspecified)
board.processor.cores.core.lsq0.loadToUse::90-99 1 0.08% 92.92% # Distribution of cycle latency between the first time a load is issued and its completion (Unspecified)
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board.processor.cores.core.lsq0.loadToUse::100-109          2     0.16%     93.08% # Distribution of cycle latency between the first time a load is issued and its completion (Unspecified)
board.processor.cores.core.lsq0.loadToUse::110-119          1     0.08%     93.16% # Distribution of cycle latency between the first time a load is issued and its completion (Unspecified)
board.processor.cores.core.lsq0.loadToUse::120-129          3     0.24%     93.40% # Distribution of cycle latency between the first time a load is issued and its completion (Unspecified)
board.processor.cores.core.lsq0.loadToUse::130-139          2     0.16%     93.55% # Distribution of cycle latency between the first time a load is issued and its completion (Unspecified)
board.processor.cores.core.lsq0.loadToUse::140-149          18     1.42%     94.97% # Distribution of cycle latency between the first time a load is issued and its completion (Unspecified)
board.processor.cores.core.lsq0.loadToUse::150-159          8     0.63%     95.60% # Distribution of cycle latency between the first time a load is issued and its completion (Unspecified)
board.processor.cores.core.lsq0.loadToUse::160-169          5     0.39%     95.99% # Distribution of cycle latency between the first time a load is issued and its completion (Unspecified)
board.processor.cores.core.lsq0.loadToUse::170-179          3     0.24%     96.23% # Distribution of cycle latency between the first time a load is issued and its completion (Unspecified)
board.processor.cores.core.lsq0.loadToUse::180-189          4     0.31%     96.54% # Distribution of cycle latency between the first time a load is issued and its completion (Unspecified)
board.processor.cores.core.lsq0.loadToUse::200-209          2     0.16%     96.70% # Distribution of cycle latency between the first time a load is issued and its completion (Unspecified)
board.processor.cores.core.lsq0.loadToUse::220-229          16     1.26%     97.96% # Distribution of cycle latency between the first time a load is issued and its completion (Unspecified)
board.processor.cores.core.lsq0.loadToUse::230-239          7     0.55%     98.51% # Distribution of cycle latency between the first time a load is issued and its completion (Unspecified)
board.processor.cores.core.lsq0.loadToUse::240-249          7     0.55%     99.06% # Distribution of cycle latency between the first time a load is issued and its completion (Unspecified)
board.processor.cores.core.lsq0.loadToUse::250-259          1     0.08%     99.14% # Distribution of cycle latency between the first time a load is issued and its completion (Unspecified)
board.processor.cores.core.lsq0.loadToUse::260-269          2     0.16%     99.29% # Distribution of cycle latency between the first time a load is issued and its completion (Unspecified)
board.processor.cores.core.lsq0.loadToUse::270-279          1     0.08%     99.37% # Distribution of cycle latency between the first time a load is issued and its completion (Unspecified)
board.processor.cores.core.lsq0.loadToUse::overflows          8     0.63%    100.00% # Distribution of cycle latency between the first time a load is issued and its completion (Unspecified)
board.processor.cores.core.lsq0.loadToUse::min_value          2                     # Distribution of cycle latency between the first time a load is issued and its completion (Unspecified)
board.processor.cores.core.lsq0.loadToUse::max_value          447                     # Distribution of cycle latency between the first time a load is issued and its completion (Unspecified)
board.processor.cores.core.lsq0.loadToUse::total          1272                     # Distribution of cycle latency between the first time a load is issued and its completion (Unspecified)
board.processor.cores.core.mmu.dtb.rdAccesses          2029                     # TLB accesses on read requests (Count)
board.processor.cores.core.mmu.dtb.wrAccesses          1824                     # TLB accesses on write requests (Count)
board.processor.cores.core.mmu.dtb.rdMisses          62                     # TLB misses on read requests (Count)
board.processor.cores.core.mmu.dtb.wrMisses          22                     # TLB misses on write requests (Count)
board.processor.cores.core.mmu.dtb.walker.power_state.pwrStateResidencyTicks::UNDEFINED          19514466                     # Cumulative time (in ticks) in various power states (Tick)
board.processor.cores.core.mmu.itb.rdAccesses          0                     # TLB accesses on read requests (Count)
board.processor.cores.core.mmu.itb.wrAccesses          2605                     # TLB accesses on write requests (Count)
board.processor.cores.core.mmu.itb.rdMisses          0                     # TLB misses on read requests (Count)
board.processor.cores.core.mmu.itb.wrMisses          47                     # TLB misses on write requests (Count)
board.processor.cores.core.mmu.itb.walker.power_state.pwrStateResidencyTicks::UNDEFINED          19514466                     # Cumulative time (in ticks) in various power states (Tick)
board.processor.cores.core.power_state.pwrStateResidencyTicks::ON          19514466                     # Cumulative time (in ticks) in various power states (Tick)
board.processor.cores.core.rename.squashCycles          437                     # Number of cycles rename is squashing (Cycle)
board.processor.cores.core.rename.idleCycles          13730                     # Number of cycles rename is idle (Cycle)
board.processor.cores.core.rename.blockCycles          1870                     # Number of cycles rename is blocking (Cycle)
board.processor.cores.core.rename.serializeStallCycles          1207                     # count of cycles rename stalled for serializing inst (Cycle)
board.processor.cores.core.rename.runCycles          3781                     # Number of cycles rename is running (Cycle)
board.processor.cores.core.rename.unblockCycles          1740                     # Number of cycles rename is unblocking (Cycle)
board.processor.cores.core.rename.renamedInsts          26518                     # Number of instructions processed by rename (Count)
board.processor.cores.core.rename.ROBFullEvents          21                     # Number of times rename has blocked due to ROB full (Count)
board.processor.cores.core.rename.IQFullEvents          44                     # Number of times rename has blocked due to IQ full (Count)
board.processor.cores.core.rename.SQFullEvents          1646                     # Number of times rename has blocked due to SQ full (Count)
board.processor.cores.core.rename.renamedOperands          41804                     # Number of destination operands rename has renamed (Count)
board.processor.cores.core.rename.lookups          85149                     # Number of register rename lookups that rename has made (Count)
board.processor.cores.core.rename.intLookups          29152                     # Number of integer rename lookups (Count)
board.processor.cores.core.rename.fpLookups          638                     # Number of floating rename lookups (Count)
board.processor.cores.core.rename.committedMaps          20771                     # Number of HB maps that are committed (Count)
board.processor.cores.core.rename.undoneMaps          21013                     # Number of HB maps that are undone due to squashing (Count)
board.processor.cores.core.rename.serializing          4                     # count of serializing insts renamed (Count)
board.processor.cores.core.rename.tempSerializing          3                     # count of temporary serializing insts renamed (Count)
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board.processor.cores.core.rename.skidInsts          967                    # count of insts added to the skid buffer (Count)
board.processor.cores.core.rob.reads           43859                    # The number of ROB reads (Count)
board.processor.cores.core.rob.writes          48886                    # The number of ROB writes (Count)
board.processor.cores.core.thread_0.numInsts         6555                    # Number of Instructions committed (Count)
board.processor.cores.core.thread_0.numOps        12992                    # Number of Ops committed (Count)
board.processor.cores.core.thread_0.numMemRefs           0                    # Number of Memory References (Count)
board.processor.cores.core.workload.numSyscalls          11                    # Number of system calls (Count)
board.workload.inst.arm                      0                    # number of arm instructions executed (Count)
board.workload.inst.quiesce                  0                    # number of quiesce instructions executed (Count)

---------- End Simulation Statistics   ----------
```

ooo_nocache:

```
---------- Begin Simulation Statistics ----------
simSeconds                                   0.000079                       # Number of seconds simulated (Second)
simTicks                                    78881373                       # Number of ticks simulated (Tick)
finalTick                                   78881373                       # Number of ticks from beginning of simulation (restored from checkpoints and never reset) (Tick)
simFreq                                  1000000000000                       # The number of ticks per simulated second ((Tick/Second))
hostSeconds                                      0.10                       # Real time elapsed on the host (Second)
hostTickRate                                801167736                       # The number of ticks simulated per host second (ticks/s) ((Tick/Second))
hostMemory                                    1195828                       # Number of bytes of host memory used (Byte)
simInsts                                         6555                       # Number of instructions simulated (Count)
simOps                                          12992                       # Number of ops (including micro ops) simulated (Count)
hostInstRate                                    66517                       # Simulator instruction rate (inst/s) ((Count/Second))
hostOpRate                                     131832                       # Simulator op (including micro ops) rate (op/s) ((Count/Second))
board.cache_hierarchy.membus.transDist::ReadReq          3691                       # Transaction distribution (Count)
board.cache_hierarchy.membus.transDist::ReadResp         3689                       # Transaction distribution (Count)
board.cache_hierarchy.membus.transDist::WriteReq         1351                       # Transaction distribution (Count)
board.cache_hierarchy.membus.transDist::WriteResp        1351                       # Transaction distribution (Count)
board.cache_hierarchy.membus.pktCount_board.processor.cores.core.icache_port::board.memory.mem_ctrl.port         3844                       # Packet count per connected requestor and responder (Count)
board.cache_hierarchy.membus.pktCount_board.processor.cores.core.icache_port::total         3844                       # Packet count per connected requestor and responder (Count)
board.cache_hierarchy.membus.pktCount_board.processor.cores.core.dcache_port::board.memory.mem_ctrl.port         6238                       # Packet count per connected requestor and responder (Count)
board.cache_hierarchy.membus.pktCount_board.processor.cores.core.dcache_port::total         6238                       # Packet count per connected requestor and responder (Count)
board.cache_hierarchy.membus.pktCount::total        10082                       # Packet count per connected requestor and responder (Count)
board.cache_hierarchy.membus.pktSize_board.processor.cores.core.icache_port::board.memory.mem_ctrl.port       122944                       # Cumulative packet size per connected requestor and responder (Byte)
board.cache_hierarchy.membus.pktSize_board.processor.cores.core.icache_port::total       122944                       # Cumulative packet size per connected requestor and responder (Byte)
board.cache_hierarchy.membus.pktSize_board.processor.cores.core.dcache_port::board.memory.mem_ctrl.port        21827                       # Cumulative packet size per connected requestor and responder (Byte)
board.cache_hierarchy.membus.pktSize_board.processor.cores.core.dcache_port::total        21827                       # Cumulative packet size per connected requestor and responder (Byte)
board.cache_hierarchy.membus.pktSize::total       144771                       # Cumulative packet size per connected requestor and responder (Byte)
board.cache_hierarchy.membus.snoops                    0                       # Total snoops (Count)
board.cache_hierarchy.membus.snoopTraffic              0                       # Total snoop traffic (Byte)
board.cache_hierarchy.membus.snoopFanout::samples         5042                       # Request fanout histogram (Count)
board.cache_hierarchy.membus.snoopFanout::mean            0                       # Request fanout histogram (Count)
board.cache_hierarchy.membus.snoopFanout::stdev           0                       # Request fanout histogram (Count)
board.cache_hierarchy.membus.snoopFanout::underflows           0      0.00%      0.00% # Request fanout histogram (Count)
board.cache_hierarchy.membus.snoopFanout::0         5042    100.00%    100.00% # Request fanout histogram (Count)
board.cache_hierarchy.membus.snoopFanout::1            0      0.00%    100.00% # Request fanout histogram (Count)
board.cache_hierarchy.membus.snoopFanout::overflows           0      0.00%    100.00% # Request fanout histogram (Count)
board.cache_hierarchy.membus.snoopFanout::min_value            0                       # Request fanout histogram (Count)
board.cache_hierarchy.membus.snoopFanout::max_value            0                       # Request fanout histogram (Count)
board.cache_hierarchy.membus.snoopFanout::total         5042                       # Request fanout histogram (Count)
board.cache_hierarchy.membus.badaddr_responder.power_state.pwrStateResidencyTicks::UNDEFINED     78881373                       # Cumulative time (in ticks) in various power states (Tick)
board.cache_hierarchy.membus.power_state.pwrStateResidencyTicks::UNDEFINED     78881373                       # Cumulative time (in ticks) in various power states (Tick)
board.cache_hierarchy.membus.reqLayer2.occupancy      2128869                       # Layer occupancy (ticks) (Tick)
board.cache_hierarchy.membus.reqLayer2.utilization          0.0                       # Layer utilization (Ratio)
board.cache_hierarchy.membus.respLayer0.occupancy      1630773                       # Layer occupancy (ticks) (Tick)
board.cache_hierarchy.membus.respLayer0.utilization          0.0                       # Layer utilization (Ratio)
board.cache_hierarchy.membus.respLayer1.occupancy      2322779                       # Layer occupancy (ticks) (Tick)
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board.cache_hierarchy.membus.respLayer1.utilization 0.0 # Layer utilization (Ratio)
board.cache_hierarchy.membus.snoop_filter.totRequests 0 # Total number of requests made to the snoop filter. (Count)
board.cache_hierarchy.membus.snoop_filter.hitSingleRequests 0 # Number of requests hitting in the snoop filter with a single holder of the requested data. (Count)
board.cache_hierarchy.membus.snoop_filter.hitMultiRequests 0 # Number of requests hitting in the snoop filter with multiple (>1) holders of the requested data. (Count)
board.cache_hierarchy.membus.snoop_filter.totSnoops 0 # Total number of snoops made to the snoop filter. (Count)
board.cache_hierarchy.membus.snoop_filter.hitSingleSnoops 0 # Number of snoops hitting in the snoop filter with a single holder of the requested data. (Count)
board.cache_hierarchy.membus.snoop_filter.hitMultiSnoops 0 # Number of snoops hitting in the snoop filter with multiple (>1) holders of the requested data. (Count)
board.clk_domain.clock 333 # Clock period in ticks (Tick)
board.clk_domain.voltage_domain.voltage 1 # Voltage in Volts (Volt)
board.memory.mem_ctrl.avgPriority_processor.cores.core.inst::samples 1923.00 # Average QoS priority value for accepted requests (Count)
board.memory.mem_ctrl.avgPriority_processor.cores.core.data::samples 1732.00 # Average QoS priority value for accepted requests (Count)
board.memory.mem_ctrl.priorityMinLatency 0.000000018750 # per QoS priority minimum request to response latency (Second)
board.memory.mem_ctrl.priorityMaxLatency 0.000040203033 # per QoS priority maximum request to response latency (Second)
board.memory.mem_ctrl.numReadWriteTurnArounds 10 # Number of turnarounds from READ to WRITE (Count)
board.memory.mem_ctrl.numWriteReadTurnArounds 10 # Number of turnarounds from WRITE to READ (Count)
board.memory.mem_ctrl.numStayReadState 7798 # Number of times bus staying in READ state (Count)
board.memory.mem_ctrl.numStayWriteState 152 # Number of times bus staying in WRITE state (Count)
board.memory.mem_ctrl.readReqs 3691 # Number of read requests accepted (Count)
board.memory.mem_ctrl.writeReqs 1351 # Number of write requests accepted (Count)
board.memory.mem_ctrl.readBursts 3691 # Number of controller read bursts, including those serviced by the write queue (Count)
board.memory.mem_ctrl.writeBursts 1351 # Number of controller write bursts, including those merged in the write queue (Count)
board.memory.mem_ctrl.servicedByWrQ 222 # Number of controller read bursts serviced by the write queue (Count)
board.memory.mem_ctrl.mergedWrBursts 1165 # Number of controller write bursts merged with an existing one (Count)
board.memory.mem_ctrl.neitherReadNorWriteReqs 0 # Number of requests that are neither read nor write (Count)
board.memory.mem_ctrl.avgRdQLen 2.65 # Average read queue length when enqueuing ((Count/Tick))
board.memory.mem_ctrl.avgWrQLen 24.10 # Average write queue length when enqueuing ((Count/Tick))
board.memory.mem_ctrl.numRdRetry 0 # Number of times read queue was full causing retry (Count)
board.memory.mem_ctrl.numWrRetry 0 # Number of times write queue was full causing retry (Count)
board.memory.mem_ctrl.readPktSize::0 134 # Read request sizes (log2) (Count)
board.memory.mem_ctrl.readPktSize::1 19 # Read request sizes (log2) (Count)
board.memory.mem_ctrl.readPktSize::2 311 # Read request sizes (log2) (Count)
board.memory.mem_ctrl.readPktSize::3 1304 # Read request sizes (log2) (Count)
board.memory.mem_ctrl.readPktSize::4 0 # Read request sizes (log2) (Count)
board.memory.mem_ctrl.readPktSize::5 0 # Read request sizes (log2) (Count)
board.memory.mem_ctrl.readPktSize::6 1923 # Read request sizes (log2) (Count)
board.memory.mem_ctrl.writePktSize::0 34 # Write request sizes (log2) (Count)
board.memory.mem_ctrl.writePktSize::1 6 # Write request sizes (log2) (Count)
board.memory.mem_ctrl.writePktSize::2 137 # Write request sizes (log2) (Count)
board.memory.mem_ctrl.writePktSize::3 1174 # Write request sizes (log2) (Count)
board.memory.mem_ctrl.writePktSize::4 0 # Write request sizes (log2) (Count)
board.memory.mem_ctrl.writePktSize::5 0 # Write request sizes (log2) (Count)
board.memory.mem_ctrl.writePktSize::6 0 # Write request sizes (log2) (Count)
board.memory.mem_ctrl.rdQLenPdf::0 1318 # What read queue length does an incoming req see (Count)
board.memory.mem_ctrl.rdQLenPdf::1 910 # What read queue length does an incoming req see (Count)
board.memory.mem_ctrl.rdQLenPdf::2 589 # What read queue length does an incoming req see (Count)
board.memory.mem_ctrl.rdQLenPdf::3 309 # What read queue length does an incoming req see (Count)
board.memory.mem_ctrl.rdQLenPdf::4 145 # What read queue length does an incoming req see (Count)
board.memory.mem_ctrl.rdQLenPdf::5 78 # What read queue length does an incoming req see (Count)
board.memory.mem_ctrl.rdQLenPdf::6 45 # What read queue length does an incoming req see (Count)
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board.memory.mem_ctrl.rdQLenPdf::7	29	# What read queue length does an incoming req see (Count)
board.memory.mem_ctrl.rdQLenPdf::8	16	# What read queue length does an incoming req see (Count)
board.memory.mem_ctrl.rdQLenPdf::9	11	# What read queue length does an incoming req see (Count)
board.memory.mem_ctrl.rdQLenPdf::10	4	# What read queue length does an incoming req see (Count)
board.memory.mem_ctrl.rdQLenPdf::11	4	# What read queue length does an incoming req see (Count)
board.memory.mem_ctrl.rdQLenPdf::12	4	# What read queue length does an incoming req see (Count)
board.memory.mem_ctrl.rdQLenPdf::13	4	# What read queue length does an incoming req see (Count)
board.memory.mem_ctrl.rdQLenPdf::14	3	# What read queue length does an incoming req see (Count)
board.memory.mem_ctrl.rdQLenPdf::15	0	# What read queue length does an incoming req see (Count)
board.memory.mem_ctrl.rdQLenPdf::16	0	# What read queue length does an incoming req see (Count)
board.memory.mem_ctrl.rdQLenPdf::17	0	# What read queue length does an incoming req see (Count)
board.memory.mem_ctrl.rdQLenPdf::18	0	# What read queue length does an incoming req see (Count)
board.memory.mem_ctrl.rdQLenPdf::19	0	# What read queue length does an incoming req see (Count)
board.memory.mem_ctrl.rdQLenPdf::20	0	# What read queue length does an incoming req see (Count)
board.memory.mem_ctrl.rdQLenPdf::21	0	# What read queue length does an incoming req see (Count)
board.memory.mem_ctrl.rdQLenPdf::22	0	# What read queue length does an incoming req see (Count)
board.memory.mem_ctrl.rdQLenPdf::23	0	# What read queue length does an incoming req see (Count)
board.memory.mem_ctrl.rdQLenPdf::24	0	# What read queue length does an incoming req see (Count)
board.memory.mem_ctrl.rdQLenPdf::25	0	# What read queue length does an incoming req see (Count)
board.memory.mem_ctrl.rdQLenPdf::26	0	# What read queue length does an incoming req see (Count)
board.memory.mem_ctrl.rdQLenPdf::27	0	# What read queue length does an incoming req see (Count)
board.memory.mem_ctrl.rdQLenPdf::28	0	# What read queue length does an incoming req see (Count)
board.memory.mem_ctrl.rdQLenPdf::29	0	# What read queue length does an incoming req see (Count)
board.memory.mem_ctrl.rdQLenPdf::30	0	# What read queue length does an incoming req see (Count)
board.memory.mem_ctrl.rdQLenPdf::31	0	# What read queue length does an incoming req see (Count)
board.memory.mem_ctrl.wrQLenPdf::0	1	# What write queue length does an incoming req see (Count)
board.memory.mem_ctrl.wrQLenPdf::1	1	# What write queue length does an incoming req see (Count)
board.memory.mem_ctrl.wrQLenPdf::2	1	# What write queue length does an incoming req see (Count)
board.memory.mem_ctrl.wrQLenPdf::3	1	# What write queue length does an incoming req see (Count)
board.memory.mem_ctrl.wrQLenPdf::4	1	# What write queue length does an incoming req see (Count)
board.memory.mem_ctrl.wrQLenPdf::5	1	# What write queue length does an incoming req see (Count)
board.memory.mem_ctrl.wrQLenPdf::6	1	# What write queue length does an incoming req see (Count)
board.memory.mem_ctrl.wrQLenPdf::7	1	# What write queue length does an incoming req see (Count)
board.memory.mem_ctrl.wrQLenPdf::8	1	# What write queue length does an incoming req see (Count)
board.memory.mem_ctrl.wrQLenPdf::9	1	# What write queue length does an incoming req see (Count)
board.memory.mem_ctrl.wrQLenPdf::10	1	# What write queue length does an incoming req see (Count)
board.memory.mem_ctrl.wrQLenPdf::11	1	# What write queue length does an incoming req see (Count)
board.memory.mem_ctrl.wrQLenPdf::12	1	# What write queue length does an incoming req see (Count)
board.memory.mem_ctrl.wrQLenPdf::13	1	# What write queue length does an incoming req see (Count)
board.memory.mem_ctrl.wrQLenPdf::14	1	# What write queue length does an incoming req see (Count)
board.memory.mem_ctrl.wrQLenPdf::15	2	# What write queue length does an incoming req see (Count)
board.memory.mem_ctrl.wrQLenPdf::16	2	# What write queue length does an incoming req see (Count)
board.memory.mem_ctrl.wrQLenPdf::17	11	# What write queue length does an incoming req see (Count)
board.memory.mem_ctrl.wrQLenPdf::18	11	# What write queue length does an incoming req see (Count)
board.memory.mem_ctrl.wrQLenPdf::19	11	# What write queue length does an incoming req see (Count)
board.memory.mem_ctrl.wrQLenPdf::20	11	# What write queue length does an incoming req see (Count)
board.memory.mem_ctrl.wrQLenPdf::21	11	# What write queue length does an incoming req see (Count)
board.memory.mem_ctrl.wrQLenPdf::22	11	# What write queue length does an incoming req see (Count)
board.memory.mem_ctrl.wrQLenPdf::23	11	# What write queue length does an incoming req see (Count)

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board.memory.mem_ctrl.wrQLenPdf::24          10                    # What write queue length does an incoming req see (Count)
board.memory.mem_ctrl.wrQLenPdf::25          10                    # What write queue length does an incoming req see (Count)
board.memory.mem_ctrl.wrQLenPdf::26          10                    # What write queue length does an incoming req see (Count)
board.memory.mem_ctrl.wrQLenPdf::27          10                    # What write queue length does an incoming req see (Count)
board.memory.mem_ctrl.wrQLenPdf::28          10                    # What write queue length does an incoming req see (Count)
board.memory.mem_ctrl.wrQLenPdf::29          10                    # What write queue length does an incoming req see (Count)
board.memory.mem_ctrl.wrQLenPdf::30          10                    # What write queue length does an incoming req see (Count)
board.memory.mem_ctrl.wrQLenPdf::31          10                    # What write queue length does an incoming req see (Count)
board.memory.mem_ctrl.wrQLenPdf::32          10                    # What write queue length does an incoming req see (Count)
board.memory.mem_ctrl.wrQLenPdf::33           0                    # What write queue length does an incoming req see (Count)
board.memory.mem_ctrl.wrQLenPdf::34           0                    # What write queue length does an incoming req see (Count)
board.memory.mem_ctrl.wrQLenPdf::35           0                    # What write queue length does an incoming req see (Count)
board.memory.mem_ctrl.wrQLenPdf::36           0                    # What write queue length does an incoming req see (Count)
board.memory.mem_ctrl.wrQLenPdf::37           0                    # What write queue length does an incoming req see (Count)
board.memory.mem_ctrl.wrQLenPdf::38           0                    # What write queue length does an incoming req see (Count)
board.memory.mem_ctrl.wrQLenPdf::39           0                    # What write queue length does an incoming req see (Count)
board.memory.mem_ctrl.wrQLenPdf::40           0                    # What write queue length does an incoming req see (Count)
board.memory.mem_ctrl.wrQLenPdf::41           0                    # What write queue length does an incoming req see (Count)
board.memory.mem_ctrl.wrQLenPdf::42           0                    # What write queue length does an incoming req see (Count)
board.memory.mem_ctrl.wrQLenPdf::43           0                    # What write queue length does an incoming req see (Count)
board.memory.mem_ctrl.wrQLenPdf::44           0                    # What write queue length does an incoming req see (Count)
board.memory.mem_ctrl.wrQLenPdf::45           0                    # What write queue length does an incoming req see (Count)
board.memory.mem_ctrl.wrQLenPdf::46           0                    # What write queue length does an incoming req see (Count)
board.memory.mem_ctrl.wrQLenPdf::47           0                    # What write queue length does an incoming req see (Count)
board.memory.mem_ctrl.wrQLenPdf::48           0                    # What write queue length does an incoming req see (Count)
board.memory.mem_ctrl.wrQLenPdf::49           0                    # What write queue length does an incoming req see (Count)
board.memory.mem_ctrl.wrQLenPdf::50           0                    # What write queue length does an incoming req see (Count)
board.memory.mem_ctrl.wrQLenPdf::51           0                    # What write queue length does an incoming req see (Count)
board.memory.mem_ctrl.wrQLenPdf::52           0                    # What write queue length does an incoming req see (Count)
board.memory.mem_ctrl.wrQLenPdf::53           0                    # What write queue length does an incoming req see (Count)
board.memory.mem_ctrl.wrQLenPdf::54           0                    # What write queue length does an incoming req see (Count)
board.memory.mem_ctrl.wrQLenPdf::55           0                    # What write queue length does an incoming req see (Count)
board.memory.mem_ctrl.wrQLenPdf::56           0                    # What write queue length does an incoming req see (Count)
board.memory.mem_ctrl.wrQLenPdf::57           0                    # What write queue length does an incoming req see (Count)
board.memory.mem_ctrl.wrQLenPdf::58           0                    # What write queue length does an incoming req see (Count)
board.memory.mem_ctrl.wrQLenPdf::59           0                    # What write queue length does an incoming req see (Count)
board.memory.mem_ctrl.wrQLenPdf::60           0                    # What write queue length does an incoming req see (Count)
board.memory.mem_ctrl.wrQLenPdf::61           0                    # What write queue length does an incoming req see (Count)
board.memory.mem_ctrl.wrQLenPdf::62           0                    # What write queue length does an incoming req see (Count)
board.memory.mem_ctrl.wrQLenPdf::63           0                    # What write queue length does an incoming req see (Count)
board.memory.mem_ctrl.rdPerTurnAround::samples          10                    # Reads before turning the bus around for writes (Count)
board.memory.mem_ctrl.rdPerTurnAround::mean   325.400000                    # Reads before turning the bus around for writes (Count)
board.memory.mem_ctrl.rdPerTurnAround::gmean   249.873762                    # Reads before turning the bus around for writes (Count)
board.memory.mem_ctrl.rdPerTurnAround::stdev   259.243858                    # Reads before turning the bus around for writes (Count)
board.memory.mem_ctrl.rdPerTurnAround::96-127           1     10.00%     10.00% # Reads before turning the bus around for writes (Count)
board.memory.mem_ctrl.rdPerTurnAround::128-159          4     40.00%     50.00% # Reads before turning the bus around for writes (Count)
board.memory.mem_ctrl.rdPerTurnAround::256-287          2     20.00%     70.00% # Reads before turning the bus around for writes (Count)
board.memory.mem_ctrl.rdPerTurnAround::576-607          1     10.00%     80.00% # Reads before turning the bus around for writes (Count)
board.memory.mem_ctrl.rdPerTurnAround::672-703          1     10.00%     90.00% # Reads before turning the bus around for writes (Count)
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board.memory.mem_ctrl.rdPerTurnAround::800-831          1     10.00%    100.00% # Reads before turning the bus around for writes (Count)
board.memory.mem_ctrl.rdPerTurnAround::total          10                    # Reads before turning the bus around for writes (Count)
board.memory.mem_ctrl.wrPerTurnAround::samples          10                    # Writes before turning the bus around for reads (Count)
board.memory.mem_ctrl.wrPerTurnAround::mean    16.200000                    # Writes before turning the bus around for reads (Count)
board.memory.mem_ctrl.wrPerTurnAround::gmean    16.189567                    # Writes before turning the bus around for reads (Count)
board.memory.mem_ctrl.wrPerTurnAround::stdev     0.632456                    # Writes before turning the bus around for reads (Count)
board.memory.mem_ctrl.wrPerTurnAround::16          9     90.00%     90.00% # Writes before turning the bus around for reads (Count)
board.memory.mem_ctrl.wrPerTurnAround::18          1     10.00%    100.00% # Writes before turning the bus around for reads (Count)
board.memory.mem_ctrl.wrPerTurnAround::total          10                    # Writes before turning the bus around for reads (Count)
board.memory.mem_ctrl.bytesReadWrQ          14208                    # Total number of bytes read from write queue (Byte)
board.memory.mem_ctrl.bytesReadSys          134913                    # Total read bytes from the system interface side (Byte)
board.memory.mem_ctrl.bytesWrittenSys          9986                    # Total written bytes from the system interface side (Byte)
board.memory.mem_ctrl.avgRdBWSys          1710327734.78727341                    # Average system read bandwidth in Byte/s ((Byte/Second))
board.memory.mem_ctrl.avgWrBWSys          126595159.54419301                    # Average system write bandwidth in Byte/s ((Byte/Second))
board.memory.mem_ctrl.totGap          78879042                    # Total gap between requests (Tick)
board.memory.mem_ctrl.avgGap          15644.40                    # Average gap between requests ((Tick/Count))
board.memory.mem_ctrl.requestorReadBytes::processor.cores.core.inst          123072                    # Per-requestor bytes read from memory (Byte)
board.memory.mem_ctrl.requestorReadBytes::processor.cores.core.data          10343                    # Per-requestor bytes read from memory (Byte)
board.memory.mem_ctrl.requestorWriteBytes::processor.cores.core.data          1202                    # Per-requestor bytes write to memory (Byte)
board.memory.mem_ctrl.requestorReadRate::processor.cores.core.inst 1560216250.292701959610                    # Per-requestor bytes read from memory rate ((Byte/Second))
board.memory.mem_ctrl.requestorReadRate::processor.cores.core.data 131120942.836529970169                    # Per-requestor bytes read from memory rate ((Byte/Second))
board.memory.mem_ctrl.requestorWriteRate::processor.cores.core.data 15238071.477280192077                    # Per-requestor bytes write to memory rate ((Byte/Second))
board.memory.mem_ctrl.requestorReadAccesses::processor.cores.core.inst          1923                    # Per-requestor read serviced memory accesses (Count)
board.memory.mem_ctrl.requestorReadAccesses::processor.cores.core.data          1768                    # Per-requestor read serviced memory accesses (Count)
board.memory.mem_ctrl.requestorWriteAccesses::processor.cores.core.data          1351                    # Per-requestor write serviced memory accesses (Count)
board.memory.mem_ctrl.requestorReadTotalLat::processor.cores.core.inst     52461831                    # Per-requestor read total memory access latency (Tick)
board.memory.mem_ctrl.requestorReadTotalLat::processor.cores.core.data     64352691                    # Per-requestor read total memory access latency (Tick)
board.memory.mem_ctrl.requestorWriteTotalLat::processor.cores.core.data   1492746838                    # Per-requestor write total memory access latency (Tick)
board.memory.mem_ctrl.requestorReadAvgLat::processor.cores.core.inst     27281.24                    # Per-requestor read average memory access latency ((Tick/Count))
board.memory.mem_ctrl.requestorReadAvgLat::processor.cores.core.data     36398.58                    # Per-requestor read average memory access latency ((Tick/Count))
board.memory.mem_ctrl.requestorWriteAvgLat::processor.cores.core.data   1104919.94                    # Per-requestor write average memory access latency ((Tick/Count))
board.memory.mem_ctrl.dram.bytesRead::processor.cores.core.inst          122944                    # Number of bytes read from this memory (Byte)
board.memory.mem_ctrl.dram.bytesRead::processor.cores.core.data          11841                    # Number of bytes read from this memory (Byte)
board.memory.mem_ctrl.dram.bytesRead::total          134785                    # Number of bytes read from this memory (Byte)
board.memory.mem_ctrl.dram.bytesInstRead::processor.cores.core.inst          122944                    # Number of instructions bytes read from this memory (Byte)
board.memory.mem_ctrl.dram.bytesInstRead::total          122944                    # Number of instructions bytes read from this memory (Byte)
board.memory.mem_ctrl.dram.bytesWritten::processor.cores.core.data          9986                    # Number of bytes written to this memory (Byte)
board.memory.mem_ctrl.dram.bytesWritten::total          9986                    # Number of bytes written to this memory (Byte)
board.memory.mem_ctrl.dram.numReads::processor.cores.core.inst          1921                    # Number of read requests responded to by this memory (Count)
board.memory.mem_ctrl.dram.numReads::processor.cores.core.data          1768                    # Number of read requests responded to by this memory (Count)
board.memory.mem_ctrl.dram.numReads::total          3689                    # Number of read requests responded to by this memory (Count)
board.memory.mem_ctrl.dram.numWrites::processor.cores.core.data          1351                    # Number of write requests responded to by this memory (Count)
board.memory.mem_ctrl.dram.numWrites::total          1351                    # Number of write requests responded to by this memory (Count)
board.memory.mem_ctrl.dram.bwRead::processor.cores.core.inst   1558593560                    # Total read bandwidth from this memory ((Byte/Second))
board.memory.mem_ctrl.dram.bwRead::processor.cores.core.data    150111484                    # Total read bandwidth from this memory ((Byte/Second))
board.memory.mem_ctrl.dram.bwRead::total   1708705045                    # Total read bandwidth from this memory ((Byte/Second))
board.memory.mem_ctrl.dram.bwInstRead::processor.cores.core.inst   1558593560                    # Instruction read bandwidth from this memory ((Byte/Second))
board.memory.mem_ctrl.dram.bwInstRead::total   1558593560                    # Instruction read bandwidth from this memory ((Byte/Second))
board.memory.mem_ctrl.dram.bwWrite::processor.cores.core.data    126595160                    # Write bandwidth from this memory ((Byte/Second))
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board.memory.mem_ctrl.dram.bwWrite::total    126595160                    # Write bandwidth from this memory ((Byte/Second))
board.memory.mem_ctrl.dram.bwTotal::processor.cores.core.inst   1558593560                    # Total bandwidth to/from this memory ((Byte/Second))
board.memory.mem_ctrl.dram.bwTotal::processor.cores.core.data    276706644                    # Total bandwidth to/from this memory ((Byte/Second))
board.memory.mem_ctrl.dram.bwTotal::total   1835300205                    # Total bandwidth to/from this memory ((Byte/Second))
board.memory.mem_ctrl.dram.readBursts           3469                    # Number of DRAM read bursts (Count)
board.memory.mem_ctrl.dram.writeBursts           162                    # Number of DRAM write bursts (Count)
board.memory.mem_ctrl.dram.perBankRdBursts::0         347                    # Per bank write bursts (Count)
board.memory.mem_ctrl.dram.perBankRdBursts::1           0                    # Per bank write bursts (Count)
board.memory.mem_ctrl.dram.perBankRdBursts::2           1                    # Per bank write bursts (Count)
board.memory.mem_ctrl.dram.perBankRdBursts::3         132                    # Per bank write bursts (Count)
board.memory.mem_ctrl.dram.perBankRdBursts::4         100                    # Per bank write bursts (Count)
board.memory.mem_ctrl.dram.perBankRdBursts::5          53                    # Per bank write bursts (Count)
board.memory.mem_ctrl.dram.perBankRdBursts::6         326                    # Per bank write bursts (Count)
board.memory.mem_ctrl.dram.perBankRdBursts::7         279                    # Per bank write bursts (Count)
board.memory.mem_ctrl.dram.perBankRdBursts::8         239                    # Per bank write bursts (Count)
board.memory.mem_ctrl.dram.perBankRdBursts::9         430                    # Per bank write bursts (Count)
board.memory.mem_ctrl.dram.perBankRdBursts::10         478                    # Per bank write bursts (Count)
board.memory.mem_ctrl.dram.perBankRdBursts::11         703                    # Per bank write bursts (Count)
board.memory.mem_ctrl.dram.perBankRdBursts::12          48                    # Per bank write bursts (Count)
board.memory.mem_ctrl.dram.perBankRdBursts::13         237                    # Per bank write bursts (Count)
board.memory.mem_ctrl.dram.perBankRdBursts::14          34                    # Per bank write bursts (Count)
board.memory.mem_ctrl.dram.perBankRdBursts::15          62                    # Per bank write bursts (Count)
board.memory.mem_ctrl.dram.perBankWrBursts::0           0                    # Per bank write bursts (Count)
board.memory.mem_ctrl.dram.perBankWrBursts::1           0                    # Per bank write bursts (Count)
board.memory.mem_ctrl.dram.perBankWrBursts::2           0                    # Per bank write bursts (Count)
board.memory.mem_ctrl.dram.perBankWrBursts::3           0                    # Per bank write bursts (Count)
board.memory.mem_ctrl.dram.perBankWrBursts::4           0                    # Per bank write bursts (Count)
board.memory.mem_ctrl.dram.perBankWrBursts::5           0                    # Per bank write bursts (Count)
board.memory.mem_ctrl.dram.perBankWrBursts::6           0                    # Per bank write bursts (Count)
board.memory.mem_ctrl.dram.perBankWrBursts::7           0                    # Per bank write bursts (Count)
board.memory.mem_ctrl.dram.perBankWrBursts::8           0                    # Per bank write bursts (Count)
board.memory.mem_ctrl.dram.perBankWrBursts::9          51                    # Per bank write bursts (Count)
board.memory.mem_ctrl.dram.perBankWrBursts::10          30                    # Per bank write bursts (Count)
board.memory.mem_ctrl.dram.perBankWrBursts::11          41                    # Per bank write bursts (Count)
board.memory.mem_ctrl.dram.perBankWrBursts::12           5                    # Per bank write bursts (Count)
board.memory.mem_ctrl.dram.perBankWrBursts::13          35                    # Per bank write bursts (Count)
board.memory.mem_ctrl.dram.perBankWrBursts::14           0                    # Per bank write bursts (Count)
board.memory.mem_ctrl.dram.perBankWrBursts::15           0                    # Per bank write bursts (Count)
board.memory.mem_ctrl.dram.totQLat          51770772                    # Total ticks spent queuing (Tick)
board.memory.mem_ctrl.dram.totBusLat         17345000                    # Total ticks spent in databus transfers (Tick)
board.memory.mem_ctrl.dram.totMemAccLat     116814522                    # Total ticks spent from burst creation until serviced by the DRAM (Tick)
board.memory.mem_ctrl.dram.avgQLat          14923.83                    # Average queueing delay per DRAM burst ((Tick/Count))
board.memory.mem_ctrl.dram.avgBusLat          5000.00                    # Average bus latency per DRAM burst ((Tick/Count))
board.memory.mem_ctrl.dram.avgMemAccLat      33673.83                    # Average memory access latency per DRAM burst ((Tick/Count))
board.memory.mem_ctrl.dram.readRowHits          2992                    # Number of row buffer hits during reads (Count)
board.memory.mem_ctrl.dram.writeRowHits          157                    # Number of row buffer hits during writes (Count)
board.memory.mem_ctrl.dram.readRowHitRate        86.25                    # Row buffer hit rate for reads (Ratio)
board.memory.mem_ctrl.dram.writeRowHitRate        96.91                    # Row buffer hit rate for writes (Ratio)
board.memory.mem_ctrl.dram.bytesPerActivate::samples          478                    # Bytes accessed per row activation (Byte)
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board.memory.mem_ctrl.dram.bytesPerActivate::mean   482.543933                    # Bytes accessed per row activation (Byte)
board.memory.mem_ctrl.dram.bytesPerActivate::gmean   296.481343                    # Bytes accessed per row activation (Byte)
board.memory.mem_ctrl.dram.bytesPerActivate::stdev   397.371602                    # Bytes accessed per row activation (Byte)
board.memory.mem_ctrl.dram.bytesPerActivate::0-127          109     22.80%     22.80% # Bytes accessed per row activation (Byte)
board.memory.mem_ctrl.dram.bytesPerActivate::128-255          92     19.25%     42.05% # Bytes accessed per row activation (Byte)
board.memory.mem_ctrl.dram.bytesPerActivate::256-383          44      9.21%     51.26% # Bytes accessed per row activation (Byte)
board.memory.mem_ctrl.dram.bytesPerActivate::384-511          35      7.32%     58.58% # Bytes accessed per row activation (Byte)
board.memory.mem_ctrl.dram.bytesPerActivate::512-639          18      3.77%     62.34% # Bytes accessed per row activation (Byte)
board.memory.mem_ctrl.dram.bytesPerActivate::640-767          19      3.97%     66.32% # Bytes accessed per row activation (Byte)
board.memory.mem_ctrl.dram.bytesPerActivate::768-895          14      2.93%     69.25% # Bytes accessed per row activation (Byte)
board.memory.mem_ctrl.dram.bytesPerActivate::896-1023          11      2.30%     71.55% # Bytes accessed per row activation (Byte)
board.memory.mem_ctrl.dram.bytesPerActivate::1024-1151         136     28.45%    100.00% # Bytes accessed per row activation (Byte)
board.memory.mem_ctrl.dram.bytesPerActivate::total          478                    # Bytes accessed per row activation (Byte)
board.memory.mem_ctrl.dram.dramBytesRead       222016                    # Total bytes read (Byte)
board.memory.mem_ctrl.dram.dramBytesWritten        10368                    # Total bytes written (Byte)
board.memory.mem_ctrl.dram.avgRdBW       2814.555472                    # Average DRAM read bandwidth in MiBytes/s ((Byte/Second))
board.memory.mem_ctrl.dram.avgWrBW         131.437874                    # Average DRAM write bandwidth in MiBytes/s ((Byte/Second))
board.memory.mem_ctrl.dram.peakBW           12800.00                    # Theoretical peak bandwidth in MiByte/s ((Byte/Second))
board.memory.mem_ctrl.dram.busUtil             23.02                    # Data bus utilization in percentage (Ratio)
board.memory.mem_ctrl.dram.busUtilRead         21.99                    # Data bus utilization in percentage for reads (Ratio)
board.memory.mem_ctrl.dram.busUtilWrite         1.03                    # Data bus utilization in percentage for writes (Ratio)
board.memory.mem_ctrl.dram.pageHitRate         86.73                    # Row buffer hit rate, read and write combined (Ratio)
board.memory.mem_ctrl.dram.power_state.pwrStateResidencyTicks::UNDEFINED     78881373                    # Cumulative time (in ticks) in various power states (Tick)
board.memory.mem_ctrl.dram.rank0.actEnergy       871080                    # Energy for activate commands per rank (pJ) (Joule)
board.memory.mem_ctrl.dram.rank0.preEnergy       455400                    # Energy for precharge commands per rank (pJ) (Joule)
board.memory.mem_ctrl.dram.rank0.readEnergy      8839320                    # Energy for read commands per rank (pJ) (Joule)
board.memory.mem_ctrl.dram.rank0.writeEnergy           0                    # Energy for write commands per rank (pJ) (Joule)
board.memory.mem_ctrl.dram.rank0.refreshEnergy 6146400.000000                    # Energy for refresh commands per rank (pJ) (Joule)
board.memory.mem_ctrl.dram.rank0.actBackEnergy     33808410                    # Energy for active background per rank (pJ) (Joule)
board.memory.mem_ctrl.dram.rank0.preBackEnergy      1820640                    # Energy for precharge background per rank (pJ) (Joule)
board.memory.mem_ctrl.dram.rank0.actPowerDownEnergy           0                    # Energy for active power-down per rank (pJ) (Joule)
board.memory.mem_ctrl.dram.rank0.prePowerDownEnergy           0                    # Energy for precharge power-down per rank (pJ) (Joule)
board.memory.mem_ctrl.dram.rank0.selfRefreshEnergy           0                    # Energy for self refresh per rank (pJ) (Joule)
board.memory.mem_ctrl.dram.rank0.totalEnergy     51941250                    # Total energy per rank (pJ) (Joule)
board.memory.mem_ctrl.dram.rank0.averagePower   658.472945                    # Core power per rank (mW) (Watt)
board.memory.mem_ctrl.dram.rank0.totalIdleTime           0                    # Total Idle time Per DRAM Rank (Tick)
board.memory.mem_ctrl.dram.rank0.pwrStateTime::IDLE      4148069                    # Time in different power states (Tick)
board.memory.mem_ctrl.dram.rank0.pwrStateTime::REF      2600000                    # Time in different power states (Tick)
board.memory.mem_ctrl.dram.rank0.pwrStateTime::SREF           0                    # Time in different power states (Tick)
board.memory.mem_ctrl.dram.rank0.pwrStateTime::PRE_PDN           0                    # Time in different power states (Tick)
board.memory.mem_ctrl.dram.rank0.pwrStateTime::ACT     72133304                    # Time in different power states (Tick)
board.memory.mem_ctrl.dram.rank0.pwrStateTime::ACT_PDN           0                    # Time in different power states (Tick)
board.memory.mem_ctrl.dram.rank1.actEnergy      2570400                    # Energy for activate commands per rank (pJ) (Joule)
board.memory.mem_ctrl.dram.rank1.preEnergy      1358610                    # Energy for precharge commands per rank (pJ) (Joule)
board.memory.mem_ctrl.dram.rank1.readEnergy     15922200                    # Energy for read commands per rank (pJ) (Joule)
board.memory.mem_ctrl.dram.rank1.writeEnergy       845640                    # Energy for write commands per rank (pJ) (Joule)
board.memory.mem_ctrl.dram.rank1.refreshEnergy 6146400.000000                    # Energy for refresh commands per rank (pJ) (Joule)
board.memory.mem_ctrl.dram.rank1.actBackEnergy     34591020                    # Energy for active background per rank (pJ) (Joule)
board.memory.mem_ctrl.dram.rank1.preBackEnergy      1161600                    # Energy for precharge background per rank (pJ) (Joule)
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board.memory.mem_ctrl.dram.rank1.actPowerDownEnergy 0 # Energy for active power-down per rank (pJ) (Joule)
board.memory.mem_ctrl.dram.rank1.prePowerDownEnergy 0 # Energy for precharge power-down per rank (pJ) (Joule)
board.memory.mem_ctrl.dram.rank1.selfRefreshEnergy 0 # Energy for self refresh per rank (pJ) (Joule)
board.memory.mem_ctrl.dram.rank1.totalEnergy 62595870 # Total energy per rank (pJ) (Joule)
board.memory.mem_ctrl.dram.rank1.averagePower 793.544377 # Core power per rank (mW) (Watt)
board.memory.mem_ctrl.dram.rank1.totalIdleTime 0 # Total Idle time Per DRAM Rank (Tick)
board.memory.mem_ctrl.dram.rank1.pwrStateTime::IDLE 2720323 # Time in different power states (Tick)
board.memory.mem_ctrl.dram.rank1.pwrStateTime::REF 2600000 # Time in different power states (Tick)
board.memory.mem_ctrl.dram.rank1.pwrStateTime::SREF 0 # Time in different power states (Tick)
board.memory.mem_ctrl.dram.rank1.pwrStateTime::PRE_PDN 0 # Time in different power states (Tick)
board.memory.mem_ctrl.dram.rank1.pwrStateTime::ACT 73561050 # Time in different power states (Tick)
board.memory.mem_ctrl.dram.rank1.pwrStateTime::ACT_PDN 0 # Time in different power states (Tick)
board.memory.mem_ctrl.power_state.pwrStateResidencyTicks::UNDEFINED 78881373 # Cumulative time (in ticks) in various power states (Tick)
board.processor.cores.core.numCycles 236882 # Number of cpu cycles simulated (Cycle)
board.processor.cores.core.cpi 36.137605 # CPI: cycles per instruction (core level) ((Cycle/Count))
board.processor.cores.core.ipc 0.027672 # IPC: instructions per cycle (core level) ((Count/Cycle))
board.processor.cores.core.numWorkItemsStarted 0 # Number of work items this cpu started (Count)
board.processor.cores.core.numWorkItemsCompleted 0 # Number of work items this cpu completed (Count)
board.processor.cores.core.instsAdded 20111 # Number of instructions added to the IQ (excludes non-spec) (Count)
board.processor.cores.core.nonSpecInstsAdded 68 # Number of non-speculative instructions added to the IQ (Count)
board.processor.cores.core.instsIssued 18906 # Number of instructions issued (Count)
board.processor.cores.core.squashedInstsIssued 8 # Number of squashed instructions issued (Count)
board.processor.cores.core.squashedInstsExamined 7181 # Number of squashed instructions iterated over during squash; mainly for profiling (Count)
board.processor.cores.core.squashedOperandsExamined 10932 # Number of squashed operands that are examined and possibly removed from graph (Count)
board.processor.cores.core.squashedNonSpecRemoved 68 # Number of squashed non-spec instructions that were removed (Count)
board.processor.cores.core.numIssuedDist::samples 85403 # Number of insts issued each cycle (Count)
board.processor.cores.core.numIssuedDist::mean 0.221374 # Number of insts issued each cycle (Count)
board.processor.cores.core.numIssuedDist::stdev 0.882162 # Number of insts issued each cycle (Count)
board.processor.cores.core.numIssuedDist::underflows 0 0.00% 0.00% # Number of insts issued each cycle (Count)
board.processor.cores.core.numIssuedDist::0 77786 91.08% 91.08% # Number of insts issued each cycle (Count)
board.processor.cores.core.numIssuedDist::1 3244 3.80% 94.88% # Number of insts issued each cycle (Count)
board.processor.cores.core.numIssuedDist::2 1599 1.87% 96.75% # Number of insts issued each cycle (Count)
board.processor.cores.core.numIssuedDist::3 798 0.93% 97.69% # Number of insts issued each cycle (Count)
board.processor.cores.core.numIssuedDist::4 777 0.91% 98.60% # Number of insts issued each cycle (Count)
board.processor.cores.core.numIssuedDist::5 560 0.66% 99.25% # Number of insts issued each cycle (Count)
board.processor.cores.core.numIssuedDist::6 373 0.44% 99.69% # Number of insts issued each cycle (Count)
board.processor.cores.core.numIssuedDist::7 204 0.24% 99.93% # Number of insts issued each cycle (Count)
board.processor.cores.core.numIssuedDist::8 62 0.07% 100.00% # Number of insts issued each cycle (Count)
board.processor.cores.core.numIssuedDist::overflows 0 0.00% 100.00% # Number of insts issued each cycle (Count)
board.processor.cores.core.numIssuedDist::min_value 0 # Number of insts issued each cycle (Count)
board.processor.cores.core.numIssuedDist::max_value 8 # Number of insts issued each cycle (Count)
board.processor.cores.core.numIssuedDist::total 85403 # Number of insts issued each cycle (Count)
board.processor.cores.core.statFuBusy::No_OpClass 0 0.00% 0.00% # attempts to use FU when none available (Count)
board.processor.cores.core.statFuBusy::IntAlu 116 81.12% 81.12% # attempts to use FU when none available (Count)
board.processor.cores.core.statFuBusy::IntMult 0 0.00% 81.12% # attempts to use FU when none available (Count)
board.processor.cores.core.statFuBusy::IntDiv 0 0.00% 81.12% # attempts to use FU when none available (Count)
board.processor.cores.core.statFuBusy::FloatAdd 0 0.00% 81.12% # attempts to use FU when none available (Count)
board.processor.cores.core.statFuBusy::FloatCmp 0 0.00% 81.12% # attempts to use FU when none available (Count)
board.processor.cores.core.statFuBusy::FloatCvt 0 0.00% 81.12% # attempts to use FU when none available (Count)
```

board.processor.cores.core.statFuBusy::FloatMult 0 0.00% 81.12% # attempts to use FU when none available (Count)

board.processor.cores.core.statFuBusy::FloatMultAcc 0 0.00% 81.12% # attempts to use FU when none available (Count)

board.processor.cores.core.statFuBusy::FloatDiv 0 0.00% 81.12% # attempts to use FU when none available (Count)

board.processor.cores.core.statFuBusy::FloatMisc 0 0.00% 81.12% # attempts to use FU when none available (Count)

board.processor.cores.core.statFuBusy::FloatSqrt 0 0.00% 81.12% # attempts to use FU when none available (Count)

board.processor.cores.core.statFuBusy::SimdAdd 0 0.00% 81.12% # attempts to use FU when none available (Count)

board.processor.cores.core.statFuBusy::SimdAddAcc 0 0.00% 81.12% # attempts to use FU when none available (Count)

board.processor.cores.core.statFuBusy::SimdAlu 10 6.99% 88.11% # attempts to use FU when none available (Count)

board.processor.cores.core.statFuBusy::SimdCmp 0 0.00% 88.11% # attempts to use FU when none available (Count)

board.processor.cores.core.statFuBusy::SimdCvt 0 0.00% 88.11% # attempts to use FU when none available (Count)

board.processor.cores.core.statFuBusy::SimdMisc 0 0.00% 88.11% # attempts to use FU when none available (Count)

board.processor.cores.core.statFuBusy::SimdMult 0 0.00% 88.11% # attempts to use FU when none available (Count)

board.processor.cores.core.statFuBusy::SimdMultAcc 0 0.00% 88.11% # attempts to use FU when none available (Count)

board.processor.cores.core.statFuBusy::SimdMatMultAcc 0 0.00% 88.11% # attempts to use FU when none available (Count)

board.processor.cores.core.statFuBusy::SimdShift 0 0.00% 88.11% # attempts to use FU when none available (Count)

board.processor.cores.core.statFuBusy::SimdShiftAcc 0 0.00% 88.11% # attempts to use FU when none available (Count)

board.processor.cores.core.statFuBusy::SimdDiv 0 0.00% 88.11% # attempts to use FU when none available (Count)

board.processor.cores.core.statFuBusy::SimdSqrt 0 0.00% 88.11% # attempts to use FU when none available (Count)

board.processor.cores.core.statFuBusy::SimdFloatAdd 0 0.00% 88.11% # attempts to use FU when none available (Count)

board.processor.cores.core.statFuBusy::SimdFloatAlu 0 0.00% 88.11% # attempts to use FU when none available (Count)

board.processor.cores.core.statFuBusy::SimdFloatCmp 0 0.00% 88.11% # attempts to use FU when none available (Count)

board.processor.cores.core.statFuBusy::SimdFloatCvt 0 0.00% 88.11% # attempts to use FU when none available (Count)

board.processor.cores.core.statFuBusy::SimdFloatDiv 0 0.00% 88.11% # attempts to use FU when none available (Count)

board.processor.cores.core.statFuBusy::SimdFloatMisc 0 0.00% 88.11% # attempts to use FU when none available (Count)

board.processor.cores.core.statFuBusy::SimdFloatMult 0 0.00% 88.11% # attempts to use FU when none available (Count)

board.processor.cores.core.statFuBusy::SimdFloatMultAcc 0 0.00% 88.11% # attempts to use FU when none available (Count)

board.processor.cores.core.statFuBusy::SimdFloatMatMultAcc 0 0.00% 88.11% # attempts to use FU when none available (Count)

board.processor.cores.core.statFuBusy::SimdFloatSqrt 0 0.00% 88.11% # attempts to use FU when none available (Count)

board.processor.cores.core.statFuBusy::SimdReduceAdd 0 0.00% 88.11% # attempts to use FU when none available (Count)

board.processor.cores.core.statFuBusy::SimdReduceAlu 0 0.00% 88.11% # attempts to use FU when none available (Count)

board.processor.cores.core.statFuBusy::SimdReduceCmp 0 0.00% 88.11% # attempts to use FU when none available (Count)

board.processor.cores.core.statFuBusy::SimdFloatReduceAdd 0 0.00% 88.11% # attempts to use FU when none available (Count)

board.processor.cores.core.statFuBusy::SimdFloatReduceCmp 0 0.00% 88.11% # attempts to use FU when none available (Count)

board.processor.cores.core.statFuBusy::SimdAes 0 0.00% 88.11% # attempts to use FU when none available (Count)

board.processor.cores.core.statFuBusy::SimdAesMix 0 0.00% 88.11% # attempts to use FU when none available (Count)

board.processor.cores.core.statFuBusy::SimdSha1Hash 0 0.00% 88.11% # attempts to use FU when none available (Count)

board.processor.cores.core.statFuBusy::SimdSha1Hash2 0 0.00% 88.11% # attempts to use FU when none available (Count)

board.processor.cores.core.statFuBusy::SimdSha256Hash 0 0.00% 88.11% # attempts to use FU when none available (Count)

board.processor.cores.core.statFuBusy::SimdSha256Hash2 0 0.00% 88.11% # attempts to use FU when none available (Count)

board.processor.cores.core.statFuBusy::SimdShaSigma2 0 0.00% 88.11% # attempts to use FU when none available (Count)

board.processor.cores.core.statFuBusy::SimdShaSigma3 0 0.00% 88.11% # attempts to use FU when none available (Count)

board.processor.cores.core.statFuBusy::SimdPredAlu 0 0.00% 88.11% # attempts to use FU when none available (Count)

board.processor.cores.core.statFuBusy::Matrix 0 0.00% 88.11% # attempts to use FU when none available (Count)

board.processor.cores.core.statFuBusy::MatrixMov 0 0.00% 88.11% # attempts to use FU when none available (Count)

board.processor.cores.core.statFuBusy::MatrixOP 0 0.00% 88.11% # attempts to use FU when none available (Count)

board.processor.cores.core.statFuBusy::MemRead 4 2.80% 90.91% # attempts to use FU when none available (Count)

board.processor.cores.core.statFuBusy::MemWrite 5 3.50% 94.41% # attempts to use FU when none available (Count)

board.processor.cores.core.statFuBusy::FloatMemRead 3 2.10% 96.50% # attempts to use FU when none available (Count)

board.processor.cores.core.statFuBusy::FloatMemWrite 5 3.50% 100.00% # attempts to use FU when none available (Count)

board.processor.cores.core.statFuBusy::IprAccess 0 0.00% 100.00% # attempts to use FU when none available (Count)

board.processor.cores.core.statFuBusy::InstPrefetch 0 0.00% 100.00% # attempts to use FU when none available (Count)

board.processor.cores.core.statFuBusy::SimdUnitStrideLoad 0 0.00% 100.00% # attempts to use FU when none available (Count)

board.processor.cores.core.statFuBusy::SimdUnitStrideStore 0 0.00% 100.00% # attempts to use FU when none available (Count)

board.processor.cores.core.statFuBusy::SimdUnitStrideMaskLoad 0 0.00% 100.00% # attempts to use FU when none available (Count)

board.processor.cores.core.statFuBusy::SimdUnitStrideMaskStore 0 0.00% 100.00% # attempts to use FU when none available (Count)

board.processor.cores.core.statFuBusy::SimdStridedLoad 0 0.00% 100.00% # attempts to use FU when none available (Count)

board.processor.cores.core.statFuBusy::SimdStridedStore 0 0.00% 100.00% # attempts to use FU when none available (Count)

board.processor.cores.core.statFuBusy::SimdIndexedLoad 0 0.00% 100.00% # attempts to use FU when none available (Count)

board.processor.cores.core.statFuBusy::SimdIndexedStore 0 0.00% 100.00% # attempts to use FU when none available (Count)

board.processor.cores.core.statFuBusy::SimdWholeRegisterLoad 0 0.00% 100.00% # attempts to use FU when none available (Count)

board.processor.cores.core.statFuBusy::SimdWholeRegisterStore 0 0.00% 100.00% # attempts to use FU when none available (Count)

board.processor.cores.core.statFuBusy::SimdUnitStrideFaultOnlyFirstLoad 0 0.00% 100.00% # attempts to use FU when none available (Count)

board.processor.cores.core.statFuBusy::SimdUnitStrideSegmentedLoad 0 0.00% 100.00% # attempts to use FU when none available (Count)

board.processor.cores.core.statFuBusy::SimdUnitStrideSegmentedStore 0 0.00% 100.00% # attempts to use FU when none available (Count)

board.processor.cores.core.statFuBusy::SimdExt 0 0.00% 100.00% # attempts to use FU when none available (Count)

board.processor.cores.core.statFuBusy::SimdFloatExt 0 0.00% 100.00% # attempts to use FU when none available (Count)

board.processor.cores.core.statFuBusy::SimdConfig 0 0.00% 100.00% # attempts to use FU when none available (Count)

board.processor.cores.core.statIssuedInstType_0::No_OpClass 131 0.69% 0.69% # Number of instructions issued per FU type, per thread (Count)

board.processor.cores.core.statIssuedInstType_0::IntAlu 14487 76.63% 77.32% # Number of instructions issued per FU type, per thread (Count)

board.processor.cores.core.statIssuedInstType_0::IntMult 4 0.02% 77.34% # Number of instructions issued per FU type, per thread (Count)

board.processor.cores.core.statIssuedInstType_0::IntDiv 23 0.12% 77.46% # Number of instructions issued per FU type, per thread (Count)

board.processor.cores.core.statIssuedInstType_0::FloatAdd 2 0.01% 77.47% # Number of instructions issued per FU type, per thread (Count)

board.processor.cores.core.statIssuedInstType_0::FloatCmp 0 0.00% 77.47% # Number of instructions issued per FU type, per thread (Count)

board.processor.cores.core.statIssuedInstType_0::FloatCvt 0 0.00% 77.47% # Number of instructions issued per FU type, per thread (Count)

board.processor.cores.core.statIssuedInstType_0::FloatMult 0 0.00% 77.47% # Number of instructions issued per FU type, per thread (Count)

board.processor.cores.core.statIssuedInstType_0::FloatMultAcc 0 0.00% 77.47% # Number of instructions issued per FU type, per thread (Count)

board.processor.cores.core.statIssuedInstType_0::FloatDiv 0 0.00% 77.47% # Number of instructions issued per FU type, per thread (Count)

board.processor.cores.core.statIssuedInstType_0::FloatMisc 0 0.00% 77.47% # Number of instructions issued per FU type, per thread (Count)

board.processor.cores.core.statIssuedInstType_0::FloatSqrt 0 0.00% 77.47% # Number of instructions issued per FU type, per thread (Count)

board.processor.cores.core.statIssuedInstType_0::SimdAdd 0 0.00% 77.47% # Number of instructions issued per FU type, per thread (Count)

board.processor.cores.core.statIssuedInstType_0::SimdAddAcc 0 0.00% 77.47% # Number of instructions issued per FU type, per thread (Count)

board.processor.cores.core.statIssuedInstType_0::SimdAlu 82 0.43% 77.91% # Number of instructions issued per FU type, per thread (Count)

board.processor.cores.core.statIssuedInstType_0::SimdCmp 0 0.00% 77.91% # Number of instructions issued per FU type, per thread (Count)

board.processor.cores.core.statIssuedInstType_0::SimdCvt 48 0.25% 78.16% # Number of instructions issued per FU type, per thread (Count)

board.processor.cores.core.statIssuedInstType_0::SimdMisc 62 0.33% 78.49% # Number of instructions issued per FU type, per thread (Count)

board.processor.cores.core.statIssuedInstType_0::SimdMult 0 0.00% 78.49% # Number of instructions issued per FU type, per thread (Count)

board.processor.cores.core.statIssuedInstType_0::SimdMultAcc 0 0.00% 78.49% # Number of instructions issued per FU type, per thread (Count)

board.processor.cores.core.statIssuedInstType_0::SimdMatMultAcc 0 0.00% 78.49% # Number of instructions issued per FU type, per thread (Count)

board.processor.cores.core.statIssuedInstType_0::SimdShift 0 0.00% 78.49% # Number of instructions issued per FU type, per thread (Count)

board.processor.cores.core.statIssuedInstType_0::SimdShiftAcc 0 0.00% 78.49% # Number of instructions issued per FU type, per thread (Count)

board.processor.cores.core.statIssuedInstType_0::SimdDiv 0 0.00% 78.49% # Number of instructions issued per FU type, per thread (Count)

board.processor.cores.core.statIssuedInstType_0::SimdSqrt 0 0.00% 78.49% # Number of instructions issued per FU type, per thread (Count)

board.processor.cores.core.statIssuedInstType_0::SimdFloatAdd 0 0.00% 78.49% # Number of instructions issued per FU type, per thread (Count)

board.processor.cores.core.statIssuedInstType_0::SimdFloatAlu 0 0.00% 78.49% # Number of instructions issued per FU type, per thread (Count)

board.processor.cores.core.statIssuedInstType_0::SimdFloatCmp 0 0.00% 78.49% # Number of instructions issued per FU type, per thread (Count)

board.processor.cores.core.statIssuedInstType_0::SimdFloatCvt 0 0.00% 78.49% # Number of instructions issued per FU type, per thread (Count)

board.processor.cores.core.statIssuedInstType_0::SimdFloatDiv 0 0.00% 78.49% # Number of instructions issued per FU type, per thread (Count)

board.processor.cores.core.statIssuedInstType_0::SimdFloatMisc 0 0.00% 78.49% # Number of instructions issued per FU type, per thread (Count)

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board.processor.cores.core.statIssuedInstType_0::SimdFloatMult 0 0.00% 78.49% # Number of instructions issued per FU type, per thread (Count)
board.processor.cores.core.statIssuedInstType_0::SimdFloatMultAcc 0 0.00% 78.49% # Number of instructions issued per FU type, per thread (Count)
board.processor.cores.core.statIssuedInstType_0::SimdFloatMatMultAcc 0 0.00% 78.49% # Number of instructions issued per FU type, per thread (Count)
board.processor.cores.core.statIssuedInstType_0::SimdFloatSqrt 0 0.00% 78.49% # Number of instructions issued per FU type, per thread (Count)
board.processor.cores.core.statIssuedInstType_0::SimdReduceAdd 0 0.00% 78.49% # Number of instructions issued per FU type, per thread (Count)
board.processor.cores.core.statIssuedInstType_0::SimdReduceAlu 0 0.00% 78.49% # Number of instructions issued per FU type, per thread (Count)
board.processor.cores.core.statIssuedInstType_0::SimdReduceCmp 0 0.00% 78.49% # Number of instructions issued per FU type, per thread (Count)
board.processor.cores.core.statIssuedInstType_0::SimdFloatReduceAdd 0 0.00% 78.49% # Number of instructions issued per FU type, per thread (Count)
board.processor.cores.core.statIssuedInstType_0::SimdFloatReduceCmp 0 0.00% 78.49% # Number of instructions issued per FU type, per thread (Count)
board.processor.cores.core.statIssuedInstType_0::SimdAes 0 0.00% 78.49% # Number of instructions issued per FU type, per thread (Count)
board.processor.cores.core.statIssuedInstType_0::SimdAesMix 0 0.00% 78.49% # Number of instructions issued per FU type, per thread (Count)
board.processor.cores.core.statIssuedInstType_0::SimdSha1Hash 0 0.00% 78.49% # Number of instructions issued per FU type, per thread (Count)
board.processor.cores.core.statIssuedInstType_0::SimdSha1Hash2 0 0.00% 78.49% # Number of instructions issued per FU type, per thread (Count)
board.processor.cores.core.statIssuedInstType_0::SimdSha256Hash 0 0.00% 78.49% # Number of instructions issued per FU type, per thread (Count)
board.processor.cores.core.statIssuedInstType_0::SimdSha256Hash2 0 0.00% 78.49% # Number of instructions issued per FU type, per thread (Count)
board.processor.cores.core.statIssuedInstType_0::SimdShaSigma2 0 0.00% 78.49% # Number of instructions issued per FU type, per thread (Count)
board.processor.cores.core.statIssuedInstType_0::SimdShaSigma3 0 0.00% 78.49% # Number of instructions issued per FU type, per thread (Count)
board.processor.cores.core.statIssuedInstType_0::SimdPredAlu 0 0.00% 78.49% # Number of instructions issued per FU type, per thread (Count)
board.processor.cores.core.statIssuedInstType_0::Matrix 0 0.00% 78.49% # Number of instructions issued per FU type, per thread (Count)
board.processor.cores.core.statIssuedInstType_0::MatrixMov 0 0.00% 78.49% # Number of instructions issued per FU type, per thread (Count)
board.processor.cores.core.statIssuedInstType_0::MatrixOP 0 0.00% 78.49% # Number of instructions issued per FU type, per thread (Count)
board.processor.cores.core.statIssuedInstType_0::MemRead 2182 11.54% 90.03% # Number of instructions issued per FU type, per thread (Count)
board.processor.cores.core.statIssuedInstType_0::MemWrite 1595 8.44% 98.47% # Number of instructions issued per FU type, per thread (Count)
board.processor.cores.core.statIssuedInstType_0::FloatMemRead 122 0.65% 99.11% # Number of instructions issued per FU type, per thread (Count)
board.processor.cores.core.statIssuedInstType_0::FloatMemWrite 168 0.89% 100.00% # Number of instructions issued per FU type, per thread (Count)
board.processor.cores.core.statIssuedInstType_0::IprAccess 0 0.00% 100.00% # Number of instructions issued per FU type, per thread (Count)
board.processor.cores.core.statIssuedInstType_0::InstPrefetch 0 0.00% 100.00% # Number of instructions issued per FU type, per thread (Count)
board.processor.cores.core.statIssuedInstType_0::SimdUnitStrideLoad 0 0.00% 100.00% # Number of instructions issued per FU type, per thread (Count)
board.processor.cores.core.statIssuedInstType_0::SimdUnitStrideStore 0 0.00% 100.00% # Number of instructions issued per FU type, per thread (Count)
board.processor.cores.core.statIssuedInstType_0::SimdUnitStrideMaskLoad 0 0.00% 100.00% # Number of instructions issued per FU type, per thread (Count)
board.processor.cores.core.statIssuedInstType_0::SimdUnitStrideMaskStore 0 0.00% 100.00% # Number of instructions issued per FU type, per thread (Count)
board.processor.cores.core.statIssuedInstType_0::SimdStridedLoad 0 0.00% 100.00% # Number of instructions issued per FU type, per thread (Count)
board.processor.cores.core.statIssuedInstType_0::SimdStridedStore 0 0.00% 100.00% # Number of instructions issued per FU type, per thread (Count)
board.processor.cores.core.statIssuedInstType_0::SimdIndexedLoad 0 0.00% 100.00% # Number of instructions issued per FU type, per thread (Count)
board.processor.cores.core.statIssuedInstType_0::SimdIndexedStore 0 0.00% 100.00% # Number of instructions issued per FU type, per thread (Count)
board.processor.cores.core.statIssuedInstType_0::SimdWholeRegisterLoad 0 0.00% 100.00% # Number of instructions issued per FU type, per thread (Count)
board.processor.cores.core.statIssuedInstType_0::SimdWholeRegisterStore 0 0.00% 100.00% # Number of instructions issued per FU type, per thread (Count)
board.processor.cores.core.statIssuedInstType_0::SimdUnitStrideFaultOnlyFirstLoad 0 0.00% 100.00% # Number of instructions issued per FU type, per thread (Count)
board.processor.cores.core.statIssuedInstType_0::SimdUnitStrideSegmentedLoad 0 0.00% 100.00% # Number of instructions issued per FU type, per thread (Count)
board.processor.cores.core.statIssuedInstType_0::SimdUnitStrideSegmentedStore 0 0.00% 100.00% # Number of instructions issued per FU type, per thread (Count)
board.processor.cores.core.statIssuedInstType_0::SimdExt 0 0.00% 100.00% # Number of instructions issued per FU type, per thread (Count)
board.processor.cores.core.statIssuedInstType_0::SimdFloatExt 0 0.00% 100.00% # Number of instructions issued per FU type, per thread (Count)
board.processor.cores.core.statIssuedInstType_0::SimdConfig 0 0.00% 100.00% # Number of instructions issued per FU type, per thread (Count)
board.processor.cores.core.statIssuedInstType_0::total 18906 # Number of instructions issued per FU type, per thread (Count)
board.processor.cores.core.issueRate 0.079812 # Inst issue rate ((Count/Cycle))
board.processor.cores.core.fuBusy 143 # FU busy when requested (Count)
board.processor.cores.core.fuBusyRate 0.007564 # FU busy rate (busy events/executed inst) ((Count/Count))
board.processor.cores.core.intInstQueueReads 122364 # Number of integer instruction queue reads (Count)
board.processor.cores.core.intInstQueueWrites 26904 # Number of integer instruction queue writes (Count)
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board.processor.cores.core.intInstQueueWakeupAccesses 17362 # Number of integer instruction queue wakeup accesses (Count)

board.processor.cores.core.fpInstQueueReads 1002 # Number of floating instruction queue reads (Count)

board.processor.cores.core.fpInstQueueWrites 462 # Number of floating instruction queue writes (Count)

board.processor.cores.core.fpInstQueueWakeupAccesses 412 # Number of floating instruction queue wakeup accesses (Count)

board.processor.cores.core.vecInstQueueReads 0 # Number of vector instruction queue reads (Count)

board.processor.cores.core.vecInstQueueWrites 0 # Number of vector instruction queue writes (Count)

board.processor.cores.core.vecInstQueueWakeupAccesses 0 # Number of vector instruction queue wakeup accesses (Count)

board.processor.cores.core.intAluAccesses 18408 # Number of integer alu accesses (Count)

board.processor.cores.core.fpAluAccesses 510 # Number of floating point alu accesses (Count)

board.processor.cores.core.vecAluAccesses 0 # Number of vector alu accesses (Count)

board.processor.cores.core.numSquashedInsts 161 # Number of squashed instructions skipped in execute (Count)

board.processor.cores.core.numSwp 0 # Number of swp insts executed (Count)

board.processor.cores.core.timesIdled 2322 # Number of times that the entire CPU went into an idle state and unscheduled itself (Count)

board.processor.cores.core.idleCycles 151479 # Total number of cycles that the CPU has spent unscheduled due to idling (Cycle)

board.processor.cores.core.MemDepUnit__0.insertedLoads 2042 # Number of loads inserted to the mem dependence unit. (Count)

board.processor.cores.core.MemDepUnit__0.insertedStores 1949 # Number of stores inserted to the mem dependence unit. (Count)

board.processor.cores.core.MemDepUnit__0.conflictingLoads 38 # Number of conflicting loads. (Count)

board.processor.cores.core.MemDepUnit__0.conflictingStores 30 # Number of conflicting stores. (Count)

board.processor.cores.core.MemDepUnit__1.insertedLoads 0 # Number of loads inserted to the mem dependence unit. (Count)

board.processor.cores.core.MemDepUnit__1.insertedStores 0 # Number of stores inserted to the mem dependence unit. (Count)

board.processor.cores.core.MemDepUnit__1.conflictingLoads 0 # Number of conflicting loads. (Count)

board.processor.cores.core.MemDepUnit__1.conflictingStores 0 # Number of conflicting stores. (Count)

board.processor.cores.core.MemDepUnit__2.insertedLoads 0 # Number of loads inserted to the mem dependence unit. (Count)

board.processor.cores.core.MemDepUnit__2.insertedStores 0 # Number of stores inserted to the mem dependence unit. (Count)

board.processor.cores.core.MemDepUnit__2.conflictingLoads 0 # Number of conflicting loads. (Count)

board.processor.cores.core.MemDepUnit__2.conflictingStores 0 # Number of conflicting stores. (Count)

board.processor.cores.core.MemDepUnit__3.insertedLoads 0 # Number of loads inserted to the mem dependence unit. (Count)

board.processor.cores.core.MemDepUnit__3.insertedStores 0 # Number of stores inserted to the mem dependence unit. (Count)

board.processor.cores.core.MemDepUnit__3.conflictingLoads 0 # Number of conflicting loads. (Count)

board.processor.cores.core.MemDepUnit__3.conflictingStores 0 # Number of conflicting stores. (Count)

board.processor.cores.core.branchPred.lookups_0::NoBranch 0 0.00% 0.00% # Number of BP lookups (Count)

board.processor.cores.core.branchPred.lookups_0::Return 195 7.58% 7.58% # Number of BP lookups (Count)

board.processor.cores.core.branchPred.lookups_0::CallDirect 213 8.28% 15.85% # Number of BP lookups (Count)

board.processor.cores.core.branchPred.lookups_0::CallIndirect 66 2.56% 18.41% # Number of BP lookups (Count)

board.processor.cores.core.branchPred.lookups_0::DirectCond 1863 72.38% 90.79% # Number of BP lookups (Count)

board.processor.cores.core.branchPred.lookups_0::DirectUncond 209 8.12% 98.91% # Number of BP lookups (Count)

board.processor.cores.core.branchPred.lookups_0::IndirectCond 0 0.00% 98.91% # Number of BP lookups (Count)

board.processor.cores.core.branchPred.lookups_0::IndirectUncond 28 1.09% 100.00% # Number of BP lookups (Count)

board.processor.cores.core.branchPred.lookups_0::total 2574 # Number of BP lookups (Count)

board.processor.cores.core.branchPred.squashes_0::NoBranch 0 0.00% 0.00% # Number of branches that got squashed (completely removed) as an earlier branch was mispredicted. (Count)

board.processor.cores.core.branchPred.squashes_0::Return 73 6.70% 6.70% # Number of branches that got squashed (completely removed) as an earlier branch was mispredicted. (Count)

board.processor.cores.core.branchPred.squashes_0::CallDirect 126 11.56% 18.26% # Number of branches that got squashed (completely removed) as an earlier branch was mispredicted. (Count)

board.processor.cores.core.branchPred.squashes_0::CallIndirect 26 2.39% 20.64% # Number of branches that got squashed (completely removed) as an earlier branch was mispredicted. (Count)

board.processor.cores.core.branchPred.squashes_0::DirectCond 754 69.17% 89.82% # Number of branches that got squashed (completely removed) as an earlier branch was mispredicted. (Count)

board.processor.cores.core.branchPred.squashes_0::DirectUncond 104 9.54% 99.36% # Number of branches that got squashed (completely removed) as an earlier branch was mispredicted. (Count)

board.processor.cores.core.branchPred.squashes_0::IndirectCond 0 0.00% 99.36% # Number of branches that got squashed (completely removed) as an earlier branch was mispredicted. (Count)

board.processor.cores.core.branchPred.squashes_0::IndirectUncond 7 0.64% 100.00% # Number of branches that got squashed (completely removed) as an earlier branch was mispredicted. (Count)

board.processor.cores.core.branchPred.squashes_0::total 1090 # Number of branches that got squashed (completely removed) as an earlier branch was mispredicted. (Count)

board.processor.cores.core.branchPred.corrected_0::NoBranch 0 0.00% 0.00% # Number of branches that got corrected but not yet commited. Branches get corrected by decode or after execute. Also a branch misprediction can be detected out-of-order. Therefore, a corrected branch might not end up beeing committed in case an even earlier branch was mispredicted (Count)

board.processor.cores.core.branchPred.corrected_0::Return 1 0.16% 0.16% # Number of branches that got corrected but not yet commited. Branches get corrected by decode or after execute. Also a branch misprediction can be detected out-of-order. Therefore, a corrected branch might not end up beeing committed in case an even earlier branch was mispredicted (Count)

board.processor.cores.core.branchPred.corrected_0::CallDirect 112 17.72% 17.88% # Number of branches that got corrected but not yet commited. Branches get corrected by decode or after execute. Also a branch misprediction can be detected out-of-order. Therefore, a corrected branch might not end up beeing committed in case an even earlier branch was mispredicted (Count)

board.processor.cores.core.branchPred.corrected_0::CallIndirect 35 5.54% 23.42% # Number of branches that got corrected but not yet commited. Branches get corrected by decode or after execute. Also a branch misprediction can be detected out-of-order. Therefore, a corrected branch might not end up beeing committed in case an even earlier branch was mispredicted (Count)

board.processor.cores.core.branchPred.corrected_0::DirectCond 355 56.17% 79.59% # Number of branches that got corrected but not yet commited. Branches get corrected by decode or after execute. Also a branch misprediction can be detected out-of-order. Therefore, a corrected branch might not end up beeing committed in case an even earlier branch was mispredicted (Count)

board.processor.cores.core.branchPred.corrected_0::DirectUncond 107 16.93% 96.52% # Number of branches that got corrected but not yet commited. Branches get corrected by decode or after execute. Also a branch misprediction can be detected out-of-order. Therefore, a corrected branch might not end up beeing committed in case an even earlier branch was mispredicted (Count)

board.processor.cores.core.branchPred.corrected_0::IndirectCond 0 0.00% 96.52% # Number of branches that got corrected but not yet commited. Branches get corrected by decode or after execute. Also a branch misprediction can be detected out-of-order. Therefore, a corrected branch might not end up beeing committed in case an even earlier branch was mispredicted (Count)

board.processor.cores.core.branchPred.corrected_0::IndirectUncond 22 3.48% 100.00% # Number of branches that got corrected but not yet commited. Branches get corrected by decode or after execute. Also a branch misprediction can be detected out-of-order. Therefore, a corrected branch might not end up beeing committed in case an even earlier branch was mispredicted (Count)

board.processor.cores.core.branchPred.corrected_0::total 632 # Number of branches that got corrected but not yet commited. Branches get corrected by decode or after execute. Also a branch misprediction can be detected out-of-order. Therefore, a corrected branch might not end up beeing committed in case an even earlier branch was mispredicted (Count)

board.processor.cores.core.branchPred.earlyResteers_0::NoBranch 0 # Number of branches that got redirected after decode. (Count)

board.processor.cores.core.branchPred.earlyResteers_0::Return 0 # Number of branches that got redirected after decode. (Count)

board.processor.cores.core.branchPred.earlyResteers_0::CallDirect 0 # Number of branches that got redirected after decode. (Count)

board.processor.cores.core.branchPred.earlyResteers_0::CallIndirect 0 # Number of branches that got redirected after decode. (Count)

board.processor.cores.core.branchPred.earlyResteers_0::DirectCond 0 # Number of branches that got redirected after decode. (Count)

board.processor.cores.core.branchPred.earlyResteers_0::DirectUncond 0 # Number of branches that got redirected after decode. (Count)

board.processor.cores.core.branchPred.earlyResteers_0::IndirectCond 0 # Number of branches that got redirected after decode. (Count)

board.processor.cores.core.branchPred.earlyResteers_0::IndirectUncond 0 # Number of branches that got redirected after decode. (Count)

board.processor.cores.core.branchPred.earlyResteers_0::total 0 # Number of branches that got redirected after decode. (Count)

board.processor.cores.core.branchPred.committed_0::NoBranch 0 0.00% 0.00% # Number of branches finally committed (Count)

board.processor.cores.core.branchPred.committed_0::Return 122 8.23% 8.23% # Number of branches finally committed (Count)

board.processor.cores.core.branchPred.committed_0::CallDirect 87 5.87% 14.09% # Number of branches finally committed (Count)

board.processor.cores.core.branchPred.committed_0::CallIndirect 40 2.70% 16.79% # Number of branches finally committed (Count)

board.processor.cores.core.branchPred.committed_0::DirectCond 1108 74.71% 91.50% # Number of branches finally committed (Count)

board.processor.cores.core.branchPred.committed_0::DirectUncond 105 7.08% 98.58% # Number of branches finally committed (Count)

board.processor.cores.core.branchPred.committed_0::IndirectCond 0 0.00% 98.58% # Number of branches finally committed (Count)

board.processor.cores.core.branchPred.committed_0::IndirectUncond 21 1.42% 100.00% # Number of branches finally committed (Count)

board.processor.cores.core.branchPred.committed_0::total 1483 # Number of branches finally committed (Count)

board.processor.cores.core.branchPred.mispredicted_0::NoBranch 0 0.00% 0.00% # Number of committed branches that were mispredicted. (Count)

board.processor.cores.core.branchPred.mispredicted_0::Return 0 0.00% 0.00% # Number of committed branches that were mispredicted. (Count)

board.processor.cores.core.branchPred.mispredicted_0::CallDirect 65 14.13% 14.13% # Number of committed branches that were mispredicted. (Count)

board.processor.cores.core.branchPred.mispredicted_0::CallIndirect 32 6.96% 21.09% # Number of committed branches that were mispredicted. (Count)

board.processor.cores.core.branchPred.mispredicted_0::DirectCond 286 62.17% 83.26% # Number of committed branches that were mispredicted. (Count)

board.processor.cores.core.branchPred.mispredicted_0::DirectUncond 56 12.17% 95.43% # Number of committed branches that were mispredicted. (Count)

board.processor.cores.core.branchPred.mispredicted_0::IndirectCond 0 0.00% 95.43% # Number of committed branches that were mispredicted. (Count)

board.processor.cores.core.branchPred.mispredicted_0::IndirectUncond 21 4.57% 100.00% # Number of committed branches that were mispredicted. (Count)

board.processor.cores.core.branchPred.mispredicted_0::total 460 # Number of committed branches that were mispredicted. (Count)

board.processor.cores.core.branchPred.targetProvider_0::NoTarget 1845 71.68% 71.68% # The component providing the target for taken branches (Count)

board.processor.cores.core.branchPred.targetProvider_0::BTB 526 20.44% 92.11% # The component providing the target for taken branches (Count)

board.processor.cores.core.branchPred.targetProvider_0::RAS 195 7.58% 99.69% # The component providing the target for taken branches (Count)

board.processor.cores.core.branchPred.targetProvider_0::Indirect 8 0.31% 100.00% # The component providing the target for taken branches (Count)

board.processor.cores.core.branchPred.targetProvider_0::total 2574 # The component providing the target for taken branches (Count)

board.processor.cores.core.branchPred.targetWrong_0::NoBranch 605 95.73% 95.73% # Number of branches where the target was incorrect or not available at prediction time. (Count)

board.processor.cores.core.branchPred.targetWrong_0::Return 26 4.11% 99.84% # Number of branches where the target was incorrect or not available at prediction time. (Count)

board.processor.cores.core.branchPred.targetWrong_0::CallDirect 1 0.16% 100.00% # Number of branches where the target was incorrect or not available at prediction time. (Count)

board.processor.cores.core.branchPred.targetWrong_0::CallIndirect 0 0.00% 100.00% # Number of branches where the target was incorrect or not available at prediction time. (Count)

board.processor.cores.core.branchPred.targetWrong_0::DirectCond 0 0.00% 100.00% # Number of branches where the target was incorrect or not available at prediction time. (Count)

board.processor.cores.core.branchPred.targetWrong_0::DirectUncond 0 0.00% 100.00% # Number of branches where the target was incorrect or not available at prediction time. (Count)

board.processor.cores.core.branchPred.targetWrong_0::IndirectCond 0 0.00% 100.00% # Number of branches where the target was incorrect or not available at prediction time. (Count)

board.processor.cores.core.branchPred.targetWrong_0::IndirectUncond 0 0.00% 100.00% # Number of branches where the target was incorrect or not available at prediction time. (Count)

board.processor.cores.core.branchPred.targetWrong_0::total 632 # Number of branches where the target was incorrect or not available at prediction time. (Count)

board.processor.cores.core.branchPred.condPredicted 1863 # Number of conditional branches predicted (Count)

board.processor.cores.core.branchPred.condPredictedTaken 778 # Number of conditional branches predicted as taken (Count)

board.processor.cores.core.branchPred.condIncorrect 632 # Number of conditional branches incorrect (Count)

board.processor.cores.core.branchPred.predTakenBTBMiss 328 # Number of branches predicted taken but missed in BTB (Count)

board.processor.cores.core.branchPred.NotTakenMispredicted 606 # Number branches predicted 'not taken' but turned out to be taken (Count)

board.processor.cores.core.branchPred.TakenMispredicted 26 # Number branches predicted taken but are actually not taken (Count)

board.processor.cores.core.branchPred.BTBLookups 2574 # Number of BTB lookups (Count)

board.processor.cores.core.branchPred.BTBUpdates 548 # Number of BTB updates (Count)

board.processor.cores.core.branchPred.BTBHits 757 # Number of BTB hits (Count)

board.processor.cores.core.branchPred.BTBHitRatio 0.294095 # BTB Hit Ratio (Ratio)

board.processor.cores.core.branchPred.BTBMispredicted 447 # Number BTB mispredictions. No target found or target wrong (Count)

board.processor.cores.core.branchPred.indirectLookups 94 # Number of indirect predictor lookups. (Count)

board.processor.cores.core.branchPred.indirectHits 8 # Number of indirect target hits. (Count)

board.processor.cores.core.branchPred.indirectMisses 86 # Number of indirect misses. (Count)

board.processor.cores.core.branchPred.indirectMispredicted 0 # Number of mispredicted indirect branches. (Count)

board.processor.cores.core.branchPred.btb.lookups::NoBranch 0 0.00% 0.00% # Number of BTB lookups (Count)

board.processor.cores.core.branchPred.btb.lookups::Return 195 7.58% 7.58% # Number of BTB lookups (Count)

board.processor.cores.core.branchPred.btb.lookups::CallDirect 213 8.28% 15.85% # Number of BTB lookups (Count)

board.processor.cores.core.branchPred.btb.lookups::CallIndirect 66 2.56% 18.41% # Number of BTB lookups (Count)

board.processor.cores.core.branchPred.btb.lookups::DirectCond 1863 72.38% 90.79% # Number of BTB lookups (Count)

board.processor.cores.core.branchPred.btb.lookups::DirectUncond 209 8.12% 98.91% # Number of BTB lookups (Count)

board.processor.cores.core.branchPred.btb.lookups::IndirectCond 0 0.00% 98.91% # Number of BTB lookups (Count)

board.processor.cores.core.branchPred.btb.lookups::IndirectUncond 28 1.09% 100.00% # Number of BTB lookups (Count)

board.processor.cores.core.branchPred.btb.lookups::total 2574 # Number of BTB lookups (Count)

board.processor.cores.core.branchPred.btb.misses::NoBranch 0 0.00% 0.00% # Number of BTB misses (Count)

board.processor.cores.core.branchPred.btb.misses::Return 194 10.68% 10.68% # Number of BTB misses (Count)

board.processor.cores.core.branchPred.btb.misses::CallDirect 133 7.32% 18.00% # Number of BTB misses (Count)

board.processor.cores.core.branchPred.btb.misses::CallIndirect 66 3.63% 21.63% # Number of BTB misses (Count)

board.processor.cores.core.branchPred.btb.misses::DirectCond 1262 69.46% 91.08% # Number of BTB misses (Count)

board.processor.cores.core.branchPred.btb.misses::DirectUncond 134 7.37% 98.46% # Number of BTB misses (Count)

board.processor.cores.core.branchPred.btb.misses::IndirectCond 0 0.00% 98.46% # Number of BTB misses (Count)

board.processor.cores.core.branchPred.btb.misses::IndirectUncond 28 1.54% 100.00% # Number of BTB misses (Count)

board.processor.cores.core.branchPred.btb.misses::total 1817 # Number of BTB misses (Count)

board.processor.cores.core.branchPred.btb.updates::NoBranch 0 0.00% 0.00% # Number of BTB updates (Count)

board.processor.cores.core.branchPred.btb.updates::Return 0 0.00% 0.00% # Number of BTB updates (Count)

board.processor.cores.core.branchPred.btb.updates::CallDirect 112 20.44% 20.44% # Number of BTB updates (Count)

board.processor.cores.core.branchPred.btb.updates::CallIndirect 0 0.00% 20.44% # Number of BTB updates (Count)

board.processor.cores.core.branchPred.btb.updates::DirectCond 329 60.04% 80.47% # Number of BTB updates (Count)

board.processor.cores.core.branchPred.btb.updates::DirectUncond 107 19.53% 100.00% # Number of BTB updates (Count)

board.processor.cores.core.branchPred.btb.updates::IndirectCond 0 0.00% 100.00% # Number of BTB updates (Count)

board.processor.cores.core.branchPred.btb.updates::IndirectUncond 0 0.00% 100.00% # Number of BTB updates (Count)

board.processor.cores.core.branchPred.btb.updates::total 548 # Number of BTB updates (Count)

board.processor.cores.core.branchPred.btb.mispredict::NoBranch 0 0.00% 0.00% # Number of BTB mispredictions. No target found or target wrong. (Count)

board.processor.cores.core.branchPred.btb.mispredict::Return 0 0.00% 0.00% # Number of BTB mispredictions. No target found or target wrong. (Count)

board.processor.cores.core.branchPred.btb.mispredict::CallDirect 112 20.44% 20.44% # Number of BTB mispredictions. No target found or target wrong. (Count)

board.processor.cores.core.branchPred.btb.mispredict::CallIndirect 0 0.00% 20.44% # Number of BTB mispredictions. No target found or target wrong. (Count)

board.processor.cores.core.branchPred.btb.mispredict::DirectCond 329 60.04% 80.47% # Number of BTB mispredictions. No target found or target wrong. (Count)

board.processor.cores.core.branchPred.btb.mispredict::DirectUncond 107 19.53% 100.00% # Number of BTB mispredictions. No target found or target wrong. (Count)

board.processor.cores.core.branchPred.btb.mispredict::IndirectCond 0 0.00% 100.00% # Number of BTB mispredictions. No target found or target wrong. (Count)

board.processor.cores.core.branchPred.btb.mispredict::IndirectUncond 0 0.00% 100.00% # Number of BTB mispredictions. No target found or target wrong. (Count)

board.processor.cores.core.branchPred.btb.mispredict::total 548 # Number of BTB mispredictions. No target found or target wrong. (Count)

board.processor.cores.core.branchPred.btb.power_state.pwrStateResidencyTicks::UNDEFINED 78881373 # Cumulative time (in ticks) in various power states (Tick)

board.processor.cores.core.branchPred.indirectBranchPred.lookups 94 # Number of lookups (Count)

board.processor.cores.core.branchPred.indirectBranchPred.hits 8 # Number of hits of a tag (Count)

board.processor.cores.core.branchPred.indirectBranchPred.misses 86 # Number of misses (Count)

board.processor.cores.core.branchPred.indirectBranchPred.targetRecords 57 # Number of targets that where recorded/installed in the cache (Count)

board.processor.cores.core.branchPred.indirectBranchPred.indirectRecords 151 # Number of indirect branches/calls recorded in the indirect hist (Count)

board.processor.cores.core.branchPred.indirectBranchPred.speculativeOverflows 0 # Number of times more than the allowed capacity for speculative branches/calls where in flight and destroy the path history (Count)

board.processor.cores.core.branchPred.ras.pushes 352 # Number of times a PC was pushed onto the RAS (Count)

board.processor.cores.core.branchPred.ras.pops 347 # Number of times a PC was poped from the RAS (Count)

board.processor.cores.core.branchPred.ras.squashes 225 # Number of times the stack operation was squashed due to wrong speculation. (Count)

board.processor.cores.core.branchPred.ras.used 122 # Number of times the RAS is the provider (Count)

board.processor.cores.core.branchPred.ras.correct 122 # Number of times the RAS is the provider and the prediction is correct (Count)

board.processor.cores.core.branchPred.ras.incorrect 0 # Number of times the RAS is the provider and the prediction is wrong (Count)

board.processor.cores.core.commit.commitSquashedInsts 7176 # The number of squashed insts skipped by commit (Count)

board.processor.cores.core.commit.branchMispredicts 438 # The number of times a branch was mispredicted (Count)

board.processor.cores.core.commit.numCommittedDist::samples 84132 # Number of insts commited each cycle (Count)

board.processor.cores.core.commit.numCommittedDist::mean 0.154424 # Number of insts commited each cycle (Count)

board.processor.cores.core.commit.numCommittedDist::stdev 0.788129 # Number of insts commited each cycle (Count)

board.processor.cores.core.commit.numCommittedDist::underflows 0 0.00% 0.00% # Number of insts commited each cycle (Count)

board.processor.cores.core.commit.numCommittedDist::0 79171 94.10% 94.10% # Number of insts commited each cycle (Count)

board.processor.cores.core.commit.numCommittedDist::1 1931 2.30% 96.40% # Number of insts commited each cycle (Count)

board.processor.cores.core.commit.numCommittedDist::2 1123 1.33% 97.73% # Number of insts commited each cycle (Count)

board.processor.cores.core.commit.numCommittedDist::3 832 0.99% 98.72% # Number of insts commited each cycle (Count)

board.processor.cores.core.commit.numCommittedDist::4 407 0.48% 99.21% # Number of insts commited each cycle (Count)

board.processor.cores.core.commit.numCommittedDist::5 113 0.13% 99.34% # Number of insts commited each cycle (Count)

board.processor.cores.core.commit.numCommittedDist::6 103 0.12% 99.46% # Number of insts commited each cycle (Count)

board.processor.cores.core.commit.numCommittedDist::7 108 0.13% 99.59% # Number of insts commited each cycle (Count)

board.processor.cores.core.commit.numCommittedDist::8 344 0.41% 100.00% # Number of insts commited each cycle (Count)

board.processor.cores.core.commit.numCommittedDist::overflows 0 0.00% 100.00% # Number of insts commited each cycle (Count)

board.processor.cores.core.commit.numCommittedDist::min_value 0 # Number of insts commited each cycle (Count)

board.processor.cores.core.commit.numCommittedDist::max_value 8 # Number of insts commited each cycle (Count)

board.processor.cores.core.commit.numCommittedDist::total 84132 # Number of insts commited each cycle (Count)

board.processor.cores.core.commit.amos 0 # Number of atomic instructions committed (Count)

board.processor.cores.core.commit.membars 0 # Number of memory barriers committed (Count)

board.processor.cores.core.commit.functionCalls 127 # Number of function calls committed. (Count)

board.processor.cores.core.commit.committedInstType_0::No_OpClass 27 0.21% 0.21% # Class of committed instruction (Count)

board.processor.cores.core.commit.committedInstType_0::IntAlu 10123 77.92% 78.12% # Class of committed instruction (Count)

board.processor.cores.core.commit.committedInstType_0::IntMult 4 0.03% 78.16% # Class of committed instruction (Count)

board.processor.cores.core.commit.committedInstType_0::IntDiv 21 0.16% 78.32% # Class of committed instruction (Count)

board.processor.cores.core.commit.committedInstType_0::FloatAdd 2 0.02% 78.33% # Class of committed instruction (Count)

board.processor.cores.core.commit.committedInstType_0::FloatCmp 0 0.00% 78.33% # Class of committed instruction (Count)

board.processor.cores.core.commit.committedInstType_0::FloatCvt 0 0.00% 78.33% # Class of committed instruction (Count)

board.processor.cores.core.commit.committedInstType_0::FloatMult 0 0.00% 78.33% # Class of committed instruction (Count)

board.processor.cores.core.commit.committedInstType_0::FloatMultAcc 0 0.00% 78.33% # Class of committed instruction (Count)

board.processor.cores.core.commit.committedInstType_0::FloatDiv 0 0.00% 78.33% # Class of committed instruction (Count)

board.processor.cores.core.commit.committedInstType_0::FloatMisc 0 0.00% 78.33% # Class of committed instruction (Count)

board.processor.cores.core.commit.committedInstType_0::FloatSqrt 0 0.00% 78.33% # Class of committed instruction (Count)

board.processor.cores.core.commit.committedInstType_0::SimdAdd 0 0.00% 78.33% # Class of committed instruction (Count)

board.processor.cores.core.commit.committedInstType_0::SimdAddAcc 0 0.00% 78.33% # Class of committed instruction (Count)

board.processor.cores.core.commit.committedInstType_0::SimdAlu 82 0.63% 78.96% # Class of committed instruction (Count)

board.processor.cores.core.commit.committedInstType_0::SimdCmp 0 0.00% 78.96% # Class of committed instruction (Count)

board.processor.cores.core.commit.committedInstType_0::SimdCvt 48 0.37% 79.33% # Class of committed instruction (Count)

board.processor.cores.core.commit.committedInstType_0::SimdMisc 62 0.48% 79.81% # Class of committed instruction (Count)

board.processor.cores.core.commit.committedInstType_0::SimdMult 0 0.00% 79.81% # Class of committed instruction (Count)

board.processor.cores.core.commit.committedInstType_0::SimdMultAcc 0 0.00% 79.81% # Class of committed instruction (Count)

board.processor.cores.core.commit.committedInstType_0::SimdMatMultAcc 0 0.00% 79.81% # Class of committed instruction (Count)

board.processor.cores.core.commit.committedInstType_0::SimdShift 0 0.00% 79.81% # Class of committed instruction (Count)

board.processor.cores.core.commit.committedInstType_0::SimdShiftAcc 0 0.00% 79.81% # Class of committed instruction (Count)

board.processor.cores.core.commit.committedInstType_0::SimdDiv 0 0.00% 79.81% # Class of committed instruction (Count)

board.processor.cores.core.commit.committedInstType_0::SimdSqrt 0 0.00% 79.81% # Class of committed instruction (Count)

board.processor.cores.core.commit.committedInstType_0::SimdFloatAdd 0 0.00% 79.81% # Class of committed instruction (Count)

board.processor.cores.core.commit.committedInstType_0::SimdFloatAlu 0 0.00% 79.81% # Class of committed instruction (Count)

board.processor.cores.core.commit.committedInstType_0::SimdFloatCmp 0 0.00% 79.81% # Class of committed instruction (Count)

board.processor.cores.core.commit.committedInstType_0::SimdFloatCvt 0 0.00% 79.81% # Class of committed instruction (Count)

board.processor.cores.core.commit.committedInstType_0::SimdFloatDiv 0 0.00% 79.81% # Class of committed instruction (Count)

board.processor.cores.core.commit.committedInstType_0::SimdFloatMisc 0 0.00% 79.81% # Class of committed instruction (Count)

board.processor.cores.core.commit.committedInstType_0::SimdFloatMult 0 0.00% 79.81% # Class of committed instruction (Count)

board.processor.cores.core.commit.committedInstType_0::SimdFloatMultAcc 0 0.00% 79.81% # Class of committed instruction (Count)

board.processor.cores.core.commit.committedInstType_0::SimdFloatMatMultAcc 0 0.00% 79.81% # Class of committed instruction (Count)

board.processor.cores.core.commit.committedInstType_0::SimdFloatSqrt 0 0.00% 79.81% # Class of committed instruction (Count)

board.processor.cores.core.commit.committedInstType_0::SimdReduceAdd 0 0.00% 79.81% # Class of committed instruction (Count)

board.processor.cores.core.commit.committedInstType_0::SimdReduceAlu 0 0.00% 79.81% # Class of committed instruction (Count)

board.processor.cores.core.commit.committedInstType_0::SimdReduceCmp 0 0.00% 79.81% # Class of committed instruction (Count)

board.processor.cores.core.commit.committedInstType_0::SimdFloatReduceAdd 0 0.00% 79.81% # Class of committed instruction (Count)

board.processor.cores.core.commit.committedInstType_0::SimdFloatReduceCmp 0 0.00% 79.81% # Class of committed instruction (Count)

board.processor.cores.core.commit.committedInstType_0::SimdAes 0 0.00% 79.81% # Class of committed instruction (Count)

board.processor.cores.core.commit.committedInstType_0::SimdAesMix 0 0.00% 79.81% # Class of committed instruction (Count)

board.processor.cores.core.commit.committedInstType_0::SimdSha1Hash 0 0.00% 79.81% # Class of committed instruction (Count)

board.processor.cores.core.commit.committedInstType_0::SimdSha1Hash2 0 0.00% 79.81% # Class of committed instruction (Count)

board.processor.cores.core.commit.committedInstType_0::SimdSha256Hash 0 0.00% 79.81% # Class of committed instruction (Count)

board.processor.cores.core.commit.committedInstType_0::SimdSha256Hash2 0 0.00% 79.81% # Class of committed instruction (Count)

board.processor.cores.core.commit.committedInstType_0::SimdShaSigma2 0 0.00% 79.81% # Class of committed instruction (Count)

board.processor.cores.core.commit.committedInstType_0::SimdShaSigma3 0 0.00% 79.81% # Class of committed instruction (Count)

board.processor.cores.core.commit.committedInstType_0::SimdPredAlu 0 0.00% 79.81% # Class of committed instruction (Count)

board.processor.cores.core.commit.committedInstType_0::Matrix 0 0.00% 79.81% # Class of committed instruction (Count)

board.processor.cores.core.commit.committedInstType_0::MatrixMov 0 0.00% 79.81% # Class of committed instruction (Count)

board.processor.cores.core.commit.committedInstType_0::MatrixOP 0 0.00% 79.81% # Class of committed instruction (Count)

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board.processor.cores.core.commit.committedInstType_0::MemRead        1224      9.42%     89.23% # Class of committed instruction (Count)
board.processor.cores.core.commit.committedInstType_0::MemWrite        1191      9.17%     98.40% # Class of committed instruction (Count)
board.processor.cores.core.commit.committedInstType_0::FloatMemRead          48      0.37%     98.77% # Class of committed instruction (Count)
board.processor.cores.core.commit.committedInstType_0::FloatMemWrite         160      1.23%    100.00% # Class of committed instruction (Count)
board.processor.cores.core.commit.committedInstType_0::IprAccess           0      0.00%    100.00% # Class of committed instruction (Count)
board.processor.cores.core.commit.committedInstType_0::InstPrefetch           0      0.00%    100.00% # Class of committed instruction (Count)
board.processor.cores.core.commit.committedInstType_0::SimdUnitStrideLoad           0      0.00%    100.00% # Class of committed instruction (Count)
board.processor.cores.core.commit.committedInstType_0::SimdUnitStrideStore           0      0.00%    100.00% # Class of committed instruction (Count)
board.processor.cores.core.commit.committedInstType_0::SimdUnitStrideMaskLoad           0      0.00%    100.00% # Class of committed instruction (Count)
board.processor.cores.core.commit.committedInstType_0::SimdUnitStrideMaskStore           0      0.00%    100.00% # Class of committed instruction (Count)
board.processor.cores.core.commit.committedInstType_0::SimdStridedLoad           0      0.00%    100.00% # Class of committed instruction (Count)
board.processor.cores.core.commit.committedInstType_0::SimdStridedStore           0      0.00%    100.00% # Class of committed instruction (Count)
board.processor.cores.core.commit.committedInstType_0::SimdIndexedLoad           0      0.00%    100.00% # Class of committed instruction (Count)
board.processor.cores.core.commit.committedInstType_0::SimdIndexedStore           0      0.00%    100.00% # Class of committed instruction (Count)
board.processor.cores.core.commit.committedInstType_0::SimdWholeRegisterLoad           0      0.00%    100.00% # Class of committed instruction (Count)
board.processor.cores.core.commit.committedInstType_0::SimdWholeRegisterStore           0      0.00%    100.00% # Class of committed instruction (Count)
board.processor.cores.core.commit.committedInstType_0::SimdUnitStrideFaultOnlyFirstLoad           0      0.00%    100.00% # Class of committed instruction (Count)
board.processor.cores.core.commit.committedInstType_0::SimdUnitStrideSegmentedLoad           0      0.00%    100.00% # Class of committed instruction (Count)
board.processor.cores.core.commit.committedInstType_0::SimdUnitStrideSegmentedStore           0      0.00%    100.00% # Class of committed instruction (Count)
board.processor.cores.core.commit.committedInstType_0::SimdExt           0      0.00%    100.00% # Class of committed instruction (Count)
board.processor.cores.core.commit.committedInstType_0::SimdFloatExt           0      0.00%    100.00% # Class of committed instruction (Count)
board.processor.cores.core.commit.committedInstType_0::SimdConfig           0      0.00%    100.00% # Class of committed instruction (Count)
board.processor.cores.core.commit.committedInstType_0::total        12992                       # Class of committed instruction (Count)
board.processor.cores.core.commit.commitEligibleSamples          344                       # number cycles where commit BW limit reached (Cycle)
board.processor.cores.core.commitStats0.numInsts         6555                       # Number of instructions committed (thread level) (Count)
board.processor.cores.core.commitStats0.numOps        12992                       # Number of ops (including micro ops) committed (thread level) (Count)
board.processor.cores.core.commitStats0.numInstsNotNOP         6555                       # Number of instructions committed excluding NOPs or prefetches (Count)
board.processor.cores.core.commitStats0.numOpsNotNOP        12992                       # Number of Ops (including micro ops) Simulated (Count)
board.processor.cores.core.commitStats0.cpi    36.137605                       # CPI: cycles per instruction (thread level) ((Cycle/Count))
board.processor.cores.core.commitStats0.ipc     0.027672                       # IPC: instructions per cycle (thread level) ((Count/Cycle))
board.processor.cores.core.commitStats0.numMemRefs         2623                       # Number of memory references committed (Count)
board.processor.cores.core.commitStats0.numFpInsts          410                       # Number of float instructions (Count)
board.processor.cores.core.commitStats0.numIntInsts        12796                       # Number of integer instructions (Count)
board.processor.cores.core.commitStats0.numLoadInsts         1272                       # Number of load instructions (Count)
board.processor.cores.core.commitStats0.numStoreInsts         1351                       # Number of store instructions (Count)
board.processor.cores.core.commitStats0.numVecInsts           0                       # Number of vector instructions (Count)
board.processor.cores.core.commitStats0.committedInstType::No_OpClass           27      0.21%      0.21% # Class of committed instruction. (Count)
board.processor.cores.core.commitStats0.committedInstType::IntAlu        10123     77.92%     78.12% # Class of committed instruction. (Count)
board.processor.cores.core.commitStats0.committedInstType::IntMult            4      0.03%     78.16% # Class of committed instruction. (Count)
board.processor.cores.core.commitStats0.committedInstType::IntDiv           21      0.16%     78.32% # Class of committed instruction. (Count)
board.processor.cores.core.commitStats0.committedInstType::FloatAdd            2      0.02%     78.33% # Class of committed instruction. (Count)
board.processor.cores.core.commitStats0.committedInstType::FloatCmp            0      0.00%     78.33% # Class of committed instruction. (Count)
board.processor.cores.core.commitStats0.committedInstType::FloatCvt            0      0.00%     78.33% # Class of committed instruction. (Count)
board.processor.cores.core.commitStats0.committedInstType::FloatMult            0      0.00%     78.33% # Class of committed instruction. (Count)
board.processor.cores.core.commitStats0.committedInstType::FloatMultAcc            0      0.00%     78.33% # Class of committed instruction. (Count)
board.processor.cores.core.commitStats0.committedInstType::FloatDiv            0      0.00%     78.33% # Class of committed instruction. (Count)
board.processor.cores.core.commitStats0.committedInstType::FloatMisc            0      0.00%     78.33% # Class of committed instruction. (Count)
board.processor.cores.core.commitStats0.committedInstType::FloatSqrt            0      0.00%     78.33% # Class of committed instruction. (Count)
board.processor.cores.core.commitStats0.committedInstType::SimdAdd            0      0.00%     78.33% # Class of committed instruction. (Count)
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board.processor.cores.core.commitStats0.committedInstType::SimdAddAcc 0 0.00% 78.33% # Class of committed instruction. (Count)

board.processor.cores.core.commitStats0.committedInstType::SimdAlu 82 0.63% 78.96% # Class of committed instruction. (Count)

board.processor.cores.core.commitStats0.committedInstType::SimdCmp 0 0.00% 78.96% # Class of committed instruction. (Count)

board.processor.cores.core.commitStats0.committedInstType::SimdCvt 48 0.37% 79.33% # Class of committed instruction. (Count)

board.processor.cores.core.commitStats0.committedInstType::SimdMisc 62 0.48% 79.81% # Class of committed instruction. (Count)

board.processor.cores.core.commitStats0.committedInstType::SimdMult 0 0.00% 79.81% # Class of committed instruction. (Count)

board.processor.cores.core.commitStats0.committedInstType::SimdMultAcc 0 0.00% 79.81% # Class of committed instruction. (Count)

board.processor.cores.core.commitStats0.committedInstType::SimdMatMultAcc 0 0.00% 79.81% # Class of committed instruction. (Count)

board.processor.cores.core.commitStats0.committedInstType::SimdShift 0 0.00% 79.81% # Class of committed instruction. (Count)

board.processor.cores.core.commitStats0.committedInstType::SimdShiftAcc 0 0.00% 79.81% # Class of committed instruction. (Count)

board.processor.cores.core.commitStats0.committedInstType::SimdDiv 0 0.00% 79.81% # Class of committed instruction. (Count)

board.processor.cores.core.commitStats0.committedInstType::SimdSqrt 0 0.00% 79.81% # Class of committed instruction. (Count)

board.processor.cores.core.commitStats0.committedInstType::SimdFloatAdd 0 0.00% 79.81% # Class of committed instruction. (Count)

board.processor.cores.core.commitStats0.committedInstType::SimdFloatAlu 0 0.00% 79.81% # Class of committed instruction. (Count)

board.processor.cores.core.commitStats0.committedInstType::SimdFloatCmp 0 0.00% 79.81% # Class of committed instruction. (Count)

board.processor.cores.core.commitStats0.committedInstType::SimdFloatCvt 0 0.00% 79.81% # Class of committed instruction. (Count)

board.processor.cores.core.commitStats0.committedInstType::SimdFloatDiv 0 0.00% 79.81% # Class of committed instruction. (Count)

board.processor.cores.core.commitStats0.committedInstType::SimdFloatMisc 0 0.00% 79.81% # Class of committed instruction. (Count)

board.processor.cores.core.commitStats0.committedInstType::SimdFloatMult 0 0.00% 79.81% # Class of committed instruction. (Count)

board.processor.cores.core.commitStats0.committedInstType::SimdFloatMultAcc 0 0.00% 79.81% # Class of committed instruction. (Count)

board.processor.cores.core.commitStats0.committedInstType::SimdFloatMatMultAcc 0 0.00% 79.81% # Class of committed instruction. (Count)

board.processor.cores.core.commitStats0.committedInstType::SimdFloatSqrt 0 0.00% 79.81% # Class of committed instruction. (Count)

board.processor.cores.core.commitStats0.committedInstType::SimdReduceAdd 0 0.00% 79.81% # Class of committed instruction. (Count)

board.processor.cores.core.commitStats0.committedInstType::SimdReduceAlu 0 0.00% 79.81% # Class of committed instruction. (Count)

board.processor.cores.core.commitStats0.committedInstType::SimdReduceCmp 0 0.00% 79.81% # Class of committed instruction. (Count)

board.processor.cores.core.commitStats0.committedInstType::SimdFloatReduceAdd 0 0.00% 79.81% # Class of committed instruction. (Count)

board.processor.cores.core.commitStats0.committedInstType::SimdFloatReduceCmp 0 0.00% 79.81% # Class of committed instruction. (Count)

board.processor.cores.core.commitStats0.committedInstType::SimdAes 0 0.00% 79.81% # Class of committed instruction. (Count)

board.processor.cores.core.commitStats0.committedInstType::SimdAesMix 0 0.00% 79.81% # Class of committed instruction. (Count)

board.processor.cores.core.commitStats0.committedInstType::SimdSha1Hash 0 0.00% 79.81% # Class of committed instruction. (Count)

board.processor.cores.core.commitStats0.committedInstType::SimdSha1Hash2 0 0.00% 79.81% # Class of committed instruction. (Count)

board.processor.cores.core.commitStats0.committedInstType::SimdSha256Hash 0 0.00% 79.81% # Class of committed instruction. (Count)

board.processor.cores.core.commitStats0.committedInstType::SimdSha256Hash2 0 0.00% 79.81% # Class of committed instruction. (Count)

board.processor.cores.core.commitStats0.committedInstType::SimdShaSigma2 0 0.00% 79.81% # Class of committed instruction. (Count)

board.processor.cores.core.commitStats0.committedInstType::SimdShaSigma3 0 0.00% 79.81% # Class of committed instruction. (Count)

board.processor.cores.core.commitStats0.committedInstType::SimdPredAlu 0 0.00% 79.81% # Class of committed instruction. (Count)

board.processor.cores.core.commitStats0.committedInstType::Matrix 0 0.00% 79.81% # Class of committed instruction. (Count)

board.processor.cores.core.commitStats0.committedInstType::MatrixMov 0 0.00% 79.81% # Class of committed instruction. (Count)

board.processor.cores.core.commitStats0.committedInstType::MatrixOP 0 0.00% 79.81% # Class of committed instruction. (Count)

board.processor.cores.core.commitStats0.committedInstType::MemRead 1224 9.42% 89.23% # Class of committed instruction. (Count)

board.processor.cores.core.commitStats0.committedInstType::MemWrite 1191 9.17% 98.40% # Class of committed instruction. (Count)

board.processor.cores.core.commitStats0.committedInstType::FloatMemRead 48 0.37% 98.77% # Class of committed instruction. (Count)

board.processor.cores.core.commitStats0.committedInstType::FloatMemWrite 160 1.23% 100.00% # Class of committed instruction. (Count)

board.processor.cores.core.commitStats0.committedInstType::IprAccess 0 0.00% 100.00% # Class of committed instruction. (Count)

board.processor.cores.core.commitStats0.committedInstType::InstPrefetch 0 0.00% 100.00% # Class of committed instruction. (Count)

board.processor.cores.core.commitStats0.committedInstType::SimdUnitStrideLoad 0 0.00% 100.00% # Class of committed instruction. (Count)

board.processor.cores.core.commitStats0.committedInstType::SimdUnitStrideStore 0 0.00% 100.00% # Class of committed instruction. (Count)

board.processor.cores.core.commitStats0.committedInstType::SimdUnitStrideMaskLoad 0 0.00% 100.00% # Class of committed instruction. (Count)

board.processor.cores.core.commitStats0.committedInstType::SimdUnitStrideMaskStore 0 0.00% 100.00% # Class of committed instruction. (Count)

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board.processor.cores.core.commitStats0.committedInstType::SimdStridedLoad 0 0.00% 100.00% # Class of committed instruction. (Count)
board.processor.cores.core.commitStats0.committedInstType::SimdStridedStore 0 0.00% 100.00% # Class of committed instruction. (Count)
board.processor.cores.core.commitStats0.committedInstType::SimdIndexedLoad 0 0.00% 100.00% # Class of committed instruction. (Count)
board.processor.cores.core.commitStats0.committedInstType::SimdIndexedStore 0 0.00% 100.00% # Class of committed instruction. (Count)
board.processor.cores.core.commitStats0.committedInstType::SimdWholeRegisterLoad 0 0.00% 100.00% # Class of committed instruction. (Count)
board.processor.cores.core.commitStats0.committedInstType::SimdWholeRegisterStore 0 0.00% 100.00% # Class of committed instruction. (Count)
board.processor.cores.core.commitStats0.committedInstType::SimdUnitStrideFaultOnlyFirstLoad 0 0.00% 100.00% # Class of committed instruction. (Count)
board.processor.cores.core.commitStats0.committedInstType::SimdUnitStrideSegmentedLoad 0 0.00% 100.00% # Class of committed instruction. (Count)
board.processor.cores.core.commitStats0.committedInstType::SimdUnitStrideSegmentedStore 0 0.00% 100.00% # Class of committed instruction. (Count)
board.processor.cores.core.commitStats0.committedInstType::SimdExt 0 0.00% 100.00% # Class of committed instruction. (Count)
board.processor.cores.core.commitStats0.committedInstType::SimdFloatExt 0 0.00% 100.00% # Class of committed instruction. (Count)
board.processor.cores.core.commitStats0.committedInstType::SimdConfig 0 0.00% 100.00% # Class of committed instruction. (Count)
board.processor.cores.core.commitStats0.committedInstType::total 12992 # Class of committed instruction. (Count)
board.processor.cores.core.commitStats0.committedControl::IsControl 1483 # Class of control type instructions committed (Count)
board.processor.cores.core.commitStats0.committedControl::IsDirectControl 1300 # Class of control type instructions committed (Count)
board.processor.cores.core.commitStats0.committedControl::IsIndirectControl 183 # Class of control type instructions committed (Count)
board.processor.cores.core.commitStats0.committedControl::IsCondControl 1108 # Class of control type instructions committed (Count)
board.processor.cores.core.commitStats0.committedControl::IsUncondControl 375 # Class of control type instructions committed (Count)
board.processor.cores.core.commitStats0.committedControl::IsCall 127 # Class of control type instructions committed (Count)
board.processor.cores.core.commitStats0.committedControl::IsReturn 122 # Class of control type instructions committed (Count)
board.processor.cores.core.decode.idleCycles 74916 # Number of cycles decode is idle (Cycle)
board.processor.cores.core.decode.blockedCycles 6631 # Number of cycles decode is blocked (Cycle)
board.processor.cores.core.decode.runCycles 3344 # Number of cycles decode is running (Cycle)
board.processor.cores.core.decode.unblockCycles 54 # Number of cycles decode is unblocking (Cycle)
board.processor.cores.core.decode.squashCycles 458 # Number of cycles decode is squashing (Cycle)
board.processor.cores.core.decode.branchResolved 727 # Number of times decode resolved a branch (Count)
board.processor.cores.core.decode.branchMispred 219 # Number of times decode detected a branch misprediction (Count)
board.processor.cores.core.decode.decodedInsts 20421 # Number of instructions handled by decode (Count)
board.processor.cores.core.decode.squashedInsts 1157 # Number of squashed instructions handled by decode (Count)
board.processor.cores.core.executeStats0.numInsts 18745 # Number of executed instructions (Count)
board.processor.cores.core.executeStats0.numNop 0 # Number of nop insts executed (Count)
board.processor.cores.core.executeStats0.numBranches 1934 # Number of branches executed (Count)
board.processor.cores.core.executeStats0.numLoadInsts 2274 # Number of load instructions executed (Count)
board.processor.cores.core.executeStats0.numStoreInsts 1728 # Number of stores executed (Count)
board.processor.cores.core.executeStats0.instRate 0.079132 # Inst execution rate ((Count/Cycle))
board.processor.cores.core.executeStats0.numCCRegReads 8185 # Number of times the CC registers were read (Count)
board.processor.cores.core.executeStats0.numCCRegWrites 5477 # Number of times the CC registers were written (Count)
board.processor.cores.core.executeStats0.numFpRegReads 568 # Number of times the floating registers were read (Count)
board.processor.cores.core.executeStats0.numFpRegWrites 202 # Number of times the floating registers were written (Count)
board.processor.cores.core.executeStats0.numIntRegReads 19319 # Number of times the integer registers were read (Count)
board.processor.cores.core.executeStats0.numIntRegWrites 12468 # Number of times the integer registers were written (Count)
board.processor.cores.core.executeStats0.numMemRefs 4002 # Number of memory refs (Count)
board.processor.cores.core.executeStats0.numMiscRegReads 8466 # Number of times the Misc registers were read (Count)
board.processor.cores.core.executeStats0.numVecAluAccesses 0 # Number of vector alu accesses (Count)
board.processor.cores.core.executeStats0.numDiscardedOps 0 # Number of ops (including micro ops) which were discarded before commit (Count)
board.processor.cores.core.fetch.predictedBranches 729 # Number of branches that fetch has predicted taken (Count)
board.processor.cores.core.fetch.cycles 8598 # Number of cycles fetch has run and was not squashing or blocked (Cycle)
board.processor.cores.core.fetch.squashCycles 1354 # Number of cycles fetch has spent squashing (Cycle)
board.processor.cores.core.fetch.miscStallCycles 41 # Number of cycles fetch has spent waiting on interrupts, or bad addresses, or out of MSHRs (Cycle)
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board.processor.cores.core.fetch.pendingTrapStallCycles 177 # Number of stall cycles due to pending traps (Cycle)
board.processor.cores.core.fetch.cacheLines 1923 # Number of cache lines fetched (Count)
board.processor.cores.core.fetch.icacheSquashes 513 # Number of outstanding Icache misses that were squashed (Count)
board.processor.cores.core.fetch.nisnDist::samples 85403 # Number of instructions fetched each cycle (Total) (Count)
board.processor.cores.core.fetch.nisnDist::mean 0.261033 # Number of instructions fetched each cycle (Total) (Count)
board.processor.cores.core.fetch.nisnDist::stdev 1.351061 # Number of instructions fetched each cycle (Total) (Count)
board.processor.cores.core.fetch.nisnDist::underflows 0 0.00% 0.00% # Number of instructions fetched each cycle (Total) (Count)
board.processor.cores.core.fetch.nisnDist::0 81985 96.00% 96.00% # Number of instructions fetched each cycle (Total) (Count)
board.processor.cores.core.fetch.nisnDist::1 106 0.12% 96.12% # Number of instructions fetched each cycle (Total) (Count)
board.processor.cores.core.fetch.nisnDist::2 148 0.17% 96.30% # Number of instructions fetched each cycle (Total) (Count)
board.processor.cores.core.fetch.nisnDist::3 288 0.34% 96.63% # Number of instructions fetched each cycle (Total) (Count)
board.processor.cores.core.fetch.nisnDist::4 157 0.18% 96.82% # Number of instructions fetched each cycle (Total) (Count)
board.processor.cores.core.fetch.nisnDist::5 312 0.37% 97.18% # Number of instructions fetched each cycle (Total) (Count)
board.processor.cores.core.fetch.nisnDist::6 157 0.18% 97.37% # Number of instructions fetched each cycle (Total) (Count)
board.processor.cores.core.fetch.nisnDist::7 103 0.12% 97.49% # Number of instructions fetched each cycle (Total) (Count)
board.processor.cores.core.fetch.nisnDist::8 2147 2.51% 100.00% # Number of instructions fetched each cycle (Total) (Count)
board.processor.cores.core.fetch.nisnDist::overflows 0 0.00% 100.00% # Number of instructions fetched each cycle (Total) (Count)
board.processor.cores.core.fetch.nisnDist::min_value 0 # Number of instructions fetched each cycle (Total) (Count)
board.processor.cores.core.fetch.nisnDist::max_value 8 # Number of instructions fetched each cycle (Total) (Count)
board.processor.cores.core.fetch.nisnDist::total 85403 # Number of instructions fetched each cycle (Total) (Count)
board.processor.cores.core.fetchStats0.numInsts 11379 # Number of instructions fetched (thread level) (Count)
board.processor.cores.core.fetchStats0.numOps 0 # Number of ops (including micro ops) fetched (thread level) (Count)
board.processor.cores.core.fetchStats0.fetchRate 0.048037 # Number of inst fetches per cycle ((Count/Cycle))
board.processor.cores.core.fetchStats0.numBranches 2574 # Number of branches fetched (Count)
board.processor.cores.core.fetchStats0.branchRate 0.010866 # Number of branch fetches per cycle (Ratio)
board.processor.cores.core.fetchStats0.icacheStallCycles 75910 # ICache total stall cycles (Cycle)
board.processor.cores.core.fetchStats0.numFetchSuspends 0 # Number of times Execute suspended instruction fetching (Count)
board.processor.cores.core.iew.idleCycles 0 # Number of cycles IEW is idle (Cycle)
board.processor.cores.core.iew.squashCycles 458 # Number of cycles IEW is squashing (Cycle)
board.processor.cores.core.iew.blockCycles 813 # Number of cycles IEW is blocking (Cycle)
board.processor.cores.core.iew.unblockCycles 2033 # Number of cycles IEW is unblocking (Cycle)
board.processor.cores.core.iew.dispatchedInsts 20179 # Number of instructions dispatched to IQ (Count)
board.processor.cores.core.iew.dispSquashedInsts 4 # Number of squashed instructions skipped by dispatch (Count)
board.processor.cores.core.iew.dispLoadInsts 2042 # Number of dispatched load instructions (Count)
board.processor.cores.core.iew.dispStoreInsts 1949 # Number of dispatched store instructions (Count)
board.processor.cores.core.iew.dispNonSpecInsts 24 # Number of dispatched non-speculative instructions (Count)
board.processor.cores.core.iew.iqFullEvents 0 # Number of times the IQ has become full, causing a stall (Count)
board.processor.cores.core.iew.lsqFullEvents 2033 # Number of times the LSQ has become full, causing a stall (Count)
board.processor.cores.core.iew.memOrderViolationEvents 6 # Number of memory order violations (Count)
board.processor.cores.core.iew.predictedTakenIncorrect 29 # Number of branches that were predicted taken incorrectly (Count)
board.processor.cores.core.iew.predictedNotTakenIncorrect 434 # Number of branches that were predicted not taken incorrectly (Count)
board.processor.cores.core.iew.branchMispredicts 463 # Number of branch mispredicts detected at execute (Count)
board.processor.cores.core.iew.instsToCommit 17874 # Cumulative count of insts sent to commit (Count)
board.processor.cores.core.iew.writebackCount 17774 # Cumulative count of insts written-back (Count)
board.processor.cores.core.iew.producerInst 11743 # Number of instructions producing a value (Count)
board.processor.cores.core.iew.consumerInst 17091 # Number of instructions consuming a value (Count)
board.processor.cores.core.iew.wbRate 0.075033 # Insts written-back per cycle ((Count/Cycle))
board.processor.cores.core.iew.wbFanout 0.687087 # Average fanout of values written-back ((Count/Count))
board.processor.cores.core.interrupts.clk_domain.clock 5328 # Clock period in ticks (Tick)
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board.processor.cores.core.lsq0.forwLoads 32 # Number of loads that had data forwarded from stores (Count)
board.processor.cores.core.lsq0.squashedLoads 770 # Number of loads squashed (Count)
board.processor.cores.core.lsq0.ignoredResponses 1 # Number of memory responses ignored because the instruction is squashed (Count)
board.processor.cores.core.lsq0.memOrderViolation 6 # Number of memory ordering violations (Count)
board.processor.cores.core.lsq0.squashedStores 598 # Number of stores squashed (Count)
board.processor.cores.core.lsq0.rescheduledLoads 1 # Number of loads that were rescheduled (Count)
board.processor.cores.core.lsq0.blockedByCache 372 # Number of times an access to memory failed due to the cache being blocked (Count)
board.processor.cores.core.lsq0.loadToUse::samples 1272 # Distribution of cycle latency between the first time a load is issued and its completion (Unspecified)
board.processor.cores.core.lsq0.loadToUse::mean 167.172956 # Distribution of cycle latency between the first time a load is issued and its completion (Unspecified)
board.processor.cores.core.lsq0.loadToUse::stdev 84.833432 # Distribution of cycle latency between the first time a load is issued and its completion (Unspecified)
board.processor.cores.core.lsq0.loadToUse::0-9 8 0.63% 0.63% # Distribution of cycle latency between the first time a load is issued and its completion (Unspecified)
board.processor.cores.core.lsq0.loadToUse::40-49 171 13.44% 14.07% # Distribution of cycle latency between the first time a load is issued and its completion (Unspecified)
board.processor.cores.core.lsq0.loadToUse::120-129 38 2.99% 17.06% # Distribution of cycle latency between the first time a load is issued and its completion (Unspecified)
board.processor.cores.core.lsq0.loadToUse::130-139 129 10.14% 27.20% # Distribution of cycle latency between the first time a load is issued and its completion (Unspecified)
board.processor.cores.core.lsq0.loadToUse::140-149 214 16.82% 44.03% # Distribution of cycle latency between the first time a load is issued and its completion (Unspecified)
board.processor.cores.core.lsq0.loadToUse::150-159 93 7.31% 51.34% # Distribution of cycle latency between the first time a load is issued and its completion (Unspecified)
board.processor.cores.core.lsq0.loadToUse::160-169 114 8.96% 60.30% # Distribution of cycle latency between the first time a load is issued and its completion (Unspecified)
board.processor.cores.core.lsq0.loadToUse::170-179 60 4.72% 65.02% # Distribution of cycle latency between the first time a load is issued and its completion (Unspecified)
board.processor.cores.core.lsq0.loadToUse::180-189 64 5.03% 70.05% # Distribution of cycle latency between the first time a load is issued and its completion (Unspecified)
board.processor.cores.core.lsq0.loadToUse::190-199 28 2.20% 72.25% # Distribution of cycle latency between the first time a load is issued and its completion (Unspecified)
board.processor.cores.core.lsq0.loadToUse::200-209 21 1.65% 73.90% # Distribution of cycle latency between the first time a load is issued and its completion (Unspecified)
board.processor.cores.core.lsq0.loadToUse::210-219 73 5.74% 79.64% # Distribution of cycle latency between the first time a load is issued and its completion (Unspecified)
board.processor.cores.core.lsq0.loadToUse::220-229 51 4.01% 83.65% # Distribution of cycle latency between the first time a load is issued and its completion (Unspecified)
board.processor.cores.core.lsq0.loadToUse::230-239 43 3.38% 87.03% # Distribution of cycle latency between the first time a load is issued and its completion (Unspecified)
board.processor.cores.core.lsq0.loadToUse::240-249 47 3.69% 90.72% # Distribution of cycle latency between the first time a load is issued and its completion (Unspecified)
board.processor.cores.core.lsq0.loadToUse::250-259 15 1.18% 91.90% # Distribution of cycle latency between the first time a load is issued and its completion (Unspecified)
board.processor.cores.core.lsq0.loadToUse::260-269 27 2.12% 94.03% # Distribution of cycle latency between the first time a load is issued and its completion (Unspecified)
board.processor.cores.core.lsq0.loadToUse::270-279 20 1.57% 95.60% # Distribution of cycle latency between the first time a load is issued and its completion (Unspecified)
board.processor.cores.core.lsq0.loadToUse::280-289 4 0.31% 95.91% # Distribution of cycle latency between the first time a load is issued and its completion (Unspecified)
board.processor.cores.core.lsq0.loadToUse::290-299 12 0.94% 96.86% # Distribution of cycle latency between the first time a load is issued and its completion (Unspecified)
board.processor.cores.core.lsq0.loadToUse::overflows 40 3.14% 100.00% # Distribution of cycle latency between the first time a load is issued and its completion (Unspecified)
board.processor.cores.core.lsq0.loadToUse::min_value 2 # Distribution of cycle latency between the first time a load is issued and its completion (Unspecified)
board.processor.cores.core.lsq0.loadToUse::max_value 981 # Distribution of cycle latency between the first time a load is issued and its completion (Unspecified)
board.processor.cores.core.lsq0.loadToUse::total 1272 # Distribution of cycle latency between the first time a load is issued and its completion (Unspecified)
board.processor.cores.core.mmu.dtb.rdAccesses 1856 # TLB accesses on read requests (Count)
board.processor.cores.core.mmu.dtb.wrAccesses 1728 # TLB accesses on write requests (Count)
board.processor.cores.core.mmu.dtb.rdMisses 56 # TLB misses on read requests (Count)
board.processor.cores.core.mmu.dtb.wrMisses 18 # TLB misses on write requests (Count)
board.processor.cores.core.mmu.dtb.walker.power_state.pwrStateResidencyTicks::UNDEFINED 78881373 # Cumulative time (in ticks) in various power states (Tick)
board.processor.cores.core.mmu.itb.rdAccesses 0 # TLB accesses on read requests (Count)
board.processor.cores.core.mmu.itb.wrAccesses 1955 # TLB accesses on write requests (Count)
board.processor.cores.core.mmu.itb.rdMisses 0 # TLB misses on read requests (Count)
board.processor.cores.core.mmu.itb.wrMisses 72 # TLB misses on write requests (Count)
board.processor.cores.core.mmu.itb.walker.power_state.pwrStateResidencyTicks::UNDEFINED 78881373 # Cumulative time (in ticks) in various power states (Tick)
board.processor.cores.core.power_state.pwrStateResidencyTicks::ON 78881373 # Cumulative time (in ticks) in various power states (Tick)
board.processor.cores.core.rename.squashCycles 458 # Number of cycles rename is squashing (Cycle)
board.processor.cores.core.rename.idleCycles 75118 # Number of cycles rename is idle (Cycle)
board.processor.cores.core.rename.blockCycles 3105 # Number of cycles rename is blocking (Cycle)
board.processor.cores.core.rename.serializeStallCycles 828 # count of cycles rename stalled for serializing inst (Cycle)
```

```
board.processor.cores.core.rename.runCycles        3181                # Number of cycles rename is running (Cycle)
board.processor.cores.core.rename.unblockCycles        2713                # Number of cycles rename is unblocking (Cycle)
board.processor.cores.core.rename.renamedInsts        20331                # Number of instructions processed by rename (Count)
board.processor.cores.core.rename.ROBFullEvents        26                # Number of times rename has blocked due to ROB full (Count)
board.processor.cores.core.rename.IQFullEvents        4                # Number of times rename has blocked due to IQ full (Count)
board.processor.cores.core.rename.SQFullEvents        2662                # Number of times rename has blocked due to SQ full (Count)
board.processor.cores.core.rename.renamedOperands        32685                # Number of destination operands rename has renamed (Count)
board.processor.cores.core.rename.lookups        65777                # Number of register rename lookups that rename has made (Count)
board.processor.cores.core.rename.intLookups        22302                # Number of integer rename lookups (Count)
board.processor.cores.core.rename.fpLookups        594                # Number of floating rename lookups (Count)
board.processor.cores.core.rename.committedMaps        20771                # Number of HB maps that are committed (Count)
board.processor.cores.core.rename.undoneMaps        11905                # Number of HB maps that are undone due to squashing (Count)
board.processor.cores.core.rename.serializing        5                # count of serializing insts renamed (Count)
board.processor.cores.core.rename.tempSerializing        4                # count of temporary serializing insts renamed (Count)
board.processor.cores.core.rename.skidInsts        374                # count of insts added to the skid buffer (Count)
board.processor.cores.core.rob.reads        103787                # The number of ROB reads (Count)
board.processor.cores.core.rob.writes        41620                # The number of ROB writes (Count)
board.processor.cores.core.thread_0.numInsts        6555                # Number of Instructions committed (Count)
board.processor.cores.core.thread_0.numOps        12992                # Number of Ops committed (Count)
board.processor.cores.core.thread_0.numMemRefs        0                # Number of Memory References (Count)
board.processor.cores.core.workload.numSyscalls        11                # Number of system calls (Count)
board.workload.inst.arm        0                # number of arm instructions executed (Count)
board.workload.inst.quiesce        0                # number of quiesce instructions executed (Count)

---------- End Simulation Statistics   ----------
```

results_ooo_mshr_2:

```
---------- Begin Simulation Statistics ----------
simSeconds                    0.009059          # Number of seconds simulated (Second)
simTicks                      9058914018        # Number of ticks simulated (Tick)
finalTick                     9058914018        # Number of ticks from beginning of simulation (restored from checkpoints and never reset) (Tick)
simFreq                       1000000000000     # The number of ticks per simulated second ((Tick/Second))
hostSeconds                   72.43             # Real time elapsed on the host (Second)
hostTickRate                  125065488         # The number of ticks simulated per host second (ticks/s) ((Tick/Second))
hostMemory                    1214236           # Number of bytes of host memory used (Byte)
simInsts                      36831240          # Number of instructions simulated (Count)
simOps                        56861208          # Number of ops (including micro ops) simulated (Count)
hostInstRate                  508484            # Simulator instruction rate (inst/s) ((Count/Second))
hostOpRate                    785013            # Simulator op (including micro ops) rate (op/s) ((Count/Second))
board.cache_hierarchy.dptw_caches.blockedCycles::no_mshrs      0      # number of cycles access was blocked (Cycle)
board.cache_hierarchy.dptw_caches.blockedCycles::no_targets    0      # number of cycles access was blocked (Cycle)
board.cache_hierarchy.dptw_caches.blockedCauses::no_mshrs      0      # number of times access was blocked (Count)
board.cache_hierarchy.dptw_caches.blockedCauses::no_targets    0      # number of times access was blocked (Count)
board.cache_hierarchy.dptw_caches.avgBlocked::no_mshrs         nan    # average number of cycles each access was blocked ((Cycle/Count))
board.cache_hierarchy.dptw_caches.avgBlocked::no_targets       nan    # average number of cycles each access was blocked ((Cycle/Count))
board.cache_hierarchy.dptw_caches.replacements                 0      # number of replacements (Count)
board.cache_hierarchy.dptw_caches.power_state.pwrStateResidencyTicks::UNDEFINED   9058914018   # Cumulative time (in ticks) in various power states (Tick)
board.cache_hierarchy.dptw_caches.tags.tagsInUse               0      # Average ticks per tags in use ((Tick/Count))
board.cache_hierarchy.dptw_caches.tags.totalRefs               0      # Total number of references to valid blocks. (Count)
board.cache_hierarchy.dptw_caches.tags.sampledRefs             0      # Sample count of references to valid blocks. (Count)
board.cache_hierarchy.dptw_caches.tags.avgRefs                 nan    # Average number of references to valid blocks. ((Count/Count))
board.cache_hierarchy.dptw_caches.tags.warmupTick              0      # The tick when the warmup percentage was hit. (Tick)
board.cache_hierarchy.dptw_caches.tags.tagAccesses             0      # Number of tag accesses (Count)
board.cache_hierarchy.dptw_caches.tags.dataAccesses            0      # Number of data accesses (Count)
board.cache_hierarchy.dptw_caches.tags.power_state.pwrStateResidencyTicks::UNDEFINED   9058914018   # Cumulative time (in ticks) in various power states (Tick)
board.cache_hierarchy.iptw_caches.blockedCycles::no_mshrs      0      # number of cycles access was blocked (Cycle)
board.cache_hierarchy.iptw_caches.blockedCycles::no_targets    0      # number of cycles access was blocked (Cycle)
board.cache_hierarchy.iptw_caches.blockedCauses::no_mshrs      0      # number of times access was blocked (Count)
board.cache_hierarchy.iptw_caches.blockedCauses::no_targets    0      # number of times access was blocked (Count)
board.cache_hierarchy.iptw_caches.avgBlocked::no_mshrs         nan    # average number of cycles each access was blocked ((Cycle/Count))
board.cache_hierarchy.iptw_caches.avgBlocked::no_targets       nan    # average number of cycles each access was blocked ((Cycle/Count))
board.cache_hierarchy.iptw_caches.replacements                 0      # number of replacements (Count)
board.cache_hierarchy.iptw_caches.power_state.pwrStateResidencyTicks::UNDEFINED   9058914018   # Cumulative time (in ticks) in various power states (Tick)
board.cache_hierarchy.iptw_caches.tags.tagsInUse               0      # Average ticks per tags in use ((Tick/Count))
board.cache_hierarchy.iptw_caches.tags.totalRefs               0      # Total number of references to valid blocks. (Count)
board.cache_hierarchy.iptw_caches.tags.sampledRefs             0      # Sample count of references to valid blocks. (Count)
board.cache_hierarchy.iptw_caches.tags.avgRefs                 nan    # Average number of references to valid blocks. ((Count/Count))
board.cache_hierarchy.iptw_caches.tags.warmupTick              0      # The tick when the warmup percentage was hit. (Tick)
board.cache_hierarchy.iptw_caches.tags.tagAccesses             0      # Number of tag accesses (Count)
board.cache_hierarchy.iptw_caches.tags.dataAccesses            0      # Number of data accesses (Count)
board.cache_hierarchy.iptw_caches.tags.power_state.pwrStateResidencyTicks::UNDEFINED   9058914018   # Cumulative time (in ticks) in various power states (Tick)
board.cache_hierarchy.l1dcaches.demandHits::processor.cores.core.data   7319201   # number of demand (read+write) hits (Count)
board.cache_hierarchy.l1dcaches.demandHits::total   7319201   # number of demand (read+write) hits (Count)
```

board.cache_hierarchy.l1dcaches.overallHits::processor.cores.core.data 7319201 # number of overall hits (Count)

board.cache_hierarchy.l1dcaches.overallHits::total 7319201 # number of overall hits (Count)

board.cache_hierarchy.l1dcaches.demandMisses::processor.cores.core.data 133946 # number of demand (read+write) misses (Count)

board.cache_hierarchy.l1dcaches.demandMisses::total 133946 # number of demand (read+write) misses (Count)

board.cache_hierarchy.l1dcaches.overallMisses::processor.cores.core.data 133946 # number of overall misses (Count)

board.cache_hierarchy.l1dcaches.overallMisses::total 133946 # number of overall misses (Count)

board.cache_hierarchy.l1dcaches.demandMissLatency::processor.cores.core.data 5077328585 # number of demand (read+write) miss ticks (Tick)

board.cache_hierarchy.l1dcaches.demandMissLatency::total 5077328585 # number of demand (read+write) miss ticks (Tick)

board.cache_hierarchy.l1dcaches.overallMissLatency::processor.cores.core.data 5077328585 # number of overall miss ticks (Tick)

board.cache_hierarchy.l1dcaches.overallMissLatency::total 5077328585 # number of overall miss ticks (Tick)

board.cache_hierarchy.l1dcaches.demandAccesses::processor.cores.core.data 7453147 # number of demand (read+write) accesses (Count)

board.cache_hierarchy.l1dcaches.demandAccesses::total 7453147 # number of demand (read+write) accesses (Count)

board.cache_hierarchy.l1dcaches.overallAccesses::processor.cores.core.data 7453147 # number of overall (read+write) accesses (Count)

board.cache_hierarchy.l1dcaches.overallAccesses::total 7453147 # number of overall (read+write) accesses (Count)

board.cache_hierarchy.l1dcaches.demandMissRate::processor.cores.core.data 0.017972 # miss rate for demand accesses (Ratio)

board.cache_hierarchy.l1dcaches.demandMissRate::total 0.017972 # miss rate for demand accesses (Ratio)

board.cache_hierarchy.l1dcaches.overallMissRate::processor.cores.core.data 0.017972 # miss rate for overall accesses (Ratio)

board.cache_hierarchy.l1dcaches.overallMissRate::total 0.017972 # miss rate for overall accesses (Ratio)

board.cache_hierarchy.l1dcaches.demandAvgMissLatency::processor.cores.core.data 37905.787295 # average overall miss latency in ticks ((Tick/Count))

board.cache_hierarchy.l1dcaches.demandAvgMissLatency::total 37905.787295 # average overall miss latency in ticks ((Tick/Count))

board.cache_hierarchy.l1dcaches.overallAvgMissLatency::processor.cores.core.data 37905.787295 # average overall miss latency ((Tick/Count))

board.cache_hierarchy.l1dcaches.overallAvgMissLatency::total 37905.787295 # average overall miss latency ((Tick/Count))

board.cache_hierarchy.l1dcaches.blockedCycles::no_mshrs 27960 # number of cycles access was blocked (Cycle)

board.cache_hierarchy.l1dcaches.blockedCycles::no_targets 0 # number of cycles access was blocked (Cycle)

board.cache_hierarchy.l1dcaches.blockedCauses::no_mshrs 142 # number of times access was blocked (Count)

board.cache_hierarchy.l1dcaches.blockedCauses::no_targets 0 # number of times access was blocked (Count)

board.cache_hierarchy.l1dcaches.avgBlocked::no_mshrs 196.901408 # average number of cycles each access was blocked ((Cycle/Count))

board.cache_hierarchy.l1dcaches.avgBlocked::no_targets nan # average number of cycles each access was blocked ((Cycle/Count))

board.cache_hierarchy.l1dcaches.writebacks::writebacks 65717 # number of writebacks (Count)

board.cache_hierarchy.l1dcaches.writebacks::total 65717 # number of writebacks (Count)

board.cache_hierarchy.l1dcaches.demandMshrHits::processor.cores.core.data 1906 # number of demand (read+write) MSHR hits (Count)

board.cache_hierarchy.l1dcaches.demandMshrHits::total 1906 # number of demand (read+write) MSHR hits (Count)

board.cache_hierarchy.l1dcaches.overallMshrHits::processor.cores.core.data 1906 # number of overall MSHR hits (Count)

board.cache_hierarchy.l1dcaches.overallMshrHits::total 1906 # number of overall MSHR hits (Count)

board.cache_hierarchy.l1dcaches.demandMshrMisses::processor.cores.core.data 132040 # number of demand (read+write) MSHR misses (Count)

board.cache_hierarchy.l1dcaches.demandMshrMisses::total 132040 # number of demand (read+write) MSHR misses (Count)

board.cache_hierarchy.l1dcaches.overallMshrMisses::processor.cores.core.data 132040 # number of overall MSHR misses (Count)

board.cache_hierarchy.l1dcaches.overallMshrMisses::total 132040 # number of overall MSHR misses (Count)

board.cache_hierarchy.l1dcaches.demandMshrMissLatency::processor.cores.core.data 4827240923 # number of demand (read+write) MSHR miss ticks (Tick)

board.cache_hierarchy.l1dcaches.demandMshrMissLatency::total 4827240923 # number of demand (read+write) MSHR miss ticks (Tick)

board.cache_hierarchy.l1dcaches.overallMshrMissLatency::processor.cores.core.data 4827240923 # number of overall MSHR miss ticks (Tick)

board.cache_hierarchy.l1dcaches.overallMshrMissLatency::total 4827240923 # number of overall MSHR miss ticks (Tick)

board.cache_hierarchy.l1dcaches.demandMshrMissRate::processor.cores.core.data 0.017716 # mshr miss ratio for demand accesses (Ratio)

board.cache_hierarchy.l1dcaches.demandMshrMissRate::total 0.017716 # mshr miss ratio for demand accesses (Ratio)

board.cache_hierarchy.l1dcaches.overallMshrMissRate::processor.cores.core.data 0.017716 # mshr miss ratio for overall accesses (Ratio)

board.cache_hierarchy.l1dcaches.overallMshrMissRate::total 0.017716 # mshr miss ratio for overall accesses (Ratio)

board.cache_hierarchy.l1dcaches.demandAvgMshrMissLatency::processor.cores.core.data 36558.928529 # average overall mshr miss latency ((Tick/Count))

board.cache_hierarchy.l1dcaches.demandAvgMshrMissLatency::total 36558.928529 # average overall mshr miss latency ((Tick/Count))

board.cache_hierarchy.l1dcaches.overallAvgMshrMissLatency::processor.cores.core.data 36558.928529 # average overall mshr miss latency ((Tick/Count))

board.cache_hierarchy.l1dcaches.overallAvgMshrMissLatency::total 36558.928529 # average overall mshr miss latency ((Tick/Count))

board.cache_hierarchy.l1dcaches.replacements 131529 # number of replacements (Count)

board.cache_hierarchy.l1dcaches.LockedRMWReadReq.hits::processor.cores.core.data 26 # number of LockedRMWReadReq hits (Count)

board.cache_hierarchy.l1dcaches.LockedRMWReadReq.hits::total 26 # number of LockedRMWReadReq hits (Count)

board.cache_hierarchy.l1dcaches.LockedRMWReadReq.misses::processor.cores.core.data 4 # number of LockedRMWReadReq misses (Count)

board.cache_hierarchy.l1dcaches.LockedRMWReadReq.misses::total 4 # number of LockedRMWReadReq misses (Count)

board.cache_hierarchy.l1dcaches.LockedRMWReadReq.missLatency::processor.cores.core.data 529803 # number of LockedRMWReadReq miss ticks (Tick)

board.cache_hierarchy.l1dcaches.LockedRMWReadReq.missLatency::total 529803 # number of LockedRMWReadReq miss ticks (Tick)

board.cache_hierarchy.l1dcaches.LockedRMWReadReq.accesses::processor.cores.core.data 30 # number of LockedRMWReadReq accesses(hits+misses) (Count)

board.cache_hierarchy.l1dcaches.LockedRMWReadReq.accesses::total 30 # number of LockedRMWReadReq accesses(hits+misses) (Count)

board.cache_hierarchy.l1dcaches.LockedRMWReadReq.missRate::processor.cores.core.data 0.133333 # miss rate for LockedRMWReadReq accesses (Ratio)

board.cache_hierarchy.l1dcaches.LockedRMWReadReq.missRate::total 0.133333 # miss rate for LockedRMWReadReq accesses (Ratio)

board.cache_hierarchy.l1dcaches.LockedRMWReadReq.avgMissLatency::processor.cores.core.data 132450.750000 # average LockedRMWReadReq miss latency ((Tick/Count))

board.cache_hierarchy.l1dcaches.LockedRMWReadReq.avgMissLatency::total 132450.750000 # average LockedRMWReadReq miss latency ((Tick/Count))

board.cache_hierarchy.l1dcaches.LockedRMWReadReq.mshrMisses::processor.cores.core.data 4 # number of LockedRMWReadReq MSHR misses (Count)

board.cache_hierarchy.l1dcaches.LockedRMWReadReq.mshrMisses::total 4 # number of LockedRMWReadReq MSHR misses (Count)

board.cache_hierarchy.l1dcaches.LockedRMWReadReq.mshrMissLatency::processor.cores.core.data 1125207 # number of LockedRMWReadReq MSHR miss ticks (Tick)

board.cache_hierarchy.l1dcaches.LockedRMWReadReq.mshrMissLatency::total 1125207 # number of LockedRMWReadReq MSHR miss ticks (Tick)

board.cache_hierarchy.l1dcaches.LockedRMWReadReq.mshrMissRate::processor.cores.core.data 0.133333 # mshr miss rate for LockedRMWReadReq accesses (Ratio)

board.cache_hierarchy.l1dcaches.LockedRMWReadReq.mshrMissRate::total 0.133333 # mshr miss rate for LockedRMWReadReq accesses (Ratio)

board.cache_hierarchy.l1dcaches.LockedRMWReadReq.avgMshrMissLatency::processor.cores.core.data 281301.750000 # average LockedRMWReadReq mshr miss latency ((Tick/Count))

board.cache_hierarchy.l1dcaches.LockedRMWReadReq.avgMshrMissLatency::total 281301.750000 # average LockedRMWReadReq mshr miss latency ((Tick/Count))

board.cache_hierarchy.l1dcaches.LockedRMWWriteReq.hits::processor.cores.core.data 30 # number of LockedRMWWriteReq hits (Count)

board.cache_hierarchy.l1dcaches.LockedRMWWriteReq.hits::total 30 # number of LockedRMWWriteReq hits (Count)

board.cache_hierarchy.l1dcaches.LockedRMWWriteReq.accesses::processor.cores.core.data 30 # number of LockedRMWWriteReq accesses(hits+misses) (Count)

board.cache_hierarchy.l1dcaches.LockedRMWWriteReq.accesses::total 30 # number of LockedRMWWriteReq accesses(hits+misses) (Count)

board.cache_hierarchy.l1dcaches.ReadReq.hits::processor.cores.core.data 3179768 # number of ReadReq hits (Count)

board.cache_hierarchy.l1dcaches.ReadReq.hits::total 3179768 # number of ReadReq hits (Count)

board.cache_hierarchy.l1dcaches.ReadReq.misses::processor.cores.core.data 68243 # number of ReadReq misses (Count)

board.cache_hierarchy.l1dcaches.ReadReq.misses::total 68243 # number of ReadReq misses (Count)

board.cache_hierarchy.l1dcaches.ReadReq.missLatency::processor.cores.core.data 2653647363 # number of ReadReq miss ticks (Tick)

board.cache_hierarchy.l1dcaches.ReadReq.missLatency::total 2653647363 # number of ReadReq miss ticks (Tick)

board.cache_hierarchy.l1dcaches.ReadReq.accesses::processor.cores.core.data 3248011 # number of ReadReq accesses(hits+misses) (Count)

board.cache_hierarchy.l1dcaches.ReadReq.accesses::total 3248011 # number of ReadReq accesses(hits+misses) (Count)

board.cache_hierarchy.l1dcaches.ReadReq.missRate::processor.cores.core.data 0.021011 # miss rate for ReadReq accesses (Ratio)

board.cache_hierarchy.l1dcaches.ReadReq.missRate::total 0.021011 # miss rate for ReadReq accesses (Ratio)

board.cache_hierarchy.l1dcaches.ReadReq.avgMissLatency::processor.cores.core.data 38885.268277 # average ReadReq miss latency ((Tick/Count))

board.cache_hierarchy.l1dcaches.ReadReq.avgMissLatency::total 38885.268277 # average ReadReq miss latency ((Tick/Count))

board.cache_hierarchy.l1dcaches.ReadReq.mshrHits::processor.cores.core.data 1906 # number of ReadReq MSHR hits (Count)

board.cache_hierarchy.l1dcaches.ReadReq.mshrHits::total 1906 # number of ReadReq MSHR hits (Count)

board.cache_hierarchy.l1dcaches.ReadReq.mshrMisses::processor.cores.core.data 66337 # number of ReadReq MSHR misses (Count)

board.cache_hierarchy.l1dcaches.ReadReq.mshrMisses::total 66337 # number of ReadReq MSHR misses (Count)

board.cache_hierarchy.l1dcaches.ReadReq.mshrMissLatency::processor.cores.core.data 2425438800 # number of ReadReq MSHR miss ticks (Tick)

board.cache_hierarchy.l1dcaches.ReadReq.mshrMissLatency::total 2425438800 # number of ReadReq MSHR miss ticks (Tick)

board.cache_hierarchy.l1dcaches.ReadReq.mshrMissRate::processor.cores.core.data 0.020424 # mshr miss rate for ReadReq accesses (Ratio)

board.cache_hierarchy.l1dcaches.ReadReq.mshrMissRate::total 0.020424 # mshr miss rate for ReadReq accesses (Ratio)

board.cache_hierarchy.l1dcaches.ReadReq.avgMshrMissLatency::processor.cores.core.data 36562.382984 # average ReadReq mshr miss latency ((Tick/Count))

board.cache_hierarchy.l1dcaches.ReadReq.avgMshrMissLatency::total 36562.382984 # average ReadReq mshr miss latency ((Tick/Count))

board.cache_hierarchy.l1dcaches.WriteReq.hits::processor.cores.core.data 4139433 # number of WriteReq hits (Count)

board.cache_hierarchy.l1dcaches.WriteReq.hits::total 4139433 # number of WriteReq hits (Count)

board.cache_hierarchy.l1dcaches.WriteReq.misses::processor.cores.core.data 65703 # number of WriteReq misses (Count)

board.cache_hierarchy.l1dcaches.WriteReq.misses::total 65703 # number of WriteReq misses (Count)

board.cache_hierarchy.l1dcaches.WriteReq.missLatency::processor.cores.core.data 2423681222 # number of WriteReq miss ticks (Tick)

board.cache_hierarchy.l1dcaches.WriteReq.missLatency::total 2423681222 # number of WriteReq miss ticks (Tick)

board.cache_hierarchy.l1dcaches.WriteReq.accesses::processor.cores.core.data 4205136 # number of WriteReq accesses(hits+misses) (Count)

board.cache_hierarchy.l1dcaches.WriteReq.accesses::total 4205136 # number of WriteReq accesses(hits+misses) (Count)

board.cache_hierarchy.l1dcaches.WriteReq.missRate::processor.cores.core.data 0.015624 # miss rate for WriteReq accesses (Ratio)

board.cache_hierarchy.l1dcaches.WriteReq.missRate::total 0.015624 # miss rate for WriteReq accesses (Ratio)

board.cache_hierarchy.l1dcaches.WriteReq.avgMissLatency::processor.cores.core.data 36888.440741 # average WriteReq miss latency ((Tick/Count))

board.cache_hierarchy.l1dcaches.WriteReq.avgMissLatency::total 36888.440741 # average WriteReq miss latency ((Tick/Count))

board.cache_hierarchy.l1dcaches.WriteReq.mshrMisses::processor.cores.core.data 65703 # number of WriteReq MSHR misses (Count)

board.cache_hierarchy.l1dcaches.WriteReq.mshrMisses::total 65703 # number of WriteReq MSHR misses (Count)

board.cache_hierarchy.l1dcaches.WriteReq.mshrMissLatency::processor.cores.core.data 2401802123 # number of WriteReq MSHR miss ticks (Tick)

board.cache_hierarchy.l1dcaches.WriteReq.mshrMissLatency::total 2401802123 # number of WriteReq MSHR miss ticks (Tick)

board.cache_hierarchy.l1dcaches.WriteReq.mshrMissRate::processor.cores.core.data 0.015624 # mshr miss rate for WriteReq accesses (Ratio)

board.cache_hierarchy.l1dcaches.WriteReq.mshrMissRate::total 0.015624 # mshr miss rate for WriteReq accesses (Ratio)

board.cache_hierarchy.l1dcaches.WriteReq.avgMshrMissLatency::processor.cores.core.data 36555.440741 # average WriteReq mshr miss latency ((Tick/Count))

board.cache_hierarchy.l1dcaches.WriteReq.avgMshrMissLatency::total 36555.440741 # average WriteReq mshr miss latency ((Tick/Count))

board.cache_hierarchy.l1dcaches.power_state.pwrStateResidencyTicks::UNDEFINED 9058914018 # Cumulative time (in ticks) in various power states (Tick)

board.cache_hierarchy.l1dcaches.prefetcher.demandMshrMisses 132040 # demands not covered by prefetchs (Count)

board.cache_hierarchy.l1dcaches.prefetcher.pfIssued 0 # number of hwpf issued (Count)

board.cache_hierarchy.l1dcaches.prefetcher.pfUseful 0 # number of useful prefetch (Count)

board.cache_hierarchy.l1dcaches.prefetcher.pfUsefulButMiss 0 # number of hit on prefetch but cache block is not in an usable state (Count)

board.cache_hierarchy.l1dcaches.prefetcher.accuracy nan # accuracy of the prefetcher (Count)

board.cache_hierarchy.l1dcaches.prefetcher.coverage 0 # coverage brought by this prefetcher (Count)

board.cache_hierarchy.l1dcaches.prefetcher.pfHitInCache 0 # number of prefetches hitting in cache (Count)

board.cache_hierarchy.l1dcaches.prefetcher.pfHitInMSHR 0 # number of prefetches hitting in a MSHR (Count)

board.cache_hierarchy.l1dcaches.prefetcher.pfHitInWB 0 # number of prefetches hit in the Write Buffer (Count)

board.cache_hierarchy.l1dcaches.prefetcher.pfLate 0 # number of late prefetches (hitting in cache, MSHR or WB) (Count)

board.cache_hierarchy.l1dcaches.prefetcher.pfIdentified 504183 # number of prefetch candidates identified (Count)

board.cache_hierarchy.l1dcaches.prefetcher.pfBufferHit 247992 # number of redundant prefetches already in prefetch queue (Count)

board.cache_hierarchy.l1dcaches.prefetcher.pfInCache 0 # number of redundant prefetches already in cache/mshr dropped (Count)

board.cache_hierarchy.l1dcaches.prefetcher.pfRemovedDemand 129105 # number of prefetches dropped due to a demand for the same address (Count)

board.cache_hierarchy.l1dcaches.prefetcher.pfRemovedFull 4 # number of prefetches dropped due to prefetch queue size (Count)

board.cache_hierarchy.l1dcaches.prefetcher.pfSpanPage 20493 # number of prefetches that crossed the page (Count)

board.cache_hierarchy.l1dcaches.prefetcher.pfUsefulSpanPage 0 # number of prefetches that is useful and crossed the page (Count)

board.cache_hierarchy.l1dcaches.prefetcher.power_state.pwrStateResidencyTicks::UNDEFINED 9058914018 # Cumulative time (in ticks) in various power states (Tick)

board.cache_hierarchy.l1dcaches.tags.tagsInUse 509.489616 # Average ticks per tags in use ((Tick/Count))

board.cache_hierarchy.l1dcaches.tags.totalRefs 7451301 # Total number of references to valid blocks. (Count)

board.cache_hierarchy.l1dcaches.tags.sampledRefs 132041 # Sample count of references to valid blocks. (Count)

board.cache_hierarchy.l1dcaches.tags.avgRefs 56.431722 # Average number of references to valid blocks. ((Count/Count))

board.cache_hierarchy.l1dcaches.tags.warmupTick 249417 # The tick when the warmup percentage was hit. (Tick)

board.cache_hierarchy.l1dcaches.tags.occupancies::processor.cores.core.data 509.489616 # Average occupied blocks per tick, per requestor ((Count/Tick))

board.cache_hierarchy.l1dcaches.tags.avgOccs::processor.cores.core.data 0.995097 # Average percentage of cache occupancy ((Ratio/Tick))

board.cache_hierarchy.l1dcaches.tags.avgOccs::total 0.995097 # Average percentage of cache occupancy ((Ratio/Tick))

board.cache_hierarchy.l1dcaches.tags.occupanciesTaskId::1024 512 # Occupied blocks per task id (Count)

board.cache_hierarchy.l1dcaches.tags.ageTaskId_1024::0 37 # Occupied blocks per task id, per block age (Count)

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board.cache_hierarchy.l1dcaches.tags.ageTaskId_1024::1          474                    # Occupied blocks per task id, per block age (Count)
board.cache_hierarchy.l1dcaches.tags.ageTaskId_1024::3          1                      # Occupied blocks per task id, per block age (Count)
board.cache_hierarchy.l1dcaches.tags.ratioOccsTaskId::1024      1                      # Ratio of occupied blocks and all blocks, per task id (Ratio)
board.cache_hierarchy.l1dcaches.tags.tagAccesses     59757697                          # Number of tag accesses (Count)
board.cache_hierarchy.l1dcaches.tags.dataAccesses    59757697                          # Number of data accesses (Count)
board.cache_hierarchy.l1dcaches.tags.power_state.pwrStateResidencyTicks::UNDEFINED   9058914018                    # Cumulative time (in ticks) in various power states (Tick)
board.cache_hierarchy.l1icaches.demandHits::processor.cores.core.inst     4231623                    # number of demand (read+write) hits (Count)
board.cache_hierarchy.l1icaches.demandHits::total     4231623                    # number of demand (read+write) hits (Count)
board.cache_hierarchy.l1icaches.overallHits::processor.cores.core.inst     4231623                    # number of overall hits (Count)
board.cache_hierarchy.l1icaches.overallHits::total     4231623                    # number of overall hits (Count)
board.cache_hierarchy.l1icaches.demandMisses::processor.cores.core.inst         971                    # number of demand (read+write) misses (Count)
board.cache_hierarchy.l1icaches.demandMisses::total         971                    # number of demand (read+write) misses (Count)
board.cache_hierarchy.l1icaches.overallMisses::processor.cores.core.inst         971                    # number of overall misses (Count)
board.cache_hierarchy.l1icaches.overallMisses::total         971                    # number of overall misses (Count)
board.cache_hierarchy.l1icaches.demandMissLatency::processor.cores.core.inst    111598290                    # number of demand (read+write) miss ticks (Tick)
board.cache_hierarchy.l1icaches.demandMissLatency::total    111598290                    # number of demand (read+write) miss ticks (Tick)
board.cache_hierarchy.l1icaches.overallMissLatency::processor.cores.core.inst    111598290                    # number of overall miss ticks (Tick)
board.cache_hierarchy.l1icaches.overallMissLatency::total    111598290                    # number of overall miss ticks (Tick)
board.cache_hierarchy.l1icaches.demandAccesses::processor.cores.core.inst     4232594                    # number of demand (read+write) accesses (Count)
board.cache_hierarchy.l1icaches.demandAccesses::total     4232594                    # number of demand (read+write) accesses (Count)
board.cache_hierarchy.l1icaches.overallAccesses::processor.cores.core.inst     4232594                    # number of overall (read+write) accesses (Count)
board.cache_hierarchy.l1icaches.overallAccesses::total     4232594                    # number of overall (read+write) accesses (Count)
board.cache_hierarchy.l1icaches.demandMissRate::processor.cores.core.inst     0.000229                    # miss rate for demand accesses (Ratio)
board.cache_hierarchy.l1icaches.demandMissRate::total     0.000229                    # miss rate for demand accesses (Ratio)
board.cache_hierarchy.l1icaches.overallMissRate::processor.cores.core.inst     0.000229                    # miss rate for overall accesses (Ratio)
board.cache_hierarchy.l1icaches.overallMissRate::total     0.000229                    # miss rate for overall accesses (Ratio)
board.cache_hierarchy.l1icaches.demandAvgMissLatency::processor.cores.core.inst 114931.297631                    # average overall miss latency in ticks ((Tick/Count))
board.cache_hierarchy.l1icaches.demandAvgMissLatency::total 114931.297631                    # average overall miss latency in ticks ((Tick/Count))
board.cache_hierarchy.l1icaches.overallAvgMissLatency::processor.cores.core.inst 114931.297631                    # average overall miss latency ((Tick/Count))
board.cache_hierarchy.l1icaches.overallAvgMissLatency::total 114931.297631                    # average overall miss latency ((Tick/Count))
board.cache_hierarchy.l1icaches.blockedCycles::no_mshrs           0                    # number of cycles access was blocked (Cycle)
board.cache_hierarchy.l1icaches.blockedCycles::no_targets           0                    # number of cycles access was blocked (Cycle)
board.cache_hierarchy.l1icaches.blockedCauses::no_mshrs           0                    # number of times access was blocked (Count)
board.cache_hierarchy.l1icaches.blockedCauses::no_targets           0                    # number of times access was blocked (Count)
board.cache_hierarchy.l1icaches.avgBlocked::no_mshrs         nan                    # average number of cycles each access was blocked ((Cycle/Count))
board.cache_hierarchy.l1icaches.avgBlocked::no_targets         nan                    # average number of cycles each access was blocked ((Cycle/Count))
board.cache_hierarchy.l1icaches.demandMshrHits::processor.cores.core.inst         220                    # number of demand (read+write) MSHR hits (Count)
board.cache_hierarchy.l1icaches.demandMshrHits::total         220                    # number of demand (read+write) MSHR hits (Count)
board.cache_hierarchy.l1icaches.overallMshrHits::processor.cores.core.inst         220                    # number of overall MSHR hits (Count)
board.cache_hierarchy.l1icaches.overallMshrHits::total         220                    # number of overall MSHR hits (Count)
board.cache_hierarchy.l1icaches.demandMshrMisses::processor.cores.core.inst         751                    # number of demand (read+write) MSHR misses (Count)
board.cache_hierarchy.l1icaches.demandMshrMisses::total         751                    # number of demand (read+write) MSHR misses (Count)
board.cache_hierarchy.l1icaches.overallMshrMisses::processor.cores.core.inst         751                    # number of overall MSHR misses (Count)
board.cache_hierarchy.l1icaches.overallMshrMisses::total         751                    # number of overall MSHR misses (Count)
board.cache_hierarchy.l1icaches.demandMshrMissLatency::processor.cores.core.inst     89646597                    # number of demand (read+write) MSHR miss ticks (Tick)
board.cache_hierarchy.l1icaches.demandMshrMissLatency::total     89646597                    # number of demand (read+write) MSHR miss ticks (Tick)
board.cache_hierarchy.l1icaches.overallMshrMissLatency::processor.cores.core.inst     89646597                    # number of overall MSHR miss ticks (Tick)
board.cache_hierarchy.l1icaches.overallMshrMissLatency::total     89646597                    # number of overall MSHR miss ticks (Tick)
board.cache_hierarchy.l1icaches.demandMshrMissRate::processor.cores.core.inst     0.000177                    # mshr miss ratio for demand accesses (Ratio)
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board.cache_hierarchy.l1icaches.demandMshrMissRate::total 0.000177 # mshr miss ratio for demand accesses (Ratio)

board.cache_hierarchy.l1icaches.overallMshrMissRate::processor.cores.core.inst 0.000177 # mshr miss ratio for overall accesses (Ratio)

board.cache_hierarchy.l1icaches.overallMshrMissRate::total 0.000177 # mshr miss ratio for overall accesses (Ratio)

board.cache_hierarchy.l1icaches.demandAvgMshrMissLatency::processor.cores.core.inst 119369.636485 # average overall mshr miss latency ((Tick/Count))

board.cache_hierarchy.l1icaches.demandAvgMshrMissLatency::total 119369.636485 # average overall mshr miss latency ((Tick/Count))

board.cache_hierarchy.l1icaches.overallAvgMshrMissLatency::processor.cores.core.inst 119369.636485 # average overall mshr miss latency ((Tick/Count))

board.cache_hierarchy.l1icaches.overallAvgMshrMissLatency::total 119369.636485 # average overall mshr miss latency ((Tick/Count))

board.cache_hierarchy.l1icaches.replacements 237 # number of replacements (Count)

board.cache_hierarchy.l1icaches.ReadReq.hits::processor.cores.core.inst 4231623 # number of ReadReq hits (Count)

board.cache_hierarchy.l1icaches.ReadReq.hits::total 4231623 # number of ReadReq hits (Count)

board.cache_hierarchy.l1icaches.ReadReq.misses::processor.cores.core.inst 971 # number of ReadReq misses (Count)

board.cache_hierarchy.l1icaches.ReadReq.misses::total 971 # number of ReadReq misses (Count)

board.cache_hierarchy.l1icaches.ReadReq.missLatency::processor.cores.core.inst 111598290 # number of ReadReq miss ticks (Tick)

board.cache_hierarchy.l1icaches.ReadReq.missLatency::total 111598290 # number of ReadReq miss ticks (Tick)

board.cache_hierarchy.l1icaches.ReadReq.accesses::processor.cores.core.inst 4232594 # number of ReadReq accesses(hits+misses) (Count)

board.cache_hierarchy.l1icaches.ReadReq.accesses::total 4232594 # number of ReadReq accesses(hits+misses) (Count)

board.cache_hierarchy.l1icaches.ReadReq.missRate::processor.cores.core.inst 0.000229 # miss rate for ReadReq accesses (Ratio)

board.cache_hierarchy.l1icaches.ReadReq.missRate::total 0.000229 # miss rate for ReadReq accesses (Ratio)

board.cache_hierarchy.l1icaches.ReadReq.avgMissLatency::processor.cores.core.inst 114931.297631 # average ReadReq miss latency ((Tick/Count))

board.cache_hierarchy.l1icaches.ReadReq.avgMissLatency::total 114931.297631 # average ReadReq miss latency ((Tick/Count))

board.cache_hierarchy.l1icaches.ReadReq.mshrHits::processor.cores.core.inst 220 # number of ReadReq MSHR hits (Count)

board.cache_hierarchy.l1icaches.ReadReq.mshrHits::total 220 # number of ReadReq MSHR hits (Count)

board.cache_hierarchy.l1icaches.ReadReq.mshrMisses::processor.cores.core.inst 751 # number of ReadReq MSHR misses (Count)

board.cache_hierarchy.l1icaches.ReadReq.mshrMisses::total 751 # number of ReadReq MSHR misses (Count)

board.cache_hierarchy.l1icaches.ReadReq.mshrMissLatency::processor.cores.core.inst 89646597 # number of ReadReq MSHR miss ticks (Tick)

board.cache_hierarchy.l1icaches.ReadReq.mshrMissLatency::total 89646597 # number of ReadReq MSHR miss ticks (Tick)

board.cache_hierarchy.l1icaches.ReadReq.mshrMissRate::processor.cores.core.inst 0.000177 # mshr miss rate for ReadReq accesses (Ratio)

board.cache_hierarchy.l1icaches.ReadReq.mshrMissRate::total 0.000177 # mshr miss rate for ReadReq accesses (Ratio)

board.cache_hierarchy.l1icaches.ReadReq.avgMshrMissLatency::processor.cores.core.inst 119369.636485 # average ReadReq mshr miss latency ((Tick/Count))

board.cache_hierarchy.l1icaches.ReadReq.avgMshrMissLatency::total 119369.636485 # average ReadReq mshr miss latency ((Tick/Count))

board.cache_hierarchy.l1icaches.power_state.pwrStateResidencyTicks::UNDEFINED 9058914018 # Cumulative time (in ticks) in various power states (Tick)

board.cache_hierarchy.l1icaches.prefetcher.demandMshrMisses 751 # demands not covered by prefetchs (Count)

board.cache_hierarchy.l1icaches.prefetcher.pfIssued 0 # number of hwpf issued (Count)

board.cache_hierarchy.l1icaches.prefetcher.pfUseful 0 # number of useful prefetch (Count)

board.cache_hierarchy.l1icaches.prefetcher.pfUsefulButMiss 0 # number of hit on prefetch but cache block is not in an usable state (Count)

board.cache_hierarchy.l1icaches.prefetcher.accuracy nan # accuracy of the prefetcher (Count)

board.cache_hierarchy.l1icaches.prefetcher.coverage 0 # coverage brought by this prefetcher (Count)

board.cache_hierarchy.l1icaches.prefetcher.pfHitInCache 0 # number of prefetches hitting in cache (Count)

board.cache_hierarchy.l1icaches.prefetcher.pfHitInMSHR 0 # number of prefetches hitting in a MSHR (Count)

board.cache_hierarchy.l1icaches.prefetcher.pfHitInWB 0 # number of prefetches hit in the Write Buffer (Count)

board.cache_hierarchy.l1icaches.prefetcher.pfLate 0 # number of late prefetches (hitting in cache, MSHR or WB) (Count)

board.cache_hierarchy.l1icaches.prefetcher.pfIdentified 0 # number of prefetch candidates identified (Count)

board.cache_hierarchy.l1icaches.prefetcher.pfBufferHit 0 # number of redundant prefetches already in prefetch queue (Count)

board.cache_hierarchy.l1icaches.prefetcher.pfInCache 0 # number of redundant prefetches already in cache/mshr dropped (Count)

board.cache_hierarchy.l1icaches.prefetcher.pfRemovedDemand 0 # number of prefetches dropped due to a demand for the same address (Count)

board.cache_hierarchy.l1icaches.prefetcher.pfRemovedFull 0 # number of prefetches dropped due to prefetch queue size (Count)

board.cache_hierarchy.l1icaches.prefetcher.pfSpanPage 0 # number of prefetches that crossed the page (Count)

board.cache_hierarchy.l1icaches.prefetcher.pfUsefulSpanPage 0 # number of prefetches that is useful and crossed the page (Count)

board.cache_hierarchy.l1icaches.prefetcher.power_state.pwrStateResidencyTicks::UNDEFINED 9058914018 # Cumulative time (in ticks) in various power states (Tick)

board.cache_hierarchy.l1icaches.tags.tagsInUse 469.624057 # Average ticks per tags in use ((Tick/Count))

board.cache_hierarchy.l1icaches.tags.totalRefs 4232374 # Total number of references to valid blocks. (Count)

board.cache_hierarchy.l1icaches.tags.sampledRefs 751 # Sample count of references to valid blocks. (Count)

board.cache_hierarchy.l1icaches.tags.avgRefs 5635.651132 # Average number of references to valid blocks. ((Count/Count))

board.cache_hierarchy.l1icaches.tags.warmupTick 123543 # The tick when the warmup percentage was hit. (Tick)

board.cache_hierarchy.l1icaches.tags.occupancies::processor.cores.core.inst 469.624057 # Average occupied blocks per tick, per requestor ((Count/Tick))

board.cache_hierarchy.l1icaches.tags.avgOccs::processor.cores.core.inst 0.917234 # Average percentage of cache occupancy ((Ratio/Tick))

board.cache_hierarchy.l1icaches.tags.avgOccs::total 0.917234 # Average percentage of cache occupancy ((Ratio/Tick))

board.cache_hierarchy.l1icaches.tags.occupanciesTaskId::1024 511 # Occupied blocks per task id (Count)

board.cache_hierarchy.l1icaches.tags.ageTaskId_1024::0 103 # Occupied blocks per task id, per block age (Count)

board.cache_hierarchy.l1icaches.tags.ageTaskId_1024::1 100 # Occupied blocks per task id, per block age (Count)

board.cache_hierarchy.l1icaches.tags.ageTaskId_1024::3 308 # Occupied blocks per task id, per block age (Count)

board.cache_hierarchy.l1icaches.tags.ratioOccsTaskId::1024 0.998047 # Ratio of occupied blocks and all blocks, per task id (Ratio)

board.cache_hierarchy.l1icaches.tags.tagAccesses 33861503 # Number of tag accesses (Count)

board.cache_hierarchy.l1icaches.tags.dataAccesses 33861503 # Number of data accesses (Count)

board.cache_hierarchy.l1icaches.tags.power_state.pwrStateResidencyTicks::UNDEFINED 9058914018 # Cumulative time (in ticks) in various power states (Tick)

board.cache_hierarchy.l2bus.transDist::ReadResp 67088 # Transaction distribution (Count)

board.cache_hierarchy.l2bus.transDist::WritebackDirty 131412 # Transaction distribution (Count)

board.cache_hierarchy.l2bus.transDist::CleanEvict 129036 # Transaction distribution (Count)

board.cache_hierarchy.l2bus.transDist::HardPFReq 135267 # Transaction distribution (Count)

board.cache_hierarchy.l2bus.transDist::UpgradeReq 3 # Transaction distribution (Count)

board.cache_hierarchy.l2bus.transDist::UpgradeResp 3 # Transaction distribution (Count)

board.cache_hierarchy.l2bus.transDist::ReadExReq 65704 # Transaction distribution (Count)

board.cache_hierarchy.l2bus.transDist::ReadExResp 65704 # Transaction distribution (Count)

board.cache_hierarchy.l2bus.transDist::ReadSharedReq 67088 # Transaction distribution (Count)

board.cache_hierarchy.l2bus.pktCount_board.cache_hierarchy.l1icaches.mem_side_port::board.cache_hierarchy.l2cache.cpu_side_port 1736 # Packet count per connected requestor and responder (Count)

board.cache_hierarchy.l2bus.pktCount_board.cache_hierarchy.l1dcaches.mem_side_port::board.cache_hierarchy.l2cache.cpu_side_port 395617 # Packet count per connected requestor and responder (Count)

board.cache_hierarchy.l2bus.pktCount::total 397353 # Packet count per connected requestor and responder (Count)

board.cache_hierarchy.l2bus.pktSize_board.cache_hierarchy.l1icaches.mem_side_port::board.cache_hierarchy.l2cache.cpu_side_port 47872 # Cumulative packet size per connected requestor and responder (Byte)

board.cache_hierarchy.l2bus.pktSize_board.cache_hierarchy.l1dcaches.mem_side_port::board.cache_hierarchy.l2cache.cpu_side_port 12656512 # Cumulative packet size per connected requestor and responder (Byte)

board.cache_hierarchy.l2bus.pktSize::total 12704384 # Cumulative packet size per connected requestor and responder (Byte)

board.cache_hierarchy.l2bus.snoops 263952 # Total snoops (Count)

board.cache_hierarchy.l2bus.snoopTraffic 4204672 # Total snoop traffic (Byte)

board.cache_hierarchy.l2bus.snoopFanout::samples 396744 # Request fanout histogram (Count)

board.cache_hierarchy.l2bus.snoopFanout::mean 0.001217 # Request fanout histogram (Count)

board.cache_hierarchy.l2bus.snoopFanout::stdev 0.034870 # Request fanout histogram (Count)

board.cache_hierarchy.l2bus.snoopFanout::underflows 0 0.00% 0.00% # Request fanout histogram (Count)

board.cache_hierarchy.l2bus.snoopFanout::0 396261 99.88% 99.88% # Request fanout histogram (Count)

board.cache_hierarchy.l2bus.snoopFanout::1 483 0.12% 100.00% # Request fanout histogram (Count)

board.cache_hierarchy.l2bus.snoopFanout::2 0 0.00% 100.00% # Request fanout histogram (Count)

board.cache_hierarchy.l2bus.snoopFanout::3 0 0.00% 100.00% # Request fanout histogram (Count)

board.cache_hierarchy.l2bus.snoopFanout::4 0 0.00% 100.00% # Request fanout histogram (Count)

board.cache_hierarchy.l2bus.snoopFanout::overflows 0 0.00% 100.00% # Request fanout histogram (Count)

board.cache_hierarchy.l2bus.snoopFanout::min_value 0 # Request fanout histogram (Count)

board.cache_hierarchy.l2bus.snoopFanout::max_value 1 # Request fanout histogram (Count)

board.cache_hierarchy.l2bus.snoopFanout::total 396744 # Request fanout histogram (Count)

board.cache_hierarchy.l2bus.power_state.pwrStateResidencyTicks::UNDEFINED 9058914018 # Cumulative time (in ticks) in various power states (Tick)

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board.cache_hierarchy.l2bus.reqLayer0.occupancy    131866335                    # Layer occupancy (ticks) (Tick)
board.cache_hierarchy.l2bus.reqLayer0.utilization          0.0                    # Layer utilization (Ratio)
board.cache_hierarchy.l2bus.respLayer0.occupancy       750249                    # Layer occupancy (ticks) (Tick)
board.cache_hierarchy.l2bus.respLayer0.utilization          0.0                    # Layer utilization (Ratio)
board.cache_hierarchy.l2bus.respLayer1.occupancy    131909958                    # Layer occupancy (ticks) (Tick)
board.cache_hierarchy.l2bus.respLayer1.utilization          0.0                    # Layer utilization (Ratio)
board.cache_hierarchy.l2bus.snoop_filter.totRequests       264561                    # Total number of requests made to the snoop filter. (Count)
board.cache_hierarchy.l2bus.snoop_filter.hitSingleRequests       131767                    # Number of requests hitting in the snoop filter with a single holder of the requested data. (Count)
board.cache_hierarchy.l2bus.snoop_filter.hitMultiRequests            6                    # Number of requests hitting in the snoop filter with multiple (>1) holders of the requested data. (Count)
board.cache_hierarchy.l2bus.snoop_filter.totSnoops          476                    # Total number of snoops made to the snoop filter. (Count)
board.cache_hierarchy.l2bus.snoop_filter.hitSingleSnoops          476                    # Number of snoops hitting in the snoop filter with a single holder of the requested data. (Count)
board.cache_hierarchy.l2bus.snoop_filter.hitMultiSnoops            0                    # Number of snoops hitting in the snoop filter with multiple (>1) holders of the requested data. (Count)
board.cache_hierarchy.l2cache.demandHits::processor.cores.core.data       128046                    # number of demand (read+write) hits (Count)
board.cache_hierarchy.l2cache.demandHits::total       128046                    # number of demand (read+write) hits (Count)
board.cache_hierarchy.l2cache.overallHits::processor.cores.core.data       128046                    # number of overall hits (Count)
board.cache_hierarchy.l2cache.overallHits::total       128046                    # number of overall hits (Count)
board.cache_hierarchy.l2cache.demandMisses::processor.cores.core.inst          748                    # number of demand (read+write) misses (Count)
board.cache_hierarchy.l2cache.demandMisses::processor.cores.core.data         3995                    # number of demand (read+write) misses (Count)
board.cache_hierarchy.l2cache.demandMisses::total         4743                    # number of demand (read+write) misses (Count)
board.cache_hierarchy.l2cache.overallMisses::processor.cores.core.inst          748                    # number of overall misses (Count)
board.cache_hierarchy.l2cache.overallMisses::processor.cores.core.data         3995                    # number of overall misses (Count)
board.cache_hierarchy.l2cache.overallMisses::total         4743                    # number of overall misses (Count)
board.cache_hierarchy.l2cache.demandMissLatency::processor.cores.core.inst     89141769                    # number of demand (read+write) miss ticks (Tick)
board.cache_hierarchy.l2cache.demandMissLatency::processor.cores.core.data    433147086                    # number of demand (read+write) miss ticks (Tick)
board.cache_hierarchy.l2cache.demandMissLatency::total    522288855                    # number of demand (read+write) miss ticks (Tick)
board.cache_hierarchy.l2cache.overallMissLatency::processor.cores.core.inst     89141769                    # number of overall miss ticks (Tick)
board.cache_hierarchy.l2cache.overallMissLatency::processor.cores.core.data    433147086                    # number of overall miss ticks (Tick)
board.cache_hierarchy.l2cache.overallMissLatency::total    522288855                    # number of overall miss ticks (Tick)
board.cache_hierarchy.l2cache.demandAccesses::processor.cores.core.inst          748                    # number of demand (read+write) accesses (Count)
board.cache_hierarchy.l2cache.demandAccesses::processor.cores.core.data       132041                    # number of demand (read+write) accesses (Count)
board.cache_hierarchy.l2cache.demandAccesses::total       132789                    # number of demand (read+write) accesses (Count)
board.cache_hierarchy.l2cache.overallAccesses::processor.cores.core.inst          748                    # number of overall (read+write) accesses (Count)
board.cache_hierarchy.l2cache.overallAccesses::processor.cores.core.data       132041                    # number of overall (read+write) accesses (Count)
board.cache_hierarchy.l2cache.overallAccesses::total       132789                    # number of overall (read+write) accesses (Count)
board.cache_hierarchy.l2cache.demandMissRate::processor.cores.core.inst            1                    # miss rate for demand accesses (Ratio)
board.cache_hierarchy.l2cache.demandMissRate::processor.cores.core.data     0.030256                    # miss rate for demand accesses (Ratio)
board.cache_hierarchy.l2cache.demandMissRate::total     0.035718                    # miss rate for demand accesses (Ratio)
board.cache_hierarchy.l2cache.overallMissRate::processor.cores.core.inst            1                    # miss rate for overall accesses (Ratio)
board.cache_hierarchy.l2cache.overallMissRate::processor.cores.core.data     0.030256                    # miss rate for overall accesses (Ratio)
board.cache_hierarchy.l2cache.overallMissRate::total     0.035718                    # miss rate for overall accesses (Ratio)
board.cache_hierarchy.l2cache.demandAvgMissLatency::processor.cores.core.inst 119173.487968                    # average overall miss latency in ticks ((Tick/Count))
board.cache_hierarchy.l2cache.demandAvgMissLatency::processor.cores.core.data 108422.299374                    # average overall miss latency in ticks ((Tick/Count))
board.cache_hierarchy.l2cache.demandAvgMissLatency::total 110117.827324                    # average overall miss latency in ticks ((Tick/Count))
board.cache_hierarchy.l2cache.overallAvgMissLatency::processor.cores.core.inst 119173.487968                    # average overall miss latency ((Tick/Count))
board.cache_hierarchy.l2cache.overallAvgMissLatency::processor.cores.core.data 108422.299374                    # average overall miss latency ((Tick/Count))
board.cache_hierarchy.l2cache.overallAvgMissLatency::total 110117.827324                    # average overall miss latency ((Tick/Count))
board.cache_hierarchy.l2cache.blockedCycles::no_mshrs            0                    # number of cycles access was blocked (Cycle)
board.cache_hierarchy.l2cache.blockedCycles::no_targets            0                    # number of cycles access was blocked (Cycle)
board.cache_hierarchy.l2cache.blockedCauses::no_mshrs            0                    # number of times access was blocked (Count)
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board.cache_hierarchy.l2cache.blockedCauses::no_targets 0 # number of times access was blocked (Count)

board.cache_hierarchy.l2cache.avgBlocked::no_mshrs nan # average number of cycles each access was blocked ((Cycle/Count))

board.cache_hierarchy.l2cache.avgBlocked::no_targets nan # average number of cycles each access was blocked ((Cycle/Count))

board.cache_hierarchy.l2cache.writebacks::writebacks 65695 # number of writebacks (Count)

board.cache_hierarchy.l2cache.writebacks::total 65695 # number of writebacks (Count)

board.cache_hierarchy.l2cache.demandMshrHits::processor.cores.core.data 1059 # number of demand (read+write) MSHR hits (Count)

board.cache_hierarchy.l2cache.demandMshrHits::total 1059 # number of demand (read+write) MSHR hits (Count)

board.cache_hierarchy.l2cache.overallMshrHits::processor.cores.core.data 1059 # number of overall MSHR hits (Count)

board.cache_hierarchy.l2cache.overallMshrHits::total 1059 # number of overall MSHR hits (Count)

board.cache_hierarchy.l2cache.demandMshrMisses::processor.cores.core.inst 748 # number of demand (read+write) MSHR misses (Count)

board.cache_hierarchy.l2cache.demandMshrMisses::processor.cores.core.data 2936 # number of demand (read+write) MSHR misses (Count)

board.cache_hierarchy.l2cache.demandMshrMisses::total 3684 # number of demand (read+write) MSHR misses (Count)

board.cache_hierarchy.l2cache.overallMshrMisses::cache_hierarchy.l2cache.prefetcher 129094 # number of overall MSHR misses (Count)

board.cache_hierarchy.l2cache.overallMshrMisses::processor.cores.core.inst 748 # number of overall MSHR misses (Count)

board.cache_hierarchy.l2cache.overallMshrMisses::processor.cores.core.data 2936 # number of overall MSHR misses (Count)

board.cache_hierarchy.l2cache.overallMshrMisses::total 132778 # number of overall MSHR misses (Count)

board.cache_hierarchy.l2cache.demandMshrMissLatency::processor.cores.core.inst 64233369 # number of demand (read+write) MSHR miss ticks (Tick)

board.cache_hierarchy.l2cache.demandMshrMissLatency::processor.cores.core.data 239000427 # number of demand (read+write) MSHR miss ticks (Tick)

board.cache_hierarchy.l2cache.demandMshrMissLatency::total 303233796 # number of demand (read+write) MSHR miss ticks (Tick)

board.cache_hierarchy.l2cache.overallMshrMissLatency::cache_hierarchy.l2cache.prefetcher 7541779191 # number of overall MSHR miss ticks (Tick)

board.cache_hierarchy.l2cache.overallMshrMissLatency::processor.cores.core.inst 64233369 # number of overall MSHR miss ticks (Tick)

board.cache_hierarchy.l2cache.overallMshrMissLatency::processor.cores.core.data 239000427 # number of overall MSHR miss ticks (Tick)

board.cache_hierarchy.l2cache.overallMshrMissLatency::total 7845012987 # number of overall MSHR miss ticks (Tick)

board.cache_hierarchy.l2cache.demandMshrMissRate::processor.cores.core.inst 1 # mshr miss ratio for demand accesses (Ratio)

board.cache_hierarchy.l2cache.demandMshrMissRate::processor.cores.core.data 0.022236 # mshr miss ratio for demand accesses (Ratio)

board.cache_hierarchy.l2cache.demandMshrMissRate::total 0.027743 # mshr miss ratio for demand accesses (Ratio)

board.cache_hierarchy.l2cache.overallMshrMissRate::cache_hierarchy.l2cache.prefetcher inf # mshr miss ratio for overall accesses (Ratio)

board.cache_hierarchy.l2cache.overallMshrMissRate::processor.cores.core.inst 1 # mshr miss ratio for overall accesses (Ratio)

board.cache_hierarchy.l2cache.overallMshrMissRate::processor.cores.core.data 0.022236 # mshr miss ratio for overall accesses (Ratio)

board.cache_hierarchy.l2cache.overallMshrMissRate::total 0.999917 # mshr miss ratio for overall accesses (Ratio)

board.cache_hierarchy.l2cache.demandAvgMshrMissLatency::processor.cores.core.inst 85873.487968 # average overall mshr miss latency ((Tick/Count))

board.cache_hierarchy.l2cache.demandAvgMshrMissLatency::processor.cores.core.data 81403.415191 # average overall mshr miss latency ((Tick/Count))

board.cache_hierarchy.l2cache.demandAvgMshrMissLatency::total 82311.019544 # average overall mshr miss latency ((Tick/Count))

board.cache_hierarchy.l2cache.overallAvgMshrMissLatency::cache_hierarchy.l2cache.prefetcher 58420.834361 # average overall mshr miss latency ((Tick/Count))

board.cache_hierarchy.l2cache.overallAvgMshrMissLatency::processor.cores.core.inst 85873.487968 # average overall mshr miss latency ((Tick/Count))

board.cache_hierarchy.l2cache.overallAvgMshrMissLatency::processor.cores.core.data 81403.415191 # average overall mshr miss latency ((Tick/Count))

board.cache_hierarchy.l2cache.overallAvgMshrMissLatency::total 59083.680934 # average overall mshr miss latency ((Tick/Count))

board.cache_hierarchy.l2cache.replacements 128682 # number of replacements (Count)

board.cache_hierarchy.l2cache.CleanEvict.mshrMisses::writebacks 165 # number of CleanEvict MSHR misses (Count)

board.cache_hierarchy.l2cache.CleanEvict.mshrMisses::total 165 # number of CleanEvict MSHR misses (Count)

board.cache_hierarchy.l2cache.CleanEvict.mshrMissRate::writebacks inf # mshr miss rate for CleanEvict accesses (Ratio)

board.cache_hierarchy.l2cache.CleanEvict.mshrMissRate::total inf # mshr miss rate for CleanEvict accesses (Ratio)

board.cache_hierarchy.l2cache.HardPFReq.mshrMisses::cache_hierarchy.l2cache.prefetcher 129094 # number of HardPFReq MSHR misses (Count)

board.cache_hierarchy.l2cache.HardPFReq.mshrMisses::total 129094 # number of HardPFReq MSHR misses (Count)

board.cache_hierarchy.l2cache.HardPFReq.mshrMissLatency::cache_hierarchy.l2cache.prefetcher 7541779191 # number of HardPFReq MSHR miss ticks (Tick)

board.cache_hierarchy.l2cache.HardPFReq.mshrMissLatency::total 7541779191 # number of HardPFReq MSHR miss ticks (Tick)

board.cache_hierarchy.l2cache.HardPFReq.mshrMissRate::cache_hierarchy.l2cache.prefetcher inf # mshr miss rate for HardPFReq accesses (Ratio)

board.cache_hierarchy.l2cache.HardPFReq.mshrMissRate::total inf # mshr miss rate for HardPFReq accesses (Ratio)

board.cache_hierarchy.l2cache.HardPFReq.avgMshrMissLatency::cache_hierarchy.l2cache.prefetcher 58420.834361 # average HardPFReq mshr miss latency ((Tick/Count))

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board.cache_hierarchy.l2cache.HardPFReq.avgMshrMissLatency::total 58420.834361 # average HardPFReq mshr miss latency ((Tick/Count))
board.cache_hierarchy.l2cache.ReadExReq.hits::processor.cores.core.data 63886 # number of ReadExReq hits (Count)
board.cache_hierarchy.l2cache.ReadExReq.hits::total 63886 # number of ReadExReq hits (Count)
board.cache_hierarchy.l2cache.ReadExReq.misses::processor.cores.core.data 1818 # number of ReadExReq misses (Count)
board.cache_hierarchy.l2cache.ReadExReq.misses::total 1818 # number of ReadExReq misses (Count)
board.cache_hierarchy.l2cache.ReadExReq.missLatency::processor.cores.core.data 209788335 # number of ReadExReq miss ticks (Tick)
board.cache_hierarchy.l2cache.ReadExReq.missLatency::total 209788335 # number of ReadExReq miss ticks (Tick)
board.cache_hierarchy.l2cache.ReadExReq.accesses::processor.cores.core.data 65704 # number of ReadExReq accesses(hits+misses) (Count)
board.cache_hierarchy.l2cache.ReadExReq.accesses::total 65704 # number of ReadExReq accesses(hits+misses) (Count)
board.cache_hierarchy.l2cache.ReadExReq.missRate::processor.cores.core.data 0.027670 # miss rate for ReadExReq accesses (Ratio)
board.cache_hierarchy.l2cache.ReadExReq.missRate::total 0.027670 # miss rate for ReadExReq accesses (Ratio)
board.cache_hierarchy.l2cache.ReadExReq.avgMissLatency::processor.cores.core.data 115395.123762 # average ReadExReq miss latency ((Tick/Count))
board.cache_hierarchy.l2cache.ReadExReq.avgMissLatency::total 115395.123762 # average ReadExReq miss latency ((Tick/Count))
board.cache_hierarchy.l2cache.ReadExReq.mshrHits::processor.cores.core.data 682 # number of ReadExReq MSHR hits (Count)
board.cache_hierarchy.l2cache.ReadExReq.mshrHits::total 682 # number of ReadExReq MSHR hits (Count)
board.cache_hierarchy.l2cache.ReadExReq.mshrMisses::processor.cores.core.data 1136 # number of ReadExReq MSHR misses (Count)
board.cache_hierarchy.l2cache.ReadExReq.mshrMisses::total 1136 # number of ReadExReq MSHR misses (Count)
board.cache_hierarchy.l2cache.ReadExReq.mshrMissLatency::processor.cores.core.data 95545026 # number of ReadExReq MSHR miss ticks (Tick)
board.cache_hierarchy.l2cache.ReadExReq.mshrMissLatency::total 95545026 # number of ReadExReq MSHR miss ticks (Tick)
board.cache_hierarchy.l2cache.ReadExReq.mshrMissRate::processor.cores.core.data 0.017290 # mshr miss rate for ReadExReq accesses (Ratio)
board.cache_hierarchy.l2cache.ReadExReq.mshrMissRate::total 0.017290 # mshr miss rate for ReadExReq accesses (Ratio)
board.cache_hierarchy.l2cache.ReadExReq.avgMshrMissLatency::processor.cores.core.data 84106.536972 # average ReadExReq mshr miss latency ((Tick/Count))
board.cache_hierarchy.l2cache.ReadExReq.avgMshrMissLatency::total 84106.536972 # average ReadExReq mshr miss latency ((Tick/Count))
board.cache_hierarchy.l2cache.ReadSharedReq.hits::processor.cores.core.data 64160 # number of ReadSharedReq hits (Count)
board.cache_hierarchy.l2cache.ReadSharedReq.hits::total 64160 # number of ReadSharedReq hits (Count)
board.cache_hierarchy.l2cache.ReadSharedReq.misses::processor.cores.core.inst 748 # number of ReadSharedReq misses (Count)
board.cache_hierarchy.l2cache.ReadSharedReq.misses::processor.cores.core.data 2177 # number of ReadSharedReq misses (Count)
board.cache_hierarchy.l2cache.ReadSharedReq.misses::total 2925 # number of ReadSharedReq misses (Count)
board.cache_hierarchy.l2cache.ReadSharedReq.missLatency::processor.cores.core.inst 89141769 # number of ReadSharedReq miss ticks (Tick)
board.cache_hierarchy.l2cache.ReadSharedReq.missLatency::processor.cores.core.data 223358751 # number of ReadSharedReq miss ticks (Tick)
board.cache_hierarchy.l2cache.ReadSharedReq.missLatency::total 312500520 # number of ReadSharedReq miss ticks (Tick)
board.cache_hierarchy.l2cache.ReadSharedReq.accesses::processor.cores.core.inst 748 # number of ReadSharedReq accesses(hits+misses) (Count)
board.cache_hierarchy.l2cache.ReadSharedReq.accesses::processor.cores.core.data 66337 # number of ReadSharedReq accesses(hits+misses) (Count)
board.cache_hierarchy.l2cache.ReadSharedReq.accesses::total 67085 # number of ReadSharedReq accesses(hits+misses) (Count)
board.cache_hierarchy.l2cache.ReadSharedReq.missRate::processor.cores.core.inst 1 # miss rate for ReadSharedReq accesses (Ratio)
board.cache_hierarchy.l2cache.ReadSharedReq.missRate::processor.cores.core.data 0.032817 # miss rate for ReadSharedReq accesses (Ratio)
board.cache_hierarchy.l2cache.ReadSharedReq.missRate::total 0.043601 # miss rate for ReadSharedReq accesses (Ratio)
board.cache_hierarchy.l2cache.ReadSharedReq.avgMissLatency::processor.cores.core.inst 119173.487968 # average ReadSharedReq miss latency ((Tick/Count))
board.cache_hierarchy.l2cache.ReadSharedReq.avgMissLatency::processor.cores.core.data 102599.334405 # average ReadSharedReq miss latency ((Tick/Count))
board.cache_hierarchy.l2cache.ReadSharedReq.avgMissLatency::total 106837.784615 # average ReadSharedReq miss latency ((Tick/Count))
board.cache_hierarchy.l2cache.ReadSharedReq.mshrHits::processor.cores.core.data 377 # number of ReadSharedReq MSHR hits (Count)
board.cache_hierarchy.l2cache.ReadSharedReq.mshrHits::total 377 # number of ReadSharedReq MSHR hits (Count)
board.cache_hierarchy.l2cache.ReadSharedReq.mshrMisses::processor.cores.core.inst 748 # number of ReadSharedReq MSHR misses (Count)
board.cache_hierarchy.l2cache.ReadSharedReq.mshrMisses::processor.cores.core.data 1800 # number of ReadSharedReq MSHR misses (Count)
board.cache_hierarchy.l2cache.ReadSharedReq.mshrMisses::total 2548 # number of ReadSharedReq MSHR misses (Count)
board.cache_hierarchy.l2cache.ReadSharedReq.mshrMissLatency::processor.cores.core.inst 64233369 # number of ReadSharedReq MSHR miss ticks (Tick)
board.cache_hierarchy.l2cache.ReadSharedReq.mshrMissLatency::processor.cores.core.data 143455401 # number of ReadSharedReq MSHR miss ticks (Tick)
board.cache_hierarchy.l2cache.ReadSharedReq.mshrMissLatency::total 207688770 # number of ReadSharedReq MSHR miss ticks (Tick)
board.cache_hierarchy.l2cache.ReadSharedReq.mshrMissRate::processor.cores.core.inst 1 # mshr miss rate for ReadSharedReq accesses (Ratio)
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board.cache_hierarchy.l2cache.ReadSharedReq.mshrMissRate::processor.cores.core.data     0.027134                    # mshr miss rate for ReadSharedReq accesses (Ratio)
board.cache_hierarchy.l2cache.ReadSharedReq.mshrMissRate::total     0.037982                    # mshr miss rate for ReadSharedReq accesses (Ratio)
board.cache_hierarchy.l2cache.ReadSharedReq.avgMshrMissLatency::processor.cores.core.inst 85873.487968                    # average ReadSharedReq mshr miss latency ((Tick/Count))
board.cache_hierarchy.l2cache.ReadSharedReq.avgMshrMissLatency::processor.cores.core.data 79697.445000                    # average ReadSharedReq mshr miss latency ((Tick/Count))
board.cache_hierarchy.l2cache.ReadSharedReq.avgMshrMissLatency::total 81510.506279                    # average ReadSharedReq mshr miss latency ((Tick/Count))
board.cache_hierarchy.l2cache.UpgradeReq.hits::processor.cores.core.data           3                    # number of UpgradeReq hits (Count)
board.cache_hierarchy.l2cache.UpgradeReq.hits::total           3                    # number of UpgradeReq hits (Count)
board.cache_hierarchy.l2cache.UpgradeReq.accesses::processor.cores.core.data           3                    # number of UpgradeReq accesses(hits+misses) (Count)
board.cache_hierarchy.l2cache.UpgradeReq.accesses::total           3                    # number of UpgradeReq accesses(hits+misses) (Count)
board.cache_hierarchy.l2cache.WritebackDirty.hits::writebacks       65717                    # number of WritebackDirty hits (Count)
board.cache_hierarchy.l2cache.WritebackDirty.hits::total       65717                    # number of WritebackDirty hits (Count)
board.cache_hierarchy.l2cache.WritebackDirty.accesses::writebacks       65717                    # number of WritebackDirty accesses(hits+misses) (Count)
board.cache_hierarchy.l2cache.WritebackDirty.accesses::total       65717                    # number of WritebackDirty accesses(hits+misses) (Count)
board.cache_hierarchy.l2cache.power_state.pwrStateResidencyTicks::UNDEFINED   9058914018                    # Cumulative time (in ticks) in various power states (Tick)
board.cache_hierarchy.l2cache.prefetcher.demandMshrMisses         3684                    # demands not covered by prefetchs (Count)
board.cache_hierarchy.l2cache.prefetcher.pfIssued       504031                    # number of hwpf issued (Count)
board.cache_hierarchy.l2cache.prefetcher.pfUnused           9                    # number of HardPF blocks evicted w/o reference (Count)
board.cache_hierarchy.l2cache.prefetcher.pfUseful       128022                    # number of useful prefetch (Count)
board.cache_hierarchy.l2cache.prefetcher.pfUsefulButMiss           0                    # number of hit on prefetch but cache block is not in an usable state (Count)
board.cache_hierarchy.l2cache.prefetcher.accuracy     0.253996                    # accuracy of the prefetcher (Count)
board.cache_hierarchy.l2cache.prefetcher.coverage     0.972029                    # coverage brought by this prefetcher (Count)
board.cache_hierarchy.l2cache.prefetcher.pfHitInCache       340203                    # number of prefetches hitting in cache (Count)
board.cache_hierarchy.l2cache.prefetcher.pfHitInMSHR        34734                    # number of prefetches hitting in a MSHR (Count)
board.cache_hierarchy.l2cache.prefetcher.pfHitInWB           0                    # number of prefetches hit in the Write Buffer (Count)
board.cache_hierarchy.l2cache.prefetcher.pfLate       374937                    # number of late prefetches (hitting in cache, MSHR or WB) (Count)
board.cache_hierarchy.l2cache.prefetcher.pfIdentified       504031                    # number of prefetch candidates identified (Count)
board.cache_hierarchy.l2cache.prefetcher.pfBufferHit           0                    # number of redundant prefetches already in prefetch queue (Count)
board.cache_hierarchy.l2cache.prefetcher.pfInCache           0                    # number of redundant prefetches already in cache/mshr dropped (Count)
board.cache_hierarchy.l2cache.prefetcher.pfRemovedDemand           0                    # number of prefetches dropped due to a demand for the same address (Count)
board.cache_hierarchy.l2cache.prefetcher.pfRemovedFull           0                    # number of prefetches dropped due to prefetch queue size (Count)
board.cache_hierarchy.l2cache.prefetcher.pfSpanPage        20493                    # number of prefetches that crossed the page (Count)
board.cache_hierarchy.l2cache.prefetcher.pfUsefulSpanPage        20390                    # number of prefetches that is useful and crossed the page (Count)
board.cache_hierarchy.l2cache.prefetcher.power_state.pwrStateResidencyTicks::UNDEFINED   9058914018                    # Cumulative time (in ticks) in various power states (Tick)
board.cache_hierarchy.l2cache.tags.tagsInUse  3999.553728                    # Average ticks per tags in use ((Tick/Count))
board.cache_hierarchy.l2cache.tags.totalRefs       392428                    # Total number of references to valid blocks. (Count)
board.cache_hierarchy.l2cache.tags.sampledRefs       132778                    # Sample count of references to valid blocks. (Count)
board.cache_hierarchy.l2cache.tags.avgRefs     2.955520                    # Average number of references to valid blocks. ((Count/Count))
board.cache_hierarchy.l2cache.tags.warmupTick        89910                    # The tick when the warmup percentage was hit. (Tick)
board.cache_hierarchy.l2cache.tags.occupancies::cache_hierarchy.l2cache.prefetcher  3866.038437                    # Average occupied blocks per tick, per requestor ((Count/Tick))
board.cache_hierarchy.l2cache.tags.occupancies::processor.cores.core.inst    27.720490                    # Average occupied blocks per tick, per requestor ((Count/Tick))
board.cache_hierarchy.l2cache.tags.occupancies::processor.cores.core.data   105.794801                    # Average occupied blocks per tick, per requestor ((Count/Tick))
board.cache_hierarchy.l2cache.tags.avgOccs::cache_hierarchy.l2cache.prefetcher     0.943857                    # Average percentage of cache occupancy ((Ratio/Tick))
board.cache_hierarchy.l2cache.tags.avgOccs::processor.cores.core.inst     0.006768                    # Average percentage of cache occupancy ((Ratio/Tick))
board.cache_hierarchy.l2cache.tags.avgOccs::processor.cores.core.data     0.025829                    # Average percentage of cache occupancy ((Ratio/Tick))
board.cache_hierarchy.l2cache.tags.avgOccs::total     0.976454                    # Average percentage of cache occupancy ((Ratio/Tick))
board.cache_hierarchy.l2cache.tags.occupanciesTaskId::1022         3750                    # Occupied blocks per task id (Count)
board.cache_hierarchy.l2cache.tags.occupanciesTaskId::1024          346                    # Occupied blocks per task id (Count)
board.cache_hierarchy.l2cache.tags.ageTaskId_1022::1         1016                    # Occupied blocks per task id, per block age (Count)
board.cache_hierarchy.l2cache.tags.ageTaskId_1022::2         2734                    # Occupied blocks per task id, per block age (Count)
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board.cache_hierarchy.l2cache.tags.ageTaskId_1024::0 139 # Occupied blocks per task id, per block age (Count)
board.cache_hierarchy.l2cache.tags.ageTaskId_1024::1 166 # Occupied blocks per task id, per block age (Count)
board.cache_hierarchy.l2cache.tags.ageTaskId_1024::2 41 # Occupied blocks per task id, per block age (Count)
board.cache_hierarchy.l2cache.tags.ratioOccsTaskId::1022 0.915527 # Ratio of occupied blocks and all blocks, per task id (Ratio)
board.cache_hierarchy.l2cache.tags.ratioOccsTaskId::1024 0.084473 # Ratio of occupied blocks and all blocks, per task id (Ratio)
board.cache_hierarchy.l2cache.tags.tagAccesses 4365706 # Number of tag accesses (Count)
board.cache_hierarchy.l2cache.tags.dataAccesses 4365706 # Number of data accesses (Count)
board.cache_hierarchy.l2cache.tags.power_state.pwrStateResidencyTicks::UNDEFINED 9058914018 # Cumulative time (in ticks) in various power states (Tick)
board.cache_hierarchy.membus.transDist::ReadResp 131642 # Transaction distribution (Count)
board.cache_hierarchy.membus.transDist::WritebackDirty 65695 # Transaction distribution (Count)
board.cache_hierarchy.membus.transDist::CleanEvict 62678 # Transaction distribution (Count)
board.cache_hierarchy.membus.transDist::ReadExReq 1136 # Transaction distribution (Count)
board.cache_hierarchy.membus.transDist::ReadExResp 1136 # Transaction distribution (Count)
board.cache_hierarchy.membus.transDist::ReadSharedReq 131642 # Transaction distribution (Count)
board.cache_hierarchy.membus.pktCount_board.cache_hierarchy.l2cache.mem_side_port::board.memory.mem_ctrl.port 393929 # Packet count per connected requestor and responder (Count)
board.cache_hierarchy.membus.pktCount_board.cache_hierarchy.l2cache.mem_side_port::total 393929 # Packet count per connected requestor and responder (Count)
board.cache_hierarchy.membus.pktCount::total 393929 # Packet count per connected requestor and responder (Count)
board.cache_hierarchy.membus.pktSize_board.cache_hierarchy.l2cache.mem_side_port::board.memory.mem_ctrl.port 12702272 # Cumulative packet size per connected requestor and responder (Byte)
board.cache_hierarchy.membus.pktSize_board.cache_hierarchy.l2cache.mem_side_port::total 12702272 # Cumulative packet size per connected requestor and responder (Byte)
board.cache_hierarchy.membus.pktSize::total 12702272 # Cumulative packet size per connected requestor and responder (Byte)
board.cache_hierarchy.membus.snoops 0 # Total snoops (Count)
board.cache_hierarchy.membus.snoopTraffic 0 # Total snoop traffic (Byte)
board.cache_hierarchy.membus.snoopFanout::samples 132778 # Request fanout histogram (Count)
board.cache_hierarchy.membus.snoopFanout::mean 0 # Request fanout histogram (Count)
board.cache_hierarchy.membus.snoopFanout::stdev 0 # Request fanout histogram (Count)
board.cache_hierarchy.membus.snoopFanout::underflows 0 0.00% 0.00% # Request fanout histogram (Count)
board.cache_hierarchy.membus.snoopFanout::0 132778 100.00% 100.00% # Request fanout histogram (Count)
board.cache_hierarchy.membus.snoopFanout::1 0 0.00% 100.00% # Request fanout histogram (Count)
board.cache_hierarchy.membus.snoopFanout::overflows 0 0.00% 100.00% # Request fanout histogram (Count)
board.cache_hierarchy.membus.snoopFanout::min_value 0 # Request fanout histogram (Count)
board.cache_hierarchy.membus.snoopFanout::max_value 0 # Request fanout histogram (Count)
board.cache_hierarchy.membus.snoopFanout::total 132778 # Request fanout histogram (Count)
board.cache_hierarchy.membus.badaddr_responder.power_state.pwrStateResidencyTicks::UNDEFINED 9058914018 # Cumulative time (in ticks) in various power states (Tick)
board.cache_hierarchy.membus.power_state.pwrStateResidencyTicks::UNDEFINED 9058914018 # Cumulative time (in ticks) in various power states (Tick)
board.cache_hierarchy.membus.reqLayer0.occupancy 148724838 # Layer occupancy (ticks) (Tick)
board.cache_hierarchy.membus.reqLayer0.utilization 0.0 # Layer utilization (Ratio)
board.cache_hierarchy.membus.respLayer2.occupancy 113457759 # Layer occupancy (ticks) (Tick)
board.cache_hierarchy.membus.respLayer2.utilization 0.0 # Layer utilization (Ratio)
board.cache_hierarchy.membus.snoop_filter.totRequests 261151 # Total number of requests made to the snoop filter. (Count)
board.cache_hierarchy.membus.snoop_filter.hitSingleRequests 128373 # Number of requests hitting in the snoop filter with a single holder of the requested data. (Count)
board.cache_hierarchy.membus.snoop_filter.hitMultiRequests 0 # Number of requests hitting in the snoop filter with multiple (>1) holders of the requested data. (Count)
board.cache_hierarchy.membus.snoop_filter.totSnoops 0 # Total number of snoops made to the snoop filter. (Count)
board.cache_hierarchy.membus.snoop_filter.hitSingleSnoops 0 # Number of snoops hitting in the snoop filter with a single holder of the requested data. (Count)
board.cache_hierarchy.membus.snoop_filter.hitMultiSnoops 0 # Number of snoops hitting in the snoop filter with multiple (>1) holders of the requested data. (Count)
board.clk_domain.clock 333 # Clock period in ticks (Tick)
board.clk_domain.voltage_domain.voltage 1 # Voltage in Volts (Volt)
board.memory.mem_ctrl.avgPriority_writebacks::samples 65695.00 # Average QoS priority value for accepted requests (Count)
board.memory.mem_ctrl.avgPriority_cache_hierarchy.l2cache.prefetcher::samples 129073.00 # Average QoS priority value for accepted requests (Count)
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board.memory.mem_ctrl.avgPriority_processor.cores.core.inst::samples      748.00                # Average QoS priority value for accepted requests (Count)
board.memory.mem_ctrl.avgPriority_processor.cores.core.data::samples     2936.00                # Average QoS priority value for accepted requests (Count)
board.memory.mem_ctrl.priorityMinLatency 0.000000018750                 # per QoS priority minimum request to response latency (Second)
board.memory.mem_ctrl.priorityMaxLatency 0.000155624752                 # per QoS priority maximum request to response latency (Second)
board.memory.mem_ctrl.numReadWriteTurnArounds        4104               # Number of turnarounds from READ to WRITE (Count)
board.memory.mem_ctrl.numWriteReadTurnArounds        4104               # Number of turnarounds from WRITE to READ (Count)
board.memory.mem_ctrl.numStayReadState        316974                    # Number of times bus staying in READ state (Count)
board.memory.mem_ctrl.numStayWriteState        61724                    # Number of times bus staying in WRITE state (Count)
board.memory.mem_ctrl.readReqs                132778                    # Number of read requests accepted (Count)
board.memory.mem_ctrl.writeReqs                65695                    # Number of write requests accepted (Count)
board.memory.mem_ctrl.readBursts              132778                    # Number of controller read bursts, including those serviced by the write queue (Count)
board.memory.mem_ctrl.writeBursts              65695                    # Number of controller write bursts, including those merged in the write queue (Count)
board.memory.mem_ctrl.servicedByWrQ               21                    # Number of controller read bursts serviced by the write queue (Count)
board.memory.mem_ctrl.mergedWrBursts               0                    # Number of controller write bursts merged with an existing one (Count)
board.memory.mem_ctrl.neitherReadNorWriteReqs          0               # Number of requests that are neither read nor write (Count)
board.memory.mem_ctrl.avgRdQLen                 1.28                   # Average read queue length when enqueuing ((Count/Tick))
board.memory.mem_ctrl.avgWrQLen                22.00                   # Average write queue length when enqueuing ((Count/Tick))
board.memory.mem_ctrl.numRdRetry                   0                   # Number of times read queue was full causing retry (Count)
board.memory.mem_ctrl.numWrRetry                   0                   # Number of times write queue was full causing retry (Count)
board.memory.mem_ctrl.readPktSize::0               0                   # Read request sizes (log2) (Count)
board.memory.mem_ctrl.readPktSize::1               0                   # Read request sizes (log2) (Count)
board.memory.mem_ctrl.readPktSize::2               0                   # Read request sizes (log2) (Count)
board.memory.mem_ctrl.readPktSize::3               0                   # Read request sizes (log2) (Count)
board.memory.mem_ctrl.readPktSize::4               0                   # Read request sizes (log2) (Count)
board.memory.mem_ctrl.readPktSize::5               0                   # Read request sizes (log2) (Count)
board.memory.mem_ctrl.readPktSize::6          132778                   # Read request sizes (log2) (Count)
board.memory.mem_ctrl.writePktSize::0              0                   # Write request sizes (log2) (Count)
board.memory.mem_ctrl.writePktSize::1              0                   # Write request sizes (log2) (Count)
board.memory.mem_ctrl.writePktSize::2              0                   # Write request sizes (log2) (Count)
board.memory.mem_ctrl.writePktSize::3              0                   # Write request sizes (log2) (Count)
board.memory.mem_ctrl.writePktSize::4              0                   # Write request sizes (log2) (Count)
board.memory.mem_ctrl.writePktSize::5              0                   # Write request sizes (log2) (Count)
board.memory.mem_ctrl.writePktSize::6          65695                   # Write request sizes (log2) (Count)
board.memory.mem_ctrl.rdQLenPdf::0            109555                   # What read queue length does an incoming req see (Count)
board.memory.mem_ctrl.rdQLenPdf::1              8181                   # What read queue length does an incoming req see (Count)
board.memory.mem_ctrl.rdQLenPdf::2              7165                   # What read queue length does an incoming req see (Count)
board.memory.mem_ctrl.rdQLenPdf::3              7503                   # What read queue length does an incoming req see (Count)
board.memory.mem_ctrl.rdQLenPdf::4               351                   # What read queue length does an incoming req see (Count)
board.memory.mem_ctrl.rdQLenPdf::5                 1                   # What read queue length does an incoming req see (Count)
board.memory.mem_ctrl.rdQLenPdf::6                 1                   # What read queue length does an incoming req see (Count)
board.memory.mem_ctrl.rdQLenPdf::7                 0                   # What read queue length does an incoming req see (Count)
board.memory.mem_ctrl.rdQLenPdf::8                 0                   # What read queue length does an incoming req see (Count)
board.memory.mem_ctrl.rdQLenPdf::9                 0                   # What read queue length does an incoming req see (Count)
board.memory.mem_ctrl.rdQLenPdf::10                0                   # What read queue length does an incoming req see (Count)
board.memory.mem_ctrl.rdQLenPdf::11                0                   # What read queue length does an incoming req see (Count)
board.memory.mem_ctrl.rdQLenPdf::12                0                   # What read queue length does an incoming req see (Count)
board.memory.mem_ctrl.rdQLenPdf::13                0                   # What read queue length does an incoming req see (Count)
board.memory.mem_ctrl.rdQLenPdf::14                0                   # What read queue length does an incoming req see (Count)
board.memory.mem_ctrl.rdQLenPdf::15                0                   # What read queue length does an incoming req see (Count)
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board.memory.mem_ctrl.rdQLenPdf::16	0	# What read queue length does an incoming req see (Count)
board.memory.mem_ctrl.rdQLenPdf::17	0	# What read queue length does an incoming req see (Count)
board.memory.mem_ctrl.rdQLenPdf::18	0	# What read queue length does an incoming req see (Count)
board.memory.mem_ctrl.rdQLenPdf::19	0	# What read queue length does an incoming req see (Count)
board.memory.mem_ctrl.rdQLenPdf::20	0	# What read queue length does an incoming req see (Count)
board.memory.mem_ctrl.rdQLenPdf::21	0	# What read queue length does an incoming req see (Count)
board.memory.mem_ctrl.rdQLenPdf::22	0	# What read queue length does an incoming req see (Count)
board.memory.mem_ctrl.rdQLenPdf::23	0	# What read queue length does an incoming req see (Count)
board.memory.mem_ctrl.rdQLenPdf::24	0	# What read queue length does an incoming req see (Count)
board.memory.mem_ctrl.rdQLenPdf::25	0	# What read queue length does an incoming req see (Count)
board.memory.mem_ctrl.rdQLenPdf::26	0	# What read queue length does an incoming req see (Count)
board.memory.mem_ctrl.rdQLenPdf::27	0	# What read queue length does an incoming req see (Count)
board.memory.mem_ctrl.rdQLenPdf::28	0	# What read queue length does an incoming req see (Count)
board.memory.mem_ctrl.rdQLenPdf::29	0	# What read queue length does an incoming req see (Count)
board.memory.mem_ctrl.rdQLenPdf::30	0	# What read queue length does an incoming req see (Count)
board.memory.mem_ctrl.rdQLenPdf::31	0	# What read queue length does an incoming req see (Count)
board.memory.mem_ctrl.wrQLenPdf::0	1	# What write queue length does an incoming req see (Count)
board.memory.mem_ctrl.wrQLenPdf::1	1	# What write queue length does an incoming req see (Count)
board.memory.mem_ctrl.wrQLenPdf::2	1	# What write queue length does an incoming req see (Count)
board.memory.mem_ctrl.wrQLenPdf::3	1	# What write queue length does an incoming req see (Count)
board.memory.mem_ctrl.wrQLenPdf::4	1	# What write queue length does an incoming req see (Count)
board.memory.mem_ctrl.wrQLenPdf::5	1	# What write queue length does an incoming req see (Count)
board.memory.mem_ctrl.wrQLenPdf::6	1	# What write queue length does an incoming req see (Count)
board.memory.mem_ctrl.wrQLenPdf::7	1	# What write queue length does an incoming req see (Count)
board.memory.mem_ctrl.wrQLenPdf::8	1	# What write queue length does an incoming req see (Count)
board.memory.mem_ctrl.wrQLenPdf::9	1	# What write queue length does an incoming req see (Count)
board.memory.mem_ctrl.wrQLenPdf::10	1	# What write queue length does an incoming req see (Count)
board.memory.mem_ctrl.wrQLenPdf::11	1	# What write queue length does an incoming req see (Count)
board.memory.mem_ctrl.wrQLenPdf::12	1	# What write queue length does an incoming req see (Count)
board.memory.mem_ctrl.wrQLenPdf::13	1	# What write queue length does an incoming req see (Count)
board.memory.mem_ctrl.wrQLenPdf::14	1	# What write queue length does an incoming req see (Count)
board.memory.mem_ctrl.wrQLenPdf::15	2	# What write queue length does an incoming req see (Count)
board.memory.mem_ctrl.wrQLenPdf::16	3	# What write queue length does an incoming req see (Count)
board.memory.mem_ctrl.wrQLenPdf::17	443	# What write queue length does an incoming req see (Count)
board.memory.mem_ctrl.wrQLenPdf::18	4019	# What write queue length does an incoming req see (Count)
board.memory.mem_ctrl.wrQLenPdf::19	4042	# What write queue length does an incoming req see (Count)
board.memory.mem_ctrl.wrQLenPdf::20	6857	# What write queue length does an incoming req see (Count)
board.memory.mem_ctrl.wrQLenPdf::21	4161	# What write queue length does an incoming req see (Count)
board.memory.mem_ctrl.wrQLenPdf::22	4113	# What write queue length does an incoming req see (Count)
board.memory.mem_ctrl.wrQLenPdf::23	4812	# What write queue length does an incoming req see (Count)
board.memory.mem_ctrl.wrQLenPdf::24	4108	# What write queue length does an incoming req see (Count)
board.memory.mem_ctrl.wrQLenPdf::25	4104	# What write queue length does an incoming req see (Count)
board.memory.mem_ctrl.wrQLenPdf::26	4108	# What write queue length does an incoming req see (Count)
board.memory.mem_ctrl.wrQLenPdf::27	4107	# What write queue length does an incoming req see (Count)
board.memory.mem_ctrl.wrQLenPdf::28	4105	# What write queue length does an incoming req see (Count)
board.memory.mem_ctrl.wrQLenPdf::29	4129	# What write queue length does an incoming req see (Count)
board.memory.mem_ctrl.wrQLenPdf::30	4106	# What write queue length does an incoming req see (Count)
board.memory.mem_ctrl.wrQLenPdf::31	4110	# What write queue length does an incoming req see (Count)
board.memory.mem_ctrl.wrQLenPdf::32	4351	# What write queue length does an incoming req see (Count)

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board.memory.mem_ctrl.wrQLenPdf::33                0                       # What write queue length does an incoming req see (Count)
board.memory.mem_ctrl.wrQLenPdf::34                0                       # What write queue length does an incoming req see (Count)
board.memory.mem_ctrl.wrQLenPdf::35                0                       # What write queue length does an incoming req see (Count)
board.memory.mem_ctrl.wrQLenPdf::36                0                       # What write queue length does an incoming req see (Count)
board.memory.mem_ctrl.wrQLenPdf::37                0                       # What write queue length does an incoming req see (Count)
board.memory.mem_ctrl.wrQLenPdf::38                0                       # What write queue length does an incoming req see (Count)
board.memory.mem_ctrl.wrQLenPdf::39                0                       # What write queue length does an incoming req see (Count)
board.memory.mem_ctrl.wrQLenPdf::40                0                       # What write queue length does an incoming req see (Count)
board.memory.mem_ctrl.wrQLenPdf::41                0                       # What write queue length does an incoming req see (Count)
board.memory.mem_ctrl.wrQLenPdf::42                0                       # What write queue length does an incoming req see (Count)
board.memory.mem_ctrl.wrQLenPdf::43                0                       # What write queue length does an incoming req see (Count)
board.memory.mem_ctrl.wrQLenPdf::44                0                       # What write queue length does an incoming req see (Count)
board.memory.mem_ctrl.wrQLenPdf::45                0                       # What write queue length does an incoming req see (Count)
board.memory.mem_ctrl.wrQLenPdf::46                0                       # What write queue length does an incoming req see (Count)
board.memory.mem_ctrl.wrQLenPdf::47                0                       # What write queue length does an incoming req see (Count)
board.memory.mem_ctrl.wrQLenPdf::48                0                       # What write queue length does an incoming req see (Count)
board.memory.mem_ctrl.wrQLenPdf::49                0                       # What write queue length does an incoming req see (Count)
board.memory.mem_ctrl.wrQLenPdf::50                0                       # What write queue length does an incoming req see (Count)
board.memory.mem_ctrl.wrQLenPdf::51                0                       # What write queue length does an incoming req see (Count)
board.memory.mem_ctrl.wrQLenPdf::52                0                       # What write queue length does an incoming req see (Count)
board.memory.mem_ctrl.wrQLenPdf::53                0                       # What write queue length does an incoming req see (Count)
board.memory.mem_ctrl.wrQLenPdf::54                0                       # What write queue length does an incoming req see (Count)
board.memory.mem_ctrl.wrQLenPdf::55                0                       # What write queue length does an incoming req see (Count)
board.memory.mem_ctrl.wrQLenPdf::56                0                       # What write queue length does an incoming req see (Count)
board.memory.mem_ctrl.wrQLenPdf::57                0                       # What write queue length does an incoming req see (Count)
board.memory.mem_ctrl.wrQLenPdf::58                0                       # What write queue length does an incoming req see (Count)
board.memory.mem_ctrl.wrQLenPdf::59                0                       # What write queue length does an incoming req see (Count)
board.memory.mem_ctrl.wrQLenPdf::60                0                       # What write queue length does an incoming req see (Count)
board.memory.mem_ctrl.wrQLenPdf::61                0                       # What write queue length does an incoming req see (Count)
board.memory.mem_ctrl.wrQLenPdf::62                0                       # What write queue length does an incoming req see (Count)
board.memory.mem_ctrl.wrQLenPdf::63                0                       # What write queue length does an incoming req see (Count)
board.memory.mem_ctrl.rdPerTurnAround::samples        4104                 # Reads before turning the bus around for writes (Count)
board.memory.mem_ctrl.rdPerTurnAround::mean    17.310185                   # Reads before turning the bus around for writes (Count)
board.memory.mem_ctrl.rdPerTurnAround::gmean    16.065934                  # Reads before turning the bus around for writes (Count)
board.memory.mem_ctrl.rdPerTurnAround::stdev    71.579904                  # Reads before turning the bus around for writes (Count)
board.memory.mem_ctrl.rdPerTurnAround::0-255        4102     99.95%    99.95% # Reads before turning the bus around for writes (Count)
board.memory.mem_ctrl.rdPerTurnAround::256-511          1      0.02%    99.98% # Reads before turning the bus around for writes (Count)
board.memory.mem_ctrl.rdPerTurnAround::4352-4607        1      0.02%   100.00% # Reads before turning the bus around for writes (Count)
board.memory.mem_ctrl.rdPerTurnAround::total        4104                 # Reads before turning the bus around for writes (Count)
board.memory.mem_ctrl.wrPerTurnAround::samples        4104                 # Writes before turning the bus around for reads (Count)
board.memory.mem_ctrl.wrPerTurnAround::mean    16.002437                   # Writes before turning the bus around for reads (Count)
board.memory.mem_ctrl.wrPerTurnAround::gmean    16.002248                  # Writes before turning the bus around for reads (Count)
board.memory.mem_ctrl.wrPerTurnAround::stdev     0.082573                  # Writes before turning the bus around for reads (Count)
board.memory.mem_ctrl.wrPerTurnAround::16        4100     99.90%    99.90% # Writes before turning the bus around for reads (Count)
board.memory.mem_ctrl.wrPerTurnAround::18           3      0.07%    99.98% # Writes before turning the bus around for reads (Count)
board.memory.mem_ctrl.wrPerTurnAround::20           1      0.02%   100.00% # Writes before turning the bus around for reads (Count)
board.memory.mem_ctrl.wrPerTurnAround::total        4104                 # Writes before turning the bus around for reads (Count)
board.memory.mem_ctrl.bytesReadWrQ              1344                     # Total number of bytes read from write queue (Byte)
board.memory.mem_ctrl.bytesReadSys           8497792                     # Total read bytes from the system interface side (Byte)
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board.memory.mem_ctrl.bytesWrittenSys 4204480 # Total written bytes from the system interface side (Byte)
board.memory.mem_ctrl.avgRdBWSys 938058577.78481448 # Average system read bandwidth in Byte/s ((Byte/Second))
board.memory.mem_ctrl.avgWrBWSys 464126272.93356872 # Average system write bandwidth in Byte/s ((Byte/Second))
board.memory.mem_ctrl.totGap 9058762836 # Total gap between requests (Tick)
board.memory.mem_ctrl.avgGap 45642.29 # Average gap between requests ((Tick/Count))
board.memory.mem_ctrl.requestorReadBytes::cache_hierarchy.l2cache.prefetcher 8260672 # Per-requestor bytes read from memory (Byte)
board.memory.mem_ctrl.requestorReadBytes::processor.cores.core.inst 47872 # Per-requestor bytes read from memory (Byte)
board.memory.mem_ctrl.requestorReadBytes::processor.cores.core.data 187904 # Per-requestor bytes read from memory (Byte)
board.memory.mem_ctrl.requestorWriteBytes::writebacks 4203136 # Per-requestor bytes write to memory (Byte)
board.memory.mem_ctrl.requestorReadRate::cache_hierarchy.l2cache.prefetcher 911883254.834531068802 # Per-requestor bytes read from memory rate ((Byte/Second))
board.memory.mem_ctrl.requestorReadRate::processor.cores.core.inst 5284518.641514718533 # Per-requestor bytes read from memory rate ((Byte/Second))
board.memory.mem_ctrl.requestorReadRate::processor.cores.core.data 20742442.154394671321 # Per-requestor bytes read from memory rate ((Byte/Second))
board.memory.mem_ctrl.requestorWriteRate::writebacks 463977910.779194653034 # Per-requestor bytes write to memory rate ((Byte/Second))
board.memory.mem_ctrl.requestorReadAccesses::cache_hierarchy.l2cache.prefetcher 129094 # Per-requestor read serviced memory accesses (Count)
board.memory.mem_ctrl.requestorReadAccesses::processor.cores.core.inst 748 # Per-requestor read serviced memory accesses (Count)
board.memory.mem_ctrl.requestorReadAccesses::processor.cores.core.data 2936 # Per-requestor read serviced memory accesses (Count)
board.memory.mem_ctrl.requestorWriteAccesses::writebacks 65695 # Per-requestor write serviced memory accesses (Count)
board.memory.mem_ctrl.requestorReadTotalLat::cache_hierarchy.l2cache.prefetcher 4463847863 # Per-requestor read total memory access latency (Tick)
board.memory.mem_ctrl.requestorReadTotalLat::processor.cores.core.inst 21494086 # Per-requestor read total memory access latency (Tick)
board.memory.mem_ctrl.requestorReadTotalLat::processor.cores.core.data 71303506 # Per-requestor read total memory access latency (Tick)
board.memory.mem_ctrl.requestorWriteTotalLat::writebacks 95770221644 # Per-requestor write total memory access latency (Tick)
board.memory.mem_ctrl.requestorReadAvgLat::cache_hierarchy.l2cache.prefetcher 34578.28 # Per-requestor read average memory access latency ((Tick/Count))
board.memory.mem_ctrl.requestorReadAvgLat::processor.cores.core.inst 28735.41 # Per-requestor read average memory access latency ((Tick/Count))
board.memory.mem_ctrl.requestorReadAvgLat::processor.cores.core.data 24285.94 # Per-requestor read average memory access latency ((Tick/Count))
board.memory.mem_ctrl.requestorWriteAvgLat::writebacks 1457800.77 # Per-requestor write average memory access latency ((Tick/Count))
board.memory.mem_ctrl.dram.bytesRead::cache_hierarchy.l2cache.prefetcher 8262016 # Number of bytes read from this memory (Byte)
board.memory.mem_ctrl.dram.bytesRead::processor.cores.core.inst 47872 # Number of bytes read from this memory (Byte)
board.memory.mem_ctrl.dram.bytesRead::processor.cores.core.data 187904 # Number of bytes read from this memory (Byte)
board.memory.mem_ctrl.dram.bytesRead::total 8497792 # Number of bytes read from this memory (Byte)
board.memory.mem_ctrl.dram.bytesInstRead::processor.cores.core.inst 47872 # Number of instructions bytes read from this memory (Byte)
board.memory.mem_ctrl.dram.bytesInstRead::total 47872 # Number of instructions bytes read from this memory (Byte)
board.memory.mem_ctrl.dram.bytesWritten::writebacks 4204480 # Number of bytes written to this memory (Byte)
board.memory.mem_ctrl.dram.bytesWritten::total 4204480 # Number of bytes written to this memory (Byte)
board.memory.mem_ctrl.dram.numReads::cache_hierarchy.l2cache.prefetcher 129094 # Number of read requests responded to by this memory (Count)
board.memory.mem_ctrl.dram.numReads::processor.cores.core.inst 748 # Number of read requests responded to by this memory (Count)
board.memory.mem_ctrl.dram.numReads::processor.cores.core.data 2936 # Number of read requests responded to by this memory (Count)
board.memory.mem_ctrl.dram.numReads::total 132778 # Number of read requests responded to by this memory (Count)
board.memory.mem_ctrl.dram.numWrites::writebacks 65695 # Number of write requests responded to by this memory (Count)
board.memory.mem_ctrl.dram.numWrites::total 65695 # Number of write requests responded to by this memory (Count)
board.memory.mem_ctrl.dram.bwRead::cache_hierarchy.l2cache.prefetcher 912031617 # Total read bandwidth from this memory ((Byte/Second))
board.memory.mem_ctrl.dram.bwRead::processor.cores.core.inst 5284519 # Total read bandwidth from this memory ((Byte/Second))
board.memory.mem_ctrl.dram.bwRead::processor.cores.core.data 20742442 # Total read bandwidth from this memory ((Byte/Second))
board.memory.mem_ctrl.dram.bwRead::total 938058578 # Total read bandwidth from this memory ((Byte/Second))
board.memory.mem_ctrl.dram.bwInstRead::processor.cores.core.inst 5284519 # Instruction read bandwidth from this memory ((Byte/Second))
board.memory.mem_ctrl.dram.bwInstRead::total 5284519 # Instruction read bandwidth from this memory ((Byte/Second))
board.memory.mem_ctrl.dram.bwWrite::writebacks 464126273 # Write bandwidth from this memory ((Byte/Second))
board.memory.mem_ctrl.dram.bwWrite::total 464126273 # Write bandwidth from this memory ((Byte/Second))
board.memory.mem_ctrl.dram.bwTotal::writebacks 464126273 # Total bandwidth to/from this memory ((Byte/Second))
board.memory.mem_ctrl.dram.bwTotal::cache_hierarchy.l2cache.prefetcher 912031617 # Total bandwidth to/from this memory ((Byte/Second))
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board.memory.mem_ctrl.dram.bwTotal::processor.cores.core.inst      5284519                    # Total bandwidth to/from this memory ((Byte/Second))
board.memory.mem_ctrl.dram.bwTotal::processor.cores.core.data     20742442                    # Total bandwidth to/from this memory ((Byte/Second))
board.memory.mem_ctrl.dram.bwTotal::total   1402184851                    # Total bandwidth to/from this memory ((Byte/Second))
board.memory.mem_ctrl.dram.readBursts          132757                    # Number of DRAM read bursts (Count)
board.memory.mem_ctrl.dram.writeBursts          65674                    # Number of DRAM write bursts (Count)
board.memory.mem_ctrl.dram.perBankRdBursts::0        8464                    # Per bank write bursts (Count)
board.memory.mem_ctrl.dram.perBankRdBursts::1        8437                    # Per bank write bursts (Count)
board.memory.mem_ctrl.dram.perBankRdBursts::2        8281                    # Per bank write bursts (Count)
board.memory.mem_ctrl.dram.perBankRdBursts::3        8268                    # Per bank write bursts (Count)
board.memory.mem_ctrl.dram.perBankRdBursts::4        8267                    # Per bank write bursts (Count)
board.memory.mem_ctrl.dram.perBankRdBursts::5        8310                    # Per bank write bursts (Count)
board.memory.mem_ctrl.dram.perBankRdBursts::6        8254                    # Per bank write bursts (Count)
board.memory.mem_ctrl.dram.perBankRdBursts::7        8225                    # Per bank write bursts (Count)
board.memory.mem_ctrl.dram.perBankRdBursts::8        8275                    # Per bank write bursts (Count)
board.memory.mem_ctrl.dram.perBankRdBursts::9        8235                    # Per bank write bursts (Count)
board.memory.mem_ctrl.dram.perBankRdBursts::10        8228                    # Per bank write bursts (Count)
board.memory.mem_ctrl.dram.perBankRdBursts::11        8228                    # Per bank write bursts (Count)
board.memory.mem_ctrl.dram.perBankRdBursts::12        8235                    # Per bank write bursts (Count)
board.memory.mem_ctrl.dram.perBankRdBursts::13        8294                    # Per bank write bursts (Count)
board.memory.mem_ctrl.dram.perBankRdBursts::14        8357                    # Per bank write bursts (Count)
board.memory.mem_ctrl.dram.perBankRdBursts::15        8399                    # Per bank write bursts (Count)
board.memory.mem_ctrl.dram.perBankWrBursts::0        4096                    # Per bank write bursts (Count)
board.memory.mem_ctrl.dram.perBankWrBursts::1        4096                    # Per bank write bursts (Count)
board.memory.mem_ctrl.dram.perBankWrBursts::2        4096                    # Per bank write bursts (Count)
board.memory.mem_ctrl.dram.perBankWrBursts::3        4104                    # Per bank write bursts (Count)
board.memory.mem_ctrl.dram.perBankWrBursts::4        4102                    # Per bank write bursts (Count)
board.memory.mem_ctrl.dram.perBankWrBursts::5        4157                    # Per bank write bursts (Count)
board.memory.mem_ctrl.dram.perBankWrBursts::6        4098                    # Per bank write bursts (Count)
board.memory.mem_ctrl.dram.perBankWrBursts::7        4096                    # Per bank write bursts (Count)
board.memory.mem_ctrl.dram.perBankWrBursts::8        4116                    # Per bank write bursts (Count)
board.memory.mem_ctrl.dram.perBankWrBursts::9        4106                    # Per bank write bursts (Count)
board.memory.mem_ctrl.dram.perBankWrBursts::10        4117                    # Per bank write bursts (Count)
board.memory.mem_ctrl.dram.perBankWrBursts::11        4106                    # Per bank write bursts (Count)
board.memory.mem_ctrl.dram.perBankWrBursts::12        4096                    # Per bank write bursts (Count)
board.memory.mem_ctrl.dram.perBankWrBursts::13        4096                    # Per bank write bursts (Count)
board.memory.mem_ctrl.dram.perBankWrBursts::14        4096                    # Per bank write bursts (Count)
board.memory.mem_ctrl.dram.perBankWrBursts::15        4096                    # Per bank write bursts (Count)
board.memory.mem_ctrl.dram.totQLat        2067451705                    # Total ticks spent queuing (Tick)
board.memory.mem_ctrl.dram.totBusLat        663785000                    # Total ticks spent in databus transfers (Tick)
board.memory.mem_ctrl.dram.totMemAccLat    4556645455                    # Total ticks spent from burst creation until serviced by the DRAM (Tick)
board.memory.mem_ctrl.dram.avgQLat          15573.20                    # Average queueing delay per DRAM burst ((Tick/Count))
board.memory.mem_ctrl.dram.avgBusLat          5000.00                    # Average bus latency per DRAM burst ((Tick/Count))
board.memory.mem_ctrl.dram.avgMemAccLat      34323.20                    # Average memory access latency per DRAM burst ((Tick/Count))
board.memory.mem_ctrl.dram.readRowHits         122815                    # Number of row buffer hits during reads (Count)
board.memory.mem_ctrl.dram.writeRowHits         61008                    # Number of row buffer hits during writes (Count)
board.memory.mem_ctrl.dram.readRowHitRate        92.51                    # Row buffer hit rate for reads (Ratio)
board.memory.mem_ctrl.dram.writeRowHitRate        92.90                    # Row buffer hit rate for writes (Ratio)
board.memory.mem_ctrl.dram.bytesPerActivate::samples        14603                    # Bytes accessed per row activation (Byte)
board.memory.mem_ctrl.dram.bytesPerActivate::mean   869.568171                    # Bytes accessed per row activation (Byte)
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board.memory.mem_ctrl.dram.bytesPerActivate::gmean   799.416996                    # Bytes accessed per row activation (Byte)
board.memory.mem_ctrl.dram.bytesPerActivate::stdev   263.418182                    # Bytes accessed per row activation (Byte)
board.memory.mem_ctrl.dram.bytesPerActivate::0-127          171      1.17%      1.17% # Bytes accessed per row activation (Byte)
board.memory.mem_ctrl.dram.bytesPerActivate::128-255        194      1.33%      2.50% # Bytes accessed per row activation (Byte)
board.memory.mem_ctrl.dram.bytesPerActivate::256-383        624      4.27%      6.77% # Bytes accessed per row activation (Byte)
board.memory.mem_ctrl.dram.bytesPerActivate::384-511        802      5.49%     12.26% # Bytes accessed per row activation (Byte)
board.memory.mem_ctrl.dram.bytesPerActivate::512-639       1153      7.90%     20.16% # Bytes accessed per row activation (Byte)
board.memory.mem_ctrl.dram.bytesPerActivate::640-767        726      4.97%     25.13% # Bytes accessed per row activation (Byte)
board.memory.mem_ctrl.dram.bytesPerActivate::768-895        400      2.74%     27.87% # Bytes accessed per row activation (Byte)
board.memory.mem_ctrl.dram.bytesPerActivate::896-1023       308      2.11%     29.98% # Bytes accessed per row activation (Byte)
board.memory.mem_ctrl.dram.bytesPerActivate::1024-1151    10225     70.02%    100.00% # Bytes accessed per row activation (Byte)
board.memory.mem_ctrl.dram.bytesPerActivate::total         14603                    # Bytes accessed per row activation (Byte)
board.memory.mem_ctrl.dram.dramBytesRead      8496448                    # Total bytes read (Byte)
board.memory.mem_ctrl.dram.dramBytesWritten      4203136                    # Total bytes written (Byte)
board.memory.mem_ctrl.dram.avgRdBW        937.910216                    # Average DRAM read bandwidth in MiBytes/s ((Byte/Second))
board.memory.mem_ctrl.dram.avgWrBW        463.977911                    # Average DRAM write bandwidth in MiBytes/s ((Byte/Second))
board.memory.mem_ctrl.dram.peakBW           12800.00                    # Theoretical peak bandwidth in MiByte/s ((Byte/Second))
board.memory.mem_ctrl.dram.busUtil             10.95                    # Data bus utilization in percentage (Ratio)
board.memory.mem_ctrl.dram.busUtilRead          7.33                    # Data bus utilization in percentage for reads (Ratio)
board.memory.mem_ctrl.dram.busUtilWrite         3.62                    # Data bus utilization in percentage for writes (Ratio)
board.memory.mem_ctrl.dram.pageHitRate         92.64                    # Row buffer hit rate, read and write combined (Ratio)
board.memory.mem_ctrl.dram.power_state.pwrStateResidencyTicks::UNDEFINED   9058914018                    # Cumulative time (in ticks) in various power states (Tick)
board.memory.mem_ctrl.dram.rank0.actEnergy     52143420                    # Energy for activate commands per rank (pJ) (Joule)
board.memory.mem_ctrl.dram.rank0.preEnergy     27707295                    # Energy for precharge commands per rank (pJ) (Joule)
board.memory.mem_ctrl.dram.rank0.readEnergy    474852840                    # Energy for read commands per rank (pJ) (Joule)
board.memory.mem_ctrl.dram.rank0.writeEnergy    171450900                    # Energy for write commands per rank (pJ) (Joule)
board.memory.mem_ctrl.dram.rank0.refreshEnergy 714826320.000000                    # Energy for refresh commands per rank (pJ) (Joule)
board.memory.mem_ctrl.dram.rank0.actBackEnergy   2173757130                    # Energy for active background per rank (pJ) (Joule)
board.memory.mem_ctrl.dram.rank0.preBackEnergy   1648091040                    # Energy for precharge background per rank (pJ) (Joule)
board.memory.mem_ctrl.dram.rank0.actPowerDownEnergy           0                    # Energy for active power-down per rank (pJ) (Joule)
board.memory.mem_ctrl.dram.rank0.prePowerDownEnergy           0                    # Energy for precharge power-down per rank (pJ) (Joule)
board.memory.mem_ctrl.dram.rank0.selfRefreshEnergy           0                    # Energy for self refresh per rank (pJ) (Joule)
board.memory.mem_ctrl.dram.rank0.totalEnergy   5262828945                    # Total energy per rank (pJ) (Joule)
board.memory.mem_ctrl.dram.rank0.averagePower   580.955834                    # Core power per rank (mW) (Watt)
board.memory.mem_ctrl.dram.rank0.totalIdleTime           0                    # Total Idle time Per DRAM Rank (Tick)
board.memory.mem_ctrl.dram.rank0.pwrStateTime::IDLE   4225939288                    # Time in different power states (Tick)
board.memory.mem_ctrl.dram.rank0.pwrStateTime::REF    302380000                    # Time in different power states (Tick)
board.memory.mem_ctrl.dram.rank0.pwrStateTime::SREF           0                    # Time in different power states (Tick)
board.memory.mem_ctrl.dram.rank0.pwrStateTime::PRE_PDN           0                    # Time in different power states (Tick)
board.memory.mem_ctrl.dram.rank0.pwrStateTime::ACT   4530594730                    # Time in different power states (Tick)
board.memory.mem_ctrl.dram.rank0.pwrStateTime::ACT_PDN           0                    # Time in different power states (Tick)
board.memory.mem_ctrl.dram.rank1.actEnergy     52157700                    # Energy for activate commands per rank (pJ) (Joule)
board.memory.mem_ctrl.dram.rank1.preEnergy     27711090                    # Energy for precharge commands per rank (pJ) (Joule)
board.memory.mem_ctrl.dram.rank1.readEnergy    473032140                    # Energy for read commands per rank (pJ) (Joule)
board.memory.mem_ctrl.dram.rank1.writeEnergy    171367380                    # Energy for write commands per rank (pJ) (Joule)
board.memory.mem_ctrl.dram.rank1.refreshEnergy 714826320.000000                    # Energy for refresh commands per rank (pJ) (Joule)
board.memory.mem_ctrl.dram.rank1.actBackEnergy   2214609030                    # Energy for active background per rank (pJ) (Joule)
board.memory.mem_ctrl.dram.rank1.preBackEnergy   1613689440                    # Energy for precharge background per rank (pJ) (Joule)
board.memory.mem_ctrl.dram.rank1.actPowerDownEnergy           0                    # Energy for active power-down per rank (pJ) (Joule)
```

board.memory.mem_ctrl.dram.rank1.prePowerDownEnergy 0 # Energy for precharge power-down per rank (pJ) (Joule)

board.memory.mem_ctrl.dram.rank1.selfRefreshEnergy 0 # Energy for self refresh per rank (pJ) (Joule)

board.memory.mem_ctrl.dram.rank1.totalEnergy 5267393100 # Total energy per rank (pJ) (Joule)

board.memory.mem_ctrl.dram.rank1.averagePower 581.459664 # Core power per rank (mW) (Watt)

board.memory.mem_ctrl.dram.rank1.totalIdleTime 0 # Total Idle time Per DRAM Rank (Tick)

board.memory.mem_ctrl.dram.rank1.pwrStateTime::IDLE 4134639576 # Time in different power states (Tick)

board.memory.mem_ctrl.dram.rank1.pwrStateTime::REF 302380000 # Time in different power states (Tick)

board.memory.mem_ctrl.dram.rank1.pwrStateTime::SREF 0 # Time in different power states (Tick)

board.memory.mem_ctrl.dram.rank1.pwrStateTime::PRE_PDN 0 # Time in different power states (Tick)

board.memory.mem_ctrl.dram.rank1.pwrStateTime::ACT 4621894442 # Time in different power states (Tick)

board.memory.mem_ctrl.dram.rank1.pwrStateTime::ACT_PDN 0 # Time in different power states (Tick)

board.memory.mem_ctrl.power_state.pwrStateResidencyTicks::UNDEFINED 9058914018 # Cumulative time (in ticks) in various power states (Tick)

board.processor.cores.core.numCycles 27203947 # Number of cpu cycles simulated (Cycle)

board.processor.cores.core.cpi 0.738611 # CPI: cycles per instruction (core level) ((Cycle/Count))

board.processor.cores.core.ipc 1.353893 # IPC: instructions per cycle (core level) ((Count/Cycle))

board.processor.cores.core.numWorkItemsStarted 0 # Number of work items this cpu started (Count)

board.processor.cores.core.numWorkItemsCompleted 0 # Number of work items this cpu completed (Count)

board.processor.cores.core.instsAdded 57072822 # Number of instructions added to the IQ (excludes non-spec) (Count)

board.processor.cores.core.nonSpecInstsAdded 239 # Number of non-speculative instructions added to the IQ (Count)

board.processor.cores.core.instsIssued 57058671 # Number of instructions issued (Count)

board.processor.cores.core.squashedInstsIssued 582 # Number of squashed instructions issued (Count)

board.processor.cores.core.squashedInstsExamined 211847 # Number of squashed instructions iterated over during squash; mainly for profiling (Count)

board.processor.cores.core.squashedOperandsExamined 87596 # Number of squashed operands that are examined and possibly removed from graph (Count)

board.processor.cores.core.squashedNonSpecRemoved 149 # Number of squashed non-spec instructions that were removed (Count)

board.processor.cores.core.numIssuedDist::samples 27059826 # Number of insts issued each cycle (Count)

board.processor.cores.core.numIssuedDist::mean 2.108612 # Number of insts issued each cycle (Count)

board.processor.cores.core.numIssuedDist::stdev 2.231590 # Number of insts issued each cycle (Count)

board.processor.cores.core.numIssuedDist::underflows 0 0.00% 0.00% # Number of insts issued each cycle (Count)

board.processor.cores.core.numIssuedDist::0 9983850 36.90% 36.90% # Number of insts issued each cycle (Count)

board.processor.cores.core.numIssuedDist::1 2376794 8.78% 45.68% # Number of insts issued each cycle (Count)

board.processor.cores.core.numIssuedDist::2 5188826 19.18% 64.85% # Number of insts issued each cycle (Count)

board.processor.cores.core.numIssuedDist::3 3466064 12.81% 77.66% # Number of insts issued each cycle (Count)

board.processor.cores.core.numIssuedDist::4 1616006 5.97% 83.64% # Number of insts issued each cycle (Count)

board.processor.cores.core.numIssuedDist::5 1535874 5.68% 89.31% # Number of insts issued each cycle (Count)

board.processor.cores.core.numIssuedDist::6 1014134 3.75% 93.06% # Number of insts issued each cycle (Count)

board.processor.cores.core.numIssuedDist::7 1348389 4.98% 98.04% # Number of insts issued each cycle (Count)

board.processor.cores.core.numIssuedDist::8 529889 1.96% 100.00% # Number of insts issued each cycle (Count)

board.processor.cores.core.numIssuedDist::overflows 0 0.00% 100.00% # Number of insts issued each cycle (Count)

board.processor.cores.core.numIssuedDist::min_value 0 # Number of insts issued each cycle (Count)

board.processor.cores.core.numIssuedDist::max_value 8 # Number of insts issued each cycle (Count)

board.processor.cores.core.numIssuedDist::total 27059826 # Number of insts issued each cycle (Count)

board.processor.cores.core.statFuBusy::No_OpClass 0 0.00% 0.00% # attempts to use FU when none available (Count)

board.processor.cores.core.statFuBusy::IntAlu 1357061 99.96% 99.96% # attempts to use FU when none available (Count)

board.processor.cores.core.statFuBusy::IntMult 0 0.00% 99.96% # attempts to use FU when none available (Count)

board.processor.cores.core.statFuBusy::IntDiv 0 0.00% 99.96% # attempts to use FU when none available (Count)

board.processor.cores.core.statFuBusy::FloatAdd 0 0.00% 99.96% # attempts to use FU when none available (Count)

board.processor.cores.core.statFuBusy::FloatCmp 0 0.00% 99.96% # attempts to use FU when none available (Count)

board.processor.cores.core.statFuBusy::FloatCvt 0 0.00% 99.96% # attempts to use FU when none available (Count)

board.processor.cores.core.statFuBusy::FloatMult 0 0.00% 99.96% # attempts to use FU when none available (Count)

board.processor.cores.core.statFuBusy::FloatMultAcc 0 0.00% 99.96% # attempts to use FU when none available (Count)

board.processor.cores.core.statFuBusy::FloatDiv 0 0.00% 99.96% # attempts to use FU when none available (Count)

board.processor.cores.core.statFuBusy::FloatMisc 0 0.00% 99.96% # attempts to use FU when none available (Count)

board.processor.cores.core.statFuBusy::FloatSqrt 0 0.00% 99.96% # attempts to use FU when none available (Count)

board.processor.cores.core.statFuBusy::SimdAdd 0 0.00% 99.96% # attempts to use FU when none available (Count)

board.processor.cores.core.statFuBusy::SimdAddAcc 0 0.00% 99.96% # attempts to use FU when none available (Count)

board.processor.cores.core.statFuBusy::SimdAlu 24 0.00% 99.96% # attempts to use FU when none available (Count)

board.processor.cores.core.statFuBusy::SimdCmp 0 0.00% 99.96% # attempts to use FU when none available (Count)

board.processor.cores.core.statFuBusy::SimdCvt 0 0.00% 99.96% # attempts to use FU when none available (Count)

board.processor.cores.core.statFuBusy::SimdMisc 0 0.00% 99.96% # attempts to use FU when none available (Count)

board.processor.cores.core.statFuBusy::SimdMult 0 0.00% 99.96% # attempts to use FU when none available (Count)

board.processor.cores.core.statFuBusy::SimdMultAcc 0 0.00% 99.96% # attempts to use FU when none available (Count)

board.processor.cores.core.statFuBusy::SimdMatMultAcc 0 0.00% 99.96% # attempts to use FU when none available (Count)

board.processor.cores.core.statFuBusy::SimdShift 0 0.00% 99.96% # attempts to use FU when none available (Count)

board.processor.cores.core.statFuBusy::SimdShiftAcc 0 0.00% 99.96% # attempts to use FU when none available (Count)

board.processor.cores.core.statFuBusy::SimdDiv 0 0.00% 99.96% # attempts to use FU when none available (Count)

board.processor.cores.core.statFuBusy::SimdSqrt 0 0.00% 99.96% # attempts to use FU when none available (Count)

board.processor.cores.core.statFuBusy::SimdFloatAdd 0 0.00% 99.96% # attempts to use FU when none available (Count)

board.processor.cores.core.statFuBusy::SimdFloatAlu 0 0.00% 99.96% # attempts to use FU when none available (Count)

board.processor.cores.core.statFuBusy::SimdFloatCmp 0 0.00% 99.96% # attempts to use FU when none available (Count)

board.processor.cores.core.statFuBusy::SimdFloatCvt 0 0.00% 99.96% # attempts to use FU when none available (Count)

board.processor.cores.core.statFuBusy::SimdFloatDiv 0 0.00% 99.96% # attempts to use FU when none available (Count)

board.processor.cores.core.statFuBusy::SimdFloatMisc 0 0.00% 99.96% # attempts to use FU when none available (Count)

board.processor.cores.core.statFuBusy::SimdFloatMult 0 0.00% 99.96% # attempts to use FU when none available (Count)

board.processor.cores.core.statFuBusy::SimdFloatMultAcc 0 0.00% 99.96% # attempts to use FU when none available (Count)

board.processor.cores.core.statFuBusy::SimdFloatMatMultAcc 0 0.00% 99.96% # attempts to use FU when none available (Count)

board.processor.cores.core.statFuBusy::SimdFloatSqrt 0 0.00% 99.96% # attempts to use FU when none available (Count)

board.processor.cores.core.statFuBusy::SimdReduceAdd 0 0.00% 99.96% # attempts to use FU when none available (Count)

board.processor.cores.core.statFuBusy::SimdReduceAlu 0 0.00% 99.96% # attempts to use FU when none available (Count)

board.processor.cores.core.statFuBusy::SimdReduceCmp 0 0.00% 99.96% # attempts to use FU when none available (Count)

board.processor.cores.core.statFuBusy::SimdFloatReduceAdd 0 0.00% 99.96% # attempts to use FU when none available (Count)

board.processor.cores.core.statFuBusy::SimdFloatReduceCmp 0 0.00% 99.96% # attempts to use FU when none available (Count)

board.processor.cores.core.statFuBusy::SimdAes 0 0.00% 99.96% # attempts to use FU when none available (Count)

board.processor.cores.core.statFuBusy::SimdAesMix 0 0.00% 99.96% # attempts to use FU when none available (Count)

board.processor.cores.core.statFuBusy::SimdSha1Hash 0 0.00% 99.96% # attempts to use FU when none available (Count)

board.processor.cores.core.statFuBusy::SimdSha1Hash2 0 0.00% 99.96% # attempts to use FU when none available (Count)

board.processor.cores.core.statFuBusy::SimdSha256Hash 0 0.00% 99.96% # attempts to use FU when none available (Count)

board.processor.cores.core.statFuBusy::SimdSha256Hash2 0 0.00% 99.96% # attempts to use FU when none available (Count)

board.processor.cores.core.statFuBusy::SimdShaSigma2 0 0.00% 99.96% # attempts to use FU when none available (Count)

board.processor.cores.core.statFuBusy::SimdShaSigma3 0 0.00% 99.96% # attempts to use FU when none available (Count)

board.processor.cores.core.statFuBusy::SimdPredAlu 0 0.00% 99.96% # attempts to use FU when none available (Count)

board.processor.cores.core.statFuBusy::Matrix 0 0.00% 99.96% # attempts to use FU when none available (Count)

board.processor.cores.core.statFuBusy::MatrixMov 0 0.00% 99.96% # attempts to use FU when none available (Count)

board.processor.cores.core.statFuBusy::MatrixOP 0 0.00% 99.96% # attempts to use FU when none available (Count)

board.processor.cores.core.statFuBusy::MemRead 205 0.02% 99.98% # attempts to use FU when none available (Count)

board.processor.cores.core.statFuBusy::MemWrite 279 0.02% 100.00% # attempts to use FU when none available (Count)

board.processor.cores.core.statFuBusy::FloatMemRead 11 0.00% 100.00% # attempts to use FU when none available (Count)

board.processor.cores.core.statFuBusy::FloatMemWrite 35 0.00% 100.00% # attempts to use FU when none available (Count)

board.processor.cores.core.statFuBusy::IprAccess 0 0.00% 100.00% # attempts to use FU when none available (Count)

board.processor.cores.core.statFuBusy::InstPrefetch 0 0.00% 100.00% # attempts to use FU when none available (Count)

board.processor.cores.core.statFuBusy::SimdUnitStrideLoad 0 0.00% 100.00% # attempts to use FU when none available (Count)

board.processor.cores.core.statFuBusy::SimdUnitStrideStore 0 0.00% 100.00% # attempts to use FU when none available (Count)

board.processor.cores.core.statFuBusy::SimdUnitStrideMaskLoad 0 0.00% 100.00% # attempts to use FU when none available (Count)

board.processor.cores.core.statFuBusy::SimdUnitStrideMaskStore 0 0.00% 100.00% # attempts to use FU when none available (Count)

board.processor.cores.core.statFuBusy::SimdStridedLoad 0 0.00% 100.00% # attempts to use FU when none available (Count)

board.processor.cores.core.statFuBusy::SimdStridedStore 0 0.00% 100.00% # attempts to use FU when none available (Count)

board.processor.cores.core.statFuBusy::SimdIndexedLoad 0 0.00% 100.00% # attempts to use FU when none available (Count)

board.processor.cores.core.statFuBusy::SimdIndexedStore 0 0.00% 100.00% # attempts to use FU when none available (Count)

board.processor.cores.core.statFuBusy::SimdWholeRegisterLoad 0 0.00% 100.00% # attempts to use FU when none available (Count)

board.processor.cores.core.statFuBusy::SimdWholeRegisterStore 0 0.00% 100.00% # attempts to use FU when none available (Count)

board.processor.cores.core.statFuBusy::SimdUnitStrideFaultOnlyFirstLoad 0 0.00% 100.00% # attempts to use FU when none available (Count)

board.processor.cores.core.statFuBusy::SimdUnitStrideSegmentedLoad 0 0.00% 100.00% # attempts to use FU when none available (Count)

board.processor.cores.core.statFuBusy::SimdUnitStrideSegmentedStore 0 0.00% 100.00% # attempts to use FU when none available (Count)

board.processor.cores.core.statFuBusy::SimdExt 0 0.00% 100.00% # attempts to use FU when none available (Count)

board.processor.cores.core.statFuBusy::SimdFloatExt 0 0.00% 100.00% # attempts to use FU when none available (Count)

board.processor.cores.core.statFuBusy::SimdConfig 0 0.00% 100.00% # attempts to use FU when none available (Count)

board.processor.cores.core.statIssuedInstType_0::No_OpClass 758 0.00% 0.00% # Number of instructions issued per FU type, per thread (Count)

board.processor.cores.core.statIssuedInstType_0::IntAlu 40200797 70.46% 70.46% # Number of instructions issued per FU type, per thread (Count)

board.processor.cores.core.statIssuedInstType_0::IntMult 1054102 1.85% 72.30% # Number of instructions issued per FU type, per thread (Count)

board.processor.cores.core.statIssuedInstType_0::IntDiv 40 0.00% 72.30% # Number of instructions issued per FU type, per thread (Count)

board.processor.cores.core.statIssuedInstType_0::FloatAdd 192 0.00% 72.30% # Number of instructions issued per FU type, per thread (Count)

board.processor.cores.core.statIssuedInstType_0::FloatCmp 0 0.00% 72.30% # Number of instructions issued per FU type, per thread (Count)

board.processor.cores.core.statIssuedInstType_0::FloatCvt 0 0.00% 72.30% # Number of instructions issued per FU type, per thread (Count)

board.processor.cores.core.statIssuedInstType_0::FloatMult 0 0.00% 72.30% # Number of instructions issued per FU type, per thread (Count)

board.processor.cores.core.statIssuedInstType_0::FloatMultAcc 0 0.00% 72.30% # Number of instructions issued per FU type, per thread (Count)

board.processor.cores.core.statIssuedInstType_0::FloatDiv 0 0.00% 72.30% # Number of instructions issued per FU type, per thread (Count)

board.processor.cores.core.statIssuedInstType_0::FloatMisc 0 0.00% 72.30% # Number of instructions issued per FU type, per thread (Count)

board.processor.cores.core.statIssuedInstType_0::FloatSqrt 0 0.00% 72.30% # Number of instructions issued per FU type, per thread (Count)

board.processor.cores.core.statIssuedInstType_0::SimdAdd 10 0.00% 72.30% # Number of instructions issued per FU type, per thread (Count)

board.processor.cores.core.statIssuedInstType_0::SimdAddAcc 0 0.00% 72.30% # Number of instructions issued per FU type, per thread (Count)

board.processor.cores.core.statIssuedInstType_0::SimdAlu 287 0.00% 72.30% # Number of instructions issued per FU type, per thread (Count)

board.processor.cores.core.statIssuedInstType_0::SimdCmp 2 0.00% 72.30% # Number of instructions issued per FU type, per thread (Count)

board.processor.cores.core.statIssuedInstType_0::SimdCvt 78 0.00% 72.30% # Number of instructions issued per FU type, per thread (Count)

board.processor.cores.core.statIssuedInstType_0::SimdMisc 253 0.00% 72.31% # Number of instructions issued per FU type, per thread (Count)

board.processor.cores.core.statIssuedInstType_0::SimdMult 0 0.00% 72.31% # Number of instructions issued per FU type, per thread (Count)

board.processor.cores.core.statIssuedInstType_0::SimdMultAcc 0 0.00% 72.31% # Number of instructions issued per FU type, per thread (Count)

board.processor.cores.core.statIssuedInstType_0::SimdMatMultAcc 0 0.00% 72.31% # Number of instructions issued per FU type, per thread (Count)

board.processor.cores.core.statIssuedInstType_0::SimdShift 6 0.00% 72.31% # Number of instructions issued per FU type, per thread (Count)

board.processor.cores.core.statIssuedInstType_0::SimdShiftAcc 0 0.00% 72.31% # Number of instructions issued per FU type, per thread (Count)

board.processor.cores.core.statIssuedInstType_0::SimdDiv 0 0.00% 72.31% # Number of instructions issued per FU type, per thread (Count)

board.processor.cores.core.statIssuedInstType_0::SimdSqrt 0 0.00% 72.31% # Number of instructions issued per FU type, per thread (Count)

board.processor.cores.core.statIssuedInstType_0::SimdFloatAdd 0 0.00% 72.31% # Number of instructions issued per FU type, per thread (Count)

board.processor.cores.core.statIssuedInstType_0::SimdFloatAlu 0 0.00% 72.31% # Number of instructions issued per FU type, per thread (Count)

board.processor.cores.core.statIssuedInstType_0::SimdFloatCmp 0 0.00% 72.31% # Number of instructions issued per FU type, per thread (Count)

board.processor.cores.core.statIssuedInstType_0::SimdFloatCvt 0 0.00% 72.31% # Number of instructions issued per FU type, per thread (Count)

board.processor.cores.core.statIssuedInstType_0::SimdFloatDiv 0 0.00% 72.31% # Number of instructions issued per FU type, per thread (Count)

board.processor.cores.core.statIssuedInstType_0::SimdFloatMisc 0 0.00% 72.31% # Number of instructions issued per FU type, per thread (Count)

board.processor.cores.core.statIssuedInstType_0::SimdFloatMult 0 0.00% 72.31% # Number of instructions issued per FU type, per thread (Count)

board.processor.cores.core.statIssuedInstType_0::SimdFloatMultAcc 0 0.00% 72.31% # Number of instructions issued per FU type, per thread (Count)

board.processor.cores.core.statIssuedInstType_0::SimdFloatMatMultAcc 0 0.00% 72.31% # Number of instructions issued per FU type, per thread (Count)

board.processor.cores.core.statIssuedInstType_0::SimdFloatSqrt 0 0.00% 72.31% # Number of instructions issued per FU type, per thread (Count)

board.processor.cores.core.statIssuedInstType_0::SimdReduceAdd 0 0.00% 72.31% # Number of instructions issued per FU type, per thread (Count)

board.processor.cores.core.statIssuedInstType_0::SimdReduceAlu 0 0.00% 72.31% # Number of instructions issued per FU type, per thread (Count)

board.processor.cores.core.statIssuedInstType_0::SimdReduceCmp 0 0.00% 72.31% # Number of instructions issued per FU type, per thread (Count)

board.processor.cores.core.statIssuedInstType_0::SimdFloatReduceAdd 0 0.00% 72.31% # Number of instructions issued per FU type, per thread (Count)

board.processor.cores.core.statIssuedInstType_0::SimdFloatReduceCmp 0 0.00% 72.31% # Number of instructions issued per FU type, per thread (Count)

board.processor.cores.core.statIssuedInstType_0::SimdAes 0 0.00% 72.31% # Number of instructions issued per FU type, per thread (Count)

board.processor.cores.core.statIssuedInstType_0::SimdAesMix 0 0.00% 72.31% # Number of instructions issued per FU type, per thread (Count)

board.processor.cores.core.statIssuedInstType_0::SimdSha1Hash 0 0.00% 72.31% # Number of instructions issued per FU type, per thread (Count)

board.processor.cores.core.statIssuedInstType_0::SimdSha1Hash2 0 0.00% 72.31% # Number of instructions issued per FU type, per thread (Count)

board.processor.cores.core.statIssuedInstType_0::SimdSha256Hash 0 0.00% 72.31% # Number of instructions issued per FU type, per thread (Count)

board.processor.cores.core.statIssuedInstType_0::SimdSha256Hash2 0 0.00% 72.31% # Number of instructions issued per FU type, per thread (Count)

board.processor.cores.core.statIssuedInstType_0::SimdShaSigma2 0 0.00% 72.31% # Number of instructions issued per FU type, per thread (Count)

board.processor.cores.core.statIssuedInstType_0::SimdShaSigma3 0 0.00% 72.31% # Number of instructions issued per FU type, per thread (Count)

board.processor.cores.core.statIssuedInstType_0::SimdPredAlu 0 0.00% 72.31% # Number of instructions issued per FU type, per thread (Count)

board.processor.cores.core.statIssuedInstType_0::Matrix 0 0.00% 72.31% # Number of instructions issued per FU type, per thread (Count)

board.processor.cores.core.statIssuedInstType_0::MatrixMov 0 0.00% 72.31% # Number of instructions issued per FU type, per thread (Count)

board.processor.cores.core.statIssuedInstType_0::MatrixOP 0 0.00% 72.31% # Number of instructions issued per FU type, per thread (Count)

board.processor.cores.core.statIssuedInstType_0::MemRead 11584442 20.30% 92.61% # Number of instructions issued per FU type, per thread (Count)

board.processor.cores.core.statIssuedInstType_0::MemWrite 4216850 7.39% 100.00% # Number of instructions issued per FU type, per thread (Count)

board.processor.cores.core.statIssuedInstType_0::FloatMemRead 216 0.00% 100.00% # Number of instructions issued per FU type, per thread (Count)

board.processor.cores.core.statIssuedInstType_0::FloatMemWrite 638 0.00% 100.00% # Number of instructions issued per FU type, per thread (Count)

board.processor.cores.core.statIssuedInstType_0::IprAccess 0 0.00% 100.00% # Number of instructions issued per FU type, per thread (Count)

board.processor.cores.core.statIssuedInstType_0::InstPrefetch 0 0.00% 100.00% # Number of instructions issued per FU type, per thread (Count)

board.processor.cores.core.statIssuedInstType_0::SimdUnitStrideLoad 0 0.00% 100.00% # Number of instructions issued per FU type, per thread (Count)

board.processor.cores.core.statIssuedInstType_0::SimdUnitStrideStore 0 0.00% 100.00% # Number of instructions issued per FU type, per thread (Count)

board.processor.cores.core.statIssuedInstType_0::SimdUnitStrideMaskLoad 0 0.00% 100.00% # Number of instructions issued per FU type, per thread (Count)

board.processor.cores.core.statIssuedInstType_0::SimdUnitStrideMaskStore 0 0.00% 100.00% # Number of instructions issued per FU type, per thread (Count)

board.processor.cores.core.statIssuedInstType_0::SimdStridedLoad 0 0.00% 100.00% # Number of instructions issued per FU type, per thread (Count)

board.processor.cores.core.statIssuedInstType_0::SimdStridedStore 0 0.00% 100.00% # Number of instructions issued per FU type, per thread (Count)

board.processor.cores.core.statIssuedInstType_0::SimdIndexedLoad 0 0.00% 100.00% # Number of instructions issued per FU type, per thread (Count)

board.processor.cores.core.statIssuedInstType_0::SimdIndexedStore 0 0.00% 100.00% # Number of instructions issued per FU type, per thread (Count)

board.processor.cores.core.statIssuedInstType_0::SimdWholeRegisterLoad 0 0.00% 100.00% # Number of instructions issued per FU type, per thread (Count)

board.processor.cores.core.statIssuedInstType_0::SimdWholeRegisterStore 0 0.00% 100.00% # Number of instructions issued per FU type, per thread (Count)

board.processor.cores.core.statIssuedInstType_0::SimdUnitStrideFaultOnlyFirstLoad 0 0.00% 100.00% # Number of instructions issued per FU type, per thread (Count)

board.processor.cores.core.statIssuedInstType_0::SimdUnitStrideSegmentedLoad 0 0.00% 100.00% # Number of instructions issued per FU type, per thread (Count)

board.processor.cores.core.statIssuedInstType_0::SimdUnitStrideSegmentedStore 0 0.00% 100.00% # Number of instructions issued per FU type, per thread (Count)

board.processor.cores.core.statIssuedInstType_0::SimdExt 0 0.00% 100.00% # Number of instructions issued per FU type, per thread (Count)

board.processor.cores.core.statIssuedInstType_0::SimdFloatExt 0 0.00% 100.00% # Number of instructions issued per FU type, per thread (Count)

board.processor.cores.core.statIssuedInstType_0::SimdConfig 0 0.00% 100.00% # Number of instructions issued per FU type, per thread (Count)

board.processor.cores.core.statIssuedInstType_0::total 57058671 # Number of instructions issued per FU type, per thread (Count)

board.processor.cores.core.issueRate 2.097441 # Inst issue rate ((Count/Cycle))

board.processor.cores.core.fuBusy 1357615 # FU busy when requested (Count)

board.processor.cores.core.fuBusyRate 0.023793 # FU busy rate (busy events/executed inst) ((Count/Count))

board.processor.cores.core.intInstQueueReads 142531517 # Number of integer instruction queue reads (Count)

board.processor.cores.core.intInstQueueWrites 57282513 # Number of integer instruction queue writes (Count)

board.processor.cores.core.intInstQueueWakeupAccesses 57050933 # Number of integer instruction queue wakeup accesses (Count)

board.processor.cores.core.fpInstQueueReads 3848 # Number of floating instruction queue reads (Count)

board.processor.cores.core.fpInstQueueWrites 2503 # Number of floating instruction queue writes (Count)

board.processor.cores.core.fpInstQueueWakeupAccesses 1776 # Number of floating instruction queue wakeup accesses (Count)

board.processor.cores.core.vecInstQueueReads 0 # Number of vector instruction queue reads (Count)

board.processor.cores.core.vecInstQueueWrites 0 # Number of vector instruction queue writes (Count)

board.processor.cores.core.vecInstQueueWakeupAccesses 0 # Number of vector instruction queue wakeup accesses (Count)

board.processor.cores.core.intAluAccesses 58413595 # Number of integer alu accesses (Count)

board.processor.cores.core.fpAluAccesses 1933 # Number of floating point alu accesses (Count)

board.processor.cores.core.vecAluAccesses 0 # Number of vector alu accesses (Count)

board.processor.cores.core.numSquashedInsts 1560 # Number of squashed instructions skipped in execute (Count)

board.processor.cores.core.numSwp 0 # Number of swp insts executed (Count)

board.processor.cores.core.timesIdled 519 # Number of times that the entire CPU went into an idle state and unscheduled itself (Count)

board.processor.cores.core.idleCycles 144121 # Total number of cycles that the CPU has spent unscheduled due to idling (Cycle)

board.processor.cores.core.MemDepUnit__0.insertedLoads 11586182 # Number of loads inserted to the mem dependence unit. (Count)

board.processor.cores.core.MemDepUnit__0.insertedStores 4218844 # Number of stores inserted to the mem dependence unit. (Count)

board.processor.cores.core.MemDepUnit__0.conflictingLoads 7481305 # Number of conflicting loads. (Count)

board.processor.cores.core.MemDepUnit__0.conflictingStores 1287395 # Number of conflicting stores. (Count)

board.processor.cores.core.MemDepUnit__1.insertedLoads 0 # Number of loads inserted to the mem dependence unit. (Count)

board.processor.cores.core.MemDepUnit__1.insertedStores 0 # Number of stores inserted to the mem dependence unit. (Count)

board.processor.cores.core.MemDepUnit__1.conflictingLoads 0 # Number of conflicting loads. (Count)

board.processor.cores.core.MemDepUnit__1.conflictingStores 0 # Number of conflicting stores. (Count)

board.processor.cores.core.MemDepUnit__2.insertedLoads 0 # Number of loads inserted to the mem dependence unit. (Count)

board.processor.cores.core.MemDepUnit__2.insertedStores 0 # Number of stores inserted to the mem dependence unit. (Count)

board.processor.cores.core.MemDepUnit__2.conflictingLoads 0 # Number of conflicting loads. (Count)

board.processor.cores.core.MemDepUnit__2.conflictingStores 0 # Number of conflicting stores. (Count)

board.processor.cores.core.MemDepUnit__3.insertedLoads 0 # Number of loads inserted to the mem dependence unit. (Count)

board.processor.cores.core.MemDepUnit__3.insertedStores 0 # Number of stores inserted to the mem dependence unit. (Count)

board.processor.cores.core.MemDepUnit__3.conflictingLoads 0 # Number of conflicting loads. (Count)

board.processor.cores.core.MemDepUnit__3.conflictingStores 0 # Number of conflicting stores. (Count)

board.processor.cores.core.branchPred.lookups_0::NoBranch 4 0.00% 0.00% # Number of BP lookups (Count)

board.processor.cores.core.branchPred.lookups_0::Return 2785 0.13% 0.13% # Number of BP lookups (Count)

board.processor.cores.core.branchPred.lookups_0::CallDirect 2891 0.13% 0.27% # Number of BP lookups (Count)

board.processor.cores.core.branchPred.lookups_0::CallIndirect 187 0.01% 0.27% # Number of BP lookups (Count)

board.processor.cores.core.branchPred.lookups_0::DirectCond 2130577 99.44% 99.71% # Number of BP lookups (Count)

board.processor.cores.core.branchPred.lookups_0::DirectUncond 3758 0.18% 99.89% # Number of BP lookups (Count)

board.processor.cores.core.branchPred.lookups_0::IndirectCond 0 0.00% 99.89% # Number of BP lookups (Count)

board.processor.cores.core.branchPred.lookups_0::IndirectUncond 2455 0.11% 100.00% # Number of BP lookups (Count)

board.processor.cores.core.branchPred.lookups_0::total 2142657 # Number of BP lookups (Count)

board.processor.cores.core.branchPred.squashes_0::NoBranch 4 0.03% 0.03% # Number of branches that got squashed (completely removed) as an earlier branch was mispredicted. (Count)

board.processor.cores.core.branchPred.squashes_0::Return 416 3.41% 3.44% # Number of branches that got squashed (completely removed) as an earlier branch was mispredicted. (Count)

board.processor.cores.core.branchPred.squashes_0::CallDirect 564 4.62% 8.07% # Number of branches that got squashed (completely removed) as an earlier branch was mispredicted. (Count)

board.processor.cores.core.branchPred.squashes_0::CallIndirect 140 1.15% 9.21% # Number of branches that got squashed (completely removed) as an earlier branch was mispredicted. (Count)

board.processor.cores.core.branchPred.squashes_0::DirectCond 10319 84.59% 93.80% # Number of branches that got squashed (completely removed) as an earlier branch was mispredicted. (Count)

board.processor.cores.core.branchPred.squashes_0::DirectUncond 439 3.60% 97.40% # Number of branches that got squashed (completely removed) as an earlier branch was mispredicted. (Count)

board.processor.cores.core.branchPred.squashes_0::IndirectCond 0 0.00% 97.40% # Number of branches that got squashed (completely removed) as an earlier branch was mispredicted. (Count)

board.processor.cores.core.branchPred.squashes_0::IndirectUncond 317 2.60% 100.00% # Number of branches that got squashed (completely removed) as an earlier branch was mispredicted. (Count)

board.processor.cores.core.branchPred.squashes_0::total 12199 # Number of branches that got squashed (completely removed) as an earlier branch was mispredicted. (Count)

board.processor.cores.core.branchPred.corrected_0::NoBranch 0 0.00% 0.00% # Number of branches that got corrected but not yet commited. Branches get corrected by decode or after execute. Also a branch misprediction can be detected out-of-order. Therefore, a corrected branch might not end up beeing committed in case an even earlier branch was mispredicted (Count)

board.processor.cores.core.branchPred.corrected_0::Return 10 0.93% 0.93% # Number of branches that got corrected but not yet commited. Branches get corrected by decode or after execute. Also a branch misprediction can be detected out-of-order. Therefore, a corrected branch might not end up beeing committed in case an even earlier branch was mispredicted (Count)

board.processor.cores.core.branchPred.corrected_0::CallDirect 222 20.73% 21.66% # Number of branches that got corrected but not yet commited. Branches get corrected by decode or after execute. Also a branch misprediction can be detected out-of-order. Therefore, a corrected branch might not end up beeing committed in case an even earlier branch was mispredicted (Count)

board.processor.cores.core.branchPred.corrected_0::CallIndirect 39 3.64% 25.30% # Number of branches that got corrected but not yet commited. Branches get corrected by decode or after execute. Also a branch misprediction can be detected out-of-order. Therefore, a corrected branch might not end up beeing committed in case an even earlier branch was mispredicted (Count)

board.processor.cores.core.branchPred.corrected_0::DirectCond 600 56.02% 81.33% # Number of branches that got corrected but not yet commited. Branches get corrected by decode or after execute. Also a branch misprediction can be detected out-of-order. Therefore, a corrected branch might not end up beeing committed in case an even earlier branch was mispredicted (Count)

board.processor.cores.core.branchPred.corrected_0::DirectUncond 151 14.10% 95.42% # Number of branches that got corrected but not yet commited. Branches get corrected by decode or after execute. Also a branch misprediction can be detected out-of-order. Therefore, a corrected branch might not end up beeing committed in case an even earlier branch was mispredicted (Count)

board.processor.cores.core.branchPred.corrected_0::IndirectCond 0 0.00% 95.42% # Number of branches that got corrected but not yet commited. Branches get corrected by decode or after execute. Also a branch misprediction can be detected out-of-order. Therefore, a corrected branch might not end up beeing committed in case an even earlier branch was mispredicted (Count)

board.processor.cores.core.branchPred.corrected_0::IndirectUncond 49 4.58% 100.00% # Number of branches that got corrected but not yet commited. Branches get corrected by decode or after execute. Also a branch misprediction can be detected out-of-order. Therefore, a corrected branch might not end up beeing committed in case an even earlier branch was mispredicted (Count)

board.processor.cores.core.branchPred.corrected_0::total 1071 # Number of branches that got corrected but not yet commited. Branches get corrected by decode or after execute. Also a branch misprediction can be detected out-of-order. Therefore, a corrected branch might not end up beeing committed in case an even earlier branch was mispredicted (Count)

board.processor.cores.core.branchPred.earlyResteers_0::NoBranch 0 # Number of branches that got redirected after decode. (Count)

board.processor.cores.core.branchPred.earlyResteers_0::Return 0 # Number of branches that got redirected after decode. (Count)

board.processor.cores.core.branchPred.earlyResteers_0::CallDirect 0 # Number of branches that got redirected after decode. (Count)

board.processor.cores.core.branchPred.earlyResteers_0::CallIndirect 0 # Number of branches that got redirected after decode. (Count)

board.processor.cores.core.branchPred.earlyResteers_0::DirectCond 0 # Number of branches that got redirected after decode. (Count)

board.processor.cores.core.branchPred.earlyResteers_0::DirectUncond 0 # Number of branches that got redirected after decode. (Count)

board.processor.cores.core.branchPred.earlyResteers_0::IndirectCond 0 # Number of branches that got redirected after decode. (Count)

board.processor.cores.core.branchPred.earlyResteers_0::IndirectUncond 0 # Number of branches that got redirected after decode. (Count)

board.processor.cores.core.branchPred.earlyResteers_0::total 0 # Number of branches that got redirected after decode. (Count)

board.processor.cores.core.branchPred.committed_0::NoBranch 0 0.00% 0.00% # Number of branches finally committed (Count)

board.processor.cores.core.branchPred.committed_0::Return 2369 0.11% 0.11% # Number of branches finally committed (Count)

board.processor.cores.core.branchPred.committed_0::CallDirect 2327 0.11% 0.22% # Number of branches finally committed (Count)

board.processor.cores.core.branchPred.committed_0::CallIndirect 47 0.00% 0.22% # Number of branches finally committed (Count)

board.processor.cores.core.branchPred.committed_0::DirectCond 2120254 99.52% 99.74% # Number of branches finally committed (Count)

board.processor.cores.core.branchPred.committed_0::DirectUncond 3319 0.16% 99.90% # Number of branches finally committed (Count)

board.processor.cores.core.branchPred.committed_0::IndirectCond 0 0.00% 99.90% # Number of branches finally committed (Count)

board.processor.cores.core.branchPred.committed_0::IndirectUncond 2138 0.10% 100.00% # Number of branches finally committed (Count)

board.processor.cores.core.branchPred.committed_0::total 2130454 # Number of branches finally committed (Count)

board.processor.cores.core.branchPred.mispredicted_0::NoBranch 0 0.00% 0.00% # Number of committed branches that were mispredicted. (Count)

board.processor.cores.core.branchPred.mispredicted_0::Return 0 0.00% 0.00% # Number of committed branches that were mispredicted. (Count)

board.processor.cores.core.branchPred.mispredicted_0::CallDirect 105 12.93% 12.93% # Number of committed branches that were mispredicted. (Count)

board.processor.cores.core.branchPred.mispredicted_0::CallIndirect 38 4.68% 17.61% # Number of committed branches that were mispredicted. (Count)

board.processor.cores.core.branchPred.mispredicted_0::DirectCond 538 66.26% 83.87% # Number of committed branches that were mispredicted. (Count)

board.processor.cores.core.branchPred.mispredicted_0::DirectUncond 88 10.84% 94.70% # Number of committed branches that were mispredicted. (Count)

board.processor.cores.core.branchPred.mispredicted_0::IndirectCond 0 0.00% 94.70% # Number of committed branches that were mispredicted. (Count)

board.processor.cores.core.branchPred.mispredicted_0::IndirectUncond 43 5.30% 100.00% # Number of committed branches that were mispredicted. (Count)

board.processor.cores.core.branchPred.mispredicted_0::total 812 # Number of committed branches that were mispredicted. (Count)

board.processor.cores.core.branchPred.targetProvider_0::NoTarget 23046 1.08% 1.08% # The component providing the target for taken branches (Count)

board.processor.cores.core.branchPred.targetProvider_0::BTB 2114627 98.69% 99.77% # The component providing the target for taken branches (Count)

board.processor.cores.core.branchPred.targetProvider_0::RAS 2785 0.13% 99.90% # The component providing the target for taken branches (Count)

board.processor.cores.core.branchPred.targetProvider_0::Indirect 2199 0.10% 100.00% # The component providing the target for taken branches (Count)

board.processor.cores.core.branchPred.targetProvider_0::total 2142657 # The component providing the target for taken branches (Count)

board.processor.cores.core.branchPred.targetWrong_0::NoBranch 968 92.02% 92.02% # Number of branches where the target was incorrect or not available at prediction time. (Count)

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board.processor.cores.core.branchPred.targetWrong_0::Return          73     6.94%     98.95% # Number of branches where the target was incorrect or not available at prediction time. (Count)
board.processor.cores.core.branchPred.targetWrong_0::CallDirect          10     0.95%     99.90% # Number of branches where the target was incorrect or not available at prediction time. (Count)
board.processor.cores.core.branchPred.targetWrong_0::CallIndirect           1     0.10%    100.00% # Number of branches where the target was incorrect or not available at prediction time. (Count)
board.processor.cores.core.branchPred.targetWrong_0::DirectCond           0     0.00%    100.00% # Number of branches where the target was incorrect or not available at prediction time. (Count)
board.processor.cores.core.branchPred.targetWrong_0::DirectUncond           0     0.00%    100.00% # Number of branches where the target was incorrect or not available at prediction time. (Count)
board.processor.cores.core.branchPred.targetWrong_0::IndirectCond           0     0.00%    100.00% # Number of branches where the target was incorrect or not available at prediction time. (Count)
board.processor.cores.core.branchPred.targetWrong_0::IndirectUncond           0     0.00%    100.00% # Number of branches where the target was incorrect or not available at prediction time. (Count)
board.processor.cores.core.branchPred.targetWrong_0::total        1052                    # Number of branches where the target was incorrect or not available at prediction time. (Count)
board.processor.cores.core.branchPred.condPredicted      2130581                    # Number of conditional branches predicted (Count)
board.processor.cores.core.branchPred.condPredictedTaken      2109201                    # Number of conditional branches predicted as taken (Count)
board.processor.cores.core.branchPred.condIncorrect         1071                    # Number of conditional branches incorrect (Count)
board.processor.cores.core.branchPred.predTakenBTBMiss         484                    # Number of branches predicted taken but missed in BTB (Count)
board.processor.cores.core.branchPred.NotTakenMispredicted         1005                    # Number branches predicted 'not taken' but turned out to be taken (Count)
board.processor.cores.core.branchPred.TakenMispredicted          66                    # Number branches predicted taken but are actually not taken (Count)
board.processor.cores.core.branchPred.BTBLookups      2142657                    # Number of BTB lookups (Count)
board.processor.cores.core.branchPred.BTBUpdates         907                    # Number of BTB updates (Count)
board.processor.cores.core.branchPred.BTBHits      2117905                    # Number of BTB hits (Count)
board.processor.cores.core.branchPred.BTBHitRatio     0.988448                    # BTB Hit Ratio (Ratio)
board.processor.cores.core.branchPred.BTBMispredicted         703                    # Number BTB mispredictions. No target found or target wrong (Count)
board.processor.cores.core.branchPred.indirectLookups         2642                    # Number of indirect predictor lookups. (Count)
board.processor.cores.core.branchPred.indirectHits         2199                    # Number of indirect target hits. (Count)
board.processor.cores.core.branchPred.indirectMisses         443                    # Number of indirect misses. (Count)
board.processor.cores.core.branchPred.indirectMispredicted           0                    # Number of mispredicted indirect branches. (Count)
board.processor.cores.core.branchPred.btb.lookups::NoBranch           4     0.00%     0.00% # Number of BTB lookups (Count)
board.processor.cores.core.branchPred.btb.lookups::Return        2785     0.13%     0.13% # Number of BTB lookups (Count)
board.processor.cores.core.branchPred.btb.lookups::CallDirect        2891     0.13%     0.27% # Number of BTB lookups (Count)
board.processor.cores.core.branchPred.btb.lookups::CallIndirect         187     0.01%     0.27% # Number of BTB lookups (Count)
board.processor.cores.core.branchPred.btb.lookups::DirectCond     2130577    99.44%    99.71% # Number of BTB lookups (Count)
board.processor.cores.core.branchPred.btb.lookups::DirectUncond        3758     0.18%    99.89% # Number of BTB lookups (Count)
board.processor.cores.core.branchPred.btb.lookups::IndirectCond           0     0.00%    99.89% # Number of BTB lookups (Count)
board.processor.cores.core.branchPred.btb.lookups::IndirectUncond        2455     0.11%   100.00% # Number of BTB lookups (Count)
board.processor.cores.core.branchPred.btb.lookups::total      2142657                    # Number of BTB lookups (Count)
board.processor.cores.core.branchPred.btb.misses::NoBranch           4     0.02%     0.02% # Number of BTB misses (Count)
board.processor.cores.core.branchPred.btb.misses::Return        2781    11.24%    11.25% # Number of BTB misses (Count)
board.processor.cores.core.branchPred.btb.misses::CallDirect         375     1.52%    12.77% # Number of BTB misses (Count)
board.processor.cores.core.branchPred.btb.misses::CallIndirect         187     0.76%    13.52% # Number of BTB misses (Count)
board.processor.cores.core.branchPred.btb.misses::DirectCond       18720    75.63%    89.15% # Number of BTB misses (Count)
board.processor.cores.core.branchPred.btb.misses::DirectUncond         230     0.93%    90.08% # Number of BTB misses (Count)
board.processor.cores.core.branchPred.btb.misses::IndirectCond           0     0.00%    90.08% # Number of BTB misses (Count)
board.processor.cores.core.branchPred.btb.misses::IndirectUncond        2455     9.92%   100.00% # Number of BTB misses (Count)
board.processor.cores.core.branchPred.btb.misses::total        24752                    # Number of BTB misses (Count)
board.processor.cores.core.branchPred.btb.updates::NoBranch           0     0.00%     0.00% # Number of BTB updates (Count)
board.processor.cores.core.branchPred.btb.updates::Return           0     0.00%     0.00% # Number of BTB updates (Count)
board.processor.cores.core.branchPred.btb.updates::CallDirect         222    24.48%    24.48% # Number of BTB updates (Count)
board.processor.cores.core.branchPred.btb.updates::CallIndirect           0     0.00%    24.48% # Number of BTB updates (Count)
board.processor.cores.core.branchPred.btb.updates::DirectCond         534    58.88%    83.35% # Number of BTB updates (Count)
board.processor.cores.core.branchPred.btb.updates::DirectUncond         151    16.65%   100.00% # Number of BTB updates (Count)
board.processor.cores.core.branchPred.btb.updates::IndirectCond           0     0.00%   100.00% # Number of BTB updates (Count)
board.processor.cores.core.branchPred.btb.updates::IndirectUncond           0     0.00%   100.00% # Number of BTB updates (Count)
```

board.processor.cores.core.branchPred.btb.updates::total 907 # Number of BTB updates (Count)

board.processor.cores.core.branchPred.btb.mispredict::NoBranch 0 0.00% 0.00% # Number of BTB mispredictions. No target found or target wrong. (Count)

board.processor.cores.core.branchPred.btb.mispredict::Return 0 0.00% 0.00% # Number of BTB mispredictions. No target found or target wrong. (Count)

board.processor.cores.core.branchPred.btb.mispredict::CallDirect 222 24.48% 24.48% # Number of BTB mispredictions. No target found or target wrong. (Count)

board.processor.cores.core.branchPred.btb.mispredict::CallIndirect 0 0.00% 24.48% # Number of BTB mispredictions. No target found or target wrong. (Count)

board.processor.cores.core.branchPred.btb.mispredict::DirectCond 534 58.88% 83.35% # Number of BTB mispredictions. No target found or target wrong. (Count)

board.processor.cores.core.branchPred.btb.mispredict::DirectUncond 151 16.65% 100.00% # Number of BTB mispredictions. No target found or target wrong. (Count)

board.processor.cores.core.branchPred.btb.mispredict::IndirectCond 0 0.00% 100.00% # Number of BTB mispredictions. No target found or target wrong. (Count)

board.processor.cores.core.branchPred.btb.mispredict::IndirectUncond 0 0.00% 100.00% # Number of BTB mispredictions. No target found or target wrong. (Count)

board.processor.cores.core.branchPred.btb.mispredict::total 907 # Number of BTB mispredictions. No target found or target wrong. (Count)

board.processor.cores.core.branchPred.btb.power_state.pwrStateResidencyTicks::UNDEFINED 9058914018 # Cumulative time (in ticks) in various power states (Tick)

board.processor.cores.core.branchPred.indirectBranchPred.lookups 2642 # Number of lookups (Count)

board.processor.cores.core.branchPred.indirectBranchPred.hits 2199 # Number of hits of a tag (Count)

board.processor.cores.core.branchPred.indirectBranchPred.misses 443 # Number of misses (Count)

board.processor.cores.core.branchPred.indirectBranchPred.targetRecords 88 # Number of targets that where recorded/installed in the cache (Count)

board.processor.cores.core.branchPred.indirectBranchPred.indirectRecords 2730 # Number of indirect branches/calls recorded in the indirect hist (Count)

board.processor.cores.core.branchPred.indirectBranchPred.speculativeOverflows 2 # Number of times more than the allowed capacity for speculative branches/calls where in flight and destroy the path history (Count)

board.processor.cores.core.branchPred.ras.pushes 3494 # Number of times a PC was pushed onto the RAS (Count)

board.processor.cores.core.branchPred.ras.pops 3489 # Number of times a PC was poped from the RAS (Count)

board.processor.cores.core.branchPred.ras.squashes 1120 # Number of times the stack operation was squashed due to wrong speculation. (Count)

board.processor.cores.core.branchPred.ras.used 2369 # Number of times the RAS is the provider (Count)

board.processor.cores.core.branchPred.ras.correct 2369 # Number of times the RAS is the provider and the prediction is correct (Count)

board.processor.cores.core.branchPred.ras.incorrect 0 # Number of times the RAS is the provider and the prediction is wrong (Count)

board.processor.cores.core.commit.commitSquashedInsts 211195 # The number of squashed insts skipped by commit (Count)

board.processor.cores.core.commit.commitNonSpecStalls 90 # The number of times commit has been forced to stall to communicate backwards (Count)

board.processor.cores.core.commit.branchMispredicts 737 # The number of times a branch was mispredicted (Count)

board.processor.cores.core.commit.numCommittedDist::samples 27032757 # Number of insts commited each cycle (Count)

board.processor.cores.core.commit.numCommittedDist::mean 2.103419 # Number of insts commited each cycle (Count)

board.processor.cores.core.commit.numCommittedDist::stdev 2.941469 # Number of insts commited each cycle (Count)

board.processor.cores.core.commit.numCommittedDist::underflows 0 0.00% 0.00% # Number of insts commited each cycle (Count)

board.processor.cores.core.commit.numCommittedDist::0 10945175 40.49% 40.49% # Number of insts commited each cycle (Count)

board.processor.cores.core.commit.numCommittedDist::1 7949505 29.41% 69.90% # Number of insts commited each cycle (Count)

board.processor.cores.core.commit.numCommittedDist::2 1933818 7.15% 77.05% # Number of insts commited each cycle (Count)

board.processor.cores.core.commit.numCommittedDist::3 179830 0.67% 77.71% # Number of insts commited each cycle (Count)

board.processor.cores.core.commit.numCommittedDist::4 235636 0.87% 78.59% # Number of insts commited each cycle (Count)

board.processor.cores.core.commit.numCommittedDist::5 177530 0.66% 79.24% # Number of insts commited each cycle (Count)

board.processor.cores.core.commit.numCommittedDist::6 1050094 3.88% 83.13% # Number of insts commited each cycle (Count)

board.processor.cores.core.commit.numCommittedDist::7 115533 0.43% 83.55% # Number of insts commited each cycle (Count)

board.processor.cores.core.commit.numCommittedDist::8 4445636 16.45% 100.00% # Number of insts commited each cycle (Count)

board.processor.cores.core.commit.numCommittedDist::overflows 0 0.00% 100.00% # Number of insts commited each cycle (Count)

board.processor.cores.core.commit.numCommittedDist::min_value 0 # Number of insts commited each cycle (Count)

board.processor.cores.core.commit.numCommittedDist::max_value 8 # Number of insts commited each cycle (Count)

board.processor.cores.core.commit.numCommittedDist::total 27032757 # Number of insts commited each cycle (Count)

board.processor.cores.core.commit.amos 0 # Number of atomic instructions committed (Count)

board.processor.cores.core.commit.membars 60 # Number of memory barriers committed (Count)

board.processor.cores.core.commit.functionCalls 2374 # Number of function calls committed. (Count)

board.processor.cores.core.commit.committedInstType_0::No_OpClass 316 0.00% 0.00% # Class of committed instruction (Count)

board.processor.cores.core.commit.committedInstType_0::IntAlu 40052454 70.44% 70.44% # Class of committed instruction (Count)

board.processor.cores.core.commit.committedInstType_0::IntMult 1048624 1.84% 72.28% # Class of committed instruction (Count)

board.processor.cores.core.commit.committedInstType_0::IntDiv 28 0.00% 72.28% # Class of committed instruction (Count)

board.processor.cores.core.commit.committedInstType_0::FloatAdd 154 0.00% 72.28% # Class of committed instruction (Count)

board.processor.cores.core.commit.committedInstType_0::FloatCmp 0 0.00% 72.28% # Class of committed instruction (Count)

board.processor.cores.core.commit.committedInstType_0::FloatCvt 0 0.00% 72.28% # Class of committed instruction (Count)

board.processor.cores.core.commit.committedInstType_0::FloatMult 0 0.00% 72.28% # Class of committed instruction (Count)

board.processor.cores.core.commit.committedInstType_0::FloatMultAcc 0 0.00% 72.28% # Class of committed instruction (Count)

board.processor.cores.core.commit.committedInstType_0::FloatDiv 0 0.00% 72.28% # Class of committed instruction (Count)

board.processor.cores.core.commit.committedInstType_0::FloatMisc 0 0.00% 72.28% # Class of committed instruction (Count)

board.processor.cores.core.commit.committedInstType_0::FloatSqrt 0 0.00% 72.28% # Class of committed instruction (Count)

board.processor.cores.core.commit.committedInstType_0::SimdAdd 10 0.00% 72.28% # Class of committed instruction (Count)

board.processor.cores.core.commit.committedInstType_0::SimdAddAcc 0 0.00% 72.28% # Class of committed instruction (Count)

board.processor.cores.core.commit.committedInstType_0::SimdAlu 229 0.00% 72.28% # Class of committed instruction (Count)

board.processor.cores.core.commit.committedInstType_0::SimdCmp 0 0.00% 72.28% # Class of committed instruction (Count)

board.processor.cores.core.commit.committedInstType_0::SimdCvt 72 0.00% 72.28% # Class of committed instruction (Count)

board.processor.cores.core.commit.committedInstType_0::SimdMisc 240 0.00% 72.29% # Class of committed instruction (Count)

board.processor.cores.core.commit.committedInstType_0::SimdMult 0 0.00% 72.29% # Class of committed instruction (Count)

board.processor.cores.core.commit.committedInstType_0::SimdMultAcc 0 0.00% 72.29% # Class of committed instruction (Count)

board.processor.cores.core.commit.committedInstType_0::SimdMatMultAcc 0 0.00% 72.29% # Class of committed instruction (Count)

board.processor.cores.core.commit.committedInstType_0::SimdShift 3 0.00% 72.29% # Class of committed instruction (Count)

board.processor.cores.core.commit.committedInstType_0::SimdShiftAcc 0 0.00% 72.29% # Class of committed instruction (Count)

board.processor.cores.core.commit.committedInstType_0::SimdDiv 0 0.00% 72.29% # Class of committed instruction (Count)

board.processor.cores.core.commit.committedInstType_0::SimdSqrt 0 0.00% 72.29% # Class of committed instruction (Count)

board.processor.cores.core.commit.committedInstType_0::SimdFloatAdd 0 0.00% 72.29% # Class of committed instruction (Count)

board.processor.cores.core.commit.committedInstType_0::SimdFloatAlu 0 0.00% 72.29% # Class of committed instruction (Count)

board.processor.cores.core.commit.committedInstType_0::SimdFloatCmp 0 0.00% 72.29% # Class of committed instruction (Count)

board.processor.cores.core.commit.committedInstType_0::SimdFloatCvt 0 0.00% 72.29% # Class of committed instruction (Count)

board.processor.cores.core.commit.committedInstType_0::SimdFloatDiv 0 0.00% 72.29% # Class of committed instruction (Count)

board.processor.cores.core.commit.committedInstType_0::SimdFloatMisc 0 0.00% 72.29% # Class of committed instruction (Count)

board.processor.cores.core.commit.committedInstType_0::SimdFloatMult 0 0.00% 72.29% # Class of committed instruction (Count)

board.processor.cores.core.commit.committedInstType_0::SimdFloatMultAcc 0 0.00% 72.29% # Class of committed instruction (Count)

board.processor.cores.core.commit.committedInstType_0::SimdFloatMatMultAcc 0 0.00% 72.29% # Class of committed instruction (Count)

board.processor.cores.core.commit.committedInstType_0::SimdFloatSqrt 0 0.00% 72.29% # Class of committed instruction (Count)

board.processor.cores.core.commit.committedInstType_0::SimdReduceAdd 0 0.00% 72.29% # Class of committed instruction (Count)

board.processor.cores.core.commit.committedInstType_0::SimdReduceAlu 0 0.00% 72.29% # Class of committed instruction (Count)

board.processor.cores.core.commit.committedInstType_0::SimdReduceCmp 0 0.00% 72.29% # Class of committed instruction (Count)

board.processor.cores.core.commit.committedInstType_0::SimdFloatReduceAdd 0 0.00% 72.29% # Class of committed instruction (Count)

board.processor.cores.core.commit.committedInstType_0::SimdFloatReduceCmp 0 0.00% 72.29% # Class of committed instruction (Count)

board.processor.cores.core.commit.committedInstType_0::SimdAes 0 0.00% 72.29% # Class of committed instruction (Count)

board.processor.cores.core.commit.committedInstType_0::SimdAesMix 0 0.00% 72.29% # Class of committed instruction (Count)

board.processor.cores.core.commit.committedInstType_0::SimdSha1Hash 0 0.00% 72.29% # Class of committed instruction (Count)

board.processor.cores.core.commit.committedInstType_0::SimdSha1Hash2 0 0.00% 72.29% # Class of committed instruction (Count)

board.processor.cores.core.commit.committedInstType_0::SimdSha256Hash 0 0.00% 72.29% # Class of committed instruction (Count)

board.processor.cores.core.commit.committedInstType_0::SimdSha256Hash2 0 0.00% 72.29% # Class of committed instruction (Count)

board.processor.cores.core.commit.committedInstType_0::SimdShaSigma2 0 0.00% 72.29% # Class of committed instruction (Count)

board.processor.cores.core.commit.committedInstType_0::SimdShaSigma3 0 0.00% 72.29% # Class of committed instruction (Count)

board.processor.cores.core.commit.committedInstType_0::SimdPredAlu 0 0.00% 72.29% # Class of committed instruction (Count)

board.processor.cores.core.commit.committedInstType_0::Matrix 0 0.00% 72.29% # Class of committed instruction (Count)

board.processor.cores.core.commit.committedInstType_0::MatrixMov 0 0.00% 72.29% # Class of committed instruction (Count)

```
board.processor.cores.core.commit.committedInstType_0::MatrixOP 0 0.00% 72.29% # Class of committed instruction (Count)
board.processor.cores.core.commit.committedInstType_0::MemRead 11553787 20.32% 92.60% # Class of committed instruction (Count)
board.processor.cores.core.commit.committedInstType_0::MemWrite 4204624 7.39% 100.00% # Class of committed instruction (Count)
board.processor.cores.core.commit.committedInstType_0::FloatMemRead 125 0.00% 100.00% # Class of committed instruction (Count)
board.processor.cores.core.commit.committedInstType_0::FloatMemWrite 542 0.00% 100.00% # Class of committed instruction (Count)
board.processor.cores.core.commit.committedInstType_0::IprAccess 0 0.00% 100.00% # Class of committed instruction (Count)
board.processor.cores.core.commit.committedInstType_0::InstPrefetch 0 0.00% 100.00% # Class of committed instruction (Count)
board.processor.cores.core.commit.committedInstType_0::SimdUnitStrideLoad 0 0.00% 100.00% # Class of committed instruction (Count)
board.processor.cores.core.commit.committedInstType_0::SimdUnitStrideStore 0 0.00% 100.00% # Class of committed instruction (Count)
board.processor.cores.core.commit.committedInstType_0::SimdUnitStrideMaskLoad 0 0.00% 100.00% # Class of committed instruction (Count)
board.processor.cores.core.commit.committedInstType_0::SimdUnitStrideMaskStore 0 0.00% 100.00% # Class of committed instruction (Count)
board.processor.cores.core.commit.committedInstType_0::SimdStridedLoad 0 0.00% 100.00% # Class of committed instruction (Count)
board.processor.cores.core.commit.committedInstType_0::SimdStridedStore 0 0.00% 100.00% # Class of committed instruction (Count)
board.processor.cores.core.commit.committedInstType_0::SimdIndexedLoad 0 0.00% 100.00% # Class of committed instruction (Count)
board.processor.cores.core.commit.committedInstType_0::SimdIndexedStore 0 0.00% 100.00% # Class of committed instruction (Count)
board.processor.cores.core.commit.committedInstType_0::SimdWholeRegisterLoad 0 0.00% 100.00% # Class of committed instruction (Count)
board.processor.cores.core.commit.committedInstType_0::SimdWholeRegisterStore 0 0.00% 100.00% # Class of committed instruction (Count)
board.processor.cores.core.commit.committedInstType_0::SimdUnitStrideFaultOnlyFirstLoad 0 0.00% 100.00% # Class of committed instruction (Count)
board.processor.cores.core.commit.committedInstType_0::SimdUnitStrideSegmentedLoad 0 0.00% 100.00% # Class of committed instruction (Count)
board.processor.cores.core.commit.committedInstType_0::SimdUnitStrideSegmentedStore 0 0.00% 100.00% # Class of committed instruction (Count)
board.processor.cores.core.commit.committedInstType_0::SimdExt 0 0.00% 100.00% # Class of committed instruction (Count)
board.processor.cores.core.commit.committedInstType_0::SimdFloatExt 0 0.00% 100.00% # Class of committed instruction (Count)
board.processor.cores.core.commit.committedInstType_0::SimdConfig 0 0.00% 100.00% # Class of committed instruction (Count)
board.processor.cores.core.commit.committedInstType_0::total 56861208 # Class of committed instruction (Count)
board.processor.cores.core.commit.commitEligibleSamples 4445636 # number cycles where commit BW limit reached (Cycle)
board.processor.cores.core.commitStats0.numInsts 36831240 # Number of instructions committed (thread level) (Count)
board.processor.cores.core.commitStats0.numOps 56861208 # Number of ops (including micro ops) committed (thread level) (Count)
board.processor.cores.core.commitStats0.numInstsNotNOP 36831240 # Number of instructions committed excluding NOPs or prefetches (Count)
board.processor.cores.core.commitStats0.numOpsNotNOP 56861208 # Number of Ops (including micro ops) Simulated (Count)
board.processor.cores.core.commitStats0.cpi 0.738611 # CPI: cycles per instruction (thread level) ((Cycle/Count))
board.processor.cores.core.commitStats0.ipc 1.353893 # IPC: instructions per cycle (thread level) ((Count/Cycle))
board.processor.cores.core.commitStats0.numMemRefs 15759078 # Number of memory references committed (Count)
board.processor.cores.core.commitStats0.numFpInsts 1535 # Number of float instructions (Count)
board.processor.cores.core.commitStats0.numIntInsts 56859973 # Number of integer instructions (Count)
board.processor.cores.core.commitStats0.numLoadInsts 11553912 # Number of load instructions (Count)
board.processor.cores.core.commitStats0.numStoreInsts 4205166 # Number of store instructions (Count)
board.processor.cores.core.commitStats0.numVecInsts 0 # Number of vector instructions (Count)
board.processor.cores.core.commitStats0.committedInstType::No_OpClass 316 0.00% 0.00% # Class of committed instruction. (Count)
board.processor.cores.core.commitStats0.committedInstType::IntAlu 40052454 70.44% 70.44% # Class of committed instruction. (Count)
board.processor.cores.core.commitStats0.committedInstType::IntMult 1048624 1.84% 72.28% # Class of committed instruction. (Count)
board.processor.cores.core.commitStats0.committedInstType::IntDiv 28 0.00% 72.28% # Class of committed instruction. (Count)
board.processor.cores.core.commitStats0.committedInstType::FloatAdd 154 0.00% 72.28% # Class of committed instruction. (Count)
board.processor.cores.core.commitStats0.committedInstType::FloatCmp 0 0.00% 72.28% # Class of committed instruction. (Count)
board.processor.cores.core.commitStats0.committedInstType::FloatCvt 0 0.00% 72.28% # Class of committed instruction. (Count)
board.processor.cores.core.commitStats0.committedInstType::FloatMult 0 0.00% 72.28% # Class of committed instruction. (Count)
board.processor.cores.core.commitStats0.committedInstType::FloatMultAcc 0 0.00% 72.28% # Class of committed instruction. (Count)
board.processor.cores.core.commitStats0.committedInstType::FloatDiv 0 0.00% 72.28% # Class of committed instruction. (Count)
board.processor.cores.core.commitStats0.committedInstType::FloatMisc 0 0.00% 72.28% # Class of committed instruction. (Count)
board.processor.cores.core.commitStats0.committedInstType::FloatSqrt 0 0.00% 72.28% # Class of committed instruction. (Count)
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board.processor.cores.core.commitStats0.committedInstType::SimdAdd 10 0.00% 72.28% # Class of committed instruction. (Count)

board.processor.cores.core.commitStats0.committedInstType::SimdAddAcc 0 0.00% 72.28% # Class of committed instruction. (Count)

board.processor.cores.core.commitStats0.committedInstType::SimdAlu 229 0.00% 72.28% # Class of committed instruction. (Count)

board.processor.cores.core.commitStats0.committedInstType::SimdCmp 0 0.00% 72.28% # Class of committed instruction. (Count)

board.processor.cores.core.commitStats0.committedInstType::SimdCvt 72 0.00% 72.28% # Class of committed instruction. (Count)

board.processor.cores.core.commitStats0.committedInstType::SimdMisc 240 0.00% 72.29% # Class of committed instruction. (Count)

board.processor.cores.core.commitStats0.committedInstType::SimdMult 0 0.00% 72.29% # Class of committed instruction. (Count)

board.processor.cores.core.commitStats0.committedInstType::SimdMultAcc 0 0.00% 72.29% # Class of committed instruction. (Count)

board.processor.cores.core.commitStats0.committedInstType::SimdMatMultAcc 0 0.00% 72.29% # Class of committed instruction. (Count)

board.processor.cores.core.commitStats0.committedInstType::SimdShift 3 0.00% 72.29% # Class of committed instruction. (Count)

board.processor.cores.core.commitStats0.committedInstType::SimdShiftAcc 0 0.00% 72.29% # Class of committed instruction. (Count)

board.processor.cores.core.commitStats0.committedInstType::SimdDiv 0 0.00% 72.29% # Class of committed instruction. (Count)

board.processor.cores.core.commitStats0.committedInstType::SimdSqrt 0 0.00% 72.29% # Class of committed instruction. (Count)

board.processor.cores.core.commitStats0.committedInstType::SimdFloatAdd 0 0.00% 72.29% # Class of committed instruction. (Count)

board.processor.cores.core.commitStats0.committedInstType::SimdFloatAlu 0 0.00% 72.29% # Class of committed instruction. (Count)

board.processor.cores.core.commitStats0.committedInstType::SimdFloatCmp 0 0.00% 72.29% # Class of committed instruction. (Count)

board.processor.cores.core.commitStats0.committedInstType::SimdFloatCvt 0 0.00% 72.29% # Class of committed instruction. (Count)

board.processor.cores.core.commitStats0.committedInstType::SimdFloatDiv 0 0.00% 72.29% # Class of committed instruction. (Count)

board.processor.cores.core.commitStats0.committedInstType::SimdFloatMisc 0 0.00% 72.29% # Class of committed instruction. (Count)

board.processor.cores.core.commitStats0.committedInstType::SimdFloatMult 0 0.00% 72.29% # Class of committed instruction. (Count)

board.processor.cores.core.commitStats0.committedInstType::SimdFloatMultAcc 0 0.00% 72.29% # Class of committed instruction. (Count)

board.processor.cores.core.commitStats0.committedInstType::SimdFloatMatMultAcc 0 0.00% 72.29% # Class of committed instruction. (Count)

board.processor.cores.core.commitStats0.committedInstType::SimdFloatSqrt 0 0.00% 72.29% # Class of committed instruction. (Count)

board.processor.cores.core.commitStats0.committedInstType::SimdReduceAdd 0 0.00% 72.29% # Class of committed instruction. (Count)

board.processor.cores.core.commitStats0.committedInstType::SimdReduceAlu 0 0.00% 72.29% # Class of committed instruction. (Count)

board.processor.cores.core.commitStats0.committedInstType::SimdReduceCmp 0 0.00% 72.29% # Class of committed instruction. (Count)

board.processor.cores.core.commitStats0.committedInstType::SimdFloatReduceAdd 0 0.00% 72.29% # Class of committed instruction. (Count)

board.processor.cores.core.commitStats0.committedInstType::SimdFloatReduceCmp 0 0.00% 72.29% # Class of committed instruction. (Count)

board.processor.cores.core.commitStats0.committedInstType::SimdAes 0 0.00% 72.29% # Class of committed instruction. (Count)

board.processor.cores.core.commitStats0.committedInstType::SimdAesMix 0 0.00% 72.29% # Class of committed instruction. (Count)

board.processor.cores.core.commitStats0.committedInstType::SimdSha1Hash 0 0.00% 72.29% # Class of committed instruction. (Count)

board.processor.cores.core.commitStats0.committedInstType::SimdSha1Hash2 0 0.00% 72.29% # Class of committed instruction. (Count)

board.processor.cores.core.commitStats0.committedInstType::SimdSha256Hash 0 0.00% 72.29% # Class of committed instruction. (Count)

board.processor.cores.core.commitStats0.committedInstType::SimdSha256Hash2 0 0.00% 72.29% # Class of committed instruction. (Count)

board.processor.cores.core.commitStats0.committedInstType::SimdShaSigma2 0 0.00% 72.29% # Class of committed instruction. (Count)

board.processor.cores.core.commitStats0.committedInstType::SimdShaSigma3 0 0.00% 72.29% # Class of committed instruction. (Count)

board.processor.cores.core.commitStats0.committedInstType::SimdPredAlu 0 0.00% 72.29% # Class of committed instruction. (Count)

board.processor.cores.core.commitStats0.committedInstType::Matrix 0 0.00% 72.29% # Class of committed instruction. (Count)

board.processor.cores.core.commitStats0.committedInstType::MatrixMov 0 0.00% 72.29% # Class of committed instruction. (Count)

board.processor.cores.core.commitStats0.committedInstType::MatrixOP 0 0.00% 72.29% # Class of committed instruction. (Count)

board.processor.cores.core.commitStats0.committedInstType::MemRead 11553787 20.32% 92.60% # Class of committed instruction. (Count)

board.processor.cores.core.commitStats0.committedInstType::MemWrite 4204624 7.39% 100.00% # Class of committed instruction. (Count)

board.processor.cores.core.commitStats0.committedInstType::FloatMemRead 125 0.00% 100.00% # Class of committed instruction. (Count)

board.processor.cores.core.commitStats0.committedInstType::FloatMemWrite 542 0.00% 100.00% # Class of committed instruction. (Count)

board.processor.cores.core.commitStats0.committedInstType::IprAccess 0 0.00% 100.00% # Class of committed instruction. (Count)

board.processor.cores.core.commitStats0.committedInstType::InstPrefetch 0 0.00% 100.00% # Class of committed instruction. (Count)

board.processor.cores.core.commitStats0.committedInstType::SimdUnitStrideLoad 0 0.00% 100.00% # Class of committed instruction. (Count)

board.processor.cores.core.commitStats0.committedInstType::SimdUnitStrideStore 0 0.00% 100.00% # Class of committed instruction. (Count)

board.processor.cores.core.commitStats0.committedInstType::SimdUnitStrideMaskLoad 0 0.00% 100.00% # Class of committed instruction. (Count)

board.processor.cores.core.commitStats0.committedInstType::SimdUnitStrideMaskStore 0 0.00% 100.00% # Class of committed instruction. (Count)

board.processor.cores.core.commitStats0.committedInstType::SimdStridedLoad 0 0.00% 100.00% # Class of committed instruction. (Count)

board.processor.cores.core.commitStats0.committedInstType::SimdStridedStore 0 0.00% 100.00% # Class of committed instruction. (Count)

board.processor.cores.core.commitStats0.committedInstType::SimdIndexedLoad 0 0.00% 100.00% # Class of committed instruction. (Count)

board.processor.cores.core.commitStats0.committedInstType::SimdIndexedStore 0 0.00% 100.00% # Class of committed instruction. (Count)

board.processor.cores.core.commitStats0.committedInstType::SimdWholeRegisterLoad 0 0.00% 100.00% # Class of committed instruction. (Count)

board.processor.cores.core.commitStats0.committedInstType::SimdWholeRegisterStore 0 0.00% 100.00% # Class of committed instruction. (Count)

board.processor.cores.core.commitStats0.committedInstType::SimdUnitStrideFaultOnlyFirstLoad 0 0.00% 100.00% # Class of committed instruction. (Count)

board.processor.cores.core.commitStats0.committedInstType::SimdUnitStrideSegmentedLoad 0 0.00% 100.00% # Class of committed instruction. (Count)

board.processor.cores.core.commitStats0.committedInstType::SimdUnitStrideSegmentedStore 0 0.00% 100.00% # Class of committed instruction. (Count)

board.processor.cores.core.commitStats0.committedInstType::SimdExt 0 0.00% 100.00% # Class of committed instruction. (Count)

board.processor.cores.core.commitStats0.committedInstType::SimdFloatExt 0 0.00% 100.00% # Class of committed instruction. (Count)

board.processor.cores.core.commitStats0.committedInstType::SimdConfig 0 0.00% 100.00% # Class of committed instruction. (Count)

board.processor.cores.core.commitStats0.committedInstType::total 56861208 # Class of committed instruction. (Count)

board.processor.cores.core.commitStats0.committedControl::IsControl 2130454 # Class of control type instructions committed (Count)

board.processor.cores.core.commitStats0.committedControl::IsDirectControl 2125900 # Class of control type instructions committed (Count)

board.processor.cores.core.commitStats0.committedControl::IsIndirectControl 4554 # Class of control type instructions committed (Count)

board.processor.cores.core.commitStats0.committedControl::IsCondControl 2120254 # Class of control type instructions committed (Count)

board.processor.cores.core.commitStats0.committedControl::IsUncondControl 10200 # Class of control type instructions committed (Count)

board.processor.cores.core.commitStats0.committedControl::IsCall 2374 # Class of control type instructions committed (Count)

board.processor.cores.core.commitStats0.committedControl::IsReturn 2369 # Class of control type instructions committed (Count)

board.processor.cores.core.decode.idleCycles 1705958 # Number of cycles decode is idle (Cycle)

board.processor.cores.core.decode.blockedCycles 18209997 # Number of cycles decode is blocked (Cycle)

board.processor.cores.core.decode.runCycles 3078256 # Number of cycles decode is running (Cycle)

board.processor.cores.core.decode.unblockCycles 4063737 # Number of cycles decode is unblocking (Cycle)

board.processor.cores.core.decode.squashCycles 1878 # Number of cycles decode is squashing (Cycle)

board.processor.cores.core.decode.branchResolved 2113650 # Number of times decode resolved a branch (Count)

board.processor.cores.core.decode.branchMispred 386 # Number of times decode detected a branch misprediction (Count)

board.processor.cores.core.decode.decodedInsts 57085638 # Number of instructions handled by decode (Count)

board.processor.cores.core.decode.squashedInsts 1998 # Number of squashed instructions handled by decode (Count)

board.processor.cores.core.executeStats0.numInsts 57057111 # Number of executed instructions (Count)

board.processor.cores.core.executeStats0.numNop 0 # Number of nop insts executed (Count)

board.processor.cores.core.executeStats0.numBranches 2137398 # Number of branches executed (Count)

board.processor.cores.core.executeStats0.numLoadInsts 11584313 # Number of load instructions executed (Count)

board.processor.cores.core.executeStats0.numStoreInsts 4217252 # Number of stores executed (Count)

board.processor.cores.core.executeStats0.instRate 2.097384 # Inst execution rate ((Count/Cycle))

board.processor.cores.core.executeStats0.numCCRegReads 14869574 # Number of times the CC registers were read (Count)

board.processor.cores.core.executeStats0.numCCRegWrites 30594410 # Number of times the CC registers were written (Count)

board.processor.cores.core.executeStats0.numFpRegReads 2113 # Number of times the floating registers were read (Count)

board.processor.cores.core.executeStats0.numFpRegWrites 1119 # Number of times the floating registers were written (Count)

board.processor.cores.core.executeStats0.numIntRegReads 80249393 # Number of times the integer registers were read (Count)

board.processor.cores.core.executeStats0.numIntRegWrites 48572006 # Number of times the integer registers were written (Count)

board.processor.cores.core.executeStats0.numMemRefs 15801565 # Number of memory refs (Count)

board.processor.cores.core.executeStats0.numMiscRegReads 20076367 # Number of times the Misc registers were read (Count)

board.processor.cores.core.executeStats0.numVecAluAccesses 0 # Number of vector alu accesses (Count)

board.processor.cores.core.executeStats0.numDiscardedOps 0 # Number of ops (including micro ops) which were discarded before commit (Count)

board.processor.cores.core.fetch.predictedBranches 2119611 # Number of branches that fetch has predicted taken (Count)

board.processor.cores.core.fetch.cycles 27016339 # Number of cycles fetch has run and was not squashing or blocked (Cycle)

board.processor.cores.core.fetch.squashCycles 4508 # Number of cycles fetch has spent squashing (Cycle)

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board.processor.cores.core.fetch.miscStallCycles 45 # Number of cycles fetch has spent waiting on interrupts, or bad addresses, or out of MSHRs (Cycle)
board.processor.cores.core.fetch.pendingTrapStallCycles 256 # Number of stall cycles due to pending traps (Cycle)
board.processor.cores.core.fetch.cacheLines 4232594 # Number of cache lines fetched (Count)
board.processor.cores.core.fetch.icacheSquashes 469 # Number of outstanding Icache misses that were squashed (Count)
board.processor.cores.core.fetch.nisnDist::samples 27059826 # Number of instructions fetched each cycle (Total) (Count)
board.processor.cores.core.fetch.nisnDist::mean 2.111861 # Number of instructions fetched each cycle (Total) (Count)
board.processor.cores.core.fetch.nisnDist::stdev 3.201993 # Number of instructions fetched each cycle (Total) (Count)
board.processor.cores.core.fetch.nisnDist::underflows 0 0.00% 0.00% # Number of instructions fetched each cycle (Total) (Count)
board.processor.cores.core.fetch.nisnDist::0 17427198 64.40% 64.40% # Number of instructions fetched each cycle (Total) (Count)
board.processor.cores.core.fetch.nisnDist::1 526122 1.94% 66.35% # Number of instructions fetched each cycle (Total) (Count)
board.processor.cores.core.fetch.nisnDist::2 971779 3.59% 69.94% # Number of instructions fetched each cycle (Total) (Count)
board.processor.cores.core.fetch.nisnDist::3 494719 1.83% 71.77% # Number of instructions fetched each cycle (Total) (Count)
board.processor.cores.core.fetch.nisnDist::4 1282103 4.74% 76.50% # Number of instructions fetched each cycle (Total) (Count)
board.processor.cores.core.fetch.nisnDist::5 528122 1.95% 78.46% # Number of instructions fetched each cycle (Total) (Count)
board.processor.cores.core.fetch.nisnDist::6 326475 1.21% 79.66% # Number of instructions fetched each cycle (Total) (Count)
board.processor.cores.core.fetch.nisnDist::7 561582 2.08% 81.74% # Number of instructions fetched each cycle (Total) (Count)
board.processor.cores.core.fetch.nisnDist::8 4941726 18.26% 100.00% # Number of instructions fetched each cycle (Total) (Count)
board.processor.cores.core.fetch.nisnDist::overflows 0 0.00% 100.00% # Number of instructions fetched each cycle (Total) (Count)
board.processor.cores.core.fetch.nisnDist::min_value 0 # Number of instructions fetched each cycle (Total) (Count)
board.processor.cores.core.fetch.nisnDist::max_value 8 # Number of instructions fetched each cycle (Total) (Count)
board.processor.cores.core.fetch.nisnDist::total 27059826 # Number of instructions fetched each cycle (Total) (Count)
board.processor.cores.core.fetchStats0.numInsts 37023071 # Number of instructions fetched (thread level) (Count)
board.processor.cores.core.fetchStats0.numOps 0 # Number of ops (including micro ops) fetched (thread level) (Count)
board.processor.cores.core.fetchStats0.fetchRate 1.360945 # Number of inst fetches per cycle ((Count/Cycle))
board.processor.cores.core.fetchStats0.numBranches 2142657 # Number of branches fetched (Count)
board.processor.cores.core.fetchStats0.branchRate 0.078763 # Number of branch fetches per cycle (Ratio)
board.processor.cores.core.fetchStats0.icacheStallCycles 40932 # ICache total stall cycles (Cycle)
board.processor.cores.core.fetchStats0.numFetchSuspends 0 # Number of times Execute suspended instruction fetching (Count)
board.processor.cores.core.iew.idleCycles 0 # Number of cycles IEW is idle (Cycle)
board.processor.cores.core.iew.squashCycles 1878 # Number of cycles IEW is squashing (Cycle)
board.processor.cores.core.iew.blockCycles 36097 # Number of cycles IEW is blocking (Cycle)
board.processor.cores.core.iew.unblockCycles 1247 # Number of cycles IEW is unblocking (Cycle)
board.processor.cores.core.iew.dispatchedInsts 57073061 # Number of instructions dispatched to IQ (Count)
board.processor.cores.core.iew.dispSquashedInsts 108 # Number of squashed instructions skipped by dispatch (Count)
board.processor.cores.core.iew.dispLoadInsts 11586182 # Number of dispatched load instructions (Count)
board.processor.cores.core.iew.dispStoreInsts 4218844 # Number of dispatched store instructions (Count)
board.processor.cores.core.iew.dispNonSpecInsts 80 # Number of dispatched non-speculative instructions (Count)
board.processor.cores.core.iew.iqFullEvents 52 # Number of times the IQ has become full, causing a stall (Count)
board.processor.cores.core.iew.lsqFullEvents 1162 # Number of times the LSQ has become full, causing a stall (Count)
board.processor.cores.core.iew.memOrderViolationEvents 109 # Number of memory order violations (Count)
board.processor.cores.core.iew.predictedTakenIncorrect 85 # Number of branches that were predicted taken incorrectly (Count)
board.processor.cores.core.iew.predictedNotTakenIncorrect 774 # Number of branches that were predicted not taken incorrectly (Count)
board.processor.cores.core.iew.branchMispredicts 859 # Number of branch mispredicts detected at execute (Count)
board.processor.cores.core.iew.instsToCommit 57056122 # Cumulative count of insts sent to commit (Count)
board.processor.cores.core.iew.writebackCount 57052709 # Cumulative count of insts written-back (Count)
board.processor.cores.core.iew.producerInst 48250345 # Number of instructions producing a value (Count)
board.processor.cores.core.iew.consumerInst 72150200 # Number of instructions consuming a value (Count)
board.processor.cores.core.iew.wbRate 2.097222 # Insts written-back per cycle ((Count/Cycle))
board.processor.cores.core.iew.wbFanout 0.668749 # Average fanout of values written-back ((Count/Count))
```

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board.processor.cores.core.interrupts.clk_domain.clock 5328 # Clock period in ticks (Tick)
board.processor.cores.core.lsq0.forwLoads 8335453 # Number of loads that had data forwarded from stores (Count)
board.processor.cores.core.lsq0.squashedLoads 32270 # Number of loads squashed (Count)
board.processor.cores.core.lsq0.ignoredResponses 5 # Number of memory responses ignored because the instruction is squashed (Count)
board.processor.cores.core.lsq0.memOrderViolation 109 # Number of memory ordering violations (Count)
board.processor.cores.core.lsq0.squashedStores 13678 # Number of stores squashed (Count)
board.processor.cores.core.lsq0.rescheduledLoads 8 # Number of loads that were rescheduled (Count)
board.processor.cores.core.lsq0.blockedByCache 121 # Number of times an access to memory failed due to the cache being blocked (Count)
board.processor.cores.core.lsq0.loadToUse::samples 11553912 # Distribution of cycle latency between the first time a load is issued and its completion (Unspecified)
board.processor.cores.core.lsq0.loadToUse::mean 2.682512 # Distribution of cycle latency between the first time a load is issued and its completion (Unspecified)
board.processor.cores.core.lsq0.loadToUse::stdev 9.858632 # Distribution of cycle latency between the first time a load is issued and its completion (Unspecified)
board.processor.cores.core.lsq0.loadToUse::0-9 11485740 99.41% 99.41% # Distribution of cycle latency between the first time a load is issued and its completion (Unspecified)
board.processor.cores.core.lsq0.loadToUse::10-19 26 0.00% 99.41% # Distribution of cycle latency between the first time a load is issued and its completion (Unspecified)
board.processor.cores.core.lsq0.loadToUse::20-29 3 0.00% 99.41% # Distribution of cycle latency between the first time a load is issued and its completion (Unspecified)
board.processor.cores.core.lsq0.loadToUse::30-39 1 0.00% 99.41% # Distribution of cycle latency between the first time a load is issued and its completion (Unspecified)
board.processor.cores.core.lsq0.loadToUse::40-49 5 0.00% 99.41% # Distribution of cycle latency between the first time a load is issued and its completion (Unspecified)
board.processor.cores.core.lsq0.loadToUse::50-59 14 0.00% 99.41% # Distribution of cycle latency between the first time a load is issued and its completion (Unspecified)
board.processor.cores.core.lsq0.loadToUse::60-69 7 0.00% 99.41% # Distribution of cycle latency between the first time a load is issued and its completion (Unspecified)
board.processor.cores.core.lsq0.loadToUse::70-79 3 0.00% 99.41% # Distribution of cycle latency between the first time a load is issued and its completion (Unspecified)
board.processor.cores.core.lsq0.loadToUse::80-89 3 0.00% 99.41% # Distribution of cycle latency between the first time a load is issued and its completion (Unspecified)
board.processor.cores.core.lsq0.loadToUse::90-99 7 0.00% 99.41% # Distribution of cycle latency between the first time a load is issued and its completion (Unspecified)
board.processor.cores.core.lsq0.loadToUse::100-109 64165 0.56% 99.97% # Distribution of cycle latency between the first time a load is issued and its completion (Unspecified)
board.processor.cores.core.lsq0.loadToUse::110-119 5 0.00% 99.97% # Distribution of cycle latency between the first time a load is issued and its completion (Unspecified)
board.processor.cores.core.lsq0.loadToUse::120-129 1 0.00% 99.97% # Distribution of cycle latency between the first time a load is issued and its completion (Unspecified)
board.processor.cores.core.lsq0.loadToUse::140-149 6 0.00% 99.97% # Distribution of cycle latency between the first time a load is issued and its completion (Unspecified)
board.processor.cores.core.lsq0.loadToUse::150-159 132 0.00% 99.97% # Distribution of cycle latency between the first time a load is issued and its completion (Unspecified)
board.processor.cores.core.lsq0.loadToUse::160-169 232 0.00% 99.97% # Distribution of cycle latency between the first time a load is issued and its completion (Unspecified)
board.processor.cores.core.lsq0.loadToUse::170-179 1 0.00% 99.97% # Distribution of cycle latency between the first time a load is issued and its completion (Unspecified)
board.processor.cores.core.lsq0.loadToUse::180-189 1 0.00% 99.97% # Distribution of cycle latency between the first time a load is issued and its completion (Unspecified)
board.processor.cores.core.lsq0.loadToUse::190-199 1 0.00% 99.97% # Distribution of cycle latency between the first time a load is issued and its completion (Unspecified)
board.processor.cores.core.lsq0.loadToUse::200-209 4 0.00% 99.97% # Distribution of cycle latency between the first time a load is issued and its completion (Unspecified)
board.processor.cores.core.lsq0.loadToUse::210-219 2 0.00% 99.97% # Distribution of cycle latency between the first time a load is issued and its completion (Unspecified)
board.processor.cores.core.lsq0.loadToUse::220-229 5 0.00% 99.97% # Distribution of cycle latency between the first time a load is issued and its completion (Unspecified)
board.processor.cores.core.lsq0.loadToUse::230-239 7 0.00% 99.97% # Distribution of cycle latency between the first time a load is issued and its completion (Unspecified)
board.processor.cores.core.lsq0.loadToUse::270-279 4 0.00% 99.97% # Distribution of cycle latency between the first time a load is issued and its completion (Unspecified)
board.processor.cores.core.lsq0.loadToUse::280-289 7 0.00% 99.97% # Distribution of cycle latency between the first time a load is issued and its completion (Unspecified)
board.processor.cores.core.lsq0.loadToUse::290-299 15 0.00% 99.97% # Distribution of cycle latency between the first time a load is issued and its completion (Unspecified)
board.processor.cores.core.lsq0.loadToUse::overflows 3515 0.03% 100.00% # Distribution of cycle latency between the first time a load is issued and its completion (Unspecified)
board.processor.cores.core.lsq0.loadToUse::min_value 2 # Distribution of cycle latency between the first time a load is issued and its completion (Unspecified)
board.processor.cores.core.lsq0.loadToUse::max_value 1225 # Distribution of cycle latency between the first time a load is issued and its completion (Unspecified)
board.processor.cores.core.lsq0.loadToUse::total 11553912 # Distribution of cycle latency between the first time a load is issued and its completion (Unspecified)
board.processor.cores.core.mmu.dtb.rdAccesses 11583694 # TLB accesses on read requests (Count)
board.processor.cores.core.mmu.dtb.wrAccesses 4217252 # TLB accesses on write requests (Count)
board.processor.cores.core.mmu.dtb.rdMisses 1145 # TLB misses on read requests (Count)
board.processor.cores.core.mmu.dtb.wrMisses 3900 # TLB misses on write requests (Count)
board.processor.cores.core.mmu.dtb.walker.power_state.pwrStateResidencyTicks::UNDEFINED 9058914018 # Cumulative time (in ticks) in various power states (Tick)
board.processor.cores.core.mmu.itb.rdAccesses 0 # TLB accesses on read requests (Count)
board.processor.cores.core.mmu.itb.wrAccesses 4232639 # TLB accesses on write requests (Count)
board.processor.cores.core.mmu.itb.rdMisses 0 # TLB misses on read requests (Count)
```

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board.processor.cores.core.mmu.itb.wrMisses 117 # TLB misses on write requests (Count)
board.processor.cores.core.mmu.itb.walker.power_state.pwrStateResidencyTicks::UNDEFINED 9058914018 # Cumulative time (in ticks) in various power states (Tick)
board.processor.cores.core.power_state.pwrStateResidencyTicks::ON 9058914018 # Cumulative time (in ticks) in various power states (Tick)
board.processor.cores.core.rename.squashCycles 1878 # Number of cycles rename is squashing (Cycle)
board.processor.cores.core.rename.idleCycles 2875517 # Number of cycles rename is idle (Cycle)
board.processor.cores.core.rename.blockCycles 542052 # Number of cycles rename is blocking (Cycle)
board.processor.cores.core.rename.serializeStallCycles 2661 # count of cycles rename stalled for serializing inst (Cycle)
board.processor.cores.core.rename.runCycles 5971710 # Number of cycles rename is running (Cycle)
board.processor.cores.core.rename.unblockCycles 17666008 # Number of cycles rename is unblocking (Cycle)
board.processor.cores.core.rename.renamedInsts 57078819 # Number of instructions processed by rename (Count)
board.processor.cores.core.rename.ROBFullEvents 5571 # Number of times rename has blocked due to ROB full (Count)
board.processor.cores.core.rename.IQFullEvents 2216280 # Number of times rename has blocked due to IQ full (Count)
board.processor.cores.core.rename.LQFullEvents 11694584 # Number of times rename has blocked due to LQ full (Count)
board.processor.cores.core.rename.SQFullEvents 3222564 # Number of times rename has blocked due to SQ full (Count)
board.processor.cores.core.rename.renamedOperands 127761952 # Number of destination operands rename has renamed (Count)
board.processor.cores.core.rename.lookups 213423301 # Number of register rename lookups that rename has made (Count)
board.processor.cores.core.rename.intLookups 80282762 # Number of integer rename lookups (Count)
board.processor.cores.core.rename.fpLookups 2507 # Number of floating rename lookups (Count)
board.processor.cores.core.rename.committedMaps 127275790 # Number of HB maps that are committed (Count)
board.processor.cores.core.rename.undoneMaps 486153 # Number of HB maps that are undone due to squashing (Count)
board.processor.cores.core.rename.serializing 45 # count of serializing insts renamed (Count)
board.processor.cores.core.rename.tempSerializing 46 # count of temporary serializing insts renamed (Count)
board.processor.cores.core.rename.skidInsts 16590051 # count of insts added to the skid buffer (Count)
board.processor.cores.core.rob.reads 79645926 # The number of ROB reads (Count)
board.processor.cores.core.rob.writes 114171938 # The number of ROB writes (Count)
board.processor.cores.core.thread_0.numInsts 36831240 # Number of Instructions committed (Count)
board.processor.cores.core.thread_0.numOps 56861208 # Number of Ops committed (Count)
board.processor.cores.core.thread_0.numMemRefs 0 # Number of Memory References (Count)
board.processor.cores.core.workload.numSyscalls 18 # Number of system calls (Count)
board.workload.inst.arm 0 # number of arm instructions executed (Count)
board.workload.inst.quiesce 0 # number of quiesce instructions executed (Count)

---------- End Simulation Statistics ----------
```

results_ooo_mshr_4:

```
---------- Begin Simulation Statistics ----------
simSeconds                    0.005875                # Number of seconds simulated (Second)
simTicks                      5874741045              # Number of ticks simulated (Tick)
finalTick                     5874741045              # Number of ticks from beginning of simulation (restored from checkpoints and never reset) (Tick)
simFreq                       1000000000000           # The number of ticks per simulated second ((Tick/Second))
hostSeconds                   65.19                   # Real time elapsed on the host (Second)
hostTickRate                  90115917                # The number of ticks simulated per host second (ticks/s) ((Tick/Second))
hostMemory                    1214240                 # Number of bytes of host memory used (Byte)
simInsts                      36831240                # Number of instructions simulated (Count)
simOps                        56861208                # Number of ops (including micro ops) simulated (Count)
hostInstRate                  564974                  # Simulator instruction rate (inst/s) ((Count/Second))
hostOpRate                    872225                  # Simulator op (including micro ops) rate (op/s) ((Count/Second))
board.cache_hierarchy.dptw_caches.blockedCycles::no_mshrs      0          # number of cycles access was blocked (Cycle)
board.cache_hierarchy.dptw_caches.blockedCycles::no_targets    0          # number of cycles access was blocked (Cycle)
board.cache_hierarchy.dptw_caches.blockedCauses::no_mshrs      0          # number of times access was blocked (Count)
board.cache_hierarchy.dptw_caches.blockedCauses::no_targets    0          # number of times access was blocked (Count)
board.cache_hierarchy.dptw_caches.avgBlocked::no_mshrs         nan        # average number of cycles each access was blocked ((Cycle/Count))
board.cache_hierarchy.dptw_caches.avgBlocked::no_targets       nan        # average number of cycles each access was blocked ((Cycle/Count))
board.cache_hierarchy.dptw_caches.replacements                 0          # number of replacements (Count)
board.cache_hierarchy.dptw_caches.power_state.pwrStateResidencyTicks::UNDEFINED   5874741045   # Cumulative time (in ticks) in various power states (Tick)
board.cache_hierarchy.dptw_caches.tags.tagsInUse               0          # Average ticks per tags in use ((Tick/Count))
board.cache_hierarchy.dptw_caches.tags.totalRefs               0          # Total number of references to valid blocks. (Count)
board.cache_hierarchy.dptw_caches.tags.sampledRefs             0          # Sample count of references to valid blocks. (Count)
board.cache_hierarchy.dptw_caches.tags.avgRefs                 nan        # Average number of references to valid blocks. ((Count/Count))
board.cache_hierarchy.dptw_caches.tags.warmupTick              0          # The tick when the warmup percentage was hit. (Tick)
board.cache_hierarchy.dptw_caches.tags.tagAccesses             0          # Number of tag accesses (Count)
board.cache_hierarchy.dptw_caches.tags.dataAccesses            0          # Number of data accesses (Count)
board.cache_hierarchy.dptw_caches.tags.power_state.pwrStateResidencyTicks::UNDEFINED   5874741045   # Cumulative time (in ticks) in various power states (Tick)
board.cache_hierarchy.iptw_caches.blockedCycles::no_mshrs      0          # number of cycles access was blocked (Cycle)
board.cache_hierarchy.iptw_caches.blockedCycles::no_targets    0          # number of cycles access was blocked (Cycle)
board.cache_hierarchy.iptw_caches.blockedCauses::no_mshrs      0          # number of times access was blocked (Count)
board.cache_hierarchy.iptw_caches.blockedCauses::no_targets    0          # number of times access was blocked (Count)
board.cache_hierarchy.iptw_caches.avgBlocked::no_mshrs         nan        # average number of cycles each access was blocked ((Cycle/Count))
board.cache_hierarchy.iptw_caches.avgBlocked::no_targets       nan        # average number of cycles each access was blocked ((Cycle/Count))
board.cache_hierarchy.iptw_caches.replacements                 0          # number of replacements (Count)
board.cache_hierarchy.iptw_caches.power_state.pwrStateResidencyTicks::UNDEFINED   5874741045   # Cumulative time (in ticks) in various power states (Tick)
board.cache_hierarchy.iptw_caches.tags.tagsInUse               0          # Average ticks per tags in use ((Tick/Count))
board.cache_hierarchy.iptw_caches.tags.totalRefs               0          # Total number of references to valid blocks. (Count)
board.cache_hierarchy.iptw_caches.tags.sampledRefs             0          # Sample count of references to valid blocks. (Count)
board.cache_hierarchy.iptw_caches.tags.avgRefs                 nan        # Average number of references to valid blocks. ((Count/Count))
board.cache_hierarchy.iptw_caches.tags.warmupTick              0          # The tick when the warmup percentage was hit. (Tick)
board.cache_hierarchy.iptw_caches.tags.tagAccesses             0          # Number of tag accesses (Count)
board.cache_hierarchy.iptw_caches.tags.dataAccesses            0          # Number of data accesses (Count)
board.cache_hierarchy.iptw_caches.tags.power_state.pwrStateResidencyTicks::UNDEFINED   5874741045   # Cumulative time (in ticks) in various power states (Tick)
board.cache_hierarchy.l1dcaches.demandHits::processor.cores.core.data   7484291   # number of demand (read+write) hits (Count)
board.cache_hierarchy.l1dcaches.demandHits::total   7484291   # number of demand (read+write) hits (Count)
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board.cache_hierarchy.l1dcaches.overallHits::processor.cores.core.data 7484291 # number of overall hits (Count)

board.cache_hierarchy.l1dcaches.overallHits::total 7484291 # number of overall hits (Count)

board.cache_hierarchy.l1dcaches.demandMisses::processor.cores.core.data 5435 # number of demand (read+write) misses (Count)

board.cache_hierarchy.l1dcaches.demandMisses::total 5435 # number of demand (read+write) misses (Count)

board.cache_hierarchy.l1dcaches.overallMisses::processor.cores.core.data 5435 # number of overall misses (Count)

board.cache_hierarchy.l1dcaches.overallMisses::total 5435 # number of overall misses (Count)

board.cache_hierarchy.l1dcaches.demandMissLatency::processor.cores.core.data 588160583 # number of demand (read+write) miss ticks (Tick)

board.cache_hierarchy.l1dcaches.demandMissLatency::total 588160583 # number of demand (read+write) miss ticks (Tick)

board.cache_hierarchy.l1dcaches.overallMissLatency::processor.cores.core.data 588160583 # number of overall miss ticks (Tick)

board.cache_hierarchy.l1dcaches.overallMissLatency::total 588160583 # number of overall miss ticks (Tick)

board.cache_hierarchy.l1dcaches.demandAccesses::processor.cores.core.data 7489726 # number of demand (read+write) accesses (Count)

board.cache_hierarchy.l1dcaches.demandAccesses::total 7489726 # number of demand (read+write) accesses (Count)

board.cache_hierarchy.l1dcaches.overallAccesses::processor.cores.core.data 7489726 # number of overall (read+write) accesses (Count)

board.cache_hierarchy.l1dcaches.overallAccesses::total 7489726 # number of overall (read+write) accesses (Count)

board.cache_hierarchy.l1dcaches.demandMissRate::processor.cores.core.data 0.000726 # miss rate for demand accesses (Ratio)

board.cache_hierarchy.l1dcaches.demandMissRate::total 0.000726 # miss rate for demand accesses (Ratio)

board.cache_hierarchy.l1dcaches.overallMissRate::processor.cores.core.data 0.000726 # miss rate for overall accesses (Ratio)

board.cache_hierarchy.l1dcaches.overallMissRate::total 0.000726 # miss rate for overall accesses (Ratio)

board.cache_hierarchy.l1dcaches.demandAvgMissLatency::processor.cores.core.data 108217.218583 # average overall miss latency in ticks ((Tick/Count))

board.cache_hierarchy.l1dcaches.demandAvgMissLatency::total 108217.218583 # average overall miss latency in ticks ((Tick/Count))

board.cache_hierarchy.l1dcaches.overallAvgMissLatency::processor.cores.core.data 108217.218583 # average overall miss latency ((Tick/Count))

board.cache_hierarchy.l1dcaches.overallAvgMissLatency::total 108217.218583 # average overall miss latency ((Tick/Count))

board.cache_hierarchy.l1dcaches.blockedCycles::no_mshrs 4039 # number of cycles access was blocked (Cycle)

board.cache_hierarchy.l1dcaches.blockedCycles::no_targets 0 # number of cycles access was blocked (Cycle)

board.cache_hierarchy.l1dcaches.blockedCauses::no_mshrs 36 # number of times access was blocked (Count)

board.cache_hierarchy.l1dcaches.blockedCauses::no_targets 0 # number of times access was blocked (Count)

board.cache_hierarchy.l1dcaches.avgBlocked::no_mshrs 112.194444 # average number of cycles each access was blocked ((Cycle/Count))

board.cache_hierarchy.l1dcaches.avgBlocked::no_targets nan # average number of cycles each access was blocked ((Cycle/Count))

board.cache_hierarchy.l1dcaches.writebacks::writebacks 65717 # number of writebacks (Count)

board.cache_hierarchy.l1dcaches.writebacks::total 65717 # number of writebacks (Count)

board.cache_hierarchy.l1dcaches.demandMshrHits::processor.cores.core.data 2486 # number of demand (read+write) MSHR hits (Count)

board.cache_hierarchy.l1dcaches.demandMshrHits::total 2486 # number of demand (read+write) MSHR hits (Count)

board.cache_hierarchy.l1dcaches.overallMshrHits::processor.cores.core.data 2486 # number of overall MSHR hits (Count)

board.cache_hierarchy.l1dcaches.overallMshrHits::total 2486 # number of overall MSHR hits (Count)

board.cache_hierarchy.l1dcaches.demandMshrMisses::processor.cores.core.data 2949 # number of demand (read+write) MSHR misses (Count)

board.cache_hierarchy.l1dcaches.demandMshrMisses::total 2949 # number of demand (read+write) MSHR misses (Count)

board.cache_hierarchy.l1dcaches.overallMshrMisses::cache_hierarchy.l1dcaches.prefetcher 129118 # number of overall MSHR misses (Count)

board.cache_hierarchy.l1dcaches.overallMshrMisses::processor.cores.core.data 2949 # number of overall MSHR misses (Count)

board.cache_hierarchy.l1dcaches.overallMshrMisses::total 132067 # number of overall MSHR misses (Count)

board.cache_hierarchy.l1dcaches.demandMshrMissLatency::processor.cores.core.data 335891771 # number of demand (read+write) MSHR miss ticks (Tick)

board.cache_hierarchy.l1dcaches.demandMshrMissLatency::total 335891771 # number of demand (read+write) MSHR miss ticks (Tick)

board.cache_hierarchy.l1dcaches.overallMshrMissLatency::cache_hierarchy.l1dcaches.prefetcher 5173827691 # number of overall MSHR miss ticks (Tick)

board.cache_hierarchy.l1dcaches.overallMshrMissLatency::processor.cores.core.data 335891771 # number of overall MSHR miss ticks (Tick)

board.cache_hierarchy.l1dcaches.overallMshrMissLatency::total 5509719462 # number of overall MSHR miss ticks (Tick)

board.cache_hierarchy.l1dcaches.demandMshrMissRate::processor.cores.core.data 0.000394 # mshr miss ratio for demand accesses (Ratio)

board.cache_hierarchy.l1dcaches.demandMshrMissRate::total 0.000394 # mshr miss ratio for demand accesses (Ratio)

board.cache_hierarchy.l1dcaches.overallMshrMissRate::cache_hierarchy.l1dcaches.prefetcher inf # mshr miss ratio for overall accesses (Ratio)

board.cache_hierarchy.l1dcaches.overallMshrMissRate::processor.cores.core.data 0.000394 # mshr miss ratio for overall accesses (Ratio)

board.cache_hierarchy.l1dcaches.overallMshrMissRate::total 0.017633 # mshr miss ratio for overall accesses (Ratio)

board.cache_hierarchy.l1dcaches.demandAvgMshrMissLatency::processor.cores.core.data 113900.227535 # average overall mshr miss latency ((Tick/Count))

board.cache_hierarchy.l1dcaches.demandAvgMshrMissLatency::total 113900.227535 # average overall mshr miss latency ((Tick/Count))

board.cache_hierarchy.l1dcaches.overallAvgMshrMissLatency::cache_hierarchy.l1dcaches.prefetcher 40070.537733 # average overall mshr miss latency ((Tick/Count))

board.cache_hierarchy.l1dcaches.overallAvgMshrMissLatency::processor.cores.core.data 113900.227535 # average overall mshr miss latency ((Tick/Count))

board.cache_hierarchy.l1dcaches.overallAvgMshrMissLatency::total 41719.123339 # average overall mshr miss latency ((Tick/Count))

board.cache_hierarchy.l1dcaches.replacements 131556 # number of replacements (Count)

board.cache_hierarchy.l1dcaches.HardPFReq.mshrMisses::cache_hierarchy.l1dcaches.prefetcher 129118 # number of HardPFReq MSHR misses (Count)

board.cache_hierarchy.l1dcaches.HardPFReq.mshrMisses::total 129118 # number of HardPFReq MSHR misses (Count)

board.cache_hierarchy.l1dcaches.HardPFReq.mshrMissLatency::cache_hierarchy.l1dcaches.prefetcher 5173827691 # number of HardPFReq MSHR miss ticks (Tick)

board.cache_hierarchy.l1dcaches.HardPFReq.mshrMissLatency::total 5173827691 # number of HardPFReq MSHR miss ticks (Tick)

board.cache_hierarchy.l1dcaches.HardPFReq.mshrMissRate::cache_hierarchy.l1dcaches.prefetcher inf # mshr miss rate for HardPFReq accesses (Ratio)

board.cache_hierarchy.l1dcaches.HardPFReq.mshrMissRate::total inf # mshr miss rate for HardPFReq accesses (Ratio)

board.cache_hierarchy.l1dcaches.HardPFReq.avgMshrMissLatency::cache_hierarchy.l1dcaches.prefetcher 40070.537733 # average HardPFReq mshr miss latency ((Tick/Count))

board.cache_hierarchy.l1dcaches.HardPFReq.avgMshrMissLatency::total 40070.537733 # average HardPFReq mshr miss latency ((Tick/Count))

board.cache_hierarchy.l1dcaches.LockedRMWReadReq.hits::processor.cores.core.data 26 # number of LockedRMWReadReq hits (Count)

board.cache_hierarchy.l1dcaches.LockedRMWReadReq.hits::total 26 # number of LockedRMWReadReq hits (Count)

board.cache_hierarchy.l1dcaches.LockedRMWReadReq.misses::processor.cores.core.data 4 # number of LockedRMWReadReq misses (Count)

board.cache_hierarchy.l1dcaches.LockedRMWReadReq.misses::total 4 # number of LockedRMWReadReq misses (Count)

board.cache_hierarchy.l1dcaches.LockedRMWReadReq.missLatency::processor.cores.core.data 529803 # number of LockedRMWReadReq miss ticks (Tick)

board.cache_hierarchy.l1dcaches.LockedRMWReadReq.missLatency::total 529803 # number of LockedRMWReadReq miss ticks (Tick)

board.cache_hierarchy.l1dcaches.LockedRMWReadReq.accesses::processor.cores.core.data 30 # number of LockedRMWReadReq accesses(hits+misses) (Count)

board.cache_hierarchy.l1dcaches.LockedRMWReadReq.accesses::total 30 # number of LockedRMWReadReq accesses(hits+misses) (Count)

board.cache_hierarchy.l1dcaches.LockedRMWReadReq.missRate::processor.cores.core.data 0.133333 # miss rate for LockedRMWReadReq accesses (Ratio)

board.cache_hierarchy.l1dcaches.LockedRMWReadReq.missRate::total 0.133333 # miss rate for LockedRMWReadReq accesses (Ratio)

board.cache_hierarchy.l1dcaches.LockedRMWReadReq.avgMissLatency::processor.cores.core.data 132450.750000 # average LockedRMWReadReq miss latency ((Tick/Count))

board.cache_hierarchy.l1dcaches.LockedRMWReadReq.avgMissLatency::total 132450.750000 # average LockedRMWReadReq miss latency ((Tick/Count))

board.cache_hierarchy.l1dcaches.LockedRMWReadReq.mshrMisses::processor.cores.core.data 4 # number of LockedRMWReadReq MSHR misses (Count)

board.cache_hierarchy.l1dcaches.LockedRMWReadReq.mshrMisses::total 4 # number of LockedRMWReadReq MSHR misses (Count)

board.cache_hierarchy.l1dcaches.LockedRMWReadReq.mshrMissLatency::processor.cores.core.data 1125207 # number of LockedRMWReadReq MSHR miss ticks (Tick)

board.cache_hierarchy.l1dcaches.LockedRMWReadReq.mshrMissLatency::total 1125207 # number of LockedRMWReadReq MSHR miss ticks (Tick)

board.cache_hierarchy.l1dcaches.LockedRMWReadReq.mshrMissRate::processor.cores.core.data 0.133333 # mshr miss rate for LockedRMWReadReq accesses (Ratio)

board.cache_hierarchy.l1dcaches.LockedRMWReadReq.mshrMissRate::total 0.133333 # mshr miss rate for LockedRMWReadReq accesses (Ratio)

board.cache_hierarchy.l1dcaches.LockedRMWReadReq.avgMshrMissLatency::processor.cores.core.data 281301.750000 # average LockedRMWReadReq mshr miss latency ((Tick/Count))

board.cache_hierarchy.l1dcaches.LockedRMWReadReq.avgMshrMissLatency::total 281301.750000 # average LockedRMWReadReq mshr miss latency ((Tick/Count))

board.cache_hierarchy.l1dcaches.LockedRMWWriteReq.hits::processor.cores.core.data 30 # number of LockedRMWWriteReq hits (Count)

board.cache_hierarchy.l1dcaches.LockedRMWWriteReq.hits::total 30 # number of LockedRMWWriteReq hits (Count)

board.cache_hierarchy.l1dcaches.LockedRMWWriteReq.accesses::processor.cores.core.data 30 # number of LockedRMWWriteReq accesses(hits+misses) (Count)

board.cache_hierarchy.l1dcaches.LockedRMWWriteReq.accesses::total 30 # number of LockedRMWWriteReq accesses(hits+misses) (Count)

board.cache_hierarchy.l1dcaches.ReadReq.hits::processor.cores.core.data 3280666 # number of ReadReq hits (Count)

board.cache_hierarchy.l1dcaches.ReadReq.hits::total 3280666 # number of ReadReq hits (Count)

board.cache_hierarchy.l1dcaches.ReadReq.misses::processor.cores.core.data 3924 # number of ReadReq misses (Count)

board.cache_hierarchy.l1dcaches.ReadReq.misses::total 3924 # number of ReadReq misses (Count)

board.cache_hierarchy.l1dcaches.ReadReq.missLatency::processor.cores.core.data 421223022 # number of ReadReq miss ticks (Tick)

board.cache_hierarchy.l1dcaches.ReadReq.missLatency::total 421223022 # number of ReadReq miss ticks (Tick)

board.cache_hierarchy.l1dcaches.ReadReq.accesses::processor.cores.core.data 3284590 # number of ReadReq accesses(hits+misses) (Count)

board.cache_hierarchy.l1dcaches.ReadReq.accesses::total 3284590 # number of ReadReq accesses(hits+misses) (Count)

board.cache_hierarchy.l1dcaches.ReadReq.missRate::processor.cores.core.data 0.001195 # miss rate for ReadReq accesses (Ratio)

board.cache_hierarchy.l1dcaches.ReadReq.missRate::total 0.001195 # miss rate for ReadReq accesses (Ratio)

board.cache_hierarchy.l1dcaches.ReadReq.avgMissLatency::processor.cores.core.data 107345.316514 # average ReadReq miss latency ((Tick/Count))

board.cache_hierarchy.l1dcaches.ReadReq.avgMissLatency::total 107345.316514 # average ReadReq miss latency ((Tick/Count))

board.cache_hierarchy.l1dcaches.ReadReq.mshrHits::processor.cores.core.data 2114 # number of ReadReq MSHR hits (Count)

board.cache_hierarchy.l1dcaches.ReadReq.mshrHits::total 2114 # number of ReadReq MSHR hits (Count)

board.cache_hierarchy.l1dcaches.ReadReq.mshrMisses::processor.cores.core.data 1810 # number of ReadReq MSHR misses (Count)

board.cache_hierarchy.l1dcaches.ReadReq.mshrMisses::total 1810 # number of ReadReq MSHR misses (Count)

board.cache_hierarchy.l1dcaches.ReadReq.mshrMissLatency::processor.cores.core.data 207221904 # number of ReadReq MSHR miss ticks (Tick)

board.cache_hierarchy.l1dcaches.ReadReq.mshrMissLatency::total 207221904 # number of ReadReq MSHR miss ticks (Tick)

board.cache_hierarchy.l1dcaches.ReadReq.mshrMissRate::processor.cores.core.data 0.000551 # mshr miss rate for ReadReq accesses (Ratio)

board.cache_hierarchy.l1dcaches.ReadReq.mshrMissRate::total 0.000551 # mshr miss rate for ReadReq accesses (Ratio)

board.cache_hierarchy.l1dcaches.ReadReq.avgMshrMissLatency::processor.cores.core.data 114487.239779 # average ReadReq mshr miss latency ((Tick/Count))

board.cache_hierarchy.l1dcaches.ReadReq.avgMshrMissLatency::total 114487.239779 # average ReadReq mshr miss latency ((Tick/Count))

board.cache_hierarchy.l1dcaches.WriteReq.hits::processor.cores.core.data 4203625 # number of WriteReq hits (Count)

board.cache_hierarchy.l1dcaches.WriteReq.hits::total 4203625 # number of WriteReq hits (Count)

board.cache_hierarchy.l1dcaches.WriteReq.misses::processor.cores.core.data 1511 # number of WriteReq misses (Count)

board.cache_hierarchy.l1dcaches.WriteReq.misses::total 1511 # number of WriteReq misses (Count)

board.cache_hierarchy.l1dcaches.WriteReq.missLatency::processor.cores.core.data 166937561 # number of WriteReq miss ticks (Tick)

board.cache_hierarchy.l1dcaches.WriteReq.missLatency::total 166937561 # number of WriteReq miss ticks (Tick)

board.cache_hierarchy.l1dcaches.WriteReq.accesses::processor.cores.core.data 4205136 # number of WriteReq accesses(hits+misses) (Count)

board.cache_hierarchy.l1dcaches.WriteReq.accesses::total 4205136 # number of WriteReq accesses(hits+misses) (Count)

board.cache_hierarchy.l1dcaches.WriteReq.missRate::processor.cores.core.data 0.000359 # miss rate for WriteReq accesses (Ratio)

board.cache_hierarchy.l1dcaches.WriteReq.missRate::total 0.000359 # miss rate for WriteReq accesses (Ratio)

board.cache_hierarchy.l1dcaches.WriteReq.avgMissLatency::processor.cores.core.data 110481.509596 # average WriteReq miss latency ((Tick/Count))

board.cache_hierarchy.l1dcaches.WriteReq.avgMissLatency::total 110481.509596 # average WriteReq miss latency ((Tick/Count))

board.cache_hierarchy.l1dcaches.WriteReq.mshrHits::processor.cores.core.data 372 # number of WriteReq MSHR hits (Count)

board.cache_hierarchy.l1dcaches.WriteReq.mshrHits::total 372 # number of WriteReq MSHR hits (Count)

board.cache_hierarchy.l1dcaches.WriteReq.mshrMisses::processor.cores.core.data 1139 # number of WriteReq MSHR misses (Count)

board.cache_hierarchy.l1dcaches.WriteReq.mshrMisses::total 1139 # number of WriteReq MSHR misses (Count)

board.cache_hierarchy.l1dcaches.WriteReq.mshrMissLatency::processor.cores.core.data 128669867 # number of WriteReq MSHR miss ticks (Tick)

board.cache_hierarchy.l1dcaches.WriteReq.mshrMissLatency::total 128669867 # number of WriteReq MSHR miss ticks (Tick)

board.cache_hierarchy.l1dcaches.WriteReq.mshrMissRate::processor.cores.core.data 0.000271 # mshr miss rate for WriteReq accesses (Ratio)

board.cache_hierarchy.l1dcaches.WriteReq.mshrMissRate::total 0.000271 # mshr miss rate for WriteReq accesses (Ratio)

board.cache_hierarchy.l1dcaches.WriteReq.avgMshrMissLatency::processor.cores.core.data 112967.398595 # average WriteReq mshr miss latency ((Tick/Count))

board.cache_hierarchy.l1dcaches.WriteReq.avgMshrMissLatency::total 112967.398595 # average WriteReq mshr miss latency ((Tick/Count))

board.cache_hierarchy.l1dcaches.power_state.pwrStateResidencyTicks::UNDEFINED 5874741045 # Cumulative time (in ticks) in various power states (Tick)

board.cache_hierarchy.l1dcaches.prefetcher.demandMshrMisses 2949 # demands not covered by prefetchs (Count)

board.cache_hierarchy.l1dcaches.prefetcher.pfIssued 484285 # number of hwpf issued (Count)

board.cache_hierarchy.l1dcaches.prefetcher.pfUnused 19 # number of HardPF blocks evicted w/o reference (Count)

board.cache_hierarchy.l1dcaches.prefetcher.pfUseful 128561 # number of useful prefetch (Count)

board.cache_hierarchy.l1dcaches.prefetcher.pfUsefulButMiss 0 # number of hit on prefetch but cache block is not in an usable state (Count)

board.cache_hierarchy.l1dcaches.prefetcher.accuracy 0.265466 # accuracy of the prefetcher (Count)

board.cache_hierarchy.l1dcaches.prefetcher.coverage 0.977576 # coverage brought by this prefetcher (Count)

board.cache_hierarchy.l1dcaches.prefetcher.pfHitInCache 303717 # number of prefetches hitting in cache (Count)

board.cache_hierarchy.l1dcaches.prefetcher.pfHitInMSHR 51450 # number of prefetches hitting in a MSHR (Count)

board.cache_hierarchy.l1dcaches.prefetcher.pfHitInWB 0 # number of prefetches hit in the Write Buffer (Count)

board.cache_hierarchy.l1dcaches.prefetcher.pfLate 355167 # number of late prefetches (hitting in cache, MSHR or WB) (Count)

board.cache_hierarchy.l1dcaches.prefetcher.pfIdentified 504179 # number of prefetch candidates identified (Count)

board.cache_hierarchy.l1dcaches.prefetcher.pfBufferHit 11243 # number of redundant prefetches already in prefetch queue (Count)

board.cache_hierarchy.l1dcaches.prefetcher.pfInCache 0 # number of redundant prefetches already in cache/mshr dropped (Count)

board.cache_hierarchy.l1dcaches.prefetcher.pfRemovedDemand 4927 # number of prefetches dropped due to a demand for the same address (Count)

board.cache_hierarchy.l1dcaches.prefetcher.pfRemovedFull 0 # number of prefetches dropped due to prefetch queue size (Count)

board.cache_hierarchy.l1dcaches.prefetcher.pfSpanPage 20533 # number of prefetches that crossed the page (Count)

board.cache_hierarchy.l1dcaches.prefetcher.pfUsefulSpanPage 20264 # number of prefetches that is useful and crossed the page (Count)

board.cache_hierarchy.l1dcaches.prefetcher.power_state.pwrStateResidencyTicks::UNDEFINED 5874741045 # Cumulative time (in ticks) in various power states (Tick)

board.cache_hierarchy.l1dcaches.tags.tagsInUse 508.375680 # Average ticks per tags in use ((Tick/Count))

board.cache_hierarchy.l1dcaches.tags.totalRefs 7616420 # Total number of references to valid blocks. (Count)

board.cache_hierarchy.l1dcaches.tags.sampledRefs 132068 # Sample count of references to valid blocks. (Count)

board.cache_hierarchy.l1dcaches.tags.avgRefs 57.670442 # Average number of references to valid blocks. ((Count/Count))

board.cache_hierarchy.l1dcaches.tags.warmupTick 249417 # The tick when the warmup percentage was hit. (Tick)

board.cache_hierarchy.l1dcaches.tags.occupancies::cache_hierarchy.l1dcaches.prefetcher 490.402534 # Average occupied blocks per tick, per requestor ((Count/Tick))

board.cache_hierarchy.l1dcaches.tags.occupancies::processor.cores.core.data 17.973147 # Average occupied blocks per tick, per requestor ((Count/Tick))

board.cache_hierarchy.l1dcaches.tags.avgOccs::cache_hierarchy.l1dcaches.prefetcher 0.957817 # Average percentage of cache occupancy ((Ratio/Tick))

board.cache_hierarchy.l1dcaches.tags.avgOccs::processor.cores.core.data 0.035104 # Average percentage of cache occupancy ((Ratio/Tick))

board.cache_hierarchy.l1dcaches.tags.avgOccs::total 0.992921 # Average percentage of cache occupancy ((Ratio/Tick))

board.cache_hierarchy.l1dcaches.tags.occupanciesTaskId::1022 421 # Occupied blocks per task id (Count)

board.cache_hierarchy.l1dcaches.tags.occupanciesTaskId::1024 91 # Occupied blocks per task id (Count)

board.cache_hierarchy.l1dcaches.tags.ageTaskId_1022::0 4 # Occupied blocks per task id, per block age (Count)

board.cache_hierarchy.l1dcaches.tags.ageTaskId_1022::1 417 # Occupied blocks per task id, per block age (Count)

board.cache_hierarchy.l1dcaches.tags.ageTaskId_1024::0 43 # Occupied blocks per task id, per block age (Count)

board.cache_hierarchy.l1dcaches.tags.ageTaskId_1024::1 47 # Occupied blocks per task id, per block age (Count)

board.cache_hierarchy.l1dcaches.tags.ageTaskId_1024::3 1 # Occupied blocks per task id, per block age (Count)

board.cache_hierarchy.l1dcaches.tags.ratioOccsTaskId::1022 0.822266 # Ratio of occupied blocks and all blocks, per task id (Ratio)

board.cache_hierarchy.l1dcaches.tags.ratioOccsTaskId::1024 0.177734 # Ratio of occupied blocks and all blocks, per task id (Ratio)

board.cache_hierarchy.l1dcaches.tags.tagAccesses 60050356 # Number of tag accesses (Count)

board.cache_hierarchy.l1dcaches.tags.dataAccesses 60050356 # Number of data accesses (Count)

board.cache_hierarchy.l1dcaches.tags.power_state.pwrStateResidencyTicks::UNDEFINED 5874741045 # Cumulative time (in ticks) in various power states (Tick)

board.cache_hierarchy.l1icaches.demandHits::processor.cores.core.inst 4231554 # number of demand (read+write) hits (Count)

board.cache_hierarchy.l1icaches.demandHits::total 4231554 # number of demand (read+write) hits (Count)

board.cache_hierarchy.l1icaches.overallHits::processor.cores.core.inst 4231554 # number of overall hits (Count)

board.cache_hierarchy.l1icaches.overallHits::total 4231554 # number of overall hits (Count)

board.cache_hierarchy.l1icaches.demandMisses::processor.cores.core.inst 977 # number of demand (read+write) misses (Count)

board.cache_hierarchy.l1icaches.demandMisses::total 977 # number of demand (read+write) misses (Count)

board.cache_hierarchy.l1icaches.overallMisses::processor.cores.core.inst 977 # number of overall misses (Count)

board.cache_hierarchy.l1icaches.overallMisses::total 977 # number of overall misses (Count)

board.cache_hierarchy.l1icaches.demandMissLatency::processor.cores.core.inst 113639580 # number of demand (read+write) miss ticks (Tick)

board.cache_hierarchy.l1icaches.demandMissLatency::total 113639580 # number of demand (read+write) miss ticks (Tick)

board.cache_hierarchy.l1icaches.overallMissLatency::processor.cores.core.inst 113639580 # number of overall miss ticks (Tick)

board.cache_hierarchy.l1icaches.overallMissLatency::total 113639580 # number of overall miss ticks (Tick)

board.cache_hierarchy.l1icaches.demandAccesses::processor.cores.core.inst 4232531 # number of demand (read+write) accesses (Count)

board.cache_hierarchy.l1icaches.demandAccesses::total 4232531 # number of demand (read+write) accesses (Count)

board.cache_hierarchy.l1icaches.overallAccesses::processor.cores.core.inst 4232531 # number of overall (read+write) accesses (Count)

board.cache_hierarchy.l1icaches.overallAccesses::total 4232531 # number of overall (read+write) accesses (Count)

board.cache_hierarchy.l1icaches.demandMissRate::processor.cores.core.inst 0.000231 # miss rate for demand accesses (Ratio)

board.cache_hierarchy.l1icaches.demandMissRate::total 0.000231 # miss rate for demand accesses (Ratio)

board.cache_hierarchy.l1icaches.overallMissRate::processor.cores.core.inst 0.000231 # miss rate for overall accesses (Ratio)

board.cache_hierarchy.l1icaches.overallMissRate::total 0.000231 # miss rate for overall accesses (Ratio)

board.cache_hierarchy.l1icaches.demandAvgMissLatency::processor.cores.core.inst 116314.820880 # average overall miss latency in ticks ((Tick/Count))

board.cache_hierarchy.l1icaches.demandAvgMissLatency::total 116314.820880 # average overall miss latency in ticks ((Tick/Count))

board.cache_hierarchy.l1icaches.overallAvgMissLatency::processor.cores.core.inst 116314.820880 # average overall miss latency ((Tick/Count))

board.cache_hierarchy.l1icaches.overallAvgMissLatency::total 116314.820880 # average overall miss latency ((Tick/Count))

board.cache_hierarchy.l1icaches.blockedCycles::no_mshrs 0 # number of cycles access was blocked (Cycle)

board.cache_hierarchy.l1icaches.blockedCycles::no_targets 0 # number of cycles access was blocked (Cycle)

board.cache_hierarchy.l1icaches.blockedCauses::no_mshrs 0 # number of times access was blocked (Count)

board.cache_hierarchy.l1icaches.blockedCauses::no_targets 0 # number of times access was blocked (Count)

board.cache_hierarchy.l1icaches.avgBlocked::no_mshrs nan # average number of cycles each access was blocked ((Cycle/Count))

board.cache_hierarchy.l1icaches.avgBlocked::no_targets nan # average number of cycles each access was blocked ((Cycle/Count))

board.cache_hierarchy.l1icaches.demandMshrHits::processor.cores.core.inst 222 # number of demand (read+write) MSHR hits (Count)

board.cache_hierarchy.l1icaches.demandMshrHits::total 222 # number of demand (read+write) MSHR hits (Count)

board.cache_hierarchy.l1icaches.overallMshrHits::processor.cores.core.inst 222 # number of overall MSHR hits (Count)

board.cache_hierarchy.l1icaches.overallMshrHits::total 222 # number of overall MSHR hits (Count)

board.cache_hierarchy.l1icaches.demandMshrMisses::processor.cores.core.inst 755 # number of demand (read+write) MSHR misses (Count)

board.cache_hierarchy.l1icaches.demandMshrMisses::total 755 # number of demand (read+write) MSHR misses (Count)

board.cache_hierarchy.l1icaches.overallMshrMisses::processor.cores.core.inst 755 # number of overall MSHR misses (Count)

board.cache_hierarchy.l1icaches.overallMshrMisses::total 755 # number of overall MSHR misses (Count)

board.cache_hierarchy.l1icaches.demandMshrMissLatency::processor.cores.core.inst 90675567 # number of demand (read+write) MSHR miss ticks (Tick)

board.cache_hierarchy.l1icaches.demandMshrMissLatency::total 90675567 # number of demand (read+write) MSHR miss ticks (Tick)

board.cache_hierarchy.l1icaches.overallMshrMissLatency::processor.cores.core.inst 90675567 # number of overall MSHR miss ticks (Tick)

board.cache_hierarchy.l1icaches.overallMshrMissLatency::total 90675567 # number of overall MSHR miss ticks (Tick)

board.cache_hierarchy.l1icaches.demandMshrMissRate::processor.cores.core.inst 0.000178 # mshr miss ratio for demand accesses (Ratio)

board.cache_hierarchy.l1icaches.demandMshrMissRate::total 0.000178 # mshr miss ratio for demand accesses (Ratio)

board.cache_hierarchy.l1icaches.overallMshrMissRate::processor.cores.core.inst 0.000178 # mshr miss ratio for overall accesses (Ratio)

board.cache_hierarchy.l1icaches.overallMshrMissRate::total 0.000178 # mshr miss ratio for overall accesses (Ratio)

board.cache_hierarchy.l1icaches.demandAvgMshrMissLatency::processor.cores.core.inst 120100.088742 # average overall mshr miss latency ((Tick/Count))

board.cache_hierarchy.l1icaches.demandAvgMshrMissLatency::total 120100.088742 # average overall mshr miss latency ((Tick/Count))

board.cache_hierarchy.l1icaches.overallAvgMshrMissLatency::processor.cores.core.inst 120100.088742 # average overall mshr miss latency ((Tick/Count))

board.cache_hierarchy.l1icaches.overallAvgMshrMissLatency::total 120100.088742 # average overall mshr miss latency ((Tick/Count))

board.cache_hierarchy.l1icaches.replacements 241 # number of replacements (Count)

board.cache_hierarchy.l1icaches.ReadReq.hits::processor.cores.core.inst 4231554 # number of ReadReq hits (Count)

board.cache_hierarchy.l1icaches.ReadReq.hits::total 4231554 # number of ReadReq hits (Count)

board.cache_hierarchy.l1icaches.ReadReq.misses::processor.cores.core.inst 977 # number of ReadReq misses (Count)

board.cache_hierarchy.l1icaches.ReadReq.misses::total 977 # number of ReadReq misses (Count)

board.cache_hierarchy.l1icaches.ReadReq.missLatency::processor.cores.core.inst 113639580 # number of ReadReq miss ticks (Tick)

board.cache_hierarchy.l1icaches.ReadReq.missLatency::total 113639580 # number of ReadReq miss ticks (Tick)

board.cache_hierarchy.l1icaches.ReadReq.accesses::processor.cores.core.inst 4232531 # number of ReadReq accesses(hits+misses) (Count)

board.cache_hierarchy.l1icaches.ReadReq.accesses::total 4232531 # number of ReadReq accesses(hits+misses) (Count)

board.cache_hierarchy.l1icaches.ReadReq.missRate::processor.cores.core.inst 0.000231 # miss rate for ReadReq accesses (Ratio)

board.cache_hierarchy.l1icaches.ReadReq.missRate::total 0.000231 # miss rate for ReadReq accesses (Ratio)

board.cache_hierarchy.l1icaches.ReadReq.avgMissLatency::processor.cores.core.inst 116314.820880 # average ReadReq miss latency ((Tick/Count))

board.cache_hierarchy.l1icaches.ReadReq.avgMissLatency::total 116314.820880 # average ReadReq miss latency ((Tick/Count))

board.cache_hierarchy.l1icaches.ReadReq.mshrHits::processor.cores.core.inst 222 # number of ReadReq MSHR hits (Count)

board.cache_hierarchy.l1icaches.ReadReq.mshrHits::total 222 # number of ReadReq MSHR hits (Count)

board.cache_hierarchy.l1icaches.ReadReq.mshrMisses::processor.cores.core.inst 755 # number of ReadReq MSHR misses (Count)

board.cache_hierarchy.l1icaches.ReadReq.mshrMisses::total 755 # number of ReadReq MSHR misses (Count)

board.cache_hierarchy.l1icaches.ReadReq.mshrMissLatency::processor.cores.core.inst 90675567 # number of ReadReq MSHR miss ticks (Tick)

board.cache_hierarchy.l1icaches.ReadReq.mshrMissLatency::total 90675567 # number of ReadReq MSHR miss ticks (Tick)

board.cache_hierarchy.l1icaches.ReadReq.mshrMissRate::processor.cores.core.inst 0.000178 # mshr miss rate for ReadReq accesses (Ratio)

board.cache_hierarchy.l1icaches.ReadReq.mshrMissRate::total 0.000178 # mshr miss rate for ReadReq accesses (Ratio)

board.cache_hierarchy.l1icaches.ReadReq.avgMshrMissLatency::processor.cores.core.inst 120100.088742 # average ReadReq mshr miss latency ((Tick/Count))

board.cache_hierarchy.l1icaches.ReadReq.avgMshrMissLatency::total 120100.088742 # average ReadReq mshr miss latency ((Tick/Count))

board.cache_hierarchy.l1icaches.power_state.pwrStateResidencyTicks::UNDEFINED 5874741045 # Cumulative time (in ticks) in various power states (Tick)

board.cache_hierarchy.l1icaches.prefetcher.demandMshrMisses 755 # demands not covered by prefetchs (Count)

board.cache_hierarchy.l1icaches.prefetcher.pfIssued 0 # number of hwpf issued (Count)

board.cache_hierarchy.l1icaches.prefetcher.pfUseful 0 # number of useful prefetch (Count)

board.cache_hierarchy.l1icaches.prefetcher.pfUsefulButMiss 0 # number of hit on prefetch but cache block is not in an usable state (Count)

board.cache_hierarchy.l1icaches.prefetcher.accuracy nan # accuracy of the prefetcher (Count)

board.cache_hierarchy.l1icaches.prefetcher.coverage 0 # coverage brought by this prefetcher (Count)

board.cache_hierarchy.l1icaches.prefetcher.pfHitInCache 0 # number of prefetches hitting in cache (Count)

board.cache_hierarchy.l1icaches.prefetcher.pfHitInMSHR 0 # number of prefetches hitting in a MSHR (Count)

board.cache_hierarchy.l1icaches.prefetcher.pfHitInWB 0 # number of prefetches hit in the Write Buffer (Count)

board.cache_hierarchy.l1icaches.prefetcher.pfLate 0 # number of late prefetches (hitting in cache, MSHR or WB) (Count)

board.cache_hierarchy.l1icaches.prefetcher.pfIdentified 0 # number of prefetch candidates identified (Count)

board.cache_hierarchy.l1icaches.prefetcher.pfBufferHit 0 # number of redundant prefetches already in prefetch queue (Count)

board.cache_hierarchy.l1icaches.prefetcher.pfInCache 0 # number of redundant prefetches already in cache/mshr dropped (Count)

board.cache_hierarchy.l1icaches.prefetcher.pfRemovedDemand 0 # number of prefetches dropped due to a demand for the same address (Count)

board.cache_hierarchy.l1icaches.prefetcher.pfRemovedFull 0 # number of prefetches dropped due to prefetch queue size (Count)

board.cache_hierarchy.l1icaches.prefetcher.pfSpanPage 0 # number of prefetches that crossed the page (Count)

board.cache_hierarchy.l1icaches.prefetcher.pfUsefulSpanPage 0 # number of prefetches that is useful and crossed the page (Count)

board.cache_hierarchy.l1icaches.prefetcher.power_state.pwrStateResidencyTicks::UNDEFINED 5874741045 # Cumulative time (in ticks) in various power states (Tick)

board.cache_hierarchy.l1icaches.tags.tagsInUse 469.914048 # Average ticks per tags in use ((Tick/Count))

board.cache_hierarchy.l1icaches.tags.totalRefs 4232309 # Total number of references to valid blocks. (Count)

board.cache_hierarchy.l1icaches.tags.sampledRefs 755 # Sample count of references to valid blocks. (Count)

board.cache_hierarchy.l1icaches.tags.avgRefs 5605.707285 # Average number of references to valid blocks. ((Count/Count))

board.cache_hierarchy.l1icaches.tags.warmupTick 123543 # The tick when the warmup percentage was hit. (Tick)

board.cache_hierarchy.l1icaches.tags.occupancies::processor.cores.core.inst 469.914048 # Average occupied blocks per tick, per requestor ((Count/Tick))

board.cache_hierarchy.l1icaches.tags.avgOccs::processor.cores.core.inst 0.917801 # Average percentage of cache occupancy ((Ratio/Tick))

board.cache_hierarchy.l1icaches.tags.avgOccs::total 0.917801 # Average percentage of cache occupancy ((Ratio/Tick))

board.cache_hierarchy.l1icaches.tags.occupanciesTaskId::1024 511 # Occupied blocks per task id (Count)

board.cache_hierarchy.l1icaches.tags.ageTaskId_1024::0 106 # Occupied blocks per task id, per block age (Count)

board.cache_hierarchy.l1icaches.tags.ageTaskId_1024::1 100 # Occupied blocks per task id, per block age (Count)

board.cache_hierarchy.l1icaches.tags.ageTaskId_1024::3 305 # Occupied blocks per task id, per block age (Count)

board.cache_hierarchy.l1icaches.tags.ratioOccsTaskId::1024 0.998047 # Ratio of occupied blocks and all blocks, per task id (Ratio)

board.cache_hierarchy.l1icaches.tags.tagAccesses 33861003 # Number of tag accesses (Count)

board.cache_hierarchy.l1icaches.tags.dataAccesses 33861003 # Number of data accesses (Count)

board.cache_hierarchy.l1icaches.tags.power_state.pwrStateResidencyTicks::UNDEFINED 5874741045 # Cumulative time (in ticks) in various power states (Tick)

board.cache_hierarchy.l2bus.transDist::ReadResp 131683 # Transaction distribution (Count)

board.cache_hierarchy.l2bus.transDist::WritebackDirty 131412 # Transaction distribution (Count)

board.cache_hierarchy.l2bus.transDist::CleanEvict 129104 # Transaction distribution (Count)

board.cache_hierarchy.l2bus.transDist::HardPFReq 133282 # Transaction distribution (Count)

board.cache_hierarchy.l2bus.transDist::UpgradeReq 3 # Transaction distribution (Count)

board.cache_hierarchy.l2bus.transDist::UpgradeResp 3 # Transaction distribution (Count)

board.cache_hierarchy.l2bus.transDist::ReadExReq 1140 # Transaction distribution (Count)

board.cache_hierarchy.l2bus.transDist::ReadExResp 1140 # Transaction distribution (Count)

board.cache_hierarchy.l2bus.transDist::ReadSharedReq 131683 # Transaction distribution (Count)

board.cache_hierarchy.l2bus.pktCount_board.cache_hierarchy.l1icaches.mem_side_port::board.cache_hierarchy.l2cache.cpu_side_port 1746 # Packet count per connected requestor and responder (Count)

board.cache_hierarchy.l2bus.pktCount_board.cache_hierarchy.l1dcaches.mem_side_port::board.cache_hierarchy.l2cache.cpu_side_port 395698 # Packet count per connected requestor and responder (Count)

board.cache_hierarchy.l2bus.pktCount::total 397444 # Packet count per connected requestor and responder (Count)

board.cache_hierarchy.l2bus.pktSize_board.cache_hierarchy.l1icaches.mem_side_port::board.cache_hierarchy.l2cache.cpu_side_port 48000 # Cumulative packet size per connected requestor and responder (Byte)

board.cache_hierarchy.l2bus.pktSize_board.cache_hierarchy.l1dcaches.mem_side_port::board.cache_hierarchy.l2cache.cpu_side_port 12658240 # Cumulative packet size per connected requestor and responder (Byte)

board.cache_hierarchy.l2bus.pktSize::total 12706240 # Cumulative packet size per connected requestor and responder (Byte)

board.cache_hierarchy.l2bus.snoops 262006 # Total snoops (Count)

board.cache_hierarchy.l2bus.snoopTraffic 4204800 # Total snoop traffic (Byte)

board.cache_hierarchy.l2bus.snoopFanout::samples 394827 # Request fanout histogram (Count)

board.cache_hierarchy.l2bus.snoopFanout::mean 0.001231 # Request fanout histogram (Count)

board.cache_hierarchy.l2bus.snoopFanout::stdev 0.035063 # Request fanout histogram (Count)

board.cache_hierarchy.l2bus.snoopFanout::underflows 0 0.00% 0.00% # Request fanout histogram (Count)

board.cache_hierarchy.l2bus.snoopFanout::0 394341 99.88% 99.88% # Request fanout histogram (Count)

board.cache_hierarchy.l2bus.snoopFanout::1 486 0.12% 100.00% # Request fanout histogram (Count)

board.cache_hierarchy.l2bus.snoopFanout::2 0 0.00% 100.00% # Request fanout histogram (Count)

board.cache_hierarchy.l2bus.snoopFanout::3 0 0.00% 100.00% # Request fanout histogram (Count)

board.cache_hierarchy.l2bus.snoopFanout::4 0 0.00% 100.00% # Request fanout histogram (Count)

board.cache_hierarchy.l2bus.snoopFanout::overflows 0 0.00% 100.00% # Request fanout histogram (Count)

board.cache_hierarchy.l2bus.snoopFanout::min_value 0 # Request fanout histogram (Count)

board.cache_hierarchy.l2bus.snoopFanout::max_value 1 # Request fanout histogram (Count)

board.cache_hierarchy.l2bus.snoopFanout::total 394827 # Request fanout histogram (Count)

board.cache_hierarchy.l2bus.power_state.pwrStateResidencyTicks::UNDEFINED 5874741045 # Cumulative time (in ticks) in various power states (Tick)

board.cache_hierarchy.l2bus.reqLayer0.occupancy 173832086 # Layer occupancy (ticks) (Tick)

board.cache_hierarchy.l2bus.reqLayer0.utilization 0.0 # Layer utilization (Ratio)

board.cache_hierarchy.l2bus.respLayer0.occupancy 754245 # Layer occupancy (ticks) (Tick)

board.cache_hierarchy.l2bus.respLayer0.utilization 0.0 # Layer utilization (Ratio)

board.cache_hierarchy.l2bus.respLayer1.occupancy 131936931 # Layer occupancy (ticks) (Tick)

board.cache_hierarchy.l2bus.respLayer1.utilization 0.0 # Layer utilization (Ratio)

board.cache_hierarchy.l2bus.snoop_filter.totRequests 264623 # Total number of requests made to the snoop filter. (Count)

board.cache_hierarchy.l2bus.snoop_filter.hitSingleRequests 131800 # Number of requests hitting in the snoop filter with a single holder of the requested data. (Count)

board.cache_hierarchy.l2bus.snoop_filter.hitMultiRequests 6 # Number of requests hitting in the snoop filter with multiple (>1) holders of the requested data. (Count)

board.cache_hierarchy.l2bus.snoop_filter.totSnoops 477 # Total number of snoops made to the snoop filter. (Count)

board.cache_hierarchy.l2bus.snoop_filter.hitSingleSnoops 477 # Number of snoops hitting in the snoop filter with a single holder of the requested data. (Count)

board.cache_hierarchy.l2bus.snoop_filter.hitMultiSnoops 0 # Number of snoops hitting in the snoop filter with multiple (>1) holders of the requested data. (Count)

board.cache_hierarchy.l2cache.demandHits::cache_hierarchy.l1dcaches.prefetcher 118488 # number of demand (read+write) hits (Count)

board.cache_hierarchy.l2cache.demandHits::processor.cores.core.data 44 # number of demand (read+write) hits (Count)

board.cache_hierarchy.l2cache.demandHits::total 118532 # number of demand (read+write) hits (Count)

board.cache_hierarchy.l2cache.overallHits::cache_hierarchy.l1dcaches.prefetcher 118488 # number of overall hits (Count)

board.cache_hierarchy.l2cache.overallHits::processor.cores.core.data 44 # number of overall hits (Count)

board.cache_hierarchy.l2cache.overallHits::total 118532 # number of overall hits (Count)

board.cache_hierarchy.l2cache.demandMisses::cache_hierarchy.l1dcaches.prefetcher 10630 # number of demand (read+write) misses (Count)

board.cache_hierarchy.l2cache.demandMisses::processor.cores.core.inst 750 # number of demand (read+write) misses (Count)

board.cache_hierarchy.l2cache.demandMisses::processor.cores.core.data 2906 # number of demand (read+write) misses (Count)

board.cache_hierarchy.l2cache.demandMisses::total 14286 # number of demand (read+write) misses (Count)

board.cache_hierarchy.l2cache.overallMisses::cache_hierarchy.l1dcaches.prefetcher 10630 # number of overall misses (Count)

board.cache_hierarchy.l2cache.overallMisses::processor.cores.core.inst 750 # number of overall misses (Count)

board.cache_hierarchy.l2cache.overallMisses::processor.cores.core.data 2906 # number of overall misses (Count)

board.cache_hierarchy.l2cache.overallMisses::total 14286 # number of overall misses (Count)

board.cache_hierarchy.l2cache.demandMissLatency::cache_hierarchy.l1dcaches.prefetcher 1066465979 # number of demand (read+write) miss ticks (Tick)

board.cache_hierarchy.l2cache.demandMissLatency::processor.cores.core.inst 90166410 # number of demand (read+write) miss ticks (Tick)

board.cache_hierarchy.l2cache.demandMissLatency::processor.cores.core.data 332858142 # number of demand (read+write) miss ticks (Tick)

board.cache_hierarchy.l2cache.demandMissLatency::total 1489490531 # number of demand (read+write) miss ticks (Tick)

board.cache_hierarchy.l2cache.overallMissLatency::cache_hierarchy.l1dcaches.prefetcher 1066465979 # number of overall miss ticks (Tick)

board.cache_hierarchy.l2cache.overallMissLatency::processor.cores.core.inst 90166410 # number of overall miss ticks (Tick)

board.cache_hierarchy.l2cache.overallMissLatency::processor.cores.core.data 332858142 # number of overall miss ticks (Tick)

board.cache_hierarchy.l2cache.overallMissLatency::total 1489490531 # number of overall miss ticks (Tick)

board.cache_hierarchy.l2cache.demandAccesses::cache_hierarchy.l1dcaches.prefetcher 129118 # number of demand (read+write) accesses (Count)

board.cache_hierarchy.l2cache.demandAccesses::processor.cores.core.inst 750 # number of demand (read+write) accesses (Count)

board.cache_hierarchy.l2cache.demandAccesses::processor.cores.core.data 2950 # number of demand (read+write) accesses (Count)

board.cache_hierarchy.l2cache.demandAccesses::total 132818 # number of demand (read+write) accesses (Count)

board.cache_hierarchy.l2cache.overallAccesses::cache_hierarchy.l1dcaches.prefetcher 129118 # number of overall (read+write) accesses (Count)

board.cache_hierarchy.l2cache.overallAccesses::processor.cores.core.inst 750 # number of overall (read+write) accesses (Count)

board.cache_hierarchy.l2cache.overallAccesses::processor.cores.core.data 2950 # number of overall (read+write) accesses (Count)

board.cache_hierarchy.l2cache.overallAccesses::total 132818 # number of overall (read+write) accesses (Count)

board.cache_hierarchy.l2cache.demandMissRate::cache_hierarchy.l1dcaches.prefetcher 0.082328 # miss rate for demand accesses (Ratio)

board.cache_hierarchy.l2cache.demandMissRate::processor.cores.core.inst 1 # miss rate for demand accesses (Ratio)

board.cache_hierarchy.l2cache.demandMissRate::processor.cores.core.data 0.985085 # miss rate for demand accesses (Ratio)

board.cache_hierarchy.l2cache.demandMissRate::total 0.107561 # miss rate for demand accesses (Ratio)

board.cache_hierarchy.l2cache.overallMissRate::cache_hierarchy.l1dcaches.prefetcher 0.082328 # miss rate for overall accesses (Ratio)

board.cache_hierarchy.l2cache.overallMissRate::processor.cores.core.inst 1 # miss rate for overall accesses (Ratio)

board.cache_hierarchy.l2cache.overallMissRate::processor.cores.core.data 0.985085 # miss rate for overall accesses (Ratio)

board.cache_hierarchy.l2cache.overallMissRate::total 0.107561 # miss rate for overall accesses (Ratio)

board.cache_hierarchy.l2cache.demandAvgMissLatency::cache_hierarchy.l1dcaches.prefetcher 100326.056350 # average overall miss latency in ticks ((Tick/Count))

board.cache_hierarchy.l2cache.demandAvgMissLatency::processor.cores.core.inst 120221.880000 # average overall miss latency in ticks ((Tick/Count))

board.cache_hierarchy.l2cache.demandAvgMissLatency::processor.cores.core.data 114541.686855 # average overall miss latency in ticks ((Tick/Count))

board.cache_hierarchy.l2cache.demandAvgMissLatency::total 104262.251925 # average overall miss latency in ticks ((Tick/Count))

board.cache_hierarchy.l2cache.overallAvgMissLatency::cache_hierarchy.l1dcaches.prefetcher 100326.056350 # average overall miss latency ((Tick/Count))

board.cache_hierarchy.l2cache.overallAvgMissLatency::processor.cores.core.inst 120221.880000 # average overall miss latency ((Tick/Count))

board.cache_hierarchy.l2cache.overallAvgMissLatency::processor.cores.core.data 114541.686855 # average overall miss latency ((Tick/Count))

board.cache_hierarchy.l2cache.overallAvgMissLatency::total 104262.251925 # average overall miss latency ((Tick/Count))

board.cache_hierarchy.l2cache.blockedCycles::no_mshrs 0 # number of cycles access was blocked (Cycle)

board.cache_hierarchy.l2cache.blockedCycles::no_targets 0 # number of cycles access was blocked (Cycle)

board.cache_hierarchy.l2cache.blockedCauses::no_mshrs 0 # number of times access was blocked (Count)

board.cache_hierarchy.l2cache.blockedCauses::no_targets 0 # number of times access was blocked (Count)

board.cache_hierarchy.l2cache.avgBlocked::no_mshrs nan # average number of cycles each access was blocked ((Cycle/Count))

board.cache_hierarchy.l2cache.avgBlocked::no_targets nan # average number of cycles each access was blocked ((Cycle/Count))

board.cache_hierarchy.l2cache.writebacks::writebacks 65695 # number of writebacks (Count)

board.cache_hierarchy.l2cache.writebacks::total 65695 # number of writebacks (Count)

board.cache_hierarchy.l2cache.demandMshrHits::cache_hierarchy.l1dcaches.prefetcher 8552 # number of demand (read+write) MSHR hits (Count)

board.cache_hierarchy.l2cache.demandMshrHits::total 8552 # number of demand (read+write) MSHR hits (Count)

board.cache_hierarchy.l2cache.overallMshrHits::cache_hierarchy.l1dcaches.prefetcher 8552 # number of overall MSHR hits (Count)

board.cache_hierarchy.l2cache.overallMshrHits::total 8552 # number of overall MSHR hits (Count)

board.cache_hierarchy.l2cache.demandMshrMisses::cache_hierarchy.l1dcaches.prefetcher 2078 # number of demand (read+write) MSHR misses (Count)

board.cache_hierarchy.l2cache.demandMshrMisses::processor.cores.core.inst 750 # number of demand (read+write) MSHR misses (Count)

board.cache_hierarchy.l2cache.demandMshrMisses::processor.cores.core.data 2906 # number of demand (read+write) MSHR misses (Count)

board.cache_hierarchy.l2cache.demandMshrMisses::total 5734 # number of demand (read+write) MSHR misses (Count)

board.cache_hierarchy.l2cache.overallMshrMisses::cache_hierarchy.l1dcaches.prefetcher 2078 # number of overall MSHR misses (Count)

board.cache_hierarchy.l2cache.overallMshrMisses::cache_hierarchy.l2cache.prefetcher 127081 # number of overall MSHR misses (Count)

board.cache_hierarchy.l2cache.overallMshrMisses::processor.cores.core.inst 750 # number of overall MSHR misses (Count)

board.cache_hierarchy.l2cache.overallMshrMisses::processor.cores.core.data 2906 # number of overall MSHR misses (Count)

board.cache_hierarchy.l2cache.overallMshrMisses::total 132815 # number of overall MSHR misses (Count)

board.cache_hierarchy.l2cache.demandMshrMissLatency::cache_hierarchy.l1dcaches.prefetcher 175779689 # number of demand (read+write) MSHR miss ticks (Tick)

board.cache_hierarchy.l2cache.demandMshrMissLatency::processor.cores.core.inst 65191410 # number of demand (read+write) MSHR miss ticks (Tick)

board.cache_hierarchy.l2cache.demandMshrMissLatency::processor.cores.core.data 236088342 # number of demand (read+write) MSHR miss ticks (Tick)

board.cache_hierarchy.l2cache.demandMshrMissLatency::total 477059441 # number of demand (read+write) MSHR miss ticks (Tick)

board.cache_hierarchy.l2cache.overallMshrMissLatency::cache_hierarchy.l1dcaches.prefetcher 175779689 # number of overall MSHR miss ticks (Tick)

board.cache_hierarchy.l2cache.overallMshrMissLatency::cache_hierarchy.l2cache.prefetcher 8092709920 # number of overall MSHR miss ticks (Tick)

board.cache_hierarchy.l2cache.overallMshrMissLatency::processor.cores.core.inst 65191410 # number of overall MSHR miss ticks (Tick)

board.cache_hierarchy.l2cache.overallMshrMissLatency::processor.cores.core.data 236088342 # number of overall MSHR miss ticks (Tick)

board.cache_hierarchy.l2cache.overallMshrMissLatency::total 8569769361 # number of overall MSHR miss ticks (Tick)

board.cache_hierarchy.l2cache.demandMshrMissRate::cache_hierarchy.l1dcaches.prefetcher 0.016094 # mshr miss ratio for demand accesses (Ratio)

board.cache_hierarchy.l2cache.demandMshrMissRate::processor.cores.core.inst 1 # mshr miss ratio for demand accesses (Ratio)

board.cache_hierarchy.l2cache.demandMshrMissRate::processor.cores.core.data 0.985085 # mshr miss ratio for demand accesses (Ratio)

board.cache_hierarchy.l2cache.demandMshrMissRate::total 0.043172 # mshr miss ratio for demand accesses (Ratio)

board.cache_hierarchy.l2cache.overallMshrMissRate::cache_hierarchy.l1dcaches.prefetcher 0.016094 # mshr miss ratio for overall accesses (Ratio)

board.cache_hierarchy.l2cache.overallMshrMissRate::cache_hierarchy.l2cache.prefetcher inf # mshr miss ratio for overall accesses (Ratio)

board.cache_hierarchy.l2cache.overallMshrMissRate::processor.cores.core.inst 1 # mshr miss ratio for overall accesses (Ratio)

board.cache_hierarchy.l2cache.overallMshrMissRate::processor.cores.core.data 0.985085 # mshr miss ratio for overall accesses (Ratio)

board.cache_hierarchy.l2cache.overallMshrMissRate::total 0.999977 # mshr miss ratio for overall accesses (Ratio)

board.cache_hierarchy.l2cache.demandAvgMshrMissLatency::cache_hierarchy.l1dcaches.prefetcher 84590.803176 # average overall mshr miss latency ((Tick/Count))

board.cache_hierarchy.l2cache.demandAvgMshrMissLatency::processor.cores.core.inst 86921.880000 # average overall mshr miss latency ((Tick/Count))

board.cache_hierarchy.l2cache.demandAvgMshrMissLatency::processor.cores.core.data 81241.686855 # average overall mshr miss latency ((Tick/Count))

board.cache_hierarchy.l2cache.demandAvgMshrMissLatency::total 83198.367806 # average overall mshr miss latency ((Tick/Count))

board.cache_hierarchy.l2cache.overallAvgMshrMissLatency::cache_hierarchy.l1dcaches.prefetcher 84590.803176 # average overall mshr miss latency ((Tick/Count))

board.cache_hierarchy.l2cache.overallAvgMshrMissLatency::cache_hierarchy.l2cache.prefetcher 63681.509588 # average overall mshr miss latency ((Tick/Count))

board.cache_hierarchy.l2cache.overallAvgMshrMissLatency::processor.cores.core.inst 86921.880000 # average overall mshr miss latency ((Tick/Count))

board.cache_hierarchy.l2cache.overallAvgMshrMissLatency::processor.cores.core.data 81241.686855 # average overall mshr miss latency ((Tick/Count))

board.cache_hierarchy.l2cache.overallAvgMshrMissLatency::total 64524.107676 # average overall mshr miss latency ((Tick/Count))

board.cache_hierarchy.l2cache.replacements 128719 # number of replacements (Count)

board.cache_hierarchy.l2cache.CleanEvict.mshrMisses::writebacks 169 # number of CleanEvict MSHR misses (Count)

board.cache_hierarchy.l2cache.CleanEvict.mshrMisses::total 169 # number of CleanEvict MSHR misses (Count)

board.cache_hierarchy.l2cache.CleanEvict.mshrMissRate::writebacks inf # mshr miss rate for CleanEvict accesses (Ratio)

board.cache_hierarchy.l2cache.CleanEvict.mshrMissRate::total inf # mshr miss rate for CleanEvict accesses (Ratio)

board.cache_hierarchy.l2cache.HardPFReq.mshrMisses::cache_hierarchy.l2cache.prefetcher 127081 # number of HardPFReq MSHR misses (Count)

board.cache_hierarchy.l2cache.HardPFReq.mshrMisses::total 127081 # number of HardPFReq MSHR misses (Count)

board.cache_hierarchy.l2cache.HardPFReq.mshrMissLatency::cache_hierarchy.l2cache.prefetcher 8092709920 # number of HardPFReq MSHR miss ticks (Tick)

board.cache_hierarchy.l2cache.HardPFReq.mshrMissLatency::total 8092709920 # number of HardPFReq MSHR miss ticks (Tick)

board.cache_hierarchy.l2cache.HardPFReq.mshrMissRate::cache_hierarchy.l2cache.prefetcher inf # mshr miss rate for HardPFReq accesses (Ratio)

board.cache_hierarchy.l2cache.HardPFReq.mshrMissRate::total inf # mshr miss rate for HardPFReq accesses (Ratio)

board.cache_hierarchy.l2cache.HardPFReq.avgMshrMissLatency::cache_hierarchy.l2cache.prefetcher 63681.509588 # average HardPFReq mshr miss latency ((Tick/Count))

board.cache_hierarchy.l2cache.HardPFReq.avgMshrMissLatency::total 63681.509588 # average HardPFReq mshr miss latency ((Tick/Count))

board.cache_hierarchy.l2cache.ReadExReq.hits::processor.cores.core.data 9 # number of ReadExReq hits (Count)

board.cache_hierarchy.l2cache.ReadExReq.hits::total 9 # number of ReadExReq hits (Count)

board.cache_hierarchy.l2cache.ReadExReq.misses::processor.cores.core.data 1131 # number of ReadExReq misses (Count)

board.cache_hierarchy.l2cache.ReadExReq.misses::total 1131 # number of ReadExReq misses (Count)

board.cache_hierarchy.l2cache.ReadExReq.missLatency::processor.cores.core.data 128031840 # number of ReadExReq miss ticks (Tick)

board.cache_hierarchy.l2cache.ReadExReq.missLatency::total 128031840 # number of ReadExReq miss ticks (Tick)

board.cache_hierarchy.l2cache.ReadExReq.accesses::processor.cores.core.data 1140 # number of ReadExReq accesses(hits+misses) (Count)

board.cache_hierarchy.l2cache.ReadExReq.accesses::total 1140 # number of ReadExReq accesses(hits+misses) (Count)

board.cache_hierarchy.l2cache.ReadExReq.missRate::processor.cores.core.data 0.992105 # miss rate for ReadExReq accesses (Ratio)

board.cache_hierarchy.l2cache.ReadExReq.missRate::total 0.992105 # miss rate for ReadExReq accesses (Ratio)

board.cache_hierarchy.l2cache.ReadExReq.avgMissLatency::processor.cores.core.data 113202.334218 # average ReadExReq miss latency ((Tick/Count))

board.cache_hierarchy.l2cache.ReadExReq.avgMissLatency::total 113202.334218 # average ReadExReq miss latency ((Tick/Count))

board.cache_hierarchy.l2cache.ReadExReq.mshrMisses::processor.cores.core.data 1131 # number of ReadExReq MSHR misses (Count)

board.cache_hierarchy.l2cache.ReadExReq.mshrMisses::total 1131 # number of ReadExReq MSHR misses (Count)

board.cache_hierarchy.l2cache.ReadExReq.mshrMissLatency::processor.cores.core.data 90369540 # number of ReadExReq MSHR miss ticks (Tick)

board.cache_hierarchy.l2cache.ReadExReq.mshrMissLatency::total 90369540 # number of ReadExReq MSHR miss ticks (Tick)

board.cache_hierarchy.l2cache.ReadExReq.mshrMissRate::processor.cores.core.data 0.992105 # mshr miss rate for ReadExReq accesses (Ratio)

board.cache_hierarchy.l2cache.ReadExReq.mshrMissRate::total 0.992105 # mshr miss rate for ReadExReq accesses (Ratio)

board.cache_hierarchy.l2cache.ReadExReq.avgMshrMissLatency::processor.cores.core.data 79902.334218 # average ReadExReq mshr miss latency ((Tick/Count))

board.cache_hierarchy.l2cache.ReadExReq.avgMshrMissLatency::total 79902.334218 # average ReadExReq mshr miss latency ((Tick/Count))

board.cache_hierarchy.l2cache.ReadSharedReq.hits::cache_hierarchy.l1dcaches.prefetcher 118488 # number of ReadSharedReq hits (Count)

board.cache_hierarchy.l2cache.ReadSharedReq.hits::processor.cores.core.data 35 # number of ReadSharedReq hits (Count)

board.cache_hierarchy.l2cache.ReadSharedReq.hits::total 118523 # number of ReadSharedReq hits (Count)

board.cache_hierarchy.l2cache.ReadSharedReq.misses::cache_hierarchy.l1dcaches.prefetcher 10630 # number of ReadSharedReq misses (Count)

board.cache_hierarchy.l2cache.ReadSharedReq.misses::processor.cores.core.inst 750 # number of ReadSharedReq misses (Count)

board.cache_hierarchy.l2cache.ReadSharedReq.misses::processor.cores.core.data 1775 # number of ReadSharedReq misses (Count)

board.cache_hierarchy.l2cache.ReadSharedReq.misses::total 13155 # number of ReadSharedReq misses (Count)

board.cache_hierarchy.l2cache.ReadSharedReq.missLatency::cache_hierarchy.l1dcaches.prefetcher 1066465979 # number of ReadSharedReq miss ticks (Tick)

board.cache_hierarchy.l2cache.ReadSharedReq.missLatency::processor.cores.core.inst 90166410 # number of ReadSharedReq miss ticks (Tick)

board.cache_hierarchy.l2cache.ReadSharedReq.missLatency::processor.cores.core.data 204826302 # number of ReadSharedReq miss ticks (Tick)

board.cache_hierarchy.l2cache.ReadSharedReq.missLatency::total 1361458691 # number of ReadSharedReq miss ticks (Tick)

board.cache_hierarchy.l2cache.ReadSharedReq.accesses::cache_hierarchy.l1dcaches.prefetcher 129118 # number of ReadSharedReq accesses(hits+misses) (Count)

board.cache_hierarchy.l2cache.ReadSharedReq.accesses::processor.cores.core.inst 750 # number of ReadSharedReq accesses(hits+misses) (Count)

board.cache_hierarchy.l2cache.ReadSharedReq.accesses::processor.cores.core.data 1810 # number of ReadSharedReq accesses(hits+misses) (Count)

board.cache_hierarchy.l2cache.ReadSharedReq.accesses::total 131678 # number of ReadSharedReq accesses(hits+misses) (Count)

board.cache_hierarchy.l2cache.ReadSharedReq.missRate::cache_hierarchy.l1dcaches.prefetcher 0.082328 # miss rate for ReadSharedReq accesses (Ratio)

board.cache_hierarchy.l2cache.ReadSharedReq.missRate::processor.cores.core.inst 1 # miss rate for ReadSharedReq accesses (Ratio)

board.cache_hierarchy.l2cache.ReadSharedReq.missRate::processor.cores.core.data 0.980663 # miss rate for ReadSharedReq accesses (Ratio)

board.cache_hierarchy.l2cache.ReadSharedReq.missRate::total 0.099903 # miss rate for ReadSharedReq accesses (Ratio)

board.cache_hierarchy.l2cache.ReadSharedReq.avgMissLatency::cache_hierarchy.l1dcaches.prefetcher 100326.056350 # average ReadSharedReq miss latency ((Tick/Count))

board.cache_hierarchy.l2cache.ReadSharedReq.avgMissLatency::processor.cores.core.inst 120221.880000 # average ReadSharedReq miss latency ((Tick/Count))

board.cache_hierarchy.l2cache.ReadSharedReq.avgMissLatency::processor.cores.core.data 115395.099718 # average ReadSharedReq miss latency ((Tick/Count))

board.cache_hierarchy.l2cache.ReadSharedReq.avgMissLatency::total 103493.629114 # average ReadSharedReq miss latency ((Tick/Count))

board.cache_hierarchy.l2cache.ReadSharedReq.mshrHits::cache_hierarchy.l1dcaches.prefetcher 8552 # number of ReadSharedReq MSHR hits (Count)

board.cache_hierarchy.l2cache.ReadSharedReq.mshrHits::total 8552 # number of ReadSharedReq MSHR hits (Count)

board.cache_hierarchy.l2cache.ReadSharedReq.mshrMisses::cache_hierarchy.l1dcaches.prefetcher 2078 # number of ReadSharedReq MSHR misses (Count)

board.cache_hierarchy.l2cache.ReadSharedReq.mshrMisses::processor.cores.core.inst 750 # number of ReadSharedReq MSHR misses (Count)

board.cache_hierarchy.l2cache.ReadSharedReq.mshrMisses::processor.cores.core.data 1775 # number of ReadSharedReq MSHR misses (Count)

board.cache_hierarchy.l2cache.ReadSharedReq.mshrMisses::total 4603 # number of ReadSharedReq MSHR misses (Count)

board.cache_hierarchy.l2cache.ReadSharedReq.mshrMissLatency::cache_hierarchy.l1dcaches.prefetcher 175779689 # number of ReadSharedReq MSHR miss ticks (Tick)

board.cache_hierarchy.l2cache.ReadSharedReq.mshrMissLatency::processor.cores.core.inst 65191410 # number of ReadSharedReq MSHR miss ticks (Tick)

board.cache_hierarchy.l2cache.ReadSharedReq.mshrMissLatency::processor.cores.core.data 145718802 # number of ReadSharedReq MSHR miss ticks (Tick)

board.cache_hierarchy.l2cache.ReadSharedReq.mshrMissLatency::total 386689901 # number of ReadSharedReq MSHR miss ticks (Tick)

board.cache_hierarchy.l2cache.ReadSharedReq.mshrMissRate::cache_hierarchy.l1dcaches.prefetcher 0.016094 # mshr miss rate for ReadSharedReq accesses (Ratio)

board.cache_hierarchy.l2cache.ReadSharedReq.mshrMissRate::processor.cores.core.inst 1 # mshr miss rate for ReadSharedReq accesses (Ratio)

board.cache_hierarchy.l2cache.ReadSharedReq.mshrMissRate::processor.cores.core.data 0.980663 # mshr miss rate for ReadSharedReq accesses (Ratio)

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board.cache_hierarchy.l2cache.ReadSharedReq.mshrMissRate::total     0.034956                    # mshr miss rate for ReadSharedReq accesses (Ratio)
board.cache_hierarchy.l2cache.ReadSharedReq.avgMshrMissLatency::cache_hierarchy.l1dcaches.prefetcher 84590.803176                    # average ReadSharedReq mshr miss latency ((Tick/Count))
board.cache_hierarchy.l2cache.ReadSharedReq.avgMshrMissLatency::processor.cores.core.inst 86921.880000                    # average ReadSharedReq mshr miss latency ((Tick/Count))
board.cache_hierarchy.l2cache.ReadSharedReq.avgMshrMissLatency::processor.cores.core.data 82095.099718                    # average ReadSharedReq mshr miss latency ((Tick/Count))
board.cache_hierarchy.l2cache.ReadSharedReq.avgMshrMissLatency::total 84008.233978                    # average ReadSharedReq mshr miss latency ((Tick/Count))
board.cache_hierarchy.l2cache.UpgradeReq.hits::processor.cores.core.data           3                    # number of UpgradeReq hits (Count)
board.cache_hierarchy.l2cache.UpgradeReq.hits::total           3                    # number of UpgradeReq hits (Count)
board.cache_hierarchy.l2cache.UpgradeReq.accesses::processor.cores.core.data           3                    # number of UpgradeReq accesses(hits+misses) (Count)
board.cache_hierarchy.l2cache.UpgradeReq.accesses::total           3                    # number of UpgradeReq accesses(hits+misses) (Count)
board.cache_hierarchy.l2cache.WritebackDirty.hits::writebacks       65717                    # number of WritebackDirty hits (Count)
board.cache_hierarchy.l2cache.WritebackDirty.hits::total       65717                    # number of WritebackDirty hits (Count)
board.cache_hierarchy.l2cache.WritebackDirty.accesses::writebacks       65717                    # number of WritebackDirty accesses(hits+misses) (Count)
board.cache_hierarchy.l2cache.WritebackDirty.accesses::total       65717                    # number of WritebackDirty accesses(hits+misses) (Count)
board.cache_hierarchy.l2cache.power_state.pwrStateResidencyTicks::UNDEFINED   5874741045                    # Cumulative time (in ticks) in various power states (Tick)
board.cache_hierarchy.l2cache.prefetcher.demandMshrMisses         5734                    # demands not covered by prefetchs (Count)
board.cache_hierarchy.l2cache.prefetcher.pfIssued       495889                    # number of hwpf issued (Count)
board.cache_hierarchy.l2cache.prefetcher.pfUnused           13                    # number of HardPF blocks evicted w/o reference (Count)
board.cache_hierarchy.l2cache.prefetcher.pfUseful       118508                    # number of useful prefetch (Count)
board.cache_hierarchy.l2cache.prefetcher.pfUsefulButMiss            0                    # number of hit on prefetch but cache block is not in an usable state (Count)
board.cache_hierarchy.l2cache.prefetcher.accuracy     0.238981                    # accuracy of the prefetcher (Count)
board.cache_hierarchy.l2cache.prefetcher.coverage     0.953848                    # coverage brought by this prefetcher (Count)
board.cache_hierarchy.l2cache.prefetcher.pfHitInCache       263190                    # number of prefetches hitting in cache (Count)
board.cache_hierarchy.l2cache.prefetcher.pfHitInMSHR       105618                    # number of prefetches hitting in a MSHR (Count)
board.cache_hierarchy.l2cache.prefetcher.pfHitInWB            0                    # number of prefetches hit in the Write Buffer (Count)
board.cache_hierarchy.l2cache.prefetcher.pfLate       368808                    # number of late prefetches (hitting in cache, MSHR or WB) (Count)
board.cache_hierarchy.l2cache.prefetcher.pfIdentified       504095                    # number of prefetch candidates identified (Count)
board.cache_hierarchy.l2cache.prefetcher.pfBufferHit         4101                    # number of redundant prefetches already in prefetch queue (Count)
board.cache_hierarchy.l2cache.prefetcher.pfInCache            0                    # number of redundant prefetches already in cache/mshr dropped (Count)
board.cache_hierarchy.l2cache.prefetcher.pfRemovedDemand         2054                    # number of prefetches dropped due to a demand for the same address (Count)
board.cache_hierarchy.l2cache.prefetcher.pfRemovedFull            0                    # number of prefetches dropped due to prefetch queue size (Count)
board.cache_hierarchy.l2cache.prefetcher.pfSpanPage        20509                    # number of prefetches that crossed the page (Count)
board.cache_hierarchy.l2cache.prefetcher.pfUsefulSpanPage        19010                    # number of prefetches that is useful and crossed the page (Count)
board.cache_hierarchy.l2cache.prefetcher.power_state.pwrStateResidencyTicks::UNDEFINED   5874741045                    # Cumulative time (in ticks) in various power states (Tick)
board.cache_hierarchy.l2cache.tags.tagsInUse  3950.311175                    # Average ticks per tags in use ((Tick/Count))
board.cache_hierarchy.l2cache.tags.totalRefs       382978                    # Total number of references to valid blocks. (Count)
board.cache_hierarchy.l2cache.tags.sampledRefs       132815                    # Sample count of references to valid blocks. (Count)
board.cache_hierarchy.l2cache.tags.avgRefs     2.883545                    # Average number of references to valid blocks. ((Count/Count))
board.cache_hierarchy.l2cache.tags.warmupTick        89910                    # The tick when the warmup percentage was hit. (Tick)
board.cache_hierarchy.l2cache.tags.occupancies::cache_hierarchy.l1dcaches.prefetcher    61.866583                    # Average occupied blocks per tick, per requestor ((Count/Tick))
board.cache_hierarchy.l2cache.tags.occupancies::cache_hierarchy.l2cache.prefetcher  3720.359984                    # Average occupied blocks per tick, per requestor ((Count/Tick))
board.cache_hierarchy.l2cache.tags.occupancies::processor.cores.core.inst    42.293044                    # Average occupied blocks per tick, per requestor ((Count/Tick))
board.cache_hierarchy.l2cache.tags.occupancies::processor.cores.core.data   125.791563                    # Average occupied blocks per tick, per requestor ((Count/Tick))
board.cache_hierarchy.l2cache.tags.avgOccs::cache_hierarchy.l1dcaches.prefetcher     0.015104                    # Average percentage of cache occupancy ((Ratio/Tick))
board.cache_hierarchy.l2cache.tags.avgOccs::cache_hierarchy.l2cache.prefetcher     0.908291                    # Average percentage of cache occupancy ((Ratio/Tick))
board.cache_hierarchy.l2cache.tags.avgOccs::processor.cores.core.inst     0.010325                    # Average percentage of cache occupancy ((Ratio/Tick))
board.cache_hierarchy.l2cache.tags.avgOccs::processor.cores.core.data     0.030711                    # Average percentage of cache occupancy ((Ratio/Tick))
board.cache_hierarchy.l2cache.tags.avgOccs::total     0.964431                    # Average percentage of cache occupancy ((Ratio/Tick))
board.cache_hierarchy.l2cache.tags.occupanciesTaskId::1022         3749                    # Occupied blocks per task id (Count)
board.cache_hierarchy.l2cache.tags.occupanciesTaskId::1024          347                    # Occupied blocks per task id (Count)
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board.cache_hierarchy.l2cache.tags.ageTaskId_1022::0 8 # Occupied blocks per task id, per block age (Count)

board.cache_hierarchy.l2cache.tags.ageTaskId_1022::1 1806 # Occupied blocks per task id, per block age (Count)

board.cache_hierarchy.l2cache.tags.ageTaskId_1022::2 1935 # Occupied blocks per task id, per block age (Count)

board.cache_hierarchy.l2cache.tags.ageTaskId_1024::0 147 # Occupied blocks per task id, per block age (Count)

board.cache_hierarchy.l2cache.tags.ageTaskId_1024::1 171 # Occupied blocks per task id, per block age (Count)

board.cache_hierarchy.l2cache.tags.ageTaskId_1024::2 29 # Occupied blocks per task id, per block age (Count)

board.cache_hierarchy.l2cache.tags.ratioOccsTaskId::1022 0.915283 # Ratio of occupied blocks and all blocks, per task id (Ratio)

board.cache_hierarchy.l2cache.tags.ratioOccsTaskId::1024 0.084717 # Ratio of occupied blocks and all blocks, per task id (Ratio)

board.cache_hierarchy.l2cache.tags.tagAccesses 4366703 # Number of tag accesses (Count)

board.cache_hierarchy.l2cache.tags.dataAccesses 4366703 # Number of data accesses (Count)

board.cache_hierarchy.l2cache.tags.power_state.pwrStateResidencyTicks::UNDEFINED 5874741045 # Cumulative time (in ticks) in various power states (Tick)

board.cache_hierarchy.membus.transDist::ReadResp 131684 # Transaction distribution (Count)

board.cache_hierarchy.membus.transDist::WritebackDirty 65695 # Transaction distribution (Count)

board.cache_hierarchy.membus.transDist::CleanEvict 62718 # Transaction distribution (Count)

board.cache_hierarchy.membus.transDist::ReadExReq 1131 # Transaction distribution (Count)

board.cache_hierarchy.membus.transDist::ReadExResp 1131 # Transaction distribution (Count)

board.cache_hierarchy.membus.transDist::ReadSharedReq 131684 # Transaction distribution (Count)

board.cache_hierarchy.membus.pktCount_board.cache_hierarchy.l2cache.mem_side_port::board.memory.mem_ctrl.port 394043 # Packet count per connected requestor and responder (Count)

board.cache_hierarchy.membus.pktCount_board.cache_hierarchy.l2cache.mem_side_port::total 394043 # Packet count per connected requestor and responder (Count)

board.cache_hierarchy.membus.pktCount::total 394043 # Packet count per connected requestor and responder (Count)

board.cache_hierarchy.membus.pktSize_board.cache_hierarchy.l2cache.mem_side_port::board.memory.mem_ctrl.port 12704640 # Cumulative packet size per connected requestor and responder (Byte)

board.cache_hierarchy.membus.pktSize_board.cache_hierarchy.l2cache.mem_side_port::total 12704640 # Cumulative packet size per connected requestor and responder (Byte)

board.cache_hierarchy.membus.pktSize::total 12704640 # Cumulative packet size per connected requestor and responder (Byte)

board.cache_hierarchy.membus.snoops 0 # Total snoops (Count)

board.cache_hierarchy.membus.snoopTraffic 0 # Total snoop traffic (Byte)

board.cache_hierarchy.membus.snoopFanout::samples 132815 # Request fanout histogram (Count)

board.cache_hierarchy.membus.snoopFanout::mean 0 # Request fanout histogram (Count)

board.cache_hierarchy.membus.snoopFanout::stdev 0 # Request fanout histogram (Count)

board.cache_hierarchy.membus.snoopFanout::underflows 0 0.00% 0.00% # Request fanout histogram (Count)

board.cache_hierarchy.membus.snoopFanout::0 132815 100.00% 100.00% # Request fanout histogram (Count)

board.cache_hierarchy.membus.snoopFanout::1 0 0.00% 100.00% # Request fanout histogram (Count)

board.cache_hierarchy.membus.snoopFanout::overflows 0 0.00% 100.00% # Request fanout histogram (Count)

board.cache_hierarchy.membus.snoopFanout::min_value 0 # Request fanout histogram (Count)

board.cache_hierarchy.membus.snoopFanout::max_value 0 # Request fanout histogram (Count)

board.cache_hierarchy.membus.snoopFanout::total 132815 # Request fanout histogram (Count)

board.cache_hierarchy.membus.badaddr_responder.power_state.pwrStateResidencyTicks::UNDEFINED 5874741045 # Cumulative time (in ticks) in various power states (Tick)

board.cache_hierarchy.membus.power_state.pwrStateResidencyTicks::UNDEFINED 5874741045 # Cumulative time (in ticks) in various power states (Tick)

board.cache_hierarchy.membus.reqLayer0.occupancy 147734144 # Layer occupancy (ticks) (Tick)

board.cache_hierarchy.membus.reqLayer0.utilization 0.0 # Layer utilization (Ratio)

board.cache_hierarchy.membus.respLayer2.occupancy 114096769 # Layer occupancy (ticks) (Tick)

board.cache_hierarchy.membus.respLayer2.utilization 0.0 # Layer utilization (Ratio)

board.cache_hierarchy.membus.snoop_filter.totRequests 261228 # Total number of requests made to the snoop filter. (Count)

board.cache_hierarchy.membus.snoop_filter.hitSingleRequests 128413 # Number of requests hitting in the snoop filter with a single holder of the requested data. (Count)

board.cache_hierarchy.membus.snoop_filter.hitMultiRequests 0 # Number of requests hitting in the snoop filter with multiple (>1) holders of the requested data. (Count)

board.cache_hierarchy.membus.snoop_filter.totSnoops 0 # Total number of snoops made to the snoop filter. (Count)

board.cache_hierarchy.membus.snoop_filter.hitSingleSnoops 0 # Number of snoops hitting in the snoop filter with a single holder of the requested data. (Count)

board.cache_hierarchy.membus.snoop_filter.hitMultiSnoops 0 # Number of snoops hitting in the snoop filter with multiple (>1) holders of the requested data. (Count)

board.clk_domain.clock 333 # Clock period in ticks (Tick)

```
board.clk_domain.voltage_domain.voltage 1 # Voltage in Volts (Volt)
board.memory.mem_ctrl.avgPriority_writebacks::samples 65695.00 # Average QoS priority value for accepted requests (Count)
board.memory.mem_ctrl.avgPriority_cache_hierarchy.l1dcaches.prefetcher::samples 2078.00 # Average QoS priority value for accepted requests (Count)
board.memory.mem_ctrl.avgPriority_cache_hierarchy.l2cache.prefetcher::samples 127060.00 # Average QoS priority value for accepted requests (Count)
board.memory.mem_ctrl.avgPriority_processor.cores.core.inst::samples 750.00 # Average QoS priority value for accepted requests (Count)
board.memory.mem_ctrl.avgPriority_processor.cores.core.data::samples 2906.00 # Average QoS priority value for accepted requests (Count)
board.memory.mem_ctrl.priorityMinLatency 0.000000018750 # per QoS priority minimum request to response latency (Second)
board.memory.mem_ctrl.priorityMaxLatency 0.000151957756 # per QoS priority maximum request to response latency (Second)
board.memory.mem_ctrl.numReadWriteTurnArounds 4104 # Number of turnarounds from READ to WRITE (Count)
board.memory.mem_ctrl.numWriteReadTurnArounds 4104 # Number of turnarounds from WRITE to READ (Count)
board.memory.mem_ctrl.numStayReadState 301053 # Number of times bus staying in READ state (Count)
board.memory.mem_ctrl.numStayWriteState 61601 # Number of times bus staying in WRITE state (Count)
board.memory.mem_ctrl.readReqs 132815 # Number of read requests accepted (Count)
board.memory.mem_ctrl.writeReqs 65695 # Number of write requests accepted (Count)
board.memory.mem_ctrl.readBursts 132815 # Number of controller read bursts, including those serviced by the write queue (Count)
board.memory.mem_ctrl.writeBursts 65695 # Number of controller write bursts, including those merged in the write queue (Count)
board.memory.mem_ctrl.servicedByWrQ 21 # Number of controller read bursts serviced by the write queue (Count)
board.memory.mem_ctrl.mergedWrBursts 0 # Number of controller write bursts merged with an existing one (Count)
board.memory.mem_ctrl.neitherReadNorWriteReqs 0 # Number of requests that are neither read nor write (Count)
board.memory.mem_ctrl.avgRdQLen 1.80 # Average read queue length when enqueuing ((Count/Tick))
board.memory.mem_ctrl.avgWrQLen 21.49 # Average write queue length when enqueuing ((Count/Tick))
board.memory.mem_ctrl.numRdRetry 0 # Number of times read queue was full causing retry (Count)
board.memory.mem_ctrl.numWrRetry 0 # Number of times write queue was full causing retry (Count)
board.memory.mem_ctrl.readPktSize::0 0 # Read request sizes (log2) (Count)
board.memory.mem_ctrl.readPktSize::1 0 # Read request sizes (log2) (Count)
board.memory.mem_ctrl.readPktSize::2 0 # Read request sizes (log2) (Count)
board.memory.mem_ctrl.readPktSize::3 0 # Read request sizes (log2) (Count)
board.memory.mem_ctrl.readPktSize::4 0 # Read request sizes (log2) (Count)
board.memory.mem_ctrl.readPktSize::5 0 # Read request sizes (log2) (Count)
board.memory.mem_ctrl.readPktSize::6 132815 # Read request sizes (log2) (Count)
board.memory.mem_ctrl.writePktSize::0 0 # Write request sizes (log2) (Count)
board.memory.mem_ctrl.writePktSize::1 0 # Write request sizes (log2) (Count)
board.memory.mem_ctrl.writePktSize::2 0 # Write request sizes (log2) (Count)
board.memory.mem_ctrl.writePktSize::3 0 # Write request sizes (log2) (Count)
board.memory.mem_ctrl.writePktSize::4 0 # Write request sizes (log2) (Count)
board.memory.mem_ctrl.writePktSize::5 0 # Write request sizes (log2) (Count)
board.memory.mem_ctrl.writePktSize::6 65695 # Write request sizes (log2) (Count)
board.memory.mem_ctrl.rdQLenPdf::0 91781 # What read queue length does an incoming req see (Count)
board.memory.mem_ctrl.rdQLenPdf::1 17277 # What read queue length does an incoming req see (Count)
board.memory.mem_ctrl.rdQLenPdf::2 7705 # What read queue length does an incoming req see (Count)
board.memory.mem_ctrl.rdQLenPdf::3 7752 # What read queue length does an incoming req see (Count)
board.memory.mem_ctrl.rdQLenPdf::4 5817 # What read queue length does an incoming req see (Count)
board.memory.mem_ctrl.rdQLenPdf::5 2460 # What read queue length does an incoming req see (Count)
board.memory.mem_ctrl.rdQLenPdf::6 1 # What read queue length does an incoming req see (Count)
board.memory.mem_ctrl.rdQLenPdf::7 1 # What read queue length does an incoming req see (Count)
board.memory.mem_ctrl.rdQLenPdf::8 0 # What read queue length does an incoming req see (Count)
board.memory.mem_ctrl.rdQLenPdf::9 0 # What read queue length does an incoming req see (Count)
board.memory.mem_ctrl.rdQLenPdf::10 0 # What read queue length does an incoming req see (Count)
board.memory.mem_ctrl.rdQLenPdf::11 0 # What read queue length does an incoming req see (Count)
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board.memory.mem_ctrl.rdQLenPdf::12	0	# What read queue length does an incoming req see (Count)
board.memory.mem_ctrl.rdQLenPdf::13	0	# What read queue length does an incoming req see (Count)
board.memory.mem_ctrl.rdQLenPdf::14	0	# What read queue length does an incoming req see (Count)
board.memory.mem_ctrl.rdQLenPdf::15	0	# What read queue length does an incoming req see (Count)
board.memory.mem_ctrl.rdQLenPdf::16	0	# What read queue length does an incoming req see (Count)
board.memory.mem_ctrl.rdQLenPdf::17	0	# What read queue length does an incoming req see (Count)
board.memory.mem_ctrl.rdQLenPdf::18	0	# What read queue length does an incoming req see (Count)
board.memory.mem_ctrl.rdQLenPdf::19	0	# What read queue length does an incoming req see (Count)
board.memory.mem_ctrl.rdQLenPdf::20	0	# What read queue length does an incoming req see (Count)
board.memory.mem_ctrl.rdQLenPdf::21	0	# What read queue length does an incoming req see (Count)
board.memory.mem_ctrl.rdQLenPdf::22	0	# What read queue length does an incoming req see (Count)
board.memory.mem_ctrl.rdQLenPdf::23	0	# What read queue length does an incoming req see (Count)
board.memory.mem_ctrl.rdQLenPdf::24	0	# What read queue length does an incoming req see (Count)
board.memory.mem_ctrl.rdQLenPdf::25	0	# What read queue length does an incoming req see (Count)
board.memory.mem_ctrl.rdQLenPdf::26	0	# What read queue length does an incoming req see (Count)
board.memory.mem_ctrl.rdQLenPdf::27	0	# What read queue length does an incoming req see (Count)
board.memory.mem_ctrl.rdQLenPdf::28	0	# What read queue length does an incoming req see (Count)
board.memory.mem_ctrl.rdQLenPdf::29	0	# What read queue length does an incoming req see (Count)
board.memory.mem_ctrl.rdQLenPdf::30	0	# What read queue length does an incoming req see (Count)
board.memory.mem_ctrl.rdQLenPdf::31	0	# What read queue length does an incoming req see (Count)
board.memory.mem_ctrl.wrQLenPdf::0	1	# What write queue length does an incoming req see (Count)
board.memory.mem_ctrl.wrQLenPdf::1	1	# What write queue length does an incoming req see (Count)
board.memory.mem_ctrl.wrQLenPdf::2	1	# What write queue length does an incoming req see (Count)
board.memory.mem_ctrl.wrQLenPdf::3	1	# What write queue length does an incoming req see (Count)
board.memory.mem_ctrl.wrQLenPdf::4	1	# What write queue length does an incoming req see (Count)
board.memory.mem_ctrl.wrQLenPdf::5	1	# What write queue length does an incoming req see (Count)
board.memory.mem_ctrl.wrQLenPdf::6	1	# What write queue length does an incoming req see (Count)
board.memory.mem_ctrl.wrQLenPdf::7	1	# What write queue length does an incoming req see (Count)
board.memory.mem_ctrl.wrQLenPdf::8	1	# What write queue length does an incoming req see (Count)
board.memory.mem_ctrl.wrQLenPdf::9	1	# What write queue length does an incoming req see (Count)
board.memory.mem_ctrl.wrQLenPdf::10	1	# What write queue length does an incoming req see (Count)
board.memory.mem_ctrl.wrQLenPdf::11	1	# What write queue length does an incoming req see (Count)
board.memory.mem_ctrl.wrQLenPdf::12	1	# What write queue length does an incoming req see (Count)
board.memory.mem_ctrl.wrQLenPdf::13	1	# What write queue length does an incoming req see (Count)
board.memory.mem_ctrl.wrQLenPdf::14	1	# What write queue length does an incoming req see (Count)
board.memory.mem_ctrl.wrQLenPdf::15	3	# What write queue length does an incoming req see (Count)
board.memory.mem_ctrl.wrQLenPdf::16	4	# What write queue length does an incoming req see (Count)
board.memory.mem_ctrl.wrQLenPdf::17	161	# What write queue length does an incoming req see (Count)
board.memory.mem_ctrl.wrQLenPdf::18	731	# What write queue length does an incoming req see (Count)
board.memory.mem_ctrl.wrQLenPdf::19	2974	# What write queue length does an incoming req see (Count)
board.memory.mem_ctrl.wrQLenPdf::20	4683	# What write queue length does an incoming req see (Count)
board.memory.mem_ctrl.wrQLenPdf::21	4657	# What write queue length does an incoming req see (Count)
board.memory.mem_ctrl.wrQLenPdf::22	4816	# What write queue length does an incoming req see (Count)
board.memory.mem_ctrl.wrQLenPdf::23	4222	# What write queue length does an incoming req see (Count)
board.memory.mem_ctrl.wrQLenPdf::24	5537	# What write queue length does an incoming req see (Count)
board.memory.mem_ctrl.wrQLenPdf::25	5677	# What write queue length does an incoming req see (Count)
board.memory.mem_ctrl.wrQLenPdf::26	4629	# What write queue length does an incoming req see (Count)
board.memory.mem_ctrl.wrQLenPdf::27	5255	# What write queue length does an incoming req see (Count)
board.memory.mem_ctrl.wrQLenPdf::28	4106	# What write queue length does an incoming req see (Count)

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board.memory.mem_ctrl.wrQLenPdf::29          4208                # What write queue length does an incoming req see (Count)
board.memory.mem_ctrl.wrQLenPdf::30          4516                # What write queue length does an incoming req see (Count)
board.memory.mem_ctrl.wrQLenPdf::31          4105                # What write queue length does an incoming req see (Count)
board.memory.mem_ctrl.wrQLenPdf::32          5392                # What write queue length does an incoming req see (Count)
board.memory.mem_ctrl.wrQLenPdf::33          4                   # What write queue length does an incoming req see (Count)
board.memory.mem_ctrl.wrQLenPdf::34          0                   # What write queue length does an incoming req see (Count)
board.memory.mem_ctrl.wrQLenPdf::35          0                   # What write queue length does an incoming req see (Count)
board.memory.mem_ctrl.wrQLenPdf::36          0                   # What write queue length does an incoming req see (Count)
board.memory.mem_ctrl.wrQLenPdf::37          0                   # What write queue length does an incoming req see (Count)
board.memory.mem_ctrl.wrQLenPdf::38          0                   # What write queue length does an incoming req see (Count)
board.memory.mem_ctrl.wrQLenPdf::39          0                   # What write queue length does an incoming req see (Count)
board.memory.mem_ctrl.wrQLenPdf::40          0                   # What write queue length does an incoming req see (Count)
board.memory.mem_ctrl.wrQLenPdf::41          0                   # What write queue length does an incoming req see (Count)
board.memory.mem_ctrl.wrQLenPdf::42          0                   # What write queue length does an incoming req see (Count)
board.memory.mem_ctrl.wrQLenPdf::43          0                   # What write queue length does an incoming req see (Count)
board.memory.mem_ctrl.wrQLenPdf::44          0                   # What write queue length does an incoming req see (Count)
board.memory.mem_ctrl.wrQLenPdf::45          0                   # What write queue length does an incoming req see (Count)
board.memory.mem_ctrl.wrQLenPdf::46          0                   # What write queue length does an incoming req see (Count)
board.memory.mem_ctrl.wrQLenPdf::47          0                   # What write queue length does an incoming req see (Count)
board.memory.mem_ctrl.wrQLenPdf::48          0                   # What write queue length does an incoming req see (Count)
board.memory.mem_ctrl.wrQLenPdf::49          0                   # What write queue length does an incoming req see (Count)
board.memory.mem_ctrl.wrQLenPdf::50          0                   # What write queue length does an incoming req see (Count)
board.memory.mem_ctrl.wrQLenPdf::51          0                   # What write queue length does an incoming req see (Count)
board.memory.mem_ctrl.wrQLenPdf::52          0                   # What write queue length does an incoming req see (Count)
board.memory.mem_ctrl.wrQLenPdf::53          0                   # What write queue length does an incoming req see (Count)
board.memory.mem_ctrl.wrQLenPdf::54          0                   # What write queue length does an incoming req see (Count)
board.memory.mem_ctrl.wrQLenPdf::55          0                   # What write queue length does an incoming req see (Count)
board.memory.mem_ctrl.wrQLenPdf::56          0                   # What write queue length does an incoming req see (Count)
board.memory.mem_ctrl.wrQLenPdf::57          0                   # What write queue length does an incoming req see (Count)
board.memory.mem_ctrl.wrQLenPdf::58          0                   # What write queue length does an incoming req see (Count)
board.memory.mem_ctrl.wrQLenPdf::59          0                   # What write queue length does an incoming req see (Count)
board.memory.mem_ctrl.wrQLenPdf::60          0                   # What write queue length does an incoming req see (Count)
board.memory.mem_ctrl.wrQLenPdf::61          0                   # What write queue length does an incoming req see (Count)
board.memory.mem_ctrl.wrQLenPdf::62          0                   # What write queue length does an incoming req see (Count)
board.memory.mem_ctrl.wrQLenPdf::63          0                   # What write queue length does an incoming req see (Count)
board.memory.mem_ctrl.rdPerTurnAround::samples        4104                # Reads before turning the bus around for writes (Count)
board.memory.mem_ctrl.rdPerTurnAround::mean    17.316277                # Reads before turning the bus around for writes (Count)
board.memory.mem_ctrl.rdPerTurnAround::gmean    15.993845               # Reads before turning the bus around for writes (Count)
board.memory.mem_ctrl.rdPerTurnAround::stdev    71.924937               # Reads before turning the bus around for writes (Count)
board.memory.mem_ctrl.rdPerTurnAround::0-255        4102     99.95%     99.95% # Reads before turning the bus around for writes (Count)
board.memory.mem_ctrl.rdPerTurnAround::256-511          1      0.02%     99.98% # Reads before turning the bus around for writes (Count)
board.memory.mem_ctrl.rdPerTurnAround::4352-4607          1      0.02%    100.00% # Reads before turning the bus around for writes (Count)
board.memory.mem_ctrl.rdPerTurnAround::total        4104                # Reads before turning the bus around for writes (Count)
board.memory.mem_ctrl.wrPerTurnAround::samples        4104                # Writes before turning the bus around for reads (Count)
board.memory.mem_ctrl.wrPerTurnAround::mean    16.002437                # Writes before turning the bus around for reads (Count)
board.memory.mem_ctrl.wrPerTurnAround::gmean    16.002248               # Writes before turning the bus around for reads (Count)
board.memory.mem_ctrl.wrPerTurnAround::stdev     0.082573               # Writes before turning the bus around for reads (Count)
board.memory.mem_ctrl.wrPerTurnAround::16        4100     99.90%     99.90% # Writes before turning the bus around for reads (Count)
board.memory.mem_ctrl.wrPerTurnAround::18          3      0.07%     99.98% # Writes before turning the bus around for reads (Count)
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board.memory.mem_ctrl.wrPerTurnAround::20          1      0.02%    100.00% # Writes before turning the bus around for reads (Count)
board.memory.mem_ctrl.wrPerTurnAround::total        4104                    # Writes before turning the bus around for reads (Count)
board.memory.mem_ctrl.bytesReadWrQ              1344                    # Total number of bytes read from write queue (Byte)
board.memory.mem_ctrl.bytesReadSys           8500160                    # Total read bytes from the system interface side (Byte)
board.memory.mem_ctrl.bytesWrittenSys        4204480                    # Total written bytes from the system interface side (Byte)
board.memory.mem_ctrl.avgRdBWSys        1446899520.31749511                    # Average system read bandwidth in Byte/s ((Byte/Second))
board.memory.mem_ctrl.avgWrBWSys        715687715.89999497                    # Average system write bandwidth in Byte/s ((Byte/Second))
board.memory.mem_ctrl.totGap              5874589863                    # Total gap between requests (Tick)
board.memory.mem_ctrl.avgGap              29593.42                    # Average gap between requests ((Tick/Count))
board.memory.mem_ctrl.requestorReadBytes::cache_hierarchy.l1dcaches.prefetcher       132992                    # Per-requestor bytes read from memory (Byte)
board.memory.mem_ctrl.requestorReadBytes::cache_hierarchy.l2cache.prefetcher      8131840                    # Per-requestor bytes read from memory (Byte)
board.memory.mem_ctrl.requestorReadBytes::processor.cores.core.inst        48000                    # Per-requestor bytes read from memory (Byte)
board.memory.mem_ctrl.requestorReadBytes::processor.cores.core.data       185984                    # Per-requestor bytes read from memory (Byte)
board.memory.mem_ctrl.requestorWriteBytes::writebacks      4203136                    # Per-requestor bytes write to memory (Byte)
board.memory.mem_ctrl.requestorReadRate::cache_hierarchy.l1dcaches.prefetcher 22637933.992544174194                    # Per-requestor bytes read from memory rate ((Byte/Second))
board.memory.mem_ctrl.requestorReadRate::cache_hierarchy.l2cache.prefetcher 1384203990.901185274124                    # Per-requestor bytes read from memory rate ((Byte/Second))
board.memory.mem_ctrl.requestorReadRate::processor.cores.core.inst 8170572.903950015083                    # Per-requestor bytes read from memory rate ((Byte/Second))
board.memory.mem_ctrl.requestorReadRate::processor.cores.core.data 31658246.478504993021                    # Per-requestor bytes read from memory rate ((Byte/Second))
board.memory.mem_ctrl.requestorWriteRate::writebacks 715458939.858684420586                    # Per-requestor bytes write to memory rate ((Byte/Second))
board.memory.mem_ctrl.requestorReadAccesses::cache_hierarchy.l1dcaches.prefetcher         2078                    # Per-requestor read serviced memory accesses (Count)
board.memory.mem_ctrl.requestorReadAccesses::cache_hierarchy.l2cache.prefetcher       127081                    # Per-requestor read serviced memory accesses (Count)
board.memory.mem_ctrl.requestorReadAccesses::processor.cores.core.inst          750                    # Per-requestor read serviced memory accesses (Count)
board.memory.mem_ctrl.requestorReadAccesses::processor.cores.core.data         2906                    # Per-requestor read serviced memory accesses (Count)
board.memory.mem_ctrl.requestorWriteAccesses::writebacks        65695                    # Per-requestor write serviced memory accesses (Count)
board.memory.mem_ctrl.requestorReadTotalLat::cache_hierarchy.l1dcaches.prefetcher     57125030                    # Per-requestor read total memory access latency (Tick)
board.memory.mem_ctrl.requestorReadTotalLat::cache_hierarchy.l2cache.prefetcher   5062388219                    # Per-requestor read total memory access latency (Tick)
board.memory.mem_ctrl.requestorReadTotalLat::processor.cores.core.inst     22336514                    # Per-requestor read total memory access latency (Tick)
board.memory.mem_ctrl.requestorReadTotalLat::processor.cores.core.data     70116991                    # Per-requestor read total memory access latency (Tick)
board.memory.mem_ctrl.requestorWriteTotalLat::writebacks  68449792290                    # Per-requestor write total memory access latency (Tick)
board.memory.mem_ctrl.requestorReadAvgLat::cache_hierarchy.l1dcaches.prefetcher     27490.39                    # Per-requestor read average memory access latency ((Tick/Count))
board.memory.mem_ctrl.requestorReadAvgLat::cache_hierarchy.l2cache.prefetcher     39835.92                    # Per-requestor read average memory access latency ((Tick/Count))
board.memory.mem_ctrl.requestorReadAvgLat::processor.cores.core.inst     29782.02                    # Per-requestor read average memory access latency ((Tick/Count))
board.memory.mem_ctrl.requestorReadAvgLat::processor.cores.core.data     24128.35                    # Per-requestor read average memory access latency ((Tick/Count))
board.memory.mem_ctrl.requestorWriteAvgLat::writebacks   1041933.06                    # Per-requestor write average memory access latency ((Tick/Count))
board.memory.mem_ctrl.dram.bytesRead::cache_hierarchy.l1dcaches.prefetcher       132992                    # Number of bytes read from this memory (Byte)
board.memory.mem_ctrl.dram.bytesRead::cache_hierarchy.l2cache.prefetcher      8133184                    # Number of bytes read from this memory (Byte)
board.memory.mem_ctrl.dram.bytesRead::processor.cores.core.inst        48000                    # Number of bytes read from this memory (Byte)
board.memory.mem_ctrl.dram.bytesRead::processor.cores.core.data       185984                    # Number of bytes read from this memory (Byte)
board.memory.mem_ctrl.dram.bytesRead::total      8500160                    # Number of bytes read from this memory (Byte)
board.memory.mem_ctrl.dram.bytesInstRead::processor.cores.core.inst        48000                    # Number of instructions bytes read from this memory (Byte)
board.memory.mem_ctrl.dram.bytesInstRead::total        48000                    # Number of instructions bytes read from this memory (Byte)
board.memory.mem_ctrl.dram.bytesWritten::writebacks      4204480                    # Number of bytes written to this memory (Byte)
board.memory.mem_ctrl.dram.bytesWritten::total      4204480                    # Number of bytes written to this memory (Byte)
board.memory.mem_ctrl.dram.numReads::cache_hierarchy.l1dcaches.prefetcher         2078                    # Number of read requests responded to by this memory (Count)
board.memory.mem_ctrl.dram.numReads::cache_hierarchy.l2cache.prefetcher       127081                    # Number of read requests responded to by this memory (Count)
board.memory.mem_ctrl.dram.numReads::processor.cores.core.inst          750                    # Number of read requests responded to by this memory (Count)
board.memory.mem_ctrl.dram.numReads::processor.cores.core.data         2906                    # Number of read requests responded to by this memory (Count)
board.memory.mem_ctrl.dram.numReads::total       132815                    # Number of read requests responded to by this memory (Count)
board.memory.mem_ctrl.dram.numWrites::writebacks        65695                    # Number of write requests responded to by this memory (Count)
```

```
board.memory.mem_ctrl.dram.numWrites::total        65695                    # Number of write requests responded to by this memory (Count)
board.memory.mem_ctrl.dram.bwRead::cache_hierarchy.l1dcaches.prefetcher     22637934                    # Total read bandwidth from this memory ((Byte/Second))
board.memory.mem_ctrl.dram.bwRead::cache_hierarchy.l2cache.prefetcher   1384432767                    # Total read bandwidth from this memory ((Byte/Second))
board.memory.mem_ctrl.dram.bwRead::processor.cores.core.inst      8170573                    # Total read bandwidth from this memory ((Byte/Second))
board.memory.mem_ctrl.dram.bwRead::processor.cores.core.data     31658246                    # Total read bandwidth from this memory ((Byte/Second))
board.memory.mem_ctrl.dram.bwRead::total   1446899520                    # Total read bandwidth from this memory ((Byte/Second))
board.memory.mem_ctrl.dram.bwInstRead::processor.cores.core.inst      8170573                    # Instruction read bandwidth from this memory ((Byte/Second))
board.memory.mem_ctrl.dram.bwInstRead::total      8170573                    # Instruction read bandwidth from this memory ((Byte/Second))
board.memory.mem_ctrl.dram.bwWrite::writebacks    715687716                    # Write bandwidth from this memory ((Byte/Second))
board.memory.mem_ctrl.dram.bwWrite::total    715687716                    # Write bandwidth from this memory ((Byte/Second))
board.memory.mem_ctrl.dram.bwTotal::writebacks    715687716                    # Total bandwidth to/from this memory ((Byte/Second))
board.memory.mem_ctrl.dram.bwTotal::cache_hierarchy.l1dcaches.prefetcher     22637934                    # Total bandwidth to/from this memory ((Byte/Second))
board.memory.mem_ctrl.dram.bwTotal::cache_hierarchy.l2cache.prefetcher   1384432767                    # Total bandwidth to/from this memory ((Byte/Second))
board.memory.mem_ctrl.dram.bwTotal::processor.cores.core.inst      8170573                    # Total bandwidth to/from this memory ((Byte/Second))
board.memory.mem_ctrl.dram.bwTotal::processor.cores.core.data     31658246                    # Total bandwidth to/from this memory ((Byte/Second))
board.memory.mem_ctrl.dram.bwTotal::total   2162587236                    # Total bandwidth to/from this memory ((Byte/Second))
board.memory.mem_ctrl.dram.readBursts          132794                    # Number of DRAM read bursts (Count)
board.memory.mem_ctrl.dram.writeBursts          65674                    # Number of DRAM write bursts (Count)
board.memory.mem_ctrl.dram.perBankRdBursts::0        8468                    # Per bank write bursts (Count)
board.memory.mem_ctrl.dram.perBankRdBursts::1        8437                    # Per bank write bursts (Count)
board.memory.mem_ctrl.dram.perBankRdBursts::2        8291                    # Per bank write bursts (Count)
board.memory.mem_ctrl.dram.perBankRdBursts::3        8268                    # Per bank write bursts (Count)
board.memory.mem_ctrl.dram.perBankRdBursts::4        8267                    # Per bank write bursts (Count)
board.memory.mem_ctrl.dram.perBankRdBursts::5        8312                    # Per bank write bursts (Count)
board.memory.mem_ctrl.dram.perBankRdBursts::6        8254                    # Per bank write bursts (Count)
board.memory.mem_ctrl.dram.perBankRdBursts::7        8225                    # Per bank write bursts (Count)
board.memory.mem_ctrl.dram.perBankRdBursts::8        8274                    # Per bank write bursts (Count)
board.memory.mem_ctrl.dram.perBankRdBursts::9        8236                    # Per bank write bursts (Count)
board.memory.mem_ctrl.dram.perBankRdBursts::10        8235                    # Per bank write bursts (Count)
board.memory.mem_ctrl.dram.perBankRdBursts::11        8235                    # Per bank write bursts (Count)
board.memory.mem_ctrl.dram.perBankRdBursts::12        8235                    # Per bank write bursts (Count)
board.memory.mem_ctrl.dram.perBankRdBursts::13        8294                    # Per bank write bursts (Count)
board.memory.mem_ctrl.dram.perBankRdBursts::14        8357                    # Per bank write bursts (Count)
board.memory.mem_ctrl.dram.perBankRdBursts::15        8406                    # Per bank write bursts (Count)
board.memory.mem_ctrl.dram.perBankWrBursts::0        4096                    # Per bank write bursts (Count)
board.memory.mem_ctrl.dram.perBankWrBursts::1        4096                    # Per bank write bursts (Count)
board.memory.mem_ctrl.dram.perBankWrBursts::2        4096                    # Per bank write bursts (Count)
board.memory.mem_ctrl.dram.perBankWrBursts::3        4104                    # Per bank write bursts (Count)
board.memory.mem_ctrl.dram.perBankWrBursts::4        4102                    # Per bank write bursts (Count)
board.memory.mem_ctrl.dram.perBankWrBursts::5        4157                    # Per bank write bursts (Count)
board.memory.mem_ctrl.dram.perBankWrBursts::6        4098                    # Per bank write bursts (Count)
board.memory.mem_ctrl.dram.perBankWrBursts::7        4096                    # Per bank write bursts (Count)
board.memory.mem_ctrl.dram.perBankWrBursts::8        4116                    # Per bank write bursts (Count)
board.memory.mem_ctrl.dram.perBankWrBursts::9        4106                    # Per bank write bursts (Count)
board.memory.mem_ctrl.dram.perBankWrBursts::10        4117                    # Per bank write bursts (Count)
board.memory.mem_ctrl.dram.perBankWrBursts::11        4106                    # Per bank write bursts (Count)
board.memory.mem_ctrl.dram.perBankWrBursts::12        4096                    # Per bank write bursts (Count)
board.memory.mem_ctrl.dram.perBankWrBursts::13        4096                    # Per bank write bursts (Count)
board.memory.mem_ctrl.dram.perBankWrBursts::14        4096                    # Per bank write bursts (Count)
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board.memory.mem_ctrl.dram.perBankWrBursts::15 4096 # Per bank write bursts (Count)
board.memory.mem_ctrl.dram.totQLat 2722079254 # Total ticks spent queuing (Tick)
board.memory.mem_ctrl.dram.totBusLat 663970000 # Total ticks spent in databus transfers (Tick)
board.memory.mem_ctrl.dram.totMemAccLat 5211966754 # Total ticks spent from burst creation until serviced by the DRAM (Tick)
board.memory.mem_ctrl.dram.avgQLat 20498.51 # Average queueing delay per DRAM burst ((Tick/Count))
board.memory.mem_ctrl.dram.avgBusLat 5000.00 # Average bus latency per DRAM burst ((Tick/Count))
board.memory.mem_ctrl.dram.avgMemAccLat 39248.51 # Average memory access latency per DRAM burst ((Tick/Count))
board.memory.mem_ctrl.dram.readRowHits 122740 # Number of row buffer hits during reads (Count)
board.memory.mem_ctrl.dram.writeRowHits 61006 # Number of row buffer hits during writes (Count)
board.memory.mem_ctrl.dram.readRowHitRate 92.43 # Row buffer hit rate for reads (Ratio)
board.memory.mem_ctrl.dram.writeRowHitRate 92.89 # Row buffer hit rate for writes (Ratio)
board.memory.mem_ctrl.dram.bytesPerActivate::samples 14713 # Bytes accessed per row activation (Byte)
board.memory.mem_ctrl.dram.bytesPerActivate::mean 863.193095 # Bytes accessed per row activation (Byte)
board.memory.mem_ctrl.dram.bytesPerActivate::gmean 772.367999 # Bytes accessed per row activation (Byte)
board.memory.mem_ctrl.dram.bytesPerActivate::stdev 280.356258 # Bytes accessed per row activation (Byte)
board.memory.mem_ctrl.dram.bytesPerActivate::0-127 222 1.51% 1.51% # Bytes accessed per row activation (Byte)
board.memory.mem_ctrl.dram.bytesPerActivate::128-255 692 4.70% 6.21% # Bytes accessed per row activation (Byte)
board.memory.mem_ctrl.dram.bytesPerActivate::256-383 581 3.95% 10.16% # Bytes accessed per row activation (Byte)
board.memory.mem_ctrl.dram.bytesPerActivate::384-511 283 1.92% 12.08% # Bytes accessed per row activation (Byte)
board.memory.mem_ctrl.dram.bytesPerActivate::512-639 1150 7.82% 19.90% # Bytes accessed per row activation (Byte)
board.memory.mem_ctrl.dram.bytesPerActivate::640-767 668 4.54% 24.44% # Bytes accessed per row activation (Byte)
board.memory.mem_ctrl.dram.bytesPerActivate::768-895 313 2.13% 26.57% # Bytes accessed per row activation (Byte)
board.memory.mem_ctrl.dram.bytesPerActivate::896-1023 781 5.31% 31.88% # Bytes accessed per row activation (Byte)
board.memory.mem_ctrl.dram.bytesPerActivate::1024-1151 10023 68.12% 100.00% # Bytes accessed per row activation (Byte)
board.memory.mem_ctrl.dram.bytesPerActivate::total 14713 # Bytes accessed per row activation (Byte)
board.memory.mem_ctrl.dram.dramBytesRead 8498816 # Total bytes read (Byte)
board.memory.mem_ctrl.dram.dramBytesWritten 4203136 # Total bytes written (Byte)
board.memory.mem_ctrl.dram.avgRdBW 1446.670744 # Average DRAM read bandwidth in MiBytes/s ((Byte/Second))
board.memory.mem_ctrl.dram.avgWrBW 715.458940 # Average DRAM write bandwidth in MiBytes/s ((Byte/Second))
board.memory.mem_ctrl.dram.peakBW 12800.00 # Theoretical peak bandwidth in MiByte/s ((Byte/Second))
board.memory.mem_ctrl.dram.busUtil 16.89 # Data bus utilization in percentage (Ratio)
board.memory.mem_ctrl.dram.busUtilRead 11.30 # Data bus utilization in percentage for reads (Ratio)
board.memory.mem_ctrl.dram.busUtilWrite 5.59 # Data bus utilization in percentage for writes (Ratio)
board.memory.mem_ctrl.dram.pageHitRate 92.58 # Row buffer hit rate, read and write combined (Ratio)
board.memory.mem_ctrl.dram.power_state.pwrStateResidencyTicks::UNDEFINED 5874741045 # Cumulative time (in ticks) in various power states (Tick)
board.memory.mem_ctrl.dram.rank0.actEnergy 52736040 # Energy for activate commands per rank (pJ) (Joule)
board.memory.mem_ctrl.dram.rank0.preEnergy 28010895 # Energy for precharge commands per rank (pJ) (Joule)
board.memory.mem_ctrl.dram.rank0.readEnergy 474967080 # Energy for read commands per rank (pJ) (Joule)
board.memory.mem_ctrl.dram.rank0.writeEnergy 171450900 # Energy for write commands per rank (pJ) (Joule)
board.memory.mem_ctrl.dram.rank0.refreshEnergy 463438560.000000 # Energy for refresh commands per rank (pJ) (Joule)
board.memory.mem_ctrl.dram.rank0.actBackEnergy 1484038890 # Energy for active background per rank (pJ) (Joule)
board.memory.mem_ctrl.dram.rank0.preBackEnergy 1006183680 # Energy for precharge background per rank (pJ) (Joule)
board.memory.mem_ctrl.dram.rank0.actPowerDownEnergy 0 # Energy for active power-down per rank (pJ) (Joule)
board.memory.mem_ctrl.dram.rank0.prePowerDownEnergy 0 # Energy for precharge power-down per rank (pJ) (Joule)
board.memory.mem_ctrl.dram.rank0.selfRefreshEnergy 0 # Energy for self refresh per rank (pJ) (Joule)
board.memory.mem_ctrl.dram.rank0.totalEnergy 3680826045 # Total energy per rank (pJ) (Joule)
board.memory.mem_ctrl.dram.rank0.averagePower 626.551199 # Core power per rank (mW) (Watt)
board.memory.mem_ctrl.dram.rank0.totalIdleTime 0 # Total Idle time Per DRAM Rank (Tick)
board.memory.mem_ctrl.dram.rank0.pwrStateTime::IDLE 2573361235 # Time in different power states (Tick)
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board.memory.mem_ctrl.dram.rank0.pwrStateTime::REF    196040000                    # Time in different power states (Tick)
board.memory.mem_ctrl.dram.rank0.pwrStateTime::SREF           0                    # Time in different power states (Tick)
board.memory.mem_ctrl.dram.rank0.pwrStateTime::PRE_PDN           0                    # Time in different power states (Tick)
board.memory.mem_ctrl.dram.rank0.pwrStateTime::ACT   3105339810                    # Time in different power states (Tick)
board.memory.mem_ctrl.dram.rank0.pwrStateTime::ACT_PDN           0                    # Time in different power states (Tick)
board.memory.mem_ctrl.dram.rank1.actEnergy     52379040                    # Energy for activate commands per rank (pJ) (Joule)
board.memory.mem_ctrl.dram.rank1.preEnergy     27824940                    # Energy for precharge commands per rank (pJ) (Joule)
board.memory.mem_ctrl.dram.rank1.readEnergy    473182080                    # Energy for read commands per rank (pJ) (Joule)
board.memory.mem_ctrl.dram.rank1.writeEnergy    171367380                    # Energy for write commands per rank (pJ) (Joule)
board.memory.mem_ctrl.dram.rank1.refreshEnergy 463438560.000000                    # Energy for refresh commands per rank (pJ) (Joule)
board.memory.mem_ctrl.dram.rank1.actBackEnergy   1478823390                    # Energy for active background per rank (pJ) (Joule)
board.memory.mem_ctrl.dram.rank1.preBackEnergy   1010575680                    # Energy for precharge background per rank (pJ) (Joule)
board.memory.mem_ctrl.dram.rank1.actPowerDownEnergy           0                    # Energy for active power-down per rank (pJ) (Joule)
board.memory.mem_ctrl.dram.rank1.prePowerDownEnergy           0                    # Energy for precharge power-down per rank (pJ) (Joule)
board.memory.mem_ctrl.dram.rank1.selfRefreshEnergy           0                    # Energy for self refresh per rank (pJ) (Joule)
board.memory.mem_ctrl.dram.rank1.totalEnergy   3677591070                    # Total energy per rank (pJ) (Joule)
board.memory.mem_ctrl.dram.rank1.averagePower   626.000541                    # Core power per rank (mW) (Watt)
board.memory.mem_ctrl.dram.rank1.totalIdleTime           0                    # Total Idle time Per DRAM Rank (Tick)
board.memory.mem_ctrl.dram.rank1.pwrStateTime::IDLE   2584896654                    # Time in different power states (Tick)
board.memory.mem_ctrl.dram.rank1.pwrStateTime::REF    196040000                    # Time in different power states (Tick)
board.memory.mem_ctrl.dram.rank1.pwrStateTime::SREF           0                    # Time in different power states (Tick)
board.memory.mem_ctrl.dram.rank1.pwrStateTime::PRE_PDN           0                    # Time in different power states (Tick)
board.memory.mem_ctrl.dram.rank1.pwrStateTime::ACT   3093804391                    # Time in different power states (Tick)
board.memory.mem_ctrl.dram.rank1.pwrStateTime::ACT_PDN           0                    # Time in different power states (Tick)
board.memory.mem_ctrl.power_state.pwrStateResidencyTicks::UNDEFINED   5874741045                    # Cumulative time (in ticks) in various power states (Tick)
board.processor.cores.core.numCycles         17641866                    # Number of cpu cycles simulated (Cycle)
board.processor.cores.core.cpi              0.478992                    # CPI: cycles per instruction (core level) ((Cycle/Count))
board.processor.cores.core.ipc              2.087718                    # IPC: instructions per cycle (core level) ((Count/Cycle))
board.processor.cores.core.numWorkItemsStarted           0                    # Number of work items this cpu started (Count)
board.processor.cores.core.numWorkItemsCompleted           0                    # Number of work items this cpu completed (Count)
board.processor.cores.core.instsAdded        57074882                    # Number of instructions added to the IQ (excludes non-spec) (Count)
board.processor.cores.core.nonSpecInstsAdded          240                    # Number of non-speculative instructions added to the IQ (Count)
board.processor.cores.core.instsIssued       57028511                    # Number of instructions issued (Count)
board.processor.cores.core.squashedInstsIssued          489                    # Number of squashed instructions issued (Count)
board.processor.cores.core.squashedInstsExamined       213908                    # Number of squashed instructions iterated over during squash; mainly for profiling (Count)
board.processor.cores.core.squashedOperandsExamined       246620                    # Number of squashed operands that are examined and possibly removed from graph (Count)
board.processor.cores.core.squashedNonSpecRemoved          150                    # Number of squashed non-spec instructions that were removed (Count)
board.processor.cores.core.numIssuedDist::samples     17494407                    # Number of insts issued each cycle (Count)
board.processor.cores.core.numIssuedDist::mean     3.259814                    # Number of insts issued each cycle (Count)
board.processor.cores.core.numIssuedDist::stdev     2.155531                    # Number of insts issued each cycle (Count)
board.processor.cores.core.numIssuedDist::underflows           0      0.00%      0.00% # Number of insts issued each cycle (Count)
board.processor.cores.core.numIssuedDist::0      1510468      8.63%      8.63% # Number of insts issued each cycle (Count)
board.processor.cores.core.numIssuedDist::1      1825077     10.43%     19.07% # Number of insts issued each cycle (Count)
board.processor.cores.core.numIssuedDist::2      4651059     26.59%     45.65% # Number of insts issued each cycle (Count)
board.processor.cores.core.numIssuedDist::3      2975751     17.01%     62.66% # Number of insts issued each cycle (Count)
board.processor.cores.core.numIssuedDist::4      1555754      8.89%     71.55% # Number of insts issued each cycle (Count)
board.processor.cores.core.numIssuedDist::5      1689202      9.66%     81.21% # Number of insts issued each cycle (Count)
board.processor.cores.core.numIssuedDist::6      1177727      6.73%     87.94% # Number of insts issued each cycle (Count)
board.processor.cores.core.numIssuedDist::7      1636277      9.35%     97.30% # Number of insts issued each cycle (Count)
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board.processor.cores.core.numIssuedDist::8 473092 2.70% 100.00% # Number of insts issued each cycle (Count)

board.processor.cores.core.numIssuedDist::overflows 0 0.00% 100.00% # Number of insts issued each cycle (Count)

board.processor.cores.core.numIssuedDist::min_value 0 # Number of insts issued each cycle (Count)

board.processor.cores.core.numIssuedDist::max_value 8 # Number of insts issued each cycle (Count)

board.processor.cores.core.numIssuedDist::total 17494407 # Number of insts issued each cycle (Count)

board.processor.cores.core.statFuBusy::No_OpClass 0 0.00% 0.00% # attempts to use FU when none available (Count)

board.processor.cores.core.statFuBusy::IntAlu 1417524 99.97% 99.97% # attempts to use FU when none available (Count)

board.processor.cores.core.statFuBusy::IntMult 0 0.00% 99.97% # attempts to use FU when none available (Count)

board.processor.cores.core.statFuBusy::IntDiv 0 0.00% 99.97% # attempts to use FU when none available (Count)

board.processor.cores.core.statFuBusy::FloatAdd 0 0.00% 99.97% # attempts to use FU when none available (Count)

board.processor.cores.core.statFuBusy::FloatCmp 0 0.00% 99.97% # attempts to use FU when none available (Count)

board.processor.cores.core.statFuBusy::FloatCvt 0 0.00% 99.97% # attempts to use FU when none available (Count)

board.processor.cores.core.statFuBusy::FloatMult 0 0.00% 99.97% # attempts to use FU when none available (Count)

board.processor.cores.core.statFuBusy::FloatMultAcc 0 0.00% 99.97% # attempts to use FU when none available (Count)

board.processor.cores.core.statFuBusy::FloatDiv 0 0.00% 99.97% # attempts to use FU when none available (Count)

board.processor.cores.core.statFuBusy::FloatMisc 0 0.00% 99.97% # attempts to use FU when none available (Count)

board.processor.cores.core.statFuBusy::FloatSqrt 0 0.00% 99.97% # attempts to use FU when none available (Count)

board.processor.cores.core.statFuBusy::SimdAdd 0 0.00% 99.97% # attempts to use FU when none available (Count)

board.processor.cores.core.statFuBusy::SimdAddAcc 0 0.00% 99.97% # attempts to use FU when none available (Count)

board.processor.cores.core.statFuBusy::SimdAlu 24 0.00% 99.97% # attempts to use FU when none available (Count)

board.processor.cores.core.statFuBusy::SimdCmp 0 0.00% 99.97% # attempts to use FU when none available (Count)

board.processor.cores.core.statFuBusy::SimdCvt 0 0.00% 99.97% # attempts to use FU when none available (Count)

board.processor.cores.core.statFuBusy::SimdMisc 0 0.00% 99.97% # attempts to use FU when none available (Count)

board.processor.cores.core.statFuBusy::SimdMult 0 0.00% 99.97% # attempts to use FU when none available (Count)

board.processor.cores.core.statFuBusy::SimdMultAcc 0 0.00% 99.97% # attempts to use FU when none available (Count)

board.processor.cores.core.statFuBusy::SimdMatMultAcc 0 0.00% 99.97% # attempts to use FU when none available (Count)

board.processor.cores.core.statFuBusy::SimdShift 0 0.00% 99.97% # attempts to use FU when none available (Count)

board.processor.cores.core.statFuBusy::SimdShiftAcc 0 0.00% 99.97% # attempts to use FU when none available (Count)

board.processor.cores.core.statFuBusy::SimdDiv 0 0.00% 99.97% # attempts to use FU when none available (Count)

board.processor.cores.core.statFuBusy::SimdSqrt 0 0.00% 99.97% # attempts to use FU when none available (Count)

board.processor.cores.core.statFuBusy::SimdFloatAdd 0 0.00% 99.97% # attempts to use FU when none available (Count)

board.processor.cores.core.statFuBusy::SimdFloatAlu 0 0.00% 99.97% # attempts to use FU when none available (Count)

board.processor.cores.core.statFuBusy::SimdFloatCmp 0 0.00% 99.97% # attempts to use FU when none available (Count)

board.processor.cores.core.statFuBusy::SimdFloatCvt 0 0.00% 99.97% # attempts to use FU when none available (Count)

board.processor.cores.core.statFuBusy::SimdFloatDiv 0 0.00% 99.97% # attempts to use FU when none available (Count)

board.processor.cores.core.statFuBusy::SimdFloatMisc 0 0.00% 99.97% # attempts to use FU when none available (Count)

board.processor.cores.core.statFuBusy::SimdFloatMult 0 0.00% 99.97% # attempts to use FU when none available (Count)

board.processor.cores.core.statFuBusy::SimdFloatMultAcc 0 0.00% 99.97% # attempts to use FU when none available (Count)

board.processor.cores.core.statFuBusy::SimdFloatMatMultAcc 0 0.00% 99.97% # attempts to use FU when none available (Count)

board.processor.cores.core.statFuBusy::SimdFloatSqrt 0 0.00% 99.97% # attempts to use FU when none available (Count)

board.processor.cores.core.statFuBusy::SimdReduceAdd 0 0.00% 99.97% # attempts to use FU when none available (Count)

board.processor.cores.core.statFuBusy::SimdReduceAlu 0 0.00% 99.97% # attempts to use FU when none available (Count)

board.processor.cores.core.statFuBusy::SimdReduceCmp 0 0.00% 99.97% # attempts to use FU when none available (Count)

board.processor.cores.core.statFuBusy::SimdFloatReduceAdd 0 0.00% 99.97% # attempts to use FU when none available (Count)

board.processor.cores.core.statFuBusy::SimdFloatReduceCmp 0 0.00% 99.97% # attempts to use FU when none available (Count)

board.processor.cores.core.statFuBusy::SimdAes 0 0.00% 99.97% # attempts to use FU when none available (Count)

board.processor.cores.core.statFuBusy::SimdAesMix 0 0.00% 99.97% # attempts to use FU when none available (Count)

board.processor.cores.core.statFuBusy::SimdSha1Hash 0 0.00% 99.97% # attempts to use FU when none available (Count)

board.processor.cores.core.statFuBusy::SimdSha1Hash2 0 0.00% 99.97% # attempts to use FU when none available (Count)

board.processor.cores.core.statFuBusy::SimdSha256Hash 0 0.00% 99.97% # attempts to use FU when none available (Count)

board.processor.cores.core.statFuBusy::SimdSha256Hash2 0 0.00% 99.97% # attempts to use FU when none available (Count)

board.processor.cores.core.statFuBusy::SimdShaSigma2 0 0.00% 99.97% # attempts to use FU when none available (Count)

board.processor.cores.core.statFuBusy::SimdShaSigma3 0 0.00% 99.97% # attempts to use FU when none available (Count)

board.processor.cores.core.statFuBusy::SimdPredAlu 0 0.00% 99.97% # attempts to use FU when none available (Count)

board.processor.cores.core.statFuBusy::Matrix 0 0.00% 99.97% # attempts to use FU when none available (Count)

board.processor.cores.core.statFuBusy::MatrixMov 0 0.00% 99.97% # attempts to use FU when none available (Count)

board.processor.cores.core.statFuBusy::MatrixOP 0 0.00% 99.97% # attempts to use FU when none available (Count)

board.processor.cores.core.statFuBusy::MemRead 100 0.01% 99.98% # attempts to use FU when none available (Count)

board.processor.cores.core.statFuBusy::MemWrite 278 0.02% 100.00% # attempts to use FU when none available (Count)

board.processor.cores.core.statFuBusy::FloatMemRead 6 0.00% 100.00% # attempts to use FU when none available (Count)

board.processor.cores.core.statFuBusy::FloatMemWrite 34 0.00% 100.00% # attempts to use FU when none available (Count)

board.processor.cores.core.statFuBusy::IprAccess 0 0.00% 100.00% # attempts to use FU when none available (Count)

board.processor.cores.core.statFuBusy::InstPrefetch 0 0.00% 100.00% # attempts to use FU when none available (Count)

board.processor.cores.core.statFuBusy::SimdUnitStrideLoad 0 0.00% 100.00% # attempts to use FU when none available (Count)

board.processor.cores.core.statFuBusy::SimdUnitStrideStore 0 0.00% 100.00% # attempts to use FU when none available (Count)

board.processor.cores.core.statFuBusy::SimdUnitStrideMaskLoad 0 0.00% 100.00% # attempts to use FU when none available (Count)

board.processor.cores.core.statFuBusy::SimdUnitStrideMaskStore 0 0.00% 100.00% # attempts to use FU when none available (Count)

board.processor.cores.core.statFuBusy::SimdStridedLoad 0 0.00% 100.00% # attempts to use FU when none available (Count)

board.processor.cores.core.statFuBusy::SimdStridedStore 0 0.00% 100.00% # attempts to use FU when none available (Count)

board.processor.cores.core.statFuBusy::SimdIndexedLoad 0 0.00% 100.00% # attempts to use FU when none available (Count)

board.processor.cores.core.statFuBusy::SimdIndexedStore 0 0.00% 100.00% # attempts to use FU when none available (Count)

board.processor.cores.core.statFuBusy::SimdWholeRegisterLoad 0 0.00% 100.00% # attempts to use FU when none available (Count)

board.processor.cores.core.statFuBusy::SimdWholeRegisterStore 0 0.00% 100.00% # attempts to use FU when none available (Count)

board.processor.cores.core.statFuBusy::SimdUnitStrideFaultOnlyFirstLoad 0 0.00% 100.00% # attempts to use FU when none available (Count)

board.processor.cores.core.statFuBusy::SimdUnitStrideSegmentedLoad 0 0.00% 100.00% # attempts to use FU when none available (Count)

board.processor.cores.core.statFuBusy::SimdUnitStrideSegmentedStore 0 0.00% 100.00% # attempts to use FU when none available (Count)

board.processor.cores.core.statFuBusy::SimdExt 0 0.00% 100.00% # attempts to use FU when none available (Count)

board.processor.cores.core.statFuBusy::SimdFloatExt 0 0.00% 100.00% # attempts to use FU when none available (Count)

board.processor.cores.core.statFuBusy::SimdConfig 0 0.00% 100.00% # attempts to use FU when none available (Count)

board.processor.cores.core.statIssuedInstType_0::No_OpClass 750 0.00% 0.00% # Number of instructions issued per FU type, per thread (Count)

board.processor.cores.core.statIssuedInstType_0::IntAlu 40174934 70.45% 70.45% # Number of instructions issued per FU type, per thread (Count)

board.processor.cores.core.statIssuedInstType_0::IntMult 1053194 1.85% 72.30% # Number of instructions issued per FU type, per thread (Count)

board.processor.cores.core.statIssuedInstType_0::IntDiv 40 0.00% 72.30% # Number of instructions issued per FU type, per thread (Count)

board.processor.cores.core.statIssuedInstType_0::FloatAdd 192 0.00% 72.30% # Number of instructions issued per FU type, per thread (Count)

board.processor.cores.core.statIssuedInstType_0::FloatCmp 0 0.00% 72.30% # Number of instructions issued per FU type, per thread (Count)

board.processor.cores.core.statIssuedInstType_0::FloatCvt 0 0.00% 72.30% # Number of instructions issued per FU type, per thread (Count)

board.processor.cores.core.statIssuedInstType_0::FloatMult 0 0.00% 72.30% # Number of instructions issued per FU type, per thread (Count)

board.processor.cores.core.statIssuedInstType_0::FloatMultAcc 0 0.00% 72.30% # Number of instructions issued per FU type, per thread (Count)

board.processor.cores.core.statIssuedInstType_0::FloatDiv 0 0.00% 72.30% # Number of instructions issued per FU type, per thread (Count)

board.processor.cores.core.statIssuedInstType_0::FloatMisc 0 0.00% 72.30% # Number of instructions issued per FU type, per thread (Count)

board.processor.cores.core.statIssuedInstType_0::FloatSqrt 0 0.00% 72.30% # Number of instructions issued per FU type, per thread (Count)

board.processor.cores.core.statIssuedInstType_0::SimdAdd 10 0.00% 72.30% # Number of instructions issued per FU type, per thread (Count)

board.processor.cores.core.statIssuedInstType_0::SimdAddAcc 0 0.00% 72.30% # Number of instructions issued per FU type, per thread (Count)

board.processor.cores.core.statIssuedInstType_0::SimdAlu 281 0.00% 72.30% # Number of instructions issued per FU type, per thread (Count)

board.processor.cores.core.statIssuedInstType_0::SimdCmp 2 0.00% 72.30% # Number of instructions issued per FU type, per thread (Count)

board.processor.cores.core.statIssuedInstType_0::SimdCvt 78 0.00% 72.30% # Number of instructions issued per FU type, per thread (Count)

board.processor.cores.core.statIssuedInstType_0::SimdMisc 253 0.00% 72.30% # Number of instructions issued per FU type, per thread (Count)

board.processor.cores.core.statIssuedInstType_0::SimdMult 0 0.00% 72.30% # Number of instructions issued per FU type, per thread (Count)

board.processor.cores.core.statIssuedInstType_0::SimdMultAcc 0 0.00% 72.30% # Number of instructions issued per FU type, per thread (Count)

board.processor.cores.core.statIssuedInstType_0::SimdMatMultAcc 0 0.00% 72.30% # Number of instructions issued per FU type, per thread (Count)

board.processor.cores.core.statIssuedInstType_0::SimdShift 6 0.00% 72.30% # Number of instructions issued per FU type, per thread (Count)

board.processor.cores.core.statIssuedInstType_0::SimdShiftAcc 0 0.00% 72.30% # Number of instructions issued per FU type, per thread (Count)

board.processor.cores.core.statIssuedInstType_0::SimdDiv 0 0.00% 72.30% # Number of instructions issued per FU type, per thread (Count)

board.processor.cores.core.statIssuedInstType_0::SimdSqrt 0 0.00% 72.30% # Number of instructions issued per FU type, per thread (Count)

board.processor.cores.core.statIssuedInstType_0::SimdFloatAdd 0 0.00% 72.30% # Number of instructions issued per FU type, per thread (Count)

board.processor.cores.core.statIssuedInstType_0::SimdFloatAlu 0 0.00% 72.30% # Number of instructions issued per FU type, per thread (Count)

board.processor.cores.core.statIssuedInstType_0::SimdFloatCmp 0 0.00% 72.30% # Number of instructions issued per FU type, per thread (Count)

board.processor.cores.core.statIssuedInstType_0::SimdFloatCvt 0 0.00% 72.30% # Number of instructions issued per FU type, per thread (Count)

board.processor.cores.core.statIssuedInstType_0::SimdFloatDiv 0 0.00% 72.30% # Number of instructions issued per FU type, per thread (Count)

board.processor.cores.core.statIssuedInstType_0::SimdFloatMisc 0 0.00% 72.30% # Number of instructions issued per FU type, per thread (Count)

board.processor.cores.core.statIssuedInstType_0::SimdFloatMult 0 0.00% 72.30% # Number of instructions issued per FU type, per thread (Count)

board.processor.cores.core.statIssuedInstType_0::SimdFloatMultAcc 0 0.00% 72.30% # Number of instructions issued per FU type, per thread (Count)

board.processor.cores.core.statIssuedInstType_0::SimdFloatMatMultAcc 0 0.00% 72.30% # Number of instructions issued per FU type, per thread (Count)

board.processor.cores.core.statIssuedInstType_0::SimdFloatSqrt 0 0.00% 72.30% # Number of instructions issued per FU type, per thread (Count)

board.processor.cores.core.statIssuedInstType_0::SimdReduceAdd 0 0.00% 72.30% # Number of instructions issued per FU type, per thread (Count)

board.processor.cores.core.statIssuedInstType_0::SimdReduceAlu 0 0.00% 72.30% # Number of instructions issued per FU type, per thread (Count)

board.processor.cores.core.statIssuedInstType_0::SimdReduceCmp 0 0.00% 72.30% # Number of instructions issued per FU type, per thread (Count)

board.processor.cores.core.statIssuedInstType_0::SimdFloatReduceAdd 0 0.00% 72.30% # Number of instructions issued per FU type, per thread (Count)

board.processor.cores.core.statIssuedInstType_0::SimdFloatReduceCmp 0 0.00% 72.30% # Number of instructions issued per FU type, per thread (Count)

board.processor.cores.core.statIssuedInstType_0::SimdAes 0 0.00% 72.30% # Number of instructions issued per FU type, per thread (Count)

board.processor.cores.core.statIssuedInstType_0::SimdAesMix 0 0.00% 72.30% # Number of instructions issued per FU type, per thread (Count)

board.processor.cores.core.statIssuedInstType_0::SimdSha1Hash 0 0.00% 72.30% # Number of instructions issued per FU type, per thread (Count)

board.processor.cores.core.statIssuedInstType_0::SimdSha1Hash2 0 0.00% 72.30% # Number of instructions issued per FU type, per thread (Count)

board.processor.cores.core.statIssuedInstType_0::SimdSha256Hash 0 0.00% 72.30% # Number of instructions issued per FU type, per thread (Count)

board.processor.cores.core.statIssuedInstType_0::SimdSha256Hash2 0 0.00% 72.30% # Number of instructions issued per FU type, per thread (Count)

board.processor.cores.core.statIssuedInstType_0::SimdShaSigma2 0 0.00% 72.30% # Number of instructions issued per FU type, per thread (Count)

board.processor.cores.core.statIssuedInstType_0::SimdShaSigma3 0 0.00% 72.30% # Number of instructions issued per FU type, per thread (Count)

board.processor.cores.core.statIssuedInstType_0::SimdPredAlu 0 0.00% 72.30% # Number of instructions issued per FU type, per thread (Count)

board.processor.cores.core.statIssuedInstType_0::Matrix 0 0.00% 72.30% # Number of instructions issued per FU type, per thread (Count)

board.processor.cores.core.statIssuedInstType_0::MatrixMov 0 0.00% 72.30% # Number of instructions issued per FU type, per thread (Count)

board.processor.cores.core.statIssuedInstType_0::MatrixOP 0 0.00% 72.30% # Number of instructions issued per FU type, per thread (Count)

board.processor.cores.core.statIssuedInstType_0::MemRead 11583814 20.31% 92.61% # Number of instructions issued per FU type, per thread (Count)

board.processor.cores.core.statIssuedInstType_0::MemWrite 4214115 7.39% 100.00% # Number of instructions issued per FU type, per thread (Count)

board.processor.cores.core.statIssuedInstType_0::FloatMemRead 204 0.00% 100.00% # Number of instructions issued per FU type, per thread (Count)

board.processor.cores.core.statIssuedInstType_0::FloatMemWrite 638 0.00% 100.00% # Number of instructions issued per FU type, per thread (Count)

board.processor.cores.core.statIssuedInstType_0::IprAccess 0 0.00% 100.00% # Number of instructions issued per FU type, per thread (Count)

board.processor.cores.core.statIssuedInstType_0::InstPrefetch 0 0.00% 100.00% # Number of instructions issued per FU type, per thread (Count)

board.processor.cores.core.statIssuedInstType_0::SimdUnitStrideLoad 0 0.00% 100.00% # Number of instructions issued per FU type, per thread (Count)

board.processor.cores.core.statIssuedInstType_0::SimdUnitStrideStore 0 0.00% 100.00% # Number of instructions issued per FU type, per thread (Count)

board.processor.cores.core.statIssuedInstType_0::SimdUnitStrideMaskLoad 0 0.00% 100.00% # Number of instructions issued per FU type, per thread (Count)

board.processor.cores.core.statIssuedInstType_0::SimdUnitStrideMaskStore 0 0.00% 100.00% # Number of instructions issued per FU type, per thread (Count)

board.processor.cores.core.statIssuedInstType_0::SimdStridedLoad 0 0.00% 100.00% # Number of instructions issued per FU type, per thread (Count)

board.processor.cores.core.statIssuedInstType_0::SimdStridedStore 0 0.00% 100.00% # Number of instructions issued per FU type, per thread (Count)

board.processor.cores.core.statIssuedInstType_0::SimdIndexedLoad 0 0.00% 100.00% # Number of instructions issued per FU type, per thread (Count)

board.processor.cores.core.statIssuedInstType_0::SimdIndexedStore 0 0.00% 100.00% # Number of instructions issued per FU type, per thread (Count)

board.processor.cores.core.statIssuedInstType_0::SimdWholeRegisterLoad 0 0.00% 100.00% # Number of instructions issued per FU type, per thread (Count)

board.processor.cores.core.statIssuedInstType_0::SimdWholeRegisterStore 0 0.00% 100.00% # Number of instructions issued per FU type, per thread (Count)

board.processor.cores.core.statIssuedInstType_0::SimdUnitStrideFaultOnlyFirstLoad 0 0.00% 100.00% # Number of instructions issued per FU type, per thread (Count)

board.processor.cores.core.statIssuedInstType_0::SimdUnitStrideSegmentedLoad 0 0.00% 100.00% # Number of instructions issued per FU type, per thread (Count)

board.processor.cores.core.statIssuedInstType_0::SimdUnitStrideSegmentedStore 0 0.00% 100.00% # Number of instructions issued per FU type, per thread (Count)

board.processor.cores.core.statIssuedInstType_0::SimdExt 0 0.00% 100.00% # Number of instructions issued per FU type, per thread (Count)

board.processor.cores.core.statIssuedInstType_0::SimdFloatExt 0 0.00% 100.00% # Number of instructions issued per FU type, per thread (Count)

board.processor.cores.core.statIssuedInstType_0::SimdConfig 0 0.00% 100.00% # Number of instructions issued per FU type, per thread (Count)

board.processor.cores.core.statIssuedInstType_0::total 57028511 # Number of instructions issued per FU type, per thread (Count)

board.processor.cores.core.issueRate 3.232567 # Inst issue rate ((Count/Cycle))

board.processor.cores.core.fuBusy 1417966 # FU busy when requested (Count)

board.processor.cores.core.fuBusyRate 0.024864 # FU busy rate (busy events/executed inst) ((Count/Count))

board.processor.cores.core.intInstQueueReads 132966083 # Number of integer instruction queue reads (Count)

board.processor.cores.core.intInstQueueWrites 57286676 # Number of integer instruction queue writes (Count)

board.processor.cores.core.intInstQueueWakeupAccesses 57017767 # Number of integer instruction queue wakeup accesses (Count)

board.processor.cores.core.fpInstQueueReads 3801 # Number of floating instruction queue reads (Count)

board.processor.cores.core.fpInstQueueWrites 2463 # Number of floating instruction queue writes (Count)

board.processor.cores.core.fpInstQueueWakeupAccesses 1769 # Number of floating instruction queue wakeup accesses (Count)

board.processor.cores.core.vecInstQueueReads 0 # Number of vector instruction queue reads (Count)

board.processor.cores.core.vecInstQueueWrites 0 # Number of vector instruction queue writes (Count)

board.processor.cores.core.vecInstQueueWakeupAccesses 0 # Number of vector instruction queue wakeup accesses (Count)

board.processor.cores.core.intAluAccesses 58443818 # Number of integer alu accesses (Count)

board.processor.cores.core.fpAluAccesses 1909 # Number of floating point alu accesses (Count)

board.processor.cores.core.vecAluAccesses 0 # Number of vector alu accesses (Count)

board.processor.cores.core.numSquashedInsts 6987 # Number of squashed instructions skipped in execute (Count)

board.processor.cores.core.numSwp 0 # Number of swp insts executed (Count)

board.processor.cores.core.timesIdled 515 # Number of times that the entire CPU went into an idle state and unscheduled itself (Count)

board.processor.cores.core.idleCycles 147459 # Total number of cycles that the CPU has spent unscheduled due to idling (Cycle)

board.processor.cores.core.MemDepUnit__0.insertedLoads 11586109 # Number of loads inserted to the mem dependence unit. (Count)

board.processor.cores.core.MemDepUnit__0.insertedStores 4218802 # Number of stores inserted to the mem dependence unit. (Count)

board.processor.cores.core.MemDepUnit__0.conflictingLoads 7481482 # Number of conflicting loads. (Count)

board.processor.cores.core.MemDepUnit__0.conflictingStores 1287411 # Number of conflicting stores. (Count)

board.processor.cores.core.MemDepUnit__1.insertedLoads 0 # Number of loads inserted to the mem dependence unit. (Count)

board.processor.cores.core.MemDepUnit__1.insertedStores 0 # Number of stores inserted to the mem dependence unit. (Count)

board.processor.cores.core.MemDepUnit__1.conflictingLoads 0 # Number of conflicting loads. (Count)

board.processor.cores.core.MemDepUnit__1.conflictingStores 0 # Number of conflicting stores. (Count)

board.processor.cores.core.MemDepUnit__2.insertedLoads 0 # Number of loads inserted to the mem dependence unit. (Count)

board.processor.cores.core.MemDepUnit__2.insertedStores 0 # Number of stores inserted to the mem dependence unit. (Count)

board.processor.cores.core.MemDepUnit__2.conflictingLoads 0 # Number of conflicting loads. (Count)

board.processor.cores.core.MemDepUnit__2.conflictingStores 0 # Number of conflicting stores. (Count)

board.processor.cores.core.MemDepUnit__3.insertedLoads 0 # Number of loads inserted to the mem dependence unit. (Count)

board.processor.cores.core.MemDepUnit__3.insertedStores 0 # Number of stores inserted to the mem dependence unit. (Count)

board.processor.cores.core.MemDepUnit__3.conflictingLoads 0 # Number of conflicting loads. (Count)

board.processor.cores.core.MemDepUnit__3.conflictingStores 0 # Number of conflicting stores. (Count)

board.processor.cores.core.branchPred.lookups_0::NoBranch 4 0.00% 0.00% # Number of BP lookups (Count)

board.processor.cores.core.branchPred.lookups_0::Return 2779 0.13% 0.13% # Number of BP lookups (Count)

board.processor.cores.core.branchPred.lookups_0::CallDirect 2861 0.13% 0.26% # Number of BP lookups (Count)

board.processor.cores.core.branchPred.lookups_0::CallIndirect 174 0.01% 0.27% # Number of BP lookups (Count)

board.processor.cores.core.branchPred.lookups_0::DirectCond 2130494 99.44% 99.71% # Number of BP lookups (Count)

board.processor.cores.core.branchPred.lookups_0::DirectUncond 3750 0.18% 99.89% # Number of BP lookups (Count)

board.processor.cores.core.branchPred.lookups_0::IndirectCond 0 0.00% 99.89% # Number of BP lookups (Count)

board.processor.cores.core.branchPred.lookups_0::IndirectUncond 2440 0.11% 100.00% # Number of BP lookups (Count)

board.processor.cores.core.branchPred.lookups_0::total 2142502 # Number of BP lookups (Count)

board.processor.cores.core.branchPred.squashes_0::NoBranch 4 0.03% 0.03% # Number of branches that got squashed (completely removed) as an earlier branch was mispredicted. (Count)

board.processor.cores.core.branchPred.squashes_0::Return 410 3.40% 3.44% # Number of branches that got squashed (completely removed) as an earlier branch was mispredicted. (Count)

board.processor.cores.core.branchPred.squashes_0::CallDirect 534 4.43% 7.87% # Number of branches that got squashed (completely removed) as an earlier branch was mispredicted. (Count)

board.processor.cores.core.branchPred.squashes_0::CallIndirect 127 1.05% 8.93% # Number of branches that got squashed (completely removed) as an earlier branch was mispredicted. (Count)

board.processor.cores.core.branchPred.squashes_0::DirectCond 10236 84.99% 93.91% # Number of branches that got squashed (completely removed) as an earlier branch was mispredicted. (Count)

board.processor.cores.core.branchPred.squashes_0::DirectUncond 431 3.58% 97.49% # Number of branches that got squashed (completely removed) as an earlier branch was mispredicted. (Count)

board.processor.cores.core.branchPred.squashes_0::IndirectCond 0 0.00% 97.49% # Number of branches that got squashed (completely removed) as an earlier branch was mispredicted. (Count)

board.processor.cores.core.branchPred.squashes_0::IndirectUncond 302 2.51% 100.00% # Number of branches that got squashed (completely removed) as an earlier branch was mispredicted. (Count)

board.processor.cores.core.branchPred.squashes_0::total 12044 # Number of branches that got squashed (completely removed) as an earlier branch was mispredicted. (Count)

board.processor.cores.core.branchPred.corrected_0::NoBranch 0 0.00% 0.00% # Number of branches that got corrected but not yet commited. Branches get corrected by decode or after execute. Also a branch misprediction can be detected out-of-order. Therefore, a corrected branch might not end up beeing committed in case an even earlier branch was mispredicted (Count)

board.processor.cores.core.branchPred.corrected_0::Return 10 0.93% 0.93% # Number of branches that got corrected but not yet commited. Branches get corrected by decode or after execute. Also a branch misprediction can be detected out-of-order. Therefore, a corrected branch might not end up beeing committed in case an even earlier branch was mispredicted (Count)

board.processor.cores.core.branchPred.corrected_0::CallDirect 223 20.80% 21.74% # Number of branches that got corrected but not yet commited. Branches get corrected by decode or after execute. Also a branch misprediction can be detected out-of-order. Therefore, a corrected branch might not end up beeing committed in case an even earlier branch was mispredicted (Count)

board.processor.cores.core.branchPred.corrected_0::CallIndirect 39 3.64% 25.37% # Number of branches that got corrected but not yet commited. Branches get corrected by decode or after execute. Also a branch misprediction can be detected out-of-order. Therefore, a corrected branch might not end up beeing committed in case an even earlier branch was mispredicted (Count)

board.processor.cores.core.branchPred.corrected_0::DirectCond 601 56.06% 81.44% # Number of branches that got corrected but not yet commited. Branches get corrected by decode or after execute. Also a branch misprediction can be detected out-of-order. Therefore, a corrected branch might not end up beeing committed in case an even earlier branch was mispredicted (Count)

board.processor.cores.core.branchPred.corrected_0::DirectUncond 150 13.99% 95.43% # Number of branches that got corrected but not yet commited. Branches get corrected by decode or after execute. Also a branch misprediction can be detected out-of-order. Therefore, a corrected branch might not end up beeing committed in case an even earlier branch was mispredicted (Count)

board.processor.cores.core.branchPred.corrected_0::IndirectCond 0 0.00% 95.43% # Number of branches that got corrected but not yet commited. Branches get corrected by decode or after execute. Also a branch misprediction can be detected out-of-order. Therefore, a corrected branch might not end up beeing committed in case an even earlier branch was mispredicted (Count)

board.processor.cores.core.branchPred.corrected_0::IndirectUncond 49 4.57% 100.00% # Number of branches that got corrected but not yet commited. Branches get corrected by decode or after execute. Also a branch misprediction can be detected out-of-order. Therefore, a corrected branch might not end up beeing committed in case an even earlier branch was mispredicted (Count)

board.processor.cores.core.branchPred.corrected_0::total 1072 # Number of branches that got corrected but not yet commited. Branches get corrected by decode or after execute. Also a branch misprediction can be detected out-of-order. Therefore, a corrected branch might not end up beeing committed in case an even earlier branch was mispredicted (Count)

board.processor.cores.core.branchPred.earlyResteers_0::NoBranch 0 # Number of branches that got redirected after decode. (Count)

board.processor.cores.core.branchPred.earlyResteers_0::Return 0 # Number of branches that got redirected after decode. (Count)

board.processor.cores.core.branchPred.earlyResteers_0::CallDirect 0 # Number of branches that got redirected after decode. (Count)

board.processor.cores.core.branchPred.earlyResteers_0::CallIndirect 0 # Number of branches that got redirected after decode. (Count)

board.processor.cores.core.branchPred.earlyResteers_0::DirectCond 0 # Number of branches that got redirected after decode. (Count)

board.processor.cores.core.branchPred.earlyResteers_0::DirectUncond 0 # Number of branches that got redirected after decode. (Count)

board.processor.cores.core.branchPred.earlyResteers_0::IndirectCond 0 # Number of branches that got redirected after decode. (Count)

board.processor.cores.core.branchPred.earlyResteers_0::IndirectUncond 0 # Number of branches that got redirected after decode. (Count)

board.processor.cores.core.branchPred.earlyResteers_0::total 0 # Number of branches that got redirected after decode. (Count)

board.processor.cores.core.branchPred.committed_0::NoBranch 0 0.00% 0.00% # Number of branches finally committed (Count)

board.processor.cores.core.branchPred.committed_0::Return 2369 0.11% 0.11% # Number of branches finally committed (Count)

board.processor.cores.core.branchPred.committed_0::CallDirect 2327 0.11% 0.22% # Number of branches finally committed (Count)

board.processor.cores.core.branchPred.committed_0::CallIndirect 47 0.00% 0.22% # Number of branches finally committed (Count)

board.processor.cores.core.branchPred.committed_0::DirectCond 2120254 99.52% 99.74% # Number of branches finally committed (Count)

board.processor.cores.core.branchPred.committed_0::DirectUncond 3319 0.16% 99.90% # Number of branches finally committed (Count)

board.processor.cores.core.branchPred.committed_0::IndirectCond 0 0.00% 99.90% # Number of branches finally committed (Count)

board.processor.cores.core.branchPred.committed_0::IndirectUncond 2138 0.10% 100.00% # Number of branches finally committed (Count)

board.processor.cores.core.branchPred.committed_0::total 2130454 # Number of branches finally committed (Count)

board.processor.cores.core.branchPred.mispredicted_0::NoBranch 0 0.00% 0.00% # Number of committed branches that were mispredicted. (Count)

board.processor.cores.core.branchPred.mispredicted_0::Return 0 0.00% 0.00% # Number of committed branches that were mispredicted. (Count)

board.processor.cores.core.branchPred.mispredicted_0::CallDirect 108 13.25% 13.25% # Number of committed branches that were mispredicted. (Count)

board.processor.cores.core.branchPred.mispredicted_0::CallIndirect 38 4.66% 17.91% # Number of committed branches that were mispredicted. (Count)

board.processor.cores.core.branchPred.mispredicted_0::DirectCond 539 66.13% 84.05% # Number of committed branches that were mispredicted. (Count)

board.processor.cores.core.branchPred.mispredicted_0::DirectUncond 87 10.67% 94.72% # Number of committed branches that were mispredicted. (Count)

board.processor.cores.core.branchPred.mispredicted_0::IndirectCond 0 0.00% 94.72% # Number of committed branches that were mispredicted. (Count)

board.processor.cores.core.branchPred.mispredicted_0::IndirectUncond 43 5.28% 100.00% # Number of committed branches that were mispredicted. (Count)

board.processor.cores.core.branchPred.mispredicted_0::total 815 # Number of committed branches that were mispredicted. (Count)

board.processor.cores.core.branchPred.targetProvider_0::NoTarget 22948 1.07% 1.07% # The component providing the target for taken branches (Count)

board.processor.cores.core.branchPred.targetProvider_0::BTB 2114576 98.70% 99.77% # The component providing the target for taken branches (Count)

board.processor.cores.core.branchPred.targetProvider_0::RAS 2779 0.13% 99.90% # The component providing the target for taken branches (Count)

board.processor.cores.core.branchPred.targetProvider_0::Indirect 2199 0.10% 100.00% # The component providing the target for taken branches (Count)

board.processor.cores.core.branchPred.targetProvider_0::total 2142502 # The component providing the target for taken branches (Count)

board.processor.cores.core.branchPred.targetWrong_0::NoBranch 970 92.12% 92.12% # Number of branches where the target was incorrect or not available at prediction time. (Count)

board.processor.cores.core.branchPred.targetWrong_0::Return 72 6.84% 98.96% # Number of branches where the target was incorrect or not available at prediction time. (Count)

board.processor.cores.core.branchPred.targetWrong_0::CallDirect 10 0.95% 99.91% # Number of branches where the target was incorrect or not available at prediction time. (Count)

board.processor.cores.core.branchPred.targetWrong_0::CallIndirect 1 0.09% 100.00% # Number of branches where the target was incorrect or not available at prediction time. (Count)

board.processor.cores.core.branchPred.targetWrong_0::DirectCond 0 0.00% 100.00% # Number of branches where the target was incorrect or not available at prediction time. (Count)

board.processor.cores.core.branchPred.targetWrong_0::DirectUncond 0 0.00% 100.00% # Number of branches where the target was incorrect or not available at prediction time. (Count)

board.processor.cores.core.branchPred.targetWrong_0::IndirectCond 0 0.00% 100.00% # Number of branches where the target was incorrect or not available at prediction time. (Count)

board.processor.cores.core.branchPred.targetWrong_0::IndirectUncond 0 0.00% 100.00% # Number of branches where the target was incorrect or not available at prediction time. (Count)

board.processor.cores.core.branchPred.targetWrong_0::total 1053 # Number of branches where the target was incorrect or not available at prediction time. (Count)

board.processor.cores.core.branchPred.condPredicted 2130498 # Number of conditional branches predicted (Count)

board.processor.cores.core.branchPred.condPredictedTaken 2109182 # Number of conditional branches predicted as taken (Count)

board.processor.cores.core.branchPred.condIncorrect 1072 # Number of conditional branches incorrect (Count)

board.processor.cores.core.branchPred.predTakenBTBMiss 486 # Number of branches predicted taken but missed in BTB (Count)

board.processor.cores.core.branchPred.NotTakenMispredicted 1007 # Number branches predicted 'not taken' but turned out to be taken (Count)

board.processor.cores.core.branchPred.TakenMispredicted 65 # Number branches predicted taken but are actually not taken (Count)

board.processor.cores.core.branchPred.BTBLookups 2142502 # Number of BTB lookups (Count)

board.processor.cores.core.branchPred.BTBUpdates 909 # Number of BTB updates (Count)

board.processor.cores.core.branchPred.BTBHits 2117842 # Number of BTB hits (Count)

board.processor.cores.core.branchPred.BTBHitRatio 0.988490 # BTB Hit Ratio (Ratio)

board.processor.cores.core.branchPred.BTBMispredicted 704 # Number BTB mispredictions. No target found or target wrong (Count)

board.processor.cores.core.branchPred.indirectLookups 2614 # Number of indirect predictor lookups. (Count)

board.processor.cores.core.branchPred.indirectHits 2199 # Number of indirect target hits. (Count)

board.processor.cores.core.branchPred.indirectMisses 415 # Number of indirect misses. (Count)

board.processor.cores.core.branchPred.indirectMispredicted 0 # Number of mispredicted indirect branches. (Count)

board.processor.cores.core.branchPred.btb.lookups::NoBranch 4 0.00% 0.00% # Number of BTB lookups (Count)

board.processor.cores.core.branchPred.btb.lookups::Return 2779 0.13% 0.13% # Number of BTB lookups (Count)

board.processor.cores.core.branchPred.btb.lookups::CallDirect 2861 0.13% 0.26% # Number of BTB lookups (Count)

board.processor.cores.core.branchPred.btb.lookups::CallIndirect 174 0.01% 0.27% # Number of BTB lookups (Count)

board.processor.cores.core.branchPred.btb.lookups::DirectCond 2130494 99.44% 99.71% # Number of BTB lookups (Count)

board.processor.cores.core.branchPred.btb.lookups::DirectUncond 3750 0.18% 99.89% # Number of BTB lookups (Count)

board.processor.cores.core.branchPred.btb.lookups::IndirectCond 0 0.00% 99.89% # Number of BTB lookups (Count)

board.processor.cores.core.branchPred.btb.lookups::IndirectUncond 2440 0.11% 100.00% # Number of BTB lookups (Count)

board.processor.cores.core.branchPred.btb.lookups::total 2142502 # Number of BTB lookups (Count)

board.processor.cores.core.branchPred.btb.misses::NoBranch 4 0.02% 0.02% # Number of BTB misses (Count)

board.processor.cores.core.branchPred.btb.misses::Return 2775 11.25% 11.27% # Number of BTB misses (Count)

board.processor.cores.core.branchPred.btb.misses::CallDirect 361 1.46% 12.73% # Number of BTB misses (Count)

board.processor.cores.core.branchPred.btb.misses::CallIndirect 174 0.71% 13.44% # Number of BTB misses (Count)

board.processor.cores.core.branchPred.btb.misses::DirectCond 18677 75.74% 89.18% # Number of BTB misses (Count)

board.processor.cores.core.branchPred.btb.misses::DirectUncond 229 0.93% 90.11% # Number of BTB misses (Count)

board.processor.cores.core.branchPred.btb.misses::IndirectCond 0 0.00% 90.11% # Number of BTB misses (Count)

board.processor.cores.core.branchPred.btb.misses::IndirectUncond 2440 9.89% 100.00% # Number of BTB misses (Count)

board.processor.cores.core.branchPred.btb.misses::total 24660 # Number of BTB misses (Count)

board.processor.cores.core.branchPred.btb.updates::NoBranch 0 0.00% 0.00% # Number of BTB updates (Count)

board.processor.cores.core.branchPred.btb.updates::Return 0 0.00% 0.00% # Number of BTB updates (Count)

board.processor.cores.core.branchPred.btb.updates::CallDirect 223 24.53% 24.53% # Number of BTB updates (Count)

board.processor.cores.core.branchPred.btb.updates::CallIndirect 0 0.00% 24.53% # Number of BTB updates (Count)

board.processor.cores.core.branchPred.btb.updates::DirectCond 536 58.97% 83.50% # Number of BTB updates (Count)

board.processor.cores.core.branchPred.btb.updates::DirectUncond 150 16.50% 100.00% # Number of BTB updates (Count)

board.processor.cores.core.branchPred.btb.updates::IndirectCond 0 0.00% 100.00% # Number of BTB updates (Count)

board.processor.cores.core.branchPred.btb.updates::IndirectUncond 0 0.00% 100.00% # Number of BTB updates (Count)

board.processor.cores.core.branchPred.btb.updates::total 909 # Number of BTB updates (Count)

board.processor.cores.core.branchPred.btb.mispredict::NoBranch 0 0.00% 0.00% # Number of BTB mispredictions. No target found or target wrong. (Count)

board.processor.cores.core.branchPred.btb.mispredict::Return 0 0.00% 0.00% # Number of BTB mispredictions. No target found or target wrong. (Count)

board.processor.cores.core.branchPred.btb.mispredict::CallDirect 223 24.53% 24.53% # Number of BTB mispredictions. No target found or target wrong. (Count)

board.processor.cores.core.branchPred.btb.mispredict::CallIndirect 0 0.00% 24.53% # Number of BTB mispredictions. No target found or target wrong. (Count)

board.processor.cores.core.branchPred.btb.mispredict::DirectCond 536 58.97% 83.50% # Number of BTB mispredictions. No target found or target wrong. (Count)

board.processor.cores.core.branchPred.btb.mispredict::DirectUncond 150 16.50% 100.00% # Number of BTB mispredictions. No target found or target wrong. (Count)

board.processor.cores.core.branchPred.btb.mispredict::IndirectCond 0 0.00% 100.00% # Number of BTB mispredictions. No target found or target wrong. (Count)

board.processor.cores.core.branchPred.btb.mispredict::IndirectUncond 0 0.00% 100.00% # Number of BTB mispredictions. No target found or target wrong. (Count)

board.processor.cores.core.branchPred.btb.mispredict::total 909 # Number of BTB mispredictions. No target found or target wrong. (Count)

board.processor.cores.core.branchPred.btb.power_state.pwrStateResidencyTicks::UNDEFINED 5874741045 # Cumulative time (in ticks) in various power states (Tick)

board.processor.cores.core.branchPred.indirectBranchPred.lookups 2614 # Number of lookups (Count)

board.processor.cores.core.branchPred.indirectBranchPred.hits 2199 # Number of hits of a tag (Count)

board.processor.cores.core.branchPred.indirectBranchPred.misses 415 # Number of misses (Count)

board.processor.cores.core.branchPred.indirectBranchPred.targetRecords 88 # Number of targets that where recorded/installed in the cache (Count)

board.processor.cores.core.branchPred.indirectBranchPred.indirectRecords 2702 # Number of indirect branches/calls recorded in the indirect hist (Count)

board.processor.cores.core.branchPred.indirectBranchPred.speculativeOverflows 2 # Number of times more than the allowed capacity for speculative branches/calls where in flight and destroy the path history (Count)

board.processor.cores.core.branchPred.ras.pushes 3445 # Number of times a PC was pushed onto the RAS (Count)

board.processor.cores.core.branchPred.ras.pops 3440 # Number of times a PC was poped from the RAS (Count)

board.processor.cores.core.branchPred.ras.squashes 1071 # Number of times the stack operation was squashed due to wrong speculation. (Count)

board.processor.cores.core.branchPred.ras.used 2369 # Number of times the RAS is the provider (Count)

board.processor.cores.core.branchPred.ras.correct 2369 # Number of times the RAS is the provider and the prediction is correct (Count)

board.processor.cores.core.branchPred.ras.incorrect 0 # Number of times the RAS is the provider and the prediction is wrong (Count)

board.processor.cores.core.commit.commitSquashedInsts 147670 # The number of squashed insts skipped by commit (Count)

board.processor.cores.core.commit.commitNonSpecStalls 90 # The number of times commit has been forced to stall to communicate backwards (Count)

board.processor.cores.core.commit.branchMispredicts 738 # The number of times a branch was mispredicted (Count)

board.processor.cores.core.commit.numCommittedDist::samples 17474697 # Number of insts commited each cycle (Count)

board.processor.cores.core.commit.numCommittedDist::mean 3.253917 # Number of insts commited each cycle (Count)

board.processor.cores.core.commit.numCommittedDist::stdev 3.204519 # Number of insts commited each cycle (Count)

board.processor.cores.core.commit.numCommittedDist::underflows 0 0.00% 0.00% # Number of insts commited each cycle (Count)

board.processor.cores.core.commit.numCommittedDist::0 2225773 12.74% 12.74% # Number of insts commited each cycle (Count)

board.processor.cores.core.commit.numCommittedDist::1 7280214 41.66% 54.40% # Number of insts commited each cycle (Count)

board.processor.cores.core.commit.numCommittedDist::2 1764354 10.10% 64.50% # Number of insts commited each cycle (Count)

board.processor.cores.core.commit.numCommittedDist::3 123835 0.71% 65.20% # Number of insts commited each cycle (Count)

board.processor.cores.core.commit.numCommittedDist::4 123846 0.71% 65.91% # Number of insts commited each cycle (Count)

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board.processor.cores.core.commit.numCommittedDist::5      121574      0.70%     66.61% # Number of insts commited each cycle (Count)
board.processor.cores.core.commit.numCommittedDist::6     1049014      6.00%     72.61% # Number of insts commited each cycle (Count)
board.processor.cores.core.commit.numCommittedDist::7        5253      0.03%     72.64% # Number of insts commited each cycle (Count)
board.processor.cores.core.commit.numCommittedDist::8     4780834     27.36%    100.00% # Number of insts commited each cycle (Count)
board.processor.cores.core.commit.numCommittedDist::overflows           0      0.00%    100.00% # Number of insts commited each cycle (Count)
board.processor.cores.core.commit.numCommittedDist::min_value           0                      # Number of insts commited each cycle (Count)
board.processor.cores.core.commit.numCommittedDist::max_value           8                      # Number of insts commited each cycle (Count)
board.processor.cores.core.commit.numCommittedDist::total     17474697                      # Number of insts commited each cycle (Count)
board.processor.cores.core.commit.amos           0                      # Number of atomic instructions committed (Count)
board.processor.cores.core.commit.membars          60                      # Number of memory barriers committed (Count)
board.processor.cores.core.commit.functionCalls        2374                      # Number of function calls committed. (Count)
board.processor.cores.core.commit.committedInstType_0::No_OpClass         316      0.00%      0.00% # Class of committed instruction (Count)
board.processor.cores.core.commit.committedInstType_0::IntAlu     40052454     70.44%     70.44% # Class of committed instruction (Count)
board.processor.cores.core.commit.committedInstType_0::IntMult      1048624      1.84%     72.28% # Class of committed instruction (Count)
board.processor.cores.core.commit.committedInstType_0::IntDiv          28      0.00%     72.28% # Class of committed instruction (Count)
board.processor.cores.core.commit.committedInstType_0::FloatAdd         154      0.00%     72.28% # Class of committed instruction (Count)
board.processor.cores.core.commit.committedInstType_0::FloatCmp           0      0.00%     72.28% # Class of committed instruction (Count)
board.processor.cores.core.commit.committedInstType_0::FloatCvt           0      0.00%     72.28% # Class of committed instruction (Count)
board.processor.cores.core.commit.committedInstType_0::FloatMult           0      0.00%     72.28% # Class of committed instruction (Count)
board.processor.cores.core.commit.committedInstType_0::FloatMultAcc           0      0.00%     72.28% # Class of committed instruction (Count)
board.processor.cores.core.commit.committedInstType_0::FloatDiv           0      0.00%     72.28% # Class of committed instruction (Count)
board.processor.cores.core.commit.committedInstType_0::FloatMisc           0      0.00%     72.28% # Class of committed instruction (Count)
board.processor.cores.core.commit.committedInstType_0::FloatSqrt           0      0.00%     72.28% # Class of committed instruction (Count)
board.processor.cores.core.commit.committedInstType_0::SimdAdd          10      0.00%     72.28% # Class of committed instruction (Count)
board.processor.cores.core.commit.committedInstType_0::SimdAddAcc           0      0.00%     72.28% # Class of committed instruction (Count)
board.processor.cores.core.commit.committedInstType_0::SimdAlu         229      0.00%     72.28% # Class of committed instruction (Count)
board.processor.cores.core.commit.committedInstType_0::SimdCmp           0      0.00%     72.28% # Class of committed instruction (Count)
board.processor.cores.core.commit.committedInstType_0::SimdCvt          72      0.00%     72.28% # Class of committed instruction (Count)
board.processor.cores.core.commit.committedInstType_0::SimdMisc         240      0.00%     72.29% # Class of committed instruction (Count)
board.processor.cores.core.commit.committedInstType_0::SimdMult           0      0.00%     72.29% # Class of committed instruction (Count)
board.processor.cores.core.commit.committedInstType_0::SimdMultAcc           0      0.00%     72.29% # Class of committed instruction (Count)
board.processor.cores.core.commit.committedInstType_0::SimdMatMultAcc           0      0.00%     72.29% # Class of committed instruction (Count)
board.processor.cores.core.commit.committedInstType_0::SimdShift           3      0.00%     72.29% # Class of committed instruction (Count)
board.processor.cores.core.commit.committedInstType_0::SimdShiftAcc           0      0.00%     72.29% # Class of committed instruction (Count)
board.processor.cores.core.commit.committedInstType_0::SimdDiv           0      0.00%     72.29% # Class of committed instruction (Count)
board.processor.cores.core.commit.committedInstType_0::SimdSqrt           0      0.00%     72.29% # Class of committed instruction (Count)
board.processor.cores.core.commit.committedInstType_0::SimdFloatAdd           0      0.00%     72.29% # Class of committed instruction (Count)
board.processor.cores.core.commit.committedInstType_0::SimdFloatAlu           0      0.00%     72.29% # Class of committed instruction (Count)
board.processor.cores.core.commit.committedInstType_0::SimdFloatCmp           0      0.00%     72.29% # Class of committed instruction (Count)
board.processor.cores.core.commit.committedInstType_0::SimdFloatCvt           0      0.00%     72.29% # Class of committed instruction (Count)
board.processor.cores.core.commit.committedInstType_0::SimdFloatDiv           0      0.00%     72.29% # Class of committed instruction (Count)
board.processor.cores.core.commit.committedInstType_0::SimdFloatMisc           0      0.00%     72.29% # Class of committed instruction (Count)
board.processor.cores.core.commit.committedInstType_0::SimdFloatMult           0      0.00%     72.29% # Class of committed instruction (Count)
board.processor.cores.core.commit.committedInstType_0::SimdFloatMultAcc           0      0.00%     72.29% # Class of committed instruction (Count)
board.processor.cores.core.commit.committedInstType_0::SimdFloatMatMultAcc           0      0.00%     72.29% # Class of committed instruction (Count)
board.processor.cores.core.commit.committedInstType_0::SimdFloatSqrt           0      0.00%     72.29% # Class of committed instruction (Count)
board.processor.cores.core.commit.committedInstType_0::SimdReduceAdd           0      0.00%     72.29% # Class of committed instruction (Count)
board.processor.cores.core.commit.committedInstType_0::SimdReduceAlu           0      0.00%     72.29% # Class of committed instruction (Count)
board.processor.cores.core.commit.committedInstType_0::SimdReduceCmp           0      0.00%     72.29% # Class of committed instruction (Count)
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board.processor.cores.core.commit.committedInstType_0::SimdFloatReduceAdd          0      0.00%     72.29% # Class of committed instruction (Count)
board.processor.cores.core.commit.committedInstType_0::SimdFloatReduceCmp          0      0.00%     72.29% # Class of committed instruction (Count)
board.processor.cores.core.commit.committedInstType_0::SimdAes          0      0.00%     72.29% # Class of committed instruction (Count)
board.processor.cores.core.commit.committedInstType_0::SimdAesMix          0      0.00%     72.29% # Class of committed instruction (Count)
board.processor.cores.core.commit.committedInstType_0::SimdSha1Hash          0      0.00%     72.29% # Class of committed instruction (Count)
board.processor.cores.core.commit.committedInstType_0::SimdSha1Hash2          0      0.00%     72.29% # Class of committed instruction (Count)
board.processor.cores.core.commit.committedInstType_0::SimdSha256Hash          0      0.00%     72.29% # Class of committed instruction (Count)
board.processor.cores.core.commit.committedInstType_0::SimdSha256Hash2          0      0.00%     72.29% # Class of committed instruction (Count)
board.processor.cores.core.commit.committedInstType_0::SimdShaSigma2          0      0.00%     72.29% # Class of committed instruction (Count)
board.processor.cores.core.commit.committedInstType_0::SimdShaSigma3          0      0.00%     72.29% # Class of committed instruction (Count)
board.processor.cores.core.commit.committedInstType_0::SimdPredAlu          0      0.00%     72.29% # Class of committed instruction (Count)
board.processor.cores.core.commit.committedInstType_0::Matrix          0      0.00%     72.29% # Class of committed instruction (Count)
board.processor.cores.core.commit.committedInstType_0::MatrixMov          0      0.00%     72.29% # Class of committed instruction (Count)
board.processor.cores.core.commit.committedInstType_0::MatrixOP          0      0.00%     72.29% # Class of committed instruction (Count)
board.processor.cores.core.commit.committedInstType_0::MemRead     11553787     20.32%     92.60% # Class of committed instruction (Count)
board.processor.cores.core.commit.committedInstType_0::MemWrite      4204624      7.39%    100.00% # Class of committed instruction (Count)
board.processor.cores.core.commit.committedInstType_0::FloatMemRead          125      0.00%    100.00% # Class of committed instruction (Count)
board.processor.cores.core.commit.committedInstType_0::FloatMemWrite          542      0.00%    100.00% # Class of committed instruction (Count)
board.processor.cores.core.commit.committedInstType_0::IprAccess          0      0.00%    100.00% # Class of committed instruction (Count)
board.processor.cores.core.commit.committedInstType_0::InstPrefetch          0      0.00%    100.00% # Class of committed instruction (Count)
board.processor.cores.core.commit.committedInstType_0::SimdUnitStrideLoad          0      0.00%    100.00% # Class of committed instruction (Count)
board.processor.cores.core.commit.committedInstType_0::SimdUnitStrideStore          0      0.00%    100.00% # Class of committed instruction (Count)
board.processor.cores.core.commit.committedInstType_0::SimdUnitStrideMaskLoad          0      0.00%    100.00% # Class of committed instruction (Count)
board.processor.cores.core.commit.committedInstType_0::SimdUnitStrideMaskStore          0      0.00%    100.00% # Class of committed instruction (Count)
board.processor.cores.core.commit.committedInstType_0::SimdStridedLoad          0      0.00%    100.00% # Class of committed instruction (Count)
board.processor.cores.core.commit.committedInstType_0::SimdStridedStore          0      0.00%    100.00% # Class of committed instruction (Count)
board.processor.cores.core.commit.committedInstType_0::SimdIndexedLoad          0      0.00%    100.00% # Class of committed instruction (Count)
board.processor.cores.core.commit.committedInstType_0::SimdIndexedStore          0      0.00%    100.00% # Class of committed instruction (Count)
board.processor.cores.core.commit.committedInstType_0::SimdWholeRegisterLoad          0      0.00%    100.00% # Class of committed instruction (Count)
board.processor.cores.core.commit.committedInstType_0::SimdWholeRegisterStore          0      0.00%    100.00% # Class of committed instruction (Count)
board.processor.cores.core.commit.committedInstType_0::SimdUnitStrideFaultOnlyFirstLoad          0      0.00%    100.00% # Class of committed instruction (Count)
board.processor.cores.core.commit.committedInstType_0::SimdUnitStrideSegmentedLoad          0      0.00%    100.00% # Class of committed instruction (Count)
board.processor.cores.core.commit.committedInstType_0::SimdUnitStrideSegmentedStore          0      0.00%    100.00% # Class of committed instruction (Count)
board.processor.cores.core.commit.committedInstType_0::SimdExt          0      0.00%    100.00% # Class of committed instruction (Count)
board.processor.cores.core.commit.committedInstType_0::SimdFloatExt          0      0.00%    100.00% # Class of committed instruction (Count)
board.processor.cores.core.commit.committedInstType_0::SimdConfig          0      0.00%    100.00% # Class of committed instruction (Count)
board.processor.cores.core.commit.committedInstType_0::total     56861208                       # Class of committed instruction (Count)
board.processor.cores.core.commit.commitEligibleSamples      4780834                       # number cycles where commit BW limit reached (Cycle)
board.processor.cores.core.commitStats0.numInsts     36831240                       # Number of instructions committed (thread level) (Count)
board.processor.cores.core.commitStats0.numOps     56861208                       # Number of ops (including micro ops) committed (thread level) (Count)
board.processor.cores.core.commitStats0.numInstsNotNOP     36831240                       # Number of instructions committed excluding NOPs or prefetches (Count)
board.processor.cores.core.commitStats0.numOpsNotNOP     56861208                       # Number of Ops (including micro ops) Simulated (Count)
board.processor.cores.core.commitStats0.cpi     0.478992                       # CPI: cycles per instruction (thread level) ((Cycle/Count))
board.processor.cores.core.commitStats0.ipc     2.087718                       # IPC: instructions per cycle (thread level) ((Count/Cycle))
board.processor.cores.core.commitStats0.numMemRefs     15759078                       # Number of memory references committed (Count)
board.processor.cores.core.commitStats0.numFpInsts     1535                       # Number of float instructions (Count)
board.processor.cores.core.commitStats0.numIntInsts     56859973                       # Number of integer instructions (Count)
board.processor.cores.core.commitStats0.numLoadInsts     11553912                       # Number of load instructions (Count)
board.processor.cores.core.commitStats0.numStoreInsts     4205166                       # Number of store instructions (Count)
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board.processor.cores.core.commitStats0.numVecInsts 0 # Number of vector instructions (Count)

board.processor.cores.core.commitStats0.committedInstType::No_OpClass 316 0.00% 0.00% # Class of committed instruction. (Count)

board.processor.cores.core.commitStats0.committedInstType::IntAlu 40052454 70.44% 70.44% # Class of committed instruction. (Count)

board.processor.cores.core.commitStats0.committedInstType::IntMult 1048624 1.84% 72.28% # Class of committed instruction. (Count)

board.processor.cores.core.commitStats0.committedInstType::IntDiv 28 0.00% 72.28% # Class of committed instruction. (Count)

board.processor.cores.core.commitStats0.committedInstType::FloatAdd 154 0.00% 72.28% # Class of committed instruction. (Count)

board.processor.cores.core.commitStats0.committedInstType::FloatCmp 0 0.00% 72.28% # Class of committed instruction. (Count)

board.processor.cores.core.commitStats0.committedInstType::FloatCvt 0 0.00% 72.28% # Class of committed instruction. (Count)

board.processor.cores.core.commitStats0.committedInstType::FloatMult 0 0.00% 72.28% # Class of committed instruction. (Count)

board.processor.cores.core.commitStats0.committedInstType::FloatMultAcc 0 0.00% 72.28% # Class of committed instruction. (Count)

board.processor.cores.core.commitStats0.committedInstType::FloatDiv 0 0.00% 72.28% # Class of committed instruction. (Count)

board.processor.cores.core.commitStats0.committedInstType::FloatMisc 0 0.00% 72.28% # Class of committed instruction. (Count)

board.processor.cores.core.commitStats0.committedInstType::FloatSqrt 0 0.00% 72.28% # Class of committed instruction. (Count)

board.processor.cores.core.commitStats0.committedInstType::SimdAdd 10 0.00% 72.28% # Class of committed instruction. (Count)

board.processor.cores.core.commitStats0.committedInstType::SimdAddAcc 0 0.00% 72.28% # Class of committed instruction. (Count)

board.processor.cores.core.commitStats0.committedInstType::SimdAlu 229 0.00% 72.28% # Class of committed instruction. (Count)

board.processor.cores.core.commitStats0.committedInstType::SimdCmp 0 0.00% 72.28% # Class of committed instruction. (Count)

board.processor.cores.core.commitStats0.committedInstType::SimdCvt 72 0.00% 72.28% # Class of committed instruction. (Count)

board.processor.cores.core.commitStats0.committedInstType::SimdMisc 240 0.00% 72.29% # Class of committed instruction. (Count)

board.processor.cores.core.commitStats0.committedInstType::SimdMult 0 0.00% 72.29% # Class of committed instruction. (Count)

board.processor.cores.core.commitStats0.committedInstType::SimdMultAcc 0 0.00% 72.29% # Class of committed instruction. (Count)

board.processor.cores.core.commitStats0.committedInstType::SimdMatMultAcc 0 0.00% 72.29% # Class of committed instruction. (Count)

board.processor.cores.core.commitStats0.committedInstType::SimdShift 3 0.00% 72.29% # Class of committed instruction. (Count)

board.processor.cores.core.commitStats0.committedInstType::SimdShiftAcc 0 0.00% 72.29% # Class of committed instruction. (Count)

board.processor.cores.core.commitStats0.committedInstType::SimdDiv 0 0.00% 72.29% # Class of committed instruction. (Count)

board.processor.cores.core.commitStats0.committedInstType::SimdSqrt 0 0.00% 72.29% # Class of committed instruction. (Count)

board.processor.cores.core.commitStats0.committedInstType::SimdFloatAdd 0 0.00% 72.29% # Class of committed instruction. (Count)

board.processor.cores.core.commitStats0.committedInstType::SimdFloatAlu 0 0.00% 72.29% # Class of committed instruction. (Count)

board.processor.cores.core.commitStats0.committedInstType::SimdFloatCmp 0 0.00% 72.29% # Class of committed instruction. (Count)

board.processor.cores.core.commitStats0.committedInstType::SimdFloatCvt 0 0.00% 72.29% # Class of committed instruction. (Count)

board.processor.cores.core.commitStats0.committedInstType::SimdFloatDiv 0 0.00% 72.29% # Class of committed instruction. (Count)

board.processor.cores.core.commitStats0.committedInstType::SimdFloatMisc 0 0.00% 72.29% # Class of committed instruction. (Count)

board.processor.cores.core.commitStats0.committedInstType::SimdFloatMult 0 0.00% 72.29% # Class of committed instruction. (Count)

board.processor.cores.core.commitStats0.committedInstType::SimdFloatMultAcc 0 0.00% 72.29% # Class of committed instruction. (Count)

board.processor.cores.core.commitStats0.committedInstType::SimdFloatMatMultAcc 0 0.00% 72.29% # Class of committed instruction. (Count)

board.processor.cores.core.commitStats0.committedInstType::SimdFloatSqrt 0 0.00% 72.29% # Class of committed instruction. (Count)

board.processor.cores.core.commitStats0.committedInstType::SimdReduceAdd 0 0.00% 72.29% # Class of committed instruction. (Count)

board.processor.cores.core.commitStats0.committedInstType::SimdReduceAlu 0 0.00% 72.29% # Class of committed instruction. (Count)

board.processor.cores.core.commitStats0.committedInstType::SimdReduceCmp 0 0.00% 72.29% # Class of committed instruction. (Count)

board.processor.cores.core.commitStats0.committedInstType::SimdFloatReduceAdd 0 0.00% 72.29% # Class of committed instruction. (Count)

board.processor.cores.core.commitStats0.committedInstType::SimdFloatReduceCmp 0 0.00% 72.29% # Class of committed instruction. (Count)

board.processor.cores.core.commitStats0.committedInstType::SimdAes 0 0.00% 72.29% # Class of committed instruction. (Count)

board.processor.cores.core.commitStats0.committedInstType::SimdAesMix 0 0.00% 72.29% # Class of committed instruction. (Count)

board.processor.cores.core.commitStats0.committedInstType::SimdSha1Hash 0 0.00% 72.29% # Class of committed instruction. (Count)

board.processor.cores.core.commitStats0.committedInstType::SimdSha1Hash2 0 0.00% 72.29% # Class of committed instruction. (Count)

board.processor.cores.core.commitStats0.committedInstType::SimdSha256Hash 0 0.00% 72.29% # Class of committed instruction. (Count)

board.processor.cores.core.commitStats0.committedInstType::SimdSha256Hash2 0 0.00% 72.29% # Class of committed instruction. (Count)

board.processor.cores.core.commitStats0.committedInstType::SimdShaSigma2 0 0.00% 72.29% # Class of committed instruction. (Count)

board.processor.cores.core.commitStats0.committedInstType::SimdShaSigma3 0 0.00% 72.29% # Class of committed instruction. (Count)

board.processor.cores.core.commitStats0.committedInstType::SimdPredAlu 0 0.00% 72.29% # Class of committed instruction. (Count)

board.processor.cores.core.commitStats0.committedInstType::Matrix 0 0.00% 72.29% # Class of committed instruction. (Count)

board.processor.cores.core.commitStats0.committedInstType::MatrixMov 0 0.00% 72.29% # Class of committed instruction. (Count)

board.processor.cores.core.commitStats0.committedInstType::MatrixOP 0 0.00% 72.29% # Class of committed instruction. (Count)

board.processor.cores.core.commitStats0.committedInstType::MemRead 11553787 20.32% 92.60% # Class of committed instruction. (Count)

board.processor.cores.core.commitStats0.committedInstType::MemWrite 4204624 7.39% 100.00% # Class of committed instruction. (Count)

board.processor.cores.core.commitStats0.committedInstType::FloatMemRead 125 0.00% 100.00% # Class of committed instruction. (Count)

board.processor.cores.core.commitStats0.committedInstType::FloatMemWrite 542 0.00% 100.00% # Class of committed instruction. (Count)

board.processor.cores.core.commitStats0.committedInstType::IprAccess 0 0.00% 100.00% # Class of committed instruction. (Count)

board.processor.cores.core.commitStats0.committedInstType::InstPrefetch 0 0.00% 100.00% # Class of committed instruction. (Count)

board.processor.cores.core.commitStats0.committedInstType::SimdUnitStrideLoad 0 0.00% 100.00% # Class of committed instruction. (Count)

board.processor.cores.core.commitStats0.committedInstType::SimdUnitStrideStore 0 0.00% 100.00% # Class of committed instruction. (Count)

board.processor.cores.core.commitStats0.committedInstType::SimdUnitStrideMaskLoad 0 0.00% 100.00% # Class of committed instruction. (Count)

board.processor.cores.core.commitStats0.committedInstType::SimdUnitStrideMaskStore 0 0.00% 100.00% # Class of committed instruction. (Count)

board.processor.cores.core.commitStats0.committedInstType::SimdStridedLoad 0 0.00% 100.00% # Class of committed instruction. (Count)

board.processor.cores.core.commitStats0.committedInstType::SimdStridedStore 0 0.00% 100.00% # Class of committed instruction. (Count)

board.processor.cores.core.commitStats0.committedInstType::SimdIndexedLoad 0 0.00% 100.00% # Class of committed instruction. (Count)

board.processor.cores.core.commitStats0.committedInstType::SimdIndexedStore 0 0.00% 100.00% # Class of committed instruction. (Count)

board.processor.cores.core.commitStats0.committedInstType::SimdWholeRegisterLoad 0 0.00% 100.00% # Class of committed instruction. (Count)

board.processor.cores.core.commitStats0.committedInstType::SimdWholeRegisterStore 0 0.00% 100.00% # Class of committed instruction. (Count)

board.processor.cores.core.commitStats0.committedInstType::SimdUnitStrideFaultOnlyFirstLoad 0 0.00% 100.00% # Class of committed instruction. (Count)

board.processor.cores.core.commitStats0.committedInstType::SimdUnitStrideSegmentedLoad 0 0.00% 100.00% # Class of committed instruction. (Count)

board.processor.cores.core.commitStats0.committedInstType::SimdUnitStrideSegmentedStore 0 0.00% 100.00% # Class of committed instruction. (Count)

board.processor.cores.core.commitStats0.committedInstType::SimdExt 0 0.00% 100.00% # Class of committed instruction. (Count)

board.processor.cores.core.commitStats0.committedInstType::SimdFloatExt 0 0.00% 100.00% # Class of committed instruction. (Count)

board.processor.cores.core.commitStats0.committedInstType::SimdConfig 0 0.00% 100.00% # Class of committed instruction. (Count)

board.processor.cores.core.commitStats0.committedInstType::total 56861208 # Class of committed instruction. (Count)

board.processor.cores.core.commitStats0.committedControl::IsControl 2130454 # Class of control type instructions committed (Count)

board.processor.cores.core.commitStats0.committedControl::IsDirectControl 2125900 # Class of control type instructions committed (Count)

board.processor.cores.core.commitStats0.committedControl::IsIndirectControl 4554 # Class of control type instructions committed (Count)

board.processor.cores.core.commitStats0.committedControl::IsCondControl 2120254 # Class of control type instructions committed (Count)

board.processor.cores.core.commitStats0.committedControl::IsUncondControl 10200 # Class of control type instructions committed (Count)

board.processor.cores.core.commitStats0.committedControl::IsCall 2374 # Class of control type instructions committed (Count)

board.processor.cores.core.commitStats0.committedControl::IsReturn 2369 # Class of control type instructions committed (Count)

board.processor.cores.core.decode.idleCycles 1703029 # Number of cycles decode is idle (Cycle)

board.processor.cores.core.decode.blockedCycles 8645729 # Number of cycles decode is blocked (Cycle)

board.processor.cores.core.decode.runCycles 3136794 # Number of cycles decode is running (Cycle)

board.processor.cores.core.decode.unblockCycles 4006977 # Number of cycles decode is unblocking (Cycle)

board.processor.cores.core.decode.squashCycles 1878 # Number of cycles decode is squashing (Cycle)

board.processor.cores.core.decode.branchResolved 2113632 # Number of times decode resolved a branch (Count)

board.processor.cores.core.decode.branchMispred 388 # Number of times decode detected a branch misprediction (Count)

board.processor.cores.core.decode.decodedInsts 57099710 # Number of instructions handled by decode (Count)

board.processor.cores.core.decode.squashedInsts 2019 # Number of squashed instructions handled by decode (Count)

board.processor.cores.core.executeStats0.numInsts 57021524 # Number of executed instructions (Count)

board.processor.cores.core.executeStats0.numNop 0 # Number of nop insts executed (Count)

board.processor.cores.core.executeStats0.numBranches 2136461 # Number of branches executed (Count)

board.processor.cores.core.executeStats0.numLoadInsts 11580977 # Number of load instructions executed (Count)

board.processor.cores.core.executeStats0.numStoreInsts 4214503 # Number of stores executed (Count)

board.processor.cores.core.executeStats0.instRate 3.232171 # Inst execution rate ((Count/Cycle))

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board.processor.cores.core.executeStats0.numCCRegReads     14860396                    # Number of times the CC registers were read (Count)
board.processor.cores.core.executeStats0.numCCRegWrites     30569771                    # Number of times the CC registers were written (Count)
board.processor.cores.core.executeStats0.numFpRegReads        2102                    # Number of times the floating registers were read (Count)
board.processor.cores.core.executeStats0.numFpRegWrites        1115                    # Number of times the floating registers were written (Count)
board.processor.cores.core.executeStats0.numIntRegReads     80183059                    # Number of times the integer registers were read (Count)
board.processor.cores.core.executeStats0.numIntRegWrites     48542550                    # Number of times the integer registers were written (Count)
board.processor.cores.core.executeStats0.numMemRefs     15795480                    # Number of memory refs (Count)
board.processor.cores.core.executeStats0.numMiscRegReads     20069244                    # Number of times the Misc registers were read (Count)
board.processor.cores.core.executeStats0.numVecAluAccesses           0                    # Number of vector alu accesses (Count)
board.processor.cores.core.executeStats0.numDiscardedOps           0                    # Number of ops (including micro ops) which were discarded before commit (Count)
board.processor.cores.core.fetch.predictedBranches      2119554                    # Number of branches that fetch has predicted taken (Count)
board.processor.cores.core.fetch.cycles      17453728                    # Number of cycles fetch has run and was not squashing or blocked (Cycle)
board.processor.cores.core.fetch.squashCycles         4508                    # Number of cycles fetch has spent squashing (Cycle)
board.processor.cores.core.fetch.miscStallCycles           45                    # Number of cycles fetch has spent waiting on interrupts, or bad addresses, or out of MSHRs (Cycle)
board.processor.cores.core.fetch.pendingTrapStallCycles          256                    # Number of stall cycles due to pending traps (Cycle)
board.processor.cores.core.fetch.cacheLines      4232531                    # Number of cache lines fetched (Count)
board.processor.cores.core.fetch.icacheSquashes          472                    # Number of outstanding Icache misses that were squashed (Count)
board.processor.cores.core.fetch.nisnDist::samples     17494407                    # Number of instructions fetched each cycle (Total) (Count)
board.processor.cores.core.fetch.nisnDist::mean     3.266502                    # Number of instructions fetched each cycle (Total) (Count)
board.processor.cores.core.fetch.nisnDist::stdev     3.480153                    # Number of instructions fetched each cycle (Total) (Count)
board.processor.cores.core.fetch.nisnDist::underflows           0      0.00%      0.00% # Number of instructions fetched each cycle (Total) (Count)
board.processor.cores.core.fetch.nisnDist::0      7917892     45.26%     45.26% # Number of instructions fetched each cycle (Total) (Count)
board.processor.cores.core.fetch.nisnDist::1       526095      3.01%     48.27% # Number of instructions fetched each cycle (Total) (Count)
board.processor.cores.core.fetch.nisnDist::2       859058      4.91%     53.18% # Number of instructions fetched each cycle (Total) (Count)
board.processor.cores.core.fetch.nisnDist::3       494689      2.83%     56.00% # Number of instructions fetched each cycle (Total) (Count)
board.processor.cores.core.fetch.nisnDist::4      1338807      7.65%     63.66% # Number of instructions fetched each cycle (Total) (Count)
board.processor.cores.core.fetch.nisnDist::5       528100      3.02%     66.68% # Number of instructions fetched each cycle (Total) (Count)
board.processor.cores.core.fetch.nisnDist::6       327522      1.87%     68.55% # Number of instructions fetched each cycle (Total) (Count)
board.processor.cores.core.fetch.nisnDist::7       561568      3.21%     71.76% # Number of instructions fetched each cycle (Total) (Count)
board.processor.cores.core.fetch.nisnDist::8      4940676     28.24%    100.00% # Number of instructions fetched each cycle (Total) (Count)
board.processor.cores.core.fetch.nisnDist::overflows           0      0.00%    100.00% # Number of instructions fetched each cycle (Total) (Count)
board.processor.cores.core.fetch.nisnDist::min_value           0                    # Number of instructions fetched each cycle (Total) (Count)
board.processor.cores.core.fetch.nisnDist::max_value           8                    # Number of instructions fetched each cycle (Total) (Count)
board.processor.cores.core.fetch.nisnDist::total     17494407                    # Number of instructions fetched each cycle (Total) (Count)
board.processor.cores.core.fetchStats0.numInsts     37022533                    # Number of instructions fetched (thread level) (Count)
board.processor.cores.core.fetchStats0.numOps           0                    # Number of ops (including micro ops) fetched (thread level) (Count)
board.processor.cores.core.fetchStats0.fetchRate     2.098561                    # Number of inst fetches per cycle ((Count/Cycle))
board.processor.cores.core.fetchStats0.numBranches      2142502                    # Number of branches fetched (Count)
board.processor.cores.core.fetchStats0.branchRate     0.121444                    # Number of branch fetches per cycle (Ratio)
board.processor.cores.core.fetchStats0.icacheStallCycles        38124                    # ICache total stall cycles (Cycle)
board.processor.cores.core.fetchStats0.numFetchSuspends           0                    # Number of times Execute suspended instruction fetching (Count)
board.processor.cores.core.iew.idleCycles           0                    # Number of cycles IEW is idle (Cycle)
board.processor.cores.core.iew.squashCycles         1878                    # Number of cycles IEW is squashing (Cycle)
board.processor.cores.core.iew.blockCycles        25570                    # Number of cycles IEW is blocking (Cycle)
board.processor.cores.core.iew.unblockCycles         1037                    # Number of cycles IEW is unblocking (Cycle)
board.processor.cores.core.iew.dispatchedInsts     57075122                    # Number of instructions dispatched to IQ (Count)
board.processor.cores.core.iew.dispSquashedInsts          115                    # Number of squashed instructions skipped by dispatch (Count)
board.processor.cores.core.iew.dispLoadInsts     11586109                    # Number of dispatched load instructions (Count)
board.processor.cores.core.iew.dispStoreInsts      4218802                    # Number of dispatched store instructions (Count)
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board.processor.cores.core.iew.dispNonSpecInsts 80 # Number of dispatched non-speculative instructions (Count)
board.processor.cores.core.iew.iqFullEvents 38 # Number of times the IQ has become full, causing a stall (Count)
board.processor.cores.core.iew.lsqFullEvents 979 # Number of times the LSQ has become full, causing a stall (Count)
board.processor.cores.core.iew.memOrderViolationEvents 110 # Number of memory order violations (Count)
board.processor.cores.core.iew.predictedTakenIncorrect 86 # Number of branches that were predicted taken incorrectly (Count)
board.processor.cores.core.iew.predictedNotTakenIncorrect 780 # Number of branches that were predicted not taken incorrectly (Count)
board.processor.cores.core.iew.branchMispredicts 866 # Number of branch mispredicts detected at execute (Count)
board.processor.cores.core.iew.instsToCommit 57021141 # Cumulative count of insts sent to commit (Count)
board.processor.cores.core.iew.writebackCount 57019536 # Cumulative count of insts written-back (Count)
board.processor.cores.core.iew.producerInst 48544567 # Number of instructions producing a value (Count)
board.processor.cores.core.iew.consumerInst 72823089 # Number of instructions consuming a value (Count)
board.processor.cores.core.iew.wbRate 3.232058 # Insts written-back per cycle ((Count/Cycle))
board.processor.cores.core.iew.wbFanout 0.666610 # Average fanout of values written-back ((Count/Count))
board.processor.cores.core.interrupts.clk_domain.clock 5328 # Clock period in ticks (Tick)
board.processor.cores.core.lsq0.forwLoads 8296174 # Number of loads that had data forwarded from stores (Count)
board.processor.cores.core.lsq0.squashedLoads 32197 # Number of loads squashed (Count)
board.processor.cores.core.lsq0.ignoredResponses 5 # Number of memory responses ignored because the instruction is squashed (Count)
board.processor.cores.core.lsq0.memOrderViolation 110 # Number of memory ordering violations (Count)
board.processor.cores.core.lsq0.squashedStores 13636 # Number of stores squashed (Count)
board.processor.cores.core.lsq0.rescheduledLoads 8 # Number of loads that were rescheduled (Count)
board.processor.cores.core.lsq0.blockedByCache 30 # Number of times an access to memory failed due to the cache being blocked (Count)
board.processor.cores.core.lsq0.loadToUse::samples 11553912 # Distribution of cycle latency between the first time a load is issued and its completion (Unspecified)
board.processor.cores.core.lsq0.loadToUse::mean 2.103535 # Distribution of cycle latency between the first time a load is issued and its completion (Unspecified)
board.processor.cores.core.lsq0.loadToUse::stdev 6.106253 # Distribution of cycle latency between the first time a load is issued and its completion (Unspecified)
board.processor.cores.core.lsq0.loadToUse::0-9 11550215 99.97% 99.97% # Distribution of cycle latency between the first time a load is issued and its completion (Unspecified)
board.processor.cores.core.lsq0.loadToUse::10-19 20 0.00% 99.97% # Distribution of cycle latency between the first time a load is issued and its completion (Unspecified)
board.processor.cores.core.lsq0.loadToUse::20-29 20 0.00% 99.97% # Distribution of cycle latency between the first time a load is issued and its completion (Unspecified)
board.processor.cores.core.lsq0.loadToUse::30-39 34 0.00% 99.97% # Distribution of cycle latency between the first time a load is issued and its completion (Unspecified)
board.processor.cores.core.lsq0.loadToUse::40-49 20 0.00% 99.97% # Distribution of cycle latency between the first time a load is issued and its completion (Unspecified)
board.processor.cores.core.lsq0.loadToUse::50-59 22 0.00% 99.97% # Distribution of cycle latency between the first time a load is issued and its completion (Unspecified)
board.processor.cores.core.lsq0.loadToUse::60-69 31 0.00% 99.97% # Distribution of cycle latency between the first time a load is issued and its completion (Unspecified)
board.processor.cores.core.lsq0.loadToUse::70-79 10 0.00% 99.97% # Distribution of cycle latency between the first time a load is issued and its completion (Unspecified)
board.processor.cores.core.lsq0.loadToUse::80-89 12 0.00% 99.97% # Distribution of cycle latency between the first time a load is issued and its completion (Unspecified)
board.processor.cores.core.lsq0.loadToUse::90-99 15 0.00% 99.97% # Distribution of cycle latency between the first time a load is issued and its completion (Unspecified)
board.processor.cores.core.lsq0.loadToUse::100-109 51 0.00% 99.97% # Distribution of cycle latency between the first time a load is issued and its completion (Unspecified)
board.processor.cores.core.lsq0.loadToUse::110-119 1 0.00% 99.97% # Distribution of cycle latency between the first time a load is issued and its completion (Unspecified)
board.processor.cores.core.lsq0.loadToUse::190-199 2 0.00% 99.97% # Distribution of cycle latency between the first time a load is issued and its completion (Unspecified)
board.processor.cores.core.lsq0.loadToUse::200-209 3 0.00% 99.97% # Distribution of cycle latency between the first time a load is issued and its completion (Unspecified)
board.processor.cores.core.lsq0.loadToUse::210-219 6 0.00% 99.97% # Distribution of cycle latency between the first time a load is issued and its completion (Unspecified)
board.processor.cores.core.lsq0.loadToUse::220-229 17 0.00% 99.97% # Distribution of cycle latency between the first time a load is issued and its completion (Unspecified)
board.processor.cores.core.lsq0.loadToUse::250-259 3 0.00% 99.97% # Distribution of cycle latency between the first time a load is issued and its completion (Unspecified)
board.processor.cores.core.lsq0.loadToUse::260-269 1 0.00% 99.97% # Distribution of cycle latency between the first time a load is issued and its completion (Unspecified)
board.processor.cores.core.lsq0.loadToUse::280-289 2 0.00% 99.97% # Distribution of cycle latency between the first time a load is issued and its completion (Unspecified)
board.processor.cores.core.lsq0.loadToUse::290-299 11 0.00% 99.97% # Distribution of cycle latency between the first time a load is issued and its completion (Unspecified)
board.processor.cores.core.lsq0.loadToUse::overflows 3416 0.03% 100.00% # Distribution of cycle latency between the first time a load is issued and its completion (Unspecified)
board.processor.cores.core.lsq0.loadToUse::min_value 2 # Distribution of cycle latency between the first time a load is issued and its completion (Unspecified)
board.processor.cores.core.lsq0.loadToUse::max_value 1194 # Distribution of cycle latency between the first time a load is issued and its completion (Unspecified)
board.processor.cores.core.lsq0.loadToUse::total 11553912 # Distribution of cycle latency between the first time a load is issued and its completion (Unspecified)
board.processor.cores.core.mmu.dtb.rdAccesses 11580886 # TLB accesses on read requests (Count)
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board.processor.cores.core.mmu.dtb.wrAccesses      4214503                    # TLB accesses on write requests (Count)
board.processor.cores.core.mmu.dtb.rdMisses        1143                       # TLB misses on read requests (Count)
board.processor.cores.core.mmu.dtb.wrMisses        2081                       # TLB misses on write requests (Count)
board.processor.cores.core.mmu.dtb.walker.power_state.pwrStateResidencyTicks::UNDEFINED   5874741045   # Cumulative time (in ticks) in various power states (Tick)
board.processor.cores.core.mmu.itb.rdAccesses      0                          # TLB accesses on read requests (Count)
board.processor.cores.core.mmu.itb.wrAccesses      4232576                    # TLB accesses on write requests (Count)
board.processor.cores.core.mmu.itb.rdMisses        0                          # TLB misses on read requests (Count)
board.processor.cores.core.mmu.itb.wrMisses        116                        # TLB misses on write requests (Count)
board.processor.cores.core.mmu.itb.walker.power_state.pwrStateResidencyTicks::UNDEFINED   5874741045   # Cumulative time (in ticks) in various power states (Tick)
board.processor.cores.core.power_state.pwrStateResidencyTicks::ON   5874741045   # Cumulative time (in ticks) in various power states (Tick)
board.processor.cores.core.rename.squashCycles     1878                       # Number of cycles rename is squashing (Cycle)
board.processor.cores.core.rename.idleCycles       2874382                    # Number of cycles rename is idle (Cycle)
board.processor.cores.core.rename.blockCycles      508752                     # Number of cycles rename is blocking (Cycle)
board.processor.cores.core.rename.serializeStallCycles   2536                 # count of cycles rename stalled for serializing inst (Cycle)
board.processor.cores.core.rename.runCycles        5970814                    # Number of cycles rename is running (Cycle)
board.processor.cores.core.rename.unblockCycles    8136045                    # Number of cycles rename is unblocking (Cycle)
board.processor.cores.core.rename.renamedInsts     57095530                   # Number of instructions processed by rename (Count)
board.processor.cores.core.rename.ROBFullEvents    2778                       # Number of times rename has blocked due to ROB full (Count)
board.processor.cores.core.rename.IQFullEvents     2217259                    # Number of times rename has blocked due to IQ full (Count)
board.processor.cores.core.rename.LQFullEvents     5051098                    # Number of times rename has blocked due to LQ full (Count)
board.processor.cores.core.rename.SQFullEvents     333381                     # Number of times rename has blocked due to SQ full (Count)
board.processor.cores.core.rename.renamedOperands  127817515                  # Number of destination operands rename has renamed (Count)
board.processor.cores.core.rename.lookups          213506292                  # Number of register rename lookups that rename has made (Count)
board.processor.cores.core.rename.intLookups       80311336                   # Number of integer rename lookups (Count)
board.processor.cores.core.rename.fpLookups        2489                       # Number of floating rename lookups (Count)
board.processor.cores.core.rename.committedMaps    127275790                  # Number of HB maps that are committed (Count)
board.processor.cores.core.rename.undoneMaps       541716                     # Number of HB maps that are undone due to squashing (Count)
board.processor.cores.core.rename.serializing      45                         # count of serializing insts renamed (Count)
board.processor.cores.core.rename.tempSerializing  46                         # count of temporary serializing insts renamed (Count)
board.processor.cores.core.rename.skidInsts        16272782                   # count of insts added to the skid buffer (Count)
board.processor.cores.core.rob.reads               69689143                   # The number of ROB reads (Count)
board.processor.cores.core.rob.writes              114037532                  # The number of ROB writes (Count)
board.processor.cores.core.thread_0.numInsts       36831240                   # Number of Instructions committed (Count)
board.processor.cores.core.thread_0.numOps         56861208                   # Number of Ops committed (Count)
board.processor.cores.core.thread_0.numMemRefs     0                          # Number of Memory References (Count)
board.processor.cores.core.workload.numSyscalls    18                         # Number of system calls (Count)
board.workload.inst.arm                            0                          # number of arm instructions executed (Count)
board.workload.inst.quiesce                        0                          # number of quiesce instructions executed (Count)

---------- End Simulation Statistics   ----------
```

results_ooo_mshr_8:

```
---------- Begin Simulation Statistics ----------
simSeconds                    0.005910                # Number of seconds simulated (Second)
simTicks                      5909877540              # Number of ticks simulated (Tick)
finalTick                     5909877540              # Number of ticks from beginning of simulation (restored from checkpoints and never reset) (Tick)
simFreq                       1000000000000           # The number of ticks per simulated second ((Tick/Second))
hostSeconds                   61.10                   # Real time elapsed on the host (Second)
hostTickRate                  96718454                # The number of ticks simulated per host second (ticks/s) ((Tick/Second))
hostMemory                    1214236                 # Number of bytes of host memory used (Byte)
simInsts                      36831240                # Number of instructions simulated (Count)
simOps                        56861208                # Number of ops (including micro ops) simulated (Count)
hostInstRate                  602763                  # Simulator instruction rate (inst/s) ((Count/Second))
hostOpRate                    930564                  # Simulator op (including micro ops) rate (op/s) ((Count/Second))
board.cache_hierarchy.dptw_caches.blockedCycles::no_mshrs      0          # number of cycles access was blocked (Cycle)
board.cache_hierarchy.dptw_caches.blockedCycles::no_targets    0          # number of cycles access was blocked (Cycle)
board.cache_hierarchy.dptw_caches.blockedCauses::no_mshrs      0          # number of times access was blocked (Count)
board.cache_hierarchy.dptw_caches.blockedCauses::no_targets    0          # number of times access was blocked (Count)
board.cache_hierarchy.dptw_caches.avgBlocked::no_mshrs         nan        # average number of cycles each access was blocked ((Cycle/Count))
board.cache_hierarchy.dptw_caches.avgBlocked::no_targets       nan        # average number of cycles each access was blocked ((Cycle/Count))
board.cache_hierarchy.dptw_caches.replacements                 0          # number of replacements (Count)
board.cache_hierarchy.dptw_caches.power_state.pwrStateResidencyTicks::UNDEFINED   5909877540   # Cumulative time (in ticks) in various power states (Tick)
board.cache_hierarchy.dptw_caches.tags.tagsInUse               0          # Average ticks per tags in use ((Tick/Count))
board.cache_hierarchy.dptw_caches.tags.totalRefs               0          # Total number of references to valid blocks. (Count)
board.cache_hierarchy.dptw_caches.tags.sampledRefs             0          # Sample count of references to valid blocks. (Count)
board.cache_hierarchy.dptw_caches.tags.avgRefs                 nan        # Average number of references to valid blocks. ((Count/Count))
board.cache_hierarchy.dptw_caches.tags.warmupTick              0          # The tick when the warmup percentage was hit. (Tick)
board.cache_hierarchy.dptw_caches.tags.tagAccesses             0          # Number of tag accesses (Count)
board.cache_hierarchy.dptw_caches.tags.dataAccesses            0          # Number of data accesses (Count)
board.cache_hierarchy.dptw_caches.tags.power_state.pwrStateResidencyTicks::UNDEFINED   5909877540   # Cumulative time (in ticks) in various power states (Tick)
board.cache_hierarchy.iptw_caches.blockedCycles::no_mshrs      0          # number of cycles access was blocked (Cycle)
board.cache_hierarchy.iptw_caches.blockedCycles::no_targets    0          # number of cycles access was blocked (Cycle)
board.cache_hierarchy.iptw_caches.blockedCauses::no_mshrs      0          # number of times access was blocked (Count)
board.cache_hierarchy.iptw_caches.blockedCauses::no_targets    0          # number of times access was blocked (Count)
board.cache_hierarchy.iptw_caches.avgBlocked::no_mshrs         nan        # average number of cycles each access was blocked ((Cycle/Count))
board.cache_hierarchy.iptw_caches.avgBlocked::no_targets       nan        # average number of cycles each access was blocked ((Cycle/Count))
board.cache_hierarchy.iptw_caches.replacements                 0          # number of replacements (Count)
board.cache_hierarchy.iptw_caches.power_state.pwrStateResidencyTicks::UNDEFINED   5909877540   # Cumulative time (in ticks) in various power states (Tick)
board.cache_hierarchy.iptw_caches.tags.tagsInUse               0          # Average ticks per tags in use ((Tick/Count))
board.cache_hierarchy.iptw_caches.tags.totalRefs               0          # Total number of references to valid blocks. (Count)
board.cache_hierarchy.iptw_caches.tags.sampledRefs             0          # Sample count of references to valid blocks. (Count)
board.cache_hierarchy.iptw_caches.tags.avgRefs                 nan        # Average number of references to valid blocks. ((Count/Count))
board.cache_hierarchy.iptw_caches.tags.warmupTick              0          # The tick when the warmup percentage was hit. (Tick)
board.cache_hierarchy.iptw_caches.tags.tagAccesses             0          # Number of tag accesses (Count)
board.cache_hierarchy.iptw_caches.tags.dataAccesses            0          # Number of data accesses (Count)
board.cache_hierarchy.iptw_caches.tags.power_state.pwrStateResidencyTicks::UNDEFINED   5909877540   # Cumulative time (in ticks) in various power states (Tick)
board.cache_hierarchy.l1dcaches.demandHits::processor.cores.core.data   7484371   # number of demand (read+write) hits (Count)
board.cache_hierarchy.l1dcaches.demandHits::total   7484371   # number of demand (read+write) hits (Count)
```

board.cache_hierarchy.l1dcaches.overallHits::processor.cores.core.data 7484371 # number of overall hits (Count)

board.cache_hierarchy.l1dcaches.overallHits::total 7484371 # number of overall hits (Count)

board.cache_hierarchy.l1dcaches.demandMisses::processor.cores.core.data 5189 # number of demand (read+write) misses (Count)

board.cache_hierarchy.l1dcaches.demandMisses::total 5189 # number of demand (read+write) misses (Count)

board.cache_hierarchy.l1dcaches.overallMisses::processor.cores.core.data 5189 # number of overall misses (Count)

board.cache_hierarchy.l1dcaches.overallMisses::total 5189 # number of overall misses (Count)

board.cache_hierarchy.l1dcaches.demandMissLatency::processor.cores.core.data 626215158 # number of demand (read+write) miss ticks (Tick)

board.cache_hierarchy.l1dcaches.demandMissLatency::total 626215158 # number of demand (read+write) miss ticks (Tick)

board.cache_hierarchy.l1dcaches.overallMissLatency::processor.cores.core.data 626215158 # number of overall miss ticks (Tick)

board.cache_hierarchy.l1dcaches.overallMissLatency::total 626215158 # number of overall miss ticks (Tick)

board.cache_hierarchy.l1dcaches.demandAccesses::processor.cores.core.data 7489560 # number of demand (read+write) accesses (Count)

board.cache_hierarchy.l1dcaches.demandAccesses::total 7489560 # number of demand (read+write) accesses (Count)

board.cache_hierarchy.l1dcaches.overallAccesses::processor.cores.core.data 7489560 # number of overall (read+write) accesses (Count)

board.cache_hierarchy.l1dcaches.overallAccesses::total 7489560 # number of overall (read+write) accesses (Count)

board.cache_hierarchy.l1dcaches.demandMissRate::processor.cores.core.data 0.000693 # miss rate for demand accesses (Ratio)

board.cache_hierarchy.l1dcaches.demandMissRate::total 0.000693 # miss rate for demand accesses (Ratio)

board.cache_hierarchy.l1dcaches.overallMissRate::processor.cores.core.data 0.000693 # miss rate for overall accesses (Ratio)

board.cache_hierarchy.l1dcaches.overallMissRate::total 0.000693 # miss rate for overall accesses (Ratio)

board.cache_hierarchy.l1dcaches.demandAvgMissLatency::processor.cores.core.data 120681.279245 # average overall miss latency in ticks ((Tick/Count))

board.cache_hierarchy.l1dcaches.demandAvgMissLatency::total 120681.279245 # average overall miss latency in ticks ((Tick/Count))

board.cache_hierarchy.l1dcaches.overallAvgMissLatency::processor.cores.core.data 120681.279245 # average overall miss latency ((Tick/Count))

board.cache_hierarchy.l1dcaches.overallAvgMissLatency::total 120681.279245 # average overall miss latency ((Tick/Count))

board.cache_hierarchy.l1dcaches.blockedCycles::no_mshrs 214 # number of cycles access was blocked (Cycle)

board.cache_hierarchy.l1dcaches.blockedCycles::no_targets 0 # number of cycles access was blocked (Cycle)

board.cache_hierarchy.l1dcaches.blockedCauses::no_mshrs 1 # number of times access was blocked (Count)

board.cache_hierarchy.l1dcaches.blockedCauses::no_targets 0 # number of times access was blocked (Count)

board.cache_hierarchy.l1dcaches.avgBlocked::no_mshrs 214 # average number of cycles each access was blocked ((Cycle/Count))

board.cache_hierarchy.l1dcaches.avgBlocked::no_targets nan # average number of cycles each access was blocked ((Cycle/Count))

board.cache_hierarchy.l1dcaches.writebacks::writebacks 65717 # number of writebacks (Count)

board.cache_hierarchy.l1dcaches.writebacks::total 65717 # number of writebacks (Count)

board.cache_hierarchy.l1dcaches.demandMshrHits::processor.cores.core.data 2240 # number of demand (read+write) MSHR hits (Count)

board.cache_hierarchy.l1dcaches.demandMshrHits::total 2240 # number of demand (read+write) MSHR hits (Count)

board.cache_hierarchy.l1dcaches.overallMshrHits::processor.cores.core.data 2240 # number of overall MSHR hits (Count)

board.cache_hierarchy.l1dcaches.overallMshrHits::total 2240 # number of overall MSHR hits (Count)

board.cache_hierarchy.l1dcaches.demandMshrMisses::processor.cores.core.data 2949 # number of demand (read+write) MSHR misses (Count)

board.cache_hierarchy.l1dcaches.demandMshrMisses::total 2949 # number of demand (read+write) MSHR misses (Count)

board.cache_hierarchy.l1dcaches.overallMshrMisses::cache_hierarchy.l1dcaches.prefetcher 129118 # number of overall MSHR misses (Count)

board.cache_hierarchy.l1dcaches.overallMshrMisses::processor.cores.core.data 2949 # number of overall MSHR misses (Count)

board.cache_hierarchy.l1dcaches.overallMshrMisses::total 132067 # number of overall MSHR misses (Count)

board.cache_hierarchy.l1dcaches.demandMshrMissLatency::processor.cores.core.data 376782840 # number of demand (read+write) MSHR miss ticks (Tick)

board.cache_hierarchy.l1dcaches.demandMshrMissLatency::total 376782840 # number of demand (read+write) MSHR miss ticks (Tick)

board.cache_hierarchy.l1dcaches.overallMshrMissLatency::cache_hierarchy.l1dcaches.prefetcher 5747808154 # number of overall MSHR miss ticks (Tick)

board.cache_hierarchy.l1dcaches.overallMshrMissLatency::processor.cores.core.data 376782840 # number of overall MSHR miss ticks (Tick)

board.cache_hierarchy.l1dcaches.overallMshrMissLatency::total 6124590994 # number of overall MSHR miss ticks (Tick)

board.cache_hierarchy.l1dcaches.demandMshrMissRate::processor.cores.core.data 0.000394 # mshr miss ratio for demand accesses (Ratio)

board.cache_hierarchy.l1dcaches.demandMshrMissRate::total 0.000394 # mshr miss ratio for demand accesses (Ratio)

board.cache_hierarchy.l1dcaches.overallMshrMissRate::cache_hierarchy.l1dcaches.prefetcher inf # mshr miss ratio for overall accesses (Ratio)

board.cache_hierarchy.l1dcaches.overallMshrMissRate::processor.cores.core.data 0.000394 # mshr miss ratio for overall accesses (Ratio)

board.cache_hierarchy.l1dcaches.overallMshrMissRate::total 0.017633 # mshr miss ratio for overall accesses (Ratio)

board.cache_hierarchy.l1dcaches.demandAvgMshrMissLatency::processor.cores.core.data 127766.307223 # average overall mshr miss latency ((Tick/Count))

board.cache_hierarchy.l1dcaches.demandAvgMshrMissLatency::total 127766.307223 # average overall mshr miss latency ((Tick/Count))

board.cache_hierarchy.l1dcaches.overallAvgMshrMissLatency::cache_hierarchy.l1dcaches.prefetcher 44515.932356 # average overall mshr miss latency ((Tick/Count))

board.cache_hierarchy.l1dcaches.overallAvgMshrMissLatency::processor.cores.core.data 127766.307223 # average overall mshr miss latency ((Tick/Count))

board.cache_hierarchy.l1dcaches.overallAvgMshrMissLatency::total 46374.877857 # average overall mshr miss latency ((Tick/Count))

board.cache_hierarchy.l1dcaches.replacements 131556 # number of replacements (Count)

board.cache_hierarchy.l1dcaches.HardPFReq.mshrMisses::cache_hierarchy.l1dcaches.prefetcher 129118 # number of HardPFReq MSHR misses (Count)

board.cache_hierarchy.l1dcaches.HardPFReq.mshrMisses::total 129118 # number of HardPFReq MSHR misses (Count)

board.cache_hierarchy.l1dcaches.HardPFReq.mshrMissLatency::cache_hierarchy.l1dcaches.prefetcher 5747808154 # number of HardPFReq MSHR miss ticks (Tick)

board.cache_hierarchy.l1dcaches.HardPFReq.mshrMissLatency::total 5747808154 # number of HardPFReq MSHR miss ticks (Tick)

board.cache_hierarchy.l1dcaches.HardPFReq.mshrMissRate::cache_hierarchy.l1dcaches.prefetcher inf # mshr miss rate for HardPFReq accesses (Ratio)

board.cache_hierarchy.l1dcaches.HardPFReq.mshrMissRate::total inf # mshr miss rate for HardPFReq accesses (Ratio)

board.cache_hierarchy.l1dcaches.HardPFReq.avgMshrMissLatency::cache_hierarchy.l1dcaches.prefetcher 44515.932356 # average HardPFReq mshr miss latency ((Tick/Count))

board.cache_hierarchy.l1dcaches.HardPFReq.avgMshrMissLatency::total 44515.932356 # average HardPFReq mshr miss latency ((Tick/Count))

board.cache_hierarchy.l1dcaches.LockedRMWReadReq.hits::processor.cores.core.data 26 # number of LockedRMWReadReq hits (Count)

board.cache_hierarchy.l1dcaches.LockedRMWReadReq.hits::total 26 # number of LockedRMWReadReq hits (Count)

board.cache_hierarchy.l1dcaches.LockedRMWReadReq.misses::processor.cores.core.data 4 # number of LockedRMWReadReq misses (Count)

board.cache_hierarchy.l1dcaches.LockedRMWReadReq.misses::total 4 # number of LockedRMWReadReq misses (Count)

board.cache_hierarchy.l1dcaches.LockedRMWReadReq.missLatency::processor.cores.core.data 516150 # number of LockedRMWReadReq miss ticks (Tick)

board.cache_hierarchy.l1dcaches.LockedRMWReadReq.missLatency::total 516150 # number of LockedRMWReadReq miss ticks (Tick)

board.cache_hierarchy.l1dcaches.LockedRMWReadReq.accesses::processor.cores.core.data 30 # number of LockedRMWReadReq accesses(hits+misses) (Count)

board.cache_hierarchy.l1dcaches.LockedRMWReadReq.accesses::total 30 # number of LockedRMWReadReq accesses(hits+misses) (Count)

board.cache_hierarchy.l1dcaches.LockedRMWReadReq.missRate::processor.cores.core.data 0.133333 # miss rate for LockedRMWReadReq accesses (Ratio)

board.cache_hierarchy.l1dcaches.LockedRMWReadReq.missRate::total 0.133333 # miss rate for LockedRMWReadReq accesses (Ratio)

board.cache_hierarchy.l1dcaches.LockedRMWReadReq.avgMissLatency::processor.cores.core.data 129037.500000 # average LockedRMWReadReq miss latency ((Tick/Count))

board.cache_hierarchy.l1dcaches.LockedRMWReadReq.avgMissLatency::total 129037.500000 # average LockedRMWReadReq miss latency ((Tick/Count))

board.cache_hierarchy.l1dcaches.LockedRMWReadReq.mshrMisses::processor.cores.core.data 4 # number of LockedRMWReadReq MSHR misses (Count)

board.cache_hierarchy.l1dcaches.LockedRMWReadReq.mshrMisses::total 4 # number of LockedRMWReadReq MSHR misses (Count)

board.cache_hierarchy.l1dcaches.LockedRMWReadReq.mshrMissLatency::processor.cores.core.data 1097901 # number of LockedRMWReadReq MSHR miss ticks (Tick)

board.cache_hierarchy.l1dcaches.LockedRMWReadReq.mshrMissLatency::total 1097901 # number of LockedRMWReadReq MSHR miss ticks (Tick)

board.cache_hierarchy.l1dcaches.LockedRMWReadReq.mshrMissRate::processor.cores.core.data 0.133333 # mshr miss rate for LockedRMWReadReq accesses (Ratio)

board.cache_hierarchy.l1dcaches.LockedRMWReadReq.mshrMissRate::total 0.133333 # mshr miss rate for LockedRMWReadReq accesses (Ratio)

board.cache_hierarchy.l1dcaches.LockedRMWReadReq.avgMshrMissLatency::processor.cores.core.data 274475.250000 # average LockedRMWReadReq mshr miss latency ((Tick/Count))

board.cache_hierarchy.l1dcaches.LockedRMWReadReq.avgMshrMissLatency::total 274475.250000 # average LockedRMWReadReq mshr miss latency ((Tick/Count))

board.cache_hierarchy.l1dcaches.LockedRMWWriteReq.hits::processor.cores.core.data 30 # number of LockedRMWWriteReq hits (Count)

board.cache_hierarchy.l1dcaches.LockedRMWWriteReq.hits::total 30 # number of LockedRMWWriteReq hits (Count)

board.cache_hierarchy.l1dcaches.LockedRMWWriteReq.accesses::processor.cores.core.data 30 # number of LockedRMWWriteReq accesses(hits+misses) (Count)

board.cache_hierarchy.l1dcaches.LockedRMWWriteReq.accesses::total 30 # number of LockedRMWWriteReq accesses(hits+misses) (Count)

board.cache_hierarchy.l1dcaches.ReadReq.hits::processor.cores.core.data 3280590 # number of ReadReq hits (Count)

board.cache_hierarchy.l1dcaches.ReadReq.hits::total 3280590 # number of ReadReq hits (Count)

board.cache_hierarchy.l1dcaches.ReadReq.misses::processor.cores.core.data 3834 # number of ReadReq misses (Count)

board.cache_hierarchy.l1dcaches.ReadReq.misses::total 3834 # number of ReadReq misses (Count)

board.cache_hierarchy.l1dcaches.ReadReq.missLatency::processor.cores.core.data 436790772 # number of ReadReq miss ticks (Tick)

board.cache_hierarchy.l1dcaches.ReadReq.missLatency::total 436790772 # number of ReadReq miss ticks (Tick)

board.cache_hierarchy.l1dcaches.ReadReq.accesses::processor.cores.core.data 3284424 # number of ReadReq accesses(hits+misses) (Count)

board.cache_hierarchy.l1dcaches.ReadReq.accesses::total 3284424 # number of ReadReq accesses(hits+misses) (Count)

board.cache_hierarchy.l1dcaches.ReadReq.missRate::processor.cores.core.data 0.001167 # miss rate for ReadReq accesses (Ratio)

board.cache_hierarchy.l1dcaches.ReadReq.missRate::total 0.001167 # miss rate for ReadReq accesses (Ratio)

board.cache_hierarchy.l1dcaches.ReadReq.avgMissLatency::processor.cores.core.data 113925.605634 # average ReadReq miss latency ((Tick/Count))

board.cache_hierarchy.l1dcaches.ReadReq.avgMissLatency::total 113925.605634 # average ReadReq miss latency ((Tick/Count))

board.cache_hierarchy.l1dcaches.ReadReq.mshrHits::processor.cores.core.data 2024 # number of ReadReq MSHR hits (Count)

board.cache_hierarchy.l1dcaches.ReadReq.mshrHits::total 2024 # number of ReadReq MSHR hits (Count)

board.cache_hierarchy.l1dcaches.ReadReq.mshrMisses::processor.cores.core.data 1810 # number of ReadReq MSHR misses (Count)

board.cache_hierarchy.l1dcaches.ReadReq.mshrMisses::total 1810 # number of ReadReq MSHR misses (Count)

board.cache_hierarchy.l1dcaches.ReadReq.mshrMissLatency::processor.cores.core.data 220885893 # number of ReadReq MSHR miss ticks (Tick)

board.cache_hierarchy.l1dcaches.ReadReq.mshrMissLatency::total 220885893 # number of ReadReq MSHR miss ticks (Tick)

board.cache_hierarchy.l1dcaches.ReadReq.mshrMissRate::processor.cores.core.data 0.000551 # mshr miss rate for ReadReq accesses (Ratio)

board.cache_hierarchy.l1dcaches.ReadReq.mshrMissRate::total 0.000551 # mshr miss rate for ReadReq accesses (Ratio)

board.cache_hierarchy.l1dcaches.ReadReq.avgMshrMissLatency::processor.cores.core.data 122036.404972 # average ReadReq mshr miss latency ((Tick/Count))

board.cache_hierarchy.l1dcaches.ReadReq.avgMshrMissLatency::total 122036.404972 # average ReadReq mshr miss latency ((Tick/Count))

board.cache_hierarchy.l1dcaches.WriteReq.hits::processor.cores.core.data 4203781 # number of WriteReq hits (Count)

board.cache_hierarchy.l1dcaches.WriteReq.hits::total 4203781 # number of WriteReq hits (Count)

board.cache_hierarchy.l1dcaches.WriteReq.misses::processor.cores.core.data 1355 # number of WriteReq misses (Count)

board.cache_hierarchy.l1dcaches.WriteReq.misses::total 1355 # number of WriteReq misses (Count)

board.cache_hierarchy.l1dcaches.WriteReq.missLatency::processor.cores.core.data 189424386 # number of WriteReq miss ticks (Tick)

board.cache_hierarchy.l1dcaches.WriteReq.missLatency::total 189424386 # number of WriteReq miss ticks (Tick)

board.cache_hierarchy.l1dcaches.WriteReq.accesses::processor.cores.core.data 4205136 # number of WriteReq accesses(hits+misses) (Count)

board.cache_hierarchy.l1dcaches.WriteReq.accesses::total 4205136 # number of WriteReq accesses(hits+misses) (Count)

board.cache_hierarchy.l1dcaches.WriteReq.missRate::processor.cores.core.data 0.000322 # miss rate for WriteReq accesses (Ratio)

board.cache_hierarchy.l1dcaches.WriteReq.missRate::total 0.000322 # miss rate for WriteReq accesses (Ratio)

board.cache_hierarchy.l1dcaches.WriteReq.avgMissLatency::processor.cores.core.data 139796.594834 # average WriteReq miss latency ((Tick/Count))

board.cache_hierarchy.l1dcaches.WriteReq.avgMissLatency::total 139796.594834 # average WriteReq miss latency ((Tick/Count))

board.cache_hierarchy.l1dcaches.WriteReq.mshrHits::processor.cores.core.data 216 # number of WriteReq MSHR hits (Count)

board.cache_hierarchy.l1dcaches.WriteReq.mshrHits::total 216 # number of WriteReq MSHR hits (Count)

board.cache_hierarchy.l1dcaches.WriteReq.mshrMisses::processor.cores.core.data 1139 # number of WriteReq MSHR misses (Count)

board.cache_hierarchy.l1dcaches.WriteReq.mshrMisses::total 1139 # number of WriteReq MSHR misses (Count)

board.cache_hierarchy.l1dcaches.WriteReq.mshrMissLatency::processor.cores.core.data 155896947 # number of WriteReq MSHR miss ticks (Tick)

board.cache_hierarchy.l1dcaches.WriteReq.mshrMissLatency::total 155896947 # number of WriteReq MSHR miss ticks (Tick)

board.cache_hierarchy.l1dcaches.WriteReq.mshrMissRate::processor.cores.core.data 0.000271 # mshr miss rate for WriteReq accesses (Ratio)

board.cache_hierarchy.l1dcaches.WriteReq.mshrMissRate::total 0.000271 # mshr miss rate for WriteReq accesses (Ratio)

board.cache_hierarchy.l1dcaches.WriteReq.avgMshrMissLatency::processor.cores.core.data 136871.770852 # average WriteReq mshr miss latency ((Tick/Count))

board.cache_hierarchy.l1dcaches.WriteReq.avgMshrMissLatency::total 136871.770852 # average WriteReq mshr miss latency ((Tick/Count))

board.cache_hierarchy.l1dcaches.power_state.pwrStateResidencyTicks::UNDEFINED 5909877540 # Cumulative time (in ticks) in various power states (Tick)

board.cache_hierarchy.l1dcaches.prefetcher.demandMshrMisses 2949 # demands not covered by prefetchs (Count)

board.cache_hierarchy.l1dcaches.prefetcher.pfIssued 504094 # number of hwpf issued (Count)

board.cache_hierarchy.l1dcaches.prefetcher.pfUnused 19 # number of HardPF blocks evicted w/o reference (Count)

board.cache_hierarchy.l1dcaches.prefetcher.pfUseful 128794 # number of useful prefetch (Count)

board.cache_hierarchy.l1dcaches.prefetcher.pfUsefulButMiss 0 # number of hit on prefetch but cache block is not in an usable state (Count)

board.cache_hierarchy.l1dcaches.prefetcher.accuracy 0.255496 # accuracy of the prefetcher (Count)

board.cache_hierarchy.l1dcaches.prefetcher.coverage 0.977616 # coverage brought by this prefetcher (Count)

board.cache_hierarchy.l1dcaches.prefetcher.pfHitInCache 307508 # number of prefetches hitting in cache (Count)

board.cache_hierarchy.l1dcaches.prefetcher.pfHitInMSHR 67468 # number of prefetches hitting in a MSHR (Count)

board.cache_hierarchy.l1dcaches.prefetcher.pfHitInWB 0 # number of prefetches hit in the Write Buffer (Count)

board.cache_hierarchy.l1dcaches.prefetcher.pfLate 374976 # number of late prefetches (hitting in cache, MSHR or WB) (Count)

board.cache_hierarchy.l1dcaches.prefetcher.pfIdentified 504175 # number of prefetch candidates identified (Count)

board.cache_hierarchy.l1dcaches.prefetcher.pfBufferHit 57 # number of redundant prefetches already in prefetch queue (Count)

board.cache_hierarchy.l1dcaches.prefetcher.pfInCache 0 # number of redundant prefetches already in cache/mshr dropped (Count)

board.cache_hierarchy.l1dcaches.prefetcher.pfRemovedDemand 5 # number of prefetches dropped due to a demand for the same address (Count)

board.cache_hierarchy.l1dcaches.prefetcher.pfRemovedFull 0 # number of prefetches dropped due to prefetch queue size (Count)

board.cache_hierarchy.l1dcaches.prefetcher.pfSpanPage 20489 # number of prefetches that crossed the page (Count)

board.cache_hierarchy.l1dcaches.prefetcher.pfUsefulSpanPage 20482 # number of prefetches that is useful and crossed the page (Count)

board.cache_hierarchy.l1dcaches.prefetcher.power_state.pwrStateResidencyTicks::UNDEFINED 5909877540 # Cumulative time (in ticks) in various power states (Tick)

board.cache_hierarchy.l1dcaches.tags.tagsInUse 508.418193 # Average ticks per tags in use ((Tick/Count))

board.cache_hierarchy.l1dcaches.tags.totalRefs 7616500 # Total number of references to valid blocks. (Count)

board.cache_hierarchy.l1dcaches.tags.sampledRefs 132068 # Sample count of references to valid blocks. (Count)

board.cache_hierarchy.l1dcaches.tags.avgRefs 57.671048 # Average number of references to valid blocks. ((Count/Count))

board.cache_hierarchy.l1dcaches.tags.warmupTick 249417 # The tick when the warmup percentage was hit. (Tick)

board.cache_hierarchy.l1dcaches.tags.occupancies::cache_hierarchy.l1dcaches.prefetcher 490.482657 # Average occupied blocks per tick, per requestor ((Count/Tick))

board.cache_hierarchy.l1dcaches.tags.occupancies::processor.cores.core.data 17.935537 # Average occupied blocks per tick, per requestor ((Count/Tick))

board.cache_hierarchy.l1dcaches.tags.avgOccs::cache_hierarchy.l1dcaches.prefetcher 0.957974 # Average percentage of cache occupancy ((Ratio/Tick))

board.cache_hierarchy.l1dcaches.tags.avgOccs::processor.cores.core.data 0.035030 # Average percentage of cache occupancy ((Ratio/Tick))

board.cache_hierarchy.l1dcaches.tags.avgOccs::total 0.993004 # Average percentage of cache occupancy ((Ratio/Tick))

board.cache_hierarchy.l1dcaches.tags.occupanciesTaskId::1022 421 # Occupied blocks per task id (Count)

board.cache_hierarchy.l1dcaches.tags.occupanciesTaskId::1024 91 # Occupied blocks per task id (Count)

board.cache_hierarchy.l1dcaches.tags.ageTaskId_1022::0 4 # Occupied blocks per task id, per block age (Count)

board.cache_hierarchy.l1dcaches.tags.ageTaskId_1022::1 417 # Occupied blocks per task id, per block age (Count)

board.cache_hierarchy.l1dcaches.tags.ageTaskId_1024::0 43 # Occupied blocks per task id, per block age (Count)

board.cache_hierarchy.l1dcaches.tags.ageTaskId_1024::1 47 # Occupied blocks per task id, per block age (Count)

board.cache_hierarchy.l1dcaches.tags.ageTaskId_1024::3 1 # Occupied blocks per task id, per block age (Count)

board.cache_hierarchy.l1dcaches.tags.ratioOccsTaskId::1022 0.822266 # Ratio of occupied blocks and all blocks, per task id (Ratio)

board.cache_hierarchy.l1dcaches.tags.ratioOccsTaskId::1024 0.177734 # Ratio of occupied blocks and all blocks, per task id (Ratio)

board.cache_hierarchy.l1dcaches.tags.tagAccesses 60049028 # Number of tag accesses (Count)

board.cache_hierarchy.l1dcaches.tags.dataAccesses 60049028 # Number of data accesses (Count)

board.cache_hierarchy.l1dcaches.tags.power_state.pwrStateResidencyTicks::UNDEFINED 5909877540 # Cumulative time (in ticks) in various power states (Tick)

board.cache_hierarchy.l1icaches.demandHits::processor.cores.core.inst 4231544 # number of demand (read+write) hits (Count)

board.cache_hierarchy.l1icaches.demandHits::total 4231544 # number of demand (read+write) hits (Count)

board.cache_hierarchy.l1icaches.overallHits::processor.cores.core.inst 4231544 # number of overall hits (Count)

board.cache_hierarchy.l1icaches.overallHits::total 4231544 # number of overall hits (Count)

board.cache_hierarchy.l1icaches.demandMisses::processor.cores.core.inst 977 # number of demand (read+write) misses (Count)

board.cache_hierarchy.l1icaches.demandMisses::total 977 # number of demand (read+write) misses (Count)

board.cache_hierarchy.l1icaches.overallMisses::processor.cores.core.inst 977 # number of overall misses (Count)

board.cache_hierarchy.l1icaches.overallMisses::total 977 # number of overall misses (Count)

board.cache_hierarchy.l1icaches.demandMissLatency::processor.cores.core.inst 113613606 # number of demand (read+write) miss ticks (Tick)

board.cache_hierarchy.l1icaches.demandMissLatency::total 113613606 # number of demand (read+write) miss ticks (Tick)

board.cache_hierarchy.l1icaches.overallMissLatency::processor.cores.core.inst 113613606 # number of overall miss ticks (Tick)

board.cache_hierarchy.l1icaches.overallMissLatency::total 113613606 # number of overall miss ticks (Tick)

board.cache_hierarchy.l1icaches.demandAccesses::processor.cores.core.inst 4232521 # number of demand (read+write) accesses (Count)

board.cache_hierarchy.l1icaches.demandAccesses::total 4232521 # number of demand (read+write) accesses (Count)

board.cache_hierarchy.l1icaches.overallAccesses::processor.cores.core.inst 4232521 # number of overall (read+write) accesses (Count)

board.cache_hierarchy.l1icaches.overallAccesses::total 4232521 # number of overall (read+write) accesses (Count)

board.cache_hierarchy.l1icaches.demandMissRate::processor.cores.core.inst 0.000231 # miss rate for demand accesses (Ratio)

board.cache_hierarchy.l1icaches.demandMissRate::total 0.000231 # miss rate for demand accesses (Ratio)

board.cache_hierarchy.l1icaches.overallMissRate::processor.cores.core.inst 0.000231 # miss rate for overall accesses (Ratio)

board.cache_hierarchy.l1icaches.overallMissRate::total 0.000231 # miss rate for overall accesses (Ratio)

board.cache_hierarchy.l1icaches.demandAvgMissLatency::processor.cores.core.inst 116288.235415 # average overall miss latency in ticks ((Tick/Count))

board.cache_hierarchy.l1icaches.demandAvgMissLatency::total 116288.235415 # average overall miss latency in ticks ((Tick/Count))

board.cache_hierarchy.l1icaches.overallAvgMissLatency::processor.cores.core.inst 116288.235415 # average overall miss latency ((Tick/Count))

board.cache_hierarchy.l1icaches.overallAvgMissLatency::total 116288.235415 # average overall miss latency ((Tick/Count))

board.cache_hierarchy.l1icaches.blockedCycles::no_mshrs 0 # number of cycles access was blocked (Cycle)

board.cache_hierarchy.l1icaches.blockedCycles::no_targets 0 # number of cycles access was blocked (Cycle)

board.cache_hierarchy.l1icaches.blockedCauses::no_mshrs 0 # number of times access was blocked (Count)

board.cache_hierarchy.l1icaches.blockedCauses::no_targets 0 # number of times access was blocked (Count)

board.cache_hierarchy.l1icaches.avgBlocked::no_mshrs nan # average number of cycles each access was blocked ((Cycle/Count))

board.cache_hierarchy.l1icaches.avgBlocked::no_targets nan # average number of cycles each access was blocked ((Cycle/Count))

board.cache_hierarchy.l1icaches.demandMshrHits::processor.cores.core.inst 224 # number of demand (read+write) MSHR hits (Count)

board.cache_hierarchy.l1icaches.demandMshrHits::total 224 # number of demand (read+write) MSHR hits (Count)

board.cache_hierarchy.l1icaches.overallMshrHits::processor.cores.core.inst 224 # number of overall MSHR hits (Count)

board.cache_hierarchy.l1icaches.overallMshrHits::total 224 # number of overall MSHR hits (Count)

board.cache_hierarchy.l1icaches.demandMshrMisses::processor.cores.core.inst 753 # number of demand (read+write) MSHR misses (Count)

board.cache_hierarchy.l1icaches.demandMshrMisses::total 753 # number of demand (read+write) MSHR misses (Count)

board.cache_hierarchy.l1icaches.overallMshrMisses::processor.cores.core.inst 753 # number of overall MSHR misses (Count)

board.cache_hierarchy.l1icaches.overallMshrMisses::total 753 # number of overall MSHR misses (Count)

board.cache_hierarchy.l1icaches.demandMshrMissLatency::processor.cores.core.inst 90491751 # number of demand (read+write) MSHR miss ticks (Tick)

board.cache_hierarchy.l1icaches.demandMshrMissLatency::total 90491751 # number of demand (read+write) MSHR miss ticks (Tick)

board.cache_hierarchy.l1icaches.overallMshrMissLatency::processor.cores.core.inst 90491751 # number of overall MSHR miss ticks (Tick)

board.cache_hierarchy.l1icaches.overallMshrMissLatency::total 90491751 # number of overall MSHR miss ticks (Tick)

board.cache_hierarchy.l1icaches.demandMshrMissRate::processor.cores.core.inst 0.000178 # mshr miss ratio for demand accesses (Ratio)

board.cache_hierarchy.l1icaches.demandMshrMissRate::total 0.000178 # mshr miss ratio for demand accesses (Ratio)

board.cache_hierarchy.l1icaches.overallMshrMissRate::processor.cores.core.inst 0.000178 # mshr miss ratio for overall accesses (Ratio)

board.cache_hierarchy.l1icaches.overallMshrMissRate::total 0.000178 # mshr miss ratio for overall accesses (Ratio)

board.cache_hierarchy.l1icaches.demandAvgMshrMissLatency::processor.cores.core.inst 120174.968127 # average overall mshr miss latency ((Tick/Count))

board.cache_hierarchy.l1icaches.demandAvgMshrMissLatency::total 120174.968127 # average overall mshr miss latency ((Tick/Count))

board.cache_hierarchy.l1icaches.overallAvgMshrMissLatency::processor.cores.core.inst 120174.968127 # average overall mshr miss latency ((Tick/Count))

board.cache_hierarchy.l1icaches.overallAvgMshrMissLatency::total 120174.968127 # average overall mshr miss latency ((Tick/Count))

board.cache_hierarchy.l1icaches.replacements 239 # number of replacements (Count)

board.cache_hierarchy.l1icaches.ReadReq.hits::processor.cores.core.inst 4231544 # number of ReadReq hits (Count)

board.cache_hierarchy.l1icaches.ReadReq.hits::total 4231544 # number of ReadReq hits (Count)

board.cache_hierarchy.l1icaches.ReadReq.misses::processor.cores.core.inst 977 # number of ReadReq misses (Count)

board.cache_hierarchy.l1icaches.ReadReq.misses::total 977 # number of ReadReq misses (Count)

board.cache_hierarchy.l1icaches.ReadReq.missLatency::processor.cores.core.inst 113613606 # number of ReadReq miss ticks (Tick)

board.cache_hierarchy.l1icaches.ReadReq.missLatency::total 113613606 # number of ReadReq miss ticks (Tick)

board.cache_hierarchy.l1icaches.ReadReq.accesses::processor.cores.core.inst 4232521 # number of ReadReq accesses(hits+misses) (Count)

board.cache_hierarchy.l1icaches.ReadReq.accesses::total 4232521 # number of ReadReq accesses(hits+misses) (Count)

board.cache_hierarchy.l1icaches.ReadReq.missRate::processor.cores.core.inst 0.000231 # miss rate for ReadReq accesses (Ratio)

board.cache_hierarchy.l1icaches.ReadReq.missRate::total 0.000231 # miss rate for ReadReq accesses (Ratio)

board.cache_hierarchy.l1icaches.ReadReq.avgMissLatency::processor.cores.core.inst 116288.235415 # average ReadReq miss latency ((Tick/Count))

board.cache_hierarchy.l1icaches.ReadReq.avgMissLatency::total 116288.235415 # average ReadReq miss latency ((Tick/Count))

board.cache_hierarchy.l1icaches.ReadReq.mshrHits::processor.cores.core.inst 224 # number of ReadReq MSHR hits (Count)

board.cache_hierarchy.l1icaches.ReadReq.mshrHits::total 224 # number of ReadReq MSHR hits (Count)

board.cache_hierarchy.l1icaches.ReadReq.mshrMisses::processor.cores.core.inst 753 # number of ReadReq MSHR misses (Count)

board.cache_hierarchy.l1icaches.ReadReq.mshrMisses::total 753 # number of ReadReq MSHR misses (Count)

board.cache_hierarchy.l1icaches.ReadReq.mshrMissLatency::processor.cores.core.inst 90491751 # number of ReadReq MSHR miss ticks (Tick)

board.cache_hierarchy.l1icaches.ReadReq.mshrMissLatency::total 90491751 # number of ReadReq MSHR miss ticks (Tick)

board.cache_hierarchy.l1icaches.ReadReq.mshrMissRate::processor.cores.core.inst 0.000178 # mshr miss rate for ReadReq accesses (Ratio)

board.cache_hierarchy.l1icaches.ReadReq.mshrMissRate::total 0.000178 # mshr miss rate for ReadReq accesses (Ratio)

board.cache_hierarchy.l1icaches.ReadReq.avgMshrMissLatency::processor.cores.core.inst 120174.968127 # average ReadReq mshr miss latency ((Tick/Count))

board.cache_hierarchy.l1icaches.ReadReq.avgMshrMissLatency::total 120174.968127 # average ReadReq mshr miss latency ((Tick/Count))

board.cache_hierarchy.l1icaches.power_state.pwrStateResidencyTicks::UNDEFINED 5909877540 # Cumulative time (in ticks) in various power states (Tick)

board.cache_hierarchy.l1icaches.prefetcher.demandMshrMisses 753 # demands not covered by prefetchs (Count)

board.cache_hierarchy.l1icaches.prefetcher.pfIssued 0 # number of hwpf issued (Count)

board.cache_hierarchy.l1icaches.prefetcher.pfUseful 0 # number of useful prefetch (Count)

board.cache_hierarchy.l1icaches.prefetcher.pfUsefulButMiss 0 # number of hit on prefetch but cache block is not in an usable state (Count)

board.cache_hierarchy.l1icaches.prefetcher.accuracy nan # accuracy of the prefetcher (Count)

board.cache_hierarchy.l1icaches.prefetcher.coverage 0 # coverage brought by this prefetcher (Count)

board.cache_hierarchy.l1icaches.prefetcher.pfHitInCache 0 # number of prefetches hitting in cache (Count)

board.cache_hierarchy.l1icaches.prefetcher.pfHitInMSHR 0 # number of prefetches hitting in a MSHR (Count)

board.cache_hierarchy.l1icaches.prefetcher.pfHitInWB 0 # number of prefetches hit in the Write Buffer (Count)

board.cache_hierarchy.l1icaches.prefetcher.pfLate 0 # number of late prefetches (hitting in cache, MSHR or WB) (Count)

board.cache_hierarchy.l1icaches.prefetcher.pfIdentified 0 # number of prefetch candidates identified (Count)

board.cache_hierarchy.l1icaches.prefetcher.pfBufferHit 0 # number of redundant prefetches already in prefetch queue (Count)

board.cache_hierarchy.l1icaches.prefetcher.pfInCache 0 # number of redundant prefetches already in cache/mshr dropped (Count)

board.cache_hierarchy.l1icaches.prefetcher.pfRemovedDemand 0 # number of prefetches dropped due to a demand for the same address (Count)

board.cache_hierarchy.l1icaches.prefetcher.pfRemovedFull 0 # number of prefetches dropped due to prefetch queue size (Count)

board.cache_hierarchy.l1icaches.prefetcher.pfSpanPage 0 # number of prefetches that crossed the page (Count)

board.cache_hierarchy.l1icaches.prefetcher.pfUsefulSpanPage 0 # number of prefetches that is useful and crossed the page (Count)

board.cache_hierarchy.l1icaches.prefetcher.power_state.pwrStateResidencyTicks::UNDEFINED 5909877540 # Cumulative time (in ticks) in various power states (Tick)

board.cache_hierarchy.l1icaches.tags.tagsInUse 469.941952 # Average ticks per tags in use ((Tick/Count))

board.cache_hierarchy.l1icaches.tags.totalRefs 4232297 # Total number of references to valid blocks. (Count)

board.cache_hierarchy.l1icaches.tags.sampledRefs 753 # Sample count of references to valid blocks. (Count)

board.cache_hierarchy.l1icaches.tags.avgRefs 5620.580345 # Average number of references to valid blocks. ((Count/Count))

board.cache_hierarchy.l1icaches.tags.warmupTick 123543 # The tick when the warmup percentage was hit. (Tick)

board.cache_hierarchy.l1icaches.tags.occupancies::processor.cores.core.inst 469.941952 # Average occupied blocks per tick, per requestor ((Count/Tick))

board.cache_hierarchy.l1icaches.tags.avgOccs::processor.cores.core.inst 0.917855 # Average percentage of cache occupancy ((Ratio/Tick))

board.cache_hierarchy.l1icaches.tags.avgOccs::total 0.917855 # Average percentage of cache occupancy ((Ratio/Tick))

board.cache_hierarchy.l1icaches.tags.occupanciesTaskId::1024 511 # Occupied blocks per task id (Count)

board.cache_hierarchy.l1icaches.tags.ageTaskId_1024::0 106 # Occupied blocks per task id, per block age (Count)

board.cache_hierarchy.l1icaches.tags.ageTaskId_1024::1 98 # Occupied blocks per task id, per block age (Count)

board.cache_hierarchy.l1icaches.tags.ageTaskId_1024::3 307 # Occupied blocks per task id, per block age (Count)

board.cache_hierarchy.l1icaches.tags.ratioOccsTaskId::1024 0.998047 # Ratio of occupied blocks and all blocks, per task id (Ratio)

board.cache_hierarchy.l1icaches.tags.tagAccesses 33860921 # Number of tag accesses (Count)

board.cache_hierarchy.l1icaches.tags.dataAccesses 33860921 # Number of data accesses (Count)

board.cache_hierarchy.l1icaches.tags.power_state.pwrStateResidencyTicks::UNDEFINED 5909877540 # Cumulative time (in ticks) in various power states (Tick)

board.cache_hierarchy.l2bus.transDist::ReadResp 131681 # Transaction distribution (Count)

board.cache_hierarchy.l2bus.transDist::WritebackDirty 131412 # Transaction distribution (Count)

board.cache_hierarchy.l2bus.transDist::CleanEvict 129099 # Transaction distribution (Count)

board.cache_hierarchy.l2bus.transDist::HardPFReq 139496 # Transaction distribution (Count)

board.cache_hierarchy.l2bus.transDist::UpgradeReq 3 # Transaction distribution (Count)

board.cache_hierarchy.l2bus.transDist::UpgradeResp 3 # Transaction distribution (Count)

board.cache_hierarchy.l2bus.transDist::ReadExReq 1140 # Transaction distribution (Count)

board.cache_hierarchy.l2bus.transDist::ReadExResp 1140 # Transaction distribution (Count)

board.cache_hierarchy.l2bus.transDist::ReadSharedReq 131681 # Transaction distribution (Count)

board.cache_hierarchy.l2bus.pktCount_board.cache_hierarchy.l1icaches.mem_side_port::board.cache_hierarchy.l2cache.cpu_side_port 1742 # Packet count per connected requestor and responder (Count)

board.cache_hierarchy.l2bus.pktCount_board.cache_hierarchy.l1dcaches.mem_side_port::board.cache_hierarchy.l2cache.cpu_side_port 395698 # Packet count per connected requestor and responder (Count)

board.cache_hierarchy.l2bus.pktCount::total 397440 # Packet count per connected requestor and responder (Count)

board.cache_hierarchy.l2bus.pktSize_board.cache_hierarchy.l1icaches.mem_side_port::board.cache_hierarchy.l2cache.cpu_side_port 48000 # Cumulative packet size per connected requestor and responder (Byte)

board.cache_hierarchy.l2bus.pktSize_board.cache_hierarchy.l1dcaches.mem_side_port::board.cache_hierarchy.l2cache.cpu_side_port 12658240 # Cumulative packet size per connected requestor and responder (Byte)

board.cache_hierarchy.l2bus.pktSize::total 12706240 # Cumulative packet size per connected requestor and responder (Byte)

board.cache_hierarchy.l2bus.snoops 268215 # Total snoops (Count)

board.cache_hierarchy.l2bus.snoopTraffic 4204672 # Total snoop traffic (Byte)

board.cache_hierarchy.l2bus.snoopFanout::samples 401036 # Request fanout histogram (Count)

board.cache_hierarchy.l2bus.snoopFanout::mean 0.001207 # Request fanout histogram (Count)

board.cache_hierarchy.l2bus.snoopFanout::stdev 0.034719 # Request fanout histogram (Count)

board.cache_hierarchy.l2bus.snoopFanout::underflows 0 0.00% 0.00% # Request fanout histogram (Count)

board.cache_hierarchy.l2bus.snoopFanout::0 400552 99.88% 99.88% # Request fanout histogram (Count)

board.cache_hierarchy.l2bus.snoopFanout::1 484 0.12% 100.00% # Request fanout histogram (Count)

board.cache_hierarchy.l2bus.snoopFanout::2 0 0.00% 100.00% # Request fanout histogram (Count)

board.cache_hierarchy.l2bus.snoopFanout::3 0 0.00% 100.00% # Request fanout histogram (Count)

board.cache_hierarchy.l2bus.snoopFanout::4 0 0.00% 100.00% # Request fanout histogram (Count)

board.cache_hierarchy.l2bus.snoopFanout::overflows 0 0.00% 100.00% # Request fanout histogram (Count)

board.cache_hierarchy.l2bus.snoopFanout::min_value 0 # Request fanout histogram (Count)

board.cache_hierarchy.l2bus.snoopFanout::max_value 1 # Request fanout histogram (Count)

board.cache_hierarchy.l2bus.snoopFanout::total 401036 # Request fanout histogram (Count)

board.cache_hierarchy.l2bus.power_state.pwrStateResidencyTicks::UNDEFINED 5909877540 # Cumulative time (in ticks) in various power states (Tick)

board.cache_hierarchy.l2bus.reqLayer0.occupancy 171649348 # Layer occupancy (ticks) (Tick)

board.cache_hierarchy.l2bus.reqLayer0.utilization 0.0 # Layer utilization (Ratio)

board.cache_hierarchy.l2bus.respLayer0.occupancy 752247 # Layer occupancy (ticks) (Tick)

board.cache_hierarchy.l2bus.respLayer0.utilization 0.0 # Layer utilization (Ratio)

board.cache_hierarchy.l2bus.respLayer1.occupancy 131936931 # Layer occupancy (ticks) (Tick)

board.cache_hierarchy.l2bus.respLayer1.utilization 0.0 # Layer utilization (Ratio)

board.cache_hierarchy.l2bus.snoop_filter.totRequests 264619 # Total number of requests made to the snoop filter. (Count)

board.cache_hierarchy.l2bus.snoop_filter.hitSingleRequests 131796 # Number of requests hitting in the snoop filter with a single holder of the requested data. (Count)

board.cache_hierarchy.l2bus.snoop_filter.hitMultiRequests 6 # Number of requests hitting in the snoop filter with multiple (>1) holders of the requested data. (Count)

board.cache_hierarchy.l2bus.snoop_filter.totSnoops 477 # Total number of snoops made to the snoop filter. (Count)

board.cache_hierarchy.l2bus.snoop_filter.hitSingleSnoops 477 # Number of snoops hitting in the snoop filter with a single holder of the requested data. (Count)

board.cache_hierarchy.l2bus.snoop_filter.hitMultiSnoops 0 # Number of snoops hitting in the snoop filter with multiple (>1) holders of the requested data. (Count)

board.cache_hierarchy.l2cache.demandHits::cache_hierarchy.l1dcaches.prefetcher 107971 # number of demand (read+write) hits (Count)

board.cache_hierarchy.l2cache.demandHits::processor.cores.core.data 30 # number of demand (read+write) hits (Count)

board.cache_hierarchy.l2cache.demandHits::total 108001 # number of demand (read+write) hits (Count)

board.cache_hierarchy.l2cache.overallHits::cache_hierarchy.l1dcaches.prefetcher 107971 # number of overall hits (Count)

board.cache_hierarchy.l2cache.overallHits::processor.cores.core.data 30 # number of overall hits (Count)

board.cache_hierarchy.l2cache.overallHits::total 108001 # number of overall hits (Count)

board.cache_hierarchy.l2cache.demandMisses::cache_hierarchy.l1dcaches.prefetcher 21147 # number of demand (read+write) misses (Count)

board.cache_hierarchy.l2cache.demandMisses::processor.cores.core.inst 750 # number of demand (read+write) misses (Count)

board.cache_hierarchy.l2cache.demandMisses::processor.cores.core.data 2920 # number of demand (read+write) misses (Count)

board.cache_hierarchy.l2cache.demandMisses::total 24817 # number of demand (read+write) misses (Count)

board.cache_hierarchy.l2cache.overallMisses::cache_hierarchy.l1dcaches.prefetcher 21147 # number of overall misses (Count)

board.cache_hierarchy.l2cache.overallMisses::processor.cores.core.inst 750 # number of overall misses (Count)

board.cache_hierarchy.l2cache.overallMisses::processor.cores.core.data 2920 # number of overall misses (Count)

board.cache_hierarchy.l2cache.overallMisses::total 24817 # number of overall misses (Count)

board.cache_hierarchy.l2cache.demandMissLatency::cache_hierarchy.l1dcaches.prefetcher 1998983928 # number of demand (read+write) miss ticks (Tick)

board.cache_hierarchy.l2cache.demandMissLatency::processor.cores.core.inst 89984925 # number of demand (read+write) miss ticks (Tick)

board.cache_hierarchy.l2cache.demandMissLatency::processor.cores.core.data 374206419 # number of demand (read+write) miss ticks (Tick)

board.cache_hierarchy.l2cache.demandMissLatency::total 2463175272 # number of demand (read+write) miss ticks (Tick)

board.cache_hierarchy.l2cache.overallMissLatency::cache_hierarchy.l1dcaches.prefetcher 1998983928 # number of overall miss ticks (Tick)

board.cache_hierarchy.l2cache.overallMissLatency::processor.cores.core.inst 89984925 # number of overall miss ticks (Tick)

board.cache_hierarchy.l2cache.overallMissLatency::processor.cores.core.data 374206419 # number of overall miss ticks (Tick)

board.cache_hierarchy.l2cache.overallMissLatency::total 2463175272 # number of overall miss ticks (Tick)

board.cache_hierarchy.l2cache.demandAccesses::cache_hierarchy.l1dcaches.prefetcher 129118 # number of demand (read+write) accesses (Count)

board.cache_hierarchy.l2cache.demandAccesses::processor.cores.core.inst 750 # number of demand (read+write) accesses (Count)

board.cache_hierarchy.l2cache.demandAccesses::processor.cores.core.data 2950 # number of demand (read+write) accesses (Count)

board.cache_hierarchy.l2cache.demandAccesses::total 132818 # number of demand (read+write) accesses (Count)

board.cache_hierarchy.l2cache.overallAccesses::cache_hierarchy.l1dcaches.prefetcher 129118 # number of overall (read+write) accesses (Count)

board.cache_hierarchy.l2cache.overallAccesses::processor.cores.core.inst 750 # number of overall (read+write) accesses (Count)

board.cache_hierarchy.l2cache.overallAccesses::processor.cores.core.data 2950 # number of overall (read+write) accesses (Count)

board.cache_hierarchy.l2cache.overallAccesses::total 132818 # number of overall (read+write) accesses (Count)

board.cache_hierarchy.l2cache.demandMissRate::cache_hierarchy.l1dcaches.prefetcher 0.163780 # miss rate for demand accesses (Ratio)

board.cache_hierarchy.l2cache.demandMissRate::processor.cores.core.inst 1 # miss rate for demand accesses (Ratio)

board.cache_hierarchy.l2cache.demandMissRate::processor.cores.core.data 0.989831 # miss rate for demand accesses (Ratio)

board.cache_hierarchy.l2cache.demandMissRate::total 0.186850 # miss rate for demand accesses (Ratio)

board.cache_hierarchy.l2cache.overallMissRate::cache_hierarchy.l1dcaches.prefetcher 0.163780 # miss rate for overall accesses (Ratio)

board.cache_hierarchy.l2cache.overallMissRate::processor.cores.core.inst 1 # miss rate for overall accesses (Ratio)

board.cache_hierarchy.l2cache.overallMissRate::processor.cores.core.data 0.989831 # miss rate for overall accesses (Ratio)

board.cache_hierarchy.l2cache.overallMissRate::total 0.186850 # miss rate for overall accesses (Ratio)

board.cache_hierarchy.l2cache.demandAvgMissLatency::cache_hierarchy.l1dcaches.prefetcher 94528.014754 # average overall miss latency in ticks ((Tick/Count))

board.cache_hierarchy.l2cache.demandAvgMissLatency::processor.cores.core.inst 119979.900000 # average overall miss latency in ticks ((Tick/Count))

board.cache_hierarchy.l2cache.demandAvgMissLatency::processor.cores.core.data 128152.883219 # average overall miss latency in ticks ((Tick/Count))

board.cache_hierarchy.l2cache.demandAvgMissLatency::total 99253.546843 # average overall miss latency in ticks ((Tick/Count))

board.cache_hierarchy.l2cache.overallAvgMissLatency::cache_hierarchy.l1dcaches.prefetcher 94528.014754 # average overall miss latency ((Tick/Count))

board.cache_hierarchy.l2cache.overallAvgMissLatency::processor.cores.core.inst 119979.900000 # average overall miss latency ((Tick/Count))

board.cache_hierarchy.l2cache.overallAvgMissLatency::processor.cores.core.data 128152.883219 # average overall miss latency ((Tick/Count))

board.cache_hierarchy.l2cache.overallAvgMissLatency::total 99253.546843 # average overall miss latency ((Tick/Count))

board.cache_hierarchy.l2cache.blockedCycles::no_mshrs 0 # number of cycles access was blocked (Cycle)

board.cache_hierarchy.l2cache.blockedCycles::no_targets 0 # number of cycles access was blocked (Cycle)

board.cache_hierarchy.l2cache.blockedCauses::no_mshrs 0 # number of times access was blocked (Count)

board.cache_hierarchy.l2cache.blockedCauses::no_targets 0 # number of times access was blocked (Count)

board.cache_hierarchy.l2cache.avgBlocked::no_mshrs nan # average number of cycles each access was blocked ((Cycle/Count))

board.cache_hierarchy.l2cache.avgBlocked::no_targets nan # average number of cycles each access was blocked ((Cycle/Count))

board.cache_hierarchy.l2cache.writebacks::writebacks 65695 # number of writebacks (Count)

board.cache_hierarchy.l2cache.writebacks::total 65695 # number of writebacks (Count)

board.cache_hierarchy.l2cache.demandMshrHits::cache_hierarchy.l1dcaches.prefetcher 19062 # number of demand (read+write) MSHR hits (Count)

board.cache_hierarchy.l2cache.demandMshrHits::processor.cores.core.data 2 # number of demand (read+write) MSHR hits (Count)

board.cache_hierarchy.l2cache.demandMshrHits::total 19064 # number of demand (read+write) MSHR hits (Count)

board.cache_hierarchy.l2cache.overallMshrHits::cache_hierarchy.l1dcaches.prefetcher 19062 # number of overall MSHR hits (Count)

board.cache_hierarchy.l2cache.overallMshrHits::processor.cores.core.data 2 # number of overall MSHR hits (Count)

board.cache_hierarchy.l2cache.overallMshrHits::total 19064 # number of overall MSHR hits (Count)

board.cache_hierarchy.l2cache.demandMshrMisses::cache_hierarchy.l1dcaches.prefetcher 2085 # number of demand (read+write) MSHR misses (Count)

board.cache_hierarchy.l2cache.demandMshrMisses::processor.cores.core.inst 750 # number of demand (read+write) MSHR misses (Count)

board.cache_hierarchy.l2cache.demandMshrMisses::processor.cores.core.data 2918 # number of demand (read+write) MSHR misses (Count)

board.cache_hierarchy.l2cache.demandMshrMisses::total 5753 # number of demand (read+write) MSHR misses (Count)

board.cache_hierarchy.l2cache.overallMshrMisses::cache_hierarchy.l1dcaches.prefetcher 2085 # number of overall MSHR misses (Count)

board.cache_hierarchy.l2cache.overallMshrMisses::cache_hierarchy.l2cache.prefetcher 127059 # number of overall MSHR misses (Count)

board.cache_hierarchy.l2cache.overallMshrMisses::processor.cores.core.inst 750 # number of overall MSHR misses (Count)

board.cache_hierarchy.l2cache.overallMshrMisses::processor.cores.core.data 2918 # number of overall MSHR misses (Count)

board.cache_hierarchy.l2cache.overallMshrMisses::total 132812 # number of overall MSHR misses (Count)

board.cache_hierarchy.l2cache.demandMshrMissLatency::cache_hierarchy.l1dcaches.prefetcher 217051383 # number of demand (read+write) MSHR miss ticks (Tick)

board.cache_hierarchy.l2cache.demandMshrMissLatency::processor.cores.core.inst 65009925 # number of demand (read+write) MSHR miss ticks (Tick)

board.cache_hierarchy.l2cache.demandMshrMissLatency::processor.cores.core.data 276818571 # number of demand (read+write) MSHR miss ticks (Tick)

board.cache_hierarchy.l2cache.demandMshrMissLatency::total 558879879 # number of demand (read+write) MSHR miss ticks (Tick)

board.cache_hierarchy.l2cache.overallMshrMissLatency::cache_hierarchy.l1dcaches.prefetcher 217051383 # number of overall MSHR miss ticks (Tick)

board.cache_hierarchy.l2cache.overallMshrMissLatency::cache_hierarchy.l2cache.prefetcher 8433567703 # number of overall MSHR miss ticks (Tick)

board.cache_hierarchy.l2cache.overallMshrMissLatency::processor.cores.core.inst 65009925 # number of overall MSHR miss ticks (Tick)

board.cache_hierarchy.l2cache.overallMshrMissLatency::processor.cores.core.data 276818571 # number of overall MSHR miss ticks (Tick)

board.cache_hierarchy.l2cache.overallMshrMissLatency::total 8992447582 # number of overall MSHR miss ticks (Tick)

board.cache_hierarchy.l2cache.demandMshrMissRate::cache_hierarchy.l1dcaches.prefetcher 0.016148 # mshr miss ratio for demand accesses (Ratio)

board.cache_hierarchy.l2cache.demandMshrMissRate::processor.cores.core.inst 1 # mshr miss ratio for demand accesses (Ratio)

board.cache_hierarchy.l2cache.demandMshrMissRate::processor.cores.core.data 0.989153 # mshr miss ratio for demand accesses (Ratio)

board.cache_hierarchy.l2cache.demandMshrMissRate::total 0.043315 # mshr miss ratio for demand accesses (Ratio)

board.cache_hierarchy.l2cache.overallMshrMissRate::cache_hierarchy.l1dcaches.prefetcher 0.016148 # mshr miss ratio for overall accesses (Ratio)

board.cache_hierarchy.l2cache.overallMshrMissRate::cache_hierarchy.l2cache.prefetcher inf # mshr miss ratio for overall accesses (Ratio)

board.cache_hierarchy.l2cache.overallMshrMissRate::processor.cores.core.inst 1 # mshr miss ratio for overall accesses (Ratio)

board.cache_hierarchy.l2cache.overallMshrMissRate::processor.cores.core.data 0.989153 # mshr miss ratio for overall accesses (Ratio)

board.cache_hierarchy.l2cache.overallMshrMissRate::total 0.999955 # mshr miss ratio for overall accesses (Ratio)

board.cache_hierarchy.l2cache.demandAvgMshrMissLatency::cache_hierarchy.l1dcaches.prefetcher 104101.382734 # average overall mshr miss latency ((Tick/Count))

board.cache_hierarchy.l2cache.demandAvgMshrMissLatency::processor.cores.core.inst 86679.900000 # average overall mshr miss latency ((Tick/Count))

board.cache_hierarchy.l2cache.demandAvgMshrMissLatency::processor.cores.core.data 94865.857094 # average overall mshr miss latency ((Tick/Count))

board.cache_hierarchy.l2cache.demandAvgMshrMissLatency::total 97145.815922 # average overall mshr miss latency ((Tick/Count))

board.cache_hierarchy.l2cache.overallAvgMshrMissLatency::cache_hierarchy.l1dcaches.prefetcher 104101.382734 # average overall mshr miss latency ((Tick/Count))

board.cache_hierarchy.l2cache.overallAvgMshrMissLatency::cache_hierarchy.l2cache.prefetcher 66375.209178 # average overall mshr miss latency ((Tick/Count))

board.cache_hierarchy.l2cache.overallAvgMshrMissLatency::processor.cores.core.inst 86679.900000 # average overall mshr miss latency ((Tick/Count))

board.cache_hierarchy.l2cache.overallAvgMshrMissLatency::processor.cores.core.data 94865.857094 # average overall mshr miss latency ((Tick/Count))

board.cache_hierarchy.l2cache.overallAvgMshrMissLatency::total 67708.095518 # average overall mshr miss latency ((Tick/Count))

board.cache_hierarchy.l2cache.replacements 128716 # number of replacements (Count)

board.cache_hierarchy.l2cache.CleanEvict.mshrMisses::writebacks 167 # number of CleanEvict MSHR misses (Count)

board.cache_hierarchy.l2cache.CleanEvict.mshrMisses::total 167 # number of CleanEvict MSHR misses (Count)

board.cache_hierarchy.l2cache.CleanEvict.mshrMissRate::writebacks inf # mshr miss rate for CleanEvict accesses (Ratio)

board.cache_hierarchy.l2cache.CleanEvict.mshrMissRate::total inf # mshr miss rate for CleanEvict accesses (Ratio)

board.cache_hierarchy.l2cache.HardPFReq.mshrMisses::cache_hierarchy.l2cache.prefetcher 127059 # number of HardPFReq MSHR misses (Count)

board.cache_hierarchy.l2cache.HardPFReq.mshrMisses::total 127059 # number of HardPFReq MSHR misses (Count)

board.cache_hierarchy.l2cache.HardPFReq.mshrMissLatency::cache_hierarchy.l2cache.prefetcher 8433567703 # number of HardPFReq MSHR miss ticks (Tick)

board.cache_hierarchy.l2cache.HardPFReq.mshrMissLatency::total 8433567703 # number of HardPFReq MSHR miss ticks (Tick)

board.cache_hierarchy.l2cache.HardPFReq.mshrMissRate::cache_hierarchy.l2cache.prefetcher inf # mshr miss rate for HardPFReq accesses (Ratio)

board.cache_hierarchy.l2cache.HardPFReq.mshrMissRate::total inf # mshr miss rate for HardPFReq accesses (Ratio)

board.cache_hierarchy.l2cache.HardPFReq.avgMshrMissLatency::cache_hierarchy.l2cache.prefetcher 66375.209178 # average HardPFReq mshr miss latency ((Tick/Count))

board.cache_hierarchy.l2cache.HardPFReq.avgMshrMissLatency::total 66375.209178 # average HardPFReq mshr miss latency ((Tick/Count))

board.cache_hierarchy.l2cache.ReadExReq.hits::processor.cores.core.data 9 # number of ReadExReq hits (Count)

board.cache_hierarchy.l2cache.ReadExReq.hits::total 9 # number of ReadExReq hits (Count)

board.cache_hierarchy.l2cache.ReadExReq.misses::processor.cores.core.data 1131 # number of ReadExReq misses (Count)

board.cache_hierarchy.l2cache.ReadExReq.misses::total 1131 # number of ReadExReq misses (Count)

board.cache_hierarchy.l2cache.ReadExReq.missLatency::processor.cores.core.data 155245599 # number of ReadExReq miss ticks (Tick)

board.cache_hierarchy.l2cache.ReadExReq.missLatency::total 155245599 # number of ReadExReq miss ticks (Tick)

board.cache_hierarchy.l2cache.ReadExReq.accesses::processor.cores.core.data 1140 # number of ReadExReq accesses(hits+misses) (Count)

board.cache_hierarchy.l2cache.ReadExReq.accesses::total 1140 # number of ReadExReq accesses(hits+misses) (Count)

board.cache_hierarchy.l2cache.ReadExReq.missRate::processor.cores.core.data 0.992105 # miss rate for ReadExReq accesses (Ratio)

board.cache_hierarchy.l2cache.ReadExReq.missRate::total 0.992105 # miss rate for ReadExReq accesses (Ratio)

board.cache_hierarchy.l2cache.ReadExReq.avgMissLatency::processor.cores.core.data 137264.013263 # average ReadExReq miss latency ((Tick/Count))

board.cache_hierarchy.l2cache.ReadExReq.avgMissLatency::total 137264.013263 # average ReadExReq miss latency ((Tick/Count))

board.cache_hierarchy.l2cache.ReadExReq.mshrMisses::processor.cores.core.data 1131 # number of ReadExReq MSHR misses (Count)

board.cache_hierarchy.l2cache.ReadExReq.mshrMisses::total 1131 # number of ReadExReq MSHR misses (Count)

board.cache_hierarchy.l2cache.ReadExReq.mshrMissLatency::processor.cores.core.data 117583299 # number of ReadExReq MSHR miss ticks (Tick)

board.cache_hierarchy.l2cache.ReadExReq.mshrMissLatency::total 117583299 # number of ReadExReq MSHR miss ticks (Tick)

board.cache_hierarchy.l2cache.ReadExReq.mshrMissRate::processor.cores.core.data 0.992105 # mshr miss rate for ReadExReq accesses (Ratio)

board.cache_hierarchy.l2cache.ReadExReq.mshrMissRate::total 0.992105 # mshr miss rate for ReadExReq accesses (Ratio)

board.cache_hierarchy.l2cache.ReadExReq.avgMshrMissLatency::processor.cores.core.data 103964.013263 # average ReadExReq mshr miss latency ((Tick/Count))

board.cache_hierarchy.l2cache.ReadExReq.avgMshrMissLatency::total 103964.013263 # average ReadExReq mshr miss latency ((Tick/Count))

board.cache_hierarchy.l2cache.ReadSharedReq.hits::cache_hierarchy.l1dcaches.prefetcher 107971 # number of ReadSharedReq hits (Count)

board.cache_hierarchy.l2cache.ReadSharedReq.hits::processor.cores.core.data 21 # number of ReadSharedReq hits (Count)

board.cache_hierarchy.l2cache.ReadSharedReq.hits::total 107992 # number of ReadSharedReq hits (Count)

board.cache_hierarchy.l2cache.ReadSharedReq.misses::cache_hierarchy.l1dcaches.prefetcher 21147 # number of ReadSharedReq misses (Count)

board.cache_hierarchy.l2cache.ReadSharedReq.misses::processor.cores.core.inst 750 # number of ReadSharedReq misses (Count)

board.cache_hierarchy.l2cache.ReadSharedReq.misses::processor.cores.core.data 1789 # number of ReadSharedReq misses (Count)

board.cache_hierarchy.l2cache.ReadSharedReq.misses::total 23686 # number of ReadSharedReq misses (Count)

board.cache_hierarchy.l2cache.ReadSharedReq.missLatency::cache_hierarchy.l1dcaches.prefetcher 1998983928 # number of ReadSharedReq miss ticks (Tick)

board.cache_hierarchy.l2cache.ReadSharedReq.missLatency::processor.cores.core.inst 89984925 # number of ReadSharedReq miss ticks (Tick)

board.cache_hierarchy.l2cache.ReadSharedReq.missLatency::processor.cores.core.data 218960820 # number of ReadSharedReq miss ticks (Tick)

board.cache_hierarchy.l2cache.ReadSharedReq.missLatency::total 2307929673 # number of ReadSharedReq miss ticks (Tick)

board.cache_hierarchy.l2cache.ReadSharedReq.accesses::cache_hierarchy.l1dcaches.prefetcher 129118 # number of ReadSharedReq accesses(hits+misses) (Count)

board.cache_hierarchy.l2cache.ReadSharedReq.accesses::processor.cores.core.inst 750 # number of ReadSharedReq accesses(hits+misses) (Count)

board.cache_hierarchy.l2cache.ReadSharedReq.accesses::processor.cores.core.data 1810 # number of ReadSharedReq accesses(hits+misses) (Count)

board.cache_hierarchy.l2cache.ReadSharedReq.accesses::total 131678 # number of ReadSharedReq accesses(hits+misses) (Count)

board.cache_hierarchy.l2cache.ReadSharedReq.missRate::cache_hierarchy.l1dcaches.prefetcher 0.163780 # miss rate for ReadSharedReq accesses (Ratio)

board.cache_hierarchy.l2cache.ReadSharedReq.missRate::processor.cores.core.inst 1 # miss rate for ReadSharedReq accesses (Ratio)

board.cache_hierarchy.l2cache.ReadSharedReq.missRate::processor.cores.core.data 0.988398 # miss rate for ReadSharedReq accesses (Ratio)

board.cache_hierarchy.l2cache.ReadSharedReq.missRate::total 0.179878 # miss rate for ReadSharedReq accesses (Ratio)

board.cache_hierarchy.l2cache.ReadSharedReq.avgMissLatency::cache_hierarchy.l1dcaches.prefetcher 94528.014754 # average ReadSharedReq miss latency ((Tick/Count))

board.cache_hierarchy.l2cache.ReadSharedReq.avgMissLatency::processor.cores.core.inst 119979.900000 # average ReadSharedReq miss latency ((Tick/Count))

board.cache_hierarchy.l2cache.ReadSharedReq.avgMissLatency::processor.cores.core.data 122392.856344 # average ReadSharedReq miss latency ((Tick/Count))

board.cache_hierarchy.l2cache.ReadSharedReq.avgMissLatency::total 97438.557502 # average ReadSharedReq miss latency ((Tick/Count))

board.cache_hierarchy.l2cache.ReadSharedReq.mshrHits::cache_hierarchy.l1dcaches.prefetcher 19062 # number of ReadSharedReq MSHR hits (Count)

board.cache_hierarchy.l2cache.ReadSharedReq.mshrHits::processor.cores.core.data 2 # number of ReadSharedReq MSHR hits (Count)

board.cache_hierarchy.l2cache.ReadSharedReq.mshrHits::total 19064 # number of ReadSharedReq MSHR hits (Count)

board.cache_hierarchy.l2cache.ReadSharedReq.mshrMisses::cache_hierarchy.l1dcaches.prefetcher 2085 # number of ReadSharedReq MSHR misses (Count)

board.cache_hierarchy.l2cache.ReadSharedReq.mshrMisses::processor.cores.core.inst 750 # number of ReadSharedReq MSHR misses (Count)

board.cache_hierarchy.l2cache.ReadSharedReq.mshrMisses::processor.cores.core.data 1787 # number of ReadSharedReq MSHR misses (Count)

board.cache_hierarchy.l2cache.ReadSharedReq.mshrMisses::total 4622 # number of ReadSharedReq MSHR misses (Count)

board.cache_hierarchy.l2cache.ReadSharedReq.mshrMissLatency::cache_hierarchy.l1dcaches.prefetcher 217051383 # number of ReadSharedReq MSHR miss ticks (Tick)

board.cache_hierarchy.l2cache.ReadSharedReq.mshrMissLatency::processor.cores.core.inst 65009925 # number of ReadSharedReq MSHR miss ticks (Tick)

board.cache_hierarchy.l2cache.ReadSharedReq.mshrMissLatency::processor.cores.core.data 159235272 # number of ReadSharedReq MSHR miss ticks (Tick)

board.cache_hierarchy.l2cache.ReadSharedReq.mshrMissLatency::total 441296580 # number of ReadSharedReq MSHR miss ticks (Tick)

board.cache_hierarchy.l2cache.ReadSharedReq.mshrMissRate::cache_hierarchy.l1dcaches.prefetcher 0.016148 # mshr miss rate for ReadSharedReq accesses (Ratio)

board.cache_hierarchy.l2cache.ReadSharedReq.mshrMissRate::processor.cores.core.inst 1 # mshr miss rate for ReadSharedReq accesses (Ratio)

board.cache_hierarchy.l2cache.ReadSharedReq.mshrMissRate::processor.cores.core.data 0.987293 # mshr miss rate for ReadSharedReq accesses (Ratio)

board.cache_hierarchy.l2cache.ReadSharedReq.mshrMissRate::total 0.035101 # mshr miss rate for ReadSharedReq accesses (Ratio)

board.cache_hierarchy.l2cache.ReadSharedReq.avgMshrMissLatency::cache_hierarchy.l1dcaches.prefetcher 104101.382734 # average ReadSharedReq mshr miss latency ((Tick/Count))

board.cache_hierarchy.l2cache.ReadSharedReq.avgMshrMissLatency::processor.cores.core.inst 86679.900000 # average ReadSharedReq mshr miss latency ((Tick/Count))

board.cache_hierarchy.l2cache.ReadSharedReq.avgMshrMissLatency::processor.cores.core.data 89107.594852 # average ReadSharedReq mshr miss latency ((Tick/Count))

board.cache_hierarchy.l2cache.ReadSharedReq.avgMshrMissLatency::total 95477.408048 # average ReadSharedReq mshr miss latency ((Tick/Count))

board.cache_hierarchy.l2cache.UpgradeReq.hits::processor.cores.core.data 3 # number of UpgradeReq hits (Count)

board.cache_hierarchy.l2cache.UpgradeReq.hits::total 3 # number of UpgradeReq hits (Count)

board.cache_hierarchy.l2cache.UpgradeReq.accesses::processor.cores.core.data 3 # number of UpgradeReq accesses(hits+misses) (Count)

board.cache_hierarchy.l2cache.UpgradeReq.accesses::total 3 # number of UpgradeReq accesses(hits+misses) (Count)

board.cache_hierarchy.l2cache.WritebackDirty.hits::writebacks 65717 # number of WritebackDirty hits (Count)

board.cache_hierarchy.l2cache.WritebackDirty.hits::total 65717 # number of WritebackDirty hits (Count)

board.cache_hierarchy.l2cache.WritebackDirty.accesses::writebacks 65717 # number of WritebackDirty accesses(hits+misses) (Count)

board.cache_hierarchy.l2cache.WritebackDirty.accesses::total 65717 # number of WritebackDirty accesses(hits+misses) (Count)

board.cache_hierarchy.l2cache.power_state.pwrStateResidencyTicks::UNDEFINED 5909877540 # Cumulative time (in ticks) in various power states (Tick)

board.cache_hierarchy.l2cache.prefetcher.demandMshrMisses 5753 # demands not covered by prefetchs (Count)

board.cache_hierarchy.l2cache.prefetcher.pfIssued 479437 # number of hwpf issued (Count)

board.cache_hierarchy.l2cache.prefetcher.pfUnused 10 # number of HardPF blocks evicted w/o reference (Count)

board.cache_hierarchy.l2cache.prefetcher.pfUseful 107977 # number of useful prefetch (Count)

board.cache_hierarchy.l2cache.prefetcher.pfUsefulButMiss 0 # number of hit on prefetch but cache block is not in an usable state (Count)

board.cache_hierarchy.l2cache.prefetcher.accuracy 0.225216 # accuracy of the prefetcher (Count)

board.cache_hierarchy.l2cache.prefetcher.coverage 0.949415 # coverage brought by this prefetcher (Count)

board.cache_hierarchy.l2cache.prefetcher.pfHitInCache 254222 # number of prefetches hitting in cache (Count)

board.cache_hierarchy.l2cache.prefetcher.pfHitInMSHR 98156 # number of prefetches hitting in a MSHR (Count)

board.cache_hierarchy.l2cache.prefetcher.pfHitInWB 0 # number of prefetches hit in the Write Buffer (Count)

board.cache_hierarchy.l2cache.prefetcher.pfLate 352378 # number of late prefetches (hitting in cache, MSHR or WB) (Count)

board.cache_hierarchy.l2cache.prefetcher.pfIdentified 504051 # number of prefetch candidates identified (Count)

board.cache_hierarchy.l2cache.prefetcher.pfBufferHit 16405 # number of redundant prefetches already in prefetch queue (Count)

board.cache_hierarchy.l2cache.prefetcher.pfInCache 0 # number of redundant prefetches already in cache/mshr dropped (Count)

board.cache_hierarchy.l2cache.prefetcher.pfRemovedDemand 6157 # number of prefetches dropped due to a demand for the same address (Count)

board.cache_hierarchy.l2cache.prefetcher.pfRemovedFull 0 # number of prefetches dropped due to prefetch queue size (Count)

board.cache_hierarchy.l2cache.prefetcher.pfSpanPage 20497 # number of prefetches that crossed the page (Count)

board.cache_hierarchy.l2cache.prefetcher.pfUsefulSpanPage 19101 # number of prefetches that is useful and crossed the page (Count)

board.cache_hierarchy.l2cache.prefetcher.power_state.pwrStateResidencyTicks::UNDEFINED 5909877540 # Cumulative time (in ticks) in various power states (Tick)

board.cache_hierarchy.l2cache.tags.tagsInUse 3950.838686 # Average ticks per tags in use ((Tick/Count))

board.cache_hierarchy.l2cache.tags.totalRefs 372444 # Total number of references to valid blocks. (Count)

board.cache_hierarchy.l2cache.tags.sampledRefs 132812 # Sample count of references to valid blocks. (Count)

board.cache_hierarchy.l2cache.tags.avgRefs 2.804295 # Average number of references to valid blocks. ((Count/Count))

board.cache_hierarchy.l2cache.tags.warmupTick 89910 # The tick when the warmup percentage was hit. (Tick)

board.cache_hierarchy.l2cache.tags.occupancies::cache_hierarchy.l1dcaches.prefetcher 62.451457 # Average occupied blocks per tick, per requestor ((Count/Tick))

board.cache_hierarchy.l2cache.tags.occupancies::cache_hierarchy.l2cache.prefetcher 3719.749302 # Average occupied blocks per tick, per requestor ((Count/Tick))

board.cache_hierarchy.l2cache.tags.occupancies::processor.cores.core.inst 42.103580 # Average occupied blocks per tick, per requestor ((Count/Tick))

board.cache_hierarchy.l2cache.tags.occupancies::processor.cores.core.data 126.534347 # Average occupied blocks per tick, per requestor ((Count/Tick))

board.cache_hierarchy.l2cache.tags.avgOccs::cache_hierarchy.l1dcaches.prefetcher 0.015247 # Average percentage of cache occupancy ((Ratio/Tick))

board.cache_hierarchy.l2cache.tags.avgOccs::cache_hierarchy.l2cache.prefetcher 0.908142 # Average percentage of cache occupancy ((Ratio/Tick))

board.cache_hierarchy.l2cache.tags.avgOccs::processor.cores.core.inst 0.010279 # Average percentage of cache occupancy ((Ratio/Tick))

board.cache_hierarchy.l2cache.tags.avgOccs::processor.cores.core.data 0.030892 # Average percentage of cache occupancy ((Ratio/Tick))

board.cache_hierarchy.l2cache.tags.avgOccs::total 0.964560 # Average percentage of cache occupancy ((Ratio/Tick))

board.cache_hierarchy.l2cache.tags.occupanciesTaskId::1022 3749 # Occupied blocks per task id (Count)

board.cache_hierarchy.l2cache.tags.occupanciesTaskId::1024 347 # Occupied blocks per task id (Count)

board.cache_hierarchy.l2cache.tags.ageTaskId_1022::0 8 # Occupied blocks per task id, per block age (Count)

board.cache_hierarchy.l2cache.tags.ageTaskId_1022::1 1807 # Occupied blocks per task id, per block age (Count)

board.cache_hierarchy.l2cache.tags.ageTaskId_1022::2 1934 # Occupied blocks per task id, per block age (Count)

board.cache_hierarchy.l2cache.tags.ageTaskId_1024::0 149 # Occupied blocks per task id, per block age (Count)

board.cache_hierarchy.l2cache.tags.ageTaskId_1024::1 169 # Occupied blocks per task id, per block age (Count)

board.cache_hierarchy.l2cache.tags.ageTaskId_1024::2 29 # Occupied blocks per task id, per block age (Count)

board.cache_hierarchy.l2cache.tags.ratioOccsTaskId::1022 0.915283 # Ratio of occupied blocks and all blocks, per task id (Ratio)

board.cache_hierarchy.l2cache.tags.ratioOccsTaskId::1024 0.084717 # Ratio of occupied blocks and all blocks, per task id (Ratio)

board.cache_hierarchy.l2cache.tags.tagAccesses 4366668 # Number of tag accesses (Count)

board.cache_hierarchy.l2cache.tags.dataAccesses 4366668 # Number of data accesses (Count)

board.cache_hierarchy.l2cache.tags.power_state.pwrStateResidencyTicks::UNDEFINED 5909877540 # Cumulative time (in ticks) in various power states (Tick)

board.cache_hierarchy.membus.transDist::ReadResp 131681 # Transaction distribution (Count)

board.cache_hierarchy.membus.transDist::WritebackDirty 65695 # Transaction distribution (Count)

board.cache_hierarchy.membus.transDist::CleanEvict 62713 # Transaction distribution (Count)

board.cache_hierarchy.membus.transDist::ReadExReq 1131 # Transaction distribution (Count)

board.cache_hierarchy.membus.transDist::ReadExResp 1131 # Transaction distribution (Count)

board.cache_hierarchy.membus.transDist::ReadSharedReq 131681 # Transaction distribution (Count)

board.cache_hierarchy.membus.pktCount_board.cache_hierarchy.l2cache.mem_side_port::board.memory.mem_ctrl.port 394032 # Packet count per connected requestor and responder (Count)

board.cache_hierarchy.membus.pktCount_board.cache_hierarchy.l2cache.mem_side_port::total 394032 # Packet count per connected requestor and responder (Count)

board.cache_hierarchy.membus.pktCount::total 394032 # Packet count per connected requestor and responder (Count)

board.cache_hierarchy.membus.pktSize_board.cache_hierarchy.l2cache.mem_side_port::board.memory.mem_ctrl.port 12704448 # Cumulative packet size per connected requestor and responder (Byte)

board.cache_hierarchy.membus.pktSize_board.cache_hierarchy.l2cache.mem_side_port::total 12704448 # Cumulative packet size per connected requestor and responder (Byte)

board.cache_hierarchy.membus.pktSize::total 12704448 # Cumulative packet size per connected requestor and responder (Byte)

board.cache_hierarchy.membus.snoops 0 # Total snoops (Count)

board.cache_hierarchy.membus.snoopTraffic 0 # Total snoop traffic (Byte)

board.cache_hierarchy.membus.snoopFanout::samples 132812 # Request fanout histogram (Count)

board.cache_hierarchy.membus.snoopFanout::mean 0 # Request fanout histogram (Count)

board.cache_hierarchy.membus.snoopFanout::stdev 0 # Request fanout histogram (Count)

board.cache_hierarchy.membus.snoopFanout::underflows 0 0.00% 0.00% # Request fanout histogram (Count)

board.cache_hierarchy.membus.snoopFanout::0 132812 100.00% 100.00% # Request fanout histogram (Count)

board.cache_hierarchy.membus.snoopFanout::1 0 0.00% 100.00% # Request fanout histogram (Count)

board.cache_hierarchy.membus.snoopFanout::overflows 0 0.00% 100.00% # Request fanout histogram (Count)

board.cache_hierarchy.membus.snoopFanout::min_value 0 # Request fanout histogram (Count)

board.cache_hierarchy.membus.snoopFanout::max_value 0 # Request fanout histogram (Count)

board.cache_hierarchy.membus.snoopFanout::total 132812 # Request fanout histogram (Count)

board.cache_hierarchy.membus.badaddr_responder.power_state.pwrStateResidencyTicks::UNDEFINED 5909877540 # Cumulative time (in ticks) in various power states (Tick)

board.cache_hierarchy.membus.power_state.pwrStateResidencyTicks::UNDEFINED 5909877540 # Cumulative time (in ticks) in various power states (Tick)

board.cache_hierarchy.membus.reqLayer0.occupancy 145671984 # Layer occupancy (ticks) (Tick)

board.cache_hierarchy.membus.reqLayer0.utilization 0.0 # Layer utilization (Ratio)

board.cache_hierarchy.membus.respLayer2.occupancy 113169844 # Layer occupancy (ticks) (Tick)

board.cache_hierarchy.membus.respLayer2.utilization 0.0 # Layer utilization (Ratio)

board.cache_hierarchy.membus.snoop_filter.totRequests 261220 # Total number of requests made to the snoop filter. (Count)

board.cache_hierarchy.membus.snoop_filter.hitSingleRequests 128408 # Number of requests hitting in the snoop filter with a single holder of the requested data. (Count)

board.cache_hierarchy.membus.snoop_filter.hitMultiRequests 0 # Number of requests hitting in the snoop filter with multiple (>1) holders of the requested data. (Count)

board.cache_hierarchy.membus.snoop_filter.totSnoops 0 # Total number of snoops made to the snoop filter. (Count)

board.cache_hierarchy.membus.snoop_filter.hitSingleSnoops 0 # Number of snoops hitting in the snoop filter with a single holder of the requested data. (Count)

board.cache_hierarchy.membus.snoop_filter.hitMultiSnoops 0 # Number of snoops hitting in the snoop filter with multiple (>1) holders of the requested data. (Count)

board.clk_domain.clock 333 # Clock period in ticks (Tick)

board.clk_domain.voltage_domain.voltage 1 # Voltage in Volts (Volt)

board.memory.mem_ctrl.avgPriority_writebacks::samples 65695.00 # Average QoS priority value for accepted requests (Count)

board.memory.mem_ctrl.avgPriority_cache_hierarchy.l1dcaches.prefetcher::samples 2085.00 # Average QoS priority value for accepted requests (Count)

board.memory.mem_ctrl.avgPriority_cache_hierarchy.l2cache.prefetcher::samples 127030.00 # Average QoS priority value for accepted requests (Count)

board.memory.mem_ctrl.avgPriority_processor.cores.core.inst::samples 750.00 # Average QoS priority value for accepted requests (Count)

board.memory.mem_ctrl.avgPriority_processor.cores.core.data::samples 2918.00 # Average QoS priority value for accepted requests (Count)

board.memory.mem_ctrl.priorityMinLatency 0.000000018750 # per QoS priority minimum request to response latency (Second)

board.memory.mem_ctrl.priorityMaxLatency 0.000158809564 # per QoS priority maximum request to response latency (Second)

board.memory.mem_ctrl.numReadWriteTurnArounds 4101 # Number of turnarounds from READ to WRITE (Count)

board.memory.mem_ctrl.numWriteReadTurnArounds 4101 # Number of turnarounds from WRITE to READ (Count)

board.memory.mem_ctrl.numStayReadState 299567 # Number of times bus staying in READ state (Count)

board.memory.mem_ctrl.numStayWriteState 61924 # Number of times bus staying in WRITE state (Count)

board.memory.mem_ctrl.readReqs 132812 # Number of read requests accepted (Count)

board.memory.mem_ctrl.writeReqs 65695 # Number of write requests accepted (Count)

board.memory.mem_ctrl.readBursts 132812 # Number of controller read bursts, including those serviced by the write queue (Count)

board.memory.mem_ctrl.writeBursts 65695 # Number of controller write bursts, including those merged in the write queue (Count)

board.memory.mem_ctrl.servicedByWrQ 29 # Number of controller read bursts serviced by the write queue (Count)

board.memory.mem_ctrl.mergedWrBursts 0 # Number of controller write bursts merged with an existing one (Count)

board.memory.mem_ctrl.neitherReadNorWriteReqs 0 # Number of requests that are neither read nor write (Count)

board.memory.mem_ctrl.avgRdQLen 1.99 # Average read queue length when enqueuing ((Count/Tick))

board.memory.mem_ctrl.avgWrQLen 25.46 # Average write queue length when enqueuing ((Count/Tick))

board.memory.mem_ctrl.numRdRetry 0 # Number of times read queue was full causing retry (Count)

board.memory.mem_ctrl.numWrRetry 0 # Number of times write queue was full causing retry (Count)

board.memory.mem_ctrl.readPktSize::0 0 # Read request sizes (log2) (Count)

board.memory.mem_ctrl.readPktSize::1 0 # Read request sizes (log2) (Count)

board.memory.mem_ctrl.readPktSize::2 0 # Read request sizes (log2) (Count)

board.memory.mem_ctrl.readPktSize::3 0 # Read request sizes (log2) (Count)

board.memory.mem_ctrl.readPktSize::4 0 # Read request sizes (log2) (Count)

board.memory.mem_ctrl.readPktSize::5 0 # Read request sizes (log2) (Count)

board.memory.mem_ctrl.readPktSize::6 132812 # Read request sizes (log2) (Count)

board.memory.mem_ctrl.writePktSize::0 0 # Write request sizes (log2) (Count)

board.memory.mem_ctrl.writePktSize::1 0 # Write request sizes (log2) (Count)

board.memory.mem_ctrl.writePktSize::2 0 # Write request sizes (log2) (Count)

board.memory.mem_ctrl.writePktSize::3 0 # Write request sizes (log2) (Count)

board.memory.mem_ctrl.writePktSize::4 0 # Write request sizes (log2) (Count)

board.memory.mem_ctrl.writePktSize::5 0 # Write request sizes (log2) (Count)

board.memory.mem_ctrl.writePktSize::6 65695 # Write request sizes (log2) (Count)

board.memory.mem_ctrl.rdQLenPdf::0 90554 # What read queue length does an incoming req see (Count)

board.memory.mem_ctrl.rdQLenPdf::1 9626 # What read queue length does an incoming req see (Count)

board.memory.mem_ctrl.rdQLenPdf::2 6782 # What read queue length does an incoming req see (Count)

board.memory.mem_ctrl.rdQLenPdf::3 6823 # What read queue length does an incoming req see (Count)

board.memory.mem_ctrl.rdQLenPdf::4 7046 # What read queue length does an incoming req see (Count)

board.memory.mem_ctrl.rdQLenPdf::5 5861 # What read queue length does an incoming req see (Count)

board.memory.mem_ctrl.rdQLenPdf::6 4217 # What read queue length does an incoming req see (Count)

board.memory.mem_ctrl.rdQLenPdf::7 1610 # What read queue length does an incoming req see (Count)

board.memory.mem_ctrl.rdQLenPdf::8 259 # What read queue length does an incoming req see (Count)

board.memory.mem_ctrl.rdQLenPdf::9	3	# What read queue length does an incoming req see (Count)
board.memory.mem_ctrl.rdQLenPdf::10	2	# What read queue length does an incoming req see (Count)
board.memory.mem_ctrl.rdQLenPdf::11	0	# What read queue length does an incoming req see (Count)
board.memory.mem_ctrl.rdQLenPdf::12	0	# What read queue length does an incoming req see (Count)
board.memory.mem_ctrl.rdQLenPdf::13	0	# What read queue length does an incoming req see (Count)
board.memory.mem_ctrl.rdQLenPdf::14	0	# What read queue length does an incoming req see (Count)
board.memory.mem_ctrl.rdQLenPdf::15	0	# What read queue length does an incoming req see (Count)
board.memory.mem_ctrl.rdQLenPdf::16	0	# What read queue length does an incoming req see (Count)
board.memory.mem_ctrl.rdQLenPdf::17	0	# What read queue length does an incoming req see (Count)
board.memory.mem_ctrl.rdQLenPdf::18	0	# What read queue length does an incoming req see (Count)
board.memory.mem_ctrl.rdQLenPdf::19	0	# What read queue length does an incoming req see (Count)
board.memory.mem_ctrl.rdQLenPdf::20	0	# What read queue length does an incoming req see (Count)
board.memory.mem_ctrl.rdQLenPdf::21	0	# What read queue length does an incoming req see (Count)
board.memory.mem_ctrl.rdQLenPdf::22	0	# What read queue length does an incoming req see (Count)
board.memory.mem_ctrl.rdQLenPdf::23	0	# What read queue length does an incoming req see (Count)
board.memory.mem_ctrl.rdQLenPdf::24	0	# What read queue length does an incoming req see (Count)
board.memory.mem_ctrl.rdQLenPdf::25	0	# What read queue length does an incoming req see (Count)
board.memory.mem_ctrl.rdQLenPdf::26	0	# What read queue length does an incoming req see (Count)
board.memory.mem_ctrl.rdQLenPdf::27	0	# What read queue length does an incoming req see (Count)
board.memory.mem_ctrl.rdQLenPdf::28	0	# What read queue length does an incoming req see (Count)
board.memory.mem_ctrl.rdQLenPdf::29	0	# What read queue length does an incoming req see (Count)
board.memory.mem_ctrl.rdQLenPdf::30	0	# What read queue length does an incoming req see (Count)
board.memory.mem_ctrl.rdQLenPdf::31	0	# What read queue length does an incoming req see (Count)
board.memory.mem_ctrl.wrQLenPdf::0	1	# What write queue length does an incoming req see (Count)
board.memory.mem_ctrl.wrQLenPdf::1	1	# What write queue length does an incoming req see (Count)
board.memory.mem_ctrl.wrQLenPdf::2	1	# What write queue length does an incoming req see (Count)
board.memory.mem_ctrl.wrQLenPdf::3	1	# What write queue length does an incoming req see (Count)
board.memory.mem_ctrl.wrQLenPdf::4	1	# What write queue length does an incoming req see (Count)
board.memory.mem_ctrl.wrQLenPdf::5	1	# What write queue length does an incoming req see (Count)
board.memory.mem_ctrl.wrQLenPdf::6	1	# What write queue length does an incoming req see (Count)
board.memory.mem_ctrl.wrQLenPdf::7	1	# What write queue length does an incoming req see (Count)
board.memory.mem_ctrl.wrQLenPdf::8	1	# What write queue length does an incoming req see (Count)
board.memory.mem_ctrl.wrQLenPdf::9	1	# What write queue length does an incoming req see (Count)
board.memory.mem_ctrl.wrQLenPdf::10	1	# What write queue length does an incoming req see (Count)
board.memory.mem_ctrl.wrQLenPdf::11	1	# What write queue length does an incoming req see (Count)
board.memory.mem_ctrl.wrQLenPdf::12	1	# What write queue length does an incoming req see (Count)
board.memory.mem_ctrl.wrQLenPdf::13	1	# What write queue length does an incoming req see (Count)
board.memory.mem_ctrl.wrQLenPdf::14	1	# What write queue length does an incoming req see (Count)
board.memory.mem_ctrl.wrQLenPdf::15	7	# What write queue length does an incoming req see (Count)
board.memory.mem_ctrl.wrQLenPdf::16	8	# What write queue length does an incoming req see (Count)
board.memory.mem_ctrl.wrQLenPdf::17	191	# What write queue length does an incoming req see (Count)
board.memory.mem_ctrl.wrQLenPdf::18	1633	# What write queue length does an incoming req see (Count)
board.memory.mem_ctrl.wrQLenPdf::19	4006	# What write queue length does an incoming req see (Count)
board.memory.mem_ctrl.wrQLenPdf::20	5672	# What write queue length does an incoming req see (Count)
board.memory.mem_ctrl.wrQLenPdf::21	4297	# What write queue length does an incoming req see (Count)
board.memory.mem_ctrl.wrQLenPdf::22	4050	# What write queue length does an incoming req see (Count)
board.memory.mem_ctrl.wrQLenPdf::23	4524	# What write queue length does an incoming req see (Count)
board.memory.mem_ctrl.wrQLenPdf::24	4627	# What write queue length does an incoming req see (Count)
board.memory.mem_ctrl.wrQLenPdf::25	5794	# What write queue length does an incoming req see (Count)

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board.memory.mem_ctrl.wrQLenPdf::26 4388 # What write queue length does an incoming req see (Count)
board.memory.mem_ctrl.wrQLenPdf::27 4107 # What write queue length does an incoming req see (Count)
board.memory.mem_ctrl.wrQLenPdf::28 4108 # What write queue length does an incoming req see (Count)
board.memory.mem_ctrl.wrQLenPdf::29 4194 # What write queue length does an incoming req see (Count)
board.memory.mem_ctrl.wrQLenPdf::30 4514 # What write queue length does an incoming req see (Count)
board.memory.mem_ctrl.wrQLenPdf::31 4109 # What write queue length does an incoming req see (Count)
board.memory.mem_ctrl.wrQLenPdf::32 5421 # What write queue length does an incoming req see (Count)
board.memory.mem_ctrl.wrQLenPdf::33 16 # What write queue length does an incoming req see (Count)
board.memory.mem_ctrl.wrQLenPdf::34 7 # What write queue length does an incoming req see (Count)
board.memory.mem_ctrl.wrQLenPdf::35 2 # What write queue length does an incoming req see (Count)
board.memory.mem_ctrl.wrQLenPdf::36 2 # What write queue length does an incoming req see (Count)
board.memory.mem_ctrl.wrQLenPdf::37 1 # What write queue length does an incoming req see (Count)
board.memory.mem_ctrl.wrQLenPdf::38 1 # What write queue length does an incoming req see (Count)
board.memory.mem_ctrl.wrQLenPdf::39 1 # What write queue length does an incoming req see (Count)
board.memory.mem_ctrl.wrQLenPdf::40 0 # What write queue length does an incoming req see (Count)
board.memory.mem_ctrl.wrQLenPdf::41 0 # What write queue length does an incoming req see (Count)
board.memory.mem_ctrl.wrQLenPdf::42 0 # What write queue length does an incoming req see (Count)
board.memory.mem_ctrl.wrQLenPdf::43 0 # What write queue length does an incoming req see (Count)
board.memory.mem_ctrl.wrQLenPdf::44 0 # What write queue length does an incoming req see (Count)
board.memory.mem_ctrl.wrQLenPdf::45 0 # What write queue length does an incoming req see (Count)
board.memory.mem_ctrl.wrQLenPdf::46 0 # What write queue length does an incoming req see (Count)
board.memory.mem_ctrl.wrQLenPdf::47 0 # What write queue length does an incoming req see (Count)
board.memory.mem_ctrl.wrQLenPdf::48 0 # What write queue length does an incoming req see (Count)
board.memory.mem_ctrl.wrQLenPdf::49 0 # What write queue length does an incoming req see (Count)
board.memory.mem_ctrl.wrQLenPdf::50 0 # What write queue length does an incoming req see (Count)
board.memory.mem_ctrl.wrQLenPdf::51 0 # What write queue length does an incoming req see (Count)
board.memory.mem_ctrl.wrQLenPdf::52 0 # What write queue length does an incoming req see (Count)
board.memory.mem_ctrl.wrQLenPdf::53 0 # What write queue length does an incoming req see (Count)
board.memory.mem_ctrl.wrQLenPdf::54 0 # What write queue length does an incoming req see (Count)
board.memory.mem_ctrl.wrQLenPdf::55 0 # What write queue length does an incoming req see (Count)
board.memory.mem_ctrl.wrQLenPdf::56 0 # What write queue length does an incoming req see (Count)
board.memory.mem_ctrl.wrQLenPdf::57 0 # What write queue length does an incoming req see (Count)
board.memory.mem_ctrl.wrQLenPdf::58 0 # What write queue length does an incoming req see (Count)
board.memory.mem_ctrl.wrQLenPdf::59 0 # What write queue length does an incoming req see (Count)
board.memory.mem_ctrl.wrQLenPdf::60 0 # What write queue length does an incoming req see (Count)
board.memory.mem_ctrl.wrQLenPdf::61 0 # What write queue length does an incoming req see (Count)
board.memory.mem_ctrl.wrQLenPdf::62 0 # What write queue length does an incoming req see (Count)
board.memory.mem_ctrl.wrQLenPdf::63 0 # What write queue length does an incoming req see (Count)
board.memory.mem_ctrl.rdPerTurnAround::samples 4101 # Reads before turning the bus around for writes (Count)
board.memory.mem_ctrl.rdPerTurnAround::mean 17.326262 # Reads before turning the bus around for writes (Count)
board.memory.mem_ctrl.rdPerTurnAround::gmean 16.016560 # Reads before turning the bus around for writes (Count)
board.memory.mem_ctrl.rdPerTurnAround::stdev 71.904828 # Reads before turning the bus around for writes (Count)
board.memory.mem_ctrl.rdPerTurnAround::0-255 4099 99.95% 99.95% # Reads before turning the bus around for writes (Count)
board.memory.mem_ctrl.rdPerTurnAround::256-511 1 0.02% 99.98% # Reads before turning the bus around for writes (Count)
board.memory.mem_ctrl.rdPerTurnAround::4352-4607 1 0.02% 100.00% # Reads before turning the bus around for writes (Count)
board.memory.mem_ctrl.rdPerTurnAround::total 4101 # Reads before turning the bus around for writes (Count)
board.memory.mem_ctrl.wrPerTurnAround::samples 4101 # Writes before turning the bus around for reads (Count)
board.memory.mem_ctrl.wrPerTurnAround::mean 16.012192 # Writes before turning the bus around for reads (Count)
board.memory.mem_ctrl.wrPerTurnAround::gmean 16.010368 # Writes before turning the bus around for reads (Count)
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board.memory.mem_ctrl.wrPerTurnAround::stdev     0.273812                    # Writes before turning the bus around for reads (Count)
board.memory.mem_ctrl.wrPerTurnAround::16        4090     99.73%     99.73% # Writes before turning the bus around for reads (Count)
board.memory.mem_ctrl.wrPerTurnAround::17           1      0.02%     99.76% # Writes before turning the bus around for reads (Count)
board.memory.mem_ctrl.wrPerTurnAround::18           3      0.07%     99.83% # Writes before turning the bus around for reads (Count)
board.memory.mem_ctrl.wrPerTurnAround::19           1      0.02%     99.85% # Writes before turning the bus around for reads (Count)
board.memory.mem_ctrl.wrPerTurnAround::20           1      0.02%     99.88% # Writes before turning the bus around for reads (Count)
board.memory.mem_ctrl.wrPerTurnAround::21           1      0.02%     99.90% # Writes before turning the bus around for reads (Count)
board.memory.mem_ctrl.wrPerTurnAround::22           1      0.02%     99.93% # Writes before turning the bus around for reads (Count)
board.memory.mem_ctrl.wrPerTurnAround::24           2      0.05%     99.98% # Writes before turning the bus around for reads (Count)
board.memory.mem_ctrl.wrPerTurnAround::25           1      0.02%    100.00% # Writes before turning the bus around for reads (Count)
board.memory.mem_ctrl.wrPerTurnAround::total        4101                    # Writes before turning the bus around for reads (Count)
board.memory.mem_ctrl.bytesReadWrQ                  1856                    # Total number of bytes read from write queue (Byte)
board.memory.mem_ctrl.bytesReadSys                  8499968                 # Total read bytes from the system interface side (Byte)
board.memory.mem_ctrl.bytesWrittenSys               4204480                 # Total written bytes from the system interface side (Byte)
board.memory.mem_ctrl.avgRdBWSys                    1438264658.18782449     # Average system read bandwidth in Byte/s ((Byte/Second))
board.memory.mem_ctrl.avgWrBWSys                    711432677.16508389      # Average system write bandwidth in Byte/s ((Byte/Second))
board.memory.mem_ctrl.totGap                        5909726358              # Total gap between requests (Tick)
board.memory.mem_ctrl.avgGap                        29770.87                # Average gap between requests ((Tick/Count))
board.memory.mem_ctrl.requestorReadBytes::cache_hierarchy.l1dcaches.prefetcher      133440      # Per-requestor bytes read from memory (Byte)
board.memory.mem_ctrl.requestorReadBytes::cache_hierarchy.l2cache.prefetcher      8129920      # Per-requestor bytes read from memory (Byte)
board.memory.mem_ctrl.requestorReadBytes::processor.cores.core.inst      48000      # Per-requestor bytes read from memory (Byte)
board.memory.mem_ctrl.requestorReadBytes::processor.cores.core.data      186752      # Per-requestor bytes read from memory (Byte)
board.memory.mem_ctrl.requestorWriteBytes::writebacks      4202624      # Per-requestor bytes write to memory (Byte)
board.memory.mem_ctrl.requestorReadRate::cache_hierarchy.l1dcaches.prefetcher 22579148.061331912875      # Per-requestor bytes read from memory rate ((Byte/Second))
board.memory.mem_ctrl.requestorReadRate::cache_hierarchy.l2cache.prefetcher 1375649485.962106943130      # Per-requestor bytes read from memory rate ((Byte/Second))
board.memory.mem_ctrl.requestorReadRate::processor.cores.core.inst 8121995.705515076406      # Per-requestor bytes read from memory rate ((Byte/Second))
board.memory.mem_ctrl.requestorReadRate::processor.cores.core.data 31599977.958257324994      # Per-requestor bytes read from memory rate ((Byte/Second))
board.memory.mem_ctrl.requestorWriteRate::writebacks 711118626.664470672607      # Per-requestor bytes write to memory rate ((Byte/Second))
board.memory.mem_ctrl.requestorReadAccesses::cache_hierarchy.l1dcaches.prefetcher      2085      # Per-requestor read serviced memory accesses (Count)
board.memory.mem_ctrl.requestorReadAccesses::cache_hierarchy.l2cache.prefetcher      127059      # Per-requestor read serviced memory accesses (Count)
board.memory.mem_ctrl.requestorReadAccesses::processor.cores.core.inst      750      # Per-requestor read serviced memory accesses (Count)
board.memory.mem_ctrl.requestorReadAccesses::processor.cores.core.data      2918      # Per-requestor read serviced memory accesses (Count)
board.memory.mem_ctrl.requestorWriteAccesses::writebacks      65695      # Per-requestor write serviced memory accesses (Count)
board.memory.mem_ctrl.requestorReadTotalLat::cache_hierarchy.l1dcaches.prefetcher      98049092      # Per-requestor read total memory access latency (Tick)
board.memory.mem_ctrl.requestorReadTotalLat::cache_hierarchy.l2cache.prefetcher      5404635693      # Per-requestor read total memory access latency (Tick)
board.memory.mem_ctrl.requestorReadTotalLat::processor.cores.core.inst      22154805      # Per-requestor read total memory access latency (Tick)
board.memory.mem_ctrl.requestorReadTotalLat::processor.cores.core.data      110207986      # Per-requestor read total memory access latency (Tick)
board.memory.mem_ctrl.requestorWriteTotalLat::writebacks      68447090353      # Per-requestor write total memory access latency (Tick)
board.memory.mem_ctrl.requestorReadAvgLat::cache_hierarchy.l1dcaches.prefetcher      47025.94      # Per-requestor read average memory access latency ((Tick/Count))
board.memory.mem_ctrl.requestorReadAvgLat::cache_hierarchy.l2cache.prefetcher      42536.43      # Per-requestor read average memory access latency ((Tick/Count))
board.memory.mem_ctrl.requestorReadAvgLat::processor.cores.core.inst      29539.74      # Per-requestor read average memory access latency ((Tick/Count))
board.memory.mem_ctrl.requestorReadAvgLat::processor.cores.core.data      37768.33      # Per-requestor read average memory access latency ((Tick/Count))
board.memory.mem_ctrl.requestorWriteAvgLat::writebacks      1041891.93      # Per-requestor write average memory access latency ((Tick/Count))
board.memory.mem_ctrl.dram.bytesRead::cache_hierarchy.l1dcaches.prefetcher      133440      # Number of bytes read from this memory (Byte)
board.memory.mem_ctrl.dram.bytesRead::cache_hierarchy.l2cache.prefetcher      8131776      # Number of bytes read from this memory (Byte)
board.memory.mem_ctrl.dram.bytesRead::processor.cores.core.inst      48000      # Number of bytes read from this memory (Byte)
board.memory.mem_ctrl.dram.bytesRead::processor.cores.core.data      186752      # Number of bytes read from this memory (Byte)
board.memory.mem_ctrl.dram.bytesRead::total      8499968      # Number of bytes read from this memory (Byte)
board.memory.mem_ctrl.dram.bytesInstRead::processor.cores.core.inst      48000      # Number of instructions bytes read from this memory (Byte)
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board.memory.mem_ctrl.dram.bytesInstRead::total        48000                    # Number of instructions bytes read from this memory (Byte)
board.memory.mem_ctrl.dram.bytesWritten::writebacks      4204480                    # Number of bytes written to this memory (Byte)
board.memory.mem_ctrl.dram.bytesWritten::total      4204480                    # Number of bytes written to this memory (Byte)
board.memory.mem_ctrl.dram.numReads::cache_hierarchy.l1dcaches.prefetcher         2085                    # Number of read requests responded to by this memory (Count)
board.memory.mem_ctrl.dram.numReads::cache_hierarchy.l2cache.prefetcher       127059                    # Number of read requests responded to by this memory (Count)
board.memory.mem_ctrl.dram.numReads::processor.cores.core.inst          750                    # Number of read requests responded to by this memory (Count)
board.memory.mem_ctrl.dram.numReads::processor.cores.core.data         2918                    # Number of read requests responded to by this memory (Count)
board.memory.mem_ctrl.dram.numReads::total       132812                    # Number of read requests responded to by this memory (Count)
board.memory.mem_ctrl.dram.numWrites::writebacks        65695                    # Number of write requests responded to by this memory (Count)
board.memory.mem_ctrl.dram.numWrites::total        65695                    # Number of write requests responded to by this memory (Count)
board.memory.mem_ctrl.dram.bwRead::cache_hierarchy.l1dcaches.prefetcher     22579148                    # Total read bandwidth from this memory ((Byte/Second))
board.memory.mem_ctrl.dram.bwRead::cache_hierarchy.l2cache.prefetcher   1375963536                    # Total read bandwidth from this memory ((Byte/Second))
board.memory.mem_ctrl.dram.bwRead::processor.cores.core.inst      8121996                    # Total read bandwidth from this memory ((Byte/Second))
board.memory.mem_ctrl.dram.bwRead::processor.cores.core.data     31599978                    # Total read bandwidth from this memory ((Byte/Second))
board.memory.mem_ctrl.dram.bwRead::total   1438264658                    # Total read bandwidth from this memory ((Byte/Second))
board.memory.mem_ctrl.dram.bwInstRead::processor.cores.core.inst      8121996                    # Instruction read bandwidth from this memory ((Byte/Second))
board.memory.mem_ctrl.dram.bwInstRead::total      8121996                    # Instruction read bandwidth from this memory ((Byte/Second))
board.memory.mem_ctrl.dram.bwWrite::writebacks    711432677                    # Write bandwidth from this memory ((Byte/Second))
board.memory.mem_ctrl.dram.bwWrite::total    711432677                    # Write bandwidth from this memory ((Byte/Second))
board.memory.mem_ctrl.dram.bwTotal::writebacks    711432677                    # Total bandwidth to/from this memory ((Byte/Second))
board.memory.mem_ctrl.dram.bwTotal::cache_hierarchy.l1dcaches.prefetcher     22579148                    # Total bandwidth to/from this memory ((Byte/Second))
board.memory.mem_ctrl.dram.bwTotal::cache_hierarchy.l2cache.prefetcher   1375963536                    # Total bandwidth to/from this memory ((Byte/Second))
board.memory.mem_ctrl.dram.bwTotal::processor.cores.core.inst      8121996                    # Total bandwidth to/from this memory ((Byte/Second))
board.memory.mem_ctrl.dram.bwTotal::processor.cores.core.data     31599978                    # Total bandwidth to/from this memory ((Byte/Second))
board.memory.mem_ctrl.dram.bwTotal::total   2149697335                    # Total bandwidth to/from this memory ((Byte/Second))
board.memory.mem_ctrl.dram.readBursts          132783                    # Number of DRAM read bursts (Count)
board.memory.mem_ctrl.dram.writeBursts         65666                    # Number of DRAM write bursts (Count)
board.memory.mem_ctrl.dram.perBankRdBursts::0        8465                    # Per bank write bursts (Count)
board.memory.mem_ctrl.dram.perBankRdBursts::1        8437                    # Per bank write bursts (Count)
board.memory.mem_ctrl.dram.perBankRdBursts::2        8291                    # Per bank write bursts (Count)
board.memory.mem_ctrl.dram.perBankRdBursts::3        8268                    # Per bank write bursts (Count)
board.memory.mem_ctrl.dram.perBankRdBursts::4        8267                    # Per bank write bursts (Count)
board.memory.mem_ctrl.dram.perBankRdBursts::5        8312                    # Per bank write bursts (Count)
board.memory.mem_ctrl.dram.perBankRdBursts::6        8254                    # Per bank write bursts (Count)
board.memory.mem_ctrl.dram.perBankRdBursts::7        8225                    # Per bank write bursts (Count)
board.memory.mem_ctrl.dram.perBankRdBursts::8        8274                    # Per bank write bursts (Count)
board.memory.mem_ctrl.dram.perBankRdBursts::9        8236                    # Per bank write bursts (Count)
board.memory.mem_ctrl.dram.perBankRdBursts::10        8235                    # Per bank write bursts (Count)
board.memory.mem_ctrl.dram.perBankRdBursts::11        8227                    # Per bank write bursts (Count)
board.memory.mem_ctrl.dram.perBankRdBursts::12        8235                    # Per bank write bursts (Count)
board.memory.mem_ctrl.dram.perBankRdBursts::13        8294                    # Per bank write bursts (Count)
board.memory.mem_ctrl.dram.perBankRdBursts::14        8357                    # Per bank write bursts (Count)
board.memory.mem_ctrl.dram.perBankRdBursts::15        8406                    # Per bank write bursts (Count)
board.memory.mem_ctrl.dram.perBankWrBursts::0        4096                    # Per bank write bursts (Count)
board.memory.mem_ctrl.dram.perBankWrBursts::1        4096                    # Per bank write bursts (Count)
board.memory.mem_ctrl.dram.perBankWrBursts::2        4096                    # Per bank write bursts (Count)
board.memory.mem_ctrl.dram.perBankWrBursts::3        4104                    # Per bank write bursts (Count)
board.memory.mem_ctrl.dram.perBankWrBursts::4        4102                    # Per bank write bursts (Count)
board.memory.mem_ctrl.dram.perBankWrBursts::5        4157                    # Per bank write bursts (Count)
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board.memory.mem_ctrl.dram.perBankWrBursts::6        4098                    # Per bank write bursts (Count)
board.memory.mem_ctrl.dram.perBankWrBursts::7        4096                    # Per bank write bursts (Count)
board.memory.mem_ctrl.dram.perBankWrBursts::8        4116                    # Per bank write bursts (Count)
board.memory.mem_ctrl.dram.perBankWrBursts::9        4106                    # Per bank write bursts (Count)
board.memory.mem_ctrl.dram.perBankWrBursts::10        4117                    # Per bank write bursts (Count)
board.memory.mem_ctrl.dram.perBankWrBursts::11        4098                    # Per bank write bursts (Count)
board.memory.mem_ctrl.dram.perBankWrBursts::12        4096                    # Per bank write bursts (Count)
board.memory.mem_ctrl.dram.perBankWrBursts::13        4096                    # Per bank write bursts (Count)
board.memory.mem_ctrl.dram.perBankWrBursts::14        4096                    # Per bank write bursts (Count)
board.memory.mem_ctrl.dram.perBankWrBursts::15        4096                    # Per bank write bursts (Count)
board.memory.mem_ctrl.dram.totQLat        3145366326                    # Total ticks spent queuing (Tick)
board.memory.mem_ctrl.dram.totBusLat        663915000                    # Total ticks spent in databus transfers (Tick)
board.memory.mem_ctrl.dram.totMemAccLat    5635047576                    # Total ticks spent from burst creation until serviced by the DRAM (Tick)
board.memory.mem_ctrl.dram.avgQLat          23688.02                    # Average queueing delay per DRAM burst ((Tick/Count))
board.memory.mem_ctrl.dram.avgBusLat          5000.00                    # Average bus latency per DRAM burst ((Tick/Count))
board.memory.mem_ctrl.dram.avgMemAccLat      42438.02                    # Average memory access latency per DRAM burst ((Tick/Count))
board.memory.mem_ctrl.dram.readRowHits        122737                    # Number of row buffer hits during reads (Count)
board.memory.mem_ctrl.dram.writeRowHits        60960                    # Number of row buffer hits during writes (Count)
board.memory.mem_ctrl.dram.readRowHitRate        92.43                    # Row buffer hit rate for reads (Ratio)
board.memory.mem_ctrl.dram.writeRowHitRate        92.83                    # Row buffer hit rate for writes (Ratio)
board.memory.mem_ctrl.dram.bytesPerActivate::samples        14752                    # Bytes accessed per row activation (Byte)
board.memory.mem_ctrl.dram.bytesPerActivate::mean   860.950108                    # Bytes accessed per row activation (Byte)
board.memory.mem_ctrl.dram.bytesPerActivate::gmean   757.687033                    # Bytes accessed per row activation (Byte)
board.memory.mem_ctrl.dram.bytesPerActivate::stdev   287.216403                    # Bytes accessed per row activation (Byte)
board.memory.mem_ctrl.dram.bytesPerActivate::0-127         618      4.19%      4.19% # Bytes accessed per row activation (Byte)
board.memory.mem_ctrl.dram.bytesPerActivate::128-255         243      1.65%      5.84% # Bytes accessed per row activation (Byte)
board.memory.mem_ctrl.dram.bytesPerActivate::256-383         350      2.37%      8.21% # Bytes accessed per row activation (Byte)
board.memory.mem_ctrl.dram.bytesPerActivate::384-511        1102      7.47%     15.68% # Bytes accessed per row activation (Byte)
board.memory.mem_ctrl.dram.bytesPerActivate::512-639         718      4.87%     20.55% # Bytes accessed per row activation (Byte)
board.memory.mem_ctrl.dram.bytesPerActivate::640-767         673      4.56%     25.11% # Bytes accessed per row activation (Byte)
board.memory.mem_ctrl.dram.bytesPerActivate::768-895         229      1.55%     26.66% # Bytes accessed per row activation (Byte)
board.memory.mem_ctrl.dram.bytesPerActivate::896-1023         596      4.04%     30.70% # Bytes accessed per row activation (Byte)
board.memory.mem_ctrl.dram.bytesPerActivate::1024-1151       10223     69.30%    100.00% # Bytes accessed per row activation (Byte)
board.memory.mem_ctrl.dram.bytesPerActivate::total        14752                    # Bytes accessed per row activation (Byte)
board.memory.mem_ctrl.dram.dramBytesRead      8498112                    # Total bytes read (Byte)
board.memory.mem_ctrl.dram.dramBytesWritten      4202624                    # Total bytes written (Byte)
board.memory.mem_ctrl.dram.avgRdBW        1437.950608                    # Average DRAM read bandwidth in MiBytes/s ((Byte/Second))
board.memory.mem_ctrl.dram.avgWrBW         711.118627                    # Average DRAM write bandwidth in MiBytes/s ((Byte/Second))
board.memory.mem_ctrl.dram.peakBW          12800.00                    # Theoretical peak bandwidth in MiByte/s ((Byte/Second))
board.memory.mem_ctrl.dram.busUtil            16.79                    # Data bus utilization in percentage (Ratio)
board.memory.mem_ctrl.dram.busUtilRead         11.23                    # Data bus utilization in percentage for reads (Ratio)
board.memory.mem_ctrl.dram.busUtilWrite          5.56                    # Data bus utilization in percentage for writes (Ratio)
board.memory.mem_ctrl.dram.pageHitRate         92.57                    # Row buffer hit rate, read and write combined (Ratio)
board.memory.mem_ctrl.dram.power_state.pwrStateResidencyTicks::UNDEFINED   5909877540                    # Cumulative time (in ticks) in various power states (Tick)
board.memory.mem_ctrl.dram.rank0.actEnergy     52857420                    # Energy for activate commands per rank (pJ) (Joule)
board.memory.mem_ctrl.dram.rank0.preEnergy     28094385                    # Energy for precharge commands per rank (pJ) (Joule)
board.memory.mem_ctrl.dram.rank0.readEnergy    474945660                    # Energy for read commands per rank (pJ) (Joule)
board.memory.mem_ctrl.dram.rank0.writeEnergy    171450900                    # Energy for write commands per rank (pJ) (Joule)
board.memory.mem_ctrl.dram.rank0.refreshEnergy 466511760.000000                    # Energy for refresh commands per rank (pJ) (Joule)
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board.memory.mem_ctrl.dram.rank0.actBackEnergy   1473422640                    # Energy for active background per rank (pJ) (Joule)
board.memory.mem_ctrl.dram.rank0.preBackEnergy   1028674080                    # Energy for precharge background per rank (pJ) (Joule)
board.memory.mem_ctrl.dram.rank0.actPowerDownEnergy           0                    # Energy for active power-down per rank (pJ) (Joule)
board.memory.mem_ctrl.dram.rank0.prePowerDownEnergy           0                    # Energy for precharge power-down per rank (pJ) (Joule)
board.memory.mem_ctrl.dram.rank0.selfRefreshEnergy           0                    # Energy for self refresh per rank (pJ) (Joule)
board.memory.mem_ctrl.dram.rank0.totalEnergy   3695956845                    # Total energy per rank (pJ) (Joule)
board.memory.mem_ctrl.dram.rank0.averagePower   625.386367                    # Core power per rank (mW) (Watt)
board.memory.mem_ctrl.dram.rank0.totalIdleTime           0                    # Total Idle time Per DRAM Rank (Tick)
board.memory.mem_ctrl.dram.rank0.pwrStateTime::IDLE   2627322277                    # Time in different power states (Tick)
board.memory.mem_ctrl.dram.rank0.pwrStateTime::REF    197175966                    # Time in different power states (Tick)
board.memory.mem_ctrl.dram.rank0.pwrStateTime::SREF           0                    # Time in different power states (Tick)
board.memory.mem_ctrl.dram.rank0.pwrStateTime::PRE_PDN           0                    # Time in different power states (Tick)
board.memory.mem_ctrl.dram.rank0.pwrStateTime::ACT   3085379297                    # Time in different power states (Tick)
board.memory.mem_ctrl.dram.rank0.pwrStateTime::ACT_PDN           0                    # Time in different power states (Tick)
board.memory.mem_ctrl.dram.rank1.actEnergy     52471860                    # Energy for activate commands per rank (pJ) (Joule)
board.memory.mem_ctrl.dram.rank1.preEnergy     27889455                    # Energy for precharge commands per rank (pJ) (Joule)
board.memory.mem_ctrl.dram.rank1.readEnergy    473124960                    # Energy for read commands per rank (pJ) (Joule)
board.memory.mem_ctrl.dram.rank1.writeEnergy    171325620                    # Energy for write commands per rank (pJ) (Joule)
board.memory.mem_ctrl.dram.rank1.refreshEnergy 466511760.000000                    # Energy for refresh commands per rank (pJ) (Joule)
board.memory.mem_ctrl.dram.rank1.actBackEnergy   1462376610                    # Energy for active background per rank (pJ) (Joule)
board.memory.mem_ctrl.dram.rank1.preBackEnergy   1037974080                    # Energy for precharge background per rank (pJ) (Joule)
board.memory.mem_ctrl.dram.rank1.actPowerDownEnergy           0                    # Energy for active power-down per rank (pJ) (Joule)
board.memory.mem_ctrl.dram.rank1.prePowerDownEnergy           0                    # Energy for precharge power-down per rank (pJ) (Joule)
board.memory.mem_ctrl.dram.rank1.selfRefreshEnergy           0                    # Energy for self refresh per rank (pJ) (Joule)
board.memory.mem_ctrl.dram.rank1.totalEnergy   3691674345                    # Total energy per rank (pJ) (Joule)
board.memory.mem_ctrl.dram.rank1.averagePower   624.661733                    # Core power per rank (mW) (Watt)
board.memory.mem_ctrl.dram.rank1.totalIdleTime           0                    # Total Idle time Per DRAM Rank (Tick)
board.memory.mem_ctrl.dram.rank1.pwrStateTime::IDLE   2652067107                    # Time in different power states (Tick)
board.memory.mem_ctrl.dram.rank1.pwrStateTime::REF    197180040                    # Time in different power states (Tick)
board.memory.mem_ctrl.dram.rank1.pwrStateTime::SREF           0                    # Time in different power states (Tick)
board.memory.mem_ctrl.dram.rank1.pwrStateTime::PRE_PDN           0                    # Time in different power states (Tick)
board.memory.mem_ctrl.dram.rank1.pwrStateTime::ACT   3060630393                    # Time in different power states (Tick)
board.memory.mem_ctrl.dram.rank1.pwrStateTime::ACT_PDN           0                    # Time in different power states (Tick)
board.memory.mem_ctrl.power_state.pwrStateResidencyTicks::UNDEFINED   5909877540                    # Cumulative time (in ticks) in various power states (Tick)
board.processor.cores.core.numCycles         17747381                    # Number of cpu cycles simulated (Cycle)
board.processor.cores.core.cpi             0.481857                    # CPI: cycles per instruction (core level) ((Cycle/Count))
board.processor.cores.core.ipc             2.075306                    # IPC: instructions per cycle (core level) ((Count/Cycle))
board.processor.cores.core.numWorkItemsStarted           0                    # Number of work items this cpu started (Count)
board.processor.cores.core.numWorkItemsCompleted           0                    # Number of work items this cpu completed (Count)
board.processor.cores.core.instsAdded        57074754                    # Number of instructions added to the IQ (excludes non-spec) (Count)
board.processor.cores.core.nonSpecInstsAdded          240                    # Number of non-speculative instructions added to the IQ (Count)
board.processor.cores.core.instsIssued       57028346                    # Number of instructions issued (Count)
board.processor.cores.core.squashedInstsIssued          491                    # Number of squashed instructions issued (Count)
board.processor.cores.core.squashedInstsExamined       213780                    # Number of squashed instructions iterated over during squash; mainly for profiling (Count)
board.processor.cores.core.squashedOperandsExamined       246107                    # Number of squashed operands that are examined and possibly removed from graph (Count)
board.processor.cores.core.squashedNonSpecRemoved          150                    # Number of squashed non-spec instructions that were removed (Count)
board.processor.cores.core.numIssuedDist::samples     17601764                    # Number of insts issued each cycle (Count)
board.processor.cores.core.numIssuedDist::mean     3.239922                    # Number of insts issued each cycle (Count)
board.processor.cores.core.numIssuedDist::stdev     2.164310                    # Number of insts issued each cycle (Count)
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board.processor.cores.core.numIssuedDist::underflows 0 0.00% 0.00% # Number of insts issued each cycle (Count)

board.processor.cores.core.numIssuedDist::0 1620403 9.21% 9.21% # Number of insts issued each cycle (Count)

board.processor.cores.core.numIssuedDist::1 1824237 10.36% 19.57% # Number of insts issued each cycle (Count)

board.processor.cores.core.numIssuedDist::2 4649366 26.41% 45.98% # Number of insts issued each cycle (Count)

board.processor.cores.core.numIssuedDist::3 2974535 16.90% 62.88% # Number of insts issued each cycle (Count)

board.processor.cores.core.numIssuedDist::4 1555315 8.84% 71.72% # Number of insts issued each cycle (Count)

board.processor.cores.core.numIssuedDist::5 1689834 9.60% 81.32% # Number of insts issued each cycle (Count)

board.processor.cores.core.numIssuedDist::6 1178432 6.69% 88.01% # Number of insts issued each cycle (Count)

board.processor.cores.core.numIssuedDist::7 1636386 9.30% 97.31% # Number of insts issued each cycle (Count)

board.processor.cores.core.numIssuedDist::8 473256 2.69% 100.00% # Number of insts issued each cycle (Count)

board.processor.cores.core.numIssuedDist::overflows 0 0.00% 100.00% # Number of insts issued each cycle (Count)

board.processor.cores.core.numIssuedDist::min_value 0 # Number of insts issued each cycle (Count)

board.processor.cores.core.numIssuedDist::max_value 8 # Number of insts issued each cycle (Count)

board.processor.cores.core.numIssuedDist::total 17601764 # Number of insts issued each cycle (Count)

board.processor.cores.core.statFuBusy::No_OpClass 0 0.00% 0.00% # attempts to use FU when none available (Count)

board.processor.cores.core.statFuBusy::IntAlu 1417639 99.97% 99.97% # attempts to use FU when none available (Count)

board.processor.cores.core.statFuBusy::IntMult 0 0.00% 99.97% # attempts to use FU when none available (Count)

board.processor.cores.core.statFuBusy::IntDiv 0 0.00% 99.97% # attempts to use FU when none available (Count)

board.processor.cores.core.statFuBusy::FloatAdd 0 0.00% 99.97% # attempts to use FU when none available (Count)

board.processor.cores.core.statFuBusy::FloatCmp 0 0.00% 99.97% # attempts to use FU when none available (Count)

board.processor.cores.core.statFuBusy::FloatCvt 0 0.00% 99.97% # attempts to use FU when none available (Count)

board.processor.cores.core.statFuBusy::FloatMult 0 0.00% 99.97% # attempts to use FU when none available (Count)

board.processor.cores.core.statFuBusy::FloatMultAcc 0 0.00% 99.97% # attempts to use FU when none available (Count)

board.processor.cores.core.statFuBusy::FloatDiv 0 0.00% 99.97% # attempts to use FU when none available (Count)

board.processor.cores.core.statFuBusy::FloatMisc 0 0.00% 99.97% # attempts to use FU when none available (Count)

board.processor.cores.core.statFuBusy::FloatSqrt 0 0.00% 99.97% # attempts to use FU when none available (Count)

board.processor.cores.core.statFuBusy::SimdAdd 0 0.00% 99.97% # attempts to use FU when none available (Count)

board.processor.cores.core.statFuBusy::SimdAddAcc 0 0.00% 99.97% # attempts to use FU when none available (Count)

board.processor.cores.core.statFuBusy::SimdAlu 24 0.00% 99.97% # attempts to use FU when none available (Count)

board.processor.cores.core.statFuBusy::SimdCmp 0 0.00% 99.97% # attempts to use FU when none available (Count)

board.processor.cores.core.statFuBusy::SimdCvt 0 0.00% 99.97% # attempts to use FU when none available (Count)

board.processor.cores.core.statFuBusy::SimdMisc 0 0.00% 99.97% # attempts to use FU when none available (Count)

board.processor.cores.core.statFuBusy::SimdMult 0 0.00% 99.97% # attempts to use FU when none available (Count)

board.processor.cores.core.statFuBusy::SimdMultAcc 0 0.00% 99.97% # attempts to use FU when none available (Count)

board.processor.cores.core.statFuBusy::SimdMatMultAcc 0 0.00% 99.97% # attempts to use FU when none available (Count)

board.processor.cores.core.statFuBusy::SimdShift 0 0.00% 99.97% # attempts to use FU when none available (Count)

board.processor.cores.core.statFuBusy::SimdShiftAcc 0 0.00% 99.97% # attempts to use FU when none available (Count)

board.processor.cores.core.statFuBusy::SimdDiv 0 0.00% 99.97% # attempts to use FU when none available (Count)

board.processor.cores.core.statFuBusy::SimdSqrt 0 0.00% 99.97% # attempts to use FU when none available (Count)

board.processor.cores.core.statFuBusy::SimdFloatAdd 0 0.00% 99.97% # attempts to use FU when none available (Count)

board.processor.cores.core.statFuBusy::SimdFloatAlu 0 0.00% 99.97% # attempts to use FU when none available (Count)

board.processor.cores.core.statFuBusy::SimdFloatCmp 0 0.00% 99.97% # attempts to use FU when none available (Count)

board.processor.cores.core.statFuBusy::SimdFloatCvt 0 0.00% 99.97% # attempts to use FU when none available (Count)

board.processor.cores.core.statFuBusy::SimdFloatDiv 0 0.00% 99.97% # attempts to use FU when none available (Count)

board.processor.cores.core.statFuBusy::SimdFloatMisc 0 0.00% 99.97% # attempts to use FU when none available (Count)

board.processor.cores.core.statFuBusy::SimdFloatMult 0 0.00% 99.97% # attempts to use FU when none available (Count)

board.processor.cores.core.statFuBusy::SimdFloatMultAcc 0 0.00% 99.97% # attempts to use FU when none available (Count)

board.processor.cores.core.statFuBusy::SimdFloatMatMultAcc 0 0.00% 99.97% # attempts to use FU when none available (Count)

board.processor.cores.core.statFuBusy::SimdFloatSqrt 0 0.00% 99.97% # attempts to use FU when none available (Count)

board.processor.cores.core.statFuBusy::SimdReduceAdd 0 0.00% 99.97% # attempts to use FU when none available (Count)

board.processor.cores.core.statFuBusy::SimdReduceAlu 0 0.00% 99.97% # attempts to use FU when none available (Count)

board.processor.cores.core.statFuBusy::SimdReduceCmp 0 0.00% 99.97% # attempts to use FU when none available (Count)

board.processor.cores.core.statFuBusy::SimdFloatReduceAdd 0 0.00% 99.97% # attempts to use FU when none available (Count)

board.processor.cores.core.statFuBusy::SimdFloatReduceCmp 0 0.00% 99.97% # attempts to use FU when none available (Count)

board.processor.cores.core.statFuBusy::SimdAes 0 0.00% 99.97% # attempts to use FU when none available (Count)

board.processor.cores.core.statFuBusy::SimdAesMix 0 0.00% 99.97% # attempts to use FU when none available (Count)

board.processor.cores.core.statFuBusy::SimdSha1Hash 0 0.00% 99.97% # attempts to use FU when none available (Count)

board.processor.cores.core.statFuBusy::SimdSha1Hash2 0 0.00% 99.97% # attempts to use FU when none available (Count)

board.processor.cores.core.statFuBusy::SimdSha256Hash 0 0.00% 99.97% # attempts to use FU when none available (Count)

board.processor.cores.core.statFuBusy::SimdSha256Hash2 0 0.00% 99.97% # attempts to use FU when none available (Count)

board.processor.cores.core.statFuBusy::SimdShaSigma2 0 0.00% 99.97% # attempts to use FU when none available (Count)

board.processor.cores.core.statFuBusy::SimdShaSigma3 0 0.00% 99.97% # attempts to use FU when none available (Count)

board.processor.cores.core.statFuBusy::SimdPredAlu 0 0.00% 99.97% # attempts to use FU when none available (Count)

board.processor.cores.core.statFuBusy::Matrix 0 0.00% 99.97% # attempts to use FU when none available (Count)

board.processor.cores.core.statFuBusy::MatrixMov 0 0.00% 99.97% # attempts to use FU when none available (Count)

board.processor.cores.core.statFuBusy::MatrixOP 0 0.00% 99.97% # attempts to use FU when none available (Count)

board.processor.cores.core.statFuBusy::MemRead 87 0.01% 99.98% # attempts to use FU when none available (Count)

board.processor.cores.core.statFuBusy::MemWrite 277 0.02% 100.00% # attempts to use FU when none available (Count)

board.processor.cores.core.statFuBusy::FloatMemRead 6 0.00% 100.00% # attempts to use FU when none available (Count)

board.processor.cores.core.statFuBusy::FloatMemWrite 33 0.00% 100.00% # attempts to use FU when none available (Count)

board.processor.cores.core.statFuBusy::IprAccess 0 0.00% 100.00% # attempts to use FU when none available (Count)

board.processor.cores.core.statFuBusy::InstPrefetch 0 0.00% 100.00% # attempts to use FU when none available (Count)

board.processor.cores.core.statFuBusy::SimdUnitStrideLoad 0 0.00% 100.00% # attempts to use FU when none available (Count)

board.processor.cores.core.statFuBusy::SimdUnitStrideStore 0 0.00% 100.00% # attempts to use FU when none available (Count)

board.processor.cores.core.statFuBusy::SimdUnitStrideMaskLoad 0 0.00% 100.00% # attempts to use FU when none available (Count)

board.processor.cores.core.statFuBusy::SimdUnitStrideMaskStore 0 0.00% 100.00% # attempts to use FU when none available (Count)

board.processor.cores.core.statFuBusy::SimdStridedLoad 0 0.00% 100.00% # attempts to use FU when none available (Count)

board.processor.cores.core.statFuBusy::SimdStridedStore 0 0.00% 100.00% # attempts to use FU when none available (Count)

board.processor.cores.core.statFuBusy::SimdIndexedLoad 0 0.00% 100.00% # attempts to use FU when none available (Count)

board.processor.cores.core.statFuBusy::SimdIndexedStore 0 0.00% 100.00% # attempts to use FU when none available (Count)

board.processor.cores.core.statFuBusy::SimdWholeRegisterLoad 0 0.00% 100.00% # attempts to use FU when none available (Count)

board.processor.cores.core.statFuBusy::SimdWholeRegisterStore 0 0.00% 100.00% # attempts to use FU when none available (Count)

board.processor.cores.core.statFuBusy::SimdUnitStrideFaultOnlyFirstLoad 0 0.00% 100.00% # attempts to use FU when none available (Count)

board.processor.cores.core.statFuBusy::SimdUnitStrideSegmentedLoad 0 0.00% 100.00% # attempts to use FU when none available (Count)

board.processor.cores.core.statFuBusy::SimdUnitStrideSegmentedStore 0 0.00% 100.00% # attempts to use FU when none available (Count)

board.processor.cores.core.statFuBusy::SimdExt 0 0.00% 100.00% # attempts to use FU when none available (Count)

board.processor.cores.core.statFuBusy::SimdFloatExt 0 0.00% 100.00% # attempts to use FU when none available (Count)

board.processor.cores.core.statFuBusy::SimdConfig 0 0.00% 100.00% # attempts to use FU when none available (Count)

board.processor.cores.core.statIssuedInstType_0::No_OpClass 739 0.00% 0.00% # Number of instructions issued per FU type, per thread (Count)

board.processor.cores.core.statIssuedInstType_0::IntAlu 40174883 70.45% 70.45% # Number of instructions issued per FU type, per thread (Count)

board.processor.cores.core.statIssuedInstType_0::IntMult 1053194 1.85% 72.30% # Number of instructions issued per FU type, per thread (Count)

board.processor.cores.core.statIssuedInstType_0::IntDiv 40 0.00% 72.30% # Number of instructions issued per FU type, per thread (Count)

board.processor.cores.core.statIssuedInstType_0::FloatAdd 192 0.00% 72.30% # Number of instructions issued per FU type, per thread (Count)

board.processor.cores.core.statIssuedInstType_0::FloatCmp 0 0.00% 72.30% # Number of instructions issued per FU type, per thread (Count)

board.processor.cores.core.statIssuedInstType_0::FloatCvt 0 0.00% 72.30% # Number of instructions issued per FU type, per thread (Count)

board.processor.cores.core.statIssuedInstType_0::FloatMult 0 0.00% 72.30% # Number of instructions issued per FU type, per thread (Count)

board.processor.cores.core.statIssuedInstType_0::FloatMultAcc 0 0.00% 72.30% # Number of instructions issued per FU type, per thread (Count)

board.processor.cores.core.statIssuedInstType_0::FloatDiv 0 0.00% 72.30% # Number of instructions issued per FU type, per thread (Count)

board.processor.cores.core.statIssuedInstType_0::FloatMisc 0 0.00% 72.30% # Number of instructions issued per FU type, per thread (Count)

board.processor.cores.core.statIssuedInstType_0::FloatSqrt 0 0.00% 72.30% # Number of instructions issued per FU type, per thread (Count)

board.processor.cores.core.statIssuedInstType_0::SimdAdd 10 0.00% 72.30% # Number of instructions issued per FU type, per thread (Count)

board.processor.cores.core.statIssuedInstType_0::SimdAddAcc 0 0.00% 72.30% # Number of instructions issued per FU type, per thread (Count)

board.processor.cores.core.statIssuedInstType_0::SimdAlu 281 0.00% 72.30% # Number of instructions issued per FU type, per thread (Count)

board.processor.cores.core.statIssuedInstType_0::SimdCmp 2 0.00% 72.30% # Number of instructions issued per FU type, per thread (Count)

board.processor.cores.core.statIssuedInstType_0::SimdCvt 78 0.00% 72.30% # Number of instructions issued per FU type, per thread (Count)

board.processor.cores.core.statIssuedInstType_0::SimdMisc 253 0.00% 72.30% # Number of instructions issued per FU type, per thread (Count)

board.processor.cores.core.statIssuedInstType_0::SimdMult 0 0.00% 72.30% # Number of instructions issued per FU type, per thread (Count)

board.processor.cores.core.statIssuedInstType_0::SimdMultAcc 0 0.00% 72.30% # Number of instructions issued per FU type, per thread (Count)

board.processor.cores.core.statIssuedInstType_0::SimdMatMultAcc 0 0.00% 72.30% # Number of instructions issued per FU type, per thread (Count)

board.processor.cores.core.statIssuedInstType_0::SimdShift 6 0.00% 72.30% # Number of instructions issued per FU type, per thread (Count)

board.processor.cores.core.statIssuedInstType_0::SimdShiftAcc 0 0.00% 72.30% # Number of instructions issued per FU type, per thread (Count)

board.processor.cores.core.statIssuedInstType_0::SimdDiv 0 0.00% 72.30% # Number of instructions issued per FU type, per thread (Count)

board.processor.cores.core.statIssuedInstType_0::SimdSqrt 0 0.00% 72.30% # Number of instructions issued per FU type, per thread (Count)

board.processor.cores.core.statIssuedInstType_0::SimdFloatAdd 0 0.00% 72.30% # Number of instructions issued per FU type, per thread (Count)

board.processor.cores.core.statIssuedInstType_0::SimdFloatAlu 0 0.00% 72.30% # Number of instructions issued per FU type, per thread (Count)

board.processor.cores.core.statIssuedInstType_0::SimdFloatCmp 0 0.00% 72.30% # Number of instructions issued per FU type, per thread (Count)

board.processor.cores.core.statIssuedInstType_0::SimdFloatCvt 0 0.00% 72.30% # Number of instructions issued per FU type, per thread (Count)

board.processor.cores.core.statIssuedInstType_0::SimdFloatDiv 0 0.00% 72.30% # Number of instructions issued per FU type, per thread (Count)

board.processor.cores.core.statIssuedInstType_0::SimdFloatMisc 0 0.00% 72.30% # Number of instructions issued per FU type, per thread (Count)

board.processor.cores.core.statIssuedInstType_0::SimdFloatMult 0 0.00% 72.30% # Number of instructions issued per FU type, per thread (Count)

board.processor.cores.core.statIssuedInstType_0::SimdFloatMultAcc 0 0.00% 72.30% # Number of instructions issued per FU type, per thread (Count)

board.processor.cores.core.statIssuedInstType_0::SimdFloatMatMultAcc 0 0.00% 72.30% # Number of instructions issued per FU type, per thread (Count)

board.processor.cores.core.statIssuedInstType_0::SimdFloatSqrt 0 0.00% 72.30% # Number of instructions issued per FU type, per thread (Count)

board.processor.cores.core.statIssuedInstType_0::SimdReduceAdd 0 0.00% 72.30% # Number of instructions issued per FU type, per thread (Count)

board.processor.cores.core.statIssuedInstType_0::SimdReduceAlu 0 0.00% 72.30% # Number of instructions issued per FU type, per thread (Count)

board.processor.cores.core.statIssuedInstType_0::SimdReduceCmp 0 0.00% 72.30% # Number of instructions issued per FU type, per thread (Count)

board.processor.cores.core.statIssuedInstType_0::SimdFloatReduceAdd 0 0.00% 72.30% # Number of instructions issued per FU type, per thread (Count)

board.processor.cores.core.statIssuedInstType_0::SimdFloatReduceCmp 0 0.00% 72.30% # Number of instructions issued per FU type, per thread (Count)

board.processor.cores.core.statIssuedInstType_0::SimdAes 0 0.00% 72.30% # Number of instructions issued per FU type, per thread (Count)

board.processor.cores.core.statIssuedInstType_0::SimdAesMix 0 0.00% 72.30% # Number of instructions issued per FU type, per thread (Count)

board.processor.cores.core.statIssuedInstType_0::SimdSha1Hash 0 0.00% 72.30% # Number of instructions issued per FU type, per thread (Count)

board.processor.cores.core.statIssuedInstType_0::SimdSha1Hash2 0 0.00% 72.30% # Number of instructions issued per FU type, per thread (Count)

board.processor.cores.core.statIssuedInstType_0::SimdSha256Hash 0 0.00% 72.30% # Number of instructions issued per FU type, per thread (Count)

board.processor.cores.core.statIssuedInstType_0::SimdSha256Hash2 0 0.00% 72.30% # Number of instructions issued per FU type, per thread (Count)

board.processor.cores.core.statIssuedInstType_0::SimdShaSigma2 0 0.00% 72.30% # Number of instructions issued per FU type, per thread (Count)

board.processor.cores.core.statIssuedInstType_0::SimdShaSigma3 0 0.00% 72.30% # Number of instructions issued per FU type, per thread (Count)

board.processor.cores.core.statIssuedInstType_0::SimdPredAlu 0 0.00% 72.30% # Number of instructions issued per FU type, per thread (Count)

board.processor.cores.core.statIssuedInstType_0::Matrix 0 0.00% 72.30% # Number of instructions issued per FU type, per thread (Count)

board.processor.cores.core.statIssuedInstType_0::MatrixMov 0 0.00% 72.30% # Number of instructions issued per FU type, per thread (Count)

board.processor.cores.core.statIssuedInstType_0::MatrixOP 0 0.00% 72.30% # Number of instructions issued per FU type, per thread (Count)

board.processor.cores.core.statIssuedInstType_0::MemRead 11583707 20.31% 92.61% # Number of instructions issued per FU type, per thread (Count)

board.processor.cores.core.statIssuedInstType_0::MemWrite 4214116 7.39% 100.00% # Number of instructions issued per FU type, per thread (Count)

board.processor.cores.core.statIssuedInstType_0::FloatMemRead 204 0.00% 100.00% # Number of instructions issued per FU type, per thread (Count)

board.processor.cores.core.statIssuedInstType_0::FloatMemWrite 641 0.00% 100.00% # Number of instructions issued per FU type, per thread (Count)

board.processor.cores.core.statIssuedInstType_0::IprAccess 0 0.00% 100.00% # Number of instructions issued per FU type, per thread (Count)

board.processor.cores.core.statIssuedInstType_0::InstPrefetch 0 0.00% 100.00% # Number of instructions issued per FU type, per thread (Count)

board.processor.cores.core.statIssuedInstType_0::SimdUnitStrideLoad 0 0.00% 100.00% # Number of instructions issued per FU type, per thread (Count)

board.processor.cores.core.statIssuedInstType_0::SimdUnitStrideStore 0 0.00% 100.00% # Number of instructions issued per FU type, per thread (Count)

board.processor.cores.core.statIssuedInstType_0::SimdUnitStrideMaskLoad 0 0.00% 100.00% # Number of instructions issued per FU type, per thread (Count)

board.processor.cores.core.statIssuedInstType_0::SimdUnitStrideMaskStore 0 0.00% 100.00% # Number of instructions issued per FU type, per thread (Count)

board.processor.cores.core.statIssuedInstType_0::SimdStridedLoad 0 0.00% 100.00% # Number of instructions issued per FU type, per thread (Count)

board.processor.cores.core.statIssuedInstType_0::SimdStridedStore 0 0.00% 100.00% # Number of instructions issued per FU type, per thread (Count)

board.processor.cores.core.statIssuedInstType_0::SimdIndexedLoad 0 0.00% 100.00% # Number of instructions issued per FU type, per thread (Count)

board.processor.cores.core.statIssuedInstType_0::SimdIndexedStore 0 0.00% 100.00% # Number of instructions issued per FU type, per thread (Count)

board.processor.cores.core.statIssuedInstType_0::SimdWholeRegisterLoad 0 0.00% 100.00% # Number of instructions issued per FU type, per thread (Count)

board.processor.cores.core.statIssuedInstType_0::SimdWholeRegisterStore 0 0.00% 100.00% # Number of instructions issued per FU type, per thread (Count)

board.processor.cores.core.statIssuedInstType_0::SimdUnitStrideFaultOnlyFirstLoad 0 0.00% 100.00% # Number of instructions issued per FU type, per thread (Count)

board.processor.cores.core.statIssuedInstType_0::SimdUnitStrideSegmentedLoad 0 0.00% 100.00% # Number of instructions issued per FU type, per thread (Count)

board.processor.cores.core.statIssuedInstType_0::SimdUnitStrideSegmentedStore 0 0.00% 100.00% # Number of instructions issued per FU type, per thread (Count)

board.processor.cores.core.statIssuedInstType_0::SimdExt 0 0.00% 100.00% # Number of instructions issued per FU type, per thread (Count)

board.processor.cores.core.statIssuedInstType_0::SimdFloatExt 0 0.00% 100.00% # Number of instructions issued per FU type, per thread (Count)

board.processor.cores.core.statIssuedInstType_0::SimdConfig 0 0.00% 100.00% # Number of instructions issued per FU type, per thread (Count)

board.processor.cores.core.statIssuedInstType_0::total 57028346 # Number of instructions issued per FU type, per thread (Count)

board.processor.cores.core.issueRate 3.213339 # Inst issue rate ((Count/Cycle))

board.processor.cores.core.fuBusy 1418066 # FU busy when requested (Count)

board.processor.cores.core.fuBusyRate 0.024866 # FU busy rate (busy events/executed inst) ((Count/Count))

board.processor.cores.core.intInstQueueReads 133073207 # Number of integer instruction queue reads (Count)

board.processor.cores.core.intInstQueueWrites 57286411 # Number of integer instruction queue writes (Count)

board.processor.cores.core.intInstQueueWakeupAccesses 57017716 # Number of integer instruction queue wakeup accesses (Count)

board.processor.cores.core.fpInstQueueReads 3806 # Number of floating instruction queue reads (Count)

board.processor.cores.core.fpInstQueueWrites 2473 # Number of floating instruction queue writes (Count)

board.processor.cores.core.fpInstQueueWakeupAccesses 1770 # Number of floating instruction queue wakeup accesses (Count)

board.processor.cores.core.vecInstQueueReads 0 # Number of vector instruction queue reads (Count)

board.processor.cores.core.vecInstQueueWrites 0 # Number of vector instruction queue writes (Count)

board.processor.cores.core.vecInstQueueWakeupAccesses 0 # Number of vector instruction queue wakeup accesses (Count)

board.processor.cores.core.intAluAccesses 58443762 # Number of integer alu accesses (Count)

board.processor.cores.core.fpAluAccesses 1911 # Number of floating point alu accesses (Count)

board.processor.cores.core.vecAluAccesses 0 # Number of vector alu accesses (Count)

board.processor.cores.core.numSquashedInsts 6993 # Number of squashed instructions skipped in execute (Count)

board.processor.cores.core.numSwp 0 # Number of swp insts executed (Count)

board.processor.cores.core.timesIdled 513 # Number of times that the entire CPU went into an idle state and unscheduled itself (Count)

board.processor.cores.core.idleCycles 145617 # Total number of cycles that the CPU has spent unscheduled due to idling (Cycle)

board.processor.cores.core.MemDepUnit__0.insertedLoads 11586088 # Number of loads inserted to the mem dependence unit. (Count)

board.processor.cores.core.MemDepUnit__0.insertedStores 4218786 # Number of stores inserted to the mem dependence unit. (Count)

board.processor.cores.core.MemDepUnit__0.conflictingLoads 7481482 # Number of conflicting loads. (Count)

board.processor.cores.core.MemDepUnit__0.conflictingStores 1287156 # Number of conflicting stores. (Count)

board.processor.cores.core.MemDepUnit__1.insertedLoads 0 # Number of loads inserted to the mem dependence unit. (Count)

board.processor.cores.core.MemDepUnit__1.insertedStores 0 # Number of stores inserted to the mem dependence unit. (Count)

board.processor.cores.core.MemDepUnit__1.conflictingLoads 0 # Number of conflicting loads. (Count)

board.processor.cores.core.MemDepUnit__1.conflictingStores 0 # Number of conflicting stores. (Count)

board.processor.cores.core.MemDepUnit__2.insertedLoads 0 # Number of loads inserted to the mem dependence unit. (Count)

board.processor.cores.core.MemDepUnit__2.insertedStores 0 # Number of stores inserted to the mem dependence unit. (Count)

board.processor.cores.core.MemDepUnit__2.conflictingLoads 0 # Number of conflicting loads. (Count)

board.processor.cores.core.MemDepUnit__2.conflictingStores 0 # Number of conflicting stores. (Count)

board.processor.cores.core.MemDepUnit__3.insertedLoads 0 # Number of loads inserted to the mem dependence unit. (Count)

board.processor.cores.core.MemDepUnit__3.insertedStores 0 # Number of stores inserted to the mem dependence unit. (Count)

board.processor.cores.core.MemDepUnit__3.conflictingLoads 0 # Number of conflicting loads. (Count)

board.processor.cores.core.MemDepUnit__3.conflictingStores 0 # Number of conflicting stores. (Count)

board.processor.cores.core.branchPred.lookups_0::NoBranch 4 0.00% 0.00% # Number of BP lookups (Count)

board.processor.cores.core.branchPred.lookups_0::Return 2777 0.13% 0.13% # Number of BP lookups (Count)

board.processor.cores.core.branchPred.lookups_0::CallDirect 2859 0.13% 0.26% # Number of BP lookups (Count)

board.processor.cores.core.branchPred.lookups_0::CallIndirect 174 0.01% 0.27% # Number of BP lookups (Count)

board.processor.cores.core.branchPred.lookups_0::DirectCond 2130479 99.44% 99.71% # Number of BP lookups (Count)

board.processor.cores.core.branchPred.lookups_0::DirectUncond 3746 0.17% 99.89% # Number of BP lookups (Count)

board.processor.cores.core.branchPred.lookups_0::IndirectCond 0 0.00% 99.89% # Number of BP lookups (Count)

board.processor.cores.core.branchPred.lookups_0::IndirectUncond 2440 0.11% 100.00% # Number of BP lookups (Count)

board.processor.cores.core.branchPred.lookups_0::total 2142479 # Number of BP lookups (Count)

board.processor.cores.core.branchPred.squashes_0::NoBranch 4 0.03% 0.03% # Number of branches that got squashed (completely removed) as an earlier branch was mispredicted. (Count)

board.processor.cores.core.branchPred.squashes_0::Return 408 3.39% 3.43% # Number of branches that got squashed (completely removed) as an earlier branch was mispredicted. (Count)

board.processor.cores.core.branchPred.squashes_0::CallDirect 532 4.43% 7.85% # Number of branches that got squashed (completely removed) as an earlier branch was mispredicted. (Count)

board.processor.cores.core.branchPred.squashes_0::CallIndirect 127 1.06% 8.91% # Number of branches that got squashed (completely removed) as an earlier branch was mispredicted. (Count)

board.processor.cores.core.branchPred.squashes_0::DirectCond 10221 85.03% 93.94% # Number of branches that got squashed (completely removed) as an earlier branch was mispredicted. (Count)

board.processor.cores.core.branchPred.squashes_0::DirectUncond 427 3.55% 97.49% # Number of branches that got squashed (completely removed) as an earlier branch was mispredicted. (Count)

board.processor.cores.core.branchPred.squashes_0::IndirectCond 0 0.00% 97.49% # Number of branches that got squashed (completely removed) as an earlier branch was mispredicted. (Count)

board.processor.cores.core.branchPred.squashes_0::IndirectUncond 302 2.51% 100.00% # Number of branches that got squashed (completely removed) as an earlier branch was mispredicted. (Count)

board.processor.cores.core.branchPred.squashes_0::total 12021 # Number of branches that got squashed (completely removed) as an earlier branch was mispredicted. (Count)

board.processor.cores.core.branchPred.corrected_0::NoBranch 0 0.00% 0.00% # Number of branches that got corrected but not yet commited. Branches get corrected by decode or after execute. Also a branch misprediction can be detected out-of-order. Therefore, a corrected branch might not end up beeing committed in case an even earlier branch was mispredicted (Count)

board.processor.cores.core.branchPred.corrected_0::Return 10 0.93% 0.93% # Number of branches that got corrected but not yet commited. Branches get corrected by decode or after execute. Also a branch misprediction can be detected out-of-order. Therefore, a corrected branch might not end up beeing committed in case an even earlier branch was mispredicted (Count)

board.processor.cores.core.branchPred.corrected_0::CallDirect 223 20.84% 21.78% # Number of branches that got corrected but not yet commited. Branches get corrected by decode or after execute. Also a branch misprediction can be detected out-of-order. Therefore, a corrected branch might not end up beeing committed in case an even earlier branch was mispredicted (Count)

board.processor.cores.core.branchPred.corrected_0::CallIndirect 39 3.64% 25.42% # Number of branches that got corrected but not yet commited. Branches get corrected by decode or after execute. Also a branch misprediction can be detected out-of-order. Therefore, a corrected branch might not end up beeing committed in case an even earlier branch was mispredicted (Count)

board.processor.cores.core.branchPred.corrected_0::DirectCond 601 56.17% 81.59% # Number of branches that got corrected but not yet commited. Branches get corrected by decode or after execute. Also a branch misprediction can be detected out-of-order. Therefore, a corrected branch might not end up beeing committed in case an even earlier branch was mispredicted (Count)

board.processor.cores.core.branchPred.corrected_0::DirectUncond 148 13.83% 95.42% # Number of branches that got corrected but not yet commited. Branches get corrected by decode or after execute. Also a branch misprediction can be detected out-of-order. Therefore, a corrected branch might not end up beeing committed in case an even earlier branch was mispredicted (Count)

board.processor.cores.core.branchPred.corrected_0::IndirectCond 0 0.00% 95.42% # Number of branches that got corrected but not yet commited. Branches get corrected by decode or after execute. Also a branch misprediction can be detected out-of-order. Therefore, a corrected branch might not end up beeing committed in case an even earlier branch was mispredicted (Count)

board.processor.cores.core.branchPred.corrected_0::IndirectUncond 49 4.58% 100.00% # Number of branches that got corrected but not yet commited. Branches get corrected by decode or after execute. Also a branch misprediction can be detected out-of-order. Therefore, a corrected branch might not end up beeing committed in case an even earlier branch was mispredicted (Count)

board.processor.cores.core.branchPred.corrected_0::total 1070 # Number of branches that got corrected but not yet commited. Branches get corrected by decode or after execute. Also a branch misprediction can be detected out-of-order. Therefore, a corrected branch might not end up beeing committed in case an even earlier branch was mispredicted (Count)

board.processor.cores.core.branchPred.earlyResteers_0::NoBranch 0 # Number of branches that got redirected after decode. (Count)

board.processor.cores.core.branchPred.earlyResteers_0::Return 0 # Number of branches that got redirected after decode. (Count)

board.processor.cores.core.branchPred.earlyResteers_0::CallDirect 0 # Number of branches that got redirected after decode. (Count)

board.processor.cores.core.branchPred.earlyResteers_0::CallIndirect 0 # Number of branches that got redirected after decode. (Count)

board.processor.cores.core.branchPred.earlyResteers_0::DirectCond 0 # Number of branches that got redirected after decode. (Count)

board.processor.cores.core.branchPred.earlyResteers_0::DirectUncond 0 # Number of branches that got redirected after decode. (Count)

board.processor.cores.core.branchPred.earlyResteers_0::IndirectCond 0 # Number of branches that got redirected after decode. (Count)

board.processor.cores.core.branchPred.earlyResteers_0::IndirectUncond 0 # Number of branches that got redirected after decode. (Count)

board.processor.cores.core.branchPred.earlyResteers_0::total 0 # Number of branches that got redirected after decode. (Count)

board.processor.cores.core.branchPred.committed_0::NoBranch 0 0.00% 0.00% # Number of branches finally committed (Count)

board.processor.cores.core.branchPred.committed_0::Return 2369 0.11% 0.11% # Number of branches finally committed (Count)

```
board.processor.cores.core.branchPred.committed_0::CallDirect        2327      0.11%      0.22% # Number of branches finally committed  (Count)
board.processor.cores.core.branchPred.committed_0::CallIndirect          47      0.00%      0.22% # Number of branches finally committed  (Count)
board.processor.cores.core.branchPred.committed_0::DirectCond     2120254     99.52%     99.74% # Number of branches finally committed  (Count)
board.processor.cores.core.branchPred.committed_0::DirectUncond        3319      0.16%     99.90% # Number of branches finally committed  (Count)
board.processor.cores.core.branchPred.committed_0::IndirectCond           0      0.00%     99.90% # Number of branches finally committed  (Count)
board.processor.cores.core.branchPred.committed_0::IndirectUncond        2138      0.10%    100.00% # Number of branches finally committed  (Count)
board.processor.cores.core.branchPred.committed_0::total      2130454                          # Number of branches finally committed  (Count)
board.processor.cores.core.branchPred.mispredicted_0::NoBranch           0      0.00%      0.00% # Number of committed branches that were mispredicted. (Count)
board.processor.cores.core.branchPred.mispredicted_0::Return           0      0.00%      0.00% # Number of committed branches that were mispredicted. (Count)
board.processor.cores.core.branchPred.mispredicted_0::CallDirect         109     13.36%     13.36% # Number of committed branches that were mispredicted. (Count)
board.processor.cores.core.branchPred.mispredicted_0::CallIndirect          38      4.66%     18.01% # Number of committed branches that were mispredicted. (Count)
board.processor.cores.core.branchPred.mispredicted_0::DirectCond         539     66.05%     84.07% # Number of committed branches that were mispredicted. (Count)
board.processor.cores.core.branchPred.mispredicted_0::DirectUncond          87     10.66%     94.73% # Number of committed branches that were mispredicted. (Count)
board.processor.cores.core.branchPred.mispredicted_0::IndirectCond           0      0.00%     94.73% # Number of committed branches that were mispredicted. (Count)
board.processor.cores.core.branchPred.mispredicted_0::IndirectUncond          43      5.27%    100.00% # Number of committed branches that were mispredicted. (Count)
board.processor.cores.core.branchPred.mispredicted_0::total         816                          # Number of committed branches that were mispredicted. (Count)
board.processor.cores.core.branchPred.targetProvider_0::NoTarget       22940      1.07%      1.07% # The component providing the target for taken branches (Count)
board.processor.cores.core.branchPred.targetProvider_0::BTB     2114563     98.70%     99.77% # The component providing the target for taken branches (Count)
board.processor.cores.core.branchPred.targetProvider_0::RAS        2777      0.13%     99.90% # The component providing the target for taken branches (Count)
board.processor.cores.core.branchPred.targetProvider_0::Indirect        2199      0.10%    100.00% # The component providing the target for taken branches (Count)
board.processor.cores.core.branchPred.targetProvider_0::total      2142479                          # The component providing the target for taken branches (Count)
board.processor.cores.core.branchPred.targetWrong_0::NoBranch         968     92.10%     92.10% # Number of branches where the target was incorrect or not available at prediction time. (Count)
board.processor.cores.core.branchPred.targetWrong_0::Return          72      6.85%     98.95% # Number of branches where the target was incorrect or not available at prediction time. (Count)
board.processor.cores.core.branchPred.targetWrong_0::CallDirect          10      0.95%     99.90% # Number of branches where the target was incorrect or not available at prediction time. (Count)
board.processor.cores.core.branchPred.targetWrong_0::CallIndirect           1      0.10%    100.00% # Number of branches where the target was incorrect or not available at prediction time. (Count)
board.processor.cores.core.branchPred.targetWrong_0::DirectCond           0      0.00%    100.00% # Number of branches where the target was incorrect or not available at prediction time. (Count)
board.processor.cores.core.branchPred.targetWrong_0::DirectUncond           0      0.00%    100.00% # Number of branches where the target was incorrect or not available at prediction time. (Count)
board.processor.cores.core.branchPred.targetWrong_0::IndirectCond           0      0.00%    100.00% # Number of branches where the target was incorrect or not available at prediction time. (Count)
board.processor.cores.core.branchPred.targetWrong_0::IndirectUncond           0      0.00%    100.00% # Number of branches where the target was incorrect or not available at prediction time. (Count)
board.processor.cores.core.branchPred.targetWrong_0::total        1051                          # Number of branches where the target was incorrect or not available at prediction time. (Count)
board.processor.cores.core.branchPred.condPredicted      2130483                          # Number of conditional branches predicted (Count)
board.processor.cores.core.branchPred.condPredictedTaken      2109170                          # Number of conditional branches predicted as taken (Count)
board.processor.cores.core.branchPred.condIncorrect         1070                          # Number of conditional branches incorrect (Count)
board.processor.cores.core.branchPred.predTakenBTBMiss          483                          # Number of branches predicted taken but missed in BTB (Count)
board.processor.cores.core.branchPred.NotTakenMispredicted         1005                          # Number branches predicted 'not taken' but turned out to be taken (Count)
board.processor.cores.core.branchPred.TakenMispredicted           65                          # Number branches predicted taken but are actually not taken (Count)
board.processor.cores.core.branchPred.BTBLookups      2142479                          # Number of BTB lookups (Count)
board.processor.cores.core.branchPred.BTBUpdates          907                          # Number of BTB updates (Count)
board.processor.cores.core.branchPred.BTBHits      2117827                          # Number of BTB hits (Count)
board.processor.cores.core.branchPred.BTBHitRatio     0.988494                          # BTB Hit Ratio (Ratio)
board.processor.cores.core.branchPred.BTBMispredicted          702                          # Number BTB mispredictions. No target found or target wrong (Count)
board.processor.cores.core.branchPred.indirectLookups         2614                          # Number of indirect predictor lookups. (Count)
board.processor.cores.core.branchPred.indirectHits         2199                          # Number of indirect target hits. (Count)
board.processor.cores.core.branchPred.indirectMisses          415                          # Number of indirect misses. (Count)
board.processor.cores.core.branchPred.indirectMispredicted           0                          # Number of mispredicted indirect branches. (Count)
board.processor.cores.core.branchPred.btb.lookups::NoBranch           4      0.00%      0.00% # Number of BTB lookups (Count)
board.processor.cores.core.branchPred.btb.lookups::Return        2777      0.13%      0.13% # Number of BTB lookups (Count)
board.processor.cores.core.branchPred.btb.lookups::CallDirect        2859      0.13%      0.26% # Number of BTB lookups (Count)
board.processor.cores.core.branchPred.btb.lookups::CallIndirect         174      0.01%      0.27% # Number of BTB lookups (Count)
```

board.processor.cores.core.branchPred.btb.lookups::DirectCond 2130479 99.44% 99.71% # Number of BTB lookups (Count)

board.processor.cores.core.branchPred.btb.lookups::DirectUncond 3746 0.17% 99.89% # Number of BTB lookups (Count)

board.processor.cores.core.branchPred.btb.lookups::IndirectCond 0 0.00% 99.89% # Number of BTB lookups (Count)

board.processor.cores.core.branchPred.btb.lookups::IndirectUncond 2440 0.11% 100.00% # Number of BTB lookups (Count)

board.processor.cores.core.branchPred.btb.lookups::total 2142479 # Number of BTB lookups (Count)

board.processor.cores.core.branchPred.btb.misses::NoBranch 4 0.02% 0.02% # Number of BTB misses (Count)

board.processor.cores.core.branchPred.btb.misses::Return 2773 11.25% 11.26% # Number of BTB misses (Count)

board.processor.cores.core.branchPred.btb.misses::CallDirect 361 1.46% 12.73% # Number of BTB misses (Count)

board.processor.cores.core.branchPred.btb.misses::CallIndirect 174 0.71% 13.44% # Number of BTB misses (Count)

board.processor.cores.core.branchPred.btb.misses::DirectCond 18673 75.75% 89.18% # Number of BTB misses (Count)

board.processor.cores.core.branchPred.btb.misses::DirectUncond 227 0.92% 90.10% # Number of BTB misses (Count)

board.processor.cores.core.branchPred.btb.misses::IndirectCond 0 0.00% 90.10% # Number of BTB misses (Count)

board.processor.cores.core.branchPred.btb.misses::IndirectUncond 2440 9.90% 100.00% # Number of BTB misses (Count)

board.processor.cores.core.branchPred.btb.misses::total 24652 # Number of BTB misses (Count)

board.processor.cores.core.branchPred.btb.updates::NoBranch 0 0.00% 0.00% # Number of BTB updates (Count)

board.processor.cores.core.branchPred.btb.updates::Return 0 0.00% 0.00% # Number of BTB updates (Count)

board.processor.cores.core.branchPred.btb.updates::CallDirect 223 24.59% 24.59% # Number of BTB updates (Count)

board.processor.cores.core.branchPred.btb.updates::CallIndirect 0 0.00% 24.59% # Number of BTB updates (Count)

board.processor.cores.core.branchPred.btb.updates::DirectCond 536 59.10% 83.68% # Number of BTB updates (Count)

board.processor.cores.core.branchPred.btb.updates::DirectUncond 148 16.32% 100.00% # Number of BTB updates (Count)

board.processor.cores.core.branchPred.btb.updates::IndirectCond 0 0.00% 100.00% # Number of BTB updates (Count)

board.processor.cores.core.branchPred.btb.updates::IndirectUncond 0 0.00% 100.00% # Number of BTB updates (Count)

board.processor.cores.core.branchPred.btb.updates::total 907 # Number of BTB updates (Count)

board.processor.cores.core.branchPred.btb.mispredict::NoBranch 0 0.00% 0.00% # Number of BTB mispredictions. No target found or target wrong. (Count)

board.processor.cores.core.branchPred.btb.mispredict::Return 0 0.00% 0.00% # Number of BTB mispredictions. No target found or target wrong. (Count)

board.processor.cores.core.branchPred.btb.mispredict::CallDirect 223 24.59% 24.59% # Number of BTB mispredictions. No target found or target wrong. (Count)

board.processor.cores.core.branchPred.btb.mispredict::CallIndirect 0 0.00% 24.59% # Number of BTB mispredictions. No target found or target wrong. (Count)

board.processor.cores.core.branchPred.btb.mispredict::DirectCond 536 59.10% 83.68% # Number of BTB mispredictions. No target found or target wrong. (Count)

board.processor.cores.core.branchPred.btb.mispredict::DirectUncond 148 16.32% 100.00% # Number of BTB mispredictions. No target found or target wrong. (Count)

board.processor.cores.core.branchPred.btb.mispredict::IndirectCond 0 0.00% 100.00% # Number of BTB mispredictions. No target found or target wrong. (Count)

board.processor.cores.core.branchPred.btb.mispredict::IndirectUncond 0 0.00% 100.00% # Number of BTB mispredictions. No target found or target wrong. (Count)

board.processor.cores.core.branchPred.btb.mispredict::total 907 # Number of BTB mispredictions. No target found or target wrong. (Count)

board.processor.cores.core.branchPred.btb.power_state.pwrStateResidencyTicks::UNDEFINED 5909877540 # Cumulative time (in ticks) in various power states (Tick)

board.processor.cores.core.branchPred.indirectBranchPred.lookups 2614 # Number of lookups (Count)

board.processor.cores.core.branchPred.indirectBranchPred.hits 2199 # Number of hits of a tag (Count)

board.processor.cores.core.branchPred.indirectBranchPred.misses 415 # Number of misses (Count)

board.processor.cores.core.branchPred.indirectBranchPred.targetRecords 88 # Number of targets that where recorded/installed in the cache (Count)

board.processor.cores.core.branchPred.indirectBranchPred.indirectRecords 2702 # Number of indirect branches/calls recorded in the indirect hist (Count)

board.processor.cores.core.branchPred.indirectBranchPred.speculativeOverflows 2 # Number of times more than the allowed capacity for speculative branches/calls where in flight and destroy the path history (Count)

board.processor.cores.core.branchPred.ras.pushes 3441 # Number of times a PC was pushed onto the RAS (Count)

board.processor.cores.core.branchPred.ras.pops 3436 # Number of times a PC was poped from the RAS (Count)

board.processor.cores.core.branchPred.ras.squashes 1067 # Number of times the stack operation was squashed due to wrong speculation. (Count)

board.processor.cores.core.branchPred.ras.used 2369 # Number of times the RAS is the provider (Count)

board.processor.cores.core.branchPred.ras.correct 2369 # Number of times the RAS is the provider and the prediction is correct (Count)

board.processor.cores.core.branchPred.ras.incorrect 0 # Number of times the RAS is the provider and the prediction is wrong (Count)

board.processor.cores.core.commit.commitSquashedInsts 147566 # The number of squashed insts skipped by commit (Count)

board.processor.cores.core.commit.commitNonSpecStalls 90 # The number of times commit has been forced to stall to communicate backwards (Count)

board.processor.cores.core.commit.branchMispredicts 738 # The number of times a branch was mispredicted (Count)

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board.processor.cores.core.commit.numCommittedDist::samples     17582068                    # Number of insts commited each cycle (Count)
board.processor.cores.core.commit.numCommittedDist::mean     3.234046                    # Number of insts commited each cycle (Count)
board.processor.cores.core.commit.numCommittedDist::stdev     3.204996                    # Number of insts commited each cycle (Count)
board.processor.cores.core.commit.numCommittedDist::underflows           0      0.00%      0.00% # Number of insts commited each cycle (Count)
board.processor.cores.core.commit.numCommittedDist::0      2335294     13.28%     13.28% # Number of insts commited each cycle (Count)
board.processor.cores.core.commit.numCommittedDist::1      7278365     41.40%     54.68% # Number of insts commited each cycle (Count)
board.processor.cores.core.commit.numCommittedDist::2      1764052     10.03%     64.71% # Number of insts commited each cycle (Count)
board.processor.cores.core.commit.numCommittedDist::3       123687      0.70%     65.42% # Number of insts commited each cycle (Count)
board.processor.cores.core.commit.numCommittedDist::4       123565      0.70%     66.12% # Number of insts commited each cycle (Count)
board.processor.cores.core.commit.numCommittedDist::5       121415      0.69%     66.81% # Number of insts commited each cycle (Count)
board.processor.cores.core.commit.numCommittedDist::6      1049024      5.97%     72.78% # Number of insts commited each cycle (Count)
board.processor.cores.core.commit.numCommittedDist::7         5129      0.03%     72.80% # Number of insts commited each cycle (Count)
board.processor.cores.core.commit.numCommittedDist::8      4781537     27.20%    100.00% # Number of insts commited each cycle (Count)
board.processor.cores.core.commit.numCommittedDist::overflows           0      0.00%    100.00% # Number of insts commited each cycle (Count)
board.processor.cores.core.commit.numCommittedDist::min_value           0                    # Number of insts commited each cycle (Count)
board.processor.cores.core.commit.numCommittedDist::max_value           8                    # Number of insts commited each cycle (Count)
board.processor.cores.core.commit.numCommittedDist::total     17582068                    # Number of insts commited each cycle (Count)
board.processor.cores.core.commit.amos           0                    # Number of atomic instructions committed (Count)
board.processor.cores.core.commit.membars          60                    # Number of memory barriers committed (Count)
board.processor.cores.core.commit.functionCalls         2374                    # Number of function calls committed. (Count)
board.processor.cores.core.commit.committedInstType_0::No_OpClass         316      0.00%      0.00% # Class of committed instruction (Count)
board.processor.cores.core.commit.committedInstType_0::IntAlu     40052454     70.44%     70.44% # Class of committed instruction (Count)
board.processor.cores.core.commit.committedInstType_0::IntMult      1048624      1.84%     72.28% # Class of committed instruction (Count)
board.processor.cores.core.commit.committedInstType_0::IntDiv          28      0.00%     72.28% # Class of committed instruction (Count)
board.processor.cores.core.commit.committedInstType_0::FloatAdd         154      0.00%     72.28% # Class of committed instruction (Count)
board.processor.cores.core.commit.committedInstType_0::FloatCmp           0      0.00%     72.28% # Class of committed instruction (Count)
board.processor.cores.core.commit.committedInstType_0::FloatCvt           0      0.00%     72.28% # Class of committed instruction (Count)
board.processor.cores.core.commit.committedInstType_0::FloatMult           0      0.00%     72.28% # Class of committed instruction (Count)
board.processor.cores.core.commit.committedInstType_0::FloatMultAcc           0      0.00%     72.28% # Class of committed instruction (Count)
board.processor.cores.core.commit.committedInstType_0::FloatDiv           0      0.00%     72.28% # Class of committed instruction (Count)
board.processor.cores.core.commit.committedInstType_0::FloatMisc           0      0.00%     72.28% # Class of committed instruction (Count)
board.processor.cores.core.commit.committedInstType_0::FloatSqrt           0      0.00%     72.28% # Class of committed instruction (Count)
board.processor.cores.core.commit.committedInstType_0::SimdAdd          10      0.00%     72.28% # Class of committed instruction (Count)
board.processor.cores.core.commit.committedInstType_0::SimdAddAcc           0      0.00%     72.28% # Class of committed instruction (Count)
board.processor.cores.core.commit.committedInstType_0::SimdAlu         229      0.00%     72.28% # Class of committed instruction (Count)
board.processor.cores.core.commit.committedInstType_0::SimdCmp           0      0.00%     72.28% # Class of committed instruction (Count)
board.processor.cores.core.commit.committedInstType_0::SimdCvt          72      0.00%     72.28% # Class of committed instruction (Count)
board.processor.cores.core.commit.committedInstType_0::SimdMisc         240      0.00%     72.29% # Class of committed instruction (Count)
board.processor.cores.core.commit.committedInstType_0::SimdMult           0      0.00%     72.29% # Class of committed instruction (Count)
board.processor.cores.core.commit.committedInstType_0::SimdMultAcc           0      0.00%     72.29% # Class of committed instruction (Count)
board.processor.cores.core.commit.committedInstType_0::SimdMatMultAcc           0      0.00%     72.29% # Class of committed instruction (Count)
board.processor.cores.core.commit.committedInstType_0::SimdShift           3      0.00%     72.29% # Class of committed instruction (Count)
board.processor.cores.core.commit.committedInstType_0::SimdShiftAcc           0      0.00%     72.29% # Class of committed instruction (Count)
board.processor.cores.core.commit.committedInstType_0::SimdDiv           0      0.00%     72.29% # Class of committed instruction (Count)
board.processor.cores.core.commit.committedInstType_0::SimdSqrt           0      0.00%     72.29% # Class of committed instruction (Count)
board.processor.cores.core.commit.committedInstType_0::SimdFloatAdd           0      0.00%     72.29% # Class of committed instruction (Count)
board.processor.cores.core.commit.committedInstType_0::SimdFloatAlu           0      0.00%     72.29% # Class of committed instruction (Count)
board.processor.cores.core.commit.committedInstType_0::SimdFloatCmp           0      0.00%     72.29% # Class of committed instruction (Count)
board.processor.cores.core.commit.committedInstType_0::SimdFloatCvt           0      0.00%     72.29% # Class of committed instruction (Count)
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board.processor.cores.core.commit.committedInstType_0::SimdFloatDiv 0 0.00% 72.29% # Class of committed instruction (Count)

board.processor.cores.core.commit.committedInstType_0::SimdFloatMisc 0 0.00% 72.29% # Class of committed instruction (Count)

board.processor.cores.core.commit.committedInstType_0::SimdFloatMult 0 0.00% 72.29% # Class of committed instruction (Count)

board.processor.cores.core.commit.committedInstType_0::SimdFloatMultAcc 0 0.00% 72.29% # Class of committed instruction (Count)

board.processor.cores.core.commit.committedInstType_0::SimdFloatMatMultAcc 0 0.00% 72.29% # Class of committed instruction (Count)

board.processor.cores.core.commit.committedInstType_0::SimdFloatSqrt 0 0.00% 72.29% # Class of committed instruction (Count)

board.processor.cores.core.commit.committedInstType_0::SimdReduceAdd 0 0.00% 72.29% # Class of committed instruction (Count)

board.processor.cores.core.commit.committedInstType_0::SimdReduceAlu 0 0.00% 72.29% # Class of committed instruction (Count)

board.processor.cores.core.commit.committedInstType_0::SimdReduceCmp 0 0.00% 72.29% # Class of committed instruction (Count)

board.processor.cores.core.commit.committedInstType_0::SimdFloatReduceAdd 0 0.00% 72.29% # Class of committed instruction (Count)

board.processor.cores.core.commit.committedInstType_0::SimdFloatReduceCmp 0 0.00% 72.29% # Class of committed instruction (Count)

board.processor.cores.core.commit.committedInstType_0::SimdAes 0 0.00% 72.29% # Class of committed instruction (Count)

board.processor.cores.core.commit.committedInstType_0::SimdAesMix 0 0.00% 72.29% # Class of committed instruction (Count)

board.processor.cores.core.commit.committedInstType_0::SimdSha1Hash 0 0.00% 72.29% # Class of committed instruction (Count)

board.processor.cores.core.commit.committedInstType_0::SimdSha1Hash2 0 0.00% 72.29% # Class of committed instruction (Count)

board.processor.cores.core.commit.committedInstType_0::SimdSha256Hash 0 0.00% 72.29% # Class of committed instruction (Count)

board.processor.cores.core.commit.committedInstType_0::SimdSha256Hash2 0 0.00% 72.29% # Class of committed instruction (Count)

board.processor.cores.core.commit.committedInstType_0::SimdShaSigma2 0 0.00% 72.29% # Class of committed instruction (Count)

board.processor.cores.core.commit.committedInstType_0::SimdShaSigma3 0 0.00% 72.29% # Class of committed instruction (Count)

board.processor.cores.core.commit.committedInstType_0::SimdPredAlu 0 0.00% 72.29% # Class of committed instruction (Count)

board.processor.cores.core.commit.committedInstType_0::Matrix 0 0.00% 72.29% # Class of committed instruction (Count)

board.processor.cores.core.commit.committedInstType_0::MatrixMov 0 0.00% 72.29% # Class of committed instruction (Count)

board.processor.cores.core.commit.committedInstType_0::MatrixOP 0 0.00% 72.29% # Class of committed instruction (Count)

board.processor.cores.core.commit.committedInstType_0::MemRead 11553787 20.32% 92.60% # Class of committed instruction (Count)

board.processor.cores.core.commit.committedInstType_0::MemWrite 4204624 7.39% 100.00% # Class of committed instruction (Count)

board.processor.cores.core.commit.committedInstType_0::FloatMemRead 125 0.00% 100.00% # Class of committed instruction (Count)

board.processor.cores.core.commit.committedInstType_0::FloatMemWrite 542 0.00% 100.00% # Class of committed instruction (Count)

board.processor.cores.core.commit.committedInstType_0::IprAccess 0 0.00% 100.00% # Class of committed instruction (Count)

board.processor.cores.core.commit.committedInstType_0::InstPrefetch 0 0.00% 100.00% # Class of committed instruction (Count)

board.processor.cores.core.commit.committedInstType_0::SimdUnitStrideLoad 0 0.00% 100.00% # Class of committed instruction (Count)

board.processor.cores.core.commit.committedInstType_0::SimdUnitStrideStore 0 0.00% 100.00% # Class of committed instruction (Count)

board.processor.cores.core.commit.committedInstType_0::SimdUnitStrideMaskLoad 0 0.00% 100.00% # Class of committed instruction (Count)

board.processor.cores.core.commit.committedInstType_0::SimdUnitStrideMaskStore 0 0.00% 100.00% # Class of committed instruction (Count)

board.processor.cores.core.commit.committedInstType_0::SimdStridedLoad 0 0.00% 100.00% # Class of committed instruction (Count)

board.processor.cores.core.commit.committedInstType_0::SimdStridedStore 0 0.00% 100.00% # Class of committed instruction (Count)

board.processor.cores.core.commit.committedInstType_0::SimdIndexedLoad 0 0.00% 100.00% # Class of committed instruction (Count)

board.processor.cores.core.commit.committedInstType_0::SimdIndexedStore 0 0.00% 100.00% # Class of committed instruction (Count)

board.processor.cores.core.commit.committedInstType_0::SimdWholeRegisterLoad 0 0.00% 100.00% # Class of committed instruction (Count)

board.processor.cores.core.commit.committedInstType_0::SimdWholeRegisterStore 0 0.00% 100.00% # Class of committed instruction (Count)

board.processor.cores.core.commit.committedInstType_0::SimdUnitStrideFaultOnlyFirstLoad 0 0.00% 100.00% # Class of committed instruction (Count)

board.processor.cores.core.commit.committedInstType_0::SimdUnitStrideSegmentedLoad 0 0.00% 100.00% # Class of committed instruction (Count)

board.processor.cores.core.commit.committedInstType_0::SimdUnitStrideSegmentedStore 0 0.00% 100.00% # Class of committed instruction (Count)

board.processor.cores.core.commit.committedInstType_0::SimdExt 0 0.00% 100.00% # Class of committed instruction (Count)

board.processor.cores.core.commit.committedInstType_0::SimdFloatExt 0 0.00% 100.00% # Class of committed instruction (Count)

board.processor.cores.core.commit.committedInstType_0::SimdConfig 0 0.00% 100.00% # Class of committed instruction (Count)

board.processor.cores.core.commit.committedInstType_0::total 56861208 # Class of committed instruction (Count)

board.processor.cores.core.commit.commitEligibleSamples 4781537 # number cycles where commit BW limit reached (Cycle)

board.processor.cores.core.commitStats0.numInsts 36831240 # Number of instructions committed (thread level) (Count)

board.processor.cores.core.commitStats0.numOps 56861208 # Number of ops (including micro ops) committed (thread level) (Count)

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board.processor.cores.core.commitStats0.numInstsNotNOP     36831240                    # Number of instructions committed excluding NOPs or prefetches (Count)
board.processor.cores.core.commitStats0.numOpsNotNOP     56861208                    # Number of Ops (including micro ops) Simulated (Count)
board.processor.cores.core.commitStats0.cpi     0.481857                    # CPI: cycles per instruction (thread level) ((Cycle/Count))
board.processor.cores.core.commitStats0.ipc     2.075306                    # IPC: instructions per cycle (thread level) ((Count/Cycle))
board.processor.cores.core.commitStats0.numMemRefs     15759078                    # Number of memory references committed (Count)
board.processor.cores.core.commitStats0.numFpInsts     1535                    # Number of float instructions (Count)
board.processor.cores.core.commitStats0.numIntInsts     56859973                    # Number of integer instructions (Count)
board.processor.cores.core.commitStats0.numLoadInsts     11553912                    # Number of load instructions (Count)
board.processor.cores.core.commitStats0.numStoreInsts     4205166                    # Number of store instructions (Count)
board.processor.cores.core.commitStats0.numVecInsts     0                    # Number of vector instructions (Count)
board.processor.cores.core.commitStats0.committedInstType::No_OpClass     316     0.00%     0.00% # Class of committed instruction. (Count)
board.processor.cores.core.commitStats0.committedInstType::IntAlu     40052454     70.44%     70.44% # Class of committed instruction. (Count)
board.processor.cores.core.commitStats0.committedInstType::IntMult     1048624     1.84%     72.28% # Class of committed instruction. (Count)
board.processor.cores.core.commitStats0.committedInstType::IntDiv     28     0.00%     72.28% # Class of committed instruction. (Count)
board.processor.cores.core.commitStats0.committedInstType::FloatAdd     154     0.00%     72.28% # Class of committed instruction. (Count)
board.processor.cores.core.commitStats0.committedInstType::FloatCmp     0     0.00%     72.28% # Class of committed instruction. (Count)
board.processor.cores.core.commitStats0.committedInstType::FloatCvt     0     0.00%     72.28% # Class of committed instruction. (Count)
board.processor.cores.core.commitStats0.committedInstType::FloatMult     0     0.00%     72.28% # Class of committed instruction. (Count)
board.processor.cores.core.commitStats0.committedInstType::FloatMultAcc     0     0.00%     72.28% # Class of committed instruction. (Count)
board.processor.cores.core.commitStats0.committedInstType::FloatDiv     0     0.00%     72.28% # Class of committed instruction. (Count)
board.processor.cores.core.commitStats0.committedInstType::FloatMisc     0     0.00%     72.28% # Class of committed instruction. (Count)
board.processor.cores.core.commitStats0.committedInstType::FloatSqrt     0     0.00%     72.28% # Class of committed instruction. (Count)
board.processor.cores.core.commitStats0.committedInstType::SimdAdd     10     0.00%     72.28% # Class of committed instruction. (Count)
board.processor.cores.core.commitStats0.committedInstType::SimdAddAcc     0     0.00%     72.28% # Class of committed instruction. (Count)
board.processor.cores.core.commitStats0.committedInstType::SimdAlu     229     0.00%     72.28% # Class of committed instruction. (Count)
board.processor.cores.core.commitStats0.committedInstType::SimdCmp     0     0.00%     72.28% # Class of committed instruction. (Count)
board.processor.cores.core.commitStats0.committedInstType::SimdCvt     72     0.00%     72.28% # Class of committed instruction. (Count)
board.processor.cores.core.commitStats0.committedInstType::SimdMisc     240     0.00%     72.29% # Class of committed instruction. (Count)
board.processor.cores.core.commitStats0.committedInstType::SimdMult     0     0.00%     72.29% # Class of committed instruction. (Count)
board.processor.cores.core.commitStats0.committedInstType::SimdMultAcc     0     0.00%     72.29% # Class of committed instruction. (Count)
board.processor.cores.core.commitStats0.committedInstType::SimdMatMultAcc     0     0.00%     72.29% # Class of committed instruction. (Count)
board.processor.cores.core.commitStats0.committedInstType::SimdShift     3     0.00%     72.29% # Class of committed instruction. (Count)
board.processor.cores.core.commitStats0.committedInstType::SimdShiftAcc     0     0.00%     72.29% # Class of committed instruction. (Count)
board.processor.cores.core.commitStats0.committedInstType::SimdDiv     0     0.00%     72.29% # Class of committed instruction. (Count)
board.processor.cores.core.commitStats0.committedInstType::SimdSqrt     0     0.00%     72.29% # Class of committed instruction. (Count)
board.processor.cores.core.commitStats0.committedInstType::SimdFloatAdd     0     0.00%     72.29% # Class of committed instruction. (Count)
board.processor.cores.core.commitStats0.committedInstType::SimdFloatAlu     0     0.00%     72.29% # Class of committed instruction. (Count)
board.processor.cores.core.commitStats0.committedInstType::SimdFloatCmp     0     0.00%     72.29% # Class of committed instruction. (Count)
board.processor.cores.core.commitStats0.committedInstType::SimdFloatCvt     0     0.00%     72.29% # Class of committed instruction. (Count)
board.processor.cores.core.commitStats0.committedInstType::SimdFloatDiv     0     0.00%     72.29% # Class of committed instruction. (Count)
board.processor.cores.core.commitStats0.committedInstType::SimdFloatMisc     0     0.00%     72.29% # Class of committed instruction. (Count)
board.processor.cores.core.commitStats0.committedInstType::SimdFloatMult     0     0.00%     72.29% # Class of committed instruction. (Count)
board.processor.cores.core.commitStats0.committedInstType::SimdFloatMultAcc     0     0.00%     72.29% # Class of committed instruction. (Count)
board.processor.cores.core.commitStats0.committedInstType::SimdFloatMatMultAcc     0     0.00%     72.29% # Class of committed instruction. (Count)
board.processor.cores.core.commitStats0.committedInstType::SimdFloatSqrt     0     0.00%     72.29% # Class of committed instruction. (Count)
board.processor.cores.core.commitStats0.committedInstType::SimdReduceAdd     0     0.00%     72.29% # Class of committed instruction. (Count)
board.processor.cores.core.commitStats0.committedInstType::SimdReduceAlu     0     0.00%     72.29% # Class of committed instruction. (Count)
board.processor.cores.core.commitStats0.committedInstType::SimdReduceCmp     0     0.00%     72.29% # Class of committed instruction. (Count)
board.processor.cores.core.commitStats0.committedInstType::SimdFloatReduceAdd     0     0.00%     72.29% # Class of committed instruction. (Count)
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board.processor.cores.core.commitStats0.committedInstType::SimdFloatReduceCmp 0 0.00% 72.29% # Class of committed instruction. (Count)

board.processor.cores.core.commitStats0.committedInstType::SimdAes 0 0.00% 72.29% # Class of committed instruction. (Count)

board.processor.cores.core.commitStats0.committedInstType::SimdAesMix 0 0.00% 72.29% # Class of committed instruction. (Count)

board.processor.cores.core.commitStats0.committedInstType::SimdSha1Hash 0 0.00% 72.29% # Class of committed instruction. (Count)

board.processor.cores.core.commitStats0.committedInstType::SimdSha1Hash2 0 0.00% 72.29% # Class of committed instruction. (Count)

board.processor.cores.core.commitStats0.committedInstType::SimdSha256Hash 0 0.00% 72.29% # Class of committed instruction. (Count)

board.processor.cores.core.commitStats0.committedInstType::SimdSha256Hash2 0 0.00% 72.29% # Class of committed instruction. (Count)

board.processor.cores.core.commitStats0.committedInstType::SimdShaSigma2 0 0.00% 72.29% # Class of committed instruction. (Count)

board.processor.cores.core.commitStats0.committedInstType::SimdShaSigma3 0 0.00% 72.29% # Class of committed instruction. (Count)

board.processor.cores.core.commitStats0.committedInstType::SimdPredAlu 0 0.00% 72.29% # Class of committed instruction. (Count)

board.processor.cores.core.commitStats0.committedInstType::Matrix 0 0.00% 72.29% # Class of committed instruction. (Count)

board.processor.cores.core.commitStats0.committedInstType::MatrixMov 0 0.00% 72.29% # Class of committed instruction. (Count)

board.processor.cores.core.commitStats0.committedInstType::MatrixOP 0 0.00% 72.29% # Class of committed instruction. (Count)

board.processor.cores.core.commitStats0.committedInstType::MemRead 11553787 20.32% 92.60% # Class of committed instruction. (Count)

board.processor.cores.core.commitStats0.committedInstType::MemWrite 4204624 7.39% 100.00% # Class of committed instruction. (Count)

board.processor.cores.core.commitStats0.committedInstType::FloatMemRead 125 0.00% 100.00% # Class of committed instruction. (Count)

board.processor.cores.core.commitStats0.committedInstType::FloatMemWrite 542 0.00% 100.00% # Class of committed instruction. (Count)

board.processor.cores.core.commitStats0.committedInstType::IprAccess 0 0.00% 100.00% # Class of committed instruction. (Count)

board.processor.cores.core.commitStats0.committedInstType::InstPrefetch 0 0.00% 100.00% # Class of committed instruction. (Count)

board.processor.cores.core.commitStats0.committedInstType::SimdUnitStrideLoad 0 0.00% 100.00% # Class of committed instruction. (Count)

board.processor.cores.core.commitStats0.committedInstType::SimdUnitStrideStore 0 0.00% 100.00% # Class of committed instruction. (Count)

board.processor.cores.core.commitStats0.committedInstType::SimdUnitStrideMaskLoad 0 0.00% 100.00% # Class of committed instruction. (Count)

board.processor.cores.core.commitStats0.committedInstType::SimdUnitStrideMaskStore 0 0.00% 100.00% # Class of committed instruction. (Count)

board.processor.cores.core.commitStats0.committedInstType::SimdStridedLoad 0 0.00% 100.00% # Class of committed instruction. (Count)

board.processor.cores.core.commitStats0.committedInstType::SimdStridedStore 0 0.00% 100.00% # Class of committed instruction. (Count)

board.processor.cores.core.commitStats0.committedInstType::SimdIndexedLoad 0 0.00% 100.00% # Class of committed instruction. (Count)

board.processor.cores.core.commitStats0.committedInstType::SimdIndexedStore 0 0.00% 100.00% # Class of committed instruction. (Count)

board.processor.cores.core.commitStats0.committedInstType::SimdWholeRegisterLoad 0 0.00% 100.00% # Class of committed instruction. (Count)

board.processor.cores.core.commitStats0.committedInstType::SimdWholeRegisterStore 0 0.00% 100.00% # Class of committed instruction. (Count)

board.processor.cores.core.commitStats0.committedInstType::SimdUnitStrideFaultOnlyFirstLoad 0 0.00% 100.00% # Class of committed instruction. (Count)

board.processor.cores.core.commitStats0.committedInstType::SimdUnitStrideSegmentedLoad 0 0.00% 100.00% # Class of committed instruction. (Count)

board.processor.cores.core.commitStats0.committedInstType::SimdUnitStrideSegmentedStore 0 0.00% 100.00% # Class of committed instruction. (Count)

board.processor.cores.core.commitStats0.committedInstType::SimdExt 0 0.00% 100.00% # Class of committed instruction. (Count)

board.processor.cores.core.commitStats0.committedInstType::SimdFloatExt 0 0.00% 100.00% # Class of committed instruction. (Count)

board.processor.cores.core.commitStats0.committedInstType::SimdConfig 0 0.00% 100.00% # Class of committed instruction. (Count)

board.processor.cores.core.commitStats0.committedInstType::total 56861208 # Class of committed instruction. (Count)

board.processor.cores.core.commitStats0.committedControl::IsControl 2130454 # Class of control type instructions committed (Count)

board.processor.cores.core.commitStats0.committedControl::IsDirectControl 2125900 # Class of control type instructions committed (Count)

board.processor.cores.core.commitStats0.committedControl::IsIndirectControl 4554 # Class of control type instructions committed (Count)

board.processor.cores.core.commitStats0.committedControl::IsCondControl 2120254 # Class of control type instructions committed (Count)

board.processor.cores.core.commitStats0.committedControl::IsUncondControl 10200 # Class of control type instructions committed (Count)

board.processor.cores.core.commitStats0.committedControl::IsCall 2374 # Class of control type instructions committed (Count)

board.processor.cores.core.commitStats0.committedControl::IsReturn 2369 # Class of control type instructions committed (Count)

board.processor.cores.core.decode.idleCycles 1703391 # Number of cycles decode is idle (Cycle)

board.processor.cores.core.decode.blockedCycles 8752719 # Number of cycles decode is blocked (Cycle)

board.processor.cores.core.decode.runCycles 3137573 # Number of cycles decode is running (Cycle)

board.processor.cores.core.decode.unblockCycles 4006202 # Number of cycles decode is unblocking (Cycle)

board.processor.cores.core.decode.squashCycles 1879 # Number of cycles decode is squashing (Cycle)

board.processor.cores.core.decode.branchResolved 2113620 # Number of times decode resolved a branch (Count)

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board.processor.cores.core.decode.branchMispred 386 # Number of times decode detected a branch misprediction (Count)
board.processor.cores.core.decode.decodedInsts 57099582 # Number of instructions handled by decode (Count)
board.processor.cores.core.decode.squashedInsts 2003 # Number of squashed instructions handled by decode (Count)
board.processor.cores.core.executeStats0.numInsts 57021353 # Number of executed instructions (Count)
board.processor.cores.core.executeStats0.numNop 0 # Number of nop insts executed (Count)
board.processor.cores.core.executeStats0.numBranches 2136463 # Number of branches executed (Count)
board.processor.cores.core.executeStats0.numLoadInsts 11580866 # Number of load instructions executed (Count)
board.processor.cores.core.executeStats0.numStoreInsts 4214506 # Number of stores executed (Count)
board.processor.cores.core.executeStats0.instRate 3.212945 # Inst execution rate ((Count/Cycle))
board.processor.cores.core.executeStats0.numCCRegReads 14860454 # Number of times the CC registers were read (Count)
board.processor.cores.core.executeStats0.numCCRegWrites 30569761 # Number of times the CC registers were written (Count)
board.processor.cores.core.executeStats0.numFpRegReads 2103 # Number of times the floating registers were read (Count)
board.processor.cores.core.executeStats0.numFpRegWrites 1115 # Number of times the floating registers were written (Count)
board.processor.cores.core.executeStats0.numIntRegReads 80182954 # Number of times the integer registers were read (Count)
board.processor.cores.core.executeStats0.numIntRegWrites 48542506 # Number of times the integer registers were written (Count)
board.processor.cores.core.executeStats0.numMemRefs 15795372 # Number of memory refs (Count)
board.processor.cores.core.executeStats0.numMiscRegReads 20069123 # Number of times the Misc registers were read (Count)
board.processor.cores.core.executeStats0.numVecAluAccesses 0 # Number of vector alu accesses (Count)
board.processor.cores.core.executeStats0.numDiscardedOps 0 # Number of ops (including micro ops) which were discarded before commit (Count)
board.processor.cores.core.fetch.predictedBranches 2119539 # Number of branches that fetch has predicted taken (Count)
board.processor.cores.core.fetch.cycles 17560302 # Number of cycles fetch has run and was not squashing or blocked (Cycle)
board.processor.cores.core.fetch.squashCycles 4506 # Number of cycles fetch has spent squashing (Cycle)
board.processor.cores.core.fetch.miscStallCycles 45 # Number of cycles fetch has spent waiting on interrupts, or bad addresses, or out of MSHRs (Cycle)
board.processor.cores.core.fetch.pendingTrapStallCycles 256 # Number of stall cycles due to pending traps (Cycle)
board.processor.cores.core.fetch.cacheLines 4232521 # Number of cache lines fetched (Count)
board.processor.cores.core.fetch.icacheSquashes 475 # Number of outstanding Icache misses that were squashed (Count)
board.processor.cores.core.fetch.nisnDist::samples 17601764 # Number of instructions fetched each cycle (Total) (Count)
board.processor.cores.core.fetch.nisnDist::mean 3.246564 # Number of instructions fetched each cycle (Total) (Count)
board.processor.cores.core.fetch.nisnDist::stdev 3.478866 # Number of instructions fetched each cycle (Total) (Count)
board.processor.cores.core.fetch.nisnDist::underflows 0 0.00% 0.00% # Number of instructions fetched each cycle (Total) (Count)
board.processor.cores.core.fetch.nisnDist::0 8025495 45.59% 45.59% # Number of instructions fetched each cycle (Total) (Count)
board.processor.cores.core.fetch.nisnDist::1 526110 2.99% 48.58% # Number of instructions fetched each cycle (Total) (Count)
board.processor.cores.core.fetch.nisnDist::2 858785 4.88% 53.46% # Number of instructions fetched each cycle (Total) (Count)
board.processor.cores.core.fetch.nisnDist::3 494697 2.81% 56.27% # Number of instructions fetched each cycle (Total) (Count)
board.processor.cores.core.fetch.nisnDist::4 1338844 7.61% 63.88% # Number of instructions fetched each cycle (Total) (Count)
board.processor.cores.core.fetch.nisnDist::5 528085 3.00% 66.88% # Number of instructions fetched each cycle (Total) (Count)
board.processor.cores.core.fetch.nisnDist::6 327364 1.86% 68.74% # Number of instructions fetched each cycle (Total) (Count)
board.processor.cores.core.fetch.nisnDist::7 561572 3.19% 71.93% # Number of instructions fetched each cycle (Total) (Count)
board.processor.cores.core.fetch.nisnDist::8 4940812 28.07% 100.00% # Number of instructions fetched each cycle (Total) (Count)
board.processor.cores.core.fetch.nisnDist::overflows 0 0.00% 100.00% # Number of instructions fetched each cycle (Total) (Count)
board.processor.cores.core.fetch.nisnDist::min_value 0 # Number of instructions fetched each cycle (Total) (Count)
board.processor.cores.core.fetch.nisnDist::max_value 8 # Number of instructions fetched each cycle (Total) (Count)
board.processor.cores.core.fetch.nisnDist::total 17601764 # Number of instructions fetched each cycle (Total) (Count)
board.processor.cores.core.fetchStats0.numInsts 37022401 # Number of instructions fetched (thread level) (Count)
board.processor.cores.core.fetchStats0.numOps 0 # Number of ops (including micro ops) fetched (thread level) (Count)
board.processor.cores.core.fetchStats0.fetchRate 2.086077 # Number of inst fetches per cycle ((Count/Cycle))
board.processor.cores.core.fetchStats0.numBranches 2142479 # Number of branches fetched (Count)
board.processor.cores.core.fetchStats0.branchRate 0.120721 # Number of branch fetches per cycle (Ratio)
board.processor.cores.core.fetchStats0.icacheStallCycles 38908 # ICache total stall cycles (Cycle)
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board.processor.cores.core.fetchStats0.numFetchSuspends 0 # Number of times Execute suspended instruction fetching (Count)
board.processor.cores.core.iew.idleCycles 0 # Number of cycles IEW is idle (Cycle)
board.processor.cores.core.iew.squashCycles 1879 # Number of cycles IEW is squashing (Cycle)
board.processor.cores.core.iew.blockCycles 25751 # Number of cycles IEW is blocking (Cycle)
board.processor.cores.core.iew.unblockCycles 1042 # Number of cycles IEW is unblocking (Cycle)
board.processor.cores.core.iew.dispatchedInsts 57074994 # Number of instructions dispatched to IQ (Count)
board.processor.cores.core.iew.dispSquashedInsts 115 # Number of squashed instructions skipped by dispatch (Count)
board.processor.cores.core.iew.dispLoadInsts 11586088 # Number of dispatched load instructions (Count)
board.processor.cores.core.iew.dispStoreInsts 4218786 # Number of dispatched store instructions (Count)
board.processor.cores.core.iew.dispNonSpecInsts 80 # Number of dispatched non-speculative instructions (Count)
board.processor.cores.core.iew.iqFullEvents 34 # Number of times the IQ has become full, causing a stall (Count)
board.processor.cores.core.iew.lsqFullEvents 988 # Number of times the LSQ has become full, causing a stall (Count)
board.processor.cores.core.iew.memOrderViolationEvents 111 # Number of memory order violations (Count)
board.processor.cores.core.iew.predictedTakenIncorrect 87 # Number of branches that were predicted taken incorrectly (Count)
board.processor.cores.core.iew.predictedNotTakenIncorrect 780 # Number of branches that were predicted not taken incorrectly (Count)
board.processor.cores.core.iew.branchMispredicts 867 # Number of branch mispredicts detected at execute (Count)
board.processor.cores.core.iew.instsToCommit 57021085 # Cumulative count of insts sent to commit (Count)
board.processor.cores.core.iew.writebackCount 57019486 # Cumulative count of insts written-back (Count)
board.processor.cores.core.iew.producerInst 48545570 # Number of instructions producing a value (Count)
board.processor.cores.core.iew.consumerInst 72824734 # Number of instructions consuming a value (Count)
board.processor.cores.core.iew.wbRate 3.212839 # Insts written-back per cycle ((Count/Cycle))
board.processor.cores.core.iew.wbFanout 0.666608 # Average fanout of values written-back ((Count/Count))
board.processor.cores.core.interrupts.clk_domain.clock 5328 # Clock period in ticks (Tick)
board.processor.cores.core.lsq0.forwLoads 8296342 # Number of loads that had data forwarded from stores (Count)
board.processor.cores.core.lsq0.squashedLoads 32176 # Number of loads squashed (Count)
board.processor.cores.core.lsq0.ignoredResponses 3 # Number of memory responses ignored because the instruction is squashed (Count)
board.processor.cores.core.lsq0.memOrderViolation 111 # Number of memory ordering violations (Count)
board.processor.cores.core.lsq0.squashedStores 13620 # Number of stores squashed (Count)
board.processor.cores.core.lsq0.rescheduledLoads 8 # Number of loads that were rescheduled (Count)
board.processor.cores.core.lsq0.blockedByCache 0 # Number of times an access to memory failed due to the cache being blocked (Count)
board.processor.cores.core.lsq0.loadToUse::samples 11553912 # Distribution of cycle latency between the first time a load is issued and its completion (Unspecified)
board.processor.cores.core.lsq0.loadToUse::mean 2.106734 # Distribution of cycle latency between the first time a load is issued and its completion (Unspecified)
board.processor.cores.core.lsq0.loadToUse::stdev 6.276050 # Distribution of cycle latency between the first time a load is issued and its completion (Unspecified)
board.processor.cores.core.lsq0.loadToUse::0-9 11550321 99.97% 99.97% # Distribution of cycle latency between the first time a load is issued and its completion (Unspecified)
board.processor.cores.core.lsq0.loadToUse::10-19 23 0.00% 99.97% # Distribution of cycle latency between the first time a load is issued and its completion (Unspecified)
board.processor.cores.core.lsq0.loadToUse::20-29 44 0.00% 99.97% # Distribution of cycle latency between the first time a load is issued and its completion (Unspecified)
board.processor.cores.core.lsq0.loadToUse::30-39 1 0.00% 99.97% # Distribution of cycle latency between the first time a load is issued and its completion (Unspecified)
board.processor.cores.core.lsq0.loadToUse::40-49 4 0.00% 99.97% # Distribution of cycle latency between the first time a load is issued and its completion (Unspecified)
board.processor.cores.core.lsq0.loadToUse::50-59 4 0.00% 99.97% # Distribution of cycle latency between the first time a load is issued and its completion (Unspecified)
board.processor.cores.core.lsq0.loadToUse::60-69 3 0.00% 99.97% # Distribution of cycle latency between the first time a load is issued and its completion (Unspecified)
board.processor.cores.core.lsq0.loadToUse::70-79 1 0.00% 99.97% # Distribution of cycle latency between the first time a load is issued and its completion (Unspecified)
board.processor.cores.core.lsq0.loadToUse::80-89 1 0.00% 99.97% # Distribution of cycle latency between the first time a load is issued and its completion (Unspecified)
board.processor.cores.core.lsq0.loadToUse::90-99 2 0.00% 99.97% # Distribution of cycle latency between the first time a load is issued and its completion (Unspecified)
board.processor.cores.core.lsq0.loadToUse::100-109 25 0.00% 99.97% # Distribution of cycle latency between the first time a load is issued and its completion (Unspecified)
board.processor.cores.core.lsq0.loadToUse::170-179 1 0.00% 99.97% # Distribution of cycle latency between the first time a load is issued and its completion (Unspecified)
board.processor.cores.core.lsq0.loadToUse::210-219 2 0.00% 99.97% # Distribution of cycle latency between the first time a load is issued and its completion (Unspecified)
board.processor.cores.core.lsq0.loadToUse::250-259 3 0.00% 99.97% # Distribution of cycle latency between the first time a load is issued and its completion (Unspecified)
board.processor.cores.core.lsq0.loadToUse::260-269 1 0.00% 99.97% # Distribution of cycle latency between the first time a load is issued and its completion (Unspecified)
board.processor.cores.core.lsq0.loadToUse::280-289 2 0.00% 99.97% # Distribution of cycle latency between the first time a load is issued and its completion (Unspecified)
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board.processor.cores.core.lsq0.loadToUse::290-299          12      0.00%     99.97% # Distribution of cycle latency between the first time a load is issued and its completion (Unspecified)
board.processor.cores.core.lsq0.loadToUse::overflows        3462      0.03%    100.00% # Distribution of cycle latency between the first time a load is issued and its completion (Unspecified)
board.processor.cores.core.lsq0.loadToUse::min_value           2                    # Distribution of cycle latency between the first time a load is issued and its completion (Unspecified)
board.processor.cores.core.lsq0.loadToUse::max_value        1193                    # Distribution of cycle latency between the first time a load is issued and its completion (Unspecified)
board.processor.cores.core.lsq0.loadToUse::total     11553912                    # Distribution of cycle latency between the first time a load is issued and its completion (Unspecified)
board.processor.cores.core.mmu.dtb.rdAccesses     11580873                    # TLB accesses on read requests (Count)
board.processor.cores.core.mmu.dtb.wrAccesses      4214506                    # TLB accesses on write requests (Count)
board.processor.cores.core.mmu.dtb.rdMisses        1137                    # TLB misses on read requests (Count)
board.processor.cores.core.mmu.dtb.wrMisses        2081                    # TLB misses on write requests (Count)
board.processor.cores.core.mmu.dtb.walker.power_state.pwrStateResidencyTicks::UNDEFINED   5909877540                    # Cumulative time (in ticks) in various power states (Tick)
board.processor.cores.core.mmu.itb.rdAccesses           0                    # TLB accesses on read requests (Count)
board.processor.cores.core.mmu.itb.wrAccesses      4232566                    # TLB accesses on write requests (Count)
board.processor.cores.core.mmu.itb.rdMisses           0                    # TLB misses on read requests (Count)
board.processor.cores.core.mmu.itb.wrMisses         116                    # TLB misses on write requests (Count)
board.processor.cores.core.mmu.itb.walker.power_state.pwrStateResidencyTicks::UNDEFINED   5909877540                    # Cumulative time (in ticks) in various power states (Tick)
board.processor.cores.core.power_state.pwrStateResidencyTicks::ON   5909877540                    # Cumulative time (in ticks) in various power states (Tick)
board.processor.cores.core.rename.squashCycles         1879                    # Number of cycles rename is squashing (Cycle)
board.processor.cores.core.rename.idleCycles      2874569                    # Number of cycles rename is idle (Cycle)
board.processor.cores.core.rename.blockCycles       511292                    # Number of cycles rename is blocking (Cycle)
board.processor.cores.core.rename.serializeStallCycles         2460                    # count of cycles rename stalled for serializing inst (Cycle)
board.processor.cores.core.rename.runCycles      5970993                    # Number of cycles rename is running (Cycle)
board.processor.cores.core.rename.unblockCycles      8240571                    # Number of cycles rename is unblocking (Cycle)
board.processor.cores.core.rename.renamedInsts     57095431                    # Number of instructions processed by rename (Count)
board.processor.cores.core.rename.ROBFullEvents         2779                    # Number of times rename has blocked due to ROB full (Count)
board.processor.cores.core.rename.IQFullEvents      2215487                    # Number of times rename has blocked due to IQ full (Count)
board.processor.cores.core.rename.LQFullEvents      5081084                    # Number of times rename has blocked due to LQ full (Count)
board.processor.cores.core.rename.SQFullEvents       409760                    # Number of times rename has blocked due to SQ full (Count)
board.processor.cores.core.rename.renamedOperands    127817317                    # Number of destination operands rename has renamed (Count)
board.processor.cores.core.rename.lookups    213505912                    # Number of register rename lookups that rename has made (Count)
board.processor.cores.core.rename.intLookups     80311184                    # Number of integer rename lookups (Count)
board.processor.cores.core.rename.fpLookups         2495                    # Number of floating rename lookups (Count)
board.processor.cores.core.rename.committedMaps    127275790                    # Number of HB maps that are committed (Count)
board.processor.cores.core.rename.undoneMaps       541518                    # Number of HB maps that are undone due to squashing (Count)
board.processor.cores.core.rename.serializing           45                    # count of serializing insts renamed (Count)
board.processor.cores.core.rename.tempSerializing           46                    # count of temporary serializing insts renamed (Count)
board.processor.cores.core.rename.skidInsts     16268531                    # count of insts added to the skid buffer (Count)
board.processor.cores.core.rob.reads         69795707                    # The number of ROB reads (Count)
board.processor.cores.core.rob.writes       114037311                    # The number of ROB writes (Count)
board.processor.cores.core.thread_0.numInsts     36831240                    # Number of Instructions committed (Count)
board.processor.cores.core.thread_0.numOps     56861208                    # Number of Ops committed (Count)
board.processor.cores.core.thread_0.numMemRefs           0                    # Number of Memory References (Count)
board.processor.cores.core.workload.numSyscalls          18                    # Number of system calls (Count)
board.workload.inst.arm                    0                    # number of arm instructions executed (Count)
board.workload.inst.quiesce                    0                    # number of quiesce instructions executed (Count)

---------- End Simulation Statistics   ----------
```

results/_ooo_mshr_16:

```
---------- Begin Simulation Statistics ----------
simSeconds                    0.005910          # Number of seconds simulated (Second)
simTicks                      5909877540        # Number of ticks simulated (Tick)
finalTick                     5909877540        # Number of ticks from beginning of simulation (restored from checkpoints and never reset) (Tick)
simFreq                       1000000000000     # The number of ticks per simulated second ((Tick/Second))
hostSeconds                   57.66             # Real time elapsed on the host (Second)
hostTickRate                  102496670         # The number of ticks simulated per host second (ticks/s) ((Tick/Second))
hostMemory                    1214236           # Number of bytes of host memory used (Byte)
simInsts                      36831240          # Number of instructions simulated (Count)
simOps                        56861208          # Number of ops (including micro ops) simulated (Count)
hostInstRate                  638774            # Simulator instruction rate (inst/s) ((Count/Second))
hostOpRate                    986159            # Simulator op (including micro ops) rate (op/s) ((Count/Second))
board.cache_hierarchy.dptw_caches.blockedCycles::no_mshrs      0      # number of cycles access was blocked (Cycle)
board.cache_hierarchy.dptw_caches.blockedCycles::no_targets    0      # number of cycles access was blocked (Cycle)
board.cache_hierarchy.dptw_caches.blockedCauses::no_mshrs      0      # number of times access was blocked (Count)
board.cache_hierarchy.dptw_caches.blockedCauses::no_targets    0      # number of times access was blocked (Count)
board.cache_hierarchy.dptw_caches.avgBlocked::no_mshrs         nan    # average number of cycles each access was blocked ((Cycle/Count))
board.cache_hierarchy.dptw_caches.avgBlocked::no_targets       nan    # average number of cycles each access was blocked ((Cycle/Count))
board.cache_hierarchy.dptw_caches.replacements                 0      # number of replacements (Count)
board.cache_hierarchy.dptw_caches.power_state.pwrStateResidencyTicks::UNDEFINED   5909877540   # Cumulative time (in ticks) in various power states (Tick)
board.cache_hierarchy.dptw_caches.tags.tagsInUse               0      # Average ticks per tags in use ((Tick/Count))
board.cache_hierarchy.dptw_caches.tags.totalRefs               0      # Total number of references to valid blocks. (Count)
board.cache_hierarchy.dptw_caches.tags.sampledRefs             0      # Sample count of references to valid blocks. (Count)
board.cache_hierarchy.dptw_caches.tags.avgRefs                 nan    # Average number of references to valid blocks. ((Count/Count))
board.cache_hierarchy.dptw_caches.tags.warmupTick              0      # The tick when the warmup percentage was hit. (Tick)
board.cache_hierarchy.dptw_caches.tags.tagAccesses             0      # Number of tag accesses (Count)
board.cache_hierarchy.dptw_caches.tags.dataAccesses            0      # Number of data accesses (Count)
board.cache_hierarchy.dptw_caches.tags.power_state.pwrStateResidencyTicks::UNDEFINED   5909877540   # Cumulative time (in ticks) in various power states (Tick)
board.cache_hierarchy.iptw_caches.blockedCycles::no_mshrs      0      # number of cycles access was blocked (Cycle)
board.cache_hierarchy.iptw_caches.blockedCycles::no_targets    0      # number of cycles access was blocked (Cycle)
board.cache_hierarchy.iptw_caches.blockedCauses::no_mshrs      0      # number of times access was blocked (Count)
board.cache_hierarchy.iptw_caches.blockedCauses::no_targets    0      # number of times access was blocked (Count)
board.cache_hierarchy.iptw_caches.avgBlocked::no_mshrs         nan    # average number of cycles each access was blocked ((Cycle/Count))
board.cache_hierarchy.iptw_caches.avgBlocked::no_targets       nan    # average number of cycles each access was blocked ((Cycle/Count))
board.cache_hierarchy.iptw_caches.replacements                 0      # number of replacements (Count)
board.cache_hierarchy.iptw_caches.power_state.pwrStateResidencyTicks::UNDEFINED   5909877540   # Cumulative time (in ticks) in various power states (Tick)
board.cache_hierarchy.iptw_caches.tags.tagsInUse               0      # Average ticks per tags in use ((Tick/Count))
board.cache_hierarchy.iptw_caches.tags.totalRefs               0      # Total number of references to valid blocks. (Count)
board.cache_hierarchy.iptw_caches.tags.sampledRefs             0      # Sample count of references to valid blocks. (Count)
board.cache_hierarchy.iptw_caches.tags.avgRefs                 nan    # Average number of references to valid blocks. ((Count/Count))
board.cache_hierarchy.iptw_caches.tags.warmupTick              0      # The tick when the warmup percentage was hit. (Tick)
board.cache_hierarchy.iptw_caches.tags.tagAccesses             0      # Number of tag accesses (Count)
board.cache_hierarchy.iptw_caches.tags.dataAccesses            0      # Number of data accesses (Count)
board.cache_hierarchy.iptw_caches.tags.power_state.pwrStateResidencyTicks::UNDEFINED   5909877540   # Cumulative time (in ticks) in various power states (Tick)
board.cache_hierarchy.l1dcaches.demandHits::processor.cores.core.data   7484370   # number of demand (read+write) hits (Count)
board.cache_hierarchy.l1dcaches.demandHits::total   7484370   # number of demand (read+write) hits (Count)
```

board.cache_hierarchy.l1dcaches.overallHits::processor.cores.core.data 7484370 # number of overall hits (Count)

board.cache_hierarchy.l1dcaches.overallHits::total 7484370 # number of overall hits (Count)

board.cache_hierarchy.l1dcaches.demandMisses::processor.cores.core.data 5192 # number of demand (read+write) misses (Count)

board.cache_hierarchy.l1dcaches.demandMisses::total 5192 # number of demand (read+write) misses (Count)

board.cache_hierarchy.l1dcaches.overallMisses::processor.cores.core.data 5192 # number of overall misses (Count)

board.cache_hierarchy.l1dcaches.overallMisses::total 5192 # number of overall misses (Count)

board.cache_hierarchy.l1dcaches.demandMissLatency::processor.cores.core.data 626186853 # number of demand (read+write) miss ticks (Tick)

board.cache_hierarchy.l1dcaches.demandMissLatency::total 626186853 # number of demand (read+write) miss ticks (Tick)

board.cache_hierarchy.l1dcaches.overallMissLatency::processor.cores.core.data 626186853 # number of overall miss ticks (Tick)

board.cache_hierarchy.l1dcaches.overallMissLatency::total 626186853 # number of overall miss ticks (Tick)

board.cache_hierarchy.l1dcaches.demandAccesses::processor.cores.core.data 7489562 # number of demand (read+write) accesses (Count)

board.cache_hierarchy.l1dcaches.demandAccesses::total 7489562 # number of demand (read+write) accesses (Count)

board.cache_hierarchy.l1dcaches.overallAccesses::processor.cores.core.data 7489562 # number of overall (read+write) accesses (Count)

board.cache_hierarchy.l1dcaches.overallAccesses::total 7489562 # number of overall (read+write) accesses (Count)

board.cache_hierarchy.l1dcaches.demandMissRate::processor.cores.core.data 0.000693 # miss rate for demand accesses (Ratio)

board.cache_hierarchy.l1dcaches.demandMissRate::total 0.000693 # miss rate for demand accesses (Ratio)

board.cache_hierarchy.l1dcaches.overallMissRate::processor.cores.core.data 0.000693 # miss rate for overall accesses (Ratio)

board.cache_hierarchy.l1dcaches.overallMissRate::total 0.000693 # miss rate for overall accesses (Ratio)

board.cache_hierarchy.l1dcaches.demandAvgMissLatency::processor.cores.core.data 120606.096495 # average overall miss latency in ticks ((Tick/Count))

board.cache_hierarchy.l1dcaches.demandAvgMissLatency::total 120606.096495 # average overall miss latency in ticks ((Tick/Count))

board.cache_hierarchy.l1dcaches.overallAvgMissLatency::processor.cores.core.data 120606.096495 # average overall miss latency ((Tick/Count))

board.cache_hierarchy.l1dcaches.overallAvgMissLatency::total 120606.096495 # average overall miss latency ((Tick/Count))

board.cache_hierarchy.l1dcaches.blockedCycles::no_mshrs 0 # number of cycles access was blocked (Cycle)

board.cache_hierarchy.l1dcaches.blockedCycles::no_targets 0 # number of cycles access was blocked (Cycle)

board.cache_hierarchy.l1dcaches.blockedCauses::no_mshrs 0 # number of times access was blocked (Count)

board.cache_hierarchy.l1dcaches.blockedCauses::no_targets 0 # number of times access was blocked (Count)

board.cache_hierarchy.l1dcaches.avgBlocked::no_mshrs nan # average number of cycles each access was blocked ((Cycle/Count))

board.cache_hierarchy.l1dcaches.avgBlocked::no_targets nan # average number of cycles each access was blocked ((Cycle/Count))

board.cache_hierarchy.l1dcaches.writebacks::writebacks 65717 # number of writebacks (Count)

board.cache_hierarchy.l1dcaches.writebacks::total 65717 # number of writebacks (Count)

board.cache_hierarchy.l1dcaches.demandMshrHits::processor.cores.core.data 2245 # number of demand (read+write) MSHR hits (Count)

board.cache_hierarchy.l1dcaches.demandMshrHits::total 2245 # number of demand (read+write) MSHR hits (Count)

board.cache_hierarchy.l1dcaches.overallMshrHits::processor.cores.core.data 2245 # number of overall MSHR hits (Count)

board.cache_hierarchy.l1dcaches.overallMshrHits::total 2245 # number of overall MSHR hits (Count)

board.cache_hierarchy.l1dcaches.demandMshrMisses::processor.cores.core.data 2947 # number of demand (read+write) MSHR misses (Count)

board.cache_hierarchy.l1dcaches.demandMshrMisses::total 2947 # number of demand (read+write) MSHR misses (Count)

board.cache_hierarchy.l1dcaches.overallMshrMisses::cache_hierarchy.l1dcaches.prefetcher 129120 # number of overall MSHR misses (Count)

board.cache_hierarchy.l1dcaches.overallMshrMisses::processor.cores.core.data 2947 # number of overall MSHR misses (Count)

board.cache_hierarchy.l1dcaches.overallMshrMisses::total 132067 # number of overall MSHR misses (Count)

board.cache_hierarchy.l1dcaches.demandMshrMissLatency::processor.cores.core.data 376598358 # number of demand (read+write) MSHR miss ticks (Tick)

board.cache_hierarchy.l1dcaches.demandMshrMissLatency::total 376598358 # number of demand (read+write) MSHR miss ticks (Tick)

board.cache_hierarchy.l1dcaches.overallMshrMissLatency::cache_hierarchy.l1dcaches.prefetcher 5748813809 # number of overall MSHR miss ticks (Tick)

board.cache_hierarchy.l1dcaches.overallMshrMissLatency::processor.cores.core.data 376598358 # number of overall MSHR miss ticks (Tick)

board.cache_hierarchy.l1dcaches.overallMshrMissLatency::total 6125412167 # number of overall MSHR miss ticks (Tick)

board.cache_hierarchy.l1dcaches.demandMshrMissRate::processor.cores.core.data 0.000393 # mshr miss ratio for demand accesses (Ratio)

board.cache_hierarchy.l1dcaches.demandMshrMissRate::total 0.000393 # mshr miss ratio for demand accesses (Ratio)

board.cache_hierarchy.l1dcaches.overallMshrMissRate::cache_hierarchy.l1dcaches.prefetcher inf # mshr miss ratio for overall accesses (Ratio)

board.cache_hierarchy.l1dcaches.overallMshrMissRate::processor.cores.core.data 0.000393 # mshr miss ratio for overall accesses (Ratio)

board.cache_hierarchy.l1dcaches.overallMshrMissRate::total 0.017633 # mshr miss ratio for overall accesses (Ratio)

board.cache_hierarchy.l1dcaches.demandAvgMshrMissLatency::processor.cores.core.data 127790.416695 # average overall mshr miss latency ((Tick/Count))

board.cache_hierarchy.l1dcaches.demandAvgMshrMissLatency::total 127790.416695 # average overall mshr miss latency ((Tick/Count))

board.cache_hierarchy.l1dcaches.overallAvgMshrMissLatency::cache_hierarchy.l1dcaches.prefetcher 44523.031358 # average overall mshr miss latency ((Tick/Count))

board.cache_hierarchy.l1dcaches.overallAvgMshrMissLatency::processor.cores.core.data 127790.416695 # average overall mshr miss latency ((Tick/Count))

board.cache_hierarchy.l1dcaches.overallAvgMshrMissLatency::total 46381.095709 # average overall mshr miss latency ((Tick/Count))

board.cache_hierarchy.l1dcaches.replacements 131556 # number of replacements (Count)

board.cache_hierarchy.l1dcaches.HardPFReq.mshrMisses::cache_hierarchy.l1dcaches.prefetcher 129120 # number of HardPFReq MSHR misses (Count)

board.cache_hierarchy.l1dcaches.HardPFReq.mshrMisses::total 129120 # number of HardPFReq MSHR misses (Count)

board.cache_hierarchy.l1dcaches.HardPFReq.mshrMissLatency::cache_hierarchy.l1dcaches.prefetcher 5748813809 # number of HardPFReq MSHR miss ticks (Tick)

board.cache_hierarchy.l1dcaches.HardPFReq.mshrMissLatency::total 5748813809 # number of HardPFReq MSHR miss ticks (Tick)

board.cache_hierarchy.l1dcaches.HardPFReq.mshrMissRate::cache_hierarchy.l1dcaches.prefetcher inf # mshr miss rate for HardPFReq accesses (Ratio)

board.cache_hierarchy.l1dcaches.HardPFReq.mshrMissRate::total inf # mshr miss rate for HardPFReq accesses (Ratio)

board.cache_hierarchy.l1dcaches.HardPFReq.avgMshrMissLatency::cache_hierarchy.l1dcaches.prefetcher 44523.031358 # average HardPFReq mshr miss latency ((Tick/Count))

board.cache_hierarchy.l1dcaches.HardPFReq.avgMshrMissLatency::total 44523.031358 # average HardPFReq mshr miss latency ((Tick/Count))

board.cache_hierarchy.l1dcaches.LockedRMWReadReq.hits::processor.cores.core.data 26 # number of LockedRMWReadReq hits (Count)

board.cache_hierarchy.l1dcaches.LockedRMWReadReq.hits::total 26 # number of LockedRMWReadReq hits (Count)

board.cache_hierarchy.l1dcaches.LockedRMWReadReq.misses::processor.cores.core.data 4 # number of LockedRMWReadReq misses (Count)

board.cache_hierarchy.l1dcaches.LockedRMWReadReq.misses::total 4 # number of LockedRMWReadReq misses (Count)

board.cache_hierarchy.l1dcaches.LockedRMWReadReq.missLatency::processor.cores.core.data 516150 # number of LockedRMWReadReq miss ticks (Tick)

board.cache_hierarchy.l1dcaches.LockedRMWReadReq.missLatency::total 516150 # number of LockedRMWReadReq miss ticks (Tick)

board.cache_hierarchy.l1dcaches.LockedRMWReadReq.accesses::processor.cores.core.data 30 # number of LockedRMWReadReq accesses(hits+misses) (Count)

board.cache_hierarchy.l1dcaches.LockedRMWReadReq.accesses::total 30 # number of LockedRMWReadReq accesses(hits+misses) (Count)

board.cache_hierarchy.l1dcaches.LockedRMWReadReq.missRate::processor.cores.core.data 0.133333 # miss rate for LockedRMWReadReq accesses (Ratio)

board.cache_hierarchy.l1dcaches.LockedRMWReadReq.missRate::total 0.133333 # miss rate for LockedRMWReadReq accesses (Ratio)

board.cache_hierarchy.l1dcaches.LockedRMWReadReq.avgMissLatency::processor.cores.core.data 129037.500000 # average LockedRMWReadReq miss latency ((Tick/Count))

board.cache_hierarchy.l1dcaches.LockedRMWReadReq.avgMissLatency::total 129037.500000 # average LockedRMWReadReq miss latency ((Tick/Count))

board.cache_hierarchy.l1dcaches.LockedRMWReadReq.mshrMisses::processor.cores.core.data 4 # number of LockedRMWReadReq MSHR misses (Count)

board.cache_hierarchy.l1dcaches.LockedRMWReadReq.mshrMisses::total 4 # number of LockedRMWReadReq MSHR misses (Count)

board.cache_hierarchy.l1dcaches.LockedRMWReadReq.mshrMissLatency::processor.cores.core.data 1097901 # number of LockedRMWReadReq MSHR miss ticks (Tick)

board.cache_hierarchy.l1dcaches.LockedRMWReadReq.mshrMissLatency::total 1097901 # number of LockedRMWReadReq MSHR miss ticks (Tick)

board.cache_hierarchy.l1dcaches.LockedRMWReadReq.mshrMissRate::processor.cores.core.data 0.133333 # mshr miss rate for LockedRMWReadReq accesses (Ratio)

board.cache_hierarchy.l1dcaches.LockedRMWReadReq.mshrMissRate::total 0.133333 # mshr miss rate for LockedRMWReadReq accesses (Ratio)

board.cache_hierarchy.l1dcaches.LockedRMWReadReq.avgMshrMissLatency::processor.cores.core.data 274475.250000 # average LockedRMWReadReq mshr miss latency ((Tick/Count))

board.cache_hierarchy.l1dcaches.LockedRMWReadReq.avgMshrMissLatency::total 274475.250000 # average LockedRMWReadReq mshr miss latency ((Tick/Count))

board.cache_hierarchy.l1dcaches.LockedRMWWriteReq.hits::processor.cores.core.data 30 # number of LockedRMWWriteReq hits (Count)

board.cache_hierarchy.l1dcaches.LockedRMWWriteReq.hits::total 30 # number of LockedRMWWriteReq hits (Count)

board.cache_hierarchy.l1dcaches.LockedRMWWriteReq.accesses::processor.cores.core.data 30 # number of LockedRMWWriteReq accesses(hits+misses) (Count)

board.cache_hierarchy.l1dcaches.LockedRMWWriteReq.accesses::total 30 # number of LockedRMWWriteReq accesses(hits+misses) (Count)

board.cache_hierarchy.l1dcaches.ReadReq.hits::processor.cores.core.data 3280589 # number of ReadReq hits (Count)

board.cache_hierarchy.l1dcaches.ReadReq.hits::total 3280589 # number of ReadReq hits (Count)

board.cache_hierarchy.l1dcaches.ReadReq.misses::processor.cores.core.data 3837 # number of ReadReq misses (Count)

board.cache_hierarchy.l1dcaches.ReadReq.misses::total 3837 # number of ReadReq misses (Count)

board.cache_hierarchy.l1dcaches.ReadReq.missLatency::processor.cores.core.data 436806423 # number of ReadReq miss ticks (Tick)

board.cache_hierarchy.l1dcaches.ReadReq.missLatency::total 436806423 # number of ReadReq miss ticks (Tick)

board.cache_hierarchy.l1dcaches.ReadReq.accesses::processor.cores.core.data 3284426 # number of ReadReq accesses(hits+misses) (Count)

board.cache_hierarchy.l1dcaches.ReadReq.accesses::total 3284426 # number of ReadReq accesses(hits+misses) (Count)

board.cache_hierarchy.l1dcaches.ReadReq.missRate::processor.cores.core.data 0.001168 # miss rate for ReadReq accesses (Ratio)

board.cache_hierarchy.l1dcaches.ReadReq.missRate::total 0.001168 # miss rate for ReadReq accesses (Ratio)

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board.cache_hierarchy.l1dcaches.ReadReq.avgMissLatency::processor.cores.core.data 113840.610633 # average ReadReq miss latency ((Tick/Count))
board.cache_hierarchy.l1dcaches.ReadReq.avgMissLatency::total 113840.610633 # average ReadReq miss latency ((Tick/Count))
board.cache_hierarchy.l1dcaches.ReadReq.mshrHits::processor.cores.core.data 2029 # number of ReadReq MSHR hits (Count)
board.cache_hierarchy.l1dcaches.ReadReq.mshrHits::total 2029 # number of ReadReq MSHR hits (Count)
board.cache_hierarchy.l1dcaches.ReadReq.mshrMisses::processor.cores.core.data 1808 # number of ReadReq MSHR misses (Count)
board.cache_hierarchy.l1dcaches.ReadReq.mshrMisses::total 1808 # number of ReadReq MSHR misses (Count)
board.cache_hierarchy.l1dcaches.ReadReq.mshrMissLatency::processor.cores.core.data 220720392 # number of ReadReq MSHR miss ticks (Tick)
board.cache_hierarchy.l1dcaches.ReadReq.mshrMissLatency::total 220720392 # number of ReadReq MSHR miss ticks (Tick)
board.cache_hierarchy.l1dcaches.ReadReq.mshrMissRate::processor.cores.core.data 0.000550 # mshr miss rate for ReadReq accesses (Ratio)
board.cache_hierarchy.l1dcaches.ReadReq.mshrMissRate::total 0.000550 # mshr miss rate for ReadReq accesses (Ratio)
board.cache_hierarchy.l1dcaches.ReadReq.avgMshrMissLatency::processor.cores.core.data 122079.862832 # average ReadReq mshr miss latency ((Tick/Count))
board.cache_hierarchy.l1dcaches.ReadReq.avgMshrMissLatency::total 122079.862832 # average ReadReq mshr miss latency ((Tick/Count))
board.cache_hierarchy.l1dcaches.WriteReq.hits::processor.cores.core.data 4203781 # number of WriteReq hits (Count)
board.cache_hierarchy.l1dcaches.WriteReq.hits::total 4203781 # number of WriteReq hits (Count)
board.cache_hierarchy.l1dcaches.WriteReq.misses::processor.cores.core.data 1355 # number of WriteReq misses (Count)
board.cache_hierarchy.l1dcaches.WriteReq.misses::total 1355 # number of WriteReq misses (Count)
board.cache_hierarchy.l1dcaches.WriteReq.missLatency::processor.cores.core.data 189380430 # number of WriteReq miss ticks (Tick)
board.cache_hierarchy.l1dcaches.WriteReq.missLatency::total 189380430 # number of WriteReq miss ticks (Tick)
board.cache_hierarchy.l1dcaches.WriteReq.accesses::processor.cores.core.data 4205136 # number of WriteReq accesses(hits+misses) (Count)
board.cache_hierarchy.l1dcaches.WriteReq.accesses::total 4205136 # number of WriteReq accesses(hits+misses) (Count)
board.cache_hierarchy.l1dcaches.WriteReq.missRate::processor.cores.core.data 0.000322 # miss rate for WriteReq accesses (Ratio)
board.cache_hierarchy.l1dcaches.WriteReq.missRate::total 0.000322 # miss rate for WriteReq accesses (Ratio)
board.cache_hierarchy.l1dcaches.WriteReq.avgMissLatency::processor.cores.core.data 139764.154982 # average WriteReq miss latency ((Tick/Count))
board.cache_hierarchy.l1dcaches.WriteReq.avgMissLatency::total 139764.154982 # average WriteReq miss latency ((Tick/Count))
board.cache_hierarchy.l1dcaches.WriteReq.mshrHits::processor.cores.core.data 216 # number of WriteReq MSHR hits (Count)
board.cache_hierarchy.l1dcaches.WriteReq.mshrHits::total 216 # number of WriteReq MSHR hits (Count)
board.cache_hierarchy.l1dcaches.WriteReq.mshrMisses::processor.cores.core.data 1139 # number of WriteReq MSHR misses (Count)
board.cache_hierarchy.l1dcaches.WriteReq.mshrMisses::total 1139 # number of WriteReq MSHR misses (Count)
board.cache_hierarchy.l1dcaches.WriteReq.mshrMissLatency::processor.cores.core.data 155877966 # number of WriteReq MSHR miss ticks (Tick)
board.cache_hierarchy.l1dcaches.WriteReq.mshrMissLatency::total 155877966 # number of WriteReq MSHR miss ticks (Tick)
board.cache_hierarchy.l1dcaches.WriteReq.mshrMissRate::processor.cores.core.data 0.000271 # mshr miss rate for WriteReq accesses (Ratio)
board.cache_hierarchy.l1dcaches.WriteReq.mshrMissRate::total 0.000271 # mshr miss rate for WriteReq accesses (Ratio)
board.cache_hierarchy.l1dcaches.WriteReq.avgMshrMissLatency::processor.cores.core.data 136855.106234 # average WriteReq mshr miss latency ((Tick/Count))
board.cache_hierarchy.l1dcaches.WriteReq.avgMshrMissLatency::total 136855.106234 # average WriteReq mshr miss latency ((Tick/Count))
board.cache_hierarchy.l1dcaches.power_state.pwrStateResidencyTicks::UNDEFINED 5909877540 # Cumulative time (in ticks) in various power states (Tick)
board.cache_hierarchy.l1dcaches.prefetcher.demandMshrMisses 2947 # demands not covered by prefetchs (Count)
board.cache_hierarchy.l1dcaches.prefetcher.pfIssued 504148 # number of hwpf issued (Count)
board.cache_hierarchy.l1dcaches.prefetcher.pfUnused 19 # number of HardPF blocks evicted w/o reference (Count)
board.cache_hierarchy.l1dcaches.prefetcher.pfUseful 128792 # number of useful prefetch (Count)
board.cache_hierarchy.l1dcaches.prefetcher.pfUsefulButMiss 0 # number of hit on prefetch but cache block is not in an usable state (Count)
board.cache_hierarchy.l1dcaches.prefetcher.accuracy 0.255465 # accuracy of the prefetcher (Count)
board.cache_hierarchy.l1dcaches.prefetcher.coverage 0.977630 # coverage brought by this prefetcher (Count)
board.cache_hierarchy.l1dcaches.prefetcher.pfHitInCache 307514 # number of prefetches hitting in cache (Count)
board.cache_hierarchy.l1dcaches.prefetcher.pfHitInMSHR 67514 # number of prefetches hitting in a MSHR (Count)
board.cache_hierarchy.l1dcaches.prefetcher.pfHitInWB 0 # number of prefetches hit in the Write Buffer (Count)
board.cache_hierarchy.l1dcaches.prefetcher.pfLate 375028 # number of late prefetches (hitting in cache, MSHR or WB) (Count)
board.cache_hierarchy.l1dcaches.prefetcher.pfIdentified 504175 # number of prefetch candidates identified (Count)
board.cache_hierarchy.l1dcaches.prefetcher.pfBufferHit 24 # number of redundant prefetches already in prefetch queue (Count)
board.cache_hierarchy.l1dcaches.prefetcher.pfInCache 0 # number of redundant prefetches already in cache/mshr dropped (Count)
```

board.cache_hierarchy.l1dcaches.prefetcher.pfRemovedDemand 0 # number of prefetches dropped due to a demand for the same address (Count)

board.cache_hierarchy.l1dcaches.prefetcher.pfRemovedFull 0 # number of prefetches dropped due to prefetch queue size (Count)

board.cache_hierarchy.l1dcaches.prefetcher.pfSpanPage 20489 # number of prefetches that crossed the page (Count)

board.cache_hierarchy.l1dcaches.prefetcher.pfUsefulSpanPage 20482 # number of prefetches that is useful and crossed the page (Count)

board.cache_hierarchy.l1dcaches.prefetcher.power_state.pwrStateResidencyTicks::UNDEFINED 5909877540 # Cumulative time (in ticks) in various power states (Tick)

board.cache_hierarchy.l1dcaches.tags.tagsInUse 508.417902 # Average ticks per tags in use ((Tick/Count))

board.cache_hierarchy.l1dcaches.tags.totalRefs 7616499 # Total number of references to valid blocks. (Count)

board.cache_hierarchy.l1dcaches.tags.sampledRefs 132068 # Sample count of references to valid blocks. (Count)

board.cache_hierarchy.l1dcaches.tags.avgRefs 57.671041 # Average number of references to valid blocks. ((Count/Count))

board.cache_hierarchy.l1dcaches.tags.warmupTick 249417 # The tick when the warmup percentage was hit. (Tick)

board.cache_hierarchy.l1dcaches.tags.occupancies::cache_hierarchy.l1dcaches.prefetcher 490.511818 # Average occupied blocks per tick, per requestor ((Count/Tick))

board.cache_hierarchy.l1dcaches.tags.occupancies::processor.cores.core.data 17.906084 # Average occupied blocks per tick, per requestor ((Count/Tick))

board.cache_hierarchy.l1dcaches.tags.avgOccs::cache_hierarchy.l1dcaches.prefetcher 0.958031 # Average percentage of cache occupancy ((Ratio/Tick))

board.cache_hierarchy.l1dcaches.tags.avgOccs::processor.cores.core.data 0.034973 # Average percentage of cache occupancy ((Ratio/Tick))

board.cache_hierarchy.l1dcaches.tags.avgOccs::total 0.993004 # Average percentage of cache occupancy ((Ratio/Tick))

board.cache_hierarchy.l1dcaches.tags.occupanciesTaskId::1022 421 # Occupied blocks per task id (Count)

board.cache_hierarchy.l1dcaches.tags.occupanciesTaskId::1024 91 # Occupied blocks per task id (Count)

board.cache_hierarchy.l1dcaches.tags.ageTaskId_1022::0 4 # Occupied blocks per task id, per block age (Count)

board.cache_hierarchy.l1dcaches.tags.ageTaskId_1022::1 417 # Occupied blocks per task id, per block age (Count)

board.cache_hierarchy.l1dcaches.tags.ageTaskId_1024::0 43 # Occupied blocks per task id, per block age (Count)

board.cache_hierarchy.l1dcaches.tags.ageTaskId_1024::1 47 # Occupied blocks per task id, per block age (Count)

board.cache_hierarchy.l1dcaches.tags.ageTaskId_1024::3 1 # Occupied blocks per task id, per block age (Count)

board.cache_hierarchy.l1dcaches.tags.ratioOccsTaskId::1022 0.822266 # Ratio of occupied blocks and all blocks, per task id (Ratio)

board.cache_hierarchy.l1dcaches.tags.ratioOccsTaskId::1024 0.177734 # Ratio of occupied blocks and all blocks, per task id (Ratio)

board.cache_hierarchy.l1dcaches.tags.tagAccesses 60049044 # Number of tag accesses (Count)

board.cache_hierarchy.l1dcaches.tags.dataAccesses 60049044 # Number of data accesses (Count)

board.cache_hierarchy.l1dcaches.tags.power_state.pwrStateResidencyTicks::UNDEFINED 5909877540 # Cumulative time (in ticks) in various power states (Tick)

board.cache_hierarchy.l1icaches.demandHits::processor.cores.core.inst 4231540 # number of demand (read+write) hits (Count)

board.cache_hierarchy.l1icaches.demandHits::total 4231540 # number of demand (read+write) hits (Count)

board.cache_hierarchy.l1icaches.overallHits::processor.cores.core.inst 4231540 # number of overall hits (Count)

board.cache_hierarchy.l1icaches.overallHits::total 4231540 # number of overall hits (Count)

board.cache_hierarchy.l1icaches.demandMisses::processor.cores.core.inst 977 # number of demand (read+write) misses (Count)

board.cache_hierarchy.l1icaches.demandMisses::total 977 # number of demand (read+write) misses (Count)

board.cache_hierarchy.l1icaches.overallMisses::processor.cores.core.inst 977 # number of overall misses (Count)

board.cache_hierarchy.l1icaches.overallMisses::total 977 # number of overall misses (Count)

board.cache_hierarchy.l1icaches.demandMissLatency::processor.cores.core.inst 112883337 # number of demand (read+write) miss ticks (Tick)

board.cache_hierarchy.l1icaches.demandMissLatency::total 112883337 # number of demand (read+write) miss ticks (Tick)

board.cache_hierarchy.l1icaches.overallMissLatency::processor.cores.core.inst 112883337 # number of overall miss ticks (Tick)

board.cache_hierarchy.l1icaches.overallMissLatency::total 112883337 # number of overall miss ticks (Tick)

board.cache_hierarchy.l1icaches.demandAccesses::processor.cores.core.inst 4232517 # number of demand (read+write) accesses (Count)

board.cache_hierarchy.l1icaches.demandAccesses::total 4232517 # number of demand (read+write) accesses (Count)

board.cache_hierarchy.l1icaches.overallAccesses::processor.cores.core.inst 4232517 # number of overall (read+write) accesses (Count)

board.cache_hierarchy.l1icaches.overallAccesses::total 4232517 # number of overall (read+write) accesses (Count)

board.cache_hierarchy.l1icaches.demandMissRate::processor.cores.core.inst 0.000231 # miss rate for demand accesses (Ratio)

board.cache_hierarchy.l1icaches.demandMissRate::total 0.000231 # miss rate for demand accesses (Ratio)

board.cache_hierarchy.l1icaches.overallMissRate::processor.cores.core.inst 0.000231 # miss rate for overall accesses (Ratio)

board.cache_hierarchy.l1icaches.overallMissRate::total 0.000231 # miss rate for overall accesses (Ratio)

board.cache_hierarchy.l1icaches.demandAvgMissLatency::processor.cores.core.inst 115540.774821 # average overall miss latency in ticks ((Tick/Count))

board.cache_hierarchy.l1icaches.demandAvgMissLatency::total 115540.774821 # average overall miss latency in ticks ((Tick/Count))

board.cache_hierarchy.l1icaches.overallAvgMissLatency::processor.cores.core.inst 115540.774821 # average overall miss latency ((Tick/Count))

board.cache_hierarchy.l1icaches.overallAvgMissLatency::total 115540.774821 # average overall miss latency ((Tick/Count))

board.cache_hierarchy.l1icaches.blockedCycles::no_mshrs 0 # number of cycles access was blocked (Cycle)

board.cache_hierarchy.l1icaches.blockedCycles::no_targets 0 # number of cycles access was blocked (Cycle)

board.cache_hierarchy.l1icaches.blockedCauses::no_mshrs 0 # number of times access was blocked (Count)

board.cache_hierarchy.l1icaches.blockedCauses::no_targets 0 # number of times access was blocked (Count)

board.cache_hierarchy.l1icaches.avgBlocked::no_mshrs nan # average number of cycles each access was blocked ((Cycle/Count))

board.cache_hierarchy.l1icaches.avgBlocked::no_targets nan # average number of cycles each access was blocked ((Cycle/Count))

board.cache_hierarchy.l1icaches.demandMshrHits::processor.cores.core.inst 224 # number of demand (read+write) MSHR hits (Count)

board.cache_hierarchy.l1icaches.demandMshrHits::total 224 # number of demand (read+write) MSHR hits (Count)

board.cache_hierarchy.l1icaches.overallMshrHits::processor.cores.core.inst 224 # number of overall MSHR hits (Count)

board.cache_hierarchy.l1icaches.overallMshrHits::total 224 # number of overall MSHR hits (Count)

board.cache_hierarchy.l1icaches.demandMshrMisses::processor.cores.core.inst 753 # number of demand (read+write) MSHR misses (Count)

board.cache_hierarchy.l1icaches.demandMshrMisses::total 753 # number of demand (read+write) MSHR misses (Count)

board.cache_hierarchy.l1icaches.overallMshrMisses::processor.cores.core.inst 753 # number of overall MSHR misses (Count)

board.cache_hierarchy.l1icaches.overallMshrMisses::total 753 # number of overall MSHR misses (Count)

board.cache_hierarchy.l1icaches.demandMshrMissLatency::processor.cores.core.inst 90479430 # number of demand (read+write) MSHR miss ticks (Tick)

board.cache_hierarchy.l1icaches.demandMshrMissLatency::total 90479430 # number of demand (read+write) MSHR miss ticks (Tick)

board.cache_hierarchy.l1icaches.overallMshrMissLatency::processor.cores.core.inst 90479430 # number of overall MSHR miss ticks (Tick)

board.cache_hierarchy.l1icaches.overallMshrMissLatency::total 90479430 # number of overall MSHR miss ticks (Tick)

board.cache_hierarchy.l1icaches.demandMshrMissRate::processor.cores.core.inst 0.000178 # mshr miss ratio for demand accesses (Ratio)

board.cache_hierarchy.l1icaches.demandMshrMissRate::total 0.000178 # mshr miss ratio for demand accesses (Ratio)

board.cache_hierarchy.l1icaches.overallMshrMissRate::processor.cores.core.inst 0.000178 # mshr miss ratio for overall accesses (Ratio)

board.cache_hierarchy.l1icaches.overallMshrMissRate::total 0.000178 # mshr miss ratio for overall accesses (Ratio)

board.cache_hierarchy.l1icaches.demandAvgMshrMissLatency::processor.cores.core.inst 120158.605578 # average overall mshr miss latency ((Tick/Count))

board.cache_hierarchy.l1icaches.demandAvgMshrMissLatency::total 120158.605578 # average overall mshr miss latency ((Tick/Count))

board.cache_hierarchy.l1icaches.overallAvgMshrMissLatency::processor.cores.core.inst 120158.605578 # average overall mshr miss latency ((Tick/Count))

board.cache_hierarchy.l1icaches.overallAvgMshrMissLatency::total 120158.605578 # average overall mshr miss latency ((Tick/Count))

board.cache_hierarchy.l1icaches.replacements 239 # number of replacements (Count)

board.cache_hierarchy.l1icaches.ReadReq.hits::processor.cores.core.inst 4231540 # number of ReadReq hits (Count)

board.cache_hierarchy.l1icaches.ReadReq.hits::total 4231540 # number of ReadReq hits (Count)

board.cache_hierarchy.l1icaches.ReadReq.misses::processor.cores.core.inst 977 # number of ReadReq misses (Count)

board.cache_hierarchy.l1icaches.ReadReq.misses::total 977 # number of ReadReq misses (Count)

board.cache_hierarchy.l1icaches.ReadReq.missLatency::processor.cores.core.inst 112883337 # number of ReadReq miss ticks (Tick)

board.cache_hierarchy.l1icaches.ReadReq.missLatency::total 112883337 # number of ReadReq miss ticks (Tick)

board.cache_hierarchy.l1icaches.ReadReq.accesses::processor.cores.core.inst 4232517 # number of ReadReq accesses(hits+misses) (Count)

board.cache_hierarchy.l1icaches.ReadReq.accesses::total 4232517 # number of ReadReq accesses(hits+misses) (Count)

board.cache_hierarchy.l1icaches.ReadReq.missRate::processor.cores.core.inst 0.000231 # miss rate for ReadReq accesses (Ratio)

board.cache_hierarchy.l1icaches.ReadReq.missRate::total 0.000231 # miss rate for ReadReq accesses (Ratio)

board.cache_hierarchy.l1icaches.ReadReq.avgMissLatency::processor.cores.core.inst 115540.774821 # average ReadReq miss latency ((Tick/Count))

board.cache_hierarchy.l1icaches.ReadReq.avgMissLatency::total 115540.774821 # average ReadReq miss latency ((Tick/Count))

board.cache_hierarchy.l1icaches.ReadReq.mshrHits::processor.cores.core.inst 224 # number of ReadReq MSHR hits (Count)

board.cache_hierarchy.l1icaches.ReadReq.mshrHits::total 224 # number of ReadReq MSHR hits (Count)

board.cache_hierarchy.l1icaches.ReadReq.mshrMisses::processor.cores.core.inst 753 # number of ReadReq MSHR misses (Count)

board.cache_hierarchy.l1icaches.ReadReq.mshrMisses::total 753 # number of ReadReq MSHR misses (Count)

board.cache_hierarchy.l1icaches.ReadReq.mshrMissLatency::processor.cores.core.inst 90479430 # number of ReadReq MSHR miss ticks (Tick)

board.cache_hierarchy.l1icaches.ReadReq.mshrMissLatency::total 90479430 # number of ReadReq MSHR miss ticks (Tick)

board.cache_hierarchy.l1icaches.ReadReq.mshrMissRate::processor.cores.core.inst 0.000178 # mshr miss rate for ReadReq accesses (Ratio)

board.cache_hierarchy.l1icaches.ReadReq.mshrMissRate::total 0.000178 # mshr miss rate for ReadReq accesses (Ratio)

board.cache_hierarchy.l1icaches.ReadReq.avgMshrMissLatency::processor.cores.core.inst 120158.605578 # average ReadReq mshr miss latency ((Tick/Count))

board.cache_hierarchy.l1icaches.ReadReq.avgMshrMissLatency::total 120158.605578 # average ReadReq mshr miss latency ((Tick/Count))

board.cache_hierarchy.l1icaches.power_state.pwrStateResidencyTicks::UNDEFINED 5909877540 # Cumulative time (in ticks) in various power states (Tick)

board.cache_hierarchy.l1icaches.prefetcher.demandMshrMisses 753 # demands not covered by prefetchs (Count)

board.cache_hierarchy.l1icaches.prefetcher.pfIssued 0 # number of hwpf issued (Count)

board.cache_hierarchy.l1icaches.prefetcher.pfUseful 0 # number of useful prefetch (Count)

board.cache_hierarchy.l1icaches.prefetcher.pfUsefulButMiss 0 # number of hit on prefetch but cache block is not in an usable state (Count)

board.cache_hierarchy.l1icaches.prefetcher.accuracy nan # accuracy of the prefetcher (Count)

board.cache_hierarchy.l1icaches.prefetcher.coverage 0 # coverage brought by this prefetcher (Count)

board.cache_hierarchy.l1icaches.prefetcher.pfHitInCache 0 # number of prefetches hitting in cache (Count)

board.cache_hierarchy.l1icaches.prefetcher.pfHitInMSHR 0 # number of prefetches hitting in a MSHR (Count)

board.cache_hierarchy.l1icaches.prefetcher.pfHitInWB 0 # number of prefetches hit in the Write Buffer (Count)

board.cache_hierarchy.l1icaches.prefetcher.pfLate 0 # number of late prefetches (hitting in cache, MSHR or WB) (Count)

board.cache_hierarchy.l1icaches.prefetcher.pfIdentified 0 # number of prefetch candidates identified (Count)

board.cache_hierarchy.l1icaches.prefetcher.pfBufferHit 0 # number of redundant prefetches already in prefetch queue (Count)

board.cache_hierarchy.l1icaches.prefetcher.pfInCache 0 # number of redundant prefetches already in cache/mshr dropped (Count)

board.cache_hierarchy.l1icaches.prefetcher.pfRemovedDemand 0 # number of prefetches dropped due to a demand for the same address (Count)

board.cache_hierarchy.l1icaches.prefetcher.pfRemovedFull 0 # number of prefetches dropped due to prefetch queue size (Count)

board.cache_hierarchy.l1icaches.prefetcher.pfSpanPage 0 # number of prefetches that crossed the page (Count)

board.cache_hierarchy.l1icaches.prefetcher.pfUsefulSpanPage 0 # number of prefetches that is useful and crossed the page (Count)

board.cache_hierarchy.l1icaches.prefetcher.power_state.pwrStateResidencyTicks::UNDEFINED 5909877540 # Cumulative time (in ticks) in various power states (Tick)

board.cache_hierarchy.l1icaches.tags.tagsInUse 469.941787 # Average ticks per tags in use ((Tick/Count))

board.cache_hierarchy.l1icaches.tags.totalRefs 4232293 # Total number of references to valid blocks. (Count)

board.cache_hierarchy.l1icaches.tags.sampledRefs 753 # Sample count of references to valid blocks. (Count)

board.cache_hierarchy.l1icaches.tags.avgRefs 5620.575033 # Average number of references to valid blocks. ((Count/Count))

board.cache_hierarchy.l1icaches.tags.warmupTick 123543 # The tick when the warmup percentage was hit. (Tick)

board.cache_hierarchy.l1icaches.tags.occupancies::processor.cores.core.inst 469.941787 # Average occupied blocks per tick, per requestor ((Count/Tick))

board.cache_hierarchy.l1icaches.tags.avgOccs::processor.cores.core.inst 0.917855 # Average percentage of cache occupancy ((Ratio/Tick))

board.cache_hierarchy.l1icaches.tags.avgOccs::total 0.917855 # Average percentage of cache occupancy ((Ratio/Tick))

board.cache_hierarchy.l1icaches.tags.occupanciesTaskId::1024 511 # Occupied blocks per task id (Count)

board.cache_hierarchy.l1icaches.tags.ageTaskId_1024::0 106 # Occupied blocks per task id, per block age (Count)

board.cache_hierarchy.l1icaches.tags.ageTaskId_1024::1 98 # Occupied blocks per task id, per block age (Count)

board.cache_hierarchy.l1icaches.tags.ageTaskId_1024::3 307 # Occupied blocks per task id, per block age (Count)

board.cache_hierarchy.l1icaches.tags.ratioOccsTaskId::1024 0.998047 # Ratio of occupied blocks and all blocks, per task id (Ratio)

board.cache_hierarchy.l1icaches.tags.tagAccesses 33860889 # Number of tag accesses (Count)

board.cache_hierarchy.l1icaches.tags.dataAccesses 33860889 # Number of data accesses (Count)

board.cache_hierarchy.l1icaches.tags.power_state.pwrStateResidencyTicks::UNDEFINED 5909877540 # Cumulative time (in ticks) in various power states (Tick)

board.cache_hierarchy.l2bus.transDist::ReadResp 131681 # Transaction distribution (Count)

board.cache_hierarchy.l2bus.transDist::WritebackDirty 131412 # Transaction distribution (Count)

board.cache_hierarchy.l2bus.transDist::CleanEvict 129099 # Transaction distribution (Count)

board.cache_hierarchy.l2bus.transDist::HardPFReq 139497 # Transaction distribution (Count)

board.cache_hierarchy.l2bus.transDist::UpgradeReq 3 # Transaction distribution (Count)

board.cache_hierarchy.l2bus.transDist::UpgradeResp 3 # Transaction distribution (Count)

board.cache_hierarchy.l2bus.transDist::ReadExReq 1140 # Transaction distribution (Count)

board.cache_hierarchy.l2bus.transDist::ReadExResp 1140 # Transaction distribution (Count)

board.cache_hierarchy.l2bus.transDist::ReadSharedReq 131681 # Transaction distribution (Count)

board.cache_hierarchy.l2bus.pktCount_board.cache_hierarchy.l1icaches.mem_side_port::board.cache_hierarchy.l2cache.cpu_side_port 1742 # Packet count per connected requestor and responder (Count)

board.cache_hierarchy.l2bus.pktCount_board.cache_hierarchy.l1dcaches.mem_side_port::board.cache_hierarchy.l2cache.cpu_side_port 395698 # Packet count per connected requestor and responder (Count)

board.cache_hierarchy.l2bus.pktCount::total 397440 # Packet count per connected requestor and responder (Count)

board.cache_hierarchy.l2bus.pktSize_board.cache_hierarchy.l1icaches.mem_side_port::board.cache_hierarchy.l2cache.cpu_side_port 48000 # Cumulative packet size per connected requestor and responder (Byte)

board.cache_hierarchy.l2bus.pktSize_board.cache_hierarchy.l1dcaches.mem_side_port::board.cache_hierarchy.l2cache.cpu_side_port 12658240 # Cumulative packet size per connected requestor and responder (Byte)

board.cache_hierarchy.l2bus.pktSize::total 12706240 # Cumulative packet size per connected requestor and responder (Byte)

board.cache_hierarchy.l2bus.snoops 268216 # Total snoops (Count)

board.cache_hierarchy.l2bus.snoopTraffic 4204672 # Total snoop traffic (Byte)

board.cache_hierarchy.l2bus.snoopFanout::samples 401037 # Request fanout histogram (Count)

board.cache_hierarchy.l2bus.snoopFanout::mean 0.001207 # Request fanout histogram (Count)

board.cache_hierarchy.l2bus.snoopFanout::stdev 0.034719 # Request fanout histogram (Count)

board.cache_hierarchy.l2bus.snoopFanout::underflows 0 0.00% 0.00% # Request fanout histogram (Count)

board.cache_hierarchy.l2bus.snoopFanout::0 400553 99.88% 99.88% # Request fanout histogram (Count)

board.cache_hierarchy.l2bus.snoopFanout::1 484 0.12% 100.00% # Request fanout histogram (Count)

board.cache_hierarchy.l2bus.snoopFanout::2 0 0.00% 100.00% # Request fanout histogram (Count)

board.cache_hierarchy.l2bus.snoopFanout::3 0 0.00% 100.00% # Request fanout histogram (Count)

board.cache_hierarchy.l2bus.snoopFanout::4 0 0.00% 100.00% # Request fanout histogram (Count)

board.cache_hierarchy.l2bus.snoopFanout::overflows 0 0.00% 100.00% # Request fanout histogram (Count)

board.cache_hierarchy.l2bus.snoopFanout::min_value 0 # Request fanout histogram (Count)

board.cache_hierarchy.l2bus.snoopFanout::max_value 1 # Request fanout histogram (Count)

board.cache_hierarchy.l2bus.snoopFanout::total 401037 # Request fanout histogram (Count)

board.cache_hierarchy.l2bus.power_state.pwrStateResidencyTicks::UNDEFINED 5909877540 # Cumulative time (in ticks) in various power states (Tick)

board.cache_hierarchy.l2bus.reqLayer0.occupancy 171646360 # Layer occupancy (ticks) (Tick)

board.cache_hierarchy.l2bus.reqLayer0.utilization 0.0 # Layer utilization (Ratio)

board.cache_hierarchy.l2bus.respLayer0.occupancy 752247 # Layer occupancy (ticks) (Tick)

board.cache_hierarchy.l2bus.respLayer0.utilization 0.0 # Layer utilization (Ratio)

board.cache_hierarchy.l2bus.respLayer1.occupancy 131936931 # Layer occupancy (ticks) (Tick)

board.cache_hierarchy.l2bus.respLayer1.utilization 0.0 # Layer utilization (Ratio)

board.cache_hierarchy.l2bus.snoop_filter.totRequests 264619 # Total number of requests made to the snoop filter. (Count)

board.cache_hierarchy.l2bus.snoop_filter.hitSingleRequests 131796 # Number of requests hitting in the snoop filter with a single holder of the requested data. (Count)

board.cache_hierarchy.l2bus.snoop_filter.hitMultiRequests 6 # Number of requests hitting in the snoop filter with multiple (>1) holders of the requested data. (Count)

board.cache_hierarchy.l2bus.snoop_filter.totSnoops 477 # Total number of snoops made to the snoop filter. (Count)

board.cache_hierarchy.l2bus.snoop_filter.hitSingleSnoops 477 # Number of snoops hitting in the snoop filter with a single holder of the requested data. (Count)

board.cache_hierarchy.l2bus.snoop_filter.hitMultiSnoops 0 # Number of snoops hitting in the snoop filter with multiple (>1) holders of the requested data. (Count)

board.cache_hierarchy.l2cache.demandHits::cache_hierarchy.l1dcaches.prefetcher 107979 # number of demand (read+write) hits (Count)

board.cache_hierarchy.l2cache.demandHits::processor.cores.core.data 30 # number of demand (read+write) hits (Count)

board.cache_hierarchy.l2cache.demandHits::total 108009 # number of demand (read+write) hits (Count)

board.cache_hierarchy.l2cache.overallHits::cache_hierarchy.l1dcaches.prefetcher 107979 # number of overall hits (Count)

board.cache_hierarchy.l2cache.overallHits::processor.cores.core.data 30 # number of overall hits (Count)

board.cache_hierarchy.l2cache.overallHits::total 108009 # number of overall hits (Count)

board.cache_hierarchy.l2cache.demandMisses::cache_hierarchy.l1dcaches.prefetcher 21141 # number of demand (read+write) misses (Count)

board.cache_hierarchy.l2cache.demandMisses::processor.cores.core.inst 750 # number of demand (read+write) misses (Count)

board.cache_hierarchy.l2cache.demandMisses::processor.cores.core.data 2918 # number of demand (read+write) misses (Count)

board.cache_hierarchy.l2cache.demandMisses::total 24809 # number of demand (read+write) misses (Count)

board.cache_hierarchy.l2cache.overallMisses::cache_hierarchy.l1dcaches.prefetcher 21141 # number of overall misses (Count)

board.cache_hierarchy.l2cache.overallMisses::processor.cores.core.inst 750 # number of overall misses (Count)

board.cache_hierarchy.l2cache.overallMisses::processor.cores.core.data 2918 # number of overall misses (Count)

board.cache_hierarchy.l2cache.overallMisses::total 24809 # number of overall misses (Count)

board.cache_hierarchy.l2cache.demandMissLatency::cache_hierarchy.l1dcaches.prefetcher 1999711230 # number of demand (read+write) miss ticks (Tick)

board.cache_hierarchy.l2cache.demandMissLatency::processor.cores.core.inst 89972271 # number of demand (read+write) miss ticks (Tick)

board.cache_hierarchy.l2cache.demandMissLatency::processor.cores.core.data 374023269 # number of demand (read+write) miss ticks (Tick)

board.cache_hierarchy.l2cache.demandMissLatency::total 2463706770 # number of demand (read+write) miss ticks (Tick)

board.cache_hierarchy.l2cache.overallMissLatency::cache_hierarchy.l1dcaches.prefetcher 1999711230 # number of overall miss ticks (Tick)

board.cache_hierarchy.l2cache.overallMissLatency::processor.cores.core.inst 89972271 # number of overall miss ticks (Tick)

board.cache_hierarchy.l2cache.overallMissLatency::processor.cores.core.data 374023269 # number of overall miss ticks (Tick)

board.cache_hierarchy.l2cache.overallMissLatency::total 2463706770 # number of overall miss ticks (Tick)

board.cache_hierarchy.l2cache.demandAccesses::cache_hierarchy.l1dcaches.prefetcher 129120 # number of demand (read+write) accesses (Count)

board.cache_hierarchy.l2cache.demandAccesses::processor.cores.core.inst 750 # number of demand (read+write) accesses (Count)

board.cache_hierarchy.l2cache.demandAccesses::processor.cores.core.data 2948 # number of demand (read+write) accesses (Count)

board.cache_hierarchy.l2cache.demandAccesses::total 132818 # number of demand (read+write) accesses (Count)

board.cache_hierarchy.l2cache.overallAccesses::cache_hierarchy.l1dcaches.prefetcher 129120 # number of overall (read+write) accesses (Count)

board.cache_hierarchy.l2cache.overallAccesses::processor.cores.core.inst 750 # number of overall (read+write) accesses (Count)

board.cache_hierarchy.l2cache.overallAccesses::processor.cores.core.data 2948 # number of overall (read+write) accesses (Count)

board.cache_hierarchy.l2cache.overallAccesses::total 132818 # number of overall (read+write) accesses (Count)

board.cache_hierarchy.l2cache.demandMissRate::cache_hierarchy.l1dcaches.prefetcher 0.163731 # miss rate for demand accesses (Ratio)

board.cache_hierarchy.l2cache.demandMissRate::processor.cores.core.inst 1 # miss rate for demand accesses (Ratio)

board.cache_hierarchy.l2cache.demandMissRate::processor.cores.core.data 0.989824 # miss rate for demand accesses (Ratio)

board.cache_hierarchy.l2cache.demandMissRate::total 0.186789 # miss rate for demand accesses (Ratio)

board.cache_hierarchy.l2cache.overallMissRate::cache_hierarchy.l1dcaches.prefetcher 0.163731 # miss rate for overall accesses (Ratio)

board.cache_hierarchy.l2cache.overallMissRate::processor.cores.core.inst 1 # miss rate for overall accesses (Ratio)

board.cache_hierarchy.l2cache.overallMissRate::processor.cores.core.data 0.989824 # miss rate for overall accesses (Ratio)

board.cache_hierarchy.l2cache.overallMissRate::total 0.186789 # miss rate for overall accesses (Ratio)

board.cache_hierarchy.l2cache.demandAvgMissLatency::cache_hierarchy.l1dcaches.prefetcher 94589.245069 # average overall miss latency in ticks ((Tick/Count))

board.cache_hierarchy.l2cache.demandAvgMissLatency::processor.cores.core.inst 119963.028000 # average overall miss latency in ticks ((Tick/Count))

board.cache_hierarchy.l2cache.demandAvgMissLatency::processor.cores.core.data 128177.953735 # average overall miss latency in ticks ((Tick/Count))

board.cache_hierarchy.l2cache.demandAvgMissLatency::total 99306.976097 # average overall miss latency in ticks ((Tick/Count))

board.cache_hierarchy.l2cache.overallAvgMissLatency::cache_hierarchy.l1dcaches.prefetcher 94589.245069 # average overall miss latency ((Tick/Count))

board.cache_hierarchy.l2cache.overallAvgMissLatency::processor.cores.core.inst 119963.028000 # average overall miss latency ((Tick/Count))

board.cache_hierarchy.l2cache.overallAvgMissLatency::processor.cores.core.data 128177.953735 # average overall miss latency ((Tick/Count))

board.cache_hierarchy.l2cache.overallAvgMissLatency::total 99306.976097 # average overall miss latency ((Tick/Count))

board.cache_hierarchy.l2cache.blockedCycles::no_mshrs 0 # number of cycles access was blocked (Cycle)

board.cache_hierarchy.l2cache.blockedCycles::no_targets 0 # number of cycles access was blocked (Cycle)

board.cache_hierarchy.l2cache.blockedCauses::no_mshrs 0 # number of times access was blocked (Count)

board.cache_hierarchy.l2cache.blockedCauses::no_targets 0 # number of times access was blocked (Count)

board.cache_hierarchy.l2cache.avgBlocked::no_mshrs nan # average number of cycles each access was blocked ((Cycle/Count))

board.cache_hierarchy.l2cache.avgBlocked::no_targets nan # average number of cycles each access was blocked ((Cycle/Count))

board.cache_hierarchy.l2cache.writebacks::writebacks 65695 # number of writebacks (Count)

board.cache_hierarchy.l2cache.writebacks::total 65695 # number of writebacks (Count)

board.cache_hierarchy.l2cache.demandMshrHits::cache_hierarchy.l1dcaches.prefetcher 19056 # number of demand (read+write) MSHR hits (Count)

board.cache_hierarchy.l2cache.demandMshrHits::total 19056 # number of demand (read+write) MSHR hits (Count)

board.cache_hierarchy.l2cache.overallMshrHits::cache_hierarchy.l1dcaches.prefetcher 19056 # number of overall MSHR hits (Count)

board.cache_hierarchy.l2cache.overallMshrHits::total 19056 # number of overall MSHR hits (Count)

board.cache_hierarchy.l2cache.demandMshrMisses::cache_hierarchy.l1dcaches.prefetcher 2085 # number of demand (read+write) MSHR misses (Count)

board.cache_hierarchy.l2cache.demandMshrMisses::processor.cores.core.inst 750 # number of demand (read+write) MSHR misses (Count)

board.cache_hierarchy.l2cache.demandMshrMisses::processor.cores.core.data 2918 # number of demand (read+write) MSHR misses (Count)

board.cache_hierarchy.l2cache.demandMshrMisses::total 5753 # number of demand (read+write) MSHR misses (Count)

board.cache_hierarchy.l2cache.overallMshrMisses::cache_hierarchy.l1dcaches.prefetcher 2085 # number of overall MSHR misses (Count)

board.cache_hierarchy.l2cache.overallMshrMisses::cache_hierarchy.l2cache.prefetcher 127059 # number of overall MSHR misses (Count)

board.cache_hierarchy.l2cache.overallMshrMisses::processor.cores.core.inst 750 # number of overall MSHR misses (Count)

board.cache_hierarchy.l2cache.overallMshrMisses::processor.cores.core.data 2918 # number of overall MSHR misses (Count)

board.cache_hierarchy.l2cache.overallMshrMisses::total 132812 # number of overall MSHR misses (Count)

board.cache_hierarchy.l2cache.demandMshrMissLatency::cache_hierarchy.l1dcaches.prefetcher 217628811 # number of demand (read+write) MSHR miss ticks (Tick)

board.cache_hierarchy.l2cache.demandMshrMissLatency::processor.cores.core.inst 64997271 # number of demand (read+write) MSHR miss ticks (Tick)

board.cache_hierarchy.l2cache.demandMshrMissLatency::processor.cores.core.data 276853869 # number of demand (read+write) MSHR miss ticks (Tick)

board.cache_hierarchy.l2cache.demandMshrMissLatency::total 559479951 # number of demand (read+write) MSHR miss ticks (Tick)

board.cache_hierarchy.l2cache.overallMshrMissLatency::cache_hierarchy.l1dcaches.prefetcher 217628811 # number of overall MSHR miss ticks (Tick)

board.cache_hierarchy.l2cache.overallMshrMissLatency::cache_hierarchy.l2cache.prefetcher 8434146795 # number of overall MSHR miss ticks (Tick)

board.cache_hierarchy.l2cache.overallMshrMissLatency::processor.cores.core.inst 64997271 # number of overall MSHR miss ticks (Tick)

board.cache_hierarchy.l2cache.overallMshrMissLatency::processor.cores.core.data 276853869 # number of overall MSHR miss ticks (Tick)

board.cache_hierarchy.l2cache.overallMshrMissLatency::total 8993626746 # number of overall MSHR miss ticks (Tick)

board.cache_hierarchy.l2cache.demandMshrMissRate::cache_hierarchy.l1dcaches.prefetcher 0.016148 # mshr miss ratio for demand accesses (Ratio)

board.cache_hierarchy.l2cache.demandMshrMissRate::processor.cores.core.inst 1 # mshr miss ratio for demand accesses (Ratio)

board.cache_hierarchy.l2cache.demandMshrMissRate::processor.cores.core.data 0.989824 # mshr miss ratio for demand accesses (Ratio)

board.cache_hierarchy.l2cache.demandMshrMissRate::total 0.043315 # mshr miss ratio for demand accesses (Ratio)

board.cache_hierarchy.l2cache.overallMshrMissRate::cache_hierarchy.l1dcaches.prefetcher 0.016148 # mshr miss ratio for overall accesses (Ratio)

board.cache_hierarchy.l2cache.overallMshrMissRate::cache_hierarchy.l2cache.prefetcher inf # mshr miss ratio for overall accesses (Ratio)

board.cache_hierarchy.l2cache.overallMshrMissRate::processor.cores.core.inst 1 # mshr miss ratio for overall accesses (Ratio)

board.cache_hierarchy.l2cache.overallMshrMissRate::processor.cores.core.data 0.989824 # mshr miss ratio for overall accesses (Ratio)

board.cache_hierarchy.l2cache.overallMshrMissRate::total 0.999955 # mshr miss ratio for overall accesses (Ratio)

board.cache_hierarchy.l2cache.demandAvgMshrMissLatency::cache_hierarchy.l1dcaches.prefetcher 104378.326619 # average overall mshr miss latency ((Tick/Count))

board.cache_hierarchy.l2cache.demandAvgMshrMissLatency::processor.cores.core.inst 86663.028000 # average overall mshr miss latency ((Tick/Count))

board.cache_hierarchy.l2cache.demandAvgMshrMissLatency::processor.cores.core.data 94877.953735 # average overall mshr miss latency ((Tick/Count))

board.cache_hierarchy.l2cache.demandAvgMshrMissLatency::total 97250.121849 # average overall mshr miss latency ((Tick/Count))

board.cache_hierarchy.l2cache.overallAvgMshrMissLatency::cache_hierarchy.l1dcaches.prefetcher 104378.326619 # average overall mshr miss latency ((Tick/Count))

board.cache_hierarchy.l2cache.overallAvgMshrMissLatency::cache_hierarchy.l2cache.prefetcher 66379.766841 # average overall mshr miss latency ((Tick/Count))

board.cache_hierarchy.l2cache.overallAvgMshrMissLatency::processor.cores.core.inst 86663.028000 # average overall mshr miss latency ((Tick/Count))

board.cache_hierarchy.l2cache.overallAvgMshrMissLatency::processor.cores.core.data 94877.953735 # average overall mshr miss latency ((Tick/Count))

board.cache_hierarchy.l2cache.overallAvgMshrMissLatency::total 67716.973963 # average overall mshr miss latency ((Tick/Count))

board.cache_hierarchy.l2cache.replacements 128716 # number of replacements (Count)

board.cache_hierarchy.l2cache.CleanEvict.mshrMisses::writebacks 167 # number of CleanEvict MSHR misses (Count)

board.cache_hierarchy.l2cache.CleanEvict.mshrMisses::total 167 # number of CleanEvict MSHR misses (Count)

board.cache_hierarchy.l2cache.CleanEvict.mshrMissRate::writebacks inf # mshr miss rate for CleanEvict accesses (Ratio)

board.cache_hierarchy.l2cache.CleanEvict.mshrMissRate::total inf # mshr miss rate for CleanEvict accesses (Ratio)

board.cache_hierarchy.l2cache.HardPFReq.mshrMisses::cache_hierarchy.l2cache.prefetcher 127059 # number of HardPFReq MSHR misses (Count)

board.cache_hierarchy.l2cache.HardPFReq.mshrMisses::total 127059 # number of HardPFReq MSHR misses (Count)

board.cache_hierarchy.l2cache.HardPFReq.mshrMissLatency::cache_hierarchy.l2cache.prefetcher 8434146795 # number of HardPFReq MSHR miss ticks (Tick)

board.cache_hierarchy.l2cache.HardPFReq.mshrMissLatency::total 8434146795 # number of HardPFReq MSHR miss ticks (Tick)

board.cache_hierarchy.l2cache.HardPFReq.mshrMissRate::cache_hierarchy.l2cache.prefetcher inf # mshr miss rate for HardPFReq accesses (Ratio)

board.cache_hierarchy.l2cache.HardPFReq.mshrMissRate::total inf # mshr miss rate for HardPFReq accesses (Ratio)

board.cache_hierarchy.l2cache.HardPFReq.avgMshrMissLatency::cache_hierarchy.l2cache.prefetcher 66379.766841 # average HardPFReq mshr miss latency ((Tick/Count))

board.cache_hierarchy.l2cache.HardPFReq.avgMshrMissLatency::total 66379.766841 # average HardPFReq mshr miss latency ((Tick/Count))

board.cache_hierarchy.l2cache.ReadExReq.hits::processor.cores.core.data 9 # number of ReadExReq hits (Count)

board.cache_hierarchy.l2cache.ReadExReq.hits::total 9 # number of ReadExReq hits (Count)

board.cache_hierarchy.l2cache.ReadExReq.misses::processor.cores.core.data 1131 # number of ReadExReq misses (Count)

board.cache_hierarchy.l2cache.ReadExReq.misses::total 1131 # number of ReadExReq misses (Count)

board.cache_hierarchy.l2cache.ReadExReq.missLatency::processor.cores.core.data 155226618 # number of ReadExReq miss ticks (Tick)

board.cache_hierarchy.l2cache.ReadExReq.missLatency::total 155226618 # number of ReadExReq miss ticks (Tick)

board.cache_hierarchy.l2cache.ReadExReq.accesses::processor.cores.core.data 1140 # number of ReadExReq accesses(hits+misses) (Count)

board.cache_hierarchy.l2cache.ReadExReq.accesses::total 1140 # number of ReadExReq accesses(hits+misses) (Count)

board.cache_hierarchy.l2cache.ReadExReq.missRate::processor.cores.core.data 0.992105 # miss rate for ReadExReq accesses (Ratio)

board.cache_hierarchy.l2cache.ReadExReq.missRate::total 0.992105 # miss rate for ReadExReq accesses (Ratio)

board.cache_hierarchy.l2cache.ReadExReq.avgMissLatency::processor.cores.core.data 137247.230769 # average ReadExReq miss latency ((Tick/Count))

board.cache_hierarchy.l2cache.ReadExReq.avgMissLatency::total 137247.230769 # average ReadExReq miss latency ((Tick/Count))

board.cache_hierarchy.l2cache.ReadExReq.mshrMisses::processor.cores.core.data 1131 # number of ReadExReq MSHR misses (Count)

board.cache_hierarchy.l2cache.ReadExReq.mshrMisses::total 1131 # number of ReadExReq MSHR misses (Count)

board.cache_hierarchy.l2cache.ReadExReq.mshrMissLatency::processor.cores.core.data 117564318 # number of ReadExReq MSHR miss ticks (Tick)

board.cache_hierarchy.l2cache.ReadExReq.mshrMissLatency::total 117564318 # number of ReadExReq MSHR miss ticks (Tick)

board.cache_hierarchy.l2cache.ReadExReq.mshrMissRate::processor.cores.core.data 0.992105 # mshr miss rate for ReadExReq accesses (Ratio)

board.cache_hierarchy.l2cache.ReadExReq.mshrMissRate::total 0.992105 # mshr miss rate for ReadExReq accesses (Ratio)

board.cache_hierarchy.l2cache.ReadExReq.avgMshrMissLatency::processor.cores.core.data 103947.230769 # average ReadExReq mshr miss latency ((Tick/Count))

board.cache_hierarchy.l2cache.ReadExReq.avgMshrMissLatency::total 103947.230769 # average ReadExReq mshr miss latency ((Tick/Count))

board.cache_hierarchy.l2cache.ReadSharedReq.hits::cache_hierarchy.l1dcaches.prefetcher 107979 # number of ReadSharedReq hits (Count)

board.cache_hierarchy.l2cache.ReadSharedReq.hits::processor.cores.core.data 21 # number of ReadSharedReq hits (Count)

board.cache_hierarchy.l2cache.ReadSharedReq.hits::total 108000 # number of ReadSharedReq hits (Count)

board.cache_hierarchy.l2cache.ReadSharedReq.misses::cache_hierarchy.l1dcaches.prefetcher 21141 # number of ReadSharedReq misses (Count)

board.cache_hierarchy.l2cache.ReadSharedReq.misses::processor.cores.core.inst 750 # number of ReadSharedReq misses (Count)

board.cache_hierarchy.l2cache.ReadSharedReq.misses::processor.cores.core.data 1787 # number of ReadSharedReq misses (Count)

board.cache_hierarchy.l2cache.ReadSharedReq.misses::total 23678 # number of ReadSharedReq misses (Count)

board.cache_hierarchy.l2cache.ReadSharedReq.missLatency::cache_hierarchy.l1dcaches.prefetcher 1999711230 # number of ReadSharedReq miss ticks (Tick)

board.cache_hierarchy.l2cache.ReadSharedReq.missLatency::processor.cores.core.inst 89972271 # number of ReadSharedReq miss ticks (Tick)

board.cache_hierarchy.l2cache.ReadSharedReq.missLatency::processor.cores.core.data 218796651 # number of ReadSharedReq miss ticks (Tick)

board.cache_hierarchy.l2cache.ReadSharedReq.missLatency::total 2308480152 # number of ReadSharedReq miss ticks (Tick)

board.cache_hierarchy.l2cache.ReadSharedReq.accesses::cache_hierarchy.l1dcaches.prefetcher 129120 # number of ReadSharedReq accesses(hits+misses) (Count)

board.cache_hierarchy.l2cache.ReadSharedReq.accesses::processor.cores.core.inst 750 # number of ReadSharedReq accesses(hits+misses) (Count)

board.cache_hierarchy.l2cache.ReadSharedReq.accesses::processor.cores.core.data 1808 # number of ReadSharedReq accesses(hits+misses) (Count)

board.cache_hierarchy.l2cache.ReadSharedReq.accesses::total 131678 # number of ReadSharedReq accesses(hits+misses) (Count)

board.cache_hierarchy.l2cache.ReadSharedReq.missRate::cache_hierarchy.l1dcaches.prefetcher 0.163731 # miss rate for ReadSharedReq accesses (Ratio)

board.cache_hierarchy.l2cache.ReadSharedReq.missRate::processor.cores.core.inst 1 # miss rate for ReadSharedReq accesses (Ratio)

board.cache_hierarchy.l2cache.ReadSharedReq.missRate::processor.cores.core.data 0.988385 # miss rate for ReadSharedReq accesses (Ratio)

board.cache_hierarchy.l2cache.ReadSharedReq.missRate::total 0.179817 # miss rate for ReadSharedReq accesses (Ratio)

board.cache_hierarchy.l2cache.ReadSharedReq.avgMissLatency::cache_hierarchy.l1dcaches.prefetcher 94589.245069 # average ReadSharedReq miss latency ((Tick/Count))

board.cache_hierarchy.l2cache.ReadSharedReq.avgMissLatency::processor.cores.core.inst 119963.028000 # average ReadSharedReq miss latency ((Tick/Count))

board.cache_hierarchy.l2cache.ReadSharedReq.avgMissLatency::processor.cores.core.data 122437.969222 # average ReadSharedReq miss latency ((Tick/Count))

board.cache_hierarchy.l2cache.ReadSharedReq.avgMissLatency::total 97494.727257 # average ReadSharedReq miss latency ((Tick/Count))

board.cache_hierarchy.l2cache.ReadSharedReq.mshrHits::cache_hierarchy.l1dcaches.prefetcher 19056 # number of ReadSharedReq MSHR hits (Count)

board.cache_hierarchy.l2cache.ReadSharedReq.mshrHits::total 19056 # number of ReadSharedReq MSHR hits (Count)

board.cache_hierarchy.l2cache.ReadSharedReq.mshrMisses::cache_hierarchy.l1dcaches.prefetcher 2085 # number of ReadSharedReq MSHR misses (Count)

board.cache_hierarchy.l2cache.ReadSharedReq.mshrMisses::processor.cores.core.inst 750 # number of ReadSharedReq MSHR misses (Count)

board.cache_hierarchy.l2cache.ReadSharedReq.mshrMisses::processor.cores.core.data 1787 # number of ReadSharedReq MSHR misses (Count)

board.cache_hierarchy.l2cache.ReadSharedReq.mshrMisses::total 4622 # number of ReadSharedReq MSHR misses (Count)

board.cache_hierarchy.l2cache.ReadSharedReq.mshrMissLatency::cache_hierarchy.l1dcaches.prefetcher 217628811 # number of ReadSharedReq MSHR miss ticks (Tick)

board.cache_hierarchy.l2cache.ReadSharedReq.mshrMissLatency::processor.cores.core.inst 64997271 # number of ReadSharedReq MSHR miss ticks (Tick)

board.cache_hierarchy.l2cache.ReadSharedReq.mshrMissLatency::processor.cores.core.data 159289551 # number of ReadSharedReq MSHR miss ticks (Tick)

board.cache_hierarchy.l2cache.ReadSharedReq.mshrMissLatency::total 441915633 # number of ReadSharedReq MSHR miss ticks (Tick)

board.cache_hierarchy.l2cache.ReadSharedReq.mshrMissRate::cache_hierarchy.l1dcaches.prefetcher 0.016148 # mshr miss rate for ReadSharedReq accesses (Ratio)

board.cache_hierarchy.l2cache.ReadSharedReq.mshrMissRate::processor.cores.core.inst 1 # mshr miss rate for ReadSharedReq accesses (Ratio)

board.cache_hierarchy.l2cache.ReadSharedReq.mshrMissRate::processor.cores.core.data 0.988385 # mshr miss rate for ReadSharedReq accesses (Ratio)

board.cache_hierarchy.l2cache.ReadSharedReq.mshrMissRate::total 0.035101 # mshr miss rate for ReadSharedReq accesses (Ratio)

board.cache_hierarchy.l2cache.ReadSharedReq.avgMshrMissLatency::cache_hierarchy.l1dcaches.prefetcher 104378.326619 # average ReadSharedReq mshr miss latency ((Tick/Count))

board.cache_hierarchy.l2cache.ReadSharedReq.avgMshrMissLatency::processor.cores.core.inst 86663.028000 # average ReadSharedReq mshr miss latency ((Tick/Count))

board.cache_hierarchy.l2cache.ReadSharedReq.avgMshrMissLatency::processor.cores.core.data 89137.969222 # average ReadSharedReq mshr miss latency ((Tick/Count))

board.cache_hierarchy.l2cache.ReadSharedReq.avgMshrMissLatency::total 95611.344223 # average ReadSharedReq mshr miss latency ((Tick/Count))

board.cache_hierarchy.l2cache.UpgradeReq.hits::processor.cores.core.data 3 # number of UpgradeReq hits (Count)

board.cache_hierarchy.l2cache.UpgradeReq.hits::total 3 # number of UpgradeReq hits (Count)

board.cache_hierarchy.l2cache.UpgradeReq.accesses::processor.cores.core.data 3 # number of UpgradeReq accesses(hits+misses) (Count)

board.cache_hierarchy.l2cache.UpgradeReq.accesses::total 3 # number of UpgradeReq accesses(hits+misses) (Count)

board.cache_hierarchy.l2cache.WritebackDirty.hits::writebacks 65717 # number of WritebackDirty hits (Count)

board.cache_hierarchy.l2cache.WritebackDirty.hits::total 65717 # number of WritebackDirty hits (Count)

board.cache_hierarchy.l2cache.WritebackDirty.accesses::writebacks 65717 # number of WritebackDirty accesses(hits+misses) (Count)

board.cache_hierarchy.l2cache.WritebackDirty.accesses::total 65717 # number of WritebackDirty accesses(hits+misses) (Count)

board.cache_hierarchy.l2cache.power_state.pwrStateResidencyTicks::UNDEFINED 5909877540 # Cumulative time (in ticks) in various power states (Tick)

board.cache_hierarchy.l2cache.prefetcher.demandMshrMisses 5753 # demands not covered by prefetchs (Count)

board.cache_hierarchy.l2cache.prefetcher.pfIssued 479441 # number of hwpf issued (Count)

board.cache_hierarchy.l2cache.prefetcher.pfUnused 10 # number of HardPF blocks evicted w/o reference (Count)

board.cache_hierarchy.l2cache.prefetcher.pfUseful 107985 # number of useful prefetch (Count)

board.cache_hierarchy.l2cache.prefetcher.pfUsefulButMiss 0 # number of hit on prefetch but cache block is not in an usable state (Count)

board.cache_hierarchy.l2cache.prefetcher.accuracy 0.225231 # accuracy of the prefetcher (Count)

board.cache_hierarchy.l2cache.prefetcher.coverage 0.949419 # coverage brought by this prefetcher (Count)

board.cache_hierarchy.l2cache.prefetcher.pfHitInCache 254214 # number of prefetches hitting in cache (Count)

board.cache_hierarchy.l2cache.prefetcher.pfHitInMSHR 98168 # number of prefetches hitting in a MSHR (Count)

board.cache_hierarchy.l2cache.prefetcher.pfHitInWB 0 # number of prefetches hit in the Write Buffer (Count)

board.cache_hierarchy.l2cache.prefetcher.pfLate 352382 # number of late prefetches (hitting in cache, MSHR or WB) (Count)

board.cache_hierarchy.l2cache.prefetcher.pfIdentified 504059 # number of prefetch candidates identified (Count)

board.cache_hierarchy.l2cache.prefetcher.pfBufferHit 16407 # number of redundant prefetches already in prefetch queue (Count)

board.cache_hierarchy.l2cache.prefetcher.pfInCache 0 # number of redundant prefetches already in cache/mshr dropped (Count)

board.cache_hierarchy.l2cache.prefetcher.pfRemovedDemand 6158 # number of prefetches dropped due to a demand for the same address (Count)

board.cache_hierarchy.l2cache.prefetcher.pfRemovedFull 0 # number of prefetches dropped due to prefetch queue size (Count)

board.cache_hierarchy.l2cache.prefetcher.pfSpanPage 20497 # number of prefetches that crossed the page (Count)

board.cache_hierarchy.l2cache.prefetcher.pfUsefulSpanPage 19125 # number of prefetches that is useful and crossed the page (Count)

board.cache_hierarchy.l2cache.prefetcher.power_state.pwrStateResidencyTicks::UNDEFINED 5909877540 # Cumulative time (in ticks) in various power states (Tick)

board.cache_hierarchy.l2cache.tags.tagsInUse 3950.838192 # Average ticks per tags in use ((Tick/Count))

board.cache_hierarchy.l2cache.tags.totalRefs 372452 # Total number of references to valid blocks. (Count)

board.cache_hierarchy.l2cache.tags.sampledRefs 132812 # Sample count of references to valid blocks. (Count)

board.cache_hierarchy.l2cache.tags.avgRefs 2.804355 # Average number of references to valid blocks. ((Count/Count))

board.cache_hierarchy.l2cache.tags.warmupTick 89910 # The tick when the warmup percentage was hit. (Tick)

board.cache_hierarchy.l2cache.tags.occupancies::cache_hierarchy.l1dcaches.prefetcher 62.451593 # Average occupied blocks per tick, per requestor ((Count/Tick))

board.cache_hierarchy.l2cache.tags.occupancies::cache_hierarchy.l2cache.prefetcher 3719.748977 # Average occupied blocks per tick, per requestor ((Count/Tick))

board.cache_hierarchy.l2cache.tags.occupancies::processor.cores.core.inst 42.103419 # Average occupied blocks per tick, per requestor ((Count/Tick))

board.cache_hierarchy.l2cache.tags.occupancies::processor.cores.core.data 126.534204 # Average occupied blocks per tick, per requestor ((Count/Tick))

board.cache_hierarchy.l2cache.tags.avgOccs::cache_hierarchy.l1dcaches.prefetcher 0.015247 # Average percentage of cache occupancy ((Ratio/Tick))

board.cache_hierarchy.l2cache.tags.avgOccs::cache_hierarchy.l2cache.prefetcher 0.908142 # Average percentage of cache occupancy ((Ratio/Tick))

board.cache_hierarchy.l2cache.tags.avgOccs::processor.cores.core.inst 0.010279 # Average percentage of cache occupancy ((Ratio/Tick))

board.cache_hierarchy.l2cache.tags.avgOccs::processor.cores.core.data 0.030892 # Average percentage of cache occupancy ((Ratio/Tick))

board.cache_hierarchy.l2cache.tags.avgOccs::total 0.964560 # Average percentage of cache occupancy ((Ratio/Tick))

board.cache_hierarchy.l2cache.tags.occupanciesTaskId::1022 3749 # Occupied blocks per task id (Count)

board.cache_hierarchy.l2cache.tags.occupanciesTaskId::1024 347 # Occupied blocks per task id (Count)

board.cache_hierarchy.l2cache.tags.ageTaskId_1022::0 8 # Occupied blocks per task id, per block age (Count)

board.cache_hierarchy.l2cache.tags.ageTaskId_1022::1 1807 # Occupied blocks per task id, per block age (Count)

board.cache_hierarchy.l2cache.tags.ageTaskId_1022::2 1934 # Occupied blocks per task id, per block age (Count)

board.cache_hierarchy.l2cache.tags.ageTaskId_1024::0 149 # Occupied blocks per task id, per block age (Count)

board.cache_hierarchy.l2cache.tags.ageTaskId_1024::1 169 # Occupied blocks per task id, per block age (Count)

board.cache_hierarchy.l2cache.tags.ageTaskId_1024::2 29 # Occupied blocks per task id, per block age (Count)

board.cache_hierarchy.l2cache.tags.ratioOccsTaskId::1022 0.915283 # Ratio of occupied blocks and all blocks, per task id (Ratio)

board.cache_hierarchy.l2cache.tags.ratioOccsTaskId::1024 0.084717 # Ratio of occupied blocks and all blocks, per task id (Ratio)

board.cache_hierarchy.l2cache.tags.tagAccesses 4366668 # Number of tag accesses (Count)

board.cache_hierarchy.l2cache.tags.dataAccesses 4366668 # Number of data accesses (Count)

board.cache_hierarchy.l2cache.tags.power_state.pwrStateResidencyTicks::UNDEFINED 5909877540 # Cumulative time (in ticks) in various power states (Tick)

board.cache_hierarchy.membus.transDist::ReadResp 131681 # Transaction distribution (Count)

board.cache_hierarchy.membus.transDist::WritebackDirty 65695 # Transaction distribution (Count)

board.cache_hierarchy.membus.transDist::CleanEvict 62713 # Transaction distribution (Count)

board.cache_hierarchy.membus.transDist::ReadExReq 1131 # Transaction distribution (Count)

board.cache_hierarchy.membus.transDist::ReadExResp 1131 # Transaction distribution (Count)

board.cache_hierarchy.membus.transDist::ReadSharedReq 131681 # Transaction distribution (Count)

board.cache_hierarchy.membus.pktCount_board.cache_hierarchy.l2cache.mem_side_port::board.memory.mem_ctrl.port 394032 # Packet count per connected requestor and responder (Count)

board.cache_hierarchy.membus.pktCount_board.cache_hierarchy.l2cache.mem_side_port::total 394032 # Packet count per connected requestor and responder (Count)

board.cache_hierarchy.membus.pktCount::total 394032 # Packet count per connected requestor and responder (Count)

board.cache_hierarchy.membus.pktSize_board.cache_hierarchy.l2cache.mem_side_port::board.memory.mem_ctrl.port 12704448 # Cumulative packet size per connected requestor and responder (Byte)

board.cache_hierarchy.membus.pktSize_board.cache_hierarchy.l2cache.mem_side_port::total 12704448 # Cumulative packet size per connected requestor and responder (Byte)

board.cache_hierarchy.membus.pktSize::total 12704448 # Cumulative packet size per connected requestor and responder (Byte)

board.cache_hierarchy.membus.snoops 0 # Total snoops (Count)

board.cache_hierarchy.membus.snoopTraffic 0 # Total snoop traffic (Byte)

board.cache_hierarchy.membus.snoopFanout::samples 132812 # Request fanout histogram (Count)

board.cache_hierarchy.membus.snoopFanout::mean 0 # Request fanout histogram (Count)

board.cache_hierarchy.membus.snoopFanout::stdev 0 # Request fanout histogram (Count)

board.cache_hierarchy.membus.snoopFanout::underflows 0 0.00% 0.00% # Request fanout histogram (Count)

board.cache_hierarchy.membus.snoopFanout::0 132812 100.00% 100.00% # Request fanout histogram (Count)

board.cache_hierarchy.membus.snoopFanout::1 0 0.00% 100.00% # Request fanout histogram (Count)

board.cache_hierarchy.membus.snoopFanout::overflows 0 0.00% 100.00% # Request fanout histogram (Count)

board.cache_hierarchy.membus.snoopFanout::min_value 0 # Request fanout histogram (Count)

board.cache_hierarchy.membus.snoopFanout::max_value 0 # Request fanout histogram (Count)

board.cache_hierarchy.membus.snoopFanout::total 132812 # Request fanout histogram (Count)

board.cache_hierarchy.membus.badaddr_responder.power_state.pwrStateResidencyTicks::UNDEFINED 5909877540 # Cumulative time (in ticks) in various power states (Tick)

board.cache_hierarchy.membus.power_state.pwrStateResidencyTicks::UNDEFINED 5909877540 # Cumulative time (in ticks) in various power states (Tick)

board.cache_hierarchy.membus.reqLayer0.occupancy 145671327 # Layer occupancy (ticks) (Tick)

board.cache_hierarchy.membus.reqLayer0.utilization 0.0 # Layer utilization (Ratio)

board.cache_hierarchy.membus.respLayer2.occupancy 113169922 # Layer occupancy (ticks) (Tick)

board.cache_hierarchy.membus.respLayer2.utilization 0.0 # Layer utilization (Ratio)

board.cache_hierarchy.membus.snoop_filter.totRequests 261220 # Total number of requests made to the snoop filter. (Count)

board.cache_hierarchy.membus.snoop_filter.hitSingleRequests 128408 # Number of requests hitting in the snoop filter with a single holder of the requested data. (Count)

board.cache_hierarchy.membus.snoop_filter.hitMultiRequests 0 # Number of requests hitting in the snoop filter with multiple (>1) holders of the requested data. (Count)

board.cache_hierarchy.membus.snoop_filter.totSnoops 0 # Total number of snoops made to the snoop filter. (Count)

board.cache_hierarchy.membus.snoop_filter.hitSingleSnoops 0 # Number of snoops hitting in the snoop filter with a single holder of the requested data. (Count)

board.cache_hierarchy.membus.snoop_filter.hitMultiSnoops 0 # Number of snoops hitting in the snoop filter with multiple (>1) holders of the requested data. (Count)

board.clk_domain.clock 333 # Clock period in ticks (Tick)

```
board.clk_domain.voltage_domain.voltage 1 # Voltage in Volts (Volt)
board.memory.mem_ctrl.avgPriority_writebacks::samples 65695.00 # Average QoS priority value for accepted requests (Count)
board.memory.mem_ctrl.avgPriority_cache_hierarchy.l1dcaches.prefetcher::samples 2085.00 # Average QoS priority value for accepted requests (Count)
board.memory.mem_ctrl.avgPriority_cache_hierarchy.l2cache.prefetcher::samples 127030.00 # Average QoS priority value for accepted requests (Count)
board.memory.mem_ctrl.avgPriority_processor.cores.core.inst::samples 750.00 # Average QoS priority value for accepted requests (Count)
board.memory.mem_ctrl.avgPriority_processor.cores.core.data::samples 2918.00 # Average QoS priority value for accepted requests (Count)
board.memory.mem_ctrl.priorityMinLatency 0.000000018750 # per QoS priority minimum request to response latency (Second)
board.memory.mem_ctrl.priorityMaxLatency 0.000158809564 # per QoS priority maximum request to response latency (Second)
board.memory.mem_ctrl.numReadWriteTurnArounds 4100 # Number of turnarounds from READ to WRITE (Count)
board.memory.mem_ctrl.numWriteReadTurnArounds 4100 # Number of turnarounds from WRITE to READ (Count)
board.memory.mem_ctrl.numStayReadState 299536 # Number of times bus staying in READ state (Count)
board.memory.mem_ctrl.numStayWriteState 61925 # Number of times bus staying in WRITE state (Count)
board.memory.mem_ctrl.readReqs 132812 # Number of read requests accepted (Count)
board.memory.mem_ctrl.writeReqs 65695 # Number of write requests accepted (Count)
board.memory.mem_ctrl.readBursts 132812 # Number of controller read bursts, including those serviced by the write queue (Count)
board.memory.mem_ctrl.writeBursts 65695 # Number of controller write bursts, including those merged in the write queue (Count)
board.memory.mem_ctrl.servicedByWrQ 29 # Number of controller read bursts serviced by the write queue (Count)
board.memory.mem_ctrl.mergedWrBursts 0 # Number of controller write bursts merged with an existing one (Count)
board.memory.mem_ctrl.neitherReadNorWriteReqs 0 # Number of requests that are neither read nor write (Count)
board.memory.mem_ctrl.avgRdQLen 2.00 # Average read queue length when enqueuing ((Count/Tick))
board.memory.mem_ctrl.avgWrQLen 25.47 # Average write queue length when enqueuing ((Count/Tick))
board.memory.mem_ctrl.numRdRetry 0 # Number of times read queue was full causing retry (Count)
board.memory.mem_ctrl.numWrRetry 0 # Number of times write queue was full causing retry (Count)
board.memory.mem_ctrl.readPktSize::0 0 # Read request sizes (log2) (Count)
board.memory.mem_ctrl.readPktSize::1 0 # Read request sizes (log2) (Count)
board.memory.mem_ctrl.readPktSize::2 0 # Read request sizes (log2) (Count)
board.memory.mem_ctrl.readPktSize::3 0 # Read request sizes (log2) (Count)
board.memory.mem_ctrl.readPktSize::4 0 # Read request sizes (log2) (Count)
board.memory.mem_ctrl.readPktSize::5 0 # Read request sizes (log2) (Count)
board.memory.mem_ctrl.readPktSize::6 132812 # Read request sizes (log2) (Count)
board.memory.mem_ctrl.writePktSize::0 0 # Write request sizes (log2) (Count)
board.memory.mem_ctrl.writePktSize::1 0 # Write request sizes (log2) (Count)
board.memory.mem_ctrl.writePktSize::2 0 # Write request sizes (log2) (Count)
board.memory.mem_ctrl.writePktSize::3 0 # Write request sizes (log2) (Count)
board.memory.mem_ctrl.writePktSize::4 0 # Write request sizes (log2) (Count)
board.memory.mem_ctrl.writePktSize::5 0 # Write request sizes (log2) (Count)
board.memory.mem_ctrl.writePktSize::6 65695 # Write request sizes (log2) (Count)
board.memory.mem_ctrl.rdQLenPdf::0 90543 # What read queue length does an incoming req see (Count)
board.memory.mem_ctrl.rdQLenPdf::1 9620 # What read queue length does an incoming req see (Count)
board.memory.mem_ctrl.rdQLenPdf::2 6786 # What read queue length does an incoming req see (Count)
board.memory.mem_ctrl.rdQLenPdf::3 6827 # What read queue length does an incoming req see (Count)
board.memory.mem_ctrl.rdQLenPdf::4 7048 # What read queue length does an incoming req see (Count)
board.memory.mem_ctrl.rdQLenPdf::5 5860 # What read queue length does an incoming req see (Count)
board.memory.mem_ctrl.rdQLenPdf::6 4217 # What read queue length does an incoming req see (Count)
board.memory.mem_ctrl.rdQLenPdf::7 1612 # What read queue length does an incoming req see (Count)
board.memory.mem_ctrl.rdQLenPdf::8 257 # What read queue length does an incoming req see (Count)
board.memory.mem_ctrl.rdQLenPdf::9 3 # What read queue length does an incoming req see (Count)
board.memory.mem_ctrl.rdQLenPdf::10 2 # What read queue length does an incoming req see (Count)
board.memory.mem_ctrl.rdQLenPdf::11 1 # What read queue length does an incoming req see (Count)
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board.memory.mem_ctrl.rdQLenPdf::12	1	# What read queue length does an incoming req see (Count)
board.memory.mem_ctrl.rdQLenPdf::13	1	# What read queue length does an incoming req see (Count)
board.memory.mem_ctrl.rdQLenPdf::14	1	# What read queue length does an incoming req see (Count)
board.memory.mem_ctrl.rdQLenPdf::15	2	# What read queue length does an incoming req see (Count)
board.memory.mem_ctrl.rdQLenPdf::16	1	# What read queue length does an incoming req see (Count)
board.memory.mem_ctrl.rdQLenPdf::17	1	# What read queue length does an incoming req see (Count)
board.memory.mem_ctrl.rdQLenPdf::18	0	# What read queue length does an incoming req see (Count)
board.memory.mem_ctrl.rdQLenPdf::19	0	# What read queue length does an incoming req see (Count)
board.memory.mem_ctrl.rdQLenPdf::20	0	# What read queue length does an incoming req see (Count)
board.memory.mem_ctrl.rdQLenPdf::21	0	# What read queue length does an incoming req see (Count)
board.memory.mem_ctrl.rdQLenPdf::22	0	# What read queue length does an incoming req see (Count)
board.memory.mem_ctrl.rdQLenPdf::23	0	# What read queue length does an incoming req see (Count)
board.memory.mem_ctrl.rdQLenPdf::24	0	# What read queue length does an incoming req see (Count)
board.memory.mem_ctrl.rdQLenPdf::25	0	# What read queue length does an incoming req see (Count)
board.memory.mem_ctrl.rdQLenPdf::26	0	# What read queue length does an incoming req see (Count)
board.memory.mem_ctrl.rdQLenPdf::27	0	# What read queue length does an incoming req see (Count)
board.memory.mem_ctrl.rdQLenPdf::28	0	# What read queue length does an incoming req see (Count)
board.memory.mem_ctrl.rdQLenPdf::29	0	# What read queue length does an incoming req see (Count)
board.memory.mem_ctrl.rdQLenPdf::30	0	# What read queue length does an incoming req see (Count)
board.memory.mem_ctrl.rdQLenPdf::31	0	# What read queue length does an incoming req see (Count)
board.memory.mem_ctrl.wrQLenPdf::0	1	# What write queue length does an incoming req see (Count)
board.memory.mem_ctrl.wrQLenPdf::1	1	# What write queue length does an incoming req see (Count)
board.memory.mem_ctrl.wrQLenPdf::2	1	# What write queue length does an incoming req see (Count)
board.memory.mem_ctrl.wrQLenPdf::3	1	# What write queue length does an incoming req see (Count)
board.memory.mem_ctrl.wrQLenPdf::4	1	# What write queue length does an incoming req see (Count)
board.memory.mem_ctrl.wrQLenPdf::5	1	# What write queue length does an incoming req see (Count)
board.memory.mem_ctrl.wrQLenPdf::6	1	# What write queue length does an incoming req see (Count)
board.memory.mem_ctrl.wrQLenPdf::7	1	# What write queue length does an incoming req see (Count)
board.memory.mem_ctrl.wrQLenPdf::8	1	# What write queue length does an incoming req see (Count)
board.memory.mem_ctrl.wrQLenPdf::9	1	# What write queue length does an incoming req see (Count)
board.memory.mem_ctrl.wrQLenPdf::10	1	# What write queue length does an incoming req see (Count)
board.memory.mem_ctrl.wrQLenPdf::11	1	# What write queue length does an incoming req see (Count)
board.memory.mem_ctrl.wrQLenPdf::12	1	# What write queue length does an incoming req see (Count)
board.memory.mem_ctrl.wrQLenPdf::13	1	# What write queue length does an incoming req see (Count)
board.memory.mem_ctrl.wrQLenPdf::14	1	# What write queue length does an incoming req see (Count)
board.memory.mem_ctrl.wrQLenPdf::15	9	# What write queue length does an incoming req see (Count)
board.memory.mem_ctrl.wrQLenPdf::16	10	# What write queue length does an incoming req see (Count)
board.memory.mem_ctrl.wrQLenPdf::17	193	# What write queue length does an incoming req see (Count)
board.memory.mem_ctrl.wrQLenPdf::18	1634	# What write queue length does an incoming req see (Count)
board.memory.mem_ctrl.wrQLenPdf::19	4007	# What write queue length does an incoming req see (Count)
board.memory.mem_ctrl.wrQLenPdf::20	5671	# What write queue length does an incoming req see (Count)
board.memory.mem_ctrl.wrQLenPdf::21	4291	# What write queue length does an incoming req see (Count)
board.memory.mem_ctrl.wrQLenPdf::22	4048	# What write queue length does an incoming req see (Count)
board.memory.mem_ctrl.wrQLenPdf::23	4523	# What write queue length does an incoming req see (Count)
board.memory.mem_ctrl.wrQLenPdf::24	4628	# What write queue length does an incoming req see (Count)
board.memory.mem_ctrl.wrQLenPdf::25	5793	# What write queue length does an incoming req see (Count)
board.memory.mem_ctrl.wrQLenPdf::26	4388	# What write queue length does an incoming req see (Count)
board.memory.mem_ctrl.wrQLenPdf::27	4105	# What write queue length does an incoming req see (Count)
board.memory.mem_ctrl.wrQLenPdf::28	4115	# What write queue length does an incoming req see (Count)

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board.memory.mem_ctrl.wrQLenPdf::29          4193                # What write queue length does an incoming req see (Count)
board.memory.mem_ctrl.wrQLenPdf::30          4514                # What write queue length does an incoming req see (Count)
board.memory.mem_ctrl.wrQLenPdf::31          4107                # What write queue length does an incoming req see (Count)
board.memory.mem_ctrl.wrQLenPdf::32          5421                # What write queue length does an incoming req see (Count)
board.memory.mem_ctrl.wrQLenPdf::33          16                  # What write queue length does an incoming req see (Count)
board.memory.mem_ctrl.wrQLenPdf::34          7                   # What write queue length does an incoming req see (Count)
board.memory.mem_ctrl.wrQLenPdf::35          2                   # What write queue length does an incoming req see (Count)
board.memory.mem_ctrl.wrQLenPdf::36          2                   # What write queue length does an incoming req see (Count)
board.memory.mem_ctrl.wrQLenPdf::37          1                   # What write queue length does an incoming req see (Count)
board.memory.mem_ctrl.wrQLenPdf::38          1                   # What write queue length does an incoming req see (Count)
board.memory.mem_ctrl.wrQLenPdf::39          1                   # What write queue length does an incoming req see (Count)
board.memory.mem_ctrl.wrQLenPdf::40          0                   # What write queue length does an incoming req see (Count)
board.memory.mem_ctrl.wrQLenPdf::41          0                   # What write queue length does an incoming req see (Count)
board.memory.mem_ctrl.wrQLenPdf::42          0                   # What write queue length does an incoming req see (Count)
board.memory.mem_ctrl.wrQLenPdf::43          0                   # What write queue length does an incoming req see (Count)
board.memory.mem_ctrl.wrQLenPdf::44          0                   # What write queue length does an incoming req see (Count)
board.memory.mem_ctrl.wrQLenPdf::45          0                   # What write queue length does an incoming req see (Count)
board.memory.mem_ctrl.wrQLenPdf::46          0                   # What write queue length does an incoming req see (Count)
board.memory.mem_ctrl.wrQLenPdf::47          0                   # What write queue length does an incoming req see (Count)
board.memory.mem_ctrl.wrQLenPdf::48          0                   # What write queue length does an incoming req see (Count)
board.memory.mem_ctrl.wrQLenPdf::49          0                   # What write queue length does an incoming req see (Count)
board.memory.mem_ctrl.wrQLenPdf::50          0                   # What write queue length does an incoming req see (Count)
board.memory.mem_ctrl.wrQLenPdf::51          0                   # What write queue length does an incoming req see (Count)
board.memory.mem_ctrl.wrQLenPdf::52          0                   # What write queue length does an incoming req see (Count)
board.memory.mem_ctrl.wrQLenPdf::53          0                   # What write queue length does an incoming req see (Count)
board.memory.mem_ctrl.wrQLenPdf::54          0                   # What write queue length does an incoming req see (Count)
board.memory.mem_ctrl.wrQLenPdf::55          0                   # What write queue length does an incoming req see (Count)
board.memory.mem_ctrl.wrQLenPdf::56          0                   # What write queue length does an incoming req see (Count)
board.memory.mem_ctrl.wrQLenPdf::57          0                   # What write queue length does an incoming req see (Count)
board.memory.mem_ctrl.wrQLenPdf::58          0                   # What write queue length does an incoming req see (Count)
board.memory.mem_ctrl.wrQLenPdf::59          0                   # What write queue length does an incoming req see (Count)
board.memory.mem_ctrl.wrQLenPdf::60          0                   # What write queue length does an incoming req see (Count)
board.memory.mem_ctrl.wrQLenPdf::61          0                   # What write queue length does an incoming req see (Count)
board.memory.mem_ctrl.wrQLenPdf::62          0                   # What write queue length does an incoming req see (Count)
board.memory.mem_ctrl.wrQLenPdf::63          0                   # What write queue length does an incoming req see (Count)
board.memory.mem_ctrl.rdPerTurnAround::samples        4100                # Reads before turning the bus around for writes (Count)
board.memory.mem_ctrl.rdPerTurnAround::mean    17.330488                # Reads before turning the bus around for writes (Count)
board.memory.mem_ctrl.rdPerTurnAround::gmean    16.018523               # Reads before turning the bus around for writes (Count)
board.memory.mem_ctrl.rdPerTurnAround::stdev    71.913964               # Reads before turning the bus around for writes (Count)
board.memory.mem_ctrl.rdPerTurnAround::0-255        4098     99.95%     99.95% # Reads before turning the bus around for writes (Count)
board.memory.mem_ctrl.rdPerTurnAround::256-511          1      0.02%     99.98% # Reads before turning the bus around for writes (Count)
board.memory.mem_ctrl.rdPerTurnAround::4352-4607          1      0.02%    100.00% # Reads before turning the bus around for writes (Count)
board.memory.mem_ctrl.rdPerTurnAround::total        4100                # Reads before turning the bus around for writes (Count)
board.memory.mem_ctrl.wrPerTurnAround::samples        4100                # Writes before turning the bus around for reads (Count)
board.memory.mem_ctrl.wrPerTurnAround::mean    16.016098                # Writes before turning the bus around for reads (Count)
board.memory.mem_ctrl.wrPerTurnAround::gmean    16.013790               # Writes before turning the bus around for reads (Count)
board.memory.mem_ctrl.wrPerTurnAround::stdev     0.306448               # Writes before turning the bus around for reads (Count)
board.memory.mem_ctrl.wrPerTurnAround::16        4085     99.63%     99.63% # Writes before turning the bus around for reads (Count)
board.memory.mem_ctrl.wrPerTurnAround::17          1      0.02%     99.66% # Writes before turning the bus around for reads (Count)
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board.memory.mem_ctrl.wrPerTurnAround::18          4     0.10%    99.76% # Writes before turning the bus around for reads (Count)
board.memory.mem_ctrl.wrPerTurnAround::19          2     0.05%    99.80% # Writes before turning the bus around for reads (Count)
board.memory.mem_ctrl.wrPerTurnAround::20          2     0.05%    99.85% # Writes before turning the bus around for reads (Count)
board.memory.mem_ctrl.wrPerTurnAround::21          1     0.02%    99.88% # Writes before turning the bus around for reads (Count)
board.memory.mem_ctrl.wrPerTurnAround::22          1     0.02%    99.90% # Writes before turning the bus around for reads (Count)
board.memory.mem_ctrl.wrPerTurnAround::23          1     0.02%    99.93% # Writes before turning the bus around for reads (Count)
board.memory.mem_ctrl.wrPerTurnAround::24          2     0.05%    99.98% # Writes before turning the bus around for reads (Count)
board.memory.mem_ctrl.wrPerTurnAround::25          1     0.02%   100.00% # Writes before turning the bus around for reads (Count)
board.memory.mem_ctrl.wrPerTurnAround::total       4100                  # Writes before turning the bus around for reads (Count)
board.memory.mem_ctrl.bytesReadWrQ                 1856                  # Total number of bytes read from write queue (Byte)
board.memory.mem_ctrl.bytesReadSys                 8499968               # Total read bytes from the system interface side (Byte)
board.memory.mem_ctrl.bytesWrittenSys              4204480               # Total written bytes from the system interface side (Byte)
board.memory.mem_ctrl.avgRdBWSys                   1438264658.18782449   # Average system read bandwidth in Byte/s ((Byte/Second))
board.memory.mem_ctrl.avgWrBWSys                   711432677.16508389    # Average system write bandwidth in Byte/s ((Byte/Second))
board.memory.mem_ctrl.totGap                       5909726358            # Total gap between requests (Tick)
board.memory.mem_ctrl.avgGap                       29770.87              # Average gap between requests ((Tick/Count))
board.memory.mem_ctrl.requestorReadBytes::cache_hierarchy.l1dcaches.prefetcher      133440      # Per-requestor bytes read from memory (Byte)
board.memory.mem_ctrl.requestorReadBytes::cache_hierarchy.l2cache.prefetcher      8129920      # Per-requestor bytes read from memory (Byte)
board.memory.mem_ctrl.requestorReadBytes::processor.cores.core.inst      48000      # Per-requestor bytes read from memory (Byte)
board.memory.mem_ctrl.requestorReadBytes::processor.cores.core.data      186752      # Per-requestor bytes read from memory (Byte)
board.memory.mem_ctrl.requestorWriteBytes::writebacks      4202624      # Per-requestor bytes write to memory (Byte)
board.memory.mem_ctrl.requestorReadRate::cache_hierarchy.l1dcaches.prefetcher 22579148.061331912875      # Per-requestor bytes read from memory rate ((Byte/Second))
board.memory.mem_ctrl.requestorReadRate::cache_hierarchy.l2cache.prefetcher 1375649485.962106943130      # Per-requestor bytes read from memory rate ((Byte/Second))
board.memory.mem_ctrl.requestorReadRate::processor.cores.core.inst 8121995.705515076406      # Per-requestor bytes read from memory rate ((Byte/Second))
board.memory.mem_ctrl.requestorReadRate::processor.cores.core.data 31599977.958257324994      # Per-requestor bytes read from memory rate ((Byte/Second))
board.memory.mem_ctrl.requestorWriteRate::writebacks 711118626.664470672607      # Per-requestor bytes write to memory rate ((Byte/Second))
board.memory.mem_ctrl.requestorReadAccesses::cache_hierarchy.l1dcaches.prefetcher      2085      # Per-requestor read serviced memory accesses (Count)
board.memory.mem_ctrl.requestorReadAccesses::cache_hierarchy.l2cache.prefetcher      127059      # Per-requestor read serviced memory accesses (Count)
board.memory.mem_ctrl.requestorReadAccesses::processor.cores.core.inst      750      # Per-requestor read serviced memory accesses (Count)
board.memory.mem_ctrl.requestorReadAccesses::processor.cores.core.data      2918      # Per-requestor read serviced memory accesses (Count)
board.memory.mem_ctrl.requestorWriteAccesses::writebacks      65695      # Per-requestor write serviced memory accesses (Count)
board.memory.mem_ctrl.requestorReadTotalLat::cache_hierarchy.l1dcaches.prefetcher      98632230      # Per-requestor read total memory access latency (Tick)
board.memory.mem_ctrl.requestorReadTotalLat::cache_hierarchy.l2cache.prefetcher      5405216028      # Per-requestor read total memory access latency (Tick)
board.memory.mem_ctrl.requestorReadTotalLat::processor.cores.core.inst      22141575      # Per-requestor read total memory access latency (Tick)
board.memory.mem_ctrl.requestorReadTotalLat::processor.cores.core.data      110242812      # Per-requestor read total memory access latency (Tick)
board.memory.mem_ctrl.requestorWriteTotalLat::writebacks      68453891753      # Per-requestor write total memory access latency (Tick)
board.memory.mem_ctrl.requestorReadAvgLat::cache_hierarchy.l1dcaches.prefetcher      47305.63      # Per-requestor read average memory access latency ((Tick/Count))
board.memory.mem_ctrl.requestorReadAvgLat::cache_hierarchy.l2cache.prefetcher      42540.99      # Per-requestor read average memory access latency ((Tick/Count))
board.memory.mem_ctrl.requestorReadAvgLat::processor.cores.core.inst      29522.10      # Per-requestor read average memory access latency ((Tick/Count))
board.memory.mem_ctrl.requestorReadAvgLat::processor.cores.core.data      37780.26      # Per-requestor read average memory access latency ((Tick/Count))
board.memory.mem_ctrl.requestorWriteAvgLat::writebacks      1041995.46      # Per-requestor write average memory access latency ((Tick/Count))
board.memory.mem_ctrl.dram.bytesRead::cache_hierarchy.l1dcaches.prefetcher      133440      # Number of bytes read from this memory (Byte)
board.memory.mem_ctrl.dram.bytesRead::cache_hierarchy.l2cache.prefetcher      8131776      # Number of bytes read from this memory (Byte)
board.memory.mem_ctrl.dram.bytesRead::processor.cores.core.inst      48000      # Number of bytes read from this memory (Byte)
board.memory.mem_ctrl.dram.bytesRead::processor.cores.core.data      186752      # Number of bytes read from this memory (Byte)
board.memory.mem_ctrl.dram.bytesRead::total      8499968      # Number of bytes read from this memory (Byte)
board.memory.mem_ctrl.dram.bytesInstRead::processor.cores.core.inst      48000      # Number of instructions bytes read from this memory (Byte)
board.memory.mem_ctrl.dram.bytesInstRead::total      48000      # Number of instructions bytes read from this memory (Byte)
board.memory.mem_ctrl.dram.bytesWritten::writebacks      4204480      # Number of bytes written to this memory (Byte)
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board.memory.mem_ctrl.dram.bytesWritten::total      4204480                    # Number of bytes written to this memory (Byte)
board.memory.mem_ctrl.dram.numReads::cache_hierarchy.l1dcaches.prefetcher        2085                    # Number of read requests responded to by this memory (Count)
board.memory.mem_ctrl.dram.numReads::cache_hierarchy.l2cache.prefetcher       127059                    # Number of read requests responded to by this memory (Count)
board.memory.mem_ctrl.dram.numReads::processor.cores.core.inst         750                    # Number of read requests responded to by this memory (Count)
board.memory.mem_ctrl.dram.numReads::processor.cores.core.data        2918                    # Number of read requests responded to by this memory (Count)
board.memory.mem_ctrl.dram.numReads::total       132812                    # Number of read requests responded to by this memory (Count)
board.memory.mem_ctrl.dram.numWrites::writebacks        65695                    # Number of write requests responded to by this memory (Count)
board.memory.mem_ctrl.dram.numWrites::total        65695                    # Number of write requests responded to by this memory (Count)
board.memory.mem_ctrl.dram.bwRead::cache_hierarchy.l1dcaches.prefetcher     22579148                    # Total read bandwidth from this memory ((Byte/Second))
board.memory.mem_ctrl.dram.bwRead::cache_hierarchy.l2cache.prefetcher   1375963536                    # Total read bandwidth from this memory ((Byte/Second))
board.memory.mem_ctrl.dram.bwRead::processor.cores.core.inst      8121996                    # Total read bandwidth from this memory ((Byte/Second))
board.memory.mem_ctrl.dram.bwRead::processor.cores.core.data     31599978                    # Total read bandwidth from this memory ((Byte/Second))
board.memory.mem_ctrl.dram.bwRead::total   1438264658                    # Total read bandwidth from this memory ((Byte/Second))
board.memory.mem_ctrl.dram.bwInstRead::processor.cores.core.inst      8121996                    # Instruction read bandwidth from this memory ((Byte/Second))
board.memory.mem_ctrl.dram.bwInstRead::total      8121996                    # Instruction read bandwidth from this memory ((Byte/Second))
board.memory.mem_ctrl.dram.bwWrite::writebacks    711432677                    # Write bandwidth from this memory ((Byte/Second))
board.memory.mem_ctrl.dram.bwWrite::total    711432677                    # Write bandwidth from this memory ((Byte/Second))
board.memory.mem_ctrl.dram.bwTotal::writebacks    711432677                    # Total bandwidth to/from this memory ((Byte/Second))
board.memory.mem_ctrl.dram.bwTotal::cache_hierarchy.l1dcaches.prefetcher     22579148                    # Total bandwidth to/from this memory ((Byte/Second))
board.memory.mem_ctrl.dram.bwTotal::cache_hierarchy.l2cache.prefetcher   1375963536                    # Total bandwidth to/from this memory ((Byte/Second))
board.memory.mem_ctrl.dram.bwTotal::processor.cores.core.inst      8121996                    # Total bandwidth to/from this memory ((Byte/Second))
board.memory.mem_ctrl.dram.bwTotal::processor.cores.core.data     31599978                    # Total bandwidth to/from this memory ((Byte/Second))
board.memory.mem_ctrl.dram.bwTotal::total   2149697335                    # Total bandwidth to/from this memory ((Byte/Second))
board.memory.mem_ctrl.dram.readBursts          132783                    # Number of DRAM read bursts (Count)
board.memory.mem_ctrl.dram.writeBursts          65666                    # Number of DRAM write bursts (Count)
board.memory.mem_ctrl.dram.perBankRdBursts::0        8465                    # Per bank write bursts (Count)
board.memory.mem_ctrl.dram.perBankRdBursts::1        8437                    # Per bank write bursts (Count)
board.memory.mem_ctrl.dram.perBankRdBursts::2        8291                    # Per bank write bursts (Count)
board.memory.mem_ctrl.dram.perBankRdBursts::3        8268                    # Per bank write bursts (Count)
board.memory.mem_ctrl.dram.perBankRdBursts::4        8267                    # Per bank write bursts (Count)
board.memory.mem_ctrl.dram.perBankRdBursts::5        8312                    # Per bank write bursts (Count)
board.memory.mem_ctrl.dram.perBankRdBursts::6        8254                    # Per bank write bursts (Count)
board.memory.mem_ctrl.dram.perBankRdBursts::7        8225                    # Per bank write bursts (Count)
board.memory.mem_ctrl.dram.perBankRdBursts::8        8274                    # Per bank write bursts (Count)
board.memory.mem_ctrl.dram.perBankRdBursts::9        8236                    # Per bank write bursts (Count)
board.memory.mem_ctrl.dram.perBankRdBursts::10        8235                    # Per bank write bursts (Count)
board.memory.mem_ctrl.dram.perBankRdBursts::11        8227                    # Per bank write bursts (Count)
board.memory.mem_ctrl.dram.perBankRdBursts::12        8235                    # Per bank write bursts (Count)
board.memory.mem_ctrl.dram.perBankRdBursts::13        8294                    # Per bank write bursts (Count)
board.memory.mem_ctrl.dram.perBankRdBursts::14        8357                    # Per bank write bursts (Count)
board.memory.mem_ctrl.dram.perBankRdBursts::15        8406                    # Per bank write bursts (Count)
board.memory.mem_ctrl.dram.perBankWrBursts::0        4096                    # Per bank write bursts (Count)
board.memory.mem_ctrl.dram.perBankWrBursts::1        4096                    # Per bank write bursts (Count)
board.memory.mem_ctrl.dram.perBankWrBursts::2        4096                    # Per bank write bursts (Count)
board.memory.mem_ctrl.dram.perBankWrBursts::3        4104                    # Per bank write bursts (Count)
board.memory.mem_ctrl.dram.perBankWrBursts::4        4102                    # Per bank write bursts (Count)
board.memory.mem_ctrl.dram.perBankWrBursts::5        4157                    # Per bank write bursts (Count)
board.memory.mem_ctrl.dram.perBankWrBursts::6        4098                    # Per bank write bursts (Count)
board.memory.mem_ctrl.dram.perBankWrBursts::7        4096                    # Per bank write bursts (Count)
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board.memory.mem_ctrl.dram.perBankWrBursts::8        4116                    # Per bank write bursts (Count)
board.memory.mem_ctrl.dram.perBankWrBursts::9        4106                    # Per bank write bursts (Count)
board.memory.mem_ctrl.dram.perBankWrBursts::10       4117                    # Per bank write bursts (Count)
board.memory.mem_ctrl.dram.perBankWrBursts::11       4098                    # Per bank write bursts (Count)
board.memory.mem_ctrl.dram.perBankWrBursts::12       4096                    # Per bank write bursts (Count)
board.memory.mem_ctrl.dram.perBankWrBursts::13       4096                    # Per bank write bursts (Count)
board.memory.mem_ctrl.dram.perBankWrBursts::14       4096                    # Per bank write bursts (Count)
board.memory.mem_ctrl.dram.perBankWrBursts::15       4096                    # Per bank write bursts (Count)
board.memory.mem_ctrl.dram.totQLat        3146551395                         # Total ticks spent queuing (Tick)
board.memory.mem_ctrl.dram.totBusLat      663915000                          # Total ticks spent in databus transfers (Tick)
board.memory.mem_ctrl.dram.totMemAccLat   5636232645                         # Total ticks spent from burst creation until serviced by the DRAM (Tick)
board.memory.mem_ctrl.dram.avgQLat        23696.94                           # Average queueing delay per DRAM burst ((Tick/Count))
board.memory.mem_ctrl.dram.avgBusLat      5000.00                            # Average bus latency per DRAM burst ((Tick/Count))
board.memory.mem_ctrl.dram.avgMemAccLat   42446.94                           # Average memory access latency per DRAM burst ((Tick/Count))
board.memory.mem_ctrl.dram.readRowHits        122735                         # Number of row buffer hits during reads (Count)
board.memory.mem_ctrl.dram.writeRowHits       60959                          # Number of row buffer hits during writes (Count)
board.memory.mem_ctrl.dram.readRowHitRate     92.43                          # Row buffer hit rate for reads (Ratio)
board.memory.mem_ctrl.dram.writeRowHitRate    92.83                          # Row buffer hit rate for writes (Ratio)
board.memory.mem_ctrl.dram.bytesPerActivate::samples      14755              # Bytes accessed per row activation (Byte)
board.memory.mem_ctrl.dram.bytesPerActivate::mean   860.775059               # Bytes accessed per row activation (Byte)
board.memory.mem_ctrl.dram.bytesPerActivate::gmean   757.624741              # Bytes accessed per row activation (Byte)
board.memory.mem_ctrl.dram.bytesPerActivate::stdev   287.222379              # Bytes accessed per row activation (Byte)
board.memory.mem_ctrl.dram.bytesPerActivate::0-127        617     4.18%     4.18% # Bytes accessed per row activation (Byte)
board.memory.mem_ctrl.dram.bytesPerActivate::128-255      243     1.65%     5.83% # Bytes accessed per row activation (Byte)
board.memory.mem_ctrl.dram.bytesPerActivate::256-383      348     2.36%     8.19% # Bytes accessed per row activation (Byte)
board.memory.mem_ctrl.dram.bytesPerActivate::384-511      1109    7.52%    15.70% # Bytes accessed per row activation (Byte)
board.memory.mem_ctrl.dram.bytesPerActivate::512-639      720     4.88%    20.58% # Bytes accessed per row activation (Byte)
board.memory.mem_ctrl.dram.bytesPerActivate::640-767      678     4.60%    25.18% # Bytes accessed per row activation (Byte)
board.memory.mem_ctrl.dram.bytesPerActivate::768-895      226     1.53%    26.71% # Bytes accessed per row activation (Byte)
board.memory.mem_ctrl.dram.bytesPerActivate::896-1023     591     4.01%    30.72% # Bytes accessed per row activation (Byte)
board.memory.mem_ctrl.dram.bytesPerActivate::1024-1151    10223   69.28%   100.00% # Bytes accessed per row activation (Byte)
board.memory.mem_ctrl.dram.bytesPerActivate::total        14755              # Bytes accessed per row activation (Byte)
board.memory.mem_ctrl.dram.dramBytesRead      8498112                        # Total bytes read (Byte)
board.memory.mem_ctrl.dram.dramBytesWritten   4202624                        # Total bytes written (Byte)
board.memory.mem_ctrl.dram.avgRdBW        1437.950608                        # Average DRAM read bandwidth in MiBytes/s ((Byte/Second))
board.memory.mem_ctrl.dram.avgWrBW        711.118627                         # Average DRAM write bandwidth in MiBytes/s ((Byte/Second))
board.memory.mem_ctrl.dram.peakBW         12800.00                           # Theoretical peak bandwidth in MiByte/s ((Byte/Second))
board.memory.mem_ctrl.dram.busUtil        16.79                              # Data bus utilization in percentage (Ratio)
board.memory.mem_ctrl.dram.busUtilRead        11.23                          # Data bus utilization in percentage for reads (Ratio)
board.memory.mem_ctrl.dram.busUtilWrite       5.56                           # Data bus utilization in percentage for writes (Ratio)
board.memory.mem_ctrl.dram.pageHitRate        92.56                          # Row buffer hit rate, read and write combined (Ratio)
board.memory.mem_ctrl.dram.power_state.pwrStateResidencyTicks::UNDEFINED   5909877540   # Cumulative time (in ticks) in various power states (Tick)
board.memory.mem_ctrl.dram.rank0.actEnergy    52857420                       # Energy for activate commands per rank (pJ) (Joule)
board.memory.mem_ctrl.dram.rank0.preEnergy    28094385                       # Energy for precharge commands per rank (pJ) (Joule)
board.memory.mem_ctrl.dram.rank0.readEnergy   474945660                      # Energy for read commands per rank (pJ) (Joule)
board.memory.mem_ctrl.dram.rank0.writeEnergy  171450900                      # Energy for write commands per rank (pJ) (Joule)
board.memory.mem_ctrl.dram.rank0.refreshEnergy 466511760.000000              # Energy for refresh commands per rank (pJ) (Joule)
board.memory.mem_ctrl.dram.rank0.actBackEnergy   1473491610                  # Energy for active background per rank (pJ) (Joule)
board.memory.mem_ctrl.dram.rank0.preBackEnergy   1028616000                  # Energy for precharge background per rank (pJ) (Joule)
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board.memory.mem_ctrl.dram.rank0.actPowerDownEnergy 0 # Energy for active power-down per rank (pJ) (Joule)
board.memory.mem_ctrl.dram.rank0.prePowerDownEnergy 0 # Energy for precharge power-down per rank (pJ) (Joule)
board.memory.mem_ctrl.dram.rank0.selfRefreshEnergy 0 # Energy for self refresh per rank (pJ) (Joule)
board.memory.mem_ctrl.dram.rank0.totalEnergy 3695967735 # Total energy per rank (pJ) (Joule)
board.memory.mem_ctrl.dram.rank0.averagePower 625.388210 # Core power per rank (mW) (Watt)
board.memory.mem_ctrl.dram.rank0.totalIdleTime 0 # Total Idle time Per DRAM Rank (Tick)
board.memory.mem_ctrl.dram.rank0.pwrStateTime::IDLE 2627175091 # Time in different power states (Tick)
board.memory.mem_ctrl.dram.rank0.pwrStateTime::REF 197175966 # Time in different power states (Tick)
board.memory.mem_ctrl.dram.rank0.pwrStateTime::SREF 0 # Time in different power states (Tick)
board.memory.mem_ctrl.dram.rank0.pwrStateTime::PRE_PDN 0 # Time in different power states (Tick)
board.memory.mem_ctrl.dram.rank0.pwrStateTime::ACT 3085526483 # Time in different power states (Tick)
board.memory.mem_ctrl.dram.rank0.pwrStateTime::ACT_PDN 0 # Time in different power states (Tick)
board.memory.mem_ctrl.dram.rank1.actEnergy 52493280 # Energy for activate commands per rank (pJ) (Joule)
board.memory.mem_ctrl.dram.rank1.preEnergy 27900840 # Energy for precharge commands per rank (pJ) (Joule)
board.memory.mem_ctrl.dram.rank1.readEnergy 473124960 # Energy for read commands per rank (pJ) (Joule)
board.memory.mem_ctrl.dram.rank1.writeEnergy 171325620 # Energy for write commands per rank (pJ) (Joule)
board.memory.mem_ctrl.dram.rank1.refreshEnergy 466511760.000000 # Energy for refresh commands per rank (pJ) (Joule)
board.memory.mem_ctrl.dram.rank1.actBackEnergy 1462379460 # Energy for active background per rank (pJ) (Joule)
board.memory.mem_ctrl.dram.rank1.preBackEnergy 1037971680 # Energy for precharge background per rank (pJ) (Joule)
board.memory.mem_ctrl.dram.rank1.actPowerDownEnergy 0 # Energy for active power-down per rank (pJ) (Joule)
board.memory.mem_ctrl.dram.rank1.prePowerDownEnergy 0 # Energy for precharge power-down per rank (pJ) (Joule)
board.memory.mem_ctrl.dram.rank1.selfRefreshEnergy 0 # Energy for self refresh per rank (pJ) (Joule)
board.memory.mem_ctrl.dram.rank1.totalEnergy 3691707600 # Total energy per rank (pJ) (Joule)
board.memory.mem_ctrl.dram.rank1.averagePower 624.667360 # Core power per rank (mW) (Watt)
board.memory.mem_ctrl.dram.rank1.totalIdleTime 0 # Total Idle time Per DRAM Rank (Tick)
board.memory.mem_ctrl.dram.rank1.pwrStateTime::IDLE 2652087738 # Time in different power states (Tick)
board.memory.mem_ctrl.dram.rank1.pwrStateTime::REF 197180040 # Time in different power states (Tick)
board.memory.mem_ctrl.dram.rank1.pwrStateTime::SREF 0 # Time in different power states (Tick)
board.memory.mem_ctrl.dram.rank1.pwrStateTime::PRE_PDN 0 # Time in different power states (Tick)
board.memory.mem_ctrl.dram.rank1.pwrStateTime::ACT 3060609762 # Time in different power states (Tick)
board.memory.mem_ctrl.dram.rank1.pwrStateTime::ACT_PDN 0 # Time in different power states (Tick)
board.memory.mem_ctrl.power_state.pwrStateResidencyTicks::UNDEFINED 5909877540 # Cumulative time (in ticks) in various power states (Tick)
board.processor.cores.core.numCycles 17747381 # Number of cpu cycles simulated (Cycle)
board.processor.cores.core.cpi 0.481857 # CPI: cycles per instruction (core level) ((Cycle/Count))
board.processor.cores.core.ipc 2.075306 # IPC: instructions per cycle (core level) ((Count/Cycle))
board.processor.cores.core.numWorkItemsStarted 0 # Number of work items this cpu started (Count)
board.processor.cores.core.numWorkItemsCompleted 0 # Number of work items this cpu completed (Count)
board.processor.cores.core.instsAdded 57074691 # Number of instructions added to the IQ (excludes non-spec) (Count)
board.processor.cores.core.nonSpecInstsAdded 240 # Number of non-speculative instructions added to the IQ (Count)
board.processor.cores.core.instsIssued 57028291 # Number of instructions issued (Count)
board.processor.cores.core.squashedInstsIssued 487 # Number of squashed instructions issued (Count)
board.processor.cores.core.squashedInstsExamined 213717 # Number of squashed instructions iterated over during squash; mainly for profiling (Count)
board.processor.cores.core.squashedOperandsExamined 246065 # Number of squashed operands that are examined and possibly removed from graph (Count)
board.processor.cores.core.squashedNonSpecRemoved 150 # Number of squashed non-spec instructions that were removed (Count)
board.processor.cores.core.numIssuedDist::samples 17601662 # Number of insts issued each cycle (Count)
board.processor.cores.core.numIssuedDist::mean 3.239938 # Number of insts issued each cycle (Count)
board.processor.cores.core.numIssuedDist::stdev 2.164296 # Number of insts issued each cycle (Count)
board.processor.cores.core.numIssuedDist::underflows 0 0.00% 0.00% # Number of insts issued each cycle (Count)
board.processor.cores.core.numIssuedDist::0 1620274 9.21% 9.21% # Number of insts issued each cycle (Count)
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board.processor.cores.core.numIssuedDist::1      1824276     10.36%     19.57% # Number of insts issued each cycle (Count)
board.processor.cores.core.numIssuedDist::2      4649372     26.41%     45.98% # Number of insts issued each cycle (Count)
board.processor.cores.core.numIssuedDist::3      2974541     16.90%     62.88% # Number of insts issued each cycle (Count)
board.processor.cores.core.numIssuedDist::4      1555460      8.84%     71.72% # Number of insts issued each cycle (Count)
board.processor.cores.core.numIssuedDist::5      1689522      9.60%     81.32% # Number of insts issued each cycle (Count)
board.processor.cores.core.numIssuedDist::6      1178283      6.69%     88.01% # Number of insts issued each cycle (Count)
board.processor.cores.core.numIssuedDist::7      1636972      9.30%     97.31% # Number of insts issued each cycle (Count)
board.processor.cores.core.numIssuedDist::8       472962      2.69%    100.00% # Number of insts issued each cycle (Count)
board.processor.cores.core.numIssuedDist::overflows            0      0.00%    100.00% # Number of insts issued each cycle (Count)
board.processor.cores.core.numIssuedDist::min_value            0                      # Number of insts issued each cycle (Count)
board.processor.cores.core.numIssuedDist::max_value            8                      # Number of insts issued each cycle (Count)
board.processor.cores.core.numIssuedDist::total     17601662                      # Number of insts issued each cycle (Count)
board.processor.cores.core.statFuBusy::No_OpClass            0      0.00%      0.00% # attempts to use FU when none available (Count)
board.processor.cores.core.statFuBusy::IntAlu      1417779     99.97%     99.97% # attempts to use FU when none available (Count)
board.processor.cores.core.statFuBusy::IntMult            0      0.00%     99.97% # attempts to use FU when none available (Count)
board.processor.cores.core.statFuBusy::IntDiv            0      0.00%     99.97% # attempts to use FU when none available (Count)
board.processor.cores.core.statFuBusy::FloatAdd            0      0.00%     99.97% # attempts to use FU when none available (Count)
board.processor.cores.core.statFuBusy::FloatCmp            0      0.00%     99.97% # attempts to use FU when none available (Count)
board.processor.cores.core.statFuBusy::FloatCvt            0      0.00%     99.97% # attempts to use FU when none available (Count)
board.processor.cores.core.statFuBusy::FloatMult            0      0.00%     99.97% # attempts to use FU when none available (Count)
board.processor.cores.core.statFuBusy::FloatMultAcc            0      0.00%     99.97% # attempts to use FU when none available (Count)
board.processor.cores.core.statFuBusy::FloatDiv            0      0.00%     99.97% # attempts to use FU when none available (Count)
board.processor.cores.core.statFuBusy::FloatMisc            0      0.00%     99.97% # attempts to use FU when none available (Count)
board.processor.cores.core.statFuBusy::FloatSqrt            0      0.00%     99.97% # attempts to use FU when none available (Count)
board.processor.cores.core.statFuBusy::SimdAdd            0      0.00%     99.97% # attempts to use FU when none available (Count)
board.processor.cores.core.statFuBusy::SimdAddAcc            0      0.00%     99.97% # attempts to use FU when none available (Count)
board.processor.cores.core.statFuBusy::SimdAlu           24      0.00%     99.97% # attempts to use FU when none available (Count)
board.processor.cores.core.statFuBusy::SimdCmp            0      0.00%     99.97% # attempts to use FU when none available (Count)
board.processor.cores.core.statFuBusy::SimdCvt            0      0.00%     99.97% # attempts to use FU when none available (Count)
board.processor.cores.core.statFuBusy::SimdMisc            0      0.00%     99.97% # attempts to use FU when none available (Count)
board.processor.cores.core.statFuBusy::SimdMult            0      0.00%     99.97% # attempts to use FU when none available (Count)
board.processor.cores.core.statFuBusy::SimdMultAcc            0      0.00%     99.97% # attempts to use FU when none available (Count)
board.processor.cores.core.statFuBusy::SimdMatMultAcc            0      0.00%     99.97% # attempts to use FU when none available (Count)
board.processor.cores.core.statFuBusy::SimdShift            0      0.00%     99.97% # attempts to use FU when none available (Count)
board.processor.cores.core.statFuBusy::SimdShiftAcc            0      0.00%     99.97% # attempts to use FU when none available (Count)
board.processor.cores.core.statFuBusy::SimdDiv            0      0.00%     99.97% # attempts to use FU when none available (Count)
board.processor.cores.core.statFuBusy::SimdSqrt            0      0.00%     99.97% # attempts to use FU when none available (Count)
board.processor.cores.core.statFuBusy::SimdFloatAdd            0      0.00%     99.97% # attempts to use FU when none available (Count)
board.processor.cores.core.statFuBusy::SimdFloatAlu            0      0.00%     99.97% # attempts to use FU when none available (Count)
board.processor.cores.core.statFuBusy::SimdFloatCmp            0      0.00%     99.97% # attempts to use FU when none available (Count)
board.processor.cores.core.statFuBusy::SimdFloatCvt            0      0.00%     99.97% # attempts to use FU when none available (Count)
board.processor.cores.core.statFuBusy::SimdFloatDiv            0      0.00%     99.97% # attempts to use FU when none available (Count)
board.processor.cores.core.statFuBusy::SimdFloatMisc            0      0.00%     99.97% # attempts to use FU when none available (Count)
board.processor.cores.core.statFuBusy::SimdFloatMult            0      0.00%     99.97% # attempts to use FU when none available (Count)
board.processor.cores.core.statFuBusy::SimdFloatMultAcc            0      0.00%     99.97% # attempts to use FU when none available (Count)
board.processor.cores.core.statFuBusy::SimdFloatMatMultAcc            0      0.00%     99.97% # attempts to use FU when none available (Count)
board.processor.cores.core.statFuBusy::SimdFloatSqrt            0      0.00%     99.97% # attempts to use FU when none available (Count)
board.processor.cores.core.statFuBusy::SimdReduceAdd            0      0.00%     99.97% # attempts to use FU when none available (Count)
board.processor.cores.core.statFuBusy::SimdReduceAlu            0      0.00%     99.97% # attempts to use FU when none available (Count)
```

board.processor.cores.core.statFuBusy::SimdReduceCmp 0 0.00% 99.97% # attempts to use FU when none available (Count)

board.processor.cores.core.statFuBusy::SimdFloatReduceAdd 0 0.00% 99.97% # attempts to use FU when none available (Count)

board.processor.cores.core.statFuBusy::SimdFloatReduceCmp 0 0.00% 99.97% # attempts to use FU when none available (Count)

board.processor.cores.core.statFuBusy::SimdAes 0 0.00% 99.97% # attempts to use FU when none available (Count)

board.processor.cores.core.statFuBusy::SimdAesMix 0 0.00% 99.97% # attempts to use FU when none available (Count)

board.processor.cores.core.statFuBusy::SimdSha1Hash 0 0.00% 99.97% # attempts to use FU when none available (Count)

board.processor.cores.core.statFuBusy::SimdSha1Hash2 0 0.00% 99.97% # attempts to use FU when none available (Count)

board.processor.cores.core.statFuBusy::SimdSha256Hash 0 0.00% 99.97% # attempts to use FU when none available (Count)

board.processor.cores.core.statFuBusy::SimdSha256Hash2 0 0.00% 99.97% # attempts to use FU when none available (Count)

board.processor.cores.core.statFuBusy::SimdShaSigma2 0 0.00% 99.97% # attempts to use FU when none available (Count)

board.processor.cores.core.statFuBusy::SimdShaSigma3 0 0.00% 99.97% # attempts to use FU when none available (Count)

board.processor.cores.core.statFuBusy::SimdPredAlu 0 0.00% 99.97% # attempts to use FU when none available (Count)

board.processor.cores.core.statFuBusy::Matrix 0 0.00% 99.97% # attempts to use FU when none available (Count)

board.processor.cores.core.statFuBusy::MatrixMov 0 0.00% 99.97% # attempts to use FU when none available (Count)

board.processor.cores.core.statFuBusy::MatrixOP 0 0.00% 99.97% # attempts to use FU when none available (Count)

board.processor.cores.core.statFuBusy::MemRead 86 0.01% 99.98% # attempts to use FU when none available (Count)

board.processor.cores.core.statFuBusy::MemWrite 277 0.02% 100.00% # attempts to use FU when none available (Count)

board.processor.cores.core.statFuBusy::FloatMemRead 6 0.00% 100.00% # attempts to use FU when none available (Count)

board.processor.cores.core.statFuBusy::FloatMemWrite 33 0.00% 100.00% # attempts to use FU when none available (Count)

board.processor.cores.core.statFuBusy::IprAccess 0 0.00% 100.00% # attempts to use FU when none available (Count)

board.processor.cores.core.statFuBusy::InstPrefetch 0 0.00% 100.00% # attempts to use FU when none available (Count)

board.processor.cores.core.statFuBusy::SimdUnitStrideLoad 0 0.00% 100.00% # attempts to use FU when none available (Count)

board.processor.cores.core.statFuBusy::SimdUnitStrideStore 0 0.00% 100.00% # attempts to use FU when none available (Count)

board.processor.cores.core.statFuBusy::SimdUnitStrideMaskLoad 0 0.00% 100.00% # attempts to use FU when none available (Count)

board.processor.cores.core.statFuBusy::SimdUnitStrideMaskStore 0 0.00% 100.00% # attempts to use FU when none available (Count)

board.processor.cores.core.statFuBusy::SimdStridedLoad 0 0.00% 100.00% # attempts to use FU when none available (Count)

board.processor.cores.core.statFuBusy::SimdStridedStore 0 0.00% 100.00% # attempts to use FU when none available (Count)

board.processor.cores.core.statFuBusy::SimdIndexedLoad 0 0.00% 100.00% # attempts to use FU when none available (Count)

board.processor.cores.core.statFuBusy::SimdIndexedStore 0 0.00% 100.00% # attempts to use FU when none available (Count)

board.processor.cores.core.statFuBusy::SimdWholeRegisterLoad 0 0.00% 100.00% # attempts to use FU when none available (Count)

board.processor.cores.core.statFuBusy::SimdWholeRegisterStore 0 0.00% 100.00% # attempts to use FU when none available (Count)

board.processor.cores.core.statFuBusy::SimdUnitStrideFaultOnlyFirstLoad 0 0.00% 100.00% # attempts to use FU when none available (Count)

board.processor.cores.core.statFuBusy::SimdUnitStrideSegmentedLoad 0 0.00% 100.00% # attempts to use FU when none available (Count)

board.processor.cores.core.statFuBusy::SimdUnitStrideSegmentedStore 0 0.00% 100.00% # attempts to use FU when none available (Count)

board.processor.cores.core.statFuBusy::SimdExt 0 0.00% 100.00% # attempts to use FU when none available (Count)

board.processor.cores.core.statFuBusy::SimdFloatExt 0 0.00% 100.00% # attempts to use FU when none available (Count)

board.processor.cores.core.statFuBusy::SimdConfig 0 0.00% 100.00% # attempts to use FU when none available (Count)

board.processor.cores.core.statIssuedInstType_0::No_OpClass 739 0.00% 0.00% # Number of instructions issued per FU type, per thread (Count)

board.processor.cores.core.statIssuedInstType_0::IntAlu 40174839 70.45% 70.45% # Number of instructions issued per FU type, per thread (Count)

board.processor.cores.core.statIssuedInstType_0::IntMult 1053193 1.85% 72.30% # Number of instructions issued per FU type, per thread (Count)

board.processor.cores.core.statIssuedInstType_0::IntDiv 40 0.00% 72.30% # Number of instructions issued per FU type, per thread (Count)

board.processor.cores.core.statIssuedInstType_0::FloatAdd 192 0.00% 72.30% # Number of instructions issued per FU type, per thread (Count)

board.processor.cores.core.statIssuedInstType_0::FloatCmp 0 0.00% 72.30% # Number of instructions issued per FU type, per thread (Count)

board.processor.cores.core.statIssuedInstType_0::FloatCvt 0 0.00% 72.30% # Number of instructions issued per FU type, per thread (Count)

board.processor.cores.core.statIssuedInstType_0::FloatMult 0 0.00% 72.30% # Number of instructions issued per FU type, per thread (Count)

board.processor.cores.core.statIssuedInstType_0::FloatMultAcc 0 0.00% 72.30% # Number of instructions issued per FU type, per thread (Count)

board.processor.cores.core.statIssuedInstType_0::FloatDiv 0 0.00% 72.30% # Number of instructions issued per FU type, per thread (Count)

board.processor.cores.core.statIssuedInstType_0::FloatMisc 0 0.00% 72.30% # Number of instructions issued per FU type, per thread (Count)

board.processor.cores.core.statIssuedInstType_0::FloatSqrt 0 0.00% 72.30% # Number of instructions issued per FU type, per thread (Count)

board.processor.cores.core.statIssuedInstType_0::SimdAdd 10 0.00% 72.30% # Number of instructions issued per FU type, per thread (Count)

board.processor.cores.core.statIssuedInstType_0::SimdAddAcc 0 0.00% 72.30% # Number of instructions issued per FU type, per thread (Count)

board.processor.cores.core.statIssuedInstType_0::SimdAlu 281 0.00% 72.30% # Number of instructions issued per FU type, per thread (Count)

board.processor.cores.core.statIssuedInstType_0::SimdCmp 2 0.00% 72.30% # Number of instructions issued per FU type, per thread (Count)

board.processor.cores.core.statIssuedInstType_0::SimdCvt 78 0.00% 72.30% # Number of instructions issued per FU type, per thread (Count)

board.processor.cores.core.statIssuedInstType_0::SimdMisc 253 0.00% 72.30% # Number of instructions issued per FU type, per thread (Count)

board.processor.cores.core.statIssuedInstType_0::SimdMult 0 0.00% 72.30% # Number of instructions issued per FU type, per thread (Count)

board.processor.cores.core.statIssuedInstType_0::SimdMultAcc 0 0.00% 72.30% # Number of instructions issued per FU type, per thread (Count)

board.processor.cores.core.statIssuedInstType_0::SimdMatMultAcc 0 0.00% 72.30% # Number of instructions issued per FU type, per thread (Count)

board.processor.cores.core.statIssuedInstType_0::SimdShift 6 0.00% 72.30% # Number of instructions issued per FU type, per thread (Count)

board.processor.cores.core.statIssuedInstType_0::SimdShiftAcc 0 0.00% 72.30% # Number of instructions issued per FU type, per thread (Count)

board.processor.cores.core.statIssuedInstType_0::SimdDiv 0 0.00% 72.30% # Number of instructions issued per FU type, per thread (Count)

board.processor.cores.core.statIssuedInstType_0::SimdSqrt 0 0.00% 72.30% # Number of instructions issued per FU type, per thread (Count)

board.processor.cores.core.statIssuedInstType_0::SimdFloatAdd 0 0.00% 72.30% # Number of instructions issued per FU type, per thread (Count)

board.processor.cores.core.statIssuedInstType_0::SimdFloatAlu 0 0.00% 72.30% # Number of instructions issued per FU type, per thread (Count)

board.processor.cores.core.statIssuedInstType_0::SimdFloatCmp 0 0.00% 72.30% # Number of instructions issued per FU type, per thread (Count)

board.processor.cores.core.statIssuedInstType_0::SimdFloatCvt 0 0.00% 72.30% # Number of instructions issued per FU type, per thread (Count)

board.processor.cores.core.statIssuedInstType_0::SimdFloatDiv 0 0.00% 72.30% # Number of instructions issued per FU type, per thread (Count)

board.processor.cores.core.statIssuedInstType_0::SimdFloatMisc 0 0.00% 72.30% # Number of instructions issued per FU type, per thread (Count)

board.processor.cores.core.statIssuedInstType_0::SimdFloatMult 0 0.00% 72.30% # Number of instructions issued per FU type, per thread (Count)

board.processor.cores.core.statIssuedInstType_0::SimdFloatMultAcc 0 0.00% 72.30% # Number of instructions issued per FU type, per thread (Count)

board.processor.cores.core.statIssuedInstType_0::SimdFloatMatMultAcc 0 0.00% 72.30% # Number of instructions issued per FU type, per thread (Count)

board.processor.cores.core.statIssuedInstType_0::SimdFloatSqrt 0 0.00% 72.30% # Number of instructions issued per FU type, per thread (Count)

board.processor.cores.core.statIssuedInstType_0::SimdReduceAdd 0 0.00% 72.30% # Number of instructions issued per FU type, per thread (Count)

board.processor.cores.core.statIssuedInstType_0::SimdReduceAlu 0 0.00% 72.30% # Number of instructions issued per FU type, per thread (Count)

board.processor.cores.core.statIssuedInstType_0::SimdReduceCmp 0 0.00% 72.30% # Number of instructions issued per FU type, per thread (Count)

board.processor.cores.core.statIssuedInstType_0::SimdFloatReduceAdd 0 0.00% 72.30% # Number of instructions issued per FU type, per thread (Count)

board.processor.cores.core.statIssuedInstType_0::SimdFloatReduceCmp 0 0.00% 72.30% # Number of instructions issued per FU type, per thread (Count)

board.processor.cores.core.statIssuedInstType_0::SimdAes 0 0.00% 72.30% # Number of instructions issued per FU type, per thread (Count)

board.processor.cores.core.statIssuedInstType_0::SimdAesMix 0 0.00% 72.30% # Number of instructions issued per FU type, per thread (Count)

board.processor.cores.core.statIssuedInstType_0::SimdSha1Hash 0 0.00% 72.30% # Number of instructions issued per FU type, per thread (Count)

board.processor.cores.core.statIssuedInstType_0::SimdSha1Hash2 0 0.00% 72.30% # Number of instructions issued per FU type, per thread (Count)

board.processor.cores.core.statIssuedInstType_0::SimdSha256Hash 0 0.00% 72.30% # Number of instructions issued per FU type, per thread (Count)

board.processor.cores.core.statIssuedInstType_0::SimdSha256Hash2 0 0.00% 72.30% # Number of instructions issued per FU type, per thread (Count)

board.processor.cores.core.statIssuedInstType_0::SimdShaSigma2 0 0.00% 72.30% # Number of instructions issued per FU type, per thread (Count)

board.processor.cores.core.statIssuedInstType_0::SimdShaSigma3 0 0.00% 72.30% # Number of instructions issued per FU type, per thread (Count)

board.processor.cores.core.statIssuedInstType_0::SimdPredAlu 0 0.00% 72.30% # Number of instructions issued per FU type, per thread (Count)

board.processor.cores.core.statIssuedInstType_0::Matrix 0 0.00% 72.30% # Number of instructions issued per FU type, per thread (Count)

board.processor.cores.core.statIssuedInstType_0::MatrixMov 0 0.00% 72.30% # Number of instructions issued per FU type, per thread (Count)

board.processor.cores.core.statIssuedInstType_0::MatrixOP 0 0.00% 72.30% # Number of instructions issued per FU type, per thread (Count)

board.processor.cores.core.statIssuedInstType_0::MemRead 11583702 20.31% 92.61% # Number of instructions issued per FU type, per thread (Count)

board.processor.cores.core.statIssuedInstType_0::MemWrite 4214111 7.39% 100.00% # Number of instructions issued per FU type, per thread (Count)

board.processor.cores.core.statIssuedInstType_0::FloatMemRead 204 0.00% 100.00% # Number of instructions issued per FU type, per thread (Count)

board.processor.cores.core.statIssuedInstType_0::FloatMemWrite 641 0.00% 100.00% # Number of instructions issued per FU type, per thread (Count)

board.processor.cores.core.statIssuedInstType_0::IprAccess 0 0.00% 100.00% # Number of instructions issued per FU type, per thread (Count)

board.processor.cores.core.statIssuedInstType_0::InstPrefetch 0 0.00% 100.00% # Number of instructions issued per FU type, per thread (Count)

board.processor.cores.core.statIssuedInstType_0::SimdUnitStrideLoad 0 0.00% 100.00% # Number of instructions issued per FU type, per thread (Count)

board.processor.cores.core.statIssuedInstType_0::SimdUnitStrideStore 0 0.00% 100.00% # Number of instructions issued per FU type, per thread (Count)

board.processor.cores.core.statIssuedInstType_0::SimdUnitStrideMaskLoad 0 0.00% 100.00% # Number of instructions issued per FU type, per thread (Count)

board.processor.cores.core.statIssuedInstType_0::SimdUnitStrideMaskStore 0 0.00% 100.00% # Number of instructions issued per FU type, per thread (Count)

board.processor.cores.core.statIssuedInstType_0::SimdStridedLoad 0 0.00% 100.00% # Number of instructions issued per FU type, per thread (Count)

board.processor.cores.core.statIssuedInstType_0::SimdStridedStore 0 0.00% 100.00% # Number of instructions issued per FU type, per thread (Count)

board.processor.cores.core.statIssuedInstType_0::SimdIndexedLoad 0 0.00% 100.00% # Number of instructions issued per FU type, per thread (Count)

board.processor.cores.core.statIssuedInstType_0::SimdIndexedStore 0 0.00% 100.00% # Number of instructions issued per FU type, per thread (Count)

board.processor.cores.core.statIssuedInstType_0::SimdWholeRegisterLoad 0 0.00% 100.00% # Number of instructions issued per FU type, per thread (Count)

board.processor.cores.core.statIssuedInstType_0::SimdWholeRegisterStore 0 0.00% 100.00% # Number of instructions issued per FU type, per thread (Count)

board.processor.cores.core.statIssuedInstType_0::SimdUnitStrideFaultOnlyFirstLoad 0 0.00% 100.00% # Number of instructions issued per FU type, per thread (Count)

board.processor.cores.core.statIssuedInstType_0::SimdUnitStrideSegmentedLoad 0 0.00% 100.00% # Number of instructions issued per FU type, per thread (Count)

board.processor.cores.core.statIssuedInstType_0::SimdUnitStrideSegmentedStore 0 0.00% 100.00% # Number of instructions issued per FU type, per thread (Count)

board.processor.cores.core.statIssuedInstType_0::SimdExt 0 0.00% 100.00% # Number of instructions issued per FU type, per thread (Count)

board.processor.cores.core.statIssuedInstType_0::SimdFloatExt 0 0.00% 100.00% # Number of instructions issued per FU type, per thread (Count)

board.processor.cores.core.statIssuedInstType_0::SimdConfig 0 0.00% 100.00% # Number of instructions issued per FU type, per thread (Count)

board.processor.cores.core.statIssuedInstType_0::total 57028291 # Number of instructions issued per FU type, per thread (Count)

board.processor.cores.core.issueRate 3.213336 # Inst issue rate ((Count/Cycle))

board.processor.cores.core.fuBusy 1418205 # FU busy when requested (Count)

board.processor.cores.core.fuBusyRate 0.024868 # FU busy rate (busy events/executed inst) ((Count/Count))

board.processor.cores.core.intInstQueueReads 133073130 # Number of integer instruction queue reads (Count)

board.processor.cores.core.intInstQueueWrites 57286284 # Number of integer instruction queue writes (Count)

board.processor.cores.core.intInstQueueWakeupAccesses 57017662 # Number of integer instruction queue wakeup accesses (Count)

board.processor.cores.core.fpInstQueueReads 3806 # Number of floating instruction queue reads (Count)

board.processor.cores.core.fpInstQueueWrites 2473 # Number of floating instruction queue writes (Count)

board.processor.cores.core.fpInstQueueWakeupAccesses 1770 # Number of floating instruction queue wakeup accesses (Count)

board.processor.cores.core.vecInstQueueReads 0 # Number of vector instruction queue reads (Count)

board.processor.cores.core.vecInstQueueWrites 0 # Number of vector instruction queue writes (Count)

board.processor.cores.core.vecInstQueueWakeupAccesses 0 # Number of vector instruction queue wakeup accesses (Count)

board.processor.cores.core.intAluAccesses 58443846 # Number of integer alu accesses (Count)

board.processor.cores.core.fpAluAccesses 1911 # Number of floating point alu accesses (Count)

board.processor.cores.core.vecAluAccesses 0 # Number of vector alu accesses (Count)

board.processor.cores.core.numSquashedInsts 6992 # Number of squashed instructions skipped in execute (Count)

board.processor.cores.core.numSwp 0 # Number of swp insts executed (Count)

board.processor.cores.core.timesIdled 515 # Number of times that the entire CPU went into an idle state and unscheduled itself (Count)

board.processor.cores.core.idleCycles 145719 # Total number of cycles that the CPU has spent unscheduled due to idling (Cycle)

board.processor.cores.core.MemDepUnit__0.insertedLoads 11586082 # Number of loads inserted to the mem dependence unit. (Count)

board.processor.cores.core.MemDepUnit__0.insertedStores 4218782 # Number of stores inserted to the mem dependence unit. (Count)

board.processor.cores.core.MemDepUnit__0.conflictingLoads 7481485 # Number of conflicting loads. (Count)

board.processor.cores.core.MemDepUnit__0.conflictingStores 1287160 # Number of conflicting stores. (Count)

board.processor.cores.core.MemDepUnit__1.insertedLoads 0 # Number of loads inserted to the mem dependence unit. (Count)

board.processor.cores.core.MemDepUnit__1.insertedStores 0 # Number of stores inserted to the mem dependence unit. (Count)

board.processor.cores.core.MemDepUnit__1.conflictingLoads 0 # Number of conflicting loads. (Count)

board.processor.cores.core.MemDepUnit__1.conflictingStores 0 # Number of conflicting stores. (Count)

board.processor.cores.core.MemDepUnit__2.insertedLoads 0 # Number of loads inserted to the mem dependence unit. (Count)

board.processor.cores.core.MemDepUnit__2.insertedStores 0 # Number of stores inserted to the mem dependence unit. (Count)

board.processor.cores.core.MemDepUnit__2.conflictingLoads 0 # Number of conflicting loads. (Count)

board.processor.cores.core.MemDepUnit__2.conflictingStores 0 # Number of conflicting stores. (Count)

board.processor.cores.core.MemDepUnit__3.insertedLoads 0 # Number of loads inserted to the mem dependence unit. (Count)

board.processor.cores.core.MemDepUnit__3.insertedStores 0 # Number of stores inserted to the mem dependence unit. (Count)

board.processor.cores.core.MemDepUnit__3.conflictingLoads 0 # Number of conflicting loads. (Count)

board.processor.cores.core.MemDepUnit__3.conflictingStores 0 # Number of conflicting stores. (Count)

board.processor.cores.core.branchPred.lookups_0::NoBranch 4 0.00% 0.00% # Number of BP lookups (Count)

board.processor.cores.core.branchPred.lookups_0::Return 2777 0.13% 0.13% # Number of BP lookups (Count)

board.processor.cores.core.branchPred.lookups_0::CallDirect 2860 0.13% 0.26% # Number of BP lookups (Count)

board.processor.cores.core.branchPred.lookups_0::CallIndirect 174 0.01% 0.27% # Number of BP lookups (Count)

board.processor.cores.core.branchPred.lookups_0::DirectCond 2130476 99.44% 99.71% # Number of BP lookups (Count)

board.processor.cores.core.branchPred.lookups_0::DirectUncond 3746 0.17% 99.89% # Number of BP lookups (Count)

board.processor.cores.core.branchPred.lookups_0::IndirectCond 0 0.00% 99.89% # Number of BP lookups (Count)

board.processor.cores.core.branchPred.lookups_0::IndirectUncond 2440 0.11% 100.00% # Number of BP lookups (Count)

board.processor.cores.core.branchPred.lookups_0::total 2142477 # Number of BP lookups (Count)

board.processor.cores.core.branchPred.squashes_0::NoBranch 4 0.03% 0.03% # Number of branches that got squashed (completely removed) as an earlier branch was mispredicted. (Count)

board.processor.cores.core.branchPred.squashes_0::Return 408 3.39% 3.43% # Number of branches that got squashed (completely removed) as an earlier branch was mispredicted. (Count)

board.processor.cores.core.branchPred.squashes_0::CallDirect 533 4.43% 7.86% # Number of branches that got squashed (completely removed) as an earlier branch was mispredicted. (Count)

board.processor.cores.core.branchPred.squashes_0::CallIndirect 127 1.06% 8.92% # Number of branches that got squashed (completely removed) as an earlier branch was mispredicted. (Count)

board.processor.cores.core.branchPred.squashes_0::DirectCond 10218 85.02% 93.93% # Number of branches that got squashed (completely removed) as an earlier branch was mispredicted. (Count)

board.processor.cores.core.branchPred.squashes_0::DirectUncond 427 3.55% 97.49% # Number of branches that got squashed (completely removed) as an earlier branch was mispredicted. (Count)

board.processor.cores.core.branchPred.squashes_0::IndirectCond 0 0.00% 97.49% # Number of branches that got squashed (completely removed) as an earlier branch was mispredicted. (Count)

board.processor.cores.core.branchPred.squashes_0::IndirectUncond 302 2.51% 100.00% # Number of branches that got squashed (completely removed) as an earlier branch was mispredicted. (Count)

board.processor.cores.core.branchPred.squashes_0::total 12019 # Number of branches that got squashed (completely removed) as an earlier branch was mispredicted. (Count)

board.processor.cores.core.branchPred.corrected_0::NoBranch 0 0.00% 0.00% # Number of branches that got corrected but not yet commited. Branches get corrected by decode or after execute. Also a branch misprediction can be detected out-of-order. Therefore, a corrected branch might not end up beeing committed in case an even earlier branch was mispredicted (Count)

board.processor.cores.core.branchPred.corrected_0::Return 10 0.93% 0.93% # Number of branches that got corrected but not yet commited. Branches get corrected by decode or after execute. Also a branch misprediction can be detected out-of-order. Therefore, a corrected branch might not end up beeing committed in case an even earlier branch was mispredicted (Count)

board.processor.cores.core.branchPred.corrected_0::CallDirect 223 20.84% 21.78% # Number of branches that got corrected but not yet commited. Branches get corrected by decode or after execute. Also a branch misprediction can be detected out-of-order. Therefore, a corrected branch might not end up beeing committed in case an even earlier branch was mispredicted (Count)

board.processor.cores.core.branchPred.corrected_0::CallIndirect 39 3.64% 25.42% # Number of branches that got corrected but not yet commited. Branches get corrected by decode or after execute. Also a branch misprediction can be detected out-of-order. Therefore, a corrected branch might not end up beeing committed in case an even earlier branch was mispredicted (Count)

board.processor.cores.core.branchPred.corrected_0::DirectCond 601 56.17% 81.59% # Number of branches that got corrected but not yet commited. Branches get corrected by decode or after execute. Also a branch misprediction can be detected out-of-order. Therefore, a corrected branch might not end up beeing committed in case an even earlier branch was mispredicted (Count)

board.processor.cores.core.branchPred.corrected_0::DirectUncond 148 13.83% 95.42% # Number of branches that got corrected but not yet commited. Branches get corrected by decode or after execute. Also a branch misprediction can be detected out-of-order. Therefore, a corrected branch might not end up beeing committed in case an even earlier branch was mispredicted (Count)

board.processor.cores.core.branchPred.corrected_0::IndirectCond 0 0.00% 95.42% # Number of branches that got corrected but not yet commited. Branches get corrected by decode or after execute. Also a branch misprediction can be detected out-of-order. Therefore, a corrected branch might not end up beeing committed in case an even earlier branch was mispredicted (Count)

board.processor.cores.core.branchPred.corrected_0::IndirectUncond 49 4.58% 100.00% # Number of branches that got corrected but not yet commited. Branches get corrected by decode or after execute. Also a branch misprediction can be detected out-of-order. Therefore, a corrected branch might not end up beeing committed in case an even earlier branch was mispredicted (Count)

board.processor.cores.core.branchPred.corrected_0::total 1070 # Number of branches that got corrected but not yet commited. Branches get corrected by decode or after execute. Also a branch misprediction can be detected out-of-order. Therefore, a corrected branch might not end up beeing committed in case an even earlier branch was mispredicted (Count)

board.processor.cores.core.branchPred.earlyResteers_0::NoBranch 0 # Number of branches that got redirected after decode. (Count)

board.processor.cores.core.branchPred.earlyResteers_0::Return 0 # Number of branches that got redirected after decode. (Count)

board.processor.cores.core.branchPred.earlyResteers_0::CallDirect 0 # Number of branches that got redirected after decode. (Count)

board.processor.cores.core.branchPred.earlyResteers_0::CallIndirect 0 # Number of branches that got redirected after decode. (Count)

board.processor.cores.core.branchPred.earlyResteers_0::DirectCond 0 # Number of branches that got redirected after decode. (Count)

board.processor.cores.core.branchPred.earlyResteers_0::DirectUncond 0 # Number of branches that got redirected after decode. (Count)

board.processor.cores.core.branchPred.earlyResteers_0::IndirectCond 0 # Number of branches that got redirected after decode. (Count)

board.processor.cores.core.branchPred.earlyResteers_0::IndirectUncond 0 # Number of branches that got redirected after decode. (Count)

board.processor.cores.core.branchPred.earlyResteers_0::total 0 # Number of branches that got redirected after decode. (Count)

board.processor.cores.core.branchPred.committed_0::NoBranch 0 0.00% 0.00% # Number of branches finally committed (Count)

board.processor.cores.core.branchPred.committed_0::Return 2369 0.11% 0.11% # Number of branches finally committed (Count)

board.processor.cores.core.branchPred.committed_0::CallDirect 2327 0.11% 0.22% # Number of branches finally committed (Count)

board.processor.cores.core.branchPred.committed_0::CallIndirect 47 0.00% 0.22% # Number of branches finally committed (Count)

board.processor.cores.core.branchPred.committed_0::DirectCond 2120254 99.52% 99.74% # Number of branches finally committed (Count)

board.processor.cores.core.branchPred.committed_0::DirectUncond 3319 0.16% 99.90% # Number of branches finally committed (Count)

board.processor.cores.core.branchPred.committed_0::IndirectCond 0 0.00% 99.90% # Number of branches finally committed (Count)

board.processor.cores.core.branchPred.committed_0::IndirectUncond 2138 0.10% 100.00% # Number of branches finally committed (Count)

board.processor.cores.core.branchPred.committed_0::total 2130454 # Number of branches finally committed (Count)

board.processor.cores.core.branchPred.mispredicted_0::NoBranch 0 0.00% 0.00% # Number of committed branches that were mispredicted. (Count)

board.processor.cores.core.branchPred.mispredicted_0::Return 0 0.00% 0.00% # Number of committed branches that were mispredicted. (Count)

board.processor.cores.core.branchPred.mispredicted_0::CallDirect 109 13.36% 13.36% # Number of committed branches that were mispredicted. (Count)

board.processor.cores.core.branchPred.mispredicted_0::CallIndirect 38 4.66% 18.01% # Number of committed branches that were mispredicted. (Count)

board.processor.cores.core.branchPred.mispredicted_0::DirectCond 539 66.05% 84.07% # Number of committed branches that were mispredicted. (Count)

board.processor.cores.core.branchPred.mispredicted_0::DirectUncond 87 10.66% 94.73% # Number of committed branches that were mispredicted. (Count)

board.processor.cores.core.branchPred.mispredicted_0::IndirectCond 0 0.00% 94.73% # Number of committed branches that were mispredicted. (Count)

board.processor.cores.core.branchPred.mispredicted_0::IndirectUncond 43 5.27% 100.00% # Number of committed branches that were mispredicted. (Count)

board.processor.cores.core.branchPred.mispredicted_0::total 816 # Number of committed branches that were mispredicted. (Count)

board.processor.cores.core.branchPred.targetProvider_0::NoTarget 22939 1.07% 1.07% # The component providing the target for taken branches (Count)

board.processor.cores.core.branchPred.targetProvider_0::BTB 2114562 98.70% 99.77% # The component providing the target for taken branches (Count)

board.processor.cores.core.branchPred.targetProvider_0::RAS 2777 0.13% 99.90% # The component providing the target for taken branches (Count)

board.processor.cores.core.branchPred.targetProvider_0::Indirect 2199 0.10% 100.00% # The component providing the target for taken branches (Count)

board.processor.cores.core.branchPred.targetProvider_0::total 2142477 # The component providing the target for taken branches (Count)

board.processor.cores.core.branchPred.targetWrong_0::NoBranch 968 92.10% 92.10% # Number of branches where the target was incorrect or not available at prediction time. (Count)

board.processor.cores.core.branchPred.targetWrong_0::Return 72 6.85% 98.95% # Number of branches where the target was incorrect or not available at prediction time. (Count)

board.processor.cores.core.branchPred.targetWrong_0::CallDirect 10 0.95% 99.90% # Number of branches where the target was incorrect or not available at prediction time. (Count)

board.processor.cores.core.branchPred.targetWrong_0::CallIndirect 1 0.10% 100.00% # Number of branches where the target was incorrect or not available at prediction time. (Count)

board.processor.cores.core.branchPred.targetWrong_0::DirectCond 0 0.00% 100.00% # Number of branches where the target was incorrect or not available at prediction time. (Count)

board.processor.cores.core.branchPred.targetWrong_0::DirectUncond 0 0.00% 100.00% # Number of branches where the target was incorrect or not available at prediction time. (Count)

board.processor.cores.core.branchPred.targetWrong_0::IndirectCond 0 0.00% 100.00% # Number of branches where the target was incorrect or not available at prediction time. (Count)

board.processor.cores.core.branchPred.targetWrong_0::IndirectUncond 0 0.00% 100.00% # Number of branches where the target was incorrect or not available at prediction time. (Count)

board.processor.cores.core.branchPred.targetWrong_0::total 1051 # Number of branches where the target was incorrect or not available at prediction time. (Count)

board.processor.cores.core.branchPred.condPredicted 2130480 # Number of conditional branches predicted (Count)

board.processor.cores.core.branchPred.condPredictedTaken 2109169 # Number of conditional branches predicted as taken (Count)

board.processor.cores.core.branchPred.condIncorrect 1070 # Number of conditional branches incorrect (Count)

board.processor.cores.core.branchPred.predTakenBTBMiss 483 # Number of branches predicted taken but missed in BTB (Count)

board.processor.cores.core.branchPred.NotTakenMispredicted 1005 # Number branches predicted 'not taken' but turned out to be taken (Count)

board.processor.cores.core.branchPred.TakenMispredicted 65 # Number branches predicted taken but are actually not taken (Count)

board.processor.cores.core.branchPred.BTBLookups 2142477 # Number of BTB lookups (Count)

board.processor.cores.core.branchPred.BTBUpdates 907 # Number of BTB updates (Count)

board.processor.cores.core.branchPred.BTBHits 2117826 # Number of BTB hits (Count)

board.processor.cores.core.branchPred.BTBHitRatio 0.988494 # BTB Hit Ratio (Ratio)

board.processor.cores.core.branchPred.BTBMispredicted 702 # Number BTB mispredictions. No target found or target wrong (Count)

board.processor.cores.core.branchPred.indirectLookups 2614 # Number of indirect predictor lookups. (Count)

board.processor.cores.core.branchPred.indirectHits 2199 # Number of indirect target hits. (Count)

board.processor.cores.core.branchPred.indirectMisses 415 # Number of indirect misses. (Count)

board.processor.cores.core.branchPred.indirectMispredicted 0 # Number of mispredicted indirect branches. (Count)

board.processor.cores.core.branchPred.btb.lookups::NoBranch 4 0.00% 0.00% # Number of BTB lookups (Count)

board.processor.cores.core.branchPred.btb.lookups::Return 2777 0.13% 0.13% # Number of BTB lookups (Count)

board.processor.cores.core.branchPred.btb.lookups::CallDirect 2860 0.13% 0.26% # Number of BTB lookups (Count)

board.processor.cores.core.branchPred.btb.lookups::CallIndirect 174 0.01% 0.27% # Number of BTB lookups (Count)

board.processor.cores.core.branchPred.btb.lookups::DirectCond 2130476 99.44% 99.71% # Number of BTB lookups (Count)

board.processor.cores.core.branchPred.btb.lookups::DirectUncond 3746 0.17% 99.89% # Number of BTB lookups (Count)

board.processor.cores.core.branchPred.btb.lookups::IndirectCond 0 0.00% 99.89% # Number of BTB lookups (Count)

board.processor.cores.core.branchPred.btb.lookups::IndirectUncond 2440 0.11% 100.00% # Number of BTB lookups (Count)

board.processor.cores.core.branchPred.btb.lookups::total 2142477 # Number of BTB lookups (Count)

board.processor.cores.core.branchPred.btb.misses::NoBranch 4 0.02% 0.02% # Number of BTB misses (Count)

board.processor.cores.core.branchPred.btb.misses::Return 2773 11.25% 11.27% # Number of BTB misses (Count)

board.processor.cores.core.branchPred.btb.misses::CallDirect 362 1.47% 12.73% # Number of BTB misses (Count)

board.processor.cores.core.branchPred.btb.misses::CallIndirect 174 0.71% 13.44% # Number of BTB misses (Count)

board.processor.cores.core.branchPred.btb.misses::DirectCond 18671 75.74% 89.18% # Number of BTB misses (Count)

board.processor.cores.core.branchPred.btb.misses::DirectUncond 227 0.92% 90.10% # Number of BTB misses (Count)

board.processor.cores.core.branchPred.btb.misses::IndirectCond 0 0.00% 90.10% # Number of BTB misses (Count)

board.processor.cores.core.branchPred.btb.misses::IndirectUncond 2440 9.90% 100.00% # Number of BTB misses (Count)

board.processor.cores.core.branchPred.btb.misses::total 24651 # Number of BTB misses (Count)

board.processor.cores.core.branchPred.btb.updates::NoBranch 0 0.00% 0.00% # Number of BTB updates (Count)

board.processor.cores.core.branchPred.btb.updates::Return 0 0.00% 0.00% # Number of BTB updates (Count)

board.processor.cores.core.branchPred.btb.updates::CallDirect 223 24.59% 24.59% # Number of BTB updates (Count)

board.processor.cores.core.branchPred.btb.updates::CallIndirect 0 0.00% 24.59% # Number of BTB updates (Count)

board.processor.cores.core.branchPred.btb.updates::DirectCond 536 59.10% 83.68% # Number of BTB updates (Count)

board.processor.cores.core.branchPred.btb.updates::DirectUncond 148 16.32% 100.00% # Number of BTB updates (Count)

board.processor.cores.core.branchPred.btb.updates::IndirectCond 0 0.00% 100.00% # Number of BTB updates (Count)

board.processor.cores.core.branchPred.btb.updates::IndirectUncond 0 0.00% 100.00% # Number of BTB updates (Count)

board.processor.cores.core.branchPred.btb.updates::total 907 # Number of BTB updates (Count)

board.processor.cores.core.branchPred.btb.mispredict::NoBranch 0 0.00% 0.00% # Number of BTB mispredictions. No target found or target wrong. (Count)

board.processor.cores.core.branchPred.btb.mispredict::Return 0 0.00% 0.00% # Number of BTB mispredictions. No target found or target wrong. (Count)

board.processor.cores.core.branchPred.btb.mispredict::CallDirect 223 24.59% 24.59% # Number of BTB mispredictions. No target found or target wrong. (Count)

board.processor.cores.core.branchPred.btb.mispredict::CallIndirect 0 0.00% 24.59% # Number of BTB mispredictions. No target found or target wrong. (Count)

board.processor.cores.core.branchPred.btb.mispredict::DirectCond 536 59.10% 83.68% # Number of BTB mispredictions. No target found or target wrong. (Count)

board.processor.cores.core.branchPred.btb.mispredict::DirectUncond 148 16.32% 100.00% # Number of BTB mispredictions. No target found or target wrong. (Count)

board.processor.cores.core.branchPred.btb.mispredict::IndirectCond 0 0.00% 100.00% # Number of BTB mispredictions. No target found or target wrong. (Count)

board.processor.cores.core.branchPred.btb.mispredict::IndirectUncond 0 0.00% 100.00% # Number of BTB mispredictions. No target found or target wrong. (Count)

board.processor.cores.core.branchPred.btb.mispredict::total 907 # Number of BTB mispredictions. No target found or target wrong. (Count)

board.processor.cores.core.branchPred.btb.power_state.pwrStateResidencyTicks::UNDEFINED 5909877540 # Cumulative time (in ticks) in various power states (Tick)

board.processor.cores.core.branchPred.indirectBranchPred.lookups 2614 # Number of lookups (Count)

board.processor.cores.core.branchPred.indirectBranchPred.hits 2199 # Number of hits of a tag (Count)

board.processor.cores.core.branchPred.indirectBranchPred.misses 415 # Number of misses (Count)

board.processor.cores.core.branchPred.indirectBranchPred.targetRecords 88 # Number of targets that where recorded/installed in the cache (Count)

board.processor.cores.core.branchPred.indirectBranchPred.indirectRecords 2702 # Number of indirect branches/calls recorded in the indirect hist (Count)

board.processor.cores.core.branchPred.indirectBranchPred.speculativeOverflows 2 # Number of times more than the allowed capacity for speculative branches/calls where in flight and destroy the path history (Count)

board.processor.cores.core.branchPred.ras.pushes 3442 # Number of times a PC was pushed onto the RAS (Count)

board.processor.cores.core.branchPred.ras.pops 3437 # Number of times a PC was poped from the RAS (Count)

board.processor.cores.core.branchPred.ras.squashes 1068 # Number of times the stack operation was squashed due to wrong speculation. (Count)

board.processor.cores.core.branchPred.ras.used 2369 # Number of times the RAS is the provider (Count)

board.processor.cores.core.branchPred.ras.correct 2369 # Number of times the RAS is the provider and the prediction is correct (Count)

board.processor.cores.core.branchPred.ras.incorrect 0 # Number of times the RAS is the provider and the prediction is wrong (Count)

board.processor.cores.core.commit.commitSquashedInsts 147495 # The number of squashed insts skipped by commit (Count)

board.processor.cores.core.commit.commitNonSpecStalls 90 # The number of times commit has been forced to stall to communicate backwards (Count)

board.processor.cores.core.commit.branchMispredicts 738 # The number of times a branch was mispredicted (Count)

board.processor.cores.core.commit.numCommittedDist::samples 17581975 # Number of insts commited each cycle (Count)

board.processor.cores.core.commit.numCommittedDist::mean 3.234063 # Number of insts commited each cycle (Count)

board.processor.cores.core.commit.numCommittedDist::stdev 3.205009 # Number of insts commited each cycle (Count)

board.processor.cores.core.commit.numCommittedDist::underflows 0 0.00% 0.00% # Number of insts commited each cycle (Count)

board.processor.cores.core.commit.numCommittedDist::0 2335171 13.28% 13.28% # Number of insts commited each cycle (Count)

board.processor.cores.core.commit.numCommittedDist::1 7278537 41.40% 54.68% # Number of insts commited each cycle (Count)

board.processor.cores.core.commit.numCommittedDist::2 1763911 10.03% 64.71% # Number of insts commited each cycle (Count)

board.processor.cores.core.commit.numCommittedDist::3 123690 0.70% 65.42% # Number of insts commited each cycle (Count)

board.processor.cores.core.commit.numCommittedDist::4 123569 0.70% 66.12% # Number of insts commited each cycle (Count)

board.processor.cores.core.commit.numCommittedDist::5 121415 0.69% 66.81% # Number of insts commited each cycle (Count)

board.processor.cores.core.commit.numCommittedDist::6 1049024 5.97% 72.78% # Number of insts commited each cycle (Count)

board.processor.cores.core.commit.numCommittedDist::7 4980 0.03% 72.80% # Number of insts commited each cycle (Count)

board.processor.cores.core.commit.numCommittedDist::8 4781678 27.20% 100.00% # Number of insts commited each cycle (Count)

board.processor.cores.core.commit.numCommittedDist::overflows 0 0.00% 100.00% # Number of insts commited each cycle (Count)

board.processor.cores.core.commit.numCommittedDist::min_value 0 # Number of insts commited each cycle (Count)

board.processor.cores.core.commit.numCommittedDist::max_value 8 # Number of insts commited each cycle (Count)

board.processor.cores.core.commit.numCommittedDist::total 17581975 # Number of insts commited each cycle (Count)

board.processor.cores.core.commit.amos 0 # Number of atomic instructions committed (Count)

board.processor.cores.core.commit.membars 60 # Number of memory barriers committed (Count)

board.processor.cores.core.commit.functionCalls 2374 # Number of function calls committed. (Count)

board.processor.cores.core.commit.committedInstType_0::No_OpClass 316 0.00% 0.00% # Class of committed instruction (Count)

board.processor.cores.core.commit.committedInstType_0::IntAlu 40052454 70.44% 70.44% # Class of committed instruction (Count)

board.processor.cores.core.commit.committedInstType_0::IntMult 1048624 1.84% 72.28% # Class of committed instruction (Count)

board.processor.cores.core.commit.committedInstType_0::IntDiv 28 0.00% 72.28% # Class of committed instruction (Count)

board.processor.cores.core.commit.committedInstType_0::FloatAdd 154 0.00% 72.28% # Class of committed instruction (Count)

board.processor.cores.core.commit.committedInstType_0::FloatCmp 0 0.00% 72.28% # Class of committed instruction (Count)

board.processor.cores.core.commit.committedInstType_0::FloatCvt 0 0.00% 72.28% # Class of committed instruction (Count)

board.processor.cores.core.commit.committedInstType_0::FloatMult 0 0.00% 72.28% # Class of committed instruction (Count)

board.processor.cores.core.commit.committedInstType_0::FloatMultAcc 0 0.00% 72.28% # Class of committed instruction (Count)

board.processor.cores.core.commit.committedInstType_0::FloatDiv 0 0.00% 72.28% # Class of committed instruction (Count)

board.processor.cores.core.commit.committedInstType_0::FloatMisc 0 0.00% 72.28% # Class of committed instruction (Count)

board.processor.cores.core.commit.committedInstType_0::FloatSqrt 0 0.00% 72.28% # Class of committed instruction (Count)

board.processor.cores.core.commit.committedInstType_0::SimdAdd 10 0.00% 72.28% # Class of committed instruction (Count)

board.processor.cores.core.commit.committedInstType_0::SimdAddAcc 0 0.00% 72.28% # Class of committed instruction (Count)

board.processor.cores.core.commit.committedInstType_0::SimdAlu 229 0.00% 72.28% # Class of committed instruction (Count)

board.processor.cores.core.commit.committedInstType_0::SimdCmp 0 0.00% 72.28% # Class of committed instruction (Count)

board.processor.cores.core.commit.committedInstType_0::SimdCvt 72 0.00% 72.28% # Class of committed instruction (Count)

board.processor.cores.core.commit.committedInstType_0::SimdMisc 240 0.00% 72.29% # Class of committed instruction (Count)

board.processor.cores.core.commit.committedInstType_0::SimdMult 0 0.00% 72.29% # Class of committed instruction (Count)

board.processor.cores.core.commit.committedInstType_0::SimdMultAcc 0 0.00% 72.29% # Class of committed instruction (Count)

board.processor.cores.core.commit.committedInstType_0::SimdMatMultAcc 0 0.00% 72.29% # Class of committed instruction (Count)

board.processor.cores.core.commit.committedInstType_0::SimdShift 3 0.00% 72.29% # Class of committed instruction (Count)

board.processor.cores.core.commit.committedInstType_0::SimdShiftAcc 0 0.00% 72.29% # Class of committed instruction (Count)

board.processor.cores.core.commit.committedInstType_0::SimdDiv 0 0.00% 72.29% # Class of committed instruction (Count)

board.processor.cores.core.commit.committedInstType_0::SimdSqrt 0 0.00% 72.29% # Class of committed instruction (Count)

board.processor.cores.core.commit.committedInstType_0::SimdFloatAdd 0 0.00% 72.29% # Class of committed instruction (Count)

board.processor.cores.core.commit.committedInstType_0::SimdFloatAlu 0 0.00% 72.29% # Class of committed instruction (Count)

board.processor.cores.core.commit.committedInstType_0::SimdFloatCmp 0 0.00% 72.29% # Class of committed instruction (Count)

board.processor.cores.core.commit.committedInstType_0::SimdFloatCvt 0 0.00% 72.29% # Class of committed instruction (Count)

board.processor.cores.core.commit.committedInstType_0::SimdFloatDiv 0 0.00% 72.29% # Class of committed instruction (Count)

board.processor.cores.core.commit.committedInstType_0::SimdFloatMisc 0 0.00% 72.29% # Class of committed instruction (Count)

board.processor.cores.core.commit.committedInstType_0::SimdFloatMult 0 0.00% 72.29% # Class of committed instruction (Count)

board.processor.cores.core.commit.committedInstType_0::SimdFloatMultAcc 0 0.00% 72.29% # Class of committed instruction (Count)

board.processor.cores.core.commit.committedInstType_0::SimdFloatMatMultAcc 0 0.00% 72.29% # Class of committed instruction (Count)

board.processor.cores.core.commit.committedInstType_0::SimdFloatSqrt 0 0.00% 72.29% # Class of committed instruction (Count)

board.processor.cores.core.commit.committedInstType_0::SimdReduceAdd 0 0.00% 72.29% # Class of committed instruction (Count)

board.processor.cores.core.commit.committedInstType_0::SimdReduceAlu 0 0.00% 72.29% # Class of committed instruction (Count)

board.processor.cores.core.commit.committedInstType_0::SimdReduceCmp 0 0.00% 72.29% # Class of committed instruction (Count)

board.processor.cores.core.commit.committedInstType_0::SimdFloatReduceAdd 0 0.00% 72.29% # Class of committed instruction (Count)

board.processor.cores.core.commit.committedInstType_0::SimdFloatReduceCmp 0 0.00% 72.29% # Class of committed instruction (Count)

board.processor.cores.core.commit.committedInstType_0::SimdAes 0 0.00% 72.29% # Class of committed instruction (Count)

board.processor.cores.core.commit.committedInstType_0::SimdAesMix 0 0.00% 72.29% # Class of committed instruction (Count)

board.processor.cores.core.commit.committedInstType_0::SimdSha1Hash 0 0.00% 72.29% # Class of committed instruction (Count)

board.processor.cores.core.commit.committedInstType_0::SimdSha1Hash2 0 0.00% 72.29% # Class of committed instruction (Count)

board.processor.cores.core.commit.committedInstType_0::SimdSha256Hash 0 0.00% 72.29% # Class of committed instruction (Count)

board.processor.cores.core.commit.committedInstType_0::SimdSha256Hash2 0 0.00% 72.29% # Class of committed instruction (Count)

board.processor.cores.core.commit.committedInstType_0::SimdShaSigma2 0 0.00% 72.29% # Class of committed instruction (Count)

board.processor.cores.core.commit.committedInstType_0::SimdShaSigma3 0 0.00% 72.29% # Class of committed instruction (Count)

board.processor.cores.core.commit.committedInstType_0::SimdPredAlu 0 0.00% 72.29% # Class of committed instruction (Count)

board.processor.cores.core.commit.committedInstType_0::Matrix 0 0.00% 72.29% # Class of committed instruction (Count)

board.processor.cores.core.commit.committedInstType_0::MatrixMov 0 0.00% 72.29% # Class of committed instruction (Count)

board.processor.cores.core.commit.committedInstType_0::MatrixOP 0 0.00% 72.29% # Class of committed instruction (Count)

board.processor.cores.core.commit.committedInstType_0::MemRead 11553787 20.32% 92.60% # Class of committed instruction (Count)

board.processor.cores.core.commit.committedInstType_0::MemWrite 4204624 7.39% 100.00% # Class of committed instruction (Count)

board.processor.cores.core.commit.committedInstType_0::FloatMemRead 125 0.00% 100.00% # Class of committed instruction (Count)

board.processor.cores.core.commit.committedInstType_0::FloatMemWrite 542 0.00% 100.00% # Class of committed instruction (Count)

board.processor.cores.core.commit.committedInstType_0::IprAccess 0 0.00% 100.00% # Class of committed instruction (Count)

board.processor.cores.core.commit.committedInstType_0::InstPrefetch 0 0.00% 100.00% # Class of committed instruction (Count)

board.processor.cores.core.commit.committedInstType_0::SimdUnitStrideLoad 0 0.00% 100.00% # Class of committed instruction (Count)

board.processor.cores.core.commit.committedInstType_0::SimdUnitStrideStore 0 0.00% 100.00% # Class of committed instruction (Count)

board.processor.cores.core.commit.committedInstType_0::SimdUnitStrideMaskLoad 0 0.00% 100.00% # Class of committed instruction (Count)

board.processor.cores.core.commit.committedInstType_0::SimdUnitStrideMaskStore 0 0.00% 100.00% # Class of committed instruction (Count)

board.processor.cores.core.commit.committedInstType_0::SimdStridedLoad 0 0.00% 100.00% # Class of committed instruction (Count)

board.processor.cores.core.commit.committedInstType_0::SimdStridedStore 0 0.00% 100.00% # Class of committed instruction (Count)

board.processor.cores.core.commit.committedInstType_0::SimdIndexedLoad 0 0.00% 100.00% # Class of committed instruction (Count)

board.processor.cores.core.commit.committedInstType_0::SimdIndexedStore 0 0.00% 100.00% # Class of committed instruction (Count)

board.processor.cores.core.commit.committedInstType_0::SimdWholeRegisterLoad 0 0.00% 100.00% # Class of committed instruction (Count)

board.processor.cores.core.commit.committedInstType_0::SimdWholeRegisterStore 0 0.00% 100.00% # Class of committed instruction (Count)

board.processor.cores.core.commit.committedInstType_0::SimdUnitStrideFaultOnlyFirstLoad 0 0.00% 100.00% # Class of committed instruction (Count)

board.processor.cores.core.commit.committedInstType_0::SimdUnitStrideSegmentedLoad 0 0.00% 100.00% # Class of committed instruction (Count)

board.processor.cores.core.commit.committedInstType_0::SimdUnitStrideSegmentedStore 0 0.00% 100.00% # Class of committed instruction (Count)

board.processor.cores.core.commit.committedInstType_0::SimdExt 0 0.00% 100.00% # Class of committed instruction (Count)

board.processor.cores.core.commit.committedInstType_0::SimdFloatExt 0 0.00% 100.00% # Class of committed instruction (Count)

board.processor.cores.core.commit.committedInstType_0::SimdConfig 0 0.00% 100.00% # Class of committed instruction (Count)

board.processor.cores.core.commit.committedInstType_0::total 56861208 # Class of committed instruction (Count)

board.processor.cores.core.commit.commitEligibleSamples 4781678 # number cycles where commit BW limit reached (Cycle)

board.processor.cores.core.commitStats0.numInsts 36831240 # Number of instructions committed (thread level) (Count)

board.processor.cores.core.commitStats0.numOps 56861208 # Number of ops (including micro ops) committed (thread level) (Count)

board.processor.cores.core.commitStats0.numInstsNotNOP 36831240 # Number of instructions committed excluding NOPs or prefetches (Count)

board.processor.cores.core.commitStats0.numOpsNotNOP 56861208 # Number of Ops (including micro ops) Simulated (Count)

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board.processor.cores.core.commitStats0.cpi     0.481857                    # CPI: cycles per instruction (thread level) ((Cycle/Count))
board.processor.cores.core.commitStats0.ipc     2.075306                    # IPC: instructions per cycle (thread level) ((Count/Cycle))
board.processor.cores.core.commitStats0.numMemRefs     15759078                    # Number of memory references committed (Count)
board.processor.cores.core.commitStats0.numFpInsts     1535                    # Number of float instructions (Count)
board.processor.cores.core.commitStats0.numIntInsts     56859973                    # Number of integer instructions (Count)
board.processor.cores.core.commitStats0.numLoadInsts     11553912                    # Number of load instructions (Count)
board.processor.cores.core.commitStats0.numStoreInsts     4205166                    # Number of store instructions (Count)
board.processor.cores.core.commitStats0.numVecInsts     0                    # Number of vector instructions (Count)
board.processor.cores.core.commitStats0.committedInstType::No_OpClass     316     0.00%     0.00% # Class of committed instruction. (Count)
board.processor.cores.core.commitStats0.committedInstType::IntAlu     40052454     70.44%     70.44% # Class of committed instruction. (Count)
board.processor.cores.core.commitStats0.committedInstType::IntMult     1048624     1.84%     72.28% # Class of committed instruction. (Count)
board.processor.cores.core.commitStats0.committedInstType::IntDiv     28     0.00%     72.28% # Class of committed instruction. (Count)
board.processor.cores.core.commitStats0.committedInstType::FloatAdd     154     0.00%     72.28% # Class of committed instruction. (Count)
board.processor.cores.core.commitStats0.committedInstType::FloatCmp     0     0.00%     72.28% # Class of committed instruction. (Count)
board.processor.cores.core.commitStats0.committedInstType::FloatCvt     0     0.00%     72.28% # Class of committed instruction. (Count)
board.processor.cores.core.commitStats0.committedInstType::FloatMult     0     0.00%     72.28% # Class of committed instruction. (Count)
board.processor.cores.core.commitStats0.committedInstType::FloatMultAcc     0     0.00%     72.28% # Class of committed instruction. (Count)
board.processor.cores.core.commitStats0.committedInstType::FloatDiv     0     0.00%     72.28% # Class of committed instruction. (Count)
board.processor.cores.core.commitStats0.committedInstType::FloatMisc     0     0.00%     72.28% # Class of committed instruction. (Count)
board.processor.cores.core.commitStats0.committedInstType::FloatSqrt     0     0.00%     72.28% # Class of committed instruction. (Count)
board.processor.cores.core.commitStats0.committedInstType::SimdAdd     10     0.00%     72.28% # Class of committed instruction. (Count)
board.processor.cores.core.commitStats0.committedInstType::SimdAddAcc     0     0.00%     72.28% # Class of committed instruction. (Count)
board.processor.cores.core.commitStats0.committedInstType::SimdAlu     229     0.00%     72.28% # Class of committed instruction. (Count)
board.processor.cores.core.commitStats0.committedInstType::SimdCmp     0     0.00%     72.28% # Class of committed instruction. (Count)
board.processor.cores.core.commitStats0.committedInstType::SimdCvt     72     0.00%     72.28% # Class of committed instruction. (Count)
board.processor.cores.core.commitStats0.committedInstType::SimdMisc     240     0.00%     72.29% # Class of committed instruction. (Count)
board.processor.cores.core.commitStats0.committedInstType::SimdMult     0     0.00%     72.29% # Class of committed instruction. (Count)
board.processor.cores.core.commitStats0.committedInstType::SimdMultAcc     0     0.00%     72.29% # Class of committed instruction. (Count)
board.processor.cores.core.commitStats0.committedInstType::SimdMatMultAcc     0     0.00%     72.29% # Class of committed instruction. (Count)
board.processor.cores.core.commitStats0.committedInstType::SimdShift     3     0.00%     72.29% # Class of committed instruction. (Count)
board.processor.cores.core.commitStats0.committedInstType::SimdShiftAcc     0     0.00%     72.29% # Class of committed instruction. (Count)
board.processor.cores.core.commitStats0.committedInstType::SimdDiv     0     0.00%     72.29% # Class of committed instruction. (Count)
board.processor.cores.core.commitStats0.committedInstType::SimdSqrt     0     0.00%     72.29% # Class of committed instruction. (Count)
board.processor.cores.core.commitStats0.committedInstType::SimdFloatAdd     0     0.00%     72.29% # Class of committed instruction. (Count)
board.processor.cores.core.commitStats0.committedInstType::SimdFloatAlu     0     0.00%     72.29% # Class of committed instruction. (Count)
board.processor.cores.core.commitStats0.committedInstType::SimdFloatCmp     0     0.00%     72.29% # Class of committed instruction. (Count)
board.processor.cores.core.commitStats0.committedInstType::SimdFloatCvt     0     0.00%     72.29% # Class of committed instruction. (Count)
board.processor.cores.core.commitStats0.committedInstType::SimdFloatDiv     0     0.00%     72.29% # Class of committed instruction. (Count)
board.processor.cores.core.commitStats0.committedInstType::SimdFloatMisc     0     0.00%     72.29% # Class of committed instruction. (Count)
board.processor.cores.core.commitStats0.committedInstType::SimdFloatMult     0     0.00%     72.29% # Class of committed instruction. (Count)
board.processor.cores.core.commitStats0.committedInstType::SimdFloatMultAcc     0     0.00%     72.29% # Class of committed instruction. (Count)
board.processor.cores.core.commitStats0.committedInstType::SimdFloatMatMultAcc     0     0.00%     72.29% # Class of committed instruction. (Count)
board.processor.cores.core.commitStats0.committedInstType::SimdFloatSqrt     0     0.00%     72.29% # Class of committed instruction. (Count)
board.processor.cores.core.commitStats0.committedInstType::SimdReduceAdd     0     0.00%     72.29% # Class of committed instruction. (Count)
board.processor.cores.core.commitStats0.committedInstType::SimdReduceAlu     0     0.00%     72.29% # Class of committed instruction. (Count)
board.processor.cores.core.commitStats0.committedInstType::SimdReduceCmp     0     0.00%     72.29% # Class of committed instruction. (Count)
board.processor.cores.core.commitStats0.committedInstType::SimdFloatReduceAdd     0     0.00%     72.29% # Class of committed instruction. (Count)
board.processor.cores.core.commitStats0.committedInstType::SimdFloatReduceCmp     0     0.00%     72.29% # Class of committed instruction. (Count)
board.processor.cores.core.commitStats0.committedInstType::SimdAes     0     0.00%     72.29% # Class of committed instruction. (Count)
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board.processor.cores.core.commitStats0.committedInstType::SimdAesMix 0 0.00% 72.29% # Class of committed instruction. (Count)

board.processor.cores.core.commitStats0.committedInstType::SimdSha1Hash 0 0.00% 72.29% # Class of committed instruction. (Count)

board.processor.cores.core.commitStats0.committedInstType::SimdSha1Hash2 0 0.00% 72.29% # Class of committed instruction. (Count)

board.processor.cores.core.commitStats0.committedInstType::SimdSha256Hash 0 0.00% 72.29% # Class of committed instruction. (Count)

board.processor.cores.core.commitStats0.committedInstType::SimdSha256Hash2 0 0.00% 72.29% # Class of committed instruction. (Count)

board.processor.cores.core.commitStats0.committedInstType::SimdShaSigma2 0 0.00% 72.29% # Class of committed instruction. (Count)

board.processor.cores.core.commitStats0.committedInstType::SimdShaSigma3 0 0.00% 72.29% # Class of committed instruction. (Count)

board.processor.cores.core.commitStats0.committedInstType::SimdPredAlu 0 0.00% 72.29% # Class of committed instruction. (Count)

board.processor.cores.core.commitStats0.committedInstType::Matrix 0 0.00% 72.29% # Class of committed instruction. (Count)

board.processor.cores.core.commitStats0.committedInstType::MatrixMov 0 0.00% 72.29% # Class of committed instruction. (Count)

board.processor.cores.core.commitStats0.committedInstType::MatrixOP 0 0.00% 72.29% # Class of committed instruction. (Count)

board.processor.cores.core.commitStats0.committedInstType::MemRead 11553787 20.32% 92.60% # Class of committed instruction. (Count)

board.processor.cores.core.commitStats0.committedInstType::MemWrite 4204624 7.39% 100.00% # Class of committed instruction. (Count)

board.processor.cores.core.commitStats0.committedInstType::FloatMemRead 125 0.00% 100.00% # Class of committed instruction. (Count)

board.processor.cores.core.commitStats0.committedInstType::FloatMemWrite 542 0.00% 100.00% # Class of committed instruction. (Count)

board.processor.cores.core.commitStats0.committedInstType::IprAccess 0 0.00% 100.00% # Class of committed instruction. (Count)

board.processor.cores.core.commitStats0.committedInstType::InstPrefetch 0 0.00% 100.00% # Class of committed instruction. (Count)

board.processor.cores.core.commitStats0.committedInstType::SimdUnitStrideLoad 0 0.00% 100.00% # Class of committed instruction. (Count)

board.processor.cores.core.commitStats0.committedInstType::SimdUnitStrideStore 0 0.00% 100.00% # Class of committed instruction. (Count)

board.processor.cores.core.commitStats0.committedInstType::SimdUnitStrideMaskLoad 0 0.00% 100.00% # Class of committed instruction. (Count)

board.processor.cores.core.commitStats0.committedInstType::SimdUnitStrideMaskStore 0 0.00% 100.00% # Class of committed instruction. (Count)

board.processor.cores.core.commitStats0.committedInstType::SimdStridedLoad 0 0.00% 100.00% # Class of committed instruction. (Count)

board.processor.cores.core.commitStats0.committedInstType::SimdStridedStore 0 0.00% 100.00% # Class of committed instruction. (Count)

board.processor.cores.core.commitStats0.committedInstType::SimdIndexedLoad 0 0.00% 100.00% # Class of committed instruction. (Count)

board.processor.cores.core.commitStats0.committedInstType::SimdIndexedStore 0 0.00% 100.00% # Class of committed instruction. (Count)

board.processor.cores.core.commitStats0.committedInstType::SimdWholeRegisterLoad 0 0.00% 100.00% # Class of committed instruction. (Count)

board.processor.cores.core.commitStats0.committedInstType::SimdWholeRegisterStore 0 0.00% 100.00% # Class of committed instruction. (Count)

board.processor.cores.core.commitStats0.committedInstType::SimdUnitStrideFaultOnlyFirstLoad 0 0.00% 100.00% # Class of committed instruction. (Count)

board.processor.cores.core.commitStats0.committedInstType::SimdUnitStrideSegmentedLoad 0 0.00% 100.00% # Class of committed instruction. (Count)

board.processor.cores.core.commitStats0.committedInstType::SimdUnitStrideSegmentedStore 0 0.00% 100.00% # Class of committed instruction. (Count)

board.processor.cores.core.commitStats0.committedInstType::SimdExt 0 0.00% 100.00% # Class of committed instruction. (Count)

board.processor.cores.core.commitStats0.committedInstType::SimdFloatExt 0 0.00% 100.00% # Class of committed instruction. (Count)

board.processor.cores.core.commitStats0.committedInstType::SimdConfig 0 0.00% 100.00% # Class of committed instruction. (Count)

board.processor.cores.core.commitStats0.committedInstType::total 56861208 # Class of committed instruction. (Count)

board.processor.cores.core.commitStats0.committedControl::IsControl 2130454 # Class of control type instructions committed (Count)

board.processor.cores.core.commitStats0.committedControl::IsDirectControl 2125900 # Class of control type instructions committed (Count)

board.processor.cores.core.commitStats0.committedControl::IsIndirectControl 4554 # Class of control type instructions committed (Count)

board.processor.cores.core.commitStats0.committedControl::IsCondControl 2120254 # Class of control type instructions committed (Count)

board.processor.cores.core.commitStats0.committedControl::IsUncondControl 10200 # Class of control type instructions committed (Count)

board.processor.cores.core.commitStats0.committedControl::IsCall 2374 # Class of control type instructions committed (Count)

board.processor.cores.core.commitStats0.committedControl::IsReturn 2369 # Class of control type instructions committed (Count)

board.processor.cores.core.decode.idleCycles 1702458 # Number of cycles decode is idle (Cycle)

board.processor.cores.core.decode.blockedCycles 8753563 # Number of cycles decode is blocked (Cycle)

board.processor.cores.core.decode.runCycles 3137555 # Number of cycles decode is running (Cycle)

board.processor.cores.core.decode.unblockCycles 4006208 # Number of cycles decode is unblocking (Cycle)

board.processor.cores.core.decode.squashCycles 1878 # Number of cycles decode is squashing (Cycle)

board.processor.cores.core.decode.branchResolved 2113618 # Number of times decode resolved a branch (Count)

board.processor.cores.core.decode.branchMispred 386 # Number of times decode detected a branch misprediction (Count)

board.processor.cores.core.decode.decodedInsts 57099515 # Number of instructions handled by decode (Count)

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board.processor.cores.core.decode.squashedInsts 1995 # Number of squashed instructions handled by decode (Count)
board.processor.cores.core.executeStats0.numInsts 57021299 # Number of executed instructions (Count)
board.processor.cores.core.executeStats0.numNop 0 # Number of nop insts executed (Count)
board.processor.cores.core.executeStats0.numBranches 2136461 # Number of branches executed (Count)
board.processor.cores.core.executeStats0.numLoadInsts 11580861 # Number of load instructions executed (Count)
board.processor.cores.core.executeStats0.numStoreInsts 4214501 # Number of stores executed (Count)
board.processor.cores.core.executeStats0.instRate 3.212942 # Inst execution rate ((Count/Cycle))
board.processor.cores.core.executeStats0.numCCRegReads 14860422 # Number of times the CC registers were read (Count)
board.processor.cores.core.executeStats0.numCCRegWrites 30569741 # Number of times the CC registers were written (Count)
board.processor.cores.core.executeStats0.numFpRegReads 2103 # Number of times the floating registers were read (Count)
board.processor.cores.core.executeStats0.numFpRegWrites 1115 # Number of times the floating registers were written (Count)
board.processor.cores.core.executeStats0.numIntRegReads 80182872 # Number of times the integer registers were read (Count)
board.processor.cores.core.executeStats0.numIntRegWrites 48542460 # Number of times the integer registers were written (Count)
board.processor.cores.core.executeStats0.numMemRefs 15795362 # Number of memory refs (Count)
board.processor.cores.core.executeStats0.numMiscRegReads 20069105 # Number of times the Misc registers were read (Count)
board.processor.cores.core.executeStats0.numVecAluAccesses 0 # Number of vector alu accesses (Count)
board.processor.cores.core.executeStats0.numDiscardedOps 0 # Number of ops (including micro ops) which were discarded before commit (Count)
board.processor.cores.core.fetch.predictedBranches 2119538 # Number of branches that fetch has predicted taken (Count)
board.processor.cores.core.fetch.cycles 17560312 # Number of cycles fetch has run and was not squashing or blocked (Cycle)results
board.processor.cores.core.fetch.squashCycles 4504 # Number of cycles fetch has spent squashing (Cycle)
board.processor.cores.core.fetch.miscStallCycles 45 # Number of cycles fetch has spent waiting on interrupts, or bad addresses, or out of MSHRs (Cycle)
board.processor.cores.core.fetch.pendingTrapStallCycles 256 # Number of stall cycles due to pending traps (Cycle)
board.processor.cores.core.fetch.cacheLines 4232517 # Number of cache lines fetched (Count)
board.processor.cores.core.fetch.icacheSquashes 475 # Number of outstanding Icache misses that were squashed (Count)
board.processor.cores.core.fetch.nisnDist::samples 17601662 # Number of instructions fetched each cycle (Total) (Count)
board.processor.cores.core.fetch.nisnDist::mean 3.246580 # Number of instructions fetched each cycle (Total) (Count)
board.processor.cores.core.fetch.nisnDist::stdev 3.478868 # Number of instructions fetched each cycle (Total) (Count)
board.processor.cores.core.fetch.nisnDist::underflows 0 0.00% 0.00% # Number of instructions fetched each cycle (Total) (Count)
board.processor.cores.core.fetch.nisnDist::0 8025406 45.59% 45.59% # Number of instructions fetched each cycle (Total) (Count)
board.processor.cores.core.fetch.nisnDist::1 526109 2.99% 48.58% # Number of instructions fetched each cycle (Total) (Count)
board.processor.cores.core.fetch.nisnDist::2 858778 4.88% 53.46% # Number of instructions fetched each cycle (Total) (Count)
board.processor.cores.core.fetch.nisnDist::3 494694 2.81% 56.27% # Number of instructions fetched each cycle (Total) (Count)
board.processor.cores.core.fetch.nisnDist::4 1338844 7.61% 63.88% # Number of instructions fetched each cycle (Total) (Count)
board.processor.cores.core.fetch.nisnDist::5 528087 3.00% 66.88% # Number of instructions fetched each cycle (Total) (Count)
board.processor.cores.core.fetch.nisnDist::6 327363 1.86% 68.74% # Number of instructions fetched each cycle (Total) (Count)
board.processor.cores.core.fetch.nisnDist::7 561573 3.19% 71.93% # Number of instructions fetched each cycle (Total) (Count)
board.processor.cores.core.fetch.nisnDist::8 4940808 28.07% 100.00% # Number of instructions fetched each cycle (Total) (Count)
board.processor.cores.core.fetch.nisnDist::overflows 0 0.00% 100.00% # Number of instructions fetched each cycle (Total) (Count)
board.processor.cores.core.fetch.nisnDist::min_value 0 # Number of instructions fetched each cycle (Total) (Count)
board.processor.cores.core.fetch.nisnDist::max_value 8 # Number of instructions fetched each cycle (Total) (Count)
board.processor.cores.core.fetch.nisnDist::total 17601662 # Number of instructions fetched each cycle (Total) (Count)
board.processor.cores.core.fetchStats0.numInsts 37022374 # Number of instructions fetched (thread level) (Count)
board.processor.cores.core.fetchStats0.numOps 0 # Number of ops (including micro ops) fetched (thread level) (Count)
board.processor.cores.core.fetchStats0.fetchRate 2.086075 # Number of inst fetches per cycle ((Count/Cycle))
board.processor.cores.core.fetchStats0.numBranches 2142477 # Number of branches fetched (Count)
board.processor.cores.core.fetchStats0.branchRate 0.120721 # Number of branch fetches per cycle (Ratio)
board.processor.cores.core.fetchStats0.icacheStallCycles 38797 # ICache total stall cycles (Cycle)
board.processor.cores.core.fetchStats0.numFetchSuspends 0 # Number of times Execute suspended instruction fetching (Count)
board.processor.cores.core.iew.idleCycles 0 # Number of cycles IEW is idle (Cycle)
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board.processor.cores.core.iew.squashCycles 1878 # Number of cycles IEW is squashing (Cycle)
board.processor.cores.core.iew.blockCycles 25706 # Number of cycles IEW is blocking (Cycle)
board.processor.cores.core.iew.unblockCycles 1042 # Number of cycles IEW is unblocking (Cycle)
board.processor.cores.core.iew.dispatchedInsts 57074931 # Number of instructions dispatched to IQ (Count)
board.processor.cores.core.iew.dispSquashedInsts 115 # Number of squashed instructions skipped by dispatch (Count)
board.processor.cores.core.iew.dispLoadInsts 11586082 # Number of dispatched load instructions (Count)
board.processor.cores.core.iew.dispStoreInsts 4218782 # Number of dispatched store instructions (Count)
board.processor.cores.core.iew.dispNonSpecInsts 80 # Number of dispatched non-speculative instructions (Count)
board.processor.cores.core.iew.iqFullEvents 34 # Number of times the IQ has become full, causing a stall (Count)
board.processor.cores.core.iew.lsqFullEvents 988 # Number of times the LSQ has become full, causing a stall (Count)
board.processor.cores.core.iew.memOrderViolationEvents 110 # Number of memory order violations (Count)
board.processor.cores.core.iew.predictedTakenIncorrect 87 # Number of branches that were predicted taken incorrectly (Count)
board.processor.cores.core.iew.predictedNotTakenIncorrect 780 # Number of branches that were predicted not taken incorrectly (Count)
board.processor.cores.core.iew.branchMispredicts 867 # Number of branch mispredicts detected at execute (Count)
board.processor.cores.core.iew.instsToCommit 57021031 # Cumulative count of insts sent to commit (Count)
board.processor.cores.core.iew.writebackCount 57019432 # Cumulative count of insts written-back (Count)
board.processor.cores.core.iew.producerInst 48545514 # Number of instructions producing a value (Count)
board.processor.cores.core.iew.consumerInst 72824651 # Number of instructions consuming a value (Count)
board.processor.cores.core.iew.wbRate 3.212836 # Insts written-back per cycle ((Count/Cycle))
board.processor.cores.core.iew.wbFanout 0.666608 # Average fanout of values written-back ((Count/Count))
board.processor.cores.core.interrupts.clk_domain.clock 5328 # Clock period in ticks (Tick)
board.processor.cores.core.lsq0.forwLoads 8296335 # Number of loads that had data forwarded from stores (Count)
board.processor.cores.core.lsq0.squashedLoads 32170 # Number of loads squashed (Count)
board.processor.cores.core.lsq0.ignoredResponses 3 # Number of memory responses ignored because the instruction is squashed (Count)
board.processor.cores.core.lsq0.memOrderViolation 110 # Number of memory ordering violations (Count)
board.processor.cores.core.lsq0.squashedStores 13616 # Number of stores squashed (Count)
board.processor.cores.core.lsq0.rescheduledLoads 8 # Number of loads that were rescheduled (Count)
board.processor.cores.core.lsq0.blockedByCache 0 # Number of times an access to memory failed due to the cache being blocked (Count)
board.processor.cores.core.lsq0.loadToUse::samples 11553912 # Distribution of cycle latency between the first time a load is issued and its completion (Unspecified)
board.processor.cores.core.lsq0.loadToUse::mean 2.106723 # Distribution of cycle latency between the first time a load is issued and its completion (Unspecified)
board.processor.cores.core.lsq0.loadToUse::stdev 6.275810 # Distribution of cycle latency between the first time a load is issued and its completion (Unspecified)
board.processor.cores.core.lsq0.loadToUse::0-9 11550319 99.97% 99.97% # Distribution of cycle latency between the first time a load is issued and its completion (Unspecified)
board.processor.cores.core.lsq0.loadToUse::10-19 25 0.00% 99.97% # Distribution of cycle latency between the first time a load is issued and its completion (Unspecified)
board.processor.cores.core.lsq0.loadToUse::20-29 44 0.00% 99.97% # Distribution of cycle latency between the first time a load is issued and its completion (Unspecified)
board.processor.cores.core.lsq0.loadToUse::30-39 1 0.00% 99.97% # Distribution of cycle latency between the first time a load is issued and its completion (Unspecified)
board.processor.cores.core.lsq0.loadToUse::40-49 4 0.00% 99.97% # Distribution of cycle latency between the first time a load is issued and its completion (Unspecified)
board.processor.cores.core.lsq0.loadToUse::50-59 3 0.00% 99.97% # Distribution of cycle latency between the first time a load is issued and its completion (Unspecified)
board.processor.cores.core.lsq0.loadToUse::60-69 5 0.00% 99.97% # Distribution of cycle latency between the first time a load is issued and its completion (Unspecified)
board.processor.cores.core.lsq0.loadToUse::70-79 1 0.00% 99.97% # Distribution of cycle latency between the first time a load is issued and its completion (Unspecified)
board.processor.cores.core.lsq0.loadToUse::80-89 1 0.00% 99.97% # Distribution of cycle latency between the first time a load is issued and its completion (Unspecified)
board.processor.cores.core.lsq0.loadToUse::90-99 2 0.00% 99.97% # Distribution of cycle latency between the first time a load is issued and its completion (Unspecified)
board.processor.cores.core.lsq0.loadToUse::100-109 25 0.00% 99.97% # Distribution of cycle latency between the first time a load is issued and its completion (Unspecified)
board.processor.cores.core.lsq0.loadToUse::170-179 1 0.00% 99.97% # Distribution of cycle latency between the first time a load is issued and its completion (Unspecified)
board.processor.cores.core.lsq0.loadToUse::210-219 2 0.00% 99.97% # Distribution of cycle latency between the first time a load is issued and its completion (Unspecified)
board.processor.cores.core.lsq0.loadToUse::250-259 3 0.00% 99.97% # Distribution of cycle latency between the first time a load is issued and its completion (Unspecified)
board.processor.cores.core.lsq0.loadToUse::260-269 1 0.00% 99.97% # Distribution of cycle latency between the first time a load is issued and its completion (Unspecified)
board.processor.cores.core.lsq0.loadToUse::280-289 2 0.00% 99.97% # Distribution of cycle latency between the first time a load is issued and its completion (Unspecified)
board.processor.cores.core.lsq0.loadToUse::290-299 12 0.00% 99.97% # Distribution of cycle latency between the first time a load is issued and its completion (Unspecified)
board.processor.cores.core.lsq0.loadToUse::overflows 3461 0.03% 100.00% # Distribution of cycle latency between the first time a load is issued and its completion (Unspecified)
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board.processor.cores.core.lsq0.loadToUse::min_value 2 # Distribution of cycle latency between the first time a load is issued and its completion (Unspecified)
board.processor.cores.core.lsq0.loadToUse::max_value 1193 # Distribution of cycle latency between the first time a load is issued and its completion (Unspecified)
board.processor.cores.core.lsq0.loadToUse::total 11553912 # Distribution of cycle latency between the first time a load is issued and its completion (Unspecified)
board.processor.cores.core.mmu.dtb.rdAccesses 11580868 # TLB accesses on read requests (Count)
board.processor.cores.core.mmu.dtb.wrAccesses 4214501 # TLB accesses on write requests (Count)
board.processor.cores.core.mmu.dtb.rdMisses 1137 # TLB misses on read requests (Count)
board.processor.cores.core.mmu.dtb.wrMisses 2081 # TLB misses on write requests (Count)
board.processor.cores.core.mmu.dtb.walker.power_state.pwrStateResidencyTicks::UNDEFINED 5909877540 # Cumulative time (in ticks) in various power states (Tick)
board.processor.cores.core.mmu.itb.rdAccesses 0 # TLB accesses on read requests (Count)
board.processor.cores.core.mmu.itb.wrAccesses 4232562 # TLB accesses on write requests (Count)
board.processor.cores.core.mmu.itb.rdMisses 0 # TLB misses on read requests (Count)
board.processor.cores.core.mmu.itb.wrMisses 116 # TLB misses on write requests (Count)
board.processor.cores.core.mmu.itb.walker.power_state.pwrStateResidencyTicks::UNDEFINED 5909877540 # Cumulative time (in ticks) in various power states (Tick)
board.processor.cores.core.power_state.pwrStateResidencyTicks::ON 5909877540 # Cumulative time (in ticks) in various power states (Tick)
board.processor.cores.core.rename.squashCycles 1878 # Number of cycles rename is squashing (Cycle)
board.processor.cores.core.rename.idleCycles 2873636 # Number of cycles rename is idle (Cycle)
board.processor.cores.core.rename.blockCycles 511252 # Number of cycles rename is blocking (Cycle)
board.processor.cores.core.rename.serializeStallCycles 2460 # count of cycles rename stalled for serializing inst (Cycle)
board.processor.cores.core.rename.runCycles 5970982 # Number of cycles rename is running (Cycle)
board.processor.cores.core.rename.unblockCycles 8241454 # Number of cycles rename is unblocking (Cycle)
board.processor.cores.core.rename.renamedInsts 57095363 # Number of instructions processed by rename (Count)
board.processor.cores.core.rename.ROBFullEvents 2779 # Number of times rename has blocked due to ROB full (Count)
board.processor.cores.core.rename.IQFullEvents 2215517 # Number of times rename has blocked due to IQ full (Count)
board.processor.cores.core.rename.LQFullEvents 5081085 # Number of times rename has blocked due to LQ full (Count)
board.processor.cores.core.rename.SQFullEvents 410617 # Number of times rename has blocked due to SQ full (Count)
board.processor.cores.core.rename.renamedOperands 127817195 # Number of destination operands rename has renamed (Count)
board.processor.cores.core.rename.lookups 213505662 # Number of register rename lookups that rename has made (Count)
board.processor.cores.core.rename.intLookups 80311084 # Number of integer rename lookups (Count)
board.processor.cores.core.rename.fpLookups 2495 # Number of floating rename lookups (Count)
board.processor.cores.core.rename.committedMaps 127275790 # Number of HB maps that are committed (Count)
board.processor.cores.core.rename.undoneMaps 541396 # Number of HB maps that are undone due to squashing (Count)
board.processor.cores.core.rename.serializing 45 # count of serializing insts renamed (Count)
board.processor.cores.core.rename.tempSerializing 46 # count of temporary serializing insts renamed (Count)
board.processor.cores.core.rename.skidInsts 16268434 # count of insts added to the skid buffer (Count)
board.processor.cores.core.rob.reads 69795402 # The number of ROB reads (Count)
board.processor.cores.core.rob.writes 114037160 # The number of ROB writes (Count)
board.processor.cores.core.thread_0.numInsts 36831240 # Number of Instructions committed (Count)
board.processor.cores.core.thread_0.numOps 56861208 # Number of Ops committed (Count)
board.processor.cores.core.thread_0.numMemRefs 0 # Number of Memory References (Count)
board.processor.cores.core.workload.numSyscalls 18 # Number of system calls (Count)
board.workload.inst.arm 0 # number of arm instructions executed (Count)
board.workload.inst.quiesce 0 # number of quiesce instructions executed (Count)

---------- End Simulation Statistics ----------
```

results_ooo_test:

```
---------- Begin Simulation Statistics ----------
simSeconds                    0.005910                # Number of seconds simulated (Second)
simTicks                      5909877540              # Number of ticks simulated (Tick)
finalTick                     5909877540              # Number of ticks from beginning of simulation (restored from checkpoints and never reset) (Tick)
simFreq                       1000000000000           # The number of ticks per simulated second ((Tick/Second))
hostSeconds                   70.79                   # Real time elapsed on the host (Second)
hostTickRate                  83489168                # The number of ticks simulated per host second (ticks/s) ((Tick/Second))
hostMemory                    1214236                 # Number of bytes of host memory used (Byte)
simInsts                      36831240                # Number of instructions simulated (Count)
simOps                        56861208                # Number of ops (including micro ops) simulated (Count)
hostInstRate                  520316                  # Simulator instruction rate (inst/s) ((Count/Second))
hostOpRate                    803281                  # Simulator op (including micro ops) rate (op/s) ((Count/Second))
board.cache_hierarchy.dptw_caches.blockedCycles::no_mshrs      0       # number of cycles access was blocked (Cycle)
board.cache_hierarchy.dptw_caches.blockedCycles::no_targets    0       # number of cycles access was blocked (Cycle)
board.cache_hierarchy.dptw_caches.blockedCauses::no_mshrs      0       # number of times access was blocked (Count)
board.cache_hierarchy.dptw_caches.blockedCauses::no_targets    0       # number of times access was blocked (Count)
board.cache_hierarchy.dptw_caches.avgBlocked::no_mshrs         nan     # average number of cycles each access was blocked ((Cycle/Count))
board.cache_hierarchy.dptw_caches.avgBlocked::no_targets       nan     # average number of cycles each access was blocked ((Cycle/Count))
board.cache_hierarchy.dptw_caches.replacements                 0       # number of replacements (Count)
board.cache_hierarchy.dptw_caches.power_state.pwrStateResidencyTicks::UNDEFINED   5909877540   # Cumulative time (in ticks) in various power states (Tick)
board.cache_hierarchy.dptw_caches.tags.tagsInUse               0       # Average ticks per tags in use ((Tick/Count))
board.cache_hierarchy.dptw_caches.tags.totalRefs               0       # Total number of references to valid blocks. (Count)
board.cache_hierarchy.dptw_caches.tags.sampledRefs             0       # Sample count of references to valid blocks. (Count)
board.cache_hierarchy.dptw_caches.tags.avgRefs                 nan     # Average number of references to valid blocks. ((Count/Count))
board.cache_hierarchy.dptw_caches.tags.warmupTick              0       # The tick when the warmup percentage was hit. (Tick)
board.cache_hierarchy.dptw_caches.tags.tagAccesses             0       # Number of tag accesses (Count)
board.cache_hierarchy.dptw_caches.tags.dataAccesses            0       # Number of data accesses (Count)
board.cache_hierarchy.dptw_caches.tags.power_state.pwrStateResidencyTicks::UNDEFINED   5909877540   # Cumulative time (in ticks) in various power states (Tick)
board.cache_hierarchy.iptw_caches.blockedCycles::no_mshrs      0       # number of cycles access was blocked (Cycle)
board.cache_hierarchy.iptw_caches.blockedCycles::no_targets    0       # number of cycles access was blocked (Cycle)
board.cache_hierarchy.iptw_caches.blockedCauses::no_mshrs      0       # number of times access was blocked (Count)
board.cache_hierarchy.iptw_caches.blockedCauses::no_targets    0       # number of times access was blocked (Count)
board.cache_hierarchy.iptw_caches.avgBlocked::no_mshrs         nan     # average number of cycles each access was blocked ((Cycle/Count))
board.cache_hierarchy.iptw_caches.avgBlocked::no_targets       nan     # average number of cycles each access was blocked ((Cycle/Count))
board.cache_hierarchy.iptw_caches.replacements                 0       # number of replacements (Count)
board.cache_hierarchy.iptw_caches.power_state.pwrStateResidencyTicks::UNDEFINED   5909877540   # Cumulative time (in ticks) in various power states (Tick)
board.cache_hierarchy.iptw_caches.tags.tagsInUse               0       # Average ticks per tags in use ((Tick/Count))
board.cache_hierarchy.iptw_caches.tags.totalRefs               0       # Total number of references to valid blocks. (Count)
board.cache_hierarchy.iptw_caches.tags.sampledRefs             0       # Sample count of references to valid blocks. (Count)
board.cache_hierarchy.iptw_caches.tags.avgRefs                 nan     # Average number of references to valid blocks. ((Count/Count))
board.cache_hierarchy.iptw_caches.tags.warmupTick              0       # The tick when the warmup percentage was hit. (Tick)
board.cache_hierarchy.iptw_caches.tags.tagAccesses             0       # Number of tag accesses (Count)
board.cache_hierarchy.iptw_caches.tags.dataAccesses            0       # Number of data accesses (Count)
board.cache_hierarchy.iptw_caches.tags.power_state.pwrStateResidencyTicks::UNDEFINED   5909877540   # Cumulative time (in ticks) in various power states (Tick)
board.cache_hierarchy.l1dcaches.demandHits::processor.cores.core.data   7484370   # number of demand (read+write) hits (Count)
board.cache_hierarchy.l1dcaches.demandHits::total   7484370   # number of demand (read+write) hits (Count)
```

board.cache_hierarchy.l1dcaches.overallHits::processor.cores.core.data 7484370 # number of overall hits (Count)

board.cache_hierarchy.l1dcaches.overallHits::total 7484370 # number of overall hits (Count)

board.cache_hierarchy.l1dcaches.demandMisses::processor.cores.core.data 5192 # number of demand (read+write) misses (Count)

board.cache_hierarchy.l1dcaches.demandMisses::total 5192 # number of demand (read+write) misses (Count)

board.cache_hierarchy.l1dcaches.overallMisses::processor.cores.core.data 5192 # number of overall misses (Count)

board.cache_hierarchy.l1dcaches.overallMisses::total 5192 # number of overall misses (Count)

board.cache_hierarchy.l1dcaches.demandMissLatency::processor.cores.core.data 626186853 # number of demand (read+write) miss ticks (Tick)

board.cache_hierarchy.l1dcaches.demandMissLatency::total 626186853 # number of demand (read+write) miss ticks (Tick)

board.cache_hierarchy.l1dcaches.overallMissLatency::processor.cores.core.data 626186853 # number of overall miss ticks (Tick)

board.cache_hierarchy.l1dcaches.overallMissLatency::total 626186853 # number of overall miss ticks (Tick)

board.cache_hierarchy.l1dcaches.demandAccesses::processor.cores.core.data 7489562 # number of demand (read+write) accesses (Count)

board.cache_hierarchy.l1dcaches.demandAccesses::total 7489562 # number of demand (read+write) accesses (Count)

board.cache_hierarchy.l1dcaches.overallAccesses::processor.cores.core.data 7489562 # number of overall (read+write) accesses (Count)

board.cache_hierarchy.l1dcaches.overallAccesses::total 7489562 # number of overall (read+write) accesses (Count)

board.cache_hierarchy.l1dcaches.demandMissRate::processor.cores.core.data 0.000693 # miss rate for demand accesses (Ratio)

board.cache_hierarchy.l1dcaches.demandMissRate::total 0.000693 # miss rate for demand accesses (Ratio)

board.cache_hierarchy.l1dcaches.overallMissRate::processor.cores.core.data 0.000693 # miss rate for overall accesses (Ratio)

board.cache_hierarchy.l1dcaches.overallMissRate::total 0.000693 # miss rate for overall accesses (Ratio)

board.cache_hierarchy.l1dcaches.demandAvgMissLatency::processor.cores.core.data 120606.096495 # average overall miss latency in ticks ((Tick/Count))

board.cache_hierarchy.l1dcaches.demandAvgMissLatency::total 120606.096495 # average overall miss latency in ticks ((Tick/Count))

board.cache_hierarchy.l1dcaches.overallAvgMissLatency::processor.cores.core.data 120606.096495 # average overall miss latency ((Tick/Count))

board.cache_hierarchy.l1dcaches.overallAvgMissLatency::total 120606.096495 # average overall miss latency ((Tick/Count))

board.cache_hierarchy.l1dcaches.blockedCycles::no_mshrs 0 # number of cycles access was blocked (Cycle)

board.cache_hierarchy.l1dcaches.blockedCycles::no_targets 0 # number of cycles access was blocked (Cycle)

board.cache_hierarchy.l1dcaches.blockedCauses::no_mshrs 0 # number of times access was blocked (Count)

board.cache_hierarchy.l1dcaches.blockedCauses::no_targets 0 # number of times access was blocked (Count)

board.cache_hierarchy.l1dcaches.avgBlocked::no_mshrs nan # average number of cycles each access was blocked ((Cycle/Count))

board.cache_hierarchy.l1dcaches.avgBlocked::no_targets nan # average number of cycles each access was blocked ((Cycle/Count))

board.cache_hierarchy.l1dcaches.writebacks::writebacks 65717 # number of writebacks (Count)

board.cache_hierarchy.l1dcaches.writebacks::total 65717 # number of writebacks (Count)

board.cache_hierarchy.l1dcaches.demandMshrHits::processor.cores.core.data 2245 # number of demand (read+write) MSHR hits (Count)

board.cache_hierarchy.l1dcaches.demandMshrHits::total 2245 # number of demand (read+write) MSHR hits (Count)

board.cache_hierarchy.l1dcaches.overallMshrHits::processor.cores.core.data 2245 # number of overall MSHR hits (Count)

board.cache_hierarchy.l1dcaches.overallMshrHits::total 2245 # number of overall MSHR hits (Count)

board.cache_hierarchy.l1dcaches.demandMshrMisses::processor.cores.core.data 2947 # number of demand (read+write) MSHR misses (Count)

board.cache_hierarchy.l1dcaches.demandMshrMisses::total 2947 # number of demand (read+write) MSHR misses (Count)

board.cache_hierarchy.l1dcaches.overallMshrMisses::cache_hierarchy.l1dcaches.prefetcher 129120 # number of overall MSHR misses (Count)

board.cache_hierarchy.l1dcaches.overallMshrMisses::processor.cores.core.data 2947 # number of overall MSHR misses (Count)

board.cache_hierarchy.l1dcaches.overallMshrMisses::total 132067 # number of overall MSHR misses (Count)

board.cache_hierarchy.l1dcaches.demandMshrMissLatency::processor.cores.core.data 376598358 # number of demand (read+write) MSHR miss ticks (Tick)

board.cache_hierarchy.l1dcaches.demandMshrMissLatency::total 376598358 # number of demand (read+write) MSHR miss ticks (Tick)

board.cache_hierarchy.l1dcaches.overallMshrMissLatency::cache_hierarchy.l1dcaches.prefetcher 5748813809 # number of overall MSHR miss ticks (Tick)

board.cache_hierarchy.l1dcaches.overallMshrMissLatency::processor.cores.core.data 376598358 # number of overall MSHR miss ticks (Tick)

board.cache_hierarchy.l1dcaches.overallMshrMissLatency::total 6125412167 # number of overall MSHR miss ticks (Tick)

board.cache_hierarchy.l1dcaches.demandMshrMissRate::processor.cores.core.data 0.000393 # mshr miss ratio for demand accesses (Ratio)

board.cache_hierarchy.l1dcaches.demandMshrMissRate::total 0.000393 # mshr miss ratio for demand accesses (Ratio)

board.cache_hierarchy.l1dcaches.overallMshrMissRate::cache_hierarchy.l1dcaches.prefetcher inf # mshr miss ratio for overall accesses (Ratio)

board.cache_hierarchy.l1dcaches.overallMshrMissRate::processor.cores.core.data 0.000393 # mshr miss ratio for overall accesses (Ratio)

board.cache_hierarchy.l1dcaches.overallMshrMissRate::total 0.017633 # mshr miss ratio for overall accesses (Ratio)

board.cache_hierarchy.l1dcaches.demandAvgMshrMissLatency::processor.cores.core.data 127790.416695 # average overall mshr miss latency ((Tick/Count))

board.cache_hierarchy.l1dcaches.demandAvgMshrMissLatency::total 127790.416695 # average overall mshr miss latency ((Tick/Count))

board.cache_hierarchy.l1dcaches.overallAvgMshrMissLatency::cache_hierarchy.l1dcaches.prefetcher 44523.031358 # average overall mshr miss latency ((Tick/Count))

board.cache_hierarchy.l1dcaches.overallAvgMshrMissLatency::processor.cores.core.data 127790.416695 # average overall mshr miss latency ((Tick/Count))

board.cache_hierarchy.l1dcaches.overallAvgMshrMissLatency::total 46381.095709 # average overall mshr miss latency ((Tick/Count))

board.cache_hierarchy.l1dcaches.replacements 131556 # number of replacements (Count)

board.cache_hierarchy.l1dcaches.HardPFReq.mshrMisses::cache_hierarchy.l1dcaches.prefetcher 129120 # number of HardPFReq MSHR misses (Count)

board.cache_hierarchy.l1dcaches.HardPFReq.mshrMisses::total 129120 # number of HardPFReq MSHR misses (Count)

board.cache_hierarchy.l1dcaches.HardPFReq.mshrMissLatency::cache_hierarchy.l1dcaches.prefetcher 5748813809 # number of HardPFReq MSHR miss ticks (Tick)

board.cache_hierarchy.l1dcaches.HardPFReq.mshrMissLatency::total 5748813809 # number of HardPFReq MSHR miss ticks (Tick)

board.cache_hierarchy.l1dcaches.HardPFReq.mshrMissRate::cache_hierarchy.l1dcaches.prefetcher inf # mshr miss rate for HardPFReq accesses (Ratio)

board.cache_hierarchy.l1dcaches.HardPFReq.mshrMissRate::total inf # mshr miss rate for HardPFReq accesses (Ratio)

board.cache_hierarchy.l1dcaches.HardPFReq.avgMshrMissLatency::cache_hierarchy.l1dcaches.prefetcher 44523.031358 # average HardPFReq mshr miss latency ((Tick/Count))

board.cache_hierarchy.l1dcaches.HardPFReq.avgMshrMissLatency::total 44523.031358 # average HardPFReq mshr miss latency ((Tick/Count))

board.cache_hierarchy.l1dcaches.LockedRMWReadReq.hits::processor.cores.core.data 26 # number of LockedRMWReadReq hits (Count)

board.cache_hierarchy.l1dcaches.LockedRMWReadReq.hits::total 26 # number of LockedRMWReadReq hits (Count)

board.cache_hierarchy.l1dcaches.LockedRMWReadReq.misses::processor.cores.core.data 4 # number of LockedRMWReadReq misses (Count)

board.cache_hierarchy.l1dcaches.LockedRMWReadReq.misses::total 4 # number of LockedRMWReadReq misses (Count)

board.cache_hierarchy.l1dcaches.LockedRMWReadReq.missLatency::processor.cores.core.data 516150 # number of LockedRMWReadReq miss ticks (Tick)

board.cache_hierarchy.l1dcaches.LockedRMWReadReq.missLatency::total 516150 # number of LockedRMWReadReq miss ticks (Tick)

board.cache_hierarchy.l1dcaches.LockedRMWReadReq.accesses::processor.cores.core.data 30 # number of LockedRMWReadReq accesses(hits+misses) (Count)

board.cache_hierarchy.l1dcaches.LockedRMWReadReq.accesses::total 30 # number of LockedRMWReadReq accesses(hits+misses) (Count)

board.cache_hierarchy.l1dcaches.LockedRMWReadReq.missRate::processor.cores.core.data 0.133333 # miss rate for LockedRMWReadReq accesses (Ratio)

board.cache_hierarchy.l1dcaches.LockedRMWReadReq.missRate::total 0.133333 # miss rate for LockedRMWReadReq accesses (Ratio)

board.cache_hierarchy.l1dcaches.LockedRMWReadReq.avgMissLatency::processor.cores.core.data 129037.500000 # average LockedRMWReadReq miss latency ((Tick/Count))

board.cache_hierarchy.l1dcaches.LockedRMWReadReq.avgMissLatency::total 129037.500000 # average LockedRMWReadReq miss latency ((Tick/Count))

board.cache_hierarchy.l1dcaches.LockedRMWReadReq.mshrMisses::processor.cores.core.data 4 # number of LockedRMWReadReq MSHR misses (Count)

board.cache_hierarchy.l1dcaches.LockedRMWReadReq.mshrMisses::total 4 # number of LockedRMWReadReq MSHR misses (Count)

board.cache_hierarchy.l1dcaches.LockedRMWReadReq.mshrMissLatency::processor.cores.core.data 1097901 # number of LockedRMWReadReq MSHR miss ticks (Tick)

board.cache_hierarchy.l1dcaches.LockedRMWReadReq.mshrMissLatency::total 1097901 # number of LockedRMWReadReq MSHR miss ticks (Tick)

board.cache_hierarchy.l1dcaches.LockedRMWReadReq.mshrMissRate::processor.cores.core.data 0.133333 # mshr miss rate for LockedRMWReadReq accesses (Ratio)

board.cache_hierarchy.l1dcaches.LockedRMWReadReq.mshrMissRate::total 0.133333 # mshr miss rate for LockedRMWReadReq accesses (Ratio)

board.cache_hierarchy.l1dcaches.LockedRMWReadReq.avgMshrMissLatency::processor.cores.core.data 274475.250000 # average LockedRMWReadReq mshr miss latency ((Tick/Count))

board.cache_hierarchy.l1dcaches.LockedRMWReadReq.avgMshrMissLatency::total 274475.250000 # average LockedRMWReadReq mshr miss latency ((Tick/Count))

board.cache_hierarchy.l1dcaches.LockedRMWWriteReq.hits::processor.cores.core.data 30 # number of LockedRMWWriteReq hits (Count)

board.cache_hierarchy.l1dcaches.LockedRMWWriteReq.hits::total 30 # number of LockedRMWWriteReq hits (Count)

board.cache_hierarchy.l1dcaches.LockedRMWWriteReq.accesses::processor.cores.core.data 30 # number of LockedRMWWriteReq accesses(hits+misses) (Count)

board.cache_hierarchy.l1dcaches.LockedRMWWriteReq.accesses::total 30 # number of LockedRMWWriteReq accesses(hits+misses) (Count)

board.cache_hierarchy.l1dcaches.ReadReq.hits::processor.cores.core.data 3280589 # number of ReadReq hits (Count)

board.cache_hierarchy.l1dcaches.ReadReq.hits::total 3280589 # number of ReadReq hits (Count)

board.cache_hierarchy.l1dcaches.ReadReq.misses::processor.cores.core.data 3837 # number of ReadReq misses (Count)

board.cache_hierarchy.l1dcaches.ReadReq.misses::total 3837 # number of ReadReq misses (Count)

board.cache_hierarchy.l1dcaches.ReadReq.missLatency::processor.cores.core.data 436806423 # number of ReadReq miss ticks (Tick)

board.cache_hierarchy.l1dcaches.ReadReq.missLatency::total 436806423 # number of ReadReq miss ticks (Tick)

board.cache_hierarchy.l1dcaches.ReadReq.accesses::processor.cores.core.data 3284426 # number of ReadReq accesses(hits+misses) (Count)

board.cache_hierarchy.l1dcaches.ReadReq.accesses::total 3284426 # number of ReadReq accesses(hits+misses) (Count)

board.cache_hierarchy.l1dcaches.ReadReq.missRate::processor.cores.core.data 0.001168 # miss rate for ReadReq accesses (Ratio)

board.cache_hierarchy.l1dcaches.ReadReq.missRate::total 0.001168 # miss rate for ReadReq accesses (Ratio)

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board.cache_hierarchy.l1dcaches.ReadReq.avgMissLatency::processor.cores.core.data 113840.610633 # average ReadReq miss latency ((Tick/Count))
board.cache_hierarchy.l1dcaches.ReadReq.avgMissLatency::total 113840.610633 # average ReadReq miss latency ((Tick/Count))
board.cache_hierarchy.l1dcaches.ReadReq.mshrHits::processor.cores.core.data 2029 # number of ReadReq MSHR hits (Count)
board.cache_hierarchy.l1dcaches.ReadReq.mshrHits::total 2029 # number of ReadReq MSHR hits (Count)
board.cache_hierarchy.l1dcaches.ReadReq.mshrMisses::processor.cores.core.data 1808 # number of ReadReq MSHR misses (Count)
board.cache_hierarchy.l1dcaches.ReadReq.mshrMisses::total 1808 # number of ReadReq MSHR misses (Count)
board.cache_hierarchy.l1dcaches.ReadReq.mshrMissLatency::processor.cores.core.data 220720392 # number of ReadReq MSHR miss ticks (Tick)
board.cache_hierarchy.l1dcaches.ReadReq.mshrMissLatency::total 220720392 # number of ReadReq MSHR miss ticks (Tick)
board.cache_hierarchy.l1dcaches.ReadReq.mshrMissRate::processor.cores.core.data 0.000550 # mshr miss rate for ReadReq accesses (Ratio)
board.cache_hierarchy.l1dcaches.ReadReq.mshrMissRate::total 0.000550 # mshr miss rate for ReadReq accesses (Ratio)
board.cache_hierarchy.l1dcaches.ReadReq.avgMshrMissLatency::processor.cores.core.data 122079.862832 # average ReadReq mshr miss latency ((Tick/Count))
board.cache_hierarchy.l1dcaches.ReadReq.avgMshrMissLatency::total 122079.862832 # average ReadReq mshr miss latency ((Tick/Count))
board.cache_hierarchy.l1dcaches.WriteReq.hits::processor.cores.core.data 4203781 # number of WriteReq hits (Count)
board.cache_hierarchy.l1dcaches.WriteReq.hits::total 4203781 # number of WriteReq hits (Count)
board.cache_hierarchy.l1dcaches.WriteReq.misses::processor.cores.core.data 1355 # number of WriteReq misses (Count)
board.cache_hierarchy.l1dcaches.WriteReq.misses::total 1355 # number of WriteReq misses (Count)
board.cache_hierarchy.l1dcaches.WriteReq.missLatency::processor.cores.core.data 189380430 # number of WriteReq miss ticks (Tick)
board.cache_hierarchy.l1dcaches.WriteReq.missLatency::total 189380430 # number of WriteReq miss ticks (Tick)
board.cache_hierarchy.l1dcaches.WriteReq.accesses::processor.cores.core.data 4205136 # number of WriteReq accesses(hits+misses) (Count)
board.cache_hierarchy.l1dcaches.WriteReq.accesses::total 4205136 # number of WriteReq accesses(hits+misses) (Count)
board.cache_hierarchy.l1dcaches.WriteReq.missRate::processor.cores.core.data 0.000322 # miss rate for WriteReq accesses (Ratio)
board.cache_hierarchy.l1dcaches.WriteReq.missRate::total 0.000322 # miss rate for WriteReq accesses (Ratio)
board.cache_hierarchy.l1dcaches.WriteReq.avgMissLatency::processor.cores.core.data 139764.154982 # average WriteReq miss latency ((Tick/Count))
board.cache_hierarchy.l1dcaches.WriteReq.avgMissLatency::total 139764.154982 # average WriteReq miss latency ((Tick/Count))
board.cache_hierarchy.l1dcaches.WriteReq.mshrHits::processor.cores.core.data 216 # number of WriteReq MSHR hits (Count)
board.cache_hierarchy.l1dcaches.WriteReq.mshrHits::total 216 # number of WriteReq MSHR hits (Count)
board.cache_hierarchy.l1dcaches.WriteReq.mshrMisses::processor.cores.core.data 1139 # number of WriteReq MSHR misses (Count)
board.cache_hierarchy.l1dcaches.WriteReq.mshrMisses::total 1139 # number of WriteReq MSHR misses (Count)
board.cache_hierarchy.l1dcaches.WriteReq.mshrMissLatency::processor.cores.core.data 155877966 # number of WriteReq MSHR miss ticks (Tick)
board.cache_hierarchy.l1dcaches.WriteReq.mshrMissLatency::total 155877966 # number of WriteReq MSHR miss ticks (Tick)
board.cache_hierarchy.l1dcaches.WriteReq.mshrMissRate::processor.cores.core.data 0.000271 # mshr miss rate for WriteReq accesses (Ratio)
board.cache_hierarchy.l1dcaches.WriteReq.mshrMissRate::total 0.000271 # mshr miss rate for WriteReq accesses (Ratio)
board.cache_hierarchy.l1dcaches.WriteReq.avgMshrMissLatency::processor.cores.core.data 136855.106234 # average WriteReq mshr miss latency ((Tick/Count))
board.cache_hierarchy.l1dcaches.WriteReq.avgMshrMissLatency::total 136855.106234 # average WriteReq mshr miss latency ((Tick/Count))
board.cache_hierarchy.l1dcaches.power_state.pwrStateResidencyTicks::UNDEFINED 5909877540 # Cumulative time (in ticks) in various power states (Tick)
board.cache_hierarchy.l1dcaches.prefetcher.demandMshrMisses 2947 # demands not covered by prefetchs (Count)
board.cache_hierarchy.l1dcaches.prefetcher.pfIssued 504148 # number of hwpf issued (Count)
board.cache_hierarchy.l1dcaches.prefetcher.pfUnused 19 # number of HardPF blocks evicted w/o reference (Count)
board.cache_hierarchy.l1dcaches.prefetcher.pfUseful 128792 # number of useful prefetch (Count)
board.cache_hierarchy.l1dcaches.prefetcher.pfUsefulButMiss 0 # number of hit on prefetch but cache block is not in an usable state (Count)
board.cache_hierarchy.l1dcaches.prefetcher.accuracy 0.255465 # accuracy of the prefetcher (Count)
board.cache_hierarchy.l1dcaches.prefetcher.coverage 0.977630 # coverage brought by this prefetcher (Count)
board.cache_hierarchy.l1dcaches.prefetcher.pfHitInCache 307514 # number of prefetches hitting in cache (Count)
board.cache_hierarchy.l1dcaches.prefetcher.pfHitInMSHR 67514 # number of prefetches hitting in a MSHR (Count)
board.cache_hierarchy.l1dcaches.prefetcher.pfHitInWB 0 # number of prefetches hit in the Write Buffer (Count)
board.cache_hierarchy.l1dcaches.prefetcher.pfLate 375028 # number of late prefetches (hitting in cache, MSHR or WB) (Count)
board.cache_hierarchy.l1dcaches.prefetcher.pfIdentified 504175 # number of prefetch candidates identified (Count)
board.cache_hierarchy.l1dcaches.prefetcher.pfBufferHit 24 # number of redundant prefetches already in prefetch queue (Count)
board.cache_hierarchy.l1dcaches.prefetcher.pfInCache 0 # number of redundant prefetches already in cache/mshr dropped (Count)
```

board.cache_hierarchy.l1dcaches.prefetcher.pfRemovedDemand 0 # number of prefetches dropped due to a demand for the same address (Count)

board.cache_hierarchy.l1dcaches.prefetcher.pfRemovedFull 0 # number of prefetches dropped due to prefetch queue size (Count)

board.cache_hierarchy.l1dcaches.prefetcher.pfSpanPage 20489 # number of prefetches that crossed the page (Count)

board.cache_hierarchy.l1dcaches.prefetcher.pfUsefulSpanPage 20482 # number of prefetches that is useful and crossed the page (Count)

board.cache_hierarchy.l1dcaches.prefetcher.power_state.pwrStateResidencyTicks::UNDEFINED 5909877540 # Cumulative time (in ticks) in various power states (Tick)

board.cache_hierarchy.l1dcaches.tags.tagsInUse 508.417902 # Average ticks per tags in use ((Tick/Count))

board.cache_hierarchy.l1dcaches.tags.totalRefs 7616499 # Total number of references to valid blocks. (Count)

board.cache_hierarchy.l1dcaches.tags.sampledRefs 132068 # Sample count of references to valid blocks. (Count)

board.cache_hierarchy.l1dcaches.tags.avgRefs 57.671041 # Average number of references to valid blocks. ((Count/Count))

board.cache_hierarchy.l1dcaches.tags.warmupTick 249417 # The tick when the warmup percentage was hit. (Tick)

board.cache_hierarchy.l1dcaches.tags.occupancies::cache_hierarchy.l1dcaches.prefetcher 490.511818 # Average occupied blocks per tick, per requestor ((Count/Tick))

board.cache_hierarchy.l1dcaches.tags.occupancies::processor.cores.core.data 17.906084 # Average occupied blocks per tick, per requestor ((Count/Tick))

board.cache_hierarchy.l1dcaches.tags.avgOccs::cache_hierarchy.l1dcaches.prefetcher 0.958031 # Average percentage of cache occupancy ((Ratio/Tick))

board.cache_hierarchy.l1dcaches.tags.avgOccs::processor.cores.core.data 0.034973 # Average percentage of cache occupancy ((Ratio/Tick))

board.cache_hierarchy.l1dcaches.tags.avgOccs::total 0.993004 # Average percentage of cache occupancy ((Ratio/Tick))

board.cache_hierarchy.l1dcaches.tags.occupanciesTaskId::1022 421 # Occupied blocks per task id (Count)

board.cache_hierarchy.l1dcaches.tags.occupanciesTaskId::1024 91 # Occupied blocks per task id (Count)

board.cache_hierarchy.l1dcaches.tags.ageTaskId_1022::0 4 # Occupied blocks per task id, per block age (Count)

board.cache_hierarchy.l1dcaches.tags.ageTaskId_1022::1 417 # Occupied blocks per task id, per block age (Count)

board.cache_hierarchy.l1dcaches.tags.ageTaskId_1024::0 43 # Occupied blocks per task id, per block age (Count)

board.cache_hierarchy.l1dcaches.tags.ageTaskId_1024::1 47 # Occupied blocks per task id, per block age (Count)

board.cache_hierarchy.l1dcaches.tags.ageTaskId_1024::3 1 # Occupied blocks per task id, per block age (Count)

board.cache_hierarchy.l1dcaches.tags.ratioOccsTaskId::1022 0.822266 # Ratio of occupied blocks and all blocks, per task id (Ratio)

board.cache_hierarchy.l1dcaches.tags.ratioOccsTaskId::1024 0.177734 # Ratio of occupied blocks and all blocks, per task id (Ratio)

board.cache_hierarchy.l1dcaches.tags.tagAccesses 60049044 # Number of tag accesses (Count)

board.cache_hierarchy.l1dcaches.tags.dataAccesses 60049044 # Number of data accesses (Count)

board.cache_hierarchy.l1dcaches.tags.power_state.pwrStateResidencyTicks::UNDEFINED 5909877540 # Cumulative time (in ticks) in various power states (Tick)

board.cache_hierarchy.l1icaches.demandHits::processor.cores.core.inst 4231540 # number of demand (read+write) hits (Count)

board.cache_hierarchy.l1icaches.demandHits::total 4231540 # number of demand (read+write) hits (Count)

board.cache_hierarchy.l1icaches.overallHits::processor.cores.core.inst 4231540 # number of overall hits (Count)

board.cache_hierarchy.l1icaches.overallHits::total 4231540 # number of overall hits (Count)

board.cache_hierarchy.l1icaches.demandMisses::processor.cores.core.inst 977 # number of demand (read+write) misses (Count)

board.cache_hierarchy.l1icaches.demandMisses::total 977 # number of demand (read+write) misses (Count)

board.cache_hierarchy.l1icaches.overallMisses::processor.cores.core.inst 977 # number of overall misses (Count)

board.cache_hierarchy.l1icaches.overallMisses::total 977 # number of overall misses (Count)

board.cache_hierarchy.l1icaches.demandMissLatency::processor.cores.core.inst 112883337 # number of demand (read+write) miss ticks (Tick)

board.cache_hierarchy.l1icaches.demandMissLatency::total 112883337 # number of demand (read+write) miss ticks (Tick)

board.cache_hierarchy.l1icaches.overallMissLatency::processor.cores.core.inst 112883337 # number of overall miss ticks (Tick)

board.cache_hierarchy.l1icaches.overallMissLatency::total 112883337 # number of overall miss ticks (Tick)

board.cache_hierarchy.l1icaches.demandAccesses::processor.cores.core.inst 4232517 # number of demand (read+write) accesses (Count)

board.cache_hierarchy.l1icaches.demandAccesses::total 4232517 # number of demand (read+write) accesses (Count)

board.cache_hierarchy.l1icaches.overallAccesses::processor.cores.core.inst 4232517 # number of overall (read+write) accesses (Count)

board.cache_hierarchy.l1icaches.overallAccesses::total 4232517 # number of overall (read+write) accesses (Count)

board.cache_hierarchy.l1icaches.demandMissRate::processor.cores.core.inst 0.000231 # miss rate for demand accesses (Ratio)

board.cache_hierarchy.l1icaches.demandMissRate::total 0.000231 # miss rate for demand accesses (Ratio)

board.cache_hierarchy.l1icaches.overallMissRate::processor.cores.core.inst 0.000231 # miss rate for overall accesses (Ratio)

board.cache_hierarchy.l1icaches.overallMissRate::total 0.000231 # miss rate for overall accesses (Ratio)

board.cache_hierarchy.l1icaches.demandAvgMissLatency::processor.cores.core.inst 115540.774821 # average overall miss latency in ticks ((Tick/Count))

board.cache_hierarchy.l1icaches.demandAvgMissLatency::total 115540.774821 # average overall miss latency in ticks ((Tick/Count))

board.cache_hierarchy.l1icaches.overallAvgMissLatency::processor.cores.core.inst 115540.774821 # average overall miss latency ((Tick/Count))

board.cache_hierarchy.l1icaches.overallAvgMissLatency::total 115540.774821 # average overall miss latency ((Tick/Count))

board.cache_hierarchy.l1icaches.blockedCycles::no_mshrs 0 # number of cycles access was blocked (Cycle)

board.cache_hierarchy.l1icaches.blockedCycles::no_targets 0 # number of cycles access was blocked (Cycle)

board.cache_hierarchy.l1icaches.blockedCauses::no_mshrs 0 # number of times access was blocked (Count)

board.cache_hierarchy.l1icaches.blockedCauses::no_targets 0 # number of times access was blocked (Count)

board.cache_hierarchy.l1icaches.avgBlocked::no_mshrs nan # average number of cycles each access was blocked ((Cycle/Count))

board.cache_hierarchy.l1icaches.avgBlocked::no_targets nan # average number of cycles each access was blocked ((Cycle/Count))

board.cache_hierarchy.l1icaches.demandMshrHits::processor.cores.core.inst 224 # number of demand (read+write) MSHR hits (Count)

board.cache_hierarchy.l1icaches.demandMshrHits::total 224 # number of demand (read+write) MSHR hits (Count)

board.cache_hierarchy.l1icaches.overallMshrHits::processor.cores.core.inst 224 # number of overall MSHR hits (Count)

board.cache_hierarchy.l1icaches.overallMshrHits::total 224 # number of overall MSHR hits (Count)

board.cache_hierarchy.l1icaches.demandMshrMisses::processor.cores.core.inst 753 # number of demand (read+write) MSHR misses (Count)

board.cache_hierarchy.l1icaches.demandMshrMisses::total 753 # number of demand (read+write) MSHR misses (Count)

board.cache_hierarchy.l1icaches.overallMshrMisses::processor.cores.core.inst 753 # number of overall MSHR misses (Count)

board.cache_hierarchy.l1icaches.overallMshrMisses::total 753 # number of overall MSHR misses (Count)

board.cache_hierarchy.l1icaches.demandMshrMissLatency::processor.cores.core.inst 90479430 # number of demand (read+write) MSHR miss ticks (Tick)

board.cache_hierarchy.l1icaches.demandMshrMissLatency::total 90479430 # number of demand (read+write) MSHR miss ticks (Tick)

board.cache_hierarchy.l1icaches.overallMshrMissLatency::processor.cores.core.inst 90479430 # number of overall MSHR miss ticks (Tick)

board.cache_hierarchy.l1icaches.overallMshrMissLatency::total 90479430 # number of overall MSHR miss ticks (Tick)

board.cache_hierarchy.l1icaches.demandMshrMissRate::processor.cores.core.inst 0.000178 # mshr miss ratio for demand accesses (Ratio)

board.cache_hierarchy.l1icaches.demandMshrMissRate::total 0.000178 # mshr miss ratio for demand accesses (Ratio)

board.cache_hierarchy.l1icaches.overallMshrMissRate::processor.cores.core.inst 0.000178 # mshr miss ratio for overall accesses (Ratio)

board.cache_hierarchy.l1icaches.overallMshrMissRate::total 0.000178 # mshr miss ratio for overall accesses (Ratio)

board.cache_hierarchy.l1icaches.demandAvgMshrMissLatency::processor.cores.core.inst 120158.605578 # average overall mshr miss latency ((Tick/Count))

board.cache_hierarchy.l1icaches.demandAvgMshrMissLatency::total 120158.605578 # average overall mshr miss latency ((Tick/Count))

board.cache_hierarchy.l1icaches.overallAvgMshrMissLatency::processor.cores.core.inst 120158.605578 # average overall mshr miss latency ((Tick/Count))

board.cache_hierarchy.l1icaches.overallAvgMshrMissLatency::total 120158.605578 # average overall mshr miss latency ((Tick/Count))

board.cache_hierarchy.l1icaches.replacements 239 # number of replacements (Count)

board.cache_hierarchy.l1icaches.ReadReq.hits::processor.cores.core.inst 4231540 # number of ReadReq hits (Count)

board.cache_hierarchy.l1icaches.ReadReq.hits::total 4231540 # number of ReadReq hits (Count)

board.cache_hierarchy.l1icaches.ReadReq.misses::processor.cores.core.inst 977 # number of ReadReq misses (Count)

board.cache_hierarchy.l1icaches.ReadReq.misses::total 977 # number of ReadReq misses (Count)

board.cache_hierarchy.l1icaches.ReadReq.missLatency::processor.cores.core.inst 112883337 # number of ReadReq miss ticks (Tick)

board.cache_hierarchy.l1icaches.ReadReq.missLatency::total 112883337 # number of ReadReq miss ticks (Tick)

board.cache_hierarchy.l1icaches.ReadReq.accesses::processor.cores.core.inst 4232517 # number of ReadReq accesses(hits+misses) (Count)

board.cache_hierarchy.l1icaches.ReadReq.accesses::total 4232517 # number of ReadReq accesses(hits+misses) (Count)

board.cache_hierarchy.l1icaches.ReadReq.missRate::processor.cores.core.inst 0.000231 # miss rate for ReadReq accesses (Ratio)

board.cache_hierarchy.l1icaches.ReadReq.missRate::total 0.000231 # miss rate for ReadReq accesses (Ratio)

board.cache_hierarchy.l1icaches.ReadReq.avgMissLatency::processor.cores.core.inst 115540.774821 # average ReadReq miss latency ((Tick/Count))

board.cache_hierarchy.l1icaches.ReadReq.avgMissLatency::total 115540.774821 # average ReadReq miss latency ((Tick/Count))

board.cache_hierarchy.l1icaches.ReadReq.mshrHits::processor.cores.core.inst 224 # number of ReadReq MSHR hits (Count)

board.cache_hierarchy.l1icaches.ReadReq.mshrHits::total 224 # number of ReadReq MSHR hits (Count)

board.cache_hierarchy.l1icaches.ReadReq.mshrMisses::processor.cores.core.inst 753 # number of ReadReq MSHR misses (Count)

board.cache_hierarchy.l1icaches.ReadReq.mshrMisses::total 753 # number of ReadReq MSHR misses (Count)

board.cache_hierarchy.l1icaches.ReadReq.mshrMissLatency::processor.cores.core.inst 90479430 # number of ReadReq MSHR miss ticks (Tick)

board.cache_hierarchy.l1icaches.ReadReq.mshrMissLatency::total 90479430 # number of ReadReq MSHR miss ticks (Tick)

board.cache_hierarchy.l1icaches.ReadReq.mshrMissRate::processor.cores.core.inst 0.000178 # mshr miss rate for ReadReq accesses (Ratio)

board.cache_hierarchy.l1icaches.ReadReq.mshrMissRate::total 0.000178 # mshr miss rate for ReadReq accesses (Ratio)

board.cache_hierarchy.l1icaches.ReadReq.avgMshrMissLatency::processor.cores.core.inst 120158.605578 # average ReadReq mshr miss latency ((Tick/Count))

board.cache_hierarchy.l1icaches.ReadReq.avgMshrMissLatency::total 120158.605578 # average ReadReq mshr miss latency ((Tick/Count))

board.cache_hierarchy.l1icaches.power_state.pwrStateResidencyTicks::UNDEFINED 5909877540 # Cumulative time (in ticks) in various power states (Tick)

board.cache_hierarchy.l1icaches.prefetcher.demandMshrMisses 753 # demands not covered by prefetchs (Count)

board.cache_hierarchy.l1icaches.prefetcher.pfIssued 0 # number of hwpf issued (Count)

board.cache_hierarchy.l1icaches.prefetcher.pfUseful 0 # number of useful prefetch (Count)

board.cache_hierarchy.l1icaches.prefetcher.pfUsefulButMiss 0 # number of hit on prefetch but cache block is not in an usable state (Count)

board.cache_hierarchy.l1icaches.prefetcher.accuracy nan # accuracy of the prefetcher (Count)

board.cache_hierarchy.l1icaches.prefetcher.coverage 0 # coverage brought by this prefetcher (Count)

board.cache_hierarchy.l1icaches.prefetcher.pfHitInCache 0 # number of prefetches hitting in cache (Count)

board.cache_hierarchy.l1icaches.prefetcher.pfHitInMSHR 0 # number of prefetches hitting in a MSHR (Count)

board.cache_hierarchy.l1icaches.prefetcher.pfHitInWB 0 # number of prefetches hit in the Write Buffer (Count)

board.cache_hierarchy.l1icaches.prefetcher.pfLate 0 # number of late prefetches (hitting in cache, MSHR or WB) (Count)

board.cache_hierarchy.l1icaches.prefetcher.pfIdentified 0 # number of prefetch candidates identified (Count)

board.cache_hierarchy.l1icaches.prefetcher.pfBufferHit 0 # number of redundant prefetches already in prefetch queue (Count)

board.cache_hierarchy.l1icaches.prefetcher.pfInCache 0 # number of redundant prefetches already in cache/mshr dropped (Count)

board.cache_hierarchy.l1icaches.prefetcher.pfRemovedDemand 0 # number of prefetches dropped due to a demand for the same address (Count)

board.cache_hierarchy.l1icaches.prefetcher.pfRemovedFull 0 # number of prefetches dropped due to prefetch queue size (Count)

board.cache_hierarchy.l1icaches.prefetcher.pfSpanPage 0 # number of prefetches that crossed the page (Count)

board.cache_hierarchy.l1icaches.prefetcher.pfUsefulSpanPage 0 # number of prefetches that is useful and crossed the page (Count)

board.cache_hierarchy.l1icaches.prefetcher.power_state.pwrStateResidencyTicks::UNDEFINED 5909877540 # Cumulative time (in ticks) in various power states (Tick)

board.cache_hierarchy.l1icaches.tags.tagsInUse 469.941787 # Average ticks per tags in use ((Tick/Count))

board.cache_hierarchy.l1icaches.tags.totalRefs 4232293 # Total number of references to valid blocks. (Count)

board.cache_hierarchy.l1icaches.tags.sampledRefs 753 # Sample count of references to valid blocks. (Count)

board.cache_hierarchy.l1icaches.tags.avgRefs 5620.575033 # Average number of references to valid blocks. ((Count/Count))

board.cache_hierarchy.l1icaches.tags.warmupTick 123543 # The tick when the warmup percentage was hit. (Tick)

board.cache_hierarchy.l1icaches.tags.occupancies::processor.cores.core.inst 469.941787 # Average occupied blocks per tick, per requestor ((Count/Tick))

board.cache_hierarchy.l1icaches.tags.avgOccs::processor.cores.core.inst 0.917855 # Average percentage of cache occupancy ((Ratio/Tick))

board.cache_hierarchy.l1icaches.tags.avgOccs::total 0.917855 # Average percentage of cache occupancy ((Ratio/Tick))

board.cache_hierarchy.l1icaches.tags.occupanciesTaskId::1024 511 # Occupied blocks per task id (Count)

board.cache_hierarchy.l1icaches.tags.ageTaskId_1024::0 106 # Occupied blocks per task id, per block age (Count)

board.cache_hierarchy.l1icaches.tags.ageTaskId_1024::1 98 # Occupied blocks per task id, per block age (Count)

board.cache_hierarchy.l1icaches.tags.ageTaskId_1024::3 307 # Occupied blocks per task id, per block age (Count)

board.cache_hierarchy.l1icaches.tags.ratioOccsTaskId::1024 0.998047 # Ratio of occupied blocks and all blocks, per task id (Ratio)

board.cache_hierarchy.l1icaches.tags.tagAccesses 33860889 # Number of tag accesses (Count)

board.cache_hierarchy.l1icaches.tags.dataAccesses 33860889 # Number of data accesses (Count)

board.cache_hierarchy.l1icaches.tags.power_state.pwrStateResidencyTicks::UNDEFINED 5909877540 # Cumulative time (in ticks) in various power states (Tick)

board.cache_hierarchy.l2bus.transDist::ReadResp 131681 # Transaction distribution (Count)

board.cache_hierarchy.l2bus.transDist::WritebackDirty 131412 # Transaction distribution (Count)

board.cache_hierarchy.l2bus.transDist::CleanEvict 129099 # Transaction distribution (Count)

board.cache_hierarchy.l2bus.transDist::HardPFReq 139497 # Transaction distribution (Count)

board.cache_hierarchy.l2bus.transDist::UpgradeReq 3 # Transaction distribution (Count)

board.cache_hierarchy.l2bus.transDist::UpgradeResp 3 # Transaction distribution (Count)

board.cache_hierarchy.l2bus.transDist::ReadExReq 1140 # Transaction distribution (Count)

board.cache_hierarchy.l2bus.transDist::ReadExResp 1140 # Transaction distribution (Count)

board.cache_hierarchy.l2bus.transDist::ReadSharedReq 131681 # Transaction distribution (Count)

board.cache_hierarchy.l2bus.pktCount_board.cache_hierarchy.l1icaches.mem_side_port::board.cache_hierarchy.l2cache.cpu_side_port 1742 # Packet count per connected requestor and responder (Count)

board.cache_hierarchy.l2bus.pktCount_board.cache_hierarchy.l1dcaches.mem_side_port::board.cache_hierarchy.l2cache.cpu_side_port 395698 # Packet count per connected requestor and responder (Count)

board.cache_hierarchy.l2bus.pktCount::total 397440 # Packet count per connected requestor and responder (Count)

board.cache_hierarchy.l2bus.pktSize_board.cache_hierarchy.l1icaches.mem_side_port::board.cache_hierarchy.l2cache.cpu_side_port 48000 # Cumulative packet size per connected requestor and responder (Byte)

board.cache_hierarchy.l2bus.pktSize_board.cache_hierarchy.l1dcaches.mem_side_port::board.cache_hierarchy.l2cache.cpu_side_port 12658240 # Cumulative packet size per connected requestor and responder (Byte)

board.cache_hierarchy.l2bus.pktSize::total 12706240 # Cumulative packet size per connected requestor and responder (Byte)

board.cache_hierarchy.l2bus.snoops 268216 # Total snoops (Count)

board.cache_hierarchy.l2bus.snoopTraffic 4204672 # Total snoop traffic (Byte)

board.cache_hierarchy.l2bus.snoopFanout::samples 401037 # Request fanout histogram (Count)

board.cache_hierarchy.l2bus.snoopFanout::mean 0.001207 # Request fanout histogram (Count)

board.cache_hierarchy.l2bus.snoopFanout::stdev 0.034719 # Request fanout histogram (Count)

board.cache_hierarchy.l2bus.snoopFanout::underflows 0 0.00% 0.00% # Request fanout histogram (Count)

board.cache_hierarchy.l2bus.snoopFanout::0 400553 99.88% 99.88% # Request fanout histogram (Count)

board.cache_hierarchy.l2bus.snoopFanout::1 484 0.12% 100.00% # Request fanout histogram (Count)

board.cache_hierarchy.l2bus.snoopFanout::2 0 0.00% 100.00% # Request fanout histogram (Count)

board.cache_hierarchy.l2bus.snoopFanout::3 0 0.00% 100.00% # Request fanout histogram (Count)

board.cache_hierarchy.l2bus.snoopFanout::4 0 0.00% 100.00% # Request fanout histogram (Count)

board.cache_hierarchy.l2bus.snoopFanout::overflows 0 0.00% 100.00% # Request fanout histogram (Count)

board.cache_hierarchy.l2bus.snoopFanout::min_value 0 # Request fanout histogram (Count)

board.cache_hierarchy.l2bus.snoopFanout::max_value 1 # Request fanout histogram (Count)

board.cache_hierarchy.l2bus.snoopFanout::total 401037 # Request fanout histogram (Count)

board.cache_hierarchy.l2bus.power_state.pwrStateResidencyTicks::UNDEFINED 5909877540 # Cumulative time (in ticks) in various power states (Tick)

board.cache_hierarchy.l2bus.reqLayer0.occupancy 171646360 # Layer occupancy (ticks) (Tick)

board.cache_hierarchy.l2bus.reqLayer0.utilization 0.0 # Layer utilization (Ratio)

board.cache_hierarchy.l2bus.respLayer0.occupancy 752247 # Layer occupancy (ticks) (Tick)

board.cache_hierarchy.l2bus.respLayer0.utilization 0.0 # Layer utilization (Ratio)

board.cache_hierarchy.l2bus.respLayer1.occupancy 131936931 # Layer occupancy (ticks) (Tick)

board.cache_hierarchy.l2bus.respLayer1.utilization 0.0 # Layer utilization (Ratio)

board.cache_hierarchy.l2bus.snoop_filter.totRequests 264619 # Total number of requests made to the snoop filter. (Count)

board.cache_hierarchy.l2bus.snoop_filter.hitSingleRequests 131796 # Number of requests hitting in the snoop filter with a single holder of the requested data. (Count)

board.cache_hierarchy.l2bus.snoop_filter.hitMultiRequests 6 # Number of requests hitting in the snoop filter with multiple (>1) holders of the requested data. (Count)

board.cache_hierarchy.l2bus.snoop_filter.totSnoops 477 # Total number of snoops made to the snoop filter. (Count)

board.cache_hierarchy.l2bus.snoop_filter.hitSingleSnoops 477 # Number of snoops hitting in the snoop filter with a single holder of the requested data. (Count)

board.cache_hierarchy.l2bus.snoop_filter.hitMultiSnoops 0 # Number of snoops hitting in the snoop filter with multiple (>1) holders of the requested data. (Count)

board.cache_hierarchy.l2cache.demandHits::cache_hierarchy.l1dcaches.prefetcher 107979 # number of demand (read+write) hits (Count)

board.cache_hierarchy.l2cache.demandHits::processor.cores.core.data 30 # number of demand (read+write) hits (Count)

board.cache_hierarchy.l2cache.demandHits::total 108009 # number of demand (read+write) hits (Count)

board.cache_hierarchy.l2cache.overallHits::cache_hierarchy.l1dcaches.prefetcher 107979 # number of overall hits (Count)

board.cache_hierarchy.l2cache.overallHits::processor.cores.core.data 30 # number of overall hits (Count)

board.cache_hierarchy.l2cache.overallHits::total 108009 # number of overall hits (Count)

board.cache_hierarchy.l2cache.demandMisses::cache_hierarchy.l1dcaches.prefetcher 21141 # number of demand (read+write) misses (Count)

board.cache_hierarchy.l2cache.demandMisses::processor.cores.core.inst 750 # number of demand (read+write) misses (Count)

board.cache_hierarchy.l2cache.demandMisses::processor.cores.core.data 2918 # number of demand (read+write) misses (Count)

board.cache_hierarchy.l2cache.demandMisses::total 24809 # number of demand (read+write) misses (Count)

board.cache_hierarchy.l2cache.overallMisses::cache_hierarchy.l1dcaches.prefetcher 21141 # number of overall misses (Count)

board.cache_hierarchy.l2cache.overallMisses::processor.cores.core.inst 750 # number of overall misses (Count)

board.cache_hierarchy.l2cache.overallMisses::processor.cores.core.data 2918 # number of overall misses (Count)

board.cache_hierarchy.l2cache.overallMisses::total 24809 # number of overall misses (Count)

board.cache_hierarchy.l2cache.demandMissLatency::cache_hierarchy.l1dcaches.prefetcher 1999711230 # number of demand (read+write) miss ticks (Tick)

board.cache_hierarchy.l2cache.demandMissLatency::processor.cores.core.inst 89972271 # number of demand (read+write) miss ticks (Tick)

board.cache_hierarchy.l2cache.demandMissLatency::processor.cores.core.data 374023269 # number of demand (read+write) miss ticks (Tick)

board.cache_hierarchy.l2cache.demandMissLatency::total 2463706770 # number of demand (read+write) miss ticks (Tick)

board.cache_hierarchy.l2cache.overallMissLatency::cache_hierarchy.l1dcaches.prefetcher 1999711230 # number of overall miss ticks (Tick)

board.cache_hierarchy.l2cache.overallMissLatency::processor.cores.core.inst 89972271 # number of overall miss ticks (Tick)

board.cache_hierarchy.l2cache.overallMissLatency::processor.cores.core.data 374023269 # number of overall miss ticks (Tick)

board.cache_hierarchy.l2cache.overallMissLatency::total 2463706770 # number of overall miss ticks (Tick)

board.cache_hierarchy.l2cache.demandAccesses::cache_hierarchy.l1dcaches.prefetcher 129120 # number of demand (read+write) accesses (Count)

board.cache_hierarchy.l2cache.demandAccesses::processor.cores.core.inst 750 # number of demand (read+write) accesses (Count)

board.cache_hierarchy.l2cache.demandAccesses::processor.cores.core.data 2948 # number of demand (read+write) accesses (Count)

board.cache_hierarchy.l2cache.demandAccesses::total 132818 # number of demand (read+write) accesses (Count)

board.cache_hierarchy.l2cache.overallAccesses::cache_hierarchy.l1dcaches.prefetcher 129120 # number of overall (read+write) accesses (Count)

board.cache_hierarchy.l2cache.overallAccesses::processor.cores.core.inst 750 # number of overall (read+write) accesses (Count)

board.cache_hierarchy.l2cache.overallAccesses::processor.cores.core.data 2948 # number of overall (read+write) accesses (Count)

board.cache_hierarchy.l2cache.overallAccesses::total 132818 # number of overall (read+write) accesses (Count)

board.cache_hierarchy.l2cache.demandMissRate::cache_hierarchy.l1dcaches.prefetcher 0.163731 # miss rate for demand accesses (Ratio)

board.cache_hierarchy.l2cache.demandMissRate::processor.cores.core.inst 1 # miss rate for demand accesses (Ratio)

board.cache_hierarchy.l2cache.demandMissRate::processor.cores.core.data 0.989824 # miss rate for demand accesses (Ratio)

board.cache_hierarchy.l2cache.demandMissRate::total 0.186789 # miss rate for demand accesses (Ratio)

board.cache_hierarchy.l2cache.overallMissRate::cache_hierarchy.l1dcaches.prefetcher 0.163731 # miss rate for overall accesses (Ratio)

board.cache_hierarchy.l2cache.overallMissRate::processor.cores.core.inst 1 # miss rate for overall accesses (Ratio)

board.cache_hierarchy.l2cache.overallMissRate::processor.cores.core.data 0.989824 # miss rate for overall accesses (Ratio)

board.cache_hierarchy.l2cache.overallMissRate::total 0.186789 # miss rate for overall accesses (Ratio)

board.cache_hierarchy.l2cache.demandAvgMissLatency::cache_hierarchy.l1dcaches.prefetcher 94589.245069 # average overall miss latency in ticks ((Tick/Count))

board.cache_hierarchy.l2cache.demandAvgMissLatency::processor.cores.core.inst 119963.028000 # average overall miss latency in ticks ((Tick/Count))

board.cache_hierarchy.l2cache.demandAvgMissLatency::processor.cores.core.data 128177.953735 # average overall miss latency in ticks ((Tick/Count))

board.cache_hierarchy.l2cache.demandAvgMissLatency::total 99306.976097 # average overall miss latency in ticks ((Tick/Count))

board.cache_hierarchy.l2cache.overallAvgMissLatency::cache_hierarchy.l1dcaches.prefetcher 94589.245069 # average overall miss latency ((Tick/Count))

board.cache_hierarchy.l2cache.overallAvgMissLatency::processor.cores.core.inst 119963.028000 # average overall miss latency ((Tick/Count))

board.cache_hierarchy.l2cache.overallAvgMissLatency::processor.cores.core.data 128177.953735 # average overall miss latency ((Tick/Count))

board.cache_hierarchy.l2cache.overallAvgMissLatency::total 99306.976097 # average overall miss latency ((Tick/Count))

board.cache_hierarchy.l2cache.blockedCycles::no_mshrs 0 # number of cycles access was blocked (Cycle)

board.cache_hierarchy.l2cache.blockedCycles::no_targets 0 # number of cycles access was blocked (Cycle)

board.cache_hierarchy.l2cache.blockedCauses::no_mshrs 0 # number of times access was blocked (Count)

board.cache_hierarchy.l2cache.blockedCauses::no_targets 0 # number of times access was blocked (Count)

board.cache_hierarchy.l2cache.avgBlocked::no_mshrs nan # average number of cycles each access was blocked ((Cycle/Count))

board.cache_hierarchy.l2cache.avgBlocked::no_targets nan # average number of cycles each access was blocked ((Cycle/Count))

board.cache_hierarchy.l2cache.writebacks::writebacks 65695 # number of writebacks (Count)

board.cache_hierarchy.l2cache.writebacks::total 65695 # number of writebacks (Count)

board.cache_hierarchy.l2cache.demandMshrHits::cache_hierarchy.l1dcaches.prefetcher 19056 # number of demand (read+write) MSHR hits (Count)

board.cache_hierarchy.l2cache.demandMshrHits::total 19056 # number of demand (read+write) MSHR hits (Count)

board.cache_hierarchy.l2cache.overallMshrHits::cache_hierarchy.l1dcaches.prefetcher 19056 # number of overall MSHR hits (Count)

board.cache_hierarchy.l2cache.overallMshrHits::total 19056 # number of overall MSHR hits (Count)

board.cache_hierarchy.l2cache.demandMshrMisses::cache_hierarchy.l1dcaches.prefetcher 2085 # number of demand (read+write) MSHR misses (Count)

board.cache_hierarchy.l2cache.demandMshrMisses::processor.cores.core.inst 750 # number of demand (read+write) MSHR misses (Count)

board.cache_hierarchy.l2cache.demandMshrMisses::processor.cores.core.data 2918 # number of demand (read+write) MSHR misses (Count)

board.cache_hierarchy.l2cache.demandMshrMisses::total 5753 # number of demand (read+write) MSHR misses (Count)

board.cache_hierarchy.l2cache.overallMshrMisses::cache_hierarchy.l1dcaches.prefetcher 2085 # number of overall MSHR misses (Count)

board.cache_hierarchy.l2cache.overallMshrMisses::cache_hierarchy.l2cache.prefetcher 127059 # number of overall MSHR misses (Count)

board.cache_hierarchy.l2cache.overallMshrMisses::processor.cores.core.inst 750 # number of overall MSHR misses (Count)

board.cache_hierarchy.l2cache.overallMshrMisses::processor.cores.core.data 2918 # number of overall MSHR misses (Count)

board.cache_hierarchy.l2cache.overallMshrMisses::total 132812 # number of overall MSHR misses (Count)

board.cache_hierarchy.l2cache.demandMshrMissLatency::cache_hierarchy.l1dcaches.prefetcher 217628811 # number of demand (read+write) MSHR miss ticks (Tick)

board.cache_hierarchy.l2cache.demandMshrMissLatency::processor.cores.core.inst 64997271 # number of demand (read+write) MSHR miss ticks (Tick)

board.cache_hierarchy.l2cache.demandMshrMissLatency::processor.cores.core.data 276853869 # number of demand (read+write) MSHR miss ticks (Tick)

board.cache_hierarchy.l2cache.demandMshrMissLatency::total 559479951 # number of demand (read+write) MSHR miss ticks (Tick)

board.cache_hierarchy.l2cache.overallMshrMissLatency::cache_hierarchy.l1dcaches.prefetcher 217628811 # number of overall MSHR miss ticks (Tick)

board.cache_hierarchy.l2cache.overallMshrMissLatency::cache_hierarchy.l2cache.prefetcher 8434146795 # number of overall MSHR miss ticks (Tick)

board.cache_hierarchy.l2cache.overallMshrMissLatency::processor.cores.core.inst 64997271 # number of overall MSHR miss ticks (Tick)

board.cache_hierarchy.l2cache.overallMshrMissLatency::processor.cores.core.data 276853869 # number of overall MSHR miss ticks (Tick)

board.cache_hierarchy.l2cache.overallMshrMissLatency::total 8993626746 # number of overall MSHR miss ticks (Tick)

board.cache_hierarchy.l2cache.demandMshrMissRate::cache_hierarchy.l1dcaches.prefetcher 0.016148 # mshr miss ratio for demand accesses (Ratio)

board.cache_hierarchy.l2cache.demandMshrMissRate::processor.cores.core.inst 1 # mshr miss ratio for demand accesses (Ratio)

board.cache_hierarchy.l2cache.demandMshrMissRate::processor.cores.core.data 0.989824 # mshr miss ratio for demand accesses (Ratio)

board.cache_hierarchy.l2cache.demandMshrMissRate::total 0.043315 # mshr miss ratio for demand accesses (Ratio)

board.cache_hierarchy.l2cache.overallMshrMissRate::cache_hierarchy.l1dcaches.prefetcher 0.016148 # mshr miss ratio for overall accesses (Ratio)

board.cache_hierarchy.l2cache.overallMshrMissRate::cache_hierarchy.l2cache.prefetcher inf # mshr miss ratio for overall accesses (Ratio)

board.cache_hierarchy.l2cache.overallMshrMissRate::processor.cores.core.inst 1 # mshr miss ratio for overall accesses (Ratio)

board.cache_hierarchy.l2cache.overallMshrMissRate::processor.cores.core.data 0.989824 # mshr miss ratio for overall accesses (Ratio)

board.cache_hierarchy.l2cache.overallMshrMissRate::total 0.999955 # mshr miss ratio for overall accesses (Ratio)

board.cache_hierarchy.l2cache.demandAvgMshrMissLatency::cache_hierarchy.l1dcaches.prefetcher 104378.326619 # average overall mshr miss latency ((Tick/Count))

board.cache_hierarchy.l2cache.demandAvgMshrMissLatency::processor.cores.core.inst 86663.028000 # average overall mshr miss latency ((Tick/Count))

board.cache_hierarchy.l2cache.demandAvgMshrMissLatency::processor.cores.core.data 94877.953735 # average overall mshr miss latency ((Tick/Count))

board.cache_hierarchy.l2cache.demandAvgMshrMissLatency::total 97250.121849 # average overall mshr miss latency ((Tick/Count))

board.cache_hierarchy.l2cache.overallAvgMshrMissLatency::cache_hierarchy.l1dcaches.prefetcher 104378.326619 # average overall mshr miss latency ((Tick/Count))

board.cache_hierarchy.l2cache.overallAvgMshrMissLatency::cache_hierarchy.l2cache.prefetcher 66379.766841 # average overall mshr miss latency ((Tick/Count))

board.cache_hierarchy.l2cache.overallAvgMshrMissLatency::processor.cores.core.inst 86663.028000 # average overall mshr miss latency ((Tick/Count))

board.cache_hierarchy.l2cache.overallAvgMshrMissLatency::processor.cores.core.data 94877.953735 # average overall mshr miss latency ((Tick/Count))

board.cache_hierarchy.l2cache.overallAvgMshrMissLatency::total 67716.973963 # average overall mshr miss latency ((Tick/Count))

board.cache_hierarchy.l2cache.replacements 128716 # number of replacements (Count)

board.cache_hierarchy.l2cache.CleanEvict.mshrMisses::writebacks 167 # number of CleanEvict MSHR misses (Count)

board.cache_hierarchy.l2cache.CleanEvict.mshrMisses::total 167 # number of CleanEvict MSHR misses (Count)

board.cache_hierarchy.l2cache.CleanEvict.mshrMissRate::writebacks inf # mshr miss rate for CleanEvict accesses (Ratio)

board.cache_hierarchy.l2cache.CleanEvict.mshrMissRate::total inf # mshr miss rate for CleanEvict accesses (Ratio)

board.cache_hierarchy.l2cache.HardPFReq.mshrMisses::cache_hierarchy.l2cache.prefetcher 127059 # number of HardPFReq MSHR misses (Count)

board.cache_hierarchy.l2cache.HardPFReq.mshrMisses::total 127059 # number of HardPFReq MSHR misses (Count)

board.cache_hierarchy.l2cache.HardPFReq.mshrMissLatency::cache_hierarchy.l2cache.prefetcher 8434146795 # number of HardPFReq MSHR miss ticks (Tick)

board.cache_hierarchy.l2cache.HardPFReq.mshrMissLatency::total 8434146795 # number of HardPFReq MSHR miss ticks (Tick)

board.cache_hierarchy.l2cache.HardPFReq.mshrMissRate::cache_hierarchy.l2cache.prefetcher inf # mshr miss rate for HardPFReq accesses (Ratio)

board.cache_hierarchy.l2cache.HardPFReq.mshrMissRate::total inf # mshr miss rate for HardPFReq accesses (Ratio)

board.cache_hierarchy.l2cache.HardPFReq.avgMshrMissLatency::cache_hierarchy.l2cache.prefetcher 66379.766841 # average HardPFReq mshr miss latency ((Tick/Count))

board.cache_hierarchy.l2cache.HardPFReq.avgMshrMissLatency::total 66379.766841 # average HardPFReq mshr miss latency ((Tick/Count))

board.cache_hierarchy.l2cache.ReadExReq.hits::processor.cores.core.data 9 # number of ReadExReq hits (Count)

board.cache_hierarchy.l2cache.ReadExReq.hits::total 9 # number of ReadExReq hits (Count)

board.cache_hierarchy.l2cache.ReadExReq.misses::processor.cores.core.data 1131 # number of ReadExReq misses (Count)

board.cache_hierarchy.l2cache.ReadExReq.misses::total 1131 # number of ReadExReq misses (Count)

board.cache_hierarchy.l2cache.ReadExReq.missLatency::processor.cores.core.data 155226618 # number of ReadExReq miss ticks (Tick)

board.cache_hierarchy.l2cache.ReadExReq.missLatency::total 155226618 # number of ReadExReq miss ticks (Tick)

board.cache_hierarchy.l2cache.ReadExReq.accesses::processor.cores.core.data 1140 # number of ReadExReq accesses(hits+misses) (Count)

board.cache_hierarchy.l2cache.ReadExReq.accesses::total 1140 # number of ReadExReq accesses(hits+misses) (Count)

board.cache_hierarchy.l2cache.ReadExReq.missRate::processor.cores.core.data 0.992105 # miss rate for ReadExReq accesses (Ratio)

board.cache_hierarchy.l2cache.ReadExReq.missRate::total 0.992105 # miss rate for ReadExReq accesses (Ratio)

board.cache_hierarchy.l2cache.ReadExReq.avgMissLatency::processor.cores.core.data 137247.230769 # average ReadExReq miss latency ((Tick/Count))

board.cache_hierarchy.l2cache.ReadExReq.avgMissLatency::total 137247.230769 # average ReadExReq miss latency ((Tick/Count))

board.cache_hierarchy.l2cache.ReadExReq.mshrMisses::processor.cores.core.data 1131 # number of ReadExReq MSHR misses (Count)

board.cache_hierarchy.l2cache.ReadExReq.mshrMisses::total 1131 # number of ReadExReq MSHR misses (Count)

board.cache_hierarchy.l2cache.ReadExReq.mshrMissLatency::processor.cores.core.data 117564318 # number of ReadExReq MSHR miss ticks (Tick)

board.cache_hierarchy.l2cache.ReadExReq.mshrMissLatency::total 117564318 # number of ReadExReq MSHR miss ticks (Tick)

board.cache_hierarchy.l2cache.ReadExReq.mshrMissRate::processor.cores.core.data 0.992105 # mshr miss rate for ReadExReq accesses (Ratio)

board.cache_hierarchy.l2cache.ReadExReq.mshrMissRate::total 0.992105 # mshr miss rate for ReadExReq accesses (Ratio)

board.cache_hierarchy.l2cache.ReadExReq.avgMshrMissLatency::processor.cores.core.data 103947.230769 # average ReadExReq mshr miss latency ((Tick/Count))

board.cache_hierarchy.l2cache.ReadExReq.avgMshrMissLatency::total 103947.230769 # average ReadExReq mshr miss latency ((Tick/Count))

board.cache_hierarchy.l2cache.ReadSharedReq.hits::cache_hierarchy.l1dcaches.prefetcher 107979 # number of ReadSharedReq hits (Count)

board.cache_hierarchy.l2cache.ReadSharedReq.hits::processor.cores.core.data 21 # number of ReadSharedReq hits (Count)

board.cache_hierarchy.l2cache.ReadSharedReq.hits::total 108000 # number of ReadSharedReq hits (Count)

board.cache_hierarchy.l2cache.ReadSharedReq.misses::cache_hierarchy.l1dcaches.prefetcher 21141 # number of ReadSharedReq misses (Count)

board.cache_hierarchy.l2cache.ReadSharedReq.misses::processor.cores.core.inst 750 # number of ReadSharedReq misses (Count)

board.cache_hierarchy.l2cache.ReadSharedReq.misses::processor.cores.core.data 1787 # number of ReadSharedReq misses (Count)

board.cache_hierarchy.l2cache.ReadSharedReq.misses::total 23678 # number of ReadSharedReq misses (Count)

board.cache_hierarchy.l2cache.ReadSharedReq.missLatency::cache_hierarchy.l1dcaches.prefetcher 1999711230 # number of ReadSharedReq miss ticks (Tick)

board.cache_hierarchy.l2cache.ReadSharedReq.missLatency::processor.cores.core.inst 89972271 # number of ReadSharedReq miss ticks (Tick)

board.cache_hierarchy.l2cache.ReadSharedReq.missLatency::processor.cores.core.data 218796651 # number of ReadSharedReq miss ticks (Tick)

board.cache_hierarchy.l2cache.ReadSharedReq.missLatency::total 2308480152 # number of ReadSharedReq miss ticks (Tick)

board.cache_hierarchy.l2cache.ReadSharedReq.accesses::cache_hierarchy.l1dcaches.prefetcher 129120 # number of ReadSharedReq accesses(hits+misses) (Count)

board.cache_hierarchy.l2cache.ReadSharedReq.accesses::processor.cores.core.inst 750 # number of ReadSharedReq accesses(hits+misses) (Count)

board.cache_hierarchy.l2cache.ReadSharedReq.accesses::processor.cores.core.data 1808 # number of ReadSharedReq accesses(hits+misses) (Count)

board.cache_hierarchy.l2cache.ReadSharedReq.accesses::total 131678 # number of ReadSharedReq accesses(hits+misses) (Count)

board.cache_hierarchy.l2cache.ReadSharedReq.missRate::cache_hierarchy.l1dcaches.prefetcher 0.163731 # miss rate for ReadSharedReq accesses (Ratio)

board.cache_hierarchy.l2cache.ReadSharedReq.missRate::processor.cores.core.inst 1 # miss rate for ReadSharedReq accesses (Ratio)

board.cache_hierarchy.l2cache.ReadSharedReq.missRate::processor.cores.core.data 0.988385 # miss rate for ReadSharedReq accesses (Ratio)

board.cache_hierarchy.l2cache.ReadSharedReq.missRate::total 0.179817 # miss rate for ReadSharedReq accesses (Ratio)

board.cache_hierarchy.l2cache.ReadSharedReq.avgMissLatency::cache_hierarchy.l1dcaches.prefetcher 94589.245069 # average ReadSharedReq miss latency ((Tick/Count))

board.cache_hierarchy.l2cache.ReadSharedReq.avgMissLatency::processor.cores.core.inst 119963.028000 # average ReadSharedReq miss latency ((Tick/Count))

board.cache_hierarchy.l2cache.ReadSharedReq.avgMissLatency::processor.cores.core.data 122437.969222 # average ReadSharedReq miss latency ((Tick/Count))

board.cache_hierarchy.l2cache.ReadSharedReq.avgMissLatency::total 97494.727257 # average ReadSharedReq miss latency ((Tick/Count))

board.cache_hierarchy.l2cache.ReadSharedReq.mshrHits::cache_hierarchy.l1dcaches.prefetcher 19056 # number of ReadSharedReq MSHR hits (Count)

board.cache_hierarchy.l2cache.ReadSharedReq.mshrHits::total 19056 # number of ReadSharedReq MSHR hits (Count)

board.cache_hierarchy.l2cache.ReadSharedReq.mshrMisses::cache_hierarchy.l1dcaches.prefetcher 2085 # number of ReadSharedReq MSHR misses (Count)

board.cache_hierarchy.l2cache.ReadSharedReq.mshrMisses::processor.cores.core.inst 750 # number of ReadSharedReq MSHR misses (Count)

board.cache_hierarchy.l2cache.ReadSharedReq.mshrMisses::processor.cores.core.data 1787 # number of ReadSharedReq MSHR misses (Count)

board.cache_hierarchy.l2cache.ReadSharedReq.mshrMisses::total 4622 # number of ReadSharedReq MSHR misses (Count)

board.cache_hierarchy.l2cache.ReadSharedReq.mshrMissLatency::cache_hierarchy.l1dcaches.prefetcher 217628811 # number of ReadSharedReq MSHR miss ticks (Tick)

board.cache_hierarchy.l2cache.ReadSharedReq.mshrMissLatency::processor.cores.core.inst 64997271 # number of ReadSharedReq MSHR miss ticks (Tick)

board.cache_hierarchy.l2cache.ReadSharedReq.mshrMissLatency::processor.cores.core.data 159289551 # number of ReadSharedReq MSHR miss ticks (Tick)

board.cache_hierarchy.l2cache.ReadSharedReq.mshrMissLatency::total 441915633 # number of ReadSharedReq MSHR miss ticks (Tick)

board.cache_hierarchy.l2cache.ReadSharedReq.mshrMissRate::cache_hierarchy.l1dcaches.prefetcher 0.016148 # mshr miss rate for ReadSharedReq accesses (Ratio)

board.cache_hierarchy.l2cache.ReadSharedReq.mshrMissRate::processor.cores.core.inst 1 # mshr miss rate for ReadSharedReq accesses (Ratio)

board.cache_hierarchy.l2cache.ReadSharedReq.mshrMissRate::processor.cores.core.data 0.988385 # mshr miss rate for ReadSharedReq accesses (Ratio)

board.cache_hierarchy.l2cache.ReadSharedReq.mshrMissRate::total 0.035101 # mshr miss rate for ReadSharedReq accesses (Ratio)

board.cache_hierarchy.l2cache.ReadSharedReq.avgMshrMissLatency::cache_hierarchy.l1dcaches.prefetcher 104378.326619 # average ReadSharedReq mshr miss latency ((Tick/Count))

board.cache_hierarchy.l2cache.ReadSharedReq.avgMshrMissLatency::processor.cores.core.inst 86663.028000 # average ReadSharedReq mshr miss latency ((Tick/Count))

board.cache_hierarchy.l2cache.ReadSharedReq.avgMshrMissLatency::processor.cores.core.data 89137.969222 # average ReadSharedReq mshr miss latency ((Tick/Count))

board.cache_hierarchy.l2cache.ReadSharedReq.avgMshrMissLatency::total 95611.344223 # average ReadSharedReq mshr miss latency ((Tick/Count))

board.cache_hierarchy.l2cache.UpgradeReq.hits::processor.cores.core.data 3 # number of UpgradeReq hits (Count)

board.cache_hierarchy.l2cache.UpgradeReq.hits::total 3 # number of UpgradeReq hits (Count)

board.cache_hierarchy.l2cache.UpgradeReq.accesses::processor.cores.core.data 3 # number of UpgradeReq accesses(hits+misses) (Count)

board.cache_hierarchy.l2cache.UpgradeReq.accesses::total 3 # number of UpgradeReq accesses(hits+misses) (Count)

board.cache_hierarchy.l2cache.WritebackDirty.hits::writebacks 65717 # number of WritebackDirty hits (Count)

board.cache_hierarchy.l2cache.WritebackDirty.hits::total 65717 # number of WritebackDirty hits (Count)

board.cache_hierarchy.l2cache.WritebackDirty.accesses::writebacks 65717 # number of WritebackDirty accesses(hits+misses) (Count)

board.cache_hierarchy.l2cache.WritebackDirty.accesses::total 65717 # number of WritebackDirty accesses(hits+misses) (Count)

board.cache_hierarchy.l2cache.power_state.pwrStateResidencyTicks::UNDEFINED 5909877540 # Cumulative time (in ticks) in various power states (Tick)

board.cache_hierarchy.l2cache.prefetcher.demandMshrMisses 5753 # demands not covered by prefetchs (Count)

board.cache_hierarchy.l2cache.prefetcher.pfIssued 479441 # number of hwpf issued (Count)

board.cache_hierarchy.l2cache.prefetcher.pfUnused 10 # number of HardPF blocks evicted w/o reference (Count)

board.cache_hierarchy.l2cache.prefetcher.pfUseful 107985 # number of useful prefetch (Count)

board.cache_hierarchy.l2cache.prefetcher.pfUsefulButMiss 0 # number of hit on prefetch but cache block is not in an usable state (Count)

board.cache_hierarchy.l2cache.prefetcher.accuracy 0.225231 # accuracy of the prefetcher (Count)

board.cache_hierarchy.l2cache.prefetcher.coverage 0.949419 # coverage brought by this prefetcher (Count)

board.cache_hierarchy.l2cache.prefetcher.pfHitInCache 254214 # number of prefetches hitting in cache (Count)

board.cache_hierarchy.l2cache.prefetcher.pfHitInMSHR 98168 # number of prefetches hitting in a MSHR (Count)

board.cache_hierarchy.l2cache.prefetcher.pfHitInWB 0 # number of prefetches hit in the Write Buffer (Count)

board.cache_hierarchy.l2cache.prefetcher.pfLate 352382 # number of late prefetches (hitting in cache, MSHR or WB) (Count)

board.cache_hierarchy.l2cache.prefetcher.pfIdentified 504059 # number of prefetch candidates identified (Count)

board.cache_hierarchy.l2cache.prefetcher.pfBufferHit 16407 # number of redundant prefetches already in prefetch queue (Count)

board.cache_hierarchy.l2cache.prefetcher.pfInCache 0 # number of redundant prefetches already in cache/mshr dropped (Count)

board.cache_hierarchy.l2cache.prefetcher.pfRemovedDemand 6158 # number of prefetches dropped due to a demand for the same address (Count)

board.cache_hierarchy.l2cache.prefetcher.pfRemovedFull 0 # number of prefetches dropped due to prefetch queue size (Count)

board.cache_hierarchy.l2cache.prefetcher.pfSpanPage 20497 # number of prefetches that crossed the page (Count)

board.cache_hierarchy.l2cache.prefetcher.pfUsefulSpanPage 19125 # number of prefetches that is useful and crossed the page (Count)

board.cache_hierarchy.l2cache.prefetcher.power_state.pwrStateResidencyTicks::UNDEFINED 5909877540 # Cumulative time (in ticks) in various power states (Tick)

board.cache_hierarchy.l2cache.tags.tagsInUse 3950.838192 # Average ticks per tags in use ((Tick/Count))

board.cache_hierarchy.l2cache.tags.totalRefs 372452 # Total number of references to valid blocks. (Count)

board.cache_hierarchy.l2cache.tags.sampledRefs 132812 # Sample count of references to valid blocks. (Count)

board.cache_hierarchy.l2cache.tags.avgRefs 2.804355 # Average number of references to valid blocks. ((Count/Count))

board.cache_hierarchy.l2cache.tags.warmupTick 89910 # The tick when the warmup percentage was hit. (Tick)

board.cache_hierarchy.l2cache.tags.occupancies::cache_hierarchy.l1dcaches.prefetcher 62.451593 # Average occupied blocks per tick, per requestor ((Count/Tick))

board.cache_hierarchy.l2cache.tags.occupancies::cache_hierarchy.l2cache.prefetcher 3719.748977 # Average occupied blocks per tick, per requestor ((Count/Tick))

board.cache_hierarchy.l2cache.tags.occupancies::processor.cores.core.inst 42.103419 # Average occupied blocks per tick, per requestor ((Count/Tick))

board.cache_hierarchy.l2cache.tags.occupancies::processor.cores.core.data 126.534204 # Average occupied blocks per tick, per requestor ((Count/Tick))

board.cache_hierarchy.l2cache.tags.avgOccs::cache_hierarchy.l1dcaches.prefetcher 0.015247 # Average percentage of cache occupancy ((Ratio/Tick))

board.cache_hierarchy.l2cache.tags.avgOccs::cache_hierarchy.l2cache.prefetcher 0.908142 # Average percentage of cache occupancy ((Ratio/Tick))

board.cache_hierarchy.l2cache.tags.avgOccs::processor.cores.core.inst 0.010279 # Average percentage of cache occupancy ((Ratio/Tick))

board.cache_hierarchy.l2cache.tags.avgOccs::processor.cores.core.data 0.030892 # Average percentage of cache occupancy ((Ratio/Tick))

board.cache_hierarchy.l2cache.tags.avgOccs::total 0.964560 # Average percentage of cache occupancy ((Ratio/Tick))

board.cache_hierarchy.l2cache.tags.occupanciesTaskId::1022 3749 # Occupied blocks per task id (Count)

board.cache_hierarchy.l2cache.tags.occupanciesTaskId::1024 347 # Occupied blocks per task id (Count)

board.cache_hierarchy.l2cache.tags.ageTaskId_1022::0 8 # Occupied blocks per task id, per block age (Count)

board.cache_hierarchy.l2cache.tags.ageTaskId_1022::1 1807 # Occupied blocks per task id, per block age (Count)

board.cache_hierarchy.l2cache.tags.ageTaskId_1022::2 1934 # Occupied blocks per task id, per block age (Count)

board.cache_hierarchy.l2cache.tags.ageTaskId_1024::0 149 # Occupied blocks per task id, per block age (Count)

board.cache_hierarchy.l2cache.tags.ageTaskId_1024::1 169 # Occupied blocks per task id, per block age (Count)

board.cache_hierarchy.l2cache.tags.ageTaskId_1024::2 29 # Occupied blocks per task id, per block age (Count)

board.cache_hierarchy.l2cache.tags.ratioOccsTaskId::1022 0.915283 # Ratio of occupied blocks and all blocks, per task id (Ratio)

board.cache_hierarchy.l2cache.tags.ratioOccsTaskId::1024 0.084717 # Ratio of occupied blocks and all blocks, per task id (Ratio)

board.cache_hierarchy.l2cache.tags.tagAccesses 4366668 # Number of tag accesses (Count)

board.cache_hierarchy.l2cache.tags.dataAccesses 4366668 # Number of data accesses (Count)

board.cache_hierarchy.l2cache.tags.power_state.pwrStateResidencyTicks::UNDEFINED 5909877540 # Cumulative time (in ticks) in various power states (Tick)

board.cache_hierarchy.membus.transDist::ReadResp 131681 # Transaction distribution (Count)

board.cache_hierarchy.membus.transDist::WritebackDirty 65695 # Transaction distribution (Count)

board.cache_hierarchy.membus.transDist::CleanEvict 62713 # Transaction distribution (Count)

board.cache_hierarchy.membus.transDist::ReadExReq 1131 # Transaction distribution (Count)

board.cache_hierarchy.membus.transDist::ReadExResp 1131 # Transaction distribution (Count)

board.cache_hierarchy.membus.transDist::ReadSharedReq 131681 # Transaction distribution (Count)

board.cache_hierarchy.membus.pktCount_board.cache_hierarchy.l2cache.mem_side_port::board.memory.mem_ctrl.port 394032 # Packet count per connected requestor and responder (Count)

board.cache_hierarchy.membus.pktCount_board.cache_hierarchy.l2cache.mem_side_port::total 394032 # Packet count per connected requestor and responder (Count)

board.cache_hierarchy.membus.pktCount::total 394032 # Packet count per connected requestor and responder (Count)

board.cache_hierarchy.membus.pktSize_board.cache_hierarchy.l2cache.mem_side_port::board.memory.mem_ctrl.port 12704448 # Cumulative packet size per connected requestor and responder (Byte)

board.cache_hierarchy.membus.pktSize_board.cache_hierarchy.l2cache.mem_side_port::total 12704448 # Cumulative packet size per connected requestor and responder (Byte)

board.cache_hierarchy.membus.pktSize::total 12704448 # Cumulative packet size per connected requestor and responder (Byte)

board.cache_hierarchy.membus.snoops 0 # Total snoops (Count)

board.cache_hierarchy.membus.snoopTraffic 0 # Total snoop traffic (Byte)

board.cache_hierarchy.membus.snoopFanout::samples 132812 # Request fanout histogram (Count)

board.cache_hierarchy.membus.snoopFanout::mean 0 # Request fanout histogram (Count)

board.cache_hierarchy.membus.snoopFanout::stdev 0 # Request fanout histogram (Count)

board.cache_hierarchy.membus.snoopFanout::underflows 0 0.00% 0.00% # Request fanout histogram (Count)

board.cache_hierarchy.membus.snoopFanout::0 132812 100.00% 100.00% # Request fanout histogram (Count)

board.cache_hierarchy.membus.snoopFanout::1 0 0.00% 100.00% # Request fanout histogram (Count)

board.cache_hierarchy.membus.snoopFanout::overflows 0 0.00% 100.00% # Request fanout histogram (Count)

board.cache_hierarchy.membus.snoopFanout::min_value 0 # Request fanout histogram (Count)

board.cache_hierarchy.membus.snoopFanout::max_value 0 # Request fanout histogram (Count)

board.cache_hierarchy.membus.snoopFanout::total 132812 # Request fanout histogram (Count)

board.cache_hierarchy.membus.badaddr_responder.power_state.pwrStateResidencyTicks::UNDEFINED 5909877540 # Cumulative time (in ticks) in various power states (Tick)

board.cache_hierarchy.membus.power_state.pwrStateResidencyTicks::UNDEFINED 5909877540 # Cumulative time (in ticks) in various power states (Tick)

board.cache_hierarchy.membus.reqLayer0.occupancy 145671327 # Layer occupancy (ticks) (Tick)

board.cache_hierarchy.membus.reqLayer0.utilization 0.0 # Layer utilization (Ratio)

board.cache_hierarchy.membus.respLayer2.occupancy 113169922 # Layer occupancy (ticks) (Tick)

board.cache_hierarchy.membus.respLayer2.utilization 0.0 # Layer utilization (Ratio)

board.cache_hierarchy.membus.snoop_filter.totRequests 261220 # Total number of requests made to the snoop filter. (Count)

board.cache_hierarchy.membus.snoop_filter.hitSingleRequests 128408 # Number of requests hitting in the snoop filter with a single holder of the requested data. (Count)

board.cache_hierarchy.membus.snoop_filter.hitMultiRequests 0 # Number of requests hitting in the snoop filter with multiple (>1) holders of the requested data. (Count)

board.cache_hierarchy.membus.snoop_filter.totSnoops 0 # Total number of snoops made to the snoop filter. (Count)

board.cache_hierarchy.membus.snoop_filter.hitSingleSnoops 0 # Number of snoops hitting in the snoop filter with a single holder of the requested data. (Count)

board.cache_hierarchy.membus.snoop_filter.hitMultiSnoops 0 # Number of snoops hitting in the snoop filter with multiple (>1) holders of the requested data. (Count)

board.clk_domain.clock 333 # Clock period in ticks (Tick)

```
board.clk_domain.voltage_domain.voltage 1 # Voltage in Volts (Volt)
board.memory.mem_ctrl.avgPriority_writebacks::samples 65695.00 # Average QoS priority value for accepted requests (Count)
board.memory.mem_ctrl.avgPriority_cache_hierarchy.l1dcaches.prefetcher::samples 2085.00 # Average QoS priority value for accepted requests (Count)
board.memory.mem_ctrl.avgPriority_cache_hierarchy.l2cache.prefetcher::samples 127030.00 # Average QoS priority value for accepted requests (Count)
board.memory.mem_ctrl.avgPriority_processor.cores.core.inst::samples 750.00 # Average QoS priority value for accepted requests (Count)
board.memory.mem_ctrl.avgPriority_processor.cores.core.data::samples 2918.00 # Average QoS priority value for accepted requests (Count)
board.memory.mem_ctrl.priorityMinLatency 0.000000018750 # per QoS priority minimum request to response latency (Second)
board.memory.mem_ctrl.priorityMaxLatency 0.000158809564 # per QoS priority maximum request to response latency (Second)
board.memory.mem_ctrl.numReadWriteTurnArounds 4100 # Number of turnarounds from READ to WRITE (Count)
board.memory.mem_ctrl.numWriteReadTurnArounds 4100 # Number of turnarounds from WRITE to READ (Count)
board.memory.mem_ctrl.numStayReadState 299536 # Number of times bus staying in READ state (Count)
board.memory.mem_ctrl.numStayWriteState 61925 # Number of times bus staying in WRITE state (Count)
board.memory.mem_ctrl.readReqs 132812 # Number of read requests accepted (Count)
board.memory.mem_ctrl.writeReqs 65695 # Number of write requests accepted (Count)
board.memory.mem_ctrl.readBursts 132812 # Number of controller read bursts, including those serviced by the write queue (Count)
board.memory.mem_ctrl.writeBursts 65695 # Number of controller write bursts, including those merged in the write queue (Count)
board.memory.mem_ctrl.servicedByWrQ 29 # Number of controller read bursts serviced by the write queue (Count)
board.memory.mem_ctrl.mergedWrBursts 0 # Number of controller write bursts merged with an existing one (Count)
board.memory.mem_ctrl.neitherReadNorWriteReqs 0 # Number of requests that are neither read nor write (Count)
board.memory.mem_ctrl.avgRdQLen 2.00 # Average read queue length when enqueuing ((Count/Tick))
board.memory.mem_ctrl.avgWrQLen 25.47 # Average write queue length when enqueuing ((Count/Tick))
board.memory.mem_ctrl.numRdRetry 0 # Number of times read queue was full causing retry (Count)
board.memory.mem_ctrl.numWrRetry 0 # Number of times write queue was full causing retry (Count)
board.memory.mem_ctrl.readPktSize::0 0 # Read request sizes (log2) (Count)
board.memory.mem_ctrl.readPktSize::1 0 # Read request sizes (log2) (Count)
board.memory.mem_ctrl.readPktSize::2 0 # Read request sizes (log2) (Count)
board.memory.mem_ctrl.readPktSize::3 0 # Read request sizes (log2) (Count)
board.memory.mem_ctrl.readPktSize::4 0 # Read request sizes (log2) (Count)
board.memory.mem_ctrl.readPktSize::5 0 # Read request sizes (log2) (Count)
board.memory.mem_ctrl.readPktSize::6 132812 # Read request sizes (log2) (Count)
board.memory.mem_ctrl.writePktSize::0 0 # Write request sizes (log2) (Count)
board.memory.mem_ctrl.writePktSize::1 0 # Write request sizes (log2) (Count)
board.memory.mem_ctrl.writePktSize::2 0 # Write request sizes (log2) (Count)
board.memory.mem_ctrl.writePktSize::3 0 # Write request sizes (log2) (Count)
board.memory.mem_ctrl.writePktSize::4 0 # Write request sizes (log2) (Count)
board.memory.mem_ctrl.writePktSize::5 0 # Write request sizes (log2) (Count)
board.memory.mem_ctrl.writePktSize::6 65695 # Write request sizes (log2) (Count)
board.memory.mem_ctrl.rdQLenPdf::0 90543 # What read queue length does an incoming req see (Count)
board.memory.mem_ctrl.rdQLenPdf::1 9620 # What read queue length does an incoming req see (Count)
board.memory.mem_ctrl.rdQLenPdf::2 6786 # What read queue length does an incoming req see (Count)
board.memory.mem_ctrl.rdQLenPdf::3 6827 # What read queue length does an incoming req see (Count)
board.memory.mem_ctrl.rdQLenPdf::4 7048 # What read queue length does an incoming req see (Count)
board.memory.mem_ctrl.rdQLenPdf::5 5860 # What read queue length does an incoming req see (Count)
board.memory.mem_ctrl.rdQLenPdf::6 4217 # What read queue length does an incoming req see (Count)
board.memory.mem_ctrl.rdQLenPdf::7 1612 # What read queue length does an incoming req see (Count)
board.memory.mem_ctrl.rdQLenPdf::8 257 # What read queue length does an incoming req see (Count)
board.memory.mem_ctrl.rdQLenPdf::9 3 # What read queue length does an incoming req see (Count)
board.memory.mem_ctrl.rdQLenPdf::10 2 # What read queue length does an incoming req see (Count)
board.memory.mem_ctrl.rdQLenPdf::11 1 # What read queue length does an incoming req see (Count)
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board.memory.mem_ctrl.rdQLenPdf::12	1	# What read queue length does an incoming req see (Count)
board.memory.mem_ctrl.rdQLenPdf::13	1	# What read queue length does an incoming req see (Count)
board.memory.mem_ctrl.rdQLenPdf::14	1	# What read queue length does an incoming req see (Count)
board.memory.mem_ctrl.rdQLenPdf::15	2	# What read queue length does an incoming req see (Count)
board.memory.mem_ctrl.rdQLenPdf::16	1	# What read queue length does an incoming req see (Count)
board.memory.mem_ctrl.rdQLenPdf::17	1	# What read queue length does an incoming req see (Count)
board.memory.mem_ctrl.rdQLenPdf::18	0	# What read queue length does an incoming req see (Count)
board.memory.mem_ctrl.rdQLenPdf::19	0	# What read queue length does an incoming req see (Count)
board.memory.mem_ctrl.rdQLenPdf::20	0	# What read queue length does an incoming req see (Count)
board.memory.mem_ctrl.rdQLenPdf::21	0	# What read queue length does an incoming req see (Count)
board.memory.mem_ctrl.rdQLenPdf::22	0	# What read queue length does an incoming req see (Count)
board.memory.mem_ctrl.rdQLenPdf::23	0	# What read queue length does an incoming req see (Count)
board.memory.mem_ctrl.rdQLenPdf::24	0	# What read queue length does an incoming req see (Count)
board.memory.mem_ctrl.rdQLenPdf::25	0	# What read queue length does an incoming req see (Count)
board.memory.mem_ctrl.rdQLenPdf::26	0	# What read queue length does an incoming req see (Count)
board.memory.mem_ctrl.rdQLenPdf::27	0	# What read queue length does an incoming req see (Count)
board.memory.mem_ctrl.rdQLenPdf::28	0	# What read queue length does an incoming req see (Count)
board.memory.mem_ctrl.rdQLenPdf::29	0	# What read queue length does an incoming req see (Count)
board.memory.mem_ctrl.rdQLenPdf::30	0	# What read queue length does an incoming req see (Count)
board.memory.mem_ctrl.rdQLenPdf::31	0	# What read queue length does an incoming req see (Count)
board.memory.mem_ctrl.wrQLenPdf::0	1	# What write queue length does an incoming req see (Count)
board.memory.mem_ctrl.wrQLenPdf::1	1	# What write queue length does an incoming req see (Count)
board.memory.mem_ctrl.wrQLenPdf::2	1	# What write queue length does an incoming req see (Count)
board.memory.mem_ctrl.wrQLenPdf::3	1	# What write queue length does an incoming req see (Count)
board.memory.mem_ctrl.wrQLenPdf::4	1	# What write queue length does an incoming req see (Count)
board.memory.mem_ctrl.wrQLenPdf::5	1	# What write queue length does an incoming req see (Count)
board.memory.mem_ctrl.wrQLenPdf::6	1	# What write queue length does an incoming req see (Count)
board.memory.mem_ctrl.wrQLenPdf::7	1	# What write queue length does an incoming req see (Count)
board.memory.mem_ctrl.wrQLenPdf::8	1	# What write queue length does an incoming req see (Count)
board.memory.mem_ctrl.wrQLenPdf::9	1	# What write queue length does an incoming req see (Count)
board.memory.mem_ctrl.wrQLenPdf::10	1	# What write queue length does an incoming req see (Count)
board.memory.mem_ctrl.wrQLenPdf::11	1	# What write queue length does an incoming req see (Count)
board.memory.mem_ctrl.wrQLenPdf::12	1	# What write queue length does an incoming req see (Count)
board.memory.mem_ctrl.wrQLenPdf::13	1	# What write queue length does an incoming req see (Count)
board.memory.mem_ctrl.wrQLenPdf::14	1	# What write queue length does an incoming req see (Count)
board.memory.mem_ctrl.wrQLenPdf::15	9	# What write queue length does an incoming req see (Count)
board.memory.mem_ctrl.wrQLenPdf::16	10	# What write queue length does an incoming req see (Count)
board.memory.mem_ctrl.wrQLenPdf::17	193	# What write queue length does an incoming req see (Count)
board.memory.mem_ctrl.wrQLenPdf::18	1634	# What write queue length does an incoming req see (Count)
board.memory.mem_ctrl.wrQLenPdf::19	4007	# What write queue length does an incoming req see (Count)
board.memory.mem_ctrl.wrQLenPdf::20	5671	# What write queue length does an incoming req see (Count)
board.memory.mem_ctrl.wrQLenPdf::21	4291	# What write queue length does an incoming req see (Count)
board.memory.mem_ctrl.wrQLenPdf::22	4048	# What write queue length does an incoming req see (Count)
board.memory.mem_ctrl.wrQLenPdf::23	4523	# What write queue length does an incoming req see (Count)
board.memory.mem_ctrl.wrQLenPdf::24	4628	# What write queue length does an incoming req see (Count)
board.memory.mem_ctrl.wrQLenPdf::25	5793	# What write queue length does an incoming req see (Count)
board.memory.mem_ctrl.wrQLenPdf::26	4388	# What write queue length does an incoming req see (Count)
board.memory.mem_ctrl.wrQLenPdf::27	4105	# What write queue length does an incoming req see (Count)
board.memory.mem_ctrl.wrQLenPdf::28	4115	# What write queue length does an incoming req see (Count)

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board.memory.mem_ctrl.wrQLenPdf::29          4193                # What write queue length does an incoming req see (Count)
board.memory.mem_ctrl.wrQLenPdf::30          4514                # What write queue length does an incoming req see (Count)
board.memory.mem_ctrl.wrQLenPdf::31          4107                # What write queue length does an incoming req see (Count)
board.memory.mem_ctrl.wrQLenPdf::32          5421                # What write queue length does an incoming req see (Count)
board.memory.mem_ctrl.wrQLenPdf::33          16                  # What write queue length does an incoming req see (Count)
board.memory.mem_ctrl.wrQLenPdf::34          7                   # What write queue length does an incoming req see (Count)
board.memory.mem_ctrl.wrQLenPdf::35          2                   # What write queue length does an incoming req see (Count)
board.memory.mem_ctrl.wrQLenPdf::36          2                   # What write queue length does an incoming req see (Count)
board.memory.mem_ctrl.wrQLenPdf::37          1                   # What write queue length does an incoming req see (Count)
board.memory.mem_ctrl.wrQLenPdf::38          1                   # What write queue length does an incoming req see (Count)
board.memory.mem_ctrl.wrQLenPdf::39          1                   # What write queue length does an incoming req see (Count)
board.memory.mem_ctrl.wrQLenPdf::40          0                   # What write queue length does an incoming req see (Count)
board.memory.mem_ctrl.wrQLenPdf::41          0                   # What write queue length does an incoming req see (Count)
board.memory.mem_ctrl.wrQLenPdf::42          0                   # What write queue length does an incoming req see (Count)
board.memory.mem_ctrl.wrQLenPdf::43          0                   # What write queue length does an incoming req see (Count)
board.memory.mem_ctrl.wrQLenPdf::44          0                   # What write queue length does an incoming req see (Count)
board.memory.mem_ctrl.wrQLenPdf::45          0                   # What write queue length does an incoming req see (Count)
board.memory.mem_ctrl.wrQLenPdf::46          0                   # What write queue length does an incoming req see (Count)
board.memory.mem_ctrl.wrQLenPdf::47          0                   # What write queue length does an incoming req see (Count)
board.memory.mem_ctrl.wrQLenPdf::48          0                   # What write queue length does an incoming req see (Count)
board.memory.mem_ctrl.wrQLenPdf::49          0                   # What write queue length does an incoming req see (Count)
board.memory.mem_ctrl.wrQLenPdf::50          0                   # What write queue length does an incoming req see (Count)
board.memory.mem_ctrl.wrQLenPdf::51          0                   # What write queue length does an incoming req see (Count)
board.memory.mem_ctrl.wrQLenPdf::52          0                   # What write queue length does an incoming req see (Count)
board.memory.mem_ctrl.wrQLenPdf::53          0                   # What write queue length does an incoming req see (Count)
board.memory.mem_ctrl.wrQLenPdf::54          0                   # What write queue length does an incoming req see (Count)
board.memory.mem_ctrl.wrQLenPdf::55          0                   # What write queue length does an incoming req see (Count)
board.memory.mem_ctrl.wrQLenPdf::56          0                   # What write queue length does an incoming req see (Count)
board.memory.mem_ctrl.wrQLenPdf::57          0                   # What write queue length does an incoming req see (Count)
board.memory.mem_ctrl.wrQLenPdf::58          0                   # What write queue length does an incoming req see (Count)
board.memory.mem_ctrl.wrQLenPdf::59          0                   # What write queue length does an incoming req see (Count)
board.memory.mem_ctrl.wrQLenPdf::60          0                   # What write queue length does an incoming req see (Count)
board.memory.mem_ctrl.wrQLenPdf::61          0                   # What write queue length does an incoming req see (Count)
board.memory.mem_ctrl.wrQLenPdf::62          0                   # What write queue length does an incoming req see (Count)
board.memory.mem_ctrl.wrQLenPdf::63          0                   # What write queue length does an incoming req see (Count)
board.memory.mem_ctrl.rdPerTurnAround::samples        4100                # Reads before turning the bus around for writes (Count)
board.memory.mem_ctrl.rdPerTurnAround::mean    17.330488                # Reads before turning the bus around for writes (Count)
board.memory.mem_ctrl.rdPerTurnAround::gmean    16.018523               # Reads before turning the bus around for writes (Count)
board.memory.mem_ctrl.rdPerTurnAround::stdev    71.913964               # Reads before turning the bus around for writes (Count)
board.memory.mem_ctrl.rdPerTurnAround::0-255        4098     99.95%     99.95% # Reads before turning the bus around for writes (Count)
board.memory.mem_ctrl.rdPerTurnAround::256-511          1      0.02%     99.98% # Reads before turning the bus around for writes (Count)
board.memory.mem_ctrl.rdPerTurnAround::4352-4607          1      0.02%    100.00% # Reads before turning the bus around for writes (Count)
board.memory.mem_ctrl.rdPerTurnAround::total        4100                # Reads before turning the bus around for writes (Count)
board.memory.mem_ctrl.wrPerTurnAround::samples        4100                # Writes before turning the bus around for reads (Count)
board.memory.mem_ctrl.wrPerTurnAround::mean    16.016098                # Writes before turning the bus around for reads (Count)
board.memory.mem_ctrl.wrPerTurnAround::gmean    16.013790               # Writes before turning the bus around for reads (Count)
board.memory.mem_ctrl.wrPerTurnAround::stdev     0.306448               # Writes before turning the bus around for reads (Count)
board.memory.mem_ctrl.wrPerTurnAround::16        4085     99.63%     99.63% # Writes before turning the bus around for reads (Count)
board.memory.mem_ctrl.wrPerTurnAround::17          1      0.02%     99.66% # Writes before turning the bus around for reads (Count)
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board.memory.mem_ctrl.wrPerTurnAround::18          4     0.10%    99.76% # Writes before turning the bus around for reads (Count)
board.memory.mem_ctrl.wrPerTurnAround::19          2     0.05%    99.80% # Writes before turning the bus around for reads (Count)
board.memory.mem_ctrl.wrPerTurnAround::20          2     0.05%    99.85% # Writes before turning the bus around for reads (Count)
board.memory.mem_ctrl.wrPerTurnAround::21          1     0.02%    99.88% # Writes before turning the bus around for reads (Count)
board.memory.mem_ctrl.wrPerTurnAround::22          1     0.02%    99.90% # Writes before turning the bus around for reads (Count)
board.memory.mem_ctrl.wrPerTurnAround::23          1     0.02%    99.93% # Writes before turning the bus around for reads (Count)
board.memory.mem_ctrl.wrPerTurnAround::24          2     0.05%    99.98% # Writes before turning the bus around for reads (Count)
board.memory.mem_ctrl.wrPerTurnAround::25          1     0.02%   100.00% # Writes before turning the bus around for reads (Count)
board.memory.mem_ctrl.wrPerTurnAround::total       4100                  # Writes before turning the bus around for reads (Count)
board.memory.mem_ctrl.bytesReadWrQ                 1856                  # Total number of bytes read from write queue (Byte)
board.memory.mem_ctrl.bytesReadSys                 8499968               # Total read bytes from the system interface side (Byte)
board.memory.mem_ctrl.bytesWrittenSys              4204480               # Total written bytes from the system interface side (Byte)
board.memory.mem_ctrl.avgRdBWSys                   1438264658.18782449   # Average system read bandwidth in Byte/s ((Byte/Second))
board.memory.mem_ctrl.avgWrBWSys                   711432677.16508389    # Average system write bandwidth in Byte/s ((Byte/Second))
board.memory.mem_ctrl.totGap                       5909726358            # Total gap between requests (Tick)
board.memory.mem_ctrl.avgGap                       29770.87              # Average gap between requests ((Tick/Count))
board.memory.mem_ctrl.requestorReadBytes::cache_hierarchy.l1dcaches.prefetcher      133440      # Per-requestor bytes read from memory (Byte)
board.memory.mem_ctrl.requestorReadBytes::cache_hierarchy.l2cache.prefetcher      8129920      # Per-requestor bytes read from memory (Byte)
board.memory.mem_ctrl.requestorReadBytes::processor.cores.core.inst      48000      # Per-requestor bytes read from memory (Byte)
board.memory.mem_ctrl.requestorReadBytes::processor.cores.core.data      186752      # Per-requestor bytes read from memory (Byte)
board.memory.mem_ctrl.requestorWriteBytes::writebacks      4202624      # Per-requestor bytes write to memory (Byte)
board.memory.mem_ctrl.requestorReadRate::cache_hierarchy.l1dcaches.prefetcher 22579148.061331912875      # Per-requestor bytes read from memory rate ((Byte/Second))
board.memory.mem_ctrl.requestorReadRate::cache_hierarchy.l2cache.prefetcher 1375649485.962106943130      # Per-requestor bytes read from memory rate ((Byte/Second))
board.memory.mem_ctrl.requestorReadRate::processor.cores.core.inst 8121995.705515076406      # Per-requestor bytes read from memory rate ((Byte/Second))
board.memory.mem_ctrl.requestorReadRate::processor.cores.core.data 31599977.958257324994      # Per-requestor bytes read from memory rate ((Byte/Second))
board.memory.mem_ctrl.requestorWriteRate::writebacks 711118626.664470672607      # Per-requestor bytes write to memory rate ((Byte/Second))
board.memory.mem_ctrl.requestorReadAccesses::cache_hierarchy.l1dcaches.prefetcher      2085      # Per-requestor read serviced memory accesses (Count)
board.memory.mem_ctrl.requestorReadAccesses::cache_hierarchy.l2cache.prefetcher      127059      # Per-requestor read serviced memory accesses (Count)
board.memory.mem_ctrl.requestorReadAccesses::processor.cores.core.inst      750      # Per-requestor read serviced memory accesses (Count)
board.memory.mem_ctrl.requestorReadAccesses::processor.cores.core.data      2918      # Per-requestor read serviced memory accesses (Count)
board.memory.mem_ctrl.requestorWriteAccesses::writebacks      65695      # Per-requestor write serviced memory accesses (Count)
board.memory.mem_ctrl.requestorReadTotalLat::cache_hierarchy.l1dcaches.prefetcher      98632230      # Per-requestor read total memory access latency (Tick)
board.memory.mem_ctrl.requestorReadTotalLat::cache_hierarchy.l2cache.prefetcher      5405216028      # Per-requestor read total memory access latency (Tick)
board.memory.mem_ctrl.requestorReadTotalLat::processor.cores.core.inst      22141575      # Per-requestor read total memory access latency (Tick)
board.memory.mem_ctrl.requestorReadTotalLat::processor.cores.core.data      110242812      # Per-requestor read total memory access latency (Tick)
board.memory.mem_ctrl.requestorWriteTotalLat::writebacks      68453891753      # Per-requestor write total memory access latency (Tick)
board.memory.mem_ctrl.requestorReadAvgLat::cache_hierarchy.l1dcaches.prefetcher      47305.63      # Per-requestor read average memory access latency ((Tick/Count))
board.memory.mem_ctrl.requestorReadAvgLat::cache_hierarchy.l2cache.prefetcher      42540.99      # Per-requestor read average memory access latency ((Tick/Count))
board.memory.mem_ctrl.requestorReadAvgLat::processor.cores.core.inst      29522.10      # Per-requestor read average memory access latency ((Tick/Count))
board.memory.mem_ctrl.requestorReadAvgLat::processor.cores.core.data      37780.26      # Per-requestor read average memory access latency ((Tick/Count))
board.memory.mem_ctrl.requestorWriteAvgLat::writebacks      1041995.46      # Per-requestor write average memory access latency ((Tick/Count))
board.memory.mem_ctrl.dram.bytesRead::cache_hierarchy.l1dcaches.prefetcher      133440      # Number of bytes read from this memory (Byte)
board.memory.mem_ctrl.dram.bytesRead::cache_hierarchy.l2cache.prefetcher      8131776      # Number of bytes read from this memory (Byte)
board.memory.mem_ctrl.dram.bytesRead::processor.cores.core.inst      48000      # Number of bytes read from this memory (Byte)
board.memory.mem_ctrl.dram.bytesRead::processor.cores.core.data      186752      # Number of bytes read from this memory (Byte)
board.memory.mem_ctrl.dram.bytesRead::total      8499968      # Number of bytes read from this memory (Byte)
board.memory.mem_ctrl.dram.bytesInstRead::processor.cores.core.inst      48000      # Number of instructions bytes read from this memory (Byte)
board.memory.mem_ctrl.dram.bytesInstRead::total      48000      # Number of instructions bytes read from this memory (Byte)
board.memory.mem_ctrl.dram.bytesWritten::writebacks      4204480      # Number of bytes written to this memory (Byte)
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board.memory.mem_ctrl.dram.bytesWritten::total      4204480                    # Number of bytes written to this memory (Byte)
board.memory.mem_ctrl.dram.numReads::cache_hierarchy.l1dcaches.prefetcher        2085                    # Number of read requests responded to by this memory (Count)
board.memory.mem_ctrl.dram.numReads::cache_hierarchy.l2cache.prefetcher       127059                    # Number of read requests responded to by this memory (Count)
board.memory.mem_ctrl.dram.numReads::processor.cores.core.inst         750                    # Number of read requests responded to by this memory (Count)
board.memory.mem_ctrl.dram.numReads::processor.cores.core.data        2918                    # Number of read requests responded to by this memory (Count)
board.memory.mem_ctrl.dram.numReads::total       132812                    # Number of read requests responded to by this memory (Count)
board.memory.mem_ctrl.dram.numWrites::writebacks        65695                    # Number of write requests responded to by this memory (Count)
board.memory.mem_ctrl.dram.numWrites::total        65695                    # Number of write requests responded to by this memory (Count)
board.memory.mem_ctrl.dram.bwRead::cache_hierarchy.l1dcaches.prefetcher     22579148                    # Total read bandwidth from this memory ((Byte/Second))
board.memory.mem_ctrl.dram.bwRead::cache_hierarchy.l2cache.prefetcher   1375963536                    # Total read bandwidth from this memory ((Byte/Second))
board.memory.mem_ctrl.dram.bwRead::processor.cores.core.inst      8121996                    # Total read bandwidth from this memory ((Byte/Second))
board.memory.mem_ctrl.dram.bwRead::processor.cores.core.data     31599978                    # Total read bandwidth from this memory ((Byte/Second))
board.memory.mem_ctrl.dram.bwRead::total   1438264658                    # Total read bandwidth from this memory ((Byte/Second))
board.memory.mem_ctrl.dram.bwInstRead::processor.cores.core.inst      8121996                    # Instruction read bandwidth from this memory ((Byte/Second))
board.memory.mem_ctrl.dram.bwInstRead::total      8121996                    # Instruction read bandwidth from this memory ((Byte/Second))
board.memory.mem_ctrl.dram.bwWrite::writebacks    711432677                    # Write bandwidth from this memory ((Byte/Second))
board.memory.mem_ctrl.dram.bwWrite::total    711432677                    # Write bandwidth from this memory ((Byte/Second))
board.memory.mem_ctrl.dram.bwTotal::writebacks    711432677                    # Total bandwidth to/from this memory ((Byte/Second))
board.memory.mem_ctrl.dram.bwTotal::cache_hierarchy.l1dcaches.prefetcher     22579148                    # Total bandwidth to/from this memory ((Byte/Second))
board.memory.mem_ctrl.dram.bwTotal::cache_hierarchy.l2cache.prefetcher   1375963536                    # Total bandwidth to/from this memory ((Byte/Second))
board.memory.mem_ctrl.dram.bwTotal::processor.cores.core.inst      8121996                    # Total bandwidth to/from this memory ((Byte/Second))
board.memory.mem_ctrl.dram.bwTotal::processor.cores.core.data     31599978                    # Total bandwidth to/from this memory ((Byte/Second))
board.memory.mem_ctrl.dram.bwTotal::total   2149697335                    # Total bandwidth to/from this memory ((Byte/Second))
board.memory.mem_ctrl.dram.readBursts          132783                    # Number of DRAM read bursts (Count)
board.memory.mem_ctrl.dram.writeBursts          65666                    # Number of DRAM write bursts (Count)
board.memory.mem_ctrl.dram.perBankRdBursts::0        8465                    # Per bank write bursts (Count)
board.memory.mem_ctrl.dram.perBankRdBursts::1        8437                    # Per bank write bursts (Count)
board.memory.mem_ctrl.dram.perBankRdBursts::2        8291                    # Per bank write bursts (Count)
board.memory.mem_ctrl.dram.perBankRdBursts::3        8268                    # Per bank write bursts (Count)
board.memory.mem_ctrl.dram.perBankRdBursts::4        8267                    # Per bank write bursts (Count)
board.memory.mem_ctrl.dram.perBankRdBursts::5        8312                    # Per bank write bursts (Count)
board.memory.mem_ctrl.dram.perBankRdBursts::6        8254                    # Per bank write bursts (Count)
board.memory.mem_ctrl.dram.perBankRdBursts::7        8225                    # Per bank write bursts (Count)
board.memory.mem_ctrl.dram.perBankRdBursts::8        8274                    # Per bank write bursts (Count)
board.memory.mem_ctrl.dram.perBankRdBursts::9        8236                    # Per bank write bursts (Count)
board.memory.mem_ctrl.dram.perBankRdBursts::10        8235                    # Per bank write bursts (Count)
board.memory.mem_ctrl.dram.perBankRdBursts::11        8227                    # Per bank write bursts (Count)
board.memory.mem_ctrl.dram.perBankRdBursts::12        8235                    # Per bank write bursts (Count)
board.memory.mem_ctrl.dram.perBankRdBursts::13        8294                    # Per bank write bursts (Count)
board.memory.mem_ctrl.dram.perBankRdBursts::14        8357                    # Per bank write bursts (Count)
board.memory.mem_ctrl.dram.perBankRdBursts::15        8406                    # Per bank write bursts (Count)
board.memory.mem_ctrl.dram.perBankWrBursts::0        4096                    # Per bank write bursts (Count)
board.memory.mem_ctrl.dram.perBankWrBursts::1        4096                    # Per bank write bursts (Count)
board.memory.mem_ctrl.dram.perBankWrBursts::2        4096                    # Per bank write bursts (Count)
board.memory.mem_ctrl.dram.perBankWrBursts::3        4104                    # Per bank write bursts (Count)
board.memory.mem_ctrl.dram.perBankWrBursts::4        4102                    # Per bank write bursts (Count)
board.memory.mem_ctrl.dram.perBankWrBursts::5        4157                    # Per bank write bursts (Count)
board.memory.mem_ctrl.dram.perBankWrBursts::6        4098                    # Per bank write bursts (Count)
board.memory.mem_ctrl.dram.perBankWrBursts::7        4096                    # Per bank write bursts (Count)
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board.memory.mem_ctrl.dram.perBankWrBursts::8        4116                    # Per bank write bursts (Count)
board.memory.mem_ctrl.dram.perBankWrBursts::9        4106                    # Per bank write bursts (Count)
board.memory.mem_ctrl.dram.perBankWrBursts::10       4117                    # Per bank write bursts (Count)
board.memory.mem_ctrl.dram.perBankWrBursts::11       4098                    # Per bank write bursts (Count)
board.memory.mem_ctrl.dram.perBankWrBursts::12       4096                    # Per bank write bursts (Count)
board.memory.mem_ctrl.dram.perBankWrBursts::13       4096                    # Per bank write bursts (Count)
board.memory.mem_ctrl.dram.perBankWrBursts::14       4096                    # Per bank write bursts (Count)
board.memory.mem_ctrl.dram.perBankWrBursts::15       4096                    # Per bank write bursts (Count)
board.memory.mem_ctrl.dram.totQLat        3146551395                         # Total ticks spent queuing (Tick)
board.memory.mem_ctrl.dram.totBusLat      663915000                          # Total ticks spent in databus transfers (Tick)
board.memory.mem_ctrl.dram.totMemAccLat   5636232645                         # Total ticks spent from burst creation until serviced by the DRAM (Tick)
board.memory.mem_ctrl.dram.avgQLat        23696.94                           # Average queueing delay per DRAM burst ((Tick/Count))
board.memory.mem_ctrl.dram.avgBusLat      5000.00                            # Average bus latency per DRAM burst ((Tick/Count))
board.memory.mem_ctrl.dram.avgMemAccLat   42446.94                           # Average memory access latency per DRAM burst ((Tick/Count))
board.memory.mem_ctrl.dram.readRowHits        122735                         # Number of row buffer hits during reads (Count)
board.memory.mem_ctrl.dram.writeRowHits       60959                          # Number of row buffer hits during writes (Count)
board.memory.mem_ctrl.dram.readRowHitRate     92.43                          # Row buffer hit rate for reads (Ratio)
board.memory.mem_ctrl.dram.writeRowHitRate    92.83                          # Row buffer hit rate for writes (Ratio)
board.memory.mem_ctrl.dram.bytesPerActivate::samples      14755              # Bytes accessed per row activation (Byte)
board.memory.mem_ctrl.dram.bytesPerActivate::mean   860.775059               # Bytes accessed per row activation (Byte)
board.memory.mem_ctrl.dram.bytesPerActivate::gmean   757.624741              # Bytes accessed per row activation (Byte)
board.memory.mem_ctrl.dram.bytesPerActivate::stdev   287.222379              # Bytes accessed per row activation (Byte)
board.memory.mem_ctrl.dram.bytesPerActivate::0-127        617     4.18%     4.18% # Bytes accessed per row activation (Byte)
board.memory.mem_ctrl.dram.bytesPerActivate::128-255      243     1.65%     5.83% # Bytes accessed per row activation (Byte)
board.memory.mem_ctrl.dram.bytesPerActivate::256-383      348     2.36%     8.19% # Bytes accessed per row activation (Byte)
board.memory.mem_ctrl.dram.bytesPerActivate::384-511      1109    7.52%    15.70% # Bytes accessed per row activation (Byte)
board.memory.mem_ctrl.dram.bytesPerActivate::512-639      720     4.88%    20.58% # Bytes accessed per row activation (Byte)
board.memory.mem_ctrl.dram.bytesPerActivate::640-767      678     4.60%    25.18% # Bytes accessed per row activation (Byte)
board.memory.mem_ctrl.dram.bytesPerActivate::768-895      226     1.53%    26.71% # Bytes accessed per row activation (Byte)
board.memory.mem_ctrl.dram.bytesPerActivate::896-1023     591     4.01%    30.72% # Bytes accessed per row activation (Byte)
board.memory.mem_ctrl.dram.bytesPerActivate::1024-1151    10223   69.28%   100.00% # Bytes accessed per row activation (Byte)
board.memory.mem_ctrl.dram.bytesPerActivate::total        14755              # Bytes accessed per row activation (Byte)
board.memory.mem_ctrl.dram.dramBytesRead      8498112                        # Total bytes read (Byte)
board.memory.mem_ctrl.dram.dramBytesWritten   4202624                        # Total bytes written (Byte)
board.memory.mem_ctrl.dram.avgRdBW        1437.950608                        # Average DRAM read bandwidth in MiBytes/s ((Byte/Second))
board.memory.mem_ctrl.dram.avgWrBW        711.118627                         # Average DRAM write bandwidth in MiBytes/s ((Byte/Second))
board.memory.mem_ctrl.dram.peakBW         12800.00                           # Theoretical peak bandwidth in MiByte/s ((Byte/Second))
board.memory.mem_ctrl.dram.busUtil        16.79                              # Data bus utilization in percentage (Ratio)
board.memory.mem_ctrl.dram.busUtilRead        11.23                          # Data bus utilization in percentage for reads (Ratio)
board.memory.mem_ctrl.dram.busUtilWrite       5.56                           # Data bus utilization in percentage for writes (Ratio)
board.memory.mem_ctrl.dram.pageHitRate        92.56                          # Row buffer hit rate, read and write combined (Ratio)
board.memory.mem_ctrl.dram.power_state.pwrStateResidencyTicks::UNDEFINED   5909877540   # Cumulative time (in ticks) in various power states (Tick)
board.memory.mem_ctrl.dram.rank0.actEnergy    52857420                       # Energy for activate commands per rank (pJ) (Joule)
board.memory.mem_ctrl.dram.rank0.preEnergy    28094385                       # Energy for precharge commands per rank (pJ) (Joule)
board.memory.mem_ctrl.dram.rank0.readEnergy   474945660                      # Energy for read commands per rank (pJ) (Joule)
board.memory.mem_ctrl.dram.rank0.writeEnergy  171450900                      # Energy for write commands per rank (pJ) (Joule)
board.memory.mem_ctrl.dram.rank0.refreshEnergy 466511760.000000              # Energy for refresh commands per rank (pJ) (Joule)
board.memory.mem_ctrl.dram.rank0.actBackEnergy   1473491610                  # Energy for active background per rank (pJ) (Joule)
board.memory.mem_ctrl.dram.rank0.preBackEnergy   1028616000                  # Energy for precharge background per rank (pJ) (Joule)
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board.memory.mem_ctrl.dram.rank0.actPowerDownEnergy 0 # Energy for active power-down per rank (pJ) (Joule)
board.memory.mem_ctrl.dram.rank0.prePowerDownEnergy 0 # Energy for precharge power-down per rank (pJ) (Joule)
board.memory.mem_ctrl.dram.rank0.selfRefreshEnergy 0 # Energy for self refresh per rank (pJ) (Joule)
board.memory.mem_ctrl.dram.rank0.totalEnergy 3695967735 # Total energy per rank (pJ) (Joule)
board.memory.mem_ctrl.dram.rank0.averagePower 625.388210 # Core power per rank (mW) (Watt)
board.memory.mem_ctrl.dram.rank0.totalIdleTime 0 # Total Idle time Per DRAM Rank (Tick)
board.memory.mem_ctrl.dram.rank0.pwrStateTime::IDLE 2627175091 # Time in different power states (Tick)
board.memory.mem_ctrl.dram.rank0.pwrStateTime::REF 197175966 # Time in different power states (Tick)
board.memory.mem_ctrl.dram.rank0.pwrStateTime::SREF 0 # Time in different power states (Tick)
board.memory.mem_ctrl.dram.rank0.pwrStateTime::PRE_PDN 0 # Time in different power states (Tick)
board.memory.mem_ctrl.dram.rank0.pwrStateTime::ACT 3085526483 # Time in different power states (Tick)
board.memory.mem_ctrl.dram.rank0.pwrStateTime::ACT_PDN 0 # Time in different power states (Tick)
board.memory.mem_ctrl.dram.rank1.actEnergy 52493280 # Energy for activate commands per rank (pJ) (Joule)
board.memory.mem_ctrl.dram.rank1.preEnergy 27900840 # Energy for precharge commands per rank (pJ) (Joule)
board.memory.mem_ctrl.dram.rank1.readEnergy 473124960 # Energy for read commands per rank (pJ) (Joule)
board.memory.mem_ctrl.dram.rank1.writeEnergy 171325620 # Energy for write commands per rank (pJ) (Joule)
board.memory.mem_ctrl.dram.rank1.refreshEnergy 466511760.000000 # Energy for refresh commands per rank (pJ) (Joule)
board.memory.mem_ctrl.dram.rank1.actBackEnergy 1462379460 # Energy for active background per rank (pJ) (Joule)
board.memory.mem_ctrl.dram.rank1.preBackEnergy 1037971680 # Energy for precharge background per rank (pJ) (Joule)
board.memory.mem_ctrl.dram.rank1.actPowerDownEnergy 0 # Energy for active power-down per rank (pJ) (Joule)
board.memory.mem_ctrl.dram.rank1.prePowerDownEnergy 0 # Energy for precharge power-down per rank (pJ) (Joule)
board.memory.mem_ctrl.dram.rank1.selfRefreshEnergy 0 # Energy for self refresh per rank (pJ) (Joule)
board.memory.mem_ctrl.dram.rank1.totalEnergy 3691707600 # Total energy per rank (pJ) (Joule)
board.memory.mem_ctrl.dram.rank1.averagePower 624.667360 # Core power per rank (mW) (Watt)
board.memory.mem_ctrl.dram.rank1.totalIdleTime 0 # Total Idle time Per DRAM Rank (Tick)
board.memory.mem_ctrl.dram.rank1.pwrStateTime::IDLE 2652087738 # Time in different power states (Tick)
board.memory.mem_ctrl.dram.rank1.pwrStateTime::REF 197180040 # Time in different power states (Tick)
board.memory.mem_ctrl.dram.rank1.pwrStateTime::SREF 0 # Time in different power states (Tick)
board.memory.mem_ctrl.dram.rank1.pwrStateTime::PRE_PDN 0 # Time in different power states (Tick)
board.memory.mem_ctrl.dram.rank1.pwrStateTime::ACT 3060609762 # Time in different power states (Tick)
board.memory.mem_ctrl.dram.rank1.pwrStateTime::ACT_PDN 0 # Time in different power states (Tick)
board.memory.mem_ctrl.power_state.pwrStateResidencyTicks::UNDEFINED 5909877540 # Cumulative time (in ticks) in various power states (Tick)
board.processor.cores.core.numCycles 17747381 # Number of cpu cycles simulated (Cycle)
board.processor.cores.core.cpi 0.481857 # CPI: cycles per instruction (core level) ((Cycle/Count))
board.processor.cores.core.ipc 2.075306 # IPC: instructions per cycle (core level) ((Count/Cycle))
board.processor.cores.core.numWorkItemsStarted 0 # Number of work items this cpu started (Count)
board.processor.cores.core.numWorkItemsCompleted 0 # Number of work items this cpu completed (Count)
board.processor.cores.core.instsAdded 57074691 # Number of instructions added to the IQ (excludes non-spec) (Count)
board.processor.cores.core.nonSpecInstsAdded 240 # Number of non-speculative instructions added to the IQ (Count)
board.processor.cores.core.instsIssued 57028291 # Number of instructions issued (Count)
board.processor.cores.core.squashedInstsIssued 487 # Number of squashed instructions issued (Count)
board.processor.cores.core.squashedInstsExamined 213717 # Number of squashed instructions iterated over during squash; mainly for profiling (Count)
board.processor.cores.core.squashedOperandsExamined 246065 # Number of squashed operands that are examined and possibly removed from graph (Count)
board.processor.cores.core.squashedNonSpecRemoved 150 # Number of squashed non-spec instructions that were removed (Count)
board.processor.cores.core.numIssuedDist::samples 17601662 # Number of insts issued each cycle (Count)
board.processor.cores.core.numIssuedDist::mean 3.239938 # Number of insts issued each cycle (Count)
board.processor.cores.core.numIssuedDist::stdev 2.164296 # Number of insts issued each cycle (Count)
board.processor.cores.core.numIssuedDist::underflows 0 0.00% 0.00% # Number of insts issued each cycle (Count)
board.processor.cores.core.numIssuedDist::0 1620274 9.21% 9.21% # Number of insts issued each cycle (Count)
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board.processor.cores.core.numIssuedDist::1      1824276     10.36%     19.57% # Number of insts issued each cycle (Count)
board.processor.cores.core.numIssuedDist::2      4649372     26.41%     45.98% # Number of insts issued each cycle (Count)
board.processor.cores.core.numIssuedDist::3      2974541     16.90%     62.88% # Number of insts issued each cycle (Count)
board.processor.cores.core.numIssuedDist::4      1555460      8.84%     71.72% # Number of insts issued each cycle (Count)
board.processor.cores.core.numIssuedDist::5      1689522      9.60%     81.32% # Number of insts issued each cycle (Count)
board.processor.cores.core.numIssuedDist::6      1178283      6.69%     88.01% # Number of insts issued each cycle (Count)
board.processor.cores.core.numIssuedDist::7      1636972      9.30%     97.31% # Number of insts issued each cycle (Count)
board.processor.cores.core.numIssuedDist::8       472962      2.69%    100.00% # Number of insts issued each cycle (Count)
board.processor.cores.core.numIssuedDist::overflows            0      0.00%    100.00% # Number of insts issued each cycle (Count)
board.processor.cores.core.numIssuedDist::min_value            0                      # Number of insts issued each cycle (Count)
board.processor.cores.core.numIssuedDist::max_value            8                      # Number of insts issued each cycle (Count)
board.processor.cores.core.numIssuedDist::total     17601662                      # Number of insts issued each cycle (Count)
board.processor.cores.core.statFuBusy::No_OpClass            0      0.00%      0.00% # attempts to use FU when none available (Count)
board.processor.cores.core.statFuBusy::IntAlu      1417779     99.97%     99.97% # attempts to use FU when none available (Count)
board.processor.cores.core.statFuBusy::IntMult            0      0.00%     99.97% # attempts to use FU when none available (Count)
board.processor.cores.core.statFuBusy::IntDiv            0      0.00%     99.97% # attempts to use FU when none available (Count)
board.processor.cores.core.statFuBusy::FloatAdd            0      0.00%     99.97% # attempts to use FU when none available (Count)
board.processor.cores.core.statFuBusy::FloatCmp            0      0.00%     99.97% # attempts to use FU when none available (Count)
board.processor.cores.core.statFuBusy::FloatCvt            0      0.00%     99.97% # attempts to use FU when none available (Count)
board.processor.cores.core.statFuBusy::FloatMult            0      0.00%     99.97% # attempts to use FU when none available (Count)
board.processor.cores.core.statFuBusy::FloatMultAcc            0      0.00%     99.97% # attempts to use FU when none available (Count)
board.processor.cores.core.statFuBusy::FloatDiv            0      0.00%     99.97% # attempts to use FU when none available (Count)
board.processor.cores.core.statFuBusy::FloatMisc            0      0.00%     99.97% # attempts to use FU when none available (Count)
board.processor.cores.core.statFuBusy::FloatSqrt            0      0.00%     99.97% # attempts to use FU when none available (Count)
board.processor.cores.core.statFuBusy::SimdAdd            0      0.00%     99.97% # attempts to use FU when none available (Count)
board.processor.cores.core.statFuBusy::SimdAddAcc            0      0.00%     99.97% # attempts to use FU when none available (Count)
board.processor.cores.core.statFuBusy::SimdAlu           24      0.00%     99.97% # attempts to use FU when none available (Count)
board.processor.cores.core.statFuBusy::SimdCmp            0      0.00%     99.97% # attempts to use FU when none available (Count)
board.processor.cores.core.statFuBusy::SimdCvt            0      0.00%     99.97% # attempts to use FU when none available (Count)
board.processor.cores.core.statFuBusy::SimdMisc            0      0.00%     99.97% # attempts to use FU when none available (Count)
board.processor.cores.core.statFuBusy::SimdMult            0      0.00%     99.97% # attempts to use FU when none available (Count)
board.processor.cores.core.statFuBusy::SimdMultAcc            0      0.00%     99.97% # attempts to use FU when none available (Count)
board.processor.cores.core.statFuBusy::SimdMatMultAcc            0      0.00%     99.97% # attempts to use FU when none available (Count)
board.processor.cores.core.statFuBusy::SimdShift            0      0.00%     99.97% # attempts to use FU when none available (Count)
board.processor.cores.core.statFuBusy::SimdShiftAcc            0      0.00%     99.97% # attempts to use FU when none available (Count)
board.processor.cores.core.statFuBusy::SimdDiv            0      0.00%     99.97% # attempts to use FU when none available (Count)
board.processor.cores.core.statFuBusy::SimdSqrt            0      0.00%     99.97% # attempts to use FU when none available (Count)
board.processor.cores.core.statFuBusy::SimdFloatAdd            0      0.00%     99.97% # attempts to use FU when none available (Count)
board.processor.cores.core.statFuBusy::SimdFloatAlu            0      0.00%     99.97% # attempts to use FU when none available (Count)
board.processor.cores.core.statFuBusy::SimdFloatCmp            0      0.00%     99.97% # attempts to use FU when none available (Count)
board.processor.cores.core.statFuBusy::SimdFloatCvt            0      0.00%     99.97% # attempts to use FU when none available (Count)
board.processor.cores.core.statFuBusy::SimdFloatDiv            0      0.00%     99.97% # attempts to use FU when none available (Count)
board.processor.cores.core.statFuBusy::SimdFloatMisc            0      0.00%     99.97% # attempts to use FU when none available (Count)
board.processor.cores.core.statFuBusy::SimdFloatMult            0      0.00%     99.97% # attempts to use FU when none available (Count)
board.processor.cores.core.statFuBusy::SimdFloatMultAcc            0      0.00%     99.97% # attempts to use FU when none available (Count)
board.processor.cores.core.statFuBusy::SimdFloatMatMultAcc            0      0.00%     99.97% # attempts to use FU when none available (Count)
board.processor.cores.core.statFuBusy::SimdFloatSqrt            0      0.00%     99.97% # attempts to use FU when none available (Count)
board.processor.cores.core.statFuBusy::SimdReduceAdd            0      0.00%     99.97% # attempts to use FU when none available (Count)
board.processor.cores.core.statFuBusy::SimdReduceAlu            0      0.00%     99.97% # attempts to use FU when none available (Count)
```

board.processor.cores.core.statFuBusy::SimdReduceCmp 0 0.00% 99.97% # attempts to use FU when none available (Count)

board.processor.cores.core.statFuBusy::SimdFloatReduceAdd 0 0.00% 99.97% # attempts to use FU when none available (Count)

board.processor.cores.core.statFuBusy::SimdFloatReduceCmp 0 0.00% 99.97% # attempts to use FU when none available (Count)

board.processor.cores.core.statFuBusy::SimdAes 0 0.00% 99.97% # attempts to use FU when none available (Count)

board.processor.cores.core.statFuBusy::SimdAesMix 0 0.00% 99.97% # attempts to use FU when none available (Count)

board.processor.cores.core.statFuBusy::SimdSha1Hash 0 0.00% 99.97% # attempts to use FU when none available (Count)

board.processor.cores.core.statFuBusy::SimdSha1Hash2 0 0.00% 99.97% # attempts to use FU when none available (Count)

board.processor.cores.core.statFuBusy::SimdSha256Hash 0 0.00% 99.97% # attempts to use FU when none available (Count)

board.processor.cores.core.statFuBusy::SimdSha256Hash2 0 0.00% 99.97% # attempts to use FU when none available (Count)

board.processor.cores.core.statFuBusy::SimdShaSigma2 0 0.00% 99.97% # attempts to use FU when none available (Count)

board.processor.cores.core.statFuBusy::SimdShaSigma3 0 0.00% 99.97% # attempts to use FU when none available (Count)

board.processor.cores.core.statFuBusy::SimdPredAlu 0 0.00% 99.97% # attempts to use FU when none available (Count)

board.processor.cores.core.statFuBusy::Matrix 0 0.00% 99.97% # attempts to use FU when none available (Count)

board.processor.cores.core.statFuBusy::MatrixMov 0 0.00% 99.97% # attempts to use FU when none available (Count)

board.processor.cores.core.statFuBusy::MatrixOP 0 0.00% 99.97% # attempts to use FU when none available (Count)

board.processor.cores.core.statFuBusy::MemRead 86 0.01% 99.98% # attempts to use FU when none available (Count)

board.processor.cores.core.statFuBusy::MemWrite 277 0.02% 100.00% # attempts to use FU when none available (Count)

board.processor.cores.core.statFuBusy::FloatMemRead 6 0.00% 100.00% # attempts to use FU when none available (Count)

board.processor.cores.core.statFuBusy::FloatMemWrite 33 0.00% 100.00% # attempts to use FU when none available (Count)

board.processor.cores.core.statFuBusy::IprAccess 0 0.00% 100.00% # attempts to use FU when none available (Count)

board.processor.cores.core.statFuBusy::InstPrefetch 0 0.00% 100.00% # attempts to use FU when none available (Count)

board.processor.cores.core.statFuBusy::SimdUnitStrideLoad 0 0.00% 100.00% # attempts to use FU when none available (Count)

board.processor.cores.core.statFuBusy::SimdUnitStrideStore 0 0.00% 100.00% # attempts to use FU when none available (Count)

board.processor.cores.core.statFuBusy::SimdUnitStrideMaskLoad 0 0.00% 100.00% # attempts to use FU when none available (Count)

board.processor.cores.core.statFuBusy::SimdUnitStrideMaskStore 0 0.00% 100.00% # attempts to use FU when none available (Count)

board.processor.cores.core.statFuBusy::SimdStridedLoad 0 0.00% 100.00% # attempts to use FU when none available (Count)

board.processor.cores.core.statFuBusy::SimdStridedStore 0 0.00% 100.00% # attempts to use FU when none available (Count)

board.processor.cores.core.statFuBusy::SimdIndexedLoad 0 0.00% 100.00% # attempts to use FU when none available (Count)

board.processor.cores.core.statFuBusy::SimdIndexedStore 0 0.00% 100.00% # attempts to use FU when none available (Count)

board.processor.cores.core.statFuBusy::SimdWholeRegisterLoad 0 0.00% 100.00% # attempts to use FU when none available (Count)

board.processor.cores.core.statFuBusy::SimdWholeRegisterStore 0 0.00% 100.00% # attempts to use FU when none available (Count)

board.processor.cores.core.statFuBusy::SimdUnitStrideFaultOnlyFirstLoad 0 0.00% 100.00% # attempts to use FU when none available (Count)

board.processor.cores.core.statFuBusy::SimdUnitStrideSegmentedLoad 0 0.00% 100.00% # attempts to use FU when none available (Count)

board.processor.cores.core.statFuBusy::SimdUnitStrideSegmentedStore 0 0.00% 100.00% # attempts to use FU when none available (Count)

board.processor.cores.core.statFuBusy::SimdExt 0 0.00% 100.00% # attempts to use FU when none available (Count)

board.processor.cores.core.statFuBusy::SimdFloatExt 0 0.00% 100.00% # attempts to use FU when none available (Count)

board.processor.cores.core.statFuBusy::SimdConfig 0 0.00% 100.00% # attempts to use FU when none available (Count)

board.processor.cores.core.statIssuedInstType_0::No_OpClass 739 0.00% 0.00% # Number of instructions issued per FU type, per thread (Count)

board.processor.cores.core.statIssuedInstType_0::IntAlu 40174839 70.45% 70.45% # Number of instructions issued per FU type, per thread (Count)

board.processor.cores.core.statIssuedInstType_0::IntMult 1053193 1.85% 72.30% # Number of instructions issued per FU type, per thread (Count)

board.processor.cores.core.statIssuedInstType_0::IntDiv 40 0.00% 72.30% # Number of instructions issued per FU type, per thread (Count)

board.processor.cores.core.statIssuedInstType_0::FloatAdd 192 0.00% 72.30% # Number of instructions issued per FU type, per thread (Count)

board.processor.cores.core.statIssuedInstType_0::FloatCmp 0 0.00% 72.30% # Number of instructions issued per FU type, per thread (Count)

board.processor.cores.core.statIssuedInstType_0::FloatCvt 0 0.00% 72.30% # Number of instructions issued per FU type, per thread (Count)

board.processor.cores.core.statIssuedInstType_0::FloatMult 0 0.00% 72.30% # Number of instructions issued per FU type, per thread (Count)

board.processor.cores.core.statIssuedInstType_0::FloatMultAcc 0 0.00% 72.30% # Number of instructions issued per FU type, per thread (Count)

board.processor.cores.core.statIssuedInstType_0::FloatDiv 0 0.00% 72.30% # Number of instructions issued per FU type, per thread (Count)

board.processor.cores.core.statIssuedInstType_0::FloatMisc 0 0.00% 72.30% # Number of instructions issued per FU type, per thread (Count)

board.processor.cores.core.statIssuedInstType_0::FloatSqrt 0 0.00% 72.30% # Number of instructions issued per FU type, per thread (Count)

board.processor.cores.core.statIssuedInstType_0::SimdAdd 10 0.00% 72.30% # Number of instructions issued per FU type, per thread (Count)

board.processor.cores.core.statIssuedInstType_0::SimdAddAcc 0 0.00% 72.30% # Number of instructions issued per FU type, per thread (Count)

board.processor.cores.core.statIssuedInstType_0::SimdAlu 281 0.00% 72.30% # Number of instructions issued per FU type, per thread (Count)

board.processor.cores.core.statIssuedInstType_0::SimdCmp 2 0.00% 72.30% # Number of instructions issued per FU type, per thread (Count)

board.processor.cores.core.statIssuedInstType_0::SimdCvt 78 0.00% 72.30% # Number of instructions issued per FU type, per thread (Count)

board.processor.cores.core.statIssuedInstType_0::SimdMisc 253 0.00% 72.30% # Number of instructions issued per FU type, per thread (Count)

board.processor.cores.core.statIssuedInstType_0::SimdMult 0 0.00% 72.30% # Number of instructions issued per FU type, per thread (Count)

board.processor.cores.core.statIssuedInstType_0::SimdMultAcc 0 0.00% 72.30% # Number of instructions issued per FU type, per thread (Count)

board.processor.cores.core.statIssuedInstType_0::SimdMatMultAcc 0 0.00% 72.30% # Number of instructions issued per FU type, per thread (Count)

board.processor.cores.core.statIssuedInstType_0::SimdShift 6 0.00% 72.30% # Number of instructions issued per FU type, per thread (Count)

board.processor.cores.core.statIssuedInstType_0::SimdShiftAcc 0 0.00% 72.30% # Number of instructions issued per FU type, per thread (Count)

board.processor.cores.core.statIssuedInstType_0::SimdDiv 0 0.00% 72.30% # Number of instructions issued per FU type, per thread (Count)

board.processor.cores.core.statIssuedInstType_0::SimdSqrt 0 0.00% 72.30% # Number of instructions issued per FU type, per thread (Count)

board.processor.cores.core.statIssuedInstType_0::SimdFloatAdd 0 0.00% 72.30% # Number of instructions issued per FU type, per thread (Count)

board.processor.cores.core.statIssuedInstType_0::SimdFloatAlu 0 0.00% 72.30% # Number of instructions issued per FU type, per thread (Count)

board.processor.cores.core.statIssuedInstType_0::SimdFloatCmp 0 0.00% 72.30% # Number of instructions issued per FU type, per thread (Count)

board.processor.cores.core.statIssuedInstType_0::SimdFloatCvt 0 0.00% 72.30% # Number of instructions issued per FU type, per thread (Count)

board.processor.cores.core.statIssuedInstType_0::SimdFloatDiv 0 0.00% 72.30% # Number of instructions issued per FU type, per thread (Count)

board.processor.cores.core.statIssuedInstType_0::SimdFloatMisc 0 0.00% 72.30% # Number of instructions issued per FU type, per thread (Count)

board.processor.cores.core.statIssuedInstType_0::SimdFloatMult 0 0.00% 72.30% # Number of instructions issued per FU type, per thread (Count)

board.processor.cores.core.statIssuedInstType_0::SimdFloatMultAcc 0 0.00% 72.30% # Number of instructions issued per FU type, per thread (Count)

board.processor.cores.core.statIssuedInstType_0::SimdFloatMatMultAcc 0 0.00% 72.30% # Number of instructions issued per FU type, per thread (Count)

board.processor.cores.core.statIssuedInstType_0::SimdFloatSqrt 0 0.00% 72.30% # Number of instructions issued per FU type, per thread (Count)

board.processor.cores.core.statIssuedInstType_0::SimdReduceAdd 0 0.00% 72.30% # Number of instructions issued per FU type, per thread (Count)

board.processor.cores.core.statIssuedInstType_0::SimdReduceAlu 0 0.00% 72.30% # Number of instructions issued per FU type, per thread (Count)

board.processor.cores.core.statIssuedInstType_0::SimdReduceCmp 0 0.00% 72.30% # Number of instructions issued per FU type, per thread (Count)

board.processor.cores.core.statIssuedInstType_0::SimdFloatReduceAdd 0 0.00% 72.30% # Number of instructions issued per FU type, per thread (Count)

board.processor.cores.core.statIssuedInstType_0::SimdFloatReduceCmp 0 0.00% 72.30% # Number of instructions issued per FU type, per thread (Count)

board.processor.cores.core.statIssuedInstType_0::SimdAes 0 0.00% 72.30% # Number of instructions issued per FU type, per thread (Count)

board.processor.cores.core.statIssuedInstType_0::SimdAesMix 0 0.00% 72.30% # Number of instructions issued per FU type, per thread (Count)

board.processor.cores.core.statIssuedInstType_0::SimdSha1Hash 0 0.00% 72.30% # Number of instructions issued per FU type, per thread (Count)

board.processor.cores.core.statIssuedInstType_0::SimdSha1Hash2 0 0.00% 72.30% # Number of instructions issued per FU type, per thread (Count)

board.processor.cores.core.statIssuedInstType_0::SimdSha256Hash 0 0.00% 72.30% # Number of instructions issued per FU type, per thread (Count)

board.processor.cores.core.statIssuedInstType_0::SimdSha256Hash2 0 0.00% 72.30% # Number of instructions issued per FU type, per thread (Count)

board.processor.cores.core.statIssuedInstType_0::SimdShaSigma2 0 0.00% 72.30% # Number of instructions issued per FU type, per thread (Count)

board.processor.cores.core.statIssuedInstType_0::SimdShaSigma3 0 0.00% 72.30% # Number of instructions issued per FU type, per thread (Count)

board.processor.cores.core.statIssuedInstType_0::SimdPredAlu 0 0.00% 72.30% # Number of instructions issued per FU type, per thread (Count)

board.processor.cores.core.statIssuedInstType_0::Matrix 0 0.00% 72.30% # Number of instructions issued per FU type, per thread (Count)

board.processor.cores.core.statIssuedInstType_0::MatrixMov 0 0.00% 72.30% # Number of instructions issued per FU type, per thread (Count)

board.processor.cores.core.statIssuedInstType_0::MatrixOP 0 0.00% 72.30% # Number of instructions issued per FU type, per thread (Count)

board.processor.cores.core.statIssuedInstType_0::MemRead 11583702 20.31% 92.61% # Number of instructions issued per FU type, per thread (Count)

board.processor.cores.core.statIssuedInstType_0::MemWrite 4214111 7.39% 100.00% # Number of instructions issued per FU type, per thread (Count)

board.processor.cores.core.statIssuedInstType_0::FloatMemRead 204 0.00% 100.00% # Number of instructions issued per FU type, per thread (Count)

board.processor.cores.core.statIssuedInstType_0::FloatMemWrite 641 0.00% 100.00% # Number of instructions issued per FU type, per thread (Count)

board.processor.cores.core.statIssuedInstType_0::IprAccess 0 0.00% 100.00% # Number of instructions issued per FU type, per thread (Count)

board.processor.cores.core.statIssuedInstType_0::InstPrefetch 0 0.00% 100.00% # Number of instructions issued per FU type, per thread (Count)

board.processor.cores.core.statIssuedInstType_0::SimdUnitStrideLoad 0 0.00% 100.00% # Number of instructions issued per FU type, per thread (Count)

board.processor.cores.core.statIssuedInstType_0::SimdUnitStrideStore 0 0.00% 100.00% # Number of instructions issued per FU type, per thread (Count)

board.processor.cores.core.statIssuedInstType_0::SimdUnitStrideMaskLoad 0 0.00% 100.00% # Number of instructions issued per FU type, per thread (Count)

board.processor.cores.core.statIssuedInstType_0::SimdUnitStrideMaskStore 0 0.00% 100.00% # Number of instructions issued per FU type, per thread (Count)

board.processor.cores.core.statIssuedInstType_0::SimdStridedLoad 0 0.00% 100.00% # Number of instructions issued per FU type, per thread (Count)

board.processor.cores.core.statIssuedInstType_0::SimdStridedStore 0 0.00% 100.00% # Number of instructions issued per FU type, per thread (Count)

board.processor.cores.core.statIssuedInstType_0::SimdIndexedLoad 0 0.00% 100.00% # Number of instructions issued per FU type, per thread (Count)

board.processor.cores.core.statIssuedInstType_0::SimdIndexedStore 0 0.00% 100.00% # Number of instructions issued per FU type, per thread (Count)

board.processor.cores.core.statIssuedInstType_0::SimdWholeRegisterLoad 0 0.00% 100.00% # Number of instructions issued per FU type, per thread (Count)

board.processor.cores.core.statIssuedInstType_0::SimdWholeRegisterStore 0 0.00% 100.00% # Number of instructions issued per FU type, per thread (Count)

board.processor.cores.core.statIssuedInstType_0::SimdUnitStrideFaultOnlyFirstLoad 0 0.00% 100.00% # Number of instructions issued per FU type, per thread (Count)

board.processor.cores.core.statIssuedInstType_0::SimdUnitStrideSegmentedLoad 0 0.00% 100.00% # Number of instructions issued per FU type, per thread (Count)

board.processor.cores.core.statIssuedInstType_0::SimdUnitStrideSegmentedStore 0 0.00% 100.00% # Number of instructions issued per FU type, per thread (Count)

board.processor.cores.core.statIssuedInstType_0::SimdExt 0 0.00% 100.00% # Number of instructions issued per FU type, per thread (Count)

board.processor.cores.core.statIssuedInstType_0::SimdFloatExt 0 0.00% 100.00% # Number of instructions issued per FU type, per thread (Count)

board.processor.cores.core.statIssuedInstType_0::SimdConfig 0 0.00% 100.00% # Number of instructions issued per FU type, per thread (Count)

board.processor.cores.core.statIssuedInstType_0::total 57028291 # Number of instructions issued per FU type, per thread (Count)

board.processor.cores.core.issueRate 3.213336 # Inst issue rate ((Count/Cycle))

board.processor.cores.core.fuBusy 1418205 # FU busy when requested (Count)

board.processor.cores.core.fuBusyRate 0.024868 # FU busy rate (busy events/executed inst) ((Count/Count))

board.processor.cores.core.intInstQueueReads 133073130 # Number of integer instruction queue reads (Count)

board.processor.cores.core.intInstQueueWrites 57286284 # Number of integer instruction queue writes (Count)

board.processor.cores.core.intInstQueueWakeupAccesses 57017662 # Number of integer instruction queue wakeup accesses (Count)

board.processor.cores.core.fpInstQueueReads 3806 # Number of floating instruction queue reads (Count)

board.processor.cores.core.fpInstQueueWrites 2473 # Number of floating instruction queue writes (Count)

board.processor.cores.core.fpInstQueueWakeupAccesses 1770 # Number of floating instruction queue wakeup accesses (Count)

board.processor.cores.core.vecInstQueueReads 0 # Number of vector instruction queue reads (Count)

board.processor.cores.core.vecInstQueueWrites 0 # Number of vector instruction queue writes (Count)

board.processor.cores.core.vecInstQueueWakeupAccesses 0 # Number of vector instruction queue wakeup accesses (Count)

board.processor.cores.core.intAluAccesses 58443846 # Number of integer alu accesses (Count)

board.processor.cores.core.fpAluAccesses 1911 # Number of floating point alu accesses (Count)

board.processor.cores.core.vecAluAccesses 0 # Number of vector alu accesses (Count)

board.processor.cores.core.numSquashedInsts 6992 # Number of squashed instructions skipped in execute (Count)

board.processor.cores.core.numSwp 0 # Number of swp insts executed (Count)

board.processor.cores.core.timesIdled 515 # Number of times that the entire CPU went into an idle state and unscheduled itself (Count)

board.processor.cores.core.idleCycles 145719 # Total number of cycles that the CPU has spent unscheduled due to idling (Cycle)

board.processor.cores.core.MemDepUnit__0.insertedLoads 11586082 # Number of loads inserted to the mem dependence unit. (Count)

board.processor.cores.core.MemDepUnit__0.insertedStores 4218782 # Number of stores inserted to the mem dependence unit. (Count)

board.processor.cores.core.MemDepUnit__0.conflictingLoads 7481485 # Number of conflicting loads. (Count)

board.processor.cores.core.MemDepUnit__0.conflictingStores 1287160 # Number of conflicting stores. (Count)

board.processor.cores.core.MemDepUnit__1.insertedLoads 0 # Number of loads inserted to the mem dependence unit. (Count)

board.processor.cores.core.MemDepUnit__1.insertedStores 0 # Number of stores inserted to the mem dependence unit. (Count)

board.processor.cores.core.MemDepUnit__1.conflictingLoads 0 # Number of conflicting loads. (Count)

board.processor.cores.core.MemDepUnit__1.conflictingStores 0 # Number of conflicting stores. (Count)

board.processor.cores.core.MemDepUnit__2.insertedLoads 0 # Number of loads inserted to the mem dependence unit. (Count)

board.processor.cores.core.MemDepUnit__2.insertedStores 0 # Number of stores inserted to the mem dependence unit. (Count)

board.processor.cores.core.MemDepUnit__2.conflictingLoads 0 # Number of conflicting loads. (Count)

board.processor.cores.core.MemDepUnit__2.conflictingStores 0 # Number of conflicting stores. (Count)

board.processor.cores.core.MemDepUnit__3.insertedLoads 0 # Number of loads inserted to the mem dependence unit. (Count)

board.processor.cores.core.MemDepUnit__3.insertedStores 0 # Number of stores inserted to the mem dependence unit. (Count)

board.processor.cores.core.MemDepUnit__3.conflictingLoads 0 # Number of conflicting loads. (Count)

board.processor.cores.core.MemDepUnit__3.conflictingStores 0 # Number of conflicting stores. (Count)

board.processor.cores.core.branchPred.lookups_0::NoBranch 4 0.00% 0.00% # Number of BP lookups (Count)

board.processor.cores.core.branchPred.lookups_0::Return 2777 0.13% 0.13% # Number of BP lookups (Count)

board.processor.cores.core.branchPred.lookups_0::CallDirect 2860 0.13% 0.26% # Number of BP lookups (Count)

board.processor.cores.core.branchPred.lookups_0::CallIndirect 174 0.01% 0.27% # Number of BP lookups (Count)

board.processor.cores.core.branchPred.lookups_0::DirectCond 2130476 99.44% 99.71% # Number of BP lookups (Count)

board.processor.cores.core.branchPred.lookups_0::DirectUncond 3746 0.17% 99.89% # Number of BP lookups (Count)

board.processor.cores.core.branchPred.lookups_0::IndirectCond 0 0.00% 99.89% # Number of BP lookups (Count)

board.processor.cores.core.branchPred.lookups_0::IndirectUncond 2440 0.11% 100.00% # Number of BP lookups (Count)

board.processor.cores.core.branchPred.lookups_0::total 2142477 # Number of BP lookups (Count)

board.processor.cores.core.branchPred.squashes_0::NoBranch 4 0.03% 0.03% # Number of branches that got squashed (completely removed) as an earlier branch was mispredicted. (Count)

board.processor.cores.core.branchPred.squashes_0::Return 408 3.39% 3.43% # Number of branches that got squashed (completely removed) as an earlier branch was mispredicted. (Count)

board.processor.cores.core.branchPred.squashes_0::CallDirect 533 4.43% 7.86% # Number of branches that got squashed (completely removed) as an earlier branch was mispredicted. (Count)

board.processor.cores.core.branchPred.squashes_0::CallIndirect 127 1.06% 8.92% # Number of branches that got squashed (completely removed) as an earlier branch was mispredicted. (Count)

board.processor.cores.core.branchPred.squashes_0::DirectCond 10218 85.02% 93.93% # Number of branches that got squashed (completely removed) as an earlier branch was mispredicted. (Count)

board.processor.cores.core.branchPred.squashes_0::DirectUncond 427 3.55% 97.49% # Number of branches that got squashed (completely removed) as an earlier branch was mispredicted. (Count)

board.processor.cores.core.branchPred.squashes_0::IndirectCond 0 0.00% 97.49% # Number of branches that got squashed (completely removed) as an earlier branch was mispredicted. (Count)

board.processor.cores.core.branchPred.squashes_0::IndirectUncond 302 2.51% 100.00% # Number of branches that got squashed (completely removed) as an earlier branch was mispredicted. (Count)

board.processor.cores.core.branchPred.squashes_0::total 12019 # Number of branches that got squashed (completely removed) as an earlier branch was mispredicted. (Count)

board.processor.cores.core.branchPred.corrected_0::NoBranch 0 0.00% 0.00% # Number of branches that got corrected but not yet commited. Branches get corrected by decode or after execute. Also a branch misprediction can be detected out-of-order. Therefore, a corrected branch might not end up beeing committed in case an even earlier branch was mispredicted (Count)

board.processor.cores.core.branchPred.corrected_0::Return 10 0.93% 0.93% # Number of branches that got corrected but not yet commited. Branches get corrected by decode or after execute. Also a branch misprediction can be detected out-of-order. Therefore, a corrected branch might not end up beeing committed in case an even earlier branch was mispredicted (Count)

board.processor.cores.core.branchPred.corrected_0::CallDirect 223 20.84% 21.78% # Number of branches that got corrected but not yet commited. Branches get corrected by decode or after execute. Also a branch misprediction can be detected out-of-order. Therefore, a corrected branch might not end up beeing committed in case an even earlier branch was mispredicted (Count)

board.processor.cores.core.branchPred.corrected_0::CallIndirect 39 3.64% 25.42% # Number of branches that got corrected but not yet commited. Branches get corrected by decode or after execute. Also a branch misprediction can be detected out-of-order. Therefore, a corrected branch might not end up beeing committed in case an even earlier branch was mispredicted (Count)

board.processor.cores.core.branchPred.corrected_0::DirectCond 601 56.17% 81.59% # Number of branches that got corrected but not yet commited. Branches get corrected by decode or after execute. Also a branch misprediction can be detected out-of-order. Therefore, a corrected branch might not end up beeing committed in case an even earlier branch was mispredicted (Count)

board.processor.cores.core.branchPred.corrected_0::DirectUncond 148 13.83% 95.42% # Number of branches that got corrected but not yet commited. Branches get corrected by decode or after execute. Also a branch misprediction can be detected out-of-order. Therefore, a corrected branch might not end up beeing committed in case an even earlier branch was mispredicted (Count)

board.processor.cores.core.branchPred.corrected_0::IndirectCond 0 0.00% 95.42% # Number of branches that got corrected but not yet commited. Branches get corrected by decode or after execute. Also a branch misprediction can be detected out-of-order. Therefore, a corrected branch might not end up beeing committed in case an even earlier branch was mispredicted (Count)

board.processor.cores.core.branchPred.corrected_0::IndirectUncond 49 4.58% 100.00% # Number of branches that got corrected but not yet commited. Branches get corrected by decode or after execute. Also a branch misprediction can be detected out-of-order. Therefore, a corrected branch might not end up beeing committed in case an even earlier branch was mispredicted (Count)

board.processor.cores.core.branchPred.corrected_0::total 1070 # Number of branches that got corrected but not yet commited. Branches get corrected by decode or after execute. Also a branch misprediction can be detected out-of-order. Therefore, a corrected branch might not end up beeing committed in case an even earlier branch was mispredicted (Count)

board.processor.cores.core.branchPred.earlyResteers_0::NoBranch 0 # Number of branches that got redirected after decode. (Count)

board.processor.cores.core.branchPred.earlyResteers_0::Return 0 # Number of branches that got redirected after decode. (Count)

board.processor.cores.core.branchPred.earlyResteers_0::CallDirect 0 # Number of branches that got redirected after decode. (Count)

board.processor.cores.core.branchPred.earlyResteers_0::CallIndirect 0 # Number of branches that got redirected after decode. (Count)

board.processor.cores.core.branchPred.earlyResteers_0::DirectCond 0 # Number of branches that got redirected after decode. (Count)

board.processor.cores.core.branchPred.earlyResteers_0::DirectUncond 0 # Number of branches that got redirected after decode. (Count)

board.processor.cores.core.branchPred.earlyResteers_0::IndirectCond 0 # Number of branches that got redirected after decode. (Count)

board.processor.cores.core.branchPred.earlyResteers_0::IndirectUncond 0 # Number of branches that got redirected after decode. (Count)

board.processor.cores.core.branchPred.earlyResteers_0::total 0 # Number of branches that got redirected after decode. (Count)

board.processor.cores.core.branchPred.committed_0::NoBranch 0 0.00% 0.00% # Number of branches finally committed (Count)

board.processor.cores.core.branchPred.committed_0::Return 2369 0.11% 0.11% # Number of branches finally committed (Count)

board.processor.cores.core.branchPred.committed_0::CallDirect 2327 0.11% 0.22% # Number of branches finally committed (Count)

board.processor.cores.core.branchPred.committed_0::CallIndirect 47 0.00% 0.22% # Number of branches finally committed (Count)

board.processor.cores.core.branchPred.committed_0::DirectCond 2120254 99.52% 99.74% # Number of branches finally committed (Count)

board.processor.cores.core.branchPred.committed_0::DirectUncond 3319 0.16% 99.90% # Number of branches finally committed (Count)

board.processor.cores.core.branchPred.committed_0::IndirectCond 0 0.00% 99.90% # Number of branches finally committed (Count)

board.processor.cores.core.branchPred.committed_0::IndirectUncond 2138 0.10% 100.00% # Number of branches finally committed (Count)

board.processor.cores.core.branchPred.committed_0::total 2130454 # Number of branches finally committed (Count)

board.processor.cores.core.branchPred.mispredicted_0::NoBranch 0 0.00% 0.00% # Number of committed branches that were mispredicted. (Count)

board.processor.cores.core.branchPred.mispredicted_0::Return 0 0.00% 0.00% # Number of committed branches that were mispredicted. (Count)

board.processor.cores.core.branchPred.mispredicted_0::CallDirect 109 13.36% 13.36% # Number of committed branches that were mispredicted. (Count)

board.processor.cores.core.branchPred.mispredicted_0::CallIndirect 38 4.66% 18.01% # Number of committed branches that were mispredicted. (Count)

board.processor.cores.core.branchPred.mispredicted_0::DirectCond 539 66.05% 84.07% # Number of committed branches that were mispredicted. (Count)

board.processor.cores.core.branchPred.mispredicted_0::DirectUncond 87 10.66% 94.73% # Number of committed branches that were mispredicted. (Count)

board.processor.cores.core.branchPred.mispredicted_0::IndirectCond 0 0.00% 94.73% # Number of committed branches that were mispredicted. (Count)

board.processor.cores.core.branchPred.mispredicted_0::IndirectUncond 43 5.27% 100.00% # Number of committed branches that were mispredicted. (Count)

board.processor.cores.core.branchPred.mispredicted_0::total 816 # Number of committed branches that were mispredicted. (Count)

board.processor.cores.core.branchPred.targetProvider_0::NoTarget 22939 1.07% 1.07% # The component providing the target for taken branches (Count)

board.processor.cores.core.branchPred.targetProvider_0::BTB 2114562 98.70% 99.77% # The component providing the target for taken branches (Count)

board.processor.cores.core.branchPred.targetProvider_0::RAS 2777 0.13% 99.90% # The component providing the target for taken branches (Count)

board.processor.cores.core.branchPred.targetProvider_0::Indirect 2199 0.10% 100.00% # The component providing the target for taken branches (Count)

board.processor.cores.core.branchPred.targetProvider_0::total 2142477 # The component providing the target for taken branches (Count)

board.processor.cores.core.branchPred.targetWrong_0::NoBranch 968 92.10% 92.10% # Number of branches where the target was incorrect or not available at prediction time. (Count)

board.processor.cores.core.branchPred.targetWrong_0::Return 72 6.85% 98.95% # Number of branches where the target was incorrect or not available at prediction time. (Count)

board.processor.cores.core.branchPred.targetWrong_0::CallDirect 10 0.95% 99.90% # Number of branches where the target was incorrect or not available at prediction time. (Count)

board.processor.cores.core.branchPred.targetWrong_0::CallIndirect 1 0.10% 100.00% # Number of branches where the target was incorrect or not available at prediction time. (Count)

board.processor.cores.core.branchPred.targetWrong_0::DirectCond 0 0.00% 100.00% # Number of branches where the target was incorrect or not available at prediction time. (Count)

board.processor.cores.core.branchPred.targetWrong_0::DirectUncond 0 0.00% 100.00% # Number of branches where the target was incorrect or not available at prediction time. (Count)

board.processor.cores.core.branchPred.targetWrong_0::IndirectCond 0 0.00% 100.00% # Number of branches where the target was incorrect or not available at prediction time. (Count)

board.processor.cores.core.branchPred.targetWrong_0::IndirectUncond 0 0.00% 100.00% # Number of branches where the target was incorrect or not available at prediction time. (Count)

board.processor.cores.core.branchPred.targetWrong_0::total 1051 # Number of branches where the target was incorrect or not available at prediction time. (Count)

board.processor.cores.core.branchPred.condPredicted 2130480 # Number of conditional branches predicted (Count)

board.processor.cores.core.branchPred.condPredictedTaken 2109169 # Number of conditional branches predicted as taken (Count)

board.processor.cores.core.branchPred.condIncorrect 1070 # Number of conditional branches incorrect (Count)

board.processor.cores.core.branchPred.predTakenBTBMiss 483 # Number of branches predicted taken but missed in BTB (Count)

board.processor.cores.core.branchPred.NotTakenMispredicted 1005 # Number branches predicted 'not taken' but turned out to be taken (Count)

board.processor.cores.core.branchPred.TakenMispredicted 65 # Number branches predicted taken but are actually not taken (Count)

board.processor.cores.core.branchPred.BTBLookups 2142477 # Number of BTB lookups (Count)

board.processor.cores.core.branchPred.BTBUpdates 907 # Number of BTB updates (Count)

board.processor.cores.core.branchPred.BTBHits 2117826 # Number of BTB hits (Count)

board.processor.cores.core.branchPred.BTBHitRatio 0.988494 # BTB Hit Ratio (Ratio)

board.processor.cores.core.branchPred.BTBMispredicted 702 # Number BTB mispredictions. No target found or target wrong (Count)

board.processor.cores.core.branchPred.indirectLookups 2614 # Number of indirect predictor lookups. (Count)

board.processor.cores.core.branchPred.indirectHits 2199 # Number of indirect target hits. (Count)

board.processor.cores.core.branchPred.indirectMisses 415 # Number of indirect misses. (Count)

board.processor.cores.core.branchPred.indirectMispredicted 0 # Number of mispredicted indirect branches. (Count)

board.processor.cores.core.branchPred.btb.lookups::NoBranch 4 0.00% 0.00% # Number of BTB lookups (Count)

board.processor.cores.core.branchPred.btb.lookups::Return 2777 0.13% 0.13% # Number of BTB lookups (Count)

board.processor.cores.core.branchPred.btb.lookups::CallDirect 2860 0.13% 0.26% # Number of BTB lookups (Count)

board.processor.cores.core.branchPred.btb.lookups::CallIndirect 174 0.01% 0.27% # Number of BTB lookups (Count)

board.processor.cores.core.branchPred.btb.lookups::DirectCond 2130476 99.44% 99.71% # Number of BTB lookups (Count)

board.processor.cores.core.branchPred.btb.lookups::DirectUncond 3746 0.17% 99.89% # Number of BTB lookups (Count)

board.processor.cores.core.branchPred.btb.lookups::IndirectCond 0 0.00% 99.89% # Number of BTB lookups (Count)

board.processor.cores.core.branchPred.btb.lookups::IndirectUncond 2440 0.11% 100.00% # Number of BTB lookups (Count)

board.processor.cores.core.branchPred.btb.lookups::total 2142477 # Number of BTB lookups (Count)

board.processor.cores.core.branchPred.btb.misses::NoBranch 4 0.02% 0.02% # Number of BTB misses (Count)

board.processor.cores.core.branchPred.btb.misses::Return 2773 11.25% 11.27% # Number of BTB misses (Count)

board.processor.cores.core.branchPred.btb.misses::CallDirect 362 1.47% 12.73% # Number of BTB misses (Count)

board.processor.cores.core.branchPred.btb.misses::CallIndirect 174 0.71% 13.44% # Number of BTB misses (Count)

board.processor.cores.core.branchPred.btb.misses::DirectCond 18671 75.74% 89.18% # Number of BTB misses (Count)

board.processor.cores.core.branchPred.btb.misses::DirectUncond 227 0.92% 90.10% # Number of BTB misses (Count)

board.processor.cores.core.branchPred.btb.misses::IndirectCond 0 0.00% 90.10% # Number of BTB misses (Count)

board.processor.cores.core.branchPred.btb.misses::IndirectUncond 2440 9.90% 100.00% # Number of BTB misses (Count)

board.processor.cores.core.branchPred.btb.misses::total 24651 # Number of BTB misses (Count)

board.processor.cores.core.branchPred.btb.updates::NoBranch 0 0.00% 0.00% # Number of BTB updates (Count)

board.processor.cores.core.branchPred.btb.updates::Return 0 0.00% 0.00% # Number of BTB updates (Count)

board.processor.cores.core.branchPred.btb.updates::CallDirect 223 24.59% 24.59% # Number of BTB updates (Count)

board.processor.cores.core.branchPred.btb.updates::CallIndirect 0 0.00% 24.59% # Number of BTB updates (Count)

board.processor.cores.core.branchPred.btb.updates::DirectCond 536 59.10% 83.68% # Number of BTB updates (Count)

board.processor.cores.core.branchPred.btb.updates::DirectUncond 148 16.32% 100.00% # Number of BTB updates (Count)

board.processor.cores.core.branchPred.btb.updates::IndirectCond 0 0.00% 100.00% # Number of BTB updates (Count)

board.processor.cores.core.branchPred.btb.updates::IndirectUncond 0 0.00% 100.00% # Number of BTB updates (Count)

board.processor.cores.core.branchPred.btb.updates::total 907 # Number of BTB updates (Count)

board.processor.cores.core.branchPred.btb.mispredict::NoBranch 0 0.00% 0.00% # Number of BTB mispredictions. No target found or target wrong. (Count)

board.processor.cores.core.branchPred.btb.mispredict::Return 0 0.00% 0.00% # Number of BTB mispredictions. No target found or target wrong. (Count)

board.processor.cores.core.branchPred.btb.mispredict::CallDirect 223 24.59% 24.59% # Number of BTB mispredictions. No target found or target wrong. (Count)

board.processor.cores.core.branchPred.btb.mispredict::CallIndirect 0 0.00% 24.59% # Number of BTB mispredictions. No target found or target wrong. (Count)

board.processor.cores.core.branchPred.btb.mispredict::DirectCond 536 59.10% 83.68% # Number of BTB mispredictions. No target found or target wrong. (Count)

board.processor.cores.core.branchPred.btb.mispredict::DirectUncond 148 16.32% 100.00% # Number of BTB mispredictions. No target found or target wrong. (Count)

board.processor.cores.core.branchPred.btb.mispredict::IndirectCond 0 0.00% 100.00% # Number of BTB mispredictions. No target found or target wrong. (Count)

board.processor.cores.core.branchPred.btb.mispredict::IndirectUncond 0 0.00% 100.00% # Number of BTB mispredictions. No target found or target wrong. (Count)

board.processor.cores.core.branchPred.btb.mispredict::total 907 # Number of BTB mispredictions. No target found or target wrong. (Count)

board.processor.cores.core.branchPred.btb.power_state.pwrStateResidencyTicks::UNDEFINED 5909877540 # Cumulative time (in ticks) in various power states (Tick)

board.processor.cores.core.branchPred.indirectBranchPred.lookups 2614 # Number of lookups (Count)

board.processor.cores.core.branchPred.indirectBranchPred.hits 2199 # Number of hits of a tag (Count)

board.processor.cores.core.branchPred.indirectBranchPred.misses 415 # Number of misses (Count)

board.processor.cores.core.branchPred.indirectBranchPred.targetRecords 88 # Number of targets that where recorded/installed in the cache (Count)

board.processor.cores.core.branchPred.indirectBranchPred.indirectRecords 2702 # Number of indirect branches/calls recorded in the indirect hist (Count)

board.processor.cores.core.branchPred.indirectBranchPred.speculativeOverflows 2 # Number of times more than the allowed capacity for speculative branches/calls where in flight and destroy the path history (Count)

board.processor.cores.core.branchPred.ras.pushes 3442 # Number of times a PC was pushed onto the RAS (Count)

board.processor.cores.core.branchPred.ras.pops 3437 # Number of times a PC was poped from the RAS (Count)

board.processor.cores.core.branchPred.ras.squashes 1068 # Number of times the stack operation was squashed due to wrong speculation. (Count)

board.processor.cores.core.branchPred.ras.used 2369 # Number of times the RAS is the provider (Count)

board.processor.cores.core.branchPred.ras.correct 2369 # Number of times the RAS is the provider and the prediction is correct (Count)

board.processor.cores.core.branchPred.ras.incorrect 0 # Number of times the RAS is the provider and the prediction is wrong (Count)

board.processor.cores.core.commit.commitSquashedInsts 147495 # The number of squashed insts skipped by commit (Count)

board.processor.cores.core.commit.commitNonSpecStalls 90 # The number of times commit has been forced to stall to communicate backwards (Count)

board.processor.cores.core.commit.branchMispredicts 738 # The number of times a branch was mispredicted (Count)

board.processor.cores.core.commit.numCommittedDist::samples 17581975 # Number of insts commited each cycle (Count)

board.processor.cores.core.commit.numCommittedDist::mean 3.234063 # Number of insts commited each cycle (Count)

board.processor.cores.core.commit.numCommittedDist::stdev 3.205009 # Number of insts commited each cycle (Count)

board.processor.cores.core.commit.numCommittedDist::underflows 0 0.00% 0.00% # Number of insts commited each cycle (Count)

board.processor.cores.core.commit.numCommittedDist::0 2335171 13.28% 13.28% # Number of insts commited each cycle (Count)

board.processor.cores.core.commit.numCommittedDist::1 7278537 41.40% 54.68% # Number of insts commited each cycle (Count)

board.processor.cores.core.commit.numCommittedDist::2 1763911 10.03% 64.71% # Number of insts commited each cycle (Count)

board.processor.cores.core.commit.numCommittedDist::3 123690 0.70% 65.42% # Number of insts commited each cycle (Count)

board.processor.cores.core.commit.numCommittedDist::4 123569 0.70% 66.12% # Number of insts commited each cycle (Count)

board.processor.cores.core.commit.numCommittedDist::5 121415 0.69% 66.81% # Number of insts commited each cycle (Count)

board.processor.cores.core.commit.numCommittedDist::6 1049024 5.97% 72.78% # Number of insts commited each cycle (Count)

board.processor.cores.core.commit.numCommittedDist::7 4980 0.03% 72.80% # Number of insts commited each cycle (Count)

board.processor.cores.core.commit.numCommittedDist::8 4781678 27.20% 100.00% # Number of insts commited each cycle (Count)

board.processor.cores.core.commit.numCommittedDist::overflows 0 0.00% 100.00% # Number of insts commited each cycle (Count)

board.processor.cores.core.commit.numCommittedDist::min_value 0 # Number of insts commited each cycle (Count)

board.processor.cores.core.commit.numCommittedDist::max_value 8 # Number of insts commited each cycle (Count)

board.processor.cores.core.commit.numCommittedDist::total 17581975 # Number of insts commited each cycle (Count)

board.processor.cores.core.commit.amos 0 # Number of atomic instructions committed (Count)

board.processor.cores.core.commit.membars 60 # Number of memory barriers committed (Count)

board.processor.cores.core.commit.functionCalls 2374 # Number of function calls committed. (Count)

board.processor.cores.core.commit.committedInstType_0::No_OpClass 316 0.00% 0.00% # Class of committed instruction (Count)

board.processor.cores.core.commit.committedInstType_0::IntAlu 40052454 70.44% 70.44% # Class of committed instruction (Count)

board.processor.cores.core.commit.committedInstType_0::IntMult 1048624 1.84% 72.28% # Class of committed instruction (Count)

board.processor.cores.core.commit.committedInstType_0::IntDiv 28 0.00% 72.28% # Class of committed instruction (Count)

board.processor.cores.core.commit.committedInstType_0::FloatAdd 154 0.00% 72.28% # Class of committed instruction (Count)

board.processor.cores.core.commit.committedInstType_0::FloatCmp 0 0.00% 72.28% # Class of committed instruction (Count)

board.processor.cores.core.commit.committedInstType_0::FloatCvt 0 0.00% 72.28% # Class of committed instruction (Count)

board.processor.cores.core.commit.committedInstType_0::FloatMult 0 0.00% 72.28% # Class of committed instruction (Count)

board.processor.cores.core.commit.committedInstType_0::FloatMultAcc 0 0.00% 72.28% # Class of committed instruction (Count)

board.processor.cores.core.commit.committedInstType_0::FloatDiv 0 0.00% 72.28% # Class of committed instruction (Count)

board.processor.cores.core.commit.committedInstType_0::FloatMisc 0 0.00% 72.28% # Class of committed instruction (Count)

board.processor.cores.core.commit.committedInstType_0::FloatSqrt 0 0.00% 72.28% # Class of committed instruction (Count)

board.processor.cores.core.commit.committedInstType_0::SimdAdd 10 0.00% 72.28% # Class of committed instruction (Count)

board.processor.cores.core.commit.committedInstType_0::SimdAddAcc 0 0.00% 72.28% # Class of committed instruction (Count)

board.processor.cores.core.commit.committedInstType_0::SimdAlu 229 0.00% 72.28% # Class of committed instruction (Count)

board.processor.cores.core.commit.committedInstType_0::SimdCmp 0 0.00% 72.28% # Class of committed instruction (Count)

board.processor.cores.core.commit.committedInstType_0::SimdCvt 72 0.00% 72.28% # Class of committed instruction (Count)

board.processor.cores.core.commit.committedInstType_0::SimdMisc 240 0.00% 72.29% # Class of committed instruction (Count)

board.processor.cores.core.commit.committedInstType_0::SimdMult 0 0.00% 72.29% # Class of committed instruction (Count)

board.processor.cores.core.commit.committedInstType_0::SimdMultAcc 0 0.00% 72.29% # Class of committed instruction (Count)

board.processor.cores.core.commit.committedInstType_0::SimdMatMultAcc 0 0.00% 72.29% # Class of committed instruction (Count)

board.processor.cores.core.commit.committedInstType_0::SimdShift 3 0.00% 72.29% # Class of committed instruction (Count)

board.processor.cores.core.commit.committedInstType_0::SimdShiftAcc 0 0.00% 72.29% # Class of committed instruction (Count)

board.processor.cores.core.commit.committedInstType_0::SimdDiv 0 0.00% 72.29% # Class of committed instruction (Count)

board.processor.cores.core.commit.committedInstType_0::SimdSqrt 0 0.00% 72.29% # Class of committed instruction (Count)

board.processor.cores.core.commit.committedInstType_0::SimdFloatAdd 0 0.00% 72.29% # Class of committed instruction (Count)

board.processor.cores.core.commit.committedInstType_0::SimdFloatAlu 0 0.00% 72.29% # Class of committed instruction (Count)

board.processor.cores.core.commit.committedInstType_0::SimdFloatCmp 0 0.00% 72.29% # Class of committed instruction (Count)

board.processor.cores.core.commit.committedInstType_0::SimdFloatCvt 0 0.00% 72.29% # Class of committed instruction (Count)

board.processor.cores.core.commit.committedInstType_0::SimdFloatDiv 0 0.00% 72.29% # Class of committed instruction (Count)

board.processor.cores.core.commit.committedInstType_0::SimdFloatMisc 0 0.00% 72.29% # Class of committed instruction (Count)

board.processor.cores.core.commit.committedInstType_0::SimdFloatMult 0 0.00% 72.29% # Class of committed instruction (Count)

board.processor.cores.core.commit.committedInstType_0::SimdFloatMultAcc 0 0.00% 72.29% # Class of committed instruction (Count)

board.processor.cores.core.commit.committedInstType_0::SimdFloatMatMultAcc 0 0.00% 72.29% # Class of committed instruction (Count)

board.processor.cores.core.commit.committedInstType_0::SimdFloatSqrt 0 0.00% 72.29% # Class of committed instruction (Count)

board.processor.cores.core.commit.committedInstType_0::SimdReduceAdd 0 0.00% 72.29% # Class of committed instruction (Count)

board.processor.cores.core.commit.committedInstType_0::SimdReduceAlu 0 0.00% 72.29% # Class of committed instruction (Count)

board.processor.cores.core.commit.committedInstType_0::SimdReduceCmp 0 0.00% 72.29% # Class of committed instruction (Count)

board.processor.cores.core.commit.committedInstType_0::SimdFloatReduceAdd 0 0.00% 72.29% # Class of committed instruction (Count)

board.processor.cores.core.commit.committedInstType_0::SimdFloatReduceCmp 0 0.00% 72.29% # Class of committed instruction (Count)

board.processor.cores.core.commit.committedInstType_0::SimdAes 0 0.00% 72.29% # Class of committed instruction (Count)

board.processor.cores.core.commit.committedInstType_0::SimdAesMix 0 0.00% 72.29% # Class of committed instruction (Count)

board.processor.cores.core.commit.committedInstType_0::SimdSha1Hash 0 0.00% 72.29% # Class of committed instruction (Count)

board.processor.cores.core.commit.committedInstType_0::SimdSha1Hash2 0 0.00% 72.29% # Class of committed instruction (Count)

board.processor.cores.core.commit.committedInstType_0::SimdSha256Hash 0 0.00% 72.29% # Class of committed instruction (Count)

board.processor.cores.core.commit.committedInstType_0::SimdSha256Hash2 0 0.00% 72.29% # Class of committed instruction (Count)

board.processor.cores.core.commit.committedInstType_0::SimdShaSigma2 0 0.00% 72.29% # Class of committed instruction (Count)

board.processor.cores.core.commit.committedInstType_0::SimdShaSigma3 0 0.00% 72.29% # Class of committed instruction (Count)

board.processor.cores.core.commit.committedInstType_0::SimdPredAlu 0 0.00% 72.29% # Class of committed instruction (Count)

board.processor.cores.core.commit.committedInstType_0::Matrix 0 0.00% 72.29% # Class of committed instruction (Count)

board.processor.cores.core.commit.committedInstType_0::MatrixMov 0 0.00% 72.29% # Class of committed instruction (Count)

board.processor.cores.core.commit.committedInstType_0::MatrixOP 0 0.00% 72.29% # Class of committed instruction (Count)

board.processor.cores.core.commit.committedInstType_0::MemRead 11553787 20.32% 92.60% # Class of committed instruction (Count)

board.processor.cores.core.commit.committedInstType_0::MemWrite 4204624 7.39% 100.00% # Class of committed instruction (Count)

board.processor.cores.core.commit.committedInstType_0::FloatMemRead 125 0.00% 100.00% # Class of committed instruction (Count)

board.processor.cores.core.commit.committedInstType_0::FloatMemWrite 542 0.00% 100.00% # Class of committed instruction (Count)

board.processor.cores.core.commit.committedInstType_0::IprAccess 0 0.00% 100.00% # Class of committed instruction (Count)

board.processor.cores.core.commit.committedInstType_0::InstPrefetch 0 0.00% 100.00% # Class of committed instruction (Count)

board.processor.cores.core.commit.committedInstType_0::SimdUnitStrideLoad 0 0.00% 100.00% # Class of committed instruction (Count)

board.processor.cores.core.commit.committedInstType_0::SimdUnitStrideStore 0 0.00% 100.00% # Class of committed instruction (Count)

board.processor.cores.core.commit.committedInstType_0::SimdUnitStrideMaskLoad 0 0.00% 100.00% # Class of committed instruction (Count)

board.processor.cores.core.commit.committedInstType_0::SimdUnitStrideMaskStore 0 0.00% 100.00% # Class of committed instruction (Count)

board.processor.cores.core.commit.committedInstType_0::SimdStridedLoad 0 0.00% 100.00% # Class of committed instruction (Count)

board.processor.cores.core.commit.committedInstType_0::SimdStridedStore 0 0.00% 100.00% # Class of committed instruction (Count)

board.processor.cores.core.commit.committedInstType_0::SimdIndexedLoad 0 0.00% 100.00% # Class of committed instruction (Count)

board.processor.cores.core.commit.committedInstType_0::SimdIndexedStore 0 0.00% 100.00% # Class of committed instruction (Count)

board.processor.cores.core.commit.committedInstType_0::SimdWholeRegisterLoad 0 0.00% 100.00% # Class of committed instruction (Count)

board.processor.cores.core.commit.committedInstType_0::SimdWholeRegisterStore 0 0.00% 100.00% # Class of committed instruction (Count)

board.processor.cores.core.commit.committedInstType_0::SimdUnitStrideFaultOnlyFirstLoad 0 0.00% 100.00% # Class of committed instruction (Count)

board.processor.cores.core.commit.committedInstType_0::SimdUnitStrideSegmentedLoad 0 0.00% 100.00% # Class of committed instruction (Count)

board.processor.cores.core.commit.committedInstType_0::SimdUnitStrideSegmentedStore 0 0.00% 100.00% # Class of committed instruction (Count)

board.processor.cores.core.commit.committedInstType_0::SimdExt 0 0.00% 100.00% # Class of committed instruction (Count)

board.processor.cores.core.commit.committedInstType_0::SimdFloatExt 0 0.00% 100.00% # Class of committed instruction (Count)

board.processor.cores.core.commit.committedInstType_0::SimdConfig 0 0.00% 100.00% # Class of committed instruction (Count)

board.processor.cores.core.commit.committedInstType_0::total 56861208 # Class of committed instruction (Count)

board.processor.cores.core.commit.commitEligibleSamples 4781678 # number cycles where commit BW limit reached (Cycle)

board.processor.cores.core.commitStats0.numInsts 36831240 # Number of instructions committed (thread level) (Count)

board.processor.cores.core.commitStats0.numOps 56861208 # Number of ops (including micro ops) committed (thread level) (Count)

board.processor.cores.core.commitStats0.numInstsNotNOP 36831240 # Number of instructions committed excluding NOPs or prefetches (Count)

board.processor.cores.core.commitStats0.numOpsNotNOP 56861208 # Number of Ops (including micro ops) Simulated (Count)

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board.processor.cores.core.commitStats0.cpi     0.481857                    # CPI: cycles per instruction (thread level) ((Cycle/Count))
board.processor.cores.core.commitStats0.ipc     2.075306                    # IPC: instructions per cycle (thread level) ((Count/Cycle))
board.processor.cores.core.commitStats0.numMemRefs     15759078                    # Number of memory references committed (Count)
board.processor.cores.core.commitStats0.numFpInsts     1535                    # Number of float instructions (Count)
board.processor.cores.core.commitStats0.numIntInsts     56859973                    # Number of integer instructions (Count)
board.processor.cores.core.commitStats0.numLoadInsts     11553912                    # Number of load instructions (Count)
board.processor.cores.core.commitStats0.numStoreInsts     4205166                    # Number of store instructions (Count)
board.processor.cores.core.commitStats0.numVecInsts     0                    # Number of vector instructions (Count)
board.processor.cores.core.commitStats0.committedInstType::No_OpClass     316     0.00%     0.00% # Class of committed instruction. (Count)
board.processor.cores.core.commitStats0.committedInstType::IntAlu     40052454     70.44%     70.44% # Class of committed instruction. (Count)
board.processor.cores.core.commitStats0.committedInstType::IntMult     1048624     1.84%     72.28% # Class of committed instruction. (Count)
board.processor.cores.core.commitStats0.committedInstType::IntDiv     28     0.00%     72.28% # Class of committed instruction. (Count)
board.processor.cores.core.commitStats0.committedInstType::FloatAdd     154     0.00%     72.28% # Class of committed instruction. (Count)
board.processor.cores.core.commitStats0.committedInstType::FloatCmp     0     0.00%     72.28% # Class of committed instruction. (Count)
board.processor.cores.core.commitStats0.committedInstType::FloatCvt     0     0.00%     72.28% # Class of committed instruction. (Count)
board.processor.cores.core.commitStats0.committedInstType::FloatMult     0     0.00%     72.28% # Class of committed instruction. (Count)
board.processor.cores.core.commitStats0.committedInstType::FloatMultAcc     0     0.00%     72.28% # Class of committed instruction. (Count)
board.processor.cores.core.commitStats0.committedInstType::FloatDiv     0     0.00%     72.28% # Class of committed instruction. (Count)
board.processor.cores.core.commitStats0.committedInstType::FloatMisc     0     0.00%     72.28% # Class of committed instruction. (Count)
board.processor.cores.core.commitStats0.committedInstType::FloatSqrt     0     0.00%     72.28% # Class of committed instruction. (Count)
board.processor.cores.core.commitStats0.committedInstType::SimdAdd     10     0.00%     72.28% # Class of committed instruction. (Count)
board.processor.cores.core.commitStats0.committedInstType::SimdAddAcc     0     0.00%     72.28% # Class of committed instruction. (Count)
board.processor.cores.core.commitStats0.committedInstType::SimdAlu     229     0.00%     72.28% # Class of committed instruction. (Count)
board.processor.cores.core.commitStats0.committedInstType::SimdCmp     0     0.00%     72.28% # Class of committed instruction. (Count)
board.processor.cores.core.commitStats0.committedInstType::SimdCvt     72     0.00%     72.28% # Class of committed instruction. (Count)
board.processor.cores.core.commitStats0.committedInstType::SimdMisc     240     0.00%     72.29% # Class of committed instruction. (Count)
board.processor.cores.core.commitStats0.committedInstType::SimdMult     0     0.00%     72.29% # Class of committed instruction. (Count)
board.processor.cores.core.commitStats0.committedInstType::SimdMultAcc     0     0.00%     72.29% # Class of committed instruction. (Count)
board.processor.cores.core.commitStats0.committedInstType::SimdMatMultAcc     0     0.00%     72.29% # Class of committed instruction. (Count)
board.processor.cores.core.commitStats0.committedInstType::SimdShift     3     0.00%     72.29% # Class of committed instruction. (Count)
board.processor.cores.core.commitStats0.committedInstType::SimdShiftAcc     0     0.00%     72.29% # Class of committed instruction. (Count)
board.processor.cores.core.commitStats0.committedInstType::SimdDiv     0     0.00%     72.29% # Class of committed instruction. (Count)
board.processor.cores.core.commitStats0.committedInstType::SimdSqrt     0     0.00%     72.29% # Class of committed instruction. (Count)
board.processor.cores.core.commitStats0.committedInstType::SimdFloatAdd     0     0.00%     72.29% # Class of committed instruction. (Count)
board.processor.cores.core.commitStats0.committedInstType::SimdFloatAlu     0     0.00%     72.29% # Class of committed instruction. (Count)
board.processor.cores.core.commitStats0.committedInstType::SimdFloatCmp     0     0.00%     72.29% # Class of committed instruction. (Count)
board.processor.cores.core.commitStats0.committedInstType::SimdFloatCvt     0     0.00%     72.29% # Class of committed instruction. (Count)
board.processor.cores.core.commitStats0.committedInstType::SimdFloatDiv     0     0.00%     72.29% # Class of committed instruction. (Count)
board.processor.cores.core.commitStats0.committedInstType::SimdFloatMisc     0     0.00%     72.29% # Class of committed instruction. (Count)
board.processor.cores.core.commitStats0.committedInstType::SimdFloatMult     0     0.00%     72.29% # Class of committed instruction. (Count)
board.processor.cores.core.commitStats0.committedInstType::SimdFloatMultAcc     0     0.00%     72.29% # Class of committed instruction. (Count)
board.processor.cores.core.commitStats0.committedInstType::SimdFloatMatMultAcc     0     0.00%     72.29% # Class of committed instruction. (Count)
board.processor.cores.core.commitStats0.committedInstType::SimdFloatSqrt     0     0.00%     72.29% # Class of committed instruction. (Count)
board.processor.cores.core.commitStats0.committedInstType::SimdReduceAdd     0     0.00%     72.29% # Class of committed instruction. (Count)
board.processor.cores.core.commitStats0.committedInstType::SimdReduceAlu     0     0.00%     72.29% # Class of committed instruction. (Count)
board.processor.cores.core.commitStats0.committedInstType::SimdReduceCmp     0     0.00%     72.29% # Class of committed instruction. (Count)
board.processor.cores.core.commitStats0.committedInstType::SimdFloatReduceAdd     0     0.00%     72.29% # Class of committed instruction. (Count)
board.processor.cores.core.commitStats0.committedInstType::SimdFloatReduceCmp     0     0.00%     72.29% # Class of committed instruction. (Count)
board.processor.cores.core.commitStats0.committedInstType::SimdAes     0     0.00%     72.29% # Class of committed instruction. (Count)
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board.processor.cores.core.commitStats0.committedInstType::SimdAesMix 0 0.00% 72.29% # Class of committed instruction. (Count)

board.processor.cores.core.commitStats0.committedInstType::SimdSha1Hash 0 0.00% 72.29% # Class of committed instruction. (Count)

board.processor.cores.core.commitStats0.committedInstType::SimdSha1Hash2 0 0.00% 72.29% # Class of committed instruction. (Count)

board.processor.cores.core.commitStats0.committedInstType::SimdSha256Hash 0 0.00% 72.29% # Class of committed instruction. (Count)

board.processor.cores.core.commitStats0.committedInstType::SimdSha256Hash2 0 0.00% 72.29% # Class of committed instruction. (Count)

board.processor.cores.core.commitStats0.committedInstType::SimdShaSigma2 0 0.00% 72.29% # Class of committed instruction. (Count)

board.processor.cores.core.commitStats0.committedInstType::SimdShaSigma3 0 0.00% 72.29% # Class of committed instruction. (Count)

board.processor.cores.core.commitStats0.committedInstType::SimdPredAlu 0 0.00% 72.29% # Class of committed instruction. (Count)

board.processor.cores.core.commitStats0.committedInstType::Matrix 0 0.00% 72.29% # Class of committed instruction. (Count)

board.processor.cores.core.commitStats0.committedInstType::MatrixMov 0 0.00% 72.29% # Class of committed instruction. (Count)

board.processor.cores.core.commitStats0.committedInstType::MatrixOP 0 0.00% 72.29% # Class of committed instruction. (Count)

board.processor.cores.core.commitStats0.committedInstType::MemRead 11553787 20.32% 92.60% # Class of committed instruction. (Count)

board.processor.cores.core.commitStats0.committedInstType::MemWrite 4204624 7.39% 100.00% # Class of committed instruction. (Count)

board.processor.cores.core.commitStats0.committedInstType::FloatMemRead 125 0.00% 100.00% # Class of committed instruction. (Count)

board.processor.cores.core.commitStats0.committedInstType::FloatMemWrite 542 0.00% 100.00% # Class of committed instruction. (Count)

board.processor.cores.core.commitStats0.committedInstType::IprAccess 0 0.00% 100.00% # Class of committed instruction. (Count)

board.processor.cores.core.commitStats0.committedInstType::InstPrefetch 0 0.00% 100.00% # Class of committed instruction. (Count)

board.processor.cores.core.commitStats0.committedInstType::SimdUnitStrideLoad 0 0.00% 100.00% # Class of committed instruction. (Count)

board.processor.cores.core.commitStats0.committedInstType::SimdUnitStrideStore 0 0.00% 100.00% # Class of committed instruction. (Count)

board.processor.cores.core.commitStats0.committedInstType::SimdUnitStrideMaskLoad 0 0.00% 100.00% # Class of committed instruction. (Count)

board.processor.cores.core.commitStats0.committedInstType::SimdUnitStrideMaskStore 0 0.00% 100.00% # Class of committed instruction. (Count)

board.processor.cores.core.commitStats0.committedInstType::SimdStridedLoad 0 0.00% 100.00% # Class of committed instruction. (Count)

board.processor.cores.core.commitStats0.committedInstType::SimdStridedStore 0 0.00% 100.00% # Class of committed instruction. (Count)

board.processor.cores.core.commitStats0.committedInstType::SimdIndexedLoad 0 0.00% 100.00% # Class of committed instruction. (Count)

board.processor.cores.core.commitStats0.committedInstType::SimdIndexedStore 0 0.00% 100.00% # Class of committed instruction. (Count)

board.processor.cores.core.commitStats0.committedInstType::SimdWholeRegisterLoad 0 0.00% 100.00% # Class of committed instruction. (Count)

board.processor.cores.core.commitStats0.committedInstType::SimdWholeRegisterStore 0 0.00% 100.00% # Class of committed instruction. (Count)

board.processor.cores.core.commitStats0.committedInstType::SimdUnitStrideFaultOnlyFirstLoad 0 0.00% 100.00% # Class of committed instruction. (Count)

board.processor.cores.core.commitStats0.committedInstType::SimdUnitStrideSegmentedLoad 0 0.00% 100.00% # Class of committed instruction. (Count)

board.processor.cores.core.commitStats0.committedInstType::SimdUnitStrideSegmentedStore 0 0.00% 100.00% # Class of committed instruction. (Count)

board.processor.cores.core.commitStats0.committedInstType::SimdExt 0 0.00% 100.00% # Class of committed instruction. (Count)

board.processor.cores.core.commitStats0.committedInstType::SimdFloatExt 0 0.00% 100.00% # Class of committed instruction. (Count)

board.processor.cores.core.commitStats0.committedInstType::SimdConfig 0 0.00% 100.00% # Class of committed instruction. (Count)

board.processor.cores.core.commitStats0.committedInstType::total 56861208 # Class of committed instruction. (Count)

board.processor.cores.core.commitStats0.committedControl::IsControl 2130454 # Class of control type instructions committed (Count)

board.processor.cores.core.commitStats0.committedControl::IsDirectControl 2125900 # Class of control type instructions committed (Count)

board.processor.cores.core.commitStats0.committedControl::IsIndirectControl 4554 # Class of control type instructions committed (Count)

board.processor.cores.core.commitStats0.committedControl::IsCondControl 2120254 # Class of control type instructions committed (Count)

board.processor.cores.core.commitStats0.committedControl::IsUncondControl 10200 # Class of control type instructions committed (Count)

board.processor.cores.core.commitStats0.committedControl::IsCall 2374 # Class of control type instructions committed (Count)

board.processor.cores.core.commitStats0.committedControl::IsReturn 2369 # Class of control type instructions committed (Count)

board.processor.cores.core.decode.idleCycles 1702458 # Number of cycles decode is idle (Cycle)

board.processor.cores.core.decode.blockedCycles 8753563 # Number of cycles decode is blocked (Cycle)

board.processor.cores.core.decode.runCycles 3137555 # Number of cycles decode is running (Cycle)

board.processor.cores.core.decode.unblockCycles 4006208 # Number of cycles decode is unblocking (Cycle)

board.processor.cores.core.decode.squashCycles 1878 # Number of cycles decode is squashing (Cycle)

board.processor.cores.core.decode.branchResolved 2113618 # Number of times decode resolved a branch (Count)

board.processor.cores.core.decode.branchMispred 386 # Number of times decode detected a branch misprediction (Count)

board.processor.cores.core.decode.decodedInsts 57099515 # Number of instructions handled by decode (Count)

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board.processor.cores.core.decode.squashedInsts 1995 # Number of squashed instructions handled by decode (Count)
board.processor.cores.core.executeStats0.numInsts 57021299 # Number of executed instructions (Count)
board.processor.cores.core.executeStats0.numNop 0 # Number of nop insts executed (Count)
board.processor.cores.core.executeStats0.numBranches 2136461 # Number of branches executed (Count)
board.processor.cores.core.executeStats0.numLoadInsts 11580861 # Number of load instructions executed (Count)
board.processor.cores.core.executeStats0.numStoreInsts 4214501 # Number of stores executed (Count)
board.processor.cores.core.executeStats0.instRate 3.212942 # Inst execution rate ((Count/Cycle))
board.processor.cores.core.executeStats0.numCCRegReads 14860422 # Number of times the CC registers were read (Count)
board.processor.cores.core.executeStats0.numCCRegWrites 30569741 # Number of times the CC registers were written (Count)
board.processor.cores.core.executeStats0.numFpRegReads 2103 # Number of times the floating registers were read (Count)
board.processor.cores.core.executeStats0.numFpRegWrites 1115 # Number of times the floating registers were written (Count)
board.processor.cores.core.executeStats0.numIntRegReads 80182872 # Number of times the integer registers were read (Count)
board.processor.cores.core.executeStats0.numIntRegWrites 48542460 # Number of times the integer registers were written (Count)
board.processor.cores.core.executeStats0.numMemRefs 15795362 # Number of memory refs (Count)
board.processor.cores.core.executeStats0.numMiscRegReads 20069105 # Number of times the Misc registers were read (Count)
board.processor.cores.core.executeStats0.numVecAluAccesses 0 # Number of vector alu accesses (Count)
board.processor.cores.core.executeStats0.numDiscardedOps 0 # Number of ops (including micro ops) which were discarded before commit (Count)
board.processor.cores.core.fetch.predictedBranches 2119538 # Number of branches that fetch has predicted taken (Count)
board.processor.cores.core.fetch.cycles 17560312 # Number of cycles fetch has run and was not squashing or blocked (Cycle)
board.processor.cores.core.fetch.squashCycles 4504 # Number of cycles fetch has spent squashing (Cycle)
board.processor.cores.core.fetch.miscStallCycles 45 # Number of cycles fetch has spent waiting on interrupts, or bad addresses, or out of MSHRs (Cycle)
board.processor.cores.core.fetch.pendingTrapStallCycles 256 # Number of stall cycles due to pending traps (Cycle)
board.processor.cores.core.fetch.cacheLines 4232517 # Number of cache lines fetched (Count)
board.processor.cores.core.fetch.icacheSquashes 475 # Number of outstanding Icache misses that were squashed (Count)
board.processor.cores.core.fetch.nisnDist::samples 17601662 # Number of instructions fetched each cycle (Total) (Count)
board.processor.cores.core.fetch.nisnDist::mean 3.246580 # Number of instructions fetched each cycle (Total) (Count)
board.processor.cores.core.fetch.nisnDist::stdev 3.478868 # Number of instructions fetched each cycle (Total) (Count)
board.processor.cores.core.fetch.nisnDist::underflows 0 0.00% 0.00% # Number of instructions fetched each cycle (Total) (Count)
board.processor.cores.core.fetch.nisnDist::0 8025406 45.59% 45.59% # Number of instructions fetched each cycle (Total) (Count)
board.processor.cores.core.fetch.nisnDist::1 526109 2.99% 48.58% # Number of instructions fetched each cycle (Total) (Count)
board.processor.cores.core.fetch.nisnDist::2 858778 4.88% 53.46% # Number of instructions fetched each cycle (Total) (Count)
board.processor.cores.core.fetch.nisnDist::3 494694 2.81% 56.27% # Number of instructions fetched each cycle (Total) (Count)
board.processor.cores.core.fetch.nisnDist::4 1338844 7.61% 63.88% # Number of instructions fetched each cycle (Total) (Count)
board.processor.cores.core.fetch.nisnDist::5 528087 3.00% 66.88% # Number of instructions fetched each cycle (Total) (Count)
board.processor.cores.core.fetch.nisnDist::6 327363 1.86% 68.74% # Number of instructions fetched each cycle (Total) (Count)
board.processor.cores.core.fetch.nisnDist::7 561573 3.19% 71.93% # Number of instructions fetched each cycle (Total) (Count)
board.processor.cores.core.fetch.nisnDist::8 4940808 28.07% 100.00% # Number of instructions fetched each cycle (Total) (Count)
board.processor.cores.core.fetch.nisnDist::overflows 0 0.00% 100.00% # Number of instructions fetched each cycle (Total) (Count)
board.processor.cores.core.fetch.nisnDist::min_value 0 # Number of instructions fetched each cycle (Total) (Count)
board.processor.cores.core.fetch.nisnDist::max_value 8 # Number of instructions fetched each cycle (Total) (Count)
board.processor.cores.core.fetch.nisnDist::total 17601662 # Number of instructions fetched each cycle (Total) (Count)
board.processor.cores.core.fetchStats0.numInsts 37022374 # Number of instructions fetched (thread level) (Count)
board.processor.cores.core.fetchStats0.numOps 0 # Number of ops (including micro ops) fetched (thread level) (Count)
board.processor.cores.core.fetchStats0.fetchRate 2.086075 # Number of inst fetches per cycle ((Count/Cycle))
board.processor.cores.core.fetchStats0.numBranches 2142477 # Number of branches fetched (Count)
board.processor.cores.core.fetchStats0.branchRate 0.120721 # Number of branch fetches per cycle (Ratio)
board.processor.cores.core.fetchStats0.icacheStallCycles 38797 # ICache total stall cycles (Cycle)
board.processor.cores.core.fetchStats0.numFetchSuspends 0 # Number of times Execute suspended instruction fetching (Count)
board.processor.cores.core.iew.idleCycles 0 # Number of cycles IEW is idle (Cycle)
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board.processor.cores.core.iew.squashCycles 1878 # Number of cycles IEW is squashing (Cycle)
board.processor.cores.core.iew.blockCycles 25706 # Number of cycles IEW is blocking (Cycle)
board.processor.cores.core.iew.unblockCycles 1042 # Number of cycles IEW is unblocking (Cycle)
board.processor.cores.core.iew.dispatchedInsts 57074931 # Number of instructions dispatched to IQ (Count)
board.processor.cores.core.iew.dispSquashedInsts 115 # Number of squashed instructions skipped by dispatch (Count)
board.processor.cores.core.iew.dispLoadInsts 11586082 # Number of dispatched load instructions (Count)
board.processor.cores.core.iew.dispStoreInsts 4218782 # Number of dispatched store instructions (Count)
board.processor.cores.core.iew.dispNonSpecInsts 80 # Number of dispatched non-speculative instructions (Count)
board.processor.cores.core.iew.iqFullEvents 34 # Number of times the IQ has become full, causing a stall (Count)
board.processor.cores.core.iew.lsqFullEvents 988 # Number of times the LSQ has become full, causing a stall (Count)
board.processor.cores.core.iew.memOrderViolationEvents 110 # Number of memory order violations (Count)
board.processor.cores.core.iew.predictedTakenIncorrect 87 # Number of branches that were predicted taken incorrectly (Count)
board.processor.cores.core.iew.predictedNotTakenIncorrect 780 # Number of branches that were predicted not taken incorrectly (Count)
board.processor.cores.core.iew.branchMispredicts 867 # Number of branch mispredicts detected at execute (Count)
board.processor.cores.core.iew.instsToCommit 57021031 # Cumulative count of insts sent to commit (Count)
board.processor.cores.core.iew.writebackCount 57019432 # Cumulative count of insts written-back (Count)
board.processor.cores.core.iew.producerInst 48545514 # Number of instructions producing a value (Count)
board.processor.cores.core.iew.consumerInst 72824651 # Number of instructions consuming a value (Count)
board.processor.cores.core.iew.wbRate 3.212836 # Insts written-back per cycle ((Count/Cycle))
board.processor.cores.core.iew.wbFanout 0.666608 # Average fanout of values written-back ((Count/Count))
board.processor.cores.core.interrupts.clk_domain.clock 5328 # Clock period in ticks (Tick)
board.processor.cores.core.lsq0.forwLoads 8296335 # Number of loads that had data forwarded from stores (Count)
board.processor.cores.core.lsq0.squashedLoads 32170 # Number of loads squashed (Count)
board.processor.cores.core.lsq0.ignoredResponses 3 # Number of memory responses ignored because the instruction is squashed (Count)
board.processor.cores.core.lsq0.memOrderViolation 110 # Number of memory ordering violations (Count)
board.processor.cores.core.lsq0.squashedStores 13616 # Number of stores squashed (Count)
board.processor.cores.core.lsq0.rescheduledLoads 8 # Number of loads that were rescheduled (Count)
board.processor.cores.core.lsq0.blockedByCache 0 # Number of times an access to memory failed due to the cache being blocked (Count)
board.processor.cores.core.lsq0.loadToUse::samples 11553912 # Distribution of cycle latency between the first time a load is issued and its completion (Unspecified)
board.processor.cores.core.lsq0.loadToUse::mean 2.106723 # Distribution of cycle latency between the first time a load is issued and its completion (Unspecified)
board.processor.cores.core.lsq0.loadToUse::stdev 6.275810 # Distribution of cycle latency between the first time a load is issued and its completion (Unspecified)
board.processor.cores.core.lsq0.loadToUse::0-9 11550319 99.97% 99.97% # Distribution of cycle latency between the first time a load is issued and its completion (Unspecified)
board.processor.cores.core.lsq0.loadToUse::10-19 25 0.00% 99.97% # Distribution of cycle latency between the first time a load is issued and its completion (Unspecified)
board.processor.cores.core.lsq0.loadToUse::20-29 44 0.00% 99.97% # Distribution of cycle latency between the first time a load is issued and its completion (Unspecified)
board.processor.cores.core.lsq0.loadToUse::30-39 1 0.00% 99.97% # Distribution of cycle latency between the first time a load is issued and its completion (Unspecified)
board.processor.cores.core.lsq0.loadToUse::40-49 4 0.00% 99.97% # Distribution of cycle latency between the first time a load is issued and its completion (Unspecified)
board.processor.cores.core.lsq0.loadToUse::50-59 3 0.00% 99.97% # Distribution of cycle latency between the first time a load is issued and its completion (Unspecified)
board.processor.cores.core.lsq0.loadToUse::60-69 5 0.00% 99.97% # Distribution of cycle latency between the first time a load is issued and its completion (Unspecified)
board.processor.cores.core.lsq0.loadToUse::70-79 1 0.00% 99.97% # Distribution of cycle latency between the first time a load is issued and its completion (Unspecified)
board.processor.cores.core.lsq0.loadToUse::80-89 1 0.00% 99.97% # Distribution of cycle latency between the first time a load is issued and its completion (Unspecified)
board.processor.cores.core.lsq0.loadToUse::90-99 2 0.00% 99.97% # Distribution of cycle latency between the first time a load is issued and its completion (Unspecified)
board.processor.cores.core.lsq0.loadToUse::100-109 25 0.00% 99.97% # Distribution of cycle latency between the first time a load is issued and its completion (Unspecified)
board.processor.cores.core.lsq0.loadToUse::170-179 1 0.00% 99.97% # Distribution of cycle latency between the first time a load is issued and its completion (Unspecified)
board.processor.cores.core.lsq0.loadToUse::210-219 2 0.00% 99.97% # Distribution of cycle latency between the first time a load is issued and its completion (Unspecified)
board.processor.cores.core.lsq0.loadToUse::250-259 3 0.00% 99.97% # Distribution of cycle latency between the first time a load is issued and its completion (Unspecified)
board.processor.cores.core.lsq0.loadToUse::260-269 1 0.00% 99.97% # Distribution of cycle latency between the first time a load is issued and its completion (Unspecified)
board.processor.cores.core.lsq0.loadToUse::280-289 2 0.00% 99.97% # Distribution of cycle latency between the first time a load is issued and its completion (Unspecified)
board.processor.cores.core.lsq0.loadToUse::290-299 12 0.00% 99.97% # Distribution of cycle latency between the first time a load is issued and its completion (Unspecified)
board.processor.cores.core.lsq0.loadToUse::overflows 3461 0.03% 100.00% # Distribution of cycle latency between the first time a load is issued and its completion (Unspecified)
```

```
board.processor.cores.core.lsq0.loadToUse::min_value 2 # Distribution of cycle latency between the first time a load is issued and its completion (Unspecified)
board.processor.cores.core.lsq0.loadToUse::max_value 1193 # Distribution of cycle latency between the first time a load is issued and its completion (Unspecified)
board.processor.cores.core.lsq0.loadToUse::total 11553912 # Distribution of cycle latency between the first time a load is issued and its completion (Unspecified)
board.processor.cores.core.mmu.dtb.rdAccesses 11580868 # TLB accesses on read requests (Count)
board.processor.cores.core.mmu.dtb.wrAccesses 4214501 # TLB accesses on write requests (Count)
board.processor.cores.core.mmu.dtb.rdMisses 1137 # TLB misses on read requests (Count)
board.processor.cores.core.mmu.dtb.wrMisses 2081 # TLB misses on write requests (Count)
board.processor.cores.core.mmu.dtb.walker.power_state.pwrStateResidencyTicks::UNDEFINED 5909877540 # Cumulative time (in ticks) in various power states (Tick)
board.processor.cores.core.mmu.itb.rdAccesses 0 # TLB accesses on read requests (Count)
board.processor.cores.core.mmu.itb.wrAccesses 4232562 # TLB accesses on write requests (Count)
board.processor.cores.core.mmu.itb.rdMisses 0 # TLB misses on read requests (Count)
board.processor.cores.core.mmu.itb.wrMisses 116 # TLB misses on write requests (Count)
board.processor.cores.core.mmu.itb.walker.power_state.pwrStateResidencyTicks::UNDEFINED 5909877540 # Cumulative time (in ticks) in various power states (Tick)
board.processor.cores.core.power_state.pwrStateResidencyTicks::ON 5909877540 # Cumulative time (in ticks) in various power states (Tick)
board.processor.cores.core.rename.squashCycles 1878 # Number of cycles rename is squashing (Cycle)
board.processor.cores.core.rename.idleCycles 2873636 # Number of cycles rename is idle (Cycle)
board.processor.cores.core.rename.blockCycles 511252 # Number of cycles rename is blocking (Cycle)
board.processor.cores.core.rename.serializeStallCycles 2460 # count of cycles rename stalled for serializing inst (Cycle)
board.processor.cores.core.rename.runCycles 5970982 # Number of cycles rename is running (Cycle)
board.processor.cores.core.rename.unblockCycles 8241454 # Number of cycles rename is unblocking (Cycle)
board.processor.cores.core.rename.renamedInsts 57095363 # Number of instructions processed by rename (Count)
board.processor.cores.core.rename.ROBFullEvents 2779 # Number of times rename has blocked due to ROB full (Count)
board.processor.cores.core.rename.IQFullEvents 2215517 # Number of times rename has blocked due to IQ full (Count)
board.processor.cores.core.rename.LQFullEvents 5081085 # Number of times rename has blocked due to LQ full (Count)
board.processor.cores.core.rename.SQFullEvents 410617 # Number of times rename has blocked due to SQ full (Count)
board.processor.cores.core.rename.renamedOperands 127817195 # Number of destination operands rename has renamed (Count)
board.processor.cores.core.rename.lookups 213505662 # Number of register rename lookups that rename has made (Count)
board.processor.cores.core.rename.intLookups 80311084 # Number of integer rename lookups (Count)
board.processor.cores.core.rename.fpLookups 2495 # Number of floating rename lookups (Count)
board.processor.cores.core.rename.committedMaps 127275790 # Number of HB maps that are committed (Count)
board.processor.cores.core.rename.undoneMaps 541396 # Number of HB maps that are undone due to squashing (Count)
board.processor.cores.core.rename.serializing 45 # count of serializing insts renamed (Count)
board.processor.cores.core.rename.tempSerializing 46 # count of temporary serializing insts renamed (Count)
board.processor.cores.core.rename.skidInsts 16268434 # count of insts added to the skid buffer (Count)
board.processor.cores.core.rob.reads 69795402 # The number of ROB reads (Count)
board.processor.cores.core.rob.writes 114037160 # The number of ROB writes (Count)
board.processor.cores.core.thread_0.numInsts 36831240 # Number of Instructions committed (Count)
board.processor.cores.core.thread_0.numOps 56861208 # Number of Ops committed (Count)
board.processor.cores.core.thread_0.numMemRefs 0 # Number of Memory References (Count)
board.processor.cores.core.workload.numSyscalls 18 # Number of system calls (Count)
board.workload.inst.arm 0 # number of arm instructions executed (Count)
board.workload.inst.quiesce 0 # number of quiesce instructions executed (Count)

---------- End Simulation Statistics ----------
```

results_secure_fast:

```
---------- Begin Simulation Statistics ----------
simSeconds                    10952.226215                # Number of seconds simulated (Second)
simTicks                      10952226214626250           # Number of ticks simulated (Tick)
finalTick                     10952226214626250           # Number of ticks from beginning of simulation (restored from checkpoints and never reset) (Tick)
simFreq                       1000000000000               # The number of ticks per simulated second ((Tick/Second))
hostSeconds                   6306.82                     # Real time elapsed on the host (Second)
hostTickRate                  1736569821081               # The number of ticks simulated per host second (ticks/s) ((Tick/Second))
hostMemory                    1198868                     # Number of bytes of host memory used (Byte)
simInsts                      4141                        # Number of instructions simulated (Count)
simOps                        7867                        # Number of ops (including micro ops) simulated (Count)
hostInstRate                  1                           # Simulator instruction rate (inst/s) ((Count/Second))
hostOpRate                    1                           # Simulator op (including micro ops) rate (op/s) ((Count/Second))
board.cache_hierarchy.dptw_caches.blockedCycles::no_mshrs      0          # number of cycles access was blocked (Cycle)
board.cache_hierarchy.dptw_caches.blockedCycles::no_targets    0          # number of cycles access was blocked (Cycle)
board.cache_hierarchy.dptw_caches.blockedCauses::no_mshrs      0          # number of times access was blocked (Count)
board.cache_hierarchy.dptw_caches.blockedCauses::no_targets    0          # number of times access was blocked (Count)
board.cache_hierarchy.dptw_caches.avgBlocked::no_mshrs         nan        # average number of cycles each access was blocked ((Cycle/Count))
board.cache_hierarchy.dptw_caches.avgBlocked::no_targets       nan        # average number of cycles each access was blocked ((Cycle/Count))
board.cache_hierarchy.dptw_caches.replacements                 0          # number of replacements (Count)
board.cache_hierarchy.dptw_caches.power_state.pwrStateResidencyTicks::UNDEFINED 10952226214626250   # Cumulative time (in ticks) in various power states (Tick)
board.cache_hierarchy.dptw_caches.tags.tagsInUse               0          # Average ticks per tags in use ((Tick/Count))
board.cache_hierarchy.dptw_caches.tags.totalRefs               0          # Total number of references to valid blocks. (Count)
board.cache_hierarchy.dptw_caches.tags.sampledRefs             0          # Sample count of references to valid blocks. (Count)
board.cache_hierarchy.dptw_caches.tags.avgRefs                 nan        # Average number of references to valid blocks. ((Count/Count))
board.cache_hierarchy.dptw_caches.tags.warmupTick              0          # The tick when the warmup percentage was hit. (Tick)
board.cache_hierarchy.dptw_caches.tags.tagAccesses             0          # Number of tag accesses (Count)
board.cache_hierarchy.dptw_caches.tags.dataAccesses            0          # Number of data accesses (Count)
board.cache_hierarchy.dptw_caches.tags.power_state.pwrStateResidencyTicks::UNDEFINED 10952226214626250   # Cumulative time (in ticks) in various power states (Tick)
board.cache_hierarchy.iptw_caches.blockedCycles::no_mshrs      0          # number of cycles access was blocked (Cycle)
board.cache_hierarchy.iptw_caches.blockedCycles::no_targets    0          # number of cycles access was blocked (Cycle)
board.cache_hierarchy.iptw_caches.blockedCauses::no_mshrs      0          # number of times access was blocked (Count)
board.cache_hierarchy.iptw_caches.blockedCauses::no_targets    0          # number of times access was blocked (Count)
board.cache_hierarchy.iptw_caches.avgBlocked::no_mshrs         nan        # average number of cycles each access was blocked ((Cycle/Count))
board.cache_hierarchy.iptw_caches.avgBlocked::no_targets       nan        # average number of cycles each access was blocked ((Cycle/Count))
board.cache_hierarchy.iptw_caches.replacements                 0          # number of replacements (Count)
board.cache_hierarchy.iptw_caches.power_state.pwrStateResidencyTicks::UNDEFINED 10952226214626250   # Cumulative time (in ticks) in various power states (Tick)
board.cache_hierarchy.iptw_caches.tags.tagsInUse               0          # Average ticks per tags in use ((Tick/Count))
board.cache_hierarchy.iptw_caches.tags.totalRefs               0          # Total number of references to valid blocks. (Count)
board.cache_hierarchy.iptw_caches.tags.sampledRefs             0          # Sample count of references to valid blocks. (Count)
board.cache_hierarchy.iptw_caches.tags.avgRefs                 nan        # Average number of references to valid blocks. ((Count/Count))
board.cache_hierarchy.iptw_caches.tags.warmupTick              0          # The tick when the warmup percentage was hit. (Tick)
board.cache_hierarchy.iptw_caches.tags.tagAccesses             0          # Number of tag accesses (Count)
board.cache_hierarchy.iptw_caches.tags.dataAccesses            0          # Number of data accesses (Count)
board.cache_hierarchy.iptw_caches.tags.power_state.pwrStateResidencyTicks::UNDEFINED 10952226214626250   # Cumulative time (in ticks) in various power states (Tick)
board.cache_hierarchy.l1dcaches.demandHits::processor.cores.core.data   1300   # number of demand (read+write) hits (Count)
board.cache_hierarchy.l1dcaches.demandHits::total              1300       # number of demand (read+write) hits (Count)
```

board.cache_hierarchy.l1dcaches.overallHits::processor.cores.core.data 1300 # number of overall hits (Count)

board.cache_hierarchy.l1dcaches.overallHits::total 1300 # number of overall hits (Count)

board.cache_hierarchy.l1dcaches.demandMisses::processor.cores.core.data 185 # number of demand (read+write) misses (Count)

board.cache_hierarchy.l1dcaches.demandMisses::total 185 # number of demand (read+write) misses (Count)

board.cache_hierarchy.l1dcaches.overallMisses::processor.cores.core.data 185 # number of overall misses (Count)

board.cache_hierarchy.l1dcaches.overallMisses::total 185 # number of overall misses (Count)

board.cache_hierarchy.l1dcaches.demandMissLatency::processor.cores.core.data 9377946 # number of demand (read+write) miss ticks (Tick)

board.cache_hierarchy.l1dcaches.demandMissLatency::total 9377946 # number of demand (read+write) miss ticks (Tick)

board.cache_hierarchy.l1dcaches.overallMissLatency::processor.cores.core.data 9377946 # number of overall miss ticks (Tick)

board.cache_hierarchy.l1dcaches.overallMissLatency::total 9377946 # number of overall miss ticks (Tick)

board.cache_hierarchy.l1dcaches.demandAccesses::processor.cores.core.data 1485 # number of demand (read+write) accesses (Count)

board.cache_hierarchy.l1dcaches.demandAccesses::total 1485 # number of demand (read+write) accesses (Count)

board.cache_hierarchy.l1dcaches.overallAccesses::processor.cores.core.data 1485 # number of overall (read+write) accesses (Count)

board.cache_hierarchy.l1dcaches.overallAccesses::total 1485 # number of overall (read+write) accesses (Count)

board.cache_hierarchy.l1dcaches.demandMissRate::processor.cores.core.data 0.124579 # miss rate for demand accesses (Ratio)

board.cache_hierarchy.l1dcaches.demandMissRate::total 0.124579 # miss rate for demand accesses (Ratio)

board.cache_hierarchy.l1dcaches.overallMissRate::processor.cores.core.data 0.124579 # miss rate for overall accesses (Ratio)

board.cache_hierarchy.l1dcaches.overallMissRate::total 0.124579 # miss rate for overall accesses (Ratio)

board.cache_hierarchy.l1dcaches.demandAvgMissLatency::processor.cores.core.data 50691.600000 # average overall miss latency in ticks ((Tick/Count))

board.cache_hierarchy.l1dcaches.demandAvgMissLatency::total 50691.600000 # average overall miss latency in ticks ((Tick/Count))

board.cache_hierarchy.l1dcaches.overallAvgMissLatency::processor.cores.core.data 50691.600000 # average overall miss latency ((Tick/Count))

board.cache_hierarchy.l1dcaches.overallAvgMissLatency::total 50691.600000 # average overall miss latency ((Tick/Count))

board.cache_hierarchy.l1dcaches.blockedCycles::no_mshrs 27977 # number of cycles access was blocked (Cycle)

board.cache_hierarchy.l1dcaches.blockedCycles::no_targets 0 # number of cycles access was blocked (Cycle)

board.cache_hierarchy.l1dcaches.blockedCauses::no_mshrs 186 # number of times access was blocked (Count)

board.cache_hierarchy.l1dcaches.blockedCauses::no_targets 0 # number of times access was blocked (Count)

board.cache_hierarchy.l1dcaches.avgBlocked::no_mshrs 150.413978 # average number of cycles each access was blocked ((Cycle/Count))

board.cache_hierarchy.l1dcaches.avgBlocked::no_targets nan # average number of cycles each access was blocked ((Cycle/Count))

board.cache_hierarchy.l1dcaches.demandMshrMisses::processor.cores.core.data 185 # number of demand (read+write) MSHR misses (Count)

board.cache_hierarchy.l1dcaches.demandMshrMisses::total 185 # number of demand (read+write) MSHR misses (Count)

board.cache_hierarchy.l1dcaches.overallMshrMisses::processor.cores.core.data 185 # number of overall MSHR misses (Count)

board.cache_hierarchy.l1dcaches.overallMshrMisses::total 185 # number of overall MSHR misses (Count)

board.cache_hierarchy.l1dcaches.demandMshrMissLatency::processor.cores.core.data 9316341 # number of demand (read+write) MSHR miss ticks (Tick)

board.cache_hierarchy.l1dcaches.demandMshrMissLatency::total 9316341 # number of demand (read+write) MSHR miss ticks (Tick)

board.cache_hierarchy.l1dcaches.overallMshrMissLatency::processor.cores.core.data 9316341 # number of overall MSHR miss ticks (Tick)

board.cache_hierarchy.l1dcaches.overallMshrMissLatency::total 9316341 # number of overall MSHR miss ticks (Tick)

board.cache_hierarchy.l1dcaches.demandMshrMissRate::processor.cores.core.data 0.124579 # mshr miss ratio for demand accesses (Ratio)

board.cache_hierarchy.l1dcaches.demandMshrMissRate::total 0.124579 # mshr miss ratio for demand accesses (Ratio)

board.cache_hierarchy.l1dcaches.overallMshrMissRate::processor.cores.core.data 0.124579 # mshr miss ratio for overall accesses (Ratio)

board.cache_hierarchy.l1dcaches.overallMshrMissRate::total 0.124579 # mshr miss ratio for overall accesses (Ratio)

board.cache_hierarchy.l1dcaches.demandAvgMshrMissLatency::processor.cores.core.data 50358.600000 # average overall mshr miss latency ((Tick/Count))

board.cache_hierarchy.l1dcaches.demandAvgMshrMissLatency::total 50358.600000 # average overall mshr miss latency ((Tick/Count))

board.cache_hierarchy.l1dcaches.overallAvgMshrMissLatency::processor.cores.core.data 50358.600000 # average overall mshr miss latency ((Tick/Count))

board.cache_hierarchy.l1dcaches.overallAvgMshrMissLatency::total 50358.600000 # average overall mshr miss latency ((Tick/Count))

board.cache_hierarchy.l1dcaches.replacements 0 # number of replacements (Count)

board.cache_hierarchy.l1dcaches.LockedRMWReadReq.hits::processor.cores.core.data 1 # number of LockedRMWReadReq hits (Count)

board.cache_hierarchy.l1dcaches.LockedRMWReadReq.hits::total 1 # number of LockedRMWReadReq hits (Count)

board.cache_hierarchy.l1dcaches.LockedRMWReadReq.accesses::processor.cores.core.data 1 # number of LockedRMWReadReq accesses(hits+misses) (Count)

board.cache_hierarchy.l1dcaches.LockedRMWReadReq.accesses::total 1 # number of LockedRMWReadReq accesses(hits+misses) (Count)

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board.cache_hierarchy.l1dcaches.ReadReq.hits::processor.cores.core.data 541 # number of ReadReq hits (Count)
board.cache_hierarchy.l1dcaches.ReadReq.hits::total 541 # number of ReadReq hits (Count)
board.cache_hierarchy.l1dcaches.ReadReq.misses::processor.cores.core.data 99 # number of ReadReq misses (Count)
board.cache_hierarchy.l1dcaches.ReadReq.misses::total 99 # number of ReadReq misses (Count)
board.cache_hierarchy.l1dcaches.ReadReq.missLatency::processor.cores.core.data 5741586 # number of ReadReq miss ticks (Tick)
board.cache_hierarchy.l1dcaches.ReadReq.missLatency::total 5741586 # number of ReadReq miss ticks (Tick)
board.cache_hierarchy.l1dcaches.ReadReq.accesses::processor.cores.core.data 640 # number of ReadReq accesses(hits+misses) (Count)
board.cache_hierarchy.l1dcaches.ReadReq.accesses::total 640 # number of ReadReq accesses(hits+misses) (Count)
board.cache_hierarchy.l1dcaches.ReadReq.missRate::processor.cores.core.data 0.154688 # miss rate for ReadReq accesses (Ratio)
board.cache_hierarchy.l1dcaches.ReadReq.missRate::total 0.154688 # miss rate for ReadReq accesses (Ratio)
board.cache_hierarchy.l1dcaches.ReadReq.avgMissLatency::processor.cores.core.data 57995.818182 # average ReadReq miss latency ((Tick/Count))
board.cache_hierarchy.l1dcaches.ReadReq.avgMissLatency::total 57995.818182 # average ReadReq miss latency ((Tick/Count))
board.cache_hierarchy.l1dcaches.ReadReq.mshrMisses::processor.cores.core.data 99 # number of ReadReq MSHR misses (Count)
board.cache_hierarchy.l1dcaches.ReadReq.mshrMisses::total 99 # number of ReadReq MSHR misses (Count)
board.cache_hierarchy.l1dcaches.ReadReq.mshrMissLatency::processor.cores.core.data 5708619 # number of ReadReq MSHR miss ticks (Tick)
board.cache_hierarchy.l1dcaches.ReadReq.mshrMissLatency::total 5708619 # number of ReadReq MSHR miss ticks (Tick)
board.cache_hierarchy.l1dcaches.ReadReq.mshrMissRate::processor.cores.core.data 0.154688 # mshr miss rate for ReadReq accesses (Ratio)
board.cache_hierarchy.l1dcaches.ReadReq.mshrMissRate::total 0.154688 # mshr miss rate for ReadReq accesses (Ratio)
board.cache_hierarchy.l1dcaches.ReadReq.avgMshrMissLatency::processor.cores.core.data 57662.818182 # average ReadReq mshr miss latency ((Tick/Count))
board.cache_hierarchy.l1dcaches.ReadReq.avgMshrMissLatency::total 57662.818182 # average ReadReq mshr miss latency ((Tick/Count))
board.cache_hierarchy.l1dcaches.WriteReq.hits::processor.cores.core.data 759 # number of WriteReq hits (Count)
board.cache_hierarchy.l1dcaches.WriteReq.hits::total 759 # number of WriteReq hits (Count)
board.cache_hierarchy.l1dcaches.WriteReq.misses::processor.cores.core.data 86 # number of WriteReq misses (Count)
board.cache_hierarchy.l1dcaches.WriteReq.misses::total 86 # number of WriteReq misses (Count)
board.cache_hierarchy.l1dcaches.WriteReq.missLatency::processor.cores.core.data 3636360 # number of WriteReq miss ticks (Tick)
board.cache_hierarchy.l1dcaches.WriteReq.missLatency::total 3636360 # number of WriteReq miss ticks (Tick)
board.cache_hierarchy.l1dcaches.WriteReq.accesses::processor.cores.core.data 845 # number of WriteReq accesses(hits+misses) (Count)
board.cache_hierarchy.l1dcaches.WriteReq.accesses::total 845 # number of WriteReq accesses(hits+misses) (Count)
board.cache_hierarchy.l1dcaches.WriteReq.missRate::processor.cores.core.data 0.101775 # miss rate for WriteReq accesses (Ratio)
board.cache_hierarchy.l1dcaches.WriteReq.missRate::total 0.101775 # miss rate for WriteReq accesses (Ratio)
board.cache_hierarchy.l1dcaches.WriteReq.avgMissLatency::processor.cores.core.data 42283.255814 # average WriteReq miss latency ((Tick/Count))
board.cache_hierarchy.l1dcaches.WriteReq.avgMissLatency::total 42283.255814 # average WriteReq miss latency ((Tick/Count))
board.cache_hierarchy.l1dcaches.WriteReq.mshrMisses::processor.cores.core.data 86 # number of WriteReq MSHR misses (Count)
board.cache_hierarchy.l1dcaches.WriteReq.mshrMisses::total 86 # number of WriteReq MSHR misses (Count)
board.cache_hierarchy.l1dcaches.WriteReq.mshrMissLatency::processor.cores.core.data 3607722 # number of WriteReq MSHR miss ticks (Tick)
board.cache_hierarchy.l1dcaches.WriteReq.mshrMissLatency::total 3607722 # number of WriteReq MSHR miss ticks (Tick)
board.cache_hierarchy.l1dcaches.WriteReq.mshrMissRate::processor.cores.core.data 0.101775 # mshr miss rate for WriteReq accesses (Ratio)
board.cache_hierarchy.l1dcaches.WriteReq.mshrMissRate::total 0.101775 # mshr miss rate for WriteReq accesses (Ratio)
board.cache_hierarchy.l1dcaches.WriteReq.avgMshrMissLatency::processor.cores.core.data 41950.255814 # average WriteReq mshr miss latency ((Tick/Count))
board.cache_hierarchy.l1dcaches.WriteReq.avgMshrMissLatency::total 41950.255814 # average WriteReq mshr miss latency ((Tick/Count))
board.cache_hierarchy.l1dcaches.power_state.pwrStateResidencyTicks::UNDEFINED 10952226214626250 # Cumulative time (in ticks) in various power states (Tick)
board.cache_hierarchy.l1dcaches.prefetcher.demandMshrMisses 185 # demands not covered by prefetchs (Count)
board.cache_hierarchy.l1dcaches.prefetcher.pfIssued 0 # number of hwpf issued (Count)
board.cache_hierarchy.l1dcaches.prefetcher.pfUseful 0 # number of useful prefetch (Count)
board.cache_hierarchy.l1dcaches.prefetcher.pfUsefulButMiss 0 # number of hit on prefetch but cache block is not in an usable state (Count)
board.cache_hierarchy.l1dcaches.prefetcher.accuracy nan # accuracy of the prefetcher (Count)
board.cache_hierarchy.l1dcaches.prefetcher.coverage 0 # coverage brought by this prefetcher (Count)
board.cache_hierarchy.l1dcaches.prefetcher.pfHitInCache 0 # number of prefetches hitting in cache (Count)
board.cache_hierarchy.l1dcaches.prefetcher.pfHitInMSHR 0 # number of prefetches hitting in a MSHR (Count)
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board.cache_hierarchy.l1dcaches.prefetcher.pfHitInWB 0 # number of prefetches hit in the Write Buffer (Count)
board.cache_hierarchy.l1dcaches.prefetcher.pfLate 0 # number of late prefetches (hitting in cache, MSHR or WB) (Count)
board.cache_hierarchy.l1dcaches.prefetcher.pfIdentified 146 # number of prefetch candidates identified (Count)
board.cache_hierarchy.l1dcaches.prefetcher.pfBufferHit 63 # number of redundant prefetches already in prefetch queue (Count)
board.cache_hierarchy.l1dcaches.prefetcher.pfInCache 0 # number of redundant prefetches already in cache/mshr dropped (Count)
board.cache_hierarchy.l1dcaches.prefetcher.pfRemovedDemand 35 # number of prefetches dropped due to a demand for the same address (Count)
board.cache_hierarchy.l1dcaches.prefetcher.pfRemovedFull 0 # number of prefetches dropped due to prefetch queue size (Count)
board.cache_hierarchy.l1dcaches.prefetcher.pfSpanPage 6 # number of prefetches that crossed the page (Count)
board.cache_hierarchy.l1dcaches.prefetcher.pfUsefulSpanPage 0 # number of prefetches that is useful and crossed the page (Count)
board.cache_hierarchy.l1dcaches.prefetcher.power_state.pwrStateResidencyTicks::UNDEFINED 10952226214626250 # Cumulative time (in ticks) in various power states (Tick)
board.cache_hierarchy.l1dcaches.tags.tagsInUse 185.000000 # Average ticks per tags in use ((Tick/Count))
board.cache_hierarchy.l1dcaches.tags.totalRefs 0 # Total number of references to valid blocks. (Count)
board.cache_hierarchy.l1dcaches.tags.sampledRefs 0 # Sample count of references to valid blocks. (Count)
board.cache_hierarchy.l1dcaches.tags.avgRefs nan # Average number of references to valid blocks. ((Count/Count))
board.cache_hierarchy.l1dcaches.tags.warmupTick 149184 # The tick when the warmup percentage was hit. (Tick)
board.cache_hierarchy.l1dcaches.tags.occupancies::processor.cores.core.data 185.000000 # Average occupied blocks per tick, per requestor ((Count/Tick))
board.cache_hierarchy.l1dcaches.tags.avgOccs::processor.cores.core.data 0.361328 # Average percentage of cache occupancy ((Ratio/Tick))
board.cache_hierarchy.l1dcaches.tags.avgOccs::total 0.361328 # Average percentage of cache occupancy ((Ratio/Tick))
board.cache_hierarchy.l1dcaches.tags.occupanciesTaskId::1024 185 # Occupied blocks per task id (Count)
board.cache_hierarchy.l1dcaches.tags.ageTaskId_1024::4 185 # Occupied blocks per task id, per block age (Count)
board.cache_hierarchy.l1dcaches.tags.ratioOccsTaskId::1024 0.361328 # Ratio of occupied blocks and all blocks, per task id (Ratio)
board.cache_hierarchy.l1dcaches.tags.tagAccesses 12073 # Number of tag accesses (Count)
board.cache_hierarchy.l1dcaches.tags.dataAccesses 12073 # Number of data accesses (Count)
board.cache_hierarchy.l1dcaches.tags.power_state.pwrStateResidencyTicks::UNDEFINED 10952226214626250 # Cumulative time (in ticks) in various power states (Tick)
board.cache_hierarchy.l1icaches.demandHits::processor.cores.core.inst 5149 # number of demand (read+write) hits (Count)
board.cache_hierarchy.l1icaches.demandHits::total 5149 # number of demand (read+write) hits (Count)
board.cache_hierarchy.l1icaches.overallHits::processor.cores.core.inst 5149 # number of overall hits (Count)
board.cache_hierarchy.l1icaches.overallHits::total 5149 # number of overall hits (Count)
board.cache_hierarchy.l1icaches.demandMisses::processor.cores.core.inst 196 # number of demand (read+write) misses (Count)
board.cache_hierarchy.l1icaches.demandMisses::total 196 # number of demand (read+write) misses (Count)
board.cache_hierarchy.l1icaches.overallMisses::processor.cores.core.inst 196 # number of overall misses (Count)
board.cache_hierarchy.l1icaches.overallMisses::total 196 # number of overall misses (Count)
board.cache_hierarchy.l1icaches.demandMissLatency::processor.cores.core.inst 12525129 # number of demand (read+write) miss ticks (Tick)
board.cache_hierarchy.l1icaches.demandMissLatency::total 12525129 # number of demand (read+write) miss ticks (Tick)
board.cache_hierarchy.l1icaches.overallMissLatency::processor.cores.core.inst 12525129 # number of overall miss ticks (Tick)
board.cache_hierarchy.l1icaches.overallMissLatency::total 12525129 # number of overall miss ticks (Tick)
board.cache_hierarchy.l1icaches.demandAccesses::processor.cores.core.inst 5345 # number of demand (read+write) accesses (Count)
board.cache_hierarchy.l1icaches.demandAccesses::total 5345 # number of demand (read+write) accesses (Count)
board.cache_hierarchy.l1icaches.overallAccesses::processor.cores.core.inst 5345 # number of overall (read+write) accesses (Count)
board.cache_hierarchy.l1icaches.overallAccesses::total 5345 # number of overall (read+write) accesses (Count)
board.cache_hierarchy.l1icaches.demandMissRate::processor.cores.core.inst 0.036670 # miss rate for demand accesses (Ratio)
board.cache_hierarchy.l1icaches.demandMissRate::total 0.036670 # miss rate for demand accesses (Ratio)
board.cache_hierarchy.l1icaches.overallMissRate::processor.cores.core.inst 0.036670 # miss rate for overall accesses (Ratio)
board.cache_hierarchy.l1icaches.overallMissRate::total 0.036670 # miss rate for overall accesses (Ratio)
board.cache_hierarchy.l1icaches.demandAvgMissLatency::processor.cores.core.inst 63903.719388 # average overall miss latency in ticks ((Tick/Count))
board.cache_hierarchy.l1icaches.demandAvgMissLatency::total 63903.719388 # average overall miss latency in ticks ((Tick/Count))
board.cache_hierarchy.l1icaches.overallAvgMissLatency::processor.cores.core.inst 63903.719388 # average overall miss latency ((Tick/Count))
board.cache_hierarchy.l1icaches.overallAvgMissLatency::total 63903.719388 # average overall miss latency ((Tick/Count))
board.cache_hierarchy.l1icaches.blockedCycles::no_mshrs 0 # number of cycles access was blocked (Cycle)
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board.cache_hierarchy.l1icaches.blockedCycles::no_targets 0 # number of cycles access was blocked (Cycle)

board.cache_hierarchy.l1icaches.blockedCauses::no_mshrs 0 # number of times access was blocked (Count)

board.cache_hierarchy.l1icaches.blockedCauses::no_targets 0 # number of times access was blocked (Count)

board.cache_hierarchy.l1icaches.avgBlocked::no_mshrs nan # average number of cycles each access was blocked ((Cycle/Count))

board.cache_hierarchy.l1icaches.avgBlocked::no_targets nan # average number of cycles each access was blocked ((Cycle/Count))

board.cache_hierarchy.l1icaches.demandMshrMisses::processor.cores.core.inst 196 # number of demand (read+write) MSHR misses (Count)

board.cache_hierarchy.l1icaches.demandMshrMisses::total 196 # number of demand (read+write) MSHR misses (Count)

board.cache_hierarchy.l1icaches.overallMshrMisses::processor.cores.core.inst 196 # number of overall MSHR misses (Count)

board.cache_hierarchy.l1icaches.overallMshrMisses::total 196 # number of overall MSHR misses (Count)

board.cache_hierarchy.l1icaches.demandMshrMissLatency::processor.cores.core.inst 12459861 # number of demand (read+write) MSHR miss ticks (Tick)

board.cache_hierarchy.l1icaches.demandMshrMissLatency::total 12459861 # number of demand (read+write) MSHR miss ticks (Tick)

board.cache_hierarchy.l1icaches.overallMshrMissLatency::processor.cores.core.inst 12459861 # number of overall MSHR miss ticks (Tick)

board.cache_hierarchy.l1icaches.overallMshrMissLatency::total 12459861 # number of overall MSHR miss ticks (Tick)

board.cache_hierarchy.l1icaches.demandMshrMissRate::processor.cores.core.inst 0.036670 # mshr miss ratio for demand accesses (Ratio)

board.cache_hierarchy.l1icaches.demandMshrMissRate::total 0.036670 # mshr miss ratio for demand accesses (Ratio)

board.cache_hierarchy.l1icaches.overallMshrMissRate::processor.cores.core.inst 0.036670 # mshr miss ratio for overall accesses (Ratio)

board.cache_hierarchy.l1icaches.overallMshrMissRate::total 0.036670 # mshr miss ratio for overall accesses (Ratio)

board.cache_hierarchy.l1icaches.demandAvgMshrMissLatency::processor.cores.core.inst 63570.719388 # average overall mshr miss latency ((Tick/Count))

board.cache_hierarchy.l1icaches.demandAvgMshrMissLatency::total 63570.719388 # average overall mshr miss latency ((Tick/Count))

board.cache_hierarchy.l1icaches.overallAvgMshrMissLatency::processor.cores.core.inst 63570.719388 # average overall mshr miss latency ((Tick/Count))

board.cache_hierarchy.l1icaches.overallAvgMshrMissLatency::total 63570.719388 # average overall mshr miss latency ((Tick/Count))

board.cache_hierarchy.l1icaches.replacements 0 # number of replacements (Count)

board.cache_hierarchy.l1icaches.ReadReq.hits::processor.cores.core.inst 5149 # number of ReadReq hits (Count)

board.cache_hierarchy.l1icaches.ReadReq.hits::total 5149 # number of ReadReq hits (Count)

board.cache_hierarchy.l1icaches.ReadReq.misses::processor.cores.core.inst 196 # number of ReadReq misses (Count)

board.cache_hierarchy.l1icaches.ReadReq.misses::total 196 # number of ReadReq misses (Count)

board.cache_hierarchy.l1icaches.ReadReq.missLatency::processor.cores.core.inst 12525129 # number of ReadReq miss ticks (Tick)

board.cache_hierarchy.l1icaches.ReadReq.missLatency::total 12525129 # number of ReadReq miss ticks (Tick)

board.cache_hierarchy.l1icaches.ReadReq.accesses::processor.cores.core.inst 5345 # number of ReadReq accesses(hits+misses) (Count)

board.cache_hierarchy.l1icaches.ReadReq.accesses::total 5345 # number of ReadReq accesses(hits+misses) (Count)

board.cache_hierarchy.l1icaches.ReadReq.missRate::processor.cores.core.inst 0.036670 # miss rate for ReadReq accesses (Ratio)

board.cache_hierarchy.l1icaches.ReadReq.missRate::total 0.036670 # miss rate for ReadReq accesses (Ratio)

board.cache_hierarchy.l1icaches.ReadReq.avgMissLatency::processor.cores.core.inst 63903.719388 # average ReadReq miss latency ((Tick/Count))

board.cache_hierarchy.l1icaches.ReadReq.avgMissLatency::total 63903.719388 # average ReadReq miss latency ((Tick/Count))

board.cache_hierarchy.l1icaches.ReadReq.mshrMisses::processor.cores.core.inst 196 # number of ReadReq MSHR misses (Count)

board.cache_hierarchy.l1icaches.ReadReq.mshrMisses::total 196 # number of ReadReq MSHR misses (Count)

board.cache_hierarchy.l1icaches.ReadReq.mshrMissLatency::processor.cores.core.inst 12459861 # number of ReadReq MSHR miss ticks (Tick)

board.cache_hierarchy.l1icaches.ReadReq.mshrMissLatency::total 12459861 # number of ReadReq MSHR miss ticks (Tick)

board.cache_hierarchy.l1icaches.ReadReq.mshrMissRate::processor.cores.core.inst 0.036670 # mshr miss rate for ReadReq accesses (Ratio)

board.cache_hierarchy.l1icaches.ReadReq.mshrMissRate::total 0.036670 # mshr miss rate for ReadReq accesses (Ratio)

board.cache_hierarchy.l1icaches.ReadReq.avgMshrMissLatency::processor.cores.core.inst 63570.719388 # average ReadReq mshr miss latency ((Tick/Count))

board.cache_hierarchy.l1icaches.ReadReq.avgMshrMissLatency::total 63570.719388 # average ReadReq mshr miss latency ((Tick/Count))

board.cache_hierarchy.l1icaches.power_state.pwrStateResidencyTicks::UNDEFINED 10952226214626250 # Cumulative time (in ticks) in various power states (Tick)

board.cache_hierarchy.l1icaches.prefetcher.demandMshrMisses 196 # demands not covered by prefetchs (Count)

board.cache_hierarchy.l1icaches.prefetcher.pfIssued 0 # number of hwpf issued (Count)

board.cache_hierarchy.l1icaches.prefetcher.pfUseful 0 # number of useful prefetch (Count)

board.cache_hierarchy.l1icaches.prefetcher.pfUsefulButMiss 0 # number of hit on prefetch but cache block is not in an usable state (Count)

board.cache_hierarchy.l1icaches.prefetcher.accuracy nan # accuracy of the prefetcher (Count)

board.cache_hierarchy.l1icaches.prefetcher.coverage 0 # coverage brought by this prefetcher (Count)

board.cache_hierarchy.l1icaches.prefetcher.pfHitInCache 0 # number of prefetches hitting in cache (Count)

board.cache_hierarchy.l1icaches.prefetcher.pfHitInMSHR 0 # number of prefetches hitting in a MSHR (Count)

board.cache_hierarchy.l1icaches.prefetcher.pfHitInWB 0 # number of prefetches hit in the Write Buffer (Count)

board.cache_hierarchy.l1icaches.prefetcher.pfLate 0 # number of late prefetches (hitting in cache, MSHR or WB) (Count)

board.cache_hierarchy.l1icaches.prefetcher.pfIdentified 0 # number of prefetch candidates identified (Count)

board.cache_hierarchy.l1icaches.prefetcher.pfBufferHit 0 # number of redundant prefetches already in prefetch queue (Count)

board.cache_hierarchy.l1icaches.prefetcher.pfInCache 0 # number of redundant prefetches already in cache/mshr dropped (Count)

board.cache_hierarchy.l1icaches.prefetcher.pfRemovedDemand 0 # number of prefetches dropped due to a demand for the same address (Count)

board.cache_hierarchy.l1icaches.prefetcher.pfRemovedFull 0 # number of prefetches dropped due to prefetch queue size (Count)

board.cache_hierarchy.l1icaches.prefetcher.pfSpanPage 0 # number of prefetches that crossed the page (Count)

board.cache_hierarchy.l1icaches.prefetcher.pfUsefulSpanPage 0 # number of prefetches that is useful and crossed the page (Count)

board.cache_hierarchy.l1icaches.prefetcher.power_state.pwrStateResidencyTicks::UNDEFINED 10952226214626250 # Cumulative time (in ticks) in various power states (Tick)

board.cache_hierarchy.l1icaches.tags.tagsInUse 196.000000 # Average ticks per tags in use ((Tick/Count))

board.cache_hierarchy.l1icaches.tags.totalRefs 0 # Total number of references to valid blocks. (Count)

board.cache_hierarchy.l1icaches.tags.sampledRefs 0 # Sample count of references to valid blocks. (Count)

board.cache_hierarchy.l1icaches.tags.avgRefs nan # Average number of references to valid blocks. ((Count/Count))

board.cache_hierarchy.l1icaches.tags.warmupTick 76590 # The tick when the warmup percentage was hit. (Tick)

board.cache_hierarchy.l1icaches.tags.occupancies::processor.cores.core.inst 196.000000 # Average occupied blocks per tick, per requestor ((Count/Tick))

board.cache_hierarchy.l1icaches.tags.avgOccs::processor.cores.core.inst 0.382812 # Average percentage of cache occupancy ((Ratio/Tick))

board.cache_hierarchy.l1icaches.tags.avgOccs::total 0.382812 # Average percentage of cache occupancy ((Ratio/Tick))

board.cache_hierarchy.l1icaches.tags.occupanciesTaskId::1024 196 # Occupied blocks per task id (Count)

board.cache_hierarchy.l1icaches.tags.ageTaskId_1024::4 196 # Occupied blocks per task id, per block age (Count)

board.cache_hierarchy.l1icaches.tags.ratioOccsTaskId::1024 0.382812 # Ratio of occupied blocks and all blocks, per task id (Ratio)

board.cache_hierarchy.l1icaches.tags.tagAccesses 42956 # Number of tag accesses (Count)

board.cache_hierarchy.l1icaches.tags.dataAccesses 42956 # Number of data accesses (Count)

board.cache_hierarchy.l1icaches.tags.power_state.pwrStateResidencyTicks::UNDEFINED 10952226214626250 # Cumulative time (in ticks) in various power states (Tick)

board.cache_hierarchy.l2bus.transDist::ReadResp 295 # Transaction distribution (Count)

board.cache_hierarchy.l2bus.transDist::HardPFReq 57 # Transaction distribution (Count)

board.cache_hierarchy.l2bus.transDist::ReadExReq 86 # Transaction distribution (Count)

board.cache_hierarchy.l2bus.transDist::ReadExResp 86 # Transaction distribution (Count)

board.cache_hierarchy.l2bus.transDist::ReadSharedReq 295 # Transaction distribution (Count)

board.cache_hierarchy.l2bus.pktCount_board.cache_hierarchy.l1icaches.mem_side_port::board.cache_hierarchy.l2cache.cpu_side_port 392 # Packet count per connected requestor and responder (Count)

board.cache_hierarchy.l2bus.pktCount_board.cache_hierarchy.l1dcaches.mem_side_port::board.cache_hierarchy.l2cache.cpu_side_port 370 # Packet count per connected requestor and responder (Count)

board.cache_hierarchy.l2bus.pktCount::total 762 # Packet count per connected requestor and responder (Count)

board.cache_hierarchy.l2bus.pktSize_board.cache_hierarchy.l1icaches.mem_side_port::board.cache_hierarchy.l2cache.cpu_side_port 12544 # Cumulative packet size per connected requestor and responder (Byte)

board.cache_hierarchy.l2bus.pktSize_board.cache_hierarchy.l1dcaches.mem_side_port::board.cache_hierarchy.l2cache.cpu_side_port 11840 # Cumulative packet size per connected requestor and responder (Byte)

board.cache_hierarchy.l2bus.pktSize::total 24384 # Cumulative packet size per connected requestor and responder (Byte)

board.cache_hierarchy.l2bus.snoops 57 # Total snoops (Count)

board.cache_hierarchy.l2bus.snoopTraffic 0 # Total snoop traffic (Byte)

board.cache_hierarchy.l2bus.snoopFanout::samples 438 # Request fanout histogram (Count)

board.cache_hierarchy.l2bus.snoopFanout::mean 0 # Request fanout histogram (Count)

board.cache_hierarchy.l2bus.snoopFanout::stdev 0 # Request fanout histogram (Count)

board.cache_hierarchy.l2bus.snoopFanout::underflows 0 0.00% 0.00% # Request fanout histogram (Count)

board.cache_hierarchy.l2bus.snoopFanout::0 438 100.00% 100.00% # Request fanout histogram (Count)

board.cache_hierarchy.l2bus.snoopFanout::1 0 0.00% 100.00% # Request fanout histogram (Count)

board.cache_hierarchy.l2bus.snoopFanout::2 0 0.00% 100.00% # Request fanout histogram (Count)

board.cache_hierarchy.l2bus.snoopFanout::3 0 0.00% 100.00% # Request fanout histogram (Count)

board.cache_hierarchy.l2bus.snoopFanout::4 0 0.00% 100.00% # Request fanout histogram (Count)

board.cache_hierarchy.l2bus.snoopFanout::overflows 0 0.00% 100.00% # Request fanout histogram (Count)

board.cache_hierarchy.l2bus.snoopFanout::min_value 0 # Request fanout histogram (Count)

board.cache_hierarchy.l2bus.snoopFanout::max_value 0 # Request fanout histogram (Count)

board.cache_hierarchy.l2bus.snoopFanout::total 438 # Request fanout histogram (Count)

board.cache_hierarchy.l2bus.power_state.pwrStateResidencyTicks::UNDEFINED 10952226214626250 # Cumulative time (in ticks) in various power states (Tick)

board.cache_hierarchy.l2bus.reqLayer0.occupancy 126873 # Layer occupancy (ticks) (Tick)

board.cache_hierarchy.l2bus.reqLayer0.utilization 0.0 # Layer utilization (Ratio)

board.cache_hierarchy.l2bus.respLayer0.occupancy 195804 # Layer occupancy (ticks) (Tick)

board.cache_hierarchy.l2bus.respLayer0.utilization 0.0 # Layer utilization (Ratio)

board.cache_hierarchy.l2bus.respLayer1.occupancy 184815 # Layer occupancy (ticks) (Tick)

board.cache_hierarchy.l2bus.respLayer1.utilization 0.0 # Layer utilization (Ratio)

board.cache_hierarchy.l2bus.snoop_filter.totRequests 381 # Total number of requests made to the snoop filter. (Count)

board.cache_hierarchy.l2bus.snoop_filter.hitSingleRequests 0 # Number of requests hitting in the snoop filter with a single holder of the requested data. (Count)

board.cache_hierarchy.l2bus.snoop_filter.hitMultiRequests 0 # Number of requests hitting in the snoop filter with multiple (>1) holders of the requested data. (Count)

board.cache_hierarchy.l2bus.snoop_filter.totSnoops 0 # Total number of snoops made to the snoop filter. (Count)

board.cache_hierarchy.l2bus.snoop_filter.hitSingleSnoops 0 # Number of snoops hitting in the snoop filter with a single holder of the requested data. (Count)

board.cache_hierarchy.l2bus.snoop_filter.hitMultiSnoops 0 # Number of snoops hitting in the snoop filter with multiple (>1) holders of the requested data. (Count)

board.cache_hierarchy.l2cache.demandHits::processor.cores.core.data 35 # number of demand (read+write) hits (Count)

board.cache_hierarchy.l2cache.demandHits::total 35 # number of demand (read+write) hits (Count)

board.cache_hierarchy.l2cache.overallHits::processor.cores.core.data 35 # number of overall hits (Count)

board.cache_hierarchy.l2cache.overallHits::total 35 # number of overall hits (Count)

board.cache_hierarchy.l2cache.demandMisses::processor.cores.core.inst 196 # number of demand (read+write) misses (Count)

board.cache_hierarchy.l2cache.demandMisses::processor.cores.core.data 150 # number of demand (read+write) misses (Count)

board.cache_hierarchy.l2cache.demandMisses::total 346 # number of demand (read+write) misses (Count)

board.cache_hierarchy.l2cache.overallMisses::processor.cores.core.inst 196 # number of overall misses (Count)

board.cache_hierarchy.l2cache.overallMisses::processor.cores.core.data 150 # number of overall misses (Count)

board.cache_hierarchy.l2cache.overallMisses::total 346 # number of overall misses (Count)

board.cache_hierarchy.l2cache.demandMissLatency::processor.cores.core.inst 12329325 # number of demand (read+write) miss ticks (Tick)

board.cache_hierarchy.l2cache.demandMissLatency::processor.cores.core.data 8948376 # number of demand (read+write) miss ticks (Tick)

board.cache_hierarchy.l2cache.demandMissLatency::total 21277701 # number of demand (read+write) miss ticks (Tick)

board.cache_hierarchy.l2cache.overallMissLatency::processor.cores.core.inst 12329325 # number of overall miss ticks (Tick)

board.cache_hierarchy.l2cache.overallMissLatency::processor.cores.core.data 8948376 # number of overall miss ticks (Tick)

board.cache_hierarchy.l2cache.overallMissLatency::total 21277701 # number of overall miss ticks (Tick)

board.cache_hierarchy.l2cache.demandAccesses::processor.cores.core.inst 196 # number of demand (read+write) accesses (Count)

board.cache_hierarchy.l2cache.demandAccesses::processor.cores.core.data 185 # number of demand (read+write) accesses (Count)

board.cache_hierarchy.l2cache.demandAccesses::total 381 # number of demand (read+write) accesses (Count)

board.cache_hierarchy.l2cache.overallAccesses::processor.cores.core.inst 196 # number of overall (read+write) accesses (Count)

board.cache_hierarchy.l2cache.overallAccesses::processor.cores.core.data 185 # number of overall (read+write) accesses (Count)

board.cache_hierarchy.l2cache.overallAccesses::total 381 # number of overall (read+write) accesses (Count)

board.cache_hierarchy.l2cache.demandMissRate::processor.cores.core.inst 1 # miss rate for demand accesses (Ratio)

board.cache_hierarchy.l2cache.demandMissRate::processor.cores.core.data 0.810811 # miss rate for demand accesses (Ratio)

board.cache_hierarchy.l2cache.demandMissRate::total 0.908136 # miss rate for demand accesses (Ratio)

board.cache_hierarchy.l2cache.overallMissRate::processor.cores.core.inst 1 # miss rate for overall accesses (Ratio)

board.cache_hierarchy.l2cache.overallMissRate::processor.cores.core.data 0.810811 # miss rate for overall accesses (Ratio)

board.cache_hierarchy.l2cache.overallMissRate::total 0.908136 # miss rate for overall accesses (Ratio)

board.cache_hierarchy.l2cache.demandAvgMissLatency::processor.cores.core.inst 62904.719388 # average overall miss latency in ticks ((Tick/Count))

board.cache_hierarchy.l2cache.demandAvgMissLatency::processor.cores.core.data 59655.840000 # average overall miss latency in ticks ((Tick/Count))

board.cache_hierarchy.l2cache.demandAvgMissLatency::total 61496.245665 # average overall miss latency in ticks ((Tick/Count))

board.cache_hierarchy.l2cache.overallAvgMissLatency::processor.cores.core.inst 62904.719388 # average overall miss latency ((Tick/Count))

board.cache_hierarchy.l2cache.overallAvgMissLatency::processor.cores.core.data 59655.840000 # average overall miss latency ((Tick/Count))

board.cache_hierarchy.l2cache.overallAvgMissLatency::total 61496.245665 # average overall miss latency ((Tick/Count))

board.cache_hierarchy.l2cache.blockedCycles::no_mshrs 0 # number of cycles access was blocked (Cycle)

board.cache_hierarchy.l2cache.blockedCycles::no_targets 0 # number of cycles access was blocked (Cycle)

board.cache_hierarchy.l2cache.blockedCauses::no_mshrs 0 # number of times access was blocked (Count)

board.cache_hierarchy.l2cache.blockedCauses::no_targets 0 # number of times access was blocked (Count)

board.cache_hierarchy.l2cache.avgBlocked::no_mshrs nan # average number of cycles each access was blocked ((Cycle/Count))

board.cache_hierarchy.l2cache.avgBlocked::no_targets nan # average number of cycles each access was blocked ((Cycle/Count))

board.cache_hierarchy.l2cache.demandMshrMisses::processor.cores.core.inst 196 # number of demand (read+write) MSHR misses (Count)

board.cache_hierarchy.l2cache.demandMshrMisses::processor.cores.core.data 150 # number of demand (read+write) MSHR misses (Count)

board.cache_hierarchy.l2cache.demandMshrMisses::total 346 # number of demand (read+write) MSHR misses (Count)

board.cache_hierarchy.l2cache.overallMshrMisses::cache_hierarchy.l2cache.prefetcher 45 # number of overall MSHR misses (Count)

board.cache_hierarchy.l2cache.overallMshrMisses::processor.cores.core.inst 196 # number of overall MSHR misses (Count)

board.cache_hierarchy.l2cache.overallMshrMisses::processor.cores.core.data 150 # number of overall MSHR misses (Count)

board.cache_hierarchy.l2cache.overallMshrMisses::total 391 # number of overall MSHR misses (Count)

board.cache_hierarchy.l2cache.demandMshrMissLatency::processor.cores.core.inst 11023965 # number of demand (read+write) MSHR miss ticks (Tick)

board.cache_hierarchy.l2cache.demandMshrMissLatency::processor.cores.core.data 7949376 # number of demand (read+write) MSHR miss ticks (Tick)

board.cache_hierarchy.l2cache.demandMshrMissLatency::total 18973341 # number of demand (read+write) MSHR miss ticks (Tick)

board.cache_hierarchy.l2cache.overallMshrMissLatency::cache_hierarchy.l2cache.prefetcher 2050185 # number of overall MSHR miss ticks (Tick)

board.cache_hierarchy.l2cache.overallMshrMissLatency::processor.cores.core.inst 11023965 # number of overall MSHR miss ticks (Tick)

board.cache_hierarchy.l2cache.overallMshrMissLatency::processor.cores.core.data 7949376 # number of overall MSHR miss ticks (Tick)

board.cache_hierarchy.l2cache.overallMshrMissLatency::total 21023526 # number of overall MSHR miss ticks (Tick)

board.cache_hierarchy.l2cache.demandMshrMissRate::processor.cores.core.inst 1 # mshr miss ratio for demand accesses (Ratio)

board.cache_hierarchy.l2cache.demandMshrMissRate::processor.cores.core.data 0.810811 # mshr miss ratio for demand accesses (Ratio)

board.cache_hierarchy.l2cache.demandMshrMissRate::total 0.908136 # mshr miss ratio for demand accesses (Ratio)

board.cache_hierarchy.l2cache.overallMshrMissRate::cache_hierarchy.l2cache.prefetcher inf # mshr miss ratio for overall accesses (Ratio)

board.cache_hierarchy.l2cache.overallMshrMissRate::processor.cores.core.inst 1 # mshr miss ratio for overall accesses (Ratio)

board.cache_hierarchy.l2cache.overallMshrMissRate::processor.cores.core.data 0.810811 # mshr miss ratio for overall accesses (Ratio)

board.cache_hierarchy.l2cache.overallMshrMissRate::total 1.026247 # mshr miss ratio for overall accesses (Ratio)

board.cache_hierarchy.l2cache.demandAvgMshrMissLatency::processor.cores.core.inst 56244.719388 # average overall mshr miss latency ((Tick/Count))

board.cache_hierarchy.l2cache.demandAvgMshrMissLatency::processor.cores.core.data 52995.840000 # average overall mshr miss latency ((Tick/Count))

board.cache_hierarchy.l2cache.demandAvgMshrMissLatency::total 54836.245665 # average overall mshr miss latency ((Tick/Count))

board.cache_hierarchy.l2cache.overallAvgMshrMissLatency::cache_hierarchy.l2cache.prefetcher 45559.666667 # average overall mshr miss latency ((Tick/Count))

board.cache_hierarchy.l2cache.overallAvgMshrMissLatency::processor.cores.core.inst 56244.719388 # average overall mshr miss latency ((Tick/Count))

board.cache_hierarchy.l2cache.overallAvgMshrMissLatency::processor.cores.core.data 52995.840000 # average overall mshr miss latency ((Tick/Count))

board.cache_hierarchy.l2cache.overallAvgMshrMissLatency::total 53768.608696 # average overall mshr miss latency ((Tick/Count))

board.cache_hierarchy.l2cache.replacements 0 # number of replacements (Count)

board.cache_hierarchy.l2cache.HardPFReq.mshrMisses::cache_hierarchy.l2cache.prefetcher 45 # number of HardPFReq MSHR misses (Count)

board.cache_hierarchy.l2cache.HardPFReq.mshrMisses::total 45 # number of HardPFReq MSHR misses (Count)

board.cache_hierarchy.l2cache.HardPFReq.mshrMissLatency::cache_hierarchy.l2cache.prefetcher 2050185 # number of HardPFReq MSHR miss ticks (Tick)

board.cache_hierarchy.l2cache.HardPFReq.mshrMissLatency::total 2050185 # number of HardPFReq MSHR miss ticks (Tick)

board.cache_hierarchy.l2cache.HardPFReq.mshrMissRate::cache_hierarchy.l2cache.prefetcher inf # mshr miss rate for HardPFReq accesses (Ratio)

board.cache_hierarchy.l2cache.HardPFReq.mshrMissRate::total inf # mshr miss rate for HardPFReq accesses (Ratio)

board.cache_hierarchy.l2cache.HardPFReq.avgMshrMissLatency::cache_hierarchy.l2cache.prefetcher 45559.666667 # average HardPFReq mshr miss latency ((Tick/Count))

board.cache_hierarchy.l2cache.HardPFReq.avgMshrMissLatency::total 45559.666667 # average HardPFReq mshr miss latency ((Tick/Count))

board.cache_hierarchy.l2cache.ReadExReq.hits::processor.cores.core.data 30 # number of ReadExReq hits (Count)

board.cache_hierarchy.l2cache.ReadExReq.hits::total 30 # number of ReadExReq hits (Count)

board.cache_hierarchy.l2cache.ReadExReq.misses::processor.cores.core.data 56 # number of ReadExReq misses (Count)

board.cache_hierarchy.l2cache.ReadExReq.misses::total 56 # number of ReadExReq misses (Count)

board.cache_hierarchy.l2cache.ReadExReq.missLatency::processor.cores.core.data 3340656 # number of ReadExReq miss ticks (Tick)

board.cache_hierarchy.l2cache.ReadExReq.missLatency::total 3340656 # number of ReadExReq miss ticks (Tick)

board.cache_hierarchy.l2cache.ReadExReq.accesses::processor.cores.core.data 86 # number of ReadExReq accesses(hits+misses) (Count)

board.cache_hierarchy.l2cache.ReadExReq.accesses::total 86 # number of ReadExReq accesses(hits+misses) (Count)

board.cache_hierarchy.l2cache.ReadExReq.missRate::processor.cores.core.data 0.651163 # miss rate for ReadExReq accesses (Ratio)

board.cache_hierarchy.l2cache.ReadExReq.missRate::total 0.651163 # miss rate for ReadExReq accesses (Ratio)

board.cache_hierarchy.l2cache.ReadExReq.avgMissLatency::processor.cores.core.data 59654.571429 # average ReadExReq miss latency ((Tick/Count))

board.cache_hierarchy.l2cache.ReadExReq.avgMissLatency::total 59654.571429 # average ReadExReq miss latency ((Tick/Count))

board.cache_hierarchy.l2cache.ReadExReq.mshrMisses::processor.cores.core.data 56 # number of ReadExReq MSHR misses (Count)

board.cache_hierarchy.l2cache.ReadExReq.mshrMisses::total 56 # number of ReadExReq MSHR misses (Count)

board.cache_hierarchy.l2cache.ReadExReq.mshrMissLatency::processor.cores.core.data 2967696 # number of ReadExReq MSHR miss ticks (Tick)

board.cache_hierarchy.l2cache.ReadExReq.mshrMissLatency::total 2967696 # number of ReadExReq MSHR miss ticks (Tick)

board.cache_hierarchy.l2cache.ReadExReq.mshrMissRate::processor.cores.core.data 0.651163 # mshr miss rate for ReadExReq accesses (Ratio)

board.cache_hierarchy.l2cache.ReadExReq.mshrMissRate::total 0.651163 # mshr miss rate for ReadExReq accesses (Ratio)

board.cache_hierarchy.l2cache.ReadExReq.avgMshrMissLatency::processor.cores.core.data 52994.571429 # average ReadExReq mshr miss latency ((Tick/Count))

board.cache_hierarchy.l2cache.ReadExReq.avgMshrMissLatency::total 52994.571429 # average ReadExReq mshr miss latency ((Tick/Count))

board.cache_hierarchy.l2cache.ReadSharedReq.hits::processor.cores.core.data 5 # number of ReadSharedReq hits (Count)

board.cache_hierarchy.l2cache.ReadSharedReq.hits::total 5 # number of ReadSharedReq hits (Count)

board.cache_hierarchy.l2cache.ReadSharedReq.misses::processor.cores.core.inst 196 # number of ReadSharedReq misses (Count)

board.cache_hierarchy.l2cache.ReadSharedReq.misses::processor.cores.core.data 94 # number of ReadSharedReq misses (Count)

board.cache_hierarchy.l2cache.ReadSharedReq.misses::total 290 # number of ReadSharedReq misses (Count)

board.cache_hierarchy.l2cache.ReadSharedReq.missLatency::processor.cores.core.inst 12329325 # number of ReadSharedReq miss ticks (Tick)

board.cache_hierarchy.l2cache.ReadSharedReq.missLatency::processor.cores.core.data 5607720 # number of ReadSharedReq miss ticks (Tick)

board.cache_hierarchy.l2cache.ReadSharedReq.missLatency::total 17937045 # number of ReadSharedReq miss ticks (Tick)

board.cache_hierarchy.l2cache.ReadSharedReq.accesses::processor.cores.core.inst 196 # number of ReadSharedReq accesses(hits+misses) (Count)

board.cache_hierarchy.l2cache.ReadSharedReq.accesses::processor.cores.core.data 99 # number of ReadSharedReq accesses(hits+misses) (Count)

board.cache_hierarchy.l2cache.ReadSharedReq.accesses::total 295 # number of ReadSharedReq accesses(hits+misses) (Count)

board.cache_hierarchy.l2cache.ReadSharedReq.missRate::processor.cores.core.inst 1 # miss rate for ReadSharedReq accesses (Ratio)

board.cache_hierarchy.l2cache.ReadSharedReq.missRate::processor.cores.core.data 0.949495 # miss rate for ReadSharedReq accesses (Ratio)

board.cache_hierarchy.l2cache.ReadSharedReq.missRate::total 0.983051 # miss rate for ReadSharedReq accesses (Ratio)

board.cache_hierarchy.l2cache.ReadSharedReq.avgMissLatency::processor.cores.core.inst 62904.719388 # average ReadSharedReq miss latency ((Tick/Count))

board.cache_hierarchy.l2cache.ReadSharedReq.avgMissLatency::processor.cores.core.data 59656.595745 # average ReadSharedReq miss latency ((Tick/Count))

board.cache_hierarchy.l2cache.ReadSharedReq.avgMissLatency::total 61851.879310 # average ReadSharedReq miss latency ((Tick/Count))

board.cache_hierarchy.l2cache.ReadSharedReq.mshrMisses::processor.cores.core.inst 196 # number of ReadSharedReq MSHR misses (Count)

board.cache_hierarchy.l2cache.ReadSharedReq.mshrMisses::processor.cores.core.data 94 # number of ReadSharedReq MSHR misses (Count)

board.cache_hierarchy.l2cache.ReadSharedReq.mshrMisses::total 290 # number of ReadSharedReq MSHR misses (Count)

board.cache_hierarchy.l2cache.ReadSharedReq.mshrMissLatency::processor.cores.core.inst 11023965 # number of ReadSharedReq MSHR miss ticks (Tick)

board.cache_hierarchy.l2cache.ReadSharedReq.mshrMissLatency::processor.cores.core.data 4981680 # number of ReadSharedReq MSHR miss ticks (Tick)

board.cache_hierarchy.l2cache.ReadSharedReq.mshrMissLatency::total 16005645 # number of ReadSharedReq MSHR miss ticks (Tick)

board.cache_hierarchy.l2cache.ReadSharedReq.mshrMissRate::processor.cores.core.inst 1 # mshr miss rate for ReadSharedReq accesses (Ratio)

board.cache_hierarchy.l2cache.ReadSharedReq.mshrMissRate::processor.cores.core.data 0.949495 # mshr miss rate for ReadSharedReq accesses (Ratio)

board.cache_hierarchy.l2cache.ReadSharedReq.mshrMissRate::total 0.983051 # mshr miss rate for ReadSharedReq accesses (Ratio)

board.cache_hierarchy.l2cache.ReadSharedReq.avgMshrMissLatency::processor.cores.core.inst 56244.719388 # average ReadSharedReq mshr miss latency ((Tick/Count))

board.cache_hierarchy.l2cache.ReadSharedReq.avgMshrMissLatency::processor.cores.core.data 52996.595745 # average ReadSharedReq mshr miss latency ((Tick/Count))

board.cache_hierarchy.l2cache.ReadSharedReq.avgMshrMissLatency::total 55191.879310 # average ReadSharedReq mshr miss latency ((Tick/Count))

board.cache_hierarchy.l2cache.power_state.pwrStateResidencyTicks::UNDEFINED 10952226214626250 # Cumulative time (in ticks) in various power states (Tick)

board.cache_hierarchy.l2cache.prefetcher.demandMshrMisses 346 # demands not covered by prefetchs (Count)

board.cache_hierarchy.l2cache.prefetcher.pfIssued 146 # number of hwpf issued (Count)

board.cache_hierarchy.l2cache.prefetcher.pfUseful 35 # number of useful prefetch (Count)

board.cache_hierarchy.l2cache.prefetcher.pfUsefulButMiss 0 # number of hit on prefetch but cache block is not in an usable state (Count)

board.cache_hierarchy.l2cache.prefetcher.accuracy 0.239726 # accuracy of the prefetcher (Count)

board.cache_hierarchy.l2cache.prefetcher.coverage 0.091864 # coverage brought by this prefetcher (Count)

board.cache_hierarchy.l2cache.prefetcher.pfHitInCache 50 # number of prefetches hitting in cache (Count)

board.cache_hierarchy.l2cache.prefetcher.pfHitInMSHR 51 # number of prefetches hitting in a MSHR (Count)

board.cache_hierarchy.l2cache.prefetcher.pfHitInWB 0 # number of prefetches hit in the Write Buffer (Count)

board.cache_hierarchy.l2cache.prefetcher.pfLate 101 # number of late prefetches (hitting in cache, MSHR or WB) (Count)

board.cache_hierarchy.l2cache.prefetcher.pfIdentified 146 # number of prefetch candidates identified (Count)

board.cache_hierarchy.l2cache.prefetcher.pfBufferHit 0 # number of redundant prefetches already in prefetch queue (Count)

board.cache_hierarchy.l2cache.prefetcher.pfInCache 0 # number of redundant prefetches already in cache/mshr dropped (Count)

board.cache_hierarchy.l2cache.prefetcher.pfRemovedDemand 0 # number of prefetches dropped due to a demand for the same address (Count)

board.cache_hierarchy.l2cache.prefetcher.pfRemovedFull 0 # number of prefetches dropped due to prefetch queue size (Count)

board.cache_hierarchy.l2cache.prefetcher.pfSpanPage 6 # number of prefetches that crossed the page (Count)

board.cache_hierarchy.l2cache.prefetcher.pfUsefulSpanPage 5 # number of prefetches that is useful and crossed the page (Count)

board.cache_hierarchy.l2cache.prefetcher.power_state.pwrStateResidencyTicks::UNDEFINED 10952226214626250 # Cumulative time (in ticks) in various power states (Tick)

board.cache_hierarchy.l2cache.tags.tagsInUse 391.000000 # Average ticks per tags in use ((Tick/Count))

board.cache_hierarchy.l2cache.tags.totalRefs 0 # Total number of references to valid blocks. (Count)

board.cache_hierarchy.l2cache.tags.sampledRefs 0 # Sample count of references to valid blocks. (Count)

board.cache_hierarchy.l2cache.tags.avgRefs nan # Average number of references to valid blocks. ((Count/Count))

board.cache_hierarchy.l2cache.tags.warmupTick 69597 # The tick when the warmup percentage was hit. (Tick)

board.cache_hierarchy.l2cache.tags.occupancies::cache_hierarchy.l2cache.prefetcher 45.000000 # Average occupied blocks per tick, per requestor ((Count/Tick))

board.cache_hierarchy.l2cache.tags.occupancies::processor.cores.core.inst 196.000000 # Average occupied blocks per tick, per requestor ((Count/Tick))

board.cache_hierarchy.l2cache.tags.occupancies::processor.cores.core.data 150.000000 # Average occupied blocks per tick, per requestor ((Count/Tick))

board.cache_hierarchy.l2cache.tags.avgOccs::cache_hierarchy.l2cache.prefetcher 0.010986 # Average percentage of cache occupancy ((Ratio/Tick))

board.cache_hierarchy.l2cache.tags.avgOccs::processor.cores.core.inst 0.047852 # Average percentage of cache occupancy ((Ratio/Tick))

board.cache_hierarchy.l2cache.tags.avgOccs::processor.cores.core.data 0.036621 # Average percentage of cache occupancy ((Ratio/Tick))

board.cache_hierarchy.l2cache.tags.avgOccs::total 0.095459 # Average percentage of cache occupancy ((Ratio/Tick))

board.cache_hierarchy.l2cache.tags.occupanciesTaskId::1022 45 # Occupied blocks per task id (Count)

board.cache_hierarchy.l2cache.tags.occupanciesTaskId::1024 346 # Occupied blocks per task id (Count)

board.cache_hierarchy.l2cache.tags.ageTaskId_1022::4 45 # Occupied blocks per task id, per block age (Count)

board.cache_hierarchy.l2cache.tags.ageTaskId_1024::4 346 # Occupied blocks per task id, per block age (Count)

board.cache_hierarchy.l2cache.tags.ratioOccsTaskId::1022 0.010986 # Ratio of occupied blocks and all blocks, per task id (Ratio)

board.cache_hierarchy.l2cache.tags.ratioOccsTaskId::1024 0.084473 # Ratio of occupied blocks and all blocks, per task id (Ratio)

board.cache_hierarchy.l2cache.tags.tagAccesses 6487 # Number of tag accesses (Count)

board.cache_hierarchy.l2cache.tags.dataAccesses 6487 # Number of data accesses (Count)

board.cache_hierarchy.l2cache.tags.power_state.pwrStateResidencyTicks::UNDEFINED 10952226214626250 # Cumulative time (in ticks) in various power states (Tick)

board.cache_hierarchy.membus.transDist::ReadResp 335 # Transaction distribution (Count)

board.cache_hierarchy.membus.transDist::ReadExReq 56 # Transaction distribution (Count)

board.cache_hierarchy.membus.transDist::ReadExResp 56 # Transaction distribution (Count)

board.cache_hierarchy.membus.transDist::ReadSharedReq 335 # Transaction distribution (Count)

board.cache_hierarchy.membus.pktCount_board.cache_hierarchy.l2cache.mem_side_port::board.memory.mem_ctrl.port 782 # Packet count per connected requestor and responder (Count)

board.cache_hierarchy.membus.pktCount_board.cache_hierarchy.l2cache.mem_side_port::total 782 # Packet count per connected requestor and responder (Count)

board.cache_hierarchy.membus.pktCount::total 782 # Packet count per connected requestor and responder (Count)

board.cache_hierarchy.membus.pktSize_board.cache_hierarchy.l2cache.mem_side_port::board.memory.mem_ctrl.port 25024 # Cumulative packet size per connected requestor and responder (Byte)

board.cache_hierarchy.membus.pktSize_board.cache_hierarchy.l2cache.mem_side_port::total 25024 # Cumulative packet size per connected requestor and responder (Byte)

board.cache_hierarchy.membus.pktSize::total 25024 # Cumulative packet size per connected requestor and responder (Byte)

board.cache_hierarchy.membus.snoops 0 # Total snoops (Count)

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board.cache_hierarchy.membus.snoopTraffic 0 # Total snoop traffic (Byte)
board.cache_hierarchy.membus.snoopFanout::samples 391 # Request fanout histogram (Count)
board.cache_hierarchy.membus.snoopFanout::mean 0 # Request fanout histogram (Count)
board.cache_hierarchy.membus.snoopFanout::stdev 0 # Request fanout histogram (Count)
board.cache_hierarchy.membus.snoopFanout::underflows 0 0.00% 0.00% # Request fanout histogram (Count)
board.cache_hierarchy.membus.snoopFanout::0 391 100.00% 100.00% # Request fanout histogram (Count)
board.cache_hierarchy.membus.snoopFanout::1 0 0.00% 100.00% # Request fanout histogram (Count)
board.cache_hierarchy.membus.snoopFanout::overflows 0 0.00% 100.00% # Request fanout histogram (Count)
board.cache_hierarchy.membus.snoopFanout::min_value 0 # Request fanout histogram (Count)
board.cache_hierarchy.membus.snoopFanout::max_value 0 # Request fanout histogram (Count)
board.cache_hierarchy.membus.snoopFanout::total 391 # Request fanout histogram (Count)
board.cache_hierarchy.membus.badaddr_responder.power_state.pwrStateResidencyTicks::UNDEFINED 10952226214626250 # Cumulative time (in ticks) in various power states (Tick)
board.cache_hierarchy.membus.power_state.pwrStateResidencyTicks::UNDEFINED 10952226214626250 # Cumulative time (in ticks) in various power states (Tick)
board.cache_hierarchy.membus.reqLayer0.occupancy 139443 # Layer occupancy (ticks) (Tick)
board.cache_hierarchy.membus.reqLayer0.utilization 0.0 # Layer utilization (Ratio)
board.cache_hierarchy.membus.respLayer2.occupancy 333977 # Layer occupancy (ticks) (Tick)
board.cache_hierarchy.membus.respLayer2.utilization 0.0 # Layer utilization (Ratio)
board.cache_hierarchy.membus.snoop_filter.totRequests 391 # Total number of requests made to the snoop filter. (Count)
board.cache_hierarchy.membus.snoop_filter.hitSingleRequests 0 # Number of requests hitting in the snoop filter with a single holder of the requested data. (Count)
board.cache_hierarchy.membus.snoop_filter.hitMultiRequests 0 # Number of requests hitting in the snoop filter with multiple (>1) holders of the requested data. (Count)
board.cache_hierarchy.membus.snoop_filter.totSnoops 0 # Total number of snoops made to the snoop filter. (Count)
board.cache_hierarchy.membus.snoop_filter.hitSingleSnoops 0 # Number of snoops hitting in the snoop filter with a single holder of the requested data. (Count)
board.cache_hierarchy.membus.snoop_filter.hitMultiSnoops 0 # Number of snoops hitting in the snoop filter with multiple (>1) holders of the requested data. (Count)
board.clk_domain.clock 333 # Clock period in ticks (Tick)
board.clk_domain.voltage_domain.voltage 1 # Voltage in Volts (Volt)
board.memory.mem_ctrl.avgPriority_cache_hierarchy.l2cache.prefetcher::samples 45.00 # Average QoS priority value for accepted requests (Count)
board.memory.mem_ctrl.avgPriority_processor.cores.core.inst::samples 196.00 # Average QoS priority value for accepted requests (Count)
board.memory.mem_ctrl.avgPriority_processor.cores.core.data::samples 150.00 # Average QoS priority value for accepted requests (Count)
board.memory.mem_ctrl.priorityMinLatency 0.000000018750 # per QoS priority minimum request to response latency (Second)
board.memory.mem_ctrl.priorityMaxLatency 0.000000471124 # per QoS priority maximum request to response latency (Second)
board.memory.mem_ctrl.numReadWriteTurnArounds 0 # Number of turnarounds from READ to WRITE (Count)
board.memory.mem_ctrl.numWriteReadTurnArounds 0 # Number of turnarounds from WRITE to READ (Count)
board.memory.mem_ctrl.numStayReadState 2813223118 # Number of times bus staying in READ state (Count)
board.memory.mem_ctrl.numStayWriteState 0 # Number of times bus staying in WRITE state (Count)
board.memory.mem_ctrl.readReqs 391 # Number of read requests accepted (Count)
board.memory.mem_ctrl.writeReqs 0 # Number of write requests accepted (Count)
board.memory.mem_ctrl.readBursts 391 # Number of controller read bursts, including those serviced by the write queue (Count)
board.memory.mem_ctrl.writeBursts 0 # Number of controller write bursts, including those merged in the write queue (Count)
board.memory.mem_ctrl.servicedByWrQ 0 # Number of controller read bursts serviced by the write queue (Count)
board.memory.mem_ctrl.mergedWrBursts 0 # Number of controller write bursts merged with an existing one (Count)
board.memory.mem_ctrl.neitherReadNorWriteReqs 0 # Number of requests that are neither read nor write (Count)
board.memory.mem_ctrl.avgRdQLen 1.00 # Average read queue length when enqueuing ((Count/Tick))
board.memory.mem_ctrl.avgWrQLen 0.00 # Average write queue length when enqueuing ((Count/Tick))
board.memory.mem_ctrl.numRdRetry 0 # Number of times read queue was full causing retry (Count)
board.memory.mem_ctrl.numWrRetry 0 # Number of times write queue was full causing retry (Count)
board.memory.mem_ctrl.readPktSize::0 0 # Read request sizes (log2) (Count)
board.memory.mem_ctrl.readPktSize::1 0 # Read request sizes (log2) (Count)
board.memory.mem_ctrl.readPktSize::2 0 # Read request sizes (log2) (Count)
board.memory.mem_ctrl.readPktSize::3 0 # Read request sizes (log2) (Count)
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board.memory.mem_ctrl.readPktSize::4	0	# Read request sizes (log2) (Count)
board.memory.mem_ctrl.readPktSize::5	0	# Read request sizes (log2) (Count)
board.memory.mem_ctrl.readPktSize::6	391	# Read request sizes (log2) (Count)
board.memory.mem_ctrl.writePktSize::0	0	# Write request sizes (log2) (Count)
board.memory.mem_ctrl.writePktSize::1	0	# Write request sizes (log2) (Count)
board.memory.mem_ctrl.writePktSize::2	0	# Write request sizes (log2) (Count)
board.memory.mem_ctrl.writePktSize::3	0	# Write request sizes (log2) (Count)
board.memory.mem_ctrl.writePktSize::4	0	# Write request sizes (log2) (Count)
board.memory.mem_ctrl.writePktSize::5	0	# Write request sizes (log2) (Count)
board.memory.mem_ctrl.writePktSize::6	0	# Write request sizes (log2) (Count)
board.memory.mem_ctrl.rdQLenPdf::0	373	# What read queue length does an incoming req see (Count)
board.memory.mem_ctrl.rdQLenPdf::1	6	# What read queue length does an incoming req see (Count)
board.memory.mem_ctrl.rdQLenPdf::2	5	# What read queue length does an incoming req see (Count)
board.memory.mem_ctrl.rdQLenPdf::3	4	# What read queue length does an incoming req see (Count)
board.memory.mem_ctrl.rdQLenPdf::4	3	# What read queue length does an incoming req see (Count)
board.memory.mem_ctrl.rdQLenPdf::5	0	# What read queue length does an incoming req see (Count)
board.memory.mem_ctrl.rdQLenPdf::6	0	# What read queue length does an incoming req see (Count)
board.memory.mem_ctrl.rdQLenPdf::7	0	# What read queue length does an incoming req see (Count)
board.memory.mem_ctrl.rdQLenPdf::8	0	# What read queue length does an incoming req see (Count)
board.memory.mem_ctrl.rdQLenPdf::9	0	# What read queue length does an incoming req see (Count)
board.memory.mem_ctrl.rdQLenPdf::10	0	# What read queue length does an incoming req see (Count)
board.memory.mem_ctrl.rdQLenPdf::11	0	# What read queue length does an incoming req see (Count)
board.memory.mem_ctrl.rdQLenPdf::12	0	# What read queue length does an incoming req see (Count)
board.memory.mem_ctrl.rdQLenPdf::13	0	# What read queue length does an incoming req see (Count)
board.memory.mem_ctrl.rdQLenPdf::14	0	# What read queue length does an incoming req see (Count)
board.memory.mem_ctrl.rdQLenPdf::15	0	# What read queue length does an incoming req see (Count)
board.memory.mem_ctrl.rdQLenPdf::16	0	# What read queue length does an incoming req see (Count)
board.memory.mem_ctrl.rdQLenPdf::17	0	# What read queue length does an incoming req see (Count)
board.memory.mem_ctrl.rdQLenPdf::18	0	# What read queue length does an incoming req see (Count)
board.memory.mem_ctrl.rdQLenPdf::19	0	# What read queue length does an incoming req see (Count)
board.memory.mem_ctrl.rdQLenPdf::20	0	# What read queue length does an incoming req see (Count)
board.memory.mem_ctrl.rdQLenPdf::21	0	# What read queue length does an incoming req see (Count)
board.memory.mem_ctrl.rdQLenPdf::22	0	# What read queue length does an incoming req see (Count)
board.memory.mem_ctrl.rdQLenPdf::23	0	# What read queue length does an incoming req see (Count)
board.memory.mem_ctrl.rdQLenPdf::24	0	# What read queue length does an incoming req see (Count)
board.memory.mem_ctrl.rdQLenPdf::25	0	# What read queue length does an incoming req see (Count)
board.memory.mem_ctrl.rdQLenPdf::26	0	# What read queue length does an incoming req see (Count)
board.memory.mem_ctrl.rdQLenPdf::27	0	# What read queue length does an incoming req see (Count)
board.memory.mem_ctrl.rdQLenPdf::28	0	# What read queue length does an incoming req see (Count)
board.memory.mem_ctrl.rdQLenPdf::29	0	# What read queue length does an incoming req see (Count)
board.memory.mem_ctrl.rdQLenPdf::30	0	# What read queue length does an incoming req see (Count)
board.memory.mem_ctrl.rdQLenPdf::31	0	# What read queue length does an incoming req see (Count)
board.memory.mem_ctrl.wrQLenPdf::0	0	# What write queue length does an incoming req see (Count)
board.memory.mem_ctrl.wrQLenPdf::1	0	# What write queue length does an incoming req see (Count)
board.memory.mem_ctrl.wrQLenPdf::2	0	# What write queue length does an incoming req see (Count)
board.memory.mem_ctrl.wrQLenPdf::3	0	# What write queue length does an incoming req see (Count)
board.memory.mem_ctrl.wrQLenPdf::4	0	# What write queue length does an incoming req see (Count)
board.memory.mem_ctrl.wrQLenPdf::5	0	# What write queue length does an incoming req see (Count)
board.memory.mem_ctrl.wrQLenPdf::6	0	# What write queue length does an incoming req see (Count)

board.memory.mem_ctrl.wrQLenPdf::7	0	# What write queue length does an incoming req see (Count)
board.memory.mem_ctrl.wrQLenPdf::8	0	# What write queue length does an incoming req see (Count)
board.memory.mem_ctrl.wrQLenPdf::9	0	# What write queue length does an incoming req see (Count)
board.memory.mem_ctrl.wrQLenPdf::10	0	# What write queue length does an incoming req see (Count)
board.memory.mem_ctrl.wrQLenPdf::11	0	# What write queue length does an incoming req see (Count)
board.memory.mem_ctrl.wrQLenPdf::12	0	# What write queue length does an incoming req see (Count)
board.memory.mem_ctrl.wrQLenPdf::13	0	# What write queue length does an incoming req see (Count)
board.memory.mem_ctrl.wrQLenPdf::14	0	# What write queue length does an incoming req see (Count)
board.memory.mem_ctrl.wrQLenPdf::15	0	# What write queue length does an incoming req see (Count)
board.memory.mem_ctrl.wrQLenPdf::16	0	# What write queue length does an incoming req see (Count)
board.memory.mem_ctrl.wrQLenPdf::17	0	# What write queue length does an incoming req see (Count)
board.memory.mem_ctrl.wrQLenPdf::18	0	# What write queue length does an incoming req see (Count)
board.memory.mem_ctrl.wrQLenPdf::19	0	# What write queue length does an incoming req see (Count)
board.memory.mem_ctrl.wrQLenPdf::20	0	# What write queue length does an incoming req see (Count)
board.memory.mem_ctrl.wrQLenPdf::21	0	# What write queue length does an incoming req see (Count)
board.memory.mem_ctrl.wrQLenPdf::22	0	# What write queue length does an incoming req see (Count)
board.memory.mem_ctrl.wrQLenPdf::23	0	# What write queue length does an incoming req see (Count)
board.memory.mem_ctrl.wrQLenPdf::24	0	# What write queue length does an incoming req see (Count)
board.memory.mem_ctrl.wrQLenPdf::25	0	# What write queue length does an incoming req see (Count)
board.memory.mem_ctrl.wrQLenPdf::26	0	# What write queue length does an incoming req see (Count)
board.memory.mem_ctrl.wrQLenPdf::27	0	# What write queue length does an incoming req see (Count)
board.memory.mem_ctrl.wrQLenPdf::28	0	# What write queue length does an incoming req see (Count)
board.memory.mem_ctrl.wrQLenPdf::29	0	# What write queue length does an incoming req see (Count)
board.memory.mem_ctrl.wrQLenPdf::30	0	# What write queue length does an incoming req see (Count)
board.memory.mem_ctrl.wrQLenPdf::31	0	# What write queue length does an incoming req see (Count)
board.memory.mem_ctrl.wrQLenPdf::32	0	# What write queue length does an incoming req see (Count)
board.memory.mem_ctrl.wrQLenPdf::33	0	# What write queue length does an incoming req see (Count)
board.memory.mem_ctrl.wrQLenPdf::34	0	# What write queue length does an incoming req see (Count)
board.memory.mem_ctrl.wrQLenPdf::35	0	# What write queue length does an incoming req see (Count)
board.memory.mem_ctrl.wrQLenPdf::36	0	# What write queue length does an incoming req see (Count)
board.memory.mem_ctrl.wrQLenPdf::37	0	# What write queue length does an incoming req see (Count)
board.memory.mem_ctrl.wrQLenPdf::38	0	# What write queue length does an incoming req see (Count)
board.memory.mem_ctrl.wrQLenPdf::39	0	# What write queue length does an incoming req see (Count)
board.memory.mem_ctrl.wrQLenPdf::40	0	# What write queue length does an incoming req see (Count)
board.memory.mem_ctrl.wrQLenPdf::41	0	# What write queue length does an incoming req see (Count)
board.memory.mem_ctrl.wrQLenPdf::42	0	# What write queue length does an incoming req see (Count)
board.memory.mem_ctrl.wrQLenPdf::43	0	# What write queue length does an incoming req see (Count)
board.memory.mem_ctrl.wrQLenPdf::44	0	# What write queue length does an incoming req see (Count)
board.memory.mem_ctrl.wrQLenPdf::45	0	# What write queue length does an incoming req see (Count)
board.memory.mem_ctrl.wrQLenPdf::46	0	# What write queue length does an incoming req see (Count)
board.memory.mem_ctrl.wrQLenPdf::47	0	# What write queue length does an incoming req see (Count)
board.memory.mem_ctrl.wrQLenPdf::48	0	# What write queue length does an incoming req see (Count)
board.memory.mem_ctrl.wrQLenPdf::49	0	# What write queue length does an incoming req see (Count)
board.memory.mem_ctrl.wrQLenPdf::50	0	# What write queue length does an incoming req see (Count)
board.memory.mem_ctrl.wrQLenPdf::51	0	# What write queue length does an incoming req see (Count)
board.memory.mem_ctrl.wrQLenPdf::52	0	# What write queue length does an incoming req see (Count)
board.memory.mem_ctrl.wrQLenPdf::53	0	# What write queue length does an incoming req see (Count)
board.memory.mem_ctrl.wrQLenPdf::54	0	# What write queue length does an incoming req see (Count)
board.memory.mem_ctrl.wrQLenPdf::55	0	# What write queue length does an incoming req see (Count)

board.memory.mem_ctrl.wrQLenPdf::56 0 # What write queue length does an incoming req see (Count)

board.memory.mem_ctrl.wrQLenPdf::57 0 # What write queue length does an incoming req see (Count)

board.memory.mem_ctrl.wrQLenPdf::58 0 # What write queue length does an incoming req see (Count)

board.memory.mem_ctrl.wrQLenPdf::59 0 # What write queue length does an incoming req see (Count)

board.memory.mem_ctrl.wrQLenPdf::60 0 # What write queue length does an incoming req see (Count)

board.memory.mem_ctrl.wrQLenPdf::61 0 # What write queue length does an incoming req see (Count)

board.memory.mem_ctrl.wrQLenPdf::62 0 # What write queue length does an incoming req see (Count)

board.memory.mem_ctrl.wrQLenPdf::63 0 # What write queue length does an incoming req see (Count)

board.memory.mem_ctrl.bytesReadWrQ 0 # Total number of bytes read from write queue (Byte)

board.memory.mem_ctrl.bytesReadSys 25024 # Total read bytes from the system interface side (Byte)

board.memory.mem_ctrl.bytesWrittenSys 0 # Total written bytes from the system interface side (Byte)

board.memory.mem_ctrl.avgRdBWSys 2.28483228 # Average system read bandwidth in Byte/s ((Byte/Second))

board.memory.mem_ctrl.avgWrBWSys 0.00000000 # Average system write bandwidth in Byte/s ((Byte/Second))

board.memory.mem_ctrl.totGap 24062247 # Total gap between requests (Tick)

board.memory.mem_ctrl.avgGap 61540.27 # Average gap between requests ((Tick/Count))

board.memory.mem_ctrl.requestorReadBytes::cache_hierarchy.l2cache.prefetcher 2880 # Per-requestor bytes read from memory (Byte)

board.memory.mem_ctrl.requestorReadBytes::processor.cores.core.inst 12544 # Per-requestor bytes read from memory (Byte)

board.memory.mem_ctrl.requestorReadBytes::processor.cores.core.data 9600 # Per-requestor bytes read from memory (Byte)

board.memory.mem_ctrl.requestorReadRate::cache_hierarchy.l2cache.prefetcher 0.262960236902 # Per-requestor bytes read from memory rate ((Byte/Second))

board.memory.mem_ctrl.requestorReadRate::processor.cores.core.inst 1.145337920728 # Per-requestor bytes read from memory rate ((Byte/Second))

board.memory.mem_ctrl.requestorReadRate::processor.cores.core.data 0.876534123006 # Per-requestor bytes read from memory rate ((Byte/Second))

board.memory.mem_ctrl.requestorReadAccesses::cache_hierarchy.l2cache.prefetcher 45 # Per-requestor read serviced memory accesses (Count)

board.memory.mem_ctrl.requestorReadAccesses::processor.cores.core.inst 196 # Per-requestor read serviced memory accesses (Count)

board.memory.mem_ctrl.requestorReadAccesses::processor.cores.core.data 150 # Per-requestor read serviced memory accesses (Count)

board.memory.mem_ctrl.requestorReadTotalLat::cache_hierarchy.l2cache.prefetcher 977836 # Per-requestor read total memory access latency (Tick)

board.memory.mem_ctrl.requestorReadTotalLat::processor.cores.core.inst 5044337 # Per-requestor read total memory access latency (Tick)

board.memory.mem_ctrl.requestorReadTotalLat::processor.cores.core.data 3372950 # Per-requestor read total memory access latency (Tick)

board.memory.mem_ctrl.requestorReadAvgLat::cache_hierarchy.l2cache.prefetcher 21729.69 # Per-requestor read average memory access latency ((Tick/Count))

board.memory.mem_ctrl.requestorReadAvgLat::processor.cores.core.inst 25736.41 # Per-requestor read average memory access latency ((Tick/Count))

board.memory.mem_ctrl.requestorReadAvgLat::processor.cores.core.data 22486.33 # Per-requestor read average memory access latency ((Tick/Count))

board.memory.mem_ctrl.dram.bytesRead::cache_hierarchy.l2cache.prefetcher 2880 # Number of bytes read from this memory (Byte)

board.memory.mem_ctrl.dram.bytesRead::processor.cores.core.inst 12544 # Number of bytes read from this memory (Byte)

board.memory.mem_ctrl.dram.bytesRead::processor.cores.core.data 9600 # Number of bytes read from this memory (Byte)

board.memory.mem_ctrl.dram.bytesRead::total 25024 # Number of bytes read from this memory (Byte)

board.memory.mem_ctrl.dram.bytesInstRead::processor.cores.core.inst 12544 # Number of instructions bytes read from this memory (Byte)

board.memory.mem_ctrl.dram.bytesInstRead::total 12544 # Number of instructions bytes read from this memory (Byte)

board.memory.mem_ctrl.dram.numReads::cache_hierarchy.l2cache.prefetcher 45 # Number of read requests responded to by this memory (Count)

board.memory.mem_ctrl.dram.numReads::processor.cores.core.inst 196 # Number of read requests responded to by this memory (Count)

board.memory.mem_ctrl.dram.numReads::processor.cores.core.data 150 # Number of read requests responded to by this memory (Count)

board.memory.mem_ctrl.dram.numReads::total 391 # Number of read requests responded to by this memory (Count)

board.memory.mem_ctrl.dram.bwRead::cache_hierarchy.l2cache.prefetcher 0 # Total read bandwidth from this memory ((Byte/Second))

board.memory.mem_ctrl.dram.bwRead::processor.cores.core.inst 1 # Total read bandwidth from this memory ((Byte/Second))

board.memory.mem_ctrl.dram.bwRead::processor.cores.core.data 1 # Total read bandwidth from this memory ((Byte/Second))

board.memory.mem_ctrl.dram.bwRead::total 2 # Total read bandwidth from this memory ((Byte/Second))

board.memory.mem_ctrl.dram.bwInstRead::processor.cores.core.inst 1 # Instruction read bandwidth from this memory ((Byte/Second))

board.memory.mem_ctrl.dram.bwInstRead::total 1 # Instruction read bandwidth from this memory ((Byte/Second))

board.memory.mem_ctrl.dram.bwTotal::cache_hierarchy.l2cache.prefetcher 0 # Total bandwidth to/from this memory ((Byte/Second))

board.memory.mem_ctrl.dram.bwTotal::processor.cores.core.inst 1 # Total bandwidth to/from this memory ((Byte/Second))

board.memory.mem_ctrl.dram.bwTotal::processor.cores.core.data 1 # Total bandwidth to/from this memory ((Byte/Second))

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board.memory.mem_ctrl.dram.bwTotal::total          2                    # Total bandwidth to/from this memory ((Byte/Second))
board.memory.mem_ctrl.dram.readBursts            391                    # Number of DRAM read bursts (Count)
board.memory.mem_ctrl.dram.writeBursts             0                    # Number of DRAM write bursts (Count)
board.memory.mem_ctrl.dram.perBankRdBursts::0         68                    # Per bank write bursts (Count)
board.memory.mem_ctrl.dram.perBankRdBursts::1         62                    # Per bank write bursts (Count)
board.memory.mem_ctrl.dram.perBankRdBursts::2         19                    # Per bank write bursts (Count)
board.memory.mem_ctrl.dram.perBankRdBursts::3         37                    # Per bank write bursts (Count)
board.memory.mem_ctrl.dram.perBankRdBursts::4         46                    # Per bank write bursts (Count)
board.memory.mem_ctrl.dram.perBankRdBursts::5         66                    # Per bank write bursts (Count)
board.memory.mem_ctrl.dram.perBankRdBursts::6          3                    # Per bank write bursts (Count)
board.memory.mem_ctrl.dram.perBankRdBursts::7          3                    # Per bank write bursts (Count)
board.memory.mem_ctrl.dram.perBankRdBursts::8         38                    # Per bank write bursts (Count)
board.memory.mem_ctrl.dram.perBankRdBursts::9         17                    # Per bank write bursts (Count)
board.memory.mem_ctrl.dram.perBankRdBursts::10         9                    # Per bank write bursts (Count)
board.memory.mem_ctrl.dram.perBankRdBursts::11         2                    # Per bank write bursts (Count)
board.memory.mem_ctrl.dram.perBankRdBursts::12         0                    # Per bank write bursts (Count)
board.memory.mem_ctrl.dram.perBankRdBursts::13         2                    # Per bank write bursts (Count)
board.memory.mem_ctrl.dram.perBankRdBursts::14         2                    # Per bank write bursts (Count)
board.memory.mem_ctrl.dram.perBankRdBursts::15        17                    # Per bank write bursts (Count)
board.memory.mem_ctrl.dram.perBankWrBursts::0          0                    # Per bank write bursts (Count)
board.memory.mem_ctrl.dram.perBankWrBursts::1          0                    # Per bank write bursts (Count)
board.memory.mem_ctrl.dram.perBankWrBursts::2          0                    # Per bank write bursts (Count)
board.memory.mem_ctrl.dram.perBankWrBursts::3          0                    # Per bank write bursts (Count)
board.memory.mem_ctrl.dram.perBankWrBursts::4          0                    # Per bank write bursts (Count)
board.memory.mem_ctrl.dram.perBankWrBursts::5          0                    # Per bank write bursts (Count)
board.memory.mem_ctrl.dram.perBankWrBursts::6          0                    # Per bank write bursts (Count)
board.memory.mem_ctrl.dram.perBankWrBursts::7          0                    # Per bank write bursts (Count)
board.memory.mem_ctrl.dram.perBankWrBursts::8          0                    # Per bank write bursts (Count)
board.memory.mem_ctrl.dram.perBankWrBursts::9          0                    # Per bank write bursts (Count)
board.memory.mem_ctrl.dram.perBankWrBursts::10         0                    # Per bank write bursts (Count)
board.memory.mem_ctrl.dram.perBankWrBursts::11         0                    # Per bank write bursts (Count)
board.memory.mem_ctrl.dram.perBankWrBursts::12         0                    # Per bank write bursts (Count)
board.memory.mem_ctrl.dram.perBankWrBursts::13         0                    # Per bank write bursts (Count)
board.memory.mem_ctrl.dram.perBankWrBursts::14         0                    # Per bank write bursts (Count)
board.memory.mem_ctrl.dram.perBankWrBursts::15         0                    # Per bank write bursts (Count)
board.memory.mem_ctrl.dram.totQLat           2063873                    # Total ticks spent queuing (Tick)
board.memory.mem_ctrl.dram.totBusLat          1955000                    # Total ticks spent in databus transfers (Tick)
board.memory.mem_ctrl.dram.totMemAccLat       9395123                    # Total ticks spent from burst creation until serviced by the DRAM (Tick)
board.memory.mem_ctrl.dram.avgQLat            5278.45                    # Average queueing delay per DRAM burst ((Tick/Count))
board.memory.mem_ctrl.dram.avgBusLat          5000.00                    # Average bus latency per DRAM burst ((Tick/Count))
board.memory.mem_ctrl.dram.avgMemAccLat      24028.45                    # Average memory access latency per DRAM burst ((Tick/Count))
board.memory.mem_ctrl.dram.readRowHits            323                    # Number of row buffer hits during reads (Count)
board.memory.mem_ctrl.dram.writeRowHits             0                    # Number of row buffer hits during writes (Count)
board.memory.mem_ctrl.dram.readRowHitRate        82.61                    # Row buffer hit rate for reads (Ratio)
board.memory.mem_ctrl.dram.writeRowHitRate         nan                    # Row buffer hit rate for writes (Ratio)
board.memory.mem_ctrl.dram.bytesPerActivate::samples          68                    # Bytes accessed per row activation (Byte)
board.memory.mem_ctrl.dram.bytesPerActivate::mean          368                    # Bytes accessed per row activation (Byte)
board.memory.mem_ctrl.dram.bytesPerActivate::gmean   229.264367                    # Bytes accessed per row activation (Byte)
board.memory.mem_ctrl.dram.bytesPerActivate::stdev   338.362695                    # Bytes accessed per row activation (Byte)
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board.memory.mem_ctrl.dram.bytesPerActivate::0-127          18     26.47%     26.47% # Bytes accessed per row activation (Byte)
board.memory.mem_ctrl.dram.bytesPerActivate::128-255        16     23.53%     50.00% # Bytes accessed per row activation (Byte)
board.memory.mem_ctrl.dram.bytesPerActivate::256-383         6      8.82%     58.82% # Bytes accessed per row activation (Byte)
board.memory.mem_ctrl.dram.bytesPerActivate::384-511         7     10.29%     69.12% # Bytes accessed per row activation (Byte)
board.memory.mem_ctrl.dram.bytesPerActivate::512-639         6      8.82%     77.94% # Bytes accessed per row activation (Byte)
board.memory.mem_ctrl.dram.bytesPerActivate::640-767         2      2.94%     80.88% # Bytes accessed per row activation (Byte)
board.memory.mem_ctrl.dram.bytesPerActivate::768-895         3      4.41%     85.29% # Bytes accessed per row activation (Byte)
board.memory.mem_ctrl.dram.bytesPerActivate::896-1023        2      2.94%     88.24% # Bytes accessed per row activation (Byte)
board.memory.mem_ctrl.dram.bytesPerActivate::1024-1151       8     11.76%    100.00% # Bytes accessed per row activation (Byte)
board.memory.mem_ctrl.dram.bytesPerActivate::total          68                      # Bytes accessed per row activation (Byte)
board.memory.mem_ctrl.dram.dramBytesRead        25024                               # Total bytes read (Byte)
board.memory.mem_ctrl.dram.dramBytesWritten         0                               # Total bytes written (Byte)
board.memory.mem_ctrl.dram.avgRdBW          0.000002                                # Average DRAM read bandwidth in MiBytes/s ((Byte/Second))
board.memory.mem_ctrl.dram.avgWrBW                0                                 # Average DRAM write bandwidth in MiBytes/s ((Byte/Second))
board.memory.mem_ctrl.dram.peakBW           12800.00                                # Theoretical peak bandwidth in MiByte/s ((Byte/Second))
board.memory.mem_ctrl.dram.busUtil              0.00                                # Data bus utilization in percentage (Ratio)
board.memory.mem_ctrl.dram.busUtilRead          0.00                                # Data bus utilization in percentage for reads (Ratio)
board.memory.mem_ctrl.dram.busUtilWrite         0.00                                # Data bus utilization in percentage for writes (Ratio)
board.memory.mem_ctrl.dram.pageHitRate          82.61                               # Row buffer hit rate, read and write combined (Ratio)
board.memory.mem_ctrl.dram.power_state.pwrStateResidencyTicks::UNDEFINED 10952226214626250   # Cumulative time (in ticks) in various power states (Tick)
board.memory.mem_ctrl.dram.rank0.actEnergy       342720                             # Energy for activate commands per rank (pJ) (Joule)
board.memory.mem_ctrl.dram.rank0.preEnergy       182160                             # Energy for precharge commands per rank (pJ) (Joule)
board.memory.mem_ctrl.dram.rank0.readEnergy      2170560                            # Energy for read commands per rank (pJ) (Joule)
board.memory.mem_ctrl.dram.rank0.writeEnergy           0                            # Energy for write commands per rank (pJ) (Joule)
board.memory.mem_ctrl.dram.rank0.refreshEnergy 864559488914160.125000              # Energy for refresh commands per rank (pJ) (Joule)
board.memory.mem_ctrl.dram.rank0.actBackEnergy 157948381991220                     # Energy for active background per rank (pJ) (Joule)
board.memory.mem_ctrl.dram.rank0.preBackEnergy 4072645702728960                    # Energy for precharge background per rank (pJ) (Joule)
board.memory.mem_ctrl.dram.rank0.actPowerDownEnergy           0                    # Energy for active power-down per rank (pJ) (Joule)
board.memory.mem_ctrl.dram.rank0.prePowerDownEnergy           0                    # Energy for precharge power-down per rank (pJ) (Joule)
board.memory.mem_ctrl.dram.rank0.selfRefreshEnergy            0                    # Energy for self refresh per rank (pJ) (Joule)
board.memory.mem_ctrl.dram.rank0.totalEnergy 5095153576329780                      # Total energy per rank (pJ) (Joule)
board.memory.mem_ctrl.dram.rank0.averagePower   465.216247                         # Core power per rank (mW) (Watt)
board.memory.mem_ctrl.dram.rank0.totalIdleTime           0                          # Total Idle time Per DRAM Rank (Tick)
board.memory.mem_ctrl.dram.rank0.pwrStateTime::IDLE 10586507280574826              # Time in different power states (Tick)
board.memory.mem_ctrl.dram.rank0.pwrStateTime::REF 365718903680000                 # Time in different power states (Tick)
board.memory.mem_ctrl.dram.rank0.pwrStateTime::SREF           0                    # Time in different power states (Tick)
board.memory.mem_ctrl.dram.rank0.pwrStateTime::PRE_PDN           0                # Time in different power states (Tick)
board.memory.mem_ctrl.dram.rank0.pwrStateTime::ACT     30371424                    # Time in different power states (Tick)
board.memory.mem_ctrl.dram.rank0.pwrStateTime::ACT_PDN           0                # Time in different power states (Tick)
board.memory.mem_ctrl.dram.rank1.actEnergy       142800                             # Energy for activate commands per rank (pJ) (Joule)
board.memory.mem_ctrl.dram.rank1.preEnergy        75900                             # Energy for precharge commands per rank (pJ) (Joule)
board.memory.mem_ctrl.dram.rank1.readEnergy       621180                            # Energy for read commands per rank (pJ) (Joule)
board.memory.mem_ctrl.dram.rank1.writeEnergy           0                            # Energy for write commands per rank (pJ) (Joule)
board.memory.mem_ctrl.dram.rank1.refreshEnergy 864559488914160.125000              # Energy for refresh commands per rank (pJ) (Joule)
board.memory.mem_ctrl.dram.rank1.actBackEnergy 157948381661190                     # Energy for active background per rank (pJ) (Joule)
board.memory.mem_ctrl.dram.rank1.preBackEnergy 4072645703006880                    # Energy for precharge background per rank (pJ) (Joule)
board.memory.mem_ctrl.dram.rank1.actPowerDownEnergy           0                    # Energy for active power-down per rank (pJ) (Joule)
board.memory.mem_ctrl.dram.rank1.prePowerDownEnergy           0                    # Energy for precharge power-down per rank (pJ) (Joule)
board.memory.mem_ctrl.dram.rank1.selfRefreshEnergy            0                    # Energy for self refresh per rank (pJ) (Joule)
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board.memory.mem_ctrl.dram.rank1.totalEnergy 5095153574422110                    # Total energy per rank (pJ) (Joule)
board.memory.mem_ctrl.dram.rank1.averagePower   465.216247                    # Core power per rank (mW) (Watt)
board.memory.mem_ctrl.dram.rank1.totalIdleTime           0                    # Total Idle time Per DRAM Rank (Tick)
board.memory.mem_ctrl.dram.rank1.pwrStateTime::IDLE 10586507281282782                    # Time in different power states (Tick)
board.memory.mem_ctrl.dram.rank1.pwrStateTime::REF 365718903680000                    # Time in different power states (Tick)
board.memory.mem_ctrl.dram.rank1.pwrStateTime::SREF           0                    # Time in different power states (Tick)
board.memory.mem_ctrl.dram.rank1.pwrStateTime::PRE_PDN           0                    # Time in different power states (Tick)
board.memory.mem_ctrl.dram.rank1.pwrStateTime::ACT     29663469                    # Time in different power states (Tick)
board.memory.mem_ctrl.dram.rank1.pwrStateTime::ACT_PDN           0                    # Time in different power states (Tick)
board.memory.mem_ctrl.power_state.pwrStateResidencyTicks::UNDEFINED 10952226214626250                    # Cumulative time (in ticks) in various power states (Tick)
board.processor.cores.core.numCycles           72487                    # Number of cpu cycles simulated (Cycle)
board.processor.cores.core.cpi           17.466747                    # CPI: cycles per instruction (core level) ((Cycle/Count))
board.processor.cores.core.ipc           0.057252                    # IPC: instructions per cycle (core level) ((Count/Cycle))
board.processor.cores.core.numWorkItemsStarted           0                    # Number of work items this cpu started (Count)
board.processor.cores.core.numWorkItemsCompleted           0                    # Number of work items this cpu completed (Count)
board.processor.cores.core.commitStats0.numInsts         4150                    # Number of instructions committed (thread level) (Count)
board.processor.cores.core.commitStats0.numOps         7878                    # Number of ops (including micro ops) committed (thread level) (Count)
board.processor.cores.core.commitStats0.numInstsNotNOP           0                    # Number of instructions committed excluding NOPs or prefetches (Count)
board.processor.cores.core.commitStats0.numOpsNotNOP           0                    # Number of Ops (including micro ops) Simulated (Count)
board.processor.cores.core.commitStats0.cpi     17.466747                    # CPI: cycles per instruction (thread level) ((Cycle/Count))
board.processor.cores.core.commitStats0.ipc     0.057252                    # IPC: instructions per cycle (thread level) ((Count/Cycle))
board.processor.cores.core.commitStats0.numMemRefs           0                    # Number of memory references committed (Count)
board.processor.cores.core.commitStats0.numFpInsts          703                    # Number of float instructions (Count)
board.processor.cores.core.commitStats0.numIntInsts         7438                    # Number of integer instructions (Count)
board.processor.cores.core.commitStats0.numLoadInsts          642                    # Number of load instructions (Count)
board.processor.cores.core.commitStats0.numStoreInsts          846                    # Number of store instructions (Count)
board.processor.cores.core.commitStats0.numVecInsts           0                    # Number of vector instructions (Count)
board.processor.cores.core.commitStats0.committedInstType::No_OpClass           90      1.14%      1.14% # Class of committed instruction. (Count)
board.processor.cores.core.commitStats0.committedInstType::IntAlu         5994     76.09%     77.23% # Class of committed instruction. (Count)
board.processor.cores.core.commitStats0.committedInstType::IntMult            1      0.01%     77.24% # Class of committed instruction. (Count)
board.processor.cores.core.commitStats0.committedInstType::IntDiv           21      0.27%     77.51% # Class of committed instruction. (Count)
board.processor.cores.core.commitStats0.committedInstType::FloatAdd          136      1.73%     79.23% # Class of committed instruction. (Count)
board.processor.cores.core.commitStats0.committedInstType::FloatCmp            0      0.00%     79.23% # Class of committed instruction. (Count)
board.processor.cores.core.commitStats0.committedInstType::FloatCvt            0      0.00%     79.23% # Class of committed instruction. (Count)
board.processor.cores.core.commitStats0.committedInstType::FloatMult            0      0.00%     79.23% # Class of committed instruction. (Count)
board.processor.cores.core.commitStats0.committedInstType::FloatMultAcc            0      0.00%     79.23% # Class of committed instruction. (Count)
board.processor.cores.core.commitStats0.committedInstType::FloatDiv            0      0.00%     79.23% # Class of committed instruction. (Count)
board.processor.cores.core.commitStats0.committedInstType::FloatMisc            0      0.00%     79.23% # Class of committed instruction. (Count)
board.processor.cores.core.commitStats0.committedInstType::FloatSqrt            0      0.00%     79.23% # Class of committed instruction. (Count)
board.processor.cores.core.commitStats0.committedInstType::SimdAdd            0      0.00%     79.23% # Class of committed instruction. (Count)
board.processor.cores.core.commitStats0.committedInstType::SimdAddAcc            0      0.00%     79.23% # Class of committed instruction. (Count)
board.processor.cores.core.commitStats0.committedInstType::SimdAlu           10      0.13%     79.36% # Class of committed instruction. (Count)
board.processor.cores.core.commitStats0.committedInstType::SimdCmp            0      0.00%     79.36% # Class of committed instruction. (Count)
board.processor.cores.core.commitStats0.committedInstType::SimdCvt            2      0.03%     79.39% # Class of committed instruction. (Count)
board.processor.cores.core.commitStats0.committedInstType::SimdMisc          136      1.73%     81.11% # Class of committed instruction. (Count)
board.processor.cores.core.commitStats0.committedInstType::SimdMult            0      0.00%     81.11% # Class of committed instruction. (Count)
board.processor.cores.core.commitStats0.committedInstType::SimdMultAcc            0      0.00%     81.11% # Class of committed instruction. (Count)
board.processor.cores.core.commitStats0.committedInstType::SimdMatMultAcc            0      0.00%     81.11% # Class of committed instruction. (Count)
board.processor.cores.core.commitStats0.committedInstType::SimdShift            0      0.00%     81.11% # Class of committed instruction. (Count)
```

board.processor.cores.core.commitStats0.committedInstType::SimdShiftAcc 0 0.00% 81.11% # Class of committed instruction. (Count)

board.processor.cores.core.commitStats0.committedInstType::SimdDiv 0 0.00% 81.11% # Class of committed instruction. (Count)

board.processor.cores.core.commitStats0.committedInstType::SimdSqrt 0 0.00% 81.11% # Class of committed instruction. (Count)

board.processor.cores.core.commitStats0.committedInstType::SimdFloatAdd 0 0.00% 81.11% # Class of committed instruction. (Count)

board.processor.cores.core.commitStats0.committedInstType::SimdFloatAlu 0 0.00% 81.11% # Class of committed instruction. (Count)

board.processor.cores.core.commitStats0.committedInstType::SimdFloatCmp 0 0.00% 81.11% # Class of committed instruction. (Count)

board.processor.cores.core.commitStats0.committedInstType::SimdFloatCvt 0 0.00% 81.11% # Class of committed instruction. (Count)

board.processor.cores.core.commitStats0.committedInstType::SimdFloatDiv 0 0.00% 81.11% # Class of committed instruction. (Count)

board.processor.cores.core.commitStats0.committedInstType::SimdFloatMisc 0 0.00% 81.11% # Class of committed instruction. (Count)

board.processor.cores.core.commitStats0.committedInstType::SimdFloatMult 0 0.00% 81.11% # Class of committed instruction. (Count)

board.processor.cores.core.commitStats0.committedInstType::SimdFloatMultAcc 0 0.00% 81.11% # Class of committed instruction. (Count)

board.processor.cores.core.commitStats0.committedInstType::SimdFloatMatMultAcc 0 0.00% 81.11% # Class of committed instruction. (Count)

board.processor.cores.core.commitStats0.committedInstType::SimdFloatSqrt 0 0.00% 81.11% # Class of committed instruction. (Count)

board.processor.cores.core.commitStats0.committedInstType::SimdReduceAdd 0 0.00% 81.11% # Class of committed instruction. (Count)

board.processor.cores.core.commitStats0.committedInstType::SimdReduceAlu 0 0.00% 81.11% # Class of committed instruction. (Count)

board.processor.cores.core.commitStats0.committedInstType::SimdReduceCmp 0 0.00% 81.11% # Class of committed instruction. (Count)

board.processor.cores.core.commitStats0.committedInstType::SimdFloatReduceAdd 0 0.00% 81.11% # Class of committed instruction. (Count)

board.processor.cores.core.commitStats0.committedInstType::SimdFloatReduceCmp 0 0.00% 81.11% # Class of committed instruction. (Count)

board.processor.cores.core.commitStats0.committedInstType::SimdAes 0 0.00% 81.11% # Class of committed instruction. (Count)

board.processor.cores.core.commitStats0.committedInstType::SimdAesMix 0 0.00% 81.11% # Class of committed instruction. (Count)

board.processor.cores.core.commitStats0.committedInstType::SimdSha1Hash 0 0.00% 81.11% # Class of committed instruction. (Count)

board.processor.cores.core.commitStats0.committedInstType::SimdSha1Hash2 0 0.00% 81.11% # Class of committed instruction. (Count)

board.processor.cores.core.commitStats0.committedInstType::SimdSha256Hash 0 0.00% 81.11% # Class of committed instruction. (Count)

board.processor.cores.core.commitStats0.committedInstType::SimdSha256Hash2 0 0.00% 81.11% # Class of committed instruction. (Count)

board.processor.cores.core.commitStats0.committedInstType::SimdShaSigma2 0 0.00% 81.11% # Class of committed instruction. (Count)

board.processor.cores.core.commitStats0.committedInstType::SimdShaSigma3 0 0.00% 81.11% # Class of committed instruction. (Count)

board.processor.cores.core.commitStats0.committedInstType::SimdPredAlu 0 0.00% 81.11% # Class of committed instruction. (Count)

board.processor.cores.core.commitStats0.committedInstType::Matrix 0 0.00% 81.11% # Class of committed instruction. (Count)

board.processor.cores.core.commitStats0.committedInstType::MatrixMov 0 0.00% 81.11% # Class of committed instruction. (Count)

board.processor.cores.core.commitStats0.committedInstType::MatrixOP 0 0.00% 81.11% # Class of committed instruction. (Count)

board.processor.cores.core.commitStats0.committedInstType::MemRead 629 7.98% 89.10% # Class of committed instruction. (Count)

board.processor.cores.core.commitStats0.committedInstType::MemWrite 573 7.27% 96.37% # Class of committed instruction. (Count)

board.processor.cores.core.commitStats0.committedInstType::FloatMemRead 13 0.17% 96.53% # Class of committed instruction. (Count)

board.processor.cores.core.commitStats0.committedInstType::FloatMemWrite 273 3.47% 100.00% # Class of committed instruction. (Count)

board.processor.cores.core.commitStats0.committedInstType::IprAccess 0 0.00% 100.00% # Class of committed instruction. (Count)

board.processor.cores.core.commitStats0.committedInstType::InstPrefetch 0 0.00% 100.00% # Class of committed instruction. (Count)

board.processor.cores.core.commitStats0.committedInstType::SimdUnitStrideLoad 0 0.00% 100.00% # Class of committed instruction. (Count)

board.processor.cores.core.commitStats0.committedInstType::SimdUnitStrideStore 0 0.00% 100.00% # Class of committed instruction. (Count)

board.processor.cores.core.commitStats0.committedInstType::SimdUnitStrideMaskLoad 0 0.00% 100.00% # Class of committed instruction. (Count)

board.processor.cores.core.commitStats0.committedInstType::SimdUnitStrideMaskStore 0 0.00% 100.00% # Class of committed instruction. (Count)

board.processor.cores.core.commitStats0.committedInstType::SimdStridedLoad 0 0.00% 100.00% # Class of committed instruction. (Count)

board.processor.cores.core.commitStats0.committedInstType::SimdStridedStore 0 0.00% 100.00% # Class of committed instruction. (Count)

board.processor.cores.core.commitStats0.committedInstType::SimdIndexedLoad 0 0.00% 100.00% # Class of committed instruction. (Count)

board.processor.cores.core.commitStats0.committedInstType::SimdIndexedStore 0 0.00% 100.00% # Class of committed instruction. (Count)

board.processor.cores.core.commitStats0.committedInstType::SimdWholeRegisterLoad 0 0.00% 100.00% # Class of committed instruction. (Count)

board.processor.cores.core.commitStats0.committedInstType::SimdWholeRegisterStore 0 0.00% 100.00% # Class of committed instruction. (Count)

board.processor.cores.core.commitStats0.committedInstType::SimdUnitStrideFaultOnlyFirstLoad 0 0.00% 100.00% # Class of committed instruction. (Count)

board.processor.cores.core.commitStats0.committedInstType::SimdUnitStrideSegmentedLoad 0 0.00% 100.00% # Class of committed instruction. (Count)

board.processor.cores.core.commitStats0.committedInstType::SimdUnitStrideSegmentedStore 0 0.00% 100.00% # Class of committed instruction. (Count)

```
board.processor.cores.core.commitStats0.committedInstType::SimdExt 0 0.00% 100.00% # Class of committed instruction. (Count)
board.processor.cores.core.commitStats0.committedInstType::SimdFloatExt 0 0.00% 100.00% # Class of committed instruction. (Count)
board.processor.cores.core.commitStats0.committedInstType::SimdConfig 0 0.00% 100.00% # Class of committed instruction. (Count)
board.processor.cores.core.commitStats0.committedInstType::total 7878 # Class of committed instruction. (Count)
board.processor.cores.core.commitStats0.committedControl::IsControl 836 # Class of control type instructions committed (Count)
board.processor.cores.core.commitStats0.committedControl::IsDirectControl 710 # Class of control type instructions committed (Count)
board.processor.cores.core.commitStats0.committedControl::IsIndirectControl 126 # Class of control type instructions committed (Count)
board.processor.cores.core.commitStats0.committedControl::IsCondControl 596 # Class of control type instructions committed (Count)
board.processor.cores.core.commitStats0.committedControl::IsUncondControl 240 # Class of control type instructions committed (Count)
board.processor.cores.core.commitStats0.committedControl::IsCall 97 # Class of control type instructions committed (Count)
board.processor.cores.core.commitStats0.committedControl::IsReturn 91 # Class of control type instructions committed (Count)
board.processor.cores.core.exec_context.thread_0.numMatAluAccesses 0 # Number of matrix alu accesses (Count)
board.processor.cores.core.exec_context.thread_0.numCallsReturns 188 # Number of times a function call or return occured (Count)
board.processor.cores.core.exec_context.thread_0.numMatInsts 0 # Number of matrix instructions (Count)
board.processor.cores.core.exec_context.thread_0.numIdleCycles 0 # Number of idle cycles (Cycle)
board.processor.cores.core.exec_context.thread_0.numBusyCycles 72487 # Number of busy cycles (Cycle)
board.processor.cores.core.exec_context.thread_0.notIdleFraction 1 # Percentage of non-idle cycles (Ratio)
board.processor.cores.core.exec_context.thread_0.idleFraction 0 # Percentage of idle cycles (Ratio)
board.processor.cores.core.executeStats0.numInsts 0 # Number of executed instructions (Count)
board.processor.cores.core.executeStats0.numNop 0 # Number of nop insts executed (Count)
board.processor.cores.core.executeStats0.numBranches 0 # Number of branches executed (Count)
board.processor.cores.core.executeStats0.numLoadInsts 0 # Number of load instructions executed (Count)
board.processor.cores.core.executeStats0.numStoreInsts 1488 # Number of stores executed (Count)
board.processor.cores.core.executeStats0.instRate 0 # Inst execution rate ((Count/Cycle))
board.processor.cores.core.executeStats0.numFpAluAccesses 703 # Number of float alu accesses (Count)
board.processor.cores.core.executeStats0.numFpRegReads 703 # Number of times the floating registers were read (Count)
board.processor.cores.core.executeStats0.numFpRegWrites 428 # Number of times the floating registers were written (Count)
board.processor.cores.core.executeStats0.numIntAluAccesses 7438 # Number of integer alu accesses (Count)
board.processor.cores.core.executeStats0.numIntRegReads 8403 # Number of times the integer registers were read (Count)
board.processor.cores.core.executeStats0.numIntRegWrites 5237 # Number of times the integer registers were written (Count)
board.processor.cores.core.executeStats0.numMemRefs 1488 # Number of memory refs (Count)
board.processor.cores.core.executeStats0.numMiscRegReads 3307 # Number of times the Misc registers were read (Count)
board.processor.cores.core.executeStats0.numVecAluAccesses 0 # Number of vector alu accesses (Count)
board.processor.cores.core.executeStats0.numDiscardedOps 0 # Number of ops (including micro ops) which were discarded before commit (Count)
board.processor.cores.core.fetchStats0.numInsts 4150 # Number of instructions fetched (thread level) (Count)
board.processor.cores.core.fetchStats0.numOps 7879 # Number of ops (including micro ops) fetched (thread level) (Count)
board.processor.cores.core.fetchStats0.fetchRate 0.057252 # Number of inst fetches per cycle ((Count/Cycle))
board.processor.cores.core.fetchStats0.numBranches 836 # Number of branches fetched (Count)
board.processor.cores.core.fetchStats0.branchRate 0.011533 # Number of branch fetches per cycle (Ratio)
board.processor.cores.core.fetchStats0.numFetchSuspends 0 # Number of times Execute suspended instruction fetching (Count)
board.processor.cores.core.interrupts.clk_domain.clock 5328 # Clock period in ticks (Tick)
board.processor.cores.core.mmu.dtb.rdAccesses 642 # TLB accesses on read requests (Count)
board.processor.cores.core.mmu.dtb.wrAccesses 847 # TLB accesses on write requests (Count)
board.processor.cores.core.mmu.dtb.rdMisses 12 # TLB misses on read requests (Count)
board.processor.cores.core.mmu.dtb.wrMisses 6 # TLB misses on write requests (Count)
board.processor.cores.core.mmu.dtb.walker.power_state.pwrStateResidencyTicks::UNDEFINED 10952226214626250 # Cumulative time (in ticks) in various power states (Tick)
board.processor.cores.core.mmu.itb.rdAccesses 0 # TLB accesses on read requests (Count)
board.processor.cores.core.mmu.itb.wrAccesses 5345 # TLB accesses on write requests (Count)
board.processor.cores.core.mmu.itb.rdMisses 0 # TLB misses on read requests (Count)
```

```
board.processor.cores.core.mmu.itb.wrMisses 22 # TLB misses on write requests (Count)
board.processor.cores.core.mmu.itb.walker.power_state.pwrStateResidencyTicks::UNDEFINED 10952226214626250 # Cumulative time (in ticks) in various power states (Tick)
board.processor.cores.core.power_state.pwrStateResidencyTicks::ON 10952226214626250 # Cumulative time (in ticks) in various power states (Tick)
board.processor.cores.core.thread_0.numInsts 0 # Number of Instructions committed (Count)
board.processor.cores.core.thread_0.numOps 0 # Number of Ops committed (Count)
board.processor.cores.core.thread_0.numMemRefs 0 # Number of Memory References (Count)
board.processor.cores.core.workload.numSyscalls 9 # Number of system calls (Count)
board.workload.inst.arm 0 # number of arm instructions executed (Count)
board.workload.inst.quiesce 0 # number of quiesce instructions executed (Count)

---------- End Simulation Statistics ----------
```

results_baseline:

```
---------- Begin Simulation Statistics ----------
simSeconds                    73837.531900                # Number of seconds simulated (Second)
simTicks                      73837531900351248           # Number of ticks simulated (Tick)
finalTick                     73837531900351248           # Number of ticks from beginning of simulation (restored from checkpoints and never reset) (Tick)
simFreq                       1000000000000               # The number of ticks per simulated second ((Tick/Second))
hostSeconds                   43355.13                    # Real time elapsed on the host (Second)
hostTickRate                  1703086364570               # The number of ticks simulated per host second (ticks/s) ((Tick/Second))
hostMemory                    1198868                     # Number of bytes of host memory used (Byte)
simInsts                      4141                        # Number of instructions simulated (Count)
simOps                        7867                        # Number of ops (including micro ops) simulated (Count)
hostInstRate                  0                           # Simulator instruction rate (inst/s) ((Count/Second))
hostOpRate                    0                           # Simulator op (including micro ops) rate (op/s) ((Count/Second))
board.cache_hierarchy.dptw_caches.blockedCycles::no_mshrs      0          # number of cycles access was blocked (Cycle)
board.cache_hierarchy.dptw_caches.blockedCycles::no_targets    0          # number of cycles access was blocked (Cycle)
board.cache_hierarchy.dptw_caches.blockedCauses::no_mshrs      0          # number of times access was blocked (Count)
board.cache_hierarchy.dptw_caches.blockedCauses::no_targets    0          # number of times access was blocked (Count)
board.cache_hierarchy.dptw_caches.avgBlocked::no_mshrs     nan           # average number of cycles each access was blocked ((Cycle/Count))
board.cache_hierarchy.dptw_caches.avgBlocked::no_targets   nan           # average number of cycles each access was blocked ((Cycle/Count))
board.cache_hierarchy.dptw_caches.replacements      0          # number of replacements (Count)
board.cache_hierarchy.dptw_caches.power_state.pwrStateResidencyTicks::UNDEFINED 73837531900351248          # Cumulative time (in ticks) in various power states (Tick)
board.cache_hierarchy.dptw_caches.tags.tagsInUse    0          # Average ticks per tags in use ((Tick/Count))
board.cache_hierarchy.dptw_caches.tags.totalRefs    0          # Total number of references to valid blocks. (Count)
board.cache_hierarchy.dptw_caches.tags.sampledRefs  0          # Sample count of references to valid blocks. (Count)
board.cache_hierarchy.dptw_caches.tags.avgRefs      nan        # Average number of references to valid blocks. ((Count/Count))
board.cache_hierarchy.dptw_caches.tags.warmupTick   0          # The tick when the warmup percentage was hit. (Tick)
board.cache_hierarchy.dptw_caches.tags.tagAccesses  0          # Number of tag accesses (Count)
board.cache_hierarchy.dptw_caches.tags.dataAccesses 0          # Number of data accesses (Count)
board.cache_hierarchy.dptw_caches.tags.power_state.pwrStateResidencyTicks::UNDEFINED 73837531900351248          # Cumulative time (in ticks) in various power states (Tick)
board.cache_hierarchy.iptw_caches.blockedCycles::no_mshrs      0          # number of cycles access was blocked (Cycle)
board.cache_hierarchy.iptw_caches.blockedCycles::no_targets    0          # number of cycles access was blocked (Cycle)
board.cache_hierarchy.iptw_caches.blockedCauses::no_mshrs      0          # number of times access was blocked (Count)
board.cache_hierarchy.iptw_caches.blockedCauses::no_targets    0          # number of times access was blocked (Count)
board.cache_hierarchy.iptw_caches.avgBlocked::no_mshrs     nan           # average number of cycles each access was blocked ((Cycle/Count))
board.cache_hierarchy.iptw_caches.avgBlocked::no_targets   nan           # average number of cycles each access was blocked ((Cycle/Count))
board.cache_hierarchy.iptw_caches.replacements      0          # number of replacements (Count)
board.cache_hierarchy.iptw_caches.power_state.pwrStateResidencyTicks::UNDEFINED 73837531900351248          # Cumulative time (in ticks) in various power states (Tick)
board.cache_hierarchy.iptw_caches.tags.tagsInUse    0          # Average ticks per tags in use ((Tick/Count))
board.cache_hierarchy.iptw_caches.tags.totalRefs    0          # Total number of references to valid blocks. (Count)
board.cache_hierarchy.iptw_caches.tags.sampledRefs  0          # Sample count of references to valid blocks. (Count)
board.cache_hierarchy.iptw_caches.tags.avgRefs      nan        # Average number of references to valid blocks. ((Count/Count))
board.cache_hierarchy.iptw_caches.tags.warmupTick   0          # The tick when the warmup percentage was hit. (Tick)
board.cache_hierarchy.iptw_caches.tags.tagAccesses  0          # Number of tag accesses (Count)
board.cache_hierarchy.iptw_caches.tags.dataAccesses 0          # Number of data accesses (Count)
board.cache_hierarchy.iptw_caches.tags.power_state.pwrStateResidencyTicks::UNDEFINED 73837531900351248          # Cumulative time (in ticks) in various power states (Tick)
board.cache_hierarchy.l1dcaches.demandHits::processor.cores.core.data      1300          # number of demand (read+write) hits (Count)
board.cache_hierarchy.l1dcaches.demandHits::total      1300          # number of demand (read+write) hits (Count)
```

board.cache_hierarchy.l1dcaches.overallHits::processor.cores.core.data 1300 # number of overall hits (Count)

board.cache_hierarchy.l1dcaches.overallHits::total 1300 # number of overall hits (Count)

board.cache_hierarchy.l1dcaches.demandMisses::processor.cores.core.data 185 # number of demand (read+write) misses (Count)

board.cache_hierarchy.l1dcaches.demandMisses::total 185 # number of demand (read+write) misses (Count)

board.cache_hierarchy.l1dcaches.overallMisses::processor.cores.core.data 185 # number of overall misses (Count)

board.cache_hierarchy.l1dcaches.overallMisses::total 185 # number of overall misses (Count)

board.cache_hierarchy.l1dcaches.demandMissLatency::processor.cores.core.data 9377946 # number of demand (read+write) miss ticks (Tick)

board.cache_hierarchy.l1dcaches.demandMissLatency::total 9377946 # number of demand (read+write) miss ticks (Tick)

board.cache_hierarchy.l1dcaches.overallMissLatency::processor.cores.core.data 9377946 # number of overall miss ticks (Tick)

board.cache_hierarchy.l1dcaches.overallMissLatency::total 9377946 # number of overall miss ticks (Tick)

board.cache_hierarchy.l1dcaches.demandAccesses::processor.cores.core.data 1485 # number of demand (read+write) accesses (Count)

board.cache_hierarchy.l1dcaches.demandAccesses::total 1485 # number of demand (read+write) accesses (Count)

board.cache_hierarchy.l1dcaches.overallAccesses::processor.cores.core.data 1485 # number of overall (read+write) accesses (Count)

board.cache_hierarchy.l1dcaches.overallAccesses::total 1485 # number of overall (read+write) accesses (Count)

board.cache_hierarchy.l1dcaches.demandMissRate::processor.cores.core.data 0.124579 # miss rate for demand accesses (Ratio)

board.cache_hierarchy.l1dcaches.demandMissRate::total 0.124579 # miss rate for demand accesses (Ratio)

board.cache_hierarchy.l1dcaches.overallMissRate::processor.cores.core.data 0.124579 # miss rate for overall accesses (Ratio)

board.cache_hierarchy.l1dcaches.overallMissRate::total 0.124579 # miss rate for overall accesses (Ratio)

board.cache_hierarchy.l1dcaches.demandAvgMissLatency::processor.cores.core.data 50691.600000 # average overall miss latency in ticks ((Tick/Count))

board.cache_hierarchy.l1dcaches.demandAvgMissLatency::total 50691.600000 # average overall miss latency in ticks ((Tick/Count))

board.cache_hierarchy.l1dcaches.overallAvgMissLatency::processor.cores.core.data 50691.600000 # average overall miss latency ((Tick/Count))

board.cache_hierarchy.l1dcaches.overallAvgMissLatency::total 50691.600000 # average overall miss latency ((Tick/Count))

board.cache_hierarchy.l1dcaches.blockedCycles::no_mshrs 27977 # number of cycles access was blocked (Cycle)

board.cache_hierarchy.l1dcaches.blockedCycles::no_targets 0 # number of cycles access was blocked (Cycle)

board.cache_hierarchy.l1dcaches.blockedCauses::no_mshrs 186 # number of times access was blocked (Count)

board.cache_hierarchy.l1dcaches.blockedCauses::no_targets 0 # number of times access was blocked (Count)

board.cache_hierarchy.l1dcaches.avgBlocked::no_mshrs 150.413978 # average number of cycles each access was blocked ((Cycle/Count))

board.cache_hierarchy.l1dcaches.avgBlocked::no_targets nan # average number of cycles each access was blocked ((Cycle/Count))

board.cache_hierarchy.l1dcaches.demandMshrMisses::processor.cores.core.data 185 # number of demand (read+write) MSHR misses (Count)

board.cache_hierarchy.l1dcaches.demandMshrMisses::total 185 # number of demand (read+write) MSHR misses (Count)

board.cache_hierarchy.l1dcaches.overallMshrMisses::processor.cores.core.data 185 # number of overall MSHR misses (Count)

board.cache_hierarchy.l1dcaches.overallMshrMisses::total 185 # number of overall MSHR misses (Count)

board.cache_hierarchy.l1dcaches.demandMshrMissLatency::processor.cores.core.data 9316341 # number of demand (read+write) MSHR miss ticks (Tick)

board.cache_hierarchy.l1dcaches.demandMshrMissLatency::total 9316341 # number of demand (read+write) MSHR miss ticks (Tick)

board.cache_hierarchy.l1dcaches.overallMshrMissLatency::processor.cores.core.data 9316341 # number of overall MSHR miss ticks (Tick)

board.cache_hierarchy.l1dcaches.overallMshrMissLatency::total 9316341 # number of overall MSHR miss ticks (Tick)

board.cache_hierarchy.l1dcaches.demandMshrMissRate::processor.cores.core.data 0.124579 # mshr miss ratio for demand accesses (Ratio)

board.cache_hierarchy.l1dcaches.demandMshrMissRate::total 0.124579 # mshr miss ratio for demand accesses (Ratio)

board.cache_hierarchy.l1dcaches.overallMshrMissRate::processor.cores.core.data 0.124579 # mshr miss ratio for overall accesses (Ratio)

board.cache_hierarchy.l1dcaches.overallMshrMissRate::total 0.124579 # mshr miss ratio for overall accesses (Ratio)

board.cache_hierarchy.l1dcaches.demandAvgMshrMissLatency::processor.cores.core.data 50358.600000 # average overall mshr miss latency ((Tick/Count))

board.cache_hierarchy.l1dcaches.demandAvgMshrMissLatency::total 50358.600000 # average overall mshr miss latency ((Tick/Count))

board.cache_hierarchy.l1dcaches.overallAvgMshrMissLatency::processor.cores.core.data 50358.600000 # average overall mshr miss latency ((Tick/Count))

board.cache_hierarchy.l1dcaches.overallAvgMshrMissLatency::total 50358.600000 # average overall mshr miss latency ((Tick/Count))

board.cache_hierarchy.l1dcaches.replacements 0 # number of replacements (Count)

board.cache_hierarchy.l1dcaches.LockedRMWReadReq.hits::processor.cores.core.data 1 # number of LockedRMWReadReq hits (Count)

board.cache_hierarchy.l1dcaches.LockedRMWReadReq.hits::total 1 # number of LockedRMWReadReq hits (Count)

board.cache_hierarchy.l1dcaches.LockedRMWReadReq.accesses::processor.cores.core.data 1 # number of LockedRMWReadReq accesses(hits+misses) (Count)

board.cache_hierarchy.l1dcaches.LockedRMWReadReq.accesses::total 1 # number of LockedRMWReadReq accesses(hits+misses) (Count)

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board.cache_hierarchy.l1dcaches.ReadReq.hits::processor.cores.core.data 541 # number of ReadReq hits (Count)
board.cache_hierarchy.l1dcaches.ReadReq.hits::total 541 # number of ReadReq hits (Count)
board.cache_hierarchy.l1dcaches.ReadReq.misses::processor.cores.core.data 99 # number of ReadReq misses (Count)
board.cache_hierarchy.l1dcaches.ReadReq.misses::total 99 # number of ReadReq misses (Count)
board.cache_hierarchy.l1dcaches.ReadReq.missLatency::processor.cores.core.data 5741586 # number of ReadReq miss ticks (Tick)
board.cache_hierarchy.l1dcaches.ReadReq.missLatency::total 5741586 # number of ReadReq miss ticks (Tick)
board.cache_hierarchy.l1dcaches.ReadReq.accesses::processor.cores.core.data 640 # number of ReadReq accesses(hits+misses) (Count)
board.cache_hierarchy.l1dcaches.ReadReq.accesses::total 640 # number of ReadReq accesses(hits+misses) (Count)
board.cache_hierarchy.l1dcaches.ReadReq.missRate::processor.cores.core.data 0.154688 # miss rate for ReadReq accesses (Ratio)
board.cache_hierarchy.l1dcaches.ReadReq.missRate::total 0.154688 # miss rate for ReadReq accesses (Ratio)
board.cache_hierarchy.l1dcaches.ReadReq.avgMissLatency::processor.cores.core.data 57995.818182 # average ReadReq miss latency ((Tick/Count))
board.cache_hierarchy.l1dcaches.ReadReq.avgMissLatency::total 57995.818182 # average ReadReq miss latency ((Tick/Count))
board.cache_hierarchy.l1dcaches.ReadReq.mshrMisses::processor.cores.core.data 99 # number of ReadReq MSHR misses (Count)
board.cache_hierarchy.l1dcaches.ReadReq.mshrMisses::total 99 # number of ReadReq MSHR misses (Count)
board.cache_hierarchy.l1dcaches.ReadReq.mshrMissLatency::processor.cores.core.data 5708619 # number of ReadReq MSHR miss ticks (Tick)
board.cache_hierarchy.l1dcaches.ReadReq.mshrMissLatency::total 5708619 # number of ReadReq MSHR miss ticks (Tick)
board.cache_hierarchy.l1dcaches.ReadReq.mshrMissRate::processor.cores.core.data 0.154688 # mshr miss rate for ReadReq accesses (Ratio)
board.cache_hierarchy.l1dcaches.ReadReq.mshrMissRate::total 0.154688 # mshr miss rate for ReadReq accesses (Ratio)
board.cache_hierarchy.l1dcaches.ReadReq.avgMshrMissLatency::processor.cores.core.data 57662.818182 # average ReadReq mshr miss latency ((Tick/Count))
board.cache_hierarchy.l1dcaches.ReadReq.avgMshrMissLatency::total 57662.818182 # average ReadReq mshr miss latency ((Tick/Count))
board.cache_hierarchy.l1dcaches.WriteReq.hits::processor.cores.core.data 759 # number of WriteReq hits (Count)
board.cache_hierarchy.l1dcaches.WriteReq.hits::total 759 # number of WriteReq hits (Count)
board.cache_hierarchy.l1dcaches.WriteReq.misses::processor.cores.core.data 86 # number of WriteReq misses (Count)
board.cache_hierarchy.l1dcaches.WriteReq.misses::total 86 # number of WriteReq misses (Count)
board.cache_hierarchy.l1dcaches.WriteReq.missLatency::processor.cores.core.data 3636360 # number of WriteReq miss ticks (Tick)
board.cache_hierarchy.l1dcaches.WriteReq.missLatency::total 3636360 # number of WriteReq miss ticks (Tick)
board.cache_hierarchy.l1dcaches.WriteReq.accesses::processor.cores.core.data 845 # number of WriteReq accesses(hits+misses) (Count)
board.cache_hierarchy.l1dcaches.WriteReq.accesses::total 845 # number of WriteReq accesses(hits+misses) (Count)
board.cache_hierarchy.l1dcaches.WriteReq.missRate::processor.cores.core.data 0.101775 # miss rate for WriteReq accesses (Ratio)
board.cache_hierarchy.l1dcaches.WriteReq.missRate::total 0.101775 # miss rate for WriteReq accesses (Ratio)
board.cache_hierarchy.l1dcaches.WriteReq.avgMissLatency::processor.cores.core.data 42283.255814 # average WriteReq miss latency ((Tick/Count))
board.cache_hierarchy.l1dcaches.WriteReq.avgMissLatency::total 42283.255814 # average WriteReq miss latency ((Tick/Count))
board.cache_hierarchy.l1dcaches.WriteReq.mshrMisses::processor.cores.core.data 86 # number of WriteReq MSHR misses (Count)
board.cache_hierarchy.l1dcaches.WriteReq.mshrMisses::total 86 # number of WriteReq MSHR misses (Count)
board.cache_hierarchy.l1dcaches.WriteReq.mshrMissLatency::processor.cores.core.data 3607722 # number of WriteReq MSHR miss ticks (Tick)
board.cache_hierarchy.l1dcaches.WriteReq.mshrMissLatency::total 3607722 # number of WriteReq MSHR miss ticks (Tick)
board.cache_hierarchy.l1dcaches.WriteReq.mshrMissRate::processor.cores.core.data 0.101775 # mshr miss rate for WriteReq accesses (Ratio)
board.cache_hierarchy.l1dcaches.WriteReq.mshrMissRate::total 0.101775 # mshr miss rate for WriteReq accesses (Ratio)
board.cache_hierarchy.l1dcaches.WriteReq.avgMshrMissLatency::processor.cores.core.data 41950.255814 # average WriteReq mshr miss latency ((Tick/Count))
board.cache_hierarchy.l1dcaches.WriteReq.avgMshrMissLatency::total 41950.255814 # average WriteReq mshr miss latency ((Tick/Count))
board.cache_hierarchy.l1dcaches.power_state.pwrStateResidencyTicks::UNDEFINED 73837531900351248 # Cumulative time (in ticks) in various power states (Tick)
board.cache_hierarchy.l1dcaches.prefetcher.demandMshrMisses 185 # demands not covered by prefetchs (Count)
board.cache_hierarchy.l1dcaches.prefetcher.pfIssued 0 # number of hwpf issued (Count)
board.cache_hierarchy.l1dcaches.prefetcher.pfUseful 0 # number of useful prefetch (Count)
board.cache_hierarchy.l1dcaches.prefetcher.pfUsefulButMiss 0 # number of hit on prefetch but cache block is not in an usable state (Count)
board.cache_hierarchy.l1dcaches.prefetcher.accuracy nan # accuracy of the prefetcher (Count)
board.cache_hierarchy.l1dcaches.prefetcher.coverage 0 # coverage brought by this prefetcher (Count)
board.cache_hierarchy.l1dcaches.prefetcher.pfHitInCache 0 # number of prefetches hitting in cache (Count)
board.cache_hierarchy.l1dcaches.prefetcher.pfHitInMSHR 0 # number of prefetches hitting in a MSHR (Count)
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board.cache_hierarchy.l1dcaches.prefetcher.pfHitInWB 0 # number of prefetches hit in the Write Buffer (Count)

board.cache_hierarchy.l1dcaches.prefetcher.pfLate 0 # number of late prefetches (hitting in cache, MSHR or WB) (Count)

board.cache_hierarchy.l1dcaches.prefetcher.pfIdentified 146 # number of prefetch candidates identified (Count)

board.cache_hierarchy.l1dcaches.prefetcher.pfBufferHit 63 # number of redundant prefetches already in prefetch queue (Count)

board.cache_hierarchy.l1dcaches.prefetcher.pfInCache 0 # number of redundant prefetches already in cache/mshr dropped (Count)

board.cache_hierarchy.l1dcaches.prefetcher.pfRemovedDemand 35 # number of prefetches dropped due to a demand for the same address (Count)

board.cache_hierarchy.l1dcaches.prefetcher.pfRemovedFull 0 # number of prefetches dropped due to prefetch queue size (Count)

board.cache_hierarchy.l1dcaches.prefetcher.pfSpanPage 6 # number of prefetches that crossed the page (Count)

board.cache_hierarchy.l1dcaches.prefetcher.pfUsefulSpanPage 0 # number of prefetches that is useful and crossed the page (Count)

board.cache_hierarchy.l1dcaches.prefetcher.power_state.pwrStateResidencyTicks::UNDEFINED 73837531900351248 # Cumulative time (in ticks) in various power states (Tick)

board.cache_hierarchy.l1dcaches.tags.tagsInUse 185.000000 # Average ticks per tags in use ((Tick/Count))

board.cache_hierarchy.l1dcaches.tags.totalRefs 0 # Total number of references to valid blocks. (Count)

board.cache_hierarchy.l1dcaches.tags.sampledRefs 0 # Sample count of references to valid blocks. (Count)

board.cache_hierarchy.l1dcaches.tags.avgRefs nan # Average number of references to valid blocks. ((Count/Count))

board.cache_hierarchy.l1dcaches.tags.warmupTick 149184 # The tick when the warmup percentage was hit. (Tick)

board.cache_hierarchy.l1dcaches.tags.occupancies::processor.cores.core.data 185.000000 # Average occupied blocks per tick, per requestor ((Count/Tick))

board.cache_hierarchy.l1dcaches.tags.avgOccs::processor.cores.core.data 0.361328 # Average percentage of cache occupancy ((Ratio/Tick))

board.cache_hierarchy.l1dcaches.tags.avgOccs::total 0.361328 # Average percentage of cache occupancy ((Ratio/Tick))

board.cache_hierarchy.l1dcaches.tags.occupanciesTaskId::1024 185 # Occupied blocks per task id (Count)

board.cache_hierarchy.l1dcaches.tags.ageTaskId_1024::4 185 # Occupied blocks per task id, per block age (Count)

board.cache_hierarchy.l1dcaches.tags.ratioOccsTaskId::1024 0.361328 # Ratio of occupied blocks and all blocks, per task id (Ratio)

board.cache_hierarchy.l1dcaches.tags.tagAccesses 12073 # Number of tag accesses (Count)

board.cache_hierarchy.l1dcaches.tags.dataAccesses 12073 # Number of data accesses (Count)

board.cache_hierarchy.l1dcaches.tags.power_state.pwrStateResidencyTicks::UNDEFINED 73837531900351248 # Cumulative time (in ticks) in various power states (Tick)

board.cache_hierarchy.l1icaches.demandHits::processor.cores.core.inst 5149 # number of demand (read+write) hits (Count)

board.cache_hierarchy.l1icaches.demandHits::total 5149 # number of demand (read+write) hits (Count)

board.cache_hierarchy.l1icaches.overallHits::processor.cores.core.inst 5149 # number of overall hits (Count)

board.cache_hierarchy.l1icaches.overallHits::total 5149 # number of overall hits (Count)

board.cache_hierarchy.l1icaches.demandMisses::processor.cores.core.inst 196 # number of demand (read+write) misses (Count)

board.cache_hierarchy.l1icaches.demandMisses::total 196 # number of demand (read+write) misses (Count)

board.cache_hierarchy.l1icaches.overallMisses::processor.cores.core.inst 196 # number of overall misses (Count)

board.cache_hierarchy.l1icaches.overallMisses::total 196 # number of overall misses (Count)

board.cache_hierarchy.l1icaches.demandMissLatency::processor.cores.core.inst 12525129 # number of demand (read+write) miss ticks (Tick)

board.cache_hierarchy.l1icaches.demandMissLatency::total 12525129 # number of demand (read+write) miss ticks (Tick)

board.cache_hierarchy.l1icaches.overallMissLatency::processor.cores.core.inst 12525129 # number of overall miss ticks (Tick)

board.cache_hierarchy.l1icaches.overallMissLatency::total 12525129 # number of overall miss ticks (Tick)

board.cache_hierarchy.l1icaches.demandAccesses::processor.cores.core.inst 5345 # number of demand (read+write) accesses (Count)

board.cache_hierarchy.l1icaches.demandAccesses::total 5345 # number of demand (read+write) accesses (Count)

board.cache_hierarchy.l1icaches.overallAccesses::processor.cores.core.inst 5345 # number of overall (read+write) accesses (Count)

board.cache_hierarchy.l1icaches.overallAccesses::total 5345 # number of overall (read+write) accesses (Count)

board.cache_hierarchy.l1icaches.demandMissRate::processor.cores.core.inst 0.036670 # miss rate for demand accesses (Ratio)

board.cache_hierarchy.l1icaches.demandMissRate::total 0.036670 # miss rate for demand accesses (Ratio)

board.cache_hierarchy.l1icaches.overallMissRate::processor.cores.core.inst 0.036670 # miss rate for overall accesses (Ratio)

board.cache_hierarchy.l1icaches.overallMissRate::total 0.036670 # miss rate for overall accesses (Ratio)

board.cache_hierarchy.l1icaches.demandAvgMissLatency::processor.cores.core.inst 63903.719388 # average overall miss latency in ticks ((Tick/Count))

board.cache_hierarchy.l1icaches.demandAvgMissLatency::total 63903.719388 # average overall miss latency in ticks ((Tick/Count))

board.cache_hierarchy.l1icaches.overallAvgMissLatency::processor.cores.core.inst 63903.719388 # average overall miss latency ((Tick/Count))

board.cache_hierarchy.l1icaches.overallAvgMissLatency::total 63903.719388 # average overall miss latency ((Tick/Count))

board.cache_hierarchy.l1icaches.blockedCycles::no_mshrs 0 # number of cycles access was blocked (Cycle)

board.cache_hierarchy.l1icaches.blockedCycles::no_targets 0 # number of cycles access was blocked (Cycle)

board.cache_hierarchy.l1icaches.blockedCauses::no_mshrs 0 # number of times access was blocked (Count)

board.cache_hierarchy.l1icaches.blockedCauses::no_targets 0 # number of times access was blocked (Count)

board.cache_hierarchy.l1icaches.avgBlocked::no_mshrs nan # average number of cycles each access was blocked ((Cycle/Count))

board.cache_hierarchy.l1icaches.avgBlocked::no_targets nan # average number of cycles each access was blocked ((Cycle/Count))

board.cache_hierarchy.l1icaches.demandMshrMisses::processor.cores.core.inst 196 # number of demand (read+write) MSHR misses (Count)

board.cache_hierarchy.l1icaches.demandMshrMisses::total 196 # number of demand (read+write) MSHR misses (Count)

board.cache_hierarchy.l1icaches.overallMshrMisses::processor.cores.core.inst 196 # number of overall MSHR misses (Count)

board.cache_hierarchy.l1icaches.overallMshrMisses::total 196 # number of overall MSHR misses (Count)

board.cache_hierarchy.l1icaches.demandMshrMissLatency::processor.cores.core.inst 12459861 # number of demand (read+write) MSHR miss ticks (Tick)

board.cache_hierarchy.l1icaches.demandMshrMissLatency::total 12459861 # number of demand (read+write) MSHR miss ticks (Tick)

board.cache_hierarchy.l1icaches.overallMshrMissLatency::processor.cores.core.inst 12459861 # number of overall MSHR miss ticks (Tick)

board.cache_hierarchy.l1icaches.overallMshrMissLatency::total 12459861 # number of overall MSHR miss ticks (Tick)

board.cache_hierarchy.l1icaches.demandMshrMissRate::processor.cores.core.inst 0.036670 # mshr miss ratio for demand accesses (Ratio)

board.cache_hierarchy.l1icaches.demandMshrMissRate::total 0.036670 # mshr miss ratio for demand accesses (Ratio)

board.cache_hierarchy.l1icaches.overallMshrMissRate::processor.cores.core.inst 0.036670 # mshr miss ratio for overall accesses (Ratio)

board.cache_hierarchy.l1icaches.overallMshrMissRate::total 0.036670 # mshr miss ratio for overall accesses (Ratio)

board.cache_hierarchy.l1icaches.demandAvgMshrMissLatency::processor.cores.core.inst 63570.719388 # average overall mshr miss latency ((Tick/Count))

board.cache_hierarchy.l1icaches.demandAvgMshrMissLatency::total 63570.719388 # average overall mshr miss latency ((Tick/Count))

board.cache_hierarchy.l1icaches.overallAvgMshrMissLatency::processor.cores.core.inst 63570.719388 # average overall mshr miss latency ((Tick/Count))

board.cache_hierarchy.l1icaches.overallAvgMshrMissLatency::total 63570.719388 # average overall mshr miss latency ((Tick/Count))

board.cache_hierarchy.l1icaches.replacements 0 # number of replacements (Count)

board.cache_hierarchy.l1icaches.ReadReq.hits::processor.cores.core.inst 5149 # number of ReadReq hits (Count)

board.cache_hierarchy.l1icaches.ReadReq.hits::total 5149 # number of ReadReq hits (Count)

board.cache_hierarchy.l1icaches.ReadReq.misses::processor.cores.core.inst 196 # number of ReadReq misses (Count)

board.cache_hierarchy.l1icaches.ReadReq.misses::total 196 # number of ReadReq misses (Count)

board.cache_hierarchy.l1icaches.ReadReq.missLatency::processor.cores.core.inst 12525129 # number of ReadReq miss ticks (Tick)

board.cache_hierarchy.l1icaches.ReadReq.missLatency::total 12525129 # number of ReadReq miss ticks (Tick)

board.cache_hierarchy.l1icaches.ReadReq.accesses::processor.cores.core.inst 5345 # number of ReadReq accesses(hits+misses) (Count)

board.cache_hierarchy.l1icaches.ReadReq.accesses::total 5345 # number of ReadReq accesses(hits+misses) (Count)

board.cache_hierarchy.l1icaches.ReadReq.missRate::processor.cores.core.inst 0.036670 # miss rate for ReadReq accesses (Ratio)

board.cache_hierarchy.l1icaches.ReadReq.missRate::total 0.036670 # miss rate for ReadReq accesses (Ratio)

board.cache_hierarchy.l1icaches.ReadReq.avgMissLatency::processor.cores.core.inst 63903.719388 # average ReadReq miss latency ((Tick/Count))

board.cache_hierarchy.l1icaches.ReadReq.avgMissLatency::total 63903.719388 # average ReadReq miss latency ((Tick/Count))

board.cache_hierarchy.l1icaches.ReadReq.mshrMisses::processor.cores.core.inst 196 # number of ReadReq MSHR misses (Count)

board.cache_hierarchy.l1icaches.ReadReq.mshrMisses::total 196 # number of ReadReq MSHR misses (Count)

board.cache_hierarchy.l1icaches.ReadReq.mshrMissLatency::processor.cores.core.inst 12459861 # number of ReadReq MSHR miss ticks (Tick)

board.cache_hierarchy.l1icaches.ReadReq.mshrMissLatency::total 12459861 # number of ReadReq MSHR miss ticks (Tick)

board.cache_hierarchy.l1icaches.ReadReq.mshrMissRate::processor.cores.core.inst 0.036670 # mshr miss rate for ReadReq accesses (Ratio)

board.cache_hierarchy.l1icaches.ReadReq.mshrMissRate::total 0.036670 # mshr miss rate for ReadReq accesses (Ratio)

board.cache_hierarchy.l1icaches.ReadReq.avgMshrMissLatency::processor.cores.core.inst 63570.719388 # average ReadReq mshr miss latency ((Tick/Count))

board.cache_hierarchy.l1icaches.ReadReq.avgMshrMissLatency::total 63570.719388 # average ReadReq mshr miss latency ((Tick/Count))

board.cache_hierarchy.l1icaches.power_state.pwrStateResidencyTicks::UNDEFINED 73837531900351248 # Cumulative time (in ticks) in various power states (Tick)

board.cache_hierarchy.l1icaches.prefetcher.demandMshrMisses 196 # demands not covered by prefetchs (Count)

board.cache_hierarchy.l1icaches.prefetcher.pfIssued 0 # number of hwpf issued (Count)

board.cache_hierarchy.l1icaches.prefetcher.pfUseful 0 # number of useful prefetch (Count)

board.cache_hierarchy.l1icaches.prefetcher.pfUsefulButMiss 0 # number of hit on prefetch but cache block is not in an usable state (Count)

board.cache_hierarchy.l1icaches.prefetcher.accuracy nan # accuracy of the prefetcher (Count)

board.cache_hierarchy.l1icaches.prefetcher.coverage 0 # coverage brought by this prefetcher (Count)

board.cache_hierarchy.l1icaches.prefetcher.pfHitInCache 0 # number of prefetches hitting in cache (Count)

board.cache_hierarchy.l1icaches.prefetcher.pfHitInMSHR 0 # number of prefetches hitting in a MSHR (Count)

board.cache_hierarchy.l1icaches.prefetcher.pfHitInWB 0 # number of prefetches hit in the Write Buffer (Count)

board.cache_hierarchy.l1icaches.prefetcher.pfLate 0 # number of late prefetches (hitting in cache, MSHR or WB) (Count)

board.cache_hierarchy.l1icaches.prefetcher.pfIdentified 0 # number of prefetch candidates identified (Count)

board.cache_hierarchy.l1icaches.prefetcher.pfBufferHit 0 # number of redundant prefetches already in prefetch queue (Count)

board.cache_hierarchy.l1icaches.prefetcher.pfInCache 0 # number of redundant prefetches already in cache/mshr dropped (Count)

board.cache_hierarchy.l1icaches.prefetcher.pfRemovedDemand 0 # number of prefetches dropped due to a demand for the same address (Count)

board.cache_hierarchy.l1icaches.prefetcher.pfRemovedFull 0 # number of prefetches dropped due to prefetch queue size (Count)

board.cache_hierarchy.l1icaches.prefetcher.pfSpanPage 0 # number of prefetches that crossed the page (Count)

board.cache_hierarchy.l1icaches.prefetcher.pfUsefulSpanPage 0 # number of prefetches that is useful and crossed the page (Count)

board.cache_hierarchy.l1icaches.prefetcher.power_state.pwrStateResidencyTicks::UNDEFINED 73837531900351248 # Cumulative time (in ticks) in various power states (Tick)

board.cache_hierarchy.l1icaches.tags.tagsInUse 196.000000 # Average ticks per tags in use ((Tick/Count))

board.cache_hierarchy.l1icaches.tags.totalRefs 0 # Total number of references to valid blocks. (Count)

board.cache_hierarchy.l1icaches.tags.sampledRefs 0 # Sample count of references to valid blocks. (Count)

board.cache_hierarchy.l1icaches.tags.avgRefs nan # Average number of references to valid blocks. ((Count/Count))

board.cache_hierarchy.l1icaches.tags.warmupTick 76590 # The tick when the warmup percentage was hit. (Tick)

board.cache_hierarchy.l1icaches.tags.occupancies::processor.cores.core.inst 196.000000 # Average occupied blocks per tick, per requestor ((Count/Tick))

board.cache_hierarchy.l1icaches.tags.avgOccs::processor.cores.core.inst 0.382812 # Average percentage of cache occupancy ((Ratio/Tick))

board.cache_hierarchy.l1icaches.tags.avgOccs::total 0.382812 # Average percentage of cache occupancy ((Ratio/Tick))

board.cache_hierarchy.l1icaches.tags.occupanciesTaskId::1024 196 # Occupied blocks per task id (Count)

board.cache_hierarchy.l1icaches.tags.ageTaskId_1024::4 196 # Occupied blocks per task id, per block age (Count)

board.cache_hierarchy.l1icaches.tags.ratioOccsTaskId::1024 0.382812 # Ratio of occupied blocks and all blocks, per task id (Ratio)

board.cache_hierarchy.l1icaches.tags.tagAccesses 42956 # Number of tag accesses (Count)

board.cache_hierarchy.l1icaches.tags.dataAccesses 42956 # Number of data accesses (Count)

board.cache_hierarchy.l1icaches.tags.power_state.pwrStateResidencyTicks::UNDEFINED 73837531900351248 # Cumulative time (in ticks) in various power states (Tick)

board.cache_hierarchy.l2bus.transDist::ReadResp 295 # Transaction distribution (Count)

board.cache_hierarchy.l2bus.transDist::HardPFReq 57 # Transaction distribution (Count)

board.cache_hierarchy.l2bus.transDist::ReadExReq 86 # Transaction distribution (Count)

board.cache_hierarchy.l2bus.transDist::ReadExResp 86 # Transaction distribution (Count)

board.cache_hierarchy.l2bus.transDist::ReadSharedReq 295 # Transaction distribution (Count)

board.cache_hierarchy.l2bus.pktCount_board.cache_hierarchy.l1icaches.mem_side_port::board.cache_hierarchy.l2cache.cpu_side_port 392 # Packet count per connected requestor and responder (Count)

board.cache_hierarchy.l2bus.pktCount_board.cache_hierarchy.l1dcaches.mem_side_port::board.cache_hierarchy.l2cache.cpu_side_port 370 # Packet count per connected requestor and responder (Count)

board.cache_hierarchy.l2bus.pktCount::total 762 # Packet count per connected requestor and responder (Count)

board.cache_hierarchy.l2bus.pktSize_board.cache_hierarchy.l1icaches.mem_side_port::board.cache_hierarchy.l2cache.cpu_side_port 12544 # Cumulative packet size per connected requestor and responder (Byte)

board.cache_hierarchy.l2bus.pktSize_board.cache_hierarchy.l1dcaches.mem_side_port::board.cache_hierarchy.l2cache.cpu_side_port 11840 # Cumulative packet size per connected requestor and responder (Byte)

board.cache_hierarchy.l2bus.pktSize::total 24384 # Cumulative packet size per connected requestor and responder (Byte)

board.cache_hierarchy.l2bus.snoops 57 # Total snoops (Count)

board.cache_hierarchy.l2bus.snoopTraffic 0 # Total snoop traffic (Byte)

board.cache_hierarchy.l2bus.snoopFanout::samples 438 # Request fanout histogram (Count)

board.cache_hierarchy.l2bus.snoopFanout::mean 0 # Request fanout histogram (Count)

board.cache_hierarchy.l2bus.snoopFanout::stdev 0 # Request fanout histogram (Count)

board.cache_hierarchy.l2bus.snoopFanout::underflows 0 0.00% 0.00% # Request fanout histogram (Count)

board.cache_hierarchy.l2bus.snoopFanout::0 438 100.00% 100.00% # Request fanout histogram (Count)

board.cache_hierarchy.l2bus.snoopFanout::1 0 0.00% 100.00% # Request fanout histogram (Count)

board.cache_hierarchy.l2bus.snoopFanout::2 0 0.00% 100.00% # Request fanout histogram (Count)

board.cache_hierarchy.l2bus.snoopFanout::3 0 0.00% 100.00% # Request fanout histogram (Count)

board.cache_hierarchy.l2bus.snoopFanout::4 0 0.00% 100.00% # Request fanout histogram (Count)

board.cache_hierarchy.l2bus.snoopFanout::overflows 0 0.00% 100.00% # Request fanout histogram (Count)

board.cache_hierarchy.l2bus.snoopFanout::min_value 0 # Request fanout histogram (Count)

board.cache_hierarchy.l2bus.snoopFanout::max_value 0 # Request fanout histogram (Count)

board.cache_hierarchy.l2bus.snoopFanout::total 438 # Request fanout histogram (Count)

board.cache_hierarchy.l2bus.power_state.pwrStateResidencyTicks::UNDEFINED 73837531900351248 # Cumulative time (in ticks) in various power states (Tick)

board.cache_hierarchy.l2bus.reqLayer0.occupancy 126873 # Layer occupancy (ticks) (Tick)

board.cache_hierarchy.l2bus.reqLayer0.utilization 0.0 # Layer utilization (Ratio)

board.cache_hierarchy.l2bus.respLayer0.occupancy 195804 # Layer occupancy (ticks) (Tick)

board.cache_hierarchy.l2bus.respLayer0.utilization 0.0 # Layer utilization (Ratio)

board.cache_hierarchy.l2bus.respLayer1.occupancy 184815 # Layer occupancy (ticks) (Tick)

board.cache_hierarchy.l2bus.respLayer1.utilization 0.0 # Layer utilization (Ratio)

board.cache_hierarchy.l2bus.snoop_filter.totRequests 381 # Total number of requests made to the snoop filter. (Count)

board.cache_hierarchy.l2bus.snoop_filter.hitSingleRequests 0 # Number of requests hitting in the snoop filter with a single holder of the requested data. (Count)

board.cache_hierarchy.l2bus.snoop_filter.hitMultiRequests 0 # Number of requests hitting in the snoop filter with multiple (>1) holders of the requested data. (Count)

board.cache_hierarchy.l2bus.snoop_filter.totSnoops 0 # Total number of snoops made to the snoop filter. (Count)

board.cache_hierarchy.l2bus.snoop_filter.hitSingleSnoops 0 # Number of snoops hitting in the snoop filter with a single holder of the requested data. (Count)

board.cache_hierarchy.l2bus.snoop_filter.hitMultiSnoops 0 # Number of snoops hitting in the snoop filter with multiple (>1) holders of the requested data. (Count)

board.cache_hierarchy.l2cache.demandHits::processor.cores.core.data 35 # number of demand (read+write) hits (Count)

board.cache_hierarchy.l2cache.demandHits::total 35 # number of demand (read+write) hits (Count)

board.cache_hierarchy.l2cache.overallHits::processor.cores.core.data 35 # number of overall hits (Count)

board.cache_hierarchy.l2cache.overallHits::total 35 # number of overall hits (Count)

board.cache_hierarchy.l2cache.demandMisses::processor.cores.core.inst 196 # number of demand (read+write) misses (Count)

board.cache_hierarchy.l2cache.demandMisses::processor.cores.core.data 150 # number of demand (read+write) misses (Count)

board.cache_hierarchy.l2cache.demandMisses::total 346 # number of demand (read+write) misses (Count)

board.cache_hierarchy.l2cache.overallMisses::processor.cores.core.inst 196 # number of overall misses (Count)

board.cache_hierarchy.l2cache.overallMisses::processor.cores.core.data 150 # number of overall misses (Count)

board.cache_hierarchy.l2cache.overallMisses::total 346 # number of overall misses (Count)

board.cache_hierarchy.l2cache.demandMissLatency::processor.cores.core.inst 12329325 # number of demand (read+write) miss ticks (Tick)

board.cache_hierarchy.l2cache.demandMissLatency::processor.cores.core.data 8948376 # number of demand (read+write) miss ticks (Tick)

board.cache_hierarchy.l2cache.demandMissLatency::total 21277701 # number of demand (read+write) miss ticks (Tick)

board.cache_hierarchy.l2cache.overallMissLatency::processor.cores.core.inst 12329325 # number of overall miss ticks (Tick)

board.cache_hierarchy.l2cache.overallMissLatency::processor.cores.core.data 8948376 # number of overall miss ticks (Tick)

board.cache_hierarchy.l2cache.overallMissLatency::total 21277701 # number of overall miss ticks (Tick)

board.cache_hierarchy.l2cache.demandAccesses::processor.cores.core.inst 196 # number of demand (read+write) accesses (Count)

board.cache_hierarchy.l2cache.demandAccesses::processor.cores.core.data 185 # number of demand (read+write) accesses (Count)

board.cache_hierarchy.l2cache.demandAccesses::total 381 # number of demand (read+write) accesses (Count)

board.cache_hierarchy.l2cache.overallAccesses::processor.cores.core.inst 196 # number of overall (read+write) accesses (Count)

board.cache_hierarchy.l2cache.overallAccesses::processor.cores.core.data 185 # number of overall (read+write) accesses (Count)

board.cache_hierarchy.l2cache.overallAccesses::total 381 # number of overall (read+write) accesses (Count)

board.cache_hierarchy.l2cache.demandMissRate::processor.cores.core.inst 1 # miss rate for demand accesses (Ratio)

board.cache_hierarchy.l2cache.demandMissRate::processor.cores.core.data 0.810811 # miss rate for demand accesses (Ratio)

board.cache_hierarchy.l2cache.demandMissRate::total 0.908136 # miss rate for demand accesses (Ratio)

board.cache_hierarchy.l2cache.overallMissRate::processor.cores.core.inst 1 # miss rate for overall accesses (Ratio)

board.cache_hierarchy.l2cache.overallMissRate::processor.cores.core.data 0.810811 # miss rate for overall accesses (Ratio)

board.cache_hierarchy.l2cache.overallMissRate::total 0.908136 # miss rate for overall accesses (Ratio)

board.cache_hierarchy.l2cache.demandAvgMissLatency::processor.cores.core.inst 62904.719388 # average overall miss latency in ticks ((Tick/Count))

board.cache_hierarchy.l2cache.demandAvgMissLatency::processor.cores.core.data 59655.840000 # average overall miss latency in ticks ((Tick/Count))

board.cache_hierarchy.l2cache.demandAvgMissLatency::total 61496.245665 # average overall miss latency in ticks ((Tick/Count))

board.cache_hierarchy.l2cache.overallAvgMissLatency::processor.cores.core.inst 62904.719388 # average overall miss latency ((Tick/Count))

board.cache_hierarchy.l2cache.overallAvgMissLatency::processor.cores.core.data 59655.840000 # average overall miss latency ((Tick/Count))

board.cache_hierarchy.l2cache.overallAvgMissLatency::total 61496.245665 # average overall miss latency ((Tick/Count))

board.cache_hierarchy.l2cache.blockedCycles::no_mshrs 0 # number of cycles access was blocked (Cycle)

board.cache_hierarchy.l2cache.blockedCycles::no_targets 0 # number of cycles access was blocked (Cycle)

board.cache_hierarchy.l2cache.blockedCauses::no_mshrs 0 # number of times access was blocked (Count)

board.cache_hierarchy.l2cache.blockedCauses::no_targets 0 # number of times access was blocked (Count)

board.cache_hierarchy.l2cache.avgBlocked::no_mshrs nan # average number of cycles each access was blocked ((Cycle/Count))

board.cache_hierarchy.l2cache.avgBlocked::no_targets nan # average number of cycles each access was blocked ((Cycle/Count))

board.cache_hierarchy.l2cache.demandMshrMisses::processor.cores.core.inst 196 # number of demand (read+write) MSHR misses (Count)

board.cache_hierarchy.l2cache.demandMshrMisses::processor.cores.core.data 150 # number of demand (read+write) MSHR misses (Count)

board.cache_hierarchy.l2cache.demandMshrMisses::total 346 # number of demand (read+write) MSHR misses (Count)

board.cache_hierarchy.l2cache.overallMshrMisses::cache_hierarchy.l2cache.prefetcher 45 # number of overall MSHR misses (Count)

board.cache_hierarchy.l2cache.overallMshrMisses::processor.cores.core.inst 196 # number of overall MSHR misses (Count)

board.cache_hierarchy.l2cache.overallMshrMisses::processor.cores.core.data 150 # number of overall MSHR misses (Count)

board.cache_hierarchy.l2cache.overallMshrMisses::total 391 # number of overall MSHR misses (Count)

board.cache_hierarchy.l2cache.demandMshrMissLatency::processor.cores.core.inst 11023965 # number of demand (read+write) MSHR miss ticks (Tick)

board.cache_hierarchy.l2cache.demandMshrMissLatency::processor.cores.core.data 7949376 # number of demand (read+write) MSHR miss ticks (Tick)

board.cache_hierarchy.l2cache.demandMshrMissLatency::total 18973341 # number of demand (read+write) MSHR miss ticks (Tick)

board.cache_hierarchy.l2cache.overallMshrMissLatency::cache_hierarchy.l2cache.prefetcher 2050185 # number of overall MSHR miss ticks (Tick)

board.cache_hierarchy.l2cache.overallMshrMissLatency::processor.cores.core.inst 11023965 # number of overall MSHR miss ticks (Tick)

board.cache_hierarchy.l2cache.overallMshrMissLatency::processor.cores.core.data 7949376 # number of overall MSHR miss ticks (Tick)

board.cache_hierarchy.l2cache.overallMshrMissLatency::total 21023526 # number of overall MSHR miss ticks (Tick)

board.cache_hierarchy.l2cache.demandMshrMissRate::processor.cores.core.inst 1 # mshr miss ratio for demand accesses (Ratio)

board.cache_hierarchy.l2cache.demandMshrMissRate::processor.cores.core.data 0.810811 # mshr miss ratio for demand accesses (Ratio)

board.cache_hierarchy.l2cache.demandMshrMissRate::total 0.908136 # mshr miss ratio for demand accesses (Ratio)

board.cache_hierarchy.l2cache.overallMshrMissRate::cache_hierarchy.l2cache.prefetcher inf # mshr miss ratio for overall accesses (Ratio)

board.cache_hierarchy.l2cache.overallMshrMissRate::processor.cores.core.inst 1 # mshr miss ratio for overall accesses (Ratio)

board.cache_hierarchy.l2cache.overallMshrMissRate::processor.cores.core.data 0.810811 # mshr miss ratio for overall accesses (Ratio)

board.cache_hierarchy.l2cache.overallMshrMissRate::total 1.026247 # mshr miss ratio for overall accesses (Ratio)

board.cache_hierarchy.l2cache.demandAvgMshrMissLatency::processor.cores.core.inst 56244.719388 # average overall mshr miss latency ((Tick/Count))

board.cache_hierarchy.l2cache.demandAvgMshrMissLatency::processor.cores.core.data 52995.840000 # average overall mshr miss latency ((Tick/Count))

board.cache_hierarchy.l2cache.demandAvgMshrMissLatency::total 54836.245665 # average overall mshr miss latency ((Tick/Count))

board.cache_hierarchy.l2cache.overallAvgMshrMissLatency::cache_hierarchy.l2cache.prefetcher 45559.666667 # average overall mshr miss latency ((Tick/Count))

board.cache_hierarchy.l2cache.overallAvgMshrMissLatency::processor.cores.core.inst 56244.719388 # average overall mshr miss latency ((Tick/Count))

board.cache_hierarchy.l2cache.overallAvgMshrMissLatency::processor.cores.core.data 52995.840000 # average overall mshr miss latency ((Tick/Count))

board.cache_hierarchy.l2cache.overallAvgMshrMissLatency::total 53768.608696 # average overall mshr miss latency ((Tick/Count))

board.cache_hierarchy.l2cache.replacements 0 # number of replacements (Count)

board.cache_hierarchy.l2cache.HardPFReq.mshrMisses::cache_hierarchy.l2cache.prefetcher 45 # number of HardPFReq MSHR misses (Count)

board.cache_hierarchy.l2cache.HardPFReq.mshrMisses::total 45 # number of HardPFReq MSHR misses (Count)

board.cache_hierarchy.l2cache.HardPFReq.mshrMissLatency::cache_hierarchy.l2cache.prefetcher 2050185 # number of HardPFReq MSHR miss ticks (Tick)

board.cache_hierarchy.l2cache.HardPFReq.mshrMissLatency::total 2050185 # number of HardPFReq MSHR miss ticks (Tick)

board.cache_hierarchy.l2cache.HardPFReq.mshrMissRate::cache_hierarchy.l2cache.prefetcher inf # mshr miss rate for HardPFReq accesses (Ratio)

board.cache_hierarchy.l2cache.HardPFReq.mshrMissRate::total inf # mshr miss rate for HardPFReq accesses (Ratio)

board.cache_hierarchy.l2cache.HardPFReq.avgMshrMissLatency::cache_hierarchy.l2cache.prefetcher 45559.666667 # average HardPFReq mshr miss latency ((Tick/Count))

board.cache_hierarchy.l2cache.HardPFReq.avgMshrMissLatency::total 45559.666667 # average HardPFReq mshr miss latency ((Tick/Count))

board.cache_hierarchy.l2cache.ReadExReq.hits::processor.cores.core.data 30 # number of ReadExReq hits (Count)

board.cache_hierarchy.l2cache.ReadExReq.hits::total 30 # number of ReadExReq hits (Count)

board.cache_hierarchy.l2cache.ReadExReq.misses::processor.cores.core.data 56 # number of ReadExReq misses (Count)

board.cache_hierarchy.l2cache.ReadExReq.misses::total 56 # number of ReadExReq misses (Count)

board.cache_hierarchy.l2cache.ReadExReq.missLatency::processor.cores.core.data 3340656 # number of ReadExReq miss ticks (Tick)

board.cache_hierarchy.l2cache.ReadExReq.missLatency::total 3340656 # number of ReadExReq miss ticks (Tick)

board.cache_hierarchy.l2cache.ReadExReq.accesses::processor.cores.core.data 86 # number of ReadExReq accesses(hits+misses) (Count)

board.cache_hierarchy.l2cache.ReadExReq.accesses::total 86 # number of ReadExReq accesses(hits+misses) (Count)

board.cache_hierarchy.l2cache.ReadExReq.missRate::processor.cores.core.data 0.651163 # miss rate for ReadExReq accesses (Ratio)

board.cache_hierarchy.l2cache.ReadExReq.missRate::total 0.651163 # miss rate for ReadExReq accesses (Ratio)

board.cache_hierarchy.l2cache.ReadExReq.avgMissLatency::processor.cores.core.data 59654.571429 # average ReadExReq miss latency ((Tick/Count))

board.cache_hierarchy.l2cache.ReadExReq.avgMissLatency::total 59654.571429 # average ReadExReq miss latency ((Tick/Count))

board.cache_hierarchy.l2cache.ReadExReq.mshrMisses::processor.cores.core.data 56 # number of ReadExReq MSHR misses (Count)

board.cache_hierarchy.l2cache.ReadExReq.mshrMisses::total 56 # number of ReadExReq MSHR misses (Count)

board.cache_hierarchy.l2cache.ReadExReq.mshrMissLatency::processor.cores.core.data 2967696 # number of ReadExReq MSHR miss ticks (Tick)

board.cache_hierarchy.l2cache.ReadExReq.mshrMissLatency::total 2967696 # number of ReadExReq MSHR miss ticks (Tick)

board.cache_hierarchy.l2cache.ReadExReq.mshrMissRate::processor.cores.core.data 0.651163 # mshr miss rate for ReadExReq accesses (Ratio)

board.cache_hierarchy.l2cache.ReadExReq.mshrMissRate::total 0.651163 # mshr miss rate for ReadExReq accesses (Ratio)

board.cache_hierarchy.l2cache.ReadExReq.avgMshrMissLatency::processor.cores.core.data 52994.571429 # average ReadExReq mshr miss latency ((Tick/Count))

board.cache_hierarchy.l2cache.ReadExReq.avgMshrMissLatency::total 52994.571429 # average ReadExReq mshr miss latency ((Tick/Count))

board.cache_hierarchy.l2cache.ReadSharedReq.hits::processor.cores.core.data 5 # number of ReadSharedReq hits (Count)

board.cache_hierarchy.l2cache.ReadSharedReq.hits::total 5 # number of ReadSharedReq hits (Count)

board.cache_hierarchy.l2cache.ReadSharedReq.misses::processor.cores.core.inst 196 # number of ReadSharedReq misses (Count)

board.cache_hierarchy.l2cache.ReadSharedReq.misses::processor.cores.core.data 94 # number of ReadSharedReq misses (Count)

board.cache_hierarchy.l2cache.ReadSharedReq.misses::total 290 # number of ReadSharedReq misses (Count)

board.cache_hierarchy.l2cache.ReadSharedReq.missLatency::processor.cores.core.inst 12329325 # number of ReadSharedReq miss ticks (Tick)

board.cache_hierarchy.l2cache.ReadSharedReq.missLatency::processor.cores.core.data 5607720 # number of ReadSharedReq miss ticks (Tick)

board.cache_hierarchy.l2cache.ReadSharedReq.missLatency::total 17937045 # number of ReadSharedReq miss ticks (Tick)

board.cache_hierarchy.l2cache.ReadSharedReq.accesses::processor.cores.core.inst 196 # number of ReadSharedReq accesses(hits+misses) (Count)

board.cache_hierarchy.l2cache.ReadSharedReq.accesses::processor.cores.core.data 99 # number of ReadSharedReq accesses(hits+misses) (Count)

board.cache_hierarchy.l2cache.ReadSharedReq.accesses::total 295 # number of ReadSharedReq accesses(hits+misses) (Count)

board.cache_hierarchy.l2cache.ReadSharedReq.missRate::processor.cores.core.inst 1 # miss rate for ReadSharedReq accesses (Ratio)

board.cache_hierarchy.l2cache.ReadSharedReq.missRate::processor.cores.core.data 0.949495 # miss rate for ReadSharedReq accesses (Ratio)

board.cache_hierarchy.l2cache.ReadSharedReq.missRate::total 0.983051 # miss rate for ReadSharedReq accesses (Ratio)

board.cache_hierarchy.l2cache.ReadSharedReq.avgMissLatency::processor.cores.core.inst 62904.719388 # average ReadSharedReq miss latency ((Tick/Count))

board.cache_hierarchy.l2cache.ReadSharedReq.avgMissLatency::processor.cores.core.data 59656.595745 # average ReadSharedReq miss latency ((Tick/Count))

board.cache_hierarchy.l2cache.ReadSharedReq.avgMissLatency::total 61851.879310 # average ReadSharedReq miss latency ((Tick/Count))

board.cache_hierarchy.l2cache.ReadSharedReq.mshrMisses::processor.cores.core.inst 196 # number of ReadSharedReq MSHR misses (Count)

board.cache_hierarchy.l2cache.ReadSharedReq.mshrMisses::processor.cores.core.data 94 # number of ReadSharedReq MSHR misses (Count)

board.cache_hierarchy.l2cache.ReadSharedReq.mshrMisses::total 290 # number of ReadSharedReq MSHR misses (Count)

board.cache_hierarchy.l2cache.ReadSharedReq.mshrMissLatency::processor.cores.core.inst 11023965 # number of ReadSharedReq MSHR miss ticks (Tick)

board.cache_hierarchy.l2cache.ReadSharedReq.mshrMissLatency::processor.cores.core.data 4981680 # number of ReadSharedReq MSHR miss ticks (Tick)

board.cache_hierarchy.l2cache.ReadSharedReq.mshrMissLatency::total 16005645 # number of ReadSharedReq MSHR miss ticks (Tick)

board.cache_hierarchy.l2cache.ReadSharedReq.mshrMissRate::processor.cores.core.inst 1 # mshr miss rate for ReadSharedReq accesses (Ratio)

board.cache_hierarchy.l2cache.ReadSharedReq.mshrMissRate::processor.cores.core.data 0.949495 # mshr miss rate for ReadSharedReq accesses (Ratio)

board.cache_hierarchy.l2cache.ReadSharedReq.mshrMissRate::total 0.983051 # mshr miss rate for ReadSharedReq accesses (Ratio)

board.cache_hierarchy.l2cache.ReadSharedReq.avgMshrMissLatency::processor.cores.core.inst 56244.719388 # average ReadSharedReq mshr miss latency ((Tick/Count))

board.cache_hierarchy.l2cache.ReadSharedReq.avgMshrMissLatency::processor.cores.core.data 52996.595745 # average ReadSharedReq mshr miss latency ((Tick/Count))

board.cache_hierarchy.l2cache.ReadSharedReq.avgMshrMissLatency::total 55191.879310 # average ReadSharedReq mshr miss latency ((Tick/Count))

board.cache_hierarchy.l2cache.power_state.pwrStateResidencyTicks::UNDEFINED 73837531900351248 # Cumulative time (in ticks) in various power states (Tick)

board.cache_hierarchy.l2cache.prefetcher.demandMshrMisses 346 # demands not covered by prefetchs (Count)

board.cache_hierarchy.l2cache.prefetcher.pfIssued 146 # number of hwpf issued (Count)

board.cache_hierarchy.l2cache.prefetcher.pfUseful 35 # number of useful prefetch (Count)

board.cache_hierarchy.l2cache.prefetcher.pfUsefulButMiss 0 # number of hit on prefetch but cache block is not in an usable state (Count)

board.cache_hierarchy.l2cache.prefetcher.accuracy 0.239726 # accuracy of the prefetcher (Count)

board.cache_hierarchy.l2cache.prefetcher.coverage 0.091864 # coverage brought by this prefetcher (Count)

board.cache_hierarchy.l2cache.prefetcher.pfHitInCache 50 # number of prefetches hitting in cache (Count)

board.cache_hierarchy.l2cache.prefetcher.pfHitInMSHR 51 # number of prefetches hitting in a MSHR (Count)

board.cache_hierarchy.l2cache.prefetcher.pfHitInWB 0 # number of prefetches hit in the Write Buffer (Count)

board.cache_hierarchy.l2cache.prefetcher.pfLate 101 # number of late prefetches (hitting in cache, MSHR or WB) (Count)

board.cache_hierarchy.l2cache.prefetcher.pfIdentified 146 # number of prefetch candidates identified (Count)

board.cache_hierarchy.l2cache.prefetcher.pfBufferHit 0 # number of redundant prefetches already in prefetch queue (Count)

board.cache_hierarchy.l2cache.prefetcher.pfInCache 0 # number of redundant prefetches already in cache/mshr dropped (Count)

board.cache_hierarchy.l2cache.prefetcher.pfRemovedDemand 0 # number of prefetches dropped due to a demand for the same address (Count)

board.cache_hierarchy.l2cache.prefetcher.pfRemovedFull 0 # number of prefetches dropped due to prefetch queue size (Count)

board.cache_hierarchy.l2cache.prefetcher.pfSpanPage 6 # number of prefetches that crossed the page (Count)

board.cache_hierarchy.l2cache.prefetcher.pfUsefulSpanPage 5 # number of prefetches that is useful and crossed the page (Count)

board.cache_hierarchy.l2cache.prefetcher.power_state.pwrStateResidencyTicks::UNDEFINED 73837531900351248 # Cumulative time (in ticks) in various power states (Tick)

board.cache_hierarchy.l2cache.tags.tagsInUse 391.000000 # Average ticks per tags in use ((Tick/Count))

board.cache_hierarchy.l2cache.tags.totalRefs 0 # Total number of references to valid blocks. (Count)

board.cache_hierarchy.l2cache.tags.sampledRefs 0 # Sample count of references to valid blocks. (Count)

board.cache_hierarchy.l2cache.tags.avgRefs nan # Average number of references to valid blocks. ((Count/Count))

board.cache_hierarchy.l2cache.tags.warmupTick 69597 # The tick when the warmup percentage was hit. (Tick)

board.cache_hierarchy.l2cache.tags.occupancies::cache_hierarchy.l2cache.prefetcher 45.000000 # Average occupied blocks per tick, per requestor ((Count/Tick))

board.cache_hierarchy.l2cache.tags.occupancies::processor.cores.core.inst 196.000000 # Average occupied blocks per tick, per requestor ((Count/Tick))

board.cache_hierarchy.l2cache.tags.occupancies::processor.cores.core.data 150.000000 # Average occupied blocks per tick, per requestor ((Count/Tick))

board.cache_hierarchy.l2cache.tags.avgOccs::cache_hierarchy.l2cache.prefetcher 0.010986 # Average percentage of cache occupancy ((Ratio/Tick))

board.cache_hierarchy.l2cache.tags.avgOccs::processor.cores.core.inst 0.047852 # Average percentage of cache occupancy ((Ratio/Tick))

board.cache_hierarchy.l2cache.tags.avgOccs::processor.cores.core.data 0.036621 # Average percentage of cache occupancy ((Ratio/Tick))

board.cache_hierarchy.l2cache.tags.avgOccs::total 0.095459 # Average percentage of cache occupancy ((Ratio/Tick))

board.cache_hierarchy.l2cache.tags.occupanciesTaskId::1022 45 # Occupied blocks per task id (Count)

board.cache_hierarchy.l2cache.tags.occupanciesTaskId::1024 346 # Occupied blocks per task id (Count)

board.cache_hierarchy.l2cache.tags.ageTaskId_1022::4 45 # Occupied blocks per task id, per block age (Count)

board.cache_hierarchy.l2cache.tags.ageTaskId_1024::4 346 # Occupied blocks per task id, per block age (Count)

board.cache_hierarchy.l2cache.tags.ratioOccsTaskId::1022 0.010986 # Ratio of occupied blocks and all blocks, per task id (Ratio)

board.cache_hierarchy.l2cache.tags.ratioOccsTaskId::1024 0.084473 # Ratio of occupied blocks and all blocks, per task id (Ratio)

board.cache_hierarchy.l2cache.tags.tagAccesses 6487 # Number of tag accesses (Count)

board.cache_hierarchy.l2cache.tags.dataAccesses 6487 # Number of data accesses (Count)

board.cache_hierarchy.l2cache.tags.power_state.pwrStateResidencyTicks::UNDEFINED 73837531900351248 # Cumulative time (in ticks) in various power states (Tick)

board.cache_hierarchy.membus.transDist::ReadResp 335 # Transaction distribution (Count)

board.cache_hierarchy.membus.transDist::ReadExReq 56 # Transaction distribution (Count)

board.cache_hierarchy.membus.transDist::ReadExResp 56 # Transaction distribution (Count)

board.cache_hierarchy.membus.transDist::ReadSharedReq 335 # Transaction distribution (Count)

board.cache_hierarchy.membus.pktCount_board.cache_hierarchy.l2cache.mem_side_port::board.memory.mem_ctrl.port 782 # Packet count per connected requestor and responder (Count)

board.cache_hierarchy.membus.pktCount_board.cache_hierarchy.l2cache.mem_side_port::total 782 # Packet count per connected requestor and responder (Count)

board.cache_hierarchy.membus.pktCount::total 782 # Packet count per connected requestor and responder (Count)

board.cache_hierarchy.membus.pktSize_board.cache_hierarchy.l2cache.mem_side_port::board.memory.mem_ctrl.port 25024 # Cumulative packet size per connected requestor and responder (Byte)

board.cache_hierarchy.membus.pktSize_board.cache_hierarchy.l2cache.mem_side_port::total 25024 # Cumulative packet size per connected requestor and responder (Byte)

board.cache_hierarchy.membus.pktSize::total 25024 # Cumulative packet size per connected requestor and responder (Byte)

board.cache_hierarchy.membus.snoops 0 # Total snoops (Count)

```
board.cache_hierarchy.membus.snoopTraffic 0 # Total snoop traffic (Byte)
board.cache_hierarchy.membus.snoopFanout::samples 391 # Request fanout histogram (Count)
board.cache_hierarchy.membus.snoopFanout::mean 0 # Request fanout histogram (Count)
board.cache_hierarchy.membus.snoopFanout::stdev 0 # Request fanout histogram (Count)
board.cache_hierarchy.membus.snoopFanout::underflows 0 0.00% 0.00% # Request fanout histogram (Count)
board.cache_hierarchy.membus.snoopFanout::0 391 100.00% 100.00% # Request fanout histogram (Count)
board.cache_hierarchy.membus.snoopFanout::1 0 0.00% 100.00% # Request fanout histogram (Count)
board.cache_hierarchy.membus.snoopFanout::overflows 0 0.00% 100.00% # Request fanout histogram (Count)
board.cache_hierarchy.membus.snoopFanout::min_value 0 # Request fanout histogram (Count)
board.cache_hierarchy.membus.snoopFanout::max_value 0 # Request fanout histogram (Count)
board.cache_hierarchy.membus.snoopFanout::total 391 # Request fanout histogram (Count)
board.cache_hierarchy.membus.badaddr_responder.power_state.pwrStateResidencyTicks::UNDEFINED 73837531900351248 # Cumulative time (in ticks) in various power states (Tick)
board.cache_hierarchy.membus.power_state.pwrStateResidencyTicks::UNDEFINED 73837531900351248 # Cumulative time (in ticks) in various power states (Tick)
board.cache_hierarchy.membus.reqLayer0.occupancy 139443 # Layer occupancy (ticks) (Tick)
board.cache_hierarchy.membus.reqLayer0.utilization 0.0 # Layer utilization (Ratio)
board.cache_hierarchy.membus.respLayer2.occupancy 333977 # Layer occupancy (ticks) (Tick)
board.cache_hierarchy.membus.respLayer2.utilization 0.0 # Layer utilization (Ratio)
board.cache_hierarchy.membus.snoop_filter.totRequests 391 # Total number of requests made to the snoop filter. (Count)
board.cache_hierarchy.membus.snoop_filter.hitSingleRequests 0 # Number of requests hitting in the snoop filter with a single holder of the requested data. (Count)
board.cache_hierarchy.membus.snoop_filter.hitMultiRequests 0 # Number of requests hitting in the snoop filter with multiple (>1) holders of the requested data. (Count)
board.cache_hierarchy.membus.snoop_filter.totSnoops 0 # Total number of snoops made to the snoop filter. (Count)
board.cache_hierarchy.membus.snoop_filter.hitSingleSnoops 0 # Number of snoops hitting in the snoop filter with a single holder of the requested data. (Count)
board.cache_hierarchy.membus.snoop_filter.hitMultiSnoops 0 # Number of snoops hitting in the snoop filter with multiple (>1) holders of the requested data. (Count)
board.clk_domain.clock 333 # Clock period in ticks (Tick)
board.clk_domain.voltage_domain.voltage 1 # Voltage in Volts (Volt)
board.memory.mem_ctrl.avgPriority_cache_hierarchy.l2cache.prefetcher::samples 45.00 # Average QoS priority value for accepted requests (Count)
board.memory.mem_ctrl.avgPriority_processor.cores.core.inst::samples 196.00 # Average QoS priority value for accepted requests (Count)
board.memory.mem_ctrl.avgPriority_processor.cores.core.data::samples 150.00 # Average QoS priority value for accepted requests (Count)
board.memory.mem_ctrl.priorityMinLatency 0.000000018750 # per QoS priority minimum request to response latency (Second)
board.memory.mem_ctrl.priorityMaxLatency 0.000000471124 # per QoS priority maximum request to response latency (Second)
board.memory.mem_ctrl.numReadWriteTurnArounds 0 # Number of turnarounds from READ to WRITE (Count)
board.memory.mem_ctrl.numWriteReadTurnArounds 0 # Number of turnarounds from WRITE to READ (Count)
board.memory.mem_ctrl.numStayReadState 18966135158 # Number of times bus staying in READ state (Count)
board.memory.mem_ctrl.numStayWriteState 0 # Number of times bus staying in WRITE state (Count)
board.memory.mem_ctrl.readReqs 391 # Number of read requests accepted (Count)
board.memory.mem_ctrl.writeReqs 0 # Number of write requests accepted (Count)
board.memory.mem_ctrl.readBursts 391 # Number of controller read bursts, including those serviced by the write queue (Count)
board.memory.mem_ctrl.writeBursts 0 # Number of controller write bursts, including those merged in the write queue (Count)
board.memory.mem_ctrl.servicedByWrQ 0 # Number of controller read bursts serviced by the write queue (Count)
board.memory.mem_ctrl.mergedWrBursts 0 # Number of controller write bursts merged with an existing one (Count)
board.memory.mem_ctrl.neitherReadNorWriteReqs 0 # Number of requests that are neither read nor write (Count)
board.memory.mem_ctrl.avgRdQLen 1.00 # Average read queue length when enqueuing ((Count/Tick))
board.memory.mem_ctrl.avgWrQLen 0.00 # Average write queue length when enqueuing ((Count/Tick))
board.memory.mem_ctrl.numRdRetry 0 # Number of times read queue was full causing retry (Count)
board.memory.mem_ctrl.numWrRetry 0 # Number of times write queue was full causing retry (Count)
board.memory.mem_ctrl.readPktSize::0 0 # Read request sizes (log2) (Count)
board.memory.mem_ctrl.readPktSize::1 0 # Read request sizes (log2) (Count)
board.memory.mem_ctrl.readPktSize::2 0 # Read request sizes (log2) (Count)
board.memory.mem_ctrl.readPktSize::3 0 # Read request sizes (log2) (Count)
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board.memory.mem_ctrl.readPktSize::4	0	# Read request sizes (log2) (Count)
board.memory.mem_ctrl.readPktSize::5	0	# Read request sizes (log2) (Count)
board.memory.mem_ctrl.readPktSize::6	391	# Read request sizes (log2) (Count)
board.memory.mem_ctrl.writePktSize::0	0	# Write request sizes (log2) (Count)
board.memory.mem_ctrl.writePktSize::1	0	# Write request sizes (log2) (Count)
board.memory.mem_ctrl.writePktSize::2	0	# Write request sizes (log2) (Count)
board.memory.mem_ctrl.writePktSize::3	0	# Write request sizes (log2) (Count)
board.memory.mem_ctrl.writePktSize::4	0	# Write request sizes (log2) (Count)
board.memory.mem_ctrl.writePktSize::5	0	# Write request sizes (log2) (Count)
board.memory.mem_ctrl.writePktSize::6	0	# Write request sizes (log2) (Count)
board.memory.mem_ctrl.rdQLenPdf::0	373	# What read queue length does an incoming req see (Count)
board.memory.mem_ctrl.rdQLenPdf::1	6	# What read queue length does an incoming req see (Count)
board.memory.mem_ctrl.rdQLenPdf::2	5	# What read queue length does an incoming req see (Count)
board.memory.mem_ctrl.rdQLenPdf::3	4	# What read queue length does an incoming req see (Count)
board.memory.mem_ctrl.rdQLenPdf::4	3	# What read queue length does an incoming req see (Count)
board.memory.mem_ctrl.rdQLenPdf::5	0	# What read queue length does an incoming req see (Count)
board.memory.mem_ctrl.rdQLenPdf::6	0	# What read queue length does an incoming req see (Count)
board.memory.mem_ctrl.rdQLenPdf::7	0	# What read queue length does an incoming req see (Count)
board.memory.mem_ctrl.rdQLenPdf::8	0	# What read queue length does an incoming req see (Count)
board.memory.mem_ctrl.rdQLenPdf::9	0	# What read queue length does an incoming req see (Count)
board.memory.mem_ctrl.rdQLenPdf::10	0	# What read queue length does an incoming req see (Count)
board.memory.mem_ctrl.rdQLenPdf::11	0	# What read queue length does an incoming req see (Count)
board.memory.mem_ctrl.rdQLenPdf::12	0	# What read queue length does an incoming req see (Count)
board.memory.mem_ctrl.rdQLenPdf::13	0	# What read queue length does an incoming req see (Count)
board.memory.mem_ctrl.rdQLenPdf::14	0	# What read queue length does an incoming req see (Count)
board.memory.mem_ctrl.rdQLenPdf::15	0	# What read queue length does an incoming req see (Count)
board.memory.mem_ctrl.rdQLenPdf::16	0	# What read queue length does an incoming req see (Count)
board.memory.mem_ctrl.rdQLenPdf::17	0	# What read queue length does an incoming req see (Count)
board.memory.mem_ctrl.rdQLenPdf::18	0	# What read queue length does an incoming req see (Count)
board.memory.mem_ctrl.rdQLenPdf::19	0	# What read queue length does an incoming req see (Count)
board.memory.mem_ctrl.rdQLenPdf::20	0	# What read queue length does an incoming req see (Count)
board.memory.mem_ctrl.rdQLenPdf::21	0	# What read queue length does an incoming req see (Count)
board.memory.mem_ctrl.rdQLenPdf::22	0	# What read queue length does an incoming req see (Count)
board.memory.mem_ctrl.rdQLenPdf::23	0	# What read queue length does an incoming req see (Count)
board.memory.mem_ctrl.rdQLenPdf::24	0	# What read queue length does an incoming req see (Count)
board.memory.mem_ctrl.rdQLenPdf::25	0	# What read queue length does an incoming req see (Count)
board.memory.mem_ctrl.rdQLenPdf::26	0	# What read queue length does an incoming req see (Count)
board.memory.mem_ctrl.rdQLenPdf::27	0	# What read queue length does an incoming req see (Count)
board.memory.mem_ctrl.rdQLenPdf::28	0	# What read queue length does an incoming req see (Count)
board.memory.mem_ctrl.rdQLenPdf::29	0	# What read queue length does an incoming req see (Count)
board.memory.mem_ctrl.rdQLenPdf::30	0	# What read queue length does an incoming req see (Count)
board.memory.mem_ctrl.rdQLenPdf::31	0	# What read queue length does an incoming req see (Count)
board.memory.mem_ctrl.wrQLenPdf::0	0	# What write queue length does an incoming req see (Count)
board.memory.mem_ctrl.wrQLenPdf::1	0	# What write queue length does an incoming req see (Count)
board.memory.mem_ctrl.wrQLenPdf::2	0	# What write queue length does an incoming req see (Count)
board.memory.mem_ctrl.wrQLenPdf::3	0	# What write queue length does an incoming req see (Count)
board.memory.mem_ctrl.wrQLenPdf::4	0	# What write queue length does an incoming req see (Count)
board.memory.mem_ctrl.wrQLenPdf::5	0	# What write queue length does an incoming req see (Count)
board.memory.mem_ctrl.wrQLenPdf::6	0	# What write queue length does an incoming req see (Count)

board.memory.mem_ctrl.wrQLenPdf::7	0	# What write queue length does an incoming req see (Count)
board.memory.mem_ctrl.wrQLenPdf::8	0	# What write queue length does an incoming req see (Count)
board.memory.mem_ctrl.wrQLenPdf::9	0	# What write queue length does an incoming req see (Count)
board.memory.mem_ctrl.wrQLenPdf::10	0	# What write queue length does an incoming req see (Count)
board.memory.mem_ctrl.wrQLenPdf::11	0	# What write queue length does an incoming req see (Count)
board.memory.mem_ctrl.wrQLenPdf::12	0	# What write queue length does an incoming req see (Count)
board.memory.mem_ctrl.wrQLenPdf::13	0	# What write queue length does an incoming req see (Count)
board.memory.mem_ctrl.wrQLenPdf::14	0	# What write queue length does an incoming req see (Count)
board.memory.mem_ctrl.wrQLenPdf::15	0	# What write queue length does an incoming req see (Count)
board.memory.mem_ctrl.wrQLenPdf::16	0	# What write queue length does an incoming req see (Count)
board.memory.mem_ctrl.wrQLenPdf::17	0	# What write queue length does an incoming req see (Count)
board.memory.mem_ctrl.wrQLenPdf::18	0	# What write queue length does an incoming req see (Count)
board.memory.mem_ctrl.wrQLenPdf::19	0	# What write queue length does an incoming req see (Count)
board.memory.mem_ctrl.wrQLenPdf::20	0	# What write queue length does an incoming req see (Count)
board.memory.mem_ctrl.wrQLenPdf::21	0	# What write queue length does an incoming req see (Count)
board.memory.mem_ctrl.wrQLenPdf::22	0	# What write queue length does an incoming req see (Count)
board.memory.mem_ctrl.wrQLenPdf::23	0	# What write queue length does an incoming req see (Count)
board.memory.mem_ctrl.wrQLenPdf::24	0	# What write queue length does an incoming req see (Count)
board.memory.mem_ctrl.wrQLenPdf::25	0	# What write queue length does an incoming req see (Count)
board.memory.mem_ctrl.wrQLenPdf::26	0	# What write queue length does an incoming req see (Count)
board.memory.mem_ctrl.wrQLenPdf::27	0	# What write queue length does an incoming req see (Count)
board.memory.mem_ctrl.wrQLenPdf::28	0	# What write queue length does an incoming req see (Count)
board.memory.mem_ctrl.wrQLenPdf::29	0	# What write queue length does an incoming req see (Count)
board.memory.mem_ctrl.wrQLenPdf::30	0	# What write queue length does an incoming req see (Count)
board.memory.mem_ctrl.wrQLenPdf::31	0	# What write queue length does an incoming req see (Count)
board.memory.mem_ctrl.wrQLenPdf::32	0	# What write queue length does an incoming req see (Count)
board.memory.mem_ctrl.wrQLenPdf::33	0	# What write queue length does an incoming req see (Count)
board.memory.mem_ctrl.wrQLenPdf::34	0	# What write queue length does an incoming req see (Count)
board.memory.mem_ctrl.wrQLenPdf::35	0	# What write queue length does an incoming req see (Count)
board.memory.mem_ctrl.wrQLenPdf::36	0	# What write queue length does an incoming req see (Count)
board.memory.mem_ctrl.wrQLenPdf::37	0	# What write queue length does an incoming req see (Count)
board.memory.mem_ctrl.wrQLenPdf::38	0	# What write queue length does an incoming req see (Count)
board.memory.mem_ctrl.wrQLenPdf::39	0	# What write queue length does an incoming req see (Count)
board.memory.mem_ctrl.wrQLenPdf::40	0	# What write queue length does an incoming req see (Count)
board.memory.mem_ctrl.wrQLenPdf::41	0	# What write queue length does an incoming req see (Count)
board.memory.mem_ctrl.wrQLenPdf::42	0	# What write queue length does an incoming req see (Count)
board.memory.mem_ctrl.wrQLenPdf::43	0	# What write queue length does an incoming req see (Count)
board.memory.mem_ctrl.wrQLenPdf::44	0	# What write queue length does an incoming req see (Count)
board.memory.mem_ctrl.wrQLenPdf::45	0	# What write queue length does an incoming req see (Count)
board.memory.mem_ctrl.wrQLenPdf::46	0	# What write queue length does an incoming req see (Count)
board.memory.mem_ctrl.wrQLenPdf::47	0	# What write queue length does an incoming req see (Count)
board.memory.mem_ctrl.wrQLenPdf::48	0	# What write queue length does an incoming req see (Count)
board.memory.mem_ctrl.wrQLenPdf::49	0	# What write queue length does an incoming req see (Count)
board.memory.mem_ctrl.wrQLenPdf::50	0	# What write queue length does an incoming req see (Count)
board.memory.mem_ctrl.wrQLenPdf::51	0	# What write queue length does an incoming req see (Count)
board.memory.mem_ctrl.wrQLenPdf::52	0	# What write queue length does an incoming req see (Count)
board.memory.mem_ctrl.wrQLenPdf::53	0	# What write queue length does an incoming req see (Count)
board.memory.mem_ctrl.wrQLenPdf::54	0	# What write queue length does an incoming req see (Count)
board.memory.mem_ctrl.wrQLenPdf::55	0	# What write queue length does an incoming req see (Count)

board.memory.mem_ctrl.wrQLenPdf::56 0 # What write queue length does an incoming req see (Count)

board.memory.mem_ctrl.wrQLenPdf::57 0 # What write queue length does an incoming req see (Count)

board.memory.mem_ctrl.wrQLenPdf::58 0 # What write queue length does an incoming req see (Count)

board.memory.mem_ctrl.wrQLenPdf::59 0 # What write queue length does an incoming req see (Count)

board.memory.mem_ctrl.wrQLenPdf::60 0 # What write queue length does an incoming req see (Count)

board.memory.mem_ctrl.wrQLenPdf::61 0 # What write queue length does an incoming req see (Count)

board.memory.mem_ctrl.wrQLenPdf::62 0 # What write queue length does an incoming req see (Count)

board.memory.mem_ctrl.wrQLenPdf::63 0 # What write queue length does an incoming req see (Count)

board.memory.mem_ctrl.bytesReadWrQ 0 # Total number of bytes read from write queue (Byte)

board.memory.mem_ctrl.bytesReadSys 25024 # Total read bytes from the system interface side (Byte)

board.memory.mem_ctrl.bytesWrittenSys 0 # Total written bytes from the system interface side (Byte)

board.memory.mem_ctrl.avgRdBWSys 0.33890624 # Average system read bandwidth in Byte/s ((Byte/Second))

board.memory.mem_ctrl.avgWrBWSys 0.00000000 # Average system write bandwidth in Byte/s ((Byte/Second))

board.memory.mem_ctrl.totGap 24062247 # Total gap between requests (Tick)

board.memory.mem_ctrl.avgGap 61540.27 # Average gap between requests ((Tick/Count))

board.memory.mem_ctrl.requestorReadBytes::cache_hierarchy.l2cache.prefetcher 2880 # Per-requestor bytes read from memory (Byte)

board.memory.mem_ctrl.requestorReadBytes::processor.cores.core.inst 12544 # Per-requestor bytes read from memory (Byte)

board.memory.mem_ctrl.requestorReadBytes::processor.cores.core.data 9600 # Per-requestor bytes read from memory (Byte)

board.memory.mem_ctrl.requestorReadRate::cache_hierarchy.l2cache.prefetcher 0.039004553997 # Per-requestor bytes read from memory rate ((Byte/Second))

board.memory.mem_ctrl.requestorReadRate::processor.cores.core.inst 0.169886501853 # Per-requestor bytes read from memory rate ((Byte/Second))

board.memory.mem_ctrl.requestorReadRate::processor.cores.core.data 0.130015179989 # Per-requestor bytes read from memory rate ((Byte/Second))

board.memory.mem_ctrl.requestorReadAccesses::cache_hierarchy.l2cache.prefetcher 45 # Per-requestor read serviced memory accesses (Count)

board.memory.mem_ctrl.requestorReadAccesses::processor.cores.core.inst 196 # Per-requestor read serviced memory accesses (Count)

board.memory.mem_ctrl.requestorReadAccesses::processor.cores.core.data 150 # Per-requestor read serviced memory accesses (Count)

board.memory.mem_ctrl.requestorReadTotalLat::cache_hierarchy.l2cache.prefetcher 977836 # Per-requestor read total memory access latency (Tick)

board.memory.mem_ctrl.requestorReadTotalLat::processor.cores.core.inst 5044337 # Per-requestor read total memory access latency (Tick)

board.memory.mem_ctrl.requestorReadTotalLat::processor.cores.core.data 3372950 # Per-requestor read total memory access latency (Tick)

board.memory.mem_ctrl.requestorReadAvgLat::cache_hierarchy.l2cache.prefetcher 21729.69 # Per-requestor read average memory access latency ((Tick/Count))

board.memory.mem_ctrl.requestorReadAvgLat::processor.cores.core.inst 25736.41 # Per-requestor read average memory access latency ((Tick/Count))

board.memory.mem_ctrl.requestorReadAvgLat::processor.cores.core.data 22486.33 # Per-requestor read average memory access latency ((Tick/Count))

board.memory.mem_ctrl.dram.bytesRead::cache_hierarchy.l2cache.prefetcher 2880 # Number of bytes read from this memory (Byte)

board.memory.mem_ctrl.dram.bytesRead::processor.cores.core.inst 12544 # Number of bytes read from this memory (Byte)

board.memory.mem_ctrl.dram.bytesRead::processor.cores.core.data 9600 # Number of bytes read from this memory (Byte)

board.memory.mem_ctrl.dram.bytesRead::total 25024 # Number of bytes read from this memory (Byte)

board.memory.mem_ctrl.dram.bytesInstRead::processor.cores.core.inst 12544 # Number of instructions bytes read from this memory (Byte)

board.memory.mem_ctrl.dram.bytesInstRead::total 12544 # Number of instructions bytes read from this memory (Byte)

board.memory.mem_ctrl.dram.numReads::cache_hierarchy.l2cache.prefetcher 45 # Number of read requests responded to by this memory (Count)

board.memory.mem_ctrl.dram.numReads::processor.cores.core.inst 196 # Number of read requests responded to by this memory (Count)

board.memory.mem_ctrl.dram.numReads::processor.cores.core.data 150 # Number of read requests responded to by this memory (Count)

board.memory.mem_ctrl.dram.numReads::total 391 # Number of read requests responded to by this memory (Count)

board.memory.mem_ctrl.dram.bwRead::cache_hierarchy.l2cache.prefetcher 0 # Total read bandwidth from this memory ((Byte/Second))

board.memory.mem_ctrl.dram.bwRead::processor.cores.core.inst 0 # Total read bandwidth from this memory ((Byte/Second))

board.memory.mem_ctrl.dram.bwRead::processor.cores.core.data 0 # Total read bandwidth from this memory ((Byte/Second))

board.memory.mem_ctrl.dram.bwRead::total 0 # Total read bandwidth from this memory ((Byte/Second))

board.memory.mem_ctrl.dram.bwInstRead::processor.cores.core.inst 0 # Instruction read bandwidth from this memory ((Byte/Second))

board.memory.mem_ctrl.dram.bwInstRead::total 0 # Instruction read bandwidth from this memory ((Byte/Second))

board.memory.mem_ctrl.dram.bwTotal::cache_hierarchy.l2cache.prefetcher 0 # Total bandwidth to/from this memory ((Byte/Second))

board.memory.mem_ctrl.dram.bwTotal::processor.cores.core.inst 0 # Total bandwidth to/from this memory ((Byte/Second))

board.memory.mem_ctrl.dram.bwTotal::processor.cores.core.data 0 # Total bandwidth to/from this memory ((Byte/Second))

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board.memory.mem_ctrl.dram.bwTotal::total            0                    # Total bandwidth to/from this memory ((Byte/Second))
board.memory.mem_ctrl.dram.readBursts              391                    # Number of DRAM read bursts (Count)
board.memory.mem_ctrl.dram.writeBursts               0                    # Number of DRAM write bursts (Count)
board.memory.mem_ctrl.dram.perBankRdBursts::0       68                    # Per bank write bursts (Count)
board.memory.mem_ctrl.dram.perBankRdBursts::1       62                    # Per bank write bursts (Count)
board.memory.mem_ctrl.dram.perBankRdBursts::2       19                    # Per bank write bursts (Count)
board.memory.mem_ctrl.dram.perBankRdBursts::3       37                    # Per bank write bursts (Count)
board.memory.mem_ctrl.dram.perBankRdBursts::4       46                    # Per bank write bursts (Count)
board.memory.mem_ctrl.dram.perBankRdBursts::5       66                    # Per bank write bursts (Count)
board.memory.mem_ctrl.dram.perBankRdBursts::6        3                    # Per bank write bursts (Count)
board.memory.mem_ctrl.dram.perBankRdBursts::7        3                    # Per bank write bursts (Count)
board.memory.mem_ctrl.dram.perBankRdBursts::8       38                    # Per bank write bursts (Count)
board.memory.mem_ctrl.dram.perBankRdBursts::9       17                    # Per bank write bursts (Count)
board.memory.mem_ctrl.dram.perBankRdBursts::10       9                    # Per bank write bursts (Count)
board.memory.mem_ctrl.dram.perBankRdBursts::11       2                    # Per bank write bursts (Count)
board.memory.mem_ctrl.dram.perBankRdBursts::12       0                    # Per bank write bursts (Count)
board.memory.mem_ctrl.dram.perBankRdBursts::13       2                    # Per bank write bursts (Count)
board.memory.mem_ctrl.dram.perBankRdBursts::14       2                    # Per bank write bursts (Count)
board.memory.mem_ctrl.dram.perBankRdBursts::15      17                    # Per bank write bursts (Count)
board.memory.mem_ctrl.dram.perBankWrBursts::0        0                    # Per bank write bursts (Count)
board.memory.mem_ctrl.dram.perBankWrBursts::1        0                    # Per bank write bursts (Count)
board.memory.mem_ctrl.dram.perBankWrBursts::2        0                    # Per bank write bursts (Count)
board.memory.mem_ctrl.dram.perBankWrBursts::3        0                    # Per bank write bursts (Count)
board.memory.mem_ctrl.dram.perBankWrBursts::4        0                    # Per bank write bursts (Count)
board.memory.mem_ctrl.dram.perBankWrBursts::5        0                    # Per bank write bursts (Count)
board.memory.mem_ctrl.dram.perBankWrBursts::6        0                    # Per bank write bursts (Count)
board.memory.mem_ctrl.dram.perBankWrBursts::7        0                    # Per bank write bursts (Count)
board.memory.mem_ctrl.dram.perBankWrBursts::8        0                    # Per bank write bursts (Count)
board.memory.mem_ctrl.dram.perBankWrBursts::9        0                    # Per bank write bursts (Count)
board.memory.mem_ctrl.dram.perBankWrBursts::10       0                    # Per bank write bursts (Count)
board.memory.mem_ctrl.dram.perBankWrBursts::11       0                    # Per bank write bursts (Count)
board.memory.mem_ctrl.dram.perBankWrBursts::12       0                    # Per bank write bursts (Count)
board.memory.mem_ctrl.dram.perBankWrBursts::13       0                    # Per bank write bursts (Count)
board.memory.mem_ctrl.dram.perBankWrBursts::14       0                    # Per bank write bursts (Count)
board.memory.mem_ctrl.dram.perBankWrBursts::15       0                    # Per bank write bursts (Count)
board.memory.mem_ctrl.dram.totQLat             2063873                    # Total ticks spent queuing (Tick)
board.memory.mem_ctrl.dram.totBusLat           1955000                    # Total ticks spent in databus transfers (Tick)
board.memory.mem_ctrl.dram.totMemAccLat        9395123                    # Total ticks spent from burst creation until serviced by the DRAM (Tick)
board.memory.mem_ctrl.dram.avgQLat             5278.45                    # Average queueing delay per DRAM burst ((Tick/Count))
board.memory.mem_ctrl.dram.avgBusLat           5000.00                    # Average bus latency per DRAM burst ((Tick/Count))
board.memory.mem_ctrl.dram.avgMemAccLat       24028.45                    # Average memory access latency per DRAM burst ((Tick/Count))
board.memory.mem_ctrl.dram.readRowHits            323                    # Number of row buffer hits during reads (Count)
board.memory.mem_ctrl.dram.writeRowHits             0                    # Number of row buffer hits during writes (Count)
board.memory.mem_ctrl.dram.readRowHitRate        82.61                    # Row buffer hit rate for reads (Ratio)
board.memory.mem_ctrl.dram.writeRowHitRate         nan                    # Row buffer hit rate for writes (Ratio)
board.memory.mem_ctrl.dram.bytesPerActivate::samples          68                    # Bytes accessed per row activation (Byte)
board.memory.mem_ctrl.dram.bytesPerActivate::mean         368                    # Bytes accessed per row activation (Byte)
board.memory.mem_ctrl.dram.bytesPerActivate::gmean   229.264367                    # Bytes accessed per row activation (Byte)
board.memory.mem_ctrl.dram.bytesPerActivate::stdev   338.362695                    # Bytes accessed per row activation (Byte)
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board.memory.mem_ctrl.dram.bytesPerActivate::0-127          18     26.47%     26.47% # Bytes accessed per row activation (Byte)
board.memory.mem_ctrl.dram.bytesPerActivate::128-255          16     23.53%     50.00% # Bytes accessed per row activation (Byte)
board.memory.mem_ctrl.dram.bytesPerActivate::256-383          6      8.82%     58.82% # Bytes accessed per row activation (Byte)
board.memory.mem_ctrl.dram.bytesPerActivate::384-511          7     10.29%     69.12% # Bytes accessed per row activation (Byte)
board.memory.mem_ctrl.dram.bytesPerActivate::512-639          6      8.82%     77.94% # Bytes accessed per row activation (Byte)
board.memory.mem_ctrl.dram.bytesPerActivate::640-767          2      2.94%     80.88% # Bytes accessed per row activation (Byte)
board.memory.mem_ctrl.dram.bytesPerActivate::768-895          3      4.41%     85.29% # Bytes accessed per row activation (Byte)
board.memory.mem_ctrl.dram.bytesPerActivate::896-1023          2      2.94%     88.24% # Bytes accessed per row activation (Byte)
board.memory.mem_ctrl.dram.bytesPerActivate::1024-1151          8     11.76%    100.00% # Bytes accessed per row activation (Byte)
board.memory.mem_ctrl.dram.bytesPerActivate::total          68                     # Bytes accessed per row activation (Byte)
board.memory.mem_ctrl.dram.dramBytesRead        25024                     # Total bytes read (Byte)
board.memory.mem_ctrl.dram.dramBytesWritten           0                     # Total bytes written (Byte)
board.memory.mem_ctrl.dram.avgRdBW          0.000000                     # Average DRAM read bandwidth in MiBytes/s ((Byte/Second))
board.memory.mem_ctrl.dram.avgWrBW                 0                     # Average DRAM write bandwidth in MiBytes/s ((Byte/Second))
board.memory.mem_ctrl.dram.peakBW           12800.00                     # Theoretical peak bandwidth in MiByte/s ((Byte/Second))
board.memory.mem_ctrl.dram.busUtil              0.00                     # Data bus utilization in percentage (Ratio)
board.memory.mem_ctrl.dram.busUtilRead          0.00                     # Data bus utilization in percentage for reads (Ratio)
board.memory.mem_ctrl.dram.busUtilWrite         0.00                     # Data bus utilization in percentage for writes (Ratio)
board.memory.mem_ctrl.dram.pageHitRate         82.61                     # Row buffer hit rate, read and write combined (Ratio)
board.memory.mem_ctrl.dram.power_state.pwrStateResidencyTicks::UNDEFINED 73837531900351248                     # Cumulative time (in ticks) in various power states (Tick)
board.memory.mem_ctrl.dram.rank0.actEnergy       342720                     # Energy for activate commands per rank (pJ) (Joule)
board.memory.mem_ctrl.dram.rank0.preEnergy       182160                     # Energy for precharge commands per rank (pJ) (Joule)
board.memory.mem_ctrl.dram.rank0.readEnergy      2170560                     # Energy for read commands per rank (pJ) (Joule)
board.memory.mem_ctrl.dram.rank0.writeEnergy           0                     # Energy for write commands per rank (pJ) (Joule)
board.memory.mem_ctrl.dram.rank0.refreshEnergy 5828672417046961                     # Energy for refresh commands per rank (pJ) (Joule)
board.memory.mem_ctrl.dram.rank0.actBackEnergy 1064853628477020                     # Energy for active background per rank (pJ) (Joule)
board.memory.mem_ctrl.dram.rank0.preBackEnergy 27456893404796160                     # Energy for precharge background per rank (pJ) (Joule)
board.memory.mem_ctrl.dram.rank0.actPowerDownEnergy           0                     # Energy for active power-down per rank (pJ) (Joule)
board.memory.mem_ctrl.dram.rank0.prePowerDownEnergy           0                     # Energy for precharge power-down per rank (pJ) (Joule)
board.memory.mem_ctrl.dram.rank0.selfRefreshEnergy           0                     # Energy for self refresh per rank (pJ) (Joule)
board.memory.mem_ctrl.dram.rank0.totalEnergy 34350410433295488                     # Total energy per rank (pJ) (Joule)
board.memory.mem_ctrl.dram.rank0.averagePower   465.216124                     # Core power per rank (mW) (Watt)
board.memory.mem_ctrl.dram.rank0.totalIdleTime           0                     # Total Idle time Per DRAM Rank (Tick)
board.memory.mem_ctrl.dram.rank0.pwrStateTime::IDLE 71371920222050264                     # Time in different power states (Tick)
board.memory.mem_ctrl.dram.rank0.pwrStateTime::REF 2465597468880000                     # Time in different power states (Tick)
board.memory.mem_ctrl.dram.rank0.pwrStateTime::SREF           0                     # Time in different power states (Tick)
board.memory.mem_ctrl.dram.rank0.pwrStateTime::PRE_PDN           0                     # Time in different power states (Tick)
board.memory.mem_ctrl.dram.rank0.pwrStateTime::ACT     30371424                     # Time in different power states (Tick)
board.memory.mem_ctrl.dram.rank0.pwrStateTime::ACT_PDN           0                     # Time in different power states (Tick)
board.memory.mem_ctrl.dram.rank1.actEnergy       142800                     # Energy for activate commands per rank (pJ) (Joule)
board.memory.mem_ctrl.dram.rank1.preEnergy        75900                     # Energy for precharge commands per rank (pJ) (Joule)
board.memory.mem_ctrl.dram.rank1.readEnergy       621180                     # Energy for read commands per rank (pJ) (Joule)
board.memory.mem_ctrl.dram.rank1.writeEnergy           0                     # Energy for write commands per rank (pJ) (Joule)
board.memory.mem_ctrl.dram.rank1.refreshEnergy 5828672417046961                     # Energy for refresh commands per rank (pJ) (Joule)
board.memory.mem_ctrl.dram.rank1.actBackEnergy 1064853628146990                     # Energy for active background per rank (pJ) (Joule)
board.memory.mem_ctrl.dram.rank1.preBackEnergy 27456893405074080                     # Energy for precharge background per rank (pJ) (Joule)
board.memory.mem_ctrl.dram.rank1.actPowerDownEnergy           0                     # Energy for active power-down per rank (pJ) (Joule)
board.memory.mem_ctrl.dram.rank1.prePowerDownEnergy           0                     # Energy for precharge power-down per rank (pJ) (Joule)
board.memory.mem_ctrl.dram.rank1.selfRefreshEnergy           0                     # Energy for self refresh per rank (pJ) (Joule)
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board.memory.mem_ctrl.dram.rank1.totalEnergy 34350410431387824                    # Total energy per rank (pJ) (Joule)
board.memory.mem_ctrl.dram.rank1.averagePower   465.216124                    # Core power per rank (mW) (Watt)
board.memory.mem_ctrl.dram.rank1.totalIdleTime           0                    # Total Idle time Per DRAM Rank (Tick)
board.memory.mem_ctrl.dram.rank1.pwrStateTime::IDLE 71371920222758216                    # Time in different power states (Tick)
board.memory.mem_ctrl.dram.rank1.pwrStateTime::REF 2465597468880000                    # Time in different power states (Tick)
board.memory.mem_ctrl.dram.rank1.pwrStateTime::SREF           0                    # Time in different power states (Tick)
board.memory.mem_ctrl.dram.rank1.pwrStateTime::PRE_PDN           0                    # Time in different power states (Tick)
board.memory.mem_ctrl.dram.rank1.pwrStateTime::ACT     29663469                    # Time in different power states (Tick)
board.memory.mem_ctrl.dram.rank1.pwrStateTime::ACT_PDN           0                    # Time in different power states (Tick)
board.memory.mem_ctrl.power_state.pwrStateResidencyTicks::UNDEFINED 73837531900351248                    # Cumulative time (in ticks) in various power states (Tick)
board.processor.cores.core.numCycles           72487                    # Number of cpu cycles simulated (Cycle)
board.processor.cores.core.cpi           17.466747                    # CPI: cycles per instruction (core level) ((Cycle/Count))
board.processor.cores.core.ipc           0.057252                    # IPC: instructions per cycle (core level) ((Count/Cycle))
board.processor.cores.core.numWorkItemsStarted           0                    # Number of work items this cpu started (Count)
board.processor.cores.core.numWorkItemsCompleted           0                    # Number of work items this cpu completed (Count)
board.processor.cores.core.commitStats0.numInsts         4150                    # Number of instructions committed (thread level) (Count)
board.processor.cores.core.commitStats0.numOps         7878                    # Number of ops (including micro ops) committed (thread level) (Count)
board.processor.cores.core.commitStats0.numInstsNotNOP           0                    # Number of instructions committed excluding NOPs or prefetches (Count)
board.processor.cores.core.commitStats0.numOpsNotNOP           0                    # Number of Ops (including micro ops) Simulated (Count)
board.processor.cores.core.commitStats0.cpi     17.466747                    # CPI: cycles per instruction (thread level) ((Cycle/Count))
board.processor.cores.core.commitStats0.ipc     0.057252                    # IPC: instructions per cycle (thread level) ((Count/Cycle))
board.processor.cores.core.commitStats0.numMemRefs           0                    # Number of memory references committed (Count)
board.processor.cores.core.commitStats0.numFpInsts          703                    # Number of float instructions (Count)
board.processor.cores.core.commitStats0.numIntInsts         7438                    # Number of integer instructions (Count)
board.processor.cores.core.commitStats0.numLoadInsts          642                    # Number of load instructions (Count)
board.processor.cores.core.commitStats0.numStoreInsts          846                    # Number of store instructions (Count)
board.processor.cores.core.commitStats0.numVecInsts           0                    # Number of vector instructions (Count)
board.processor.cores.core.commitStats0.committedInstType::No_OpClass          90      1.14%      1.14% # Class of committed instruction. (Count)
board.processor.cores.core.commitStats0.committedInstType::IntAlu        5994     76.09%     77.23% # Class of committed instruction. (Count)
board.processor.cores.core.commitStats0.committedInstType::IntMult           1      0.01%     77.24% # Class of committed instruction. (Count)
board.processor.cores.core.commitStats0.committedInstType::IntDiv          21      0.27%     77.51% # Class of committed instruction. (Count)
board.processor.cores.core.commitStats0.committedInstType::FloatAdd         136      1.73%     79.23% # Class of committed instruction. (Count)
board.processor.cores.core.commitStats0.committedInstType::FloatCmp           0      0.00%     79.23% # Class of committed instruction. (Count)
board.processor.cores.core.commitStats0.committedInstType::FloatCvt           0      0.00%     79.23% # Class of committed instruction. (Count)
board.processor.cores.core.commitStats0.committedInstType::FloatMult           0      0.00%     79.23% # Class of committed instruction. (Count)
board.processor.cores.core.commitStats0.committedInstType::FloatMultAcc           0      0.00%     79.23% # Class of committed instruction. (Count)
board.processor.cores.core.commitStats0.committedInstType::FloatDiv           0      0.00%     79.23% # Class of committed instruction. (Count)
board.processor.cores.core.commitStats0.committedInstType::FloatMisc           0      0.00%     79.23% # Class of committed instruction. (Count)
board.processor.cores.core.commitStats0.committedInstType::FloatSqrt           0      0.00%     79.23% # Class of committed instruction. (Count)
board.processor.cores.core.commitStats0.committedInstType::SimdAdd           0      0.00%     79.23% # Class of committed instruction. (Count)
board.processor.cores.core.commitStats0.committedInstType::SimdAddAcc           0      0.00%     79.23% # Class of committed instruction. (Count)
board.processor.cores.core.commitStats0.committedInstType::SimdAlu          10      0.13%     79.36% # Class of committed instruction. (Count)
board.processor.cores.core.commitStats0.committedInstType::SimdCmp           0      0.00%     79.36% # Class of committed instruction. (Count)
board.processor.cores.core.commitStats0.committedInstType::SimdCvt           2      0.03%     79.39% # Class of committed instruction. (Count)
board.processor.cores.core.commitStats0.committedInstType::SimdMisc         136      1.73%     81.11% # Class of committed instruction. (Count)
board.processor.cores.core.commitStats0.committedInstType::SimdMult           0      0.00%     81.11% # Class of committed instruction. (Count)
board.processor.cores.core.commitStats0.committedInstType::SimdMultAcc           0      0.00%     81.11% # Class of committed instruction. (Count)
board.processor.cores.core.commitStats0.committedInstType::SimdMatMultAcc           0      0.00%     81.11% # Class of committed instruction. (Count)
board.processor.cores.core.commitStats0.committedInstType::SimdShift           0      0.00%     81.11% # Class of committed instruction. (Count)
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board.processor.cores.core.commitStats0.committedInstType::SimdShiftAcc 0 0.00% 81.11% # Class of committed instruction. (Count)

board.processor.cores.core.commitStats0.committedInstType::SimdDiv 0 0.00% 81.11% # Class of committed instruction. (Count)

board.processor.cores.core.commitStats0.committedInstType::SimdSqrt 0 0.00% 81.11% # Class of committed instruction. (Count)

board.processor.cores.core.commitStats0.committedInstType::SimdFloatAdd 0 0.00% 81.11% # Class of committed instruction. (Count)

board.processor.cores.core.commitStats0.committedInstType::SimdFloatAlu 0 0.00% 81.11% # Class of committed instruction. (Count)

board.processor.cores.core.commitStats0.committedInstType::SimdFloatCmp 0 0.00% 81.11% # Class of committed instruction. (Count)

board.processor.cores.core.commitStats0.committedInstType::SimdFloatCvt 0 0.00% 81.11% # Class of committed instruction. (Count)

board.processor.cores.core.commitStats0.committedInstType::SimdFloatDiv 0 0.00% 81.11% # Class of committed instruction. (Count)

board.processor.cores.core.commitStats0.committedInstType::SimdFloatMisc 0 0.00% 81.11% # Class of committed instruction. (Count)

board.processor.cores.core.commitStats0.committedInstType::SimdFloatMult 0 0.00% 81.11% # Class of committed instruction. (Count)

board.processor.cores.core.commitStats0.committedInstType::SimdFloatMultAcc 0 0.00% 81.11% # Class of committed instruction. (Count)

board.processor.cores.core.commitStats0.committedInstType::SimdFloatMatMultAcc 0 0.00% 81.11% # Class of committed instruction. (Count)

board.processor.cores.core.commitStats0.committedInstType::SimdFloatSqrt 0 0.00% 81.11% # Class of committed instruction. (Count)

board.processor.cores.core.commitStats0.committedInstType::SimdReduceAdd 0 0.00% 81.11% # Class of committed instruction. (Count)

board.processor.cores.core.commitStats0.committedInstType::SimdReduceAlu 0 0.00% 81.11% # Class of committed instruction. (Count)

board.processor.cores.core.commitStats0.committedInstType::SimdReduceCmp 0 0.00% 81.11% # Class of committed instruction. (Count)

board.processor.cores.core.commitStats0.committedInstType::SimdFloatReduceAdd 0 0.00% 81.11% # Class of committed instruction. (Count)

board.processor.cores.core.commitStats0.committedInstType::SimdFloatReduceCmp 0 0.00% 81.11% # Class of committed instruction. (Count)

board.processor.cores.core.commitStats0.committedInstType::SimdAes 0 0.00% 81.11% # Class of committed instruction. (Count)

board.processor.cores.core.commitStats0.committedInstType::SimdAesMix 0 0.00% 81.11% # Class of committed instruction. (Count)

board.processor.cores.core.commitStats0.committedInstType::SimdSha1Hash 0 0.00% 81.11% # Class of committed instruction. (Count)

board.processor.cores.core.commitStats0.committedInstType::SimdSha1Hash2 0 0.00% 81.11% # Class of committed instruction. (Count)

board.processor.cores.core.commitStats0.committedInstType::SimdSha256Hash 0 0.00% 81.11% # Class of committed instruction. (Count)

board.processor.cores.core.commitStats0.committedInstType::SimdSha256Hash2 0 0.00% 81.11% # Class of committed instruction. (Count)

board.processor.cores.core.commitStats0.committedInstType::SimdShaSigma2 0 0.00% 81.11% # Class of committed instruction. (Count)

board.processor.cores.core.commitStats0.committedInstType::SimdShaSigma3 0 0.00% 81.11% # Class of committed instruction. (Count)

board.processor.cores.core.commitStats0.committedInstType::SimdPredAlu 0 0.00% 81.11% # Class of committed instruction. (Count)

board.processor.cores.core.commitStats0.committedInstType::Matrix 0 0.00% 81.11% # Class of committed instruction. (Count)

board.processor.cores.core.commitStats0.committedInstType::MatrixMov 0 0.00% 81.11% # Class of committed instruction. (Count)

board.processor.cores.core.commitStats0.committedInstType::MatrixOP 0 0.00% 81.11% # Class of committed instruction. (Count)

board.processor.cores.core.commitStats0.committedInstType::MemRead 629 7.98% 89.10% # Class of committed instruction. (Count)

board.processor.cores.core.commitStats0.committedInstType::MemWrite 573 7.27% 96.37% # Class of committed instruction. (Count)

board.processor.cores.core.commitStats0.committedInstType::FloatMemRead 13 0.17% 96.53% # Class of committed instruction. (Count)

board.processor.cores.core.commitStats0.committedInstType::FloatMemWrite 273 3.47% 100.00% # Class of committed instruction. (Count)

board.processor.cores.core.commitStats0.committedInstType::IprAccess 0 0.00% 100.00% # Class of committed instruction. (Count)

board.processor.cores.core.commitStats0.committedInstType::InstPrefetch 0 0.00% 100.00% # Class of committed instruction. (Count)

board.processor.cores.core.commitStats0.committedInstType::SimdUnitStrideLoad 0 0.00% 100.00% # Class of committed instruction. (Count)

board.processor.cores.core.commitStats0.committedInstType::SimdUnitStrideStore 0 0.00% 100.00% # Class of committed instruction. (Count)

board.processor.cores.core.commitStats0.committedInstType::SimdUnitStrideMaskLoad 0 0.00% 100.00% # Class of committed instruction. (Count)

board.processor.cores.core.commitStats0.committedInstType::SimdUnitStrideMaskStore 0 0.00% 100.00% # Class of committed instruction. (Count)

board.processor.cores.core.commitStats0.committedInstType::SimdStridedLoad 0 0.00% 100.00% # Class of committed instruction. (Count)

board.processor.cores.core.commitStats0.committedInstType::SimdStridedStore 0 0.00% 100.00% # Class of committed instruction. (Count)

board.processor.cores.core.commitStats0.committedInstType::SimdIndexedLoad 0 0.00% 100.00% # Class of committed instruction. (Count)

board.processor.cores.core.commitStats0.committedInstType::SimdIndexedStore 0 0.00% 100.00% # Class of committed instruction. (Count)

board.processor.cores.core.commitStats0.committedInstType::SimdWholeRegisterLoad 0 0.00% 100.00% # Class of committed instruction. (Count)

board.processor.cores.core.commitStats0.committedInstType::SimdWholeRegisterStore 0 0.00% 100.00% # Class of committed instruction. (Count)

board.processor.cores.core.commitStats0.committedInstType::SimdUnitStrideFaultOnlyFirstLoad 0 0.00% 100.00% # Class of committed instruction. (Count)

board.processor.cores.core.commitStats0.committedInstType::SimdUnitStrideSegmentedLoad 0 0.00% 100.00% # Class of committed instruction. (Count)

board.processor.cores.core.commitStats0.committedInstType::SimdUnitStrideSegmentedStore 0 0.00% 100.00% # Class of committed instruction. (Count)

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board.processor.cores.core.commitStats0.committedInstType::SimdExt 0 0.00% 100.00% # Class of committed instruction. (Count)
board.processor.cores.core.commitStats0.committedInstType::SimdFloatExt 0 0.00% 100.00% # Class of committed instruction. (Count)
board.processor.cores.core.commitStats0.committedInstType::SimdConfig 0 0.00% 100.00% # Class of committed instruction. (Count)
board.processor.cores.core.commitStats0.committedInstType::total 7878 # Class of committed instruction. (Count)
board.processor.cores.core.commitStats0.committedControl::IsControl 836 # Class of control type instructions committed (Count)
board.processor.cores.core.commitStats0.committedControl::IsDirectControl 710 # Class of control type instructions committed (Count)
board.processor.cores.core.commitStats0.committedControl::IsIndirectControl 126 # Class of control type instructions committed (Count)
board.processor.cores.core.commitStats0.committedControl::IsCondControl 596 # Class of control type instructions committed (Count)
board.processor.cores.core.commitStats0.committedControl::IsUncondControl 240 # Class of control type instructions committed (Count)
board.processor.cores.core.commitStats0.committedControl::IsCall 97 # Class of control type instructions committed (Count)
board.processor.cores.core.commitStats0.committedControl::IsReturn 91 # Class of control type instructions committed (Count)
board.processor.cores.core.exec_context.thread_0.numMatAluAccesses 0 # Number of matrix alu accesses (Count)
board.processor.cores.core.exec_context.thread_0.numCallsReturns 188 # Number of times a function call or return occured (Count)
board.processor.cores.core.exec_context.thread_0.numMatInsts 0 # Number of matrix instructions (Count)
board.processor.cores.core.exec_context.thread_0.numIdleCycles 0 # Number of idle cycles (Cycle)
board.processor.cores.core.exec_context.thread_0.numBusyCycles 72487 # Number of busy cycles (Cycle)
board.processor.cores.core.exec_context.thread_0.notIdleFraction 1 # Percentage of non-idle cycles (Ratio)
board.processor.cores.core.exec_context.thread_0.idleFraction 0 # Percentage of idle cycles (Ratio)
board.processor.cores.core.executeStats0.numInsts 0 # Number of executed instructions (Count)
board.processor.cores.core.executeStats0.numNop 0 # Number of nop insts executed (Count)
board.processor.cores.core.executeStats0.numBranches 0 # Number of branches executed (Count)
board.processor.cores.core.executeStats0.numLoadInsts 0 # Number of load instructions executed (Count)
board.processor.cores.core.executeStats0.numStoreInsts 1488 # Number of stores executed (Count)
board.processor.cores.core.executeStats0.instRate 0 # Inst execution rate ((Count/Cycle))
board.processor.cores.core.executeStats0.numFpAluAccesses 703 # Number of float alu accesses (Count)
board.processor.cores.core.executeStats0.numFpRegReads 703 # Number of times the floating registers were read (Count)
board.processor.cores.core.executeStats0.numFpRegWrites 428 # Number of times the floating registers were written (Count)
board.processor.cores.core.executeStats0.numIntAluAccesses 7438 # Number of integer alu accesses (Count)
board.processor.cores.core.executeStats0.numIntRegReads 8403 # Number of times the integer registers were read (Count)
board.processor.cores.core.executeStats0.numIntRegWrites 5237 # Number of times the integer registers were written (Count)
board.processor.cores.core.executeStats0.numMemRefs 1488 # Number of memory refs (Count)
board.processor.cores.core.executeStats0.numMiscRegReads 3307 # Number of times the Misc registers were read (Count)
board.processor.cores.core.executeStats0.numVecAluAccesses 0 # Number of vector alu accesses (Count)
board.processor.cores.core.executeStats0.numDiscardedOps 0 # Number of ops (including micro ops) which were discarded before commit (Count)
board.processor.cores.core.fetchStats0.numInsts 4150 # Number of instructions fetched (thread level) (Count)
board.processor.cores.core.fetchStats0.numOps 7879 # Number of ops (including micro ops) fetched (thread level) (Count)
board.processor.cores.core.fetchStats0.fetchRate 0.057252 # Number of inst fetches per cycle ((Count/Cycle))
board.processor.cores.core.fetchStats0.numBranches 836 # Number of branches fetched (Count)
board.processor.cores.core.fetchStats0.branchRate 0.011533 # Number of branch fetches per cycle (Ratio)
board.processor.cores.core.fetchStats0.numFetchSuspends 0 # Number of times Execute suspended instruction fetching (Count)
board.processor.cores.core.interrupts.clk_domain.clock 5328 # Clock period in ticks (Tick)
board.processor.cores.core.mmu.dtb.rdAccesses 642 # TLB accesses on read requests (Count)
board.processor.cores.core.mmu.dtb.wrAccesses 847 # TLB accesses on write requests (Count)
board.processor.cores.core.mmu.dtb.rdMisses 12 # TLB misses on read requests (Count)
board.processor.cores.core.mmu.dtb.wrMisses 6 # TLB misses on write requests (Count)
board.processor.cores.core.mmu.dtb.walker.power_state.pwrStateResidencyTicks::UNDEFINED 73837531900351248 # Cumulative time (in ticks) in various power states (Tick)
board.processor.cores.core.mmu.itb.rdAccesses 0 # TLB accesses on read requests (Count)
board.processor.cores.core.mmu.itb.wrAccesses 5345 # TLB accesses on write requests (Count)
board.processor.cores.core.mmu.itb.rdMisses 0 # TLB misses on read requests (Count)
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board.processor.cores.core.mmu.itb.wrMisses 22 # TLB misses on write requests (Count)

board.processor.cores.core.mmu.itb.walker.power_state.pwrStateResidencyTicks::UNDEFINED 73837531900351248 # Cumulative time (in ticks) in various power states (Tick)

board.processor.cores.core.power_state.pwrStateResidencyTicks::ON 73837531900351248 # Cumulative time (in ticks) in various power states (Tick)

board.processor.cores.core.thread_0.numInsts 0 # Number of Instructions committed (Count)

board.processor.cores.core.thread_0.numOps 0 # Number of Ops committed (Count)

board.processor.cores.core.thread_0.numMemRefs 0 # Number of Memory References (Count)

board.processor.cores.core.workload.numSyscalls 9 # Number of system calls (Count)

board.workload.inst.arm 0 # number of arm instructions executed (Count)

board.workload.inst.quiesce 0 # number of quiesce instructions executed (Count)

---------- End Simulation Statistics ----------

---------- Begin Simulation Statistics ----------

simSeconds 84926.990561 # Number of seconds simulated (Second)

simTicks 84926990560998752 # Number of ticks simulated (Tick)

finalTick 84926990560998752 # Number of ticks from beginning of simulation (restored from checkpoints and never reset) (Tick)

simFreq 1000000000000 # The number of ticks per simulated second ((Tick/Second))

hostSeconds 49769.97 # Real time elapsed on the host (Second)

hostTickRate 1706390089302 # The number of ticks simulated per host second (ticks/s) ((Tick/Second))

hostMemory 1198868 # Number of bytes of host memory used (Byte)

simInsts 4141 # Number of instructions simulated (Count)

simOps 7867 # Number of ops (including micro ops) simulated (Count)

hostInstRate 0 # Simulator instruction rate (inst/s) ((Count/Second))

hostOpRate 0 # Simulator op (including micro ops) rate (op/s) ((Count/Second))

board.cache_hierarchy.dptw_caches.blockedCycles::no_mshrs 0 # number of cycles access was blocked (Cycle)

board.cache_hierarchy.dptw_caches.blockedCycles::no_targets 0 # number of cycles access was blocked (Cycle)

board.cache_hierarchy.dptw_caches.blockedCauses::no_mshrs 0 # number of times access was blocked (Count)

board.cache_hierarchy.dptw_caches.blockedCauses::no_targets 0 # number of times access was blocked (Count)

board.cache_hierarchy.dptw_caches.avgBlocked::no_mshrs nan # average number of cycles each access was blocked ((Cycle/Count))

board.cache_hierarchy.dptw_caches.avgBlocked::no_targets nan # average number of cycles each access was blocked ((Cycle/Count))

board.cache_hierarchy.dptw_caches.replacements 0 # number of replacements (Count)

board.cache_hierarchy.dptw_caches.power_state.pwrStateResidencyTicks::UNDEFINED 84926990560998752 # Cumulative time (in ticks) in various power states (Tick)

board.cache_hierarchy.dptw_caches.tags.tagsInUse 0 # Average ticks per tags in use ((Tick/Count))

board.cache_hierarchy.dptw_caches.tags.totalRefs 0 # Total number of references to valid blocks. (Count)

board.cache_hierarchy.dptw_caches.tags.sampledRefs 0 # Sample count of references to valid blocks. (Count)

board.cache_hierarchy.dptw_caches.tags.avgRefs nan # Average number of references to valid blocks. ((Count/Count))

board.cache_hierarchy.dptw_caches.tags.warmupTick 0 # The tick when the warmup percentage was hit. (Tick)

board.cache_hierarchy.dptw_caches.tags.tagAccesses 0 # Number of tag accesses (Count)

board.cache_hierarchy.dptw_caches.tags.dataAccesses 0 # Number of data accesses (Count)

board.cache_hierarchy.dptw_caches.tags.power_state.pwrStateResidencyTicks::UNDEFINED 84926990560998752 # Cumulative time (in ticks) in various power states (Tick)

board.cache_hierarchy.iptw_caches.blockedCycles::no_mshrs 0 # number of cycles access was blocked (Cycle)

board.cache_hierarchy.iptw_caches.blockedCycles::no_targets 0 # number of cycles access was blocked (Cycle)

board.cache_hierarchy.iptw_caches.blockedCauses::no_mshrs 0 # number of times access was blocked (Count)

board.cache_hierarchy.iptw_caches.blockedCauses::no_targets 0 # number of times access was blocked (Count)

board.cache_hierarchy.iptw_caches.avgBlocked::no_mshrs nan # average number of cycles each access was blocked ((Cycle/Count))

board.cache_hierarchy.iptw_caches.avgBlocked::no_targets nan # average number of cycles each access was blocked ((Cycle/Count))

board.cache_hierarchy.iptw_caches.replacements 0 # number of replacements (Count)

board.cache_hierarchy.iptw_caches.power_state.pwrStateResidencyTicks::UNDEFINED 84926990560998752 # Cumulative time (in ticks) in various power states (Tick)

board.cache_hierarchy.iptw_caches.tags.tagsInUse 0 # Average ticks per tags in use ((Tick/Count))

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board.cache_hierarchy.iptw_caches.tags.totalRefs 0 # Total number of references to valid blocks. (Count)
board.cache_hierarchy.iptw_caches.tags.sampledRefs 0 # Sample count of references to valid blocks. (Count)
board.cache_hierarchy.iptw_caches.tags.avgRefs nan # Average number of references to valid blocks. ((Count/Count))
board.cache_hierarchy.iptw_caches.tags.warmupTick 0 # The tick when the warmup percentage was hit. (Tick)
board.cache_hierarchy.iptw_caches.tags.tagAccesses 0 # Number of tag accesses (Count)
board.cache_hierarchy.iptw_caches.tags.dataAccesses 0 # Number of data accesses (Count)
board.cache_hierarchy.iptw_caches.tags.power_state.pwrStateResidencyTicks::UNDEFINED 84926990560998752 # Cumulative time (in ticks) in various power states (Tick)
board.cache_hierarchy.l1dcaches.demandHits::processor.cores.core.data 1300 # number of demand (read+write) hits (Count)
board.cache_hierarchy.l1dcaches.demandHits::total 1300 # number of demand (read+write) hits (Count)
board.cache_hierarchy.l1dcaches.overallHits::processor.cores.core.data 1300 # number of overall hits (Count)
board.cache_hierarchy.l1dcaches.overallHits::total 1300 # number of overall hits (Count)
board.cache_hierarchy.l1dcaches.demandMisses::processor.cores.core.data 185 # number of demand (read+write) misses (Count)
board.cache_hierarchy.l1dcaches.demandMisses::total 185 # number of demand (read+write) misses (Count)
board.cache_hierarchy.l1dcaches.overallMisses::processor.cores.core.data 185 # number of overall misses (Count)
board.cache_hierarchy.l1dcaches.overallMisses::total 185 # number of overall misses (Count)
board.cache_hierarchy.l1dcaches.demandMissLatency::processor.cores.core.data 9377946 # number of demand (read+write) miss ticks (Tick)
board.cache_hierarchy.l1dcaches.demandMissLatency::total 9377946 # number of demand (read+write) miss ticks (Tick)
board.cache_hierarchy.l1dcaches.overallMissLatency::processor.cores.core.data 9377946 # number of overall miss ticks (Tick)
board.cache_hierarchy.l1dcaches.overallMissLatency::total 9377946 # number of overall miss ticks (Tick)
board.cache_hierarchy.l1dcaches.demandAccesses::processor.cores.core.data 1485 # number of demand (read+write) accesses (Count)
board.cache_hierarchy.l1dcaches.demandAccesses::total 1485 # number of demand (read+write) accesses (Count)
board.cache_hierarchy.l1dcaches.overallAccesses::processor.cores.core.data 1485 # number of overall (read+write) accesses (Count)
board.cache_hierarchy.l1dcaches.overallAccesses::total 1485 # number of overall (read+write) accesses (Count)
board.cache_hierarchy.l1dcaches.demandMissRate::processor.cores.core.data 0.124579 # miss rate for demand accesses (Ratio)
board.cache_hierarchy.l1dcaches.demandMissRate::total 0.124579 # miss rate for demand accesses (Ratio)
board.cache_hierarchy.l1dcaches.overallMissRate::processor.cores.core.data 0.124579 # miss rate for overall accesses (Ratio)
board.cache_hierarchy.l1dcaches.overallMissRate::total 0.124579 # miss rate for overall accesses (Ratio)
board.cache_hierarchy.l1dcaches.demandAvgMissLatency::processor.cores.core.data 50691.600000 # average overall miss latency in ticks ((Tick/Count))
board.cache_hierarchy.l1dcaches.demandAvgMissLatency::total 50691.600000 # average overall miss latency in ticks ((Tick/Count))
board.cache_hierarchy.l1dcaches.overallAvgMissLatency::processor.cores.core.data 50691.600000 # average overall miss latency ((Tick/Count))
board.cache_hierarchy.l1dcaches.overallAvgMissLatency::total 50691.600000 # average overall miss latency ((Tick/Count))
board.cache_hierarchy.l1dcaches.blockedCycles::no_mshrs 27977 # number of cycles access was blocked (Cycle)
board.cache_hierarchy.l1dcaches.blockedCycles::no_targets 0 # number of cycles access was blocked (Cycle)
board.cache_hierarchy.l1dcaches.blockedCauses::no_mshrs 186 # number of times access was blocked (Count)
board.cache_hierarchy.l1dcaches.blockedCauses::no_targets 0 # number of times access was blocked (Count)
board.cache_hierarchy.l1dcaches.avgBlocked::no_mshrs 150.413978 # average number of cycles each access was blocked ((Cycle/Count))
board.cache_hierarchy.l1dcaches.avgBlocked::no_targets nan # average number of cycles each access was blocked ((Cycle/Count))
board.cache_hierarchy.l1dcaches.demandMshrMisses::processor.cores.core.data 185 # number of demand (read+write) MSHR misses (Count)
board.cache_hierarchy.l1dcaches.demandMshrMisses::total 185 # number of demand (read+write) MSHR misses (Count)
board.cache_hierarchy.l1dcaches.overallMshrMisses::processor.cores.core.data 185 # number of overall MSHR misses (Count)
board.cache_hierarchy.l1dcaches.overallMshrMisses::total 185 # number of overall MSHR misses (Count)
board.cache_hierarchy.l1dcaches.demandMshrMissLatency::processor.cores.core.data 9316341 # number of demand (read+write) MSHR miss ticks (Tick)
board.cache_hierarchy.l1dcaches.demandMshrMissLatency::total 9316341 # number of demand (read+write) MSHR miss ticks (Tick)
board.cache_hierarchy.l1dcaches.overallMshrMissLatency::processor.cores.core.data 9316341 # number of overall MSHR miss ticks (Tick)
board.cache_hierarchy.l1dcaches.overallMshrMissLatency::total 9316341 # number of overall MSHR miss ticks (Tick)
board.cache_hierarchy.l1dcaches.demandMshrMissRate::processor.cores.core.data 0.124579 # mshr miss ratio for demand accesses (Ratio)
board.cache_hierarchy.l1dcaches.demandMshrMissRate::total 0.124579 # mshr miss ratio for demand accesses (Ratio)
board.cache_hierarchy.l1dcaches.overallMshrMissRate::processor.cores.core.data 0.124579 # mshr miss ratio for overall accesses (Ratio)
board.cache_hierarchy.l1dcaches.overallMshrMissRate::total 0.124579 # mshr miss ratio for overall accesses (Ratio)
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board.cache_hierarchy.l1dcaches.demandAvgMshrMissLatency::processor.cores.core.data 50358.600000 # average overall mshr miss latency ((Tick/Count))
board.cache_hierarchy.l1dcaches.demandAvgMshrMissLatency::total 50358.600000 # average overall mshr miss latency ((Tick/Count))
board.cache_hierarchy.l1dcaches.overallAvgMshrMissLatency::processor.cores.core.data 50358.600000 # average overall mshr miss latency ((Tick/Count))
board.cache_hierarchy.l1dcaches.overallAvgMshrMissLatency::total 50358.600000 # average overall mshr miss latency ((Tick/Count))
board.cache_hierarchy.l1dcaches.replacements 0 # number of replacements (Count)
board.cache_hierarchy.l1dcaches.LockedRMWReadReq.hits::processor.cores.core.data 1 # number of LockedRMWReadReq hits (Count)
board.cache_hierarchy.l1dcaches.LockedRMWReadReq.hits::total 1 # number of LockedRMWReadReq hits (Count)
board.cache_hierarchy.l1dcaches.LockedRMWReadReq.accesses::processor.cores.core.data 1 # number of LockedRMWReadReq accesses(hits+misses) (Count)
board.cache_hierarchy.l1dcaches.LockedRMWReadReq.accesses::total 1 # number of LockedRMWReadReq accesses(hits+misses) (Count)
board.cache_hierarchy.l1dcaches.ReadReq.hits::processor.cores.core.data 541 # number of ReadReq hits (Count)
board.cache_hierarchy.l1dcaches.ReadReq.hits::total 541 # number of ReadReq hits (Count)
board.cache_hierarchy.l1dcaches.ReadReq.misses::processor.cores.core.data 99 # number of ReadReq misses (Count)
board.cache_hierarchy.l1dcaches.ReadReq.misses::total 99 # number of ReadReq misses (Count)
board.cache_hierarchy.l1dcaches.ReadReq.missLatency::processor.cores.core.data 5741586 # number of ReadReq miss ticks (Tick)
board.cache_hierarchy.l1dcaches.ReadReq.missLatency::total 5741586 # number of ReadReq miss ticks (Tick)
board.cache_hierarchy.l1dcaches.ReadReq.accesses::processor.cores.core.data 640 # number of ReadReq accesses(hits+misses) (Count)
board.cache_hierarchy.l1dcaches.ReadReq.accesses::total 640 # number of ReadReq accesses(hits+misses) (Count)
board.cache_hierarchy.l1dcaches.ReadReq.missRate::processor.cores.core.data 0.154688 # miss rate for ReadReq accesses (Ratio)
board.cache_hierarchy.l1dcaches.ReadReq.missRate::total 0.154688 # miss rate for ReadReq accesses (Ratio)
board.cache_hierarchy.l1dcaches.ReadReq.avgMissLatency::processor.cores.core.data 57995.818182 # average ReadReq miss latency ((Tick/Count))
board.cache_hierarchy.l1dcaches.ReadReq.avgMissLatency::total 57995.818182 # average ReadReq miss latency ((Tick/Count))
board.cache_hierarchy.l1dcaches.ReadReq.mshrMisses::processor.cores.core.data 99 # number of ReadReq MSHR misses (Count)
board.cache_hierarchy.l1dcaches.ReadReq.mshrMisses::total 99 # number of ReadReq MSHR misses (Count)
board.cache_hierarchy.l1dcaches.ReadReq.mshrMissLatency::processor.cores.core.data 5708619 # number of ReadReq MSHR miss ticks (Tick)
board.cache_hierarchy.l1dcaches.ReadReq.mshrMissLatency::total 5708619 # number of ReadReq MSHR miss ticks (Tick)
board.cache_hierarchy.l1dcaches.ReadReq.mshrMissRate::processor.cores.core.data 0.154688 # mshr miss rate for ReadReq accesses (Ratio)
board.cache_hierarchy.l1dcaches.ReadReq.mshrMissRate::total 0.154688 # mshr miss rate for ReadReq accesses (Ratio)
board.cache_hierarchy.l1dcaches.ReadReq.avgMshrMissLatency::processor.cores.core.data 57662.818182 # average ReadReq mshr miss latency ((Tick/Count))
board.cache_hierarchy.l1dcaches.ReadReq.avgMshrMissLatency::total 57662.818182 # average ReadReq mshr miss latency ((Tick/Count))
board.cache_hierarchy.l1dcaches.WriteReq.hits::processor.cores.core.data 759 # number of WriteReq hits (Count)
board.cache_hierarchy.l1dcaches.WriteReq.hits::total 759 # number of WriteReq hits (Count)
board.cache_hierarchy.l1dcaches.WriteReq.misses::processor.cores.core.data 86 # number of WriteReq misses (Count)
board.cache_hierarchy.l1dcaches.WriteReq.misses::total 86 # number of WriteReq misses (Count)
board.cache_hierarchy.l1dcaches.WriteReq.missLatency::processor.cores.core.data 3636360 # number of WriteReq miss ticks (Tick)
board.cache_hierarchy.l1dcaches.WriteReq.missLatency::total 3636360 # number of WriteReq miss ticks (Tick)
board.cache_hierarchy.l1dcaches.WriteReq.accesses::processor.cores.core.data 845 # number of WriteReq accesses(hits+misses) (Count)
board.cache_hierarchy.l1dcaches.WriteReq.accesses::total 845 # number of WriteReq accesses(hits+misses) (Count)
board.cache_hierarchy.l1dcaches.WriteReq.missRate::processor.cores.core.data 0.101775 # miss rate for WriteReq accesses (Ratio)
board.cache_hierarchy.l1dcaches.WriteReq.missRate::total 0.101775 # miss rate for WriteReq accesses (Ratio)
board.cache_hierarchy.l1dcaches.WriteReq.avgMissLatency::processor.cores.core.data 42283.255814 # average WriteReq miss latency ((Tick/Count))
board.cache_hierarchy.l1dcaches.WriteReq.avgMissLatency::total 42283.255814 # average WriteReq miss latency ((Tick/Count))
board.cache_hierarchy.l1dcaches.WriteReq.mshrMisses::processor.cores.core.data 86 # number of WriteReq MSHR misses (Count)
board.cache_hierarchy.l1dcaches.WriteReq.mshrMisses::total 86 # number of WriteReq MSHR misses (Count)
board.cache_hierarchy.l1dcaches.WriteReq.mshrMissLatency::processor.cores.core.data 3607722 # number of WriteReq MSHR miss ticks (Tick)
board.cache_hierarchy.l1dcaches.WriteReq.mshrMissLatency::total 3607722 # number of WriteReq MSHR miss ticks (Tick)
board.cache_hierarchy.l1dcaches.WriteReq.mshrMissRate::processor.cores.core.data 0.101775 # mshr miss rate for WriteReq accesses (Ratio)
board.cache_hierarchy.l1dcaches.WriteReq.mshrMissRate::total 0.101775 # mshr miss rate for WriteReq accesses (Ratio)
board.cache_hierarchy.l1dcaches.WriteReq.avgMshrMissLatency::processor.cores.core.data 41950.255814 # average WriteReq mshr miss latency ((Tick/Count))
board.cache_hierarchy.l1dcaches.WriteReq.avgMshrMissLatency::total 41950.255814 # average WriteReq mshr miss latency ((Tick/Count))
```

board.cache_hierarchy.l1dcaches.power_state.pwrStateResidencyTicks::UNDEFINED 84926990560998752 # Cumulative time (in ticks) in various power states (Tick)

board.cache_hierarchy.l1dcaches.prefetcher.demandMshrMisses 185 # demands not covered by prefetchs (Count)

board.cache_hierarchy.l1dcaches.prefetcher.pfIssued 0 # number of hwpf issued (Count)

board.cache_hierarchy.l1dcaches.prefetcher.pfUseful 0 # number of useful prefetch (Count)

board.cache_hierarchy.l1dcaches.prefetcher.pfUsefulButMiss 0 # number of hit on prefetch but cache block is not in an usable state (Count)

board.cache_hierarchy.l1dcaches.prefetcher.accuracy nan # accuracy of the prefetcher (Count)

board.cache_hierarchy.l1dcaches.prefetcher.coverage 0 # coverage brought by this prefetcher (Count)

board.cache_hierarchy.l1dcaches.prefetcher.pfHitInCache 0 # number of prefetches hitting in cache (Count)

board.cache_hierarchy.l1dcaches.prefetcher.pfHitInMSHR 0 # number of prefetches hitting in a MSHR (Count)

board.cache_hierarchy.l1dcaches.prefetcher.pfHitInWB 0 # number of prefetches hit in the Write Buffer (Count)

board.cache_hierarchy.l1dcaches.prefetcher.pfLate 0 # number of late prefetches (hitting in cache, MSHR or WB) (Count)

board.cache_hierarchy.l1dcaches.prefetcher.pfIdentified 146 # number of prefetch candidates identified (Count)

board.cache_hierarchy.l1dcaches.prefetcher.pfBufferHit 63 # number of redundant prefetches already in prefetch queue (Count)

board.cache_hierarchy.l1dcaches.prefetcher.pfInCache 0 # number of redundant prefetches already in cache/mshr dropped (Count)

board.cache_hierarchy.l1dcaches.prefetcher.pfRemovedDemand 35 # number of prefetches dropped due to a demand for the same address (Count)

board.cache_hierarchy.l1dcaches.prefetcher.pfRemovedFull 0 # number of prefetches dropped due to prefetch queue size (Count)

board.cache_hierarchy.l1dcaches.prefetcher.pfSpanPage 6 # number of prefetches that crossed the page (Count)

board.cache_hierarchy.l1dcaches.prefetcher.pfUsefulSpanPage 0 # number of prefetches that is useful and crossed the page (Count)

board.cache_hierarchy.l1dcaches.prefetcher.power_state.pwrStateResidencyTicks::UNDEFINED 84926990560998752 # Cumulative time (in ticks) in various power states (Tick)

board.cache_hierarchy.l1dcaches.tags.tagsInUse 185.000000 # Average ticks per tags in use ((Tick/Count))

board.cache_hierarchy.l1dcaches.tags.totalRefs 1486 # Total number of references to valid blocks. (Count)

board.cache_hierarchy.l1dcaches.tags.sampledRefs 185 # Sample count of references to valid blocks. (Count)

board.cache_hierarchy.l1dcaches.tags.avgRefs 8.032432 # Average number of references to valid blocks. ((Count/Count))

board.cache_hierarchy.l1dcaches.tags.warmupTick 149184 # The tick when the warmup percentage was hit. (Tick)

board.cache_hierarchy.l1dcaches.tags.occupancies::processor.cores.core.data 185.000000 # Average occupied blocks per tick, per requestor ((Count/Tick))

board.cache_hierarchy.l1dcaches.tags.avgOccs::processor.cores.core.data 0.361328 # Average percentage of cache occupancy ((Ratio/Tick))

board.cache_hierarchy.l1dcaches.tags.avgOccs::total 0.361328 # Average percentage of cache occupancy ((Ratio/Tick))

board.cache_hierarchy.l1dcaches.tags.occupanciesTaskId::1024 185 # Occupied blocks per task id (Count)

board.cache_hierarchy.l1dcaches.tags.ageTaskId_1024::4 185 # Occupied blocks per task id, per block age (Count)

board.cache_hierarchy.l1dcaches.tags.ratioOccsTaskId::1024 0.361328 # Ratio of occupied blocks and all blocks, per task id (Ratio)

board.cache_hierarchy.l1dcaches.tags.tagAccesses 12073 # Number of tag accesses (Count)

board.cache_hierarchy.l1dcaches.tags.dataAccesses 12073 # Number of data accesses (Count)

board.cache_hierarchy.l1dcaches.tags.power_state.pwrStateResidencyTicks::UNDEFINED 84926990560998752 # Cumulative time (in ticks) in various power states (Tick)

board.cache_hierarchy.l1icaches.demandHits::processor.cores.core.inst 5149 # number of demand (read+write) hits (Count)

board.cache_hierarchy.l1icaches.demandHits::total 5149 # number of demand (read+write) hits (Count)

board.cache_hierarchy.l1icaches.overallHits::processor.cores.core.inst 5149 # number of overall hits (Count)

board.cache_hierarchy.l1icaches.overallHits::total 5149 # number of overall hits (Count)

board.cache_hierarchy.l1icaches.demandMisses::processor.cores.core.inst 196 # number of demand (read+write) misses (Count)

board.cache_hierarchy.l1icaches.demandMisses::total 196 # number of demand (read+write) misses (Count)

board.cache_hierarchy.l1icaches.overallMisses::processor.cores.core.inst 196 # number of overall misses (Count)

board.cache_hierarchy.l1icaches.overallMisses::total 196 # number of overall misses (Count)

board.cache_hierarchy.l1icaches.demandMissLatency::processor.cores.core.inst 12525129 # number of demand (read+write) miss ticks (Tick)

board.cache_hierarchy.l1icaches.demandMissLatency::total 12525129 # number of demand (read+write) miss ticks (Tick)

board.cache_hierarchy.l1icaches.overallMissLatency::processor.cores.core.inst 12525129 # number of overall miss ticks (Tick)

board.cache_hierarchy.l1icaches.overallMissLatency::total 12525129 # number of overall miss ticks (Tick)

board.cache_hierarchy.l1icaches.demandAccesses::processor.cores.core.inst 5345 # number of demand (read+write) accesses (Count)

board.cache_hierarchy.l1icaches.demandAccesses::total 5345 # number of demand (read+write) accesses (Count)

board.cache_hierarchy.l1icaches.overallAccesses::processor.cores.core.inst 5345 # number of overall (read+write) accesses (Count)

board.cache_hierarchy.l1icaches.overallAccesses::total 5345 # number of overall (read+write) accesses (Count)

board.cache_hierarchy.l1icaches.demandMissRate::processor.cores.core.inst 0.036670 # miss rate for demand accesses (Ratio)

board.cache_hierarchy.l1icaches.demandMissRate::total 0.036670 # miss rate for demand accesses (Ratio)

board.cache_hierarchy.l1icaches.overallMissRate::processor.cores.core.inst 0.036670 # miss rate for overall accesses (Ratio)

board.cache_hierarchy.l1icaches.overallMissRate::total 0.036670 # miss rate for overall accesses (Ratio)

board.cache_hierarchy.l1icaches.demandAvgMissLatency::processor.cores.core.inst 63903.719388 # average overall miss latency in ticks ((Tick/Count))

board.cache_hierarchy.l1icaches.demandAvgMissLatency::total 63903.719388 # average overall miss latency in ticks ((Tick/Count))

board.cache_hierarchy.l1icaches.overallAvgMissLatency::processor.cores.core.inst 63903.719388 # average overall miss latency ((Tick/Count))

board.cache_hierarchy.l1icaches.overallAvgMissLatency::total 63903.719388 # average overall miss latency ((Tick/Count))

board.cache_hierarchy.l1icaches.blockedCycles::no_mshrs 0 # number of cycles access was blocked (Cycle)

board.cache_hierarchy.l1icaches.blockedCycles::no_targets 0 # number of cycles access was blocked (Cycle)

board.cache_hierarchy.l1icaches.blockedCauses::no_mshrs 0 # number of times access was blocked (Count)

board.cache_hierarchy.l1icaches.blockedCauses::no_targets 0 # number of times access was blocked (Count)

board.cache_hierarchy.l1icaches.avgBlocked::no_mshrs nan # average number of cycles each access was blocked ((Cycle/Count))

board.cache_hierarchy.l1icaches.avgBlocked::no_targets nan # average number of cycles each access was blocked ((Cycle/Count))

board.cache_hierarchy.l1icaches.demandMshrMisses::processor.cores.core.inst 196 # number of demand (read+write) MSHR misses (Count)

board.cache_hierarchy.l1icaches.demandMshrMisses::total 196 # number of demand (read+write) MSHR misses (Count)

board.cache_hierarchy.l1icaches.overallMshrMisses::processor.cores.core.inst 196 # number of overall MSHR misses (Count)

board.cache_hierarchy.l1icaches.overallMshrMisses::total 196 # number of overall MSHR misses (Count)

board.cache_hierarchy.l1icaches.demandMshrMissLatency::processor.cores.core.inst 12459861 # number of demand (read+write) MSHR miss ticks (Tick)

board.cache_hierarchy.l1icaches.demandMshrMissLatency::total 12459861 # number of demand (read+write) MSHR miss ticks (Tick)

board.cache_hierarchy.l1icaches.overallMshrMissLatency::processor.cores.core.inst 12459861 # number of overall MSHR miss ticks (Tick)

board.cache_hierarchy.l1icaches.overallMshrMissLatency::total 12459861 # number of overall MSHR miss ticks (Tick)

board.cache_hierarchy.l1icaches.demandMshrMissRate::processor.cores.core.inst 0.036670 # mshr miss ratio for demand accesses (Ratio)

board.cache_hierarchy.l1icaches.demandMshrMissRate::total 0.036670 # mshr miss ratio for demand accesses (Ratio)

board.cache_hierarchy.l1icaches.overallMshrMissRate::processor.cores.core.inst 0.036670 # mshr miss ratio for overall accesses (Ratio)

board.cache_hierarchy.l1icaches.overallMshrMissRate::total 0.036670 # mshr miss ratio for overall accesses (Ratio)

board.cache_hierarchy.l1icaches.demandAvgMshrMissLatency::processor.cores.core.inst 63570.719388 # average overall mshr miss latency ((Tick/Count))

board.cache_hierarchy.l1icaches.demandAvgMshrMissLatency::total 63570.719388 # average overall mshr miss latency ((Tick/Count))

board.cache_hierarchy.l1icaches.overallAvgMshrMissLatency::processor.cores.core.inst 63570.719388 # average overall mshr miss latency ((Tick/Count))

board.cache_hierarchy.l1icaches.overallAvgMshrMissLatency::total 63570.719388 # average overall mshr miss latency ((Tick/Count))

board.cache_hierarchy.l1icaches.replacements 0 # number of replacements (Count)

board.cache_hierarchy.l1icaches.ReadReq.hits::processor.cores.core.inst 5149 # number of ReadReq hits (Count)

board.cache_hierarchy.l1icaches.ReadReq.hits::total 5149 # number of ReadReq hits (Count)

board.cache_hierarchy.l1icaches.ReadReq.misses::processor.cores.core.inst 196 # number of ReadReq misses (Count)

board.cache_hierarchy.l1icaches.ReadReq.misses::total 196 # number of ReadReq misses (Count)

board.cache_hierarchy.l1icaches.ReadReq.missLatency::processor.cores.core.inst 12525129 # number of ReadReq miss ticks (Tick)

board.cache_hierarchy.l1icaches.ReadReq.missLatency::total 12525129 # number of ReadReq miss ticks (Tick)

board.cache_hierarchy.l1icaches.ReadReq.accesses::processor.cores.core.inst 5345 # number of ReadReq accesses(hits+misses) (Count)

board.cache_hierarchy.l1icaches.ReadReq.accesses::total 5345 # number of ReadReq accesses(hits+misses) (Count)

board.cache_hierarchy.l1icaches.ReadReq.missRate::processor.cores.core.inst 0.036670 # miss rate for ReadReq accesses (Ratio)

board.cache_hierarchy.l1icaches.ReadReq.missRate::total 0.036670 # miss rate for ReadReq accesses (Ratio)

board.cache_hierarchy.l1icaches.ReadReq.avgMissLatency::processor.cores.core.inst 63903.719388 # average ReadReq miss latency ((Tick/Count))

board.cache_hierarchy.l1icaches.ReadReq.avgMissLatency::total 63903.719388 # average ReadReq miss latency ((Tick/Count))

board.cache_hierarchy.l1icaches.ReadReq.mshrMisses::processor.cores.core.inst 196 # number of ReadReq MSHR misses (Count)

board.cache_hierarchy.l1icaches.ReadReq.mshrMisses::total 196 # number of ReadReq MSHR misses (Count)

board.cache_hierarchy.l1icaches.ReadReq.mshrMissLatency::processor.cores.core.inst 12459861 # number of ReadReq MSHR miss ticks (Tick)

board.cache_hierarchy.l1icaches.ReadReq.mshrMissLatency::total 12459861 # number of ReadReq MSHR miss ticks (Tick)

board.cache_hierarchy.l1icaches.ReadReq.mshrMissRate::processor.cores.core.inst 0.036670 # mshr miss rate for ReadReq accesses (Ratio)

board.cache_hierarchy.l1icaches.ReadReq.mshrMissRate::total 0.036670 # mshr miss rate for ReadReq accesses (Ratio)

board.cache_hierarchy.l1icaches.ReadReq.avgMshrMissLatency::processor.cores.core.inst 63570.719388 # average ReadReq mshr miss latency ((Tick/Count))

board.cache_hierarchy.l1icaches.ReadReq.avgMshrMissLatency::total 63570.719388 # average ReadReq mshr miss latency ((Tick/Count))

board.cache_hierarchy.l1icaches.power_state.pwrStateResidencyTicks::UNDEFINED 84926990560998752 # Cumulative time (in ticks) in various power states (Tick)

board.cache_hierarchy.l1icaches.prefetcher.demandMshrMisses 196 # demands not covered by prefetchs (Count)

board.cache_hierarchy.l1icaches.prefetcher.pfIssued 0 # number of hwpf issued (Count)

board.cache_hierarchy.l1icaches.prefetcher.pfUseful 0 # number of useful prefetch (Count)

board.cache_hierarchy.l1icaches.prefetcher.pfUsefulButMiss 0 # number of hit on prefetch but cache block is not in an usable state (Count)

board.cache_hierarchy.l1icaches.prefetcher.accuracy nan # accuracy of the prefetcher (Count)

board.cache_hierarchy.l1icaches.prefetcher.coverage 0 # coverage brought by this prefetcher (Count)

board.cache_hierarchy.l1icaches.prefetcher.pfHitInCache 0 # number of prefetches hitting in cache (Count)

board.cache_hierarchy.l1icaches.prefetcher.pfHitInMSHR 0 # number of prefetches hitting in a MSHR (Count)

board.cache_hierarchy.l1icaches.prefetcher.pfHitInWB 0 # number of prefetches hit in the Write Buffer (Count)

board.cache_hierarchy.l1icaches.prefetcher.pfLate 0 # number of late prefetches (hitting in cache, MSHR or WB) (Count)

board.cache_hierarchy.l1icaches.prefetcher.pfIdentified 0 # number of prefetch candidates identified (Count)

board.cache_hierarchy.l1icaches.prefetcher.pfBufferHit 0 # number of redundant prefetches already in prefetch queue (Count)

board.cache_hierarchy.l1icaches.prefetcher.pfInCache 0 # number of redundant prefetches already in cache/mshr dropped (Count)

board.cache_hierarchy.l1icaches.prefetcher.pfRemovedDemand 0 # number of prefetches dropped due to a demand for the same address (Count)

board.cache_hierarchy.l1icaches.prefetcher.pfRemovedFull 0 # number of prefetches dropped due to prefetch queue size (Count)

board.cache_hierarchy.l1icaches.prefetcher.pfSpanPage 0 # number of prefetches that crossed the page (Count)

board.cache_hierarchy.l1icaches.prefetcher.pfUsefulSpanPage 0 # number of prefetches that is useful and crossed the page (Count)

board.cache_hierarchy.l1icaches.prefetcher.power_state.pwrStateResidencyTicks::UNDEFINED 84926990560998752 # Cumulative time (in ticks) in various power states (Tick)

board.cache_hierarchy.l1icaches.tags.tagsInUse 196.000000 # Average ticks per tags in use ((Tick/Count))

board.cache_hierarchy.l1icaches.tags.totalRefs 5345 # Total number of references to valid blocks. (Count)

board.cache_hierarchy.l1icaches.tags.sampledRefs 196 # Sample count of references to valid blocks. (Count)

board.cache_hierarchy.l1icaches.tags.avgRefs 27.270408 # Average number of references to valid blocks. ((Count/Count))

board.cache_hierarchy.l1icaches.tags.warmupTick 76590 # The tick when the warmup percentage was hit. (Tick)

board.cache_hierarchy.l1icaches.tags.occupancies::processor.cores.core.inst 196.000000 # Average occupied blocks per tick, per requestor ((Count/Tick))

board.cache_hierarchy.l1icaches.tags.avgOccs::processor.cores.core.inst 0.382812 # Average percentage of cache occupancy ((Ratio/Tick))

board.cache_hierarchy.l1icaches.tags.avgOccs::total 0.382812 # Average percentage of cache occupancy ((Ratio/Tick))

board.cache_hierarchy.l1icaches.tags.occupanciesTaskId::1024 196 # Occupied blocks per task id (Count)

board.cache_hierarchy.l1icaches.tags.ageTaskId_1024::4 196 # Occupied blocks per task id, per block age (Count)

board.cache_hierarchy.l1icaches.tags.ratioOccsTaskId::1024 0.382812 # Ratio of occupied blocks and all blocks, per task id (Ratio)

board.cache_hierarchy.l1icaches.tags.tagAccesses 42956 # Number of tag accesses (Count)

board.cache_hierarchy.l1icaches.tags.dataAccesses 42956 # Number of data accesses (Count)

board.cache_hierarchy.l1icaches.tags.power_state.pwrStateResidencyTicks::UNDEFINED 84926990560998752 # Cumulative time (in ticks) in various power states (Tick)

board.cache_hierarchy.l2bus.transDist::ReadResp 295 # Transaction distribution (Count)

board.cache_hierarchy.l2bus.transDist::HardPFReq 57 # Transaction distribution (Count)

board.cache_hierarchy.l2bus.transDist::ReadExReq 86 # Transaction distribution (Count)

board.cache_hierarchy.l2bus.transDist::ReadExResp 86 # Transaction distribution (Count)

board.cache_hierarchy.l2bus.transDist::ReadSharedReq 295 # Transaction distribution (Count)

board.cache_hierarchy.l2bus.pktCount_board.cache_hierarchy.l1icaches.mem_side_port::board.cache_hierarchy.l2cache.cpu_side_port 392 # Packet count per connected requestor and responder (Count)

board.cache_hierarchy.l2bus.pktCount_board.cache_hierarchy.l1dcaches.mem_side_port::board.cache_hierarchy.l2cache.cpu_side_port 370 # Packet count per connected requestor and responder (Count)

board.cache_hierarchy.l2bus.pktCount::total 762 # Packet count per connected requestor and responder (Count)

board.cache_hierarchy.l2bus.pktSize_board.cache_hierarchy.l1icaches.mem_side_port::board.cache_hierarchy.l2cache.cpu_side_port 12544 # Cumulative packet size per connected requestor and responder (Byte)

board.cache_hierarchy.l2bus.pktSize_board.cache_hierarchy.l1dcaches.mem_side_port::board.cache_hierarchy.l2cache.cpu_side_port 11840 # Cumulative packet size per connected requestor and responder (Byte)

board.cache_hierarchy.l2bus.pktSize::total 24384 # Cumulative packet size per connected requestor and responder (Byte)

board.cache_hierarchy.l2bus.snoops 57 # Total snoops (Count)

board.cache_hierarchy.l2bus.snoopTraffic 0 # Total snoop traffic (Byte)

board.cache_hierarchy.l2bus.snoopFanout::samples 438 # Request fanout histogram (Count)

board.cache_hierarchy.l2bus.snoopFanout::mean 0 # Request fanout histogram (Count)

board.cache_hierarchy.l2bus.snoopFanout::stdev 0 # Request fanout histogram (Count)

board.cache_hierarchy.l2bus.snoopFanout::underflows 0 0.00% 0.00% # Request fanout histogram (Count)

board.cache_hierarchy.l2bus.snoopFanout::0 438 100.00% 100.00% # Request fanout histogram (Count)

board.cache_hierarchy.l2bus.snoopFanout::1 0 0.00% 100.00% # Request fanout histogram (Count)

board.cache_hierarchy.l2bus.snoopFanout::2 0 0.00% 100.00% # Request fanout histogram (Count)

board.cache_hierarchy.l2bus.snoopFanout::3 0 0.00% 100.00% # Request fanout histogram (Count)

board.cache_hierarchy.l2bus.snoopFanout::4 0 0.00% 100.00% # Request fanout histogram (Count)

board.cache_hierarchy.l2bus.snoopFanout::overflows 0 0.00% 100.00% # Request fanout histogram (Count)

board.cache_hierarchy.l2bus.snoopFanout::min_value 0 # Request fanout histogram (Count)

board.cache_hierarchy.l2bus.snoopFanout::max_value 0 # Request fanout histogram (Count)

board.cache_hierarchy.l2bus.snoopFanout::total 438 # Request fanout histogram (Count)

board.cache_hierarchy.l2bus.power_state.pwrStateResidencyTicks::UNDEFINED 84926990560998752 # Cumulative time (in ticks) in various power states (Tick)

board.cache_hierarchy.l2bus.reqLayer0.occupancy 126873 # Layer occupancy (ticks) (Tick)

board.cache_hierarchy.l2bus.reqLayer0.utilization 0.0 # Layer utilization (Ratio)

board.cache_hierarchy.l2bus.respLayer0.occupancy 195804 # Layer occupancy (ticks) (Tick)

board.cache_hierarchy.l2bus.respLayer0.utilization 0.0 # Layer utilization (Ratio)

board.cache_hierarchy.l2bus.respLayer1.occupancy 184815 # Layer occupancy (ticks) (Tick)

board.cache_hierarchy.l2bus.respLayer1.utilization 0.0 # Layer utilization (Ratio)

board.cache_hierarchy.l2bus.snoop_filter.totRequests 381 # Total number of requests made to the snoop filter. (Count)

board.cache_hierarchy.l2bus.snoop_filter.hitSingleRequests 0 # Number of requests hitting in the snoop filter with a single holder of the requested data. (Count)

board.cache_hierarchy.l2bus.snoop_filter.hitMultiRequests 0 # Number of requests hitting in the snoop filter with multiple (>1) holders of the requested data. (Count)

board.cache_hierarchy.l2bus.snoop_filter.totSnoops 0 # Total number of snoops made to the snoop filter. (Count)

board.cache_hierarchy.l2bus.snoop_filter.hitSingleSnoops 0 # Number of snoops hitting in the snoop filter with a single holder of the requested data. (Count)

board.cache_hierarchy.l2bus.snoop_filter.hitMultiSnoops 0 # Number of snoops hitting in the snoop filter with multiple (>1) holders of the requested data. (Count)

board.cache_hierarchy.l2cache.demandHits::processor.cores.core.data 35 # number of demand (read+write) hits (Count)

board.cache_hierarchy.l2cache.demandHits::total 35 # number of demand (read+write) hits (Count)

board.cache_hierarchy.l2cache.overallHits::processor.cores.core.data 35 # number of overall hits (Count)

board.cache_hierarchy.l2cache.overallHits::total 35 # number of overall hits (Count)

board.cache_hierarchy.l2cache.demandMisses::processor.cores.core.inst 196 # number of demand (read+write) misses (Count)

board.cache_hierarchy.l2cache.demandMisses::processor.cores.core.data 150 # number of demand (read+write) misses (Count)

board.cache_hierarchy.l2cache.demandMisses::total 346 # number of demand (read+write) misses (Count)

board.cache_hierarchy.l2cache.overallMisses::processor.cores.core.inst 196 # number of overall misses (Count)

board.cache_hierarchy.l2cache.overallMisses::processor.cores.core.data 150 # number of overall misses (Count)

board.cache_hierarchy.l2cache.overallMisses::total 346 # number of overall misses (Count)

board.cache_hierarchy.l2cache.demandMissLatency::processor.cores.core.inst 12329325 # number of demand (read+write) miss ticks (Tick)

board.cache_hierarchy.l2cache.demandMissLatency::processor.cores.core.data 8948376 # number of demand (read+write) miss ticks (Tick)

board.cache_hierarchy.l2cache.demandMissLatency::total 21277701 # number of demand (read+write) miss ticks (Tick)

board.cache_hierarchy.l2cache.overallMissLatency::processor.cores.core.inst 12329325 # number of overall miss ticks (Tick)

board.cache_hierarchy.l2cache.overallMissLatency::processor.cores.core.data 8948376 # number of overall miss ticks (Tick)

board.cache_hierarchy.l2cache.overallMissLatency::total 21277701 # number of overall miss ticks (Tick)

board.cache_hierarchy.l2cache.demandAccesses::processor.cores.core.inst 196 # number of demand (read+write) accesses (Count)

board.cache_hierarchy.l2cache.demandAccesses::processor.cores.core.data 185 # number of demand (read+write) accesses (Count)

board.cache_hierarchy.l2cache.demandAccesses::total 381 # number of demand (read+write) accesses (Count)

board.cache_hierarchy.l2cache.overallAccesses::processor.cores.core.inst 196 # number of overall (read+write) accesses (Count)

board.cache_hierarchy.l2cache.overallAccesses::processor.cores.core.data 185 # number of overall (read+write) accesses (Count)

board.cache_hierarchy.l2cache.overallAccesses::total 381 # number of overall (read+write) accesses (Count)

board.cache_hierarchy.l2cache.demandMissRate::processor.cores.core.inst 1 # miss rate for demand accesses (Ratio)

board.cache_hierarchy.l2cache.demandMissRate::processor.cores.core.data 0.810811 # miss rate for demand accesses (Ratio)

board.cache_hierarchy.l2cache.demandMissRate::total 0.908136 # miss rate for demand accesses (Ratio)

board.cache_hierarchy.l2cache.overallMissRate::processor.cores.core.inst 1 # miss rate for overall accesses (Ratio)

board.cache_hierarchy.l2cache.overallMissRate::processor.cores.core.data 0.810811 # miss rate for overall accesses (Ratio)

board.cache_hierarchy.l2cache.overallMissRate::total 0.908136 # miss rate for overall accesses (Ratio)

board.cache_hierarchy.l2cache.demandAvgMissLatency::processor.cores.core.inst 62904.719388 # average overall miss latency in ticks ((Tick/Count))

board.cache_hierarchy.l2cache.demandAvgMissLatency::processor.cores.core.data 59655.840000 # average overall miss latency in ticks ((Tick/Count))

board.cache_hierarchy.l2cache.demandAvgMissLatency::total 61496.245665 # average overall miss latency in ticks ((Tick/Count))

board.cache_hierarchy.l2cache.overallAvgMissLatency::processor.cores.core.inst 62904.719388 # average overall miss latency ((Tick/Count))

board.cache_hierarchy.l2cache.overallAvgMissLatency::processor.cores.core.data 59655.840000 # average overall miss latency ((Tick/Count))

board.cache_hierarchy.l2cache.overallAvgMissLatency::total 61496.245665 # average overall miss latency ((Tick/Count))

board.cache_hierarchy.l2cache.blockedCycles::no_mshrs 0 # number of cycles access was blocked (Cycle)

board.cache_hierarchy.l2cache.blockedCycles::no_targets 0 # number of cycles access was blocked (Cycle)

board.cache_hierarchy.l2cache.blockedCauses::no_mshrs 0 # number of times access was blocked (Count)

board.cache_hierarchy.l2cache.blockedCauses::no_targets 0 # number of times access was blocked (Count)

board.cache_hierarchy.l2cache.avgBlocked::no_mshrs nan # average number of cycles each access was blocked ((Cycle/Count))

board.cache_hierarchy.l2cache.avgBlocked::no_targets nan # average number of cycles each access was blocked ((Cycle/Count))

board.cache_hierarchy.l2cache.demandMshrMisses::processor.cores.core.inst 196 # number of demand (read+write) MSHR misses (Count)

board.cache_hierarchy.l2cache.demandMshrMisses::processor.cores.core.data 150 # number of demand (read+write) MSHR misses (Count)

board.cache_hierarchy.l2cache.demandMshrMisses::total 346 # number of demand (read+write) MSHR misses (Count)

board.cache_hierarchy.l2cache.overallMshrMisses::cache_hierarchy.l2cache.prefetcher 45 # number of overall MSHR misses (Count)

board.cache_hierarchy.l2cache.overallMshrMisses::processor.cores.core.inst 196 # number of overall MSHR misses (Count)

board.cache_hierarchy.l2cache.overallMshrMisses::processor.cores.core.data 150 # number of overall MSHR misses (Count)

board.cache_hierarchy.l2cache.overallMshrMisses::total 391 # number of overall MSHR misses (Count)

board.cache_hierarchy.l2cache.demandMshrMissLatency::processor.cores.core.inst 11023965 # number of demand (read+write) MSHR miss ticks (Tick)

board.cache_hierarchy.l2cache.demandMshrMissLatency::processor.cores.core.data 7949376 # number of demand (read+write) MSHR miss ticks (Tick)

board.cache_hierarchy.l2cache.demandMshrMissLatency::total 18973341 # number of demand (read+write) MSHR miss ticks (Tick)

board.cache_hierarchy.l2cache.overallMshrMissLatency::cache_hierarchy.l2cache.prefetcher 2050185 # number of overall MSHR miss ticks (Tick)

board.cache_hierarchy.l2cache.overallMshrMissLatency::processor.cores.core.inst 11023965 # number of overall MSHR miss ticks (Tick)

board.cache_hierarchy.l2cache.overallMshrMissLatency::processor.cores.core.data 7949376 # number of overall MSHR miss ticks (Tick)

board.cache_hierarchy.l2cache.overallMshrMissLatency::total 21023526 # number of overall MSHR miss ticks (Tick)

board.cache_hierarchy.l2cache.demandMshrMissRate::processor.cores.core.inst 1 # mshr miss ratio for demand accesses (Ratio)

board.cache_hierarchy.l2cache.demandMshrMissRate::processor.cores.core.data 0.810811 # mshr miss ratio for demand accesses (Ratio)

board.cache_hierarchy.l2cache.demandMshrMissRate::total 0.908136 # mshr miss ratio for demand accesses (Ratio)

board.cache_hierarchy.l2cache.overallMshrMissRate::cache_hierarchy.l2cache.prefetcher inf # mshr miss ratio for overall accesses (Ratio)

board.cache_hierarchy.l2cache.overallMshrMissRate::processor.cores.core.inst 1 # mshr miss ratio for overall accesses (Ratio)

board.cache_hierarchy.l2cache.overallMshrMissRate::processor.cores.core.data 0.810811 # mshr miss ratio for overall accesses (Ratio)

board.cache_hierarchy.l2cache.overallMshrMissRate::total 1.026247 # mshr miss ratio for overall accesses (Ratio)

board.cache_hierarchy.l2cache.demandAvgMshrMissLatency::processor.cores.core.inst 56244.719388 # average overall mshr miss latency ((Tick/Count))

board.cache_hierarchy.l2cache.demandAvgMshrMissLatency::processor.cores.core.data 52995.840000 # average overall mshr miss latency ((Tick/Count))

board.cache_hierarchy.l2cache.demandAvgMshrMissLatency::total 54836.245665 # average overall mshr miss latency ((Tick/Count))

board.cache_hierarchy.l2cache.overallAvgMshrMissLatency::cache_hierarchy.l2cache.prefetcher 45559.666667 # average overall mshr miss latency ((Tick/Count))

board.cache_hierarchy.l2cache.overallAvgMshrMissLatency::processor.cores.core.inst 56244.719388 # average overall mshr miss latency ((Tick/Count))

board.cache_hierarchy.l2cache.overallAvgMshrMissLatency::processor.cores.core.data 52995.840000 # average overall mshr miss latency ((Tick/Count))

board.cache_hierarchy.l2cache.overallAvgMshrMissLatency::total 53768.608696 # average overall mshr miss latency ((Tick/Count))

board.cache_hierarchy.l2cache.replacements 0 # number of replacements (Count)

board.cache_hierarchy.l2cache.HardPFReq.mshrMisses::cache_hierarchy.l2cache.prefetcher 45 # number of HardPFReq MSHR misses (Count)

board.cache_hierarchy.l2cache.HardPFReq.mshrMisses::total 45 # number of HardPFReq MSHR misses (Count)

board.cache_hierarchy.l2cache.HardPFReq.mshrMissLatency::cache_hierarchy.l2cache.prefetcher 2050185 # number of HardPFReq MSHR miss ticks (Tick)

board.cache_hierarchy.l2cache.HardPFReq.mshrMissLatency::total 2050185 # number of HardPFReq MSHR miss ticks (Tick)

board.cache_hierarchy.l2cache.HardPFReq.mshrMissRate::cache_hierarchy.l2cache.prefetcher inf # mshr miss rate for HardPFReq accesses (Ratio)

board.cache_hierarchy.l2cache.HardPFReq.mshrMissRate::total inf # mshr miss rate for HardPFReq accesses (Ratio)

board.cache_hierarchy.l2cache.HardPFReq.avgMshrMissLatency::cache_hierarchy.l2cache.prefetcher 45559.666667 # average HardPFReq mshr miss latency ((Tick/Count))

board.cache_hierarchy.l2cache.HardPFReq.avgMshrMissLatency::total 45559.666667 # average HardPFReq mshr miss latency ((Tick/Count))

board.cache_hierarchy.l2cache.ReadExReq.hits::processor.cores.core.data 30 # number of ReadExReq hits (Count)

board.cache_hierarchy.l2cache.ReadExReq.hits::total 30 # number of ReadExReq hits (Count)

board.cache_hierarchy.l2cache.ReadExReq.misses::processor.cores.core.data 56 # number of ReadExReq misses (Count)

board.cache_hierarchy.l2cache.ReadExReq.misses::total 56 # number of ReadExReq misses (Count)

board.cache_hierarchy.l2cache.ReadExReq.missLatency::processor.cores.core.data 3340656 # number of ReadExReq miss ticks (Tick)

board.cache_hierarchy.l2cache.ReadExReq.missLatency::total 3340656 # number of ReadExReq miss ticks (Tick)

board.cache_hierarchy.l2cache.ReadExReq.accesses::processor.cores.core.data 86 # number of ReadExReq accesses(hits+misses) (Count)

board.cache_hierarchy.l2cache.ReadExReq.accesses::total 86 # number of ReadExReq accesses(hits+misses) (Count)

board.cache_hierarchy.l2cache.ReadExReq.missRate::processor.cores.core.data 0.651163 # miss rate for ReadExReq accesses (Ratio)

board.cache_hierarchy.l2cache.ReadExReq.missRate::total 0.651163 # miss rate for ReadExReq accesses (Ratio)

board.cache_hierarchy.l2cache.ReadExReq.avgMissLatency::processor.cores.core.data 59654.571429 # average ReadExReq miss latency ((Tick/Count))

board.cache_hierarchy.l2cache.ReadExReq.avgMissLatency::total 59654.571429 # average ReadExReq miss latency ((Tick/Count))

board.cache_hierarchy.l2cache.ReadExReq.mshrMisses::processor.cores.core.data 56 # number of ReadExReq MSHR misses (Count)

board.cache_hierarchy.l2cache.ReadExReq.mshrMisses::total 56 # number of ReadExReq MSHR misses (Count)

board.cache_hierarchy.l2cache.ReadExReq.mshrMissLatency::processor.cores.core.data 2967696 # number of ReadExReq MSHR miss ticks (Tick)

board.cache_hierarchy.l2cache.ReadExReq.mshrMissLatency::total 2967696 # number of ReadExReq MSHR miss ticks (Tick)

board.cache_hierarchy.l2cache.ReadExReq.mshrMissRate::processor.cores.core.data 0.651163 # mshr miss rate for ReadExReq accesses (Ratio)

board.cache_hierarchy.l2cache.ReadExReq.mshrMissRate::total 0.651163 # mshr miss rate for ReadExReq accesses (Ratio)

board.cache_hierarchy.l2cache.ReadExReq.avgMshrMissLatency::processor.cores.core.data 52994.571429 # average ReadExReq mshr miss latency ((Tick/Count))

board.cache_hierarchy.l2cache.ReadExReq.avgMshrMissLatency::total 52994.571429 # average ReadExReq mshr miss latency ((Tick/Count))

board.cache_hierarchy.l2cache.ReadSharedReq.hits::processor.cores.core.data 5 # number of ReadSharedReq hits (Count)

board.cache_hierarchy.l2cache.ReadSharedReq.hits::total 5 # number of ReadSharedReq hits (Count)

board.cache_hierarchy.l2cache.ReadSharedReq.misses::processor.cores.core.inst 196 # number of ReadSharedReq misses (Count)

board.cache_hierarchy.l2cache.ReadSharedReq.misses::processor.cores.core.data 94 # number of ReadSharedReq misses (Count)

board.cache_hierarchy.l2cache.ReadSharedReq.misses::total 290 # number of ReadSharedReq misses (Count)

board.cache_hierarchy.l2cache.ReadSharedReq.missLatency::processor.cores.core.inst 12329325 # number of ReadSharedReq miss ticks (Tick)

board.cache_hierarchy.l2cache.ReadSharedReq.missLatency::processor.cores.core.data 5607720 # number of ReadSharedReq miss ticks (Tick)

board.cache_hierarchy.l2cache.ReadSharedReq.missLatency::total 17937045 # number of ReadSharedReq miss ticks (Tick)

board.cache_hierarchy.l2cache.ReadSharedReq.accesses::processor.cores.core.inst 196 # number of ReadSharedReq accesses(hits+misses) (Count)

board.cache_hierarchy.l2cache.ReadSharedReq.accesses::processor.cores.core.data 99 # number of ReadSharedReq accesses(hits+misses) (Count)

board.cache_hierarchy.l2cache.ReadSharedReq.accesses::total 295 # number of ReadSharedReq accesses(hits+misses) (Count)

board.cache_hierarchy.l2cache.ReadSharedReq.missRate::processor.cores.core.inst 1 # miss rate for ReadSharedReq accesses (Ratio)

board.cache_hierarchy.l2cache.ReadSharedReq.missRate::processor.cores.core.data 0.949495 # miss rate for ReadSharedReq accesses (Ratio)

board.cache_hierarchy.l2cache.ReadSharedReq.missRate::total 0.983051 # miss rate for ReadSharedReq accesses (Ratio)

board.cache_hierarchy.l2cache.ReadSharedReq.avgMissLatency::processor.cores.core.inst 62904.719388 # average ReadSharedReq miss latency ((Tick/Count))

board.cache_hierarchy.l2cache.ReadSharedReq.avgMissLatency::processor.cores.core.data 59656.595745 # average ReadSharedReq miss latency ((Tick/Count))

board.cache_hierarchy.l2cache.ReadSharedReq.avgMissLatency::total 61851.879310 # average ReadSharedReq miss latency ((Tick/Count))

board.cache_hierarchy.l2cache.ReadSharedReq.mshrMisses::processor.cores.core.inst 196 # number of ReadSharedReq MSHR misses (Count)

board.cache_hierarchy.l2cache.ReadSharedReq.mshrMisses::processor.cores.core.data 94 # number of ReadSharedReq MSHR misses (Count)

board.cache_hierarchy.l2cache.ReadSharedReq.mshrMisses::total 290 # number of ReadSharedReq MSHR misses (Count)

board.cache_hierarchy.l2cache.ReadSharedReq.mshrMissLatency::processor.cores.core.inst 11023965 # number of ReadSharedReq MSHR miss ticks (Tick)

board.cache_hierarchy.l2cache.ReadSharedReq.mshrMissLatency::processor.cores.core.data 4981680 # number of ReadSharedReq MSHR miss ticks (Tick)

board.cache_hierarchy.l2cache.ReadSharedReq.mshrMissLatency::total 16005645 # number of ReadSharedReq MSHR miss ticks (Tick)

board.cache_hierarchy.l2cache.ReadSharedReq.mshrMissRate::processor.cores.core.inst 1 # mshr miss rate for ReadSharedReq accesses (Ratio)

board.cache_hierarchy.l2cache.ReadSharedReq.mshrMissRate::processor.cores.core.data 0.949495 # mshr miss rate for ReadSharedReq accesses (Ratio)

board.cache_hierarchy.l2cache.ReadSharedReq.mshrMissRate::total 0.983051 # mshr miss rate for ReadSharedReq accesses (Ratio)

board.cache_hierarchy.l2cache.ReadSharedReq.avgMshrMissLatency::processor.cores.core.inst 56244.719388 # average ReadSharedReq mshr miss latency ((Tick/Count))

board.cache_hierarchy.l2cache.ReadSharedReq.avgMshrMissLatency::processor.cores.core.data 52996.595745 # average ReadSharedReq mshr miss latency ((Tick/Count))

board.cache_hierarchy.l2cache.ReadSharedReq.avgMshrMissLatency::total 55191.879310 # average ReadSharedReq mshr miss latency ((Tick/Count))

board.cache_hierarchy.l2cache.power_state.pwrStateResidencyTicks::UNDEFINED 84926990560998752 # Cumulative time (in ticks) in various power states (Tick)

board.cache_hierarchy.l2cache.prefetcher.demandMshrMisses 346 # demands not covered by prefetchs (Count)

board.cache_hierarchy.l2cache.prefetcher.pfIssued 146 # number of hwpf issued (Count)

board.cache_hierarchy.l2cache.prefetcher.pfUseful 35 # number of useful prefetch (Count)

board.cache_hierarchy.l2cache.prefetcher.pfUsefulButMiss 0 # number of hit on prefetch but cache block is not in an usable state (Count)

board.cache_hierarchy.l2cache.prefetcher.accuracy 0.239726 # accuracy of the prefetcher (Count)

board.cache_hierarchy.l2cache.prefetcher.coverage 0.091864 # coverage brought by this prefetcher (Count)

board.cache_hierarchy.l2cache.prefetcher.pfHitInCache 50 # number of prefetches hitting in cache (Count)

board.cache_hierarchy.l2cache.prefetcher.pfHitInMSHR 51 # number of prefetches hitting in a MSHR (Count)

board.cache_hierarchy.l2cache.prefetcher.pfHitInWB 0 # number of prefetches hit in the Write Buffer (Count)

board.cache_hierarchy.l2cache.prefetcher.pfLate 101 # number of late prefetches (hitting in cache, MSHR or WB) (Count)

board.cache_hierarchy.l2cache.prefetcher.pfIdentified 146 # number of prefetch candidates identified (Count)

board.cache_hierarchy.l2cache.prefetcher.pfBufferHit 0 # number of redundant prefetches already in prefetch queue (Count)

board.cache_hierarchy.l2cache.prefetcher.pfInCache 0 # number of redundant prefetches already in cache/mshr dropped (Count)

board.cache_hierarchy.l2cache.prefetcher.pfRemovedDemand 0 # number of prefetches dropped due to a demand for the same address (Count)

board.cache_hierarchy.l2cache.prefetcher.pfRemovedFull 0 # number of prefetches dropped due to prefetch queue size (Count)

board.cache_hierarchy.l2cache.prefetcher.pfSpanPage 6 # number of prefetches that crossed the page (Count)

board.cache_hierarchy.l2cache.prefetcher.pfUsefulSpanPage 5 # number of prefetches that is useful and crossed the page (Count)

board.cache_hierarchy.l2cache.prefetcher.power_state.pwrStateResidencyTicks::UNDEFINED 84926990560998752 # Cumulative time (in ticks) in various power states (Tick)

board.cache_hierarchy.l2cache.tags.tagsInUse 391.000000 # Average ticks per tags in use ((Tick/Count))

board.cache_hierarchy.l2cache.tags.totalRefs 426 # Total number of references to valid blocks. (Count)

board.cache_hierarchy.l2cache.tags.sampledRefs 391 # Sample count of references to valid blocks. (Count)

board.cache_hierarchy.l2cache.tags.avgRefs 1.089514 # Average number of references to valid blocks. ((Count/Count))

board.cache_hierarchy.l2cache.tags.warmupTick 69597 # The tick when the warmup percentage was hit. (Tick)

board.cache_hierarchy.l2cache.tags.occupancies::cache_hierarchy.l2cache.prefetcher 45.000000 # Average occupied blocks per tick, per requestor ((Count/Tick))

board.cache_hierarchy.l2cache.tags.occupancies::processor.cores.core.inst 196.000000 # Average occupied blocks per tick, per requestor ((Count/Tick))

board.cache_hierarchy.l2cache.tags.occupancies::processor.cores.core.data 150.000000 # Average occupied blocks per tick, per requestor ((Count/Tick))

board.cache_hierarchy.l2cache.tags.avgOccs::cache_hierarchy.l2cache.prefetcher 0.010986 # Average percentage of cache occupancy ((Ratio/Tick))

board.cache_hierarchy.l2cache.tags.avgOccs::processor.cores.core.inst 0.047852 # Average percentage of cache occupancy ((Ratio/Tick))

board.cache_hierarchy.l2cache.tags.avgOccs::processor.cores.core.data 0.036621 # Average percentage of cache occupancy ((Ratio/Tick))

board.cache_hierarchy.l2cache.tags.avgOccs::total 0.095459 # Average percentage of cache occupancy ((Ratio/Tick))

board.cache_hierarchy.l2cache.tags.occupanciesTaskId::1022 45 # Occupied blocks per task id (Count)

board.cache_hierarchy.l2cache.tags.occupanciesTaskId::1024 346 # Occupied blocks per task id (Count)

board.cache_hierarchy.l2cache.tags.ageTaskId_1022::4 45 # Occupied blocks per task id, per block age (Count)

board.cache_hierarchy.l2cache.tags.ageTaskId_1024::4 346 # Occupied blocks per task id, per block age (Count)

board.cache_hierarchy.l2cache.tags.ratioOccsTaskId::1022 0.010986 # Ratio of occupied blocks and all blocks, per task id (Ratio)

board.cache_hierarchy.l2cache.tags.ratioOccsTaskId::1024 0.084473 # Ratio of occupied blocks and all blocks, per task id (Ratio)

board.cache_hierarchy.l2cache.tags.tagAccesses 6487 # Number of tag accesses (Count)

board.cache_hierarchy.l2cache.tags.dataAccesses 6487 # Number of data accesses (Count)

board.cache_hierarchy.l2cache.tags.power_state.pwrStateResidencyTicks::UNDEFINED 84926990560998752 # Cumulative time (in ticks) in various power states (Tick)

board.cache_hierarchy.membus.transDist::ReadResp 335 # Transaction distribution (Count)

board.cache_hierarchy.membus.transDist::ReadExReq 56 # Transaction distribution (Count)

board.cache_hierarchy.membus.transDist::ReadExResp 56 # Transaction distribution (Count)

board.cache_hierarchy.membus.transDist::ReadSharedReq 335 # Transaction distribution (Count)

board.cache_hierarchy.membus.pktCount_board.cache_hierarchy.l2cache.mem_side_port::board.memory.mem_ctrl.port 782 # Packet count per connected requestor and responder (Count)

board.cache_hierarchy.membus.pktCount_board.cache_hierarchy.l2cache.mem_side_port::total 782 # Packet count per connected requestor and responder (Count)

board.cache_hierarchy.membus.pktCount::total 782 # Packet count per connected requestor and responder (Count)

board.cache_hierarchy.membus.pktSize_board.cache_hierarchy.l2cache.mem_side_port::board.memory.mem_ctrl.port 25024 # Cumulative packet size per connected requestor and responder (Byte)

board.cache_hierarchy.membus.pktSize_board.cache_hierarchy.l2cache.mem_side_port::total 25024 # Cumulative packet size per connected requestor and responder (Byte)

board.cache_hierarchy.membus.pktSize::total 25024 # Cumulative packet size per connected requestor and responder (Byte)

board.cache_hierarchy.membus.snoops 0 # Total snoops (Count)

board.cache_hierarchy.membus.snoopTraffic 0 # Total snoop traffic (Byte)

board.cache_hierarchy.membus.snoopFanout::samples 391 # Request fanout histogram (Count)

board.cache_hierarchy.membus.snoopFanout::mean 0 # Request fanout histogram (Count)

board.cache_hierarchy.membus.snoopFanout::stdev 0 # Request fanout histogram (Count)

board.cache_hierarchy.membus.snoopFanout::underflows 0 0.00% 0.00% # Request fanout histogram (Count)

board.cache_hierarchy.membus.snoopFanout::0 391 100.00% 100.00% # Request fanout histogram (Count)

board.cache_hierarchy.membus.snoopFanout::1 0 0.00% 100.00% # Request fanout histogram (Count)

board.cache_hierarchy.membus.snoopFanout::overflows 0 0.00% 100.00% # Request fanout histogram (Count)

board.cache_hierarchy.membus.snoopFanout::min_value 0 # Request fanout histogram (Count)

board.cache_hierarchy.membus.snoopFanout::max_value 0 # Request fanout histogram (Count)

board.cache_hierarchy.membus.snoopFanout::total 391 # Request fanout histogram (Count)

board.cache_hierarchy.membus.badaddr_responder.power_state.pwrStateResidencyTicks::UNDEFINED 84926990560998752 # Cumulative time (in ticks) in various power states (Tick)

board.cache_hierarchy.membus.power_state.pwrStateResidencyTicks::UNDEFINED 84926990560998752 # Cumulative time (in ticks) in various power states (Tick)

board.cache_hierarchy.membus.reqLayer0.occupancy 139443 # Layer occupancy (ticks) (Tick)

board.cache_hierarchy.membus.reqLayer0.utilization 0.0 # Layer utilization (Ratio)

board.cache_hierarchy.membus.respLayer2.occupancy 333977 # Layer occupancy (ticks) (Tick)

board.cache_hierarchy.membus.respLayer2.utilization 0.0 # Layer utilization (Ratio)

board.cache_hierarchy.membus.snoop_filter.totRequests 391 # Total number of requests made to the snoop filter. (Count)

board.cache_hierarchy.membus.snoop_filter.hitSingleRequests 0 # Number of requests hitting in the snoop filter with a single holder of the requested data. (Count)

board.cache_hierarchy.membus.snoop_filter.hitMultiRequests 0 # Number of requests hitting in the snoop filter with multiple (>1) holders of the requested data. (Count)

board.cache_hierarchy.membus.snoop_filter.totSnoops 0 # Total number of snoops made to the snoop filter. (Count)

board.cache_hierarchy.membus.snoop_filter.hitSingleSnoops 0 # Number of snoops hitting in the snoop filter with a single holder of the requested data. (Count)

board.cache_hierarchy.membus.snoop_filter.hitMultiSnoops 0 # Number of snoops hitting in the snoop filter with multiple (>1) holders of the requested data. (Count)

board.clk_domain.clock 333 # Clock period in ticks (Tick)

board.clk_domain.voltage_domain.voltage 1 # Voltage in Volts (Volt)

board.memory.mem_ctrl.avgPriority_cache_hierarchy.l2cache.prefetcher::samples 45.00 # Average QoS priority value for accepted requests (Count)

board.memory.mem_ctrl.avgPriority_processor.cores.core.inst::samples 196.00 # Average QoS priority value for accepted requests (Count)

board.memory.mem_ctrl.avgPriority_processor.cores.core.data::samples 150.00 # Average QoS priority value for accepted requests (Count)

board.memory.mem_ctrl.priorityMinLatency 0.000000018750 # per QoS priority minimum request to response latency (Second)

board.memory.mem_ctrl.priorityMaxLatency 0.000000471124 # per QoS priority maximum request to response latency (Second)

board.memory.mem_ctrl.numReadWriteTurnArounds 0 # Number of turnarounds from READ to WRITE (Count)

board.memory.mem_ctrl.numWriteReadTurnArounds 0 # Number of turnarounds from WRITE to READ (Count)

board.memory.mem_ctrl.numStayReadState 21814607442 # Number of times bus staying in READ state (Count)

board.memory.mem_ctrl.numStayWriteState 0 # Number of times bus staying in WRITE state (Count)

board.memory.mem_ctrl.readReqs 391 # Number of read requests accepted (Count)

board.memory.mem_ctrl.writeReqs 0 # Number of write requests accepted (Count)

board.memory.mem_ctrl.readBursts 391 # Number of controller read bursts, including those serviced by the write queue (Count)

board.memory.mem_ctrl.writeBursts 0 # Number of controller write bursts, including those merged in the write queue (Count)

board.memory.mem_ctrl.servicedByWrQ 0 # Number of controller read bursts serviced by the write queue (Count)

board.memory.mem_ctrl.mergedWrBursts 0 # Number of controller write bursts merged with an existing one (Count)

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board.memory.mem_ctrl.neitherReadNorWriteReqs 0 # Number of requests that are neither read nor write (Count)
board.memory.mem_ctrl.avgRdQLen 1.00 # Average read queue length when enqueuing ((Count/Tick))
board.memory.mem_ctrl.avgWrQLen 0.00 # Average write queue length when enqueuing ((Count/Tick))
board.memory.mem_ctrl.numRdRetry 0 # Number of times read queue was full causing retry (Count)
board.memory.mem_ctrl.numWrRetry 0 # Number of times write queue was full causing retry (Count)
board.memory.mem_ctrl.readPktSize::0 0 # Read request sizes (log2) (Count)
board.memory.mem_ctrl.readPktSize::1 0 # Read request sizes (log2) (Count)
board.memory.mem_ctrl.readPktSize::2 0 # Read request sizes (log2) (Count)
board.memory.mem_ctrl.readPktSize::3 0 # Read request sizes (log2) (Count)
board.memory.mem_ctrl.readPktSize::4 0 # Read request sizes (log2) (Count)
board.memory.mem_ctrl.readPktSize::5 0 # Read request sizes (log2) (Count)
board.memory.mem_ctrl.readPktSize::6 391 # Read request sizes (log2) (Count)
board.memory.mem_ctrl.writePktSize::0 0 # Write request sizes (log2) (Count)
board.memory.mem_ctrl.writePktSize::1 0 # Write request sizes (log2) (Count)
board.memory.mem_ctrl.writePktSize::2 0 # Write request sizes (log2) (Count)
board.memory.mem_ctrl.writePktSize::3 0 # Write request sizes (log2) (Count)
board.memory.mem_ctrl.writePktSize::4 0 # Write request sizes (log2) (Count)
board.memory.mem_ctrl.writePktSize::5 0 # Write request sizes (log2) (Count)
board.memory.mem_ctrl.writePktSize::6 0 # Write request sizes (log2) (Count)
board.memory.mem_ctrl.rdQLenPdf::0 373 # What read queue length does an incoming req see (Count)
board.memory.mem_ctrl.rdQLenPdf::1 6 # What read queue length does an incoming req see (Count)
board.memory.mem_ctrl.rdQLenPdf::2 5 # What read queue length does an incoming req see (Count)
board.memory.mem_ctrl.rdQLenPdf::3 4 # What read queue length does an incoming req see (Count)
board.memory.mem_ctrl.rdQLenPdf::4 3 # What read queue length does an incoming req see (Count)
board.memory.mem_ctrl.rdQLenPdf::5 0 # What read queue length does an incoming req see (Count)
board.memory.mem_ctrl.rdQLenPdf::6 0 # What read queue length does an incoming req see (Count)
board.memory.mem_ctrl.rdQLenPdf::7 0 # What read queue length does an incoming req see (Count)
board.memory.mem_ctrl.rdQLenPdf::8 0 # What read queue length does an incoming req see (Count)
board.memory.mem_ctrl.rdQLenPdf::9 0 # What read queue length does an incoming req see (Count)
board.memory.mem_ctrl.rdQLenPdf::10 0 # What read queue length does an incoming req see (Count)
board.memory.mem_ctrl.rdQLenPdf::11 0 # What read queue length does an incoming req see (Count)
board.memory.mem_ctrl.rdQLenPdf::12 0 # What read queue length does an incoming req see (Count)
board.memory.mem_ctrl.rdQLenPdf::13 0 # What read queue length does an incoming req see (Count)
board.memory.mem_ctrl.rdQLenPdf::14 0 # What read queue length does an incoming req see (Count)
board.memory.mem_ctrl.rdQLenPdf::15 0 # What read queue length does an incoming req see (Count)
board.memory.mem_ctrl.rdQLenPdf::16 0 # What read queue length does an incoming req see (Count)
board.memory.mem_ctrl.rdQLenPdf::17 0 # What read queue length does an incoming req see (Count)
board.memory.mem_ctrl.rdQLenPdf::18 0 # What read queue length does an incoming req see (Count)
board.memory.mem_ctrl.rdQLenPdf::19 0 # What read queue length does an incoming req see (Count)
board.memory.mem_ctrl.rdQLenPdf::20 0 # What read queue length does an incoming req see (Count)
board.memory.mem_ctrl.rdQLenPdf::21 0 # What read queue length does an incoming req see (Count)
board.memory.mem_ctrl.rdQLenPdf::22 0 # What read queue length does an incoming req see (Count)
board.memory.mem_ctrl.rdQLenPdf::23 0 # What read queue length does an incoming req see (Count)
board.memory.mem_ctrl.rdQLenPdf::24 0 # What read queue length does an incoming req see (Count)
board.memory.mem_ctrl.rdQLenPdf::25 0 # What read queue length does an incoming req see (Count)
board.memory.mem_ctrl.rdQLenPdf::26 0 # What read queue length does an incoming req see (Count)
board.memory.mem_ctrl.rdQLenPdf::27 0 # What read queue length does an incoming req see (Count)
board.memory.mem_ctrl.rdQLenPdf::28 0 # What read queue length does an incoming req see (Count)
board.memory.mem_ctrl.rdQLenPdf::29 0 # What read queue length does an incoming req see (Count)
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board.memory.mem_ctrl.rdQLenPdf::30	0	# What read queue length does an incoming req see (Count)
board.memory.mem_ctrl.rdQLenPdf::31	0	# What read queue length does an incoming req see (Count)
board.memory.mem_ctrl.wrQLenPdf::0	0	# What write queue length does an incoming req see (Count)
board.memory.mem_ctrl.wrQLenPdf::1	0	# What write queue length does an incoming req see (Count)
board.memory.mem_ctrl.wrQLenPdf::2	0	# What write queue length does an incoming req see (Count)
board.memory.mem_ctrl.wrQLenPdf::3	0	# What write queue length does an incoming req see (Count)
board.memory.mem_ctrl.wrQLenPdf::4	0	# What write queue length does an incoming req see (Count)
board.memory.mem_ctrl.wrQLenPdf::5	0	# What write queue length does an incoming req see (Count)
board.memory.mem_ctrl.wrQLenPdf::6	0	# What write queue length does an incoming req see (Count)
board.memory.mem_ctrl.wrQLenPdf::7	0	# What write queue length does an incoming req see (Count)
board.memory.mem_ctrl.wrQLenPdf::8	0	# What write queue length does an incoming req see (Count)
board.memory.mem_ctrl.wrQLenPdf::9	0	# What write queue length does an incoming req see (Count)
board.memory.mem_ctrl.wrQLenPdf::10	0	# What write queue length does an incoming req see (Count)
board.memory.mem_ctrl.wrQLenPdf::11	0	# What write queue length does an incoming req see (Count)
board.memory.mem_ctrl.wrQLenPdf::12	0	# What write queue length does an incoming req see (Count)
board.memory.mem_ctrl.wrQLenPdf::13	0	# What write queue length does an incoming req see (Count)
board.memory.mem_ctrl.wrQLenPdf::14	0	# What write queue length does an incoming req see (Count)
board.memory.mem_ctrl.wrQLenPdf::15	0	# What write queue length does an incoming req see (Count)
board.memory.mem_ctrl.wrQLenPdf::16	0	# What write queue length does an incoming req see (Count)
board.memory.mem_ctrl.wrQLenPdf::17	0	# What write queue length does an incoming req see (Count)
board.memory.mem_ctrl.wrQLenPdf::18	0	# What write queue length does an incoming req see (Count)
board.memory.mem_ctrl.wrQLenPdf::19	0	# What write queue length does an incoming req see (Count)
board.memory.mem_ctrl.wrQLenPdf::20	0	# What write queue length does an incoming req see (Count)
board.memory.mem_ctrl.wrQLenPdf::21	0	# What write queue length does an incoming req see (Count)
board.memory.mem_ctrl.wrQLenPdf::22	0	# What write queue length does an incoming req see (Count)
board.memory.mem_ctrl.wrQLenPdf::23	0	# What write queue length does an incoming req see (Count)
board.memory.mem_ctrl.wrQLenPdf::24	0	# What write queue length does an incoming req see (Count)
board.memory.mem_ctrl.wrQLenPdf::25	0	# What write queue length does an incoming req see (Count)
board.memory.mem_ctrl.wrQLenPdf::26	0	# What write queue length does an incoming req see (Count)
board.memory.mem_ctrl.wrQLenPdf::27	0	# What write queue length does an incoming req see (Count)
board.memory.mem_ctrl.wrQLenPdf::28	0	# What write queue length does an incoming req see (Count)
board.memory.mem_ctrl.wrQLenPdf::29	0	# What write queue length does an incoming req see (Count)
board.memory.mem_ctrl.wrQLenPdf::30	0	# What write queue length does an incoming req see (Count)
board.memory.mem_ctrl.wrQLenPdf::31	0	# What write queue length does an incoming req see (Count)
board.memory.mem_ctrl.wrQLenPdf::32	0	# What write queue length does an incoming req see (Count)
board.memory.mem_ctrl.wrQLenPdf::33	0	# What write queue length does an incoming req see (Count)
board.memory.mem_ctrl.wrQLenPdf::34	0	# What write queue length does an incoming req see (Count)
board.memory.mem_ctrl.wrQLenPdf::35	0	# What write queue length does an incoming req see (Count)
board.memory.mem_ctrl.wrQLenPdf::36	0	# What write queue length does an incoming req see (Count)
board.memory.mem_ctrl.wrQLenPdf::37	0	# What write queue length does an incoming req see (Count)
board.memory.mem_ctrl.wrQLenPdf::38	0	# What write queue length does an incoming req see (Count)
board.memory.mem_ctrl.wrQLenPdf::39	0	# What write queue length does an incoming req see (Count)
board.memory.mem_ctrl.wrQLenPdf::40	0	# What write queue length does an incoming req see (Count)
board.memory.mem_ctrl.wrQLenPdf::41	0	# What write queue length does an incoming req see (Count)
board.memory.mem_ctrl.wrQLenPdf::42	0	# What write queue length does an incoming req see (Count)
board.memory.mem_ctrl.wrQLenPdf::43	0	# What write queue length does an incoming req see (Count)
board.memory.mem_ctrl.wrQLenPdf::44	0	# What write queue length does an incoming req see (Count)
board.memory.mem_ctrl.wrQLenPdf::45	0	# What write queue length does an incoming req see (Count)
board.memory.mem_ctrl.wrQLenPdf::46	0	# What write queue length does an incoming req see (Count)

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board.memory.mem_ctrl.wrQLenPdf::47 0 # What write queue length does an incoming req see (Count)
board.memory.mem_ctrl.wrQLenPdf::48 0 # What write queue length does an incoming req see (Count)
board.memory.mem_ctrl.wrQLenPdf::49 0 # What write queue length does an incoming req see (Count)
board.memory.mem_ctrl.wrQLenPdf::50 0 # What write queue length does an incoming req see (Count)
board.memory.mem_ctrl.wrQLenPdf::51 0 # What write queue length does an incoming req see (Count)
board.memory.mem_ctrl.wrQLenPdf::52 0 # What write queue length does an incoming req see (Count)
board.memory.mem_ctrl.wrQLenPdf::53 0 # What write queue length does an incoming req see (Count)
board.memory.mem_ctrl.wrQLenPdf::54 0 # What write queue length does an incoming req see (Count)
board.memory.mem_ctrl.wrQLenPdf::55 0 # What write queue length does an incoming req see (Count)
board.memory.mem_ctrl.wrQLenPdf::56 0 # What write queue length does an incoming req see (Count)
board.memory.mem_ctrl.wrQLenPdf::57 0 # What write queue length does an incoming req see (Count)
board.memory.mem_ctrl.wrQLenPdf::58 0 # What write queue length does an incoming req see (Count)
board.memory.mem_ctrl.wrQLenPdf::59 0 # What write queue length does an incoming req see (Count)
board.memory.mem_ctrl.wrQLenPdf::60 0 # What write queue length does an incoming req see (Count)
board.memory.mem_ctrl.wrQLenPdf::61 0 # What write queue length does an incoming req see (Count)
board.memory.mem_ctrl.wrQLenPdf::62 0 # What write queue length does an incoming req see (Count)
board.memory.mem_ctrl.wrQLenPdf::63 0 # What write queue length does an incoming req see (Count)
board.memory.mem_ctrl.bytesReadWrQ 0 # Total number of bytes read from write queue (Byte)
board.memory.mem_ctrl.bytesReadSys 25024 # Total read bytes from the system interface side (Byte)
board.memory.mem_ctrl.bytesWrittenSys 0 # Total written bytes from the system interface side (Byte)
board.memory.mem_ctrl.avgRdBWSys 0.29465309 # Average system read bandwidth in Byte/s ((Byte/Second))
board.memory.mem_ctrl.avgWrBWSys 0.00000000 # Average system write bandwidth in Byte/s ((Byte/Second))
board.memory.mem_ctrl.totGap 24062247 # Total gap between requests (Tick)
board.memory.mem_ctrl.avgGap 61540.27 # Average gap between requests ((Tick/Count))
board.memory.mem_ctrl.requestorReadBytes::cache_hierarchy.l2cache.prefetcher 2880 # Per-requestor bytes read from memory (Byte)
board.memory.mem_ctrl.requestorReadBytes::processor.cores.core.inst 12544 # Per-requestor bytes read from memory (Byte)
board.memory.mem_ctrl.requestorReadBytes::processor.cores.core.data 9600 # Per-requestor bytes read from memory (Byte)
board.memory.mem_ctrl.requestorReadRate::cache_hierarchy.l2cache.prefetcher 0.033911480684 # Per-requestor bytes read from memory rate ((Byte/Second))
board.memory.mem_ctrl.requestorReadRate::processor.cores.core.inst 0.147703338092 # Per-requestor bytes read from memory rate ((Byte/Second))
board.memory.mem_ctrl.requestorReadRate::processor.cores.core.data 0.113038268948 # Per-requestor bytes read from memory rate ((Byte/Second))
board.memory.mem_ctrl.requestorReadAccesses::cache_hierarchy.l2cache.prefetcher 45 # Per-requestor read serviced memory accesses (Count)
board.memory.mem_ctrl.requestorReadAccesses::processor.cores.core.inst 196 # Per-requestor read serviced memory accesses (Count)
board.memory.mem_ctrl.requestorReadAccesses::processor.cores.core.data 150 # Per-requestor read serviced memory accesses (Count)
board.memory.mem_ctrl.requestorReadTotalLat::cache_hierarchy.l2cache.prefetcher 977836 # Per-requestor read total memory access latency (Tick)
board.memory.mem_ctrl.requestorReadTotalLat::processor.cores.core.inst 5044337 # Per-requestor read total memory access latency (Tick)
board.memory.mem_ctrl.requestorReadTotalLat::processor.cores.core.data 3372950 # Per-requestor read total memory access latency (Tick)
board.memory.mem_ctrl.requestorReadAvgLat::cache_hierarchy.l2cache.prefetcher 21729.69 # Per-requestor read average memory access latency ((Tick/Count))
board.memory.mem_ctrl.requestorReadAvgLat::processor.cores.core.inst 25736.41 # Per-requestor read average memory access latency ((Tick/Count))
board.memory.mem_ctrl.requestorReadAvgLat::processor.cores.core.data 22486.33 # Per-requestor read average memory access latency ((Tick/Count))
board.memory.mem_ctrl.dram.bytesRead::cache_hierarchy.l2cache.prefetcher 2880 # Number of bytes read from this memory (Byte)
board.memory.mem_ctrl.dram.bytesRead::processor.cores.core.inst 12544 # Number of bytes read from this memory (Byte)
board.memory.mem_ctrl.dram.bytesRead::processor.cores.core.data 9600 # Number of bytes read from this memory (Byte)
board.memory.mem_ctrl.dram.bytesRead::total 25024 # Number of bytes read from this memory (Byte)
board.memory.mem_ctrl.dram.bytesInstRead::processor.cores.core.inst 12544 # Number of instructions bytes read from this memory (Byte)
board.memory.mem_ctrl.dram.bytesInstRead::total 12544 # Number of instructions bytes read from this memory (Byte)
board.memory.mem_ctrl.dram.numReads::cache_hierarchy.l2cache.prefetcher 45 # Number of read requests responded to by this memory (Count)
board.memory.mem_ctrl.dram.numReads::processor.cores.core.inst 196 # Number of read requests responded to by this memory (Count)
board.memory.mem_ctrl.dram.numReads::processor.cores.core.data 150 # Number of read requests responded to by this memory (Count)
board.memory.mem_ctrl.dram.numReads::total 391 # Number of read requests responded to by this memory (Count)
```

```
board.memory.mem_ctrl.dram.bwRead::cache_hierarchy.l2cache.prefetcher 0 # Total read bandwidth from this memory ((Byte/Second))
board.memory.mem_ctrl.dram.bwRead::processor.cores.core.inst 0 # Total read bandwidth from this memory ((Byte/Second))
board.memory.mem_ctrl.dram.bwRead::processor.cores.core.data 0 # Total read bandwidth from this memory ((Byte/Second))
board.memory.mem_ctrl.dram.bwRead::total 0 # Total read bandwidth from this memory ((Byte/Second))
board.memory.mem_ctrl.dram.bwInstRead::processor.cores.core.inst 0 # Instruction read bandwidth from this memory ((Byte/Second))
board.memory.mem_ctrl.dram.bwInstRead::total 0 # Instruction read bandwidth from this memory ((Byte/Second))
board.memory.mem_ctrl.dram.bwTotal::cache_hierarchy.l2cache.prefetcher 0 # Total bandwidth to/from this memory ((Byte/Second))
board.memory.mem_ctrl.dram.bwTotal::processor.cores.core.inst 0 # Total bandwidth to/from this memory ((Byte/Second))
board.memory.mem_ctrl.dram.bwTotal::processor.cores.core.data 0 # Total bandwidth to/from this memory ((Byte/Second))
board.memory.mem_ctrl.dram.bwTotal::total 0 # Total bandwidth to/from this memory ((Byte/Second))
board.memory.mem_ctrl.dram.readBursts 391 # Number of DRAM read bursts (Count)
board.memory.mem_ctrl.dram.writeBursts 0 # Number of DRAM write bursts (Count)
board.memory.mem_ctrl.dram.perBankRdBursts::0 68 # Per bank write bursts (Count)
board.memory.mem_ctrl.dram.perBankRdBursts::1 62 # Per bank write bursts (Count)
board.memory.mem_ctrl.dram.perBankRdBursts::2 19 # Per bank write bursts (Count)
board.memory.mem_ctrl.dram.perBankRdBursts::3 37 # Per bank write bursts (Count)
board.memory.mem_ctrl.dram.perBankRdBursts::4 46 # Per bank write bursts (Count)
board.memory.mem_ctrl.dram.perBankRdBursts::5 66 # Per bank write bursts (Count)
board.memory.mem_ctrl.dram.perBankRdBursts::6 3 # Per bank write bursts (Count)
board.memory.mem_ctrl.dram.perBankRdBursts::7 3 # Per bank write bursts (Count)
board.memory.mem_ctrl.dram.perBankRdBursts::8 38 # Per bank write bursts (Count)
board.memory.mem_ctrl.dram.perBankRdBursts::9 17 # Per bank write bursts (Count)
board.memory.mem_ctrl.dram.perBankRdBursts::10 9 # Per bank write bursts (Count)
board.memory.mem_ctrl.dram.perBankRdBursts::11 2 # Per bank write bursts (Count)
board.memory.mem_ctrl.dram.perBankRdBursts::12 0 # Per bank write bursts (Count)
board.memory.mem_ctrl.dram.perBankRdBursts::13 2 # Per bank write bursts (Count)
board.memory.mem_ctrl.dram.perBankRdBursts::14 2 # Per bank write bursts (Count)
board.memory.mem_ctrl.dram.perBankRdBursts::15 17 # Per bank write bursts (Count)
board.memory.mem_ctrl.dram.perBankWrBursts::0 0 # Per bank write bursts (Count)
board.memory.mem_ctrl.dram.perBankWrBursts::1 0 # Per bank write bursts (Count)
board.memory.mem_ctrl.dram.perBankWrBursts::2 0 # Per bank write bursts (Count)
board.memory.mem_ctrl.dram.perBankWrBursts::3 0 # Per bank write bursts (Count)
board.memory.mem_ctrl.dram.perBankWrBursts::4 0 # Per bank write bursts (Count)
board.memory.mem_ctrl.dram.perBankWrBursts::5 0 # Per bank write bursts (Count)
board.memory.mem_ctrl.dram.perBankWrBursts::6 0 # Per bank write bursts (Count)
board.memory.mem_ctrl.dram.perBankWrBursts::7 0 # Per bank write bursts (Count)
board.memory.mem_ctrl.dram.perBankWrBursts::8 0 # Per bank write bursts (Count)
board.memory.mem_ctrl.dram.perBankWrBursts::9 0 # Per bank write bursts (Count)
board.memory.mem_ctrl.dram.perBankWrBursts::10 0 # Per bank write bursts (Count)
board.memory.mem_ctrl.dram.perBankWrBursts::11 0 # Per bank write bursts (Count)
board.memory.mem_ctrl.dram.perBankWrBursts::12 0 # Per bank write bursts (Count)
board.memory.mem_ctrl.dram.perBankWrBursts::13 0 # Per bank write bursts (Count)
board.memory.mem_ctrl.dram.perBankWrBursts::14 0 # Per bank write bursts (Count)
board.memory.mem_ctrl.dram.perBankWrBursts::15 0 # Per bank write bursts (Count)
board.memory.mem_ctrl.dram.totQLat 2063873 # Total ticks spent queuing (Tick)
board.memory.mem_ctrl.dram.totBusLat 1955000 # Total ticks spent in databus transfers (Tick)
board.memory.mem_ctrl.dram.totMemAccLat 9395123 # Total ticks spent from burst creation until serviced by the DRAM (Tick)
board.memory.mem_ctrl.dram.avgQLat 5278.45 # Average queueing delay per DRAM burst ((Tick/Count))
board.memory.mem_ctrl.dram.avgBusLat 5000.00 # Average bus latency per DRAM burst ((Tick/Count))
```

```
board.memory.mem_ctrl.dram.avgMemAccLat      24028.45                    # Average memory access latency per DRAM burst ((Tick/Count))
board.memory.mem_ctrl.dram.readRowHits           323                     # Number of row buffer hits during reads (Count)
board.memory.mem_ctrl.dram.writeRowHits            0                     # Number of row buffer hits during writes (Count)
board.memory.mem_ctrl.dram.readRowHitRate        82.61                   # Row buffer hit rate for reads (Ratio)
board.memory.mem_ctrl.dram.writeRowHitRate         nan                   # Row buffer hit rate for writes (Ratio)
board.memory.mem_ctrl.dram.bytesPerActivate::samples          68         # Bytes accessed per row activation (Byte)
board.memory.mem_ctrl.dram.bytesPerActivate::mean          368           # Bytes accessed per row activation (Byte)
board.memory.mem_ctrl.dram.bytesPerActivate::gmean   229.264367          # Bytes accessed per row activation (Byte)
board.memory.mem_ctrl.dram.bytesPerActivate::stdev   338.362695          # Bytes accessed per row activation (Byte)
board.memory.mem_ctrl.dram.bytesPerActivate::0-127           18     26.47%     26.47% # Bytes accessed per row activation (Byte)
board.memory.mem_ctrl.dram.bytesPerActivate::128-255          16     23.53%     50.00% # Bytes accessed per row activation (Byte)
board.memory.mem_ctrl.dram.bytesPerActivate::256-383           6      8.82%     58.82% # Bytes accessed per row activation (Byte)
board.memory.mem_ctrl.dram.bytesPerActivate::384-511           7     10.29%     69.12% # Bytes accessed per row activation (Byte)
board.memory.mem_ctrl.dram.bytesPerActivate::512-639           6      8.82%     77.94% # Bytes accessed per row activation (Byte)
board.memory.mem_ctrl.dram.bytesPerActivate::640-767           2      2.94%     80.88% # Bytes accessed per row activation (Byte)
board.memory.mem_ctrl.dram.bytesPerActivate::768-895           3      4.41%     85.29% # Bytes accessed per row activation (Byte)
board.memory.mem_ctrl.dram.bytesPerActivate::896-1023          2      2.94%     88.24% # Bytes accessed per row activation (Byte)
board.memory.mem_ctrl.dram.bytesPerActivate::1024-1151          8     11.76%    100.00% # Bytes accessed per row activation (Byte)
board.memory.mem_ctrl.dram.bytesPerActivate::total          68         # Bytes accessed per row activation (Byte)
board.memory.mem_ctrl.dram.dramBytesRead        25024                   # Total bytes read (Byte)
board.memory.mem_ctrl.dram.dramBytesWritten            0                # Total bytes written (Byte)
board.memory.mem_ctrl.dram.avgRdBW           0.000000                  # Average DRAM read bandwidth in MiBytes/s ((Byte/Second))
board.memory.mem_ctrl.dram.avgWrBW                   0                 # Average DRAM write bandwidth in MiBytes/s ((Byte/Second))
board.memory.mem_ctrl.dram.peakBW           12800.00                    # Theoretical peak bandwidth in MiByte/s ((Byte/Second))
board.memory.mem_ctrl.dram.busUtil               0.00                   # Data bus utilization in percentage (Ratio)
board.memory.mem_ctrl.dram.busUtilRead           0.00                   # Data bus utilization in percentage for reads (Ratio)
board.memory.mem_ctrl.dram.busUtilWrite          0.00                   # Data bus utilization in percentage for writes (Ratio)
board.memory.mem_ctrl.dram.pageHitRate          82.61                   # Row buffer hit rate, read and write combined (Ratio)
board.memory.mem_ctrl.dram.power_state.pwrStateResidencyTicks::UNDEFINED 84926990560998752          # Cumulative time (in ticks) in various power states (Tick)
board.memory.mem_ctrl.dram.rank0.actEnergy       342720                 # Energy for activate commands per rank (pJ) (Joule)
board.memory.mem_ctrl.dram.rank0.preEnergy       182160                 # Energy for precharge commands per rank (pJ) (Joule)
board.memory.mem_ctrl.dram.rank0.readEnergy      2170560                # Energy for read commands per rank (pJ) (Joule)
board.memory.mem_ctrl.dram.rank0.writeEnergy           0                # Energy for write commands per rank (pJ) (Joule)
board.memory.mem_ctrl.dram.rank0.refreshEnergy 6704064919365841        # Energy for refresh commands per rank (pJ) (Joule)
board.memory.mem_ctrl.dram.rank0.actBackEnergy 1224781104862200        # Energy for active background per rank (pJ) (Joule)
board.memory.mem_ctrl.dram.rank0.preBackEnergy 31580569760897280       # Energy for precharge background per rank (pJ) (Joule)
board.memory.mem_ctrl.dram.rank0.actPowerDownEnergy           0        # Energy for active power-down per rank (pJ) (Joule)
board.memory.mem_ctrl.dram.rank0.prePowerDownEnergy           0        # Energy for precharge power-down per rank (pJ) (Joule)
board.memory.mem_ctrl.dram.rank0.selfRefreshEnergy           0         # Energy for self refresh per rank (pJ) (Joule)
board.memory.mem_ctrl.dram.rank0.totalEnergy 39509403919628384         # Total energy per rank (pJ) (Joule)
board.memory.mem_ctrl.dram.rank0.averagePower   465.216107              # Core power per rank (mW) (Watt)
board.memory.mem_ctrl.dram.rank0.totalIdleTime           0              # Total Idle time Per DRAM Rank (Tick)
board.memory.mem_ctrl.dram.rank0.pwrStateTime::IDLE 82091085302359936  # Time in different power states (Tick)
board.memory.mem_ctrl.dram.rank0.pwrStateTime::REF 2835898865800000    # Time in different power states (Tick)
board.memory.mem_ctrl.dram.rank0.pwrStateTime::SREF           0        # Time in different power states (Tick)
board.memory.mem_ctrl.dram.rank0.pwrStateTime::PRE_PDN           0     # Time in different power states (Tick)
board.memory.mem_ctrl.dram.rank0.pwrStateTime::ACT     30371424        # Time in different power states (Tick)
board.memory.mem_ctrl.dram.rank0.pwrStateTime::ACT_PDN           0     # Time in different power states (Tick)
board.memory.mem_ctrl.dram.rank1.actEnergy       142800                 # Energy for activate commands per rank (pJ) (Joule)
```

board.memory.mem_ctrl.dram.rank1.preEnergy 75900 # Energy for precharge commands per rank (pJ) (Joule)

board.memory.mem_ctrl.dram.rank1.readEnergy 621180 # Energy for read commands per rank (pJ) (Joule)

board.memory.mem_ctrl.dram.rank1.writeEnergy 0 # Energy for write commands per rank (pJ) (Joule)

board.memory.mem_ctrl.dram.rank1.refreshEnergy 6704064919365841 # Energy for refresh commands per rank (pJ) (Joule)

board.memory.mem_ctrl.dram.rank1.actBackEnergy 1224781104532170 # Energy for active background per rank (pJ) (Joule)

board.memory.mem_ctrl.dram.rank1.preBackEnergy 31580569761175200 # Energy for precharge background per rank (pJ) (Joule)

board.memory.mem_ctrl.dram.rank1.actPowerDownEnergy 0 # Energy for active power-down per rank (pJ) (Joule)

board.memory.mem_ctrl.dram.rank1.prePowerDownEnergy 0 # Energy for precharge power-down per rank (pJ) (Joule)

board.memory.mem_ctrl.dram.rank1.selfRefreshEnergy 0 # Energy for self refresh per rank (pJ) (Joule)

board.memory.mem_ctrl.dram.rank1.totalEnergy 39509403917720720 # Total energy per rank (pJ) (Joule)

board.memory.mem_ctrl.dram.rank1.averagePower 465.216107 # Core power per rank (mW) (Watt)

board.memory.mem_ctrl.dram.rank1.totalIdleTime 0 # Total Idle time Per DRAM Rank (Tick)

board.memory.mem_ctrl.dram.rank1.pwrStateTime::IDLE 82091085303067888 # Time in different power states (Tick)

board.memory.mem_ctrl.dram.rank1.pwrStateTime::REF 2835898865800000 # Time in different power states (Tick)

board.memory.mem_ctrl.dram.rank1.pwrStateTime::SREF 0 # Time in different power states (Tick)

board.memory.mem_ctrl.dram.rank1.pwrStateTime::PRE_PDN 0 # Time in different power states (Tick)

board.memory.mem_ctrl.dram.rank1.pwrStateTime::ACT 29663469 # Time in different power states (Tick)

board.memory.mem_ctrl.dram.rank1.pwrStateTime::ACT_PDN 0 # Time in different power states (Tick)

board.memory.mem_ctrl.power_state.pwrStateResidencyTicks::UNDEFINED 84926990560998752 # Cumulative time (in ticks) in various power states (Tick)

board.processor.cores.core.numCycles 72487 # Number of cpu cycles simulated (Cycle)

board.processor.cores.core.cpi 17.466747 # CPI: cycles per instruction (core level) ((Cycle/Count))

board.processor.cores.core.ipc 0.057252 # IPC: instructions per cycle (core level) ((Count/Cycle))

board.processor.cores.core.numWorkItemsStarted 0 # Number of work items this cpu started (Count)

board.processor.cores.core.numWorkItemsCompleted 0 # Number of work items this cpu completed (Count)

board.processor.cores.core.commitStats0.numInsts 4150 # Number of instructions committed (thread level) (Count)

board.processor.cores.core.commitStats0.numOps 7878 # Number of ops (including micro ops) committed (thread level) (Count)

board.processor.cores.core.commitStats0.numInstsNotNOP 0 # Number of instructions committed excluding NOPs or prefetches (Count)

board.processor.cores.core.commitStats0.numOpsNotNOP 0 # Number of Ops (including micro ops) Simulated (Count)

board.processor.cores.core.commitStats0.cpi 17.466747 # CPI: cycles per instruction (thread level) ((Cycle/Count))

board.processor.cores.core.commitStats0.ipc 0.057252 # IPC: instructions per cycle (thread level) ((Count/Cycle))

board.processor.cores.core.commitStats0.numMemRefs 0 # Number of memory references committed (Count)

board.processor.cores.core.commitStats0.numFpInsts 703 # Number of float instructions (Count)

board.processor.cores.core.commitStats0.numIntInsts 7438 # Number of integer instructions (Count)

board.processor.cores.core.commitStats0.numLoadInsts 642 # Number of load instructions (Count)

board.processor.cores.core.commitStats0.numStoreInsts 846 # Number of store instructions (Count)

board.processor.cores.core.commitStats0.numVecInsts 0 # Number of vector instructions (Count)

board.processor.cores.core.commitStats0.committedInstType::No_OpClass 90 1.14% 1.14% # Class of committed instruction. (Count)

board.processor.cores.core.commitStats0.committedInstType::IntAlu 5994 76.09% 77.23% # Class of committed instruction. (Count)

board.processor.cores.core.commitStats0.committedInstType::IntMult 1 0.01% 77.24% # Class of committed instruction. (Count)

board.processor.cores.core.commitStats0.committedInstType::IntDiv 21 0.27% 77.51% # Class of committed instruction. (Count)

board.processor.cores.core.commitStats0.committedInstType::FloatAdd 136 1.73% 79.23% # Class of committed instruction. (Count)

board.processor.cores.core.commitStats0.committedInstType::FloatCmp 0 0.00% 79.23% # Class of committed instruction. (Count)

board.processor.cores.core.commitStats0.committedInstType::FloatCvt 0 0.00% 79.23% # Class of committed instruction. (Count)

board.processor.cores.core.commitStats0.committedInstType::FloatMult 0 0.00% 79.23% # Class of committed instruction. (Count)

board.processor.cores.core.commitStats0.committedInstType::FloatMultAcc 0 0.00% 79.23% # Class of committed instruction. (Count)

board.processor.cores.core.commitStats0.committedInstType::FloatDiv 0 0.00% 79.23% # Class of committed instruction. (Count)

board.processor.cores.core.commitStats0.committedInstType::FloatMisc 0 0.00% 79.23% # Class of committed instruction. (Count)

board.processor.cores.core.commitStats0.committedInstType::FloatSqrt 0 0.00% 79.23% # Class of committed instruction. (Count)

board.processor.cores.core.commitStats0.committedInstType::SimdAdd 0 0.00% 79.23% # Class of committed instruction. (Count)

board.processor.cores.core.commitStats0.committedInstType::SimdAddAcc 0 0.00% 79.23% # Class of committed instruction. (Count)

board.processor.cores.core.commitStats0.committedInstType::SimdAlu 10 0.13% 79.36% # Class of committed instruction. (Count)

board.processor.cores.core.commitStats0.committedInstType::SimdCmp 0 0.00% 79.36% # Class of committed instruction. (Count)

board.processor.cores.core.commitStats0.committedInstType::SimdCvt 2 0.03% 79.39% # Class of committed instruction. (Count)

board.processor.cores.core.commitStats0.committedInstType::SimdMisc 136 1.73% 81.11% # Class of committed instruction. (Count)

board.processor.cores.core.commitStats0.committedInstType::SimdMult 0 0.00% 81.11% # Class of committed instruction. (Count)

board.processor.cores.core.commitStats0.committedInstType::SimdMultAcc 0 0.00% 81.11% # Class of committed instruction. (Count)

board.processor.cores.core.commitStats0.committedInstType::SimdMatMultAcc 0 0.00% 81.11% # Class of committed instruction. (Count)

board.processor.cores.core.commitStats0.committedInstType::SimdShift 0 0.00% 81.11% # Class of committed instruction. (Count)

board.processor.cores.core.commitStats0.committedInstType::SimdShiftAcc 0 0.00% 81.11% # Class of committed instruction. (Count)

board.processor.cores.core.commitStats0.committedInstType::SimdDiv 0 0.00% 81.11% # Class of committed instruction. (Count)

board.processor.cores.core.commitStats0.committedInstType::SimdSqrt 0 0.00% 81.11% # Class of committed instruction. (Count)

board.processor.cores.core.commitStats0.committedInstType::SimdFloatAdd 0 0.00% 81.11% # Class of committed instruction. (Count)

board.processor.cores.core.commitStats0.committedInstType::SimdFloatAlu 0 0.00% 81.11% # Class of committed instruction. (Count)

board.processor.cores.core.commitStats0.committedInstType::SimdFloatCmp 0 0.00% 81.11% # Class of committed instruction. (Count)

board.processor.cores.core.commitStats0.committedInstType::SimdFloatCvt 0 0.00% 81.11% # Class of committed instruction. (Count)

board.processor.cores.core.commitStats0.committedInstType::SimdFloatDiv 0 0.00% 81.11% # Class of committed instruction. (Count)

board.processor.cores.core.commitStats0.committedInstType::SimdFloatMisc 0 0.00% 81.11% # Class of committed instruction. (Count)

board.processor.cores.core.commitStats0.committedInstType::SimdFloatMult 0 0.00% 81.11% # Class of committed instruction. (Count)

board.processor.cores.core.commitStats0.committedInstType::SimdFloatMultAcc 0 0.00% 81.11% # Class of committed instruction. (Count)

board.processor.cores.core.commitStats0.committedInstType::SimdFloatMatMultAcc 0 0.00% 81.11% # Class of committed instruction. (Count)

board.processor.cores.core.commitStats0.committedInstType::SimdFloatSqrt 0 0.00% 81.11% # Class of committed instruction. (Count)

board.processor.cores.core.commitStats0.committedInstType::SimdReduceAdd 0 0.00% 81.11% # Class of committed instruction. (Count)

board.processor.cores.core.commitStats0.committedInstType::SimdReduceAlu 0 0.00% 81.11% # Class of committed instruction. (Count)

board.processor.cores.core.commitStats0.committedInstType::SimdReduceCmp 0 0.00% 81.11% # Class of committed instruction. (Count)

board.processor.cores.core.commitStats0.committedInstType::SimdFloatReduceAdd 0 0.00% 81.11% # Class of committed instruction. (Count)

board.processor.cores.core.commitStats0.committedInstType::SimdFloatReduceCmp 0 0.00% 81.11% # Class of committed instruction. (Count)

board.processor.cores.core.commitStats0.committedInstType::SimdAes 0 0.00% 81.11% # Class of committed instruction. (Count)

board.processor.cores.core.commitStats0.committedInstType::SimdAesMix 0 0.00% 81.11% # Class of committed instruction. (Count)

board.processor.cores.core.commitStats0.committedInstType::SimdSha1Hash 0 0.00% 81.11% # Class of committed instruction. (Count)

board.processor.cores.core.commitStats0.committedInstType::SimdSha1Hash2 0 0.00% 81.11% # Class of committed instruction. (Count)

board.processor.cores.core.commitStats0.committedInstType::SimdSha256Hash 0 0.00% 81.11% # Class of committed instruction. (Count)

board.processor.cores.core.commitStats0.committedInstType::SimdSha256Hash2 0 0.00% 81.11% # Class of committed instruction. (Count)

board.processor.cores.core.commitStats0.committedInstType::SimdShaSigma2 0 0.00% 81.11% # Class of committed instruction. (Count)

board.processor.cores.core.commitStats0.committedInstType::SimdShaSigma3 0 0.00% 81.11% # Class of committed instruction. (Count)

board.processor.cores.core.commitStats0.committedInstType::SimdPredAlu 0 0.00% 81.11% # Class of committed instruction. (Count)

board.processor.cores.core.commitStats0.committedInstType::Matrix 0 0.00% 81.11% # Class of committed instruction. (Count)

board.processor.cores.core.commitStats0.committedInstType::MatrixMov 0 0.00% 81.11% # Class of committed instruction. (Count)

board.processor.cores.core.commitStats0.committedInstType::MatrixOP 0 0.00% 81.11% # Class of committed instruction. (Count)

board.processor.cores.core.commitStats0.committedInstType::MemRead 629 7.98% 89.10% # Class of committed instruction. (Count)

board.processor.cores.core.commitStats0.committedInstType::MemWrite 573 7.27% 96.37% # Class of committed instruction. (Count)

board.processor.cores.core.commitStats0.committedInstType::FloatMemRead 13 0.17% 96.53% # Class of committed instruction. (Count)

board.processor.cores.core.commitStats0.committedInstType::FloatMemWrite 273 3.47% 100.00% # Class of committed instruction. (Count)

board.processor.cores.core.commitStats0.committedInstType::IprAccess 0 0.00% 100.00% # Class of committed instruction. (Count)

board.processor.cores.core.commitStats0.committedInstType::InstPrefetch 0 0.00% 100.00% # Class of committed instruction. (Count)

board.processor.cores.core.commitStats0.committedInstType::SimdUnitStrideLoad 0 0.00% 100.00% # Class of committed instruction. (Count)

board.processor.cores.core.commitStats0.committedInstType::SimdUnitStrideStore 0 0.00% 100.00% # Class of committed instruction. (Count)

board.processor.cores.core.commitStats0.committedInstType::SimdUnitStrideMaskLoad 0 0.00% 100.00% # Class of committed instruction. (Count)

board.processor.cores.core.commitStats0.committedInstType::SimdUnitStrideMaskStore 0 0.00% 100.00% # Class of committed instruction. (Count)

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board.processor.cores.core.commitStats0.committedInstType::SimdStridedLoad 0 0.00% 100.00% # Class of committed instruction. (Count)
board.processor.cores.core.commitStats0.committedInstType::SimdStridedStore 0 0.00% 100.00% # Class of committed instruction. (Count)
board.processor.cores.core.commitStats0.committedInstType::SimdIndexedLoad 0 0.00% 100.00% # Class of committed instruction. (Count)
board.processor.cores.core.commitStats0.committedInstType::SimdIndexedStore 0 0.00% 100.00% # Class of committed instruction. (Count)
board.processor.cores.core.commitStats0.committedInstType::SimdWholeRegisterLoad 0 0.00% 100.00% # Class of committed instruction. (Count)
board.processor.cores.core.commitStats0.committedInstType::SimdWholeRegisterStore 0 0.00% 100.00% # Class of committed instruction. (Count)
board.processor.cores.core.commitStats0.committedInstType::SimdUnitStrideFaultOnlyFirstLoad 0 0.00% 100.00% # Class of committed instruction. (Count)
board.processor.cores.core.commitStats0.committedInstType::SimdUnitStrideSegmentedLoad 0 0.00% 100.00% # Class of committed instruction. (Count)
board.processor.cores.core.commitStats0.committedInstType::SimdUnitStrideSegmentedStore 0 0.00% 100.00% # Class of committed instruction. (Count)
board.processor.cores.core.commitStats0.committedInstType::SimdExt 0 0.00% 100.00% # Class of committed instruction. (Count)
board.processor.cores.core.commitStats0.committedInstType::SimdFloatExt 0 0.00% 100.00% # Class of committed instruction. (Count)
board.processor.cores.core.commitStats0.committedInstType::SimdConfig 0 0.00% 100.00% # Class of committed instruction. (Count)
board.processor.cores.core.commitStats0.committedInstType::total 7878 # Class of committed instruction. (Count)
board.processor.cores.core.commitStats0.committedControl::IsControl 836 # Class of control type instructions committed (Count)
board.processor.cores.core.commitStats0.committedControl::IsDirectControl 710 # Class of control type instructions committed (Count)
board.processor.cores.core.commitStats0.committedControl::IsIndirectControl 126 # Class of control type instructions committed (Count)
board.processor.cores.core.commitStats0.committedControl::IsCondControl 596 # Class of control type instructions committed (Count)
board.processor.cores.core.commitStats0.committedControl::IsUncondControl 240 # Class of control type instructions committed (Count)
board.processor.cores.core.commitStats0.committedControl::IsCall 97 # Class of control type instructions committed (Count)
board.processor.cores.core.commitStats0.committedControl::IsReturn 91 # Class of control type instructions committed (Count)
board.processor.cores.core.exec_context.thread_0.numMatAluAccesses 0 # Number of matrix alu accesses (Count)
board.processor.cores.core.exec_context.thread_0.numCallsReturns 188 # Number of times a function call or return occured (Count)
board.processor.cores.core.exec_context.thread_0.numMatInsts 0 # Number of matrix instructions (Count)
board.processor.cores.core.exec_context.thread_0.numIdleCycles 0 # Number of idle cycles (Cycle)
board.processor.cores.core.exec_context.thread_0.numBusyCycles 72487 # Number of busy cycles (Cycle)
board.processor.cores.core.exec_context.thread_0.notIdleFraction 1 # Percentage of non-idle cycles (Ratio)
board.processor.cores.core.exec_context.thread_0.idleFraction 0 # Percentage of idle cycles (Ratio)
board.processor.cores.core.executeStats0.numInsts 0 # Number of executed instructions (Count)
board.processor.cores.core.executeStats0.numNop 0 # Number of nop insts executed (Count)
board.processor.cores.core.executeStats0.numBranches 0 # Number of branches executed (Count)
board.processor.cores.core.executeStats0.numLoadInsts 0 # Number of load instructions executed (Count)
board.processor.cores.core.executeStats0.numStoreInsts 1488 # Number of stores executed (Count)
board.processor.cores.core.executeStats0.instRate 0 # Inst execution rate ((Count/Cycle))
board.processor.cores.core.executeStats0.numFpAluAccesses 703 # Number of float alu accesses (Count)
board.processor.cores.core.executeStats0.numFpRegReads 703 # Number of times the floating registers were read (Count)
board.processor.cores.core.executeStats0.numFpRegWrites 428 # Number of times the floating registers were written (Count)
board.processor.cores.core.executeStats0.numIntAluAccesses 7438 # Number of integer alu accesses (Count)
board.processor.cores.core.executeStats0.numIntRegReads 8403 # Number of times the integer registers were read (Count)
board.processor.cores.core.executeStats0.numIntRegWrites 5237 # Number of times the integer registers were written (Count)
board.processor.cores.core.executeStats0.numMemRefs 1488 # Number of memory refs (Count)
board.processor.cores.core.executeStats0.numMiscRegReads 3307 # Number of times the Misc registers were read (Count)
board.processor.cores.core.executeStats0.numVecAluAccesses 0 # Number of vector alu accesses (Count)
board.processor.cores.core.executeStats0.numDiscardedOps 0 # Number of ops (including micro ops) which were discarded before commit (Count)
board.processor.cores.core.fetchStats0.numInsts 4150 # Number of instructions fetched (thread level) (Count)
board.processor.cores.core.fetchStats0.numOps 7879 # Number of ops (including micro ops) fetched (thread level) (Count)
board.processor.cores.core.fetchStats0.fetchRate 0.057252 # Number of inst fetches per cycle ((Count/Cycle))
board.processor.cores.core.fetchStats0.numBranches 836 # Number of branches fetched (Count)
board.processor.cores.core.fetchStats0.branchRate 0.011533 # Number of branch fetches per cycle (Ratio)
board.processor.cores.core.fetchStats0.numFetchSuspends 0 # Number of times Execute suspended instruction fetching (Count)
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board.processor.cores.core.interrupts.clk_domain.clock 5328 # Clock period in ticks (Tick)
board.processor.cores.core.mmu.dtb.rdAccesses 642 # TLB accesses on read requests (Count)
board.processor.cores.core.mmu.dtb.wrAccesses 847 # TLB accesses on write requests (Count)
board.processor.cores.core.mmu.dtb.rdMisses 12 # TLB misses on read requests (Count)
board.processor.cores.core.mmu.dtb.wrMisses 6 # TLB misses on write requests (Count)
board.processor.cores.core.mmu.dtb.walker.power_state.pwrStateResidencyTicks::UNDEFINED 84926990560998752 # Cumulative time (in ticks) in various power states (Tick)
board.processor.cores.core.mmu.itb.rdAccesses 0 # TLB accesses on read requests (Count)
board.processor.cores.core.mmu.itb.wrAccesses 5345 # TLB accesses on write requests (Count)
board.processor.cores.core.mmu.itb.rdMisses 0 # TLB misses on read requests (Count)
board.processor.cores.core.mmu.itb.wrMisses 22 # TLB misses on write requests (Count)
board.processor.cores.core.mmu.itb.walker.power_state.pwrStateResidencyTicks::UNDEFINED 84926990560998752 # Cumulative time (in ticks) in various power states (Tick)
board.processor.cores.core.power_state.pwrStateResidencyTicks::ON 84926990560998752 # Cumulative time (in ticks) in various power states (Tick)
board.processor.cores.core.thread_0.numInsts 0 # Number of Instructions committed (Count)
board.processor.cores.core.thread_0.numOps 0 # Number of Ops committed (Count)
board.processor.cores.core.thread_0.numMemRefs 0 # Number of Memory References (Count)
board.processor.cores.core.workload.numSyscalls 9 # Number of system calls (Count)
board.workload.inst.arm 0 # number of arm instructions executed (Count)
board.workload.inst.quiesce 0 # number of quiesce instructions executed (Count)

---------- End Simulation Statistics ----------
```